Identification, Assumptions, and Fragility

A Rigorous, Self-Study Textbook in Econometrics, Stated-Preference Valuation, Welfare Analysis, and Cost–Benefit Appraisal

Prepared for Mattia Galante

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About this volume

This is a complete, self-contained textbook of **52 chapters across 14 parts**, built around a single intellectual spine — the question posed in every chapter's epigraph:

> *Given this experimental design and this model, what economic object am I identifying, under which assumptions, and how fragile is the result with respect to specification, distributions, incentives, survey design, sampling, computation, and interpretation?*

It runs from the mathematical foundations of econometrics (Part I) through probability and inference, maximum likelihood, core econometrics, random-utility and discrete-choice theory, contingent valuation, identification and treatment effects, welfare foundations, stated-preference practice, multiple-discreteness (MDCEV), cost–benefit analysis, evidence synthesis and benefit transfer, to the synthesizing *fragility audit* (Ch 50) and the practical disciplines of survey design (Ch 51) and computational diagnostics (Ch 52). Every worked numerical example in the volume was verified computationally, and independently re-verified in the editorial pass recorded in Appendix B.

Each chapter is woven throughout with the reader's own research program: the SBDC/payment-card contingent valuation with real/hypothetical and BRIS treatments; the error-components mixed logit in WTP space; the MDCEV allocation of ecosystem services; the meta-analysis of 18 finfish-WTP discrete choice experiments; and the cost–benefit appraisal of fish-restoration devices linked to blue finance.

Every chapter follows the same 17-device pedagogical structure: conceptual motivation, prerequisites, core exposition with full derivations, worked examples, applied intuition, failure modes, diagnostic questions, active-recall and Socratic checks, three-tier exercises with detailed solutions, spaced-review prompts, cross-chapter connections, a research-translation task, and a three-tier bibliography (with unverified citations flagged "verify").

A note on rendering. This master file carries YAML front matter for the reader's own RMarkdown/pandoc pipeline; each chapter's individual YAML block has been removed during assembly. Internal cross-references cite chapters by number and have been audited against the final chapter order in the editorial pass recorded in Appendix B.

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Volume 0 — Book Architecture and Learning Philosophy

A Rigorous, Self-Study Textbook in Econometrics, Stated-Preference Valuation, Welfare Analysis, and Cost–Benefit Appraisal

Book architecture and study guide — Volume 0

Prepared for Mattia Galante, built directly from `piano_studio_argomenti.pdf`.

How to read this document

This is not the textbook. It is the **blueprint** for the textbook: the title, the preface written to you, the learning philosophy, the complete table of contents, the learning objectives for all 52 chapters, a dependency map, a study schedule keyed to the six blocks in your plan, a description of the pedagogical machinery each chapter will run, the cross-disciplinary connections that hold the book together, and the catalogue of high-risk mistakes the book is designed to inoculate you against. At the very end I ask you which chapter to write first.

Everything below is in English, as requested, and everything is traceable to your PDF. Where I reorganized your plan, I say so and explain why the reorganization serves learning.

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1. Proposed title

Primary title: *Identification, Assumptions, and Fragility*

Subtitle: *A Rigorous, Self-Study Textbook in Econometrics, Stated-Preference Valuation, Welfare Analysis, and Cost–Benefit Appraisal*

Why this title. Your PDF ends with a single sentence that it calls the "sintesi estrema" — the extreme synthesis — of your entire training:

> *Given this experimental design and this model, what economic object am I identifying, under which assumptions, and how fragile is the result with respect to specification, distributions, incentives, survey design, and sampling?*

That sentence is the intellectual spine of the book, so the title names its three load-bearing beams. **Identification** is the question of *what economic object* the data can pin down. **Assumptions** is the question of *what you had to believe* to get there. **Fragility** is the question of *how much the answer moves* when specification, distributions, incentives, design, sampling, or computation move. Every chapter, no matter how mathematical, is ultimately in service of letting you answer those three questions about your own papers.

An alternative working title, if you prefer something that foregrounds the field: *What Am I Identifying? Econometrics and Welfare Analysis for Stated Preference Research.* I will use the primary title throughout unless you tell me otherwise.

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2. Preface (written to you, the learner)

Mattia,

You did something unusual when you wrote your study plan: instead of listing topics you "should know," you listed the *judgments* you cannot yet make with confidence. Read your PDF again and you will notice that almost every section has two halves — a *devi studiare* half (what to study) and a *da sapere giudicare criticamente* half (what to be able to judge critically). The second half is the real curriculum. You do not want to learn maximum likelihood so that you can run `maxLik`; you already run it. You want to learn it so that when the optimizer reports convergence you can tell whether the convergence is real or cosmetic, whether the Hessian is trustworthy, whether the maximum is global, and whether your parameters are separately identified. That is a different and much harder skill, and it is the skill this book teaches.

This book is written for exactly your situation: a doctoral researcher with genuine research momentum — a contingent valuation paper built on single-bounded dichotomous choice and payment cards with real and hypothetical treatments and a BRIS mechanism; a discrete choice experiment estimated as an error-components mixed logit in WTP space with a cost-vector manipulation and a familiarity dimension; a meta-analysis of finfish valuations drawn from eighteen discrete choice experiments; an MDCEV proposal for allocating ecosystem services; and a cost-benefit analysis of fish restoration devices linked to blue finance — but with foundations that are, in your own honest phrase, *uneven*. You have used every method in your plan. What you are missing is not exposure; it is the connective tissue that lets you *defend* each method against the specific ways it can mislead you.

So I have written the book around your projects, not around a generic syllabus. When Part III derives the single-bounded dichotomous choice likelihood, it derives *your* likelihood — latent normally-distributed willingness to pay, censored by a yes/no response, probit-like — and then asks you whether normality is defensible when it puts positive mass on negative WTP for a good you believe has non-negative value. When Part V builds the mixed logit, it builds toward *your* EC-MXL in WTP space and confronts the exact instability you flagged: WTP as a ratio of coefficients, with a cost coefficient that may drift toward zero and drag a fat tail behind it. When Part VII teaches identification, the running exercise is *your* worry — that the incentivized cost-vector effect identifies an average object tied to preference formation, while the hypothetical-versus-incentivized contrast cannot separate anchoring from hypothetical bias without an untestable restriction. The book keeps returning to your question about whether your design can distinguish κ from λ, because that is precisely the kind of question a committed empiricist must answer and a careless one never asks.

A word on tone and pace. You asked me not to oversimplify, and I will not. But not oversimplifying is not the same as being terse. When a derivation is central — OLS and its variance, the score and Hessian of a likelihood, the probit and logit likelihoods, the SBDC and interval-censored likelihoods, WTP as a ratio of marginal utilities and why its tails misbehave, the compensating variation, the Kuhn-Tucker conditions behind MDCEV, the simulated likelihood of the mixed logit, the role of the Hessian in standard errors, clustered inference, robust variance estimation in meta-analysis — I will show every step, and I will show it patiently, because a skipped step in a central derivation is exactly where uneven foundations hide. Where a topic is genuinely peripheral to your work (measure theory, advanced optimal control, deep learning), the book says so and moves on, honoring the priority ordering you set out in sections 7 and 8 of your plan.

Finally, a promise about intellectual honesty. This book will teach you methods that are useful and dangerous at the same time — WTP-space estimation, weighted likelihoods, post-randomization exclusions, benefit transfer, meta-regression over dependent estimates — and for each it will explain both the use and the danger, because a method's danger is usually invisible in the software output. It will emphasize the correct

interpretation of results that are routinely misinterpreted. It will state assumptions explicitly wherever a result depends on them. And when I am unsure about a reference or a fact, I will say so and ask you to verify, rather than inventing a citation. The goal is not for you to trust the book. The goal is for you to become the kind of researcher who trusts nothing — including your own beautiful models — until you can say what they identify, under what assumptions, and how fragile the answer is.

Let's build the machinery to do that.

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3. Learning philosophy

The book rests on six commitments. Each is a design decision, and each shapes how every chapter is written.

3.1 The spine question governs everything. Every chapter, including the purely mathematical ones, closes the loop back to the spine question — *what object, which assumptions, how fragile.* Linear algebra is taught not as abstract structure but as the language in which identification failures, ill-conditioning, and unstable standard errors first become visible. This keeps the mathematics from feeling like a detour.

3.2 Identification is prior to estimation. A recurring, deliberately repetitive discipline runs through the book: before we estimate anything, we ask what the data can and cannot pin down. Estimation is what software does; identification is what the design and the assumptions allow. The book trains the reflex of separating the two, because the most expensive errors in stated-preference research are estimation successes attached to identification failures — a model that converges beautifully to the wrong object.

3.3 Assumptions are made visible, always. Results are never stated free-floating. Each proposition carries its assumptions on its face, and each chapter includes a "what breaks if this assumption fails" analysis. This is the habit that lets you read your own results as *conditional* claims rather than facts.

3.4 Fragility is a first-class object of study, not an afterthought. Robustness, sensitivity, distributional dependence, simulation noise, seed dependence, and computational pathology are treated as core content, not as an appendix labelled "robustness checks." You will learn to *design* fragility diagnostics, not just run them.

3.5 Evidence-informed pedagogy is built into the structure. The book uses the learning techniques with the strongest empirical support for durable, transferable understanding: **retrieval practice** (active recall prompts embedded mid-chapter, not just at the end), **spaced repetition** (explicit 1-day / 1-week / 1-month review prompts), **interleaving** (deliberate connections backward and forward, so linear algebra is revisited when we hit Hessians and covariance matrices, and constrained optimization is revisited at MDCEV and compensating variation), **worked-example-to-independent-problem fading** (fully worked examples first, then scaffolded exercises, then open critique), **elaborative interrogation and self-explanation** ("Stop and think" Socratic checks where you predict an implication before reading it), and **concreteness fading** (starting from your own concrete research cases and abstracting upward, rather than the reverse). These are not decorations; they are the reason the material will still be usable months later when a referee asks a hard question.

3.6 The book is honest about danger and about its own limits. Useful-but-dangerous methods are taught with both faces shown. Competing conventions in the literature are named as conventions, not presented as the truth. And references are recommended in three tiers (core / applied / advanced) with a one-line justification each, never invented; where I am unsure of a citation I flag it for your verification.

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4. Complete table of contents

The book is organized into fourteen parts and fifty-two chapters, following the fourteen-part structure you specified, with the internal chapter breakdown designed for a clean dependency chain. Assessment blocks

close each part.

PART I — Mathematical Foundations for Econometrics and Welfare Analysis

- **Ch 1.** Applied Linear Algebra for Econometrics — vectors, matrices, the matrix product as a data operation, rank, inverse, determinant, quadratic forms, positive (semi)definite matrices, the variance–covariance matrix and its decomposition, correlation among estimates, matrix conditioning, the Hessian and the information matrix.
- **Ch 2.** Multivariate Calculus for Likelihood and Welfare — partial derivatives, the gradient, the Hessian, Taylor expansion, convexity and concavity, local versus global optima, the implicit function theorem, marginal effects and elasticities.
- **Ch 3.** Constrained Optimization and Duality — the Lagrangian, equality and inequality constraints, Kuhn–Tucker conditions, interior versus corner solutions, duality, utility maximization, expenditure minimization, indirect utility. (The mathematical engine of MDCEV and of compensating variation.)
- **Ch 4.** The Mathematics of Discounting — present value, net present value, IRR, private versus social discount rates, real versus nominal flows, sensitivity to the rate. (The engine of CBA.)
- *Part I assessment: cumulative problem set, research critique, red-flag diagnosis, mock oral, integration mini-project.*

PART II — Probability and Statistical Inference

- **Ch 5.** Random Variables and the Distributions of Stated Preference — discrete and continuous random variables; the normal, logistic, and Type-I extreme value (Gumbel) distributions and why each one shows up where it does; conditional distributions, conditional expectation, conditional variance.
- **Ch 6.** Asymptotic Theory You Actually Use — laws of large numbers, the central limit theorem, convergence in probability and in distribution, and what "asymptotic" licenses and forbids.
- **Ch 7.** Estimators and Inference — bias, consistency, efficiency, variance, standard errors, confidence intervals, p-values, one- and two-tailed tests, and the delta method.
- **Ch 8.** Power, Minimum Detectable Effects, and Equivalence — statistical power, minimum detectable effect, multiple testing, and the central discipline that *"not significant" does not mean "equal"*: equivalence testing and TOST, confidence intervals around differences, minimum relevant effect size.
- **Ch 9.** Simulation-Based Inference — why some integrals have no closed form, Monte Carlo and quasi-Monte Carlo integration, Halton draws, the draws–precision–time trade-off, the bootstrap, the parametric bootstrap, Krinsky–Robb, and uncertainty propagation.
- **Ch 10.** Nonparametric and Randomization Inference — Mann–Whitney, Kolmogorov–Smirnov, permutation tests, randomization inference, the Poe test, and the Turnbull estimator as a distribution-free benchmark.
- *Part II assessment (five components).*

PART III — Maximum Likelihood and Latent Variable Models

- **Ch 11.** The Likelihood Framework — likelihood and log-likelihood, the score, the Hessian, Fisher information, the MLE and its asymptotic properties, the curvature interpretation.
- **Ch 12.** The Testing Trinity — Wald, likelihood-ratio, and score tests; constrained versus unconstrained likelihoods; when the three disagree and why.
- **Ch 13.** Computing the MLE, and Distrusting It — convergence (real versus cosmetic), local maxima, starting values, Hessian pathologies, invertibility, separate identification of parameters, sensitivity to starting values.
- **Ch 14.** Latent Variables and Censored Data — latent-index models, probit and logit as latent-variable models, left/right/interval censoring, the survival function.
- **Ch 15.** Contingent Valuation Likelihoods from Scratch — the SBDC likelihood derived as censored latent WTP, the double-bounded likelihood, the payment-card interval-data likelihood, mean versus median WTP under alternative distributions, and Turnbull as a nonparametric robustness check.
- *Part III assessment (five components), including: derive and estimate a simple SBDC model.*

PART IV — Core Econometrics

- **Ch 16.** OLS from First Principles — the linear model, the OLS estimator derived, exogeneity, omitted-variable bias, multicollinearity, prediction.
- **Ch 17.** Inference That Survives Reality — heteroskedasticity, robust (sandwich) standard errors, clustered standard errors derived from first principles, interaction terms, marginal effects, residual diagnostics.
- **Ch 18.** Binary Response Models — logit and probit as latent-utility models, marginal effects, odds ratios, normal versus logistic, identification of scale, predicted probabilities, pseudo-R^2, likelihood-ratio tests.
- **Ch 19.** Repeated Observations — clustered, panel, and multilevel inference; fixed effects; random effects; within/between variation; the pseudo-panel logic behind repeated choice tasks and multiple estimates per study.
- *Part IV assessment (five components).*

PART V — Random Utility, Discrete Choice, and Stated Preference Econometrics

- **Ch 20.** Lancaster and Random Utility Theory — characteristics-based demand, deterministic and stochastic utility, the additive random utility model, and what "random" means.
- **Ch 21.** Multinomial Logit and IIA — iid extreme-value errors leading to the logit choice probability (full derivation), the scale parameter, independence of irrelevant alternatives, and substitution patterns.
- **Ch 22.** Relaxing IIA — nested logit, mixed logit, random coefficients, the panel mixed logit, and error components with an economic interpretation.
- **Ch 23.** Simulated Maximum Likelihood — the simulated likelihood, Halton/Sobol/pseudo-random draws, simulation noise, convergence, and model comparison via AIC, BIC, log-likelihood, and LR tests.
- **Ch 24.** Preference Space versus WTP Space — WTP = $-\beta$_attribute / β_cost, the pathologies of a ratio of random coefficients, heavy tails, the near-zero cost coefficient, direct estimation in WTP space, constrained distributions, and interpretation. (Built toward your EC-MXL.)
- **Ch 25.** Scale Heterogeneity versus Preference Heterogeneity — the book's central discrimination problem: how differences in error variance, attention, comprehension, familiarity, and the cost coefficient masquerade as differences in preference, and what can and cannot separate them.
- *Part V assessment (five components), including: compare WTP in preference space and WTP space; diagnose a mixed logit convergence problem.*

PART VI — Contingent Valuation and Stated Preference Design

- **Ch 26.** Contingent Valuation Formats — open-ended, single-bounded and double-bounded dichotomous choice, payment card, referendum; interval-data models, Turnbull, and spike models.
- **Ch 27.** The Behavioral Foundations of Stated Preference — constructed preferences, coherent arbitrariness, anchoring, internal versus external anchors, yea-saying, preference uncertainty, and hypothetical bias.
- **Ch 28.** Incentive Compatibility and Mechanism Design — incentive compatibility, dominant-strategy truth-telling, the referendum, BDM, random incentive systems, BRIS, pay-all versus pay-some, consequentiality, cheap talk, and budget reminders.
- **Ch 29.** Bid Design, Anchoring, and Cost-Vector Effects — bid-vector effects, scope sensitivity, protest responses, payment vehicles, and the question at the heart of your work: does the cost vector move *preferences* or only *reporting*?
- *Part VI assessment (five components), including: critique a contingent valuation design; write an "assumption → risk → test" table for the CV paper.*

PART VII — Identification and Treatment Effects in Survey Experiments

- **Ch 30.** Estimation Is Not Identification — identifying variation, exclusion restrictions, overidentification, partial identification, observational equivalence, parametric versus nonparametric identification.
- **Ch 31.** Treatment Effects in Randomized Survey Experiments — random assignment, factorial designs, interaction effects, ATE and ATT, difference in means, regression-adjusted effects, heterogeneous treatment effects.
- **Ch 32.** Threats to Identification — post-treatment variables and bad controls, DAGs and confounding, mediation, selection, attrition, exclusions after randomization, protest responders and speeders, manipulation checks, balance checks and why they do *not* prove successful randomization, and the weighted likelihood and how it changes the estimand.
- **Ch 33.** Multiplicity, Equivalence, and Randomization Inference in Experiments — multiple hypothesis testing, equivalence testing applied to convergent validity, randomization inference, and power for interactions.
- *Part VII assessment (five components), including: design a power analysis for a survey experiment; explain whether a treatment contrast identifies preference formation, reporting bias, or both.*

PART VIII — Welfare Foundations

- **Ch 34.** Welfare Measurement Foundations — Marshallian and Hicksian demand, indirect utility and the expenditure function, the duality that links them, Roy's identity and Shephard's lemma; compensating variation, equivalent variation and Marshallian consumer surplus, their ordering and the Willig bounds; the quasilinear case in which all three coincide; and the Small–Rosen log-sum that makes a discrete-choice WTP an *exact* welfare measure.
- **Ch 35.** Public Goods, Non-Use Values, and the Total Economic Value Framework — non-rivalry and non-excludability, free-riding and strategic behaviour, the TEV decomposition into use and non-use (option, bequest, existence) value, warm glow and embedding, scope sensitivity, and aggregation of WTP across a population.
- *Part VIII assessment (five components).*

PART IX — Stated Preference in Practice

- **Ch 36.** Contingent Valuation: Formats, Elicitation, and the Single-Bounded Dichotomous Choice — open-ended, payment card, SBDC, DBDC and referendum formats; the SBDC as a RUM referendum; interval and spike models; Turnbull as the distribution-free anchor.
- **Ch 37.** Discrete Choice Experiments: Design, Attributes, and Experimental Efficiency — attribute and level selection, D-efficient and Bayesian-efficient design, priors, blocking, balance, orthogonality, overlap, dominance, cost-vector construction, choke prices, opt-out design, cognitive burden, and attribute non-attendance.
- **Ch 38.** Estimating the Mixed Logit: Error Components, WTP Space, and Simulated Likelihood — the EC-MXL specification, simulated maximum likelihood, draws and convergence, and estimation directly in WTP space.
- **Ch 39.** Validity, Hypothetical Bias, and the Real-versus-Hypothetical Comparison — content, criterion and convergent validity; hypothetical bias and its 1.2–3× signature; cheap talk, oath scripts and consequentiality; calibration; and the behavioural-welfare question of whether attention, familiarity, learning and non-attendance belong *inside* or *outside* the welfare measure.
- *Part IX assessment (five components), including: critique a contingent valuation design; write an "assumption → risk → test" table for the CV paper.*

PART X — Multiple Discreteness

- **Ch 40.** The MDCEV Framework: Multiple Discreteness and Allocation — the translated additive utility, satiation and translation parameters, the outside good and the numeraire, Kuhn–Tucker demand and corner solutions, the closed-form likelihood, the location normalization, and the α–γ identification ridge.

- **Ch 41.** Heterogeneity and Welfare in MDCEV — random parameters over baseline utilities and satiation, exact compensating variation, and the sensitivity of welfare estimates to design.
- **Ch 42.** Forecasting, Policy Simulation, and Model Validation in MDCEV — the forecasting algorithm, policy scenarios, and out-of-sample validation.
- *Part X assessment (five components), including: derive the compensating variation in the MDCEV proposal and verify which parameters identify it.*

PART XI — Cost–Benefit Analysis

- **Ch 43.** Foundations of Cost–Benefit Analysis — the Kaldor–Hicks criterion and its limits, financial versus economic analysis, FNPV versus ENPV, shadow pricing, standing, and distributional weights.
- **Ch 44.** Discounting, Time, and Intergenerational Equity — the Ramsey formula, descriptive versus prescriptive social discount rates, declining rate schedules, and the social cost of carbon. *(The arithmetic of discounting, NPV/IRR and the stock–flow double-counting trap is Ch 4.)*
- **Ch 45.** Risk, Uncertainty, and Sensitivity Analysis in Appraisal — switching values, scenario analysis, Monte Carlo uncertainty propagation, option and quasi-option value, and scalability and saturation of benefits.
- **Ch 46.** Transparent Reporting, Reproducibility, and the Appraisal Document — the specification multiverse, preregistration, open code and data, and the structure of a defensible appraisal document.
- *Part XI assessment (five components), including: build a CBA sensitivity table.*

PART XII — Evidence Synthesis

- **Ch 47.** Meta-Analysis of Valuation Studies — systematic search and PRISMA, the coding protocol and inter-coder reliability, unit/PPP/inflation harmonization; effect sizes, fixed versus random effects, Cochran's Q, I^2, τ^2 and the prediction interval; meta-regression; publication bias (funnel plots, Egger, PET-PEESE); and dependence among estimates with clustering and robust variance estimation.
- **Ch 48.** Advanced Synthesis: Bayesian, Multivariate, and Structural Approaches — Bayesian random-effects meta-analysis and shrinkage, multivariate and network synthesis, and structural approaches to pooling.
- **Ch 49.** Benefit Transfer: Using Existing Values at New Policy Sites — unit-value versus function transfer, the meta-regression as a transfer function, transfer error and its measurement, and the prediction interval as the honest transfer uncertainty.
- *Part XII assessment (five components), including: a small meta-regression with clustered inference.*

PART XIII — Synthesis, Survey Methodology, and Implementation

- **Ch 50.** The Fragility Audit: Synthesis of the Identification–Assumptions–Fragility Spine — the method that traces any empirical conclusion back through its chain of assumptions, locates the weakest link, and reports the conclusion as honestly conditional.
- **Ch 51.** Survey and Questionnaire Design — payment vehicles, scenario credibility, consequentiality scripts, protest detection, speeding, straight-lining, comprehension checks, visual aids, representativeness, quota sampling, panel quality, order effects, priming, and fatigue.
- *Part XIII assessment (five components).*

PART XIV — Computational Diagnostics and Reproducible Research

- **Ch 52.** Computational Diagnostics and the Reproducible Research Pipeline — convergence real versus cosmetic, multiple starting values, seed and draw sensitivity, gradient and Hessian checks, conditioning and rescaling, detecting local maxima; and the end-to-end reproducible pipeline in R (targets, renv, Git, Quarto).
- *Part XIV assessment (five components).*

Capstone (after Part XIV): an integrated re-analysis of one of your own papers that forces every part of the book to speak at once — stating, for each empirical claim, the identifying variation, the assumptions,

the fragility, and the welfare interpretation.

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5. Chapter-by-chapter learning objectives

For each chapter, "after this chapter you can…" statements. These are the assessable targets; the end-of-part mock orals test them directly.

Ch 1 — Applied Linear Algebra. Read a variance–covariance matrix and diagnose ill-conditioning and near-non-identification from it; explain why a quadratic form appears in every standard-error formula; connect positive definiteness of the Hessian to a genuine maximum; explain what a decomposition of a covariance matrix is doing inside Krinsky–Robb.

Ch 2 — Multivariate Calculus. Differentiate a log-likelihood by hand; interpret the gradient (score) and Hessian geometrically; use a second-order Taylor expansion to explain asymptotic normality of the MLE; derive marginal WTP as a ratio of partial derivatives and explain why direct WTP-space estimation differs from ex-post computation.

Ch 3 — Constrained Optimization. Write and solve a Lagrangian; state the Kuhn–Tucker conditions and read them as "shadow price" statements; distinguish a true corner solution from a survey artifact; derive indirect utility and the expenditure function as dual problems; explain why constrained optimization is the mathematical core of MDCEV.

Ch 4 — Discounting. Compute NPV and IRR; distinguish private from social discount rates and real from nominal flows; run a sensitivity analysis over the rate; explain the stakes of the discount rate in intergenerational restoration valuation.

Ch 5 — Random Variables and Distributions. Motivate the normal, logistic, and Gumbel distributions from the models that use them; manipulate conditional expectations and variances; explain why the difference of two iid Gumbel errors is logistic (the logit's origin).

Ch 6 — Asymptotics. State the LLN and CLT precisely; distinguish convergence in probability from convergence in distribution; explain what "consistent but biased in finite samples" means for your estimates.

Ch 7 — Estimators and Inference. Define bias, consistency, and efficiency without hand-waving; construct and interpret a confidence interval and a p-value correctly; apply the delta method to a nonlinear function of estimates (e.g., a WTP ratio).

Ch 8 — Power and Equivalence. Compute power and a minimum detectable effect for a two-arm comparison; run a TOST; articulate precisely why failing to reject equality is not evidence of equality; design an equivalence test for convergent validity.

Ch 9 — Simulation. Explain why the mixed logit likelihood is an integral without closed form; implement Monte Carlo and Halton-based simulation; propagate parameter uncertainty into a WTP distribution via Krinsky–Robb and via the parametric bootstrap; judge whether a result is robust to the number of draws and the seed.

Ch 10 — Nonparametric/Randomization. Run and interpret permutation and randomization-inference tests; apply the Poe test to compare two simulated WTP distributions; compute a Turnbull lower bound and use it as a distribution-free check.

Ch 11 — Likelihood Framework. Write a log-likelihood, derive its score and Hessian, and connect Fisher information to the asymptotic variance; interpret likelihood curvature as precision.

Ch 12 — Testing Trinity. Derive and choose among Wald, LR, and score tests; explain why they can disagree and which to trust when they do.

Ch 13 — Computing the MLE. Diagnose cosmetic convergence; check Hessian invertibility and eigenvalues; run a multi-start search for global maxima; recognize when large standard errors signal a structural

(identification) problem rather than merely small data.

Ch 14 — Latent Variables and Censoring. Represent probit/logit as latent-index models; write likelihoods for left-, right-, and interval-censored data; use the survival function fluently.

Ch 15 — CV Likelihoods. Derive the SBDC, DBDC, and payment-card interval likelihoods from scratch; compute mean and median WTP under alternative distributions and show how heavy tails and negative-WTP mass change the mean; justify or reject the normality assumption for your good.

Ch 16 — OLS. Derive the OLS estimator and its variance; state the exogeneity assumption and the direction of omitted-variable bias; explain multicollinearity's effect on standard errors without conflating it with bias.

Ch 17 — Robust/Clustered Inference. Derive the sandwich estimator and the cluster-robust variance from first principles; explain what clustering assumes; interpret interaction terms and marginal effects correctly.

Ch 18 — Binary Response. Derive the logit and probit likelihoods; interpret marginal effects and odds ratios; explain scale identification and why coefficients across binary models are not directly comparable.

Ch 19 — Repeated Observations. Justify clustering choice-task observations within respondents; distinguish fixed from random effects and within from between variation; set up a multilevel model for repeated estimates.

Ch 20 — Lancaster and RUT. State the characteristics model and the additive random utility model; explain what is deterministic and what is stochastic and why the split matters for interpretation.

Ch 21 — MNL and IIA. Derive the multinomial logit probability from Gumbel errors; state IIA and construct a case where it fails; explain the scale parameter's confound with utility coefficients.

Ch 22 — Beyond IIA. Specify nested logit, mixed logit, and error-components models; give an economic interpretation of an error component; explain the panel structure of repeated choices.

Ch 23 — Simulated ML. Write the simulated log-likelihood; compare Halton, Sobol, and random draws; quantify simulation noise; compare models with AIC/BIC/LR and know each criterion's assumptions.

Ch 24 — Preference vs WTP Space. Derive WTP as $-\beta_\text{attr}/\beta_\text{cost}$; show analytically why the ratio has heavy tails when the cost coefficient's distribution has mass near zero; estimate directly in WTP space and state exactly which restrictions that imposes; interpret an EC-MXL in WTP space.

Ch 25 — Scale vs Preference Heterogeneity. Given a between-group difference in estimated coefficients, enumerate the scale, attention, comprehension, familiarity, and cost-coefficient explanations that are observationally equivalent to a preference difference, and state what evidence could separate them.

Ch 26 — CV Formats. Compare open-ended, SBDC, DBDC, payment card, and referendum on incentive properties and statistical efficiency; fit interval-data and spike models; use Turnbull as a benchmark.

Ch 27 — Behavioral Foundations. Explain constructed preferences and coherent arbitrariness; distinguish internal from external anchors; connect anchoring to bid-vector effects in your data.

Ch 28 — Incentive Compatibility. State the conditions for incentive compatibility and dominant-strategy truth-telling; analyze BDM, RIS, BRIS, and referendum mechanisms; defend your payment-card-via-RIS and BRIS choices, and name their vulnerabilities.

Ch 29 — Bid Design and Cost-Vector Effects. Explain how the bid/cost vector affects both identification and reporting; design a test that attempts to separate a preference-formation effect from a reporting effect; handle protest responses defensibly.

Ch 30 — Estimation vs Identification. For any given model+design, state what is identified, by what variation, under what assumptions, and what observationally equivalent alternative remains; distinguish parametric from nonparametric identification and partial from point identification.

Ch 31 — Treatment Effects. Define ATE and ATT; analyze factorial designs and interaction effects; estimate regression-adjusted effects and heterogeneous effects; explain what random assignment buys you.

Ch 32 — Threats to Identification. Identify post-treatment variables and bad controls using DAGs; explain why balance checks do not prove randomization; show how post-randomization exclusions and a weighted likelihood each change the estimand; bound results under exclusion.

Ch 33 — Multiplicity and Equivalence in Experiments. Correct for multiple testing without destroying power; apply equivalence testing to convergent-validity claims; run randomization inference; compute power for interaction contrasts.

Ch 34 — Welfare Measurement Foundations. Derive Marshallian and Hicksian demand, indirect utility, and the expenditure function, and move between them via duality, Roy's identity and Shephard's lemma; define CV, EV and consumer surplus and rank them; apply the Willig bounds; state the quasilinear condition under which an estimated WTP *is* an exact welfare measure; compute discrete-choice welfare with the Small–Rosen log-sum.

Ch 35 — Public Goods, Non-Use Values, and TEV. Explain non-rivalry, non-excludability, free-riding and strategic behaviour; decompose total economic value into use and non-use (option, bequest, existence) components; separate warm glow and embedding from genuine value; test scope sensitivity; aggregate WTP across a population correctly.

Ch 36 — Contingent Valuation in Practice. Compare open-ended, payment card, SBDC, DBDC and referendum formats on incentive properties and statistical efficiency; derive the SBDC as a RUM referendum; fit interval and spike models; use Turnbull as the distribution-free benchmark.

Ch 37 — DCE Design and Efficiency. Select attributes and levels; construct D-efficient and Bayesian-efficient designs with priors; manage blocking, balance, orthogonality, overlap, dominance, choke prices and opt-out; manage cognitive burden and diagnose attribute non-attendance; explain how design choices govern identification and power.

Ch 38 — Estimating the Mixed Logit. Specify and estimate an error-components mixed logit; write the simulated log-likelihood; choose draws and diagnose simulation noise; estimate directly in WTP space and interpret the result.

Ch 39 — Validity and Hypothetical Bias. Distinguish content, criterion and convergent validity; quantify hypothetical bias and interpret the real-versus-hypothetical contrast; apply cheap talk, oath scripts and consequentiality; calibrate defensibly; and decide, for each behavioural phenomenon (social desirability, attention, familiarity, learning, non-attendance), whether it belongs inside or outside the welfare measure.

Ch 40 — MDCEV Framework. Set up the MDCEV utility with satiation and translation; derive Kuhn–Tucker demand and the closed-form likelihood; explain the outside good and numeraire; distinguish a corner solution from a design artifact; recognize the location normalization and the α–γ ridge.

Ch 41 — MDCEV Heterogeneity and Welfare. Introduce random parameters over baseline utilities and satiation; compute exact compensating variation; assess the sensitivity of welfare estimates to design.

Ch 42 — MDCEV Forecasting and Validation. Run the forecasting algorithm; build policy simulations; validate the model out of sample.

Ch 43 — CBA Foundations. Apply the Kaldor–Hicks criterion and state its limits; separate financial from economic analysis and FNPV from ENPV; apply shadow pricing; decide standing; explain private-versus-social return gaps like the one in your FRD paper.

Ch 44 — Discounting and Intergenerational Equity. Decompose a social discount rate with the Ramsey formula; argue the descriptive-versus-prescriptive choice; apply a declining schedule; explain why the social cost of carbon inherits the rate. *(NPV/IRR mechanics and the stock–flow double-counting trap are Ch 4.)*

Ch 45 — Risk and Uncertainty in Appraisal. Build sensitivity and scenario tables; compute switching values; run Monte Carlo uncertainty propagation; reason about option and quasi-option value, and about

scalability and saturation of benefits.

Ch 46 — Transparent Reporting. Expose a specification multiverse; preregister; publish code and data reproducibly; structure an appraisal document a reviewer can audit.

Ch 47 — Meta-Analysis of Valuation Studies. Run a PRISMA-compliant search and coding protocol and measure inter-coder reliability; harmonize units, PPP and inflation; choose among FE, RE, REML and WLS and state each one's assumptions; quantify heterogeneity with Q, I^2 and τ^2 and report the prediction interval; fit a meta-regression; read funnel plots skeptically and run Egger and PET-PEESE; model dependence among estimates via clustering and robust variance estimation; recognize too-few-studies-for-too-many-covariates.

Ch 48 — Advanced Synthesis. Fit a Bayesian random-effects meta-analysis and interpret shrinkage; handle multivariate and network structures; state when a structural synthesis is warranted.

Ch 49 — Benefit Transfer. Distinguish unit-value from function transfer; use a meta-regression as a transfer function; measure transfer error; carry the prediction interval — not the confidence interval — as the honest transfer uncertainty.

Ch 50 — The Fragility Audit. Trace any empirical conclusion back through its chain of assumptions; locate the weakest link; state the conclusion as honestly conditional; produce the fragility dossier for your own paper.

Ch 51 — Survey Design. Design payment vehicles and consequentiality scripts; detect protests, speeders, and straight-liners; manage order effects, priming, fatigue, and representativeness; connect each choice to identification, power, and external validity.

Ch 52 — Computational Diagnostics and the Reproducible Pipeline. Run convergence, gradient, and Hessian diagnostics; test sensitivity to starting values, seeds, draws, and scaling; decide, on evidence, when a converged model should not be believed; and build an end-to-end pipeline in R with renv, targets, Git, and Quarto so results regenerate on one command.

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6. Chapter dependency map

The arrow `A -> B` reads "B assumes A." The graph is a directed acyclic structure; the spine question runs through all of it.

Foundational spine (everything depends on these):

```
Ch 1 (Linear Algebra) -+-> Ch 2 (Calculus) --> Ch 3 (Constrained Opt) --> Ch 34 (Welfare Foundations) --
                       |                    \-> Ch 35 (Public Goods/TEV) --> Ch 41 (MDCEV Welfare)
                       \-> Ch 11 (Likelihood) [also needs Ch 2]
```

Probability/inference feeder:

```
Ch 5 (Distributions) --> Ch 6 (Asymptotics) --> Ch 7 (Inference) --> Ch 8 (Power/Equivalence)
Ch 7 --> Ch 9 (Simulation) --> Ch 10 (Nonparametric/Randomization)
Ch 5 --> Ch 21 (MNL, needs Gumbel)
```

Maximum likelihood core:

```
Ch 2, Ch 5, Ch 7 --> Ch 11 (Likelihood) --> Ch 12 (Testing)
Ch 11 --> Ch 13 (Computation) --> Ch 52 (Distrusting Output)
Ch 11 --> Ch 14 (Latent/Censored) --> Ch 15 (CV Likelihoods)
Ch 1 --> Ch 11 (Hessian/information matrix)
```

Econometrics core:

```
Ch 1, Ch 7 --> Ch 16 (OLS) --> Ch 17 (Robust/Clustered) --> Ch 19 (Repeated Obs)
Ch 14 --> Ch 18 (Binary Response)
Ch 17, Ch 19 --> Ch 47 (Meta-Regression)
```

Discrete choice / stated preference:

```
Ch 3 --> Ch 20 (Lancaster/RUT) --> Ch 21 (MNL/IIA) --> Ch 22 (Beyond IIA) --> Ch 23 (Simulated ML)
Ch 9 --> Ch 23
Ch 23 --> Ch 24 (Preference vs WTP Space) --> Ch 25 (Scale vs Preference Het.)
Ch 15, Ch 18 --> Ch 26 (CV Formats)
Ch 26 --> Ch 27 (Behavioral) --> Ch 28 (Incentive Compatibility) --> Ch 29 (Bid/Cost-Vector)
Ch 34, Ch 35 --> Ch 28 (mechanism design needs welfare + duality)
```

Identification and treatment effects:

```
Ch 7, Ch 16 --> Ch 30 (Estimation vs Identification) --> Ch 31 (Treatment Effects) --> Ch 32 (Threats)
Ch 8 --> Ch 33
Ch 25 --> Ch 30 (observational equivalence recurs)
```

Welfare, stated preference in practice, MDCEV, CBA:

```
Ch 34 (Welfare Foundations) --> Ch 35 (Public Goods/TEV) --> Ch 36 (CV in Practice)
Ch 37 (DCE Design) --> Ch 38 (Mixed Logit) --> Ch 39 (Validity/Hypothetical Bias)
Ch 3, Ch 40 (MDCEV) --> Ch 41 (Heterogeneity/Welfare) --> Ch 42 (Forecasting/Validation)
Ch 4, Ch 35 --> Ch 43 (CBA) --> Ch 44 (Discounting/Equity) --> Ch 45 (Risk) --> Ch 46 (Reporting)
Ch 34 --> Ch 43 (ENPV needs welfare)
```

Evidence synthesis:

```
Ch 17, Ch 19 --> Ch 47 (Meta-Analysis) --> Ch 48 (Advanced Synthesis) --> Ch 49 (Benefit Transfer)
Ch 24 --> Ch 47 (WTP effect sizes come from choice models)
Ch 47 --> Ch 49 (the meta-regression is the transfer function)
```

Design and computation (support the whole book, best consolidated late):

```
Ch 21, Ch 30, Ch 37 (DCE Design) <--> Ch 31 (design <-> identification loop)
Ch 51 (Survey Design) draws on Ch 27, Ch 28, Ch 32
Ch 50 (Fragility Audit) draws on everything; Ch 52 (Diagnostics + Pipeline) draws on Ch 13, Ch 23, Ch 4
```

The three chapters with the most inbound arrows — the ones to over-learn because everything leans on them — are **Ch 11 (Likelihood), Ch 24 (Preference vs WTP Space), and Ch 30 (Estimation vs Identification)**. If you master those three, most of the rest becomes application.

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7. Suggested study schedule

This schedule honors the six blocks and the priority tiers in sections 7 and 8 of your PDF. It is **priority-ordered, not strictly dependency-ordered**: your plan sensibly front-loads maximum likelihood and choice modelling because they are your bottleneck, and pulls in mathematics *just in time* rather than all up front. I mark where a math foundation chapter should be read "on demand" the week before you need it. Each block keeps your PDF's stated practical output.

Block 0 — Optional warm-up (1 week, only if foundations feel shaky). Skim **Ch 1** and **Ch 2**. Do not try to master them; just prime the vocabulary of gradients, Hessians, and covariance matrices so Block 1 lands. Return to them properly on demand.

Block 1 — Weeks 1–6: Statistics, probability, MLE. *Objective (yours): understand every likelihood you use.* Read **Ch 5, 6, 7, 9, 11, 12, 13**, pulling **Ch 1** and **Ch 2** in as needed for the Hessian and information-matrix material. Add **Ch 14** at the end as the bridge to censored data. **Practical output**

(yours): re-derive the SBDC and payment-card likelihoods of your CV paper from scratch — which is exactly **Ch 15**, so finish Block 1 by reading Ch 15 and doing its worked derivation on your own data structure.

Block 2 — Weeks 7–14: Choice modelling. *Objective (yours): understand DCE and mixed logit without depending on Apollo.* Read **Ch 20, 21, 22, 23, 24, 25**. Keep **Ch 9** (simulation) open beside Ch 23. **Practical output (yours): estimate an MNL, then a mixed logit, then an EC-MXL on a simple dataset and compare preference space versus WTP space** — the capstone exercise of Ch 24 and the assessment of Part V.

Block 3 — Weeks 15–20: Stated preference and CV. *Objective (yours): defend your survey experiments theoretically.* Read **Ch 26, 27, 28, 29**, and read **Ch 34–35** in parallel because incentive compatibility (Ch 28) leans on welfare and duality. **Practical output (yours): write the "assumption → risk → test/robustness" table for the CV paper** — the Part VI assessment.

Block 4 — Weeks 21–26: Identification, treatments, power. *Objective (yours): know what you can and cannot claim.* Read **Ch 30, 31, 32, 33**, revisiting **Ch 8** (power/equivalence) and **Ch 25** (observational equivalence) as needed. **Practical output (yours): a full power analysis for every principal comparison in the CV paper** — high vs low cost vector, hypothetical vs incentivized, PC vs SBDC, BRIS vs full payment, experienced vs non-experienced — plus equivalence tests for the convergent-validity claims.

Block 5 — Weeks 27–34: Welfare, stated preference in practice, MDCEV, and CBA. *Objective (yours): connect models to real welfare.* Read **Ch 36, 37, 38, 39** (the applied stated-preference arc, finishing the welfare foundations begun in Block 3), then **Ch 3** on demand if constrained optimization needs refreshing, then **Ch 40, 41, 42**, then **Ch 4, 43, 44, 45, 46**. **Practical output (yours): derive the compensating variation in your MDCEV proposal and verify which parameters identify it** — the Part X assessment — and **build the CBA sensitivity table** for the FRD paper.

Block 6 — Weeks 35–38: Evidence synthesis. *Objective (yours): make the finfish paper robust.* Read **Ch 47, 48, 49**, revisiting **Ch 17** and **Ch 19** for clustered/robust inference. **Practical output (yours): leave-one-study-out, clustered/RVE standard errors, and a WLS-versus-REML comparison** for the finfish synthesis — plus the prediction interval you will need if any of it is transferred (Ch 49).

Continuous, not a block — Design, synthesis, and computation. Ch 37 (DCE design) and **Ch 51 (survey methodology)**, together with **Ch 46 (transparent reporting)** and **Ch 52 (diagnostics and the reproducible pipeline)**, are best read *alongside* the blocks whose work they support rather than in sequence. Read Ch 37 and Ch 51 during Blocks 2–3 when you are thinking about your DCE and CV designs; read Ch 46 and Ch 52 continuously, because the diagnostics in Ch 52 are what turn every other block's estimation into something you can trust. Read **Ch 50 (the fragility audit)** last, as the synthesis. Reserve the final 2 weeks for the **capstone re-analysis** of one of your own papers.

Total: roughly a 40-week program matching your PDF's 38 scheduled weeks plus warm-up and capstone. It is intense but it is *your* intensity — every block ends in a deliverable you already need for your dissertation.

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8. Recurring pedagogical devices

Every chapter runs the same seventeen-part machine, so that by the third chapter the structure is invisible and you are simply learning. The devices, and how each is implemented:

1. **Conceptual motivation** — opens each chapter by naming the research mistake the topic prevents, tied to one of your papers.
2. **Prerequisites** — an explicit "you should be able to…" checklist pointing back to specific earlier chapters (this is the dependency map made local).
3. **Core exposition from first principles** — theory built step by step, no skipped steps in central derivations.

4. **Mathematical detail** — numbered Definitions, Assumptions, Propositions, and Derivations, with proofs or proof sketches where they aid understanding, and each result carrying its assumptions on its face.
5. **Applied intuition** — an "in plain economic terms" pass that translates every equation into meaning.
6. **Worked examples** — fully solved, drawn from SBDC, payment cards, DCE, EC-MXL, WTP, MDCEV, welfare, meta-analysis, or CBA, with realistic stylized data where needed.
7. **Research interpretation** — how the concept surfaces in actual outputs: likelihood values, standard-error columns, WTP tables, diagnostics, treatment contrasts, welfare estimates.
8. **Failure modes** — a dedicated section on the mistakes, misinterpretations, identification problems, computational pathologies, and red flags specific to the topic.
9. **Diagnostic questions** — the questions to ask *your own* papers, phrased as a checklist.
10. **Active recall** — short retrieval prompts placed *mid-chapter*, right after the material they test, not saved for the end.
11. **Spaced review prompts** — three explicit prompts at each chapter's end for +1 day, +1 week, +1 month.
12. **Interleaving** — a running thread of backward and forward connections (linear algebra ↔ Hessians and covariance matrices; constrained optimization ↔ MDCEV and compensating variation; MLE ↔ probit, mixed logit, censored WTP; welfare theory ↔ WTP estimation; experimental design ↔ identification and power).
13. **Socratic "Stop and think" checks** — boxed moments where you must predict an implication before the text reveals it.
14. **Three-tier exercises** — (a) basic comprehension, (b) mathematical/statistical derivation, (c) research application or critique.
15. **Solutions** — detailed, appearing after the exercises unless you ask me to hide them for a given chapter.
16. **"How this chapter connects to the rest of the book"** — a cumulative-integration section naming the specific earlier and later chapters in play.
17. **Research translation task** — a closing task that makes you apply the chapter to your own PhD work.

Two further devices run at the **part** level, per your assessment structure: each part closes with a **cumulative problem set, a research-critique exercise, a red-flag-diagnosis exercise, a mock oral-exam section, and an integration mini-project**. And each chapter carries a **three-tier bibliography** — core / applied / advanced — with a one-line reason for each entry, no invented citations, and an explicit flag whenever I am not certain of a reference and you should verify it.

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9. The most important cross-disciplinary connections

The book is built so that the seams between disciplines become the most instructive places in it. The connections it most insists on:

Linear algebra ⟷ inference. The variance–covariance matrix, matrix conditioning, and positive definiteness are not linear-algebra trivia; they are how identification failure and untrustworthy standard errors first show up. The Hessian is simultaneously a calculus object, the curvature of the likelihood, and — inverted — the covariance of your estimates. One object, three disciplines.

Calculus ⟷ maximum likelihood ⟷ welfare. The gradient is the score; the Hessian is the information; the Taylor expansion is why the MLE is asymptotically normal. The *same* differentiation skill, applied to a utility function under a budget constraint, produces marginal WTP as a ratio of marginal utilities — which is why WTP inherits the pathologies of a ratio.

Constrained optimization ⟷ MDCEV ⟷ compensating variation. Kuhn–Tucker conditions are where microeconomic theory, the MDCEV likelihood, and exact welfare measurement literally coincide. Corner solutions in the theory are zero allocations in the data are censored observations in the likelihood.

Probability ⟷ discrete choice. The Gumbel distribution is why logit has a closed form; the *absence* of a closed form for the mixed logit is why simulation (Halton draws, simulated ML) enters; simulation noise then becomes a fragility to diagnose. Distributional assumptions in Part II return as WTP-tail behavior in Part V.

Maximum likelihood ⟷ CV econometrics ⟷ meta-analysis effect sizes. The SBDC likelihood, the interval-censored payment-card likelihood, and the choice-model likelihoods that generate the WTP estimates you later meta-analyze are all one framework. The dependence among multiple WTP estimates from one study (a clustering problem in Part XII) is the same repeated-observations logic as multiple choice tasks per respondent (Part IV).

Welfare theory ⟷ WTP estimation ⟷ CBA. Whether an estimated WTP is a genuine welfare measure or merely a stated-preference indicator is a *theory* question that governs whether it can enter an ENPV. The book refuses to let you move a number from a choice model into a cost-benefit table without first passing it through the welfare-conditions gate of Ch 35.

Experimental design ⟷ identification ⟷ power. Design is not logistics; it is what determines the identifying variation, and therefore what is identified, and therefore whether you have the power to support the claim. The D-efficiency of a design and the identification of a treatment effect are the same question asked from two directions.

Mechanism design/behavioral economics ⟷ stated-preference validity. Incentive compatibility (mechanism design) and anchoring/constructed preferences (behavioral economics) are not "context"; they are the conditions under which your survey responses mean what your welfare interpretation claims they mean. The cost-vector effect sits exactly on this seam — is it preference formation (behavioral), reporting (measurement), or design artifact (identification)?

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10. The highest-risk conceptual mistakes the book will train you to avoid

These are the errors the book is engineered to make you incapable of committing without noticing. The first nine are the "priorità massima" mistakes from section 6 of your PDF; the remainder are the recurring themes you asked to run throughout. Each is tagged with the chapters that drill it.

1. **"Not significant" does not mean "equal."** Failing to reject equality is not evidence of equivalence. Convergent-validity and equality claims require equivalence testing (TOST), not a large p-value. *(Ch 8, Ch 33.)*
2. **Estimation is not identification.** Software estimates whatever you specify; whether that corresponds to the economic object you care about is a separate, prior question. *(Ch 30, and reinforced everywhere.)*
3. **WTP ratios are unstable when the cost coefficient is weak or near zero.** The ratio $-\beta_\text{attr}/\beta_\text{cost}$ has heavy tails driven by cost-coefficient mass near zero; the mean can be dominated by the upper tail. *(Ch 24, Ch 15.)*
4. **Scale heterogeneity and preference heterogeneity are not the same.** Between-group differences in coefficients can come from error-variance, attention, comprehension, or familiarity differences, not preferences. *(Ch 25, Ch 21.)*
5. **Post-randomization exclusions can change the estimand.** Dropping protesters, speeders, or comprehension failures after assignment can break the randomization and silently redefine what you are estimating. *(Ch 32.)*
6. **A weighted likelihood changes the target population.** Weighting respondents (e.g., by coastal-recreation exposure) estimates an exposure-weighted WTP, not the sample-mean WTP; you must say which population that is and whether a few individuals dominate. *(Ch 32, Ch 19.)*
7. **Multiple WTP estimates from the same study are not independent.** Treating them as independent overstates precision; dependence needs clustering, random intercepts, or robust variance

estimation. *(Ch 47, Ch 19.)*

8. **Welfare interpretation requires economic assumptions, not just statistical estimates.** A coefficient is not a welfare measure until the theory says it is. *(Ch 34, and the welfare-gate discipline throughout Parts VIII–XI.)*
9. **Computational convergence is not proof of a valid model.** Convergence can be cosmetic; the Hessian can be non-invertible; the maximum can be local; parameters can be non-identified while the optimizer reports success. *(Ch 13, Ch 52.)*
10. **A beautiful model can identify the wrong object if the design does not support the claim.** Elegance is not identification; the design is what licenses the interpretation. *(Ch 30, Ch 37.)*
11. **CBA requires stock–flow consistency and no double counting.** Do not add a biomass *stock* value to an annual *flow* value; do not count the same benefit through fisheries, biodiversity, tourism, and habitat channels at once. *(Ch 4 §4.11, Ch 43.)*
12. **Mechanism design and behavioral economics are essential to stated-preference validity, not optional context.** Incentive compatibility and anchoring determine whether responses mean what you claim. *(Ch 28, Ch 27, Ch 29.)*
13. **Every empirical claim must be tied to a source of identifying variation.** If you cannot name the variation that identifies a parameter, you cannot defend the parameter. *(Ch 30, recurring exercise.)*
14. **Every model must be interpreted through its assumptions, not only its coefficients.** A coefficient's meaning is conditional on the assumptions that produced it. *(All chapters; enforced by the "assumptions on the face of every proposition" rule.)*

Two more that your PDF flags specifically and the book will drill as named diagnostics: **incomplete identification of behavioral parameters** (whether your design can separate κ from λ, anchoring from hypothetical bias, preference formation from reporting — Ch 29, Ch 30, Ch 32) and **too many cells with too little power** (a multi-dimensional CV design that dilutes per-arm sample size below what the comparisons require — Ch 8, Ch 33, Ch 37).

End of Volume 0 (architecture). The next volumes are the chapters themselves.

Part I — Mathematical Foundations for Econometrics and Welfare Analysis

Chapter 1 — Applied Linear Algebra for Econometrics

Part I: Mathematical Foundations for Econometrics and Welfare Analysis

> **The spine question, applied to this chapter.** Every model you estimate — the SBDC probit, the payment-card interval likelihood, the EC-MXL in WTP space, the meta-regression on your finfish estimates — ultimately hands you two things: a vector of numbers you call "the estimates," and a matrix you usually ignore, the variance–covariance matrix. This chapter is about that matrix and the linear-algebra machinery around it, because *that matrix is where identification, precision, and fragility become visible.* By the end you will be able to look at a covariance matrix and say: these two coefficients are barely separable; this standard error is enormous because the design is nearly singular, not because the sample is small; this optimizer converged but the curvature says the maximum is not real. That is the linear-algebra half of "what am I identifying, and how fragile is it."

1.0 Conceptual motivation: the matrices hiding in your models

Open any of your papers and you will not see a matrix written down. You will see a table of coefficients with stars, a column of standard errors, a WTP estimate with a confidence interval, maybe a log-likelihood at the bottom. Every one of those numbers is the output of a matrix operation you did not perform by hand because the software did it for you. Your PDF says this exactly: *"I tuoi modelli sono pieni di oggetti matriciali anche quando non li vedi esplicitamente."* Let us make the hidden objects explicit, because you cannot judge what you cannot see.

- The **coefficient standard errors** in your SBDC model are the square roots of the diagonal of a matrix obtained by inverting the (negative) Hessian of the log-likelihood. If that matrix is nearly non-invertible, the standard errors explode — and this can happen for structural reasons (the design cannot separate two effects) that have nothing to do with sample size.
- The **Krinsky–Robb simulation** you use in the DCE to build a WTP distribution draws coefficient vectors from a multivariate normal centred at your estimates with covariance equal to that same inverse-Hessian matrix. To *draw* from a multivariate normal you must **decompose** the covariance matrix (a Cholesky factorization). The heavy tails you worry about in WTP come partly from what that covariance matrix looks like in the cost direction.
- The **delta-method** standard error for a WTP ratio is a **quadratic form** in the coefficient covariance matrix. When the cost coefficient is imprecisely estimated, that quadratic form inflates, and the WTP standard error blows up. This is the linear-algebra face of your recurring worry "WTP instabile per colpa del coefficiente di costo."
- The **correlated random coefficients** in an EC-MXL are generated by multiplying independent draws by a lower-triangular matrix — again a Cholesky factor, this time of the covariance of the random parameters. The error components you add to relax IIA *are* a structured covariance matrix on the

utilities.
- **Clustered and robust ("sandwich") standard errors**, **REML** in your meta-analysis, **random effects** — all are statements about the covariance structure of your data, written in matrix form.

So this chapter is not a mathematical detour before the "real" econometrics. It is the real econometrics, seen at the level where its failures first appear. We build the objects — vectors, matrices, the matrix product, rank, inverse, determinant, quadratic forms, definiteness, the covariance matrix and its decomposition, conditioning, and finally the Hessian and information matrix — and at every step we connect them to a place where, in your own work, they decide whether a result can be trusted.

1.1 Prerequisites

This is the first chapter of the book, so the prerequisites are light and mostly conceptual. You should be comfortable with:

- Basic algebra and summation notation ($\sum_{i=1}^{n} x_i$).
- The idea of a function of several variables (we will differentiate one informally here and rigorously in Chapter 2).
- The intuition that an estimator is a random quantity with a sampling distribution (developed fully in Chapters 6–7). Here we only need the idea that "the estimate has a variance."

You do **not** need any prior linear algebra. Everything is built from scratch. If you have seen matrices before, treat the early sections as calibration and slow down at §1.6 (quadratic forms) onward, where the econometric payoff begins.

Notation conventions used throughout the book. Scalars are lowercase italic (x, β). Vectors are lowercase bold ($\mathbf{x}$, $\boldsymbol{\beta}$) and are **column** vectors unless stated otherwise. Matrices are uppercase bold ($\mathbf{X}$, $\mathbf{V}$). The transpose is $\mathbf{X}'$ (some literatures write $\mathbf{X}^\top$; we use the prime, the dominant convention in econometrics — a first example of the "competing conventions" the book will always flag). The (i, j) entry of $\mathbf{A}$ is a_{ij} or $[\mathbf{A}]_{ij}$. An estimator carries a hat: $\widehat{\beta}$. Expectation is $\mathbb{E}[\cdot]$, variance $\mathrm{Var}(\cdot)$, covariance $\mathrm{Cov}(\cdot, \cdot)$. The identity matrix is $\mathbf{I}$ (or $\mathbf{I}_k$ when the size matters).

1.2 Vectors, matrices, and the data matrix

Definition 1.1 (Vector). A (column) vector $\mathbf{x} \in \mathbb{R}^n$ is an ordered list of n real numbers written vertically:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \qquad \mathbf{x}' = (x_1, x_2, \dots, x_n).$$

Its transpose $\mathbf{x}'$ is the same list written horizontally (a row vector).

Definition 1.2 (Matrix). A matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ is a rectangular array of real numbers with m rows and n columns; a_{ij} is the entry in row i, column j. Its transpose $\mathbf{A}'$ is the $n \times m$ matrix with $[\mathbf{A}']_{ij} = a_{ji}$ (rows and columns swapped). A matrix is **square** if $m = n$, and **symmetric** if $\mathbf{A} = \mathbf{A}'$ (so $a_{ij} = a_{ji}$).

Symmetry is not a technicality you can ignore: **every covariance matrix and every Hessian in this book is symmetric**, and symmetry is what makes their eigenvalues real and their behaviour interpretable. Keep it in mind.

The data matrix. In econometrics the central object is the **data matrix $\mathbf{X}$**, of dimension $n \times k$: one row per observation (n of them), one column per variable/regressor (k of them). For a stated-preference dataset an "observation" might be one respondent's answer to one choice task, and the columns might be

an intercept, attribute levels, and the cost. For example, a tiny contingent-valuation-style design with an intercept and a bid amount (in euros) offered to four respondents:

$$\mathbf{X} = \begin{pmatrix} 1 & 10 \\ 1 & 20 \\ 1 & 30 \\ 1 & 40 \end{pmatrix}, \qquad n = 4,\ k = 2.$$

The first column is a column of ones (the intercept); the second is the bid vector. We will carry this little matrix through several sections because it is small enough to compute by hand and rich enough to illustrate rank, inversion, conditioning, and the effect of rescaling — a preview of your "bid vector / cost vector" concerns.

> **Active recall 1.A.** Without looking back: what are the dimensions of $\mathbf{X}'$ for the matrix above? What does the entry x_{32} equal? *(Answer at the end of the section.)*

Vector operations. Addition is entrywise (only for equal dimensions). Scalar multiplication multiplies every entry. The single most important scalar built from two vectors is the **inner (dot) product**:

Definition 1.3 (Inner product). For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$,

$$\mathbf{a}'\mathbf{b} = \sum_{i=1}^{n} a_i b_i.$$

This single number is the atom from which matrix multiplication, sums of squares, variances, and quadratic forms are all built. Two special cases you will use constantly:

- $\mathbf{a}'\mathbf{a} = \sum_i a_i^2$ is the **sum of squares** of $\mathbf{a}$ (its squared length). It is always ≥ 0, and $= 0$ only if $\mathbf{a} = \mathbf{0}$.
- If $\mathbf{a}'\mathbf{b} = 0$, the vectors are **orthogonal**. Orthogonality of columns of a design matrix is exactly the "no correlation between attributes" property you chase in D-efficient design (Chapter 37). Orthogonality is a linear-algebra word for "these two things carry independent information."

Answer to 1.A: $\mathbf{X}'$ is 2×4; $x_{32} = 30$ (row 3, column 2).

1.3 The matrix product as a data operation

Matrix multiplication looks like an arbitrary rule when you first meet it. It is not arbitrary: it is *designed* so that $\mathbf{X}'\mathbf{X}$ produces the table of cross-products that every regression and every likelihood needs. Learn the rule, then immediately learn what it *computes* for you.

Definition 1.4 (Matrix product). If $\mathbf{A}$ is $m \times p$ and $\mathbf{B}$ is $p \times n$ (inner dimensions must match), their product $\mathbf{C} = \mathbf{AB}$ is $m \times n$ with

$$c_{ij} = \sum_{\ell=1}^{p} a_{i\ell}\, b_{\ell j}.$$

That is: entry (i, j) of the product is the inner product of **row i of $\mathbf{A}$** with **column j of $\mathbf{B}$**.

Three properties matter and one non-property matters even more:

1. **Associative:** $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$.
2. **Distributive:** $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$.
3. **Transpose of a product reverses order:** $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$. (Memorize this; it is used in every variance derivation.)
4. **NOT commutative:** in general $\mathbf{AB} \neq \mathbf{BA}$. Order is meaning. Writing $\mathbf{X}\beta$ (an n-vector of fitted values) is completely different from $\beta\mathbf{X}$ (usually undefined).

Why $\mathbf{X}'\mathbf{X}$ is the object you actually care about. Let $\mathbf{X}$ be $n \times k$. Then $\mathbf{X}'\mathbf{X}$ is $k \times k$, and its (j, ℓ) entry is

$$[\mathbf{X}'\mathbf{X}]_{j\ell} = \sum_{i=1}^{n} x_{ij}\, x_{i\ell},$$

the sum over observations of the product of variable j and variable ℓ. So $\mathbf{X}'\mathbf{X}$ is the **matrix of (uncentred) cross-products** of your regressors. Its diagonal holds the sums of squares $\sum_i x_{ij}^2$; its off-diagonals hold the co-movement of pairs of regressors. Divide by n and it is (essentially) the sample second-moment matrix. This one matrix is the informational backbone of OLS (Chapter 16) and, in generalized form, of every likelihood curvature you will meet.

Worked Example 1.1 (Compute $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}'\mathbf{y}$). Take the bid design from §1.2 and suppose the four respondents' (latent, for illustration) responses are $\mathbf{y} = (0, 0, 1, 1)'$ (say, "yes" to the higher bids).

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} \sum 1 & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix} = \begin{pmatrix} 4 & 100 \\ 100 & 3000 \end{pmatrix},$$

because $\sum 1 = 4$, $\sum x_i = 10{+}20{+}30{+}40 = 100$, and $\sum x_i^2 = 100{+}400{+}900{+}1600 = 3000$. Likewise

$$\mathbf{X}'\mathbf{y} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix} = \begin{pmatrix} 2 \\ 30 + 40 \end{pmatrix} = \begin{pmatrix} 2 \\ 70 \end{pmatrix}.$$

Hold onto $\mathbf{X}'\mathbf{X} = \begin{pmatrix} 4 & 100 \\ 100 & 3000 \end{pmatrix}$; the OLS estimator is $\widehat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$, so to estimate anything we must be able to *invert* this matrix — which is only possible if it has full rank. That is the next two sections.

> **Stop and think 1.1.** Before reading on: in Worked Example 1.1 the off-diagonal of $\mathbf{X}'\mathbf{X}$ is 100, which is large relative to the diagonal entry 4. Guess what that large off-diagonal will do to the *precision* with which we can separately estimate the intercept and the bid slope. Write your guess down. *(We resolve it in §1.10 on conditioning — and the answer is the seed of why rescaling your cost variable matters.)*

1.4 Rank, linear independence, and identification

This is the first section where linear algebra becomes identification, and it deserves slow reading, because "my design does not identify this parameter" and "this matrix is rank-deficient" are the same sentence in two languages.

Definition 1.5 (Linear independence). Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are **linearly independent** if the only scalars $c_1, \ldots, c_p$ satisfying

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$$

are $c_1 = \cdots = c_p = 0$. If some non-trivial combination gives $\mathbf{0}$, the vectors are **linearly dependent**: at least one of them is a combination of the others and therefore carries no information the others do not already carry.

Definition 1.6 (Rank). The **rank** of a matrix $\mathbf{A}$ is the number of linearly independent columns it has (equivalently, the number of linearly independent rows — these are always equal). A matrix $\mathbf{A} \in \mathbb{R}^{n \times k}$ with $n \geq k$ has **full column rank** if $\operatorname{rank}(\mathbf{A}) = k$: no column is a linear combination of the others.

The identification theorem for OLS, in words and symbols.

Proposition 1.1. For the $n \times k$ design matrix $\mathbf{X}$ with $n \geq k$, the following are equivalent:

$$\mathbf{X} \text{ has full column rank } k \iff \mathbf{X}'\mathbf{X} \text{ is invertible} \iff \det(\mathbf{X}'\mathbf{X}) \neq 0 \iff \beta \text{ is identified.}$$

Proof sketch (the important direction). Suppose $\mathbf{X}$ is **not** full rank: some non-zero vector $\mathbf{c}$ satisfies $\mathbf{Xc} = \mathbf{0}$ (a column is redundant). Then $\mathbf{X}'\mathbf{Xc} = \mathbf{X}'(\mathbf{Xc}) = \mathbf{X}'\mathbf{0} = \mathbf{0}$, so $\mathbf{X}'\mathbf{X}$ maps a non-zero vector to zero and therefore cannot be inverted (an invertible matrix sends only $\mathbf{0}$ to $\mathbf{0}$). Contrapositively, invertibility forces full rank. And when $\mathbf{X}'\mathbf{X}$ is not invertible, the normal equations $\mathbf{X}'\mathbf{X}\widehat{\beta} = \mathbf{X}'\mathbf{y}$ have infinitely many solutions $\widehat{\beta}$ that fit the data identically — the very definition of a parameter that is **not identified**: the data cannot tell the candidate values apart. ■

What rank deficiency looks like in your work. Rank deficiency is never abstract in practice; it is a modeling mistake or a design limitation:

- **The dummy-variable trap.** Include an intercept plus a full set of category dummies (e.g., dummies for *every* familiarity group) and the dummies sum to the intercept column: a column is redundant, rank drops, the model is not identified. The software either drops a column silently or fails. You must know *which* normalization was imposed, because it defines what your coefficients mean.
- **Scale non-identification in discrete choice.** In a logit, utility is only identified up to scale: multiply all coefficients and the error scale by the same constant and the choice probabilities are unchanged (Chapter 21). This is an identification restriction you *impose* (usually fixing the error variance), not something the data give you. It is a "rank-like" deficiency in the utility parameters, and it is why coefficients are not comparable across models with different scale (Chapter 18, Chapter 25).
- **A design that cannot separate two effects.** If, in your CV design, "high cost vector" is perfectly confounded with "incentivized" (every high-cost respondent was also incentivized), then the two columns are collinear, rank drops, and no amount of data separates the cost-vector effect from the incentive effect. This is the identification worry from your PDF (§6.2, separating κ from λ) in its starkest, *perfect*-collinearity form. The subtler and more dangerous case is **near**-collinearity, where rank is technically full but the matrix is nearly singular — treated in §1.10.

  Active recall 1.B. State, in one sentence each, the two directions of "estimation is not identification" that Proposition 1.1 makes precise. *(Answer: (i) if the design is rank-deficient, the parameter is not identified no matter how you estimate — the data admit many observationally equivalent values; (ii) full rank is what lets estimation recover a unique value. Software can still return "an estimate" under near-deficiency, which is the trap.)*

1.5 The inverse and the determinant

Definition 1.7 (Inverse). For a square matrix $\mathbf{A} \in \mathbb{R}^{k\times k}$, the **inverse** $\mathbf{A}^{-1}$ (if it exists) is the unique matrix with $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_k$. A matrix that has an inverse is **nonsingular** (equivalently, invertible); one that does not is **singular**.

The inverse is how we "solve" matrix equations: the normal equations $\mathbf{X}'\mathbf{X}\widehat{\beta} = \mathbf{X}'\mathbf{y}$ give $\widehat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$, and — crucially for us — the covariance matrix of the OLS estimator will turn out to be proportional to $(\mathbf{X}'\mathbf{X})^{-1}$. So **the inverse of the cross-product matrix is, up to a scalar, the covariance matrix of your estimates**. This is the single most important sentence in the chapter for reading standard errors, and we return to it in §1.8 and §1.11.

Definition 1.8 (Determinant). The **determinant** $\det(\mathbf{A})$ is a scalar summarizing $\mathbf{A}$; geometrically, $|\det(\mathbf{A})|$ is the factor by which $\mathbf{A}$ scales volumes. For a 2×2 matrix,

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \det(\mathbf{A}) = ad - bc.$$

Key facts: $\mathbf{A}$ is invertible **iff** $\det(\mathbf{A}) \neq 0$; $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$; $\det(\mathbf{A}') = \det(\mathbf{A})$. A determinant *near* zero (not exactly zero) signals a matrix that is *nearly* singular — invertible in principle but numerically treacherous. Hold that thought for conditioning.

The 2×2 inverse (worth memorizing).

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Notice the determinant in the denominator: **as $\det(\mathbf{A}) \to 0$, every entry of $\mathbf{A}^{-1}$ blows up.** Since the covariance of your estimates is (proportional to) an inverse, a near-zero determinant of the cross-product/information matrix means **enormous standard errors**. This is the mechanism behind your PDF's red flag: *"standard errors enormi segnalano un problema strutturale, non solo pochi dati."* Big SEs are often a small determinant, i.e., near-non-identification — not a sample-size problem.

Worked Example 1.2 (Invert $\mathbf{X}'\mathbf{X}$). From Worked Example 1.1,

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} 4 & 100 \\ 100 & 3000 \end{pmatrix}, \quad \det = 4 \cdot 3000 - 100 \cdot 100 = 12000 - 10000 = 2000 \neq 0,$$

so it is invertible (the design has full rank; intercept and bid are not collinear). Then

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{2000} \begin{pmatrix} 3000 & -100 \\ -100 & 4 \end{pmatrix} = \begin{pmatrix} 1.5 & -0.05 \\ -0.05 & 0.002 \end{pmatrix}.$$

Read this covariance-shaped object: the $(2,2)$ entry 0.002 is tiny, so (once scaled by the error variance) the **bid slope** will be estimated relatively precisely; the $(1,1)$ entry 1.5 is large, so the **intercept** is estimated imprecisely; and the negative off-diagonal -0.05 says the two estimates are **negatively correlated** — if we revise the slope up, we revise the intercept down to keep the fit. All three readings come straight from the inverse, and all three have direct analogues when the matrix is the inverse-Hessian of your SBDC likelihood.

> **Stop and think 1.2.** The determinant here is 2000, comfortably away from zero, so the inverse is well-behaved. What single change to the *bid design* (the second column of $\mathbf{X}$) would push $\det(\mathbf{X}'\mathbf{X})$ toward zero and make the inverse — and hence the standard errors — explode? *(Hint: think about what happens if all four bids are nearly the same number. Answer in §1.10.)*

A practical warning you will meet again in Chapter 52. We just inverted a matrix by hand using the determinant formula. **Do not do this numerically in real work.** Explicitly forming an inverse via cofactors/determinants is numerically unstable and unnecessary; good software solves the linear system directly (LU, QR, or Cholesky factorizations). When your PDF lists "l'Hessiano è invertibile?" as a diagnostic, the practical check is not "did the determinant come out non-zero" but "is the matrix *well*-conditioned enough that the inverse is trustworthy" — §1.10.

1.6 Quadratic forms

We now reach the construct that appears, quietly, in **every standard error, every Wald test, every confidence ellipse, and the exponent of every multivariate normal** you will ever write. If you understand quadratic forms, a huge amount of inferential machinery stops being mysterious.

Definition 1.9 (Quadratic form). Given a symmetric matrix $\mathbf{A} \in \mathbb{R}^{k \times k}$ and a vector $\mathbf{x} \in \mathbb{R}^k$, the **quadratic form** is the scalar

$$Q(\mathbf{x}) = \mathbf{x}'\mathbf{A}\mathbf{x} = \sum_{i=1}^{k} \sum_{j=1}^{k} a_{ij}\, x_i x_j.$$

For $k = 2$ this expands to $Q(\mathbf{x}) = a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2$ (using $a_{12} = a_{21}$). It is the matrix generalization of "ax^2": a single number, quadratic in the entries of $\mathbf{x}$, with the "cross" term $2a_{12}x_1x_2$ carrying the interaction.

The one derivation to internalize: variance of a linear combination. This is where quadratic forms enter inference, and it is worth doing carefully because you will reuse it constantly — for treatment contrasts, for linear WTP combinations, and (via the delta method) for nonlinear WTP ratios.

Proposition 1.2 (Variance of a linear combination). Let $\hat{\beta}$ be a random k-vector of estimators with covariance matrix $\mathbf{V} = \mathrm{Var}(\hat{\beta})$, and let $\mathbf{a} \in \mathbb{R}^k$ be a fixed vector of constants. Then the scalar $\mathbf{a}'\hat{\beta}$ has variance

$$\mathrm{Var}(\mathbf{a}'\hat{\beta}) = \mathbf{a}'\mathbf{V}\mathbf{a},$$

a **quadratic form** in $\mathbf{V}$ with the vector $\mathbf{a}$.

Derivation. By definition of variance for a scalar random variable $z = \mathbf{a}'\hat{\beta} = \sum_j a_j\hat{\beta}_j$,

$$\mathrm{Var}(z) = \mathrm{Var}\Big(\sum_j a_j\hat{\beta}_j\Big) = \sum_j\sum_\ell a_j a_\ell \,\mathrm{Cov}(\hat{\beta}_j, \hat{\beta}_\ell) = \sum_j\sum_\ell a_j a_\ell V_{j\ell} = \mathbf{a}'\mathbf{V}\mathbf{a}.$$

The middle step is just "the variance of a sum is the sum of all pairwise covariances," and the last step recognizes that double sum as the quadratic form. Notice what the **off-diagonal** covariances do: they enter with the *products* $a_j a_\ell$, so the correlation between estimates can either inflate or deflate the variance of a combination depending on the signs. This is not a curiosity — it is exactly why a treatment contrast can be estimated far more (or less) precisely than the individual coefficients suggest. ■

Worked Example 1.3 (A treatment contrast, tied to your CV design). Suppose in a specification you have two coefficients, $\hat{\beta}_{\text{high}}$ and $\hat{\beta}_{\text{low}}$, capturing the effect of the high- and low-cost-vector arms, with estimated covariance matrix

$$\mathbf{V} = \begin{pmatrix} 4 & -1.5 \\ -1.5 & 1 \end{pmatrix} \quad \Rightarrow \quad \mathrm{SE}(\hat{\beta}_{\text{high}}) = \sqrt{4} = 2, \;\; \mathrm{SE}(\hat{\beta}_{\text{low}}) = \sqrt{1} = 1.$$

You want to test whether the two arms differ, i.e., you care about the contrast $\delta = \beta_{\text{high}} - \beta_{\text{low}}$, which is $\mathbf{a}'\beta$ with $\mathbf{a} = (1, -1)'$. By Proposition 1.2,

$$\mathrm{Var}(\hat{\delta}) = \mathbf{a}'\mathbf{V}\mathbf{a} = V_{11} - 2V_{12} + V_{22} = 4 - 2(-1.5) + 1 = 8, \qquad \mathrm{SE}(\hat{\delta}) = \sqrt{8} \approx 2.83.$$

Here the negative covariance $V_{12} = -1.5$ **increased** the variance of the difference (the $-2V_{12}$ term added $+3$). Had the estimates been positively correlated, the difference would have been estimated *more* precisely. Two lessons, both operational: (i) you cannot get the standard error of a difference by combining the two individual standard errors as if independent — $\sqrt{2^2 + 1^2} = \sqrt{5} \approx 2.24 \neq 2.83$; you **must** use the full covariance matrix; and (ii) whether a "high vs low cost vector" contrast is well-powered depends on the *off-diagonal* of $\mathbf{V}$, not just the diagonal. This is the linear-algebra substrate of the power analyses you will run in Chapter 8 and Chapter 33.

Where else the quadratic form appears (a preview map). - **Wald statistic:** $W = (\hat{\beta} - \beta_0)'\mathbf{V}^{-1}(\hat{\beta} - \beta_0)$ — a quadratic form in $\mathbf{V}^{-1}$; asymptotically χ^2 (Chapter 12). - **Multivariate normal density:** its exponent is $-\frac{1}{2}(\mathbf{x} - \mu)'\Sigma^{-1}(\mathbf{x} - \mu)$ — a quadratic form; the "Σ^{-1}" is why the covariance's *inverse* governs the shape (Chapter 5). - **Mahalanobis distance / confidence ellipses:** the same quadratic form defines the elliptical contours you would draw around $(\hat{\beta}_{\text{high}}, \hat{\beta}_{\text{low}})$; the correlation tilts the ellipse. - **Delta-method WTP variance (Chapter 24):** $\mathrm{Var}(\widehat{\mathrm{WTP}}) \approx \mathbf{g}'\mathbf{V}\mathbf{g}$ where $\mathbf{g}$ is the gradient of the WTP ratio — again a quadratic form, and the vehicle through which an imprecise cost coefficient detonates the WTP standard error.

1.7 Positive definite and positive semidefinite matrices

Quadratic forms lead directly to **definiteness**, the property that separates a genuine covariance matrix from an impossible one, and a genuine maximum of the likelihood from a saddle point or a numerical mirage. Your PDF flags "quando il software converge ma l'Hessiano è sospetto" — definiteness is the concept that makes "suspect Hessian" precise.

Definition 1.10 (Definiteness). A symmetric matrix $\mathbf{A} \in \mathbb{R}^{k\times k}$ is: - **positive definite (PD)** if $\mathbf{x}'\mathbf{A}\mathbf{x} > 0$ for **every** non-zero $\mathbf{x}$; - **positive semidefinite (PSD)** if $\mathbf{x}'\mathbf{A}\mathbf{x} \geq 0$ for every $\mathbf{x}$ (allowing $= 0$ for some $\mathbf{x} \neq \mathbf{0}$); - **negative (semi)definite** if $-\mathbf{A}$ is positive (semi)definite; **indefinite** if $\mathbf{x}'\mathbf{A}\mathbf{x}$ takes both signs.

Proposition 1.3 (Equivalent characterizations of PD). For symmetric $\mathbf{A}$, the following are equivalent: (i) $\mathbf{A}$ is positive definite; (ii) all eigenvalues of $\mathbf{A}$ are strictly positive; (iii) all leading principal minors (the determinants of the top-left $1\times 1, 2\times 2, \ldots, k\times k$ submatrices) are strictly positive (Sylvester's criterion); (iv) $\mathbf{A}$ has a Cholesky factorization $\mathbf{A} = \mathbf{L}\mathbf{L}'$ with $\mathbf{L}$ lower-triangular and strictly positive diagonal. PSD is the same with "≥ 0" / "non-negative" and a non-unique Cholesky.

You do not need to prove these now; you need to be able to *use* them as checks. Two of them (eigenvalues, Cholesky) reappear as the actual diagnostics your software runs.

Why covariance matrices are PSD (and why an "estimated covariance" that is not PSD is a bug or a warning). Any covariance matrix $\mathbf{V} = \mathrm{Var}(\hat{\beta})$ satisfies, for any constants $\mathbf{a}$,

$$\mathbf{a}'\mathbf{V}\mathbf{a} = \mathrm{Var}(\mathbf{a}'\hat{\beta}) \geq 0$$

by Proposition 1.2 and the fact that a variance cannot be negative. So **every true covariance matrix is PSD**. If a program reports an estimated covariance matrix that is *not* PSD (a negative variance for some linear combination, a correlation outside $[-1, 1]$, a negative eigenvalue), something has gone wrong: a non-converged optimizer, a Hessian evaluated away from the true maximum, or a near-non-identified model. This is a concrete Chapter 52 diagnostic you can run on any model output.

Why the maximum of a log-likelihood requires a negative (semi)definite Hessian. This is the payoff that ties definiteness to "is my model really estimated?" Near a maximum $\hat{\theta}$ of the log-likelihood $\ell(\theta)$, a second-order Taylor expansion (Chapter 2 makes this rigorous) gives

$$\ell(\theta) \approx \ell(\hat{\theta}) + \underbrace{\nabla\ell(\hat{\theta})'}_{=\mathbf{0}\text{ at a max}}(\theta - \hat{\theta}) + \tfrac{1}{2}(\theta - \hat{\theta})'\mathbf{H}(\hat{\theta})(\theta - \hat{\theta}),$$

where $\mathbf{H}$ is the **Hessian** (matrix of second derivatives, §1.11). The gradient vanishes at a maximum, so whether $\hat{\theta}$ is really a *maximum* is decided by the quadratic form $(\theta - \hat{\theta})'\mathbf{H}(\theta - \hat{\theta})$: for ℓ to *decrease* in every direction away from $\hat{\theta}$ (a maximum), we need this quadratic form < 0 in every direction, i.e., **H negative definite**. If $\mathbf{H}$ is merely negative *semidefinite* (some direction with zero curvature), the likelihood is *flat* in that direction — the signature of a **non-identified or weakly-identified parameter**: the data cannot distinguish nearby values. If $\mathbf{H}$ is indefinite, $\hat{\theta}$ is a **saddle point, not a maximum at all**, and any standard errors reported there are meaningless.

This is the rigorous content of "il massimo è globale? l'Hessiano è invertibile? i parametri sono identificati separatamente?" from your PDF. A flat direction of the log-likelihood $\Leftrightarrow$ a zero eigenvalue of $\mathbf{H}$ $\Leftrightarrow$ a singular information matrix $\Leftrightarrow$ an infinite variance for some parameter combination $\Leftrightarrow$ non-identification. Same phenomenon, five vocabularies, all in this chapter.

> **Active recall 1.C.** A colleague's mixed logit "converged" and reports finite standard errors, but the smallest eigenvalue of the negative Hessian is 10^{-9} while the largest is 10^{3}. In definiteness language, what is going on, and why should you distrust at least one of the reported standard errors? *(Answer: the negative Hessian is essentially singular — nearly PSD with a near-zero eigenvalue — so there is a near-flat direction of the log-likelihood; a parameter combination is nearly non-identified, and its variance (from the inverse) is near-infinite. "Converged with finite SEs" is cosmetic; §1.10 and §1.11 explain the mechanism.)*

1.8 The variance–covariance matrix: reading your results

We now assemble the covariance matrix as the *central deliverable* of estimation — the object you should read as carefully as the coefficients themselves, and usually do not.

Definition 1.11 (Covariance matrix of a random vector). For a random vector $\mathbf{z} = (z_1, \dots, z_k)'$ with mean $\mu = \mathbb{E}[\mathbf{z}]$, the **variance–covariance matrix** is

$$\Sigma = \mathrm{Var}(\mathbf{z}) = \mathbb{E}\big[(\mathbf{z} - \mu)(\mathbf{z} - \mu)'\big],$$

the $k \times k$ symmetric matrix with $\Sigma_{jj} = \mathrm{Var}(z_j)$ on the diagonal and $\Sigma_{j\ell} = \mathrm{Cov}(z_j, z_\ell)$ off the diagonal. It is always symmetric and PSD (§1.7).

Applied to the estimator $\hat{\beta}$, the covariance matrix $\mathbf{V} = \mathrm{Var}(\hat{\beta})$ is what the software reports (often called the "vcov"). From it you read:

1. **Standard errors** = $\sqrt{V_{jj}}$ (square roots of the diagonal). The stars in your tables are $\hat{\beta}_j / \sqrt{V_{jj}}$ compared to critical values.
2. **The correlation matrix of the estimates**, obtained by rescaling: with $\mathbf{D} = \mathrm{diag}(\sqrt{V_{11}}, \dots, \sqrt{V_{kk}})$ (the diagonal matrix of standard errors),

$$\mathbf{R} = \mathbf{D}^{-1}\mathbf{V}\mathbf{D}^{-1}, \qquad R_{j\ell} = \frac{V_{j\ell}}{\sqrt{V_{jj}}\sqrt{V_{\ell\ell}}} \in [-1, 1].$$

   The correlation of *estimates* is a different object from the correlation of *variables* — a distinction we return to in the failure modes (§1.12), because confusing them is common.

Worked Example 1.4 (Read a covariance matrix like a referee). Take the same

$$\mathbf{V} = \begin{pmatrix} 4 & -1.5 \\ -1.5 & 1 \end{pmatrix}$$

for $(\hat{\beta}_{\text{attribute}}, \hat{\beta}_{\text{cost}})$. Then $\mathrm{SE}(\hat{\beta}_{\text{attribute}}) = 2$, $\mathrm{SE}(\hat{\beta}_{\text{cost}}) = 1$, and the correlation of the two estimates is

$$R_{12} = \frac{-1.5}{2 \cdot 1} = -0.75.$$

A correlation of -0.75 between the attribute and cost coefficient estimates is a **warning light for WTP**. Recall WTP $= -\beta_{\text{attribute}}/\beta_{\text{cost}}$ (Chapter 24). The estimates move together strongly and inversely, and — as we will see in §1.9 — that correlation, together with an imprecise cost coefficient, is what generates the fat WTP tails your PDF worries about (§6.4). You can diagnose the *risk* here, before you ever compute a WTP, just by reading the off-diagonal of $\mathbf{V}$.

> **Stop and think 1.3.** Two coefficients each have a comfortably "significant" t-statistic of 3. Their estimates are correlated at $+0.98$. Predict whether the *difference* between them can be precisely estimated, and whether the *sum* can. *(Use Proposition 1.2. Answer: the difference has variance $V_{11} - 2V_{12} + V_{22}$, and the large positive V_{12} makes this small — the difference is precise; the sum has variance $V_{11} + 2V_{12} + V_{22}$, inflated by the positive covariance — the sum is imprecise. High correlation of estimates means the data pin down some combinations tightly and others barely: exactly the situation where "each coefficient is significant" tells you almost nothing about the contrast you care about.)*

1.9 Decomposing a covariance matrix: Cholesky, simulation, and Krinsky–Robb

Here the chapter earns one of its stated learning objectives: **explain what a decomposition of a covariance matrix is doing inside Krinsky–Robb**, and inside the correlated random coefficients of your EC-MXL. The tool is the Cholesky factorization.

Definition 1.12 (Cholesky factorization). Every symmetric positive definite matrix Σ has a unique factorization

$$\Sigma = \mathbf{L}\mathbf{L}',$$

where $\mathbf{L}$ is **lower-triangular** with strictly positive diagonal. Think of $\mathbf{L}$ as the "matrix square root" of Σ: it is what you multiply independent unit-variance noise by to *induce* the covariance structure Σ.

Proposition 1.4 (Simulating a correlated normal vector). Let $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_k)$ be a vector of independent standard normals, let $\Sigma = \mathbf{L}\mathbf{L}'$, and set

$$\mathbf{x} = \mu + \mathbf{L}\mathbf{z}.$$

Then $\mathbf{x} \sim N(\mu, \Sigma)$.

Derivation. $\mathbf{x}$ is an affine transform of a normal vector, hence normal, with mean $\mathbb{E}[\mathbf{x}] = \mu + \mathbf{L}\,\mathbb{E}[\mathbf{z}] = \mu$, and covariance

$$\mathrm{Var}(\mathbf{x}) = \mathrm{Var}(\mathbf{L}\mathbf{z}) = \mathbf{L}\,\mathrm{Var}(\mathbf{z})\mathbf{L}' = \mathbf{L}\,\mathbf{I}_k\,\mathbf{L}' = \mathbf{L}\mathbf{L}' = \Sigma.$$

The middle equality uses the matrix rule $\mathrm{Var}(\mathbf{A}\mathbf{z}) = \mathbf{A}\,\mathrm{Var}(\mathbf{z})\mathbf{A}'$ — itself a consequence of Proposition 1.2 applied row by row, and worth committing to memory. ■

This *is* Krinsky–Robb. Your DCE (and CV) work uses Krinsky–Robb to build a simulated distribution of WTP from the estimated model. The recipe is precisely Proposition 1.4 with $\mu = \widehat{\beta}$ (your point estimates) and $\Sigma = \widehat{\mathbf{V}}(\widehat{\beta})$ (your estimated coefficient covariance, i.e., the inverse-Hessian of §1.11):

1. Cholesky-factor the estimated covariance: $\widehat{\mathbf{V}} = \mathbf{L}\mathbf{L}'$.
2. For each replication $r = 1, \ldots, R$: draw $\mathbf{z}^{(r)} \sim N(\mathbf{0}, \mathbf{I})$, form $\beta^{(r)} = \widehat{\beta} + \mathbf{L}\mathbf{z}^{(r)}$.
3. Compute the quantity of interest for each draw — for WTP, $\mathrm{WTP}^{(r)} = -\beta^{(r)}_{\text{attribute}}/\beta^{(r)}_{\text{cost}}$.
4. The empirical distribution of $\{\mathrm{WTP}^{(r)}\}$ is your simulated sampling distribution; its percentiles give a confidence interval (and its spread across treatments feeds the **Poe test**, Chapter 10, that you use to compare WTP distributions between arms).

So the Cholesky factor is not a numerical detail — it is *the object that transports your coefficient uncertainty into WTP uncertainty*, correlations and all. And now we can see, concretely, where the fat tails come from.

Worked Example 1.5 (Cholesky, one draw, and the birth of a fat WTP tail). Take $\widehat{\beta} = (\widehat{\beta}_{\text{attr}}, \widehat{\beta}_{\text{cost}}) = (8, -2)$ with covariance $\mathbf{V} = \begin{pmatrix} 4 & -1.5 \\ -1.5 & 1 \end{pmatrix}$ (so SEs 2 and 1, correlation -0.75; the cost coefficient is only 2 SEs from zero — deliberately, to mimic a *weak* cost coefficient). The Cholesky factor is

$$\mathbf{L} = \begin{pmatrix} 2 & 0 \\ -0.75 & 0.6614 \end{pmatrix},$$

obtained from $L_{11} = \sqrt{V_{11}} = 2$, $L_{21} = V_{21}/L_{11} = -1.5/2 = -0.75$, $L_{22} = \sqrt{V_{22} - L_{21}^2} = \sqrt{1 - 0.5625} = \sqrt{0.4375} = 0.6614$. (Check: $\mathbf{L}\mathbf{L}' = \mathbf{V}$ — verify it yourself as a two-minute exercise.)

A "central" draw: $\mathbf{z} = (0.5, -1.0)'$ gives $\mathbf{L}\mathbf{z} = (1.0, -1.036)'$, so $\beta = (9.0, -3.036)'$ and $\mathrm{WTP} = -9.0/(-3.036) = 2.96$. A reasonable number.

A "tail" draw: $\mathbf{z} = (0, 3)'$ (a cost coefficient drawn 3 standard-normal units up) gives $\mathbf{L}\mathbf{z} = (0, 1.984)'$, so $\beta = (8.0, -0.016)'$ — the cost coefficient has wandered *almost to zero* — and $\mathrm{WTP} = -8.0/(-0.016) = 510$. One draw, and WTP explodes by two orders of magnitude, purely because β_{cost} landed near zero.

The full simulation. Drawing 200,000 replications from this $N(\widehat{\beta}, \mathbf{V})$ and computing WTP each time yields (verified numerically): **median** ≈ 3.92, a 95% interval of about [2.03, 16.65], and about **2.2% of draws with a *positive* cost coefficient** (economically nonsensical, and each such draw sends WTP to the wrong sign or to a huge magnitude). Notice three fingerprints of the pathology your PDF names:

- **The sample mean does not converge.** Repeat the same 200,000-draw simulation under different seeds and the sample mean of WTP wanders — 3.5, 4.9, 24.0, 46.7 — while the median holds at 3.92 and the interval endpoints barely move (verified across seeds). This is not sampling noise you can average away by taking more draws: because $\beta_{\text{cost}} \sim N(-2, 1)$ places *positive density at zero*, the expectation $\mathbb{E}[\beta_{\text{attr}}/\beta_{\text{cost}}]$ **does not exist** (the ratio is Cauchy-like in the tail, Chapter 6 §6.9). A "mean simulated WTP" computed here is an artefact of the particular draws, not an estimate of anything. Report the median and percentile interval, which are well defined and stable.

- **A wildly asymmetric interval** (2.03 to 16.65 around a point estimate of $-8/-2 = 4$): the upper limit is far from the point estimate because near-zero cost draws inflate it.
- **Wrong-sign draws** (2.2%): the covariance matrix places non-negligible probability on $\beta_{\text{cost}} \geq 0$, which is both an economic red flag and a signal that a *constrained* distribution for the cost coefficient (Chapter 24) — e.g., forcing it negative via a log-normal in WTP space — may be warranted.

This is the linear-algebra origin of "WTP instabile per colpa del coefficiente di costo." The instability is not a bug in Krinsky–Robb; Krinsky–Robb is faithfully reporting that a ratio with a near-zero, imprecisely-estimated denominator has heavy tails. The fix is upstream — a better-identified cost coefficient (design, §1.10 and Chapter 37), or a reparameterization (WTP space, Chapter 24) — not downstream in the simulation.

The same decomposition builds your EC-MXL random coefficients. In a mixed logit with correlated random parameters, the individual-specific coefficients are generated as $\beta_n = \bar{\beta} + \mathbf{L}\eta_n$, with η_n independent draws (often Halton, Chapter 9) and $\mathbf{L}$ the lower-triangular Cholesky factor of the random-coefficient covariance — *the same construction as Proposition 1.4*. The error components you add to relax IIA are additional structured entries in that covariance. When you estimate an EC-MXL "in WTP space," you are estimating the entries of $\mathbf{L}$ directly in willingness-to-pay units. So the Cholesky factor appears twice in your DCE: once inside the model (generating heterogeneity) and once after it (propagating uncertainty via Krinsky–Robb). Seeing them as the same object is a genuine consolidation of your "uneven foundations."

1.10 Matrix conditioning: why rescaling your cost variable matters

We have twice deferred a question: what happens when a matrix is *nearly* singular — full rank in theory, but on the edge? The answer is the theory of **conditioning**, and it is the bridge between "near-multicollinearity," "huge standard errors," and the humble practical fix of rescaling variables that your PDF lists under diagnostics.

Definition 1.13 (Condition number). For a symmetric positive definite matrix $\mathbf{A}$ with eigenvalues $\lambda_{\max} \geq \cdots \geq \lambda_{\min} > 0$, the **condition number** is

$$\kappa(\mathbf{A}) = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

(More generally, $\kappa = \sigma_{\max}/\sigma_{\min}$ using singular values.) It measures how much the *solution* of a linear system $\mathbf{Ab} = \mathbf{c}$ can change when $\mathbf{c}$ (or $\mathbf{A}$) is perturbed slightly. A small κ (near 1) is **well-conditioned**: stable. A large κ (say $> 10^3$–10^4, and catastrophically $> 10^8$) is **ill-conditioned**: tiny perturbations — including the finite-precision rounding inside your computer — produce large swings in the estimates and in the inverse. Because your covariance matrix *is* an inverse (of the information/cross-product matrix), ill-conditioning shows up directly as **unstable, inflated standard errors**.

Worked Example 1.6 (The bid design, and the payoff of Stop-and-think 1.1 and 1.2). Return to

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} 4 & 100 \\ 100 & 3000 \end{pmatrix}.$$

Its eigenvalues are (verified) $\lambda_{\max} \approx 3003.3$ and $\lambda_{\min} \approx 0.666$, so

$$\kappa(\mathbf{X}'\mathbf{X}) = \frac{3003.3}{0.666} \approx 4510.$$

That is *badly* conditioned — and nothing is "wrong" with the data. The problem is purely one of **scale**: the intercept column has entries of size 1, while the bid column has entries of size 10–40, so $\sum x_i^2 = 3000$ dwarfs $\sum 1 = 4$, stretching one eigenvalue far beyond the other. This is the resolution of **Stop-and-think 1.1**: the large off-diagonal (100) relative to the diagonal (4) was a symptom of this scale mismatch, and it *does* damage the precision with which the intercept and slope can be jointly resolved.

Now **rescale** the bid to units of ten euros, so the bid column becomes $(1, 2, 3, 4)'$:

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}, \qquad \lambda_{\max} \approx 33.4, \;\; \lambda_{\min} \approx 0.60, \qquad \kappa \approx 55.8.$$

The condition number drops from 4510 to 56 — an eighty-fold improvement — *without changing the model*, only the units. And if you additionally **center** the bid (subtract its mean before forming the design), the cross-product matrix becomes diagonal, $\begin{pmatrix} 4 & 0 \\ 0 & 500 \end{pmatrix}$, which removes the *correlation* between the intercept and slope estimates entirely (though scale still matters — $\kappa = 125$ here — so the fully clean move is to **standardize**: center *and* scale). The general lesson, which resolves **Stop-and-think 1.2**: if all bids were nearly identical (say $30.0, 30.1, 29.9, 30.05$), the bid column would be nearly a constant multiple of the intercept column, $\det(\mathbf{X}'\mathbf{X}) \to 0$, $\kappa \to \infty$, and standard errors would explode — the design would nearly fail to identify the slope. **Bid/cost-vector spread is identification.** A cost vector with little variation, or on an awkward scale, is an ill-conditioning problem masquerading as a modeling problem.

Operational takeaways (each a Chapter 52 habit): - **Rescale regressors to comparable magnitudes** before estimation — cost in tens or hundreds of euros rather than euros, attributes on similar ranges. This is not cosmetic; it changes the conditioning of the matrix you invert and can be the difference between convergence and failure in a mixed logit. - **A large reported standard error can be a conditioning symptom**, not a data-quantity symptom. Before concluding "we need a bigger sample," check the condition number of the (negative) Hessian / cross-product matrix. Your PDF's red flag — *"standard errors enormi segnalano un problema strutturale, non solo pochi dati"* — is literally a statement about conditioning. - **Design for spread and orthogonality.** D-efficient design (Chapter 37) is, in part, the art of choosing attribute/cost levels so that the information matrix is well-conditioned — large, balanced eigenvalues — which is why efficient design and precise estimation are the same goal seen from two sides.

> **Active recall 1.D.** Your optimizer fails to converge on an EC-MXL. Name two *scale-related* changes to the data (not the model) that you would try first, and say which matrix property they improve. *(Answer: (i) rescale the cost/price variable to tens or hundreds so its magnitude matches the other regressors; (ii) center/standardize continuous attributes. Both reduce the condition number of the Hessian/information matrix, stabilizing the inversion that produces the covariance and often restoring convergence.)*

1.11 The Hessian and the information matrix: where all of it converges

Everything so far — cross-products, inverses, quadratic forms, definiteness, covariance, decomposition, conditioning — now assembles into the single object from which your standard errors are actually computed: the **Hessian of the log-likelihood**, and its expectation, the **Fisher information matrix**. This section is the destination of the chapter.

Definition 1.14 (Gradient and Hessian). For a scalar function $\ell(\theta)$ of a k-vector θ (here, the log-likelihood as a function of the parameters), the **gradient** (score, when ℓ is a log-likelihood) is the k-vector of first partial derivatives,

$$\mathbf{g}(\theta) = \nabla \ell(\theta) = \left(\tfrac{\partial \ell}{\partial \theta_1}, \dots, \tfrac{\partial \ell}{\partial \theta_k} \right)',$$

and the **Hessian** is the $k \times k$ symmetric matrix of second partial derivatives,

$$\mathbf{H}(\theta) = \nabla^2 \ell(\theta), \qquad [\mathbf{H}]_{j\ell} = \frac{\partial^2 \ell}{\partial \theta_j \, \partial \theta_\ell}.$$

(Chapter 2 develops gradients and Hessians as calculus; here we need only their linear-algebraic role. Chapter 11 derives them for likelihoods in general.)

Definition 1.15 (Fisher information and observed information). The **Fisher (expected) information matrix** is $\mathbf{I}(\theta) = -\mathbb{E}[\mathbf{H}(\theta)]$, and the **observed information** is $-\mathbf{H}(\hat{\theta})$, the negative Hessian evaluated at the MLE. Both are symmetric; at a genuine maximum, both are positive (semi)definite (because $\mathbf{H}$ is negative (semi)definite there, §1.7).

The result that produces your standard errors.

Proposition 1.5 (Asymptotic covariance of the MLE). Under regularity conditions (Chapter 11), the maximum likelihood estimator satisfies, approximately in large samples,

$$\hat{\theta} \overset{\text{approx}}{\sim} N(\theta_0,\ \mathbf{I}(\theta_0)^{-1}),$$

and in practice we estimate the covariance by inverting the observed information:

$$\widehat{\mathrm{Var}}(\hat{\theta}) = \big[-\mathbf{H}(\hat{\theta})\big]^{-1}.$$

The standard errors you read off your regression table are $\sqrt{\text{diagonal of } [-\mathbf{H}(\hat{\theta})]^{-1}}$.

Read Proposition 1.5 slowly, because it makes every earlier section pay off at once:

- The covariance of your estimates is an **inverse** (§1.5) of the **information/Hessian** matrix. A near-singular Hessian (small determinant, near-zero eigenvalue) inverts to a **huge covariance** — enormous standard errors — even with a large sample. *(This is the mechanism your PDF keeps pointing at.)*
- That inverse must be **PSD** to be a legitimate covariance (§1.7); the Hessian being **negative definite** (a real maximum) is exactly what guarantees it. If the optimizer stopped at a saddle or a flat ridge, the "covariance" is not one.
- The **conditioning** of the information matrix (§1.10) governs how trustworthy the inversion — and hence the standard errors — are. Ill-conditioning from a poorly scaled cost variable degrades every standard error in the table.
- To turn this covariance into **WTP uncertainty**, you Cholesky-factor it and run Krinsky–Robb (§1.9). To turn it into a **treatment-contrast standard error**, you form a quadratic form $\mathbf{a}'\hat{\mathbf{V}}\mathbf{a}$ (§1.6). Same matrix, different downstream use.

Worked Example 1.7 (Information from first principles: the Bernoulli/"yes-no" case — your SBDC atom). A single-bounded dichotomous-choice response is a yes/no at a given bid: a **Bernoulli** trial. Consider the simplest version, n i.i.d. responses $y_i \in \{0, 1\}$ with $\Pr(y_i = 1) = p$ (a constant probability — the real SBDC lets p depend on the bid through a probit/logit link, Chapter 15, but the information logic is identical and clearest here). The log-likelihood is

$$\ell(p) = \sum_{i=1}^{n} \big[y_i \ln p + (1 - y_i) \ln(1 - p)\big].$$

Differentiate once (the score):

$$\frac{d\ell}{dp} = \sum_{i=1}^{n} \Big[\frac{y_i}{p} - \frac{1 - y_i}{1 - p}\Big] = \frac{\sum_i y_i - np}{p(1 - p)}.$$

Setting it to zero gives the MLE $\hat{p} = \bar{y} = \frac{1}{n}\sum_i y_i$, the sample proportion of "yes." Differentiate again (the Hessian, here a 1×1 "matrix"):

$$\frac{d^2\ell}{dp^2} = \sum_{i=1}^{n} \Big[-\frac{y_i}{p^2} - \frac{1 - y_i}{(1 - p)^2}\Big] < 0 \quad \text{for all } p \in (0, 1),$$

which is **negative** everywhere — the log-likelihood is concave, so the maximum is unique and global (a clean case; your mixed logit will not be so lucky, Chapter 13). Take expectations, using $\mathbb{E}[y_i] = p$:

$$\mathbf{I}(p) = -\mathbb{E}\Big[\frac{d^2\ell}{dp^2}\Big] = \sum_{i=1}^{n} \Big[\frac{p}{p^2} + \frac{1 - p}{(1 - p)^2}\Big] = n\Big[\frac{1}{p} + \frac{1}{1 - p}\Big] = \frac{n}{p(1 - p)}.$$

Invert to get the asymptotic variance (Proposition 1.5):

$$\widehat{\mathrm{Var}}(\hat{p}) = \mathbf{I}(p)^{-1} = \frac{p(1-p)}{n}, \qquad \mathrm{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

You have just re-derived the familiar standard error of a proportion **from the Hessian of a log-likelihood** — and in doing so you have seen, in miniature and with no matrices bigger than 1×1, the entire pipeline of Proposition 1.5: write the log-likelihood, take its second derivative (Hessian), take the (negative) expectation (information), invert it (covariance), square-root the diagonal (standard error). Your actual SBDC model runs this exact pipeline with a k-vector θ (bid coefficient, constant, and any covariates), a genuine $k \times k$ Hessian, and a matrix inverse — and every concept you need to read and diagnose that computation is now in hand.

> **Stop and think 1.4.** In the Bernoulli information $\mathbf{I}(p) = n/[p(1-p)]$, at what value of p is information *largest*, and at what values does it *collapse*? Translate the answer into advice about where to place bids in an SBDC design. *(Answer: information is maximized at $p = 0.5$ and collapses as $p \to 0$ or $p \to 1$. A bid so high that almost everyone says "no" (or so low that almost everyone says "yes") carries little information about WTP — the response is nearly deterministic. This is a first-principles argument for placing bids where the "yes" probability is intermediate, i.e., near the center of the WTP distribution: a bid-design implication, Chapter 15 and Chapter 37, derived purely from the shape of the information matrix.)*

1.12 Failure modes: how linear algebra misleads in practice

Your PDF asks, for every topic, to know the red flags. Here are the linear-algebra failure modes that recur in stated-preference and welfare work, each with its symptom and its fix.

F1. Perfect collinearity (rank deficiency). *Symptom:* the software drops a variable, reports `NA`, or refuses to invert. *Cause:* a redundant column — a full dummy set with an intercept, a variable that is a linear combination of others, or a treatment perfectly confounded with another. *Fix:* find the linear dependence (which normalization to impose is a *modeling* decision that determines coefficient meaning); never let the software choose it silently.

F2. Near-collinearity / ill-conditioning (the dangerous case). *Symptom:* the model runs and returns estimates, but standard errors are huge, signs flip across specifications, and small data changes move results a lot. *Cause:* a large condition number of the information matrix — often from a poorly scaled cost variable, near-duplicate attributes, or a low-spread bid vector. *Fix:* rescale/standardize regressors (§1.10); increase design spread (Chapter 49); check the condition number and eigenvalues as a routine diagnostic (Chapter 52). **Do not** mistake this for a pure sample-size problem.

F3. Trusting standard errors from a suspect Hessian. *Symptom:* "converged," finite SEs, but a near-zero eigenvalue of the negative Hessian, or a covariance matrix that is not PSD. *Cause:* the optimizer stopped at a flat ridge (weak/near-non-identification) or a saddle, or the Hessian was computed numerically with error. *Fix:* inspect eigenvalues of $-\mathbf{H}$ (all comfortably positive?); re-estimate from multiple starting values (Chapter 13); check that the reported covariance is PSD (§1.7).

F4. Scale-induced convergence failure. *Symptom:* a mixed logit or MDCEV that will not converge, or converges to different points depending on tiny changes. *Cause:* variables on wildly different scales make the Hessian ill-conditioned, so the optimizer's steps are unreliable. *Fix:* rescale before estimating; this is often the *single most effective* intervention, and it changes no economics (§1.10, Chapter 52).

F5. Confusing correlation of *estimates* with correlation of *variables*. *Symptom:* misreading the coefficient correlation matrix $\mathbf{R} = \mathbf{D}^{-1}\mathbf{V}\mathbf{D}^{-1}$ as if it described the raw data. *Clarification:* $\mathrm{Cov}(\hat{\beta}_j, \hat{\beta}_\ell)$ is about how the *sampling errors of the estimates* co-move; two coefficients can be strongly correlated as estimates even when their underlying regressors are only mildly correlated, and this is what governs the

precision of *contrasts* (Stop-and-think 1.3). Keep the two matrices — the covariance of the *data* ($\propto \mathbf{X}'\mathbf{X}/n$) and the covariance of the *estimates* ($\propto (\mathbf{X}'\mathbf{X})^{-1}$) — mentally separate; note they are inverses of each other up to scale, which is exactly why *high* data collinearity produces *high* estimate correlation.

F6. The WTP-ratio tail from a weak cost coefficient. *Symptom:* Krinsky–Robb WTP with an enormous upper tail, a sample mean that shifts with the seed, some wrong-sign cost draws (Worked Example 1.5). *Cause:* the estimated covariance places non-negligible mass on β_{cost} near zero; the ratio's denominator can vanish. *Fix:* upstream — better cost-coefficient identification (design spread, §1.10) or reparameterization to WTP space with a sign-constrained cost coefficient (Chapter 24); report the median and percentile interval, not just the mean.

F7. Inverting when you should be solving. *Symptom:* numerical instability from explicitly forming $(\mathbf{X}'\mathbf{X})^{-1}$ or a matrix inverse in code. *Fix:* solve the linear system directly (QR/Cholesky); reserve the explicit inverse for the covariance matrix you actually need to report (§1.5). A Chapter 52 coding habit.

1.13 Diagnostic questions for your own papers

Ask these of every model in your CV, DCE, MDCEV, and meta-analysis work:

1. **Rank/identification.** Can I name a source of variation in my design that separately identifies each coefficient? Is any treatment perfectly (or nearly) confounded with another? *(§1.4, forward to Chapter 30.)*
2. **Conditioning.** What is the condition number of my information/Hessian matrix? Are my regressors on comparable scales? Would rescaling the cost variable change convergence or the standard errors? *(§1.10.)*
3. **Definiteness.** Are all eigenvalues of the negative Hessian comfortably positive? Is the reported coefficient covariance matrix PSD? *(§1.7, §1.11.)*
4. **Reading the covariance.** What are the *correlations among my estimates*, especially between each attribute coefficient and the cost coefficient? Which *contrasts* (differences) are precise and which are not, given the off-diagonals? *(§1.6, §1.8.)*
5. **WTP propagation.** When I Cholesky-factor my covariance for Krinsky–Robb, does the cost direction place mass near zero? Is my WTP mean being driven by an upper tail? Should the cost coefficient be sign-constrained? *(§1.9, §1.12/F6.)*
6. **Standard-error interpretation.** If a standard error is large, have I checked whether it is a *conditioning/identification* problem before concluding it is a *sample-size* problem? *(§1.5, §1.10, §1.12/F2.)*

1.14 Exercises

Exercises come at three levels: **(A) comprehension**, **(B) derivation**, **(C) research application/critique**. Full solutions follow in §1.15. Try each before reading the solution; the retrieval effort is where the learning happens.

A1. For $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, compute $\det(\mathbf{A})$, $\mathbf{A}^{-1}$, and verify $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$. Is $\mathbf{A}$ positive definite? Justify using two different criteria from Proposition 1.3.

A2. A design matrix has columns: an intercept, a dummy for "experienced," a dummy for "not experienced." Explain in one paragraph why $\mathbf{X}'\mathbf{X}$ is singular, and state two different ways to make the model identified. What does each normalization make the coefficients mean?

A3. Given $\mathbf{V} = \begin{pmatrix} 0.25 & 0.20 \\ 0.20 & 0.25 \end{pmatrix}$ for $(\hat{\beta}_1, \hat{\beta}_2)$, compute the two standard errors and the correlation of the estimates. Then, using Proposition 1.2, compute the standard error of $\hat{\beta}_1 - \hat{\beta}_2$ and of $\hat{\beta}_1 + \hat{\beta}_2$. Which

contrast is more precisely estimated, and why?

B1. Prove that for any matrix $\mathbf{X}$ (of full column rank), $\mathbf{X}'\mathbf{X}$ is positive definite. *(Hint: consider the quadratic form* $\mathbf{c}'(\mathbf{X}'\mathbf{X})\mathbf{c}$ *and rewrite it as a sum of squares.)*

B2. Derive the Cholesky factor $\mathbf{L}$ of $\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$ (the general 2×2 covariance with correlation ρ). Then write the explicit formula for a simulated draw $(\beta_1, \beta_2) = (\bar{\beta}_1, \bar{\beta}_2) + \mathbf{L}(z_1, z_2)'$ with $z_1, z_2 \sim N(0,1)$ independent, and confirm that $\mathrm{Var}(\beta_2) = \sigma_2^2$ and $\mathrm{Corr}(\beta_1, \beta_2) = \rho$.

B3. For the one-parameter Bernoulli log-likelihood of Worked Example 1.7, derive the Fisher information $\mathbf{I}(p) = n/[p(1-p)]$ starting from the score, and use it to show $\mathrm{Var}(\hat{p}) = p(1-p)/n$. Then, treating p as depending on a bid b via a logit link $p(b) = 1/(1+e^{-(\alpha+\gamma b)})$, explain qualitatively (no full derivation) why the information about γ is largest for bids where $p(b) \approx 0.5$.

C1. You estimate an EC-MXL in WTP space. The reported coefficient covariance matrix has a condition number of 2×10^7, and the cost-related standard error is very large, but every attribute coefficient is "significant." A co-author concludes the model is fine because the coefficients have stars. Write a one-paragraph critique: what is your first hypothesis about what is happening, what three diagnostics would you run (naming the linear-algebra property each checks), and what upstream fixes would you consider?

C2. In your finfish meta-analysis you have 18 studies but many more WTP estimates, and you build a design matrix of study-level moderators. You find that two moderators (e.g., "coastal" and "uses a particular attribute") are present in almost exactly the same subset of studies. Explain, in linear-algebra terms, the risk to your meta-regression, connect it to the condition number, and state what you would check and report. *(Ties to Chapter 47; here give the linear-algebra diagnosis only.)*

C3. A referee asks you to report willingness-to-pay as a *mean* across Krinsky–Robb draws. Your simulated WTP has median ≈ 3.9 with a 97.5th percentile of ≈ 16.7, and the sample mean moves substantially when you change the seed. Using the concepts of §1.9 and §1.12, write a two-to-three sentence response explaining what the unstable mean reveals about your cost coefficient's covariance, and what you propose to report instead and why.

1.15 Solutions

A1. $\det(\mathbf{A}) = 2 \cdot 3 - 1 \cdot 1 = 5$. $\mathbf{A}^{-1} = \frac{1}{5}\begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 0.6 & -0.2 \\ -0.2 & 0.4 \end{pmatrix}$. Check: $\mathbf{A}\mathbf{A}^{-1} = \begin{pmatrix} 2\cdot 0.6 + 1\cdot(-0.2) & 2\cdot(-0.2) + 1\cdot 0.4 \\ 1\cdot 0.6 + 3\cdot(-0.2) & 1\cdot(-0.2) + 3\cdot 0.4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. ✓ Positive definite by **(iii) Sylvester**: leading minors $2 > 0$ and $\det = 5 > 0$. Also by **(ii) eigenvalues**: $\lambda = \frac{5 \pm \sqrt{25-20}}{2} = \frac{5 \pm \sqrt{5}}{2} \approx \{3.62, 1.38\}$, both positive. ✓

A2. The three columns satisfy intercept = "experienced" + "not experienced" (the two dummies sum to a column of ones), so the columns are linearly dependent, $\mathrm{rank}(\mathbf{X}) < 3$, and $\mathbf{X}'\mathbf{X}$ is singular (Proposition 1.1) — the dummy-variable trap. Two fixes: **(i)** drop one dummy (say "not experienced"), keeping the intercept; then the intercept is the mean for the omitted (reference) "not experienced" group and the "experienced" coefficient is the *difference* experienced − not experienced. **(ii)** Drop the intercept, keep both dummies; then each dummy's coefficient is the *level* (group mean) for that group directly. Both identify the same underlying content; they differ only in what each coefficient *means*, which you must state.

A3. $\mathrm{SE}(\hat{\beta}_1) = \mathrm{SE}(\hat{\beta}_2) = \sqrt{0.25} = 0.5$. Correlation $= 0.20/(0.5 \cdot 0.5) = 0.8$. Difference: $\mathrm{Var}(\hat{\beta}_1 - \hat{\beta}_2) = 0.25 - 2(0.20) + 0.25 = 0.10$, $\mathrm{SE} = \sqrt{0.10} \approx 0.316$. Sum: $\mathrm{Var}(\hat{\beta}_1 + \hat{\beta}_2) = 0.25 + 2(0.20) + 0.25 = 0.90$, $\mathrm{SE} = \sqrt{0.90} \approx 0.949$. The **difference** is far more precise (SE 0.316 vs 0.949) because the strong *positive* estimate correlation (0.8) means the two estimates' sampling errors move together and largely cancel in the difference but compound in the sum. Lesson: with positively correlated estimates, contrasts (differences) can

be estimated much better than the individual coefficients — and you would never see this from the standard errors alone.

B1. Take any non-zero $\mathbf{c} \in \mathbb{R}^k$. Then

$$\mathbf{c}'(\mathbf{X}'\mathbf{X})\mathbf{c} = (\mathbf{X}\mathbf{c})'(\mathbf{X}\mathbf{c}) = \|\mathbf{X}\mathbf{c}\|^2 = \sum_{i=1}^{n}(\mathbf{X}\mathbf{c})_i^2 \geq 0,$$

using $(\mathbf{X}\mathbf{c})'(\mathbf{X}\mathbf{c}) = \mathbf{c}'\mathbf{X}'\mathbf{X}\mathbf{c}$ (transpose-of-product rule) and that a sum of squares is non-negative. This shows $\mathbf{X}'\mathbf{X}$ is *at least* PSD. For strict positivity (PD), note the sum of squares is zero only if $\mathbf{X}\mathbf{c} = \mathbf{0}$; but full column rank means $\mathbf{X}\mathbf{c} = \mathbf{0} \Rightarrow \mathbf{c} = \mathbf{0}$. So for $\mathbf{c} \neq \mathbf{0}$ the form is strictly positive: $\mathbf{X}'\mathbf{X}$ is positive definite. ■ *(This is also why the information matrix — a generalized $\mathbf{X}'\mathbf{X}$ — is PD exactly when the model is identified.)*

B2. Solve $\Sigma = \mathbf{L}\mathbf{L}'$ with $\mathbf{L} = \begin{pmatrix} L_{11} & 0 \\ L_{21} & L_{22} \end{pmatrix}$: $L_{11} = \sqrt{\sigma_1^2} = \sigma_1$; $L_{21} = (\rho\sigma_1\sigma_2)/L_{11} = \rho\sigma_2$; $L_{22} = \sqrt{\sigma_2^2 - L_{21}^2} = \sqrt{\sigma_2^2 - \rho^2\sigma_2^2} = \sigma_2\sqrt{1-\rho^2}$. So

$$\mathbf{L} = \begin{pmatrix} \sigma_1 & 0 \\ \rho\sigma_2 & \sigma_2\sqrt{1-\rho^2} \end{pmatrix}, \qquad \begin{aligned} \beta_1 &= \bar{\beta}_1 + \sigma_1 z_1, \\ \beta_2 &= \bar{\beta}_2 + \rho\sigma_2 z_1 + \sigma_2\sqrt{1-\rho^2}\, z_2. \end{aligned}$$

Check: $\mathrm{Var}(\beta_2) = \rho^2\sigma_2^2\,\mathrm{Var}(z_1) + \sigma_2^2(1-\rho^2)\,\mathrm{Var}(z_2) = \rho^2\sigma_2^2 + \sigma_2^2(1-\rho^2) = \sigma_2^2$. ✓ And $\mathrm{Cov}(\beta_1, \beta_2) = \mathrm{Cov}(\sigma_1 z_1,\ \rho\sigma_2 z_1 + \cdots) = \sigma_1\rho\sigma_2\,\mathrm{Var}(z_1) = \rho\sigma_1\sigma_2$, so Corr $= \rho\sigma_1\sigma_2/(\sigma_1\sigma_2) = \rho$. ✓ Notice how z_1 (the first, "shared" draw) injects the correlation and z_2 adds the independent residual scaled by $\sqrt{1-\rho^2}$ — the geometry of correlated draws in one formula.

B3. Score: $\ell'(p) = \sum_i [y_i/p - (1-y_i)/(1-p)]$. Second derivative: $\ell''(p) = \sum_i [-y_i/p^2 - (1-y_i)/(1-p)^2]$. Information: $\mathbf{I}(p) = -\mathbb{E}[\ell''(p)] = \sum_i [\mathbb{E}(y_i)/p^2 + (1-\mathbb{E}(y_i))/(1-p)^2] = \sum_i [p/p^2 + (1-p)/(1-p)^2] = n[1/p + 1/(1-p)] = n/[p(1-p)]$. Hence $\mathrm{Var}(\widehat{p}) = \mathbf{I}(p)^{-1} = p(1-p)/n$. For the logit link, information about γ is (heuristically) the bid variance weighted by $p(b)[1-p(b)]$; that weight peaks at $p = 0.5$ and vanishes as $p \to 0, 1$, so bids landing responses near a coin-flip are the most informative about the slope γ (and thus about WTP) — the design implication of Stop-and-think 1.4.

C1. *First hypothesis:* the model is near-non-identified in the cost direction — the information matrix is severely ill-conditioned ($\kappa = 2\times10^7$ is dangerous), so the covariance (its inverse) is unstable and the large cost SE is a *structural* symptom, not a data-quantity one; the attribute "stars" say nothing about whether the cost coefficient — and hence every WTP — is trustworthy. *Three diagnostics:* (1) eigenvalues of the negative Hessian (definiteness/near-singularity — a near-zero eigenvalue confirms a flat direction); (2) the condition number of the information matrix (conditioning); (3) re-estimation from multiple starting values plus a check that the reported covariance is PSD (genuine maximum vs. saddle/ridge). *Upstream fixes:* rescale the cost/price variable and other regressors to comparable magnitudes; consider WTP-space reparameterization with a sign-constrained cost coefficient; and, if the design lacks cost-vector spread, recognize that no estimator can rescue an unidentified object — the fix is in the design (Chapter 37).

C2. Two moderators present in nearly the same studies are nearly collinear columns of the meta-regression design matrix: $\mathbf{X}'\mathbf{X}$ (or the weighted analogue) is nearly singular, its condition number is large, and the two coefficients cannot be separately identified — their individual estimates will have huge, unstable standard errors and may flip sign across specifications, even though *some combination* of them is well estimated. Diagnose by computing the condition number and the pairwise correlation of the two moderators (and of their estimates); check variance-inflation-style measures. Report the problem honestly: either combine the two moderators, drop one (stating it), or present the identified combination, and be explicit that with 18 studies you have limited degrees of freedom for many correlated moderators (the "pochi studi per troppe covariate" red flag of your PDF, Chapter 47).

C3. The seed-dependent mean and the far-out upper percentile reveal a **WTP distribution generated by a cost coefficient whose covariance places non-negligible mass near zero** — the ratio $-\beta_{\text{attr}}/\beta_{\text{cost}}$ has a heavy upper tail whenever the denominator can approach zero, and when the denominator's density at zero is strictly positive the population mean **does not exist at all**, so the simulated mean is a property of

the draws rather than an estimate (§1.9, F6). I would report the **median and a percentile-based (e.g., 2.5–97.5%) interval** rather than the mean, because these are well defined and stable while the mean is not; and I would note (and ideally address) the underlying cause by reparameterizing in WTP space with a sign-constrained cost coefficient so the denominator cannot cross zero (Chapter 24).

1.16 Spaced review prompts

Use these on a schedule; the point is retrieval from memory, not re-reading.

+1 day. From memory, write the chain: log-likelihood → (second derivative) → Hessian → (negative expectation) → information → (inverse) → covariance → (square root of diagonal) → standard error. Then state, for each arrow, one thing that can go wrong (definiteness, conditioning, invertibility).

+1 week. Without notes, reconstruct Worked Example 1.5: given $\hat{\beta} = (8, -2)$ and $\mathbf{V} = \begin{pmatrix} 4 & -1.5 \\ -1.5 & 1 \end{pmatrix}$, compute the Cholesky factor and one WTP draw, and explain in two sentences why a draw with the cost coefficient near zero produces an enormous WTP. Connect to which of your papers this matters for.

+1 month. Take one real model from your own CV or DCE work. Extract its coefficient covariance matrix, compute the correlation of the attribute and cost estimates and the condition number of the information matrix, and write a paragraph diagnosing whether any standard error is a *conditioning/identification* symptom rather than a sample-size symptom. This is the chapter's learning objective, executed on your data.

1.17 How this chapter connects to the rest of the book

- **Chapter 2 (Multivariate Calculus)** turns the gradient and Hessian from linear-algebra objects into calculus you compute: it defines the score as a gradient and derives the Taylor expansion we used informally in §1.7 to justify why a maximum needs a negative-definite Hessian.
- **Chapter 3 (Constrained Optimization)** and **Chapter 34 (Welfare Measurement Foundations)** use quadratic forms and definiteness for second-order conditions and for the concavity of utility; the expenditure function and compensating variation live on the optimization machinery seeded here.
- **Chapters 11–13 (Maximum Likelihood)** are the full-dimensional version of Worked Example 1.7: the score, Hessian, Fisher information, and their inversion, now for k-vectors, with the definiteness and conditioning diagnostics of §1.7 and §1.10 becoming the convergence checks of Chapter 13.
- **Chapters 16–17 (OLS, robust/clustered inference)** make $(\mathbf{X}'\mathbf{X})^{-1}$ and the sandwich estimator explicit; clustered standard errors are a particular structured covariance matrix, read exactly as in §1.8.
- **Chapters 21–25 (discrete choice)** build the mixed-logit covariance of random coefficients via the Cholesky factor of §1.9, and Chapter 24's WTP-tail pathology is the F6 failure mode of §1.12 seen at full strength; Chapter 25's scale-vs-preference problem is, in part, an identification/conditioning question about the information matrix.
- **Chapter 9 (Simulation)** and **Chapter 10 (Poe test)** run the Krinsky–Robb machinery of §1.9 to propagate the covariance into WTP distributions and compare them across your treatment arms.
- **Chapters 47–48 (meta-analysis)** re-use rank, conditioning, and covariance-reading when you regress WTP estimates on correlated study-level moderators (Exercise C2).
- **Chapters 37, 52 (design, diagnostics)** are where §1.10's conditioning and §1.11's information matrix become, respectively, the *goal* of D-efficient design and the *object* of your computational trust checks.

The three highest-dependency chapters flagged in the architecture — **Ch 11 (Likelihood), Ch 24 (Preference vs WTP Space), Ch 30 (Identification)** — each stand directly on this chapter's foundation.

1.18 Research translation task

Pick **one** estimated model from your current work — the SBDC model from the CV paper is the ideal candidate, or the EC-MXL from the DCE paper. Do the following and write it up in one page:

1. Extract the estimated coefficient covariance matrix $\hat{\mathbf{V}}$ (in R: `vcov(model)`; in Apollo, the robust or classical covariance from the output).
2. Report the standard errors (diagonal) and the **full correlation matrix of the estimates** $\mathbf{R} = \mathbf{D}^{-1}\hat{\mathbf{V}}\mathbf{D}^{-1}$. Flag any $|R_{j\ell}| > 0.8$, especially involving the cost coefficient.
3. Compute the **condition number** of the information matrix (or of $\hat{\mathbf{V}}^{-1}$). State whether any large standard error is plausibly a conditioning/identification symptom, and test the hypothesis by rescaling the cost variable and re-estimating.
4. Cholesky-factor $\hat{\mathbf{V}}$, run a small Krinsky–Robb (say $R = 10{,}000$) for one WTP, and report the **mean, median, and 2.5–97.5% interval**. If mean $\gg$ median, diagnose the cost-coefficient tail per §1.12/F6 and state what you will report and why.
5. Conclude with one sentence in the spine format: *"In this model, [parameter] is identified by [variation]; its precision is governed by [conditioning/covariance feature]; the WTP built from it is fragile with respect to [near-zero cost draws / scale / design spread]."*

Bring the write-up to Chapter 11, where we will redo steps 1–3 from the likelihood's score and Hessian rather than from software output.

1.19 Bibliography

Recommendations in three tiers, with a one-line reason each. I have flagged where you should verify details (edition, exact title) yourself.

Core (foundational; know these eventually). - Strang, *Introduction to Linear Algebra* (MIT). The clearest development of rank, the four fundamental subspaces, and positive definiteness with genuine intuition; read the chapters on $\mathbf{X}'\mathbf{X}$, projections, and eigenvalues. *(Verify current edition.)* - Greene, *Econometric Analysis*, the matrix-algebra appendix and the chapter introducing the linear model. This is the econometrician's standard reference and uses exactly the prime-transpose notation of this book. *(Verify edition/appendix letter.)* - Abadir & Magnus, *Matrix Algebra* (Econometric Exercises series). A rigorous, econometrics-oriented treatment; excellent for the definiteness, quadratic-form, and differentiation results you will lean on in Chapters 2 and 11. *(Verify.)*

Applied (help you implement). - Train, *Discrete Choice Methods with Simulation*. The reference for how the coefficient covariance, Cholesky factors of random-coefficient covariances, and simulation come together in mixed logit; read alongside §1.9 and again in Part V. Freely available from the author's page — *verify the current link.* - The **Apollo** manual/primer (Hess & Palma). Shows in practice where the covariance matrix, robust standard errors, and draws enter estimation; map its outputs onto §1.8 and §1.11. *(Verify current version.)* - Krinsky & Robb (1986), "On approximating the statistical properties of elasticities," *Review of Economics and Statistics*. The original simulation method you use for WTP intervals; the §1.9 recipe is this paper. *(Verify volume/pages.)*

Advanced (for later, or for depth). - Golub & Van Loan, *Matrix Computations*. The definitive treatment of conditioning, factorizations (Cholesky, QR), and numerical stability — the rigorous backing for §1.10 and the F7 failure mode, and useful when you write your own estimation code. *(Verify edition.)* - Magnus & Neudecker, *Matrix Differential Calculus with Applications in Statistics and Econometrics*. The systematic machinery for differentiating log-likelihoods and WTP functions with respect to vectors and matrices — the tool behind Chapters 2, 11, and the delta method of Chapter 24. *(Verify edition.)*

Where I am unsure of exact editions, page numbers, or links above, treat the citation as a pointer and verify before citing it in your own work — consistent with the book's rule never to present an unverified reference as settled.

End of Chapter 1. Next: Chapter 2 — Multivariate Calculus for Likelihood and Welfare, which makes rigorous the gradient, Hessian, and Taylor expansion we used here, and derives marginal WTP as a ratio of partial derivatives — the calculus that sets up Chapter 24's central pathology. Tell me when to write it, or ask for a different chapter.

Chapter 2 — Multivariate Calculus for Likelihood and Welfare

Part I: Mathematical Foundations for Econometrics and Welfare Analysis

> **The spine question, applied to this chapter.** Chapter 1 gave you the matrices; this chapter gives you the derivatives that *fill* those matrices with content. The score is a gradient. The information is a Hessian. The reason your maximum likelihood estimator is approximately normal is a Taylor expansion. The marginal effect in your logit table is not the coefficient — it is a derivative evaluated at a point. And willingness to pay is a *ratio of two partial derivatives*, which is exactly why it inherits the instability of a ratio. Calculus is the language in which "what am I identifying" (the shape of the likelihood), "under what assumptions" (concavity, smoothness), and "how fragile" (curvature, near-zero denominators) are all written. By the end you will differentiate a log-likelihood by hand, read its curvature, and derive marginal WTP from first principles — and understand why estimating in WTP space is a genuinely different act from computing WTP after the fact.

2.0 Conceptual motivation: why an applied economist must actually differentiate

You can run `apollo_estimate()` or `glm(..., family=binomial)` without ever writing a derivative. So why spend a chapter on calculus? Because every judgment your PDF asks you to make — *"derivare una log-likelihood; capire il ruolo di score e Hessiano; interpretare la curvatura della likelihood; capire perché alcune stime sono instabili; derivare la marginal WTP come rapporto tra coefficienti; capire perché stimare direttamente in WTP space è diverso da calcolare WTP ex post"* — is a statement about derivatives. Let me make that concrete before we build the machinery.

- When the optimizer "finds the maximum," it is finding where the **gradient** of the log-likelihood is zero. Whether that point is a genuine maximum, and how *precisely* the parameters are pinned down there, is decided by the **Hessian** — the matrix of second derivatives whose negative you invert to get standard errors (Chapter 1, §1.11). If you cannot picture what a gradient and a Hessian are, "the model converged" and "the standard errors are trustworthy" remain black boxes.
- When you report a **marginal effect** from your SBDC probit or your choice model, you are reporting a derivative $\partial P / \partial x_j$ — and in any nonlinear model that derivative is *not* the coefficient β_j; it depends on where you evaluate it. Confusing the two is one of the most common misreadings in applied stated-preference tables.
- When you write WTP $= -\beta_{\text{attribute}} / \beta_{\text{cost}}$, you are writing a **ratio of marginal utilities**, i.e., a ratio of partial derivatives of the utility function. The calculus tells you immediately why the ratio misbehaves when the denominator (the marginal utility of cost) is near zero — the same fat-tail pathology you met numerically in Chapter 1, now understood at its source.

- When you claim your MLE is approximately normal so you can put a confidence interval around it, you are invoking a **second-order Taylor expansion** of the log-likelihood around the truth. This is not decoration; it is *the* argument, and Chapter 11 will make it rigorous. This chapter gives you the sketch so the rigor later has somewhere to land.
- When you decide whether a model has a unique, well-behaved optimum or a treacherous landscape of **local maxima** (your PDF's "il massimo è globale?"), you are asking whether the log-likelihood is **concave** — a second-derivative property. The logit log-likelihood is globally concave (we will prove it). The mixed-logit simulated log-likelihood is *not*, which is why Chapter 13 exists.

So this chapter is the calculus an applied welfare and choice econometrician actually uses, taught at the level where it decides whether your results mean what you say they mean. We build partial derivatives, the gradient, the Hessian, the chain rule, Taylor expansion, concavity, optimization conditions, the implicit function theorem, marginal effects, and elasticities — and at every step we tie the object to a place in your CV, DCE, MDCEV, or CBA work where it earns its keep.

2.1 Prerequisites

From **Chapter 1** you should be able to:

- read a vector and a matrix, form $\mathbf{X}'\mathbf{X}$, and transpose a product (§1.2–1.3);
- write and evaluate a **quadratic form** $\mathbf{x}'\mathbf{A}\mathbf{x}$ (§1.6);
- state what **positive/negative definiteness** means and connect a negative-definite Hessian to a maximum (§1.7);
- recall that the **information matrix is the negative expected Hessian** and its **inverse is the covariance of the MLE** (§1.11).

From basic calculus you need single-variable differentiation (power, product, quotient, chain rules) and the idea of a derivative as a slope. Everything multivariate is built from scratch. We keep the same **notation conventions** as Chapter 1: scalars italic, vectors lowercase bold (columns), matrices uppercase bold, transpose with a prime, estimators hatted. New this chapter: ∂ for partial derivatives, ∇ for the gradient, ∇^2 for the Hessian.

2.2 Partial derivatives and the gradient

Definition 2.1 (Partial derivative). Let $f(\theta) = f(\theta_1, \dots, \theta_k)$ be a scalar function of k variables. The **partial derivative** of f with respect to θ_j is

$$\frac{\partial f}{\partial \theta_j} = \lim_{h \to 0} \frac{f(\dots, \theta_j + h, \dots) - f(\dots, \theta_j, \dots)}{h},$$

the ordinary derivative of f in the direction of θ_j **holding all other variables fixed**. The phrase "holding all else fixed" is the entire economic content: a partial derivative is a *ceteris paribus* rate of change, which is why partial derivatives and marginal effects are the same idea.

Definition 2.2 (Gradient). The **gradient** of f is the column vector of all first partial derivatives:

$$\nabla f(\theta) = \left(\frac{\partial f}{\partial \theta_1}, \ \dots, \ \frac{\partial f}{\partial \theta_k} \right)'.$$

Geometrically, ∇f points in the direction of **steepest ascent** of f, and its length is the steepness. Two facts we will use constantly:

1. **At an interior maximum or minimum, $\nabla f = \mathbf{0}$.** Every partial slope is zero — you cannot go up (or down) in any coordinate direction. This is the **first-order condition (FOC)**, and it is what "the optimizer converged" means operationally.
2. **When f is the log-likelihood $\ell(\theta)$, the gradient is called the *score*, $\mathbf{s}(\theta) = \nabla\ell(\theta)$.** The MLE $\hat{\theta}$ solves $\mathbf{s}(\hat{\theta}) = \mathbf{0}$. So "finding the MLE" is "finding the root of the score."

Applied intuition. Think of $\ell(\theta)$ as a landscape over the parameter space, with height = log-likelihood. The score is the local uphill direction. The optimizer is a hiker climbing until the ground is flat in every direction (score = $\mathbf{0}$). Whether that flat spot is a summit, a mountain pass (saddle), or a flat plateau (non-identification) is a question about the *second* derivatives — the next section.

> **Active recall 2.A.** Your DCE model reports it has "converged." In the language of this section, what equation has the software (approximately) solved, and what does convergence *not* by itself guarantee? *(Answer: it has solved $\mathbf{s}(\hat{\theta}) = \nabla\ell(\hat{\theta}) \approx \mathbf{0}$, the first-order condition; it does not guarantee the point is a maximum rather than a saddle/plateau, nor that it is the global maximum — those are second-order and global properties, §2.3, §2.7.)*

2.3 The Hessian: curvature in every direction

Definition 2.3 (Hessian). The **Hessian** of f is the $k \times k$ symmetric matrix of all second partial derivatives:

$$\nabla^2 f(\theta) = \mathbf{H}(\theta), \qquad [\mathbf{H}]_{j\ell} = \frac{\partial^2 f}{\partial\theta_j\,\partial\theta_\ell}.$$

The diagonal entries $\partial^2 f/\partial\theta_j^2$ measure curvature along each axis; the off-diagonal entries $\partial^2 f/\partial\theta_j\partial\theta_\ell$ measure how the slope in direction j changes as you move in direction ℓ — the **interaction** of the parameters. Symmetry ($[\mathbf{H}]_{j\ell} = [\mathbf{H}]_{\ell j}$) holds whenever the second partials are continuous (Young/Schwarz theorem), which is true for every likelihood in this book.

The curvature–precision link (the reason you care). Curvature of the log-likelihood *is* information. A log-likelihood that is sharply peaked (large negative curvature, i.e., a very negative-definite Hessian) pins the parameters down tightly — small standard errors. A log-likelihood that is nearly flat near its maximum (curvature close to zero, Hessian near-singular) barely distinguishes nearby parameter values — huge standard errors, and, in the limit, non-identification. This is Chapter 1's Proposition 1.5 seen from the calculus side:

$$\widehat{\mathrm{Var}}(\hat{\theta}) = [-\mathbf{H}(\hat{\theta})]^{-1}.$$

Flat likelihood ⇒ small $|\mathbf{H}|$ ⇒ large inverse ⇒ large standard errors. Your PDF's instruction "interpretare la curvatura della likelihood" is exactly this sentence.

Off-diagonals and correlated estimates. The off-diagonal Hessian entries determine the *off-diagonal* of the covariance (after inversion), i.e., the **correlation between your coefficient estimates** (Chapter 1, §1.8). When the log-likelihood has a curved ridge — high in one tilted direction, flat along it — the estimates of the two parameters are strongly correlated, and some contrasts are precise while others are not (Chapter 1, Stop-and-think 1.3). The Hessian is where that ridge lives.

> **Stop and think 2.1.** Picture a log-likelihood for two parameters whose contour map is a set of long, thin, diagonal ellipses (a narrow ridge running northeast). Before reading on, predict: (a) will the two estimates be strongly correlated? (b) will their *individual* standard errors be large or small? (c) will the *sum* or the *difference* be better identified? *(Answer: (a) yes, strongly — the ridge tilts the ellipse; (b) large individually — the likelihood is flat along the ridge; (c) the combination* along* the ridge is poorly identified, the combination *across* it is well identified. Which is "sum" vs "difference" depends on the ridge's orientation. This is the calculus picture of Chapter 1's covariance reading.)*

2.4 The chain rule and total derivatives

Nonlinear models — logit, probit, mixed logit, MDCEV — are compositions: a probability is a *function* (the link) of a linear index $\mathbf{x}'\beta$, which is itself a function of the parameters. To differentiate them you need the **chain rule**, and to interpret marginal effects you need **total** versus **partial** derivatives.

Chain rule (composite functions). If $g(\theta) = h(u(\theta))$ with u scalar-valued, then

$$\frac{\partial g}{\partial \theta_j} = h'(u(\theta))\,\frac{\partial u}{\partial \theta_j}.$$

For the linear index $u = \mathbf{x}'\beta = \sum_m x_m\beta_m$, we have $\partial u/\partial \beta_j = x_j$, so

$$\frac{\partial\, h(\mathbf{x}'\beta)}{\partial \beta_j} = h'(\mathbf{x}'\beta)\,x_j.$$

This single formula is the engine behind every score and every marginal effect in a discrete choice model. Keep it visible; we use it in §2.8 (score) and §2.10 (marginal effects).

Total derivative and the total differential. For a function $f(\theta_1, \dots, \theta_k)$, the **total differential** collects the effect of simultaneously nudging every argument:

$$df = \sum_{j=1}^{k} \frac{\partial f}{\partial \theta_j}\, d\theta_j = \nabla f(\theta)'\, d\theta.$$

The total differential is the tool that turns "utility is held constant along an indifference curve" ($df = 0$) into the marginal-rate-of-substitution formula, and it is how WTP will drop out as a ratio of partials in §2.11. It is also the basis of the **delta method** (Chapter 7): the variance of a smooth function $g(\hat{\theta})$ is approximated by propagating $\hat{\theta}$'s covariance through g's gradient, $\mathrm{Var}(g(\hat{\theta})) \approx \nabla g'\,\mathbf{V}\,\nabla g$ — a quadratic form (Chapter 1, §1.6) in the gradient you compute with the chain rule.

2.5 Taylor expansion: approximating a function by a quadratic

Almost everything asymptotic in econometrics rests on replacing a complicated function by its **local quadratic approximation**. That approximation is the Taylor expansion.

Definition 2.4 (Second-order Taylor expansion, multivariate). For a twice-differentiable f, near a point θ_0,

$$f(\theta) \approx f(\theta_0) + \nabla f(\theta_0)'(\theta - \theta_0) + \tfrac{1}{2}(\theta - \theta_0)'\mathbf{H}(\theta_0)(\theta - \theta_0),$$

with an error that shrinks faster than $\|\theta - \theta_0\|^2$. The three terms are: the height at the base point, a **linear** term (the gradient dotted with the displacement), and a **quadratic** term (a quadratic form in the Hessian). This is the multivariate generalization of $f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2$.

Applied to the log-likelihood at its maximum. Evaluate the expansion of ℓ at $\theta_0 = \hat{\theta}$ (the MLE), where $\nabla\ell(\hat{\theta}) = \mathbf{0}$:

$$\ell(\theta) \approx \ell(\hat{\theta}) + \tfrac{1}{2}(\theta - \hat{\theta})'\mathbf{H}(\hat{\theta})(\theta - \hat{\theta}).$$

The linear term vanishes, so **near its peak the log-likelihood is approximately a quadratic bowl (opening downward) whose shape is the Hessian.** A downward bowl in every direction requires $\mathbf{H}$ negative definite (Chapter 1, §1.7) — this is why the second-order condition for a maximum is exactly negative-definiteness, and why a flat direction (zero curvature) signals non-identification. Everything Chapter 1 asserted about definiteness is *proved* by this expansion.

2.6 Convexity and concavity

Concavity is the property that turns a *local* maximum into a *global* one and makes optimization safe. It matters twice in your work: for **log-likelihoods** (is the optimum unique?) and for **utility functions** (is consumer behavior well-behaved, are interior/corner solutions sensible — Chapter 3, MDCEV).

Definition 2.5 (Concave and convex functions). A function f on a convex set is **concave** if for all θ_1, θ_2 and $\lambda \in [0, 1]$,

$$f(\lambda\theta_1 + (1-\lambda)\theta_2) \geq \lambda f(\theta_1) + (1-\lambda) f(\theta_2)$$

(the graph lies **above** every chord). It is **convex** if the inequality reverses (graph below every chord). **Strictly** concave/convex if the inequality is strict for $\theta_1 \neq \theta_2$ and $\lambda \in (0, 1)$.

Proposition 2.1 (Second-derivative test for concavity). For a twice-differentiable f:

$$f \text{ concave} \iff \mathbf{H}(\theta) \text{ is negative semidefinite for all } \theta;$$

$$\mathbf{H}(\theta) \text{ negative definite for all } \theta \implies f \text{ strictly concave.}$$

(Convexity is the mirror image with positive semidefinite/definite.) The proof is the Taylor expansion of §2.5 applied along the segment between any two points; the quadratic-form sign controls the chord inequality.

Why a concave log-likelihood is a gift. If $\ell(\theta)$ is concave, any point with $\nabla\ell = \mathbf{0}$ is a **global** maximum, and if strictly concave it is **unique**. There are no local maxima to trap the optimizer, starting values do not matter for the answer, and "converged" really does mean "found the MLE." The plain **logit** log-likelihood has this property (we prove it in §2.8). The plain **probit** likelihood is also globally concave. This is why single-level SBDC and MNL estimation is reliable.

Why the loss of concavity is where danger begins. The **mixed logit** simulated log-likelihood is generally **not** globally concave: it can have multiple local maxima, flat regions, and simulation-induced wiggles (Chapter 9, Chapter 23). Your EC-MXL lives here. That is precisely why your PDF elevates "local maxima," "starting values," and "il massimo è globale?" to first-order concerns — they are non-issues under concavity and serious issues without it. Concavity is the dividing line between models you can trust on one estimation and models you must interrogate with multiple starts (Chapter 13, Chapter 52).

Utility concavity and economics. A concave utility function encodes **diminishing marginal utility**: each additional unit of an attribute adds less. Concavity (and quasi-concavity) of utility is what guarantees well-behaved, convex indifference curves and sensible demand — and, in MDCEV, the **satiation** that makes an interior allocation optimal rather than spending everything on one good (Chapter 3, Chapter 40). So the same mathematical property governs the reliability of your *estimation* and the plausibility of your *economic model*.

> **Active recall 2.B.** Give one econometric consequence and one economic consequence of concavity. *(Answer: econometric — a concave log-likelihood has a unique global maximum, so estimation is reliable and start-value-independent; economic — a concave utility function has diminishing marginal utility and convex indifference curves, giving well-behaved demand and interior optima.)*

2.7 Local versus global optima, and the conditions that separate them

First-order condition (FOC). At any interior optimum (max or min), $\nabla f(\theta^*) = \mathbf{0}$. Necessary, not sufficient — a saddle also has zero gradient.

Second-order condition (SOC). At a point with $\nabla f = \mathbf{0}$: - **H negative definite** ⇒ strict local **maximum**; - **H positive definite** ⇒ strict local **minimum**; - **H indefinite** ⇒ **saddle point** (not an optimum); - **H singular** (a zero eigenvalue) ⇒ **test inconclusive** — a flat direction; in likelihood terms, weak or non-identification.

Global guarantee. If f is concave *everywhere*, any FOC point is a global maximum (§2.6). Without global concavity, an FOC point is only *local*, and there may be others.

The research payoff. This is the theory behind your PDF's diagnostic list for MLE. "La convergenza è vera o cosmetica?" — check the FOC (gradient near zero) *and* the SOC (negative-definite Hessian, all eigenvalues comfortably negative). "Il massimo è globale?" — under concavity, yes automatically; otherwise, re-estimate from multiple starting values and compare the log-likelihood values (Chapter 13). "I parametri sono identificati separatamente?" — check for near-singular Hessian (a near-zero eigenvalue = a flat direction = a parameter combination the data cannot resolve). All three questions are FOC/SOC/definiteness questions, and you now have the calculus to ask them precisely.

2.8 Differentiating a log-likelihood by hand: the logit (and the SBDC probit)

This is the chapter's central derivation, and per the book's rule we skip no steps. We derive the score and Hessian of the binary **logit**, prove its concavity, and then state the **probit** analogue that underlies your SBDC contingent-valuation model.

Setup. For observation i with covariate vector $\mathbf{x}_i$ (which, in an SBDC, includes a constant and the bid) and binary outcome $y_i \in \{0, 1\}$, the logit model sets

$$\Pr(y_i = 1 \mid \mathbf{x}_i) = \Lambda(\mathbf{x}_i'\beta), \qquad \Lambda(z) = \frac{1}{1 + e^{-z}},$$

where Λ is the logistic CDF. A key, clean derivative property (verify it):

$$\Lambda'(z) = \Lambda(z)(1 - \Lambda(z)).$$

Writing $\Lambda_i \equiv \Lambda(\mathbf{x}_i'\beta)$, the contribution of observation i to the log-likelihood is

$$\ell_i(\beta) = y_i \ln \Lambda_i + (1 - y_i) \ln(1 - \Lambda_i).$$

Step 1 — the score (gradient). Differentiate ℓ_i with respect to β using the chain rule (§2.4). First the derivative of ℓ_i with respect to the probability Λ_i:

$$\frac{\partial \ell_i}{\partial \Lambda_i} = \frac{y_i}{\Lambda_i} - \frac{1 - y_i}{1 - \Lambda_i}.$$

Then $\partial \Lambda_i / \partial \beta = \Lambda_i(1 - \Lambda_i)\mathbf{x}_i$ (chain rule with $\Lambda' = \Lambda(1 - \Lambda)$ and $\partial(\mathbf{x}_i'\beta)/\partial\beta = \mathbf{x}_i$). Multiply:

$$\frac{\partial \ell_i}{\partial \beta} = \left[\frac{y_i}{\Lambda_i} - \frac{1 - y_i}{1 - \Lambda_i}\right] \Lambda_i(1 - \Lambda_i)\, \mathbf{x}_i.$$

Now simplify the bracket times $\Lambda_i(1 - \Lambda_i)$:

$$\frac{y_i}{\Lambda_i}\Lambda_i(1 - \Lambda_i) - \frac{1 - y_i}{1 - \Lambda_i}\Lambda_i(1 - \Lambda_i) = y_i(1 - \Lambda_i) - (1 - y_i)\Lambda_i = y_i - y_i\Lambda_i - \Lambda_i + y_i\Lambda_i = y_i - \Lambda_i.$$

Therefore the score for the full sample is

$$\boxed{\mathbf{s}(\beta) = \sum_{i=1}^{n} (y_i - \Lambda_i)\, \mathbf{x}_i}.$$

Read this: the score is the sum of covariate vectors weighted by the **residual** $y_i - \Lambda_i$ (actual minus predicted probability). At the MLE, $\sum_i (y_i - \widehat{\Lambda}_i)\mathbf{x}_i = \mathbf{0}$: the residuals are orthogonal to the regressors — the same

orthogonality that defines OLS (Chapter 16), here in a nonlinear guise. This is a deep unity worth pausing on: *maximum likelihood makes the prediction errors uncorrelated with the covariates.*

Step 2 — the Hessian. Differentiate the score again. Since $\partial(y_i - \Lambda_i)/\partial\beta' = -\Lambda_i(1-\Lambda_i)\mathbf{x}_i'$ (chain rule again),

$$\mathbf{H}(\beta) = \frac{\partial \mathbf{s}}{\partial \beta'} = -\sum_{i=1}^{n} \Lambda_i(1-\Lambda_i)\,\mathbf{x}_i\mathbf{x}_i'.$$

Each term is a positive scalar $\Lambda_i(1-\Lambda_i) > 0$ times the outer product $\mathbf{x}_i\mathbf{x}_i'$ (which is positive semidefinite), with an overall minus sign.

Step 3 — concavity (a proof, using Chapter 1). For any non-zero $\mathbf{c}$,

$$\mathbf{c}'\mathbf{H}(\beta)\mathbf{c} = -\sum_i \Lambda_i(1-\Lambda_i)\,\mathbf{c}'\mathbf{x}_i\mathbf{x}_i'\mathbf{c} = -\sum_i \Lambda_i(1-\Lambda_i)\,(\mathbf{x}_i'\mathbf{c})^2 \le 0,$$

because it is a negative sum of non-negative terms. So $\mathbf{H}$ is negative semidefinite **everywhere**, hence (Proposition 2.1) the logit log-likelihood is **globally concave**; and it is strictly negative — strictly concave — as long as the $\mathbf{x}_i$ have full rank (so that $\mathbf{x}_i'\mathbf{c} = 0$ for all i forces $\mathbf{c} = \mathbf{0}$). This *proves* the reliability claim of §2.6: the logit has a unique global maximum, and estimation does not depend on starting values. ■

Numerical check (verified). For a simulated logit with 200 observations and a two-column design (constant, a regressor on $[1, 4]$), evaluating the analytic formulas at $\beta = (0.8, -0.25)$ gives score $\approx (9.39,\ 29.57)'$ (non-zero — we are not at the MLE) and Hessian $\begin{pmatrix} -49.1 & -123.5 \\ -123.5 & -346.2 \end{pmatrix}$ with eigenvalues $\{-390.9,\ -4.4\}$ — **both negative**, confirming negative-definiteness and hence local concavity at that point, exactly as the proof guarantees globally.

The SBDC probit (your CV model). Your single-bounded contingent valuation is modelled as latent WTP that is normally distributed, so the "yes" probability uses the **normal** CDF Φ rather than Λ: $\Pr(y_i = 1) = \Phi(\mathbf{x}_i'\beta)$ (with the bid entering $\mathbf{x}_i$). The derivation is structurally identical but the derivative of Φ is the normal density ϕ, so the score is

$$\mathbf{s}(\beta) = \sum_i \left[\frac{y_i\,\phi_i}{\Phi_i} - \frac{(1-y_i)\,\phi_i}{1-\Phi_i}\right]\mathbf{x}_i,$$

where the bracket is the **generalized residual** (the probit analogue of $y_i - \Lambda_i$). The probit log-likelihood is also globally concave (a classic result), so your SBDC model, like the logit, has a unique maximum — a reassuring fact you can now *justify* rather than merely trust. Chapter 15 builds the full SBDC likelihood, including how the bid coefficient maps to mean and median WTP; here the point is that you can differentiate it by hand and know its optimum is well-behaved.

> **Stop and think 2.2.** The logit Hessian is $-\sum_i \Lambda_i(1-\Lambda_i)\mathbf{x}_i\mathbf{x}_i'$. The weight $\Lambda_i(1-\Lambda_i)$ is largest when $\Lambda_i = 0.5$ and vanishes as $\Lambda_i \to 0$ or 1. What does this say about which respondents/bids contribute most *information* (curvature) to your estimates — and therefore where you should place bids in an SBDC design? *(Answer: observations with predicted probability near* 0.5 *contribute the most curvature, hence the most information; bids so high that almost everyone says "no," or so low that almost everyone says "yes," contribute almost nothing. Place bids where the yes-probability is intermediate — the same conclusion Chapter 1's Bernoulli information gave, now from the Hessian weight. A design implication straight from calculus.)*

2.9 Taylor expansion and the asymptotic normality of the MLE

We can now sketch *why* your estimator is approximately normal — the fact that licenses every standard error and confidence interval you report. The full theorem is Chapter 11; here is the argument in three lines, built entirely from this chapter's tools.

Start from the FOC $\mathbf{s}(\hat{\theta}) = \mathbf{0}$ and Taylor-expand the *score* around the true value θ_0 (a first-order expansion of a vector function, using the Hessian as the derivative of the score):

$$\mathbf{0} = \mathbf{s}(\hat{\theta}) \approx \mathbf{s}(\theta_0) + \mathbf{H}(\theta_0)(\hat{\theta} - \theta_0).$$

Solve for the estimation error:

$$\hat{\theta} - \theta_0 \approx -\mathbf{H}(\theta_0)^{-1}\mathbf{s}(\theta_0).$$

Rescale by $\sqrt{n}$ and split the Hessian and score by n:

$$\sqrt{n}(\hat{\theta} - \theta_0) \approx \Big[-\tfrac{1}{n}\mathbf{H}(\theta_0) \Big]^{-1} \cdot \tfrac{1}{\sqrt{n}}\mathbf{s}(\theta_0).$$

Now two limit theorems from Part II do the work (Chapter 6): by the **law of large numbers**, $-\frac{1}{n}\mathbf{H}(\theta_0) \to \mathbf{I}_1(\theta_0)$, the per-observation information; by the **central limit theorem**, the score $\frac{1}{\sqrt{n}}\mathbf{s}(\theta_0)$ — a sum of mean-zero terms — converges to $N(\mathbf{0}, \mathbf{I}_1(\theta_0))$. Combining,

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(\mathbf{0},\ \mathbf{I}_1(\theta_0)^{-1}),$$

which is Chapter 1's Proposition 1.5. **The Hessian appears twice** — once as the curvature that gets inverted, once (as its expectation) as the variance of the score — and that is why the asymptotic covariance is the *inverse information*. You do not need the full rigor yet; you need to see that the normal approximation is a **Taylor expansion of the score plus the LLN and CLT**, so that when Chapter 11 makes it precise, and when Chapter 13 warns that it *fails* if the Hessian is near-singular or the maximum is only local, you understand exactly which step breaks.

2.10 Marginal effects and elasticities

A coefficient in a nonlinear model is not, in general, the effect of the variable. The *effect* is a derivative — a marginal effect — and this section is about computing and reporting it correctly, a routine but routinely-botched task in stated-preference tables.

Definition 2.6 (Marginal effect). The **marginal effect** of covariate x_j on an outcome $m(\mathbf{x})$ is the partial derivative $\partial m(\mathbf{x})/\partial x_j$, the *ceteris paribus* change in the outcome per unit change in x_j.

Linear model — the easy case. If $\mathbb{E}[y \mid \mathbf{x}] = \mathbf{x}'\beta$, then $\partial \mathbb{E}[y]/\partial x_j = \beta_j$: the coefficient *is* the marginal effect, constant everywhere. This is why OLS coefficients read directly.

Nonlinear model — the coefficient is not the effect. For the logit, using the chain rule (§2.4) on $P(\mathbf{x}) = \Lambda(\mathbf{x}'\beta)$:

$$\frac{\partial P(\mathbf{x})}{\partial x_j} = \Lambda(\mathbf{x}'\beta)(1 - \Lambda(\mathbf{x}'\beta))\,\beta_j = P(\mathbf{x})(1 - P(\mathbf{x}))\,\beta_j.$$

The marginal effect is the coefficient **scaled by** $P(1-P)$, which depends on where you evaluate it. Two consequences you must respect when reporting:

1. **The marginal effect varies across respondents.** It is largest at $P = 0.5$ (scale 0.25) and shrinks toward the extremes. There is no single "the marginal effect"; there is a marginal effect *at a point*. Conventions: the **Average Marginal Effect (AME)** averages $\partial P/\partial x_j$ over the sample; the **Marginal Effect at the Mean (MEM)** evaluates it at $\bar{\mathbf{x}}$. These differ, and the literature uses both — state which you report (a "competing conventions" flag).
2. **Only the *sign* and *relative* magnitude of coefficients are directly interpretable; levels require the scaling.** Comparing β_j across two logit models with different error scales is invalid (Chapter 18, Chapter 25), but the marginal effect, being a probability derivative, is more comparable.

Worked Example 2.1 (Marginal effect of a bid, verified). Suppose an SBDC-style logit has bid coefficient $\beta_{\text{bid}} = -0.03$ (per euro). The marginal effect of raising the bid by one euro on the probability of "yes" is $P(1-P)(-0.03)$: - at a respondent with $P = 0.5$: $0.25 \times (-0.03) = -0.0075$ — a one-euro higher bid lowers the "yes" probability by 0.75 percentage points; - at a respondent with $P = 0.9$ (a low bid they will likely accept): $0.09 \times (-0.03) = -0.0027$ — only 0.27 points. Same coefficient, marginal effect nearly three times larger in the middle of the distribution than in the tail. Reporting "$\beta_{\text{bid}} = -0.03$" as "the effect of the bid" is meaningless without saying *where*.

Definition 2.7 (Elasticity). The **elasticity** of y with respect to x is the ratio of *proportional* changes,

$$\varepsilon_{y,x} = \frac{\partial y/y}{\partial x/x} = \frac{\partial y}{\partial x} \cdot \frac{x}{y} = \frac{\partial \ln y}{\partial \ln x},$$

a unit-free "percent change in y per percent change in x." Elasticities are the natural currency for scale-free comparison (demand elasticities, WTP elasticities with respect to income or scope). **Worked check (verified):** for $P(\text{bid}) = \Lambda(1 - 0.03 \cdot \text{bid})$ at a bid of 30, $P \approx 0.525$, $\partial P/\partial \text{bid} \approx -0.0075$, so the elasticity of the "yes" probability with respect to the bid is $(-0.0075)(30)/0.525 \approx -0.43$: a one-percent higher bid lowers the acceptance probability by about 0.43 percent, at that point.

> **Active recall 2.C.** A referee writes: "Your logit says the cost coefficient is -0.04, so a €10 cost increase reduces choice probability by 0.4." What is wrong, and how do you correctly compute the effect? *(Answer: the referee treated the coefficient as a linear marginal effect. The correct effect of a €10 change is not* 10×0.04*; it is (approximately)* $P(1-P) \times (-0.04) \times 10$ *evaluated at the relevant point — smaller, and point-dependent — or better, the difference* $P(\text{cost}+10) - P(\text{cost})$ *for a discrete change. Report an AME or a discrete change, and state the convention.)*

2.11 Marginal WTP as a ratio of partial derivatives — and why WTP space differs

This is the section your PDF flags most insistently: *"derivare la marginal WTP come rapporto tra coefficienti; capire perché stimare direttamente in WTP space è diverso da calcolare WTP ex post."* We derive it cleanly, then explain the reparameterization that defines WTP space, and connect both to the fragility you saw numerically in Chapter 1.

Setup. Let a respondent's (deterministic) utility depend on an attribute q (say, an environmental quality level) and on the cost c they must pay:

$$V(q, c) = \beta_q \, q - \beta_c \, c + (\text{other terms}), \qquad \beta_c > 0,$$

where β_c is the **marginal (dis)utility of cost** — equivalently, the marginal utility of income, since paying c reduces income. (Signs: more q is good, paying more is bad.)

Definition 2.8 (Marginal willingness to pay). The **marginal WTP** for attribute q is the amount of money that exactly offsets, in utility, a marginal increase in q — i.e., the change in c that keeps utility constant when q rises by a unit. Formally, it is the marginal rate of substitution between q and money.

Derivation via the total differential (using §2.4). Hold utility constant along an indifference set: $dV = 0$. The total differential is

$$dV = \frac{\partial V}{\partial q}\, dq + \frac{\partial V}{\partial c}\, dc = 0.$$

Solve for the trade-off dc/dq that keeps V fixed:

$$\left.\frac{dc}{dq}\right|_{V=\text{const}} = -\frac{\partial V/\partial q}{\partial V/\partial c}.$$

With the linear specification, $\partial V/\partial q = \beta_q$ and $\partial V/\partial c = -\beta_c$, so

$$\boxed{\text{WTP}_q = \left.\frac{dc}{dq}\right|_{V=\text{const}} = -\frac{\beta_q}{-\beta_c} = \frac{\beta_q}{\beta_c}}.$$

Marginal WTP is a **ratio of two partial derivatives of utility** — the marginal utility of the attribute over the marginal utility of money. (Chapter 24 writes it as $-\beta_{\text{attr}}/\beta_{\text{cost}}$ when the cost coefficient is defined with its own negative sign; the content is identical.) The indifference-curve slope in (q, c) space *is* WTP — a geometric reading we develop with the implicit function theorem in §2.12.

Why the ratio is fragile — the calculus of the fat tail. Since WTP $= \beta_q/\beta_c$, its sensitivity to the denominator is, by the quotient rule,

$$\frac{\partial\,\text{WTP}}{\partial\beta_c} = -\frac{\beta_q}{\beta_c^2}.$$

As $\beta_c \to 0$, this derivative blows up like $1/\beta_c^2$: **small errors in a near-zero cost coefficient produce enormous swings in WTP.** This is the analytic explanation of the heavy WTP tails you simulated in Chapter 1 (Worked Example 1.5) and the pathology your PDF lists in §6.4. The delta-method standard error of WTP is $\text{Var}(\widehat{\text{WTP}}) \approx \nabla g'\,\mathbf{V}\,\nabla g$ with $g = \beta_q/\beta_c$ and gradient $\nabla g = (1/\beta_c,\ -\beta_q/\beta_c^2)'$; the second component's $1/\beta_c^2$ is exactly why an imprecise cost coefficient detonates the WTP standard error. Calculus and covariance tell the same story.

Preference space vs WTP space — why they are different acts. In **preference space** you estimate β_q and β_c (and, in a mixed logit, distributions for each), then form the ratio β_q/β_c *after* estimation. The distribution of a ratio of two random coefficients is analytically nasty — heavy-tailed, possibly without a finite mean — which is the source of the instability. In **WTP space** you *reparameterize the utility function before estimating.* Factor out the cost coefficient:

$$V(q,c) = \beta_c\Big(\frac{\beta_q}{\beta_c}\,q - c\Big) = \beta_c(\omega_q\,q - c), \qquad \omega_q \equiv \frac{\beta_q}{\beta_c} = \text{WTP}_q.$$

Now the parameters are $(\beta_c,\ \omega_q)$: a **scale** β_c multiplying everything, and the WTP ω_q *itself* as a primitive parameter. You put a distribution directly on ω_q (the object you care about) instead of inferring it from a ratio. The benefits: WTP has an interpretable, directly-modelled distribution, and you can *constrain* it (e.g., keep it positive, or bound its tails) — addressing the fragility at its root. The cost (Chapter 24 develops this): WTP space **imposes restrictions** — the factorization assumes a particular way the cost coefficient scales all attributes, and the choice of distribution for ω_q and β_c has real consequences that a preference-space model would express differently. So estimating in WTP space is not "computing WTP more conveniently"; it is fitting a *different model* with different assumptions, whose likelihood, though built from the same data, has a different shape. That is the precise sense of "diverso da calcolare WTP ex post" — and the reason your EC-MXL's choice of space is a modeling decision to defend, not a formatting choice.

> **Stop and think 2.3.** In WTP space, β_c multiplies the whole utility. Recalling that in a logit utility is only identified up to scale (Chapter 1 §1.4, Chapter 21), predict what role β_c plays and why a *large* estimated β_c tends to accompany *tighter* choice probabilities. *(Answer: β_c acts as the scale parameter of the model — it is inversely related to the error variance. A larger β_c means less relative noise, so choices are more deterministic and probabilities more extreme. This is why "scale" and "the cost coefficient" are entangled, foreshadowing the scale-vs-preference-heterogeneity problem of Chapter 25: a treatment that changes apparent WTP might be changing β_c/scale, not the ω_q you meant to measure.)*

2.12 The implicit function theorem: WTP, indifference curves, and comparative statics

Often the object you care about is not written as an explicit function but is defined *implicitly* by an equation — "the c that makes utility equal to $\bar{U}$," "the bid at which the yes-probability is 0.5," "how the optimal choice shifts when a parameter changes." The **implicit function theorem (IFT)** tells you when such an implicit relationship defines a genuine function and gives you its derivative without solving explicitly.

Theorem 2.1 (Implicit function theorem, scalar version). Suppose $F(x, c) = 0$ defines c as a function of x near a point (x_0, c_0) where $F(x_0, c_0) = 0$. If F is continuously differentiable and $\partial F/\partial c \neq 0$ at that point, then there is a differentiable function $c = c(x)$ locally, with

$$\frac{dc}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial c}.$$

The condition $\partial F/\partial c \neq 0$ is doing exactly what a non-zero denominator does everywhere in this chapter: it guarantees the implicit relationship is well-defined and the derivative is finite.

Application 1 — WTP as an indifference-curve slope. Let $F(q, c) = V(q, c) - \bar{U} = 0$ define the indifference curve (the combinations of quality and cost giving fixed utility $\bar{U}$). The IFT gives its slope:

$$\frac{dc}{dq}\Big|_{V=\bar{U}} = -\frac{\partial V/\partial q}{\partial V/\partial c} = \frac{\beta_q}{\beta_c} = \text{WTP}_q,$$

reproducing §2.11 and adding a picture: **WTP is the slope of the indifference curve in attribute–money space.** The IFT's non-vanishing-denominator condition here is $\partial V/\partial c \neq 0$, i.e., $\beta_c \neq 0$ — the very condition whose *near*-failure produces WTP instability. The mathematics warns you at the level of existence: if the marginal utility of money is zero, WTP is not even well-defined, let alone estimable.

Application 2 — the median-WTP bid in SBDC. In your CV model, the "median WTP" is the bid b^* at which the yes-probability equals one half: $\Phi(\alpha + \gamma b^*) = 0.5$, i.e. $F(b^*; \alpha, \gamma) = \alpha + \gamma b^* = 0$, giving $b^* = -\alpha/\gamma$. The IFT (or direct differentiation) tells you how the estimated median WTP moves with the parameters, $\partial b^*/\partial \alpha = -1/\gamma$ and $\partial b^*/\partial \gamma = \alpha/\gamma^2$ — again a $1/\gamma$ and $1/\gamma^2$ sensitivity to the bid coefficient γ (the cost-side parameter), the same fragility signature. Chapter 15 uses exactly this to derive mean and median WTP and their standard errors; you now see the IFT underneath it.

Application 3 — comparative statics. The IFT is the general tool for "how does the optimum move when a parameter changes." If an optimal choice $q^*(\alpha)$ is defined by a first-order condition $\partial V(q, \alpha)/\partial q = 0$, differentiating the FOC implicitly gives $dq^*/d\alpha = -(\partial^2 V/\partial q \partial \alpha)/(\partial^2 V/\partial q^2)$ — the denominator is the *second-order condition* (curvature), which must be non-zero (a real optimum, §2.7) for the comparative static to exist. This is the machinery behind every "how does the optimal ecosystem-service allocation respond to a price change" question in your MDCEV welfare analysis (Chapters 40–42), and behind the Kuhn-Tucker comparative statics of Chapter 3.

> **Active recall 2.D.** State the one condition the implicit function theorem needs in order for "the cost that holds utility constant" to define WTP as a well-behaved function, and connect it to a fragility your PDF warns about. *(Answer: $\partial V/\partial c \neq 0$, i.e., the marginal utility of money/cost is non-zero, $\beta_c \neq 0$. Its near-failure — a near-zero cost coefficient — is exactly the source of unstable, heavy-tailed WTP, §6.4 of the plan.)*

2.13 Failure modes: how calculus misleads in practice

F1. Reading a nonlinear coefficient as a marginal effect. In logit/probit/MDCEV, β_j is not $\partial(\text{outcome})/\partial x_j$; the marginal effect is β_j scaled by a point-dependent factor ($P(1-P)$ for logit; ϕ for probit). *Symptom:* "a unit of x changes the probability by β_j." *Fix:* report AME or MEM (state which), or a discrete change; never present the raw coefficient as an effect size.

F2. WTP-ratio instability from a near-zero denominator. WTP $= \beta_q/\beta_c$ has $\partial \text{WTP}/\partial\beta_c = -\beta_q/\beta_c^2$, exploding as $\beta_c \to 0$. *Symptom:* enormous WTP point estimates or standard errors, fat simulated tails. *Fix:* diagnose cost-coefficient precision (Chapter 1 conditioning); consider WTP-space reparameterization with a sign-constrained/bounded cost coefficient; report median and percentile intervals (Chapter 24).

F3. Trusting a maximum without checking curvature. A zero gradient (FOC) is necessary, not sufficient. *Symptom:* "converged" reported at a saddle or a flat ridge. *Fix:* verify the SOC — negative-definite Hessian, all eigenvalues of $-\mathbf{H}$ comfortably positive (Chapter 1 §1.7; Chapter 13).

F4. Assuming global concavity where it does not hold. Logit and probit log-likelihoods are globally concave; **mixed logit / MDCEV simulated log-likelihoods are not.** *Symptom:* treating a single converged run of an EC-MXL as definitive. *Fix:* multiple starting values, compare log-likelihoods, expect local maxima (Chapter 13, Chapter 52).

F5. Confusing marginal effect and elasticity. A marginal effect is a level change per unit; an elasticity is a percent change per percent. *Symptom:* "a 1% increase" reported when a "1-unit increase" was computed, or vice versa. *Fix:* be explicit about units; use elasticities for scale-free comparison, marginal effects for unit-specific policy statements.

F6. Ignoring cross-partials / interactions. The off-diagonal Hessian entries carry the interaction between parameters (and, in a model with interaction terms, the marginal effect of x_j depends on x_ℓ). *Symptom:* interpreting a main-effect coefficient in a model with interactions as the total effect. *Fix:* compute the full derivative $\partial m/\partial x_j$ including interaction terms (Chapter 17).

F7. Extrapolating a local (Taylor) approximation too far. The normal approximation to the MLE and the delta method are *local* second-order approximations; they degrade for highly nonlinear functions (like a WTP ratio) or far from the estimate. *Symptom:* symmetric delta-method intervals for a skewed WTP. *Fix:* use simulation (Krinsky–Robb, Chapter 9) or the bootstrap when the function is strongly nonlinear.

2.14 Diagnostic questions for your own papers

1. **Score/FOC.** Can I write the score of my model and confirm the software's "converged" corresponds to $\mathbf{s}(\widehat{\theta}) \approx \mathbf{0}$? *(§2.2, §2.8.)*
2. **Curvature/SOC.** Is the Hessian negative definite at my optimum? Are any eigenvalues near zero (flat direction = weak identification)? *(§2.3, §2.7.)*
3. **Concavity.** Is my model in the globally-concave family (logit/probit) or the non-concave family (mixed logit/MDCEV)? If the latter, have I used multiple starts? *(§2.6, F4.)*
4. **Marginal effects.** Am I reporting marginal effects (AME/MEM) rather than raw nonlinear coefficients? Did I state the convention and the evaluation point? *(§2.10, F1.)*
5. **WTP as a ratio.** Have I checked the precision of my cost coefficient before trusting WTP magnitudes? Is β_c safely away from zero? *(§2.11, F2.)*
6. **Space choice.** If I estimate in WTP space, can I state the restriction the reparameterization imposes, and why it is defensible for my attributes? *(§2.11, Stop-and-think 2.3.)*
7. **Approximation validity.** For skewed quantities (WTP), am I using simulation rather than a symmetric delta-method interval? *(§2.9, F7.)*

2.15 Exercises

(A) Comprehension.

A1. For $f(\theta_1, \theta_2) = 3\theta_1^2 + 2\theta_1\theta_2 + \theta_2^2$, compute the gradient and Hessian. Is f convex, concave, or neither? Justify with the Hessian.

A2. A logit has cost coefficient $\beta_c = -0.05$. At a respondent with predicted choice probability $P = 0.2$, compute the marginal effect of a one-unit cost increase, and the marginal effect at $P = 0.5$. Which respondent's choice is more sensitive to cost, and why?

A3. Utility is $V = 0.8\,q - 0.04\,c$. Compute the marginal WTP for q. If instead β_c were estimated as -0.005 (with the same $\beta_q = 0.8$), what is WTP, and what does the change illustrate?

(B) Derivation.

B1. Derive the score and Hessian of the **probit** log-likelihood $\ell_i = y_i \ln \Phi(\mathbf{x}_i'\beta) + (1 - y_i)\ln(1 - \Phi(\mathbf{x}_i'\beta))$. Show the score has the "generalized residual" form. *(You may leave the Hessian in terms of ϕ, Φ, and $\mathbf{x}_i$.)*

B2. Starting from $V(q, c) = \beta_q q - \beta_c c$, derive WTP$= \beta_q/\beta_c$ two ways: (i) via the total differential with $dV = 0$; (ii) via the implicit function theorem applied to $V(q, c) = \bar{U}$. Then reparameterize V into WTP space and identify the scale parameter.

B3. Using a second-order Taylor expansion of the log-likelihood at the MLE, show that the log-likelihood is locally approximately quadratic and explain, in terms of the Hessian's definiteness, the difference between (i) a well-identified parameter, (ii) a weakly-identified parameter, and (iii) a saddle point.

(C) Research application / critique.

C1. A DCE paper estimates a mixed logit in preference space and reports a mean WTP of €340 for an attribute, with a 95% delta-method interval of €[−120, €800]. The cost coefficient is "significant at 10%." Critique this in terms of the calculus of §2.11: what is likely happening, why is the interval symmetric and wide, and what two changes would you recommend?

C2. Your co-author reports, from a probit SBDC, that "the bid coefficient is −0.03, so willingness to pay falls by 3 cents per euro of bid." Explain what is conceptually confused here (relate the bid coefficient to median WTP $= -\alpha/\gamma$, §2.12), and state what the bid coefficient actually governs.

C3. In your EC-MXL you must choose between preference space and WTP space. Write a short defense of estimating in WTP space for a *cost-vector* experiment (where you manipulate the cost levels across arms), explicitly naming (i) the fragility WTP space addresses, (ii) the restriction it imposes, and (iii) how scale (Stop-and-think 2.3) could confound a cost-vector treatment effect regardless of space.

2.16 Solutions

A1. $\nabla f = (6\theta_1 + 2\theta_2,\ 2\theta_1 + 2\theta_2)'$. Hessian $\mathbf{H} = \begin{pmatrix} 6 & 2 \\ 2 & 2 \end{pmatrix}$ (constant). Leading minors: $6 > 0$ and $\det = 12 - 4 = 8 > 0$, so $\mathbf{H}$ is positive definite everywhere $\Rightarrow$ f is **strictly convex** (a bowl opening up). Not concave.

A2. Marginal effect $= P(1 - P)\beta_c$. At $P = 0.2$: $0.16 \times (-0.05) = -0.008$. At $P = 0.5$: $0.25 \times (-0.05) = -0.0125$. The $P = 0.5$ respondent is **more** cost-sensitive: near the middle of the distribution the choice is on a knife-edge, so a cost change moves the probability more; near the extremes the respondent has largely made up their mind.

A3. WTP $= \beta_q/\beta_c = 0.8/0.04 =$ €20. With $\beta_c = 0.005$: WTP $= 0.8/0.005 =$ €160. An eight-fold change in WTP from a small change in the (imprecisely estimated, near-zero) cost coefficient — the $1/\beta_c$ sensitivity of §2.11 in action. It illustrates why WTP magnitudes are untrustworthy when the cost coefficient is weak.

B1. With $P_i = \Phi(\mathbf{x}_i'\beta)$ and $\Phi' = \phi$: $\partial\ell_i/\partial\beta = [y_i/\Phi_i - (1 - y_i)/(1 - \Phi_i)]\,\phi_i\,\mathbf{x}_i$. Define the **generalized residual** $u_i = \phi_i[y_i - \Phi_i]/[\Phi_i(1 - \Phi_i)]$; then $\partial\ell_i/\partial\beta = u_i\mathbf{x}_i$ (check: combine the two terms over the common denominator $\Phi_i(1 - \Phi_i)$, numerator $y_i(1 - \Phi_i)\phi_i - (1 - y_i)\Phi_i\phi_i = \phi_i(y_i - \Phi_i)$). So the score is $\sum_i u_i\mathbf{x}_i$, the probit analogue of $\sum_i(y_i - \Lambda_i)\mathbf{x}_i$. The Hessian is $\sum_i \partial u_i/\partial(\mathbf{x}_i'\beta)\,\mathbf{x}_i\mathbf{x}_i'$; differentiating u_i (product/quotient rules, using $\phi'(z) = -z\phi(z)$) yields a negative term that establishes concavity — the standard result that probit is globally concave. *(Leaving the Hessian in ϕ, Φ form is acceptable; the sign is what matters.)*

B2. (i) $dV = \beta_q\,dq - \beta_c\,dc = 0 \Rightarrow dc/dq = \beta_q/\beta_c$. (ii) $F(q,c) = \beta_q q - \beta_c c - \bar{U}$; IFT gives $dc/dq = -F_q/F_c = -\beta_q/(-\beta_c) = \beta_q/\beta_c$. Both give WTP $= \beta_q/\beta_c$. Reparameterization: $V = \beta_c(\frac{\beta_q}{\beta_c}q - c) = \beta_c(\omega_q q - c)$ with $\omega_q = \beta_q/\beta_c$ the WTP; β_c is the **scale** parameter multiplying the whole utility.

B3. At the MLE, $\nabla\ell(\hat{\theta}) = \mathbf{0}$, so $\ell(\theta) \approx \ell(\hat{\theta}) + \frac{1}{2}(\theta - \hat{\theta})'\mathbf{H}(\hat{\theta})(\theta - \hat{\theta})$ — a quadratic in the displacement. (i) A well-identified parameter has strong negative curvature along its axis (large |eigenvalue|): the quadratic bowl is steep, the parameter is pinned down, small SE. (ii) A weakly-identified parameter corresponds to a near-zero eigenvalue: the bowl is nearly flat in that direction, the likelihood barely changes, SE is huge. (iii) A saddle has **H** indefinite (mixed-sign eigenvalues): the "bowl" goes up in some directions and down in others, so $\hat{\theta}$ is not a maximum at all and reported SEs are invalid.

C1. The symmetric, very wide interval spanning zero, plus a barely-significant cost coefficient, is the signature of the WTP-ratio pathology: WTP $= \beta_q/\beta_c$ with β_c near zero has $\partial\text{WTP}/\partial\beta_c = -\beta_q/\beta_c^2$ exploding, so the delta-method variance $\nabla g'\mathbf{V}\nabla g$ is enormous, and because the delta method is a *local linear* approximation it produces a *symmetric* interval even though the true WTP distribution is skewed and heavy-tailed. Recommendations: (i) reduce the fragility at the source — reparameterize in WTP space with a sign-constrained cost coefficient, or improve cost identification via design spread/conditioning (Chapter 1 §1.10); (ii) report a **simulation-based** (Krinsky–Robb) median and percentile interval instead of the symmetric delta-method one (§2.9 F7, Chapter 9).

C2. The bid coefficient γ is not "WTP falling per euro of bid"; it is the slope of the latent-utility index in the probability of "yes," and it enters median WTP as $b^* = -\alpha/\gamma$ (§2.12). A larger $|\gamma|$ makes the acceptance curve steeper (responses more sharply determined by the bid) but does not itself equal a rate of change of WTP. WTP is a *level* recovered from the whole model (Chapter 15); the coefficient governs the *steepness/scale* of the response function and, jointly with the constant, the *location* of median WTP.

C3. (i) WTP space directly models the WTP parameter ω_q and lets you constrain its sign and tails, addressing the ratio instability that afflicts a cost-vector experiment where the cost coefficient is central and its estimate may be imprecise across arms. (ii) The restriction: factoring out β_c assumes the cost coefficient scales *all* attribute WTPs multiplicatively and imposes a chosen distributional form on ω_q and on the scale β_c; a preference-space model would allow a different, ratio-implied WTP distribution. (iii) Because β_c *is* the scale in WTP space, a cost-vector treatment that changes the error variance (scale) rather than true preferences would show up as a change in β_c and hence in apparent WTP — so a "cost-vector effect" could be scale heterogeneity, not preference formation, regardless of the space you estimate in. Separating these requires the tools of Chapter 25, not a change of parameterization.

2.17 Spaced review prompts

+1 day. From memory, differentiate the logit log-likelihood to get the score $\sum_i (y_i - \Lambda_i)\mathbf{x}_i$ and the Hessian $-\sum_i \Lambda_i(1-\Lambda_i)\mathbf{x}_i\mathbf{x}_i'$, and state in one sentence why the Hessian's sign proves global concavity.

+1 week. Reconstruct the WTP derivation two ways (total differential and IFT), then explain in three sentences why estimating in WTP space is a different model from computing WTP from a preference-space fit — naming the restriction and the fragility.

+1 month. On one of your own models, compute an average marginal effect for the cost variable by hand from the coefficient and the fitted probabilities, and compare it to the raw coefficient. Then write one paragraph on whether your reported WTP magnitudes are safe given your cost coefficient's precision (tie to §2.11 and Chapter 1's covariance reading).

2.18 How this chapter connects to the rest of the book

- **Chapter 1** supplied the gradient/Hessian as linear-algebra objects and the covariance as an inverse-Hessian; this chapter *computes* them and gives the Taylor argument behind Proposition 1.5.
- **Chapter 3 (Constrained Optimization)** takes the FOC/SOC and Lagrangian machinery to constrained problems — utility maximization, expenditure minimization, and the Kuhn-Tucker conditions of MDCEV; the implicit function theorem here is the comparative-statics tool there.
- **Chapters 11–13 (Maximum Likelihood)** make §2.8–§2.9 rigorous: the score, information, asymptotic normality, and the computational consequences of losing concavity (local maxima, starting values).
- **Chapter 15 (CV Likelihoods)** builds the SBDC/probit and payment-card likelihoods whose score you derived in §2.8/B1, and uses the IFT median-WTP logic of §2.12 to get mean and median WTP.
- **Chapters 18, 21, 24, 25 (binary response, MNL, WTP space, scale vs preference)** rest on the marginal-effect and WTP-ratio calculus of §2.10–§2.11; Chapter 24 is the full treatment of the preference-vs-WTP-space distinction sketched here, and Chapter 25 confronts the scale entanglement of Stop-and-think 2.3.
- **Chapters 40–42 (MDCEV, welfare)** use concavity/satiation (§2.6), comparative statics (§2.12), and the ratio-of-marginal-utilities logic to derive demand and compensating variation.
- **Chapter 7 (delta method)** and **Chapter 9 (simulation)** operationalize §2.9 and §2.11-F7: propagating parameter uncertainty into WTP either by the gradient (delta) or by simulation (Krinsky–Robb).

The chapter feeds directly into the three highest-dependency chapters — **Ch 11**, **Ch 24**, **Ch 30** — by supplying, respectively, the likelihood calculus, the WTP-ratio derivation, and the curvature-based notion of weak identification.

2.19 Research translation task

Take the **same model** you used in Chapter 1's translation task (ideally the SBDC probit or the EC-MXL). Produce a two-page note:

1. **Write the score by hand.** For the SBDC probit, write $\sum_i u_i \mathbf{x}_i$ with the generalized residual explicitly; for the choice model, write the score of one choice observation. Confirm the software's converged gradient is (near) zero.
2. **Report the marginal effect of cost/bid**, both as the raw coefficient and as an average marginal effect computed from your fitted probabilities. Comment on the gap (§2.10).
3. **Derive your WTP** as a ratio of the relevant partial derivatives, state the value, and compute $\partial \text{WTP}/\partial \beta_c$ at your estimate to quantify its sensitivity to the cost coefficient (§2.11). Decide whether your cost coefficient is "safely away from zero."
4. **If you estimate in WTP space**, write the reparameterized utility, identify your scale parameter, and state in two sentences the restriction you are imposing and why it is defensible for your attributes.
5. **Close** with the spine sentence: *"WTP for [attribute] is identified as the ratio [marginal utility of attribute]/[marginal utility of cost]; its precision is governed by the curvature of the likelihood in the cost direction; it is fragile because [1/β_c² sensitivity / near-zero cost draws / scale entanglement]."*

Carry this into Chapter 11 (where the score/Hessian become the full ML apparatus) and Chapter 24 (where the space choice is decided).

2.20 Bibliography

Core. - Simon & Blume, *Mathematics for Economists*. The standard graduate reference for multivariate calculus, the implicit function theorem, concavity, and optimization, written for exactly your purposes;

read the chapters on derivatives, the IFT, and concave/convex functions. *(Verify edition.)* - Chiang & Wainwright, *Fundamental Methods of Mathematical Economics*. Gentler on comparative statics and the total-differential/IFT technique used in §2.11–§2.12; good for building fluency. *(Verify edition.)* - Greene, *Econometric Analysis*, chapters on the binary choice models and the appendix on estimation/differentiation. Where the logit/probit score and Hessian of §2.8 appear in econometric context. *(Verify edition/chapter.)*

Applied. - Train, *Discrete Choice Methods with Simulation*. Chapters on logit and mixed logit derive exactly the score/Hessian and the WTP-space vs preference-space distinction of §2.8 and §2.11; the single most relevant applied source for this chapter. Freely available from the author — *verify link.* - Cameron & Trivedi, *Microeconometrics: Methods and Applications*, on marginal effects (AME vs MEM), elasticities, and their correct reporting — the §2.10 material at implementation depth. *(Verify edition.)* - Hole, "Calculating Murphy–Topel and delta-method standard errors" / the `margins`-style tooling documentation in R/Stata for marginal effects. Practical computation of §2.10 and the delta-method interval of §2.9. *(Verify exact reference/package; I am not certain of the precise citation — confirm before citing.)*

Advanced. - Magnus & Neudecker, *Matrix Differential Calculus with Applications in Statistics and Econometrics*. The systematic apparatus for differentiating likelihoods and WTP functions with respect to vectors/matrices — the rigorous backing for §2.8 and the delta method. *(Verify edition.)* - Train & Weeks, "Discrete Choice Models in Preference Space and Willingness-to-Pay Space," in *Applications of Simulation Methods in Environmental and Resource Economics* (eds. Scarpa & Alberini). The canonical treatment of the §2.11 distinction; essential before Chapter 24. *(Verify exact title/editors/year.)* - Rockafellar, *Convex Analysis*. The deep reference on concavity/convexity if you later need it for optimization theory; well beyond current needs but the definitive source. *(Verify.)*

Where editions, page numbers, or exact titles are marked "verify," treat the entry as a pointer and confirm before citing — the book never presents an unverified reference as settled.

End of Chapter 2. Next in Part I: Chapter 3 — Constrained Optimization and Duality, which takes the FOC/SOC and implicit-function machinery into constrained problems (Lagrangians, Kuhn-Tucker conditions, corner vs interior solutions, expenditure minimization, indirect utility) — the mathematical engine of MDCEV and compensating variation. Tell me when to write it, or request a different chapter.

Chapter 3 — Constrained Optimization and Duality

Part I: Mathematical Foundations for Econometrics and Welfare Analysis

> **The spine question, applied to this chapter.** Chapter 2 gave you unconstrained optimization — set the gradient to zero. But no economic agent optimizes freely: a consumer maximizes utility *subject to a budget*, a respondent allocates a fixed sum *across ecosystem services*, a planner maximizes net benefit *subject to resources*. The mathematics of "optimize subject to a constraint" is the Lagrangian and its Kuhn-Tucker generalization, and it is, quite literally, the engine of the two objects at the center of your welfare work: the **MDCEV allocation model** (a Kuhn-Tucker demand system with corner solutions) and the **compensating variation** (defined through expenditure minimization). This chapter builds that engine. By the end you will read a Lagrange multiplier as a shadow price, tell a genuine economic corner from a survey artifact, derive indirect utility and the expenditure function as two sides of one problem, and see exactly why "constrained optimization is the mathematical core of MDCEV" is not a slogan but a description. The spine question here becomes: *when I interpret a zero allocation, a WTP, or a compensating variation, what optimization problem am I assuming the respondent solved, is that problem well-posed, and how fragile is my welfare number to that assumption?*

3.0 Conceptual motivation: why every welfare number is a constrained optimum

Your PDF is unusually explicit about this chapter's stakes. Under *Ottimizzazione vincolata* it writes: *"Ti serve direttamente per MDCEV, BEBCE, budget allocation, outside good, compensating variation e CBA. Nel tuo progetto MDCEV il cuore del problema è modellare scelte in cui l'individuo può allocare zero a certi ecosystem services e importi positivi ad altri."* And then it lists the judgments you must be able to make: *when is a zero a true corner and not a questionnaire artifact; when is the budget constraint economically credible; when is the outside good really a numeraire; when is a compensating variation well-defined; when does the model force substitutability or complementarity the data do not identify.* Every one of those judgments is a statement about a constrained optimization problem. Let me make the stakes concrete before the machinery.

- **Consumer choice** is $\max U(\mathbf{x})$ subject to $\mathbf{p}'\mathbf{x} \leq m$. The first-order conditions of *this* problem — not of free maximization — define demand, and the multiplier on the budget is the **marginal utility of income**, the quantity that converts utils into euros and therefore makes WTP a money measure at all (Chapter 2, §2.11; Chapter 34).
- **MDCEV** is the constrained problem in which a respondent divides a budget across many goods (your ecosystem services) plus an outside good, and *chooses zero of some goods.* The zeros are **corner solutions**, and they are governed by the **Kuhn-Tucker conditions** with **complementary slackness**. The entire likelihood of MDCEV (Chapter 41) is the probability, under an extreme-value error, that a particular pattern of interior and corner solutions satisfies those conditions. If you do not understand

the KKT conditions, MDCEV is a black box, and you cannot answer your own proposal's questions about identification, the numeraire, and complementarity/substitution.
- **Compensating variation** — the exact welfare measure you need for MDCEV welfare analysis and for the benefit side of your restoration CBA — is *defined* as a difference of **expenditure functions**, and the expenditure function is the solution of a constrained *minimization*: the cheapest bundle achieving a target utility. WTP for an environmental improvement is a compensating variation. So the number your DCE and CV papers ultimately produce is, at its foundation, a constrained-optimization object.
- **Cost-benefit analysis** uses **shadow prices** — the social value of relaxing a constraint by one unit — which are Lagrange multipliers (Chapter 43). "Shadow price" and "Lagrange multiplier" are the same concept in two vocabularies.

So this chapter is where microeconomic theory, your MDCEV proposal, and your welfare/CBA work share a single mathematical spine. We build it carefully: the constrained problem, the Lagrangian for equality constraints, the multiplier as a shadow price (via the envelope theorem), the consumer problem, the Kuhn-Tucker conditions for inequality constraints, the interior-versus-corner distinction, and finally duality — indirect utility and expenditure minimization — which delivers the compensating variation. Every abstract result is tied to a decision you must make in your own papers.

3.1 Prerequisites

From **Chapter 2** you need:

- gradients and first-order conditions (§2.2, §2.7): an interior optimum sets $\nabla f = \mathbf{0}$;
- the Hessian and second-order conditions (§2.3, §2.7): a maximum needs a negative (semi)definite Hessian;
- **concavity/quasi-concavity** (§2.6): why it makes an optimum global and demand well-behaved;
- the **marginal rate of substitution** as a ratio of marginal utilities and **WTP** $= \beta_q/\beta_c$ (§2.11);
- the **implicit function theorem** for comparative statics (§2.12).

From **Chapter 1**: definiteness and quadratic forms (§1.6–1.7). Same **notation** as before; new symbols: λ, μ for Lagrange/KKT multipliers, $\mathcal{L}$ for the Lagrangian, $V(\mathbf{p}, m)$ for indirect utility, $e(\mathbf{p}, u)$ for the expenditure function, $\mathbf{x}^M$ and $\mathbf{x}^H$ for Marshallian and Hicksian demand.

3.2 The constrained optimization problem

Definition 3.1 (Constrained optimization problem). A general constrained problem is

$$\max_{\mathbf{x}\in\mathbb{R}^n} f(\mathbf{x}) \quad \text{subject to} \quad g_i(\mathbf{x}) = 0 \ (i = 1, \dots, r) \ \text{ and } \ h_j(\mathbf{x}) \geq 0 \ (j = 1, \dots, s),$$

where f is the **objective**, the g_i are **equality constraints**, and the h_j are **inequality constraints**. The set of $\mathbf{x}$ satisfying all constraints is the **feasible set**. A point is a **constrained maximum** if it gives the largest f among feasible points.

Two features distinguish constrained from unconstrained optimization, and both matter economically:

1. **The optimum need not have $\nabla f = \mathbf{0}$.** At the top of a hill you would keep climbing, but the constraint stops you. So the FOC changes: at the optimum the objective's gradient is *balanced against* the constraints' gradients, not zero. The Lagrangian encodes that balance.
2. **Inequality constraints may or may not "bind."** A binding (active) constraint holds with equality at the optimum ($h_j = 0$); a slack (inactive) one holds strictly ($h_j > 0$) and is locally irrelevant. Which constraints bind is *part of the solution*, and for the nonnegativity constraints $x_k \geq 0$ this is exactly the **interior-versus-corner** question that defines MDCEV.

The consumer problem is the canonical case: $f = U(\mathbf{x})$ (utility), one equality/inequality constraint $g = m - \mathbf{p}'\mathbf{x} \geq 0$ (budget), and n nonnegativity constraints $x_k \geq 0$. We build up to it.

3.3 Equality constraints and the Lagrangian

Start with a single equality constraint: $\max f(\mathbf{x})$ subject to $g(\mathbf{x}) = 0$.

The geometric idea. At a constrained maximum, you cannot increase f by moving *along* the constraint surface. Moving along the surface means moving perpendicular to ∇g (the constraint's gradient). "Cannot increase f along the surface" means ∇f has no component along the surface — i.e., ∇f points entirely *perpendicular* to the surface, parallel to ∇g. So at the optimum,

$$\nabla f(\mathbf{x}^*) = \lambda \nabla g(\mathbf{x}^*)$$

for some scalar λ: the objective's gradient is a multiple of the constraint's gradient. This tangency condition is the heart of the method.

Definition 3.2 (Lagrangian). For $\max f(\mathbf{x})$ s.t. $g(\mathbf{x}) = 0$, the **Lagrangian** is

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda\, g(\mathbf{x}),$$

where λ is the **Lagrange multiplier**. Setting the gradient of $\mathcal{L}$ (with respect to *both* $\mathbf{x}$ and λ) to zero reproduces the tangency condition and the constraint:

$$\frac{\partial \mathcal{L}}{\partial x_k} = \frac{\partial f}{\partial x_k} + \lambda \frac{\partial g}{\partial x_k} = 0 \ (k = 1, \dots, n), \qquad \frac{\partial \mathcal{L}}{\partial \lambda} = g(\mathbf{x}) = 0.$$

The first n equations are the tangency condition $\nabla f = -\lambda \nabla g$ (the sign of λ is a convention; with $\mathcal{L} = f + \lambda g$ we get $\nabla f = -\lambda \nabla g$, and one adjusts signs to taste — a "competing conventions" flag: some texts write $\mathcal{L} = f - \lambda g$). The last equation just restores the constraint. So the Lagrangian is a device that turns a *constrained* problem in n variables into an *unconstrained* stationarity problem in $n + 1$ variables $(\mathbf{x}, \lambda)$ — a genuinely beautiful trick.

Second-order conditions. As in Chapter 2, the first-order conditions are necessary, not sufficient. For a constrained *maximum* one checks the **bordered Hessian** (the Hessian of $\mathcal{L}$ bordered by the constraint gradient) has the right definiteness on the tangent space. In practice, if f is concave and the constraint is linear (the consumer's budget), the FOC point is a global max and you need not fuss — another dividend of the concavity you studied in §2.6. We flag the bordered-Hessian check where it matters (MDCEV, §3.7) and otherwise lean on concavity.

> **Active recall 3.A.** In one sentence, what does the tangency condition $\nabla f = \lambda \nabla g$ say an optimizer cannot do at the constrained optimum? *(Answer: it cannot increase the objective by any feasible move along the constraint surface — the objective's gradient has no component in any direction the constraint allows.)*

3.4 The multiplier is a shadow price

The Lagrange multiplier is not a bookkeeping device; it is an economic quantity — the marginal value of the constraint. This is the single most useful interpretation in the chapter, and it comes from the **envelope theorem**.

Theorem 3.1 (Envelope theorem, constrained form — statement). Let $\mathbf{x}^*(c)$ solve $\max f(\mathbf{x})$ s.t. $g(\mathbf{x}) = c$, and let $f^*(c) = f(\mathbf{x}^*(c))$ be the **value function** (the optimized objective as a function of the constraint level c). Then

$$\frac{df^*}{dc} = \lambda^*,$$

the optimal multiplier. That is, λ **measures how much the best achievable objective rises when the constraint is relaxed by one unit.**

Why (sketch). Totally differentiate $f^*(c) = f(\mathbf{x}^*(c))$ and use the FOC $\nabla f = \lambda \nabla g$ together with $g(\mathbf{x}^*(c)) = c$: the direct effect of c through the constraint, weighted by how the objective responds, collapses to λ. The terms involving $d\mathbf{x}^*/dc$ cancel precisely because we are *at* an optimum (the FOC holds) — the same "envelope" cancellation you will see again for Roy's identity and Shephard's lemma in §3.9. The full proof is in §3.10.

The shadow-price reading, everywhere in your work.

- In the **consumer problem**, the budget multiplier is the **marginal utility of income**: one extra euro of budget buys λ extra utils. This λ is the denominator that turns marginal utility of an attribute into WTP (Chapter 2, §2.11). When your PDF asks whether "il budget constraint è economicamente credibile," it is asking whether this λ — the price of money in utils — is meaningful for the respondent; if the budget is not real, λ is not either, and the WTP built on it is not a welfare number.
- In **CBA**, the multiplier on a resource constraint is the **shadow price** of that resource — its social value at the margin, which may differ from its market price when markets are distorted (Chapter 43). "Shadow pricing" in your restoration CBA is literally the computation of these multipliers.
- In **MDCEV**, the budget multiplier λ is the common "marginal utility of income" that every consumed good's marginal-utility-per-euro is driven to equal — and that the corner goods fail to reach (§3.7). It is the hinge of the whole model.

Stop and think 3.1. A respondent is given a hypothetical budget to allocate across ecosystem services, but faces no real consequence for the allocation. Predict what happens to the interpretation of the budget multiplier λ, and hence to any welfare measure derived from the exercise. *(Answer: with no real trade-off, the respondent's marginal utility of "income" in the exercise is not tied to genuine scarcity, so λ loses its meaning as a util-to-euro conversion; welfare measures built on it — WTP, CV — are stated-preference indicators, not money-metric welfare. This is the consequentiality/incentive-compatibility problem of Chapters 28 and 37, showing up already in the multiplier.)*

3.5 The consumer problem: utility maximization

Now the canonical application, worked in general so the FOC's economic content is unmistakable.

The problem. A consumer chooses a bundle $\mathbf{x} = (x_1, \dots, x_n) \geq \mathbf{0}$ to

$$\max_{\mathbf{x}} U(\mathbf{x}) \quad \text{s.t.} \quad \mathbf{p}'\mathbf{x} = m,$$

with prices $\mathbf{p}$ and income m. (We use the budget with equality, assuming nonsatiation so all income is spent; nonnegativity is handled in §3.6–3.7.) The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \lambda) = U(\mathbf{x}) + \lambda\,(m - \mathbf{p}'\mathbf{x}).$$

First-order conditions. For each good k (assuming an interior solution, $x_k > 0$):

$$\frac{\partial \mathcal{L}}{\partial x_k} = \frac{\partial U}{\partial x_k} - \lambda p_k = 0 \quad \Longrightarrow \quad \frac{\partial U/\partial x_k}{p_k} = \lambda \ \text{ for all } k.$$

Read this: **at the optimum, the marginal utility per euro is equalized across all goods**, and the common value is λ, the marginal utility of income. If good k delivered more utility per euro than good ℓ, you would shift a euro from ℓ to k and gain — so at the optimum no such shift helps. Dividing the FOC for k by that for ℓ removes λ:

$$\frac{\partial U/\partial x_k}{\partial U/\partial x_\ell} = \frac{p_k}{p_\ell},$$

the **marginal rate of substitution equals the price ratio** — the tangency of the indifference curve to the budget line, and the exact object you met as WTP in Chapter 2 (with p_ℓ the price of the numeraire/money). The Marshallian (uncompensated) demand $\mathbf{x}^M(\mathbf{p}, m)$ is the solution of these equations.

Why this matters for stated preferences. Your DCE and CV models *are* implementations of this problem: the respondent's choice reveals the MRS between an attribute and cost, and the ratio of estimated marginal utilities recovers WTP precisely because of the tangency condition above. The budget multiplier λ (marginal utility of income) is the cost coefficient (up to scale) — which is why a near-zero cost coefficient is a near-zero λ, and why WTP explodes (Chapter 2, §2.11). The constrained-optimization FOC is the theoretical origin of the econometric ratio.

3.6 Inequality constraints and the Kuhn-Tucker conditions

The consumer cannot buy negative quantities, and — crucially for MDCEV — may optimally buy *zero* of some goods. Nonnegativity constraints $x_k \geq 0$ are **inequality** constraints, and handling them requires the **Kuhn-Tucker (KKT)** generalization of the Lagrangian.

The problem with inequality constraints.

$$\max_{\mathbf{x}} f(\mathbf{x}) \quad \text{s.t.} \quad h_j(\mathbf{x}) \geq 0 \ (j = 1, \ldots, s).$$

Form the Lagrangian $\mathcal{L} = f(\mathbf{x}) + \sum_j \mu_j h_j(\mathbf{x})$ with multipliers μ_j.

Theorem 3.2 (Kuhn-Tucker necessary conditions). Under a constraint qualification (see the failure modes, §3.12), if $\mathbf{x}^*$ is a local maximum then there exist multipliers $\mu_j^* \geq 0$ such that:

1. **Stationarity:** $\dfrac{\partial f}{\partial x_k}(\mathbf{x}^*) + \displaystyle\sum_j \mu_j^* \frac{\partial h_j}{\partial x_k}(\mathbf{x}^*) = 0$ for all k.
2. **Primal feasibility:** $h_j(\mathbf{x}^*) \geq 0$ for all j.
3. **Dual feasibility:** $\mu_j^* \geq 0$ for all j.
4. **Complementary slackness:** $\mu_j^* h_j(\mathbf{x}^*) = 0$ for all j.

The new and vital condition is **complementary slackness** (4): for each constraint, *either* the multiplier is zero *or* the constraint binds (or both). Its contrapositive is the workhorse: **if a constraint is slack ($h_j > 0$), its multiplier is zero; if a multiplier is positive, the constraint binds ($h_j = 0$).** Economically, a shadow price is positive only for a constraint that is actually limiting you; a resource you are not fully using has zero marginal value.

Reading dual feasibility. The sign restriction $\mu_j \geq 0$ says relaxing an inequality constraint can only *help* (or not hurt) the maximum — you would never pay to tighten a $\geq$ constraint. This is why, unlike the equality-constraint multiplier (which can have either sign), KKT multipliers on inequality constraints are nonnegative.

3.7 Interior versus corner solutions: the mathematical heart of MDCEV

Apply KKT to the consumer's nonnegativity constraints $x_k \geq 0$ (i.e., $h_k = x_k \geq 0$) alongside the budget. The result is the interior-versus-corner dichotomy that *is* MDCEV.

Combine the budget FOC and the nonnegativity KKT conditions. For each good k, complementary slackness on $x_k \geq 0$ gives two mutually exclusive cases:

- **Interior solution ($x_k > 0$):** the nonnegativity constraint is slack, its multiplier is zero, and the ordinary FOC holds with *equality*:
$$\frac{\partial U/\partial x_k}{p_k} = \lambda.$$
The good is consumed up to the point where its marginal utility per euro equals the common marginal utility of income.

- **Corner solution ($x_k = 0$):** the nonnegativity constraint binds, its multiplier is nonnegative, and the FOC holds as an *inequality*:
$$\left.\frac{\partial U/\partial x_k}{p_k}\right|_{x_k=0} \leq \lambda.$$
Even the *first* unit of good k delivers less utility per euro than λ — it is not worth buying at all. The good sits at a corner of the feasible set (hence "corner solution").

This is the entire logic of MDCEV. In your ecosystem-services allocation, a respondent consumes ("allocates budget to") a service if and only if its marginal utility per euro *at zero* exceeds the marginal utility of income λ; otherwise they allocate zero — a corner. Whether a good is at a corner is determined by the same λ that ties all the interior goods together, which is why MDCEV must be solved as a *system*: you cannot decide each good's fate in isolation. The presence of **satiation** (diminishing marginal utility, from a concave sub-utility) is what makes interior goods stop at finite quantities rather than absorbing the whole budget, and the **translation/satiation parameters** γ_k (Chapter 40) shift the marginal-utility-at-zero and hence govern which goods hit corners. The extreme-value random error on each good's marginal utility (Chapter 41) turns this deterministic pattern into a *probability* of any given interior/corner configuration — and that probability is the MDCEV likelihood.

Worked Example 3.1 (Ecosystem-service allocation with a corner — an MDCEV in miniature). Let a respondent value three ecosystem services with an additively separable, translated-log sub-utility plus a quasi-linear outside good x_0 (the numeraire money left over, $p_0 = 1$):
$$U(\mathbf{x}, x_0) = \sum_{k\in\{A,B,C\}} \psi_k \ln\Big(\frac{x_k}{\gamma_k} + 1\Big) + x_0, \qquad \text{s.t.} \quad \sum_k p_k x_k + x_0 = m.$$
The marginal utility of service k is $\partial U/\partial x_k = \psi_k/(x_k + \gamma_k)$, and of the outside good is $\partial U/\partial x_0 = 1$. Because the outside good is the numeraire and quasi-linear, the marginal utility of income is pinned at $\lambda = 1$. Parameters:

Service	ψ_k (baseline utility)	γ_k (satiation)	p_k (price)	$\psi_k/(\gamma_k p_k)$ = MU/€ at x_k=0
A (water quality)	6	2	2	$6/4 = 1.5$
B (biodiversity)	4	1	1	$4/1 = 4.0$
C (carbon)	1	3	2	$1/6 \approx 0.17$

Apply the corner rule (compare MU/€ at zero to $\lambda = 1$):

- **A:** $1.5 > 1 \Rightarrow$ interior. Set $\psi_A/((x_A+\gamma_A)p_A) = 1 \Rightarrow x_A+\gamma_A = \psi_A/p_A = 3 \Rightarrow x_A = 3-2 = 1$. Spends $p_A x_A = 2$.
- **B:** $4 > 1 \Rightarrow$ interior. $x_B + \gamma_B = \psi_B/p_B = 4 \Rightarrow x_B = 4 - 1 = 3$. Spends $p_B x_B = 3$.
- **C:** $0.17 < 1 \Rightarrow$ **corner**, $x_C = 0$. Even the first unit of carbon gives only 0.17 utils per euro, below the 1 util per euro of just keeping the money.

Outside good: $x_0 = m-(2+3) = m-5 = 5$ when $m = 10$. Check the interior FOCs: A gives $6/((1+2)\cdot 2) = 1 = \lambda$ ✓; B gives $4/((3+1)\cdot 1) = 1 = \lambda$ ✓; C at zero gives $0.17 \leq 1$ ✓. Total utility ≈ 12.98 (verified numerically).

The lesson for your proposal. Service C is at a corner *for a genuine economic reason*: its satiation-adjusted marginal utility at zero is below the marginal utility of income. That is a **true corner**. But —

and this is the judgment your PDF demands — a zero in your *data* might instead be a **survey artifact**: the respondent protested, misread the task, was never shown the option, or was anchored by the budget framing. MDCEV's likelihood will happily interpret every zero as a true corner and estimate satiation parameters accordingly. Distinguishing the two is not a computation; it is a design-and-diagnostics judgment (§3.12, Chapters 29, 32, 50). The mathematics tells you what a true corner *is*; it cannot tell you whether your zeros are true corners.

> **Active recall 3.B.** Write the two-line complementary-slackness statement of the corner rule for good k, and translate each line into plain economics. *(Answer: if $x_k > 0$ then $MU_k/p_k = \lambda$ — a consumed good's marginal utility per euro equals the marginal utility of income; if $x_k = 0$ then $MU_k/p_k|_0 \leq \lambda$ — an unconsumed good's very first unit is not worth the money. Exactly one holds for each good.)*

3.8 Duality: indirect utility and the expenditure function

We now build the pair of "value functions" that convert the consumer's optimization into the welfare measures your papers report. This is **duality**: the same preferences seen through two dual problems.

Definition 3.3 (Indirect utility function). The **indirect utility function** is the optimized utility as a function of prices and income:

$$V(\mathbf{p}, m) = \max_{\mathbf{x} \geq \mathbf{0}} \; U(\mathbf{x}) \;\; \text{s.t.} \;\; \mathbf{p}'\mathbf{x} = m.$$

It answers "how well off is the consumer facing prices $\mathbf{p}$ with income m?" — the value function of the utility-maximization problem (§3.5), and the object a discrete choice model implicitly compares across alternatives (Chapter 20: the deterministic utility of an alternative is an indirect utility conditional on choosing it).

Definition 3.4 (Expenditure function). The **expenditure function** is the solution of the **dual** problem — the *minimum* cost of reaching a target utility u:

$$e(\mathbf{p}, u) = \min_{\mathbf{x} \geq \mathbf{0}} \; \mathbf{p}'\mathbf{x} \;\; \text{s.t.} \;\; U(\mathbf{x}) \geq u.$$

Its solution $\mathbf{x}^H(\mathbf{p}, u)$ is the **Hicksian (compensated) demand** — the cheapest bundle achieving utility u. The expenditure function answers "how much money does the consumer *need* to be as well off as u?" — which is exactly the question a welfare measure asks.

Duality relations. The two problems are inverses of each other:

$$V(\mathbf{p}, e(\mathbf{p}, u)) = u \qquad \text{and} \qquad e(\mathbf{p}, V(\mathbf{p}, m)) = m.$$

The first says: give the consumer just enough money to reach u (namely $e(\mathbf{p}, u)$), and their indirect utility is u. The second says: the cost of reaching the utility they actually achieve with income m is m. These identities are the bridge on which compensating variation is built (§3.9).

Two classical results (envelope theorem in action). Both come from differentiating a value function and using the FOC-cancellation of the envelope theorem (§3.10):

- **Roy's identity** recovers Marshallian demand from indirect utility:

$$x_k^M(\mathbf{p}, m) = -\frac{\partial V / \partial p_k}{\partial V / \partial m}.$$

  Note the denominator $\partial V/\partial m = \lambda$, the marginal utility of income — the multiplier again. Roy's identity is a ratio of partials, structurally the same shape as WTP (Chapter 2).
- **Shephard's lemma** recovers Hicksian demand from the expenditure function:

$$x_k^H(\mathbf{p}, u) = \frac{\partial e(\mathbf{p}, u)}{\partial p_k}.$$

The derivative of minimum expenditure with respect to a price is the compensated quantity demanded — the cleanest envelope result in economics.

Worked Example 3.2 (Cobb-Douglas duality and a compensating variation, verified). Let $U(x_1, x_2) = x_1^a x_2^{1-a}$ with $a = 0.5$, $p_2 = 1$, income $m = 100$. Standard Cobb-Douglas results (derive them as Exercise B2):

$$V(\mathbf{p}, m) = m\,\frac{a^a(1-a)^{1-a}}{p_1^a p_2^{1-a}}, \qquad e(\mathbf{p}, u) = u\,\frac{p_1^a p_2^{1-a}}{a^a(1-a)^{1-a}}.$$

With $a = 0.5$ the constant $a^a(1-a)^{1-a} = 0.5$. At baseline $p_1 = 1$: $u_0 = V((1,1), 100) = 100 \cdot 0.5/1 = 50$, and as a check $e((1,1), 50) = 50/0.5 = 100 = m$ ✓ (duality relation).

Now let p_1 rise from 1 to 2 (say the cost of an environmental good increases). How much money would the consumer need to be as well off as before? That is the **compensating variation** of the price change, defined via the expenditure function:

$$\text{CV} = e(\mathbf{p}^1, u_0) - e(\mathbf{p}^0, u_0) = e((2,1), 50) - e((1,1), 50).$$

Compute $e((2,1), 50) = 50 \cdot (2^{0.5} \cdot 1)/0.5 = 50 \cdot 1.4142/0.5 = 141.42$, and $e((1,1), 50) = 100$. So

$$\text{CV} = 141.42 - 100 = €41.42.$$

The consumer would need €41.42 of extra income to reach their original utility after the price rise — a money measure of the welfare loss, computed entirely from the expenditure function. (For contrast, the **equivalent variation**, using the *new* utility $u_1 = V((2,1), 100) = 35.36$ as the reference, is $EV = m - e((1,1), u_1) = 100 - 70.71 = €29.29$ — verified. CV and EV differ because they hold *different* utility levels constant, a distinction Chapter 34 develops and that matters for which welfare measure your CBA should use.)

Why this is the foundation of your WTP. A willingness to pay for an environmental *improvement* is a compensating variation: the maximum money you could take from the consumer after the improvement while leaving them as well off as before. Your DCE and CV estimates are, in theory, exactly these expenditure-function differences. When your PDF asks "quando una compensating variation è ben definita," it is asking whether the expenditure function is well-behaved for your good — which requires the underlying optimization to be well-posed (a real budget, a genuine numeraire, sensible preferences). Chapter 35 makes the WTP-as-CV identification precise; Chapter 42 computes *exact* CV inside MDCEV using the estimated utility. Here you have the definition and the machinery.

3.9 The envelope theorem, proved

Because the envelope theorem underlies §3.4 (shadow prices), Roy's identity, and Shephard's lemma, here is the clean statement and proof for the constrained case — a short, worthwhile derivation.

Theorem 3.3 (Envelope theorem). Consider $\max_{\mathbf{x}} f(\mathbf{x}, \alpha)$ s.t. $g(\mathbf{x}, \alpha) = 0$, with solution $\mathbf{x}^*(\alpha)$, multiplier $\lambda^*(\alpha)$, and value function $f^*(\alpha) = f(\mathbf{x}^*(\alpha), \alpha)$. Then

$$\frac{df^*}{d\alpha} = \left.\frac{\partial \mathcal{L}}{\partial \alpha}\right|_{\mathbf{x}^*, \lambda^*} = \frac{\partial f}{\partial \alpha} + \lambda^* \frac{\partial g}{\partial \alpha},$$

evaluated at the optimum. **Only the *direct* effect of the parameter matters; the indirect effect through the optimal $\mathbf{x}^*(\alpha)$ vanishes.**

Proof. Along the optimum the constraint holds, $g(\mathbf{x}^*(\alpha), \alpha) = 0$, so $f^*(\alpha) = \mathcal{L}(\mathbf{x}^*(\alpha), \lambda^*(\alpha), \alpha)$ (the added term λg is zero). Differentiate totally with respect to α:

$$\frac{df^*}{d\alpha} = \underbrace{\sum_k \frac{\partial \mathcal{L}}{\partial x_k}}_{=0 \text{ by FOC}} \frac{dx_k^*}{d\alpha} + \underbrace{\frac{\partial \mathcal{L}}{\partial \lambda}}_{=g=0} \frac{d\lambda^*}{d\alpha} + \frac{\partial \mathcal{L}}{\partial \alpha}.$$

The first sum vanishes because each $\partial\mathcal{L}/\partial x_k = 0$ at the optimum (stationarity); the second term vanishes because $\partial\mathcal{L}/\partial\lambda = g = 0$. Only $\partial\mathcal{L}/\partial\alpha$ survives. ■

Corollaries you just used. Take $\alpha = c$ the constraint level (with $g = c - (\text{something})$): $df^*/dc = \lambda^*$ — the shadow price (§3.4). Take $\alpha = p_k$ in the expenditure problem: $\partial e/\partial p_k = x_k^H$ — Shephard's lemma (the objective $\mathbf{p}'\mathbf{x}$ has direct derivative x_k). Take $\alpha = p_k$ and $\alpha = m$ in the utility-max problem and form the ratio: Roy's identity. Three central results, one cancellation.

3.10 Compensating variation as a constrained-optimization object (bridge to welfare)

Collecting the pieces, here is the definition your welfare chapters will build on, stated so you can see it is *pure constrained optimization*.

Definition 3.5 (Compensating variation). For a change in the environment from $(\mathbf{p}^0, q^0)$ to $(\mathbf{p}^1, q^1)$ — where q is an environmental quality that enters utility — with income m, the **compensating variation** is the money adjustment that restores baseline utility $u_0 = V(\mathbf{p}^0, q^0, m)$ under the new conditions:

$$\text{CV} = e(\mathbf{p}^1, q^1, u_0) - e(\mathbf{p}^0, q^0, u_0) = e(\mathbf{p}^1, q^1, u_0) - m.$$

The expenditure function does all the work, and the expenditure function is a constrained minimization — hence the chapter.

> **A competing convention, flagged.** Two sign conventions for CV coexist in the literature, and mixing them is a standard source of confusion. **This book uses the expenditure-difference convention above**, in which CV > 0 measures the *extra money needed* to restore baseline utility — so CV is positive for a **welfare loss** (a price rise, a degradation), and for an **improvement** it is $-$CV that equals the **maximum WTP**. The alternative convention, common in the environmental-valuation literature, defines CV with the opposite sign so that CV > 0 *is* WTP for an improvement. Neither is wrong; what matters is that you state which you use and never switch mid-paper. Chapter 34 §34.3 develops the same convention formally alongside EV, and Worked Example 3.2 below applies it to a price increase.

Three judgments this definition forces (all in your PDF).

1. **Is the numeraire a true numeraire?** CV is money-metric only if there is a well-defined outside good against which utility is denominated (the x_0 of §3.7). If your MDCEV "outside good" is not a genuine residual money good, $e(\cdot)$ and hence CV are ill-defined. *(Your PDF: "quando l'outside good è davvero un numeraire.")*
2. **Is CV well-defined?** It requires the expenditure function to exist and be single-valued — guaranteed by well-behaved (locally nonsatiated, continuous) preferences and a real budget. With large income effects, CV and EV diverge sharply and WTP $\neq$ WTA, so *which* measure you report matters (Chapter 34). *(Your PDF: "quando una compensating variation è ben definita.")*
3. **Does the model force substitution/complementarity the data cannot identify?** The cross-price behavior of Hicksian demand (the substitution matrix, the derivatives of $\mathbf{x}^H$) encodes substitutability/complementarity. In MDCEV, the functional form imposes patterns of these that your design may not actually identify — a corner-heavy design with little price variation cannot pin down substitution among rarely-chosen services. *(Your PDF: "quando il modello forza sostituibilità o complementarità che i dati non identificano.")* Chapter 40 §40.6–§40.7 makes this an explicit identification analysis.

Full welfare treatment is Chapter 34 (measures) and Chapter 41 (exact CV in MDCEV). This section's job is to show you the welfare number is a constrained optimum, so that "how fragile is my CV?" becomes "how fragile is the optimization problem I assumed?"

3.11 Failure modes: how constrained optimization misleads in practice

F1. A corner that is a survey artifact, not a true corner. MDCEV interprets every $x_k = 0$ as a genuine economic corner (MU/€ at zero below λ). But a zero can come from protest, non-attendance, misunderstanding, an unshown option, or budget-framing anchoring. *Symptom:* implausible satiation parameters, many zeros concentrated in a subgroup. *Fix:* diagnose zeros against protest/comprehension indicators (Chapters 29, 32, 50); test sensitivity to excluding suspect zeros; never let the likelihood's willingness to rationalize a zero substitute for design judgment. *(Your PDF's first bullet under this topic.)*

F2. A budget constraint that is not economically credible. If the budget is hypothetical with no real trade-off, the multiplier λ (marginal utility of income) is not a genuine util-to-euro rate, and every WTP/CV built on it is a stated-preference indicator, not welfare. *Fix:* build consequentiality and real (or incentivized) trade-offs into the design (Chapters 28, 37); state explicitly when your welfare numbers are conditional on a credible budget.

F3. An "outside good" that is not a numeraire. Money-metric welfare requires a residual money good. If your outside option is itself a valued environmental good with its own satiation, or is mis-specified, the expenditure function and CV are ill-defined. *Fix:* justify the numeraire; check robustness to its specification (Chapter 42).

F4. Treating a corner as interior (ignoring complementary slackness). Solving the interior FOCs for a good that is actually at a corner produces a negative "demand," which naive code may not flag. *Symptom:* negative fitted allocations. *Fix:* always impose the KKT inequality; check MU/€ at zero against λ before assuming interiority.

F5. Non-concave / non-quasi-concave utility → KKT necessary but not sufficient. If preferences are not quasi-concave, the FOC/KKT point may be a local, not global, optimum, and demand can be multivalued. *Fix:* verify quasi-concavity (or use functional forms, like MDCEV's, that guarantee it); check the bordered Hessian / second-order conditions where concavity is not assured.

F6. Constraint qualification failures. The KKT *necessity* theorem needs a constraint qualification (e.g., the active constraints' gradients are linearly independent). At awkward points (kinks, redundant constraints) KKT can fail to hold at a true optimum, or hold at a non-optimum. *Fix:* be alert at boundaries and with redundant/overlapping constraints; this rarely bites in smooth demand systems but can in hand-built constrained problems.

F7. Large income effects → CV, EV, WTP, WTA all diverge. When the good is a large share of budget or income effects are strong, CV $\neq$ EV and WTP $\neq$ WTA, and choosing the wrong one biases welfare. *Fix:* report the appropriate measure for the policy question (CV for the proposed change referenced to the status quo; EV referenced to the new state), and acknowledge the divergence (Chapter 34).

F8. Forcing substitution/complementarity the design cannot identify. The utility form imposes cross-good substitution patterns; a design with little price variation or few interior choices cannot identify them, so the "estimates" are functional-form artifacts. *Fix:* an explicit identification check (Chapter 40 §40.7); design for price/quantity variation (Chapter 37).

3.12 Diagnostic questions for your own papers

1. **The assumed problem.** For each welfare number I report, what constrained optimization am I assuming the respondent solved? Is the objective (utility), the constraint (budget), and the feasible set (nonnegativity, numeraire) all specified and credible? *(§3.5, §3.10.)*
2. **The multiplier.** Is my budget multiplier λ (marginal utility of income / cost coefficient) meaningful — i.e., is the budget real and consequential? *(§3.4, F2.)*

3. **Corners.** Are the zeros in my MDCEV data true corners (economic) or artifacts (survey)? What evidence separates them, and how sensitive are my satiation estimates to that call? *(§3.7, F1.)*
4. **The numeraire.** Is my outside good a genuine residual money good, so that CV is money-metric? *(§3.10, F3.)*
5. **CV well-posedness.** Is my compensating variation well-defined? Are income effects small enough that CV, EV, WTP, WTA roughly coincide, or must I choose carefully? *(§3.10, F7.)*
6. **Identification of substitution.** Does my design actually identify the substitution/complementarity my MDCEV form imposes, or am I reading functional form as evidence? *(§3.10, F8; Chapter 41.)*
7. **Optimum validity.** Is my utility (quasi-)concave so KKT points are global optima, and have I checked second-order/constraint-qualification conditions where concavity is not guaranteed? *(§3.6, F5–F6.)*

3.13 Exercises

(A) Comprehension.

A1. A consumer maximizes $U = x_1^{0.5} x_2^{0.5}$ subject to $2x_1 + x_2 = 40$. Write the Lagrangian, derive the FOCs, and solve for x_1, x_2, λ. Interpret λ in words.

A2. In Worked Example 3.1, suppose the price of service C falls from $p_C = 2$ to $p_C = 0.5$ (all else equal). Does C remain a corner? Show the calculation and, if it becomes interior, compute x_C.

A3. State the four Kuhn-Tucker conditions for $\max f(\mathbf{x})$ s.t. $x_k \geq 0$, and explain complementary slackness in one sentence using the phrase "shadow price."

(B) Derivation.

B1. For the general consumer problem $\max U(\mathbf{x})$ s.t. $\mathbf{p}'\mathbf{x} = m$, derive the condition $\text{MRS}_{k\ell} = p_k/p_\ell$ from the Lagrangian FOCs, and explain why the multiplier λ is the marginal utility of income (use the envelope theorem).

B2. Derive the Cobb-Douglas indirect utility $V(\mathbf{p}, m)$ and expenditure function $e(\mathbf{p}, u)$ used in Worked Example 3.2, starting from $U = x_1^a x_2^{1-a}$. Verify the duality relation $e(\mathbf{p}, V(\mathbf{p}, m)) = m$.

B3. Using the translated-log MDCEV sub-utility $\psi_k \ln(x_k/\gamma_k + 1)$ with a numeraire outside good, derive the corner rule: good k is consumed iff $\psi_k/(\gamma_k p_k) > \lambda$, and if consumed, $x_k = \psi_k/(\lambda p_k) - \gamma_k$. Interpret the role of γ_k (satiation).

(C) Research application / critique.

C1. Your MDCEV of ecosystem-service allocation finds that 60% of respondents allocate zero to "carbon sequestration." A co-author concludes carbon has low satiation-adjusted marginal utility. Write a critique: what alternative (non-economic) explanations for the zeros exist, why does MDCEV conflate them with true corners, and what three checks would you run before accepting the satiation interpretation?

C2. A referee objects that your reported WTP for a wetland improvement "is not a real welfare measure." Using the constrained-optimization definition of compensating variation (§3.10), explain the conditions under which your WTP *is* a valid CV, and which of those conditions your stated-preference design most plausibly violates.

C3. In your restoration CBA you need a shadow price for a scarce input (say, restored habitat area). Explain, using the multiplier-as-shadow-price result (§3.4), what the shadow price represents, why it can differ from the market price, and how you would connect it to the Lagrange multiplier of a constrained social-planner problem.

3.14 Solutions

A1. $\mathcal{L} = x_1^{0.5}x_2^{0.5} + \lambda(40 - 2x_1 - x_2)$. FOCs: $0.5x_1^{-0.5}x_2^{0.5} = 2\lambda$; $0.5x_1^{0.5}x_2^{-0.5} = \lambda$; $2x_1 + x_2 = 40$. Divide the first by the second: $x_2/x_1 = 2 \Rightarrow x_2 = 2x_1$. Budget: $2x_1 + 2x_1 = 40 \Rightarrow x_1 = 10$, $x_2 = 20$. From the second FOC, $\lambda = 0.5(10)^{0.5}(20)^{-0.5} = 0.5\sqrt{10/20} = 0.5/\sqrt{2} \approx 0.354$. Interpretation: $\lambda \approx 0.354$ is the marginal utility of income — one extra euro of budget raises maximized utility by about 0.354 utils.

A2. Corner rule: consumed iff $\psi_C/(\gamma_C p_C) > \lambda = 1$. With $p_C = 0.5$: $\psi_C/(\gamma_C p_C) = 1/(3 \cdot 0.5) = 1/1.5 = 0.667 < 1$. Still a **corner** — even at the lower price, carbon's first unit yields 0.667 utils per euro, below the marginal utility of income. (It would take $p_C < \psi_C/\gamma_C = 1/3$ to bring carbon in.) A nice illustration that a low satiation-adjusted baseline utility can keep a good at a corner across a wide price range.

A3. Conditions: (1) stationarity $\partial f/\partial x_k + \mu_k = 0$; (2) primal feasibility $x_k \geq 0$; (3) dual feasibility $\mu_k \geq 0$; (4) complementary slackness $\mu_k x_k = 0$. Complementary slackness: the shadow price μ_k of the nonnegativity constraint is positive only when the constraint binds ($x_k = 0$) — i.e., a good has a positive "shadow price of being forced to zero" exactly when you would want to consume negative amounts, which you cannot.

B1. FOCs: $\partial U/\partial x_k = \lambda p_k$ for all k. Ratio for goods k, ℓ: $(\partial U/\partial x_k)/(\partial U/\partial x_\ell) = p_k/p_\ell$, i.e., MRS = price ratio. Envelope: $V(\mathbf{p}, m) = U(\mathbf{x}^*)$ with the budget as the constraint at level m; by Theorem 3.3 with $\alpha = m$, $\partial V/\partial m = \partial \mathcal{L}/\partial m = \lambda$ (the budget term $\lambda(m - \mathbf{p}'\mathbf{x})$ has direct m-derivative λ). So $\lambda = \partial V/\partial m$ is the marginal utility of income.

B2. With $U = x_1^a x_2^{1-a}$, the FOC ratio gives $ax_2/((1-a)x_1) = p_1/p_2$, so $p_1x_1/(p_2x_2) = a/(1-a)$; combined with $p_1x_1 + p_2x_2 = m$ this yields Marshallian demands $x_1^M = am/p_1$, $x_2^M = (1-a)m/p_2$. Substitute into U: $V = (am/p_1)^a((1-a)m/p_2)^{1-a} = m\,a^a(1-a)^{1-a}/(p_1^a p_2^{1-a})$. Invert $u = V$ for m to get expenditure: $e(\mathbf{p}, u) = u\,p_1^a p_2^{1-a}/(a^a(1-a)^{1-a})$. Duality check: $e(\mathbf{p}, V(\mathbf{p}, m)) = V \cdot p_1^a p_2^{1-a}/(a^a(1-a)^{1-a}) = m\,a^a(1-a)^{1-a}/(p_1^a p_2^{1-a}) \cdot p_1^a p_2^{1-a}/(a^a(1-a)^{1-a}) = m$ ✓.

B3. Marginal utility of good k: $\partial U/\partial x_k = \psi_k/(x_k + \gamma_k)$ (chain rule on $\psi_k \ln(x_k/\gamma_k + 1)$). Interior KKT: $\psi_k/((x_k + \gamma_k)p_k) = \lambda \Rightarrow x_k + \gamma_k = \psi_k/(\lambda p_k) \Rightarrow x_k = \psi_k/(\lambda p_k) - \gamma_k$, which is positive iff $\psi_k/(\lambda p_k) > \gamma_k$, i.e., $\psi_k/(\gamma_k p_k) > \lambda$ — the corner rule. If this fails, $x_k = 0$. Role of γ_k: it is the satiation/translation parameter — larger γ_k lowers the marginal utility at zero (ψ_k/γ_k) making a corner more likely, and flattens the marginal-utility decline so that, once interior, the good satiates more slowly. It sets both the extensive margin (in or out) and the intensive margin (how much).

C1. Alternative explanations for the zeros: protest responses (rejecting the payment vehicle or the exercise), attribute non-attendance (ignoring carbon), misunderstanding of an abstract good, the good never being salient, or budget-framing anchoring that pushes allocations toward familiar services. MDCEV conflates these with true corners because its likelihood attributes *every* zero to the economic condition MU/€ at zero $\leq \lambda$, absorbing artifacts into the satiation parameters ψ_C, γ_C. Three checks: (i) cross-tabulate zeros against protest/comprehension/non-attendance indicators and test whether zeros concentrate among flagged respondents; (ii) re-estimate excluding suspect zeros and report parameter sensitivity (bounding); (iii) examine whether design features (position, framing, information about carbon) predict the zeros — if they do, the zeros are partly artifactual. Only after these would the satiation interpretation be defensible.

C2. WTP is a valid compensating variation when: (i) it derives from a genuine constrained optimization with a real, binding budget (so λ is a true marginal utility of income); (ii) there is a proper numeraire against which utility is money-denominated; (iii) preferences are stable and well-behaved so $e(\mathbf{p}, q, u)$ exists and is single-valued; and (iv) the mechanism is incentive-compatible/consequential so stated choices reveal those preferences. A stated-preference design most plausibly violates (iv) (and often (i)): hypothetical payments with no real consequence mean the budget is not truly binding and truth-telling is not guaranteed, so the "WTP" may reflect stated-preference construction rather than a money-metric CV. The referee is right *unless* you can defend consequentiality and incentive compatibility (Chapters 28, 37) — which is precisely why those chapters exist.

C3. The shadow price is the Lagrange multiplier on the input constraint in the social planner's constrained optimization (max social net benefit s.t. habitat-area availability): by the envelope theorem it equals the marginal increase in maximized social welfare from one more unit of the input — its social value at the margin.

It can differ from the market price when markets are distorted, missing, or the input generates externalities (a hallmark of environmental goods): the market price reflects private valuation, the shadow price reflects social valuation. To connect them, you set up the planner's problem with the resource constraint, and the shadow price is the multiplier λ; in practice you estimate it from the marginal social benefit (e.g., the CV-based value of the ecosystem services the habitat produces), which is exactly why your non-market valuation feeds your CBA shadow prices (Chapters 43–44).

3.15 Spaced review prompts

+1 day. From memory, write the consumer's Lagrangian, derive "marginal utility per euro equalized across goods," and state what λ equals (marginal utility of income) and why (envelope theorem).

+1 week. Reconstruct the corner rule from complementary slackness: for good k, state the interior condition and the corner condition in terms of MU/€ versus λ, and explain in two sentences why this is "the mathematical core of MDCEV."

+1 month. Take your MDCEV proposal (or a stylized version). Write the assumed constrained optimization explicitly — objective, budget, numeraire, nonnegativity — and for one service work out whether a representative respondent is at an interior or corner solution. Then write one paragraph on whether your data's zeros are true corners or artifacts, and how you would defend the distinction (tie to §3.7, F1).

3.16 How this chapter connects to the rest of the book

- **Chapter 2** supplied the FOC/SOC, concavity, and the MRS-as-ratio-of-partials that this chapter constrains; the implicit function theorem there is the comparative-statics engine for KKT here.
- **Chapter 4 (Discounting)** and **Chapters 43–45 (CBA)** use the shadow-price reading of the multiplier (§3.4) for social valuation and constrained social-planner problems.
- **Chapters 20–24 (discrete choice)** rest on indirect utility (§3.8): an alternative's deterministic utility is an indirect utility, and the WTP ratio is the MRS/tangency condition of §3.5. The cost coefficient is the empirical counterpart of λ.
- **Chapter 34 (Welfare Measurement Foundations)** revisits Marshallian/Hicksian demand, Roy, and Shephard in depth and builds CV, EV, WTP, WTA on the expenditure function of §3.8–§3.10.
- **Chapters 40–42 (MDCEV)** are the full development of §3.7 and §3.10: Chapter 40 the utility and KKT demand, Chapter 40 the likelihood and identification of the corner-generating parameters, Chapter 41 exact compensating variation from the estimated utility.
- **Chapters 29, 32, 50 (bid/cost-vector effects, threats to identification, survey design)** carry the F1 judgment — telling a true corner from a survey artifact — into design and diagnostics.

This chapter is the mathematical hinge between Part I and the welfare/MDCEV parts, and it directly serves the high-dependency chapter **Ch 24** (via the tangency origin of WTP) and the welfare spine of Parts VIII and X.

3.17 Research translation task

For your **MDCEV proposal**, produce a three-page note:

1. **Write the assumed optimization.** State the utility function (with satiation/translation parameters), the budget constraint, the outside good/numeraire, and the nonnegativity constraints. Make explicit what "allocating to an ecosystem service" means as a constrained choice.

2. **Derive the corner rule** for your specification (as in Exercise B3): the condition under which a service is at a corner, in terms of its parameters, prices, and the marginal utility of income.
3. **Interrogate the numeraire and budget.** Argue whether your outside good is a genuine numeraire and whether the budget is economically credible/consequential; note the consequence for whether your welfare measures are money-metric (§3.10, F2–F3).
4. **Confront the corners.** For the service most likely to show many zeros, list the true-corner and artifact explanations, and specify the checks (F1) you will run to defend a satiation interpretation.
5. **Set up the compensating variation.** Write CV for a specific ecosystem-service improvement as an expenditure-function difference, and note which parameters must be identified for it to be computable — a preview of the identification analysis you will do in Chapter 40–41.
6. **Close** with the spine sentence: *"My MDCEV welfare estimate is the compensating variation of a constrained allocation problem; it is identified under [budget credibility, numeraire, design variation]; it is fragile to [artifact corners, weak price variation, numeraire misspecification]."*

Carry this into Chapters 40–42, where the likelihood and exact CV are built on exactly this setup.

3.18 Bibliography

Core. - Mas-Colell, Whinston & Green, *Microeconomic Theory*, chapters on consumer choice, duality, and welfare. The definitive graduate treatment of utility maximization, expenditure minimization, indirect utility, Roy, Shephard, and CV/EV — everything in §3.5–§3.10 at full rigor. *(Verify chapter numbers.)* - Varian, *Microeconomic Analysis*. A cleaner, more compact development of duality and the envelope theorem; excellent as the first pass before MWG. *(Verify edition.)* - Simon & Blume, *Mathematics for Economists*, chapters on constrained optimization and the Kuhn-Tucker theorem. The mathematical backing for §3.3–§3.6, including constraint qualifications. *(Verify edition/chapters.)*

Applied. - Bhat (2008), "The multiple discrete-continuous extreme value (MDCEV) model: Role of utility function parameters, identification considerations, and model extensions," *Transportation Research Part B*. The canonical MDCEV reference; §3.7 and §3.10 are the microeconomic foundations of this paper, and it directly addresses your proposal's questions on satiation, the outside good, and identification. *(Verify volume/pages.)* - von Haefen & Phaneuf, and the Kuhn-Tucker demand-system literature in environmental economics (e.g., Phaneuf, Kling & Herriges). Shows KKT corner-solution demand estimation for recreation/environmental goods — the applied lineage of your MDCEV. *(Verify exact citations before citing — I am not certain of the precise references.)* - Freeman, Herriges & Kling, *The Measurement of Environmental and Resource Values*. The standard applied reference tying compensating variation and the expenditure function to non-market valuation — bridges §3.10 to your CV/DCE welfare interpretation. *(Verify edition.)*

Advanced. - Deaton & Muellbauer, *Economics and Consumer Behavior*. Deep treatment of duality, demand systems, and welfare, including the substitution matrix and integrability — relevant to the "does the design identify substitution?" question (§3.10, F8). *(Verify.)* - Boadway & Bruce, *Welfare Economics*, or MWG's welfare chapters, for the CV/EV/consumer-surplus relationships and the conditions under which they coincide — the theory behind F7 and Chapter 34. *(Verify.)* - Luenberger, *Microeconomic Theory* (benefit function / duality), for a unified optimization view of welfare measurement, if you later want the most general framework. *(Verify.)*

Entries marked "verify" are pointers; confirm exact editions, pages, and — for the Kuhn-Tucker environmental-demand references — the precise citations before using them, per the book's no-unverified-citation rule.

End of Chapter 3. Next in Part I: Chapter 4 — The Mathematics of Discounting (present value, NPV, IRR, real vs nominal, private vs social discount rates), the engine of the CBA in your restoration and blue-finance work. Tell me when to write it, or request a different chapter.

Chapter 4 — The Mathematics of Discounting

Part I: Mathematical Foundations for Econometrics and Welfare Analysis

> **The spine question, applied to this chapter.** A restoration project spends money now and delivers ecosystem services — recovered fish biomass, sequestered carbon, biodiversity, coastal protection — over decades. To decide whether it is worthwhile you must compare euros today with euros in year 30, and that comparison is *discounting*. But the discount rate is not a technical constant you look up; it is the single most consequential assumption in the entire appraisal, and at long horizons it can swing the answer by orders of magnitude. This chapter builds the arithmetic (present value, NPV, IRR) and then confronts the judgment (real vs nominal, private vs social, the intergenerational rate, and the stock-versus-flow trap your PDF flags as a top-priority error). The spine question becomes: *when I report that a restoration project has a positive net present value, what discount rate and what treatment of stocks, flows, inflation, and horizon did I assume, and how fragile is the sign of the answer to those choices?*

4.0 Conceptual motivation: the discount rate decides everything

Your PDF places discounting under "Importante: da consolidare," but do not let that ranking mislead you about its weight in your applied work. For the fish-restoration-device (FRD) paper and any blue-finance appraisal, discounting is where the whole cost-benefit verdict is decided, and your plan is explicit about the stakes: *"Ti serve per interpretare FNPV, ENPV, IRR e scenari nei lavori su restoration ed ecosystem services,"* and, under the top-priority error list, *"CBA: stock vs flow e double counting … distinguere stock di biomassa da flusso sostenibile; non sommare valore di stock e valore annuale come se fossero la stessa cosa; … giustificare tasso di sconto e social cost of carbon."* Both of those are, at root, discounting problems. Let me set out why this modest-looking arithmetic carries so much responsibility.

- **The sign of NPV can flip on the discount rate.** A restoration project with costs concentrated early and benefits spread over decades looks *good* at a low rate and *bad* at a high one, because a high rate crushes distant benefits more than near costs. Whether your FRD project "passes" can be entirely an artifact of choosing 3% versus 7%. If you cannot compute and defend the crossover rate at which the verdict flips, you cannot defend the verdict.
- **Private and social appraisal give different answers on purpose.** Your PDF notes that in the FRD paper "emerge proprio la distinzione tra non profittabilità privata e benefici sociali positivi." That gap is a *discount-rate and scope* statement: the **financial** NPV (FNPV) uses market prices and a private rate and may be negative; the **economic** NPV (ENPV) adds non-market ecosystem-service benefits valued by your DCE/CV work and uses a social discount rate, and may be positive. Understanding when and why FNPV and ENPV diverge is understanding this chapter.
- **Long horizons make the rate dominate.** For the social cost of carbon and for blue-carbon sequestration that accrues over a century, the discount rate does not merely adjust the answer — it *is* the

answer. As you will compute, a benefit a century out is worth about 320 times more at a 1% rate than at 7%. The Stern–Nordhaus climate debate is, mathematically, an argument about this one number.

- **The stock–flow trap is a discounting error.** Adding the market value of a standing biomass *stock* to the discounted stream of the sustainable harvest *flow* double-counts, because — as we will prove — the capitalized value of a perpetual sustainable flow *is* the asset value of the stock. They are two views of one thing. This is your PDF's §6.9 red flag, and it lives inside the mathematics of the perpetuity.

So this is a short chapter with high leverage. We build present value and the annuity/perpetuity formulas from a geometric series, define NPV and IRR and expose IRR's pitfalls, get the real-versus-nominal consistency rule exactly right, then turn to the genuinely hard part — the social discount rate, intergenerational discounting, and the stock–flow distinction — where the mathematics is easy but the judgment is where careers and policies are made or lost. Full CBA machinery (shadow pricing, benefit transfer, Monte Carlo uncertainty) is Part XI; this chapter is its engine.

4.1 Prerequisites

Light, by design. You need:

- from **Chapter 2**: functions and derivatives (for the sensitivity analysis in §4.10, where we differentiate NPV with respect to the rate), and comfort with a geometric series (which we derive in §4.3);
- from **Chapter 3**: the shadow-price reading of a Lagrange multiplier (§3.4) — useful when we connect the social discount rate to a constrained social-planner problem (§4.8) — and the compensating-variation definition of the *benefits* that get discounted (§3.10);
- basic algebra of exponents and logarithms.

Same **notation** as before. New symbols: r (discount rate, real unless noted), i (nominal rate), π (inflation), δ (pure rate of time preference), η (elasticity of marginal utility of consumption), g (consumption growth), B_t, C_t (benefit and cost in year t), DF_t (discount factor), $\mathrm{PV}, \mathrm{NPV}$.

4.2 The time value of money: compounding and the discount factor

The primitive fact. A euro today is worth more than a euro next year, for three reasons that the discount rate bundles together: it could be invested and earn a return (opportunity cost of capital); prices may rise so a future euro buys less (inflation — handled separately in §4.7); and people (and societies) generally prefer benefits sooner (time preference). Discounting is the arithmetic of converting future euros into today's equivalent.

Definition 4.1 (Future and present value, discrete compounding). With a per-period rate r compounded once per period, an amount X today grows to a **future value** after t periods of

$$\mathrm{FV}_t = X\,(1+r)^t,$$

and, inverting, an amount received in period t has a **present value** today of

$$\mathrm{PV} = \frac{X}{(1+r)^t} = X \cdot \mathrm{DF}_t, \qquad \mathrm{DF}_t \equiv \frac{1}{(1+r)^t},$$

where DF_t is the **discount factor** — the price today of one euro delivered in year t. The discount factor falls geometrically in t: this geometric decay is the entire mathematical content of discounting, and it is why distant flows are compressed so hard.

Compounding frequency and the continuous limit. If interest compounds k times per period at nominal annual rate r, one euro becomes $(1 + r/k)^{kt}$ after t periods. Letting $k \to \infty$ gives **continuous**

compounding:

$$\lim_{k\to\infty}\left(1+\frac{r}{k}\right)^{kt}=e^{rt},\qquad \text{so PV}=Xe^{-rt}.$$

Continuous discounting (e^{-rt}) is analytically convenient — it turns products into sums via logs and makes calculus clean — and is standard in climate-economics and theoretical work. Discrete annual discounting ($1/(1+r)^t$) is standard in applied CBA guidance (the UK Green Book, EU appraisal). The two are near-identical for small r (since $e^{-r}\approx 1/(1+r)$) but not exactly equal, and mixing them is a small, avoidable inconsistency (§4.12, F-notation). A "competing conventions" flag: state which you use.

> **Active recall 4.A.** One euro of ecosystem-service benefit arrives in year 25. At $r=4\%$, write its present value and, in one sentence, say what the discount factor $1/(1.04)^{25}$ represents. *(Answer: $PV=1/(1.04)^{25}\approx$ €0.375; the discount factor is the amount you would pay today to receive one euro in 25 years, i.e., today's price of that future euro.)*

4.3 Present value of a stream: annuities and perpetuities

Projects deliver *streams* of cash, so we need the present value of a whole sequence.

Definition 4.2 (Present value of a stream). For cash flows $X_0, X_1, \dots, X_T$ (with X_0 occurring now, at $t=0$),

$$\text{PV}=\sum_{t=0}^{T}\frac{X_t}{(1+r)^t}.$$

Two special streams have closed forms worth deriving, because they recur constantly and because the derivation is the geometric series you will reuse (and which underlies the stock–flow result of §4.11).

Derivation — the annuity formula. Suppose a constant amount X arrives each year from $t=1$ to $t=T$. Then

$$\text{PV}=X\sum_{t=1}^{T}\frac{1}{(1+r)^t}.$$

Let $v=1/(1+r)$ (the one-year discount factor), so the sum is $X(v+v^2+\cdots+v^T)$. This is a finite geometric series. Multiply the sum $S=v+v^2+\cdots+v^T$ by v: $vS=v^2+\cdots+v^{T+1}$; subtract: $S-vS=v-v^{T+1}$, so $S(1-v)=v(1-v^T)$, giving $S=v(1-v^T)/(1-v)$. Substitute $v=1/(1+r)$ and simplify (note $1-v=r/(1+r)$):

$$\boxed{\text{PV}_{\text{annuity}}=X\cdot\frac{1-(1+r)^{-T}}{r}\equiv X\cdot A(r,T)}.$$

$A(r,T)=[1-(1+r)^{-T}]/r$ is the **annuity factor**: the present value of one euro per year for T years. It is decreasing in both r and (for its marginal contributions) diminishing in T.

Perpetuity — the $T\to\infty$ limit. As $T\to\infty$, $(1+r)^{-T}\to 0$ (for $r>0$), so the annuity factor converges:

$$\boxed{\text{PV}_{\text{perpetuity}}=\frac{X}{r}}.$$

A perpetual stream of X per year is worth X/r today. This tiny formula is the hinge of the stock–flow argument in §4.11: **the capitalized value of a permanent sustainable flow is the flow divided by the discount rate.** A growing perpetuity (flow growing at rate $\gamma<r$) generalizes to $X/(r-\gamma)$ — the Gordon growth formula — useful when ecosystem-service benefits are projected to grow.

> **Stop and think 4.1.** A wetland delivers €20/year of ecosystem services "in perpetuity." Before reading on, compute its present value at $r=2\%$ and at $r=5\%$, and note the ratio. *(Answer: 20/0.02 = €1000 vs 20/0.05 = €400 — a factor of 2.5. Halving the rate more than doubles the capitalized value. The perpetuity formula makes the rate's leverage stark: value is inversely proportional to r.)*

4.4 Net present value

Definition 4.3 (Net present value). For a project with benefits B_t and costs C_t in year t over horizon T, the **net present value** at rate r is

$$\mathrm{NPV}(r) = \sum_{t=0}^{T} \frac{B_t - C_t}{(1+r)^t}.$$

Decision rule: undertake the project if NPV > 0 (its discounted benefits exceed its discounted costs); among mutually exclusive alternatives, choose the highest NPV (subject to the caveats of §4.6). NPV is the theoretically correct criterion because it measures the net addition to present-valued welfare, is additive across independent projects, and — unlike IRR — always gives a coherent ranking.

Worked Example 4.1 (NPV of a restoration project, verified). An FRD-style restoration project costs €100 (say, €100k) up front and yields €10/year of net ecosystem-service benefits for 20 years. At a social discount rate $r = 3.5\%$ (the UK Green Book's central rate for this horizon):

$$\mathrm{NPV} = -100 + 10 \cdot A(0.035, 20), \qquad A(0.035, 20) = \frac{1 - (1.035)^{-20}}{0.035} = \frac{1 - 0.50257}{0.035} = 14.2124.$$

So PV of benefits $= 10 \times 14.2124 = 142.12$, and

$$\mathrm{NPV} = -100 + 142.12 = €42.12 > 0.$$

The project passes at 3.5%. But hold the verdict lightly: §4.10 shows it fails above ~7.75%, and whether 3.5% or 7% is "right" is the judgment of §4.8. The arithmetic is settled; the assumption is not.

FNPV versus ENPV — the financial/economic distinction (your FRD paper). The same NPV formula is applied twice with different contents:

- **Financial NPV (FNPV)** uses *market* cash flows only — the revenues and costs an investor actually receives and pays — discounted at a *private* (market/financial) rate. For a restoration project, market revenues (e.g., harvestable fish) may be small, so FNPV can be **negative**: privately unprofitable.
- **Economic NPV (ENPV)** uses *social* values — market flows *plus* the non-market ecosystem-service benefits (carbon, biodiversity, coastal protection, non-use value) that your DCE/CV work monetizes, with market prices replaced by **shadow prices** where markets are distorted (Chapter 3, §3.4; Chapter 43) — discounted at a *social* discount rate. ENPV can be **positive** even when FNPV is negative.

Your FRD paper's core message — private non-profitability alongside positive social returns — *is* the statement FNPV $< 0 <$ ENPV. The gap is the wedge between private and social valuation, and it is exactly what justifies public support or blue-finance instruments. This chapter gives you the discounting half; the valuation half (what goes into the ENPV benefit stream) is your stated-preference work, and the shadow-pricing half is Chapter 43.

> **Active recall 4.B.** Give two distinct reasons FNPV and ENPV can have opposite signs for the same restoration project. *(Answer: (i) ENPV includes non-market ecosystem-service benefits — carbon, biodiversity, non-use value — absent from FNPV's market-only flows; (ii) they use different discount rates (private/market vs social) and may replace market prices with shadow prices. Either can flip the sign.)*

4.5 The internal rate of return

Definition 4.4 (Internal rate of return). The **internal rate of return** (IRR) is the discount rate that makes NPV exactly zero:

$$\text{NPV}(\text{IRR}) = \sum_{t=0}^{T} \frac{B_t - C_t}{(1+\text{IRR})^t} = 0.$$

Decision rule (for a conventional project): accept if IRR exceeds the required rate (the hurdle rate / discount rate). Intuitively, the IRR is the project's own "break-even" rate of return; if the cost of capital is below it, the project earns a surplus.

Worked Example 4.2 (IRR of the restoration project, verified). For the project of Worked Example 4.1, the IRR solves $-100 + 10 \cdot A(r, 20) = 0$, i.e. $A(r, 20) = 10$. There is no closed form, so it is found numerically: IRR = 7.75% (verified). Reading it against §4.10: since the crossover is 7.75%, the project is worthwhile at any social discount rate below 7.75% and not above it. The IRR is therefore the same number as the **switching value** of the discount rate — the sensitivity result of §4.10 — which is the most useful way to think about IRR: *not* as a return to report in isolation, but as the threshold rate at which the verdict flips.

Relationship to the NPV curve. NPV(r) is a smooth, generally decreasing function of r for a conventional project (costs early, benefits later): high r shrinks the later benefits more, so NPV falls as r rises, crossing zero once at the IRR. The IRR is the unique root of a downward-sloping NPV curve — *when* the curve is well-behaved. It is not always well-behaved, which is the next section.

4.6 Why IRR misleads: the pitfalls

IRR is intuitive and beloved by practitioners, and it is treacherous. Because your PDF asks you to *interpret* IRR in restoration appraisals, you must know exactly when it breaks. NPV is the criterion of record; IRR is a supplementary summary that fails in identifiable ways.

P1. Multiple IRRs (non-conventional cash flows). The equation NPV(r) = 0 is a polynomial in $v = 1/(1+r)$ of degree T. By Descartes' rule of signs, the number of positive real roots is bounded by the number of **sign changes** in the cash-flow sequence. A conventional project ($-, +, +, \ldots, +$: one sign change) has one IRR. But restoration projects often have **later costs** — decommissioning an FRD, replanting, a maintenance spike, or end-of-life remediation — giving a sequence like $-, +, +, \ldots, -$ with two sign changes and therefore possibly **two IRRs**, neither of which is meaningful as "the return." *Symptom:* the software reports an IRR that changes with the starting guess, or two candidate rates. *Fix:* use NPV; if you must summarize with a rate, use the modified IRR (MIRR) or simply report the full NPV(r) curve.

P2. No real IRR. Some cash-flow patterns give NPV(r) = 0 no real solution (all-positive or all-negative net flows, or complex roots only). *Fix:* NPV is still perfectly well-defined; IRR simply does not exist.

P3. Ranking mutually exclusive projects. IRR ignores *scale*: a small project with a high IRR can add less present value than a large project with a lower IRR. Choosing the higher-IRR project can leave welfare on the table. *Example:* a €10 project with 50% IRR versus a €10,000 project with 12% IRR at an 8% cost of capital — the second adds far more NPV. *Fix:* rank by NPV, not IRR (or by incremental IRR on the difference of the projects, done carefully).

P4. The reinvestment assumption. IRR implicitly assumes interim cash flows are reinvested *at the IRR itself*, which is unrealistic when the IRR is far from the cost of capital. NPV assumes reinvestment at the discount rate, which is the economically correct opportunity cost. *Fix:* MIRR makes the reinvestment rate explicit; NPV sidesteps the issue.

P5. IRR says nothing about the discount-rate controversy. Because IRR is a property of the cash flows alone, it can seduce you into ignoring the social-discount-rate question (§4.8) that IRR was supposed

to help you finesse. The right use of IRR is precisely to *frame* that question: "the project passes iff society's discount rate is below X%." That reframing is valuable; treating IRR as an assumption-free verdict is not.

Bottom line for your work: report **NPV at a defensible social discount rate**, report the **NPV**(r) **sensitivity curve** and its **crossover rate** (which coincides with the IRR for conventional projects), and treat a bare IRR — especially for a project with later costs — with suspicion.

> **Stop and think 4.2.** Your FRD project has an installation cost in year 0, benefits in years 1–19, and a decommissioning cost in year 20. How many sign changes does the net cash-flow sequence have, and what does that imply about the IRR? *(Answer: two sign changes $(-,+,\dots,+,-)$, so up to two real IRRs — the IRR is potentially non-unique and unreliable; use the NPV curve instead.)*

4.7 Real versus nominal: getting the consistency rule exactly right

Inflation is the most common source of silent discounting errors, and the fix is a single discipline: **discount real flows at a real rate, and nominal flows at a nominal rate — never mix.**

Definitions. A **nominal** cash flow is measured in the actual (future) euros paid or received; a **real** cash flow is measured in constant (today's) purchasing power. The **nominal rate** i compensates for both the real opportunity cost and expected inflation; the **real rate** r strips inflation out.

The Fisher equation. The exact relationship between nominal rate i, real rate r, and expected inflation π is

$$(1+i) = (1+r)(1+\pi) \quad \Longrightarrow \quad i = r + \pi + r\pi,$$

often approximated by $i \approx r + \pi$ (dropping the small cross term $r\pi$). Use the *exact* form in appraisal; the approximation error grows with the horizon.

The consistency theorem (why mixing is wrong). Suppose a benefit is €B in today's prices and grows only with inflation, so its nominal value in year t is $B(1+\pi)^t$. Discount it at the nominal rate:

$$\mathrm{PV} = \frac{B(1+\pi)^t}{(1+i)^t} = \frac{B(1+\pi)^t}{[(1+r)(1+\pi)]^t} = \frac{B}{(1+r)^t},$$

which is exactly the real flow B discounted at the real rate. **The inflation terms cancel** — provided you are consistent. Mix them and they do not cancel, producing a systematic bias.

Worked Example 4.3 (The mismatch error, verified). Let a benefit be €10 in real terms, expected inflation $\pi = 2\%$, real rate $r = 3.5\%$, so the nominal rate is $i = (1.035)(1.02) - 1 = 5.57\%$. Value the year-10 benefit four ways:

Treatment	Computation	PV	Verdict
Real flow @ real rate	$10/(1.035)^{10}$	**€7.09**	correct
Nominal flow @ nominal rate	$10(1.02)^{10}/(1.0557)^{10}$	**€7.09**	correct (identical)
Real flow @ nominal rate	$10/(1.0557)^{10}$	€5.82	**wrong — understates by 18%**
Nominal flow @ real rate	$10(1.02)^{10}/(1.035)^{10}$	€8.64	**wrong — overstates by 22%**

The two consistent treatments agree exactly (€7.09); the two mismatches are off by roughly a fifth *at just ten years*, and the error compounds with horizon. For a restoration project with century-scale carbon benefits, a real/nominal mismatch is not a rounding error — it can dominate the result. *In practice:* environmental

CBA is almost always done in **real** terms (constant prices) discounted at a **real** social rate, because inflation forecasts over decades are meaningless; just make sure your benefit *growth* assumptions are real growth, not inflation.

4.8 Private versus social discount rates

Here the mathematics is trivial and the economics is contested. *Which* rate belongs in the ENPV of a public restoration project is one of the most consequential and debated choices in applied welfare economics, and your PDF rightly asks you to be able to "giustificare tasso di sconto."

Why the social rate differs from the market rate. A private investor discounts at their opportunity cost of capital (a market rate), reflecting private risk and return. A *social* appraisal discounts at the **social discount rate (SDR)**, which differs because: society's time horizon is longer than any individual's; public projects pool risk across the population; capital markets are distorted (taxes, externalities) so market rates are not the true social opportunity cost; and society may weigh future generations differently than markets do. The SDR is typically *lower* than private rates, which is one reason socially valuable long-horizon restoration can be worthwhile even when privately unprofitable (the FNPV $<$ ENPV wedge of §4.4).

Two philosophies for setting the SDR.

1. **Descriptive (positive) approach:** the SDR should reflect *observed* market returns — the real opportunity cost of capital — because that is what displaced investment actually earns. This tends to give higher rates (say 4–7%).
2. **Prescriptive (normative) approach:** the SDR should reflect *how society ought to* weigh future welfare, derived from ethical parameters via the Ramsey formula. This tends to give lower rates, especially for the intergenerational component.

The Ramsey formula. The prescriptive social discount rate on *consumption* is

$$r = \delta + \eta\, g,$$

where δ is the **pure rate of time preference** (how much we discount future *utility* simply because it is future — an ethical parameter, arguably near zero for intergenerational problems), η is the **elasticity of marginal utility of consumption** (how fast marginal utility falls as consumption rises — capturing both inequality aversion and consumption smoothing), and g is the expected growth rate of per-capita consumption. The intuition: we discount future benefits partly from impatience (δ) and partly because, if growth continues ($g > 0$), future people will be richer and an extra euro means less to them (ηg).

Worked Example 4.4 (Ramsey rates and the Stern–Nordhaus gap, verified).

Stance	δ	η	g	$r = \delta + \eta g$
Central	1.0%	1.0	2.0%	**3.0%**
Stern-like (low pure time preference)	0.1%	1.0	1.3%	**1.4%**
Descriptive / market-calibrated	3.0%	1.5	2.0%	**6.0%**

The **Stern Review** argued, on ethical grounds, that discounting future generations' *utility* is unjustified ($\delta \approx 0$), yielding a low rate (~1.4%) and therefore a *high* present value of avoided climate damage — a case for aggressive action. **Nordhaus** and others argued the rate should be disciplined by observed market returns, yielding a higher rate (~4.5–6%) and a lower present value of distant damages — a case for a more gradual ramp. **This is not a disagreement about arithmetic; it is a disagreement about δ and η — about how to weigh the welfare of people not yet born.** As an author you should present the Ramsey decomposition, state your parameter choices and their ethical content, and — crucially — show how your conclusion depends on them (§4.10). You are not obliged to adjudicate Stern versus Nordhaus, but you are obliged to be transparent that your restoration NPV inherits their disagreement.

Declining discount rates for long horizons. A growing consensus (and official guidance in the UK and France) uses a **declining** SDR schedule: a rate around 3.5% for the first few decades, falling toward ~1% beyond a century. The rationale is partly uncertainty about future rates (under uncertainty, the *certainty-equivalent* discount factor is dominated by the lowest plausible rate at long horizons — the "Weitzman–Gollier" argument) and partly intergenerational ethics. For your blue-carbon and long-lived restoration benefits, a declining schedule can materially raise the present value of century-scale benefits relative to a constant high rate. Flag which schedule you use.

> **Active recall 4.C.** In the Ramsey formula $r = \delta + \eta g$, which single parameter most sharply separates the Stern and Nordhaus positions, and what does it represent ethically? *(Answer: δ, the pure rate of time preference — whether, and how much, to discount future generations' welfare simply because it lies in the future. Stern takes $\delta \approx 0$ on ethical grounds; Nordhaus takes it higher to match observed behavior/markets.)*

4.9 Intergenerational discounting: why the rate dominates at long horizons

For benefits accruing over a century — the social cost of carbon, blue-carbon sequestration, slow ecosystem recovery — the choice of rate is not a sensitivity issue among others; it *is* the result. The geometric decay of the discount factor makes this unavoidable.

Worked Example 4.5 (The tyranny of the long horizon, verified). The discount factor for a benefit 100 years out:

Rate r	$\mathrm{DF}_{100} = 1/(1+r)^{100}$	€1 in 100 years is worth today
1.0%	0.370	37.0 cents
2.0%	0.138	13.8 cents
3.5%	0.032	3.2 cents
4.0%	0.0198	2.0 cents
7.0%	0.00115	0.1 cents

A benefit a century out is worth **321 times more at 1% than at 7%** (verified: $0.370/0.00115 \approx 321$). No other appraisal assumption comes close to this leverage. The practical consequence for your work: the social cost of carbon — essentially the present value of a stream of future climate damages — is *dominated* by the discount rate, which is why published SCC values range over an order of magnitude largely because of rate choices. Any blue-carbon or long-horizon restoration benefit you monetize inherits this sensitivity. You must (i) state your rate and schedule, (ii) show the result at low, central, and high rates, and (iii) never present a single-rate NPV of a century-scale benefit as if it were precise.

Exponential versus hyperbolic discounting. Standard CBA uses **exponential** discounting ($\mathrm{DF}_t = (1+r)^{-t}$), which is *time-consistent*: the trade-off between year t and $t+1$ is the same whenever you evaluate it. Behavioral evidence shows people often discount **hyperbolically** (e.g., $\mathrm{DF}_t = 1/(1+kt)$), which discounts the near future steeply but the far future shallowly and is *time-inconsistent* (preferences reverse as time passes — the source of present bias). Hyperbolic discounting matters descriptively (for how respondents in your surveys value delayed environmental benefits) and connects to the declining-rate schedules of §4.8, which can be read as a time-consistent approximation to society's hyperbolic-like concern for the far future. For prescriptive CBA, exponential (possibly declining) discounting remains standard; be aware of the behavioral alternative when you interpret stated valuations of delayed benefits.

4.10 Sensitivity analysis over the discount rate

Because the rate is contested and consequential, a single-rate NPV is never an adequate deliverable. The minimum standard is to present NPV as a *function* of the rate and to identify the **switching value** — the rate at which the verdict flips.

Definition 4.5 (Switching value / crossover rate). The **switching value** of the discount rate is the rate r^* at which $\mathrm{NPV}(r^*) = 0$. For a conventional project this is exactly the IRR (§4.5). It answers the decision-relevant question directly: "for what range of defensible social discount rates does this project pass?"

Worked Example 4.6 (Sensitivity table for the restoration project, verified). For the project of Worked Example 4.1 (−100 now, +10/year for 20 years):

Discount rate	NPV	Verdict
1.0%	€80.46	pass (strongly)
3.5%	€42.12	pass
5.0%	€24.62	pass
7.75%	€0.03 ≈ 0	**switching value (= IRR)**
10.0%	−€14.86	fail

The verdict flips at ~7.75%. Now the interpretation is honest and decision-useful: *the project is worthwhile at any social discount rate below about 7.75%.* Whether it passes is then a transparent function of the §4.8 debate — at Stern-like (1.4%) or central (3.0–3.5%) social rates it passes comfortably; only at high descriptive rates near 8% does it fail. This is exactly how you should present an FRD appraisal: not "NPV = €42.12" but "NPV is positive for all social discount rates below 7.75%, and equals €42.12 at the 3.5% central rate."

The calculus of sensitivity (why NPV falls in r). Differentiating NPV with respect to the rate,

$$\frac{d\,\mathrm{NPV}}{dr} = -\sum_{t=1}^{T} \frac{t\,(B_t - C_t)}{(1+r)^{t+1}},$$

which is negative for a conventional project (net benefits positive and later): each net inflow is weighted by its *timing* t, so later flows are penalized more heavily as r rises. The magnitude of this derivative is large precisely when benefits are far in the future — the mathematical statement of "long-horizon projects are rate-sensitive," and a direct link to Chapter 2's marginal-effect thinking applied to appraisal. Full probabilistic treatment (Monte Carlo over the rate and other uncertain inputs) is Chapter 45; the deterministic switching value is the indispensable first step.

4.11 Stock versus flow, and the double-counting trap

This section addresses your PDF's §6.9 top-priority error directly, and it turns out to be a theorem about the perpetuity formula. The distinction between a **stock** and a **flow** is where restoration valuations most often go wrong, and discounting is the tool that both creates and resolves the confusion.

Definitions. A **stock** is a quantity existing at a point in time (a standing biomass of fish, a hectare of restored habitat, tonnes of stored blue carbon). A **flow** is a quantity *per unit time* (the annual sustainable harvest, the yearly carbon sequestered, the annual value of ecosystem services). A stock is measured in "units"; a flow in "units per year." They are not the same kind of object and cannot be added.

The theorem that prevents the error. Consider a renewable resource — say a fish population — managed for a **sustainable** annual harvest worth F euros per year, in perpetuity. Its **asset (capital)**

value is the present value of that perpetual flow, which by §4.3 is

$$\text{Asset value} = \frac{F}{r}.$$

Now suppose the standing biomass, if liquidated, has a market value V_{stock}. **The claim is that a correct valuation counts *either* the asset value of the sustainable flow *or* the stock value — not their sum — because they are two accountings of the same underlying natural capital.** The two are *conceptually* the same asset seen twice: the natural capital is the stock, and what makes it valuable is the stream of services it yields.

A caveat worth stating precisely. It is **not** a theorem that the liquidation value of the standing biomass equals the capitalized sustainable rent F/r. The two coincide only under strong conditions — a genuine steady state, harvest at the rent-maximizing level, and a market price for the stock that reflects its productive rather than its scrap value. In general they differ, and the fisheries-economics literature (Clark–Munro) and the forestry analogue (Faustmann) are precisely about how and why they diverge. The discipline the section teaches is therefore the *accounting* rule — **count the asset or the discounted flow, never both** — not a claim that the two numbers are equal. In the worked example below they are equal *by construction*, which is what makes the double count visible.

Worked Example 4.7 (Double counting a fish stock, verified). Suppose an FRD-restored fish population yields a sustainable harvest worth $F = €20$/year, and the discount rate is $r = 4\%$. The asset value of the sustainable flow is

$$\frac{F}{r} = \frac{20}{0.04} = €500.$$

Suppose the standing biomass also has a "stock value" of €500 (its capitalized productive value, by construction). A careless appraisal writes:

$$\text{WRONG: } \underbrace{500}_{\text{stock}} + \underbrace{500}_{\text{PV of flow}} = €1000,$$

double-counting the same natural capital and doubling the apparent benefit. The correct value is €500: the stock's value simply *is* the capitalized flow it produces. You count the tree or the apples-in-perpetuity it bears, never both.

How the trap operates in ecosystem-service CBA (and how to avoid it). The danger multiplies across the benefit categories your PDF lists — "evitare double counting tra fisheries, biodiversity, tourism e habitat value." Overlaps are pervasive: the fishery value and the biodiversity value may both be capturing the same fish; the tourism value and the habitat value may both reflect the same reef. The disciplines that prevent it:

- **Decide stock or flow, and be consistent.** Either value the natural-capital *asset* (a stock, = PV of its service flows) or value the annual *service flows* (and discount them), but never add an asset value to a discounted flow of the same service.
- **Map each euro to one service and one channel.** Build an explicit benefit-accounting table where every benefit is attributed to exactly one category, and check that no physical quantity (a fish, a tonne of carbon, a visitor-day) is monetized twice through different attributes — including across the attributes of your own DCE, where "biodiversity" and "fish abundance" attributes may overlap in respondents' minds (a link to attribute design, Chapter 37, and to what your WTP estimates actually represent, Chapter 35).
- **Watch the sustainability assumption.** F/r is the value of a *sustainable* flow. If the "flow" is actually drawing down the stock (unsustainable harvest), capitalizing it overstates value and ignores the stock depletion — a stock–flow error in the opposite direction.

Active recall 4.D. State the perpetuity identity that makes "stock value + PV of sustainable flow" a double count, and give the one-sentence rule it implies. *(Answer: the asset value of a perpetual sustainable flow F is F/r, which equals the value of the stock that produces it; the rule: count the natural-capital asset value **or** the discounted service flow, never both.)*

4.12 Failure modes: how discounting misleads in practice

F1. Undefended discount rate. Reporting a single-rate NPV without justifying the rate or showing sensitivity. *Fix:* state the rate and its basis (§4.8), report the NPV(r) curve and switching value (§4.10).

F2. Real/nominal mismatch. Discounting real flows at a nominal rate (understates) or nominal flows at a real rate (overstates). *Symptom:* results sensitive to inflation assumptions that should cancel. *Fix:* work in real terms with a real rate; use the exact Fisher relation if converting (§4.7).

F3. IRR misuse. Reporting a single IRR for a project with later costs (multiple/no IRR), or ranking mutually exclusive projects by IRR (ignores scale and reinvestment). *Fix:* use NPV; treat the crossover rate as the honest summary (§4.6).

F4. Stock–flow double counting. Adding a stock value to the discounted flow of the same natural capital, or across overlapping ecosystem-service categories. *Fix:* one-service-one-euro accounting; asset value *or* flow, not both (§4.11).

F5. Discount-rate sleight of hand at long horizons. Using a high constant rate that silently zeroes out century-scale climate/blue-carbon benefits, or a very low rate that inflates them, without flagging the leverage. *Fix:* show low/central/high rates (and a declining schedule) and state the 100-year discount factors explicitly (§4.9).

F6. Discounting benefits that are not yet welfare measures. Discounting a stream of "benefits" that are stated-preference indicators rather than valid compensating variations (Chapter 3, §3.10; Chapter 34) launders a valuation problem into a finance calculation. *Fix:* ensure the flows entering the ENPV are defensible welfare measures before discounting them.

F7. Ignoring risk and the risk-free/risky rate distinction. Discounting risky ecosystem-service benefits at a risk-free rate (or vice versa) without adjusting for risk. *Fix:* handle risk explicitly (certainty-equivalents or a risk-adjusted rate; Monte Carlo in Chapter 45), not by an ad hoc rate bump.

F8. Compounding-convention inconsistency. Mixing continuous (e^{-rt}) and discrete ($(1+r)^{-t}$) discounting, or annual and sub-annual compounding, across a single appraisal. *Fix:* pick one convention and apply it throughout (§4.2).

4.13 Diagnostic questions for your own papers

1. **The rate.** What social discount rate did I use, on what basis (descriptive/prescriptive, Ramsey parameters), and did I show sensitivity and the switching value? *(§4.8, §4.10.)*
2. **Sign robustness.** For what range of discount rates is my NPV positive? Is the sign of my conclusion an artifact of the rate? *(§4.10.)*
3. **FNPV vs ENPV.** Have I clearly separated financial and economic NPV, and is my "private unprofitable, socially valuable" claim a transparent FNPV $< 0 <$ ENPV statement? *(§4.4.)*
4. **Real/nominal.** Are all my flows and my rate consistently real (or consistently nominal)? *(§4.7.)*
5. **Stock vs flow.** Have I counted each item of natural capital once — asset value *or* discounted flow — with no double counting across fisheries, biodiversity, tourism, habitat, and carbon? *(§4.11.)*
6. **Long horizon.** For century-scale benefits (carbon, blue carbon), did I report results across rates and consider a declining schedule, given the 100-year discount factor's extreme sensitivity? *(§4.9.)*
7. **IRR hygiene.** If I report an IRR, is the project conventional (one sign change)? If not, did I avoid IRR and use NPV? *(§4.6.)*
8. **Welfare validity of the flows.** Are the benefits I discount valid welfare measures (compensating variations), not raw stated-preference indicators? *(§4.4, F6.)*

4.14 Exercises

(A) Comprehension.

A1. A blue-carbon project sequesters carbon worth €5/year for 30 years, starting next year, with an upfront cost of €70. At a real social discount rate of 3%, compute the annuity factor, the present value of benefits, and the NPV. Does it pass?

A2. A wetland yields €40/year of ecosystem services in perpetuity. Compute its present value at 2%, 3.5%, and 6%. By what factor does the value change between 2% and 6%, and why?

A3. Expected inflation is 2.5% and the real social discount rate is 3.5%. Compute the exact nominal rate via the Fisher equation. If a colleague discounts a stream of *real* benefits at this nominal rate, will they over- or under-state the present value? Explain.

(B) Derivation.

B1. Derive the annuity factor $A(r,T) = [1-(1+r)^{-T}]/r$ from the finite geometric series, and take the limit $T \to \infty$ to obtain the perpetuity value X/r. State the condition on r needed for the limit to exist.

B2. Show that a real cash flow B growing at inflation π, discounted at the nominal rate $i = (1+r)(1+\pi)-1$, has the same present value as the real flow B discounted at the real rate r. Then show algebraically the sign of the error from discounting the real flow at the nominal rate.

B3. For a renewable resource yielding a sustainable annual rent F in perpetuity at discount rate r, derive the asset value and explain, using this result, why adding the standing-stock value to the discounted harvest flow double-counts. State the condition (sustainability) under which the identity holds.

(C) Research application / critique.

C1. Your FRD cost-benefit analysis reports "NPV = €2.3M at a 3% discount rate, IRR = 9%." A reviewer asks whether the project is robust to the discount rate. Write a response that (i) explains what the 9% IRR already tells you about robustness, (ii) notes one reason the IRR might be unreliable for a restoration project, and (iii) states what additional sensitivity output you would provide.

C2. A restoration appraisal values the recovered fish population two ways and adds them: the market value of the standing biomass (€1.2M) plus the present value of 25 years of harvest revenue (€1.0M), for a "total fishery benefit" of €2.2M. Diagnose the error using §4.11, and explain how you would correct the accounting.

C3. Your blue-carbon benefits accrue over 100 years. Explain, with reference to the 100-year discount factors of §4.9 and the Stern–Nordhaus debate of §4.8, why you cannot report a single-rate present value as if it were precise, and describe exactly how you would present the carbon-benefit result to a policy audience.

4.15 Solutions

A1. $A(0.03, 30) = [1-(1.03)^{-30}]/0.03$. $(1.03)^{30} \approx 2.4273$, so $(1.03)^{-30} \approx 0.4120$; $A = (1-0.4120)/0.03 = 0.5880/0.03 = 19.60$. PV of benefits $= 5 \times 19.60 =$ €98.0; NPV $= -70 + 98.0 =$ €28.0 > 0. It passes at 3%.

A2. Perpetuity $= 40/r$: at 2%, $40/0.02 =$ €2000; at 3.5%, $40/0.035 =$ €1143; at 6%, $40/0.06 =$ €667. Between 2% and 6% the value changes by a factor of $2000/667 = 3.0$ — exactly the ratio $0.06/0.02 = 3$, because for a perpetuity value is inversely proportional to r. Tripling the rate divides the value by three.

A3. Fisher: $i = (1.035)(1.025) - 1 = 1.061375 - 1 = 6.14\%$. Discounting *real* benefits at this *nominal* rate over-discounts (the denominator carries an inflation component the numerator lacks), so the present value is **understated**, increasingly so with horizon.

B1. From §4.3: with $v = 1/(1+r)$, $S = \sum_{t=1}^{T} v^t = v(1-v^T)/(1-v)$. Substituting $v = 1/(1+r)$ and $1-v = r/(1+r)$: $S = \frac{1/(1+r)\cdot(1-(1+r)^{-T})}{r/(1+r)} = \frac{1-(1+r)^{-T}}{r}$, so PV $= XS = X \cdot A(r,T)$. As $T \to \infty$, $(1+r)^{-T} \to 0$

provided $r > 0$, giving $A \to 1/r$ and PV $\to X/r$. The condition $r > 0$ (more precisely $|v| < 1$) is required for the geometric series to converge.

B2. Nominal flow in year t is $B(1+\pi)^t$; discount at i: PV $= B(1+\pi)^t/(1+i)^t$. With $1+i = (1+r)(1+\pi)$, $(1+i)^t = (1+r)^t(1+\pi)^t$, so PV $= B(1+\pi)^t/[(1+r)^t(1+\pi)^t] = B/(1+r)^t$ — the real flow at the real rate. Discounting the *real* flow B at the nominal rate gives $B/(1+i)^t = B/[(1+r)^t(1+\pi)^t] < B/(1+r)^t$ since $(1+\pi)^t > 1$; the error is negative (understatement), growing with t.

B3. Asset value $= \sum_{t=1}^{\infty} F/(1+r)^t = F/r$ (perpetuity, §4.3, requires $r > 0$ and the flow sustainable/perpetual). This F/r *is* the value of the natural-capital stock, since the stock's economic worth is the capitalized value of the rents it yields in perpetuity. Adding a separately-assessed stock value to F/r therefore counts the same natural capital twice. The identity requires the harvest to be **sustainable** (steady-state): if the flow depletes the stock, capitalizing it overstates value and ignores the declining stock — a different stock–flow error.

C1. (i) An IRR of 9% means NPV is positive for every discount rate below 9%; since plausible social discount rates (1.4–6%) all lie below 9%, the project is robust to the rate across the entire defensible range — that is genuinely reassuring. (ii) But if the FRD has end-of-life decommissioning or a maintenance cost spike, the cash flows have more than one sign change, so the 9% IRR may be non-unique or unreliable; I would verify the sign-change count. (iii) I would provide the full NPV(r) curve from ~1% to ~9%, report the switching value explicitly, and give the NPV at low/central/high social rates so the reader sees the sign is stable throughout — not just a point IRR.

C2. The appraisal double-counts the fish population: the €1.2M standing-biomass value and the €1.0M PV of harvest revenue are two accountings of the *same* natural capital (§4.11). If the harvest is sustainable, the capitalized harvest flow already *is* the asset value of the stock, so they should not be summed. Correction: count *either* the asset value of the sustainable fishery (PV of the perpetual sustainable rent) *or* the discounted flow of harvest benefits over the appraisal horizon plus a terminal (residual) stock value — but consistently, with no overlap. Reporting €2.2M overstates the fishery benefit by roughly a factor of two; the defensible figure is on the order of €1.0–1.2M, not their sum.

C3. At 100 years the discount factor is 0.370 at 1% but 0.00115 at 7% — a 321-fold difference — so the present value of century-scale carbon benefits is dominated by the rate, and the rate is exactly what Stern and Nordhaus disagree about (via δ and η in the Ramsey formula). A single-rate PV therefore encodes a contested ethical choice as if it were a fact. I would present the carbon benefit as a *range*: PV at a Stern-like (~1.4%), central (~3–3.5%), and Nordhaus-like (~6%) rate, plus a declining-rate schedule, with the Ramsey parameters and their ethical interpretation stated explicitly, and I would show how the overall project verdict does or does not change across that range — letting the policy audience see the dependence rather than hiding it behind one number.

4.16 Spaced review prompts

+1 day. From memory, derive the annuity factor and the perpetuity value from the geometric series, and compute the NPV of a project costing 100 now with 10/year for 20 years at 3.5% (target: €42).

+1 week. Write the Ramsey formula, plug in Stern-like and Nordhaus-like parameters, and explain in three sentences why the same restoration project can pass under one and fail under the other. Then state the real/nominal consistency rule and why the inflation terms cancel.

+1 month. Take your FRD (or a stylized restoration) cash-flow stream. Build the NPV(r) sensitivity table, find the switching value, and write one paragraph checking for stock–flow double counting across your benefit categories. Close with the spine sentence for this project.

4.17 How this chapter connects to the rest of the book

- **Chapter 3** supplied the benefits that get discounted: compensating variation (§3.10) is the welfare measure whose *stream* forms the ENPV numerator, and the shadow-price reading of the multiplier (§3.4) is how market prices become the social prices used in ENPV.
- **Chapter 2** supplied the geometric-series and derivative tools; the sensitivity derivative $d\text{NPV}/dr$ (§4.10) is Chapter 2's marginal-effect reasoning applied to appraisal.
- **Chapters 35, 39 (public goods and TEV, behavioural welfare)** provide the ecosystem-service benefits, the social cost of carbon, and the behavioral (hyperbolic) discounting that §4.9 contrasts with prescriptive CBA.
- **Chapters 43–45 (cost-benefit analysis)** are the full development of this engine: Chapter 43 (FNPV/ENPV, shadow pricing), Chapter 4 §4.11 (double counting, stock–flow consistency — the full treatment of §4.11), Chapter 45 (Monte Carlo over the rate and other inputs — the probabilistic extension of §4.10's deterministic sensitivity).
- **Chapter 34 (welfare measures)** enforces that the discounted flows are valid welfare measures, closing failure mode F6.
- Discounting also connects back to your **stated-preference** work: how respondents value *delayed* environmental benefits (Chapters 27, 39) is an empirical discounting question, and the benefit-transfer that feeds a CBA (Chapters 35, 49) delivers the values you then discount here.

This chapter completes **Part I**: you now hold the mathematical foundations — linear algebra, multivariate calculus, constrained optimization, and discounting — on which the statistics, econometrics, and welfare analysis of Parts II onward are built.

4.18 Research translation task

For your **FRD / restoration CBA**, produce a three-page note:

1. **Lay out the cash-flow stream** in real terms: upfront and ongoing costs, and the annual benefit stream separated into market (financial) flows and non-market ecosystem-service flows (from your DCE/CV valuation), with any later costs (decommissioning) explicit.
2. **Compute FNPV and ENPV** at a stated central social discount rate, showing clearly how the non-market benefits move the verdict from private unprofitability to social value (§4.4).
3. **Build the NPV(r) sensitivity table** across a Stern-like, central, and Nordhaus-like rate (and a declining schedule if horizons are long), and report the switching value (§4.8–§4.10).
4. **Audit for stock–flow double counting** across fisheries, biodiversity, tourism, habitat, and carbon: build a one-service-one-euro benefit table and confirm no natural-capital item is counted as both a stock and a discounted flow (§4.11).
5. **State your discounting conventions** (real terms, discrete vs continuous, constant vs declining rate) and your Ramsey parameters with their ethical content.
6. **Close** with the spine sentence: *"This restoration project has ENPV [value] at [rate]; it is worthwhile for all social discount rates below [switching value]; the result is fragile to [the discount-rate debate / long-horizon carbon benefits / stock–flow accounting], and I have shown that dependence rather than hidden it."*

Carry this into Chapters 43–45, where shadow pricing, full double-counting analysis, and Monte Carlo uncertainty complete the appraisal.

4.19 Bibliography

Core. - Boardman, Greenberg, Vining & Weimer, *Cost-Benefit Analysis: Concepts and Practice.* The standard applied CBA textbook; chapters on discounting, NPV/IRR, and the social discount rate cover §4.4–§4.10

thoroughly and at the right level for your work. *(Verify edition.)* - HM Treasury, *The Green Book: Central Government Guidance on Appraisal and Evaluation*. The authoritative practitioner guidance on the social discount rate (including the declining schedule and the Ramsey decomposition) and on real-terms appraisal; directly relevant to how you should set and defend your rate. Freely available — *verify current version.* - Pearce, Atkinson & Mourato, *Cost-Benefit Analysis and the Environment: Recent Developments* (OECD). CBA specialized to environmental and non-market contexts — discounting, ecosystem-service valuation, and the FNPV/ENPV distinction for exactly your kind of project. *(Verify edition/year.)*

Applied. - EU / European Commission, *Guide to Cost-Benefit Analysis of Investment Projects*. Operational guidance on FNPV vs ENPV, shadow pricing, and sensitivity/switching values as used in European appraisal — a practical template for the FRD note. *(Verify current edition.)* - Groom, Hepburn, Koundouri & Pearce, "Declining Discount Rates: The Long and the Short of It," *Environmental and Resource Economics*. The accessible treatment of declining discount rates and long-horizon environmental appraisal underpinning §4.8–§4.9. *(Verify volume/year.)* - The blue-carbon / natural-capital valuation literature (e.g., relevant TEEB and Natural Capital Committee reports) for how carbon and ecosystem-service flows are monetized before discounting. *(I am not certain of the exact best citations here — verify and select those matching your FRD context.)*

Advanced. - Stern, *The Economics of Climate Change: The Stern Review*, and Nordhaus's responses (e.g., *A Question of Balance*). Read both sides on the intergenerational discount rate; §4.8's Ramsey debate is drawn from this exchange, and reading the primary sources is the best way to form your own defensible stance. *(Verify editions.)* - Weitzman, "Why the Far-Distant Future Should Be Discounted at Its Lowest Possible Rate," *Journal of Environmental Economics and Management*, and Gollier's work on discounting under uncertainty. The theoretical basis for declining certainty-equivalent discount rates (§4.8–§4.9). *(Verify volume/year.)* - Arrow et al., "Determining Benefits and Costs for Future Generations," *Science*, on intergenerational discounting and its ethical foundations — a concise expert synthesis. *(Verify year/volume.)*

Entries marked "verify" are pointers; confirm editions, versions, and — for the blue-carbon and Ramsey-debate sources — the precise citations before using them, per the book's no-unverified-citation rule.

End of Chapter 4 and of Part I. You now hold the mathematical foundations — applied linear algebra, multivariate calculus, constrained optimization and duality, and the mathematics of discounting — that Parts II–XIV build on. The natural next step is Part II, Chapter 5 (Random Variables and the Distributions of Stated Preference), which opens the probability and inference block (your PDF's Blocco 1) and feeds directly into maximum likelihood. Tell me when to write it, or request a different chapter — for example, if you want to start the high-priority MLE block immediately, Chapter 11.

Part I Assessment — Mathematical Foundations

Covers Chapters 1–4. Five components, in the order they are best attempted. Allow a full working day; the mock oral is the part that matters most for your viva.

1. Cumulative problem set

P1. Given $\mathbf{X}'\mathbf{X} = \begin{pmatrix} 4 & 100 \\ 100 & 3000 \end{pmatrix}$, compute the determinant, the inverse, both eigenvalues and the condition number. Rescale the second column to units of ten and recompute. State in one sentence what the rescaling did and did not change.

P2. For $\mathbf{V} = \begin{pmatrix} 9 & -2 \\ -2 & 1 \end{pmatrix}$ on $(\hat{\beta}_{\text{attr}}, \hat{\beta}_{\text{cost}})$, report both standard errors, the correlation of the estimates, and the standard error of the contrast $\hat{\beta}_{\text{attr}} - \hat{\beta}_{\text{cost}}$. Then Cholesky-factor $\mathbf{V}$ and write the explicit formula for one simulated draw.

P3. Differentiate the binary probit log-likelihood by hand to obtain the score. Show that the generalized residual reduces to $y_i - \Lambda_i$ when Φ is replaced by Λ.

P4. A respondent has $U = \sum_k \psi_k \ln(x_k/\gamma_k + 1) + x_0$ with $\psi = (8, 3, 1)$, $\gamma = (2, 2, 4)$, $p = (2, 1, 2)$, $m = 12$. Solve the allocation by hand using the corner rule, and verify the interior first-order conditions.

P5. A project costs €250 now and returns €22/year for 25 years. Compute the NPV at 2%, 3.5% and 6%, find the switching value, and state the verdict as a function of the social discount rate rather than as a single number.

2. Research critique

Take the appraisal section of your FRD paper. For every monetary figure in it, write one line stating (i) whether it is a stock or a flow, (ii) whether it is real or nominal, (iii) which discount rate it was discounted at and why, and (iv) which other benefit category it might overlap with. Then write a paragraph identifying the single figure whose justification is weakest and say what evidence would strengthen it.

3. Red-flag diagnosis

A colleague's EC-MXL output reports every attribute coefficient significant at 1%, a cost coefficient with a standard error four times its point estimate, and a note that the model "converged normally." Write the diagnosis: what is your first hypothesis, which three linear-algebra quantities would you inspect, what would each tell you, and what upstream design change would you propose? State explicitly why "collect more data" is probably the wrong recommendation.

4. Mock oral examination

Answer aloud, without notes, in under three minutes each.

1. Why does the covariance matrix of your estimates involve an *inverse*, and what follows when the matrix being inverted is nearly singular?
2. What is the difference between a rank-deficient design and an ill-conditioned one, and which is more dangerous in practice?
3. Why does a maximum of the log-likelihood require a negative-definite Hessian, and what does a zero eigenvalue mean economically?
4. Derive marginal WTP as a ratio of partial derivatives, and explain from the calculus why it misbehaves.
5. Your CBA passes at 3.5% and fails at 8%. What do you report?

5. Integration mini-project

Take one estimated model from your own work. Extract $\hat{\mathbf{V}}$; report the standard errors, the full correlation matrix of the estimates, and the condition number of the information matrix. Rescale your cost variable and re-estimate. Write one page reporting what changed, what did not, and — in the spine format — what is identified, by what variation, and how fragile the WTP built from it is.

Part II — Probability and Statistical Inference

Chapter 5 — Random Variables and the Distributions of Stated Preference

Part II: Probability and Statistical Inference

> **The spine question, applied to this chapter.** Your PDF says it plainly: *"i tuoi modelli sono tutti probabilistici: probit per SBDC, logit e mixed logit per DCE, random coefficients, WTP distributions, simulated maximum likelihood, meta-regression e publication bias tests."* Every model you estimate rests on a distributional assumption, and those assumptions are not decoration — they *are* the model. Why is your SBDC willingness to pay assumed **normal**? Why does the logit have a **closed form** while the probit does not? Why does the same choice model appear with **Gumbel** errors, and why does that particular distribution make multinomial logit tractable? Why does a normal latent WTP put embarrassing probability mass on *negative* values for a good you believe has non-negative worth? This chapter answers those questions from first principles, because the ability to *judge* a distributional assumption — the second half of every section of your plan — begins with knowing where the distribution came from and what it commits you to. The spine question here: *which distribution did I assume, why that one, what does it imply about the object I care about (WTP, choice probability, welfare), and how fragile is my result to the choice?*

5.0 Conceptual motivation: distributions are assumptions, not defaults

There is a temptation, when you run `apollo_estimate()` or a probit, to treat the distributional choice as a technical setting — "logit because that's what the DCE literature uses," "normal because that's the default for a latent variable." This chapter exists to dismantle that temptation. The three distributions at the heart of your work — the **normal**, the **logistic**, and the **Type-I extreme value (Gumbel)** — each carry a specific economic and statistical meaning, and each generates specific, checkable implications. Consider what is at stake in your own papers:

- Your **SBDC contingent valuation** models WTP as a latent, normally distributed variable, and a yes/no answer censors it (Chapter 15). But the normal is symmetric with support on all of $\mathbb{R}$ — so it necessarily assigns positive probability to *negative* WTP. For a good you believe has non-negative value, is that defensible? Your PDF asks exactly this: *"la normalità della WTP è plausibile? accetti WTP negative? perché?"* You cannot answer without understanding what the normal commits you to.
- Your **DCE** uses logit and mixed logit, whose entire tractability rests on the **Gumbel** error. The reason the multinomial logit has the elegant "softmax" choice probability — and the reason the mixed logit does *not* have a closed form and needs simulation (Chapter 9, 23) — is a fact about these distributions that we will *derive*: the difference of two independent Gumbels is logistic. If you do not know this derivation, the logit is a black box and its famous limitation (independence of irrelevant alternatives, Chapter 21) is a mystery.

- Your worry about **heavy WTP tails** (Chapters 1, 2, 24) is a statement about the tails of a distribution — specifically, what happens when you form a ratio of coefficients, and how the tail behavior of the chosen distribution propagates. You cannot reason about tails without reasoning about distributions.
- The **scale parameter** that entangles with your cost coefficient and drives the scale-versus-preference-heterogeneity problem (Chapter 25) is nothing but the standard deviation of the error distribution. The normalization "fix the error variance to $\pi^2/6$" that your software imposes silently is a statement about the Gumbel, and knowing it is knowing why logit and probit coefficients are not directly comparable (Chapter 18).

So this chapter builds the probabilistic vocabulary of the whole book: random variables and their CDFs (the CDF, not the density, turns out to be the fundamental object for censored and discrete-choice data), the three key distributions and the precise reasons each arises, the central derivation that gives logit its closed form, and finally conditional expectation and variance — the tools for integrating over heterogeneity (mixed logit, random effects) and for decomposing variation into "between" and "within" components (the mathematical skeleton of the scale-versus-preference-heterogeneity distinction). Every abstract object is tied to a distributional judgment you must make in your own work.

5.1 Prerequisites

From **Chapter 2** you need integration (expectations are integrals), the exponential and logarithm, and comfort with a change of variables in an integral (used in the Gumbel derivation, §5.6). From **Chapter 1**, the covariance matrix (§1.8) reappears when we discuss the multivariate normal. No prior probability theory is assumed beyond the intuition that a random variable has a distribution; we build the rest.

Same **notation** as before. New symbols: X, Y, ε for random variables; $f_X(\cdot)$ for a density (PDF); $p_X(\cdot)$ for a probability mass function (PMF); $F_X(\cdot)$ for a cumulative distribution function (CDF); $\mathbb{E}[\cdot]$, $\mathrm{Var}(\cdot)$; Φ, ϕ for the standard normal CDF and density; Λ for the logistic CDF; $\gamma \approx 0.5772$ for the Euler–Mascheroni constant. We write $X \sim \mathcal{D}$ for "X is distributed as $\mathcal{D}$."

5.2 Random variables: discrete, continuous, and the primacy of the CDF

Definition 5.1 (Random variable). A **random variable** X is a function assigning a real number to each outcome of a random experiment. It is **discrete** if it takes values in a countable set (a "yes/no" SBDC response; a choice among J alternatives; a count) and **continuous** if it takes values in a continuum (a latent WTP; an underlying utility).

Discrete random variables. A discrete X is described by its **probability mass function (PMF)** $p_X(x) = \Pr(X = x)$, with $p_X(x) \geq 0$ and $\sum_x p_X(x) = 1$. Your yes/no SBDC answer is **Bernoulli**: $\Pr(Y = 1) = p$, $\Pr(Y = 0) = 1 - p$, the atom you met in Chapters 1–2. A respondent's choice among J alternatives is **categorical (multinomial)**: $\Pr(\text{choose } j) = P_j$, $\sum_j P_j = 1$.

Continuous random variables. A continuous X is described by its **probability density function (PDF)** $f_X(x) \geq 0$, with $\int f_X(x)\,dx = 1$. The density is *not* a probability (it can exceed 1); probabilities are areas: $\Pr(a \leq X \leq b) = \int_a^b f_X(x)\,dx$, and $\Pr(X = x) = 0$ for any single point.

Definition 5.2 (Cumulative distribution function). For any random variable, the **CDF** is

$$F_X(x) = \Pr(X \leq x),$$

non-decreasing, right-continuous, with $F_X(-\infty) = 0$ and $F_X(+\infty) = 1$. For a continuous variable $F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$ and $f_X = F_X'$; the **survival function** is $S_X(x) = 1 - F_X(x) = \Pr(X > x)$.

Why the CDF is the fundamental object for you (not the density). In your work the quantities you actually observe are almost always *probabilities of a latent variable lying in a region*, and those are read off the CDF:

- **SBDC:** you offer a bid b; the respondent says "yes" iff their latent WTP exceeds b. So $\Pr(\text{yes}) = \Pr(\text{WTP} > b) = S_{\text{WTP}}(b) = 1 - F_{\text{WTP}}(b)$. The likelihood is built from the **CDF of WTP**, not its density (Chapter 15). This is why the *shape* of the assumed WTP distribution — normal, logistic, log-normal — matters so directly.
- **Payment card / interval data:** the respondent's answer places WTP in an interval $(\ell, u]$; the probability is $F_{\text{WTP}}(u) - F_{\text{WTP}}(\ell)$ — again the CDF (Chapter 14, 15).
- **Discrete choice:** the choice probability is $\Pr(\text{utility of } i \text{ is largest})$, an integral of a joint density that, for the Gumbel, collapses to a CDF expression (§5.6).

Because the CDF is what the data touch, distributional assumptions bite through the CDF's shape — especially its **tails** (how fast $F \to 0$ and $\to 1$) and its **support** (where F rises from 0). Keep this in mind: when we ask "is normal WTP plausible?", we are asking about the shape and support of F_{WTP}.

Expectation, variance, moments. The **expectation** (mean) is $\mathbb{E}[X] = \sum_x x\, p_X(x)$ (discrete) or $\int x f_X(x)\, dx$ (continuous); the **variance** is $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$. Two rules we use constantly, for constants a, b: $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$ and $\text{Var}(aX + b) = a^2 \text{Var}(X)$. The variance scaling by a^2 is why *scale* (standard deviation) and *variance* differ by squaring — relevant when we normalize error variances below.

> **Active recall 5.A.** In an SBDC design, you offer bid $b = €30$. Write the probability of a "yes" in terms of the CDF of latent WTP, and say which feature of that CDF (tail or support) governs how many "yes" answers you expect at a very high bid. *(Answer:* $\Pr(yes) = 1 - F_{WTP}(30)$*; at a very high bid the* upper tail* of F_{WTP} governs the (small) yes-probability — a heavy upper tail means more people still say yes at high bids, which is exactly how bid design probes the WTP distribution.)*

5.3 The normal distribution

Definition 5.3 (Normal distribution). $X \sim \mathcal{N}(\mu, \sigma^2)$ has density

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R},$$

with mean μ, variance σ^2. The **standard normal** $Z \sim \mathcal{N}(0, 1)$ has density $\phi(z)$ and CDF $\Phi(z)$ (no closed form — it is an integral, which is why probit likelihoods require numerical evaluation of Φ). Any normal is an affine transform of the standard: $X = \mu + \sigma Z$.

Key properties. (i) *Symmetry* about μ: $f(\mu + t) = f(\mu - t)$. (ii) *Thin tails*: density decays like e^{-x^2}, so extreme values are very rare. (iii) *Closed under affine transforms and sums*: if $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, then $aX_1 + bX_2 \sim \mathcal{N}(a\mu_1 + b\mu_2,\ a^2\sigma_1^2 + b^2\sigma_2^2)$. The multivariate version, $\mathbf{X} \sim \mathcal{N}(\mu, \Sigma)$, has the density whose exponent is the quadratic form $-\frac{1}{2}(\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)$ you met in Chapter 1 (§1.6) — and it is generated from independent standard normals via the Cholesky factor (Chapter 1, §1.9), the exact mechanism of Krinsky–Robb and correlated random coefficients.

Why the normal arises. The deep reason is the **central limit theorem** (Chapter 6): a sum of many small, independent influences is approximately normal. So whenever a latent quantity is plausibly the aggregate of many unmodeled factors — the "true" WTP as the sum of many considerations, a measurement error as the sum of many small perturbations — the normal is a principled default. This is the standard justification for **probit** (normal latent error) and for the **normally distributed latent WTP** in your SBDC model.

Why the normal is *dangerous* for WTP. Its virtues are also its liability here. Because the normal is **symmetric with support on all of** $\mathbb{R}$, a normal latent WTP with mean μ and standard deviation σ assigns

probability $\Phi(-\mu/\sigma) > 0$ to *negative* WTP. If your estimated mean WTP is €20 with standard deviation €25, then $\Phi(-20/25) = \Phi(-0.8) \approx 0.21$ — the model says **21% of respondents have negative WTP** for an environmental improvement you believe no one is harmed by. That may be acceptable (some people genuinely dislike the change) or it may be an artifact of forcing a symmetric distribution onto a non-negative quantity. Your PDF's questions — *"accetti WTP negative? perché? il supporto della WTP è coerente con il bene valutato?"* — are precisely about this support mismatch. The alternatives (a log-normal, which has support $(0, \infty)$; a spike-at-zero model; a normal truncated at zero) each embody a different belief about the good, and Chapter 15 develops them. The point here: **the normal's support is an assumption about the economics, and you must own it.**

> **Stop and think 5.1.** Your SBDC model estimates mean WTP €20, standard deviation €25, under normality. Before reading on, predict what happens to the estimated *mean* WTP if you instead impose a distribution with support only on $(0, \infty)$ (e.g., log-normal), holding the data fixed. *(Answer: removing the negative mass generally shifts and reshapes the fitted distribution; the mean can move substantially, and — because non-negative distributions like the log-normal are right-skewed — the mean can be pulled* up* by an upper tail even as the median stays modest. The lesson: reported mean WTP is partly a distributional artifact, which is why you report the distributional assumption and check robustness across distributions, Chapter 15.)*

5.4 The logistic distribution

Definition 5.4 (Logistic distribution). X has a standard **logistic** distribution if its CDF is the **logistic function**

$$\Lambda(x) = \frac{1}{1 + e^{-x}} = \frac{e^{x}}{1 + e^{x}},$$

with density $\Lambda'(x) = \Lambda(x)(1 - \Lambda(x)) = \dfrac{e^{-x}}{(1 + e^{-x})^2}$. It is symmetric about 0, has mean 0, and **variance** $\pi^2/3 \approx 3.29$ (standard deviation $\pi/\sqrt{3} \approx 1.814$). A location-scale version shifts and stretches: $\mu + sX$.

Normal versus logistic. The two look almost identical near the center — both symmetric, bell-shaped — but the logistic has **heavier tails** (its density decays like $e^{-|x|}$, exponentially, versus the normal's e^{-x^2}). Practically this means the logistic assigns more probability to extreme values, so in the middle of the data logit and probit fit almost interchangeably, while they can differ in the tails (relevant when your bids reach far into the WTP distribution). The logistic's decisive practical advantage is that its CDF Λ is a **closed-form elementary function**, whereas the normal's Φ is not. That is why the **logit** likelihood is computationally cheaper than probit and why logit dominates large discrete-choice applications.

The variance normalization (and why logit ≠ probit coefficients). Here is a fact with real consequences for reading your tables. The logistic has standard deviation $\pi/\sqrt{3} \approx 1.814$, while the standard normal has standard deviation 1. In a binary choice model, only the ratio of the utility coefficients to the error scale is identified (Chapter 18, §5.8 below), so the coefficients are estimated *relative to the error's standard deviation*. Because the logistic's scale is ≈ 1.814 times the standard normal's, **logit coefficients are roughly** 1.8 **times the corresponding probit coefficients** — not because the effects differ, but because the yardstick (error scale) differs. Marginal effects and predicted probabilities, being scale-free, are comparable; raw coefficients are not. This is your first concrete encounter with **scale identification**, the theme that culminates in Chapter 25.

> **Active recall 5.B.** A referee compares a coefficient from your logit ($\hat{\beta}_{\text{logit}} = 0.9$) to one from a colleague's probit ($\hat{\beta}_{\text{probit}} = 0.5$) and concludes your effect is "much larger." What is the error, and roughly what would you expect $\hat{\beta}_{\text{logit}}$ to be if the underlying effect were identical? *(Answer: the comparison ignores the error-scale normalization; logit coefficients are ~1.8× probit coefficients for the same effect, so an identical effect would give* $\hat{\beta}_{logit} \approx 1.8 \times 0.5 = 0.9$ *— exactly*

your value. The effects are the same size; only the yardsticks differ. Compare marginal effects, not raw coefficients.)

5.5 The Gumbel (Type-I extreme value) distribution

Definition 5.5 (Gumbel / Type-I Extreme Value distribution). A random variable ε has a standard **Gumbel** (EV1) distribution if its CDF and density are

$$F(z) = \exp(-e^{-z}), \qquad f(z) = e^{-z}\exp(-e^{-z}), \qquad z \in \mathbb{R}.$$

Its mean is the **Euler–Mascheroni constant** $\gamma \approx 0.5772$ (not zero — it is *right-skewed*), and its variance is $\pi^2/6 \approx 1.645$. A location-scale version has CDF $\exp(-e^{-(z-\eta)/\sigma})$ with location η and scale σ; its variance is $\sigma^2\pi^2/6$.

Why "extreme value," and why it is the natural RUM error. The Gumbel is one of the limiting distributions of the **maximum** of a large sample (extreme value theory: just as sums tend to normal by the CLT, maxima tend to one of three extreme-value forms, of which Gumbel is the relevant one for many distributions). This is the deep reason it belongs in random utility theory. In a RUM, the decision is **choose the alternative with the highest utility** — an *argmax*. The mathematics of choice is therefore the mathematics of maxima, and the Gumbel is the distribution built for maxima. Concretely, it has a property no other common distribution has in this form:

Max-stability. If $\varepsilon_1, \dots, \varepsilon_J$ are independent Gumbel$(\eta_j, 1)$, then their maximum is *also* Gumbel:

$$\max_j \varepsilon_j \;\sim\; \text{Gumbel}\Big(\log \textstyle\sum_j e^{\eta_j},\; 1\Big).$$

The location of the maximum is the **log-sum-exp** of the locations. This closure-under-maximization is exactly what makes multinomial logit tractable: the distribution of the best utility stays in the same family, so the choice probabilities and the *expected maximum utility* have closed forms. That expected maximum — $\mathbb{E}[\max_j(V_j + \varepsilon_j)] = \log \sum_j e^{V_j} + \gamma$ (verified numerically below) — is the **logsum / inclusive value** that reappears as the nesting structure of nested logit (Chapter 22) and, crucially for you, as the basis of **welfare measurement in choice models** (the expected-utility change from a policy, Chapter 42). The Gumbel is not an arbitrary convenience; it is the distribution under which "the value of the best option" has a clean formula.

> **Stop and think 5.2.** The Gumbel is right-skewed with mean $\gamma \approx 0.58$, not 0. Yet logit models are written as if the error has mean 0. Predict how this is reconciled and whether it affects the choice probabilities. *(Answer: the non-zero mean is a location shift common to all alternatives, and since choice depends only on utility* differences*, a common location cancels — it is absorbed into the alternative-specific constants and does not affect choice probabilities. Only the* scale* ($\pi^2/6$) matters for identification. This is why we can normalize freely; §5.6 shows the cancellation explicitly.)*

5.6 The central derivation: why the difference of two Gumbels is logistic (and the logit is born)

This is the most important derivation in the probabilistic foundations of your DCE work. It explains why the logit has a closed form, why it is "logistic," and — by extension — where the multinomial softmax comes from. We do it in full.

Setup (binary choice). A respondent chooses between alternatives 0 and 1 with utilities

$$U_0 = V_0 + \varepsilon_0, \qquad U_1 = V_1 + \varepsilon_1,$$

where V_0, V_1 are the deterministic (observed) parts and $\varepsilon_0, \varepsilon_1$ are independent standard Gumbel errors. The respondent chooses 1 iff $U_1 > U_0$, i.e. iff $\varepsilon_1 - \varepsilon_0 > V_0 - V_1$. So the choice probability depends on the distribution of the **difference** $\varepsilon_1 - \varepsilon_0$.

Claim. The difference of two independent standard Gumbels is standard **logistic**:

$$\Pr(\varepsilon_1 - \varepsilon_0 \le t) = \frac{1}{1 + e^{-t}} = \Lambda(t).$$

Derivation. Condition on $\varepsilon_0 = s$ and integrate. Writing F and f for the Gumbel CDF and density,

$$\Pr(\varepsilon_1 - \varepsilon_0 \le t) = \int_{-\infty}^{\infty} \Pr(\varepsilon_1 \le t + s)\, f(s)\, ds = \int_{-\infty}^{\infty} F(t + s)\, f(s)\, ds.$$

Substitute the Gumbel forms $F(t+s) = \exp(-e^{-(t+s)}) = \exp(-e^{-t}e^{-s})$ and $f(s) = e^{-s}\exp(-e^{-s})$:

$$= \int_{-\infty}^{\infty} \exp(-e^{-t}e^{-s})\, e^{-s} \exp(-e^{-s})\, ds = \int_{-\infty}^{\infty} e^{-s} \exp\Big(-e^{-s}(1 + e^{-t})\Big)\, ds.$$

Now change variables: let $u = e^{-s}$, so $du = -e^{-s}\, ds$, i.e. $e^{-s}\, ds = -du$; as $s : -\infty \to \infty$, $u : \infty \to 0$. The integral becomes

$$= \int_{\infty}^{0} \exp(-u(1 + e^{-t}))\, (-du) = \int_{0}^{\infty} \exp(-u(1 + e^{-t}))\, du = \frac{1}{1 + e^{-t}}.$$

The last step is the elementary integral $\int_0^\infty e^{-au} du = 1/a$ with $a = 1 + e^{-t}$. So $\varepsilon_1 - \varepsilon_0 \sim \text{Logistic}(0, 1)$, and its variance is $\pi^2/6 + \pi^2/6 = \pi^2/3$ (two independent Gumbel variances add) — consistent with the logistic variance of §5.4. ■ *(Verified by simulation: the empirical CDF of $\varepsilon_1 - \varepsilon_0$ matches $1/(1 + e^{-t})$ at $t = -1, 0, 1, 2$ to three decimals.)*

The binary logit falls out immediately. Using the claim,

$$\Pr(\text{choose } 1) = \Pr(\varepsilon_1 - \varepsilon_0 > V_0 - V_1) = 1 - \Lambda(V_0 - V_1) = \Lambda(V_1 - V_0) = \frac{e^{V_1 - V_0}}{1 + e^{V_1 - V_0}} = \frac{e^{V_1}}{e^{V_0} + e^{V_1}},$$

using the symmetry $1 - \Lambda(-x) = \Lambda(x)$. **This is the binary logit**, and it is now demystified: it is logistic because the difference of the Gumbel errors is logistic; it has a closed form because that integral evaluated cleanly; and it depends only on the utility *difference* $V_1 - V_0$, so any common component (including the Gumbel's non-zero mean γ) cancels — the point of Stop-and-think 5.2. *(Verified: with $V_1 - V_0 = 0.7$, the closed form gives $\Pr = 0.668$, matching simulation.)*

The multinomial logit (softmax), by the same logic. With J alternatives, $U_j = V_j + \varepsilon_j$, ε_j iid Gumbel, the probability that i has the largest utility is

$$\Pr(\text{choose } i) = \int_{-\infty}^{\infty} \Big[\prod_{j \ne i} \Pr(\varepsilon_j \le s + V_i - V_j)\Big] f(s)\, ds$$

(conditioning on $\varepsilon_i = s$ and requiring every other utility to be below U_i). Substituting the Gumbel CDF, the product becomes $\exp(-e^{-s}e^{-V_i}\sum_{j\ne i} e^{V_j})$; multiplying by $f(s) = e^{-s}\exp(-e^{-s})$ and using $1 + e^{-V_i}\sum_{j\ne i} e^{V_j} = e^{-V_i}\sum_j e^{V_j}$, the same $u = e^{-s}$ substitution gives

$$\boxed{\Pr(\text{choose } i) = \frac{e^{V_i}}{\sum_{j=1}^{J} e^{V_j}}}.$$

This **softmax** formula is the multinomial logit, the workhorse of your DCE (Chapter 21). *(Verified: for $V = (0.5, 1.0, 0.2)$ the softmax gives $(0.295, 0.486, 0.219)$, matching simulated argmax choice frequencies.)*

Two features to carry forward: (i) the probability depends on utilities only through e^{V_j}, so **only differences matter** and one alternative's utility can be normalized; (ii) the ratio $\Pr(i)/\Pr(k) = e^{V_i - V_k}$ depends only on i and k — the **independence of irrelevant alternatives (IIA)**, the logit's signature strength and weakness (Chapter 21–22). Both are consequences of the Gumbel that you have now derived rather than been told.

The scale parameter, made explicit. If instead ε_j has scale σ (variance $\sigma^2\pi^2/6$), the same derivation yields $\Pr(i) = e^{V_i/\sigma}/\sum_j e^{V_j/\sigma}$. Define $\mu = 1/\sigma$ (the "scale parameter"); then utilities are multiplied by μ. Since $V_j = \mathbf{x}_j'\beta$, what the data identify is $\mu\beta$ — the coefficients **times the scale** — never β and μ separately. Software normalizes σ (usually to give variance $\pi^2/6$, i.e. $\mu = 1$). This is the origin of everything in Chapter 25: a treatment that changes the error variance changes μ and hence *apparent* coefficients and WTP, even with identical true preferences.

> **Active recall 5.C.** In one line each: why does the logit have a closed form, and why does that closed form imply IIA? *(Answer: closed form — because the difference of iid Gumbel errors is logistic and the maximization integrates in closed form (max-stability); IIA — because* $\Pr(i)/\Pr(k) = e^{V_i - V_k}$ *involves only i and k, so adding or removing other alternatives cannot change their relative odds.)*

5.7 Conditional distributions, conditional expectation, and conditional variance

The last pillar of this chapter is conditioning — the tool for models with heterogeneity (mixed logit, random effects) and for decomposing variation, which is the mathematical skeleton of the scale-versus-preference-heterogeneity problem.

Definition 5.6 (Conditional expectation and variance). For random variables Y and X, the **conditional expectation** $\mathbb{E}[Y \mid X = x]$ is the mean of Y among units with $X = x$; as x varies it is a function of X, written $\mathbb{E}[Y \mid X]$ (itself a random variable). The **conditional variance** $\mathrm{Var}(Y \mid X)$ is the variance of Y within $X = x$. In econometrics $\mathbb{E}[Y \mid X]$ is the **regression function** — the object OLS approximates (Chapter 16) and the thing "controlling for X" refers to.

Theorem 5.1 (Law of iterated expectations, LIE).

$$\mathbb{E}[Y] = \mathbb{E}\big[\mathbb{E}[Y \mid X]\big].$$

The overall mean is the average of the conditional means. *Why it matters:* this is exactly how a **mixed logit** produces its choice probabilities. Each individual has a choice probability *conditional* on their random coefficients β, namely the logit formula $L_i(\beta)$; the model's (unconditional) choice probability integrates over the coefficient distribution:

$$P_i = \mathbb{E}_\beta[L_i(\beta)] = \int L_i(\beta)\, f(\beta)\, d\beta.$$

This integral has **no closed form** (a logit averaged over a normal is not a logit) — which is precisely why mixed logit needs **simulation** (Chapter 9, 23). The LIE is the reason the mixed-logit likelihood is an expectation, and simulation is how you approximate that expectation.

Theorem 5.2 (Law of total variance).

$$\mathrm{Var}(Y) = \underbrace{\mathbb{E}\big[\mathrm{Var}(Y \mid X)\big]}_{\text{within-group (residual)}} + \underbrace{\mathrm{Var}\big(\mathbb{E}[Y \mid X]\big)}_{\text{between-group (explained)}}.$$

Total variation splits into variation *within* groups (averaged) plus variation *between* group means. *Why it matters for you:* this decomposition is the formal skeleton of the distinction your PDF flags as critical (§6.3) — differences across respondents can come from genuine differences in preference (the *between* term, variation

in $\mathbb{E}[\text{WTP} \mid \text{type}]$) or from noise/scale within groups (the *within* term). It is also the logic of **random effects and multilevel models** (Chapter 19): the between-respondent variance of $\mathbb{E}[Y \mid \text{respondent}]$ is the random-intercept variance, and the within-respondent variance is the residual. And it is the logic of **clustered inference**: multiple choice tasks per respondent share the between-respondent component, so they are not independent (Chapter 17, 19).

Worked Example 5.1 (Variance decomposition of WTP across familiarity groups, verified). Suppose your DCE sample splits into two familiarity groups. Experienced respondents have WTP with mean €30 and variance 100; inexperienced respondents have mean €20 and variance 64; the groups are equal-sized (so X = group is 50/50). Then:

$$\mathbb{E}[\text{WTP}] = 0.5(30) + 0.5(20) = €25 \quad \text{(LIE)}.$$

The **between-group** (preference-heterogeneity) variance is the variance of the group means around 25:

$$\text{Var}\left(\mathbb{E}[\text{WTP} \mid \text{group}]\right) = 0.5(30-25)^2 + 0.5(20-25)^2 = 0.5(25) + 0.5(25) = 25.$$

The **within-group** (residual/scale) variance is the average of the group variances:

$$\mathbb{E}\big[\,\text{Var}(\text{WTP} \mid \text{group})\big] = 0.5(100) + 0.5(64) = 82.$$

Total variance = 82+25 = 107 (verified). Read the decomposition: only 25/107 ≈ 23% of the WTP variation is *between* familiarity groups (the part a "familiarity effect" could explain); 77% is *within* groups. If you observed a raw difference in WTP between groups, this decomposition warns you how much of the total spread is genuinely group-level versus individual noise — and, critically, whether an apparent "familiarity effect" on WTP is a shift in the between-group means (a preference effect) or partly a difference in *within*-group variance (a scale effect). Separating those is Chapter 25; the arithmetic that makes the separation *thinkable* is the law of total variance.

> **Stop and think 5.3.** You find experienced respondents have both a higher mean WTP *and* a smaller within-group variance than inexperienced respondents. Using the law of total variance and the scale intuition of §5.6, name two distinct stories consistent with this, and say why they are hard to separate. *(Answer: (i) a genuine preference difference — experience raises the mean valuation (between-group shift); (ii) a scale difference — experienced respondents choose more consistently (lower error variance, higher scale μ), which mechanically sharpens their responses and can inflate apparent coefficients/WTP. Both produce "higher, tighter" WTP for the experienced group; distinguishing a shift in true preference from a shift in error scale is the scale-vs-preference-heterogeneity problem, and it requires models that let scale differ across groups, Chapter 25.)*

5.8 Synthesis: which distribution, where, and why it matters

Pulling the chapter together, here is the map of the three distributions across your models, and the judgment each entails.

Distribution	Where it appears in your work	Why this distribution	The judgment it forces
Normal $\mathcal{N}(\mu, \sigma^2)$	Probit; latent WTP in SBDC; measurement error	CLT (sum of many small factors); analytic tractability of the multivariate form (Cholesky, Krinsky–Robb)	Symmetric, full-real support ⇒ mass on **negative WTP**; is that economically defensible for your good? (Ch 15)

Distribution	Where it appears in your work	Why this distribution	The judgment it forces
Logistic Λ	Binary/multinomial logit choice probabilities	It is the distribution of the **difference of Gumbel errors** (§5.6); closed-form CDF	Heavier tails than normal; coefficients on a $\pi/\sqrt{3}$ scale $\Rightarrow$ **not comparable** to probit coefficients (Ch 18)
Gumbel (EV1)	The **error** in random utility models (logit, MNL, mixed logit, MDCEV)	Max-stability: the distribution built for the argmax that defines choice; gives closed-form choice probs and expected-max (logsum) welfare	The **scale** ($\pi^2/6$) is normalized, entangling with coefficients $\Rightarrow$ **scale identification** and the scale-vs-preference problem (Ch 25)

The one distinction to never blur: error distribution vs. object-of-interest distribution. The Gumbel/normal is the distribution of the *error* in utility; the *WTP* is a *derived* quantity — a ratio of coefficients (preference space) or a directly parameterized quantity (WTP space), and *its* distribution is a separate modeling choice (Chapter 24). In a mixed logit you assume a distribution for the **random coefficients** (normal, log-normal, triangular) *and* a Gumbel for the **choice error** — two different distributional assumptions doing two different jobs. Confusing "the error is logistic/Gumbel" with "WTP is logistic/Gumbel" is a category error that leads to nonsense. The error distribution governs the *choice probabilities*; the coefficient/WTP distribution governs the *heterogeneity and the welfare object*. Keep them in separate mental boxes.

5.9 Failure modes: how distributional assumptions mislead

F1. Symmetric WTP distribution forcing negative mass. A normal (or logistic) latent WTP puts probability on negative values; for a non-negative good this may be an artifact, and it distorts the mean. *Fix:* justify the support; compare to a non-negative distribution (log-normal), a truncated normal, or a spike model; report robustness (Ch 15).

F2. Reading logit and probit coefficients as comparable. They differ by the error-scale ratio (≈ 1.8). *Symptom:* "my effect is bigger than the probit study's." *Fix:* compare marginal effects or predicted probabilities, never raw coefficients (§5.4, Ch 18).

F3. Confusing the error distribution with the WTP/coefficient distribution. Assuming "WTP is Gumbel because the model is logit," or that the random-coefficient distribution is the same as the error distribution. *Fix:* keep the two assumptions separate (§5.8); state each explicitly.

F4. Ignoring the scale normalization. Treating estimated coefficients as pure preferences when they are preferences $\times$ scale. *Symptom:* attributing a between-group coefficient difference entirely to preferences. *Fix:* recognize $\mu\beta$ is identified, not β; model scale where groups may differ (§5.6, Ch 25).

F5. Mistaking heavy tails for signal. The logistic's and (especially) a mis-specified WTP distribution's tails can produce extreme fitted values (huge WTP); these may be distributional artifacts, not real respondents. *Fix:* inspect the fitted distribution's tails; prefer robust summaries (median, percentile intervals) for skewed WTP (Ch 24).

F6. Assuming independence across correlated observations. Multiple choice tasks per respondent share a between-respondent component (law of total variance), so they are not independent; treating them as independent understates standard errors. *Fix:* cluster or use panel/mixed structures (Ch 17, 19).

F7. Forgetting that only utility differences are identified. Trying to interpret an alternative-specific constant's *level*, or the Gumbel's mean, as meaningful. *Fix:* recall choice depends only on differences; normalize a reference alternative (§5.6).

5.10 Diagnostic questions for your own papers

1. **Support.** For every latent quantity (WTP, utility), does the assumed distribution's *support* match the economics (can WTP be negative for this good)? *(§5.3, F1.)*
2. **Tails.** Are my results sensitive to the *tails* of the assumed distribution — especially fitted mean WTP and extreme predictions? *(§5.4, §5.8, F5.)*
3. **Which distribution, and why.** Can I state *why* I chose normal vs logistic vs Gumbel in each place, rather than "it's the default"? *(§5.8.)*
4. **Error vs object.** Have I kept the error distribution (Gumbel/normal) separate from the coefficient/WTP distribution (normal/log-normal), and stated both? *(§5.8, F3.)*
5. **Scale.** Am I aware that my coefficients are identified only up to the error scale, and have I checked whether group differences could be scale rather than preference? *(§5.6, F4, F2.)*
6. **Dependence.** Do my repeated observations share a between-unit component that breaks independence, and have I accounted for it? *(§5.7, F6.)*
7. **Comparability.** When comparing to other studies, am I comparing scale-free quantities (marginal effects, WTP) rather than raw coefficients from possibly different link functions? *(§5.4, F2.)*

5.11 Exercises

(A) Comprehension.

A1. Latent WTP is assumed $\mathcal{N}(\mu, \sigma^2)$ with $\mu = 15$, $\sigma = 20$. Compute the probability of negative WTP. Is it negligible? What does this suggest about the normality assumption for a good you believe is weakly desirable?

A2. A standard Gumbel has mean $\gamma \approx 0.577$ and variance $\pi^2/6$. In a logit, why does the non-zero mean not affect the choice probabilities? State the property of choice that makes this so.

A3. A colleague's probit reports a bid coefficient of -0.020; your logit on similar data reports -0.036. Are these consistent with the *same* underlying effect? Show the scale reasoning.

(B) Derivation.

B1. Derive the binary logit probability $\Pr(\text{choose } 1) = e^{V_1}/(e^{V_0} + e^{V_1})$ from the assumption that $\varepsilon_0, \varepsilon_1$ are iid standard Gumbel, citing the difference-is-logistic result. Show explicitly where the utility *difference* (not level) enters.

B2. Using the law of total variance, decompose the variance of WTP for a sample with three equal-sized groups having means $(10, 20, 30)$ and variances $(50, 50, 50)$. Report the between- and within-group components and interpret which corresponds to "preference heterogeneity."

B3. Show, from the multinomial logit formula, that $\Pr(i)/\Pr(k) = e^{V_i - V_k}$ and explain why this is the IIA property. Then give one example from a *labelled* choice set where IIA is economically implausible.

(C) Research application / critique.

C1. Your SBDC model, under normality, estimates mean WTP €18 (s.d. €30) but a large fraction of implied negative WTP. A co-author says "negative WTP just means some people dislike the improvement." Write a critique: when is that interpretation defensible, when is it an artifact, and what two alternative specifications would you compare (naming their supports)?

C2. In your DCE you find experienced respondents have larger and more precisely estimated coefficients than inexperienced respondents, and you are tempted to conclude experience *strengthens* preferences. Using §5.6 and §5.7, explain the alternative (scale) explanation and describe, in words, what modeling change would let you separate the two.

C3. Explain to a non-technical co-author why the mixed logit needs simulation while the plain logit does not, using the law of iterated expectations (§5.7) and the closed-form logit derivation (§5.6). Keep it to a short paragraph.

5.12 Solutions

A1. $\Pr(\text{WTP} < 0) = \Phi(-\mu/\sigma) = \Phi(-15/20) = \Phi(-0.75) \approx 0.227$ — about 23%, *not* negligible. For a weakly desirable good this large negative mass suggests the symmetric normal is imposing implausible structure; a non-negative or spike-at-zero specification may fit the economics better (F1, Ch 15).

A2. The Gumbel's mean γ is a common location shift across all alternatives. Choice depends only on utility *differences* $U_i - U_j$, in which any common additive term cancels. So the non-zero mean is absorbed (into alternative-specific constants) and leaves choice probabilities unchanged; only the *scale* is identified.

A3. Logit coefficients are $\approx \pi/\sqrt{3} \approx 1.814$ times probit coefficients for the same effect. $1.814 \times (-0.020) = -0.0363 \approx -0.036$. Yes — the two are consistent with an identical underlying effect; the difference is the error-scale normalization, not a difference in the effect.

B1. With $U_j = V_j + \varepsilon_j$ and iid standard Gumbel ε_j, choose 1 iff $\varepsilon_1 - \varepsilon_0 > V_0 - V_1$. By §5.6, $\varepsilon_1 - \varepsilon_0 \sim \text{Logistic}(0, 1)$, so $\Pr(\text{choose } 1) = 1 - \Lambda(V_0 - V_1) = \Lambda(V_1 - V_0) = e^{V_1 - V_0}/(1 + e^{V_1 - V_0}) = e^{V_1}/(e^{V_0} + e^{V_1})$. The difference $V_1 - V_0$ enters throughout; the levels V_0, V_1 appear only through their difference (dividing top and bottom by e^{V_0}).

B2. $\mathbb{E}[\text{WTP}] = (10+20+30)/3 = 20$. Between: $\frac{1}{3}[(10-20)^2+(20-20)^2+(30-20)^2] = \frac{1}{3}[100+0+100] = 66.7$. Within: average of variances = 50. Total = 116.7. The **between** component (66.7) corresponds to preference heterogeneity — genuine differences in group-mean WTP; the within component (50) is residual/individual variation. Here most variation (57%) is between groups, unlike Worked Example 5.1.

B3. From $\Pr(j) = e^{V_j}/\sum_m e^{V_m}$, the ratio $\Pr(i)/\Pr(k) = e^{V_i}/e^{V_k} = e^{V_i - V_k}$ — the denominator cancels, so the relative odds of i and k depend only on V_i, V_k, not on any other alternative's presence or attributes (IIA). Implausible example: adding a second "red bus" to a {car, blue bus} set should draw riders mainly from the blue bus (near-substitute), but IIA forces it to draw proportionally from car and blue bus alike — the classic red-bus/blue-bus violation, requiring nested or mixed logit (Ch 22).

C1. The "some people dislike it" reading is defensible only if there is a genuine reason for negative valuation (e.g., the change imposes real costs or disamenities on some respondents) and the *magnitude* of negative mass is plausible. It is an **artifact** when the good is unambiguously non-negative (a pure improvement) yet the symmetric normal mechanically manufactures negative WTP to fit the spread. To test: compare (i) a **log-normal** WTP (support $(0, \infty)$, no negative mass, but right-skewed — watch the mean) and (ii) a **spike-at-zero / truncated-normal** model (a point mass at zero plus positive mass), and report how mean and median WTP move across specifications (F1, F5, Ch 15). If results are stable, the negative mass was harmless; if they move a lot, it was an artifact driving your estimates.

C2. The alternative is **scale**: experienced respondents may simply choose more consistently (smaller error variance, larger scale μ), which mechanically enlarges estimated coefficients ($\mu\beta$ is identified) and tightens them, without any change in true preferences. The law of total variance (§5.7) shows a group with smaller *within* variance will look "sharper." To separate, estimate a model that lets the **scale differ by group** (a heteroskedastic/scaled logit, e.g., a scale parameter for experienced vs inexperienced, or a generalized/scaled MNL), so that a genuine preference shift (change in β) is identified apart from a scale shift (change in μ) — Chapter 25.

C3. The plain logit gives each respondent's choice probability in closed form (the softmax), because the Gumbel errors integrate out cleanly (§5.6). The mixed logit says each respondent has their *own* coefficients drawn from a distribution, so their choice probability is a logit *conditional* on those coefficients; the model's overall probability is the **average of these logits over the coefficient distribution** (law of iterated expectations). That average — a logit integrated over a normal — has no closed formula, so we approximate it by **simulation**: draw many coefficient vectors, compute the logit for each, and average (Chapter 9, 23). Simulation is just numerically doing the integral the plain logit didn't need.

5.13 Spaced review prompts

+1 day. From memory, derive that the difference of two iid standard Gumbels is standard logistic, and use it to write the binary logit probability. State in one sentence why this gives the logit a closed form.

+1 week. Write the law of total variance and apply it to a two-group WTP example of your own choosing; identify the between (preference) and within (residual/scale) components. Then state the error-scale reason logit and probit coefficients are not directly comparable.

+1 month. For one of your own models, write down every distributional assumption it makes (the choice-error distribution *and* any coefficient/WTP distribution), justify each in one line, and check the support of your WTP distribution against the economics of the good. Close with the spine sentence.

5.14 How this chapter connects to the rest of the book

- **Chapter 6 (Asymptotics)** explains *why* the normal recurs — the central limit theorem — and gives the convergence concepts behind "approximately normal."
- **Chapters 7–9 (inference, power, simulation)** build on these distributions: standard errors and confidence intervals assume approximate normality of estimators; Krinsky–Robb (Ch 9) draws from the multivariate normal via the Cholesky factor (Ch 1, §1.9); the mixed-logit integral of §5.7 is what simulation approximates.
- **Chapters 14–15 (latent variables, CV likelihoods)** use the CDF-centric view of §5.2 and the normal-support discussion of §5.3 to build the SBDC and interval likelihoods and to confront negative-WTP and distributional choice head-on.
- **Chapter 18 (binary response)** develops the logit/probit scale (non)comparability of §5.4; **Chapters 20–25 (discrete choice)** are built on the Gumbel derivation of §5.6 — IIA, nested and mixed logit, and the scale-vs-preference-heterogeneity problem that §5.6–§5.7 seed.
- **Chapter 19 (repeated observations)** and **Chapter 17 (clustered inference)** are the law-of-total-variance decomposition of §5.7 turned into estimators; **Chapter 41 (MDCEV welfare)** uses the Gumbel logsum (expected-max utility) of §5.5 for welfare measurement.

This chapter is the probabilistic gateway to Part II and to the discrete-choice heart of the book; the Gumbel derivation directly serves the high-dependency chapters **Ch 11** (likelihoods built from these distributions) and **Ch 24** (WTP distributions).

5.15 Research translation task

For your **DCE (or SBDC) model**, produce a two-page note:

1. **Inventory the distributional assumptions.** List every distribution the model uses — the choice-error distribution (Gumbel for logit) *and* every coefficient/WTP distribution (normal, log-normal, etc.) — and state, in one line each, *why* that distribution and what it commits you to.

2. **Check supports.** For your WTP (or latent WTP in SBDC), state the support of the assumed distribution and whether it matches the economics of the good; if it admits negative WTP, quantify the implied fraction and judge whether it is defensible (§5.3).
3. **Locate the scale.** Identify where the error scale is normalized and note that your coefficients are $\mu\beta$; flag any group comparison (experienced/inexperienced, high/low cost vector) that could be confounded by scale (§5.6, forward to Ch 25).
4. **Decompose a key variance.** For one attribute or WTP, use the law of total variance to describe (even roughly) how much variation is between respondents/groups versus within — the preference-vs-residual split (§5.7).
5. **Close** with the spine sentence: *"My choice probabilities rest on a [Gumbel] error; my WTP heterogeneity rests on a [chosen] distribution; the result is fragile to [negative-WTP support / tail behavior / scale entanglement], and I have stated why each distribution was chosen."*

Carry this into Chapters 14–15 (CV likelihoods and negative WTP), Chapter 21 (the Gumbel-based MNL and IIA), and Chapter 25 (scale vs preference heterogeneity).

5.16 Bibliography

Core. - Wooldridge, *Introductory Econometrics: A Modern Approach* (or *Econometric Analysis of Cross Section and Panel Data* for the graduate level), chapters on random variables, conditional expectations, and the normal/logistic in binary models. The conditional-expectation and law-of-iterated-expectations material of §5.7 at the right level. *(Verify edition/chapters.)* - Train, *Discrete Choice Methods with Simulation*, Chapters on logit and the extreme-value assumption. The canonical derivation of §5.6 (Gumbel → logit, the softmax, IIA) and the scale discussion; the single most relevant source for this chapter. Freely available from the author — *verify link.* - Casella & Berger, *Statistical Inference*, chapters on distributions, transformations, and conditional expectation. The rigorous backing for §5.2–§5.7, including the change-of-variables technique used in the Gumbel derivation. *(Verify edition.)*

Applied. - Ben-Akiva & Lerman, *Discrete Choice Analysis.* The classic that develops random utility, the Gumbel error, and the logit from first principles in a transport/choice context; complements Train on §5.5–§5.6. *(Verify edition.)* - Haab & McConnell, *Valuing Environmental and Natural Resources.* Directly addresses the distribution of WTP, negative-WTP problems, and distributional choice in contingent valuation — the §5.3 support issue in exactly your setting. *(Verify edition.)* - Hensher, Rose & Greene, *Applied Choice Analysis.* Practical treatment of logit/probit scale, marginal effects, and the (non)comparability of coefficients (§5.4), with implementation guidance. *(Verify edition.)*

Advanced. - Coles, *An Introduction to Statistical Modeling of Extreme Values.* For the extreme-value theory behind the Gumbel and max-stability (§5.5), if you want the foundations of why EV1 is *the* distribution for maxima. *(Verify.)* - McFadden, "Conditional Logit Analysis of Qualitative Choice Behavior," and "Econometric Models of Probabilistic Choice." The original derivations of the RUM logit from the extreme-value assumption — the primary sources for §5.6. *(Verify exact references/years.)* - Small & Rosen, "Applied Welfare Economics with Discrete Choice Models," *Econometrica.* The logsum/expected-max welfare result of §5.5 developed for applied welfare analysis — essential before Chapter 41. *(Verify volume/year.)*

Entries marked "verify" are pointers; confirm editions, chapters, and the McFadden/Small–Rosen citations before using them, per the book's no-unverified-citation rule.

End of Chapter 5. Next in Part II: Chapter 6 — Asymptotic Theory You Actually Use (laws of large numbers, the central limit theorem, convergence in probability and distribution), which explains why the normal recurs and gives the convergence concepts behind every standard error and confidence interval. Tell me when to write it, or request a different chapter.

Chapter 6 — Asymptotic Theory You Actually Use

Part II: Probability and Statistical Inference

> **The spine question, applied to this chapter.** Every standard error you report, every confidence interval, every p-value, every Wald and likelihood-ratio test rests on a promise: that your estimator is *consistent* (it homes in on the truth as data accumulate) and *asymptotically normal* (its sampling distribution is approximately bell-shaped). Those promises are theorems — the law of large numbers and the central limit theorem — and they come with fine print. This chapter states the theorems precisely and, just as importantly, tells you what they *forbid*: when the sample is too small, when the variance is infinite (your WTP ratio!), when observations are dependent (your repeated choice tasks, your clustered meta-estimates), or when you have too few units (eighteen studies). The spine question here: *when I attach a standard error to an estimate, am I inside the regime where the asymptotic promise holds, or am I quoting a large-sample approximation to a small-sample reality — and how would I know?*

6.0 Conceptual motivation: the fine print behind every standard error

You have already used asymptotic theory without naming it. In Chapter 2 (§2.9) we sketched why the MLE is approximately normal by Taylor-expanding the score and invoking "the LLN and the CLT." In Chapter 1 (§1.11) we read standard errors off the inverse Hessian *because* the estimator is asymptotically normal with that covariance. This chapter pays the debt: it builds the two limit theorems that justify the entire inferential apparatus, and then it does something your PDF repeatedly demands — it teaches you to judge *when the approximation can be trusted*.

That judgment is not academic for your work. Consider where the fine print bites:

- **Many treatment cells, few respondents each.** Your PDF's §6.5 warns: *"Il CV paper ha molte dimensioni: formato, incentivo, cost vector e BRIS … rischia di diluire il sample size per cella."* Asymptotic standard errors and t-tests assume n per cell is "large." When a cell has 40 respondents, the normal approximation to a t-statistic, or to a treatment contrast, may be poor — and a "not significant" result may reflect the approximation and the small n, not a true null (Chapter 8). You need to know that the CLT is an approximation that degrades as cells shrink.
- **The WTP ratio is consistent but biased — and sometimes has no mean at all.** Your recurring worry (Chapters 1, 2, 24) about WTP $= -\hat{\beta}_1/\hat{\beta}_2$ is, at its deepest level, an *asymptotic-theory* statement. A ratio of estimators is a nonlinear function, so it is biased in finite samples even when the coefficients are unbiased (Jensen's inequality). Worse: when the cost coefficient is near zero, the ratio behaves like a **Cauchy** distribution, for which the law of large numbers and the central limit theorem *simply fail* — the mean does not even exist. The heavy tails you have simulated are not a nuisance;

they are a violation of the conditions this chapter lays down.

- **Eighteen studies is not "large."** Your finfish meta-analysis (§6.8 of the plan, §2.8) has WTP estimates clustered in a small number of studies. Asymptotic inference in meta-analysis is asymptotic *in the number of studies*, and eighteen is small — which is exactly why small-sample corrections (Chapter 47 §47.7) exist. Knowing that "asymptotic" means "as the relevant count $\to \infty$" tells you which count matters and whether you have enough of it.
- **Repeated choice tasks and clustered estimates are not independent.** The plain CLT assumes independence. Multiple choice tasks per respondent, and multiple WTP estimates per study, share a common component (Chapter 5, §5.7), so the effective sample size is smaller than the raw count. Naive standard errors that ignore this are too small — a failure of the independence assumption behind the CLT, and the reason for clustered inference (Chapters 17, 19).

So this chapter is the theory of *when your inference is allowed to work*. We build the modes of convergence (in probability, in distribution), prove the law of large numbers, state and demonstrate the central limit theorem, assemble the tools (Slutsky, the continuous mapping theorem) that let you derive the asymptotic distribution of any smooth function of estimators, and then devote a full, honest section to what asymptotics licenses and what it forbids — with the WTP ratio and the meta-analysis as running cautionary tales. This is the chapter that turns "the software printed a standard error" into "I know whether to believe it."

6.1 Prerequisites

From **Chapter 5**: random variables, expectation and variance and their algebra ($\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$), the normal distribution and the CDF, and the variance-decomposition ideas of §5.7. From **Chapter 2**: the Taylor expansion (used for the delta-method preview, §6.7) and the idea of a smooth function of a random argument. From **Chapter 1**: the covariance matrix, for the multivariate versions.

Same **notation**. New symbols: $\xrightarrow{p}$ (convergence in probability), $\xrightarrow{d}$ (convergence in distribution), $\xrightarrow{a.s.}$ (almost sure, mentioned briefly), $o_p(\cdot), O_p(\cdot)$ (stochastic order, used lightly), and subscript n to denote a sequence indexed by sample size, e.g. $\hat{\theta}_n$, $\bar{X}_n$.

6.2 Estimators are sequences; convergence needs a definition

An estimator is not a single number; it is a **rule** that produces a number for each sample, and as the sample size n grows it produces a *sequence* of random variables $\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3, \ldots$. "The estimator is good in large samples" is a statement about where this sequence *goes*. But random variables do not converge the way ordinary numbers do — a sequence of random quantities can settle down in several inequivalent senses, and we need to be precise about which. Two modes carry almost all the weight in econometrics.

Definition 6.1 (Convergence in probability). A sequence of random variables Z_n **converges in probability** to a constant c, written $Z_n \xrightarrow{p} c$, if for every $\varepsilon > 0$,

$$\Pr\left(|Z_n - c| > \varepsilon\right) \longrightarrow 0 \quad \text{as } n \to \infty.$$

Intuition: for any tolerance ε, the probability that Z_n is more than ε away from c becomes negligible. The random variable piles up ever more tightly around c. This is the notion behind **consistency**.

Definition 6.2 (Convergence in distribution). A sequence Z_n **converges in distribution** to a random variable Z (with CDF F), written $Z_n \xrightarrow{d} Z$, if

$$F_{Z_n}(z) \longrightarrow F_Z(z) \quad \text{at every point } z \text{ where } F_Z \text{ is continuous.}$$

Intuition: the *whole shape* of the distribution of Z_n settles down to the shape of Z. This is the notion behind **asymptotic normality** — we do not claim the estimator equals a normal; we claim its (rescaled) distribution *looks like* a normal in large samples.

The relationship between the two (memorize this).

- Convergence in probability to a **constant** implies convergence in distribution to that constant: $Z_n \xrightarrow{p} c \Rightarrow Z_n \xrightarrow{d} c$. (Piling up at a point is a special case of the distribution settling to a point mass.)
- Convergence in distribution does **not** imply convergence in probability. Knowing the *shape* of the limit says nothing about whether any particular sample's value is close to a fixed target. A sequence can converge in distribution to $N(0,1)$ while bouncing around forever.
- Special case: $Z_n \xrightarrow{d} c$ (a constant) **does** imply $Z_n \xrightarrow{p} c$. Convergence in distribution to a constant is as strong as convergence in probability.

(A third mode, **almost sure** convergence $\xrightarrow{a.s.}$, is stronger than $\xrightarrow{p}$ and underlies the *strong* law of large numbers; you rarely need it directly, but it is why simulation averages converge "for sure," §6.4.)

> **Active recall 6.A.** Your estimator $\hat{\theta}_n$ satisfies $\hat{\theta}_n \xrightarrow{p} \theta_0$ (consistency) and $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, V)$ (asymptotic normality). In plain words, what does each statement promise, and why do you need *both*? *(Answer: consistency promises the point estimate homes in on the truth as data grow; asymptotic normality promises the* rescaled error* has an approximately normal distribution, which is what lets you build standard errors and confidence intervals. Consistency alone gives no measure of uncertainty; asymptotic normality supplies the shape needed for inference.)*

6.3 Consistency, and why it is not unbiasedness

Definition 6.3 (Consistency). An estimator $\hat{\theta}_n$ of a parameter θ_0 is **consistent** if $\hat{\theta}_n \xrightarrow{p} \theta_0$. Consistency is the minimal respectability requirement: with enough data, the estimator gets arbitrarily close to the truth with probability approaching one.

A sufficient condition (and a useful tool). Consistency follows if the **mean squared error** goes to zero, $\mathrm{MSE}(\hat{\theta}_n) = \mathrm{Bias}(\hat{\theta}_n)^2 + \mathrm{Var}(\hat{\theta}_n) \to 0$, i.e., if *both* the bias and the variance vanish as $n \to \infty$. The tool that converts this into a probability statement is **Chebyshev's inequality**.

Proposition 6.1 (Markov and Chebyshev inequalities). For a non-negative random variable X and $a > 0$, **Markov**: $\Pr(X \geq a) \leq \mathbb{E}[X]/a$. Applying Markov to $(Z-\mu)^2$ gives **Chebyshev**: for any random variable with mean μ and variance σ^2,

$$\Pr\left(|Z - \mu| \geq k\right) \leq \frac{\sigma^2}{k^2}.$$

Proof of Markov. $\mathbb{E}[X] = \int_0^\infty x f(x)\,dx \geq \int_a^\infty x f(x)\,dx \geq a\int_a^\infty f(x)\,dx = a\Pr(X \geq a)$; divide by a. Chebyshev is Markov applied to $X = (Z-\mu)^2$ with $a = k^2$, since $\mathbb{E}[(Z-\mu)^2] = \sigma^2$. ■

Chebyshev is loose (it uses only the variance, so it never exploits the distribution's shape) but universal — it holds for *any* distribution with finite variance, which makes it the right tool for general proofs. *(Numerically: for $\bar{X}_{50}$ of exponential(1) data, $\varepsilon = 0.2$, Chebyshev bounds $\Pr(|\bar{X}-1| \geq 0.2) \leq \sigma^2/(n\varepsilon^2) = 1/(50\cdot 0.04) = 0.50$; the empirical probability is only 0.15 — the bound holds but is loose, as expected.)*

The distinction that trips people up: consistent ≠ unbiased. These are different properties, and an estimator can have either without the other. *Unbiasedness* is a finite-sample statement: $\mathbb{E}[\hat{\theta}_n] = \theta_0$ for every n. *Consistency* is a large-sample statement: $\hat{\theta}_n \xrightarrow{p} \theta_0$. Neither implies the other:

- **Consistent but biased.** The maximum-likelihood variance estimator $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$ (dividing by n, not $n-1$) is biased downward in finite samples — $\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2 < \sigma^2$ — but consistent, since the bias $\frac{n-1}{n} \to 1$ vanishes. *(Verified: at $n = 5$, $\mathbb{E}[\frac{1}{n}$-variance$] = 0.80$ against a true $\sigma^2 = 1$, while the $\frac{1}{n-1}$ version averages 1.00.)* Most MLEs are of this type: consistent, asymptotically unbiased, but biased in small samples.
- **Unbiased but inconsistent** is possible too (e.g., an estimator that always uses only the first observation is unbiased but never converges).

Why this matters for your WTP. Your WTP ratio $-\hat{\beta}_1/\hat{\beta}_2$ is a **consistent but biased** estimator of the true WTP: because it is a *nonlinear* function of the coefficient estimates, $\mathbb{E}[-\hat{\beta}_1/\hat{\beta}_2] \neq -\beta_1/\beta_2$ even if $\hat{\beta}_1, \hat{\beta}_2$ are unbiased (Jensen's inequality — the expectation of a nonlinear function is not the function of the expectation). The bias shrinks as the sample grows (consistency), but in a finite sample — especially a small treatment cell — your WTP point estimate is systematically off, and the direction and size of the offset depend on the curvature of the ratio and the precision of the cost coefficient. This is a concrete, quantifiable instance of "consistent but biased," and it is why the *median* of a simulated WTP distribution (robust) is often a better summary than the *mean* (bias-and-tail-sensitive). We demonstrate it in §6.9.

> **Stop and think 6.1.** An estimator is unbiased in every sample but its variance does not shrink as n grows. Is it consistent? Conversely, an estimator is biased in every finite sample but its bias and variance both vanish as $n \to \infty$. Is it consistent? *(Answer: the first is **not** consistent — without shrinking variance it keeps bouncing and never piles up at θ_0 (MSE does not $\to 0$); the second **is** consistent — vanishing bias and variance give MSE $\to 0$, hence $\xrightarrow{p} \theta_0$. Consistency is about the whole MSE going to zero, not about unbiasedness.)*

6.4 The Law of Large Numbers

Theorem 6.1 (Weak Law of Large Numbers, WLLN). Let $X_1, \ldots, X_n$ be i.i.d. with finite mean μ and finite variance σ^2. Then the sample mean converges in probability to the population mean:

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \;\xrightarrow{p}\; \mu.$$

Proof (via Chebyshev). The sample mean has $\mathbb{E}[\bar{X}_n] = \mu$ and, by independence, $\mathrm{Var}(\bar{X}_n) = \frac{1}{n^2}\sum_i \mathrm{Var}(X_i) = \frac{\sigma^2}{n}$. Chebyshev then gives, for any $\varepsilon > 0$,

$$\Pr\left(|\bar{X}_n - \mu| \geq \varepsilon\right) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \;\longrightarrow\; 0.$$

So $\bar{X}_n \xrightarrow{p} \mu$. ■ (The **Strong** LLN strengthens $\xrightarrow{p}$ to $\xrightarrow{a.s.}$ and needs only a finite mean, not finite variance; the intuition is the same.)

Why the LLN is everywhere in your work. The WLLN is the reason *averages* recover *expectations*, and estimation is built on averages:

- **Method of moments and the score.** Sample moments converge to population moments, so moment estimators are consistent. The MLE's first-order condition sets a *sample average* of scores to zero; by the LLN this average converges to its expectation, whose root is the true parameter — the consistency half of the MLE story (Chapter 11).
- **Simulation and Monte Carlo integration.** When you approximate the mixed-logit integral (Chapter 5, §5.7) by averaging the logit over simulated coefficient draws, the LLN is *exactly* what guarantees the simulated average converges to the true integral as the number of draws grows. Halton and quasi-Monte-Carlo draws (Chapter 9, 23) are refinements that converge *faster*, but the base guarantee is

the LLN. Your PDF's concern about "sensitivity to Halton draws" is a concern about how fast this convergence is — a finite-simulation-size question layered on top of the finite-sample question.

- **Krinsky–Robb and the bootstrap** (Chapter 9) likewise rely on the LLN: the empirical distribution of many simulated/resampled statistics converges to the target sampling distribution.

The crucial precondition: finite variance (and finite mean). The WLLN requires μ and σ^2 to *exist and be finite*. When they do not — the Cauchy case of §6.9 — the sample mean does not converge at all. Hold this precondition; it is the trapdoor under the WTP ratio.

> **Active recall 6.B.** State the two ways you use the LLN in a mixed-logit estimation and in a WTP confidence-interval simulation. *(Answer: (i) in estimation, the simulated average of individual logits converges to the true choice-probability integral (simulated maximum likelihood); (ii) in inference, the empirical distribution of Krinsky–Robb / bootstrap draws converges to the sampling distribution of the WTP so its percentiles give a valid interval. Both are LLN guarantees, provided the relevant means exist.)*

6.5 Convergence in distribution, revisited

We defined $\xrightarrow{d}$ in §6.2; here is the intuition you need for the CLT. Convergence in distribution is about the *shape* of the sampling distribution stabilizing. It is exactly the right notion for inference, because a confidence interval or a p-value is computed from the *shape* of the estimator's distribution, not from any single realization. When we say "$\sqrt{n}(\widehat{\theta}_n - \theta_0)$ is asymptotically $N(0, V)$," we mean its distribution's shape approaches the normal — so we can compute tail probabilities (p-values) and quantiles (critical values) using the normal, *as an approximation whose quality improves with n*. The phrase "as an approximation" is the whole ethic of this chapter: convergence in distribution licenses using the limit shape, but it is silent on how good the approximation is at your actual n. That silence is filled by simulation, small-sample corrections, and exact/permutation methods (Chapter 10) when n is modest.

6.6 The Central Limit Theorem

Theorem 6.2 (Lindeberg–Lévy Central Limit Theorem). Let $X_1, \dots, X_n$ be i.i.d. with finite mean μ and finite variance $\sigma^2 > 0$. Then the standardized sample mean converges in distribution to a standard normal:

$$\sqrt{n}\,\frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} N(0,1), \qquad \text{equivalently} \qquad \sqrt{n}\,(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

Three features to internalize:

1. **The $\sqrt{n}$ scaling.** The raw error $\bar{X}_n - \mu$ shrinks to zero (that is the LLN). To see a non-degenerate distribution you must *magnify* it by exactly $\sqrt{n}$ — because $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$, so $\mathrm{Var}(\sqrt{n}\,\bar{X}_n) = \sigma^2$ is stabilized. The $\sqrt{n}$ rate is why standard errors scale like $1/\sqrt{n}$: to halve a standard error you need *four times* the data. This single fact governs the economics of your sample-size decisions (Chapter 8).
2. **The limit is normal regardless of the parent distribution.** The X_i can be skewed, discrete, bounded — as long as they have finite variance, the *average* is asymptotically normal. This is why the normal is ubiquitous (Chapter 5, §5.3): estimators are (approximately) averages, and averages are (approximately) normal. It is also why you can put a symmetric normal confidence interval around a mean even when the underlying data are wildly non-normal — *in large enough samples*.
3. **"Large enough" depends on the parent's skewness.** The CLT is a limit; the approximation's quality at finite n depends on how non-normal the parent is. Skewed parents need larger n.

Worked Example 6.1 (The CLT in action, and its finite-sample honesty — verified). Take $X_i \sim$ Exponential(1) — strongly right-skewed (population skewness = 2), mean 1, variance 1. Simulate the distribution of $\sqrt{n}(\bar{X}_n - 1)$, which the CLT says approaches $N(0,1)$:

n	mean of $\sqrt{n}(\bar{X}_n - 1)$	variance	skewness
1	0.00	0.99	**1.98**
5	0.00	1.00	0.90
30	0.00	1.00	0.37
100	0.00	0.99	**0.20**

The mean and variance are already correct at every n (they must be — those are exact), but the **skewness** — the departure from normality — starts at the parent's 2.0 and decays toward 0 like $1/\sqrt{n}$ (skewness of a sample mean = parent skewness$/\sqrt{n}$). At $n = 30$ it is still 0.37; at $n = 100$ it is 0.20. **This is the CLT's honesty:** convergence is real but *gradual*, and for a skewed quantity a modest n leaves visible non-normality. For your small treatment cells, this is the warning: a t-test or normal confidence interval on a skewed outcome in a 40-person cell inherits residual skewness, and the nominal 95% coverage may be off. When in doubt, use a method that does not lean on the CLT — the bootstrap, permutation, or randomization inference (Chapters 9–10).

The general CLT (why it still works when observations are not identical). The Lindeberg–Lévy CLT assumes i.i.d. Real estimators are averages of *non*-identically-distributed terms (different respondents, different covariates). The **Lindeberg–Feller CLT** extends the result to independent, non-identically distributed sequences under a condition (no single term dominates the variance). This is the version that actually justifies asymptotic normality of your regression and MLE estimators. The critical assumption it retains is **independence** — which is exactly what fails for clustered/panel data (§6.9), requiring the clustered CLT and clustered standard errors (Chapters 17, 19).

> **Stop and think 6.2.** You estimate mean WTP in a treatment cell of $n = 45$ where the WTP distribution is strongly right-skewed (a few large values). Predict whether a symmetric $\bar{X} \pm 1.96\,\text{SE}$ interval will have correct 95% coverage, and what you would do instead. *(Answer: probably not — with strong skew and small n, residual skewness (of order parent-skew$/\sqrt{45}$) makes the sampling distribution non-normal, so the symmetric normal interval mis-covers, typically too short on the long-tail side. Use a bootstrap percentile interval or a transformation, and report the median (Chapters 9, 24).)*

6.7 The workhorses: Slutsky's theorem and the continuous mapping theorem

The LLN and CLT are about sample means. Real estimators combine several converging pieces — a "numerator" that is asymptotically normal and a "denominator" that converges to a constant (like the Hessian, Chapter 2 §2.9). Two theorems let you assemble the pieces.

Theorem 6.3 (Continuous Mapping Theorem, CMT). If $Z_n \xrightarrow{d} Z$ and g is a continuous function (at the points that matter), then $g(Z_n) \xrightarrow{d} g(Z)$. Likewise for $\xrightarrow{p}$: $Z_n \xrightarrow{p} c \Rightarrow g(Z_n) \xrightarrow{p} g(c)$. *Continuous transformations preserve convergence.*

Theorem 6.4 (Slutsky's Theorem). If $Z_n \xrightarrow{d} Z$ and $A_n \xrightarrow{p} a$ (a constant), then:

$$A_n + Z_n \xrightarrow{d} a + Z, \qquad A_n Z_n \xrightarrow{d} aZ, \qquad Z_n / A_n \xrightarrow{d} Z/a \ (a \neq 0).$$

Slutsky lets you *replace a consistent estimator by its limit* inside an asymptotic-distribution argument — e.g., replace an estimated variance by the true variance when forming a t-statistic, which is why $\hat{\theta}/\mathrm{SE}(\hat{\theta})$ is asymptotically $N(0,1)$ even though the SE is itself estimated.

The delta method (preview of Chapter 7), from CMT + CLT. The single most useful application for you: the asymptotic distribution of a *smooth function* of an asymptotically normal estimator. If

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \sigma^2) \quad \text{and } g \text{ is differentiable at } \theta_0 \text{ with } g'(\theta_0) \neq 0,$$

then a first-order Taylor expansion $g(\hat{\theta}_n) \approx g(\theta_0) + g'(\theta_0)(\hat{\theta}_n - \theta_0)$ combined with Slutsky gives

$$\sqrt{n}(g(\hat{\theta}_n) - g(\theta_0)) \xrightarrow{d} N(0,\ g'(\theta_0)^2\sigma^2).$$

The multivariate version, $\mathrm{Var}(g(\hat{\theta})) \approx \nabla g'\, \mathbf{V}\, \nabla g$, is the **quadratic form** of Chapter 1 (§1.6) and the tool for the WTP standard error (Chapter 24). It also carries the warning we now make explicit: the delta method is a *first-order* (local, linear) approximation, so it produces a *symmetric* normal interval — fine for mild nonlinearity, misleading for the strongly nonlinear WTP ratio, where you should simulate instead (§6.9, Chapter 9). Everything connects: the delta method is CMT+CLT, and its failure mode is the ratio's non-normality.

6.8 Assembling the asymptotics of an estimator

We can now state the template that every estimator in the book follows, making rigorous the sketch of Chapter 2 (§2.9). For an M-estimator (including the MLE) defined by a first-order condition $\frac{1}{n}\sum_i \psi(X_i, \hat{\theta}_n) = \mathbf{0}$:

1. **Consistency (LLN + CMT).** The sample average $\frac{1}{n}\sum_i \psi(X_i, \theta)$ converges in probability (LLN) to its expectation $\mathbb{E}[\psi(X,\theta)]$, whose root is θ_0; under identification and continuity (CMT), the root of the sample equation converges to the root of the limit equation: $\hat{\theta}_n \xrightarrow{p} \theta_0$.
2. **Asymptotic normality (Taylor + CLT + Slutsky).** Taylor-expand the FOC around θ_0: $\mathbf{0} = \frac{1}{n}\sum \psi(X_i, \theta_0) + [\frac{1}{n}\sum \nabla\psi(X_i, \theta_0)](\hat{\theta}_n - \theta_0) + \ldots$. The bracket $\xrightarrow{p}$ a constant matrix (LLN); the leading term, scaled by $\sqrt{n}$, $\xrightarrow{d}$ normal (CLT, since it is a scaled average of mean-zero terms). Solving and applying Slutsky,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(\mathbf{0},\ \mathbf{V}), \qquad \mathbf{V} = \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1},$$

the **sandwich** form, with $\mathbf{A}$ the limit of the derivative (the Hessian/information) and $\mathbf{B}$ the variance of the score. For the MLE, the information equality collapses this to $\mathbf{V} = \mathbf{I}(\theta_0)^{-1}$ (Chapter 1 §1.11, Chapter 11); when the equality fails (misspecification, clustering), you keep the full sandwich — the theoretical origin of **robust and clustered standard errors** (Chapter 17). This one template — LLN for the denominator, CLT for the numerator, Slutsky to combine — generates the asymptotic distribution of OLS, logit, probit, mixed logit, MDCEV, and meta-regression estimators alike.

Active recall 6.C. In the sandwich $\mathbf{V} = \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$, which limit theorem delivers $\mathbf{A}$ and which delivers $\mathbf{B}$, and what real-world violation makes you keep the full sandwich rather than the simpler $\mathbf{I}^{-1}$? *(Answer: the LLN delivers* $\mathbf{A}$ *(the average Hessian converging to the information); the CLT delivers* $\mathbf{B}$ *(the variance of the score in the asymptotically-normal numerator). You keep the full sandwich under misspecification or dependence — e.g., clustered/panel data — where the information equality* $\mathbf{B} = \mathbf{A}$ *fails; this is robust/clustered inference.)*

6.9 What asymptotics licenses — and what it forbids

This is the chapter's ethical core. Asymptotic theory is a set of promises that hold *in the limit* and *under conditions*. Here is exactly what it lets you do and where it silently breaks — organized as the judgments your PDF demands.

What it licenses. Given consistency and asymptotic normality, you may: treat $\hat{\theta}_n$ as approximately $N(\theta_0, \hat{\mathbf{V}}/n)$; form Wald confidence intervals $\hat{\theta} \pm 1.96\,\text{SE}$ and t/z tests; use the likelihood-ratio, Wald, and score tests with their χ^2 limits (Chapter 12); and use the delta method for smooth transformations. All of these are *approximations that improve with n*.

Forbidden zone 1 — small samples. The approximations degrade as n shrinks, and the degradation is worse for skewed quantities (§6.6). *Your exposure:* CV designs with many cells and few respondents each (§6.5 of the plan). A "not significant" contrast in a 40-person cell may reflect a poor normal approximation and low power, not a true null — which is why equivalence testing and power analysis (Chapter 8) and exact/permutation methods (Chapter 10) matter precisely here.

Forbidden zone 2 — infinite variance: the WTP ratio and the Cauchy. The LLN and CLT *both require finite variance*. When they fail, inference collapses in a way no sample size cures. The canonical failure is the **Cauchy** distribution — which is exactly the ratio of two centered normals. Consider $C = Z_1/Z_2$ with $Z_1, Z_2 \sim N(0,1)$ independent: C has *no mean and no variance* (its density has tails so heavy the defining integrals diverge). Its sample mean does **not** converge — averaging more Cauchy draws does not help, because the average of Cauchys is *itself* Cauchy. *(Verified: the running mean of* 10^5 *Cauchy draws wanders* — $-0.32,\ 1.15,\ 0.65,\ -0.50$ *at* $n = 10^2, 10^3, 10^4, 10^5$ *— never settling.)*

Now the punchline for your work: **WTP $= -\hat{\beta}_1/\hat{\beta}_2$ is a ratio of (asymptotically normal) estimators**, and when the cost coefficient $\hat{\beta}_2$ has a distribution with non-negligible mass near zero, the ratio is *approximately Cauchy* — heavy-tailed, with an ill-defined or highly unstable mean. This is the deepest explanation of the pathology you have met three times (Chapters 1, 2, 24): it is not merely that the ratio is "unstable," it is that **the asymptotic machinery that justifies means and standard errors can fail for the ratio itself.** The demonstration:

Worked Example 6.2 (WTP ratio: consistent, biased, and heavy-tailed — verified). True $\beta_1 = 2$, $\beta_2 = -1$, so true WTP $= 2$. Draw $\hat{\beta}_1 \sim N(2, \text{SE}^2)$, $\hat{\beta}_2 \sim N(-1, \text{SE}^2)$ and form $-\hat{\beta}_1/\hat{\beta}_2$:

SE (shrinks as n grows)	mean of ratio	**median**	$\Pr(\hat{\beta}_2 > 0)$	99th pct
0.5	2.05 (unstable)	1.94	2.3%	**25.3**
0.3	2.27 (unstable)	2.00	0.04%	6.7
0.1	2.02	2.00	0.0%	2.7

Read all three lessons: **(consistency)** as SE $\to 0$ (large n), both mean and median $\to 2$ — the estimator is consistent; **(finite-sample bias & instability)** at SE $= 0.5$ the mean is biased and *noisy* (the near-Cauchy tail makes the sample mean itself an unreliable statistic), while the **median is stable and near the truth**; **(heavy tails)** the 99th percentile is 25 at SE $= 0.5$ — a fat upper tail driven by cost draws near zero, and 2.3% of draws even have the wrong-sign cost coefficient. The operational rules that follow directly: report the **median** WTP and **percentile** intervals (robust to the tail), not the mean and a symmetric delta-method interval; diagnose $\Pr(\hat{\beta}_2$ near $0)$ as the tail's source; and fix it upstream by identifying the cost coefficient better (design spread, Chapter 1 §1.10) or reparameterizing in WTP space with a sign-constrained cost coefficient (Chapter 24). Your PDF's §6.4 red flag is, at bottom, "the CLT may not apply to my WTP."

Forbidden zone 3 — dependence. The CLT that justifies your standard errors assumes independence (or weak dependence). Your data violate it in two places your PDF names: **repeated choice tasks per respondent** (Chapter 5, §5.7 — the tasks share a respondent-level component) and **multiple WTP estimates per study** in the finfish meta-analysis (§6.8). Dependence means the *effective* sample size is smaller than the raw count, so naive standard errors are too small and t-statistics too large. The fix

is inference that accounts for the dependence — clustered standard errors, random effects, robust variance estimation (Chapters 17, 19, 47) — each a modification of the CLT's variance term **B** to reflect within-cluster correlation.

Forbidden zone 4 — too few units: meta-analysis with 18 studies. "Asymptotic" always means "as some count $\to \infty$" — and you must ask *which* count. In your finfish meta-analysis, inference is asymptotic in the **number of studies (clusters)**, not the number of estimates. Eighteen studies is small, so cluster-robust standard errors (which are consistent as the number of clusters grows) are themselves unreliable, and downward-biased. This is precisely why small-sample corrections for cluster-robust variance exist and are essential here (Chapter 47 §47.7). Knowing that the relevant asymptotic count is the number of studies — not the number of WTP estimates — is the difference between defensible and misleading meta-inference.

Forbidden zone 5 — too many parameters. The standard asymptotics assume the number of parameters is fixed while $n \to \infty$. Rich mixed-logit or many-moderator meta-regressions push against this; with many parameters relative to observations, asymptotic approximations degrade (the "pochi studi per troppe covariate" warning, §2.8). Watch the ratio of parameters to (effective) sample size.

> **Stop and think 6.3.** Your co-author reports mean WTP and a delta-method 95% interval of $[-40, 260]$ from a mixed logit where the cost coefficient is barely significant. Using this section, name the specific asymptotic assumption most likely violated and the two-part fix. *(Answer: the near-zero, imprecise cost coefficient makes the WTP ratio approximately Cauchy — plausibly violating the finite-variance condition behind the CLT and the delta method, which is why the interval is huge and symmetric around a possibly meaningless mean. Fix: (i) report the median and a simulation/bootstrap percentile interval instead of the mean and delta-method interval; (ii) address the cause — better cost-coefficient identification or WTP-space reparameterization with a sign constraint.)*

6.10 Failure modes: how asymptotics misleads in practice

F1. Quoting asymptotic SEs in small cells. Treating $\hat{\theta} \pm 1.96\,\text{SE}$ as exact 95% coverage in a 40-person cell with a skewed outcome. *Fix:* bootstrap/permutation intervals; report that inference is small-sample (Ch 8, 10).

F2. Confusing consistency with unbiasedness. Assuming a consistent estimator (MLE, WTP ratio) is unbiased in your finite sample. *Fix:* recognize nonlinear/ML estimators are typically biased in small samples; prefer robust summaries (median) and bias-aware reporting.

F3. Trusting the mean of a heavy-tailed quantity. Reporting mean WTP when the ratio is near-Cauchy. *Symptom:* mean far from median, unstable across resamples, huge symmetric intervals. *Fix:* median and percentile intervals; diagnose $\Pr(\hat{\beta}_{\text{cost}} \approx 0)$ (§6.9).

F4. Ignoring dependence. Using i.i.d.-based standard errors for repeated choice tasks or multiple estimates per study. *Symptom:* implausibly small SEs, over-rejection. *Fix:* cluster-robust/random-effects/RVE inference (Ch 17, 19, 47).

F5. Wrong asymptotic count. Treating a meta-analysis as "large" because there are many *estimates*, when inference is asymptotic in the number of *studies*. *Fix:* count clusters, apply small-sample corrections (Ch 48).

F6. Delta method on strong nonlinearity. Using a symmetric delta-method interval for a WTP ratio or other strongly nonlinear quantity. *Fix:* simulate (Krinsky–Robb) or bootstrap; report asymmetric intervals (Ch 9, 24).

F7. Reading "not significant" as "no effect." Under low power (small cells) the CLT-based test simply lacks resolution; a large p-value is uninformative about the null. *Fix:* power analysis and equivalence testing (Ch 8) — the sequel to this chapter.

6.11 Diagnostic questions for your own papers

1. **Which count is large?** For each inference, what count must be large for the asymptotics to hold (n per cell? respondents? studies?), and is it actually large? *(§6.9.)*
2. **Finite variance?** Does the quantity I am averaging or forming a CI for have finite variance — in particular, is my WTP ratio at risk of near-Cauchy behavior from a near-zero cost coefficient? *(§6.9, F3.)*
3. **Consistency vs bias.** Am I treating a consistent-but-biased estimator (MLE, WTP ratio) as if it were unbiased in my finite sample? *(§6.3, F2.)*
4. **Independence.** Are my observations independent, or do repeated tasks / multiple estimates per study induce dependence that shrinks the effective sample size? *(§6.9, F4.)*
5. **Approximation quality.** Given my sample size and the skewness of my outcome, is the normal approximation good enough, or should I use bootstrap/permutation methods? *(§6.6, F1.)*
6. **Interval shape.** For nonlinear quantities (WTP), am I using a symmetric delta-method interval where an asymmetric simulated interval is warranted? *(§6.7, F6.)*
7. **Null interpretation.** Am I reading a non-significant result as evidence of no effect when low power (small n) makes the test uninformative? *(§6.9, F7; Ch 8.)*

6.12 Exercises

(A) Comprehension.

A1. State the difference between $\hat{\theta}_n \xrightarrow{p} \theta_0$ and $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, V)$. Which one justifies a confidence interval, and why do you need the other too?

A2. The MLE variance estimator $\frac{1}{n}\sum(X_i - \bar{X})^2$ has expectation $\frac{n-1}{n}\sigma^2$. Is it biased? Is it consistent? Reconcile the two answers.

A3. Explain why halving a standard error requires quadrupling the sample size, using the $\sqrt{n}$ rate in the CLT.

(B) Derivation.

B1. Prove the Weak Law of Large Numbers for i.i.d. data with finite variance using Chebyshev's inequality. State exactly where finite variance is used.

B2. Using the delta method (CMT + CLT), derive the asymptotic variance of $g(\hat{\theta}_n) = \log \hat{\theta}_n$ given $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \sigma^2)$. Then explain, for the WTP ratio $g(\beta_1, \beta_2) = -\beta_1/\beta_2$, why the delta-method variance blows up as $\beta_2 \to 0$ (compute $\partial g/\partial \beta_2$).

B3. Show that if $Z_1, Z_2 \sim N(0,1)$ independently, the ratio Z_1/Z_2 has heavier-than-normal tails by arguing that its variance is infinite (you may argue heuristically from the behavior near $Z_2 = 0$). Connect this to the WTP ratio and to why the LLN fails for such a variable.

(C) Research application / critique.

C1. Your CV paper compares WTP across a high- and low-cost-vector arm, each with $n \approx 50$, and reports a t-test with $p = 0.18$, concluding "no cost-vector effect." Critique this using §6.6 and §6.9: what asymptotic issues undermine the conclusion, and what three analyses would you run instead?

C2. A meta-analysis reviewer praises your finfish paper for having "over 120 WTP observations, so large-sample inference is safe." Explain why this reassurance is misplaced, identifying the correct asymptotic count and the correction it implies.

C3. Explain to a co-author, in a short paragraph, why reporting the *mean* of a Krinsky–Robb WTP simulation can be misleading while the *median* is trustworthy, grounding the explanation in the finite-variance condition for the LLN/CLT and the near-Cauchy nature of the ratio.

6.13 Solutions

A1. $\hat{\theta}_n \xrightarrow{p} \theta_0$ (consistency) says the point estimate homes in on the truth; $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, V)$ (asymptotic normality) says the rescaled error is approximately normal, supplying the *shape* needed to compute quantiles. The confidence interval is justified by asymptotic normality (it uses the normal quantiles ± 1.96); you need consistency too, because without it the interval would be centered on the wrong value — normality around a biased center is useless.

A2. It is **biased** (expectation $\frac{n-1}{n}\sigma^2 \neq \sigma^2$ for finite n) but **consistent** (the bias factor $\frac{n-1}{n} \to 1$, and the variance vanishes, so it $\xrightarrow{p} \sigma^2$). No contradiction: unbiasedness is a finite-sample property, consistency an asymptotic one; an estimator can fail the first while satisfying the second. *(Verified:* $\mathbb{E} = 0.80$ *at* $n = 5$*, converging to* 1*.)*

A3. $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$, so $\mathrm{SE} = \sigma/\sqrt{n}$. Halving the SE means dividing $\sigma/\sqrt{n}$ by 2, i.e. multiplying $\sqrt{n}$ by 2, i.e. multiplying n by 4. The $\sqrt{n}$ rate makes precision expensive: each additional digit of precision costs a hundredfold in sample size.

B1. $\bar{X}_n$ has mean μ and, by independence, variance σ^2/n. Chebyshev: $\Pr(|\bar{X}_n - \mu| \geq \varepsilon) \leq \mathrm{Var}(\bar{X}_n)/\varepsilon^2 = \sigma^2/(n\varepsilon^2) \to 0$. Hence $\bar{X}_n \xrightarrow{p} \mu$. Finite variance $\sigma^2 < \infty$ is used to bound $\mathrm{Var}(\bar{X}_n)$; without it Chebyshev gives no bound and the proof (and often the result) fails.

B2. $g(\theta) = \log\theta$, $g'(\theta) = 1/\theta$, so $\sqrt{n}(\log\hat{\theta}_n - \log\theta_0) \xrightarrow{d} N(0,\ \sigma^2/\theta_0^2)$. For WTP $g(\beta_1, \beta_2) = -\beta_1/\beta_2$: $\partial g/\partial\beta_1 = -1/\beta_2$ and $\partial g/\partial\beta_2 = \beta_1/\beta_2^2$. The delta-method variance is $\nabla g' \mathbf{V} \nabla g$; the $\partial g/\partial\beta_2 = \beta_1/\beta_2^2$ term blows up like $1/\beta_2^2$ as $\beta_2 \to 0$, so the WTP variance explodes — the calculus twin of the Cauchy heavy tail.

B3. Near $Z_2 = 0$, the ratio Z_1/Z_2 takes arbitrarily large values, and because Z_2 has positive density at 0, the probability of $|Z_2| < \epsilon$ is $\approx 2\phi(0)\epsilon$, over which the ratio is $\gtrsim 1/\epsilon$; the contribution to $\mathbb{E}[(Z_1/Z_2)^2]$ from this region behaves like $\int_0 (1/z_2)^2 dz_2$, which diverges. So the variance is infinite (in fact Z_1/Z_2 is Cauchy, with no mean either). The WTP ratio $-\hat{\beta}_1/\hat{\beta}_2$ inherits this when $\hat{\beta}_2$ has mass near zero; with infinite variance the LLN's precondition fails, so averaging WTP draws (or trusting a mean/SE) is unjustified.

C1. Issues: (i) with $n \approx 50$ per arm and likely skewed WTP, the CLT approximation behind the t-test is imperfect (§6.6); (ii) the comparison may be *underpowered*, so $p = 0.18$ is uninformative about the null — "not significant" is not "no effect" (§6.9, F7); (iii) if WTP is a near-Cauchy ratio, the mean/SE the test relies on may be unstable (§6.9, F3). Instead: (a) run a **power analysis / equivalence test** to say what effect sizes you *could* have detected and whether you can claim equivalence (Chapter 8); (b) use **randomization inference or a permutation test** for the contrast, which does not lean on the CLT (Chapter 10); (c) compare **WTP distributions** via simulation and the **Poe test** with medians/percentile intervals rather than a single mean difference (Chapters 9–10).

C2. The reassurance mistakes the asymptotic count. Meta-analytic inference is asymptotic in the number of **studies (clusters)** — here 18 — not the number of WTP estimates (120+), because estimates within a study are dependent (§6.9, F4–F5). With only 18 clusters, cluster-robust standard errors are downward-biased and unreliable, so you must apply **small-sample corrections** (e.g., the appropriate degrees-of-freedom and bias adjustments for cluster-robust/RVE variance) and treat the "120 observations" as far fewer effective independent units (Chapters 47–48).

C3. Because the WTP ratio can behave like a Cauchy when the cost coefficient is imprecise, its distribution may have very heavy tails — even an ill-defined mean and infinite variance. The LLN and CLT require finite variance, so the *sample mean* of the simulated WTP draws is not guaranteed to converge or be stable: a few

draws with the cost coefficient near zero can dominate it, making the reported mean an artifact of the tail and unreliable across re-simulations. The **median**, by contrast, depends only on the bulk of the distribution, is unaffected by the extreme tail, and remains a stable, consistent summary — which is why we report the median WTP and percentile-based intervals rather than the mean and a symmetric standard error.

6.14 Spaced review prompts

+1 day. From memory, state the WLLN and the CLT precisely (with their assumptions), prove the WLLN via Chebyshev, and explain the $\sqrt{n}$ scaling in one sentence.

+1 week. Explain, with the Cauchy example, why the WTP ratio can violate the conditions for the LLN/CLT, and state the reporting rule (median + percentile interval) and the two upstream fixes that follow. Then distinguish consistency from unbiasedness with the $1/n$-variance example.

+1 month. For one comparison in your CV or DCE paper, write down which count must be large for your standard errors to be valid, check whether it is, and note whether dependence (repeated tasks / within-study clustering) shrinks your effective sample size. Close with the spine sentence about whether you are inside the asymptotic regime.

6.15 How this chapter connects to the rest of the book

- **Chapter 5** supplied the distributions and the variance decomposition; this chapter explains *why the normal recurs* (CLT) and why sample averages recover expectations (LLN).
- **Chapter 7 (estimation and inference)** builds standard errors, confidence intervals, and the delta method on §6.6–§6.7; **Chapter 8 (power/equivalence)** is the direct sequel to §6.9's "not significant $\neq$ no effect."
- **Chapter 9 (simulation)** uses the LLN to justify Monte Carlo, Krinsky–Robb, and the bootstrap, and is the recommended tool when the CLT approximation is poor (§6.6, §6.9); **Chapter 10 (randomization/permutation)** provides the CLT-free inference for small cells.
- **Chapter 11 (maximum likelihood)** makes the estimator-asymptotics template of §6.8 fully rigorous (score, information, sandwich); **Chapters 17, 19 (robust/clustered/panel inference)** are §6.9's dependence fix.
- **Chapter 24 (WTP space)** is where the near-Cauchy WTP-ratio diagnosis of §6.9 drives the reparameterization decision; **Chapters 47–48 (meta-regression, small-sample corrections)** are §6.9's "which count is large" and few-studies corrections applied to the finfish paper.

This chapter is the theoretical license for all inference in the book, and it feeds directly into the high-dependency chapters **Ch 11** (the asymptotic template) and, via the ratio/Cauchy analysis, **Ch 24**.

6.16 Research translation task

For one inferential claim in your work — a treatment contrast in the CV paper, or a WTP in the DCE — produce a two-page note:

1. **Identify the asymptotic count.** State precisely which count must be large for your standard error / test to be valid (n per cell, number of respondents, number of studies), and report its actual value.
2. **Check finite variance.** For any ratio or nonlinear quantity (WTP), assess the risk of near-Cauchy behavior: report the cost coefficient's precision and $\Pr(\hat{\beta}_{\text{cost}} \approx 0)$, and decide whether the mean/SE is trustworthy or whether to switch to median/percentile summaries.

3. **Check independence.** Note any dependence (repeated tasks, within-study clustering) and state how it shrinks your effective sample size and what inference correction you use.
4. **Assess approximation quality.** Given your sample size and outcome skewness, judge whether the normal approximation suffices or whether you should use a bootstrap/permutation/randomization method; if feasible, run one and compare.
5. **Close** with the spine sentence: *"This inference is asymptotic in [count] = [value]; the normal approximation is [adequate/strained] because [skew/finite-variance/dependence]; I therefore report [Wald / bootstrap / permutation] and [mean / median] intervals."*

Carry this into Chapter 8 (power and equivalence for your treatment contrasts) and Chapter 9 (simulation-based inference for WTP).

6.17 Bibliography

Core. - Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, the asymptotics chapters (consistency, asymptotic normality, the sandwich form). The estimator-asymptotics template of §6.8 at exactly the level you need, in econometric notation. *(Verify edition/chapters.)* - Casella & Berger, *Statistical Inference*, chapters on convergence concepts, the LLN, the CLT, and the delta method. The rigorous backing for §6.2–§6.7, with clean proofs. *(Verify edition.)* - Hansen, *Econometrics* (Bruce Hansen's textbook), the asymptotic-theory chapters. A modern, careful, freely-available graduate treatment of exactly this material, including Slutsky, CMT, and the delta method. *(Verify current version/link.)*

Applied. - Cameron & Trivedi, *Microeconometrics: Methods and Applications*, on asymptotic inference, the delta method, and cluster-robust variance — bridges §6.7–§6.9 to practice, including the WTP-ratio delta method. *(Verify edition.)* - Cameron & Miller, "A Practitioner's Guide to Cluster-Robust Inference," *Journal of Human Resources*. The definitive practical treatment of the "how many clusters is enough" question (§6.9, forbidden zone 4) and small-sample corrections — directly relevant to your 18-study meta-analysis. *(Verify volume/year.)* - Efron & Tibshirani, *An Introduction to the Bootstrap*. The CLT-free alternative recommended when asymptotics are strained (§6.6, §6.9); foundational for Chapter 9. *(Verify edition.)*

Advanced. - van der Vaart, *Asymptotic Statistics*. The rigorous reference for modes of convergence, the delta method (including higher-order and non-differentiable cases), and M-estimation — for when you want the theory of §6.8 in full. *(Verify.)* - White, *Asymptotic Theory for Econometricians*. The classic econometric treatment of the sandwich/robust asymptotics of §6.8, including misspecification. *(Verify edition.)* - Hall, *The Bootstrap and Edgeworth Expansion*, for the finite-sample corrections and the rate at which the CLT approximation improves (the theory behind §6.6's skewness decay). *(Verify.)*

Entries marked "verify" are pointers; confirm editions, versions, and the Cameron–Miller citation before using them, per the book's no-unverified-citation rule.

End of Chapter 6. Next in Part II: Chapter 7 — Estimators and Inference (bias, consistency, efficiency, standard errors, confidence intervals, p-values, and the delta method), which turns the limit theorems of this chapter into the standard errors and intervals you report, and develops the delta method whose WTP application we previewed here. Tell me when to write it, or request a different chapter.

Chapter 7 — Estimators and Inference

Part II: Probability and Statistical Inference

> **The spine question, applied to this chapter.** Chapter 6 proved that your estimators are consistent and asymptotically normal. This chapter turns those promises into the four numbers you actually print — the standard error, the confidence interval, the p-value, and, for any derived quantity like WTP, the propagated standard error from the delta method — and it teaches you to interpret each one *correctly*, because these are the most misread objects in empirical economics. A p-value is not the probability the null is true. A 95% confidence interval does not have a 95% chance of containing the parameter. A wide interval is not "no effect." A symmetric delta-method interval around a WTP ratio can be badly wrong. The spine question here: *when I report an estimate with its uncertainty and a p-value, what exactly does each number mean, is it the right kind of interval for the quantity, and what conclusion does it license — and forbid?*

7.0 Conceptual motivation: the four numbers you report, and what they mean

Almost every empirical claim in your papers reduces to reporting an estimate together with a statement of its uncertainty: a mean WTP with a confidence interval; a treatment contrast with a p-value; a coefficient with a standard error and stars. Your PDF's inference section is a list of exactly these objects — *"stimatori; bias; consistenza; efficienza; varianza; standard error; intervalli di confidenza; p-value; test a una e due code"* — and it flags that your papers "si basano su confronti tra trattamenti: high vs low cost vector, hypothetical vs incentivized, PC vs SBDC, experienced vs non-experienced, full payment vs BRIS." Each such comparison ends in a p-value and a confidence interval, and the *interpretation* of those objects is where careful researchers and careless ones diverge.

This chapter builds and interprets the four reporting objects, with three priorities set by your work:

- **Correct interpretation of p-values and confidence intervals.** These are frequentist objects with precise, counterintuitive meanings. The most consequential error in your field — the one your PDF elevates to top priority in §6.1, *" 'non significativo' non significa 'uguale' "* — is a p-value misinterpretation, and it cannot be avoided without first understanding what a p-value *is*. This chapter lays that foundation; Chapter 8 builds the equivalence-testing machinery on top of it.
- **Standard errors for *contrasts*, not just coefficients.** When you compare high- and low-cost-vector arms, the object of interest is a *difference*, and its standard error is a quadratic form in the full covariance matrix (Chapter 1, §1.6), not a naive combination of two separate standard errors. Getting this right is the difference between a defensible and an indefensible treatment comparison.
- **The delta method for WTP, and when to abandon it.** Your central estimand — WTP — is a nonlinear function of coefficients, so its standard error must be *propagated* through that function. The

delta method does this analytically; but, as Chapter 6 warned, it is a local linear approximation that produces a *symmetric* interval, which can be badly wrong for the skewed WTP ratio. This chapter derives the delta method in full, shows exactly where it breaks, and puts it side by side with the simulation (Krinsky–Robb) and bootstrap alternatives on the same numbers.

We proceed: estimator properties (bias, variance, MSE, consistency, efficiency, and the Cramér–Rao bound that explains why the MLE is the default), standard errors and their flavors, confidence intervals and their true meaning, hypothesis tests and the anatomy of a p-value, and finally the delta method with a fully worked WTP example computed three ways. Throughout, the emphasis is on *correct interpretation* — the skill your PDF calls "sapere giudicare criticamente."

7.1 Prerequisites

From **Chapter 6**: asymptotic normality of estimators, $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, V)$; the delta method (CMT + CLT); consistency; and the warnings about small samples and near-Cauchy ratios. From **Chapter 5**: the normal distribution, its quantiles, and the CDF. From **Chapter 1**: the covariance matrix, quadratic forms (§1.6), and standard errors as square roots of the diagonal of $\hat{\mathbf{V}}$ (§1.8, §1.11). From **Chapter 2**: gradients, for the delta method.

Same **notation**. New symbols: θ_0 (true parameter), $\hat{\theta}$ (estimator), $\mathrm{SE}(\hat{\theta})$ (standard error), z_α (standard-normal quantile), H_0/H_1 (null/alternative hypotheses), α (significance level), $\mathrm{CI}_{1-\alpha}$ (confidence interval).

7.2 Estimators, sampling distributions, and the standard error

An **estimator** $\hat{\theta}$ is a rule that maps a sample to a number; because the sample is random, $\hat{\theta}$ is a random variable with a **sampling distribution** — the distribution of the values it would take across hypothetical repeated samples. Everything in inference is a statement about this sampling distribution.

Definition 7.1 (Bias, variance, mean squared error). For an estimator $\hat{\theta}$ of θ_0:

$$\mathrm{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta_0, \qquad \mathrm{Var}(\hat{\theta}) = \mathbb{E}\big[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2\big],$$

$$\mathrm{MSE}(\hat{\theta}) = \mathbb{E}\big[(\hat{\theta} - \theta_0)^2\big] = \mathrm{Bias}(\hat{\theta})^2 + \mathrm{Var}(\hat{\theta}).$$

The MSE decomposition (proved by adding and subtracting $\mathbb{E}[\hat{\theta}]$ inside the square and expanding) is the **bias–variance trade-off**: total error has a systematic part (bias) and a noise part (variance). An estimator can be unbiased but noisy, or slightly biased but much more precise; MSE trades them off. Most of the estimators you use (MLE, WTP ratio) accept a little finite-sample bias in exchange for asymptotic efficiency (§7.3).

Definition 7.2 (Standard error). The **standard error** of $\hat{\theta}$ is the standard deviation of its sampling distribution, $\mathrm{SE}(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}$ — or, in practice, an *estimate* of it, since $\mathrm{Var}(\hat{\theta})$ depends on unknowns. This is the single most important number attached to an estimate: it is the yardstick against which the estimate's magnitude is judged. Note carefully that the standard error is *itself an estimated quantity* with its own sampling variability — a fact that matters in small samples (where we use t rather than normal quantiles) and that Chapter 6's warnings (near-Cauchy ratios) push to the extreme, where the estimated SE can be wildly unstable.

The standard error is the standard deviation of the estimator, not of the data. A common confusion: the *data* have a standard deviation σ (spread of observations); the *estimator* has a standard error $\sigma/\sqrt{n}$ (spread of the estimate across samples). The $\sqrt{n}$ (Chapter 6, §6.6) separates them: as n grows, the data's spread is unchanged but the estimator's spread shrinks. When you report "mean WTP €20 (SE

€3)," the €3 is how much the *estimate of the mean* would wander across resamples — not how spread out individual respondents' WTP is.

> **Active recall 7.A.** A colleague reports "mean WTP is €25 with a standard deviation of €40" and then writes the 95% interval as €25 ± 1.96 × €40. What has gone wrong? *(Answer: they used the standard deviation of the* data* (€40, individual spread) where they needed the standard *error* of the *mean* (€40/$\sqrt{n}$, estimate spread). The interval should use the SE; with, say, $n = 400$, SE = €2 and the interval is €25 ± €3.92, not ±€78. Confusing SD of data with SE of estimator inflates the interval enormously.)*

7.3 Consistency, efficiency, and the Cramér–Rao bound

Consistency (Chapter 6, §6.3): $\hat{\theta} \xrightarrow{p} \theta_0$ — the estimator homes in on the truth as data accumulate. It is the minimal requirement. Among consistent estimators, we prefer the one that is *most precise*, which is **efficiency**.

Definition 7.3 (Relative and absolute efficiency). Between two unbiased (or consistent, asymptotically unbiased) estimators, the more **efficient** is the one with smaller variance. An estimator is **efficient** (absolutely) if no other unbiased estimator has smaller variance — i.e., it attains the theoretical lower bound on variance.

Theorem 7.1 (Cramér–Rao lower bound). For any *unbiased* estimator $\hat{\theta}$ of a scalar θ, under regularity conditions,

$$\mathrm{Var}(\hat{\theta}) \;\geq\; \frac{1}{\mathcal{J}(\theta)},$$

where $\mathcal{J}(\theta)$ is the **Fisher information** (Chapter 1, §1.11; Chapter 11). The multivariate version is $\mathrm{Var}(\hat{\theta}) \succeq \mathbf{I}(\theta)^{-1}$ (the difference is positive semidefinite). **You cannot beat the information bound.** The more information the data carry about θ (sharper likelihood curvature), the smaller the achievable variance — the same curvature-is-precision idea from Chapter 2 (§2.3), now stated as a hard limit.

Why this makes the MLE the default. A central result (Chapter 11) is that the **MLE is asymptotically efficient**: it attains the Cramér–Rao bound in large samples, so no consistent estimator is asymptotically more precise. This is the theoretical reason your SBDC probit, your logit, and your mixed logit are estimated by maximum likelihood rather than by some ad hoc method — among consistent estimators, ML squeezes the most precision out of the data. The price, as Chapter 6 noted, is finite-sample bias and (for complex models) computational difficulty (Chapter 13), but the asymptotic-efficiency guarantee is why ML is the workhorse.

Worked Example 7.1 (The Bernoulli attains the bound, verified). For n i.i.d. Bernoulli(p) (your SBDC yes/no atom), Chapter 1 (§1.11) derived $\mathcal{J}(p) = n/[p(1-p)]$, so the Cramér–Rao bound is $p(1-p)/n$. The MLE $\hat{p} = \bar{y}$ has variance exactly $p(1-p)/n$ — it **attains** the bound (verified: at $p = 0.3$, $n = 100$, both equal 0.0021). The sample proportion is the efficient estimator of a probability; no unbiased estimator of p does better. This is the cleanest illustration that the information bound is achievable and that ML achieves it.

> **Stop and think 7.1.** Two designs for your SBDC study place bids differently: Design A concentrates bids where the yes-probability is near 0.5; Design B spreads them into the tails (near 0 and 1). Using the Cramér–Rao bound and the Bernoulli information $n/[p(1-p)]$, predict which design yields a more efficient (lower-variance) estimate of the bid coefficient. *(Answer: Design A. Information per observation is $1/[p(1-p)]$, maximized at $p = 0.5$; bids landing responses near a coin flip carry the most information, tightening the Cramér–Rao bound and lowering the achievable variance. Tail bids are nearly uninformative — the same design lesson as Chapters 1–2 and 5, now framed as efficiency.)*

7.4 Standard errors and their flavors

The standard error is $\sqrt{\text{diagonal of } \hat{\mathbf{V}}(\hat{\theta})}$, but *which* $\hat{\mathbf{V}}$ you use encodes different assumptions about the data. You will meet four flavors, and choosing the right one is a substantive decision, not a default.

1. **Classical (information-based) SEs.** $\hat{\mathbf{V}} = [-\mathbf{H}(\hat{\theta})]^{-1}$ (inverse observed information; Chapter 1, §1.11). Valid when the model is correctly specified and observations are independent. This is the MLE default and attains the Cramér–Rao bound asymptotically.
2. **Robust ("sandwich") SEs.** $\hat{\mathbf{V}} = \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$ (Chapter 6, §6.8). Valid under certain misspecification (e.g., heteroskedasticity in the outcome variance) where the information equality fails. They protect against getting the variance wrong even if the model's mean structure is right (Chapter 17).
3. **Clustered SEs.** A sandwich that allows correlation *within* clusters (respondents, studies). Essential for your repeated choice tasks and your within-study meta-estimates, where independence fails (Chapter 6, §6.9; Chapters 17, 19, 47). Clustered SEs are larger than naive ones because dependence reduces the effective sample size — and they are themselves unreliable with few clusters (your 18 studies; Chapter 48).
4. **Simulation/bootstrap SEs.** The standard deviation (or, better, percentile interval) of an estimate across resamples (bootstrap) or parameter draws (Krinsky–Robb). Used when analytic SEs are unavailable or unreliable — the WTP ratio being the leading case (§7.7–§7.8, Chapter 9).

The choice is an assumption. Reporting classical SEs when observations are clustered *understates* uncertainty (over-narrow intervals, over-rejection); reporting robust SEs buys insurance at a small efficiency cost. Your PDF's insistence on clustered/robust inference for repeated observations and meta-estimates is exactly a statement about which $\hat{\mathbf{V}}$ is honest for your data.

> **Active recall 7.B.** You have 8 choice tasks per respondent in a DCE with 500 respondents. Why are classical standard errors likely too small, and which flavor do you use? *(Answer: the 8 tasks per respondent are correlated (shared respondent-level taste/scale), so the 4000 observations are not 4000 independent units — classical SEs treat them as independent and understate uncertainty. Use respondent-clustered (or panel/mixed) standard errors, which account for the within-respondent correlation, Chapters 17, 19.)*

7.5 Confidence intervals, and what they actually mean

Definition 7.4 (Confidence interval). A $100(1-\alpha)\%$ **confidence interval** for θ is a random interval $[\hat{\theta}_L, \hat{\theta}_U]$, computed from the data, with the **coverage** property

$$\Pr\left(\hat{\theta}_L \le \theta_0 \le \hat{\theta}_U\right) = 1 - \alpha,$$

where the probability is over *repeated samples* (the interval is random; θ_0 is fixed). The standard **Wald interval** from asymptotic normality is

$$\hat{\theta} \;\pm\; z_{1-\alpha/2}\,\mathrm{SE}(\hat{\theta}), \qquad z_{0.975} = 1.96 \text{ for } 95\%.$$

The correct interpretation (and the near-universal misinterpretation). The coverage statement is about the *procedure*, not about any single computed interval. Over many hypothetical repetitions of the study, 95% of the intervals the procedure produces would contain the true θ_0. It is **wrong** to say, of a specific computed interval $[0.8, 3.2]$, "there is a 95% probability that θ_0 lies in $[0.8, 3.2]$" — in the frequentist framework θ_0 is a fixed constant, so it is either in that interval or not; the 95% attaches to the long-run behavior of the method, not to this realization. This is subtle and it matters: it is why a confidence interval is a statement about *precision and consistency of the procedure*, and why the (different) Bayesian "credible interval" — which *does* make a probability statement about the parameter — requires a prior and a different framework. State CIs as "the 95% confidence interval is $[\cdot, \cdot]$," and interpret them as "the data are consistent with values in this range," not as a probability about the parameter.

Duality with hypothesis tests. A value θ^* lies *outside* the $100(1-\alpha)\%$ confidence interval **if and only if** a two-sided test of $H_0 : \theta = \theta^*$ rejects at level α. The confidence interval is the set of null values *not* rejected. This duality (developed in §7.6) means the interval carries *all* the information of the test *and more* — the range of plausible values, not just a yes/no at one null. **This is why a confidence interval is almost always more informative than a p-value**, a theme that returns forcefully in Chapter 8: a wide interval that includes both zero and large effects tells you "underpowered / inconclusive," while a narrow interval tightly around zero tells you "precisely estimated null" — two situations a single "not significant" p-value cannot distinguish.

Wald versus percentile intervals. The Wald interval $\hat{\theta} \pm z \cdot \text{SE}$ is *symmetric* and assumes approximate normality of $\hat{\theta}$. For skewed quantities — the WTP ratio above all — a **percentile interval** from simulation or the bootstrap (the 2.5th and 97.5th percentiles of the resampled/simulated distribution) respects the asymmetry and is more accurate (§7.8, Chapter 9). Choosing between them is choosing whether the normal approximation is good enough for *this* quantity.

> **Stop and think 7.2.** Two treatment contrasts are both "not significant at 5%." Contrast A has 95% CI $[-0.2, 0.3]$; Contrast B has 95% CI $[-8, 9]$. Before reading on, say what each interval tells you that the shared "not significant" label hides. *(Answer: A is a* precisely estimated near-zero* effect — the data rule out anything but a small effect, genuine evidence of near-equivalence; B is *uninformative/underpowered* — the data are consistent with large effects in either direction, so "not significant" reflects lack of power, not evidence of no effect. Same p-verdict, opposite conclusions — the reason your PDF says "not significant ≠ equal," and the reason to report CIs, Chapter 8.)*

7.6 Hypothesis testing and the anatomy of a p-value

The testing setup. A **null hypothesis** H_0 (e.g., "the high- and low-cost-vector arms have equal WTP," $\delta = 0$) is tested against an **alternative** H_1 (e.g., $\delta \neq 0$). A **test statistic** measures departure from H_0; the standard one is the z (or t) statistic

$$z = \frac{\hat{\theta} - \theta_0}{\text{SE}(\hat{\theta})},$$

which, under H_0 and asymptotic normality, is approximately $N(0, 1)$.

Definition 7.5 (p-value). The **p-value** is the probability, *computed under the assumption that H_0 is true*, of observing a test statistic at least as extreme as the one actually observed:

$$p = \Pr\left(|Z| \geq |z_{\text{obs}}| \;\middle|\; H_0\right) \quad \text{(two-sided)}.$$

Small p means the observed data would be surprising if H_0 held, which is taken as evidence against H_0.

What a p-value is NOT (memorize this list — it is where careers and papers go wrong).

- It is **not** the probability that H_0 is true. $p = \Pr(\text{data} \mid H_0)$, not $\Pr(H_0 \mid \text{data})$. Converting one to the other requires a prior (Bayes' theorem); the p-value alone cannot tell you how likely the null is.
- It is **not** the probability that the result is due to chance, nor the probability of a Type I error for *this* test.
- It is **not** an effect size or a measure of importance. A tiny p can accompany a trivial effect (large n); a large p can accompany a large but imprecisely estimated effect (small n).
- It is **not** the probability of replication.
- A **large p-value is not evidence that H_0 is true.** Failing to reject is not accepting. This is the single most consequential error for your convergent-validity claims, and Chapter 8 is devoted to the correct alternative (equivalence testing).

Type I and Type II errors. A **Type I error** (false positive) is rejecting a true H_0; its probability is the significance level α (you *choose* it, conventionally 0.05). A **Type II error** (false negative) is failing to reject

a false H_0; its probability is β, and $1-\beta$ is the **power** (Chapter 8). The two trade off: a stricter α (fewer false positives) raises β (more false negatives) for fixed sample size. Your PDF's concern with underpowered many-cell designs is a concern about large β — high false-negative risk — which no p-value reveals.

One-tailed versus two-tailed tests. A **two-tailed** test ($H_1 : \theta \neq \theta_0$) splits α across both tails; a **one-tailed** test ($H_1 : \theta > \theta_0$ or $< \theta_0$) puts all of α in one tail and so is more powerful *in that direction* — but only legitimately when the direction is specified *a priori* on substantive grounds and effects in the other direction are genuinely irrelevant or impossible. The danger: choosing one-tailed *after* seeing the data, or to nudge p below 0.05, is a form of p-hacking that inflates the true Type I rate. Default to two-tailed unless you can justify direction in advance; report which you used.

Worked Example 7.2 (A treatment contrast, tested and interval-ed, verified). Suppose the estimated WTP difference between your high- and low-cost-vector arms is $\hat{\delta} = €5$ with $\text{SE}(\hat{\delta}) = €3$. Then $z = 5/3 = 1.667$. The two-tailed p-value is

$$p = 2(1 - \Phi(1.667)) = 2(0.0478) = 0.096,$$

so you would **not** reject equality at 5%. The one-tailed p (if the direction were pre-specified) is 0.048 — which *would* cross 0.05, illustrating exactly why the one/two-tailed choice must be principled, not opportunistic. The 95% confidence interval is

$$5 \pm 1.96(3) = [-0.88,\ 10.88],$$

which **includes 0** — consistent with the two-tailed non-rejection (duality, §7.5). But read the interval, not just the verdict: it stretches from essentially no effect to a large €10.88 difference. This is **not** evidence that the arms are equivalent; it is evidence that the study cannot tell a null from a large effect — an *underpowered* contrast. The honest report is: "the difference is €5 (95% CI [−0.9, 10.9]); the data are consistent with anything from a small negative to a large positive cost-vector effect, so we cannot conclude either an effect or its absence." Chapter 8 shows how to turn "cannot conclude absence" into a rigorous equivalence claim when warranted.

> **Active recall 7.C.** Your co-author writes: "$p = 0.096$, so there is a 90% chance the cost-vector effect is real." Identify the two distinct errors in that sentence. *(Answer: (1) it inverts the conditional — the p-value is* $\Pr(\textit{data} \mid H_0)$*, not* $\Pr(H_1 \mid \textit{data})$*, so "90% chance the effect is real" does not follow; (2)* $1 - p$ *is not the probability the alternative is true. The correct reading: if there were truly no effect, we'd see a contrast this large or larger about 9.6% of the time — weak evidence against the null, nothing about the probability the effect is real.)*

7.7 The delta method: propagating uncertainty through a nonlinear function

Your estimand is rarely a raw coefficient; it is a *function* of coefficients — WTP $= -\beta_1/\beta_2$, a marginal effect $P(1-P)\beta_j$, an elasticity. To attach a standard error to such a quantity you must propagate the coefficients' uncertainty through the function. The **delta method** does this analytically.

Theorem 7.2 (Delta method). If $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{V})$ and g is continuously differentiable at θ_0 with gradient $\nabla g(\theta_0) \neq \mathbf{0}$, then

$$\sqrt{n}(g(\hat{\theta}) - g(\theta_0)) \ \xrightarrow{d}\ N(0,\ \nabla g(\theta_0)'\, \mathbf{V}\, \nabla g(\theta_0)).$$

In finite-sample notation, $\text{Var}(g(\hat{\theta})) \approx \nabla g'\, \hat{\mathbf{V}}\, \nabla g$ — a **quadratic form** in the covariance matrix (Chapter 1, §1.6). *Derivation* (Chapter 6, §6.7): first-order Taylor-expand $g(\hat{\theta}) \approx g(\theta_0) + \nabla g'(\hat{\theta} - \theta_0)$; the right side is a linear combination of an asymptotically normal vector, hence asymptotically normal with variance $\nabla g' \mathbf{V} \nabla g$ (variance of a linear combination, Chapter 1, Proposition 1.2). Slutsky lets you replace $\nabla g(\theta_0)$ by $\nabla g(\hat{\theta})$ in practice.

Full worked derivation for WTP $= -\beta_1/\beta_2$. Let $\theta = (\beta_1, \beta_2)$ (attribute and cost coefficients) with estimated covariance $\hat{\mathbf{V}} = \begin{pmatrix} V_{11} & V_{12} \\ V_{12} & V_{22} \end{pmatrix}$. The function is $g(\theta) = -\beta_1/\beta_2$. Its partial derivatives:

$$\frac{\partial g}{\partial \beta_1} = -\frac{1}{\beta_2}, \qquad \frac{\partial g}{\partial \beta_2} = \frac{\beta_1}{\beta_2^2}, \qquad \nabla g = \left(-\frac{1}{\beta_2},\ \frac{\beta_1}{\beta_2^2}\right)'.$$

Then the delta-method variance is the quadratic form

$$\mathrm{Var}(\widehat{\mathrm{WTP}}) \approx \frac{V_{11}}{\beta_2^2} - \frac{2\beta_1 V_{12}}{\beta_2^3} + \frac{\beta_1^2 V_{22}}{\beta_2^4}.$$

Read the three terms: the attribute-coefficient variance (V_{11}), the covariance term, and — decisively — the cost-coefficient variance amplified by β_1^2/β_2^4. **As $\beta_2 \to 0$, this last term explodes like $1/\beta_2^4$,** so the WTP standard error detonates when the cost coefficient is weak — the analytic twin of the near-Cauchy heavy tail (Chapter 6, §6.9) and the calculus of Chapter 2 (§2.11). The delta method *shows you the fragility in a formula*: an imprecise cost coefficient in the denominator, raised to the fourth power, dominates the WTP variance.

Worked Example 7.3 (Delta-method WTP SE, verified). Take $\hat{\beta}_1 = 2$ (SE 0.5), $\hat{\beta}_2 = -1$ (SE 0.3), correlation -0.4, so $\hat{\mathbf{V}} = \begin{pmatrix} 0.25 & -0.06 \\ -0.06 & 0.09 \end{pmatrix}$. WTP $= -2/(-1) = 2$. The gradient at the estimate is $\nabla g = (-1/(-1),\ 2/(-1)^2) = (1,\ 2)$. Then

$$\mathrm{Var}(\widehat{\mathrm{WTP}}) = (1)^2(0.25) + 2(1)(2)(-0.06) + (2)^2(0.09) = 0.25 - 0.24 + 0.36 = 0.37,$$

so $\mathrm{SE}(\widehat{\mathrm{WTP}}) = \sqrt{0.37} = 0.61$, and the delta-method 95% interval is $2 \pm 1.96(0.61) = [0.81,\ 3.19]$ — **symmetric** by construction. Hold this interval; §7.8 compares it to the simulated one and shows the symmetry is already an approximation error here.

> **Stop and think 7.3.** In the WTP delta-method variance, which term would you most want to shrink to stabilize the estimate, and what design change (recall Chapter 1, §1.10) reduces it? *(Answer: the $\beta_1^2 V_{22}/\beta_2^4$ term — driven by V_{22}, the cost-coefficient variance, amplified by the small denominator. Shrink V_{22} (a more precisely estimated cost coefficient) via a cost/bid vector with more spread and better conditioning (Chapter 1, §1.10; Chapter 37), which is the upstream fix for WTP instability.)*

7.8 WTP inference three ways: delta method, Krinsky–Robb, and bootstrap

The payoff of Chapters 6–7: three ways to attach uncertainty to a WTP, when each is appropriate, and how they diverge. All use the *same* point estimate and covariance from Worked Example 7.3.

1. Delta method (analytic, symmetric). As computed: SE = 0.61, 95% CI = [0.81, 3.19]. Fast, closed-form, and *symmetric*. Valid when g is approximately linear over the relevant range — i.e., when the cost coefficient is comfortably away from zero and the WTP distribution is not too skewed.

2. Krinsky–Robb (parametric simulation). Draw many coefficient vectors $\beta^{(r)} \sim N(\hat{\beta}, \hat{\mathbf{V}})$ (via the Cholesky factor, Chapter 1, §1.9), compute $\mathrm{WTP}^{(r)} = -\beta_1^{(r)}/\beta_2^{(r)}$ for each, and read off percentiles. *(Verified, 2M draws:* median = 2.00, 2.5th–97.5th percentile interval = [1.07, 4.31], with $\Pr(\hat{\beta}_2 > 0) = 0.0004$.) Note two things. First, the percentile interval $[1.07, 4.31]$ is **right-skewed** and shifted up relative to the symmetric delta interval $[0.81, 3.19]$ — even though the cost coefficient here is 3.3 SEs from zero, the ratio's nonlinearity already bends the distribution, and the delta method's symmetry misses it. Second, the *standard deviation* of the KR draws is huge and unstable (dominated by rare near-zero-denominator draws — the near-Cauchy

tail of Chapter 6), which is exactly why you report the **median and percentile interval, not the mean and SD**. Krinsky–Robb respects the asymmetry the delta method cannot see.

3. Bootstrap (nonparametric resampling). Resample respondents *with replacement*, re-estimate the model on each resample, compute WTP each time, and take percentiles of the resulting distribution. The bootstrap makes fewer distributional assumptions than Krinsky–Robb (it does not assume the coefficients are normal — it uses the data's own variability) and, with clustered resampling, handles dependence. It is the most robust option and the most computationally expensive (re-estimating the model thousands of times). For a mixed logit it may be prohibitive, which is why Krinsky–Robb (drawing from the estimated coefficient distribution rather than re-estimating) is the common compromise in DCE practice.

The catastrophic case — a weak cost coefficient (verified). Now weaken the cost coefficient to $\widehat{\beta}_2 = -0.5$ (SE 0.3 — only 1.67 SEs from zero). WTP $= -2/(-0.5) = 4$. The delta method gives SE $= 2.20$ and a symmetric 95% interval $[-0.31,\ 8.31]$. Krinsky–Robb gives a **wildly skewed** interval $[-15.3,\ 27.3]$ with median 3.75 and $\Pr(\widehat{\beta}_2 > 0) = 4.8\%$. The two methods now disagree violently: the delta interval is a tidy symmetric band, the simulated interval reveals a heavy-tailed, asymmetric mess reflecting the real near-Cauchy behavior. **The delta method's symmetric interval here is dangerously misleading** — it hides the fat tails and the wrong-sign risk. This is the operational rule your PDF's §6.4 demands: when the cost coefficient is weak, *do not* trust the delta-method SE; simulate, report the median and percentile interval, and — better — fix the cause (WTP-space reparameterization with a sign-constrained cost coefficient, Chapter 24; or a better-identified design, Chapter 1 §1.10).

Decision rule. Use the delta method for smooth, near-linear transformations with well-identified denominators; use Krinsky–Robb or the bootstrap for WTP whenever the cost coefficient is not comfortably away from zero or the WTP distribution is skewed — which, for stated-preference work, is most of the time. Always report *which* method and, for WTP, prefer median + percentile interval over mean ± SE.

7.9 Failure modes

F1. Confusing SD of data with SE of estimator. Building an interval with the data's standard deviation instead of the estimate's standard error. *Fix:* SE $=$ SD$/\sqrt{n}$ for a mean; always the standard deviation of the *sampling distribution* (§7.2).

F2. Misinterpreting the confidence interval. Saying "95% probability θ_0 is in this interval." *Fix:* coverage is a property of the procedure over repeated samples; a computed interval either contains θ_0 or not (§7.5).

F3. Misinterpreting the p-value. Reading p as $\Pr(H_0 \mid \text{data})$, as an effect size, or as $1 - p =$ probability the effect is real. *Fix:* $p = \Pr(\text{data as extreme} \mid H_0)$; report effect sizes and CIs alongside (§7.6).

F4. "Not significant" = "no effect." Treating a large p-value as evidence for the null. *Fix:* read the CI (precise-null vs underpowered); use equivalence testing (Chapter 8).

F5. Opportunistic one-tailed tests. Switching to one-tailed after seeing the data or to cross 0.05. *Fix:* pre-specify direction on substantive grounds; default to two-tailed; report which (§7.6).

F6. Delta method on a strongly nonlinear quantity. Symmetric delta-method interval for a WTP with a weak cost coefficient. *Symptom:* interval disagrees with simulation; ignores skew and wrong-sign risk. *Fix:* Krinsky–Robb/bootstrap; median + percentile interval (§7.8).

F7. Wrong SE flavor for the data structure. Classical SEs for clustered/panel data. *Symptom:* implausibly small SEs, over-rejection. *Fix:* clustered/robust/bootstrap SEs (§7.4; Chapters 17, 19).

F8. Ignoring that the SE is itself estimated. Using normal quantiles in small samples where the SE is imprecise. *Fix:* use t quantiles in small samples; recognize the SE's own instability for near-Cauchy quantities (§7.2).

7.10 Diagnostic questions for your own papers

1. **Right SE for the object.** For a *contrast*, did I use the full covariance (quadratic form), not a naive combination of two SEs? For WTP, did I propagate correctly (delta or simulation)? *(§7.4, §7.7.)*
2. **Right SE flavor.** Does my SE flavor (classical/robust/clustered/bootstrap) match my data's dependence structure? *(§7.4.)*
3. **CI interpretation.** Am I reading confidence intervals as ranges of plausible values (procedure coverage), not as probabilities about the parameter? *(§7.5.)*
4. **P-value interpretation.** Have I avoided reading p as the probability the null is true, as an effect size, or as evidence *for* the null? *(§7.6.)*
5. **Not-significant claims.** For every "no difference" claim, have I checked the CI width — precise null vs underpowered — rather than resting on $p > 0.05$? *(§7.5, F4; Chapter 8.)*
6. **Delta vs simulation.** For WTP, is my cost coefficient safely away from zero (delta OK), or should I report a simulated median + percentile interval? *(§7.8.)*
7. **One vs two-tailed.** Did I pre-specify any one-tailed test on substantive grounds and report the choice? *(§7.6, F5.)*

7.11 Exercises

(A) Comprehension.

A1. An estimator has bias 0.2 and variance 0.09. Compute its MSE. A competing estimator is unbiased with variance 0.16. Which has smaller MSE, and what does this illustrate about the bias–variance trade-off?

A2. A 95% CI for a WTP difference is $[-2, 12]$ (euros). State (i) the correct interpretation, (ii) two *incorrect* interpretations, and (iii) whether you can conclude the two arms are equivalent.

A3. A study reports $p = 0.04$. Which of these are correct? (a) There is a 4% chance the null is true. (b) If the null were true, data this extreme would occur 4% of the time. (c) There is a 96% chance the effect is real. (d) The effect is large. Justify each.

(B) Derivation.

B1. Derive the delta-method variance of a marginal effect $m = \Lambda(x'\beta)(1 - \Lambda(x'\beta))\beta_j$ is complex; instead, derive the delta-method variance of the simpler log-transform $g(\widehat{\theta}) = \log \widehat{\theta}$ and of the odds $g(\widehat{p}) = \widehat{p}/(1-\widehat{p})$, given the relevant variances. Interpret when each transform stabilizes inference.

B2. For WTP $= -\beta_1/\beta_2$, derive the gradient and the full delta-method variance formula, and show algebraically that the variance is dominated by the $\beta_1^2 V_{22}/\beta_2^4$ term as $\beta_2 \to 0$. State the implication for reporting.

B3. Show the duality between the two-sided Wald test and the confidence interval: prove that $H_0 : \theta = \theta^*$ is rejected at level α if and only if θ^* lies outside the $100(1-\alpha)\%$ Wald interval.

(C) Research application / critique.

C1. Your DCE reports mean WTP €120 with a delta-method 95% interval of €[−30, 270], and the cost coefficient is significant only at the 10% level. A reviewer says the wide interval "just means you need more data." Critique using §7.7–§7.8: what is the likely cause, why is the *symmetric* interval itself suspect, and what would you report instead?

C2. In your CV paper you compare BRIS and full-payment arms and find $p = 0.21$, concluding "BRIS and full payment give equivalent WTP." Write a critique grounded in §7.5–§7.6: what does $p = 0.21$ actually

license, what does the CI need to show for an equivalence claim, and what analysis (forward-referencing Chapter 8) is required?

C3. Explain to a co-author, in a short paragraph, why you will report a Krinsky–Robb median and percentile interval for WTP rather than the delta-method SE, connecting the choice to the $1/\beta_2^4$ term in the delta-method variance and the near-Cauchy behavior of the ratio.

7.12 Solutions

A1. $\text{MSE}_1 = 0.2^2 + 0.09 = 0.04 + 0.09 = 0.13$. $\text{MSE}_2 = 0 + 0.16 = 0.16$. The *biased* estimator has smaller MSE ($0.13 < 0.16$): accepting a little bias bought enough variance reduction to win on total error. This is the bias–variance trade-off — unbiasedness is not the goal; minimizing MSE (or, asymptotically, attaining efficiency) is.

A2. (i) Correct: the interval was produced by a procedure that, over repeated samples, would contain the true difference 95% of the time; the data are consistent with differences from −€2 to €12. (ii) Incorrect: "there's a 95% probability the true difference is between −2 and 12" (the parameter is fixed, not random); "95% of respondents' WTP differences fall in this range" (it is about the *estimate*, not individuals). (iii) No — the interval includes both 0 and a large €12, so it is *underpowered/inconclusive*, not evidence of equivalence; an equivalence claim needs a narrow interval inside a pre-specified margin (Chapter 8).

A3. (a) Incorrect — p is not $\Pr(H_0)$; it is computed *assuming* H_0. (b) **Correct** — this is the definition. (c) Incorrect — $1 - p$ is not the probability the effect is real; that requires a prior. (d) Incorrect — p says nothing about effect size; a tiny effect can give $p = 0.04$ with large n.

B1. For $g(\hat{\theta}) = \log \hat{\theta}$: $g' = 1/\theta$, so $\text{Var}(\log \hat{\theta}) \approx \text{Var}(\hat{\theta})/\theta^2$; the log stabilizes when the quantity is positive and multiplicatively varying (it symmetrizes right-skew, giving better-behaved intervals — e.g., for variance ratios). For the odds $g(\hat{p}) = \hat{p}/(1-\hat{p})$: $g' = 1/(1-p)^2$, so $\text{Var}(\text{odds}) \approx \text{Var}(\hat{p})/(1-p)^4$; the odds inflate variance as $p \to 1$. Working on the log-odds (logit) scale, $g(\hat{p}) = \log[\hat{p}/(1-\hat{p})]$ with $g' = 1/[p(1-p)]$, is better-behaved and is why logistic inference is done on the log-odds scale. Transforms stabilize inference by making the sampling distribution closer to normal and the variance more constant.

B2. $g = -\beta_1/\beta_2$; $\partial g/\partial \beta_1 = -1/\beta_2$, $\partial g/\partial \beta_2 = \beta_1/\beta_2^2$. Variance $= \nabla g' \mathbf{V} \nabla g = V_{11}/\beta_2^2 - 2\beta_1 V_{12}/\beta_2^3 + \beta_1^2 V_{22}/\beta_2^4$. As $\beta_2 \to 0$, the terms scale as $\beta_2^{-2}, \beta_2^{-3}, \beta_2^{-4}$; the last dominates, so $\text{Var}(\widehat{\text{WTP}}) \sim \beta_1^2 V_{22}/\beta_2^4 \to \infty$. Implication: when the cost coefficient is weak, the delta-method SE is enormous *and* the symmetric normal approximation fails — report a simulated median and percentile interval, and address the cause (§7.8).

B3. The Wald test rejects $H_0 : \theta = \theta^*$ at level α iff $|(\hat{\theta} - \theta^*)/\text{SE}| > z_{1-\alpha/2}$, i.e. iff $|\hat{\theta} - \theta^*| > z_{1-\alpha/2}\text{SE}$, i.e. iff $\theta^* < \hat{\theta} - z_{1-\alpha/2}\text{SE}$ or $\theta^* > \hat{\theta} + z_{1-\alpha/2}\text{SE}$ — exactly the condition that θ^* lies *outside* $[\hat{\theta} \pm z_{1-\alpha/2}\text{SE}]$, the $100(1-\alpha)\%$ interval. Hence reject $\iff$ θ^* outside the CI. ■

C1. The likely cause is a weak cost coefficient (significant only at 10%): $\text{WTP} = -\beta_{\text{attr}}/\beta_{\text{cost}}$ has variance dominated by $V_{\text{cost}}/\beta_{\text{cost}}^4$, so an imprecise denominator produces a huge SE — *and* the near-Cauchy behavior means the true WTP distribution is heavy-tailed and skewed, so the *symmetric* delta interval $[-30, 270]$ is itself suspect (it imposes symmetry the real distribution lacks and may understate the right tail while implying implausible negative WTP). "More data" would help only via better cost-coefficient identification; equally, a better-conditioned cost vector or WTP-space estimation would. I would report a Krinsky–Robb median and percentile interval (asymmetric, honest), diagnose $\Pr(\hat{\beta}_{\text{cost}} \approx 0)$, and consider WTP-space reparameterization (Chapters 1 §1.10, 24).

C2. $p = 0.21$ licenses only "we fail to reject equality at conventional levels" — it is **not** evidence that BRIS and full payment are equivalent (F4). An equivalence claim requires showing the confidence interval for the difference lies *entirely within* a pre-specified margin of practical equivalence (e.g., ±€X judged negligible); a wide interval consistent with large differences cannot support equivalence no matter how large p is. The

required analysis is **equivalence testing (TOST)**: pre-specify the equivalence margin, and reject non-equivalence only if the CI fits inside it (Chapter 8). Without that, "equivalent WTP" is an unsupported inference from a non-significant test.

C3. WTP is a ratio $-\beta_{\text{attr}}/\beta_{\text{cost}}$, and its delta-method variance contains a $\beta_{\text{attr}}^2 V_{\text{cost}}/\beta_{\text{cost}}^4$ term that blows up as the cost coefficient approaches zero, reflecting that the ratio behaves like a heavy-tailed near-Cauchy variable whose mean and standard deviation are unstable and whose distribution is skewed. The delta method forces a symmetric normal interval that cannot capture this skew and can imply implausible values; the Krinsky–Robb median is robust to the tail and the percentile interval respects the asymmetry, so together they give an honest picture of WTP uncertainty where mean ± delta-SE would mislead.

7.13 Spaced review prompts

+1 day. From memory, define bias, variance, MSE (with the decomposition), consistency, and efficiency; state the Cramér–Rao bound and why it makes the MLE the default. Then write the correct one-sentence interpretation of a p-value and of a confidence interval.

+1 week. Derive the delta-method variance of WTP $= -\beta_1/\beta_2$ from the gradient, and explain the $1/\beta_2^4$ blow-up. Then compute, for a contrast with $\hat{\delta} = 4$, SE $= 2.5$, the two-tailed p-value and 95% CI, and interpret both correctly.

+1 month. For one WTP in your own work, compute the delta-method SE by hand from `vcov`, run a small Krinsky–Robb, and compare the intervals; decide which to report and why. For one treatment contrast, write the correct interpretation of its p-value and CI, avoiding all the §7.9 misinterpretations. Close with the spine sentence.

7.14 How this chapter connects to the rest of the book

- **Chapter 6** supplied the asymptotic normality and the delta method's foundation; this chapter turns them into reported SEs, CIs, p-values, and propagated WTP uncertainty.
- **Chapter 8 (power and equivalence)** is the direct sequel: it builds on §7.5–§7.6's correct p-value/CI interpretation to handle "not significant ≠ equal," power, minimum detectable effects, and TOST for your convergent-validity and BRIS claims.
- **Chapter 9 (simulation)** develops Krinsky–Robb and the bootstrap of §7.8 in full; **Chapter 10 (randomization/permutation)** provides CLT-free tests for the small cells where §7.6's normal approximation is weak.
- **Chapter 11 (MLE)** proves the asymptotic efficiency (Cramér–Rao attainment) asserted in §7.3; **Chapters 12 (testing trinity)** develops Wald/LR/score tests whose Wald member is §7.6's z-test.
- **Chapters 17, 19 (robust/clustered/panel inference)** are the correct SE flavors of §7.4 for your repeated tasks and clustered estimates; **Chapter 24 (WTP space)** uses the delta-vs-simulation analysis of §7.8 to justify the space choice; **Chapters 47–48 (meta-regression)** apply §7.4's clustered/RVE SEs with small-sample corrections to the 18-study finfish data.

This chapter is the reporting-and-interpretation core of Part II, feeding the high-dependency chapters **Ch 11** (efficiency, testing) and **Ch 24** (WTP inference).

7.15 Research translation task

For one estimate in your work (ideally a WTP and a treatment contrast), produce a two-page note:

1. **Point estimate + right SE.** Report the estimate with a standard error computed with the correct *flavor* for your data (clustered for repeated tasks/studies) and, for WTP, the correct *propagation* (delta and simulation).
2. **Confidence interval, interpreted.** Give the 95% CI and write its correct interpretation (range of plausible values / procedure coverage), avoiding the F2/F4 misreadings.
3. **P-value, interpreted.** For the contrast, report the two-tailed p-value, state whether any one-tailed choice was pre-specified, and write the correct interpretation, explicitly noting what it does *not* say.
4. **WTP inference three ways.** Compute the delta-method interval and a Krinsky–Robb median + percentile interval for one WTP; compare them, judge whether the cost coefficient is safely away from zero, and decide which to report.
5. **Close** with the spine sentence: *"This estimate is [value] with [flavor] SE [·]; its 95% CI [·,·] means [plausible-value range]; the p-value [·] licenses [·] and forbids [·]; for WTP I report [delta/median+percentile] because [cost-coefficient strength / skew]."*

Carry this into Chapter 8 (turning "not significant" into a rigorous equivalence or power statement) and Chapter 9 (the simulation machinery behind the WTP interval).

7.16 Bibliography

Core. - Casella & Berger, *Statistical Inference*, chapters on point estimation (bias, MSE, efficiency, Cramér–Rao), interval estimation, and hypothesis testing. The rigorous backing for §7.2–§7.6. *(Verify edition/chapters.)* - Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, on estimation, inference, and the delta method in econometric context; the §7.4 SE-flavor discussion (robust/clustered) at the right level. *(Verify edition.)* - Wasserstein & Lazar, "The ASA Statement on p-Values: Context, Process, and Purpose," *The American Statistician* (2016). The authoritative short statement of what p-values are and are not — read it directly for §7.6. *(Verify volume/pages.)*

Applied. - Cameron & Trivedi, *Microeconometrics: Methods and Applications*, on the delta method, marginal-effect SEs, and cluster-robust inference — bridges §7.4–§7.7 to practice. *(Verify edition.)* - Krinsky & Robb (1986), *Review of Economics and Statistics*, and Haab & McConnell, *Valuing Environmental and Natural Resources*, on WTP confidence intervals via simulation vs the delta method — the §7.8 material in your exact setting. *(Verify citations.)* - Greene, *Econometric Analysis*, appendix on the delta method and the chapters on hypothesis testing; a standard reference for the WTP-ratio variance derivation. *(Verify edition.)*

Advanced. - Lehmann & Casella, *Theory of Point Estimation*, and Lehmann & Romano, *Testing Statistical Hypotheses*, for the decision-theoretic foundations of efficiency, the Cramér–Rao bound, and optimal tests. *(Verify editions.)* - Efron & Tibshirani, *An Introduction to the Bootstrap*, for the bootstrap intervals of §7.8 and when they beat Wald/delta intervals. *(Verify edition.)* - Poe, Giraud & Loomis, "Computational Methods for Measuring the Difference of Empirical Distributions" (the Poe test), for comparing WTP distributions across treatments — the tool behind your convergent-validity comparisons. *(Verify exact citation/year.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Krinsky–Robb / Poe citations before using them, per the book's no-unverified-citation rule.

End of Chapter 7. Next in Part II: Chapter 8 — Power, Minimum Detectable Effects, and Equivalence, which builds directly on this chapter's p-values and confidence intervals to answer the questions your many-armed CV design most needs: how much power do my contrasts have, what effects could I have detected, and how do I make a rigorous claim that two formats are equivalent (TOST) rather than merely "not significantly different"? Tell me when to write it, or request a different chapter.

Chapter 8 — Power, Minimum Detectable Effects, and Equivalence

Part II: Probability and Statistical Inference

> **The spine question, applied to this chapter.** Your PDF puts one error at the very top of its "priorità massima" list: *"'Non significativo' non significa 'uguale'."* It names exactly where this bites you — PC vs SBDC, full payment vs BRIS, experienced-low vs experienced-high, incentivized vs hypothetical — every place you want to claim two formats *agree*. This chapter is the cure. It teaches you to compute the **power** of a comparison (the probability it would detect a real effect), the **minimum detectable effect** (the smallest effect your design could catch), and — decisively — **equivalence testing (TOST)**, the only rigorous way to conclude that two things are the same. It also confronts the structural threat your PDF flags in §6.5: a CV design with many arms dilutes the sample per cell, gutting power. The spine question here: *when I claim two formats are equivalent, or claim an effect is absent, what could my design actually have detected — and have I tested equivalence, or merely failed to find a difference?*

8.0 Conceptual motivation: the asymmetry that ruins convergent-validity claims

Null hypothesis significance testing has a built-in logical asymmetry that trips up an enormous fraction of applied work, and it happens to sit at the center of your research program. A significance test can *reject* a null hypothesis (find evidence of a difference), but it can never *accept* one (find evidence of equality). When you run a difference test comparing WTP under a payment card and under single-bounded dichotomous choice and get $p = 0.55$, you have learned that the data are *consistent* with no difference — but they are equally consistent with a range of real differences the test simply lacked the power to detect. Concluding "PC and SBDC give equivalent WTP" from $p > 0.05$ is not a small imprecision; it is a logical error, and it is the error your PDF elevates above all others.

Why does this matter so much for you specifically? Because a large share of your intended contributions are *equivalence* or *convergence* claims dressed as difference tests:

- **Convergent validity** — the claim that two elicitation formats (PC vs SBDC) yield the same WTP — is inherently a claim of *equivalence*, and your PDF says so directly: *"Quando vuoi dire che due formati sono equivalenti o convergenti, non basta non rigettare l'ipotesi di uguaglianza. Devi conoscere equivalence testing, confidence intervals around differences, minimum relevant effect size e possibilmente TOST. Questo è cruciale per convergent validity e BRIS."*
- **BRIS as a practical substitute** for full payment is defensible only if you can show the two give *equivalent* WTP — again an equivalence claim, not a failure-to-reject.
- **The multi-armed design** (format × incentive × cost-vector × BRIS) spreads your sample thin, so each contrast may have low power, meaning your difference tests are *predisposed* to non-significance

regardless of the truth — the §6.5 problem. Low power is precisely what makes "not significant" uninformative.

This chapter builds the three tools that convert these traps into rigorous inference. **Power analysis** tells you, before you collect data, whether a contrast has a fighting chance of detecting an effect you would care about — and, after, whether a null result is meaningful or just weak. The **minimum detectable effect** translates power into a plain statement: "with this sample, I could only reliably catch effects bigger than X." And **equivalence testing (TOST)** flips the hypotheses so that "the same" becomes a claim you can actually *establish*, by pre-specifying a smallest relevant effect and showing the data rule out anything larger. Together they turn your convergent-validity and BRIS ambitions from unsupportable ("we didn't find a difference") into defensible ("we established equivalence within a margin we specified in advance"). Everything in this chapter is the mature, referee-proof version of the comparisons your papers already make.

8.1 Prerequisites

From **Chapter 7**: p-values and their correct interpretation (§7.6), confidence intervals and the test–CI duality (§7.5), Type I / Type II errors, and the standard error of a *contrast* (a difference), which is a quadratic form in the covariance matrix (§7.4, and Chapter 1 §1.6). From **Chapter 6**: the $\sqrt{n}$ rate (precision scales like $1/\sqrt{n}$) and asymptotic normality. From **Chapter 5**: the normal CDF Φ and its quantiles z_α.

Same **notation**. New symbols: α (significance level / Type I rate), β (Type II rate), $1-\beta$ (power), δ (true effect), $\hat{\delta}$ (its estimate), $\sigma_\delta = \mathrm{SE}(\hat{\delta})$, Δ (the equivalence margin / smallest relevant effect), $d = \delta/\sigma$ (standardized effect size), $z_{1-\alpha/2}$ (two-sided critical value, $= 1.96$ at $\alpha = 0.05$), $z_{1-\beta}$ (e.g. 0.8416 for 80% power).

8.2 The anatomy of a test: Type I, Type II, and power

Every test faces two ways to be wrong. Fix them precisely, because power is defined against them.

Definition 8.1 (Type I and Type II errors, power). In testing H_0 against H_1: - A **Type I error** is rejecting H_0 when it is true. Its probability is the **significance level** α — chosen by you (conventionally 0.05). - A **Type II error** is failing to reject H_0 when it is false (a specific H_1 holds). Its probability is β. - The **power** is $1-\beta$: the probability of correctly rejecting H_0 when a specific alternative is true.

The four cells:

	H_0 true	H_1 true (effect δ)
Reject H_0	Type I error (α)	Correct ✓ (power $1-\beta$)
Fail to reject	Correct ✓ ($1-\alpha$)	Type II error (β)

Power is defined against a *specific* effect. "The power of my test" is meaningless until you say "power to detect an effect of size δ." The power *function* traces power as δ varies: it is α at $\delta = 0$ (you reject a true null with probability α) and rises toward 1 as $|\delta|$ grows. Power depends on four things, and you control three:

1. the **true effect size** δ (larger effects are easier to detect) — you do not control this, but you can specify the *smallest effect worth detecting*;
2. the **significance level** α (a stricter α lowers power);
3. the **standard error** σ_δ of the estimate — which depends on the outcome variance σ^2 and the sample size n (through $\sigma_\delta \propto \sigma/\sqrt{n}$);
4. the **one- vs two-sided** choice (one-sided is more powerful in its direction).

The single most important relationship: power rises with δ/σ_δ — the true effect measured in standard-error units. Everything in power analysis is about making that ratio big enough.

> **Active recall 8.A.** Your co-author says "our test has 80% power." What is missing from that statement, and why does it make the claim meaningless without it? *(Answer: the effect size it has 80% power to detect. Power is a function of δ; "80% power" is only defined relative to a specified effect (e.g., "80% power to detect a €20 WTP difference"). Without the effect size, the number says nothing.)*

8.3 The power formula and required sample size, derived

We now derive power for the workhorse case: a two-sided test of a single contrast $\hat{\delta}$ (a treatment difference), which covers all your two-arm comparisons.

Setup. Test $H_0 : \delta = 0$ vs $H_1 : \delta \neq 0$ at level α. Reject if $|\hat{\delta}/\sigma_\delta| > z_{1-\alpha/2}$. Under a true effect δ, $\hat{\delta} \sim N(\delta, \sigma_\delta^2)$ (asymptotic normality, Chapter 6).

Derivation of the power function. Power is the probability of landing in the rejection region:

$$1 - \beta = \Pr(\hat{\delta} > z_{1-\alpha/2}\sigma_\delta) + \Pr(\hat{\delta} < -z_{1-\alpha/2}\sigma_\delta).$$

Standardize $\hat{\delta} = \delta + \sigma_\delta Z$ with $Z \sim N(0,1)$:

$$1 - \beta = \Pr\Big(Z > z_{1-\alpha/2} - \tfrac{\delta}{\sigma_\delta}\Big) + \Pr\Big(Z < -z_{1-\alpha/2} - \tfrac{\delta}{\sigma_\delta}\Big) = \Phi\Big(\tfrac{\delta}{\sigma_\delta} - z_{1-\alpha/2}\Big) + \Phi\Big(-\tfrac{\delta}{\sigma_\delta} - z_{1-\alpha/2}\Big).$$

For a positive effect the second term is negligible, giving the working formula

$$\boxed{1 - \beta \approx \Phi\Big(\frac{\delta}{\sigma_\delta} - z_{1-\alpha/2}\Big)}.$$

Read it: power is the normal CDF of "the effect in SE units, minus the critical value." To have power $1-\beta$ you need $\delta/\sigma_\delta - z_{1-\alpha/2} = z_{1-\beta}$, i.e.

$$\frac{\delta}{\sigma_\delta} = z_{1-\alpha/2} + z_{1-\beta}.$$

The effect, in standard-error units, must equal the sum of two critical values — one for the Type I rate, one for the Type II rate. At $\alpha = 0.05$ and 80% power, that sum is $1.96 + 0.8416 = 2.80$ (verified). So you need the true effect to be about **2.8 standard errors** to detect it 80% of the time.

Required sample size for a two-arm comparison. For comparing two means with per-arm variance σ^2 and n per arm, $\sigma_\delta = \sigma\sqrt{1/n + 1/n} = \sigma\sqrt{2/n}$. Substituting into $\delta/\sigma_\delta = z_{1-\alpha/2} + z_{1-\beta}$ and solving:

$$\boxed{n = \frac{2\sigma^2\,(z_{1-\alpha/2} + z_{1-\beta})^2}{\delta^2} = \frac{2(z_{1-\alpha/2} + z_{1-\beta})^2}{d^2}}, \qquad d = \frac{\delta}{\sigma}.$$

The δ^2 (equivalently d^2) in the denominator is the expensive part: **halving the effect you want to detect quadruples the required sample** — the $\sqrt{n}$ rate of Chapter 6 in disguise.

Worked Example 8.1 (Sample sizes, verified). At $\alpha = 0.05$ (two-sided) and 80% power, $(z_{1-\alpha/2} + z_{1-\beta})^2 = 2.80^2 = 7.85$, so $n = 15.70/d^2$ per arm:

Standardized effect d	n per arm (80% power)
0.8 (large)	25

Standardized effect d	n per arm (80% power)
0.5 (medium)	**63**
0.3 (small–medium)	175
0.2 (small)	393

These are the numbers to carry in your head: detecting a *medium* effect ($d = 0.5$) needs ~63 per arm; a *small* effect ($d = 0.2$) needs ~393 per arm. Now recall your multi-armed CV design: if the total sample is split across, say, eight cells, then a cell of 60–70 respondents can only reliably detect medium-to-large effects, and any genuinely *small* cost-vector or format effect will usually be missed. That is not a data problem you can interpret away — it is baked into the design, and it is exactly the §6.5 warning.

> **Stop and think 8.1.** You want to detect a cost-vector effect on WTP that you believe is "small" ($d \approx 0.2$) with 80% power, but each arm has only 100 respondents. Before computing: will you succeed? *(Answer: no — detecting $d = 0.2$ at 80% power needs ~393 per arm; with 100 you have far less power (roughly 30%), so you will fail to detect this effect most of the time even if it is real. A non-significant result here is nearly uninformative — precisely why you must report the MDE, §8.4, and why you cannot claim "no effect.")*

8.4 The minimum detectable effect

Power analysis is often run "backwards": given the sample you have (or can afford), what is the smallest effect you could reliably detect? That is the **minimum detectable effect**, and it is the single most useful number for *interpreting a null result.*

Definition 8.2 (Minimum detectable effect, MDE). The **MDE** at level α and power $1-\beta$ is the effect size for which power equals $1-\beta$:

$$\text{MDE} = (z_{1-\alpha/2} + z_{1-\beta})\,\sigma_\delta = (z_{1-\alpha/2} + z_{1-\beta})\,\sigma\sqrt{\tfrac{2}{n}} \quad \text{(two-arm case)}.$$

Any true effect smaller than the MDE will be *missed* more often than $1-\beta$ of the time. Reporting the MDE alongside a non-significant result is the honest way to say what your study *could* and *could not* have found.

Worked Example 8.2 (MDE for a CV cell, verified). With $\alpha = 0.05$, 80% power, and per-arm sizes typical of a diluted multi-armed design:

n per arm	MDE (standardized)	MDE in € (if WTP SD = €40)
30	0.72 SD	**€28.9**
50	0.56 SD	**€22.4**
100	0.40 SD	€15.8

The interpretation is bracing. With 50 respondents per arm and a WTP standard deviation of €40, your design can only reliably detect WTP differences **larger than about €22**. So if your PC-vs-SBDC difference comes back "not significant," the honest statement is not "the formats are equivalent" but "we could not detect any difference smaller than €22 — differences up to €22 remain entirely consistent with our data." Whether that is good enough depends on whether €22 is a *relevant* difference for your purposes, which is exactly the judgment the equivalence margin formalizes (§8.6). The MDE is the bridge from "not significant" to "here is what not-significant is worth."

> **Active recall 8.B.** A reviewer asks you to justify a claim of "no cost-vector effect" from a non-significant test with 40 respondents per arm. What single number should you report to address the question honestly, and what does it establish? *(Answer: the minimum detectable effect. It*

states the smallest effect your design could reliably detect; if the MDE is large relative to effects that would matter, "not significant" is uninformative and the claim of "no effect" is unsupported. Report the MDE, not just the p-value.)

8.5 Why "not significant" does not mean "equal"

We can now state the core discipline precisely, because you have the tools. The logic of a difference test is asymmetric:

- **Rejecting** $H_0 : \delta = 0$ licenses "there is evidence of a non-zero effect."
- **Failing to reject** $H_0 : \delta = 0$ licenses only "the data are consistent with $\delta = 0$" — but the *same data are consistent with every value inside the confidence interval*, including possibly large effects. Failure to reject is **absence of evidence**, not **evidence of absence**.

The confidence interval makes this concrete (test–CI duality, Chapter 7 §7.5). A non-significant result means the CI contains zero. But a CI can contain zero while stretching to large effects (underpowered) *or* while hugging zero tightly (precise near-null). These are opposite epistemic situations that share the label "not significant."

Worked Example 8.3 (A large, non-significant effect, verified). Suppose the estimated WTP difference between two arms is $\hat{\delta} = €12$ with SE = €8. Then $z = 12/8 = 1.5$, two-sided $p = 0.13$ — **not significant**. But the 95% CI is $12 \pm 1.96(8) = [-€3.7,\ €27.7]$. The data are consistent with anything from a small negative effect to a *large* €28 effect. Concluding "no difference" here would be absurd: the point estimate is €12, and the interval barely excludes zero. The non-significance reflects a **large standard error (low power)**, not a small effect. This is the anatomy of the error your PDF warns against: a real, possibly important difference, mislabeled "equivalent" because a weak test failed to reach $p < 0.05$.

Why you cannot fix this by lowering standards. You cannot rescue an equality claim by using a larger α or a one-sided test — those change the difference test, not its logic. The only correct move is to **change the hypotheses**: make equivalence the thing you try to *establish*, with the burden of proof on you. That is TOST.

> **Stop and think 8.2.** Two convergent-validity studies both report "$p > 0.05$, formats are equivalent." Study A: $\hat{\delta} = €1$, 95% CI $[-€3, €5]$. Study B: $\hat{\delta} = €1$, 95% CI $[-€40, €42]$. Which study, if either, has actually shown equivalence, and why? *(Answer: neither has* shown* equivalence by the difference test alone, but Study A has the *evidence* for it — its interval rules out anything but tiny differences, so a formal equivalence test against a reasonable margin would likely succeed; Study B's interval admits huge differences, so it is simply uninformative. Same point estimate, same p-verdict, completely different evidential value — which only the interval (and a formal TOST) reveals.)*

8.6 Equivalence testing and TOST

To *establish* equivalence you must first say what "equivalent enough" means — a **margin** of practical or economic equivalence — and then show the data rule out differences larger than that margin. This requires a pre-specified judgment and a re-arranged hypothesis.

Definition 8.3 (Equivalence margin). The **equivalence margin** $\Delta > 0$ is the smallest difference that would be *practically or economically relevant*. Differences within $(-\Delta, +\Delta)$ are treated as "equivalent for our purposes." Δ must be chosen on **substantive grounds and specified in advance** — it is where domain judgment enters, and it cannot be read off the data. For a WTP convergence claim, Δ might be "€15, because differences below that would not change any policy conclusion or benefit-transfer use."

Definition 8.4 (Equivalence hypotheses and TOST). Equivalence testing *reverses* the usual null and alternative:

$$H_0 : |\delta| \geq \Delta \quad \text{(a relevant difference exists)} \qquad \text{vs} \qquad H_1 : |\delta| < \Delta \quad \text{(equivalence)}.$$

Now the *burden of proof is on equivalence*: you conclude equivalence only if the data provide positive evidence for it. The standard procedure is **TOST — Two One-Sided Tests**:

$$\text{Lower: } H_0^- : \delta \leq -\Delta \;\text{ vs }\; H_1^- : \delta > -\Delta, \qquad \text{Upper: } H_0^+ : \delta \geq \Delta \;\text{ vs }\; H_1^+ : \delta < \Delta.$$

Compute both one-sided statistics:

$$t^- = \frac{\hat{\delta} - (-\Delta)}{\sigma_\delta}, \qquad t^+ = \frac{\hat{\delta} - \Delta}{\sigma_\delta},$$

and **conclude equivalence at level α if you reject *both*** one-sided nulls — i.e., if $t^- > z_{1-\alpha}$ *and* $t^+ < -z_{1-\alpha}$. The TOST p-value is the larger of the two one-sided p-values.

The confidence-interval version (the one to use in practice). TOST at level α is exactly equivalent to the following simple rule:

Conclude equivalence iff the $(1-2\alpha)$ confidence interval for δ lies entirely within $(-\Delta, +\Delta)$.

For $\alpha = 0.05$ that is the **90%** CI (not 95% — the two one-sided tests at 5% each correspond to a 90% two-sided interval). This is the cleanest way to run and report TOST: build the 90% CI for the difference and check whether it fits inside your pre-specified equivalence band. *(Derivation: rejecting $H_0^+ : \delta \geq \Delta$ at α means the upper $(1-\alpha)$ one-sided bound $\hat{\delta} + z_{1-\alpha}\sigma_\delta < \Delta$; rejecting H_0^- means the lower bound $\hat{\delta} - z_{1-\alpha}\sigma_\delta > -\Delta$; together these say the $(1-2\alpha)$ CI $[\hat{\delta} \pm z_{1-\alpha}\sigma_\delta]$ sits inside $(-\Delta, \Delta)$.)*

Worked Example 8.4 (TOST for convergent validity — two verified cases). You compare WTP under payment card and SBDC and pre-specify an equivalence margin of $\Delta = €15$ (differences below €15 are deemed irrelevant for benefit transfer). Two scenarios, *same point estimate* $\hat{\delta} = €3$, different precision:

Case 1 — precise (SE = €5). Difference test: $z = 3/5 = 0.6$, $p = 0.55$ (not significant). TOST: $t^- = (3+15)/5 = 3.6$ and $t^+ = (3-15)/5 = -2.4$; one-sided p-values 0.0002 and 0.008, so the TOST p-value is $\max = 0.008 < 0.05$ — **reject both, conclude equivalence.** The 90% CI is $3 \pm 1.645(5) = [-€5.2,\ €11.2]$, which lies entirely within $(-15, 15)$ ✓. You have *established* that PC and SBDC WTP are equivalent within €15.

Case 2 — imprecise (SE = €10). Difference test: $z = 0.3$, $p = 0.76$ (not significant — *identical verdict to Case 1's spirit*). TOST: $t^- = (3+15)/10 = 1.8$, $t^+ = (3-15)/10 = -1.2$; TOST p-value $= \max(0.036, 0.115) = 0.115 > 0.05$ — **cannot reject; equivalence not established.** The 90% CI is $3 \pm 1.645(10) = [-€13.5,\ €19.5]$, which spills past €15 ×. Despite the same point estimate and a "not significant" difference test, you have **not** shown equivalence — the design is too imprecise to rule out relevant differences.

The pair is the whole lesson. Both cases are "not significantly different." Only Case 1 supports an equivalence claim; Case 2 is *inconclusive*. A difference test alone cannot tell them apart — TOST can. This is precisely the machinery your PDF says is "cruciale per convergent validity e BRIS."

Equivalence testing needs power too. A crucial and often-missed point: TOST can *fail to establish equivalence* not because the things differ, but because the sample is too small to rule out the margin — exactly Case 2. So you must **power your equivalence tests** in advance: the sample size needed to establish equivalence within Δ with adequate power depends on Δ, σ, and the true δ (equivalence is hardest to show when the true difference is near the margin). Equivalence is not "free" — a convergent-validity study must be sized to *demonstrate* convergence, not merely to *fail to find* divergence.

> **Active recall 8.C.** State the CI rule for concluding equivalence at $\alpha = 0.05$ with margin Δ, and explain why it uses the 90% (not 95%) interval. *(Answer: conclude equivalence iff the 90% CI*

for the difference lies entirely within $(-\Delta, +\Delta)$. It is the 90% interval because TOST performs two one-sided 5% tests, and two one-sided 5% bounds correspond to a two-sided 90% interval, not 95%.)

8.7 The mature report: crossing difference and equivalence tests

The difference test and the equivalence test answer *different* questions — "is there a non-zero effect?" and "is any effect smaller than Δ?" — and you should run *both*. Crossing them gives a 2×2 that is the honest, complete way to report a comparison:

	Equivalence established (CI within $\pm\Delta$)	**Equivalence not established**
Difference significant	**Significant but negligible** — a real but practically unimportant effect (typical with very large n)	**Relevant effect** — a real and non-negligible difference
Difference not significant	**Equivalent** — statistically indistinguishable within the margin (your convergent-validity goal)	**Inconclusive / underpowered** — the data can neither confirm an effect nor rule out a relevant one

Read the cells against your work:

- **"Equivalent" (bottom-left)** is the cell you want for PC≈SBDC and BRIS≈full payment: not significant *and* equivalence established. This is a genuine convergent-validity finding.
- **"Inconclusive" (bottom-right)** is the trap you must avoid mislabeling: not significant but equivalence *not* established (Case 2 above). The correct report is "we cannot conclude either a difference or equivalence — the study is underpowered for this comparison," never "the formats are equivalent."
- **"Relevant effect" (top-right)** is a real, meaningful difference — report the effect and its interval.
- **"Significant but negligible" (top-left)** warns you that with a large sample even a trivial difference becomes "significant"; significance is not importance, and the equivalence test protects you from over-interpreting a tiny but precise effect.

This 2×2 is the intellectual payoff of the chapter. It replaces the impoverished "significant / not significant" dichotomy with a framework that separates **relevance** (is the effect big enough to matter? — the equivalence margin) from **precision/detectability** (can we tell? — power and the CI). Reporting where each of your key contrasts falls in this table is the single most credible upgrade you can make to your convergent-validity and BRIS analyses.

8.8 Multiple testing: the many-armed problem

Your CV design generates *many* comparisons — several formats, incentive conditions, cost vectors, BRIS, and their interactions — and each significance test carries its own Type I risk. Run enough tests and false positives become nearly inevitable.

The family-wise error rate. If you run m independent tests each at level α, the probability of *at least one* false positive (the **family-wise error rate**, FWER) is

$$\text{FWER} = 1 - (1 - \alpha)^m.$$

(Verified: at $\alpha = 0.05$, FWER is 0.23 for $m = 5$, 0.40 for $m = 10$, and 0.64 for $m = 20$.) With ten comparisons you have a 40% chance of "finding" an effect that is not there. In a design with many arms and many contrasts, uncorrected testing manufactures spurious significance.

Corrections. - **Bonferroni:** test each hypothesis at α/m (verified: 0.005 for $m = 10$). Simple, controls FWER, but *conservative* — it sacrifices power, which you can ill afford in small cells. - **Holm's step-down:** orders the p-values and applies successively less strict thresholds; uniformly more powerful than Bonferroni while still controlling FWER. Prefer it to plain Bonferroni. - **Benjamini–Hochberg (FDR):** controls the *false discovery rate* (the expected proportion of false positives *among rejections*) rather than the probability of *any* false positive. Less strict, more powerful, appropriate when you are screening many exploratory contrasts and can tolerate some false discoveries.

The tension, and the better discipline. Multiplicity correction *reduces power* — and your design is already power-starved (§8.3, §8.9). Slashing α to α/m across dozens of contrasts can leave you unable to detect anything. The mature resolution is not purely mechanical: **pre-specify a small number of *primary* comparisons** (the ones your paper is really about — say, PC vs SBDC and BRIS vs full payment), test those with minimal or no correction, and treat the rest as *secondary/exploratory* with FDR control and appropriately hedged language. Pre-registration formalizes this: it fixes which contrasts are confirmatory before you see the data, which both controls multiplicity honestly and preserves power for the questions you care about. This is far superior to running every contrast and Bonferroni-correcting the lot.

> **Stop and think 8.3.** Your design yields 15 treatment contrasts, and you find 2 significant at $p < 0.05$ uncorrected. Before celebrating, what is the expected number of false positives under a global null, and what does that imply for interpreting the 2 "findings"? *(Answer: under the global null, expected false positives $= 15 \times 0.05 = 0.75$, and $\Pr(\geq 1) = 1 - 0.95^{15} \approx 0.54$ — so finding 1–2 "significant" contrasts is entirely consistent with pure noise. The 2 findings cannot be trusted without multiplicity control or pre-specification; they may well be false positives.)*

8.9 Power in the many-armed CV design (synthesis)

Bringing it together for the design your PDF worries about in §6.5. The core tension: scientific richness (many formats, incentives, cost vectors, BRIS) versus statistical power (finite total sample split across many cells). Three specific hazards:

1. Dilution of per-cell n. If a fixed total N is split across K cells, each cell has N/K. From §8.3, per-arm power collapses as cell size falls; a design with eight cells and $N = 500$ gives ~62 per cell, enough only for medium-to-large effects. **Every cell you add lowers the power of every comparison.** This directly limits how many dimensions you can cross before your contrasts become uninformative.

2. Interactions cost far more than main effects. Much of your scientific interest is in *interactions* — does the cost-vector effect differ by experience? does BRIS behave differently under high vs low cost? A 2×2 interaction contrast is a *difference of differences*, built from four cells, with variance $4\sigma^2/n$ versus $2\sigma^2/n$ for a main-effect contrast — so its standard error is $\sqrt{2}$ times larger *for the same cell size* (verified). To detect an interaction of the *same* magnitude as a main effect thus needs **twice** the per-cell sample; and because interactions are typically *smaller* than main effects, the requirement compounds through the $1/\delta^2$ in the sample-size formula — often demanding four to sixteen times the sample of the corresponding main effect. **Interaction claims from a diluted design are usually underpowered**, and a non-significant interaction is especially uninformative. Report the interaction's MDE, and be cautious about interpreting null interactions.

3. Equivalence tests need to be powered too. Your convergent-validity and BRIS goals are *equivalence* claims, and §8.6 showed equivalence is not established for free — Case 2 failed precisely because the SE was too large. So the arms whose equivalence you most want to demonstrate (PC vs SBDC, BRIS vs full payment) must be *sized to establish equivalence within your margin* Δ, which typically requires **more**, not

fewer, respondents than a difference test, especially when Δ is small. A design that spreads the sample thinly enough to underpower these equivalence tests cannot deliver its headline convergent-validity conclusion.

Design implications (the actionable synthesis). (i) *Prioritize.* Decide which two or three contrasts are the paper's confirmatory questions and allocate sample to power *those* — including their equivalence tests — rather than spreading evenly. (ii) *Limit cells.* Resist crossing every factor; each new dimension taxes all comparisons. (iii) *Pre-specify Δ and primary contrasts* before data collection, and run a power analysis for both the difference *and* equivalence tests of the primary comparisons. (iv) *Report MDEs* for secondary contrasts so null results are interpretable. (v) *Plan multiplicity* — a short confirmatory list tested at full power, the rest exploratory with FDR. This turns the §6.5 vulnerability into a defensible design.

> **Active recall 8.D.** Why does a non-significant *interaction* in your cost-vector × experience design deserve even more caution than a non-significant main effect? *(Answer: the interaction contrast has larger variance ($\sqrt{2}\times$ the SE of a main effect at the same cell size) and interactions are typically smaller, so power for interactions is much lower; a null interaction is therefore especially likely to be a false negative — report its MDE and avoid concluding "no interaction.")*

8.10 Failure modes

F1. "Not significant" reported as "equal / equivalent / no effect." The core error. *Fix:* run TOST against a pre-specified margin; report the MDE; use the 2×2 (§8.5–§8.7).

F2. No equivalence margin, or a data-driven one. Claiming equivalence without a pre-specified, substantively justified Δ (or choosing Δ after seeing the data to get the desired verdict). *Fix:* specify Δ in advance on economic/policy grounds; pre-register it (§8.6).

F3. Underpowered design mistaken for a null finding. Interpreting non-significance from small cells as evidence of no effect. *Fix:* compute power/MDE in advance; if underpowered, say "inconclusive," not "no effect" (§8.4, §8.9).

F4. Forgetting to power the equivalence test. Assuming equivalence will be establishable without sizing for it. *Symptom:* TOST fails (Case 2) and you cannot support convergent validity. *Fix:* power the equivalence test for your Δ (§8.6, §8.9).

F5. Multiplicity ignored. Running many contrasts uncorrected and celebrating the significant few. *Fix:* pre-specify primary contrasts; control FWER/FDR for the rest (§8.8).

F6. Over-correcting into powerlessness. Bonferroni across dozens of contrasts, destroying power for the questions that matter. *Fix:* separate confirmatory (few, high-power) from exploratory (FDR); pre-register (§8.8).

F7. Significance mistaken for importance. With large n, a trivial difference is "significant." *Fix:* pair the difference test with an equivalence test / effect-size interval; use the "significant but negligible" cell (§8.7).

F8. One-sided equivalence or opportunistic margins. Testing equivalence on only one side, or widening Δ until equivalence "passes." *Fix:* TOST tests both sides; Δ is fixed in advance (§8.6).

8.11 Diagnostic questions for your own papers

1. **Equality claims.** For every claim that two formats/arms are equivalent or convergent, did I run an equivalence test (TOST) against a pre-specified margin, not just a difference test? *(§8.6.)*
2. **Margin.** Is my equivalence margin Δ specified in advance and justified on economic/policy grounds (e.g., benefit-transfer relevance)? *(§8.6, F2.)*

3. **MDE.** For every null result, have I reported the minimum detectable effect, so readers know what the null could and could not rule out? *(§8.4, F3.)*
4. **Power in design.** Did I compute power for my *primary* contrasts — both difference and equivalence — before collecting data, given realistic effect sizes and cell sizes? *(§8.3, §8.9.)*
5. **Interactions.** Are my interaction contrasts adequately powered, and have I avoided interpreting null interactions as "no interaction"? *(§8.9.)*
6. **Multiplicity.** Have I pre-specified a small set of primary comparisons and controlled multiplicity for the rest? *(§8.8.)*
7. **The 2×2.** For each key comparison, can I place it in the difference×equivalence table (relevant effect / equivalent / inconclusive / significant-but-negligible)? *(§8.7.)*

8.12 Exercises

(A) Comprehension.

A1. Define power, and state the four quantities it depends on. Your test has 80% power to detect $d = 0.5$ at $\alpha = 0.05$. If the true effect is actually $d = 0.3$, is your power higher or lower than 80%, and why?

A2. A comparison gives $\hat{\delta} = €2$, 95% CI $[-€1, €5]$, $p = 0.19$. Can you conclude the arms are equivalent? What additional ingredient do you need, and how would you use the CI to decide?

A3. With $\alpha = 0.05$ and 10 independent contrasts, compute the probability of at least one false positive if all nulls are true. What does this imply for reporting?

(B) Derivation.

B1. Derive the required per-arm sample size $n = 2(z_{1-\alpha/2} + z_{1-\beta})^2/d^2$ from the power condition $\delta/\sigma_\delta = z_{1-\alpha/2} + z_{1-\beta}$, stating each assumption. Then compute n for $d = 0.4$, 90% power, $\alpha = 0.05$.

B2. Derive the confidence-interval form of TOST: show that rejecting both one-sided nulls at level α is equivalent to the $(1 - 2\alpha)$ CI for δ lying entirely within $(-\Delta, \Delta)$.

B3. For a 2×2 factorial with cell size n and outcome variance σ^2, derive the variance of (i) a main-effect contrast and (ii) an interaction contrast, and show the interaction SE is $\sqrt{2}$ times larger. Explain the implication for powering interaction tests.

(C) Research application / critique.

C1. Your CV paper concludes "BRIS and full payment yield equivalent WTP ($p = 0.31$)." A referee objects. Write the corrected analysis plan: what margin, what test, what CI, and what would each possible outcome license?

C2. Your design crosses format (2) × incentive (2) × cost vector (2) = 8 cells with $N = 560$ total. Assess the power situation: per-cell n, the MDE for a main effect (assume WTP SD €35), and the feasibility of testing a three-way interaction. Recommend a redesign.

C3. Explain to a co-author, in a short paragraph, why running all 12 pairwise contrasts in your design and reporting the 3 that are significant at $p < 0.05$ is misleading, and propose a defensible alternative.

8.13 Solutions

A1. Power = Pr(reject H_0 | true effect); it depends on the true effect size δ, the significance level α, the standard error σ_δ (via σ and n), and the one/two-sided choice. If the true effect is smaller ($d = 0.3$ instead of 0.5), power is **lower** than 80% — smaller effects are harder to detect, so with the same n the ratio δ/σ_δ falls and power drops (here to roughly 40%).

A2. Not from the difference test alone ($p = 0.19$ is failure to reject, not evidence for equality). You need a **pre-specified equivalence margin** Δ. Then check whether the 90% CI (here roughly $[-0.5, 4.5]$) lies within $(-\Delta, \Delta)$: if Δ is, say, €6, the 90% CI fits inside and you conclude equivalence; if Δ is €3, it does not and you cannot. The CI is narrow, so equivalence against a reasonable margin is plausible — but only a formal TOST against a stated Δ establishes it.

A3. FWER $= 1 - (1 - 0.05)^{10} = 1 - 0.95^{10} = 0.40$. A 40% chance of at least one false positive means uncorrected "findings" among 10 contrasts are unreliable; report pre-specified primary contrasts and control multiplicity (FWER/FDR) for the rest.

B1. Assume asymptotic normality of $\hat{\delta}$, equal per-arm variance σ^2, per-arm size n, so $\sigma_\delta = \sigma\sqrt{2/n}$; a two-sided level-$\alpha$ test; and power evaluated at true effect δ (ignoring the negligible far-tail term). The power condition $\delta/\sigma_\delta = z_{1-\alpha/2} + z_{1-\beta}$ gives $\delta/(\sigma\sqrt{2/n}) = z_{1-\alpha/2} + z_{1-\beta}$, i.e. $\sqrt{n} = \sqrt{2}\,\sigma(z_{1-\alpha/2} + z_{1-\beta})/\delta$, so $n = 2\sigma^2(z_{1-\alpha/2} + z_{1-\beta})^2/\delta^2 = 2(z_{1-\alpha/2} + z_{1-\beta})^2/d^2$. For $d = 0.4$, 90% power ($z_{0.90} = 1.2816$), $\alpha = 0.05$: $(1.96 + 1.2816)^2 = 10.51$, $n = 2(10.51)/0.16 = 131.4$, so ~132 per arm.

B2. Rejecting $H_0^+ : \delta \geq \Delta$ at α requires $t^+ = (\hat{\delta} - \Delta)/\sigma_\delta < -z_{1-\alpha}$, i.e. $\hat{\delta} + z_{1-\alpha}\sigma_\delta < \Delta$ (the upper $(1-\alpha)$ one-sided bound is below Δ). Rejecting $H_0^- : \delta \leq -\Delta$ requires $t^- = (\hat{\delta} + \Delta)/\sigma_\delta > z_{1-\alpha}$, i.e. $\hat{\delta} - z_{1-\alpha}\sigma_\delta > -\Delta$. Together: $-\Delta < \hat{\delta} - z_{1-\alpha}\sigma_\delta$ and $\hat{\delta} + z_{1-\alpha}\sigma_\delta < \Delta$, i.e. the interval $[\hat{\delta} \pm z_{1-\alpha}\sigma_\delta]$ — the $(1-2\alpha)$ CI — lies within $(-\Delta, \Delta)$. ■

B3. Let the four cell means be $\bar{y}_{11}, \bar{y}_{10}, \bar{y}_{01}, \bar{y}_{00}$, each with variance σ^2/n and independent. (i) A main effect (e.g., average over the second factor) contrast such as $\frac{1}{2}(\bar{y}_{11} + \bar{y}_{10}) - \frac{1}{2}(\bar{y}_{01} + \bar{y}_{00})$ has variance $4 \cdot \frac{1}{4} \cdot \sigma^2/n = \sigma^2/n$; a simple two-cell difference $\bar{y}_1 - \bar{y}_0$ has variance $2\sigma^2/n$. (ii) The interaction $(\bar{y}_{11} - \bar{y}_{10}) - (\bar{y}_{01} - \bar{y}_{00})$ has variance $4\sigma^2/n$. Comparing the interaction $(4\sigma^2/n)$ to a two-cell main-effect contrast $(2\sigma^2/n)$, the SE ratio is $\sqrt{4/2} = \sqrt{2}$. Implication: to detect an interaction of the same size as a main effect needs twice the per-cell n; since interactions are usually smaller, the requirement compounds through $1/\delta^2$ — interaction tests need much larger samples and null interactions are especially uninformative.

C1. Corrected plan: (1) Pre-specify an equivalence margin Δ for WTP (e.g., €X, justified as the smallest difference that would change a benefit-transfer or policy conclusion). (2) Run **TOST** on the BRIS-vs-full-payment WTP difference (or, equivalently, check whether the 90% CI for the difference lies within $\pm\Delta$). (3) Report the point estimate, its 90% CI, the TOST p-value, and the MDE. Outcomes: if the 90% CI lies within $\pm\Delta \rightarrow$ **conclude equivalence** (supports BRIS as a substitute); if the CI spills past $\pm\Delta$ while the difference test is non-significant $\rightarrow$ **inconclusive/underpowered** (cannot claim equivalence; report as such and note the sample needed); if the difference test is significant and the effect exceeds $\Delta \rightarrow$ **relevant difference** (BRIS is not equivalent). The bare "$p = 0.31$" licenses none of these conclusions.

C2. Per-cell $n = 560/8 = 70$. Main-effect contrast (two groups of $4 \times 70 = 280$ each, or as a two-cell difference of size 70): the MDE for a simple two-cell comparison at 80% power is $(1.96 + 0.8416)\sigma\sqrt{2/70} = 2.80 \cdot 35 \cdot 0.169 =$ €16.6 (main effects averaged over other factors are better-powered; a specific two-cell contrast is as stated). A **three-way interaction** is a contrast of all 8 cells with variance $8\sigma^2/n$, SE $= \sqrt{8} \cdot \sigma/\sqrt{70} \approx \sqrt{8} \cdot 4.18 =$ €11.8, giving an MDE of $2.80 \times 11.8 \approx$ €33 — larger than most plausible interaction effects, so the three-way interaction is essentially untestable here. Recommendation: drop to a 2×2 (four cells of 140) focused on the two primary factors, or increase N; power the primary difference *and* equivalence tests; treat higher-order interactions as exploratory only.

C3. Reporting only the significant 3 of 12 contrasts is misleading because with 12 tests at $\alpha = 0.05$ the family-wise false-positive probability is $1 - 0.95^{12} \approx 0.46$ and the expected number of false positives under the global null is 0.6 — so finding a few "significant" results is expected by chance alone, and cherry-picking them inflates the true error rate (and is a form of p-hacking). A defensible alternative: pre-specify the one or two primary contrasts the paper is about, test those at full power (little/no correction), and report the remaining contrasts as exploratory with FDR (Benjamini–Hochberg) control and hedged language — ideally pre-registered so the confirmatory/exploratory split is fixed before seeing the data.

8.14 Spaced review prompts

+1 day. From memory, write the power condition $\delta/\sigma_\delta = z_{1-\alpha/2} + z_{1-\beta}$, derive the sample-size formula, and state the MDE. Then explain in two sentences why "not significant" is not "equivalent."

+1 week. Run a TOST by hand for a comparison of your choice: pick a margin, compute the two one-sided tests (or the 90% CI), and state the conclusion. Then fill in the difference×equivalence 2×2 for a "not significant, wide CI" result and a "not significant, narrow CI" result.

+1 month. For each primary contrast in your CV paper, (i) compute its MDE given the actual cell sizes, (ii) pre-specify an equivalence margin, (iii) run the difference and equivalence tests, and (iv) place the result in the 2×2. Close with the spine sentence about what your design could detect and whether you established equivalence.

8.15 How this chapter connects to the rest of the book

- **Chapter 7** supplied the p-values, confidence intervals, test–CI duality, and contrast standard errors this chapter builds on; §8.5–§8.7 are the resolution of Chapter 7's "not significant ≠ no effect."
- **Chapter 6** supplied the $\sqrt{n}$ rate behind the sample-size formula and the small-sample cautions that make power analysis essential for your diluted cells.
- **Chapter 10 (randomization/permutation inference)** provides CLT-free tests and the **Poe test** for comparing WTP distributions across arms — the small-sample companion to this chapter's power analysis; **Chapter 9 (simulation)** supplies the WTP distributions those comparisons act on.
- **Chapter 33 (multiplicity, equivalence, randomization inference in experiments)** is the full experimental-design treatment of §8.6–§8.8, applied to your survey experiments; **Chapter 31–32 (treatment effects, threats)** connect power to what is actually identified.
- **Chapters 47–48 (meta-regression, small-sample corrections)** carry the multiplicity and power concerns into the 18-study finfish synthesis, where "few studies" is the power-limiting count (Chapter 6, §6.9).

This chapter is the direct operationalization of your PDF's top-priority error (§6.1) and its power concern (§2.7, §6.5), and it feeds the experimental-design chapters (31–33) and the identification spine (Ch 30).

8.16 Research translation task

For the **convergent-validity comparisons in your CV paper** (PC vs SBDC, BRIS vs full payment, and any experienced-low vs experienced-high), produce a three-page pre-analysis-style note:

1. **Specify margins.** For each equivalence claim, pre-specify an equivalence margin Δ in WTP terms and justify it (benefit-transfer relevance, policy threshold). State it *before* looking at outcomes.
2. **Compute power and MDE.** For each primary contrast, given the actual per-cell sample sizes and an estimate of the WTP standard deviation, compute the MDE for the difference test and the sample needed to establish equivalence within Δ. Flag any contrast that is underpowered for *either* test.
3. **Run difference and equivalence tests.** For each contrast, report the point estimate, the 95% CI (for the difference test) and the 90% CI (for TOST), the difference p-value, and the TOST p-value.
4. **Place each in the 2×2.** Classify every comparison as *relevant effect*, *equivalent*, *inconclusive*, or *significant-but-negligible*, and write the one-sentence conclusion each cell licenses.
5. **Plan multiplicity.** List which contrasts are confirmatory (tested at full power) and which are exploratory (FDR-controlled).
6. **Close** with the spine sentence: *"For [contrast], my design can detect effects above [MDE]; I [did/did not] establish equivalence within [Δ]; the correct conclusion is [equivalent / inconclusive / relevant effect], not merely 'not significantly different.'"*

Carry this into Chapters 10 and 33, where randomization inference and the experimental-design treatment of multiplicity and equivalence complete the toolkit.

8.17 Bibliography

Core. - Cohen, *Statistical Power Analysis for the Behavioral Sciences.* The foundational reference for power, effect sizes, and sample-size determination; the source of the conventional d benchmarks and the sample-size logic of §8.3. *(Verify edition.)* - Lakens, "Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses," *Social Psychological and Personality Science* (2017), and the associated `TOSTER` package documentation. The clearest applied introduction to TOST and equivalence margins — read it directly for §8.6–§8.7. *(Verify volume/pages.)* - Wellek, *Testing Statistical Hypotheses of Equivalence and Noninferiority.* The definitive treatment of equivalence testing, for depth on §8.6. *(Verify edition.)*

Applied. - Gelman & Hill, *Data Analysis Using Regression and Multilevel/Hierarchical Models*, and Gelman's writing on the difficulty of estimating interactions ("you need 16 times the sample size"). The §8.9 interaction-power argument. *(Verify exact source/wording.)* - List, Sadoff & Wagner, "So you want to run an experiment, now what? Some simple rules of thumb for optimal design of experiments," *Experimental Economics.* Practical power and sample-allocation guidance for experiments with multiple arms — directly relevant to your §6.5 design problem. *(Verify volume/year.)* - Benjamini & Hochberg (1995), "Controlling the False Discovery Rate," *JRSS-B*, and Holm (1979) on step-down multiple testing. The multiplicity procedures of §8.8. *(Verify volumes/pages.)*

Advanced. - Schuirmann (1987), "A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability," *Journal of Pharmacokinetics and Biopharmaceutics.* The original TOST paper (from bioequivalence, where equivalence testing was developed) — the primary source for §8.6. *(Verify volume/pages.)* - Ioannidis, "Why Most Published Research Findings Are False," *PLoS Medicine.* On how low power and multiplicity produce unreliable literatures — the meta-scientific backdrop to this chapter and to your meta-analysis (Chapter 48). *(Verify year.)* - The stated-preference validity literature on convergent validity and criterion validity (e.g., work by Carson, Champ, or the NOAA-panel lineage) for how equivalence/convergence claims are argued in non-market valuation. *(I am not certain of the single best citation — verify and select those matching your CV context.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Lakens/Schuirmann/List citations before using them, per the book's no-unverified-citation rule.

End of Chapter 8. Next in Part II: Chapter 9 — Simulation-Based Inference (why some integrals have no closed form, Monte Carlo and quasi-Monte-Carlo, Halton draws, the bootstrap, the parametric bootstrap, Krinsky–Robb, and uncertainty propagation), which supplies the simulation machinery behind the WTP intervals of Chapter 7 and the mixed-logit likelihood, and the CLT-free inference this chapter's small cells demand. Tell me when to write it, or request a different chapter.

Chapter 9 — Simulation-Based Inference

Part II: Probability and Statistical Inference

> **The spine question, applied to this chapter.** Two of the quantities at the center of your work cannot be computed with a formula. The mixed-logit choice probability is an integral — a logit averaged over a distribution of coefficients — with no closed form, so your EC-MXL is estimated by *simulated* maximum likelihood with Halton draws. And the uncertainty around your WTP is propagated by *simulation* — Krinsky–Robb — and compared across arms by the *Poe test*. Simulation is therefore not an optional convenience in your research; it is how the numbers get made. But simulation adds a second source of error on top of sampling error — **simulation noise** — that depends on the number of draws, the type of draws (random vs Halton), and the seed. Your PDF is blunt about the stakes: *"per giudicare il modello devi sapere se il risultato è robusto al numero di draw, ai seed, alla specificazione delle distribuzioni e agli starting values."* The spine question here: *when a number came out of a simulation, how much of what I see is signal about the world and how much is an artifact of finite draws — and have I used enough draws, of the right kind, to trust it?*

9.0 Conceptual motivation: the numbers your models produce are simulated

Your PDF calls computational diagnostics "una competenza invisibile ma decisiva" — an invisible but decisive skill — and lists exactly the levers this chapter turns: *"Monte Carlo; bootstrap; parametric bootstrap; Krinsky–Robb; simulazione di distribuzioni di WTP; intervalli di confidenza simulati; propagazione dell'incertezza,"* and, on the estimation side, *"perché i mixed logit non hanno soluzione chiusa; simulazione Monte Carlo; quasi-Monte Carlo; Halton draws; trade-off tra numero di draw, precisione e tempo computazionale."* Simulation runs through your whole pipeline in two distinct roles, and it is worth separating them at the outset because they are often confused:

- **Simulation to *compute a model* (simulated maximum likelihood).** The mixed-logit and MDCEV likelihoods contain integrals with no analytic solution. To evaluate the likelihood at all — and hence to estimate the model — the software replaces each integral with a simulated average over draws (your DCE uses 1000 Halton draws). Here simulation is *inside* estimation: it is how the objective function is even computed, and the number and type of draws affect the point estimates and standard errors themselves.
- **Simulation to *propagate uncertainty* (Krinsky–Robb, the bootstrap).** Once the model is estimated, WTP — a nonlinear function of the coefficients — has a sampling distribution you cannot write down. Krinsky–Robb draws coefficient vectors from the estimated distribution and pushes each through the WTP formula; the bootstrap resamples the data and re-estimates. Here simulation is

after estimation: it is how uncertainty is quantified, and the Poe test extends it to comparing WTP distributions across your treatment arms.

Both roles introduce **simulation error** — variability that comes not from the finite sample of respondents (real uncertainty about the world) but from the finite number of draws (an artifact of computation). The central skill this chapter builds is telling these two apart and controlling the second: using enough draws, and the right kind of draws, that simulation noise becomes negligible relative to sampling noise. That is what "robusto al numero di draw, ai seed" means, and it is the difference between a result you can defend and one that changes when a co-author reruns it with a different seed.

We build the machinery in order: why the integrals arise and resist closed forms; Monte Carlo integration and its $1/\sqrt{R}$ error rate (and why that rate, unlike numerical quadrature, does not blow up in high dimensions); simulated maximum likelihood and its subtle Jensen bias; quasi-Monte-Carlo and Halton draws as variance reduction (why 1000 Halton draws beat many more random ones); the draws–precision–time trade-off and the robustness diagnostics your PDF demands; the bootstrap family; Krinsky–Robb and the Poe test; and finally the clean conceptual separation of simulation noise from sampling uncertainty. Every piece is tied to a place where, in your own DCE or CV work, a simulation produces a number you must judge.

9.1 Prerequisites

From **Chapter 6**: the law of large numbers (which justifies Monte Carlo), the $\sqrt{n}$-type rate, and the CLT (which gives the simulation standard error). From **Chapter 5**: the mixed logit as an *expectation* over the coefficient distribution (law of iterated expectations, §5.7), the distributions to draw from, and — from **Chapter 1** (§1.9) — the Cholesky factor that generates correlated normal draws. From **Chapter 7**: the delta method and the three-ways WTP inference (§7.8), of which Krinsky–Robb and the bootstrap are two; and from **Chapter 8**, the difference-vs-equivalence framing that the Poe test serves.

Same **notation**. New symbols: R (number of simulation/integration draws), B (number of bootstrap resamples), $\hat{g}_R$ (a simulated estimate of an integral/expectation), $\beta^{(r)}$ (the r-th draw), a superscript $*$ for a bootstrap resample.

9.2 Why integrals arise, and why they have no closed form

The mixed logit is an average of logits. Recall (Chapter 5, §5.7) that in a mixed logit each individual n has coefficients β_n drawn from a distribution $f(\beta \mid \theta)$ (with parameters θ — means, variances, Cholesky entries). Conditional on β_n, their choice probability is the ordinary logit $L_i(\beta_n)$. The *unconditional* probability — the one that enters the likelihood — integrates over the coefficient distribution:

$$P_i(\theta) = \int L_i(\beta)\, f(\beta \mid \theta)\, d\beta = \mathbb{E}_\beta[L_i(\beta)].$$

This integral has no closed form. A logistic function averaged over a normal (or lognormal, or triangular) distribution is not itself any elementary function — the two do not combine analytically. There is no formula to print; there is only the integral. For a panel of choices (your respondents each answer several tasks), the integrand is a *product* of logits over the tasks, integrated over the shared coefficient vector — still no closed form, and now the integrand is more peaked.

The dimension is the number of random coefficients. If you have K random coefficients (say a random price coefficient plus random attribute coefficients plus error components), the integral is K-dimensional. Your EC-MXL, with error components and random parameters, can easily have an integral of dimension 5–10 or more.

Why not just use numerical quadrature? For a one- or two-dimensional integral you could use deterministic quadrature (evaluate the integrand on a grid and sum with weights — Gaussian quadrature, Simpson's rule). But quadrature suffers the **curse of dimensionality**: a grid with m points per dimension needs m^K evaluations, which explodes — $m = 20$ points in $K = 8$ dimensions is $20^8 \approx 2.6 \times 10^{10}$ evaluations *per likelihood evaluation*, times thousands of likelihood evaluations during optimization. Quadrature is hopeless beyond a few dimensions. This is *the* reason simulation exists in your toolkit: Monte Carlo's error rate, as we will see, does not depend on dimension, so it beats quadrature exactly where you need it.

The same problem in welfare. Exact compensating variation in MDCEV, and expected-maximum-utility welfare in mixed logit, are also integrals over the coefficient/error distribution with no closed form (Chapters 42, 5 §5.5). So the simulation machinery you build here serves both your estimation and your welfare analysis.

> **Active recall 9.A.** State in one sentence why the mixed-logit likelihood cannot be written in closed form, and why numerical quadrature is not a viable substitute for your EC-MXL. *(Answer: the choice probability is a logit averaged over a continuous coefficient distribution, and a logistic averaged over a normal is not an elementary function; quadrature fails because the integral's dimension equals the number of random coefficients, and grid-based methods need points growing exponentially in dimension — the curse of dimensionality.)*

9.3 Monte Carlo integration

The escape is to recognize the integral as an **expectation** and estimate it by **averaging over random draws**.

The idea. Any integral against a density is an expectation:

$$I = \int g(\beta) f(\beta)\, d\beta = \mathbb{E}_f[g(\beta)].$$

Draw $\beta^{(1)}, \dots, \beta^{(R)}$ independently from f and estimate the expectation by the sample average:

$$\hat{I}_R = \frac{1}{R} \sum_{r=1}^{R} g(\beta^{(r)}).$$

Why it works (LLN) and how fast (CLT). By the law of large numbers (Chapter 6, §6.4), $\hat{I}_R \xrightarrow{p} I$ as $R \to \infty$ — the simulated average converges to the true integral. By the CLT, the **simulation error** is asymptotically normal with

$$\mathrm{Var}(\hat{I}_R) = \frac{\sigma_g^2}{R}, \qquad \text{simulation SE} = \frac{\sigma_g}{\sqrt{R}},$$

where $\sigma_g^2 = \mathrm{Var}_f[g(\beta)]$. Two facts about this rate are decisive:

1. **The rate is $O(R^{-1/2})$** — to halve the simulation error you need four times the draws (the same expensive $\sqrt{}$ law as sample size, Chapter 6). Precision is bought slowly.
2. **The rate is independent of the dimension K.** The $\sigma_g/\sqrt{R}$ formula has no K in it. This is the whole point: quadrature's cost explodes with dimension, Monte Carlo's does not. In high dimensions — exactly your EC-MXL — Monte Carlo wins decisively.

Worked Example 9.1 (Monte Carlo convergence, verified). Estimate $I = \int_0^1 e^x\, dx = e - 1 = 1.71828$ by averaging e^{u_r} over $u_r \sim \mathrm{Uniform}(0, 1)$. The average absolute error across many seeds:

R	Monte Carlo error (random)
100	0.0375
500	0.0175
2000	0.0088
10000	0.0037

Multiplying the draws by 100 (from 100 to 10,000) reduces the error by about a factor of 10 — exactly the $\sqrt{100} = 10$ predicted by the $1/\sqrt{R}$ rate (verified). This is the base convergence you get from plain random draws; §9.5 shows how Halton draws do dramatically better.

Applied to the mixed logit. The choice probability $P_i(\theta) = \mathbb{E}_\beta[L_i(\beta)]$ is estimated by $\hat{P}_i(\theta) = \frac{1}{R}\sum_r L_i(\beta^{(r)})$, where $\beta^{(r)} = \bar{\beta} + \mathbf{L}z^{(r)}$ are generated from draws $z^{(r)}$ via the Cholesky factor $\mathbf{L}$ (Chapter 1, §1.9). This is the engine inside `apollo_estimate()` for a mixed logit: every time the optimizer evaluates the likelihood, it runs this simulated average for every individual. The number of draws R is your "1000 Halton draws."

> **Stop and think 9.1.** Numerical quadrature achieves error $O(m^{-4})$ per dimension (very fast) but with cost m^K; Monte Carlo achieves error $O(R^{-1/2})$ (slow) but with cost independent of dimension. For a 1-dimensional WTP integral versus an 8-dimensional EC-MXL integral, which method would you use for each, and why? *(Answer: for the 1-D integral use quadrature — it is far more accurate for the same cost in low dimensions; for the 8-D EC-MXL use Monte Carlo (or quasi-MC) — quadrature's m^8 cost is prohibitive while Monte Carlo's $1/\sqrt{R}$ error is dimension-free. Match the method to the dimension.)*

9.4 Simulated maximum likelihood, and its hidden bias

Replacing the choice probability by its simulated estimate turns maximum likelihood into **simulated maximum likelihood (SML)**. This is what estimates your EC-MXL, and it has a subtlety that matters for choosing R.

The simulated log-likelihood. The true log-likelihood contribution is $\log P_i(\theta)$; SML uses

$$\log \hat{P}_i(\theta) = \log\Big(\frac{1}{R}\sum_{r=1}^{R} L_i(\beta^{(r)})\Big),$$

and maximizes the sum over individuals. There are now **two layers of approximation**: the finite sample of respondents (n, giving ordinary sampling uncertainty) and the finite number of draws (R, giving simulation error).

The Jensen bias (why more draws matter, not just for precision). Here is the subtlety. The simulated probability $\hat{P}_i$ is an *unbiased* estimate of P_i (it is an average of the right thing). But the log-likelihood takes the **log** of it, and the log is concave, so by **Jensen's inequality**,

$$\mathbb{E}\big[\log \hat{P}_i\big] \le \log \mathbb{E}[\hat{P}_i] = \log P_i.$$

The simulated log-likelihood is **biased downward**, and the bias is larger the noisier $\hat{P}_i$ is — i.e., the *smaller* R is. The bias is of order $O(1/R)$ (it shrinks as draws increase), which is why SML is **consistent only if $R \to \infty$ as $n \to \infty$** (formally, $\sqrt{n}/R \to 0$). Practically: too few draws bias your estimates, not just inflate their variance. This is a distinct reason — beyond precision — that R cannot be tiny, and a reason to prefer variance-reduced draws (§9.5) that make $\hat{P}_i$ less noisy for a given R.

The one practical rule that trips people up: hold the draws fixed. During optimization, the *same* set of draws $\{z^{(r)}\}$ must be used at every evaluation of the likelihood. If you regenerate random draws at

each iteration, the objective function becomes a *noisy, jittering surface* and the optimizer chases simulation noise instead of climbing the likelihood — it will fail to converge or converge to garbage. Fixing the draws (a fixed seed, or a deterministic Halton sequence) makes the simulated likelihood a smooth, deterministic function of θ that the optimizer can maximize. This is why "seed" and "draws" are estimation settings, not afterthoughts, and why Apollo asks you to set them.

> **Active recall 9.B.** Why is the simulated maximum likelihood estimate biased for finite R even though the simulated probability is unbiased, and what does that imply for choosing the number of draws? *(Answer: the log-likelihood applies the concave log to the unbiased simulated probability, so by Jensen's inequality $\mathbb{E}[\log \widehat{P}] \leq \log P$ — a downward bias of order $1/R$; therefore R must be large (and grow with n) to control bias, not merely variance, which favors variance-reduced draws that lower simulation noise for a given R.)*

9.5 Quasi-Monte Carlo and Halton draws: getting more from fewer draws

Plain random draws are wasteful: by chance they clump in some regions and leave gaps in others, so many draws are "redundant" and the error only falls like $1/\sqrt{R}$. **Quasi-Monte Carlo (QMC)** replaces random draws with *deterministic* sequences engineered to fill the space evenly, achieving much faster convergence. This is why your DCE uses **Halton draws** rather than pseudo-random ones.

Low-discrepancy sequences. "Discrepancy" measures how unevenly a set of points fills the unit cube. Random points have relatively high discrepancy (clumps and gaps); **low-discrepancy sequences** are constructed to spread points as uniformly as possible. The payoff is the convergence rate: for smooth integrands, QMC error falls roughly like $O((\log R)^K/R)$ — close to $1/R$ in low dimensions, versus $1/\sqrt{R}$ for random Monte Carlo. Near-$1/R$ instead of $1/\sqrt{R}$ means **dramatically fewer draws for the same accuracy**.

Halton sequences. The Halton sequence generates points via the *radical inverse* function: to get the i-th point in base b, write i in base b, then reflect its digits about the decimal point. Base 2 gives the sequence $\frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \frac{5}{8}, \ldots$ — each new point falls in the largest current gap. For a K-dimensional integral, use a *different prime base* for each dimension (2, 3, 5, 7, …). The result is a deterministic point set that fills space far more evenly than random draws.

Worked Example 9.2 (Halton beats random, verified). Re-estimate $\int_0^1 e^x\,dx = 1.71828$ using Halton (base 2) instead of random draws:

R	random MC error	**Halton (QMC) error**
100	0.0375	**0.0181**
500	0.0175	**0.0038**
2000	0.0088	**0.0010**
10000	0.0037	**0.0003**

Halton is about an order of magnitude more accurate at every R, and its error falls faster (roughly $1/R$: multiplying draws by 100 cuts the error ~60-fold, versus ~10-fold for random). So **1000 Halton draws deliver accuracy that would take tens of thousands of random draws** — the practical reason your EC-MXL uses Halton, saving enormous computation. *(Verified.)*

The crucial caveat your PDF flags: Halton degrades in high dimensions. Halton's even coverage deteriorates as dimension grows, because the sequences for *large* prime bases become highly correlated over their initial points, leaving structured gaps in high-dimensional projections. For an EC-MXL with many random parameters and error components — a high-dimensional integral — plain Halton can be *worse* than it looks, introducing subtle bias. The fixes: **scrambled Halton** (randomly permute the digits, breaking

the correlation while keeping low discrepancy), or **Sobol sequences** (better-behaved in higher dimensions), or simply more draws. This is precisely why your PDF lists "sensitivity to Halton draws" as a diagnostic: the number *and type* of draws can affect a high-dimensional EC-MXL, and you must check that they do not. The remedy is empirical (§9.6): re-estimate with more draws and with scrambled/Sobol sequences and confirm the estimates are stable.

> **Stop and think 9.2.** Your EC-MXL has 9 random dimensions and uses 1000 plain (unscrambled) Halton draws. A referee worries about draw quality. Before defending, what is the specific risk, and what two changes would you try to check robustness? *(Answer: in 9 dimensions the higher-base Halton sequences are correlated, so unscrambled Halton may cover the space poorly and bias the simulated likelihood; try (i) scrambled Halton (or Sobol) draws and (ii) a larger number of draws, and confirm the estimates, standard errors, and WTP are stable — if they move, the draws were inadequate.)*

9.6 The draws–precision–time trade-off, and the robustness diagnostics

More draws mean less simulation error but more computation — and for a mixed logit, likelihood evaluations dominate runtime, so R directly sets how long estimation takes. Choosing R is a trade-off, and — crucially — it is one you must *verify empirically*, not guess.

The trade-off. Simulation error falls like $1/\sqrt{R}$ (random) or ~$1/R$ (Halton); computation rises linearly in R. There is no universal "right" R: it depends on the model's dimension, the draw type, and how peaked the integrand is (panel likelihoods with many tasks per person are more peaked and need more draws). The 1000-Halton-draws convention is a reasonable default for moderate models, but it is a starting point, not a guarantee.

The robustness diagnostics your PDF demands. The only way to know R is large enough is to check that the answer does not depend on it. Run these, and report them:

1. **Vary the number of draws.** Re-estimate with $R = 250, 500, 1000, 2000$ (and more). Track the parameter estimates, standard errors, log-likelihood, and — most importantly — the **WTP estimates**. If they move materially as R grows, R is too small; if they stabilize, you have enough. Report the comparison.
2. **Vary the seed (for random or scrambled draws).** Re-estimate with several seeds. Seed-to-seed variation is *pure simulation noise* — it reflects nothing about the world. If estimates or WTP shift noticeably across seeds, the simulation is under-resolved (increase R or use better draws). A well-resolved model is seed-stable.
3. **Vary the draw type.** Compare plain Halton, scrambled Halton, and Sobol. Divergence signals draw-quality problems (§9.5), especially in high dimensions.

A verified illustration of seed sensitivity. Estimating a simple "logit averaged over a normal coefficient" integral, the standard deviation of the result *across seeds* was 0.030 at $R = 50$, 0.014 at $R = 200$, and 0.0067 at $R = 1000$ for random draws — shrinking like $1/\sqrt{R}$, exactly as theory predicts. Halton draws, being *deterministic* (no seed), showed **zero** seed-to-seed variation — a clean illustration that QMC removes seed sensitivity entirely (though it still has its own fixed approximation error, §9.5). The lesson: seed sensitivity is a symptom of too few random draws; if your WTP wobbles across seeds, add draws or switch to (scrambled) Halton/Sobol.

> **Active recall 9.C.** Your co-author reports that the WTP from your EC-MXL changes from €18 to €24 when they rerun with a different random seed. What does this tell you, and what do you do? *(Answer: a €6 shift from a seed change is* pure simulation noise* — the number of draws is too small to resolve the integral, so the estimate is unreliable. Increase the number of draws (and/or switch to scrambled Halton/Sobol), re-check that WTP stabilizes across seeds and draw

counts, and report the draw sensitivity; never report a seed-dependent WTP as if it were a stable estimate.)*

9.7 The bootstrap

Simulation also propagates uncertainty *after* estimation. The **bootstrap** does this by resampling the data — a remarkably general way to get standard errors and confidence intervals, especially for nonlinear statistics and small samples where the asymptotic normal approximation is poor (Chapter 6, §6.9; Chapter 8's small cells).

The nonparametric bootstrap. The idea rests on the **plug-in principle**: the empirical distribution of your data is your best estimate of the true distribution, so resampling from the data mimics drawing new samples from the population. The procedure:

1. From the observed data of n units, draw a resample of n units **with replacement**.
2. Recompute the statistic of interest (a coefficient, a WTP, a treatment contrast) on the resample.
3. Repeat B times to build the **bootstrap distribution** of the statistic.
4. The bootstrap **standard error** is the standard deviation of the B values; a **percentile confidence interval** is the 2.5th–97.5th percentiles.

Because it uses the data's own variability, the bootstrap makes no normality assumption about the statistic and captures asymmetry — valuable for a skewed WTP (Chapter 7, §7.8).

Clustered bootstrap (essential for your data). Your observations are *not* independent — multiple choice tasks per respondent, multiple WTP estimates per study (Chapter 6, §6.9). The fix is to **resample clusters, not observations**: draw respondents (or studies) with replacement and take all of a selected cluster's observations together. This preserves the within-cluster dependence and gives honest standard errors, the resampling analogue of clustered standard errors (Chapters 17, 19, 47). Resampling individual choice tasks would destroy the dependence structure and understate uncertainty — a real error to avoid.

The parametric bootstrap. Instead of resampling the data, the parametric bootstrap draws *new data from the fitted model*: simulate outcomes using the estimated parameters, re-estimate, and repeat. It assumes the model is correct (unlike the nonparametric bootstrap) but is useful when the data-generating process is well specified and you want to propagate model uncertainty — and it is the conceptual bridge to Krinsky–Robb (§9.8), which draws *parameters* rather than data.

When the bootstrap beats asymptotics. Use it when: the statistic is strongly nonlinear (WTP ratios); the sample is small (your diluted cells, Chapter 8); the asymptotic distribution is a poor approximation (skew, Chapter 6 §6.6); or an analytic standard error is unavailable. Its cost is computational — for a mixed logit, re-estimating the whole model B times can be prohibitive, which is why Krinsky–Robb (drawing parameters, not re-estimating) is the common practical compromise for DCE WTP.

> **Stop and think 9.3.** You have 500 respondents each doing 8 choice tasks (4000 rows) and want a bootstrap SE for WTP. Should you resample the 4000 rows or the 500 respondents, and what goes wrong with the other choice? *(Answer: resample the 500 respondents (clusters), taking all 8 of each selected respondent's tasks. Resampling the 4000 rows treats correlated within-respondent tasks as independent, destroying the dependence and understating the SE — the same error as using classical instead of clustered standard errors.)*

9.8 Krinsky–Robb and the Poe test: propagating and comparing WTP uncertainty

The workhorse for WTP inference in discrete-choice work is **Krinsky–Robb** — a parametric simulation of the *coefficient* distribution — and its companion the **Poe test** for comparing WTP across arms. You use both; here is what they are doing.

Krinsky–Robb (recap and placement). From the estimated coefficients $\hat{\beta}$ and their covariance $\hat{\mathbf{V}}$, draw $\beta^{(r)} \sim N(\hat{\beta}, \hat{\mathbf{V}})$ for $r = 1, \ldots, R$ (via the Cholesky factor, Chapter 1 §1.9), compute $\text{WTP}^{(r)}$ for each draw, and use the empirical distribution — median and percentile interval — as the WTP's sampling distribution (Chapter 1 §1.9, Chapter 7 §7.8). Krinsky–Robb is a **parametric bootstrap on the parameters rather than the data**: it assumes the coefficient estimator is (asymptotically) normal with the estimated covariance, then propagates that through the nonlinear WTP function, capturing the asymmetry the delta method misses. It is far cheaper than a full data bootstrap (no re-estimation), which is why it dominates DCE practice — but it inherits the normality assumption on the coefficients, so where that is doubtful (small samples, near-Cauchy ratios, Chapter 6 §6.9) a full bootstrap is the more honest, costlier alternative.

The Poe test: comparing two WTP distributions. Your convergent-validity and treatment comparisons ask whether WTP *differs across arms* — high vs low cost vector, PC vs SBDC, incentivized vs hypothetical. Since each arm's WTP is a *simulated distribution* (from Krinsky–Robb), you compare *distributions*, not point estimates. The **Poe test** (the "complete combinatorial" method) does this: given draws $\{\text{WTP}_A^{(i)}\}$ and $\{\text{WTP}_B^{(j)}\}$, it forms *all* pairwise differences $\text{WTP}_A^{(i)} - \text{WTP}_B^{(j)}$ and computes the proportion that are ≤ 0 (or ≥ 0), yielding a one-sided p-value for $H_0 : \text{WTP}_A = \text{WTP}_B$ against a directional alternative. It respects the full (possibly skewed) shape of each WTP distribution rather than reducing each to a mean and SE.

Worked Example 9.3 (Krinsky–Robb + Poe test for two arms, verified). Suppose Krinsky–Robb gives, for the low-cost-vector arm, WTP draws centered at €20 (SE €4), and for the high-cost-vector arm, WTP draws centered at €28 (SE €5). The Poe test forms all pairwise differences and computes the fraction with $\text{WTP}_{\text{low}} \geq \text{WTP}_{\text{high}}$:

$$\Pr(\text{WTP}_{\text{low}} \geq \text{WTP}_{\text{high}}) = 0.106 \quad \text{(verified, complete combinatorial)},$$

matching the analytic value $\Pr(A - B \geq 0)$ where $A - B \sim N(-8, \sqrt{4^2 + 5^2} = 6.4)$, i.e. $1 - \Phi(8/6.4) = 0.106$. Interpretation: the one-sided Poe p-value is 0.106 — the difference between arms is **not significant** at 5%. But note the connection to Chapter 8: this is a *difference* test, and "not significant" is not "equivalent." To claim the arms are equivalent you would run an *equivalence* test (TOST) against a pre-specified margin (Chapter 8, §8.6), and to interpret a null you would report the MDE. The Poe test tells you whether WTP distributions differ; Chapter 8's machinery tells you whether they are *the same*.

Krinsky–Robb, delta, and bootstrap — the map. Three tools, three assumptions: the **delta method** (Chapter 7) is analytic and local-linear (symmetric intervals, fast, breaks for strong nonlinearity); **Krinsky–Robb** draws parameters from the estimated normal and propagates through WTP (captures asymmetry, cheap, assumes coefficient normality); the **bootstrap** resamples data and re-estimates (fewest assumptions, captures everything, most expensive). For your WTP: prefer Krinsky–Robb + Poe over the delta method whenever WTP is skewed or the cost coefficient is weak; escalate to a (clustered) bootstrap when coefficient normality itself is doubtful.

> **Active recall 9.D.** What does the Poe test compute, and why is it the right tool for comparing WTP across two of your treatment arms rather than a simple t-test on the mean WTPs? *(Answer: it forms all pairwise differences between the two arms' simulated WTP distributions and reports the proportion on one side of zero — a distribution-level comparison that respects the full, possibly skewed shape of each WTP, rather than reducing each to a mean and SE as a t-test would. It is a difference test; equivalence still requires TOST, Chapter 8.)*

9.9 Simulation noise versus sampling uncertainty: the distinction to never lose

The single most important conceptual habit in this chapter is separating two sources of variability that look alike but mean opposite things.

- **Sampling uncertainty** comes from the finite sample of respondents (n). It is *real* uncertainty about the world — you genuinely do not know the population WTP, and more respondents would reduce it. It is what standard errors and confidence intervals are *supposed* to convey.
- **Simulation noise** comes from the finite number of draws (R or B). It is an *artifact of your computation* — it reflects nothing about the world and would vanish if you used infinitely many draws. Seed sensitivity is pure simulation noise.

The rule: **make simulation noise negligible relative to sampling uncertainty.** Use enough draws that the simulation-induced variability (across seeds, across draw counts) is small compared to the standard errors you report. When that holds, your intervals reflect genuine uncertainty about the world, not the accident of a particular seed. When it does not — when WTP moves across seeds by amounts comparable to its standard error — your reported uncertainty is contaminated by computation, and the fix is more/better draws, not a different interpretation.

This is why the robustness diagnostics of §9.6 are not optional box-ticking: they are how you *demonstrate* that simulation noise has been driven below the sampling signal, converting "the software gave me a number" into "the number is a property of my data, not my seed." Your PDF's "competenza invisibile ma decisiva" is exactly this discipline.

9.10 Failure modes

F1. Too few draws (biased SML). Estimating an EC-MXL with too few draws biases the estimates (Jensen, §9.4), not just their variance. *Fix:* increase R; use variance-reduced (Halton/Sobol) draws; verify estimates stabilize as R grows.

F2. Regenerating draws each iteration. Using fresh random draws at every likelihood evaluation makes the objective jitter and the optimizer chase noise. *Fix:* fix the draws (fixed seed or deterministic sequence) across the whole optimization (§9.4).

F3. Seed-dependent results reported as stable. Reporting a WTP that shifts materially across seeds. *Symptom:* co-authors get different numbers. *Fix:* increase draws / use QMC until seed-stable; report seed and draw-count sensitivity (§9.6, §9.9).

F4. Unscrambled Halton in high dimensions. Plain Halton degrades with many random dimensions, biasing the simulated likelihood. *Fix:* scrambled Halton or Sobol; compare across draw types; more draws (§9.5).

F5. Confusing simulation noise with sampling uncertainty. Treating draw-induced variability as if it were real uncertainty (or vice versa). *Fix:* keep the two separate; drive simulation noise well below the standard errors (§9.9).

F6. Bootstrapping the wrong unit. Resampling individual choice tasks instead of respondents (or estimates instead of studies), destroying dependence and understating SEs. *Fix:* clustered bootstrap — resample the clusters (§9.7).

F7. Over-trusting Krinsky–Robb where coefficient normality fails. Using KR when the coefficient estimator is far from normal (tiny cells, near-Cauchy ratios). *Fix:* escalate to a (clustered) bootstrap; report the median and percentile interval; address the cause (Chapter 24).

F8. "Not significant" from a Poe test read as equivalence. A non-significant Poe test is a difference test failing to reject, not evidence of equal WTP. *Fix:* run an equivalence test (TOST) against a margin; report the MDE (Chapter 8).

9.11 Diagnostic questions for your own papers

1. **Enough draws?** Did I check that my EC-MXL estimates, standard errors, and WTP are stable as the number of draws increases (e.g., 500 → 1000 → 2000)? *(§9.6, F1.)*
2. **Seed-stable?** Do my results (especially WTP) change across seeds? If so, are my draws adequate? *(§9.6, §9.9, F3.)*
3. **Draw type.** For a high-dimensional EC-MXL, did I use scrambled Halton or Sobol (not plain Halton), and check robustness across draw types? *(§9.5, F4.)*
4. **Draws fixed in estimation.** Are the simulation draws held fixed across optimization iterations? *(§9.4, F2.)*
5. **Right bootstrap unit.** For any bootstrap, am I resampling clusters (respondents/studies), not individual observations? *(§9.7, F6.)*
6. **KR appropriateness.** Is coefficient normality reasonable for Krinsky–Robb, or should I use a full bootstrap (weak cost coefficient, tiny cells)? *(§9.8, F7.)*
7. **Simulation vs sampling.** Have I confirmed simulation noise is negligible relative to the sampling standard errors I report? *(§9.9, F5.)*
8. **Poe test interpretation.** Am I reading a non-significant Poe test as a difference-test result, not as evidence of equivalence? *(§9.8, F8; Chapter 8.)*

9.12 Exercises

(A) Comprehension.

A1. Explain why the mixed-logit choice probability is an integral and why Monte Carlo, not quadrature, is used to evaluate it for a model with 7 random coefficients.

A2. Monte Carlo error falls like $1/\sqrt{R}$. If a simulated WTP has a simulation SE of €2 at $R = 1000$ draws, roughly how many draws would you need to cut the simulation SE to €0.5? Interpret the cost.

A3. Your EC-MXL gives WTP €22 with seed 1 and €19 with seed 2 (same data, same R). Is the €3 gap sampling uncertainty or simulation noise? What do you do?

(B) Derivation.

B1. Show that the Monte Carlo estimator $\hat{I}_R = \frac{1}{R}\sum_r g(\beta^{(r)})$ is unbiased for $I = \mathbb{E}[g]$ and has variance σ_g^2/R, and state why the error rate is independent of the integral's dimension.

B2. Using Jensen's inequality, show that the simulated log-likelihood contribution $\log \hat{P}_i$ is biased downward for the true $\log P_i$, and explain why the bias shrinks with more draws and with variance-reduced draws.

B3. Describe the Poe complete-combinatorial test as an estimator of $\Pr(\text{WTP}_A - \text{WTP}_B \leq 0)$, and show that for large draw counts it converges to the true probability under the simulated distributions. Relate it to the analytic normal result for the worked example.

(C) Research application / critique.

C1. A referee asks whether your EC-MXL's WTP of €31 is "an artifact of the 1000 Halton draws." Write a response describing the exact robustness analysis you would run and what results would reassure (or worry) you.

C2. Your co-author computed a bootstrap SE for a DCE WTP by resampling the 6,400 choice observations (800 respondents × 8 tasks). Critique this and give the correct procedure.

C3. You compare WTP in the incentivized and hypothetical arms with a Poe test and get a one-sided p of 0.09, and conclude "hypothetical and incentivized WTP are equivalent." Critique using §9.8 and Chapter 8, and state what you should do instead.

9.13 Solutions

A1. The choice probability is a logit averaged over the distribution of random coefficients: $P_i = \int L_i(\beta) f(\beta) d\beta$, which has no closed form (a logistic averaged over a normal is not elementary). With 7 random coefficients the integral is 7-dimensional; quadrature would need m^7 points per evaluation (curse of dimensionality), which is infeasible, whereas Monte Carlo's error rate $1/\sqrt{R}$ is independent of dimension — so Monte Carlo (or quasi-MC) is used.

A2. Simulation SE $\propto 1/\sqrt{R}$, so to cut it by a factor of 4 (€2 → €0.5) you need $4^2 = 16$ times the draws: $R \approx 16{,}000$. Cost: sixteen times the computation for a fourfold precision gain — the slow $\sqrt{}$ law. (Halton draws would reach the target with far fewer, since their error falls closer to $1/R$.)

A3. The €3 gap across seeds (same data, same R) is **simulation noise**, not sampling uncertainty — it reflects the finite draw count, not the world. Increase the number of draws (and/or switch to scrambled Halton/Sobol), confirm the WTP stabilizes and becomes seed-insensitive, and report the draw/seed sensitivity. Do not report a seed-dependent WTP.

B1. $\mathbb{E}[\hat{I}_R] = \frac{1}{R}\sum_r \mathbb{E}[g(\beta^{(r)})] = \mathbb{E}[g] = I$ (unbiased). By independence, $\mathrm{Var}(\hat{I}_R) = \frac{1}{R^2}\sum_r \mathrm{Var}(g) = \sigma_g^2/R$. The variance formula contains only σ_g^2 and R, with no reference to the dimension of β; hence the $O(R^{-1/2})$ error rate is dimension-independent — the reason MC beats quadrature in high dimensions.

B2. $\hat{P}_i$ is unbiased for P_i. The log is concave, so Jensen gives $\mathbb{E}[\log \hat{P}_i] \le \log \mathbb{E}[\hat{P}_i] = \log P_i$ — a downward bias. The gap in Jensen's inequality grows with the variance of $\hat{P}_i$; since $\mathrm{Var}(\hat{P}_i) = \sigma^2/R$ (or smaller with variance-reduced draws), the bias is $O(1/R)$ and shrinks as R grows or as draws are made more even (Halton/Sobol lower $\mathrm{Var}(\hat{P}_i)$ for given R, hence lower bias).

B3. The Poe test computes $\widehat{\Pr}(\mathrm{WTP}_A - \mathrm{WTP}_B \le 0) = \frac{1}{R_A R_B}\sum_{i,j} \mathbf{1}\{\mathrm{WTP}_A^{(i)} - \mathrm{WTP}_B^{(j)} \le 0\}$, the empirical fraction of all pairwise differences ≤ 0. This is a Monte Carlo estimate of $\Pr(\mathrm{WTP}_A - \mathrm{WTP}_B \le 0)$ under the simulated distributions; by the LLN it converges to that probability as $R_A, R_B \to \infty$. In the worked example the simulated distributions are $N(20, 4^2)$ and $N(28, 5^2)$, so $\mathrm{WTP}_A - \mathrm{WTP}_B \sim N(-8, 6.4^2)$ and $\Pr(\ge 0) = 1 - \Phi(8/6.4) = 0.106$ — matching the complete-combinatorial value 0.106.

C1. I would re-estimate the model at several draw counts (e.g., 250, 500, 1000, 2000, 4000) and report the WTP (and key parameters and SEs) at each; if €31 is stable as draws increase, it is not a draw artifact. I would also re-estimate across several seeds (for scrambled draws) to confirm seed-insensitivity, and compare plain Halton, scrambled Halton, and Sobol draws (important given the model's dimension). Reassurance: WTP stable across draw counts, seeds, and draw types. Worry: WTP drifting as draws increase (too few draws / Jensen bias) or varying across seeds/types (inadequate or poor-quality draws) — in which case I would increase draws and use scrambled Halton/Sobol before trusting the €31.

C2. Resampling the 6,400 individual choice observations treats the 8 correlated tasks within each respondent as independent, destroying the within-respondent dependence and understating the WTP standard error. The correct procedure is a **clustered (block) bootstrap**: resample the 800 *respondents* with replacement, taking all 8 of each selected respondent's tasks together, re-estimate the model on each resample, compute WTP, and use the distribution across resamples. (For a mixed logit this is costly, so Krinsky–Robb is the common alternative — but if bootstrapping, cluster on respondents.)

C3. A one-sided Poe p of 0.09 is a *difference* test failing to reject equality at 5% — it is **not** evidence that hypothetical and incentivized WTP are equivalent (absence of evidence ≠ evidence of absence; Chapter 8,

§8.5). To claim equivalence I would pre-specify an equivalence margin Δ (the largest WTP gap that would be irrelevant, e.g. for hypothetical-bias correction), run an equivalence test (TOST) on the two arms' WTP difference — equivalently, check whether the 90% interval of the KR difference distribution lies within $\pm\Delta$ — and report the MDE so the null is interpretable. Only if the equivalence test passes can I claim the arms give equivalent WTP.

9.14 Spaced review prompts

+1 day. From memory, explain why the mixed-logit likelihood is a no-closed-form integral, write the Monte Carlo estimator and its $\sigma_g/\sqrt{R}$ error, and state why the rate is dimension-independent. Then explain the Jensen bias of simulated maximum likelihood.

+1 week. Contrast random Monte Carlo and Halton draws (convergence rate, seed dependence, high-dimensional caveat). Then describe the three robustness diagnostics for the number and type of draws, and what each rules out.

+1 month. For your own EC-MXL, run (even a small version of) the draw-count and seed robustness checks, and a Krinsky–Robb WTP interval plus a Poe test between two arms. Write a paragraph separating simulation noise from sampling uncertainty in your results, and close with the spine sentence.

9.15 How this chapter connects to the rest of the book

- **Chapter 6** supplied the LLN (why Monte Carlo converges), the $1/\sqrt{R}$ rate, and the CLT (the simulation SE); **Chapter 5** (§5.7) supplied the mixed logit as an expectation and the Cholesky draw mechanism (Chapter 1, §1.9) this chapter simulates from.
- **Chapter 7** placed Krinsky–Robb, the bootstrap, and the delta method as three ways to get a WTP interval; this chapter builds the simulation two of them rest on. **Chapter 8** supplied the equivalence framing that the Poe test's "not significant" must be read against.
- **Chapter 10 (nonparametric/randomization inference)** develops the Poe test's cousins (permutation, randomization) and Turnbull as CLT-free methods; **Chapter 23 (simulated maximum likelihood)** is the full estimation treatment of §9.4–§9.6 for the mixed logit, and **Chapter 24 (WTP space)** uses the KR/simulation machinery to characterize WTP distributions.
- **Chapters 40–42 (MDCEV, welfare)** apply simulation to the no-closed-form welfare (exact CV) integrals; **Chapter 52 (computational diagnostics)** is the full treatment of the draw/seed/convergence robustness checks of §9.6, and **Chapters 47–48 (meta-analysis)** use the clustered bootstrap of §9.7 for dependent estimates.

This chapter is the computational backbone of the discrete-choice and welfare parts, and it operationalizes the "invisible but decisive" diagnostics your PDF places at the center of trustworthy results.

9.16 Research translation task

For your **EC-MXL and its WTP**, produce a two-to-three-page note:

1. **Draw-count robustness.** Re-estimate at several draw counts (e.g., 250/500/1000/2000) and tabulate the key parameters, standard errors, log-likelihood, and WTP; state at what point they stabilize and whether 1000 is enough.
2. **Seed and draw-type robustness.** Re-estimate across a few seeds (scrambled draws) and compare plain Halton, scrambled Halton, and Sobol; report whether WTP is seed- and type-stable, given your model's dimension.

3. **WTP uncertainty.** Compute a Krinsky–Robb median and percentile interval for a key WTP, and — if feasible — a clustered (respondent-level) bootstrap interval; compare them and note whether coefficient normality (KR's assumption) looks reasonable.
4. **Arm comparison.** Run a Poe test between two arms (e.g., high vs low cost vector) and report the one-sided p-value; then note that an *equivalence* claim requires the TOST/margin machinery of Chapter 8.
5. **Separate the two noises.** State explicitly how large your simulation noise is (seed-to-seed WTP variation) relative to your reported sampling standard errors, and confirm the former is negligible.
6. **Close** with the spine sentence: *"My EC-MXL WTP of [value] is stable across [draw counts/seeds/draw types], so simulation noise is negligible relative to its [SE]; the WTP interval comes from [KR/bootstrap]; the arm difference is [Poe result], and equivalence requires [TOST]."*

Carry this into Chapter 10 (randomization/Turnbull), Chapter 23 (full SML), and Chapter 52 (computational diagnostics).

9.17 Bibliography

Core. - Train, *Discrete Choice Methods with Simulation*, chapters on simulation, simulated maximum likelihood, Halton draws, and the accuracy–draws trade-off. The single most relevant source for §9.2–§9.6 and the EC-MXL context; read it alongside this chapter. Freely available from the author — *verify link.* - Robert & Casella, *Monte Carlo Statistical Methods.* The rigorous reference for Monte Carlo integration, variance reduction, and simulation error (§9.3–§9.5). *(Verify edition.)* - Efron & Tibshirani, *An Introduction to the Bootstrap.* The foundational bootstrap reference — nonparametric, parametric, and percentile intervals (§9.7). *(Verify edition.)*

Applied. - Bhat (2001, 2003), on quasi-random (Halton) and scrambled/randomized draws for mixed logit, *Transportation Research Part B.* The source for §9.5's Halton discussion and the high-dimensional degradation caveat. *(Verify volumes/years.)* - Krinsky & Robb (1986), *Review of Economics and Statistics*, and Poe, Giraud & Loomis (2005), "Computational Methods for Measuring the Difference of Empirical Distributions," *American Journal of Agricultural Economics.* The primary sources for §9.8 (Krinsky–Robb and the Poe test) — exactly your WTP-inference tools. *(Verify volumes/pages.)* - Hensher, Rose & Greene, *Applied Choice Analysis*, and the Apollo manual (Hess & Palma), for practical draw settings, seeds, and simulation diagnostics in DCE estimation. *(Verify editions/versions.)*

Advanced. - Cameron & Trivedi, *Microeconometrics: Methods and Applications*, on simulation-based estimation (SML, MSM) and the bootstrap in econometrics. *(Verify edition.)* - Gourieroux & Monfort, *Simulation-Based Econometric Methods.* The theoretical treatment of simulated maximum likelihood, its bias, and consistency conditions (§9.4). *(Verify edition.)* - Owen, *Monte Carlo theory, methods and examples* (freely available online). Deep, modern treatment of Monte Carlo and quasi-Monte-Carlo, including scrambled nets and the high-dimensional behavior relevant to §9.5. *(Verify link.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Bhat / Krinsky–Robb / Poe citations before using them, per the book's no-unverified-citation rule.

End of Chapter 9. Next in Part II: Chapter 10 — Nonparametric and Randomization Inference (Mann–Whitney, Kolmogorov–Smirnov, permutation tests, randomization inference, the Poe test in its distributional context, and the Turnbull estimator), which supplies the distribution-free and small-sample inference your diluted cells and skewed WTP distributions call for, and completes the inference toolkit of Part II. Tell me when to write it, or request a different chapter.

Chapter 10 — Nonparametric and Randomization Inference

Part II: Probability and Statistical Inference

> **The spine question, applied to this chapter.** Everything so far leaned on assumptions — normality of the estimator, a parametric distribution for WTP, the CLT holding in your sample. This chapter asks: what can you conclude when you are unwilling, or unable, to make those assumptions? Your PDF is explicit that your objects of interest are often *distributions* of WTP rather than single coefficients, and that you use Turnbull as a distribution-free robustness check, the Poe test to compare WTP across arms, and permutation-style logic for your survey experiments. Nonparametric and randomization methods buy robustness by trading strong distributional assumptions for weaker ones — exchangeability, or simply the randomization you actually performed. The spine question here: *when my conclusion depends on a distributional assumption (normal WTP, asymptotic normality in a small cell), does it survive when I drop that assumption and let the data — or the design — speak for themselves?*

10.0 Conceptual motivation: inference when you distrust the assumptions

Parametric inference is powerful *when its assumptions hold* and misleading when they do not. Three situations in your work make the assumptions doubtful, and your PDF names the tools for each:

- **Small treatment cells (Chapters 6, 8).** When a CV cell has 40–50 respondents and the outcome is skewed, the CLT approximation behind t-tests and normal confidence intervals is imperfect (Chapter 6, §6.6). A **permutation test** or **randomization inference** gives a valid p-value *without* the normal approximation — the inference is exact (or design-based), not asymptotic.
- **Your object is a whole distribution of WTP, not a single number.** Your PDF says it directly: *"i tuoi oggetti di interesse sono spesso distribuzioni di WTP, non solo coefficienti singoli."* Comparing two WTP *distributions* across arms (high vs low cost vector, PC vs SBDC) is what the **Poe test** and the **Kolmogorov–Smirnov** test do, and rank comparisons come from **Mann–Whitney**.
- **You do not want to assume a WTP distribution at all.** In SBDC contingent valuation, the parametric approach assumes WTP is normal or logistic (Chapter 15) — and your PDF flags exactly this worry: *"la normalità della WTP è plausibile? … come cambia la media se la distribuzione ha code pesanti?"* The **Turnbull estimator** recovers the WTP distribution *nonparametrically* from the yes/no data, giving a distribution-free **lower-bound mean** that serves as a robustness benchmark against the parametric mean. Your PDF says you use it: *"usi anche Turnbull come robustezza non parametrica."*

There is also a deeper, design-based motivation. In a randomized survey experiment, the *one* source of randomness you are certain about is the treatment assignment *you* performed. **Randomization inference**

builds its p-value entirely from that known randomization — it does not need the CLT, a distributional assumption, or even a super-population. This makes it, in a real sense, the most honest inference available for an experiment: the p-value follows from what you did. Your PDF's §6.6 concern — that post-randomization exclusions can break the randomization — is precisely a concern about preserving the foundation randomization inference rests on.

This chapter builds the toolkit: the spectrum from parametric to distribution-free; rank-based tests (Mann–Whitney, Kolmogorov–Smirnov); permutation tests and their design-based cousin, randomization inference; the Poe test placed among distributional comparisons; and — the centerpiece for your CV work — the Turnbull estimator as a distribution-free WTP benchmark. Throughout, the message is the one your PDF keeps returning to: report the parametric result *and* the nonparametric check, and trust the conclusion more when they agree.

10.1 Prerequisites

From **Chapter 6**: where asymptotics fail (small samples, skew) — the motivation for distribution-free methods. From **Chapter 7**: p-values, test statistics, and the correct interpretation of a test. From **Chapter 8**: small-cell power, and the difference-vs-equivalence distinction (nonparametric tests are still *difference* tests). From **Chapter 9**: the Poe test and simulation (the Poe test acts on Krinsky–Robb-simulated WTP distributions). From **Chapter 5**: the CDF and survival function (central to KS and Turnbull).

Same **notation**. New symbols: F_A, F_B (empirical CDFs of two samples), $S(\cdot)$ (survival function), U (Mann–Whitney statistic), D (Kolmogorov–Smirnov statistic), π_j (proportion of "no" at bid j), and b_j (bid levels).

10.2 The parametric–nonparametric spectrum, and the assumption trade-off

Definition 10.1 (Parametric vs nonparametric). A **parametric** method assumes the data come from a distribution indexed by finitely many parameters (WTP $\sim N(\mu, \sigma^2)$; the estimator $\sim$ normal). A **nonparametric** (or **distribution-free**) method makes no such assumption about the shape of the distribution; it uses only weaker structure — ranks, exchangeability, or the empirical distribution itself.

The trade-off (state it honestly). Fewer assumptions buy **robustness** but usually cost **power** *when the parametric assumption is actually true*:

- If WTP really is normal, the t-test is the most powerful test of a mean difference, and a rank test is slightly less powerful (it "wastes" some information by using ranks).
- If WTP is *not* normal — skewed, heavy-tailed, contaminated by outliers — the t-test can be biased or badly sized, while the rank/permutation test remains valid. In that case the nonparametric test is not just safer but often *more* powerful.

So the choice is not "nonparametric is better" but "nonparametric is a robustness insurance policy whose premium (lost power under normality) is small and whose payout (validity under non-normality) is large." The mature practice — and the one your PDF implies — is to **report both** and check they agree.

A subtlety about what nonparametric tests test. A nonparametric test still tests a hypothesis — usually a *distributional* one. Mann–Whitney tests (loosely) whether one distribution is stochastically larger; KS tests whether two distributions are identical *in any respect*; a permutation test on the mean tests equality of means under exchangeability. Do not read "nonparametric" as "assumption-free" — read it as "free of the *distributional-shape* assumption," while still assuming something (exchangeability, or the randomization). Naming exactly what each test tests is part of using it correctly.

Active recall 10.A. Your co-author says "let's use a nonparametric test so we don't have to make any assumptions." Correct them in one sentence. *(Answer: nonparametric tests drop the* distributional-shape* assumption (e.g., normality) but still assume *something* — exchangeability under the null, or the known randomization — and they still test a specific hypothesis (often distributional); they trade the normality assumption for robustness, not for assumption-freeness.)*

10.3 Mann–Whitney U (Wilcoxon rank-sum): comparing two samples by ranks

The most common distribution-free replacement for a two-sample t-test.

The idea. To compare samples A and B without assuming normality, pool all observations, rank them, and ask whether A's values tend to rank higher or lower than B's. Equivalently, count how often an A-value exceeds a B-value.

Definition 10.2 (Mann–Whitney U statistic). For samples $A = (a_1, \dots, a_{n_A})$ and $B = (b_1, \dots, b_{n_B})$,

$$U_A = \#\{(i,j) : a_i > b_j\} + \tfrac{1}{2}\#\{(i,j) : a_i = b_j\},$$

the number of pairs in which the A-value wins (ties count half). Its normalization estimates a meaningful quantity:

$$\frac{U_A}{n_A n_B} \;\approx\; \Pr(A > B),$$

the probability that a randomly chosen A exceeds a randomly chosen B — a natural, interpretable measure of stochastic dominance. Under the null that A and B have the same distribution, $\mathbb{E}[U_A] = n_A n_B/2$ and $\mathrm{Var}(U_A) = n_A n_B(n_A + n_B + 1)/12$, so for moderate samples $z = (U_A - \mathbb{E}[U_A])/\sqrt{\mathrm{Var}(U_A)}$ is approximately standard normal; for small samples the exact distribution is used.

What it tests, precisely. Mann–Whitney tests $H_0 : \Pr(A > B) = 1/2$ (stochastic equality). Under the *additional* assumption that the two distributions differ only by a location shift, it is a test of equal medians. Without that assumption it is a test of stochastic dominance — a difference in *any* feature that makes one sample tend to exceed the other. State which interpretation you intend.

Worked Example 10.1 (Mann–Whitney on a small WTP comparison, verified). Two small arms of WTP (€): $A = (10, 15, 22, 30, 8)$, $B = (18, 25, 33, 40, 20)$, each $n = 5$. Counting pairs where $a_i > b_j$ gives $U_A = 5$ out of 25 pairs, so the estimated $\Pr(A > B) = 5/25 = 0.20$ — the A arm tends to have *lower* WTP. The two-sided p-value is 0.15 (verified) — **not significant** at 5%. With only five per arm you have almost no power (Chapter 8), so this non-significance is uninformative, not evidence of equality. The Mann–Whitney gives a valid p-value without assuming WTP is normal, and its statistic ($\Pr(A > B) = 0.20$) is directly interpretable.

Active recall 10.B. What quantity does $U_A/(n_A n_B)$ estimate, and why is it a more robust comparison than the difference in means when WTP is skewed? *(Answer: it estimates* $\Pr(A > B)$*, the probability a random draw from A exceeds one from B — a rank-based measure unaffected by extreme values or the skew of the WTP distribution, whereas the difference in means is sensitive to outliers and heavy tails.)*

10.4 Kolmogorov–Smirnov: comparing whole distributions

Mann–Whitney is sensitive mainly to *location/dominance* differences. Sometimes you want to know whether two WTP distributions differ in *any* way — location, spread, or shape. That is the Kolmogorov–Smirnov test.

Definition 10.3 (Two-sample Kolmogorov–Smirnov statistic). Let F_A and F_B be the empirical CDFs of the two samples. The KS statistic is the largest vertical gap between them:

$$D = \sup_x \left|F_A(x) - F_B(x)\right|.$$

Under the null that the two samples come from the same distribution, D has a known (distribution-free) limiting distribution, and the 5% critical value is approximately

$$D_{0.05} \approx 1.36\sqrt{\frac{n_A + n_B}{n_A n_B}}.$$

Reject equality if D exceeds this. Because D looks at the *entire* CDF, KS detects differences in scale and shape that a location-focused test would miss — useful when a treatment might change the *spread* of WTP (e.g., more preference uncertainty in a hypothetical arm) rather than its center.

Worked Example 10.2 (KS on the same arms, verified). For A and B above, $D = 0.60$ with p-value 0.36 (verified), and the 5% critical value is $1.36\sqrt{10/25} = 0.86$. Since $D = 0.60 < 0.86$, do not reject — again, unsurprising with $n = 5$ per arm. KS complements Mann–Whitney: together they ask "do these WTP distributions differ in location (MW) or in any respect (KS)?" — a richer picture than a single mean comparison, and the natural distribution-level counterpart to the Poe test (§10.7).

> **Stop and think 10.1.** A hypothetical and an incentivized arm have nearly identical *mean* WTP, but you suspect the hypothetical arm has more dispersed responses (more preference uncertainty). Which test — Mann–Whitney or KS — is more likely to detect this, and why? *(Answer: KS, because it compares the entire CDFs and is sensitive to differences in spread/shape even when the medians coincide; Mann–Whitney focuses on stochastic dominance/location and could easily miss a pure scale difference. This matters for detecting whether hypothetical elicitation changes the* shape* of WTP, not just its center.)*

10.5 Permutation tests: exact inference from exchangeability

Rank tests are one family of distribution-free tests; **permutation tests** are the more general and more flexible idea, and they work for *any* test statistic you like — a mean difference, a median difference, a WTP difference, a difference in dispersion.

The core idea. Under the null hypothesis that the two groups have the *same* distribution, the group labels ("treatment" / "control") are **exchangeable** — arbitrary. So if the null is true, the labeling you observed is just one of the many equally likely labelings. To build the null distribution of any test statistic T, simply **re-shuffle the labels**, recompute T for each shuffle, and see where your observed T falls.

The algorithm. 1. Compute the observed statistic T_{obs} (e.g., $\bar{y}_B - \bar{y}_A$). 2. Pool all observations. For each of the $\binom{n_A+n_B}{n_A}$ ways to relabel (or a large random sample of them), recompute T. 3. The **permutation p-value** is the fraction of relabelings giving a statistic at least as extreme as T_{obs}:

$$p = \frac{\#\{T_{\text{perm}} : |T_{\text{perm}}| \geq |T_{\text{obs}}|\}}{\text{number of permutations}}.$$

When all permutations are enumerated, the p-value is **exact** (no asymptotics, no distributional assumption); when there are too many, a large random sample of permutations gives a Monte Carlo approximation to any desired accuracy. The only assumption is **exchangeability under the null** — far weaker than normality.

Worked Example 10.3 (Exact permutation test for a WTP difference, verified). For $A = (10, 15, 22, 30, 8)$ and $B = (18, 25, 33, 40, 20)$, the observed mean difference is $\bar{y}_B - \bar{y}_A = 27.2 - 17.0 = €10.2$. There are $\binom{10}{5} = 252$ ways to split the ten values into two groups of five. Recomputing the mean difference for all 252 and counting those with $|\text{diff}| \geq 10.2$ gives a two-sided p-value of 0.127 (verified) — the **exact**

p-value under exchangeability, requiring no normality. For comparison, the t-test gives $p = 0.115$: the two are close here (the data are not badly non-normal), which is itself reassuring — when the parametric and permutation p-values agree, the normal approximation was adequate; when they diverge, trust the permutation test. This is exactly the kind of robustness check your PDF calls for in small cells.

Why permutation tests are ideal for your setting. They handle any statistic (so you can permutation-test a *median* WTP difference, or a difference in WTP *dispersion*, not just a mean), they are exact in small samples where the CLT is shaky, and they make no distributional assumption. For your many-armed CV design with modest per-cell sizes, a permutation test is often the most defensible way to test a treatment contrast.

> **Active recall 10.C.** State the single assumption a permutation test needs, and why it makes the p-value "exact" rather than "asymptotic." *(Answer: exchangeability of the observations under the null (the labels are arbitrary if the groups have the same distribution). Because the null distribution is built by literally enumerating all relabelings of the observed data, the p-value is exact — computed from a finite, known reference set — not an asymptotic approximation.)*

10.6 Randomization inference: the design is the source of the p-value

Permutation tests assume exchangeability as a modeling stance. **Randomization inference** turns this into something stronger and, for an experiment, more principled: it derives the p-value from the *actual randomization you performed*, treating that — not a super-population or a distribution — as the sole source of randomness. This is the Fisherian, design-based foundation for analyzing your survey experiments.

The setup (potential outcomes). Each respondent i has two potential outcomes: $Y_i(1)$ if assigned treatment, $Y_i(0)$ if control. You observe only one, depending on the assignment Z_i you randomly chose. The **sharp null hypothesis** of no effect states

$$H_0^{\text{sharp}}: \quad Y_i(1) = Y_i(0) \quad \text{for every unit } i,$$

i.e. the treatment changes *no one's* outcome. Under this sharp null, each respondent's outcome is *fixed* regardless of assignment — so the *only* thing that varies across hypothetical repetitions of the experiment is the assignment Z, whose distribution you *know exactly* because you chose the randomization.

The procedure. Build the null distribution of the test statistic by **re-randomizing the assignment exactly as it was actually performed** (the same number to each arm, the same blocking if any), recomputing the statistic under each re-randomization while holding the observed outcomes fixed (valid under the sharp null). The p-value is the fraction of re-randomizations giving a statistic at least as extreme as observed. For a simple two-arm completely randomized design, the *mechanics* coincide with the permutation test of §10.5 — but the *justification* is different and stronger: it comes from the physical act of randomization, not from an assumed super-population or distribution.

Why this is the gold standard for survey experiments. Randomization inference:

- makes **no distributional assumption** and **no asymptotic approximation** — valid in the smallest samples;
- rests only on **the randomization you actually did**, which you control and can document — the most credible possible basis for a causal p-value;
- generalizes to complex designs (blocking, clustering, stratification) by re-randomizing *within the design's structure*, keeping inference faithful to how treatment was assigned.

The sharp vs weak null (a real distinction). Randomization inference in its classic form tests the *sharp* null (no effect for *anyone*), which is stronger than the *weak* null of no *average* effect (ATE = 0). Rejecting the sharp null tells you *some* unit was affected, not necessarily the average. For most survey-experiment purposes this is fine, and extensions exist for the weak null; just be precise about which null you rejected.

The connection to your PDF's §6.6 warning. Randomization inference is valid *because* the assignment mechanism is known and intact. **Post-randomization exclusions** — dropping protesters, speeders, or comprehension failures *after* assignment — can break this: if exclusion is related to treatment, the remaining sample is no longer a clean randomization, and the design-based p-value is no longer justified (Chapter 32). This is the deep reason your PDF insists on handling exclusions carefully: they threaten the very foundation that makes randomization inference (and the experiment's causal claim) credible. Report results with and without exclusions, and show exclusion rates by arm.

> **Stop and think 10.2.** You run randomization inference on a survey experiment and then, to "clean" the data, drop the 8% of respondents flagged as speeders — but the speeder rate is 12% in the treatment arm and 4% in control. Why does this threaten your randomization-inference p-value? *(Answer: the differential speeder rate means exclusion is related to treatment, so the retained sample is no longer the clean randomization you performed — the assignment mechanism that justifies the design-based p-value has been disturbed. The remaining groups may differ for reasons other than treatment, and the sharp-null re-randomization no longer reflects how the analyzed sample was formed. Report with and without exclusions and analyze the differential attrition, Chapter 32.)*

10.7 The Poe test in its distributional context

Chapter 9 introduced the Poe test as the way to compare two Krinsky–Robb-simulated WTP distributions. Now we can place it precisely among the methods of this chapter.

What the Poe test is. Given draws $\{\text{WTP}_A^{(i)}\}$ and $\{\text{WTP}_B^{(j)}\}$ from two arms' simulated WTP distributions, the Poe "complete combinatorial" test forms *all* pairwise differences and computes the proportion ≤ 0 (or ≥ 0), yielding a one-sided p-value for a difference in WTP. It is, in structure, a **permutation-style comparison of two empirical distributions** — closely related to Mann–Whitney (which counts pairwise wins) applied to the simulated WTP draws. Where Mann–Whitney and KS act on *raw data*, the Poe test acts on the *simulated coefficient-based WTP distributions* (from Krinsky–Robb), which is why it is the natural tool when WTP is a derived, simulated quantity rather than a directly observed one.

When to use which for comparing WTP across arms. - **Poe test** — when each arm's WTP is a *simulated distribution* from an estimated choice/CV model (your usual DCE case): it respects the full simulated shape and is standard in the stated-preference literature. - **Mann–Whitney / KS** — when you have *individual-level* WTP or a directly observed outcome per respondent: MW for location/dominance, KS for any distributional difference. - **Permutation test** — when you want an *exact* test of a *specific* statistic (mean, median, dispersion) on raw data in a small cell.

The Chapter 8 caveat, restated. All of these are *difference* tests. A non-significant Poe/MW/KS/permutation result is *not* evidence that the arms are equivalent — that requires an equivalence test (TOST) against a pre-specified margin (Chapter 8, §8.6). The nonparametric machinery tells you whether distributions *differ*; establishing that they are *the same* needs the equivalence framework. Keep the two questions distinct in every convergent-validity comparison.

> **Active recall 10.D.** Your DCE gives Krinsky–Robb WTP distributions for two arms and you want to test whether they differ. Which test, and what must you add if you want to claim they are *equivalent*? *(Answer: the Poe test (a permutation-style comparison of the two simulated WTP distributions) for the difference; to claim equivalence, add a pre-specified margin and run an equivalence test (TOST), because a non-significant Poe test is not evidence of equality — Chapter 8.)*

10.8 The Turnbull estimator: distribution-free WTP from contingent valuation

This is the centerpiece for your CV work. The parametric approach to SBDC assumes WTP follows a specific distribution (normal, logistic — Chapter 15) and estimates its parameters. The **Turnbull estimator** instead recovers the WTP distribution *nonparametrically*, letting the yes/no data determine its shape, and yields a conservative **lower-bound mean** that is the standard distribution-free robustness check against the parametric mean.

The data structure. In single-bounded dichotomous choice, respondent i is offered a bid b and says "yes" (willing to pay b) or "no." A "yes" means WTP $\geq b$; a "no" means WTP $< b$. So each response is **interval-censored**: it locates WTP on one side of the bid, nothing more. Across respondents offered different bids $b_1 < b_2 < \cdots < b_M$, you learn the *survival function* $S(b) = \Pr(\text{WTP} \geq b)$ at each bid — the proportion saying "yes."

The Turnbull idea (nonparametric MLE of the survival function). The Turnbull estimator is the nonparametric maximum-likelihood estimate of the distribution of interval-censored data. For SBDC it reduces to something intuitive: at each bid b_j, the proportion of "yes" responses estimates $S(b_j)$; the probability mass falling in the interval $[b_j, b_{j+1})$ is the *drop* in survival, $S(b_j) - S(b_{j+1})$. Stringing these together gives a step-function CDF that makes *no assumption about shape* — the data alone determine where the mass sits.

Monotonicity and pooling (PAVA). A valid survival function must be **non-increasing** in the bid. Sampling noise can produce a *higher* "yes" proportion at a higher bid — a monotonicity violation. The **Pool Adjacent Violators Algorithm (PAVA)** fixes this: wherever S increases, pool the offending adjacent bid cells and replace their survival with the (sample-size-weighted) average, repeating until S is non-increasing. This is the nonparametric MLE's way of enforcing the logical constraint that fewer people accept higher bids. Pooling merges intervals — the price of letting the data speak is that closely spaced or noisy bids may be combined.

The lower-bound mean WTP. With the (monotonized) survival function, mean WTP is the area under S. Because the data do not tell you *where within each interval* the mass sits, the **Turnbull lower bound** conservatively assigns each interval's mass to its *lower* endpoint:

$$\widehat{E}_{\text{LB}}[\text{WTP}] = \sum_j b_j \left[S(b_j) - S(b_{j+1})\right],$$

valuing the mass in $[b_j, b_{j+1})$ at b_j. This is a *lower bound* because every unit's WTP is at least the lower endpoint of its interval. It is conservative by construction — and that is its virtue as a robustness check: if even the conservative, assumption-free lower bound is substantial, your welfare estimate is not an artifact of a parametric tail.

Worked Example 10.4 (Turnbull lower-bound mean, verified). Suppose an SBDC study offers five bids with these "yes" (survival) proportions:

Bid b_j (€)	5	10	20	40	80
$S(b_j) = \Pr(\text{yes})$	0.90	0.76	0.60	0.36	0.12

The proportions are already monotone decreasing (no pooling needed), and $S(0) = 1$ by definition. The interval masses and their lower-endpoint valuations:

Interval	Mass $= S(b_j) - S(b_{j+1})$	Lower endpoint	Contribution
$[0, 5)$	$1.00 - 0.90 = 0.10$	0	0
$[5, 10)$	$0.90 - 0.76 = 0.14$	5	0.70

Interval	Mass $= S(b_j) - S(b_{j+1})$	Lower endpoint	Contribution
$[10, 20)$	$0.76 - 0.60 = 0.16$	10	1.60
$[20, 40)$	$0.60 - 0.36 = 0.24$	20	4.80
$[40, 80)$	$0.36 - 0.12 = 0.24$	40	9.60
$[80, \infty)$	0.12	80	9.60

Summing the contributions gives $\widehat{E}_{\text{LB}}[\text{WTP}] = €26.30$ (verified; masses sum to 1). This is the distribution-free lower-bound mean WTP. Its uses as a robustness check:

- **Benchmark the parametric mean.** If your fitted normal/logistic SBDC model reports a mean WTP of, say, €45, the gap to the €26.30 Turnbull lower bound is a warning: the parametric mean is being inflated by the *upper tail* the distribution extrapolates beyond the highest bid (€80). Your PDF's exact worry — *"come cambia la media se la distribuzione ha code pesanti? la media è dominata dalla coda superiore?"* — is made concrete by the Turnbull comparison. The lower bound cannot be inflated by tail assumptions, so a large parametric–Turnbull gap flags tail-driven fragility.
- **Distribution-free validity.** Reporting the Turnbull lower bound alongside the parametric mean tells a referee that your welfare estimate does not *depend* on the normality assumption — the assumption-free number is already substantial.
- **Handles negative-WTP and support issues.** Turnbull never assigns WTP below the lowest bid interval and never manufactures negative WTP the way a symmetric normal does (Chapter 5, §5.3) — so it sidesteps the negative-WTP problem entirely.

The bid design determines Turnbull's resolution. Turnbull can only place mass at the bids you offered, so the estimator's granularity — and how tight the lower bound is — depends on the bid vector (Chapters 15, 49). Sparse or poorly placed bids give a coarse step function and a looser bound; a well-designed bid vector spanning the WTP range gives a sharper distribution-free estimate. This ties the nonparametric benchmark back to your experimental-design choices.

> **Stop and think 10.3.** Your parametric SBDC model reports mean WTP €48; the Turnbull lower bound is €27. Before concluding anything, what is the most likely explanation for the gap, and what does it tell you about the robustness of the €48? *(Answer: the parametric model is extrapolating an upper tail beyond the highest bid, inflating the mean; the €48 is partly an artifact of the assumed distribution's tail, while the €27 lower bound is assumption-free. The gap signals that your headline mean is tail-sensitive — report both, consider a median (less tail-sensitive) and alternative distributions, and be cautious using €48 in a CBA without acknowledging the range, Chapters 15, 4.)*

10.9 The assumption trade-off in practice: which tool, when

A decision guide synthesizing the chapter, keyed to your work:

Situation	Tool	What it buys
Two-arm WTP contrast, small cell, skewed outcome	**Permutation test** (exact)	Valid p-value with no normality assumption
Causal p-value for a randomized survey experiment	**Randomization inference**	Design-based validity from the known assignment
Compare two arms' *location*/dominance (raw WTP)	**Mann–Whitney**	Robust, interpretable $\Pr(A > B)$
Compare two arms' *whole distribution* (raw WTP)	**Kolmogorov–Smirnov**	Detects location, scale, and shape differences

Situation	Tool	What it buys
Compare two arms' *simulated* WTP distributions	**Poe test**	Distribution-level comparison of KR-simulated WTP
Distribution-free WTP from CV yes/no data	**Turnbull**	Assumption-free mean (lower bound), tail-robustness check

Two overarching principles. **First, report parametric and nonparametric together.** The parametric result is your efficient main estimate; the nonparametric check tells you whether it survives dropping the distributional assumption. Agreement builds credibility; disagreement is a finding (it localizes where the parametric assumption mattered). **Second, keep difference and equivalence distinct.** Every test in this chapter is a *difference* test; a non-significant result is never, by itself, evidence of equivalence (Chapter 8). Your convergent-validity claims need the parametric/nonparametric difference tests *and* an equivalence test against a pre-specified margin.

10.10 Failure modes

F1. Reading a nonparametric null as "assumption-free proof of equality." A non-significant MW/KS/Poe/permutation result is a difference test failing to reject, not evidence of equivalence, and it still assumed exchangeability. *Fix:* run equivalence tests (TOST); state the assumption made (Chapter 8; §10.2).

F2. Misinterpreting Mann–Whitney as a test of means. MW tests stochastic dominance / $\Pr(A > B)$; it equals a median test only under a location-shift assumption. *Fix:* state the intended interpretation; don't call it a "test of mean WTP" without the shift assumption (§10.3).

F3. Randomization inference after broken randomization. Running design-based inference on a sample cleaned by treatment-related exclusions. *Fix:* preserve the randomization; analyze exclusions by arm; report with/without (§10.6; Chapter 32).

F4. Confusing the sharp and weak null. Claiming "no average effect" after rejecting (or failing to reject) the *sharp* null. *Fix:* be explicit about which null; use weak-null methods if the ATE is the target (§10.6).

F5. Over-pooling in Turnbull. Aggressive PAVA pooling from noisy or poorly spaced bids yielding a coarse, uninformative distribution. *Fix:* design an adequate bid vector (Chapters 15, 49); report how much pooling occurred (§10.8).

F6. Treating the Turnbull lower bound as the point estimate. It is a *lower* bound (mass valued at interval lower endpoints); the true mean is higher. *Fix:* report it as a conservative benchmark alongside the parametric mean and an upper bound, not as the estimate (§10.8).

F7. Using nonparametric tests where a parametric model is clearly right and needed. Discarding an efficient, correctly specified model for a rank test loses power and cannot deliver WTP/welfare quantities. *Fix:* use nonparametric methods as robustness checks, not replacements, when the parametric model is justified (§10.2, §10.9).

F8. Too few permutations for a Monte-Carlo permutation p-value. Reporting an imprecise permutation p from too few reshuffles. *Fix:* enumerate exactly when feasible, or use enough reshuffles that the p-value's simulation error is small (§10.5; Chapter 9).

10.11 Diagnostic questions for your own papers

1. **Robustness of contrasts.** For each treatment contrast, have I run a nonparametric (permutation/MW/KS) check alongside the parametric test, and do they agree? *(§10.2, §10.5.)*
2. **Design-based inference.** For my survey experiments, could I report a randomization-inference p-value grounded in the actual assignment, and is the randomization intact (no treatment-related exclusions)? *(§10.6.)*
3. **Distribution vs location.** When comparing arms, am I asking about location (MW), the whole distribution (KS), or simulated WTP (Poe), and using the right tool? *(§10.3–§10.7.)*
4. **Turnbull benchmark.** Have I reported a Turnbull distribution-free lower-bound mean WTP alongside my parametric SBDC mean, and examined the gap for tail-driven inflation? *(§10.8.)*
5. **Which null.** For randomization inference, am I clear about testing the sharp vs weak null? *(§10.6, F4.)*
6. **Difference vs equivalence.** Am I avoiding reading any non-significant nonparametric test as evidence of equivalence? *(§10.9, F1; Chapter 8.)*
7. **Bid design and Turnbull.** Does my bid vector give Turnbull adequate resolution, or is heavy pooling coarsening the estimate? *(§10.8, F5.)*

10.12 Exercises

(A) Comprehension.

A1. Explain the trade-off between a t-test and a permutation test for a two-arm WTP comparison. When is each preferable?

A2. In an SBDC study, "yes" at bid €30 means what about WTP? Write the relationship between the proportion of "yes" responses and the survival/CDF of WTP.

A3. A Mann–Whitney test gives $\Pr(A > B) = 0.35$, $p = 0.04$. State precisely what you can and cannot conclude about the two arms' WTP.

(B) Derivation.

B1. Show that the permutation p-value is exact under the null of exchangeability, and explain why enumerating all label assignments (rather than assuming a distribution) gives a valid reference distribution.

B2. For the SBDC survival data $S(5) = 0.85, S(10) = 0.70, S(20) = 0.50, S(40) = 0.25$ (and $S(0) = 1$, mass above €40 = 0.25), compute the Turnbull lower-bound mean WTP, showing each interval's mass and contribution.

B3. Explain, using the potential-outcomes sharp null, why re-randomizing the treatment assignment (holding observed outcomes fixed) yields a valid null distribution for a survey experiment, and identify the one design fact the validity depends on.

(C) Research application / critique.

C1. Your parametric SBDC reports mean WTP €55 (normal WTP); the Turnbull lower bound is €30. A co-author wants to report only the €55. Write a critique and a recommendation for what to report and why.

C2. You analyze a survey experiment with a t-test on WTP in cells of 45 respondents each, and a reviewer questions the normal approximation. Describe the two distribution-free analyses you would add (one design-based, one distribution-comparison) and what each would establish.

C3. You compare hypothetical and incentivized WTP with a Poe test ($p = 0.07$) and conclude "no evidence of hypothetical bias, so the formats are equivalent." Critique using §10.7 and §10.9, and state what analyses are actually required to support an equivalence claim.

10.13 Solutions

A1. The t-test is more powerful *if* WTP is (approximately) normal and the sample is large enough for the CLT; the permutation test is exact and valid *regardless* of the distribution, at a small power cost when normality holds. Prefer the t-test when normality/CLT are credible (large, roughly symmetric samples); prefer the permutation test in small cells, with skewed WTP or outliers, or as a robustness check — and if the two disagree, trust the permutation test.

A2. "Yes" at €30 means the respondent is willing to pay €30, i.e. WTP $\geq$ €30. So $\Pr(\text{yes at } b) = \Pr(\text{WTP} \geq b) = S(b) = 1 - F(b)$; the proportion of "yes" responses at bid b estimates the survival function at b, and the proportion of "no" estimates the CDF $F(b)$.

A3. You can conclude that WTP in arm A is *stochastically smaller* than in B ($\Pr(A > B) = 0.35 < 0.5$, $p = 0.04$ significant): a randomly chosen A-respondent exceeds a random B-respondent only 35% of the time, a significant dominance difference. You cannot conclude anything specifically about *means* unless you assume a pure location shift; MW is a test of stochastic dominance, not of means per se.

B1. Under exchangeability, every assignment of labels to the pooled observations is equally likely if the null (same distribution) holds. Enumerating all $\binom{n_A+n_B}{n_A}$ labelings and computing T for each yields the *exact* null distribution of T conditional on the observed data — a finite, known reference set generated by the null's own symmetry, requiring no distributional assumption. The p-value (fraction as extreme as observed) is therefore exact: it is a genuine probability under the null, not an asymptotic approximation. ■

B2. Masses (survival drops): $[0, 5) : 1 - 0.85 = 0.15$ at $0 \to 0$; $[5, 10) : 0.85 - 0.70 = 0.15$ at $5 \to 0.75$; $[10, 20) : 0.70 - 0.50 = 0.20$ at $10 \to 2.00$; $[20, 40) : 0.50 - 0.25 = 0.25$ at $20 \to 5.00$; $[40, \infty) : 0.25$ at $40 \to 10.00$. Sum of masses $= 0.15 + 0.15 + 0.20 + 0.25 + 0.25 = 1.00$ ✓. Turnbull lower-bound mean $= 0 + 0.75 + 2.00 + 5.00 + 10.00 =$ €17.75.

B3. Under the sharp null $Y_i(1) = Y_i(0)$ for all i, each respondent's outcome is the same whichever arm they are assigned to, so the observed outcomes are *fixed constants*. The only randomness is the assignment Z, whose distribution is *known* because you chose the randomization. Re-drawing Z from that known distribution and recomputing the statistic (with outcomes fixed) therefore reproduces the statistic's exact null distribution — valid with no distributional or asymptotic assumptions. The validity depends on the assignment mechanism being known and intact (the actual randomization, undisturbed by treatment-related exclusions).

C1. Reporting only €55 hides that the estimate is tail-sensitive: the €25 gap to the €30 Turnbull lower bound indicates the parametric mean is inflated by an upper tail extrapolated beyond the observed bids, and the €55 depends on the normality assumption. Recommendation: report both — the parametric mean €55 (with its distributional assumption stated), the Turnbull distribution-free lower bound €30, and ideally the median (less tail-sensitive) and results under an alternative WTP distribution (e.g., log-normal or a mixture). For any CBA use (Chapter 4), carry the range, not the single tail-driven figure. Transparency about the gap is the honest and more defensible report.

C2. (i) **Randomization inference** — re-randomize the treatment assignment as actually performed and build the null distribution of the WTP mean difference under the sharp null; this gives a design-based p-value valid in the small cells without the normal approximation, establishing that the causal p-value follows from the randomization. (ii) **A permutation test and/or KS test** — an exact permutation test of the mean (or median) difference confirms the contrast without normality, and a KS test checks whether the arms differ in the whole WTP distribution (location, spread, shape), not just the mean. Together they establish that the finding is not an artifact of the normal approximation and characterize *how* the arms differ.

C3. Two errors: (i) $p = 0.07$ is a *difference* test failing to reject at 5% — it is not evidence of equivalence (absence of evidence $\neq$ evidence of absence); (ii) "no evidence of hypothetical bias" conflates non-significance with a null effect, ignoring power. To support equivalence: pre-specify an equivalence margin Δ for the WTP difference (the largest gap that would be irrelevant for the intended use), then run an equivalence test (TOST) — equivalently, check whether the 90% interval of the Poe/KR difference distribution lies entirely within $\pm\Delta$ — and report the MDE so the null is interpretable (Chapter 8). Only if the equivalence

test passes can hypothetical and incentivized WTP be called equivalent; otherwise the correct statement is "inconclusive."

10.14 Spaced review prompts

+1 day. From memory, state what Mann–Whitney, KS, a permutation test, and randomization inference each test, and the single assumption each requires. Then write the SBDC relationship between "yes" proportions and the survival function.

+1 week. Compute a Turnbull lower-bound mean WTP from a small set of bid/survival data, including one monotonicity violation you fix by pooling. Explain in two sentences why the parametric mean can exceed the Turnbull lower bound.

+1 month. For your CV paper, compute the Turnbull lower-bound mean alongside your parametric mean and interpret the gap; for one survey-experiment contrast, run a permutation test and (conceptually) a randomization-inference test and compare to the parametric p-value. Close with the spine sentence about whether your conclusions survive dropping the distributional assumption.

10.15 How this chapter connects to the rest of the book

- **Chapter 6** motivated distribution-free methods (where asymptotics fail); **Chapter 7** supplied the p-value/test framework; **Chapter 8** supplied the difference-vs-equivalence distinction that every test here must be read against; **Chapter 9** supplied the Poe test and the simulated WTP distributions it compares.
- **Chapter 15 (CV likelihoods)** is the *parametric* counterpart to Turnbull: this chapter's distribution-free lower bound is the robustness benchmark for the normal/logistic SBDC mean WTP developed there, and the negative-WTP/tail worries of Chapter 5 (§5.3) are what Turnbull sidesteps.
- **Chapters 31–33 (treatment effects, threats, multiplicity/equivalence/randomization)** are the full experimental-design development of §10.6: randomization inference, the sharp null, and how post-randomization exclusions (Chapter 32, your PDF §6.6) threaten the design-based foundation.
- **Chapter 37 (efficient design)** determines the bid vector that sets Turnbull's resolution (§10.8); **Chapters 47–48 (meta-analysis)** use rank/nonparametric and clustered-bootstrap methods for dependent, non-normal estimates.

This chapter completes **Part II**: you now hold the full inference toolkit — distributions and asymptotics (Ch 5–6), estimation and reporting (Ch 7), power and equivalence (Ch 8), simulation (Ch 9), and distribution-free/design-based inference (Ch 10) — ready for the maximum-likelihood and latent-variable models of Part III.

10.16 Research translation task

For your **CV paper and one survey-experiment contrast**, produce a two-to-three-page note:

1. **Turnbull benchmark.** Compute the Turnbull distribution-free lower-bound mean WTP from your SBDC yes/no data (with PAVA pooling where needed), and place it beside your parametric mean; interpret the gap as a tail-robustness diagnostic and note how much pooling occurred.
2. **Nonparametric contrast checks.** For one treatment contrast, run a permutation test (exact if feasible) and, where you have individual WTP, a Mann–Whitney and a KS test; compare their p-values to the parametric test and report agreement/disagreement.

3. **Design-based inference.** Describe (and, if feasible, run) a randomization-inference p-value for the contrast, and confirm the randomization is intact (report exclusion rates by arm).
4. **Difference vs equivalence.** State explicitly that these are difference tests, and pair any convergent-validity comparison with an equivalence test against a pre-specified margin (Chapter 8).
5. **Close** with the spine sentence: *"My WTP/contrast conclusion [does/does not] survive dropping the distributional assumption: the Turnbull lower bound is [value] vs the parametric mean [value], and the nonparametric/design-based tests [agree/disagree] with the parametric one."*

This closes the Part II toolkit; carry the Turnbull benchmark into Chapter 15 (parametric CV likelihoods) and the randomization-inference material into Chapters 31–33 (survey-experiment identification).

10.17 Bibliography

Core. - Haab & McConnell, *Valuing Environmental and Natural Resources: The Econometrics of Non-Market Valuation*. The definitive treatment of the Turnbull estimator and distribution-free WTP bounds in contingent valuation (§10.8) — the single most relevant source for your CV work. *(Verify edition/chapters.)* - Hollander, Wolfe & Chicken, *Nonparametric Statistical Methods*. The standard reference for Mann–Whitney, Kolmogorov–Smirnov, and rank-based inference (§10.3–§10.4). *(Verify edition.)* - Good, *Permutation, Parametric, and Bootstrap Tests of Hypotheses*. A clear, applied treatment of permutation tests and exchangeability (§10.5). *(Verify edition.)*

Applied. - Poe, Giraud & Loomis (2005), "Computational Methods for Measuring the Difference of Empirical Distributions," *American Journal of Agricultural Economics*. The Poe test in its distributional context (§10.7) — your WTP-comparison tool. *(Verify volume/pages.)* - Turnbull (1976), "The Empirical Distribution Function with Arbitrarily Grouped, Censored and Truncated Data," *JRSS-B*. The original nonparametric MLE for interval-censored data underlying §10.8. *(Verify volume/pages.)* - Carson & Hanemann, "Contingent Valuation" (Handbook of Environmental Economics chapter), for how nonparametric bounds and distribution-free estimates are used in CV practice. *(Verify exact citation.)*

Advanced. - Imbens & Rubin, *Causal Inference for Statistics, Social, and Biomedical Sciences*, chapters on Fisherian randomization inference, the sharp null, and re-randomization. The rigorous foundation for §10.6 — read before Chapters 31–33. *(Verify edition/chapters.)* - Rosenbaum, *Observational Studies* (and his work on randomization inference and sensitivity analysis). Deep treatment of design-based inference and its robustness. *(Verify edition.)* - Barber & Robertson / PAVA and isotonic-regression references for the pool-adjacent-violators algorithm used in Turnbull monotonization (§10.8). *(I am not certain of the single best citation — verify; any standard isotonic-regression reference suffices.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Poe/Turnbull/Carson–Hanemann citations before using them, per the book's no-unverified-citation rule.

End of Chapter 10 and of Part II. You now hold the full probability-and-inference toolkit: the distributions of stated preference (Ch 5), asymptotic theory (Ch 6), estimation and inference (Ch 7), power and equivalence (Ch 8), simulation (Ch 9), and distribution-free and design-based inference (Ch 10). Part III — Maximum Likelihood and Latent Variable Models — begins with Chapter 11 (The Likelihood Framework), the highest-dependency chapter in the book, where the score, Hessian, information, and asymptotic efficiency you have met in pieces are assembled into the engine behind your SBDC, logit, mixed logit, and MDCEV estimation. Tell me when to write it, or request a different chapter.

Part II Assessment — Probability and Statistical Inference

Covers Chapters 5–10. Five components, in the order they are best attempted. Allow a full working day; the mock oral is the part that matters most for your viva.

1. Cumulative problem set

P1. Derive that the difference of two i.i.d. standard Gumbel variables is standard logistic, showing the change of variables explicitly.

P2. Latent WTP is $\mathcal{N}(\mu, \sigma^2)$ with $\mu = 22$, $\sigma = 28$. Compute the implied probability of negative WTP and state whether you would accept it for a coastal-habitat improvement, with reasons.

P3. Two groups have WTP means (24, 36) and variances (81, 144), equally sized. Decompose the total variance into between and within components and state which corresponds to preference heterogeneity.

P4. At $\alpha = 0.05$ and 80% power, compute the required n per arm for $d = 0.25$, and the MDE in euros for $n = 45$ per arm when the WTP standard deviation is €35.

P5. You observe $\hat{\delta} = €6$ with SE €7 and a pre-specified equivalence margin of €12. Run the TOST both ways (two one-sided tests and the 90% confidence interval) and state the conclusion in one sentence.

2. Research critique

Find a published stated-preference paper that reports "no significant difference" between two arms and treats it as evidence of equivalence. Reconstruct, from the reported numbers, what its minimum detectable effect was. Write a referee paragraph explaining what the paper is entitled to claim and what it is not, and specify the equivalence test it should have run.

3. Red-flag diagnosis

A Krinsky–Robb simulation of WTP is reported with a mean of €61, a median of €22, and a 95% interval of [€6, €340]. The author writes "mean WTP is €61 (95% CI 6–340)." List everything wrong with that sentence, explain the mechanism generating the pattern, and write the sentence you would report instead.

4. Mock oral examination

1. What exactly does "asymptotic" license, and what does it forbid, when you have 18 studies?
2. Why does the logit have a closed form and the mixed logit not?
3. State the law of total variance and explain why it is the skeleton of the scale-versus-preference problem.
4. Why is failing to reject equality not evidence of equality, and what would be?

5. When does a bootstrap beat a delta-method interval, and when does neither help?

5. Integration mini-project

For each principal comparison in your CV paper — high vs low cost vector, hypothetical vs incentivized, PC vs SBDC, BRIS vs full payment, experienced vs non-experienced — compute the achieved power for the effect size you consider policy-relevant, and the MDE given your realized per-arm sample sizes. Present it as one table, and write a paragraph stating which comparisons your design can actually support.

Part III — Maximum Likelihood and Latent Variable Models

Chapter 11 — The Likelihood Framework

Part III: Maximum Likelihood and Latent Variable Models

> **The spine question, applied to this chapter.** This is the highest-dependency chapter in the book. Every model you estimate — the SBDC probit, the payment-card interval model, the logit, the mixed logit, MDCEV, and your meta-models — is a **likelihood**, and every number they produce (coefficients, standard errors, test statistics, WTP, welfare) flows from that one object. Master the likelihood framework and most of the rest of the book becomes application. Your PDF gives this "priorità altissima" and names both halves of the skill: the mechanics (*"likelihood; log-likelihood; score; Hessiano; informazione di Fisher; MLE; proprietà asintotiche"*) and the judgment (*"la convergenza è vera o cosmetica? l'Hessiano è invertibile? il massimo è globale? i parametri sono identificati separatamente?"*). Both are properties of a single surface — the log-likelihood over the parameter space. The spine question here: *when I estimate a model by maximum likelihood, what surface am I climbing, does its peak identify the object I care about, and does its curvature honestly report how precisely I know it?*

11.0 Conceptual motivation: one object, everything downstream

You have met the pieces of this framework already, scattered across the book: the Hessian and Fisher information as linear-algebra objects (Chapter 1, §1.11); the score and Hessian of the logit and probit derived by hand (Chapter 2, §2.8); the sketch of asymptotic normality via a Taylor expansion of the score (Chapter 2, §2.9; Chapter 6, §6.8); the Cramér–Rao bound and the claim that the MLE attains it (Chapter 7, §7.3). This chapter assembles those pieces into a single, rigorous, self-contained framework — and adds the two deep results that hold it together: the **score has mean zero** (which makes the MLE consistent) and the **information matrix equality** (which makes classical standard errors valid, and whose failure is the reason robust standard errors exist).

Why does this deserve to be the book's keystone? Because in your work maximum likelihood is not one method among many — it is *the* method:

- The **SBDC contingent valuation** model is a likelihood: latent WTP is normally distributed, the yes/no answer censors it, and the resulting probit-like likelihood is what you maximize. Your PDF's Block 1 deliverable is literally *"riscrivere da zero la likelihood SBDC e payment card del tuo CV paper"* — and this chapter derives it from scratch (§11.8).
- The **payment-card** model is an interval-censored likelihood (Chapter 14–15); the **logit** and **mixed logit** for your DCE are likelihoods (the mixed logit a *simulated* one, Chapter 9, 23); **MDCEV** is a Kuhn–Tucker likelihood (Chapter 41); the **meta-models** for your finfish synthesis are likelihoods (Chapter 47). One framework, all of them.
- Every **judgment** your PDF demands is a statement about the log-likelihood surface. "Is convergence real or cosmetic?" — is the gradient zero *and* the curvature negative-definite? "Is the Hessian invert-

ible?" — is the surface curved in every direction, or flat somewhere? "Are the parameters separately identified?" — does the surface have a unique peak, or a ridge? "Is the maximum global?" — is the surface concave, or does it have multiple hills? You cannot answer any of these by staring at software output; you answer them by understanding the surface, which is this chapter.

So we build the framework in order: the likelihood and log-likelihood; the score and its zero-mean property; the Hessian, Fisher information, and the information matrix equality; the MLE and its four asymptotic properties (consistency, asymptotic normality, efficiency, invariance); the geometric synthesis that curvature *is* precision; a full derivation of *your* SBDC probit likelihood; and finally the framework's built-in warnings about identification, convergence, and constraints (developed computationally in Chapter 13). This is the engine room. Everything else in Parts III–XIV is an application of what is built here.

11.1 Prerequisites

This chapter assembles threads from across the book, so its prerequisites are substantial but all previously developed:

- **Chapter 1** (§1.11): the Hessian and information matrix as linear-algebra objects; positive/negative definiteness (§1.7); the covariance as an inverse (§1.5); conditioning (§1.10).
- **Chapter 2** (§2.8–2.9): differentiating a log-likelihood (score, Hessian); concavity and the second-order condition; the Taylor expansion behind asymptotic normality.
- **Chapter 5**: distributions and the CDF (the ingredients of any likelihood).
- **Chapter 6** (§6.4, 6.6, 6.8): the LLN, the CLT, Slutsky's theorem, and the estimator-asymptotics template (which this chapter specializes to ML).
- **Chapter 7** (§7.3): bias, efficiency, and the Cramér–Rao bound.

Same **notation**. New/consolidated symbols: θ (parameter vector), θ_0 (true value), $L(\theta)$ (likelihood), $\ell(\theta)$ (log-likelihood), $\mathbf{s}(\theta) = \nabla\ell$ (score), $\mathbf{H}(\theta) = \nabla^2\ell$ (Hessian), $\mathbf{I}(\theta)$ (Fisher information), $\widehat{\theta}$ (MLE). Subscript i denotes a single observation's contribution.

11.2 The likelihood function

The construction. You have data $y_1, \dots, y_n$ and a model that says the data were generated by a density (or mass function) $f(y;\theta)$ indexed by unknown parameters θ. The joint density of the whole sample, *as a function of the data*, is $\prod_i f(y_i;\theta)$ (under independence). The **likelihood** takes this same expression but reads it *as a function of the parameters*, with the data held fixed at their observed values:

Definition 11.1 (Likelihood). Given observed data $y_1, \dots, y_n$ and a model $f(y;\theta)$, the **likelihood function** is

$$L(\theta) = \prod_{i=1}^{n} f(y_i;\theta),$$

viewed as a function of θ for the fixed observed data. It measures how *plausible* each parameter value is, given what you saw: $L(\theta_A) > L(\theta_B)$ means the data were more probable under θ_A than under θ_B.

The conceptual shift — from "density of the data given parameters" to "plausibility of parameters given data" — is the whole idea. The density and the likelihood are the *same formula*; they differ only in what is treated as variable. This is the **likelihood principle**: all the evidence the data provide about θ is contained in the likelihood function.

For your models. The likelihood is built from the model's probability of what was observed:

- **SBDC:** each respondent contributes the probability of their yes/no answer, $\Pr(\text{yes})^{y_i}\,\Pr(\text{no})^{1-y_i}$ — a Bernoulli likelihood whose success probability depends on the bid and the parameters (§11.8).

- **Discrete choice:** each choice contributes the model's probability of the chosen alternative, $P_{i,\text{chosen}}(\theta)$ — the logit or mixed-logit probability (Chapters 21–23).
- **Payment card / interval data:** each answer contributes the probability that WTP falls in the reported interval, $F(u_i;\theta) - F(\ell_i;\theta)$ (Chapter 14–15).

In every case the likelihood is "the probability, under the model, of exactly what each respondent did," multiplied across respondents.

> **Active recall 11.A.** A respondent offered a bid of €40 says "no." Write their contribution to the SBDC likelihood in terms of the model probability of a "yes" at €40, and explain what a parameter value that makes this contribution large is "saying." *(Answer: their contribution is* $1-\Pr(\textit{yes at } €40;\theta)$*, i.e.* $\Pr(\textit{WTP} < 40;\theta)$*. A parameter value that makes this large is one under which it is very probable that this person's WTP is below €40 — i.e., a low mean/median WTP is more consistent with a "no" at €40.)*

11.3 The log-likelihood

We almost never work with L directly; we work with its logarithm.

Definition 11.2 (Log-likelihood). The **log-likelihood** is

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f(y_i;\theta) = \sum_{i=1}^{n} \ell_i(\theta).$$

Four reasons the log is used, all practical:

1. **Products become sums.** $\log \prod_i f_i = \sum_i \log f_i$. Sums are far easier to differentiate (the score becomes a sum of per-observation scores) and to reason about (the LLN and CLT apply to sums, Chapter 6).
2. **Numerical stability.** A product of thousands of probabilities (each < 1) underflows to zero in floating-point arithmetic; a sum of log-probabilities does not. Every estimation routine works in logs for this reason.
3. **Same maximizer.** log is strictly increasing, so $\ell(\theta)$ and $L(\theta)$ are maximized at the *same* θ. Maximizing the log-likelihood is maximizing the likelihood.
4. **Additive, hence amenable to asymptotics.** ℓ is a sum of n i.i.d.-ish terms, so $\frac{1}{n}\ell(\theta) \to \mathbb{E}[\ell_i(\theta)]$ by the LLN — the average log-likelihood converges to the *expected* log-likelihood, whose maximizer is the true θ_0 (a fact behind consistency, §11.6).

The log-likelihood is the surface you actually climb. Its **height** is goodness of fit; its **gradient** (the score) tells you which way is up; its **curvature** (the Hessian) tells you how sharply peaked it is — and therefore how precisely the data pin down the parameters. The next three sections make each of these precise.

> **Stop and think 11.1.** Two candidate parameter vectors give log-likelihoods $\ell(\theta_A) = -1520.3$ and $\ell(\theta_B) = -1524.8$. Which fits better, by how much on the *likelihood* scale, and why can you not interpret the raw difference as a probability? *(Answer:* θ_A *fits better; the likelihood ratio is* $e^{\ell_A - \ell_B} = e^{4.5} \approx 90$*, so the data are about 90 times more probable under* θ_A*. The raw log-likelihood values are not probabilities (they can be any negative number, and depend on sample size and the data's scale); only* differences*, exponentiated, are interpretable — as likelihood ratios, the basis of the LR test, Chapter 12.)*

11.4 The score, and why its mean is zero

Definition 11.3 (Score). The **score** is the gradient of the log-likelihood:

$$\mathbf{s}(\theta) = \nabla \ell(\theta) = \sum_{i=1}^{n} \nabla \ell_i(\theta) = \sum_{i=1}^{n} \mathbf{s}_i(\theta).$$

It is the sum of per-observation scores $\mathbf{s}_i = \nabla \log f(y_i;\theta)$. The MLE $\hat{\theta}$ solves the **score equation** $\mathbf{s}(\hat{\theta}) = \mathbf{0}$: the peak of the surface is where every partial slope vanishes (Chapter 2, §2.7).

Now the first deep result — the property that makes maximum likelihood *work*.

Proposition 11.1 (The score has mean zero at the truth). Under regularity conditions (the support of f does not depend on θ, and differentiation and integration can be interchanged), at the true parameter θ_0,

$$\mathbb{E}_{\theta_0}\big[\mathbf{s}_i(\theta_0)\big] = \mathbf{0}.$$

Derivation. A density integrates to one for every parameter value: $\int f(y;\theta)\,dy = 1$. Differentiate both sides with respect to θ (interchanging derivative and integral):

$$\int \nabla_\theta f(y;\theta)\,dy = \mathbf{0}.$$

Now use the identity $\nabla f = f\,\nabla \log f = f\,\mathbf{s}$ (the log-derivative trick):

$$\int \mathbf{s}(y;\theta)\, f(y;\theta)\,dy = \mathbf{0}, \qquad \text{i.e.} \qquad \mathbb{E}_\theta[\mathbf{s}(\theta)] = \mathbf{0}.$$

Evaluated at the true θ_0 (where $f(\cdot;\theta_0)$ is the actual data density), this says the expected score is zero. ■

Why this is the engine of consistency. The population score equation $\mathbb{E}[\mathbf{s}(\theta)] = \mathbf{0}$ is solved by θ_0. The MLE solves the *sample* analogue $\frac{1}{n}\sum_i \mathbf{s}_i(\hat{\theta}) = \mathbf{0}$. By the law of large numbers (Chapter 6), the sample average score converges to the population expected score, so its root $\hat{\theta}$ converges to the root θ_0 of the population equation — **the MLE is consistent** (§11.6). The zero-mean score is *why* setting the sample score to zero targets the truth rather than some other value. It is the most important single fact in the framework.

Intuition. The score $\mathbf{s}_i$ measures how sensitive the log-probability of observation i is to the parameter. At the true parameter, these sensitivities *cancel on average*: the data give no systematic push in any direction, because you are already at the value that best explains the data-generating process. Away from θ_0, the expected score is non-zero and points (on average) back toward θ_0 — which is what lets an optimizer climb to it.

> **Active recall 11.B.** State the zero-mean-score property and explain, in one sentence, how it delivers consistency of the MLE via the LLN. *(Answer: at the true parameter the expected score is zero, $\mathbb{E}_{\theta_0}[\mathbf{s}] = \mathbf{0}$; the MLE sets the* sample* average score to zero, and by the LLN the sample average score converges to the expected score, so the MLE's root converges to the truth — consistency.)*

11.5 The Hessian, Fisher information, and the information matrix equality

The curvature of the log-likelihood is the second derivative, and it turns out to carry the *precision* of the estimate. This section defines the information and proves the identity that makes classical standard errors valid.

Definition 11.4 (Hessian and information). The **Hessian** is $\mathbf{H}(\theta) = \nabla^2 \ell(\theta) = \sum_i \nabla^2 \ell_i$. The **Fisher (expected) information matrix** for one observation is

$$\mathbf{I}_1(\theta) = \mathrm{Var}_\theta\left[\mathbf{s}_i(\theta)\right] = \mathbb{E}[\mathbf{s}_i \mathbf{s}_i']$$

(the second equality because the score has mean zero, §11.4). The **observed information** is $-\mathbf{H}(\hat{\theta})$, the negative Hessian at the MLE. For the full sample, $\mathbf{I}(\theta) = n\mathbf{I}_1(\theta)$.

Now the second deep result — the identity that unifies the two ways of measuring precision.

Proposition 11.2 (Information matrix equality). Under correct specification and the regularity conditions of §11.4,

$$\mathbf{I}_1(\theta_0) = \mathrm{Var}[\mathbf{s}_i(\theta_0)] = -\mathbb{E}[\mathbf{H}_i(\theta_0)].$$

That is, **the variance of the score equals the negative expected Hessian** — the information can be computed either as the *spread of the score* (how much the evidence varies) or as the *curvature of the log-likelihood* (how sharply peaked it is), and the two coincide.

Derivation. Differentiate the zero-mean-score identity $\int \mathbf{s}(y;\theta)\, f(y;\theta)\, dy = \mathbf{0}$ (from §11.4) once more with respect to θ':

$$\int \left[\nabla_{\theta'} \mathbf{s} \cdot f + \mathbf{s} \cdot \nabla_{\theta'} f\right] dy = \mathbf{0}.$$

The first term uses $\nabla_{\theta'} \mathbf{s} = \mathbf{H}$ (the Hessian is the derivative of the score); the second uses $\nabla f = f\mathbf{s}'$ again:

$$\int \mathbf{H}\, f\, dy + \int \mathbf{s}\,\mathbf{s}'\, f\, dy = \mathbf{0} \quad \Longrightarrow \quad \mathbb{E}[\mathbf{H}] + \mathbb{E}[\mathbf{s}\mathbf{s}'] = \mathbf{0}.$$

Rearranging, $\mathbb{E}[\mathbf{s}\mathbf{s}'] = -\mathbb{E}[\mathbf{H}]$, i.e. $\mathrm{Var}[\mathbf{s}] = -\mathbb{E}[\mathbf{H}] = \mathbf{I}_1$. ■

Why this identity is the hinge of everything. It has three consequences you use constantly:

1. **It gives the MLE's asymptotic variance** (§11.6): the covariance of $\hat{\theta}$ is $\mathbf{I}(\theta_0)^{-1} = [-\mathbb{E}\mathbf{H}]^{-1}$ — the **inverse of the (negative) expected Hessian**. This is the origin of "standard errors = $\sqrt{\mathrm{diag}([-\mathbf{H}]^{-1})}$" from Chapter 1 (§1.11).
2. **It gives two estimators of that variance**, which agree under correct specification: the **observed-information** estimator $[-\mathbf{H}(\hat{\theta})]^{-1}$ and the **outer-product-of-scores (BHHH)** estimator $[\sum_i \mathbf{s}_i \mathbf{s}_i']^{-1}$. Their agreement is a *specification check*.
3. **Its failure is the reason robust ("sandwich") standard errors exist.** When the model is *mis*specified (or observations are dependent), the information equality *breaks*: $\mathrm{Var}[\mathbf{s}] \neq -\mathbb{E}[\mathbf{H}]$. Then the asymptotic variance is the **sandwich** $\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$ with $\mathbf{A} = -\mathbb{E}[\mathbf{H}]$ and $\mathbf{B} = \mathrm{Var}[\mathbf{s}]$ (Chapter 6, §6.8), which no longer collapses to $\mathbf{I}^{-1}$. Robust and clustered standard errors (Chapters 17, 19) are exactly what you use when you do not trust the information equality — for instance, when your repeated choice tasks induce dependence the simple likelihood ignores. So this abstract identity is precisely the fork in the road between classical and robust inference.

A verified illustration. In a simulated SBDC probit (§11.8), evaluated at the MLE, the negative Hessian and the outer product of scores gave nearly identical information matrices — diagonals $[1321,\ 1.46 \times 10^6]$ versus $[1321,\ 1.50 \times 10^6]$ (ratios 1.00 and 0.98). Under correct specification they agree up to finite-sample noise, exactly as Proposition 11.2 predicts; a *large* discrepancy would signal misspecification and warn you to use robust standard errors.

> **Stop and think 11.2.** In your DCE, each respondent answers 8 choice tasks, and you estimate a plain logit that assumes all observations are independent. Predict whether the information matrix equality holds, and what that implies for your standard errors. *(Answer: it fails — the 8 tasks per respondent are dependent (shared tastes/scale), so the true variance of the (summed) score exceeds what the negative Hessian implies under independence; classical SEs are too small. You must use respondent-clustered (or panel/mixed) standard errors — the sandwich form — because the information equality the classical SEs rely on is violated by the dependence, Chapters 17, 19.)*

11.6 The maximum likelihood estimator and its asymptotic properties

We can now state the payoff: what the MLE *is* and the four properties that make it the default estimator in your field. The proofs assemble Chapter 6's template and the two propositions above; I give the logic and point to the full arguments.

Definition 11.5 (Maximum likelihood estimator). The **MLE** is the parameter value that maximizes the (log-)likelihood:

$$\hat{\theta} = \arg\max_{\theta} \ \ell(\theta),$$

equivalently the solution of the score equation $\mathbf{s}(\hat{\theta}) = \mathbf{0}$ with a negative-definite Hessian (the second-order condition, Chapter 2 §2.7).

The four asymptotic properties.

(1) Consistency. $\hat{\theta} \xrightarrow{p} \theta_0$. *Why:* the average log-likelihood converges (LLN) to the expected log-likelihood, whose unique maximizer is θ_0 (a consequence of the zero-mean score, §11.4, and identification); the maximizer of the sample objective converges to the maximizer of the limit objective. **Requires identification** — the expected log-likelihood must have a *unique* maximizer, which fails if the parameters are not separately identified (§11.9).

(2) Asymptotic normality.

$$\sqrt{n}\,(\hat{\theta} - \theta_0) \ \xrightarrow{d}\ N(\mathbf{0},\ \mathbf{I}_1(\theta_0)^{-1}).$$

Why (assembling Chapter 6, §6.8): Taylor-expand the score equation around θ_0: $\mathbf{0} = \mathbf{s}(\hat{\theta}) \approx \mathbf{s}(\theta_0) + \mathbf{H}(\theta_0)(\hat{\theta} - \theta_0)$, so $\sqrt{n}(\hat{\theta} - \theta_0) \approx [-\frac{1}{n}\mathbf{H}(\theta_0)]^{-1} \cdot \frac{1}{\sqrt{n}}\mathbf{s}(\theta_0)$. By the LLN, $-\frac{1}{n}\mathbf{H} \to \mathbf{I}_1$; by the CLT, $\frac{1}{\sqrt{n}}\mathbf{s}(\theta_0) \to N(\mathbf{0}, \mathbf{I}_1)$ (the score is a sum of mean-zero terms each with variance $\mathbf{I}_1$, by §11.4–11.5). Combining, $\sqrt{n}(\hat{\theta} - \theta_0) \to \mathbf{I}_1^{-1} N(\mathbf{0}, \mathbf{I}_1) = N(\mathbf{0}, \mathbf{I}_1^{-1})$. **The information matrix equality is exactly what collapses the general sandwich $\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$ to $\mathbf{I}_1^{-1}$** (§11.5); when it fails, you keep the sandwich.

(3) Efficiency. The MLE attains the Cramér–Rao lower bound asymptotically (Chapter 7, §7.3): its asymptotic variance $\mathbf{I}_1^{-1}$ is the smallest achievable among consistent, asymptotically-unbiased estimators. **No consistent estimator is asymptotically more precise than the MLE.** This is *the* reason your SBDC, logit, mixed logit, and MDCEV models are estimated by maximum likelihood — it squeezes the most information out of the data.

(4) Invariance. If $\hat{\theta}$ is the MLE of θ, then $g(\hat{\theta})$ is the MLE of $g(\theta)$ for any function g. *Why:* maximizing the likelihood over a reparameterization gives the transformed maximizer. **This is why your WTP estimate is legitimate:** WTP $= g(\text{coefficients}) = -\hat{\beta}_{\text{attr}}/\hat{\beta}_{\text{cost}}$ is the MLE of the true WTP because the MLE is invariant to the transformation — and its standard error comes from the delta method (Chapter 7, §7.7) applied to the coefficients' $\mathbf{I}_1^{-1}$ covariance.

The finite-sample caveat. These are *asymptotic* properties. In finite samples the MLE is typically **biased** (consistent but biased, Chapter 6, §6.3) — the SBDC and mixed-logit MLEs included — and its distribution is only approximately normal. In your small treatment cells (Chapter 8), lean on this framework for point estimation but pair it with the small-sample and distribution-free tools of Chapters 6, 8, 10.

> **Active recall 11.C.** Name the four asymptotic properties of the MLE and, for each, the one thing in your work that depends on it. *(Answer: consistency — your point estimates approach the truth with more data (requires identification); asymptotic normality — your standard errors and confidence intervals; efficiency — why you use ML rather than an ad hoc estimator (most precise); invariance — why WTP $= -\hat{\beta}_1/\hat{\beta}_2$ is a legitimate MLE.)*

11.7 Curvature is precision: the geometric synthesis

Everything in this chapter reduces to one picture, and it is worth holding it explicitly because it makes the abstract results intuitive and the diagnostics obvious.

The log-likelihood is a surface over the parameter space. The MLE sits at its **peak** (score zero). The **curvature** at the peak — the negative Hessian, i.e. the observed information — determines how *precisely* the data pin down the parameters, because the asymptotic covariance is the *inverse* of that curvature:

$$\widehat{\mathrm{Var}}(\hat{\theta}) = \big[-\mathbf{H}(\hat{\theta}) \big]^{-1}.$$

- A **sharply peaked** log-likelihood (large curvature, large information) means small movements away from the peak sharply reduce fit — the data strongly distinguish nearby parameter values — so the inverse curvature is small: **tight standard errors**.
- A **flat** log-likelihood near the peak (small curvature, small information) means nearby parameter values fit almost as well — the data barely distinguish them — so the inverse curvature is large: **wide standard errors**, and in the limit of a *flat direction*, **non-identification** (infinite variance in that direction).

This single geometric fact unifies the whole book so far: it is Chapter 1's "information matrix and its inverse" (§1.11); Chapter 2's "curvature of the likelihood" (§2.3); Chapter 6's asymptotic-normality argument (§6.8); and the source of every judgment in §11.9. When your PDF says "interpretare la curvatura della likelihood," this is the sentence it means: **curvature is precision; flatness is ignorance.** Every diagnostic that follows — is the Hessian invertible? is the maximum well-defined? are parameters identified? — is a question about the shape of this surface.

> **Stop and think 11.3.** Two coefficients in your mixed logit have a joint log-likelihood whose contour map near the peak is a long, narrow, diagonal ridge (very curved across the ridge, nearly flat along it). Predict: (a) which parameter *combination* is precisely estimated and which is not; (b) what this does to the two coefficients' individual standard errors and their correlation. *(Answer: (a) the combination* across* the ridge (where the surface is sharply curved) is precisely estimated; the combination *along* the ridge (flat) is poorly identified; (b) both individual coefficients have large standard errors (each varies a lot along the ridge), and they are strongly correlated (the ridge tilts the confidence ellipse) — the inverse of a near-singular curvature. This is the geometric picture of Chapter 1's covariance reading and a near-identification warning, §11.9.)*

11.8 Worked example: the SBDC probit likelihood, from scratch

This is your PDF's Block 1 deliverable — *"riscrivere da zero la likelihood SBDC"* — and the payoff of the chapter. We build the likelihood of your single-bounded dichotomous-choice contingent valuation model, derive its score, Hessian, and information, and recover WTP, all from first principles. (The full CV treatment, including the payment card and Turnbull, is Chapter 15; here we build the ML engine.)

Step 1 — the model. Respondent i is offered a bid b_i and says "yes" ($y_i = 1$, willing to pay b_i) or "no" ($y_i = 0$). Latent willingness to pay is normally distributed,

$$\mathrm{WTP}_i = \mu + \sigma\varepsilon_i, \qquad \varepsilon_i \sim N(0,1),$$

and the respondent says "yes" iff WTP exceeds the bid: $y_i = \mathbf{1}\{\mathrm{WTP}_i \geq b_i\}$. This is the standard SBDC/probit-like model your PDF describes: latent normal WTP, censored by a yes/no answer.

Step 2 — the choice probability (a CDF, per Chapter 5, §5.2).

$$\Pr(y_i = 1 \mid b_i) = \Pr(\mu + \sigma\varepsilon_i \geq b_i) = \Pr\Big(\varepsilon_i \geq \tfrac{b_i - \mu}{\sigma}\Big) = \Phi\Big(\tfrac{\mu - b_i}{\sigma}\Big),$$

using the symmetry of the normal ($1-\Phi(z)=\Phi(-z)$). Reparameterize to a linear index by letting $\alpha=\mu/\sigma$ and $\beta=-1/\sigma$:

$$\Pr(y_i=1 \mid b_i)=\Phi(\alpha+\beta b_i)\equiv\Phi_i, \qquad \beta<0.$$

This is a **probit in the bid**: higher bids ($\beta<0$) lower the "yes" probability. The parameters you estimate are (α,β); the structural quantities are recovered as $\sigma=-1/\beta$ and $\mu=\alpha/(-\beta)=-\alpha/\beta$ — and since WTP is normal, mean = median = $\mu=-\alpha/\beta$ (a ratio of coefficients — Chapter 2 §2.11, Chapter 7 §7.7).

Step 3 — the likelihood and log-likelihood. Each response is Bernoulli with success probability Φ_i:

$$L(\alpha,\beta)=\prod_{i=1}^{n}\Phi_i^{y_i}(1-\Phi_i)^{1-y_i}, \qquad \ell(\alpha,\beta)=\sum_{i=1}^{n}\left[y_i\log\Phi_i+(1-y_i)\log(1-\Phi_i)\right].$$

Step 4 — the score (using Chapter 2, §2.8/B1). With index $v_i=\alpha+\beta b_i$, covariate vector $\mathbf{x}_i=(1,b_i)'$, and normal density $\phi_i=\phi(v_i)$, differentiating gives

$$\mathbf{s}(\alpha,\beta)=\sum_{i=1}^{n}\underbrace{\left[\frac{y_i\phi_i}{\Phi_i}-\frac{(1-y_i)\phi_i}{1-\Phi_i}\right]}_{g_i\ \text{(generalized residual)}}\mathbf{x}_i=\sum_i g_i\,\mathbf{x}_i.$$

The **generalized residual** g_i is the probit analogue of the "actual minus predicted" residual: it is positive when a "yes" was more likely than the model expected and negative otherwise, and — by the zero-mean-score property (§11.4) — at the MLE $\sum_i g_i\mathbf{x}_i=\mathbf{0}$: the generalized residuals are orthogonal to the covariates (a "yes"-rate that the bids cannot systematically predict), exactly the SBDC analogue of OLS orthogonality.

Step 5 — the Hessian and information. The Hessian is $\mathbf{H}=-\sum_i w_i\,\mathbf{x}_i\mathbf{x}_i'$ with a positive weight $w_i=g_i(g_i+v_i)$ (from differentiating g_i, using $\phi'(v)=-v\phi(v)$); the probit log-likelihood is **globally concave** (Chapter 2, §2.8), so the maximum is unique and estimation is reliable — no local-maxima worries, unlike the mixed logit (§11.9). The Fisher information is $\mathbf{I}=-\mathbb{E}[\mathbf{H}]$, and the asymptotic covariance of $(\hat\alpha,\hat\beta)$ is $\mathbf{I}^{-1}$, estimated by $[-\mathbf{H}(\hat\alpha,\hat\beta)]^{-1}$.

Step 6 — WTP and its standard error. By invariance (§11.6), the MLE of mean WTP is $\widehat{\text{WTP}}=-\hat\alpha/\hat\beta$, and its standard error comes from the delta method (Chapter 7, §7.7) applied to the covariance $\mathbf{I}^{-1}$:

$$\text{Var}(\widehat{\text{WTP}})\approx\nabla g'\,\mathbf{I}^{-1}\,\nabla g, \qquad \nabla g=\left(-\tfrac{1}{\hat\beta},\ \tfrac{\hat\alpha}{\hat\beta^2}\right)'.$$

The $1/\hat\beta^2$ term warns (as always) that a weak bid coefficient inflates the WTP standard error — the SBDC face of the WTP-ratio fragility (Chapters 2, 6, 7).

Verification (simulated, verified). With true $\mu=30$, $\sigma=20$ (so $\alpha=1.5$, $\beta=-0.05$) and $n=3000$ SBDC responses across bids $\{5,15,25,35,50,70\}$: the MLE recovered $(\hat\alpha,\hat\beta)=(1.529,-0.0513)$; the score at the MLE was $\approx\mathbf{0}$ (confirming the first-order condition); the negative Hessian and the outer product of scores agreed (confirming the information matrix equality, §11.5); and the recovered mean WTP was $\widehat{\text{WTP}}=-\hat\alpha/\hat\beta=$ €29.82 (true €30) with a delta-method SE of €0.54 and 95% CI [€28.8, €30.9], and $\hat\sigma=-1/\hat\beta=$ €19.5 (true €20). **You have re-derived and validated your CV paper's core likelihood from scratch** — the deliverable your study plan sets for the end of Block 1.

> **Active recall 11.D.** In the SBDC probit, how do you recover mean WTP from the estimated index coefficients $(\hat\alpha,\hat\beta)$, and why is that recovered quantity a legitimate maximum-likelihood estimate? *(Answer: $\widehat{WTP}=-\hat\alpha/\hat\beta$ (since $\alpha=\mu/\sigma$, $\beta=-1/\sigma$ give $\mu=-\alpha/\beta$); it is a legitimate MLE by the invariance property — the MLE of a function of the parameters is that function of the MLE — with its standard error from the delta method on the coefficients' $\mathbf{I}^{-1}$ covariance.)*

11.9 Reading the framework's warnings: identification, convergence, constraints

The second half of your PDF's likelihood section is the judgment — *"la convergenza è vera o cosmetica? l'Hessiano è invertibile? il massimo è globale? i parametri sono identificati separatamente?"* Every one of these is a property of the log-likelihood surface, and the framework makes each precise. (The *computational* handling — multiple starts, gradient/Hessian checks, rescaling — is Chapter 13; here we name the phenomena as surface properties.)

Identification = a unique peak. The parameters are **identified** if the expected log-likelihood has a *unique* maximizer θ_0. If two different parameter values give the same likelihood for all data (observational equivalence, Chapter 30), the surface has a *ridge* (a flat direction), the information matrix is *singular* (a zero eigenvalue, Chapter 1 §1.7), and the MLE is not consistent for the individual parameters — only for the identified combination. **A flat direction of the log-likelihood, a singular information matrix, an infinite standard error, and non-identification are the same phenomenon in four vocabularies.** In your DCE, the scale/coefficient confound (Chapter 5, §5.6) and near-collinear attributes are exactly this.

Convergence: real vs cosmetic. An optimizer "converges" when it can no longer improve the objective — but that is *necessary, not sufficient* for a genuine maximum. Real convergence requires **both** (i) the gradient (score) is essentially zero *and* (ii) the Hessian is negative definite (all eigenvalues of $-\mathbf{H}$ comfortably positive). Cosmetic convergence is stopping at a flat ridge (near-zero eigenvalue → non-identification) or a saddle (indefinite Hessian → not a maximum at all, Chapter 2 §2.7). Your PDF's "convergenza vera o cosmetica" is exactly the gradient-*and*-curvature check.

Global vs local maxima. If the log-likelihood is **globally concave** (SBDC probit, plain logit — Chapter 2 §2.8), any convergence point is *the* global maximum, and starting values do not matter. If it is **not** concave — the **mixed logit** and **MDCEV** simulated likelihoods (Chapters 9, 23, 41) — there can be multiple local maxima, and the optimizer may land on a suboptimal one depending on starting values. This is why your PDF elevates "il massimo è globale?" and "le stime cambiano molto con starting values diversi?" to first-order concerns for your EC-MXL: the fix is multiple starting values and comparing log-likelihoods (Chapter 13).

Parameter constraints. Some parameters must obey bounds — a standard deviation > 0, a cost coefficient < 0, a WTP > 0 in a constrained-distribution mixed logit (Chapter 24). The optimizer must respect these (via reparameterization, e.g. estimating $\log \sigma$, or bounded optimization), and a maximum *on* a boundary changes the asymptotics (the normal approximation can fail at a constrained boundary). Your PDF's "i vincoli sui parametri sono rispettati?" is this concern.

These four warnings are not separate topics; they are all the question *"is the surface I climbed well-behaved enough that its peak and curvature mean what I think?"* — the spine question of the chapter. Chapter 13 turns each into an operational diagnostic; here you have the conceptual apparatus to know *what* you are checking and *why*.

> **Stop and think 11.4.** Your EC-MXL "converges" and reports finite standard errors, but re-estimating from three different starting values gives three different log-likelihood values and materially different WTPs. What are the two distinct problems this could indicate, and how does the framework distinguish them? *(Answer: (i)* ***multiple local maxima*** *— a non-concave simulated likelihood, so different starts climb different hills; the fix is to take the start with the highest log-likelihood and confirm it replicates. (ii)* ***near-non-identification / simulation noise*** *— a flat ridge (singular information) or too-few draws making the surface jittery, so the "maxima" are ill-defined; the framework distinguishes them by checking the Hessian's eigenvalues (a near-zero eigenvalue → identification/flatness problem) versus the log-likelihood values (genuinely different peaks → local-maxima problem) and by increasing draws, Chapters 9, 13.)*

11.10 Failure modes

F1. Treating convergence as proof of a valid model. A zero gradient is necessary, not sufficient. *Fix:* also check the Hessian is negative definite (all eigenvalues of $-\mathbf{H}$ positive) and the model is identified (§11.5, §11.9; Chapter 13).

F2. Trusting classical SEs when the information equality fails. Using $[-\mathbf{H}]^{-1}$ under dependence or misspecification. *Symptom:* observed information and outer-product-of-scores disagree; SEs implausibly small. *Fix:* robust/clustered (sandwich) standard errors (§11.5; Chapters 17, 19).

F3. Ignoring non-identification (flat directions). A ridge in the likelihood → singular information → huge/undefined SEs mistaken for a data-quantity problem. *Fix:* inspect eigenvalues of $-\mathbf{H}$; identify the flat combination; fix the specification/design (§11.9; Chapter 1 §1.10).

F4. Local maxima in non-concave likelihoods. Reporting a single converged run of a mixed logit / MDCEV as definitive. *Fix:* multiple starting values; compare log-likelihoods; keep the highest and confirm replication (§11.9; Chapter 13).

F5. Over-trusting asymptotic normality in small samples. Using normal CIs and χ^2 tests in tiny cells where the MLE's finite-sample distribution is skewed. *Fix:* bootstrap/simulation/permutation (Chapters 8, 9, 10); report finite-sample caveats.

F6. Forgetting the MLE is biased in finite samples. Treating a consistent MLE (WTP ratio, variance parameters) as unbiased. *Fix:* recognize finite-sample bias; prefer robust summaries and bias-aware reporting (Chapter 6, §6.3).

F7. Boundary/constraint mishandling. Estimating an unconstrained standard deviation (going negative) or ignoring a maximum on a boundary. *Fix:* reparameterize (estimate $\log\sigma$), use bounded optimization, and treat boundary maxima's inference carefully (§11.9).

F8. Misreading raw log-likelihood values. Interpreting the level of ℓ (rather than differences) as meaningful. *Fix:* only *differences* in ℓ are interpretable (as likelihood ratios / for model comparison, Chapter 12, 23).

11.11 Diagnostic questions for your own papers

1. **Can I write the likelihood?** For each model, can I write its likelihood and log-likelihood from the model's probability of what each respondent did? *(§11.2, §11.8.)*
2. **Real convergence?** Is the gradient essentially zero *and* the Hessian negative definite (eigenvalues of $-\mathbf{H}$ all positive)? *(§11.5, §11.9, F1.)*
3. **Information equality.** Do the observed information and the outer product of scores agree? If not (dependence/misspecification), am I using robust/clustered SEs? *(§11.5, F2.)*
4. **Identification.** Is the information matrix well away from singular — no flat directions / near-zero eigenvalues? Are all parameters separately identified? *(§11.9, F3.)*
5. **Global maximum.** For non-concave models (mixed logit, MDCEV), did I use multiple starting values and compare log-likelihoods? *(§11.9, F4.)*
6. **WTP legitimacy.** Am I recovering WTP by invariance ($g(\hat{\theta})$) with a delta-method (or simulation) SE, and is the denominator coefficient strong enough? *(§11.6, §11.8, F6.)*
7. **Constraints.** Are parameter bounds (positive σ, negative cost coefficient) enforced, and is any boundary maximum handled correctly? *(§11.9, F7.)*

11.12 Exercises

(A) Comprehension.

A1. Explain the difference between a density $f(y;\theta)$ and the likelihood $L(\theta)$ — same formula, different reading — and state the likelihood principle in one sentence.

A2. Why is the log-likelihood used instead of the likelihood? Give three distinct reasons.

A3. Your SBDC probit converges with a gradient of essentially zero, but the smallest eigenvalue of the negative Hessian is 2×10^{-8}. Is this a valid, well-identified maximum? What does the eigenvalue tell you?

(B) Derivation.

B1. Derive the zero-mean-score property $\mathbb{E}_\theta[\mathbf{s}(\theta)] = \mathbf{0}$ from the fact that a density integrates to one, and explain how it delivers consistency of the MLE.

B2. Derive the information matrix equality $\mathrm{Var}[\mathbf{s}] = -\mathbb{E}[\mathbf{H}]$, stating the regularity conditions. Then explain, in terms of this identity, why robust standard errors differ from classical ones under misspecification.

B3. From scratch, derive the SBDC probit log-likelihood, its score (in generalized-residual form), and the recovery of mean WTP $= -\alpha/\beta$ from the index coefficients, stating the reparameterization $\alpha = \mu/\sigma$, $\beta = -1/\sigma$.

(C) Research application / critique.

C1. A co-author reports a mixed-logit result from a single estimation run and says "it converged, so we're done." Write a critique listing the specific likelihood-surface checks you would insist on before trusting the result, and why each matters.

C2. In a model with 8 choice tasks per respondent estimated as an independent logit, the classical standard errors seem "too good" (very small). Explain, using the information matrix equality, why they are likely wrong and what you would report instead.

C3. Explain to a co-author why your SBDC mean WTP of $-\hat{\alpha}/\hat{\beta}$ is a legitimate maximum-likelihood estimate and how you obtain its confidence interval, and note the one condition under which you would distrust the delta-method interval.

11.13 Solutions

A1. The density $f(y;\theta)$ describes the probability of data y for a *fixed* parameter θ (a function of y); the likelihood $L(\theta)$ is the *same formula* read as a function of θ with the data fixed at their observed values. The likelihood principle: all the evidence the data provide about θ is contained in the likelihood function.

A2. (i) Products become sums ($\log \prod = \sum \log$), which are easier to differentiate and to apply the LLN/CLT to; (ii) numerical stability — a product of many probabilities underflows to zero, a sum of logs does not; (iii) same maximizer — log is strictly increasing, so maximizing ℓ maximizes L.

A3. No — it is a *cosmetic* convergence. A near-zero eigenvalue of the negative Hessian means the log-likelihood is nearly *flat* in one direction (a ridge), so the information matrix is near-singular, some parameter combination is nearly non-identified, and its standard error is near-infinite. The zero gradient alone is not enough; the curvature check (negative *definite*, all eigenvalues comfortably positive) fails, signalling an identification problem to diagnose (§11.9).

B1. Since $\int f(y;\theta)dy = 1$ for all θ, differentiating gives $\int \nabla f \, dy = 0$; using $\nabla f = f \nabla \log f = f\mathbf{s}$, this is $\int \mathbf{s}\, f \, dy = \mathbb{E}_\theta[\mathbf{s}] = 0$. Consistency: the MLE solves $\frac{1}{n}\sum \mathbf{s}_i(\hat{\theta}) = 0$; by the LLN this sample average converges to $\mathbb{E}[\mathbf{s}(\theta)]$, whose root is θ_0; so $\hat{\theta} \xrightarrow{p} \theta_0$ (given identification — a unique root).

B2. Regularity: θ-independent support, and interchangeability of ∇ and $\int$. Differentiate the identity $\int \mathbf{s} f\, dy = 0$ w.r.t. θ': $\int (\mathbf{H} f + \mathbf{s}\nabla f')dy = 0$; with $\nabla f' = f\mathbf{s}'$, this is $\mathbb{E}[\mathbf{H}] + \mathbb{E}[\mathbf{s}\mathbf{s}'] = 0$, so $\mathrm{Var}[\mathbf{s}] = \mathbb{E}[\mathbf{s}\mathbf{s}'] = -\mathbb{E}[\mathbf{H}]$. Under misspecification the model's f is not the true density, so the derivation's steps (which assumed f is the truth) fail: $\mathrm{Var}[\mathbf{s}] \neq -\mathbb{E}[\mathbf{H}]$, and the asymptotic variance is the sandwich $\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$ with $\mathbf{A} = -\mathbb{E}[\mathbf{H}] \neq \mathbf{B} = \mathrm{Var}[\mathbf{s}]$; robust SEs use this sandwich, classical SEs (wrongly) use $\mathbf{A}^{-1}$ only.

B3. With μ, σ and "yes" iff WTP$\geq b$: $\Pr(\text{yes}) = \Phi((\mu - b)/\sigma) = \Phi(\alpha + \beta b)$ with $\alpha = \mu/\sigma$, $\beta = -1/\sigma$. Log-likelihood $\ell = \sum_i [y_i \log \Phi_i + (1 - y_i)\log(1 - \Phi_i)]$, $\Phi_i = \Phi(\alpha + \beta b_i)$. Score $\mathbf{s} = \sum_i [y_i \phi_i/\Phi_i - (1 - y_i)\phi_i/(1 - \Phi_i)]\mathbf{x}_i = \sum_i g_i \mathbf{x}_i$ with $\mathbf{x}_i = (1, b_i)'$ and g_i the generalized residual. WTP: $\mu = -\alpha/\beta$ (from $\alpha = \mu/\sigma$, $\beta = -1/\sigma$), so $\widehat{\mathrm{WTP}} = -\hat{\alpha}/\hat{\beta}$ (mean = median for normal WTP).

C1. Checks before trusting a single mixed-logit run: (i) **multiple starting values** — because the simulated likelihood is non-concave, confirm the run reached the *global* max by re-estimating from several starts and taking the highest log-likelihood; (ii) **gradient and Hessian at the optimum** — gradient ≈ 0 *and* $-\mathbf{H}$ negative definite (all eigenvalues positive), ruling out saddles/ridges; (iii) **identification** — no near-zero eigenvalue (flat direction); (iv) **draw robustness** — re-estimate with more/other draws and seeds so the result is not simulation noise (Chapter 9); (v) **information-equality/robust SEs** — cluster on respondents given repeated tasks. "It converged" addresses none of these; each guards a distinct way the surface can betray you.

C2. With 8 tasks per respondent, the observations are dependent (shared respondent-level tastes/scale), so the true variance of the score exceeds the negative expected Hessian under the independence assumption — the information matrix equality fails, and the classical SEs $[-\mathbf{H}]^{-1}$ understate uncertainty (they count 8 correlated tasks as 8 independent observations). Report **respondent-clustered** (or panel/mixed) standard errors — the sandwich form — which restore honesty by using the true score variance $\mathbf{B}$ (Chapters 17, 19).

C3. By the invariance property, the MLE of any function of the parameters is that function of the MLE, so $-\hat{\alpha}/\hat{\beta}$ is the MLE of mean WTP $= -\alpha/\beta$. Its confidence interval comes from the delta method: $\mathrm{Var}(\widehat{\mathrm{WTP}}) \approx \nabla g' \mathbf{I}^{-1} \nabla g$ with $\nabla g = (-1/\hat{\beta},\ \hat{\alpha}/\hat{\beta}^2)'$, giving $\widehat{\mathrm{WTP}} \pm 1.96\,\mathrm{SE}$. I would distrust the (symmetric) delta-method interval if the bid coefficient $\hat{\beta}$ is weak/near zero, because then the $1/\hat{\beta}^2$ term makes WTP a near-Cauchy ratio with a skewed, heavy-tailed distribution — in which case I would report a Krinsky–Robb median and percentile interval instead (Chapters 6, 7, 9).

11.14 Spaced review prompts

+1 day. From memory, write the likelihood, log-likelihood, score, and Hessian in general, and derive the zero-mean-score property. State in one sentence how it gives consistency.

+1 week. Derive the information matrix equality and explain its three consequences (asymptotic variance; two agreeing variance estimators; robust SEs when it fails). Then re-derive the SBDC probit log-likelihood and score from scratch.

+1 month. For one of your own models, verify: gradient ≈ 0 and $-\mathbf{H}$ negative definite at the optimum; observed information vs outer-product-of-scores agreement (information equality); and, for a non-concave model, replication across starting values. Recover a WTP by invariance with its SE. Close with the spine sentence about whether the surface's peak and curvature mean what you think.

11.15 How this chapter connects to the rest of the book

- **Chapters 1, 2, 6, 7** supplied the pieces (Hessian/information, score/Hessian derivations, LLN/CLT template, Cramér–Rao); this chapter assembled them and added the zero-mean score and information matrix equality.

- **Chapter 12 (the testing trinity)** builds the Wald, likelihood-ratio, and score tests directly on the score, Hessian, and information of this chapter; **Chapter 13 (computing the MLE)** turns §11.9's warnings (convergence, local maxima, identification, constraints) into operational diagnostics.
- **Chapter 14 (latent variables/censoring)** and **Chapter 15 (CV likelihoods)** extend §11.8 to interval-censored (payment card) and other CV likelihoods, and to mean/median WTP under alternative distributions; **Chapters 18, 21–23 (binary response, logit, mixed logit, SML)** are all specific likelihoods estimated by this framework (the mixed logit a *simulated* one, Chapter 9).
- **Chapters 17, 19 (robust/clustered/panel inference)** are the information-equality-failure fix of §11.5; **Chapter 41 (MDCEV likelihood)** is a Kuhn–Tucker likelihood built on this engine; **Chapter 47 (meta-regression)** and **Chapter 52 (computational diagnostics)** apply the estimation and diagnostic apparatus throughout.

This is the keystone of Part III and, per the architecture, one of the three highest-dependency chapters in the book: master the likelihood surface — its peak (score), its curvature (information), and its pathologies (identification, local maxima) — and the estimation of every model you use becomes an application of what is here.

11.16 Research translation task

For your **SBDC contingent valuation model** (and, if ready, one choice model), produce a three-page note that *is* your Block 1 deliverable:

1. **Write the likelihood from scratch.** State the model (latent normal WTP, yes iff WTP $\geq$ bid), derive the choice probability $\Phi(\alpha + \beta b)$, and write the log-likelihood, score (generalized-residual form), and the recovery WTP $= -\alpha/\beta$, $\sigma = -1/\beta$.
2. **Estimate and verify the first-order and second-order conditions.** Confirm the gradient is ≈ 0 at the optimum and $-\mathbf{H}$ is negative definite; report the eigenvalues.
3. **Check the information matrix equality.** Compare the observed information $[-\mathbf{H}]$ to the outer product of scores; if they disagree (or you have repeated observations), switch to robust/clustered standard errors and explain why.
4. **Recover WTP by invariance with an SE.** Compute $-\hat{\alpha}/\hat{\beta}$, its delta-method SE, and — if the bid coefficient is not strong — a Krinsky–Robb median and percentile interval; judge which to report.
5. **Interrogate the surface.** State whether the likelihood is concave (SBDC probit: yes), whether the parameters are identified, and (for any non-concave model) whether you confirmed the global maximum across starting values.
6. **Close** with the spine sentence: *"My SBDC model climbs a [concave/non-concave] log-likelihood; its peak identifies [WTP] under [normality/identification]; its curvature reports precision [SE], and I have checked convergence, identification, and the information equality."*

Carry this into Chapter 12 (tests on these parameters), Chapter 13 (computational diagnostics), and Chapter 15 (the full CV likelihood suite, including the payment card).

11.17 Bibliography

Core. - Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, the maximum-likelihood chapters (score, information, asymptotics, the information matrix equality, and robust "sandwich" inference). The single best econometric reference for this chapter. *(Verify edition/chapters.)* - Casella & Berger, *Statistical Inference*, chapters on maximum likelihood, the score, Fisher information, and the Cramér–Rao bound. The rigorous statistical backing for §11.4–§11.6. *(Verify edition.)* - Greene, *Econometric Analysis*, the estimation chapters and the appendix on maximum likelihood; the SBDC/probit likelihood of §11.8 in econometric context, and the classical/robust variance distinction. *(Verify edition.)*

Applied. - Haab & McConnell, *Valuing Environmental and Natural Resources*. The SBDC/probit likelihood, WTP recovery ($-\alpha/\beta$), and the mean/median WTP issues of §11.8 in exactly your contingent-valuation setting. *(Verify edition.)* - Train, *Discrete Choice Methods with Simulation*, the maximum-likelihood and simulated-likelihood chapters — the framework applied to logit and mixed logit, and the local-maxima/identification concerns of §11.9. Freely available — *verify link.* - Cameron & Trivedi, *Microeconometrics: Methods and Applications*, on ML estimation, the information matrix equality, and robust inference — bridges §11.5 to practice. *(Verify edition.)*

Advanced. - White, *Estimation, Inference and Specification Analysis*. The definitive treatment of quasi-maximum-likelihood, the information matrix equality and its failure, and the sandwich estimator — the theory behind §11.5's robust-SE discussion. *(Verify edition.)* - Newey & McFadden, "Large Sample Estimation and Hypothesis Testing," in the *Handbook of Econometrics*. The rigorous, general treatment of M-estimation and ML asymptotics assembled in §11.6. *(Verify volume/year.)* - Lehmann & Casella, *Theory of Point Estimation*, for the decision-theoretic foundations of efficiency and the Cramér–Rao bound. *(Verify edition.)*

Entries marked "verify" are pointers; confirm editions, chapters, and the Newey–McFadden citation before using them, per the book's no-unverified-citation rule.

End of Chapter 11. Next in Part III: Chapter 12 — The Testing Trinity (Wald, likelihood-ratio, and score tests; constrained versus unconstrained likelihoods; when the three disagree and why), which builds the hypothesis tests for your models directly on the score, Hessian, and information assembled here. Tell me when to write it, or request a different chapter.

Chapter 12 — The Testing Trinity

Part III: Maximum Likelihood and Latent Variable Models

> **The spine question, applied to this chapter.** Once you have a likelihood, you need to test hypotheses on it: is this coefficient zero? does adding random parameters improve the mixed logit? is there preference heterogeneity? All of these are answered by one of three tests built from the score, Hessian, and information of Chapter 11 — the **Wald**, **likelihood-ratio (LR)**, and **score (Lagrange multiplier)** tests. They measure the same thing three ways, agree asymptotically, and differ in finite samples — and one of them (Wald) can give different answers for the *same* hypothesis depending on how you write it. Most dangerously for your work, the standard theory *breaks* when you test whether a variance is zero — exactly the test that decides whether your DCE needs random parameters. The spine question here: *when I test a restriction on my model, which of the three tests am I using, do its assumptions hold (in particular, am I testing on a boundary?), and would a different test — or the correct null distribution — change my conclusion?*

12.0 Conceptual motivation: three tests, one gap, and a trap

Your PDF places the testing trinity under "priorità altissima" alongside the likelihood itself: *"Wald test; likelihood ratio test; score test; constrained vs unconstrained likelihood."* These are not interchangeable formalities — they are the tools by which you decide what belongs in your models, and the choice among them (and the correct use of each) shapes real conclusions in your work:

- **Is a coefficient zero?** The t-statistic you read off every table is (its square is) a **Wald test**. Testing whether the cost-vector effect, the familiarity interaction, or a treatment dummy is zero is a Wald test.
- **Does a richer model fit significantly better?** Deciding whether your **mixed logit** beats the **MNL**, whether an **error component** improves the model, or whether adding random parameters is justified, is a **likelihood-ratio test** comparing nested models — the core model-comparison tool your PDF lists under DCE (*"model comparison: AIC, BIC, log-likelihood e LR tests"*).
- **Should I add a feature I have not estimated yet?** Testing whether the data call for heteroskedasticity, or whether a parameter should be random, *before* fitting the richer model, is a **score test** — it asks whether the restricted model's likelihood is still "pulling" toward the richer one.

And there is a trap that sits precisely at the center of your DCE modeling. Testing whether a **variance parameter is zero** — "is there preference heterogeneity?", "does this random parameter have non-zero spread?" — is a test *on the boundary* of the parameter space (a variance cannot be negative). The standard chi-square theory that everyone applies by default is **wrong** here, and applying it makes you *understate* the evidence for heterogeneity: you can conclude "no random parameter needed" when the data actually support one. This chapter derives all three tests, explains their geometry and why they disagree, tells you which to use when, and — with a verified example — shows exactly how the boundary problem would make you miss heterogeneity in your mixed logit if you used the naive test.

We build: the constrained-vs-unconstrained framing; the three test statistics and their derivations; the

geometry (three views of one gap); their common χ^2_q asymptotics and asymptotic equivalence; why they disagree in finite samples and the Wald test's notorious non-invariance; the practical choice among them; nested model comparison (MNL vs mixed logit) and the AIC/BIC alternatives for non-nested models; and the boundary problem for variance tests. Everything is built on Chapter 11's score, Hessian, and information — this chapter is what those objects are *for*.

12.1 Prerequisites

From **Chapter 11**: the log-likelihood $\ell(\theta)$, the score $\mathbf{s}(\theta) = \nabla\ell$, the Hessian $\mathbf{H}$, the Fisher information $\mathbf{I} = -\mathbb{E}[\mathbf{H}]$, the information matrix equality, and the MLE's asymptotic normality $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{I}_1^{-1})$. From **Chapter 7**: p-values, the z-test, and the delta method (§7.7). From **Chapter 1**: quadratic forms (§1.6) and definiteness. From **Chapter 5**: the χ^2 distribution (a sum of squared normals — the reference distribution for all three tests).

Same **notation**. New symbols: $\hat{\theta}$ (unrestricted MLE), $\tilde{\theta}$ (restricted MLE, maximizing ℓ subject to the null restriction), $R(\theta) = \mathbf{0}$ (the q restrictions the null imposes), q (number of restrictions = degrees of freedom), W (Wald), LR (likelihood ratio), LM (score / Lagrange multiplier).

12.2 The testing problem: constrained versus unconstrained likelihoods

The setup. You have a model with parameter vector θ (dimension k) and a null hypothesis that imposes q **restrictions**:

$$H_0 : R(\theta) = \mathbf{0} \qquad (q \text{ equations}),$$

against the unrestricted alternative. The simplest case is "a coefficient is zero" ($H_0 : \theta_j = 0$, one restriction); a common case is "these several coefficients are all zero" (q restrictions); the restrictions can also be nonlinear (a ratio equals a value).

Nested models. The null corresponds to a **restricted model** nested inside the **unrestricted model**: the restricted model is what you get by imposing $R(\theta) = \mathbf{0}$. Examples in your work:

- MNL is the mixed logit with all random-parameter variances set to zero — MNL is *nested* in the mixed logit.
- A model without an error component is nested in one with it (the error-component variance = 0).
- A model that pools across arms is nested in one that lets a coefficient differ by arm (the difference = 0).

Two likelihoods. Testing H_0 compares two maximizations: - the **unrestricted** log-likelihood maximum $\ell(\hat{\theta})$, obtained by maximizing freely; - the **restricted** log-likelihood maximum $\ell(\tilde{\theta})$, obtained by maximizing subject to $R(\theta) = \mathbf{0}$.

Because the restricted model is a special case of the unrestricted, imposing the restriction can never *increase* the maximized likelihood: $\ell(\tilde{\theta}) \leq \ell(\hat{\theta})$. The whole question is whether the restriction hurts fit by *more than sampling noise would explain*. The three tests measure this "hurt" from three vantage points — and, crucially, they need different pieces: Wald needs only the unrestricted fit, score needs only the restricted fit, LR needs both.

> **Active recall 12.A.** Your MNL has log-likelihood -1300 and your mixed logit (adding two random-parameter variances) has -1285. Which is $\ell(\hat{\theta})$ and which is $\ell(\tilde{\theta})$, and how many restrictions does the null "MNL is adequate" impose? *(Answer: the mixed logit (-1285, higher) is the unrestricted $\ell(\hat{\theta})$; the MNL (-1300) is the restricted $\ell(\tilde{\theta})$ with the random-parameter variances*

set to zero. The null imposes $q = 2$ restrictions — but note this is a boundary* test, §12.9, so the naive χ^2_2 is wrong.)*

12.3 The three test statistics

Each test converts the "hurt from the restriction" into a single number that is compared to a χ^2_q distribution. Here are the three, built from Chapter 11's objects.

Definition 12.1 (Likelihood-ratio test). Estimate *both* models and compare the maximized log-likelihoods:

$$LR = 2[\ell(\hat{\theta}) - \ell(\tilde{\theta})].$$

This is (twice) the vertical gap between the unrestricted and restricted likelihood peaks. Large LR means the restriction costs a lot of fit → reject H_0. Requires estimating both models.

Definition 12.2 (Wald test). Estimate only the *unrestricted* model, and measure how far the unrestricted estimate is from satisfying the restriction, scaled by its precision:

$$W = R(\hat{\theta})' \left[\widehat{\mathrm{Var}}(R(\hat{\theta}))\right]^{-1} R(\hat{\theta}),$$

where $\widehat{\mathrm{Var}}(R(\hat{\theta})) = \mathbf{G}\,\hat{\mathbf{V}}\,\mathbf{G}'$ with $\mathbf{G} = \partial R/\partial\theta$ (the delta method, Chapter 7 §7.7) and $\hat{\mathbf{V}} = [-\mathbf{H}(\hat{\theta})]^{-1}$. Large W means the free estimate is far from the null → reject. For the single restriction $H_0 : \theta_j = 0$, this is exactly

$$W = \frac{\hat{\theta}_j^2}{\widehat{\mathrm{Var}}(\hat{\theta}_j)} = \left(\frac{\hat{\theta}_j}{\mathrm{SE}(\hat{\theta}_j)}\right)^2 = t^2,$$

the squared t-statistic — so every t-test you run is a one-restriction Wald test. Requires only the unrestricted model.

Definition 12.3 (Score / Lagrange-multiplier test). Estimate only the *restricted* model, and measure how steep the log-likelihood still is at the restricted estimate — because if the restriction were true, the score there should be near zero:

$$LM = \mathbf{s}(\tilde{\theta})' \left[\mathbf{I}(\tilde{\theta})\right]^{-1} \mathbf{s}(\tilde{\theta}),$$

a quadratic form in the (full-model) score evaluated at the restricted estimate. Large LM means the restricted point still has a strong "pull" toward the unrestricted model → reject. It is called the **Lagrange-multiplier** test because the score at the constrained optimum is proportional to the multiplier on the constraint (Chapter 3, §3.4) — the multiplier measures how much the constraint binds. Requires only the restricted model.

All three, under H_0 and regularity conditions, converge in distribution to χ^2_q, where q is the number of restrictions. Reject at level α if the statistic exceeds the χ^2_q critical value (e.g., 3.84 for $q = 1$ at 5%).

> **Active recall 12.B.** You have already estimated your full mixed logit and want to test whether one coefficient is zero. Which test is essentially free (needs no further estimation), and what is it equal to? *(Answer: the Wald test — it uses only the unrestricted (already-estimated) model and its covariance, and for a single coefficient it equals the squared t-statistic $(\hat{\theta}_j/SE)^2$. No re-estimation needed.)*

12.4 The geometry: three views of one gap

The three tests are not arbitrary; they are three ways of measuring the *same* distance between the unrestricted peak and the null, and seeing this makes the whole trinity intuitive. Picture the log-likelihood as a function of a single parameter θ_j being tested against the null value $\theta_j = 0$. The MLE $\hat{\theta}_j$ sits at the peak; the null sits at 0.

- **Wald measures the horizontal distance.** How far is the peak $\hat{\theta}_j$ from the null 0, in standard-error units? $W = (\hat{\theta}_j/\text{SE})^2$. It uses the curvature *at the peak* (the unrestricted Hessian) to convert distance into a standardized statistic.
- **The likelihood ratio measures the vertical distance.** How much lower is the log-likelihood at the null than at the peak? $LR = 2[\ell(\hat{\theta}_j) - \ell(0)]$ (with other parameters re-optimized at the null). It uses the *actual* likelihood values along the way — no local approximation.
- **The score measures the slope at the null.** How steep is the log-likelihood at $\theta_j = 0$? If 0 is far from the peak, the likelihood is still climbing steeply there. LM standardizes this slope by the curvature *at the null* (the restricted information).

For a **perfectly quadratic** (Gaussian) log-likelihood — a parabola — these three measurements are *exactly* the same: horizontal distance, vertical drop, and slope-at-a-point all encode the same parabola. That is why the three tests are asymptotically equivalent (the log-likelihood is asymptotically quadratic near the peak, by the Taylor expansion of Chapter 11 §11.6). They **differ** precisely to the extent the log-likelihood is *not* quadratic — because then the curvature at the peak (Wald), the curvature at the null (score), and the true drop (LR) are genuinely different quantities.

A derivation sketch tying them together. Taylor-expand the restricted log-likelihood around the unrestricted MLE, using $\mathbf{s}(\hat{\theta}) = \mathbf{0}$ (Chapter 11):

$$\ell(\tilde{\theta}) \approx \ell(\hat{\theta}) + \tfrac{1}{2}(\tilde{\theta} - \hat{\theta})'\mathbf{H}(\hat{\theta})(\tilde{\theta} - \hat{\theta}),$$

so $LR = 2[\ell(\hat{\theta}) - \ell(\tilde{\theta})] \approx (\tilde{\theta} - \hat{\theta})'[-\mathbf{H}](\tilde{\theta} - \hat{\theta})$ — a quadratic form in the information matrix, which is exactly the form of the Wald statistic (measuring $\hat{\theta} - \tilde{\theta}$ scaled by curvature). Under H_0, $\sqrt{n}(\hat{\theta} - \tilde{\theta})$ is asymptotically normal, so this quadratic form is asymptotically χ^2_q (a q-dimensional standardized normal squared). The same expansion from the restricted side yields the score statistic. All three are the same quadratic form evaluated slightly differently.

> **Stop and think 12.1.** For a symmetric, bell-shaped (near-quadratic) log-likelihood, the three tests nearly coincide. Predict in which situation they would diverge most, and which one you would trust. *(Answer: they diverge most when the log-likelihood is strongly* non-quadratic* near the peak — skewed or with changing curvature, as with a weak/near-boundary parameter or a small sample. The LR test uses the actual likelihood values rather than a single local curvature, is invariant to parameterization (§12.6), and is generally the most reliable when they disagree.)*

12.5 Common asymptotics and asymptotic equivalence

Proposition 12.1 (Common null distribution). Under H_0 and standard regularity conditions (the true parameter is in the *interior* of the parameter space — this caveat is the boundary problem, §12.9), each of W, LR, and LM converges in distribution to χ^2_q, and the three are **asymptotically equivalent** (they differ by terms that vanish as $n \to \infty$).

Why χ^2_q. Each statistic is asymptotically a quadratic form $\mathbf{z}'\mathbf{z}$ where $\mathbf{z}$ is a q-vector of standardized asymptotically-normal quantities (the standardized restriction $R(\hat{\theta})$ for Wald; the standardized score for LM; the equivalent for LR). A sum of q squared independent standard normals is χ^2_q by definition (Chapter 5). The degrees of freedom equal the number of restrictions because that is the dimension of the space the restriction pins down.

What "asymptotically equivalent" does and does not promise. It promises the three give the *same answer in large samples* — so in a well-behaved model with plenty of data, it does not matter which you use, and you would choose on computational convenience (§12.7). It does *not* promise they agree in *finite* samples, and it does *not* protect you when the regularity conditions fail (boundary, §12.9). The equivalence is a large-sample statement; the disagreements of §12.6 are finite-sample realities.

Verified illustration. In a logit with $n = 2000$ testing whether a coefficient is zero (true effect present), the three statistics were $W = 127.6$, $LR = 140.8$, $LM = 136.6$ — all far above the χ^2_1 critical value of 3.84, all with p-values ≈ 0, all agreeing on the conclusion (reject) — yet **numerically distinct** (127.6 vs 140.8 vs 136.6). This is exactly what the theory predicts: same asymptotic distribution and conclusion, different finite-sample values. When the truth is clear-cut the differences are immaterial; near a decision boundary ($p \approx 0.05$) they can flip a conclusion, which is where §12.6 matters.

12.6 Why the three disagree, and the Wald test's fatal flaw

The three tests disagree in finite samples for two distinct reasons, and one of them is a genuine defect you must guard against.

Reason 1 — they use curvature at different points. Wald uses the Hessian at the *unrestricted* MLE $\hat{\theta}$; the score test uses information at the *restricted* MLE $\tilde{\theta}$; LR uses the actual likelihood values along the path between them. When the log-likelihood is not quadratic, the curvature genuinely differs between $\hat{\theta}$ and $\tilde{\theta}$, so the three statistics differ. In many standard models there is a known ordering ($W \geq LR \geq LM$ in the classical linear model), but this is not universal; the reliable summary is: *they differ when the likelihood is non-quadratic, and the difference matters near $p = 0.05$.*

Reason 2 — the Wald test is not invariant to reparameterization (its fatal flaw). This is the single most important practical fact about the trinity. Consider testing the *same* economic hypothesis written two ways: $H_0 : \theta = 1$ versus the equivalent $H_0 : \log\theta = 0$. These are *identical* hypotheses. The **LR** and **score** tests give the *same* answer for both (they depend on the likelihood, which does not care how you write the parameter). But the **Wald** test can give *different* answers, because its variance term $\mathbf{G}\hat{\mathbf{V}}\mathbf{G}'$ depends on the delta-method gradient $\mathbf{G} = \partial R/\partial\theta$, which differs between the two parameterizations. A hypothesis that is "significant" written one way can be "not significant" written another — a genuinely disturbing property for a test.

This bites in your work whenever you test a *nonlinear* restriction — for example, a hypothesis about a **WTP** (a ratio of coefficients), a variance, or a scale parameter. The Wald test of "WTP = WTP_0" (i.e., $-\beta_1/\beta_2 = \text{WTP}_0$) can depend on whether you write the restriction as $\beta_1 + \text{WTP}_0\beta_2 = 0$ (linear) or $-\beta_1/\beta_2 - \text{WTP}_0 = 0$ (nonlinear), and the two Wald statistics differ. **The lesson: for nonlinear restrictions, prefer the LR test (invariant), and be suspicious of a Wald test whose conclusion sits near the threshold.** This is also why, for WTP inference, simulation (Krinsky–Robb) and the LR test are more trustworthy than a Wald/delta-method test that inherits the ratio's parameterization sensitivity (Chapters 7, 9).

> **Stop and think 12.2.** You test whether WTP equals €50 two ways: as the linear restriction $\beta_1 + 50\beta_2 = 0$ and as the nonlinear restriction $-\beta_1/\beta_2 = 50$, using Wald each time, and get $p = 0.03$ and $p = 0.11$. What has happened, and how do you resolve it? *(Answer: the Wald test's non-invariance — the two algebraically equivalent restrictions give different delta-method variances, hence different Wald statistics and p-values, even though the hypothesis is identical. Resolve it by using the **LR test** (invariant to how the restriction is written) or a simulation-based comparison, and do not report a Wald result that flips with the parameterization.)*

12.7 Which test to use, and when

Given asymptotic equivalence, the choice among the three is driven by **computational cost** and **reliability**:

Test	Needs to estimate	Best used when	Watch out for
Wald	Unrestricted only	You already have the full model; testing a single coefficient (it's the t^2)	Non-invariance for nonlinear restrictions; unreliable near boundaries
LR	Both models	Comparing nested models (MNL vs mixed logit); nonlinear restrictions; when in doubt	Must fit both; boundary problem for variance tests
Score (LM)	Restricted only	Testing whether to *add* a feature you haven't fit (heteroskedasticity, a random parameter) without estimating the richer model	Boundary problem; needs the full-model score at the restricted fit

Practical guidance for your models.

- **Testing a coefficient is zero:** use the Wald/t-test — it is free and, for a single linear restriction, unproblematic.
- **Comparing nested models (does the mixed logit beat the MNL? does the error component help?):** use the **LR test** on $2[\ell_{\text{unrestricted}} - \ell_{\text{restricted}}]$ — but *with the boundary correction* when the restriction involves a variance (§12.9). This is your primary DCE model-comparison tool.
- **Testing nonlinear restrictions (a WTP value, a scale equality):** prefer the **LR test** for its invariance; distrust a borderline Wald result.
- **Screening whether a richer model is warranted before fitting it:** the **score test** lets you test, from the restricted model alone, whether the data call for the extra structure — efficient when the unrestricted model is expensive to estimate (a large mixed logit).
- **When the three disagree near $p = 0.05$:** trust the **LR test** (uses actual likelihood values, invariant), and treat the disagreement itself as a signal that the model is non-quadratic / weakly identified / near a boundary — investigate rather than pick the answer you like.

Active recall 12.C. You want to know whether your DCE needs a particular random parameter, but the full mixed logit is slow to estimate. Which test lets you check this *before* fitting the richer model, and what does it examine? *(Answer: the score (Lagrange-multiplier) test — estimate only the restricted model (without the random parameter) and examine whether the full-model score at that restricted fit is significantly non-zero, i.e. whether the likelihood is still "pulling" toward the richer model. It avoids estimating the expensive unrestricted model. Note it is still subject to the boundary problem for a variance, §12.9.)*

12.8 Nested model comparison and the non-nested alternatives (AIC, BIC)

The LR test compares **nested** models. Much of your model selection is exactly this: MNL $\subset$ mixed logit; no-error-component $\subset$ error-component; pooled $\subset$ arm-specific. For these, the LR test (with the boundary caveat) is the right tool.

But many comparisons are *non-nested* — neither model is a restriction of the other (a lognormal vs a normal random-coefficient distribution; two different nesting structures; different attribute specifications).

The LR test does not apply. For these you use **information criteria** that trade fit against complexity:

$$\text{AIC} = -2\ell(\hat{\theta}) + 2k, \qquad \text{BIC} = -2\ell(\hat{\theta}) + k \log n,$$

where k is the number of parameters and n the sample size. Both reward higher log-likelihood and penalize more parameters; **BIC penalizes complexity more heavily** (its penalty grows with $\log n$), so it favors more parsimonious models than AIC in large samples. Lower is better; there is no formal "significance" — you compare values and prefer the model with the smaller criterion, ideally by a non-trivial margin. Your PDF lists exactly these — *"AIC, BIC, log-likelihood e LR tests"* — as the DCE model-comparison suite: LR for nested, AIC/BIC for non-nested (and as a cross-check for nested).

A caution. AIC and BIC are guides, not proofs. They can disagree (AIC favoring the richer model, BIC the simpler), and neither substitutes for economic judgment about whether a specification is *interpretable* and *identified*. A model that wins on AIC but has an implausible sign, a near-boundary variance, or a fragile WTP is not the "best" model (Chapters 11, 13, 24). Report the criteria, but let identification and interpretability govern.

> **Stop and think 12.3.** You compare a mixed logit with normal random coefficients to one with lognormal random coefficients (same attributes, same number of parameters). Can you use an LR test? What do you use instead, and what does an equal parameter count imply for the comparison? *(Answer: no — the two are* non-nested* (neither is a restriction of the other), so the LR test does not apply. Use AIC/BIC (or out-of-sample fit). With equal k, the AIC/BIC penalties are equal, so the comparison reduces to which model has the higher log-likelihood — but also weigh interpretability, sign plausibility, and WTP behavior, not just fit.)*

12.9 The boundary problem: why the naive variance test is wrong

This is the section with the highest stakes for your DCE, and the one most often gotten wrong in the applied literature. The standard χ^2_q theory of §12.5 assumes the true parameter is in the **interior** of the parameter space. Testing whether a **variance is zero** violates this, because a variance cannot be negative — the null value ($\sigma^2 = 0$) sits *on the boundary* of the parameter space.

Why it matters. The tests that decide whether your mixed logit needs random parameters are boundary tests:

- "Is there preference heterogeneity in attribute j?" = "$H_0 : \sigma_j^2 = 0$."
- "Does this error component belong?" = "H_0 : its variance $= 0$."
- "Is the random-coefficient covariance non-trivial?" — restrictions on variance/Cholesky terms.

What goes wrong. Under $H_0 : \sigma^2 = 0$ on the boundary, the LR (and score) statistic does *not* have a χ^2_q distribution. For a single variance, the correct asymptotic null distribution is a **50:50 mixture** of a point mass at zero (χ^2_0) and a χ^2_1:

$$LR \xrightarrow{d} \tfrac{1}{2}\chi^2_0 + \tfrac{1}{2}\chi^2_1.$$

The intuition: because the estimated variance is constrained to be ≥ 0, about half the time (under the null) the unconstrained maximum would want a *negative* variance and is pushed to the boundary $\hat{\sigma}^2 = 0$, giving $LR = 0$; the other half it is positive, giving a χ^2_1-like value. Using the **naive** χ^2_1 reference — which everyone does by default — makes the test **conservative**: it demands too large a statistic, so it *understates* the evidence for heterogeneity and can lead you to wrongly conclude "no random parameter needed."

The critical values, quantified (verified). For a single-variance test at the 5% level:

$$\text{naive } \chi^2_1 \text{ critical value} = 3.84, \qquad \text{correct mixture critical value} = 2.71.$$

The correct value is $\chi^2_1(0.90) = 2.71$ because $\Pr(\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1 > c) = \frac{1}{2}\Pr(\chi^2_1 > c) = 0.05$ requires $\Pr(\chi^2_1 > c) = 0.10$, i.e. $c = 2.71$. **The consequence is decisive:** a test statistic of 3.0 is "not significant" against

the naive 3.84 (naive $p = 0.083$) but *significant* against the correct 2.71 (correct $p = 0.042$). Using the wrong reference, you would conclude "no preference heterogeneity" and estimate an MNL, when the data actually support a random parameter (verified). This is a real, common error that could change the specification — and the WTP distribution and welfare conclusions — of your DCE.

What to do. When an LR (or score) test involves a variance or other boundary parameter: (i) use the correct **mixture** null distribution — for one variance, halve the naive p-value (equivalently, compare to 2.71 not 3.84); (ii) for several variances the mixture is more complex (a mixture of $\chi^2_0, \ldots, \chi^2_q$ with binomial-type weights), so use the appropriate correction (software and the references in §12.16 provide these); (iii) cross-check with AIC/BIC and with the *estimated* variance and its plausibility. Report that you used the boundary-corrected test — it signals methodological care and can be the difference between finding and missing heterogeneity.

> **Active recall 12.D.** Why is the standard χ^2_1 test wrong when you test whether a random-parameter variance is zero, and what is the correct 5% critical value for one variance? *(Answer: because the null $\sigma^2 = 0$ is on the boundary of the parameter space (variances can't be negative), violating the interior-point regularity condition; the correct null distribution is $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$, giving a 5% critical value of 2.71, not 3.84. Using 3.84 is conservative and can make you miss real heterogeneity.)*

12.10 Failure modes

F1. Naive χ^2 for a variance/boundary test. Testing $\sigma^2 = 0$ (random parameter, error component) against χ^2_q. *Symptom:* concluding "no heterogeneity" from a borderline statistic. *Fix:* use the mixture distribution (2.71, not 3.84, for one variance); halve the p-value (§12.9).

F2. Trusting a borderline Wald test on a nonlinear restriction. A WTP or ratio hypothesis whose Wald verdict flips with the parameterization. *Fix:* use the LR test (invariant) or simulation; distrust near-threshold Wald results (§12.6).

F3. Applying an LR test to non-nested models. Comparing a lognormal vs normal mixing distribution, or two nesting structures, with $2\Delta\ell$. *Fix:* AIC/BIC or out-of-sample comparison, not LR (§12.8).

F4. Wrong degrees of freedom. Miscounting the number of restrictions q (e.g., forgetting that a random-coefficient covariance adds several parameters). *Fix:* q = (unrestricted params) − (restricted params); count carefully, including Cholesky terms.

F5. Comparing log-likelihoods across different samples/weights. LR and AIC/BIC are only comparable for models fit to the *same* data (same exclusions, same weighting). *Fix:* ensure identical estimation samples before comparing (Chapters 32; your PDF §6.6–6.7 on exclusions/weights).

F6. Reading AIC/BIC as significance tests. Treating a small AIC gap as "significant." *Fix:* AIC/BIC are relative guides, not tests; weigh identification and interpretability too (§12.8).

F7. Test disagreement ignored. Picking whichever of Wald/LR/LM gives the desired answer when they diverge. *Fix:* treat disagreement as a diagnostic of non-quadratic likelihood / weak identification / boundary; trust LR and investigate (§12.6).

F8. Ignoring dependence in the test. Using classical (non-clustered) test statistics with repeated choice tasks, inflating the statistic. *Fix:* robust/clustered versions of the tests (score and Wald have robust forms); cluster on respondents (Chapters 11 §11.5, 17).

12.11 Diagnostic questions for your own papers

1. **Which test, and why.** For each hypothesis, am I using Wald, LR, or score, and is the choice justified by computation and reliability (not convenience of the answer)? *(§12.7.)*
2. **Boundary?** Does the null set a variance (or other boundary parameter) to zero? If so, am I using the mixture distribution, not the naive χ^2? *(§12.9, F1.)*
3. **Nonlinear restriction?** For a WTP/ratio/scale hypothesis, am I using the invariant LR test rather than a parameterization-dependent Wald test? *(§12.6, F2.)*
4. **Nested or not?** Is my model comparison nested (LR valid) or non-nested (AIC/BIC)? *(§12.8, F3.)*
5. **Same sample?** Are the models I compare fit to identical data (exclusions, weights)? *(§12.5, F5.)*
6. **Degrees of freedom.** Have I counted the restrictions correctly, including covariance/Cholesky parameters? *(F4.)*
7. **Disagreement.** If the three tests disagree near $p = 0.05$, have I investigated why (non-quadratic likelihood, weak identification, boundary) rather than cherry-picking? *(§12.6, F7.)*

12.12 Exercises

(A) Comprehension.

A1. State what each of the three tests requires you to estimate (restricted, unrestricted, or both), and give one situation where each is the natural choice.

A2. Your logit reports a coefficient of 0.6 with SE 0.15. Compute the Wald statistic for $H_0 : \theta = 0$ and its degrees of freedom, and relate it to the t-statistic.

A3. Explain in one paragraph why a variance test ($H_0 : \sigma^2 = 0$) needs a different null distribution than the standard χ^2_1, and what the correct 5% critical value is.

(B) Derivation.

B1. Using a second-order Taylor expansion of the restricted log-likelihood around the unrestricted MLE (with $\mathbf{s}(\hat{\theta}) = \mathbf{0}$), show that $LR \approx (\tilde{\theta} - \hat{\theta})'[-\mathbf{H}](\tilde{\theta} - \hat{\theta})$, a quadratic form in the information, and explain why this is asymptotically χ^2_q.

B2. Show that for a single restriction $H_0 : \theta_j = 0$, the Wald statistic equals the squared t-statistic, and explain why the Wald test's value depends on the parameterization of a *nonlinear* restriction while the LR test's does not.

B3. Derive the boundary-corrected 5% critical value for a single-variance LR test from the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$, and show that the naive χ^2_1 test is conservative.

(C) Research application / critique.

C1. Your co-author tests whether your DCE needs a random price coefficient with an LR test, gets $LR = 3.1$, compares it to 3.84, and concludes "no heterogeneity, use the MNL." Write a critique and the corrected conclusion.

C2. You test whether WTP equals €80 with a Wald test and get $p = 0.04$, but a reviewer re-runs it with the restriction written differently and gets $p = 0.12$. Explain what happened and what test you should report.

C3. You want to compare (i) MNL vs mixed logit and (ii) mixed logit with normal vs lognormal price coefficient. Explain which comparison uses which tool (LR, AIC/BIC, boundary correction) and why, and what non-statistical considerations should also weigh in.

12.13 Solutions

A1. Wald requires only the *unrestricted* model (natural for testing a coefficient in an already-estimated model — it's the t^2); the score/LM requires only the *restricted* model (natural for testing whether to add a feature — e.g. a random parameter — without fitting the expensive richer model); LR requires *both* (natural for nested model comparison like MNL vs mixed logit, and for nonlinear restrictions where its invariance matters).

A2. $W = (0.6/0.15)^2 = 4^2 = 16$, with $q = 1$ degree of freedom. It is the square of the t-statistic ($t = 0.6/0.15 = 4$), and $16 > 3.84$, so reject $H_0 : \theta = 0$ at 5% (equivalently $|t| = 4 > 1.96$). The Wald test for one restriction *is* the squared t-test.

A3. The standard χ^2_1 derivation assumes the true parameter is interior; a variance is bounded below by zero, so under $H_0 : \sigma^2 = 0$ the null sits on the boundary and the unconstrained estimator is truncated at zero. Roughly half the time the free maximum would want a negative variance and is pushed to $\hat{\sigma}^2 = 0$ (giving $LR = 0$), and half the time it is positive (χ^2_1-like), so the correct null is $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$; the 5% critical value is 2.71 (not 3.84).

B1. Since $\mathbf{s}(\hat{\theta}) = 0$, $\ell(\tilde{\theta}) \approx \ell(\hat{\theta}) + \frac{1}{2}(\tilde{\theta} - \hat{\theta})'\mathbf{H}(\hat{\theta})(\tilde{\theta} - \hat{\theta})$, so $LR = 2[\ell(\hat{\theta}) - \ell(\tilde{\theta})] \approx -(\tilde{\theta} - \hat{\theta})'\mathbf{H}(\tilde{\theta} - \hat{\theta}) = (\tilde{\theta} - \hat{\theta})'[-\mathbf{H}](\tilde{\theta} - \hat{\theta})$. Under H_0, $\sqrt{n}(\hat{\theta} - \tilde{\theta})$ is asymptotically $N(\mathbf{0},\ \text{cov})$ with $-\mathbf{H}/n \to \mathbf{I}_1$, so the standardized quadratic form is a sum of q squared independent standard normals $= \chi^2_q$ (q = number of restrictions = dimension of the constrained subspace).

B2. For $H_0 : \theta_j = 0$, $R(\theta) = \theta_j$, $\mathbf{G} = \partial R/\partial\theta = \mathbf{e}_j$, so $\text{Var}(R(\hat{\theta})) = \mathbf{e}_j'\hat{\mathbf{V}}\mathbf{e}_j = \hat{V}_{jj}$ and $W = \hat{\theta}_j^2/\hat{V}_{jj} = (\hat{\theta}_j/\text{SE})^2 = t^2$. For a *nonlinear* restriction R, the variance $\mathbf{G}\hat{\mathbf{V}}\mathbf{G}'$ depends on $\mathbf{G} = \partial R/\partial\theta$, which differs between algebraically equivalent forms of the restriction (e.g. $\beta_1 + c\beta_2$ vs $-\beta_1/\beta_2 - c$) — so the Wald statistic differs. The LR test depends only on the maximized log-likelihoods, which are unchanged by rewriting the restriction, hence invariant.

B3. For $c > 0$, $\Pr(\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1 > c) = \frac{1}{2}\Pr(\chi^2_0 > c) + \frac{1}{2}\Pr(\chi^2_1 > c) = \frac{1}{2} \cdot 0 + \frac{1}{2}\Pr(\chi^2_1 > c)$. Setting this to 0.05 gives $\Pr(\chi^2_1 > c) = 0.10$, so $c = \chi^2_1(0.90) = 2.71$. The naive test compares to $\chi^2_1(0.95) = 3.84 > 2.71$, so it requires a larger statistic to reject — it is conservative, understating evidence for a non-zero variance.

C1. The conclusion is wrong because testing $\sigma^2_{\text{price}} = 0$ is a **boundary test**: the correct null distribution is $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$, so the 5% critical value is 2.71, not 3.84. With $LR = 3.1 > 2.71$ (boundary p $= \frac{1}{2}\Pr(\chi^2_1 > 3.1) \approx 0.039$), the test is **significant**: the data support a random price coefficient. Corrected conclusion: there *is* evidence of preference heterogeneity in the price coefficient; use the mixed logit (and cross-check with AIC/BIC and the estimated variance's plausibility). Using the naive χ^2_1 would have wrongly discarded real heterogeneity — with downstream consequences for the WTP distribution and welfare.

C2. The Wald test is not invariant to how a nonlinear restriction (WTP = €80, i.e. $-\beta_1/\beta_2 = 80$) is written: the two algebraically equivalent forms give different delta-method variances and hence different Wald statistics and p-values (0.04 vs 0.12), even though the hypothesis is identical (§12.6). Report the **LR test** of the restriction, which is invariant to the parameterization, and/or a simulation-based (Krinsky–Robb) comparison; do not report a Wald result that flips with how the restriction is expressed.

C3. (i) MNL vs mixed logit is **nested** (MNL = mixed logit with variances zero), so use an **LR test — with the boundary correction** (mixture distribution), because the restriction sets variances to zero; cross-check with AIC/BIC. (ii) Normal vs lognormal price coefficient is **non-nested** (neither is a restriction of the other), so the LR test does not apply — use **AIC/BIC** (or out-of-sample fit); with equal parameter counts this reduces to comparing log-likelihoods. Non-statistical considerations: sign and support plausibility (a lognormal forces the price coefficient's sign, which may be desirable or not), interpretability of the WTP distribution, identification and convergence stability, and whether the richer model's WTP is robust (Chapters 11, 13, 24) — the "best" model is not merely the one with the smallest criterion.

12.14 Spaced review prompts

+1 day. From memory, define the Wald, LR, and score statistics (what each estimates and its formula), state their common χ^2_q null distribution, and draw the three-views-of-one-gap geometry.

+1 week. Explain the Wald test's non-invariance with the WTP example, and derive the boundary-corrected 5% critical value (2.71) for a single-variance test. State which test to use for (i) a coefficient, (ii) MNL vs mixed logit, (iii) a WTP restriction.

+1 month. For your own DCE, run the nested model comparison (MNL vs your mixed logit) with the boundary-corrected LR test, and cross-check with AIC/BIC; for one nonlinear (WTP) hypothesis, use an LR or simulation test rather than Wald. Close with the spine sentence about which test you used and whether the boundary/parameterization issues changed any conclusion.

12.15 How this chapter connects to the rest of the book

- **Chapter 11** supplied the score, Hessian, and information that all three tests are built from; the LR test's derivation is the Taylor expansion of §11.6, and the boundary problem is a failure of the interior-point regularity condition of §11.6.
- **Chapter 13 (computing the MLE)** handles the estimation the tests require (restricted and unrestricted fits, convergence, local maxima) — a divergence among the three tests is often a symptom of the computational pathologies Chapter 13 diagnoses.
- **Chapters 22–23 (mixed logit, SML)** are where the nested LR test (MNL vs mixed logit) and the boundary correction of §12.9 are applied in practice, and where AIC/BIC (§12.8) select among specifications; **Chapter 24 (WTP space)** is where the Wald non-invariance of §12.6 makes LR/simulation the right tools for WTP hypotheses.
- **Chapter 8 (power/equivalence)** complements this chapter: these are *difference* tests, and a non-significant LR/Wald/score result is not evidence for the null (equivalence needs TOST); **Chapters 17, 19 (robust/clustered inference)** provide the cluster-robust versions of the tests for your repeated-observation data.
- **Chapter 47 (meta-regression)** uses LR/Wald tests and information criteria for moderator selection; **Chapter 41 (MDCEV)** uses the trinity to test restrictions on the Kuhn–Tucker likelihood.

This chapter completes the *inference* half of Part III's likelihood engine: Chapter 11 built the likelihood and its estimator, Chapter 12 tests hypotheses on it, and Chapter 13 makes both computationally trustworthy.

12.16 Research translation task

For your **DCE model selection**, produce a two-to-three-page note:

1. **Nested comparison with boundary correction.** Run the LR test of MNL vs your mixed logit (and of a model with vs without an error component); count the restrictions correctly, and apply the mixture (boundary) correction for any variance restriction — report both the naive and corrected p-values and note whether the correction changes the conclusion.
2. **Non-nested comparison.** For any non-nested choice (e.g., mixing-distribution shape, nesting structure), report AIC and BIC, and state your selection with its reasoning, including interpretability.
3. **A nonlinear (WTP) hypothesis.** For one WTP restriction, use an LR (or Krinsky–Robb simulation) test rather than a Wald test, and note whether a Wald test would have given a parameterization-dependent answer.
4. **Robustness.** Ensure all compared models are fit to the identical sample (same exclusions/weights), and use cluster-robust test versions given repeated tasks.

5. **Close** with the spine sentence: *"I selected [model] using [LR with boundary correction / AIC-BIC]; the variance/boundary and parameterization issues [did/did not] change the conclusion, and I used [LR/simulation] rather than Wald for the nonlinear WTP hypothesis."*

Carry this into Chapter 13 (the estimation and convergence behind the tests) and Chapters 22–24 (mixed logit selection and WTP-space inference).

12.17 Bibliography

Core. - Engle, "Wald, Likelihood Ratio, and Lagrange Multiplier Tests in Econometrics," in the *Handbook of Econometrics*. The classic, definitive treatment of the trinity — the geometry, the asymptotic equivalence, and the finite-sample ordering; the primary source for §12.3–§12.6. *(Verify volume/year.)* - Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, the chapters on ML hypothesis testing (Wald, LR, LM, and their robust forms). The econometric reference at the right level. *(Verify edition/chapters.)* - Greene, *Econometric Analysis*, the hypothesis-testing chapters, including the Wald non-invariance discussion and information criteria. *(Verify edition.)*

Applied. - Train, *Discrete Choice Methods with Simulation*, on model comparison for logit/mixed logit (LR tests, AIC/BIC, and testing for random parameters). The §12.8 model-selection material in your DCE setting. Freely available — *verify link.* - Hensher, Rose & Greene, *Applied Choice Analysis*, on practical model comparison, nested tests, and information criteria for choice models. *(Verify edition.)* - Self & Liang (1987), "Asymptotic Properties of Maximum Likelihood Estimators and Likelihood Ratio Tests Under Nonstandard Conditions," *JASA*. The authoritative treatment of the **boundary problem** and the mixture-of-chi-squares distribution of §12.9 — essential for variance/random-parameter tests. *(Verify volume/pages.)*

Advanced. - Andrews (2001), "Testing When a Parameter Is on the Boundary of the Maintained Hypothesis," *Econometrica*. The rigorous, general theory of boundary testing behind §12.9. *(Verify volume/year.)* - Godfrey, *Misspecification Tests in Econometrics*, for the score/LM test and its many applications (testing for omitted structure without estimating it). *(Verify edition.)* - Vuong (1989), "Likelihood Ratio Tests for Model Selection and Non-Nested Hypotheses," *Econometrica*, for a *test-based* approach to non-nested comparison beyond AIC/BIC (§12.8). *(Verify volume/year.)*

Entries marked "verify" are pointers; confirm volumes, pages, and the Engle / Self–Liang / Vuong citations before using them, per the book's no-unverified-citation rule.

End of Chapter 12. Next in Part III: Chapter 13 — Computing the MLE, and Distrusting It (convergence real vs cosmetic, local maxima, starting values, Hessian pathologies, separate identification, sensitivity to starting values), which turns the likelihood-surface warnings of Chapter 11 §11.9 and the test-disagreements of this chapter into the operational computational diagnostics your PDF calls "invisible but decisive." Tell me when to write it, or request a different chapter.

Chapter 13 — Computing the MLE, and Distrusting It

Part III: Maximum Likelihood and Latent Variable Models

> **The spine question, applied to this chapter.** Chapter 11 built the likelihood surface and Chapter 12 tested hypotheses on it — both assuming the optimizer actually found the peak. This chapter confronts the uncomfortable truth your PDF states plainly: *"Molti modelli choice avanzati producono output anche quando la specificazione è fragile."* The software prints coefficients, standard errors, and a reassuring "converged" even when it has stopped on a flat ridge, climbed the wrong hill, or reported a Hessian it cannot invert. Learning to tell a trustworthy fit from a plausible-looking fake is what your PDF calls *"una competenza invisibile ma decisiva."* The spine question here: *when my optimizer says "converged," did it find a real maximum (gradient zero and negative-definite Hessian), the global one, of an identified model — or did it merely stop, leaving me output I should not believe?*

13.0 Conceptual motivation: the optimizer stops; that is not the same as being right

Your PDF's maximum-likelihood section lists a set of questions that separate a careful empiricist from a careless one, and every one of them is answered in this chapter: *"la convergenza è vera o cosmetica? l'Hessiano è invertibile? il massimo è globale? i parametri sono identificati separatamente? i vincoli sui parametri sono rispettati? le stime cambiano molto con starting values diversi?"* Its whole section 5.4 (*Diagnostica computazionale*) is a checklist — convergence diagnostics, multiple starting values, gradient and Hessian checks, rescaling, detecting local maxima, comparing log-likelihoods — and it warns that this skill is *invisible* (nothing in the default output flags the problem) but *decisive* (it determines whether your results are real).

Why does this deserve a full chapter, when Chapter 11 already named these as properties of the likelihood surface (§11.9)? Because in practice you do not *see* the surface — you see an optimizer's report, and you must reconstruct the surface's health from that report and a few extra computations. The danger is acute precisely for *your* models:

- The **SBDC probit** and **plain logit** are globally concave (Chapter 2, §2.8), so their optimization is benign — one peak, no local maxima, starting values irrelevant. If your whole toolkit were these, this chapter would be short.
- But your **EC-MXL mixed logit** and your **MDCEV** are *not* concave, and they are estimated by *simulated* maximum likelihood (Chapter 9). Their log-likelihood surface can have **multiple local maxima**, **flat ridges** (weak identification), and **simulation-induced wiggles** — and the optimizer will happily stop at any of them and report success. Your PDF elevates "il massimo è globale?" and "le stime cambiano molto con starting values diversi?" to first-order concerns *because your central models*

live in this dangerous regime.

- And the single most consequential misreading your PDF names — *"standard errors enormi segnalano un problema strutturale, non solo pochi dati"* — is a computing-the-MLE issue: a huge standard error is often a **flat direction of the likelihood** (non-identification), not a shortage of data, and no amount of extra sample will fix it.

So this chapter is the craft of *interrogating* an estimation. We cover how optimizers actually climb the likelihood; what "converged" really means and how to tell real convergence from cosmetic; how to read the Hessian's eigenvalues to diagnose identification, precision, and validity at once; how to hunt the global maximum with multiple starting values; how to use starting-value sensitivity as a diagnostic; and — the synthesis — how to distinguish the three culprits behind bad output (non-identification, local maxima, simulation noise), which demand three different fixes. The full reproducible-pipeline treatment (seeds, draw counts, the whole workflow) is Chapter 52; this chapter is the MLE-specific core: knowing when the peak you were handed is real.

13.1 Prerequisites

From **Chapter 11**: the score $\mathbf{s}(\theta) = \nabla\ell$, the Hessian $\mathbf{H}$, the information matrix, the asymptotic covariance $[-\mathbf{H}(\widehat{\theta})]^{-1}$, and especially §11.9 (identification as a unique peak; convergence as gradient-*and*-curvature; global vs local maxima; boundary constraints). From **Chapter 2**: the first- and second-order conditions (§2.7) and concavity (§2.6). From **Chapter 1**: definiteness, eigenvalues, and matrix conditioning (§1.7, §1.10). From **Chapter 9**: simulation noise and the draws–seed sensitivity of simulated likelihoods. From **Chapter 12**: test disagreements as symptoms of a non-quadratic/ill-behaved surface.

Same **notation**. New symbols: θ_k (parameter value at optimizer iteration k), $\|\mathbf{s}\|$ (gradient norm), $\lambda_{\min}, \lambda_{\max}$ (smallest/largest eigenvalues of $-\mathbf{H}$), κ (condition number).

13.2 How optimizers find the MLE

You maximize $\ell(\theta)$ by **iterative hill-climbing**: start somewhere, repeatedly step uphill, stop when you can no longer improve. The steps use the score (which way is up) and, usually, the curvature (how far to step). Three algorithms you will meet:

Newton–Raphson. Uses both the gradient and the Hessian. The update solves the local quadratic approximation exactly:

$$\theta_{k+1} = \theta_k - \left[\mathbf{H}(\theta_k)\right]^{-1}\mathbf{s}(\theta_k).$$

Near a maximum (where $-\mathbf{H}$ is positive definite) this moves uphill and converges *quadratically* (very fast). But it requires the Hessian at every step, and if $\mathbf{H}$ is near-singular or not negative definite (away from a maximum, or on a bad surface), the step can be huge or point the wrong way.

BHHH (Berndt–Hall–Hall–Hausman). Replaces the Hessian by the outer product of scores $\sum_i \mathbf{s}_i\mathbf{s}_i'$ (the BHHH information estimate, Chapter 11 §11.5). This is always positive semidefinite (so the step always points uphill) and needs only first derivatives — cheaper and more robust than Newton, at the cost of slower convergence. Common in discrete-choice software.

BFGS / quasi-Newton. Builds an *approximation* to the (inverse) Hessian from the sequence of gradients, avoiding explicit second derivatives. Robust and widely used (Apollo, `maxLik`, `optim` default for many problems). A relative, L-BFGS, handles many parameters.

Why they fail. All of these assume a reasonably well-behaved surface. They struggle when: the surface is **non-concave** (multiple hills — they climb the nearest one); it is **flat** in some direction (they wander or stop prematurely); it is **badly scaled** (steps are unreliable — Chapter 1, §1.10); or it is **noisy** from simulation

(they chase jitter — Chapter 9, §9.4). The optimizer does not know it has failed; it simply hits a **stopping rule** and reports "converged." Understanding that "converged" means "a stopping rule triggered" — not "a valid maximum was found" — is the whole point of the next section.

> **Active recall 13.A.** Newton–Raphson uses the Hessian in its update step. What happens to the step if the Hessian is near-singular (a near-zero eigenvalue), and what does that tell you about the surface? *(Answer: the update multiplies by* $\mathbf{H}^{-1}$*, so a near-zero eigenvalue makes* $\mathbf{H}^{-1}$ *enormous in that direction and the step blows up (or the algorithm stalls); this signals a near-flat direction of the likelihood — weak or non-identification — the very pathology §13.4 diagnoses.)*

13.3 Convergence: real versus cosmetic

What "converged" means. An optimizer declares convergence when a **stopping criterion** is met — typically one of: the gradient norm $\|\mathbf{s}\|$ is below a tolerance; the change in parameters between iterations is tiny; or the change in the log-likelihood is tiny. Some of these are *good* signals (small gradient) and some are *ambiguous* (a tiny step can mean "at the peak" *or* "stuck on a flat ridge, unable to move"). The maximum-iterations cap also ends the run and may be reported as (non-)convergence. **"Converged" is a statement about the optimizer, not about the model.**

The definition of real convergence. A genuine, trustworthy maximum requires *all* of:

1. **The gradient (score) is essentially zero:** $\|\mathbf{s}(\widehat{\theta})\| \approx 0$ (the first-order condition, Chapter 2 §2.7). Check the *reported gradient*, not just the "converged" flag.
2. **The Hessian is negative definite:** all eigenvalues of $-\mathbf{H}(\widehat{\theta})$ are comfortably positive (the second-order condition). This rules out saddles (indefinite $\mathbf{H}$) and flat ridges (near-zero eigenvalue).
3. **It is not an artifact:** not simulation noise (for a simulated likelihood, stable across seeds/draws — Chapter 9), and not a spurious boundary stop (a parameter pinned at a bound — §13.7).
4. **It replicates:** re-starting the optimizer *from the reported optimum* does not move it, and starting from several points reaches the same log-likelihood (§13.5).

Cosmetic convergence is any "converged" report that fails one of these — most commonly, the optimizer took a step so small (on a flat or noisy region) that the step-size criterion triggered *even though the gradient is not really zero or the Hessian is singular.*

Worked Example 13.1 (Cosmetic convergence on flat tails, verified). Consider maximizing a bimodal objective with peaks near $\theta = 1$ (height 2) and $\theta = 4$ (height 3), whose tails are essentially flat (an exponential mixture). Running a standard optimizer (BFGS) from ten starting values across $[-2, 7]$:

- starts at $\theta = -2, -1, 7$ (in the flat tails) **"converge" to the starting point** with log-likelihood ≈ 0 — the gradient is near zero because the tail is *flat*, so the optimizer stops, but this is **not a maximum at all** (cosmetic convergence to a non-maximum);
- starts at $\theta = 0, 1, 2$ converge to the *local* max at $\theta = 1$ (log-lik 2);
- starts at $\theta = 3, 4, 5, 6$ converge to the *global* max at $\theta = 4$ (log-lik 3).

(Verified.) Two lessons: (i) a near-zero gradient alone does *not* certify a maximum — the flat-tail stops had near-zero gradients yet were nowhere near a peak, which is why you must also check the Hessian (there, not negative definite) and the log-likelihood *value*; (ii) the reported result depends entirely on where you start, the subject of §13.5. In a real mixed logit the analogue is an optimizer stopping in a flat region of the simulated surface and reporting a "converged" model that is neither a maximum nor the global one.

How to check convergence in practice. Report and inspect: the **gradient norm** at the optimum (should be tiny relative to the scale of the score); the **eigenvalues of the Hessian** (§13.4); and a **re-start** from the reported optimum. In Apollo/`maxLik` these are available — the gradient and the (numerical or analytic) Hessian are part of the output; use them, do not trust the flag alone.

Stop and think 13.1. Your mixed logit reports "converged" with a maximum-gradient element of 10^{-3} and a log-likelihood of -2410. A colleague's run reports "converged" with a maximum-gradient element of 4.2 and log-likelihood -2455. Which, if either, should you trust, and what do you check next? *(Answer: neither on the flag alone, but the colleague's is clearly* not* converged — a gradient element of 4.2 is far from zero (the first-order condition fails), so it stopped for another reason (max iterations, flat step). Yours has a small gradient, which is necessary but not sufficient: check the Hessian eigenvalues (negative definite?), re-start from the optimum, and confirm it beats other starts on log-likelihood — the higher log-lik ($-2410 > -2455$) already suggests yours found a better peak, §13.5.)*

13.4 The Hessian: invertibility, eigenvalues, and conditioning

The Hessian at the optimum is where **identification, precision, and validity all become visible at once**, because the asymptotic covariance is its inverse: $\widehat{\mathrm{Var}}(\hat{\theta}) = [-\mathbf{H}(\hat{\theta})]^{-1}$ (Chapter 11, §11.5). Reading the eigenvalues of $-\mathbf{H}$ is the single most informative diagnostic in this chapter.

The eigenvalue readout. Let $-\mathbf{H}$ have eigenvalues $\lambda_1 \geq ... \geq \lambda_k$. Then:

- **All eigenvalues comfortably positive** $\rightarrow$ $-\mathbf{H}$ is positive definite $\rightarrow$ a genuine maximum, and the covariance $(-\mathbf{H})^{-1}$ is well-defined. Good.
- **A near-zero eigenvalue** ($\lambda_{\min} \approx 0$) $\rightarrow$ the likelihood is **nearly flat** in the corresponding direction $\rightarrow$ that parameter combination is **weakly or non-identified** $\rightarrow$ its variance (an eigenvalue of the inverse, $1/\lambda_{\min}$) is **enormous**. This is the structural cause of huge standard errors.
- **A negative eigenvalue** $\rightarrow$ $-\mathbf{H}$ is *not* positive definite $\rightarrow$ $\mathbf{H}$ is indefinite $\rightarrow$ the point is a **saddle, not a maximum** (Chapter 2, §2.7). Any standard errors reported there are meaningless.
- **Non-invertible Hessian** (an exactly zero eigenvalue) $\rightarrow$ the covariance cannot be computed; software either errors or returns garbage. Exact non-identification.

Conditioning (Chapter 1, §1.10). The **condition number** $\kappa = \lambda_{\max}/\lambda_{\min}$ measures how ill-behaved the inversion is. A large κ (say $> 10^4$, catastrophically $> 10^8$) means the covariance is numerically unstable and the standard errors untrustworthy — often from a poorly scaled variable (a cost in euros next to attributes on a 0–1 scale), fixable by rescaling (§13.8).

Worked Example 13.2 (Reading eigenvalues to diagnose identification, verified). Suppose the information matrix $-\mathbf{H}$ has eigenvalues $\{500,\ 0.001\}$. Then the covariance $(-\mathbf{H})^{-1}$ has eigenvalues $\{1/500,\ 1/0.001\} = \{0.002,\ 1000\}$, so the standard deviation of the estimate is $\sqrt{0.002} = 0.045$ in the sharply-curved direction but $\sqrt{1000} = 31.6$ in the flat direction — a *huge* standard error for one parameter combination (verified). The condition number is $500/0.001 = 500{,}000$ — severely ill-conditioned. **Diagnosis:** the model is nearly non-identified in one direction; the giant standard error is *structural*, not a data-quantity problem. This is exactly your PDF's red flag — *"standard errors enormi segnalano un problema strutturale, non solo pochi dati"* — read off the Hessian's eigenvalues. Collecting more data would shrink both eigenvalues' *effects* only slowly; the fix is to identify the flat direction (which parameters are trading off) and repair the *specification or design* (Chapter 30, 49), or rescale if it is merely conditioning.

Numerical vs analytic Hessian (a subtle trap). Many programs compute the Hessian *numerically* (by finite differences) at the optimum to get standard errors. Numerical Hessians can be **inaccurate**, especially for simulated likelihoods (the simulation noise contaminates the finite differences) or badly scaled problems — producing standard errors that are wrong even when the point estimate is fine. Where possible, use analytic gradients/Hessians (Apollo can, for many models); at minimum, be aware that a strange-looking standard error may be a numerical-Hessian artifact, and check robustness (rescale, tighten tolerances, more draws).

Active recall 13.B. You compute the eigenvalues of $-\mathbf{H}$ at your optimum and find $\{820, 140, 0.00003\}$. What does the tiny eigenvalue tell you, and is "collect more data" the right response? *(Answer: the near-zero eigenvalue signals a nearly flat direction of the likelihood — one parameter combination is weakly/non-identified, with a near-infinite variance ($\sim 1/0.00003$).*

This is a structural* identification problem, so "collect more data" will not fix it; you must find which parameters trade off along that flat direction and repair the specification or design (or, if it is a scaling issue, rescale variables), Chapters 1 §1.10, 30.)*

13.5 Local maxima and the global maximum

For a **concave** log-likelihood (SBDC probit, plain logit — Chapter 2 §2.8), any point the optimizer converges to is *the* global maximum, and starting values are irrelevant. For a **non-concave** log-likelihood — your **mixed logit** and **MDCEV** simulated likelihoods — the surface can have several hills, and the optimizer climbs whichever one it started nearest. It finds *a* maximum, not necessarily *the* maximum. This is the single biggest computational danger in your central models.

The multi-start method (the fix). Since you cannot see the surface, you *sample* it: run the optimizer from **many different starting values** and compare the results.

1. Estimate from, say, 10–50 starting points (random, and sensible ones like the MNL estimates, §13.6).
2. Record the final **log-likelihood** and estimates from each.
3. The run with the **highest log-likelihood** is your best candidate for the global maximum.
4. **Read the pattern.** If many starts reach the *same* top log-likelihood (and the same estimates), you have strong evidence of a well-behaved global maximum. If the top log-likelihoods **scatter** across several distinct values, the surface has **multiple local maxima** — treacherous — and you must report this, take the best, and be cautious.

Worked Example 13.3 (Multi-start finds the global max, verified). For the bimodal objective of Worked Example 13.1, the ten starts produced three outcomes: log-likelihood 0 (flat-tail non-maxima), 2 (local max at $\theta = 1$), and 3 (global max at $\theta = 4$). Taking the **best over starts** correctly identifies the global maximum ($\theta = 4$, log-lik 3) — which any *single* start below $\theta \approx 2.5$ would have missed (verified). The scatter of final log-likelihoods (0, 2, 3) is itself the diagnostic: it tells you the surface is multi-modal and that a single run cannot be trusted. In a mixed logit the analogue is running the EC-MXL from many starts; if the top log-likelihoods agree you are safe, if they scatter you have a local-maxima problem to manage.

A caution specific to simulated likelihoods. For a *simulated* likelihood, apparent "different local maxima" across starts can also be **simulation noise** (Chapter 9) rather than true multi-modality — the surface jitters, so different starts land at slightly different noisy points. Distinguishing genuine local maxima from simulation noise is part of §13.7; the key is to *hold the draws fixed* across all starts (so the surface is deterministic) and then increase draws to see whether the scatter shrinks.

> **Stop and think 13.2.** You run your EC-MXL from 20 starting values. Eighteen reach a log-likelihood of $-1994.x$ (with essentially identical estimates); two reach -2003. Which is your model, and what do the two "stragglers" tell you? *(Answer: the model is the* -1994 *solution (higher log-likelihood, reached by the large majority) — take it and confirm its estimates are stable. The two stragglers at* -2003 *are inferior* local maxima*: the surface is not concave, so a minority of starts got trapped on a lower hill. This is normal for a mixed logit; the multi-start protocol is exactly what protects you, and you should report that you used it.)*

13.6 Starting values, and their use as a diagnostic

Starting values matter for non-concave problems in two ways: they affect *which* maximum you find, and their *influence* is itself informative.

Good starting strategies. - **Start the mixed logit from the MNL estimates.** The MNL is concave and quickly gives sensible, identified means; using them as the starting point for the mixed logit's means (with small variances) is a principled, well-located start — far better than zeros or random noise. - **Use**

several random starts around the sensible point to probe for local maxima (§13.5). - **Warm-start from a simpler model** — estimate a model without the error component first, then add it starting from those estimates; build complexity incrementally. - **Respect constraints** — start variance/scale parameters at plausible positive values (via the reparameterization of §13.8), not at or below their bounds.

Sensitivity to starting values *is* a diagnostic. Your PDF asks: *"le stime cambiano molto con starting values diversi?"* The answer is a test of your model's health:

- If estimates and log-likelihood are **stable** across sensible starts → the maximum is well-behaved (effectively concave near the solution) and you can trust it.
- If estimates **move materially** across starts → you have either **multiple local maxima** (different log-likelihoods) or **weak identification / simulation noise** (similar log-likelihoods but wandering estimates). Either way, a single run is untrustworthy, and the sensitivity itself is the finding to investigate and report.

This is the practical face of §11.9's "is the maximum global / are parameters identified": you do not inspect the surface directly, you *perturb the start and watch*. Stability under perturbation is evidence of a genuine, identified peak; instability is a warning.

> **Active recall 13.C.** Why is starting your mixed logit from the MNL estimates better than starting from zeros, and what does it *not* protect against? *(Answer: the MNL is concave and gives sensible, identified estimates of the means, so starting there places the optimizer near a good region and speeds/stabilizes convergence; but it does* not* guarantee the global maximum of the non-concave mixed logit — you still need multiple starts to probe for other local maxima, §13.5.)*

13.7 The synthesis: three culprits behind bad output, three different fixes

When an estimation "looks wrong" — huge standard errors, non-convergence, unstable estimates, an uninvertible Hessian — the invisible-but-decisive skill is to diagnose *which* of three distinct problems you have, because they demand different responses and are easily confused.

Culprit A — non-identification / weak identification (structural). The likelihood has a flat direction; some parameter combination is not pinned down by the data *no matter how much you collect*. **Signature:** a near-zero eigenvalue of $-\mathbf{H}$; a huge standard error for a specific combination; estimates that trade off along a ridge (strongly correlated). **Fix:** find the flat combination and repair the *specification or design* — drop a redundant parameter, add identifying variation, reconsider what the data can support (Chapters 1 §1.10, 11 §11.9, 30, 49). *"More data" is the wrong response* — this is your PDF's red flag that big SEs are structural.

Culprit B — local maxima / optimization failure (computational, deterministic). The surface is non-concave (or badly scaled) and the optimizer stopped at a suboptimal hill or stalled. **Signature:** different starts reach different, *deterministic* log-likelihoods; the best start clearly wins; rescaling or a better optimizer helps. **Fix:** multiple starting values (take the best, §13.5); rescale variables (§13.8); a more robust optimizer; sensible warm starts (§13.6).

Culprit C — simulation noise (computational, stochastic). For a *simulated* likelihood, too few or poor draws make the surface jitter, so estimates wander with the seed or draw count. **Signature:** estimates/log-likelihood change with the *seed* or *number of draws*, even from the same start; the scatter shrinks as draws increase. **Fix:** more draws, better (scrambled Halton / Sobol) draws, fixed draws across the optimization (Chapter 9, §9.4–9.6; Chapter 52).

How to tell them apart (the decision procedure).

1. **Check the Hessian eigenvalues.** A near-zero eigenvalue → **Culprit A** (identification). Investigate the flat direction before anything else.

2. **If eigenvalues are healthy, run multiple starts with draws fixed.** Scatter in *deterministic* log-likelihoods → **Culprit B** (local maxima). Take the best; rescale.
3. **Vary the seed / number of draws.** Estimates that move with the seed or shrink with more draws → **Culprit C** (simulation noise). Add draws.

The reason this framework is "decisive" is that the three culprits produce *similar-looking symptoms* (unstable, imprecise estimates) but require *opposite* responses: A cannot be fixed by computation (it is structural), while B and C are purely computational and are made *worse* by misdiagnosing them as "not enough data." Getting this diagnosis right is the difference between fixing the real problem and spinning your wheels — or, worse, publishing a fragile model that "converged."

> **Stop and think 13.3.** Your EC-MXL gives a WTP with a standard error of €300. Walk through the three-culprit procedure: what would each culprit look like, and what would you do first? *(Answer: first check the Hessian eigenvalues — if one is near zero, it is* ***Culprit A*** *(the WTP's parameter combination, likely a weak cost coefficient, is non-identified: structural, so fix the design/specification, not the sample size). If eigenvalues are fine, run multiple starts with fixed draws — if the WTP varies across starts with different log-likelihoods, it is* ***Culprit B*** *(local maxima: take the best start, rescale cost). If it varies with the* seed* or shrinks with more draws, it is **Culprit C** (simulation noise: add scrambled/Sobol draws). Do the eigenvalue check *first*, because a €300 SE on a WTP most often signals a weak, near-non-identified cost coefficient — Culprit A — which no computation fixes.)*

13.8 Practical fixes and estimation hygiene

A checklist of the interventions that most often turn a fragile estimation into a trustworthy one. (The full reproducible-pipeline version — seeds, draw protocols, version control — is Chapter 52; these are the MLE-optimization essentials.)

- **Rescale variables to comparable magnitudes** (Chapter 1, §1.10). A cost in euros (10–100) next to attributes on a 0–1 scale gives an ill-conditioned Hessian and unreliable steps; rescaling cost to tens, or standardizing attributes, is often the *single most effective* fix for convergence failure and untrustworthy standard errors. It changes no economics.
- **Reparameterize constrained parameters.** Estimate $\log \sigma$ instead of σ (guaranteeing $\sigma > 0$), or a Cholesky factor instead of a covariance (guaranteeing positive semidefiniteness), or impose a sign on the cost coefficient (Chapter 24). This keeps the optimizer inside the valid region and avoids boundary pathologies (Chapter 11, §11.9; Chapter 12, §12.9).
- **Use analytic gradients (and Hessians) where available.** Faster and more accurate than finite differences, and they avoid the numerical-Hessian standard-error artifacts of §13.4.
- **Do a gradient check.** Compare the analytic gradient to a finite-difference numerical gradient at a few points; a mismatch reveals a coding error in the likelihood — a common, silent source of wrong results.
- **Tighten tolerances and raise the iteration cap** if convergence is marginal, and always inspect the *final gradient norm*, not just the flag.
- **Multiple starting values**, with sensible warm starts (§13.5–13.6).
- **Fix and then increase the simulation draws** (for simulated likelihoods), checking seed/draw stability (Chapter 9).
- **Compare log-likelihoods across runs and models** on the *same sample* (Chapter 12, §12.5) — the highest log-likelihood among valid, replicating runs is your model.
- **Confirm the Hessian is negative definite** and well-conditioned at the reported optimum before trusting any standard error.

None of these is exotic; together they are the difference between "the software produced output" and "I have a maximum I can defend."

Active recall 13.D. Name the single most common and most effective fix when a mixed logit fails to converge or reports untrustworthy standard errors, and why it works. *(Answer: rescale the variables (especially a large-magnitude cost/price variable) to comparable magnitudes; it reduces the condition number of the Hessian, making the optimizer's steps reliable and the covariance stable, without changing the model — Chapter 1 §1.10.)*

13.9 Failure modes

F1. Trusting the "converged" flag. Reading convergence as validity. *Fix:* check gradient norm ≈ 0 *and* negative-definite Hessian *and* replication (§13.3).

F2. Reporting a single run of a non-concave model. One mixed-logit/MDCEV estimation taken as definitive. *Fix:* multiple starting values; compare log-likelihoods; take the best and report the multi-start (§13.5).

F3. Reading huge SEs as "need more data." Treating a structural identification problem as a sample-size problem. *Fix:* check Hessian eigenvalues; a near-zero eigenvalue is non-identification, fixed by specification/design, not data (§13.4, §13.7).

F4. Trusting a numerical-Hessian standard error blindly. A finite-difference Hessian contaminated by simulation noise or bad scaling. *Fix:* analytic Hessian where possible; rescale; check robustness (§13.4).

F5. Ignoring simulation-noise instability. Estimates that move with the seed/draws mistaken for real results. *Fix:* fix draws across optimization; increase and improve draws; check seed stability (§13.7; Chapter 9).

F6. Starting/estimating unconstrained constrained parameters. A standard deviation going negative, or a boundary maximum mishandled. *Fix:* reparameterize ($\log \sigma$, Cholesky); handle boundary inference carefully (§13.8; Chapters 11–12).

F7. Not checking the gradient (coding errors). A miscoded likelihood that "converges" to the wrong answer. *Fix:* gradient check (analytic vs numerical) at several points (§13.8).

F8. Comparing log-likelihoods across different samples. Different exclusions/weights make log-likelihoods incomparable. *Fix:* identical estimation sample for all compared runs (§13.8; Chapter 12).

13.10 Diagnostic questions for your own papers

1. **Real convergence?** Is the final gradient norm essentially zero, *and* is $-\mathbf{H}$ negative definite (all eigenvalues positive), *and* does the optimum replicate on re-start? *(§13.3.)*
2. **Hessian health.** What are the eigenvalues of $-\mathbf{H}$? Any near zero (identification) or negative (saddle)? What is the condition number? *(§13.4.)*
3. **Global maximum?** For non-concave models, did I use multiple starting values, and do the top log-likelihoods agree (well-behaved) or scatter (local maxima)? *(§13.5.)*
4. **Starting-value sensitivity.** Do my estimates change materially with the start? If so, is it local maxima or weak identification? *(§13.6.)*
5. **Which culprit?** For any instability, have I diagnosed non-identification (eigenvalues) vs local maxima (multi-start scatter) vs simulation noise (seed/draw sensitivity)? *(§13.7.)*
6. **Structural vs data.** Am I about to blame "small sample" for a huge SE that is actually a flat likelihood direction? *(§13.4, §13.7, F3.)*
7. **Hygiene.** Did I rescale variables, reparameterize constrained parameters, and do a gradient check? *(§13.8.)*

13.11 Exercises

(A) Comprehension.

A1. State the difference between "the optimizer converged" and "I found a valid maximum," and list the three checks that upgrade the former to the latter.

A2. The eigenvalues of $-\mathbf{H}$ at your optimum are $\{300, 45, -2\}$. Is this a maximum? What does the negative eigenvalue mean, and are the standard errors trustworthy?

A3. Your EC-MXL gives the same estimates from 15 different starting values but a WTP standard error of €250. Is this a local-maxima problem, and what is the likely culprit?

(B) Derivation / reasoning.

B1. For the information matrix $-\mathbf{H}$ with eigenvalues $\{\lambda_1, \ldots, \lambda_k\}$, show that the largest standard deviation among parameter combinations is $1/\sqrt{\lambda_{\min}}$, and explain why a near-zero $\lambda_{\min}$ means non-identification.

B2. Explain, using the Newton–Raphson update $\theta_{k+1} = \theta_k - \mathbf{H}^{-1}\mathbf{s}$, why a near-singular Hessian produces unstable steps, and how rescaling variables (improving the condition number) helps.

B3. Construct (in words and a simple function) a two-parameter log-likelihood that is flat along one direction, show its Hessian is singular, and explain what "converged" would mean for it and why the individual parameters are not identified.

(C) Research application / critique.

C1. A co-author reports an EC-MXL from a single run: "converged, log-likelihood −2103, all parameters significant." Write the checklist of diagnostics you would require before believing it, and why each matters.

C2. Your mixed logit will not converge. Describe, in order, the practical fixes you would try, and how you would diagnose whether the underlying problem is non-identification, local maxima, or simulation noise.

C3. Your WTP has a standard error of €400. A reviewer says "your sample is too small." Using this chapter, explain why they may be wrong, what you would check, and what the real fix might be.

13.12 Solutions

A1. "Converged" means a stopping rule triggered (small gradient, tiny step, or max iterations) — a statement about the optimizer. "A valid maximum" additionally requires: (i) the gradient (score) is essentially zero; (ii) the Hessian is negative definite (all eigenvalues of $-\mathbf{H}$ positive — a maximum, not a saddle or flat ridge); (iii) it replicates (re-starting from the optimum does not move it, and it is the best of multiple starts / stable across seeds and draws). Only with all three do you have a maximum you can defend.

A2. No — it is **not** a maximum. The negative eigenvalue (-2) means $-\mathbf{H}$ is not positive definite, so $\mathbf{H}$ is indefinite and the point is a **saddle** (the log-likelihood increases in one direction and decreases in others, Chapter 2 §2.7). The standard errors are **not trustworthy**: the covariance $(-\mathbf{H})^{-1}$ is not a valid (positive-definite) covariance at a saddle. Re-estimate (better start, rescale) to find an actual maximum.

A3. No, it is **not** a local-maxima problem — 15 starts giving the *same* estimates rules out multi-modality and instability. The €250 SE with stable estimates points to **weak identification (Culprit A)**: the likelihood is flat in the WTP's direction (likely a weak cost coefficient), so that combination has a huge, structural standard error. Check the Hessian eigenvalues (expect a near-zero one); the fix is better cost-coefficient identification (design spread, WTP-space reparameterization), not more starts or more data.

B1. $-\mathbf{H} = \mathbf{Q}\Lambda\mathbf{Q}'$ (spectral decomposition), so the covariance is $(-\mathbf{H})^{-1} = \mathbf{Q}\Lambda^{-1}\mathbf{Q}'$ with eigenvalues $1/\lambda_i$. The variance of the linear combination $\mathbf{q}_i'\hat{\theta}$ (the i-th eigenvector direction) is $1/\lambda_i$, so its standard deviation is $1/\sqrt{\lambda_i}$; the largest is $1/\sqrt{\lambda_{\min}}$. As $\lambda_{\min} \to 0$, this $\to \infty$ — the likelihood is flat in that direction, the data cannot distinguish nearby values of that combination, and its variance diverges: non-identification.

B2. The Newton step multiplies the score by $\mathbf{H}^{-1}$. If $\mathbf{H}$ has a near-zero eigenvalue, $\mathbf{H}^{-1}$ has a near-infinite eigenvalue, so the component of the step in that direction is enormously amplified — the optimizer overshoots or oscillates, producing unstable, unreliable steps. Rescaling variables reduces the condition number $\lambda_{\max}/\lambda_{\min}$ (evening out the eigenvalues), so $\mathbf{H}^{-1}$ is well-behaved and the steps are stable and reliable.

B3. Take $\ell(\theta_1, \theta_2) = -(\theta_1 + \theta_2 - 1)^2$ (a ridge along $\theta_1 + \theta_2 = 1$). Its gradient is $-2(\theta_1 + \theta_2 - 1)(1, 1)'$ and its Hessian is $-2\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, with eigenvalues -4 and 0 — so $-\mathbf{H}$ has a **zero eigenvalue** (singular). "Converged" would mean the optimizer stopped at *some* point on the ridge $\theta_1 + \theta_2 = 1$, but *which* point depends entirely on the start (verified: different starts give different (θ_1, θ_2) with the same sum). Only the combination $\theta_1 + \theta_2$ is identified; θ_1 and θ_2 individually are not — the flat direction $(\theta_1 - \theta_2)$ has infinite variance.

C1. Required diagnostics: (i) **final gradient norm** ≈ 0 (first-order condition — not just the flag); (ii) **Hessian eigenvalues** all positive for $-\mathbf{H}$ (a real maximum, no saddle) and none near zero (identified) with an acceptable condition number; (iii) **multiple starting values** reaching the same top log-likelihood (global max, since the mixed logit is non-concave) — a single run cannot certify this; (iv) **seed/draw robustness** (the log-likelihood and estimates stable across seeds and increasing draws — else simulation noise); (v) **clustered standard errors** given repeated tasks (Chapter 11 §11.5); (vi) **plausibility** of the estimated variances and WTP (near-boundary variances, weak cost coefficient). Each guards a distinct way "converged" can be a mirage.

C2. In order: (1) **rescale** variables (especially cost/price) to comparable magnitudes — the most common cure for non-convergence (improves conditioning). (2) **Reparameterize** constrained parameters ($\log \sigma$, Cholesky) so the optimizer stays in the valid region. (3) **Warm-start** from the MNL (or a simpler model). (4) **Gradient check** to rule out a coding error. (5) **Multiple starts** with fixed draws. Diagnosis: check **Hessian eigenvalues** (near-zero → non-identification, Culprit A — fix specification/design); if healthy, **multi-start with fixed draws** (deterministic log-lik scatter → local maxima, Culprit B — take the best, rescale); **vary the seed/draws** (estimates move with seed or settle with more draws → simulation noise, Culprit C — add scrambled/Sobol draws).

C3. The reviewer may be wrong because a huge WTP standard error is most often **structural**, not a sample-size issue: WTP $= -\beta_{\text{attr}}/\beta_{\text{cost}}$ is imprecise when the *cost coefficient* is weakly identified — a flat direction of the likelihood (near-zero Hessian eigenvalue) — which no amount of extra data fixes efficiently. I would check: the Hessian eigenvalues (is there a near-zero one?), the cost coefficient's own precision, and whether estimates are stable across starts and draws (ruling out local maxima / simulation noise). The real fix is likely **better cost-coefficient identification**: more cost-vector spread and better conditioning (Chapter 1 §1.10, Chapter 37), or WTP-space reparameterization with a sign-constrained cost coefficient (Chapter 24) — a design/specification fix, not merely a bigger sample.

13.13 Spaced review prompts

+1 day. From memory, state the three conditions for real (vs cosmetic) convergence, and describe what the eigenvalues of $-\mathbf{H}$ tell you (positive definite, near-zero, negative, condition number).

+1 week. Describe the multi-start method for finding a global maximum and how to read the pattern of final log-likelihoods; then lay out the three-culprit decision procedure (identification vs local maxima vs simulation noise) and the fix for each.

+1 month. For your own EC-MXL, run a multi-start estimation (10+ starts), report the Hessian eigenvalues and condition number at the best optimum, and diagnose any instability using the three-culprit framework. Close with the spine sentence about whether your reported maximum is real, global, and identified.

13.14 How this chapter connects to the rest of the book

- **Chapter 11** defined the likelihood surface and named its pathologies (§11.9 — identification, convergence, global maxima, constraints); this chapter turns each into an operational diagnostic. **Chapter 12**'s test disagreements are often symptoms of the surface problems diagnosed here.
- **Chapter 1** (§1.7, §1.10) supplied the eigenvalue/definiteness/conditioning tools; **Chapter 2** (§2.6–2.7) the concavity and FOC/SOC; this chapter applies them to real estimation.
- **Chapter 9** (§9.4–9.6) is the simulation-noise culprit of §13.7 in full; **Chapters 22–24 (mixed logit, SML, WTP space)** are where the non-concavity, multi-start, and identification issues of this chapter are lived daily, and where rescaling/reparameterization (§13.8) are routine.
- **Chapter 30 (identification)** is the theoretical companion to §13.4's flat-direction diagnosis; **Chapter 41 (MDCEV)** inherits all of this chapter's non-concave-estimation care; **Chapter 52 (computational diagnostics and reproducible research)** is the full pipeline treatment — seeds, draw protocols, version control, and reproducibility — of which this chapter is the MLE-optimization core.

This chapter completes Part III's likelihood engine: Chapter 11 built the likelihood and estimator, Chapter 12 tests it, and Chapter 13 makes the computation trustworthy — the "invisible but decisive" skill that separates a defensible model from a plausible-looking artifact.

13.15 Research translation task

For your **EC-MXL (and, if relevant, your MDCEV)**, produce a two-to-three-page estimation-diagnostics note:

1. **Convergence.** Report the final gradient norm and confirm it is essentially zero; confirm the run replicates on re-start.
2. **Hessian.** Report the eigenvalues of $-\mathbf{H}$ at the optimum and its condition number; confirm negative-definiteness (all positive), flag any near-zero eigenvalue, and state what parameter combination any flat direction corresponds to.
3. **Global maximum.** Run 10+ starting values (including MNL-based warm starts); tabulate the final log-likelihoods; report whether they agree (global max) or scatter (local maxima), and take the best.
4. **Three-culprit diagnosis.** For any instability or huge standard error, diagnose whether it is non-identification (eigenvalues), local maxima (multi-start scatter), or simulation noise (seed/draw sensitivity), and state the fix.
5. **Hygiene.** Note the rescaling and reparameterizations you used, and any gradient check.
6. **Close** with the spine sentence: *"My reported maximum is [real/global/identified] because [gradient≈ 0, $-H$ positive definite, agreed across starts, seed-stable]; any large SE is [structural/computational] and the fix is [design/rescaling/more draws]."*

Carry this into Chapters 22–24 (mixed logit and WTP-space estimation) and Chapter 52 (the full reproducible-diagnostics pipeline).

13.16 Bibliography

Core. - Train, *Discrete Choice Methods with Simulation*, the chapters on numerical maximization, starting values, local maxima, and simulation in mixed-logit estimation. The single most relevant source for §13.2–§13.7 in your DCE setting. Freely available — *verify link.* - Greene, *Econometric Analysis*, the appendix/chapters on numerical optimization (Newton, BHHH, quasi-Newton) and the practicalities of ML estimation. *(Verify edition.)* - Nocedal & Wright, *Numerical Optimization.* The definitive reference on the optimization algorithms of §13.2 (Newton, quasi-Newton/BFGS, line search, conditioning) — for depth on how and why optimizers behave as they do. *(Verify edition.)*

Applied. - The **Apollo** manual (Hess & Palma) and the `maxLik` package documentation (Henningsen & Toomet). Practical guidance on convergence diagnostics, gradients/Hessians, starting values, and multi-start estimation in exactly the tools you use. *(Verify current versions.)* - Hensher, Rose & Greene, *Applied Choice Analysis*, on estimation practicalities, starting values, and diagnosing mixed-logit convergence. *(Verify edition.)* - Gill, Murray & Wright, *Practical Optimization*, for the engineering of well-conditioned, reliable optimization (scaling, stopping rules, ill-conditioning). *(Verify edition.)*

Advanced. - Bhat, on estimation of mixed logit and MDCEV, including starting values and identification in high dimensions. *(Verify exact references.)* - McCullough & Vinod, "The Numerical Reliability of Econometric Software," *Journal of Economic Literature*, on how software can silently produce wrong results and how to check — the meta-theme of this chapter. *(Verify volume/year.)* - Gourieroux & Monfort, *Simulation-Based Econometric Methods*, for the theory of simulated-likelihood estimation and its computational behavior (Culprit C). *(Verify edition.)*

Entries marked "verify" are pointers; confirm editions, versions, and the McCullough–Vinod citation before using them, per the book's no-unverified-citation rule.

End of Chapter 13 and of Part III. You now hold the maximum-likelihood engine in full: the likelihood, score, Hessian, information, and the MLE's properties (Chapter 11); the Wald, LR, and score tests, with the boundary correction for variance tests (Chapter 12); and the computational craft of finding a trustworthy maximum and knowing when not to believe the output (Chapter 13). Part IV — Core Econometrics — begins with Chapter 16 (OLS from First Principles); or, following your PDF's priority ordering, the natural next step is Chapter 14 (Latent Variables and Censored Data) and Chapter 15 (Contingent Valuation Likelihoods), which apply this engine to build your SBDC and payment-card models in full. Tell me which chapter to write next.

Chapter 14 — Latent Variables and Censored Data

Part III: Maximum Likelihood and Latent Variable Models

> **The spine question, applied to this chapter.** Your contingent-valuation data never measure willingness to pay directly. A respondent says "yes" or "no" to a bid, or points to a bracket on a payment card — coarse, thresholded signals of an underlying continuous WTP you never see. This chapter builds the framework that turns those signals into a likelihood for the latent WTP distribution: **latent-variable models** (an unobserved continuous variable generating an observed coarse outcome) and **censoring** (knowing only that the latent value falls in an interval). Your PDF calls this *"Fondamentale per CV"* and states the connection exactly: *"tratti risposte SBDC come segnali censurati e payment card come dati intervallari."* The spine question here: *what does each respondent's coarse answer actually tell me about their latent WTP — which interval does it pin down — and how do I build the likelihood that recovers the WTP distribution from these interval observations?*

14.0 Conceptual motivation: your data are shadows of a latent variable

Chapters 11–13 built the maximum-likelihood engine in general. This chapter and the next apply it to the specific structure of your CV data, and the key realization — the one that unifies your whole contingent-valuation econometrics — is this: **you observe coarsened versions of a latent willingness to pay.** No respondent tells you "my WTP is €37.20." Instead:

- In **single-bounded dichotomous choice (SBDC)**, they answer one yes/no to one bid. A "yes" at €40 tells you only that their WTP is *at least* €40; a "no" tells you it is *below* €40. Each response locates WTP on one side of a single threshold — a very coarse, one-bit signal.
- In a **payment card**, they indicate a bracket — "I would pay between €30 and €40." That locates WTP in an *interval*, a richer signal than SBDC.
- In **double-bounded** designs, a follow-up bid narrows the interval further.

Your PDF names this structure precisely and lists the tools: *"latent variable models; probit; logit; left censoring; right censoring; interval censoring; survival function; Turnbull estimator; mean e median WTP sotto diverse distribuzioni."* Every one of these is in this chapter (Turnbull was Chapter 10; mean/median WTP is Chapter 15). The framework does three things for you:

1. **It turns coarse answers into a likelihood.** Once you see a "yes" as "WTP falls in (b, ∞)" and a payment-card answer as "WTP falls in $(\ell, u]$," each response contributes a probability — a difference of CDF values — and the product over respondents is the likelihood you maximize to estimate the WTP distribution. This *is* your SBDC and payment-card econometrics.

2. **It unifies binary response with contingent valuation.** The probit and logit you use for choice, the SBDC "yes/no," and survival/duration analysis are all the *same* latent-variable-plus-censoring structure. Seeing them as one idea — rather than three separate techniques — is a genuine consolidation of the "uneven foundations" your PDF describes.
3. **It exposes the judgments.** Whether WTP can be negative (the support question, Chapter 5), how the mean depends on the tail (Chapter 15), whether a zero answer is a genuine zero or a protest (Chapter 26, 32), and whether your exclusions turn *censoring* into *truncation* (a different, biasing problem) — all become precise once you have the framework.

We build: the latent-variable idea; latent-index models (probit and logit as threshold-crossing, with the scale normalization); the types of censoring and the survival function, with their likelihood contributions; the SBDC as single-threshold censoring and the payment card / double-bounded as interval data (and why intervals are more efficient); the beautiful identity that the mean of a non-negative variable is the area under its survival function; the crucial distinction between censoring, truncation, and missing data; and the Tobit/spike-at-zero model. Chapter 15 then derives the full SBDC, double-bounded, and payment-card likelihoods and the mean/median WTP from this foundation.

14.1 Prerequisites

From **Chapter 11**: the likelihood framework, the MLE, and how a likelihood is the product of each observation's probability under the model. From **Chapter 5**: the CDF F, the **survival function** $S(t) = 1 - F(t) = \Pr(Y > t)$ (§5.2), and the normal and logistic distributions. From **Chapter 2** (§2.8) and **Chapter 11** (§11.8): probit and logit as models built on a normal/logistic error, and the SBDC probit. From **Chapter 10**: the Turnbull estimator (the *nonparametric* version of what this chapter does parametrically).

Same **notation**. New/consolidated symbols: Y^* (latent variable), Y (observed outcome), $F(\cdot;\theta)$ and $S(\cdot;\theta) = 1 - F$ (CDF and survival of the latent variable), b_i (bid), $(\ell_i, u_i]$ (observed interval for respondent i), c (a censoring threshold).

14.2 Latent variables: the idea

Definition 14.1 (Latent variable model). A **latent variable model** posits an unobserved (latent) continuous variable Y^* that follows a distribution indexed by parameters, together with an **observation rule** mapping Y^* to the observed outcome Y. You never see Y^*; you see only Y, a coarsening of it. The parameters of the latent distribution are what you estimate.

The pattern recurs across your models:

Latent variable Y^*	Observation rule	Observed Y
Utility of an alternative	choose the alternative with highest utility	chosen alternative (Chapters 20–21)
Willingness to pay	"yes" iff WTP $\geq$ bid	yes/no (SBDC)
Willingness to pay	which bracket contains WTP	interval (payment card)
A propensity/index	1 iff index > 0	binary outcome (probit/logit)

In every row the observed data are a *thresholded* or *bracketed* view of a continuous latent quantity. The econometric task is to recover the latent distribution's parameters from these coarse views — and the reason it is possible is that *different respondents face different thresholds* (different bids, different brackets), so pooling their coarse answers traces out the latent CDF. This is why **bid design matters** (Chapters 15, 49): the thresholds you offer determine how much of the WTP distribution your data can illuminate.

Active recall 14.A. In the SBDC, what is the latent variable, and what is the observation rule that maps it to the yes/no you observe? *(Answer: the latent variable is the respondent's willingness to pay, WTP*; the observation rule is "yes iff WTP* ≥ bid," so the observed yes/no is a one-bit, thresholded view of the continuous latent WTP.)*

14.3 Latent-index models: probit and logit as threshold-crossing

The most common latent-variable model is the **latent-index** (threshold-crossing) model, which unifies binary response and the SBDC.

Definition 14.2 (Latent-index model). A latent index is a linear function of covariates plus an error,

$$Y^* = \mathbf{x}'\beta + \varepsilon,$$

and the observed binary outcome is whether the index crosses a threshold (normalized to zero):

$$Y = \mathbf{1}\{Y^* > 0\} = \mathbf{1}\{\varepsilon > -\mathbf{x}'\beta\}.$$

The choice probability is then, for a symmetric error with CDF F,

$$\Pr(Y = 1 \mid \mathbf{x}) = \Pr(\varepsilon > -\mathbf{x}'\beta) = \Pr(-\varepsilon < \mathbf{x}'\beta) = F(\mathbf{x}'\beta),$$

using symmetry ($-\varepsilon$ has the same distribution as ε). Choosing the error distribution gives the model:

- $\varepsilon \sim N(0,1) \to F = \Phi \to$ **probit**;
- $\varepsilon \sim$ logistic $\to F = \Lambda \to$ **logit**.

This is exactly the binary-response model of Chapter 11 (§11.8), now seen as *threshold-crossing of a latent index* — a viewpoint that connects it directly to the SBDC and to censoring.

The scale normalization (Chapter 5, §5.6, restated in latent terms). Only the *ratio* β/σ_ε is identified, because multiplying Y^*, β, and ε by any positive constant leaves the observation rule $\mathbf{1}\{Y^* > 0\}$ unchanged. So the error scale must be normalized — $\sigma_\varepsilon = 1$ for probit, the logistic scale for logit — and the coefficients are always *relative to the error scale*. This is the latent-variable origin of "logit and probit coefficients are not directly comparable" (they normalize different scales) and of the scale-vs-preference-heterogeneity problem (Chapter 25): a threshold-crossing model can never separately identify the index and the error scale, only their ratio.

Stop and think 14.1. In the latent-index model $Y = \mathbf{1}\{\mathbf{x}'\beta + \varepsilon > 0\}$, multiplying β and ε both by 2 leaves the observed Y unchanged. What does this imply about what the data can identify, and how is it resolved? *(Answer: the data cannot identify the* level* of β and the error scale separately — only their ratio β/σ_ε — because the observed threshold-crossing is invariant to common rescaling. It is resolved by *normalizing* the error scale (probit: $\sigma = 1$; logit: the logistic scale), so all coefficients are reported relative to that scale.)*

14.4 Censoring and the survival function

The SBDC and payment card are not binary-*covariate* problems; they are **censored observations of a latent WTP**. Censoring means you know the latent value lies in an interval but not its exact value. The likelihood contribution depends on *which kind* of interval.

Definition 14.3 (Censoring types). For a latent variable Y with CDF F and survival $S = 1 - F$, an observation can be:

- **Exactly observed** ($Y = y$): contributes the density $f(y)$.

- **Right-censored** (you know only $Y > c$): contributes $\Pr(Y > c) = S(c) = 1 - F(c)$.
- **Left-censored** (you know only $Y \le c$): contributes $\Pr(Y \le c) = F(c)$.
- **Interval-censored** (you know only $\ell < Y \le u$): contributes $\Pr(\ell < Y \le u) = F(u) - F(\ell)$.

Definition 14.4 (Survival function). $S(t) = \Pr(Y > t) = 1 - F(t)$ — the probability the latent variable *exceeds* t. It is decreasing from $S(-\infty) = 1$ (or $S(0) = 1$ for a non-negative variable) to $S(\infty) = 0$. In your setting, $S(b) = \Pr(\text{WTP} > b)$ is *the probability of saying "yes" to a bid* b — the survival function and the "yes"-probability curve are the same object (Chapter 10, §10.8). The related **hazard** $\lambda(t) = f(t)/S(t)$ (the instantaneous "failure" rate) is the workhorse of duration analysis; you will not need it heavily, but it is worth knowing that CV interval data and survival/duration data are the same mathematics.

The general censored likelihood. For a sample of independent censored observations, the likelihood is the product of each observation's contribution according to its censoring type:

$$L(\theta) = \prod_{i \in \text{exact}} f(y_i) \prod_{i \in \text{right}} S(c_i) \prod_{i \in \text{left}} F(c_i) \prod_{i \in \text{interval}} [F(u_i) - F(\ell_i)].$$

Everything in CV econometrics is a special case of this. The art is recognizing *which* censoring type each response represents — the subject of the next two sections.

> **Active recall 14.B.** Write the likelihood contribution of a respondent who is (i) right-censored at €50, (ii) left-censored at €20, (iii) interval-censored in €[30, 40], in terms of the WTP CDF F and survival S. *(Answer: (i) $S(50) = 1 - F(50)$; (ii) $F(20)$; (iii) $F(40) - F(30)$. Each is a probability that latent WTP falls in the known region.)*

14.5 The SBDC as a single-threshold censored observation

Now the unification. The single-bounded dichotomous-choice response is a **censored observation of WTP with one threshold per respondent** — and this is why the SBDC likelihood *is* a binary/probit likelihood.

The mapping. Respondent i is offered bid b_i, with latent WTP_i having CDF $F(\cdot;\theta)$:

- A **"yes"** ($y_i = 1$) means $\text{WTP}_i \ge b_i$ — a **right-censored** observation: you learn only that WTP exceeds b_i. Contributes $S(b_i) = 1 - F(b_i) = \Pr(\text{WTP} \ge b_i)$.
- A **"no"** ($y_i = 0$) means $\text{WTP}_i < b_i$ — a **left-censored** observation. Contributes $F(b_i) = \Pr(\text{WTP} < b_i)$.

So the SBDC likelihood is

$$L(\theta) = \prod_{i=1}^{n} S(b_i)^{\,y_i}\, F(b_i)^{\,1-y_i},$$

which, for normally distributed WTP $\sim N(\mu, \sigma^2)$, gives $S(b_i) = 1 - \Phi(\frac{b_i - \mu}{\sigma}) = \Phi(\frac{\mu - b_i}{\sigma}) = \Phi(\alpha + \beta b_i)$ with $\alpha = \mu/\sigma$, $\beta = -1/\sigma$ — **exactly the SBDC probit of Chapter 11 (§11.8).** The two views are identical: "a Bernoulli whose probability depends on the bid" and "a single-threshold censored observation of WTP" are the same likelihood, and seeing them as one is the payoff of this section.

Why the SBDC is *coarse* (and what that costs). Each SBDC respondent contributes a *single bit* — WTP is above or below one number. That is the least informative interval possible (one endpoint at $\pm\infty$). Consequently the SBDC needs *variation in bids across respondents* to trace out the WTP distribution: if everyone faced the same bid, you would learn only one point of the CDF. This coarseness is the statistical price of the SBDC's incentive-compatibility virtues (Chapter 28) — and it motivates the richer interval formats of §14.6.

The bid design connection. Because each SBDC observation informs the CDF only near its bid, the *placement* of bids determines what you can estimate. Bids concentrated where the "yes"-probability is near 0.5 are most informative (Chapters 1, 11); bids spread across the WTP range let you estimate the whole

distribution and its tails — the design question of Chapter 37, seen here as "which thresholds do I censor at?"

> **Stop and think 14.2.** Two SBDC designs offer bids at {€30, €31, €29} (all near the median WTP) versus {€5, €20, €40, €80} (spread across the range). Which better identifies the *upper tail* of the WTP distribution, and why does this matter for mean WTP? *(Answer: the spread design — its €80 bid censors WTP at a high threshold, so the "yes"-rate there informs the upper tail $S(80)$; the clustered design tells you almost nothing above €31. This matters because mean WTP is the area under the survival function (§14.7), which is dominated by the upper tail — a design that cannot see the tail cannot pin down the mean, only the median, Chapter 15.)*

14.6 Payment card and double-bounded: interval data and its efficiency

Richer elicitation formats give **narrower intervals** per respondent, and therefore more efficient estimates.

Payment card as interval data. In a payment-card format the respondent indicates a range — e.g., "I would pay between €30 and €40" (or the highest amount they would pay, which brackets WTP between that and the next card value). This is a genuine **interval-censored** observation:

$$\text{contributes } \Pr(\ell_i < \text{WTP} \le u_i) = F(u_i) - F(\ell_i).$$

The interval $(\ell_i, u_i]$ is typically much narrower than the SBDC's (b_i, ∞) or $(-\infty, b_i)$, so each payment-card respondent pins WTP down more tightly.

Double-bounded dichotomous choice as interval data. A double-bounded design asks a follow-up bid after the first. The two yes/no answers produce one of four intervals (with a first bid b_1 and follow-ups $b_1^U > b_1$ after a "yes", $b_1^L < b_1$ after a "no"):

First / Follow-up	WTP interval	Contribution
Yes, Yes	(b_1^U, ∞)	$S(b_1^U)$
Yes, No	$(b_1, b_1^U]$	$F(b_1^U) - F(b_1)$
No, Yes	$(b_1^L, b_1]$	$F(b_1) - F(b_1^L)$
No, No	$(-\infty, b_1^L]$	$F(b_1^L)$

The two interior outcomes are genuine intervals, narrower than the SBDC's single threshold — which is why double-bounded is *statistically more efficient* than single-bounded (Chapter 15 discusses the offsetting incentive/consistency concerns). The likelihood is the product of these contributions over respondents.

Worked Example 14.1 (Interval data is more efficient, verified). Simulate WTP $\sim N(30, 15^2)$ for 2000 respondents, and estimate mean WTP two ways from the *same* latent WTPs: (i) SBDC — each respondent gets one random bid from $\{10, 20, 30, 40, 50\}$ and answers yes/no (single-threshold censoring); (ii) payment card — each respondent's WTP is bracketed into $[0, 10, 20, \dots, 80, \infty)$ cards (interval censoring). Fitting the interval likelihood $F(u) - F(\ell)$ in both cases:

- **SBDC:** $\hat{\mu} = 30.09$, $\text{SE}(\hat{\mu}) = 0.47$.
- **Payment card:** $\hat{\mu} = 30.59$, $\text{SE}(\hat{\mu}) = 0.32$.

The payment card's standard error is about **1.5 times smaller** (ratio $0.47/0.32 = 1.47$, verified) — a substantial efficiency gain from the same respondents, purely because each interval observation carries more information than a single threshold. This is the statistical case for richer elicitation, to be weighed against the incentive-compatibility and anchoring concerns of Chapters 27–29 (the payment card can induce anchoring; the SBDC's single referendum-style question has better incentive properties). The trade-off — efficiency versus incentive compatibility — is exactly the kind of judgment your CV paper must make explicit.

Active recall 14.C. Why does a payment card give a smaller standard error for mean WTP than an SBDC with the same respondents, and what is the countervailing concern? *(Answer: the payment card yields a* narrow interval* per respondent ($F(u) - F(\ell)$), pinning WTP down more tightly than the SBDC's single threshold, so each observation carries more information and the estimate is more efficient. The countervailing concern is that payment cards can induce *anchoring* / range effects and are less incentive-compatible than the single dichotomous-choice referendum, Chapters 27–29.)*

14.7 The survival function and the mean: area under the curve

There is a beautiful identity that connects the survival function directly to mean WTP, and it underlies both the parametric mean (Chapter 15) and the nonparametric Turnbull mean (Chapter 10).

Proposition 14.1 (Mean as the integral of survival). For a non-negative random variable Y with survival function $S(t) = \Pr(Y > t)$,

$$\mathbb{E}[Y] = \int_0^\infty S(t)\, dt.$$

The mean is the area under the survival curve.

Derivation. Write $y = \int_0^y dt = \int_0^\infty \mathbf{1}\{t < y\}\, dt$, so

$$\mathbb{E}[Y] = \int_0^\infty y\, f(y)\, dy = \int_0^\infty \int_0^\infty \mathbf{1}\{t < y\}\, dt\, f(y)\, dy = \int_0^\infty \Big(\int_t^\infty f(y)\, dy \Big) dt = \int_0^\infty S(t)\, dt,$$

swapping the order of integration (Fubini) and recognizing the inner integral as $S(t) = \Pr(Y > t)$. ■ *(Verified: for a lognormal WTP, the analytic mean* 22.76 *equals* $\int_0^\infty S(t)\, dt = 22.76$.*)*

Why this matters for your WTP. In your CV model, the "yes"-probability at bid b *is* the survival $S(b)$ (§14.5). So **mean WTP is the area under your fitted (or nonparametric) yes-probability curve** — literally the area under the acceptance curve as a function of the bid. Three consequences:

- **The mean is tail-sensitive.** Because it integrates $S(t)$ out to infinity, the mean depends heavily on the *upper tail* — the acceptance rate at high bids, which your bids may not reach. This is the source of the mean-vs-median discrepancy and the tail-driven inflation you saw with Turnbull (Chapter 10, §10.8) and will formalize in Chapter 15.
- **The median is robust.** The median WTP is simply the t where $S(t) = 0.5$ — the bid at which half say yes. It depends only on the center of the distribution, not the tail, which is why the median is the more stable welfare summary when the tail is uncertain.
- **Turnbull is this identity, discretized.** The Turnbull lower-bound mean (Chapter 10) is exactly $\sum_j b_j[S(b_j) - S(b_{j+1})]$ — a discrete approximation to $\int S\, dt$ using only the bids you offered. The parametric mean fills in the survival curve with an assumed distribution; the Turnbull mean uses only the observed steps and is therefore a distribution-free lower bound.

Stop and think 14.3. Your fitted SBDC survival curve $S(b)$ is well-estimated up to €60 (your highest bid) but you must extrapolate it beyond. Which is more sensitive to that extrapolation, mean or median WTP, and what does this imply for what you report? *(Answer: the* mean* is far more sensitive — it is the area under S integrated to infinity, so the extrapolated tail beyond €60 directly enters it (and different assumed distributions give different tails and different means); the *median* (where $S = 0.5$) is unaffected if it lies below €60. Report the median as the robust central estimate, report the mean with its distributional assumption stated, and use the Turnbull lower bound as a distribution-free check — Chapters 10, 15.)*

14.8 Censoring versus truncation versus missing data

A distinction that is easy to blur and biasing to get wrong — and directly relevant to how you handle protest responses and exclusions (Chapter 32).

Censoring. The unit *is* in your sample, and you know its value lies in a known interval. The likelihood contribution is the *probability of that interval* ($F(u)-F(\ell)$, etc.), as in §14.4. Your SBDC and payment-card data are censored.

Truncation. Units whose latent value falls outside a range are *never observed at all* — they are absent from the sample. This is fundamentally different: you must *renormalize* the likelihood by the probability of being sampled, because you are conditioning on selection. For a variable truncated to (a, b), the density of an observed value is $f(y)/[F(b) - F(a)]$, not $f(y)$. Ignoring truncation — treating a selected sample as if it were random — biases the parameter estimates. *Example in your work:* if you *dropped* respondents whose WTP is below some threshold (rather than recording them as "no"), you would truncate, not censor, and the naive likelihood would be wrong.

Missing data. The value is absent for reasons that may or may not relate to the latent variable (missing completely at random, at random, or not at random). Whether missingness biases estimates depends on *why* the data are missing. Protest responses and item non-response are missing-data problems whose treatment (drop, model, bound) changes the estimand.

Why this matters for exclusions (Chapter 32, your PDF §6.6). Your PDF flags that *"protest responses, speeders e comprehension failures … possono rompere la randomizzazione se non gestiti bene."* In censoring language: a "no" is a *censored* observation (record it, contribute $F(b)$); a *protest* "no" is arguably *missing* (the respondent rejected the exercise, not revealed WTP $< b$); and *dropping* respondents based on their responses can convert honest censoring into *truncation* or informative missingness, biasing your WTP estimates. Distinguishing "this respondent has low WTP" (censored, keep) from "this respondent refused to engage" (missing/protest, model or bound) is a substantive judgment with real consequences for the estimand — exactly the care your PDF demands.

> **Active recall 14.D.** A respondent says "no" to a €40 bid. Under what interpretation is this a *censored* observation (keep it, contribute $F(40)$), and under what interpretation is it *missing/protest* (requiring different handling)? *(Answer: it is* censored* if the "no" genuinely reveals WTP $<$ €40 (a real low valuation) — contribute $F(40)$ to the likelihood; it is *missing/protest* if the "no" reflects rejection of the payment vehicle or the scenario rather than a low valuation, in which case treating it as WTP $<$ €40 mis-measures the latent variable and biases estimates — handle via protest modeling, spike models, or sensitivity bounds, Chapters 26, 32.)*

14.9 Tobit and the spike at zero

The classic **censored regression** is the **Tobit** model: a latent $Y^* = \mathbf{x}'\beta + \varepsilon$ (with $\varepsilon \sim N(0, \sigma^2)$) observed as $Y = \max(0, Y^*)$ — censored at zero. Positive values are observed exactly; the mass of negative latent values piles up at zero. Its likelihood mixes a *density* for the positive observations and a *probability* for the censored (zero) ones:

$$L = \prod_{Y_i>0} \frac{1}{\sigma}\phi\Big(\tfrac{y_i-\mathbf{x}_i'\beta}{\sigma}\Big) \prod_{Y_i=0} \Phi\Big(\tfrac{-\mathbf{x}_i'\beta}{\sigma}\Big).$$

Relevance to your WTP. The Tobit's "spike at a point plus a continuous part" is the template for the **spike models** of contingent valuation (Chapter 26): some respondents have genuinely *zero* WTP (a point mass at zero), while others have positive WTP (a continuous distribution). A spike model puts probability π on WTP $= 0$ and $(1 - \pi)$ on a positive distribution, and its likelihood combines a discrete term (the spike) with the interval-censored terms (§14.4) for the positive part. This addresses two problems at once: it accommodates genuine non-participants (true zeros), and it sidesteps the negative-WTP artifact of a symmetric normal

(Chapter 5, §5.3) by giving zero-and-below its own probability mass rather than a symmetric tail. Chapter 26 develops spike models for your CV data; here the point is that they are latent-variable-plus-censoring models with an added point mass — the same framework, extended.

> **Stop and think 14.4.** Your SBDC estimates a normal WTP with mean €18 and implies 20% negative WTP — implausible for a pure environmental improvement. How does a spike-at-zero model address this within the latent-variable/censoring framework? *(Answer: instead of a symmetric normal that manufactures negative-WTP mass, the spike model places an explicit point mass π on WTP $= 0$ (genuine non-participants) and a distribution on strictly positive WTP; the likelihood combines the spike probability with interval-censored contributions for positive WTP. This gives the "negative/zero" region its own honest probability without forcing a symmetric tail, respecting the good's non-negative support — Chapters 5, 26.)*

14.10 Failure modes

F1. Treating censored data as exactly observed. Assigning WTP $= b$ (the bid) to a "yes/no" respondent, or the bracket midpoint to a payment-card respondent, and running OLS. *Fix:* use the interval likelihood ($F(u)-F(\ell)$, $S(b)$, $F(b)$); censored data are probabilities of intervals, not point values (§14.4–14.6).

F2. Wrong censoring direction. Coding a "yes" as left-censored or a "no" as right-censored. *Fix:* "yes" means WTP $\geq$ bid (right-censored, $S(b)$); "no" means WTP $<$ bid (left-censored, $F(b)$) (§14.5).

F3. Confusing censoring with truncation. Dropping respondents based on their WTP responses and using the naive (un-renormalized) likelihood. *Symptom:* biased estimates from selection. *Fix:* if units are excluded on the outcome, renormalize (truncation) or, better, keep them as censored; distinguish from honest censoring (§14.8).

F4. Mishandling protests as censored. Treating a protest "no" as a genuine WTP $< b$. *Fix:* model protests separately (spike/missing), not as low valuations (§14.8, §14.9; Chapters 26, 32).

F5. Ignoring the scale normalization. Interpreting latent-index coefficients as if the error scale were 1 when it is normalized differently, or comparing across probit/logit. *Fix:* remember only β/σ is identified; compare marginal effects/WTP, not raw coefficients (§14.3; Chapters 5, 18).

F6. Extrapolating the survival tail for the mean without acknowledging it. Reporting a mean WTP driven by an assumed tail beyond your highest bid. *Fix:* report the median (robust), the Turnbull lower bound (distribution-free), and the mean with its distributional assumption flagged (§14.7; Chapters 10, 15).

F7. Negative-WTP artifacts from a symmetric latent distribution. A normal latent WTP putting mass below zero for a non-negative good. *Fix:* consider a spike-at-zero or non-negative (lognormal) distribution; state the support (§14.9; Chapter 5).

F8. Wasting information with coarse formats when efficiency matters. Using single-bounded when a (defensible) interval format would sharpen estimates. *Fix:* weigh efficiency (interval formats) against incentive compatibility (single dichotomous choice) explicitly (§14.6; Chapters 15, 28).

14.11 Diagnostic questions for your own papers

1. **What is censored, and how?** For each response format, have I written the correct censoring contribution (right/left/interval) to the likelihood? *(§14.4–14.6.)*
2. **Direction.** Are "yes" = right-censored ($S(b)$) and "no" = left-censored ($F(b)$) coded correctly? *(§14.5, F2.)*

3. **Censoring vs truncation vs missing.** Have any exclusions turned censoring into truncation or informative missingness? Are protests handled as missing/spike, not as low WTP? *(§14.8, F3–F4.)*
4. **Scale.** Am I aware my latent-index coefficients are identified only up to the error scale? *(§14.3, F5.)*
5. **Mean vs median.** Is my mean WTP being driven by an extrapolated survival tail? Have I reported the median and the Turnbull lower bound? *(§14.7, F6.)*
6. **Support.** Does my latent WTP distribution's support match the good (no implausible negative mass), or should I use a spike/non-negative model? *(§14.9, F7.)*
7. **Efficiency vs incentives.** Have I weighed the efficiency of interval formats against their incentive-compatibility costs? *(§14.6, F8.)*

14.12 Exercises

(A) Comprehension.

A1. Write the likelihood contribution for each of these CV responses, in terms of the WTP CDF F: (i) "yes" to €25; (ii) "no" to €25; (iii) payment-card answer "between €40 and €50"; (iv) double-bounded "yes to €30, no to €60."

A2. Explain why the SBDC "yes"-probability curve as a function of the bid *is* the survival function of WTP, and what its value at the median bid equals.

A3. Give one reason a payment card is statistically more efficient than a single-bounded dichotomous choice, and one reason you might still prefer the SBDC.

(B) Derivation.

B1. Derive the latent-index choice probability $\Pr(Y = 1) = F(\mathbf{x}'\beta)$ from $Y = \mathbf{1}\{\mathbf{x}'\beta + \varepsilon > 0\}$ with symmetric ε, and explain the scale normalization.

B2. Prove that for a non-negative variable $\mathbb{E}[Y] = \int_0^\infty S(t)\,dt$, and explain why this makes mean WTP sensitive to the upper tail while the median is not.

B3. Write the double-bounded dichotomous-choice likelihood (four outcomes) as interval-censored contributions, and explain why it is more efficient than the single-bounded likelihood.

(C) Research application / critique.

C1. A colleague codes SBDC data by assigning each "yes" respondent a WTP equal to the bid and running OLS on those values. Explain everything wrong with this and give the correct approach.

C2. Your CV team drops all respondents who said "no" to the lowest bid, reasoning they have "no real WTP." Explain, using §14.8, why this can bias your estimates and what to do instead.

C3. Your SBDC (highest bid €60) gives a parametric mean WTP of €52 but a Turnbull lower bound of €30 and a median of €28. Interpret this using §14.7, and state what you would report and why.

14.13 Solutions

A1. (i) "yes" to €25: WTP ≥ 25, right-censored $\to S(25) = 1 - F(25)$. (ii) "no" to €25: WTP < 25, left-censored $\to F(25)$. (iii) payment card €40–€50: interval $\to F(50) - F(40)$. (iv) DB "yes €30, no €60": WTP in $(30, 60] \to F(60) - F(30)$.

A2. A "yes" to bid b occurs iff WTP $\geq b$, so $\Pr(\text{yes at } b) = \Pr(\text{WTP} \geq b) = S(b)$ — the yes-probability curve is the survival function. At the median bid (where half say yes), $S(b_{\text{med}}) = 0.5$, i.e. b_{med} is the median WTP.

A3. More efficient: the payment card gives a *narrow interval* per respondent ($F(u) - F(\ell)$), which carries more information than the SBDC's single threshold, so standard errors are smaller (verified ~1.5× in Worked Example 14.1). Prefer SBDC: the single dichotomous-choice (referendum-style) question has better incentive-compatibility properties and avoids the anchoring/range effects payment cards can induce (Chapters 27–28).

B1. $\Pr(Y = 1) = \Pr(\mathbf{x}'\beta + \varepsilon > 0) = \Pr(\varepsilon > -\mathbf{x}'\beta) = \Pr(-\varepsilon < \mathbf{x}'\beta) = F_{-\varepsilon}(\mathbf{x}'\beta) = F(\mathbf{x}'\beta)$ using symmetry ($-\varepsilon \sim \varepsilon$). Scale normalization: scaling Y^*, β, ε by any $c > 0$ leaves $\mathbf{1}\{Y^* > 0\}$ unchanged, so only β/σ_ε is identified; fix σ_ε (probit: 1; logit: logistic scale).

B2. $\mathbb{E}[Y] = \int_0^\infty y f(y)dy = \int_0^\infty (\int_0^y dt) f(y)dy = \int_0^\infty (\int_t^\infty f(y)dy)dt = \int_0^\infty S(t)dt$ (Fubini). Because the mean integrates S all the way to infinity, it depends on the upper tail (the high-t survival, i.e. acceptance at high bids); the median solves $S(t) = 0.5$, a single central point unaffected by the tail — hence the median is robust to tail/extrapolation while the mean is not.

B3. With first bid b_1, up-bid b_1^U, down-bid b_1^L: (Yes,Yes) → WTP $> b_1^U$ → $S(b_1^U)$; (Yes,No) → $(b_1, b_1^U]$ → $F(b_1^U) - F(b_1)$; (No,Yes) → $(b_1^L, b_1]$ → $F(b_1) - F(b_1^L)$; (No,No) → WTP $\le b_1^L$ → $F(b_1^L)$. The likelihood is the product of these over respondents. It is more efficient than single-bounded because the two interior outcomes are *bounded intervals* (both endpoints finite), pinning WTP more tightly than the single-bounded's one-sided threshold, so each respondent contributes more information.

C1. Assigning WTP = bid to "yes" respondents and running OLS is wrong on several counts: (i) it treats *censored* data as *exactly observed* (a "yes" means WTP ≥ bid, not = bid — the bid is a lower bound, not the value); (ii) it discards the "no" respondents' information; (iii) OLS on these fabricated values estimates nothing meaningful about the WTP distribution and is badly biased (systematically understating WTP, since the bid is a floor). The correct approach is maximum likelihood on the *interval-censored* likelihood: each "yes" contributes $S(b) = 1 - F(b)$, each "no" contributes $F(b)$, and you estimate the WTP distribution's parameters (Chapter 11, §11.8; Chapter 15).

C2. Dropping "no"-to-lowest-bid respondents converts honest *censoring* into *truncation/selection*: those respondents are informative left-censored observations (WTP below the lowest bid — genuinely low, but real), and removing them conditions the sample on WTP being above a threshold, biasing the estimated distribution *upward*. If instead they are protests (rejecting the exercise), that is a *missing-data* problem, not evidence of low WTP. Correct approach: keep genuine low-WTP "no" responses as left-censored contributions ($F(b_{\min})$); identify and handle protests separately (spike model or sensitivity bounds), and report results with and without any exclusions (§14.8; Chapters 26, 32).

C3. The mean (€52) far exceeds both the median (€28) and the Turnbull lower bound (€30), and the highest bid is only €60 — the classic signature of a mean **inflated by an extrapolated upper tail** (the parametric distribution puts substantial acceptance probability above €60, which the data cannot observe, and the mean integrates that tail via $\int S\,dt$). The median (€28) and Turnbull lower bound (€30) are robust and agree. Report: the **median** as the primary central estimate; the **Turnbull lower bound** as a distribution-free check; and the mean **only with the distributional assumption and its tail-sensitivity explicitly flagged** (and ideally under alternative distributions to show the range). For any CBA use (Chapter 4), carry the range, not the tail-driven €52.

14.14 Spaced review prompts

+1 day. From memory, state the four censoring types and their likelihood contributions (f, $S(c)$, $F(c)$, $F(u) - F(\ell)$), and map the SBDC "yes"/"no" onto right-/left-censoring.

+1 week. Derive $\mathbb{E}[Y] = \int_0^\infty S(t)\,dt$ and explain why mean WTP is tail-sensitive and the median is not. Then distinguish censoring, truncation, and missing data with a CV example of each.

+1 month. For your own CV data, write the correct interval-censored likelihood contribution for each response type, check that your exclusions have not created truncation/informative missingness, and compare

your mean, median, and Turnbull lower-bound WTP. Close with the spine sentence about what each response tells you about latent WTP.

14.15 How this chapter connects to the rest of the book

- **Chapter 11** supplied the likelihood/MLE engine and the SBDC probit (§11.8); this chapter reframes the SBDC as single-threshold censoring and generalizes to interval data. **Chapter 5** supplied the CDF, survival function, and the normal/logistic distributions; **Chapter 10** the *nonparametric* (Turnbull) counterpart to this chapter's parametric censoring.
- **Chapter 15 (CV likelihoods from scratch)** is the direct sequel: it derives the full SBDC, double-bounded, and payment-card likelihoods from this framework and computes mean/median WTP under alternative distributions — the tail-sensitivity of §14.7 made quantitative.
- **Chapter 18 (binary response)** develops the latent-index probit/logit and the scale normalization of §14.3; **Chapter 26 (CV formats)** develops the spike models and interval-data models of §14.6/§14.9; **Chapter 28 (incentive compatibility)** is the efficiency-vs-incentives trade-off of §14.6.
- **Chapter 32 (threats to identification)** is the censoring-vs-truncation-vs-missing distinction of §14.8 applied to protests, speeders, and post-randomization exclusions (your PDF §6.6); **Chapter 37 (efficient design)** is the bid-placement/threshold-design question of §14.5.

This chapter is the conceptual bridge from the general likelihood engine (Chapters 11–13) to your specific contingent-valuation econometrics (Chapter 15), and it unifies binary response, the SBDC, interval data, and survival analysis as one latent-variable-plus-censoring idea.

14.16 Research translation task

For your **CV data (SBDC and payment card)**, produce a two-to-three-page note:

1. **Map responses to censoring.** For each response type in your data (SBDC yes/no, payment-card bracket, any double-bounded follow-up), write the exact interval-censored likelihood contribution in terms of the WTP CDF.
2. **Check the observation scheme.** Confirm "yes" = right-censored, "no" = left-censored; verify no exclusions have converted censoring into truncation, and that protests are handled as missing/spike rather than low WTP.
3. **Efficiency.** If you have both formats, compare the standard errors on mean WTP (as in Worked Example 14.1) to quantify the efficiency of the interval format, and weigh it against incentive-compatibility considerations.
4. **Survival and the mean.** Plot your fitted (and Turnbull) survival/acceptance curve; compute mean WTP as the area under it and the median as the $S = 0.5$ crossing; assess how much the mean depends on the extrapolated tail beyond your highest bid.
5. **Support.** State your latent WTP distribution's support and whether a spike-at-zero or non-negative distribution is warranted.
6. **Close** with the spine sentence: *"Each CV response is a [right/left/interval]-censored observation of latent WTP; my likelihood recovers the WTP distribution; my mean is [tail-sensitive/robust] and I report [median/Turnbull/mean] because [tail extrapolation]."*

Carry this into Chapter 15 (the full CV likelihood derivations and mean/median WTP) and Chapter 26 (spike and interval-data CV models).

14.17 Bibliography

Core. - Haab & McConnell, *Valuing Environmental and Natural Resources: The Econometrics of Non-Market Valuation*. The definitive treatment of SBDC, double-bounded, and payment-card likelihoods as censored/interval WTP data, and mean/median WTP — the single most relevant source for this chapter and Chapter 15. *(Verify edition/chapters.)* - Cameron & Trivedi, *Microeconometrics: Methods and Applications*, chapters on censored and truncated data (Tobit), latent-variable models, and the censoring-vs-truncation distinction (§14.4, §14.8, §14.9). *(Verify edition.)* - Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, on latent-variable (index) models, censored regression, and the interval-data likelihood. *(Verify edition/chapters.)*

Applied. - Cameron & James / Cameron (1988), on interval-data (censored) estimation of WTP in contingent valuation — the interval-likelihood approach of §14.6 in your setting. *(Verify exact citation/year.)* - Klein & Moeschberger, *Survival Analysis: Techniques for Censored and Truncated Data*, for the survival-function and censoring machinery (§14.4, §14.7) presented in its native duration-analysis context — the same mathematics as CV interval data. *(Verify edition.)* - Boyle, "Contingent Valuation in Practice" (in *A Primer on Nonmarket Valuation*), for how censoring/interval formats and protest handling are treated in applied CV. *(Verify exact citation.)*

Advanced. - Amemiya, *Advanced Econometrics*, the chapters on Tobit, truncated, and censored models — the rigorous foundation for §14.9 and the truncation/censoring distinction. *(Verify edition.)* - Turnbull (1976), *JRSS-B*, for the nonparametric MLE of interval-censored data (the distribution-free counterpart, Chapter 10). *(Verify volume/pages.)* - Kriström (1990) and the spike-model literature in contingent valuation, for the point-mass-at-zero models of §14.9. *(I am not certain of the exact best citations — verify and select those matching your CV context.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Cameron / Kriström citations before using them, per the book's no-unverified-citation rule.

End of Chapter 14. Next in Part III: Chapter 15 — Contingent Valuation Likelihoods from Scratch (the SBDC likelihood as censored latent WTP, the double-bounded likelihood, the payment-card interval likelihood, mean vs median WTP under alternative distributions, and Turnbull as a nonparametric robustness check), which turns this chapter's censoring framework into your CV paper's full estimation and welfare machinery — completing your PDF's Block 1 deliverable. Tell me when to write it, or request a different chapter.

Chapter 15 — Contingent Valuation Likelihoods from Scratch

Part III: Maximum Likelihood and Latent Variable Models

> **The spine question, applied to this chapter.** This chapter completes your PDF's Block 1 deliverable — *"riscrivere da zero la likelihood SBDC e payment card del tuo CV paper"* — by turning the censoring framework of Chapter 14 into your contingent-valuation paper's full estimation and welfare machinery. You will derive the single-bounded, double-bounded, and payment-card likelihoods from scratch, and then confront the question your PDF asks most insistently: *"la normalità della WTP è plausibile? accetti WTP negative? come cambia la media se la distribuzione ha code pesanti? la media è dominata dalla coda superiore?"* Because mean WTP is the area under the survival curve (Chapter 14), it is dominated by the upper tail — which your bids may never reach — so the *distributional assumption*, not just the data, drives your headline welfare number. The spine question here: *given my CV responses and an assumed WTP distribution, what welfare object (mean? median? truncated mean?) am I estimating, how much does it depend on the tail my design cannot observe, and does it survive replacing the distribution — or dropping it entirely (Turnbull)?*

15.0 Conceptual motivation: the likelihood is the easy part; the distribution is the danger

By now you have all the pieces. Chapter 11 built the likelihood/MLE engine and derived the SBDC probit. Chapter 14 reframed every CV response as a censored observation of latent WTP and gave the mean-equals-area-under-survival identity. Chapter 10 gave the nonparametric Turnbull benchmark. This chapter assembles them into the complete CV econometrics of your paper and — crucially — draws out the judgment that separates a defensible welfare estimate from an artifact.

Your PDF is unusually specific about what this chapter must deliver and what it must warn against. It lists the formats — *"open-ended CV; single-bounded dichotomous choice; double-bounded dichotomous choice; payment card; referendum format; interval data models; Turnbull estimator"* — and it lists the critical questions, all of which are about the *distribution*, not the likelihood:

- *"la normalità della WTP è plausibile?"* — Is the symmetric normal defensible for a good with non-negative value?
- *"accetti WTP negative? perché?"* — The normal manufactures negative WTP; do you accept that or fix it?
- *"come cambia la media se la distribuzione ha code pesanti? la media è dominata dalla coda superiore?"* — The mean integrates the tail; different distributions give different tails and wildly different means.
- *"il supporto della WTP è coerente con il bene valutato?"* — Does the distribution's support match the economics?

The uncomfortable truth this chapter makes precise is that **writing the likelihood is mechanical, but choosing the WTP distribution is a consequential, partly unfalsifiable assumption that can move your mean WTP by a factor of two or three** — because your bids illuminate only the middle of the distribution, and the mean depends on the tail beyond them. Your CV paper's welfare estimates, which feed benefit transfer and CBA (Chapters 4, 44), live or die on this choice.

We proceed: the SBDC likelihood derived under normal, logistic, and lognormal WTP; the two parameterizations (utility-difference vs WTP-function — the CV ancestor of preference space vs WTP space, Chapter 24); the double-bounded likelihood and its anchoring/shift pathologies; the payment-card interval likelihood; mean vs median WTP under alternative distributions, with the tail-sensitivity made quantitative; the negative-WTP problem and truncated/spike remedies; Turnbull as the distribution-free robustness check; and WTP standard errors via the delta method and Krinsky–Robb. Every derivation is *your* CV paper's estimation, and every warning is a judgment your paper must make explicit.

15.1 Prerequisites

From **Chapter 14**: CV responses as censored observations (SBDC = single threshold, payment card = interval), the survival function $S(b) = \Pr(\text{WTP} > b) = \Pr(\text{yes at } b)$, and the mean-as-area-under-survival identity (§14.7). From **Chapter 11**: the likelihood/MLE engine and the SBDC probit (§11.8). From **Chapter 5**: the normal, logistic, and lognormal distributions and the negative-WTP support problem (§5.3). From **Chapter 10**: the Turnbull estimator (§10.8). From **Chapters 2, 7, 9**: WTP as a ratio of coefficients, the delta method, and Krinsky–Robb.

Same **notation**. Consolidated symbols: WTP (latent willingness to pay) with CDF $F(\cdot;\theta)$ and survival $S = 1 - F$; b (bid); α, β (index parameters, $\alpha + \beta b$); μ, σ (WTP location/scale); μ_ℓ, σ_ℓ (log-WTP parameters for the lognormal); C^+ (mean WTP), C^{med} (median WTP).

15.2 The single-bounded dichotomous-choice likelihood, three distributions

The SBDC likelihood is the product of each respondent's censored contribution (Chapter 14, §14.5): a "yes" at bid b_i contributes $S(b_i)$, a "no" contributes $F(b_i)$:

$$L(\theta) = \prod_{i=1}^{n} S(b_i;\theta)^{y_i}\, F(b_i;\theta)^{1-y_i}, \qquad \ell(\theta) = \sum_i \left[y_i \log S(b_i) + (1 - y_i) \log F(b_i) \right].$$

Everything hinges on the choice of F — the assumed WTP distribution. Three standard choices:

(a) Normal WTP. WTP $\sim N(\mu, \sigma^2)$. Then $S(b) = 1 - \Phi(\frac{b-\mu}{\sigma}) = \Phi(\frac{\mu-b}{\sigma}) = \Phi(\alpha + \beta b)$ with $\alpha = \mu/\sigma$, $\beta = -1/\sigma$ — a **probit in the bid** (Chapter 11, §11.8). Mean = median = $\mu = -\alpha/\beta$; scale $\sigma = -1/\beta$. *Virtue:* simple, familiar. *Vice:* symmetric with full-real support, so it puts probability $\Phi(-\mu/\sigma)$ on **negative WTP** — an artifact for a non-negative good (§15.6).

(b) Logistic WTP. WTP has a logistic distribution. Then $S(b) = \Lambda(\alpha + \beta b)$ — a **logit in the bid**. Mean = median = $\mu = -\alpha/\beta$ (logistic is symmetric). *Difference from normal:* heavier tails (Chapter 5, §5.4), so acceptance probabilities decay more slowly at high bids and the estimated mean can differ. Still admits negative WTP.

(c) Lognormal WTP. $\log(\text{WTP}) \sim N(\mu_\ell, \sigma_\ell^2)$, so WTP > 0 always. A "yes" iff WTP $\geq b$ iff $\log \text{WTP} \geq \log b$, giving $S(b) = \Phi(\frac{\mu_\ell - \log b}{\sigma_\ell}) = \Phi(\alpha + \beta \log b)$ — a **probit in the *log* bid**. Median WTP = e^{μ_ℓ}; **mean**

WTP $= e^{\mu_\ell + \sigma_\ell^2/2}$ (right-skewed, mean > median). *Virtue:* non-negative support, matching a good with non-negative value. *Vice:* the mean depends on σ_ℓ through the tail — a large σ_ℓ inflates the mean enormously (§15.5).

Worked Example 15.1 (Same data, three distributions, verified). Simulate $n = 3000$ SBDC responses from a *true* lognormal WTP (median €30, $\sigma_\ell = 0.6$, so true mean $= 30\,e^{0.18} =$ €35.9), with bids in $\{5, 10, 20, 40, 80\}$. Fitting each assumed distribution:

Assumed WTP distribution	Median WTP	Mean WTP	Note
Normal	€36.9	€36.9	mean=median; implies **2.5% negative WTP**
Logistic	€35.0	€35.0	mean=median; heavier tails
Lognormal (true)	€30.2	€35.7	recovers truth; **no negative WTP**
Turnbull (nonparametric)	—	€25.6 (lower bound)	distribution-free benchmark

(Verified.) Read the lessons. **The lognormal — the correct model here — recovers both the true median (€30) and true mean (€36).** The normal gets a mean-median of €37 but *manufactures 2.5% negative WTP* (an artifact, since the true good is non-negative) and cannot distinguish mean from median. The logistic gives €35. The distribution-free Turnbull lower bound (€25.6) is conservative (it values mass at interval lower endpoints and cannot see beyond the highest bid). **The point:** the mean WTP you report — €25.6, €35.7, or €36.9 — depends substantially on the distributional assumption, and only the lognormal here respects the good's non-negative support. Choosing the distribution is a modeling decision with a several-euro consequence, and it must be defended, not defaulted.

> **Active recall 15.A.** Under a normal WTP the SBDC is a probit in the bid; under a lognormal WTP it is a probit in the *log* bid. Why the difference, and what does each give for mean vs median WTP? *(Answer: because "yes iff WTP $\geq b$" becomes "yes iff log WTP $\geq \log b$" under lognormality, so the index is linear in $\log b$; the normal gives mean = median $= -\alpha/\beta$, while the lognormal gives median $= e^{\mu_\ell}$ and a larger mean $= e^{\mu_\ell + \sigma_\ell^2/2}$ — the right-skew separates them.)*

15.3 Two parameterizations: utility-difference versus WTP-function (the CV ancestor of WTP space)

There are two equivalent-looking but conceptually distinct ways to write the SBDC model, and the distinction is the contingent-valuation ancestor of the preference-space-vs-WTP-space choice you face in your DCE (Chapter 24). It is worth seeing them side by side.

The utility-difference parameterization (Hanemann). Model the *utility difference* between accepting and rejecting the bid as a function of the bid, e.g. $\Delta V = \alpha + \beta b$, and derive WTP as a *function* of the estimated coefficients: WTP $= -\alpha/\beta$. This is what §15.2 did — you estimate index coefficients (α, β) and *recover* WTP as a ratio. The WTP is a derived quantity, and (as always with a ratio) its distribution and standard error can misbehave when β (the bid/cost coefficient) is weak (Chapters 2, 7).

The WTP-function parameterization (Cameron). Model the *WTP distribution directly* — parameterize $F(b) = G(\frac{b-\mu}{\sigma})$ where μ is mean/median WTP and σ its scale, and estimate (μ, σ) *as* the WTP parameters. Here WTP is a *primitive* of the model, not a ratio recovered after the fact.

Why they are (mostly) the same, and where they differ. For the SBDC the two are algebraically equivalent reparameterizations ($\mu = -\alpha/\beta$, $\sigma = -1/\beta$), so they fit identically and give the same WTP. But the *conceptual* difference matters: the Cameron parameterization puts the distribution on WTP directly (so you can constrain it — e.g. force non-negativity, or bound the tail — and read its parameters as welfare quantities), whereas the Hanemann parameterization recovers WTP as a ratio (inheriting the ratio's tail fragility). **This is exactly the preference-space (estimate coefficients, form the WTP ratio) versus**

WTP-space (parameterize WTP directly) distinction of your DCE (Chapter 24), appearing first in its simpler CV form. Recognizing that your SBDC "recover WTP as $-\alpha/\beta$" and your EC-MXL "estimate in WTP space" are the *same idea* at different levels of complexity is a genuine consolidation — and it tells you that the WTP-function parameterization is the natural home for imposing the non-negativity and tail constraints your good requires.

> **Stop and think 15.1.** You want to impose that WTP is non-negative and has a bounded tail. Which parameterization makes this natural, and what is the DCE analogue? *(Answer: the WTP-function (Cameron) parameterization, because WTP is a primitive whose distribution you specify directly — you can choose a non-negative distribution (lognormal) and constrain its spread. The DCE analogue is estimating in WTP space (Chapter 24), where WTP is a directly-modeled parameter you can sign-constrain, rather than a ratio of coefficients recovered ex post from preference space.)*

15.4 The double-bounded likelihood: more efficiency, new pathologies

A double-bounded design asks a follow-up bid after the first, narrowing the WTP interval and gaining efficiency — but introducing behavioral pathologies your CV paper must address.

The likelihood (four outcomes, from Chapter 14, §14.6). With first bid b_1, an up-bid $b^U > b_1$ after "yes," and a down-bid $b^L < b_1$ after "no," the four response patterns give interval contributions:

$$L = \prod \begin{cases} S(b^U) & \text{(Yes, Yes): WTP} > b^U \\ F(b^U) - F(b_1) & \text{(Yes, No): } b_1 < \text{WTP} \le b^U \\ F(b_1) - F(b^L) & \text{(No, Yes): } b^L < \text{WTP} \le b_1 \\ F(b^L) & \text{(No, No): WTP} \le b^L \end{cases}$$

The two interior outcomes are *bounded* intervals, so each respondent pins WTP more tightly than in the SBDC — hence double-bounded is **statistically more efficient** (smaller standard errors on the WTP parameters).

The behavioral pathologies (why efficiency is not free). The follow-up question is *not* generally incentive-compatible, and it induces systematic response patterns that violate the model's assumption that both answers reflect the *same* underlying WTP:

- **Anchoring / starting-point bias.** The first bid *anchors* the respondent's stated WTP, so the follow-up response is pulled toward the first bid rather than revealing an independent valuation (Chapter 27, coherent arbitrariness). Directly relevant to your PDF's central theme that the *cost vector can enter preference formation or reporting.*
- **Shift effect.** The distribution implied by the follow-up answers is systematically *shifted* relative to the first — respondents appear to have a *lower* WTP threshold on the second question (a "yes–no" asymmetry), inconsistent with a single stable WTP.
- **Yea-saying / framing effects** on the second question.

How to handle it. The standard responses: (i) test for the shift/anchoring by allowing the second-bound parameters (mean, variance) to differ and testing equality (an LR test, Chapter 12) — a significant difference *is* the pathology; (ii) model the anchoring explicitly (a shift or anchoring parameter); (iii) fall back to a *single-bounded* analysis of just the first response, which is incentive-compatible but less efficient — trading efficiency for validity. This efficiency-versus-validity trade-off is exactly the judgment your CV paper must make and defend, and it connects to the incentive-compatibility material of Chapter 28. **The methodological point:** more information (double-bounded) is only better if the extra information is *valid*; if the follow-up induces anchoring, the efficiency gain is bought with bias, and single-bounded may be the more honest estimate.

Active recall 15.B. Double-bounded is more efficient than single-bounded, yet many CV studies report the single-bounded estimate as their preferred one. Give the two reasons. *(Answer: (i) the follow-up question is not incentive-compatible and induces anchoring/shift effects, so the second answer may not reflect the same true WTP, biasing the estimate; (ii) the first (single-bounded) dichotomous choice retains the referendum-style incentive-compatibility. The efficiency gain of the second bound is bought at the cost of validity, so many prefer single-bounded despite its larger standard errors.)*

15.5 Mean versus median WTP under alternative distributions, and tail sensitivity

This is the heart of the chapter — the quantitative version of your PDF's *"come cambia la media se la distribuzione ha code pesanti? la media è dominata dalla coda superiore?"* Recall (Chapter 14, §14.7) that mean WTP is the area under the survival curve, $C^{+} = \int_0^{\infty} S(b)\,db$, so it integrates the *entire tail*, while median WTP C^{med} solves $S(b) = 0.5$ — a single central point. This asymmetry has decisive consequences.

The mean is tail-dominated; the median is robust. Because your bids reach only to some maximum $b_{\max}$, the survival curve beyond $b_{\max}$ is *extrapolated* by the assumed distribution, and the mean depends heavily on that extrapolation. The median, if it lies below $b_{\max}$, is estimated from data and is insensitive to the tail.

Worked Example 15.2 (Tail sensitivity of the lognormal mean, verified). Fix the median WTP at €30 and vary the log-scale σ_ℓ (which governs the tail). The lognormal mean is $C^{+} = 30\,e^{\sigma_\ell^2/2}$:

σ_ℓ	Median WTP	Mean WTP
0.3	€30	€31.4
0.6	€30	€35.9
1.0	€30	€49.5
1.5	€30	**€92.4**

(Verified.) **The median is fixed at €30, but the mean ranges from €31 to €92 — a threefold swing — driven entirely by the tail parameter** σ_ℓ, which your data may pin down only weakly (it depends on acceptance at high bids you may not have). This is the precise, quantitative form of "la media è dominata dalla coda superiore": a distribution with a fatter upper tail (larger σ_ℓ) has the same middle but a far larger mean, and the CV survey often cannot distinguish these tails. The implications:

- **Report the median as the primary central estimate.** It is data-driven and robust to the tail. Many CV guidelines (and the NOAA panel lineage) favor the median for exactly this reason.
- **Report the mean with its distributional assumption and tail-sensitivity flagged**, ideally under alternative distributions (normal, logistic, lognormal) to show the range — as in Worked Example 15.1, where the mean ranged €25.6–€36.9.
- **Use the Turnbull lower bound as a distribution-free floor** (§15.7): a conservative mean that cannot be inflated by tail assumptions.
- **For CBA and benefit transfer (Chapters 4, 44), carry the range**, not a single tail-driven point — a mean WTP that doubles with the assumed tail cannot be treated as precise.

The normal's negative-WTP problem, quantified. A symmetric normal WTP with mean μ and scale σ assigns $\Phi(-\mu/\sigma)$ to negative WTP (Chapter 5, §5.3). In Worked Example 15.1 this was 2.5%; with a smaller mean-to-scale ratio it can be 10–20%. For a good you believe no one is harmed by, this mass is an artifact that both distorts the mean (pulling it down) and violates the support — a reason to prefer a non-negative distribution or a truncated/spike model (§15.6).

Stop and think 15.2. Two referees demand a single mean WTP number for your CBA. Your lognormal mean is €36 but shifts to €55 under a heavier-tailed fit that fits the observed responses almost as well; your median is €30 in both. What do you report and argue? *(Answer: report the **median (€30)** as the robust central estimate, and present the mean as a **range (€36–€55)** explicitly attributable to the poorly-identified upper tail (which your bids cannot pin down), with the Turnbull lower bound as a distribution-free floor. Argue that a single mean is not defensible when it swings with an untestable tail assumption; for the CBA, use the range and show how the verdict depends on it — Chapters 4, 44. Do not launder a tail assumption into a point estimate.)*

15.6 The negative-WTP problem and its remedies

The symmetric-distribution artifact deserves its own treatment because it recurs and has standard fixes, each embodying a different belief about the good.

The problem. A normal or logistic WTP has support on all of $\mathbb{R}$, so it necessarily places mass on WTP < 0. For a pure environmental improvement (a good that harms no one), negative WTP is economically implausible, and the mass is a modeling artifact that biases the mean and misrepresents the population.

When negative WTP is *legitimate*. Sometimes a change genuinely harms some respondents (a project with disamenities, a policy with losers), and negative WTP is real. The question your PDF poses — *"accetti WTP negative? perché?"* — is substantive: decide, on the economics of *your* good, whether negative valuations are possible, and let that govern the distribution.

The remedies (each a belief about the good).

- **Lognormal (or other non-negative) distribution:** support $(0, \infty)$, no negative mass. Belief: everyone's WTP is strictly positive. *Cost:* right-skew and tail-sensitive mean (§15.5).
- **Truncated normal:** a normal truncated at zero, renormalized. Belief: an underlying normal but no negatives. *Note:* truncation renormalizes the density (Chapter 14, §14.8).
- **Spike (or "spike-at-zero") model:** a point mass at WTP $= 0$ (genuine non-participants / indifferent respondents) plus a distribution on positive WTP (Chapter 14, §14.9; Chapter 26). Belief: some people value the good at exactly zero, the rest positively. This is often the *most defensible* for environmental goods because it accommodates true zeros without either manufacturing negatives (normal) or forbidding zeros (lognormal).
- **Bounded/mixture models:** constrain WTP to $[0, \text{income}]$ or use a mixture.

The practical protocol. Estimate under the normal (transparent, familiar) *and* under a non-negative or spike specification, report how mean and median move, and let the good's economics justify the preferred model. If the negative mass is small and the results are stable, the normal is a harmless approximation; if it is large or the mean shifts, the distribution matters and you must choose deliberately.

Active recall 15.C. Your good is a coastal-ecosystem improvement that plausibly harms no one, and a normal fit implies 15% negative WTP. Which remedy is likely most defensible, and why not simply use the lognormal? *(Answer: a spike-at-zero model is often most defensible — it lets genuine non-participants sit at exactly zero WTP (plausible for people indifferent to the improvement) while a positive distribution handles the rest, without manufacturing negatives (normal) or forbidding zeros (lognormal). The pure lognormal forbids WTP = 0, which may be implausible if a real fraction of the population is genuinely indifferent; the spike model is the more flexible, economically honest choice — Chapters 14, 26.)*

15.7 Turnbull as the nonparametric robustness check

Chapter 10 built the Turnbull estimator; here we place it as the *robustness anchor* for the parametric CV estimates of this chapter — the distribution-free number every CV paper should report alongside the parametric mean.

What it does (recap). Turnbull estimates the WTP survival function *nonparametrically* from the yes-proportions at each bid (Chapter 10, §10.8), with no assumed distribution, and yields a conservative **lower-bound mean** $\widehat{C}^{+}_{\mathrm{LB}} = \sum_j b_j[S(b_j) - S(b_{j+1})]$ (valuing each interval's mass at its lower endpoint). In Worked Example 15.1 the yes-proportions at bids $\{5, 10, 20, 40, 80\}$ were $\{0.998, 0.964, 0.738, 0.330, 0.045\}$, giving a Turnbull lower-bound mean of €25.6 (verified).

How to use it against the parametric estimates.

- **Benchmark the parametric mean.** Turnbull is distribution-free, so it *cannot* be inflated by tail assumptions. If your parametric mean (€36 lognormal, or €52 under a heavier tail) far exceeds the Turnbull lower bound (€25.6), the gap measures how much of your mean is driven by the *assumed tail beyond your highest bid* rather than the data — exactly the §15.5 diagnostic. A parametric mean close to the Turnbull lower bound is reassuring; a large gap is a tail-fragility flag.
- **Sidestep the negative-WTP and support problems.** Turnbull never assigns negative WTP and never extrapolates beyond the bids, so it is immune to §15.6's artifacts.
- **Report it as the floor.** The Turnbull lower bound is a defensible, conservative welfare number for a CBA when you want to avoid tail-driven optimism.

Its limitation. Turnbull's resolution is limited to your bid points, and it is a *lower* bound (it undervalues the mass above the highest bid, assigning it to the highest bid rather than the tail). So it is a robustness *anchor*, not a replacement for the parametric estimate — the two together (parametric mean/median for the central estimate, Turnbull for the distribution-free floor) give an honest picture. Your PDF's instruction to use *"Turnbull come robustezza non parametrica"* is exactly this pairing.

> **Stop and think 15.3.** Your lognormal mean is €36 and the Turnbull lower bound is €26; your normal mean is €37 with 3% negative WTP. What story do these three numbers tell together, and how do you report? *(Answer: the parametric means (€36–€37) agree across normal and lognormal, and the Turnbull lower bound (€26) — a distribution-free floor — is below them by the amount attributable to the tail above the highest bid; the small negative mass (3%) under the normal is a minor artifact. Story: the central estimate is robust across distributions, the mean is modestly tail-sensitive (€26 floor vs €36 estimate), and the negative-WTP artifact is negligible. Report the median as the central estimate, the parametric mean with its range, and the Turnbull lower bound as the conservative floor — a coherent, defensible package.)*

15.8 Standard errors for mean and median WTP

The point estimate is only half the deliverable; you need its uncertainty, and — because WTP is a nonlinear function of the estimated parameters — you propagate it with the delta method or, better for skewed WTP, Krinsky–Robb (Chapters 7, 9).

Median WTP standard error. For the normal SBDC, median $= \mu = -\alpha/\beta$, a ratio, so its delta-method variance is $\nabla g' \hat{\mathbf{V}} \nabla g$ with $\nabla g = (-1/\beta,\ \alpha/\beta^2)'$ (Chapter 7, §7.7). The $1/\beta^2$ term warns that a **weak bid coefficient** (few "yes"/"no" transitions, little bid spread near the median) inflates the WTP standard error — the SBDC face of the WTP-ratio fragility. In the WTP-function (Cameron) parameterization, μ is estimated directly and its standard error is read off the covariance directly — another reason that parameterization is cleaner for inference.

Mean WTP standard error. For the lognormal, mean $= e^{\mu_\ell + \sigma_\ell^2/2}$, a nonlinear function of (μ_ℓ, σ_ℓ); the delta method applies but, because the mean is skewed and tail-sensitive, a **Krinsky–Robb** simulation (draw

(μ_ℓ, σ_ℓ) from their estimated normal, compute the mean for each draw, take percentiles) gives an honest, asymmetric interval (Chapter 9, §9.8). The median–mean gap and the tail sensitivity of §15.5 mean the mean's interval is typically wide and right-skewed — report it as such, not as a symmetric ± band.

Comparing WTP across treatments (your convergent-validity work). To compare WTP between arms — SBDC vs payment card, hypothetical vs incentivized, high vs low cost vector — use the **Poe test** on the Krinsky–Robb-simulated WTP distributions (Chapters 9–10), and — because these are convergent-validity/equivalence claims — pair it with an **equivalence test (TOST)** against a pre-specified margin (Chapter 8). A non-significant WTP difference is *not* evidence of equivalence. This closes the loop back to your PDF's top-priority error.

> **Active recall 15.D.** Why report a Krinsky–Robb percentile interval rather than a symmetric delta-method interval for a lognormal mean WTP? *(Answer: the lognormal mean is a skewed, tail-sensitive nonlinear function of the parameters, so its sampling distribution is right-skewed; a symmetric delta-method interval mis-represents it (understating the right tail, possibly implying implausible values), while Krinsky–Robb draws the parameters and propagates them through the mean formula to give an honest, asymmetric percentile interval — Chapters 7, 9.)*

15.9 Failure modes

F1. Defaulting to the normal without checking negative WTP. Reporting a normal mean that implies substantial negative WTP for a non-negative good. *Fix:* quantify $\Phi(-\mu/\sigma)$; compare to lognormal/spike; justify the distribution (§15.2, §15.6).

F2. Reporting a single tail-driven mean. A mean that swings with the assumed tail, presented as precise. *Fix:* report the median (robust) and the mean as a range across distributions, with the Turnbull floor (§15.5, §15.7).

F3. Ignoring the mean-vs-median distinction. Treating "the WTP" as unambiguous. *Fix:* report both; use the median for tail-sensitive central estimates, the mean (with caveats) for aggregation (§15.5).

F4. Double-bounded efficiency taken at face value. Using the tighter double-bounded estimate without testing for anchoring/shift. *Fix:* test second-bound parameter equality (LR); model anchoring or fall back to single-bounded (§15.4).

F5. WTP-ratio fragility from a weak bid coefficient. A huge median-WTP SE from few transitions / little bid spread near the median. *Fix:* design bids around the median (Chapter 49); use the WTP-function parameterization; simulate the interval (§15.8).

F6. Symmetric delta-method interval for a skewed mean. A ± band on a lognormal/tail-sensitive mean. *Fix:* Krinsky–Robb percentile interval (§15.8; Chapter 9).

F7. No distribution-free check. Reporting only parametric WTP. *Fix:* report the Turnbull lower bound as a robustness anchor (§15.7; Chapter 10).

F8. Reading a non-significant WTP difference as convergent validity. "SBDC and payment card give the same WTP (p>0.05)." *Fix:* equivalence test (TOST) against a margin; report the MDE (§15.8; Chapter 8).

15.10 Diagnostic questions for your own papers

1. **Distribution justified?** Is my assumed WTP distribution defended on the good's economics (support, skew), not defaulted? What does the negative-WTP mass ($\Phi(-\mu/\sigma)$) come to? *(§15.2, §15.6.)*

2. **Mean vs median.** Have I reported both, and is my mean tail-driven (does it swing across distributions / with σ_ℓ)? *(§15.5.)*
3. **Turnbull floor.** Have I reported the distribution-free Turnbull lower bound, and how big is the gap to my parametric mean? *(§15.7.)*
4. **Parameterization.** Am I recovering WTP as a ratio (Hanemann) or parameterizing it directly (Cameron), and does the choice affect my ability to constrain support/tail? *(§15.3.)*
5. **Double-bounded validity.** If I use double-bounded, have I tested for anchoring/shift and considered single-bounded as the incentive-compatible fallback? *(§15.4.)*
6. **WTP uncertainty.** Am I using Krinsky–Robb (not a symmetric delta interval) for skewed mean WTP, and is my bid coefficient strong enough for a stable median SE? *(§15.8.)*
7. **Cross-arm comparison.** For convergent validity, am I using the Poe test *and* an equivalence test, not just a non-significant difference? *(§15.8; Chapter 8.)*

15.11 Exercises

(A) Comprehension.

A1. Write the SBDC log-likelihood for normally distributed WTP, and state how mean and median WTP are recovered from the index coefficients (α, β).

A2. A lognormal SBDC fit gives $\mu_\ell = 3.4$, $\sigma_\ell = 0.7$. Compute the median and mean WTP. Which is larger and why?

A3. Explain why the Turnbull lower-bound mean is (almost) always below the parametric mean, and what a large gap between them tells you.

(B) Derivation.

B1. Derive the SBDC survival function $S(b) = \Phi(\alpha + \beta \log b)$ for lognormal WTP, and show median WTP $= e^{\mu_\ell}$ and mean WTP $= e^{\mu_\ell + \sigma_\ell^2/2}$.

B2. Write the double-bounded likelihood as four interval contributions, and explain the efficiency gain and the anchoring/shift pathology, including how you would test for the latter.

B3. For a lognormal mean WTP $C^+ = e^{\mu_\ell + \sigma_\ell^2/2}$, derive the delta-method variance in terms of $\mathrm{Var}(\mu_\ell), \mathrm{Var}(\sigma_\ell), \mathrm{Cov}$, and explain why Krinsky–Robb is preferable in practice.

(C) Research application / critique.

C1. Your CV paper reports a normal SBDC mean WTP of €48 with a 95% CI, and a reviewer notes the fit implies 18% negative WTP. Write the corrected analysis: what distributions to compare, what to report, and how the negative mass affects the mean.

C2. Your team prefers the double-bounded estimate because its standard errors are half those of the single-bounded. Critique this preference and lay out the tests and the trade-off you would present.

C3. You compare SBDC and payment-card mean WTP and find €34 vs €31, Poe test p = 0.22, and conclude "the formats give equivalent WTP, confirming convergent validity." Critique using §15.8 and Chapter 8.

15.12 Solutions

A1. $\ell = \sum_i [y_i \log \Phi(\frac{\mu - b_i}{\sigma}) + (1 - y_i) \log(1 - \Phi(\frac{\mu - b_i}{\sigma}))]$, equivalently $\Phi(\alpha + \beta b_i)$ with $\alpha = \mu/\sigma$, $\beta = -1/\sigma$. Mean = median = $\mu = -\alpha/\beta$; $\sigma = -1/\beta$. (Normal is symmetric, so mean and median coincide.)

A2. Median $= e^{\mu_\ell} = e^{3.4} = €29.96$. Mean $= e^{\mu_\ell + \sigma_\ell^2/2} = e^{3.4+0.245} = e^{3.645} = €38.3$. The mean is larger because the lognormal is right-skewed — the upper tail pulls the mean above the median; the gap grows with σ_ℓ.

A3. Turnbull values each interval's probability mass at its *lower* endpoint (a conservative choice) and cannot see mass above the highest bid (it lumps it at the highest bid rather than extrapolating a tail), so it systematically *underestimates* the mean — hence a lower bound. A large gap to the parametric mean indicates that much of the parametric mean comes from the *assumed upper tail beyond your bids* rather than the data — a tail-fragility warning that your headline mean is distribution-driven.

B1. Yes iff WTP $\geq b$ iff $\log \text{WTP} \geq \log b$; with $\log \text{WTP} \sim N(\mu_\ell, \sigma_\ell^2)$, $S(b) = \Pr(\log \text{WTP} \geq \log b) = 1 - \Phi(\frac{\log b - \mu_\ell}{\sigma_\ell}) = \Phi(\frac{\mu_\ell - \log b}{\sigma_\ell}) = \Phi(\alpha + \beta \log b)$ with $\alpha = \mu_\ell/\sigma_\ell$, $\beta = -1/\sigma_\ell$. Median: $S(C^{\text{med}}) = 0.5 \Rightarrow \mu_\ell - \log C^{\text{med}} = 0 \Rightarrow C^{\text{med}} = e^{\mu_\ell}$. Mean: for a lognormal, $\mathbb{E}[\text{WTP}] = e^{\mu_\ell + \sigma_\ell^2/2}$ (the standard lognormal mean).

B2. Contributions: (YY) $S(b^U)$; (YN) $F(b^U) - F(b_1)$; (NY) $F(b_1) - F(b^L)$; (NN) $F(b^L)$; likelihood is their product over respondents. Efficiency: the two interior outcomes are *bounded* intervals, pinning WTP more tightly than the single-bounded's one-sided threshold, reducing standard errors. Anchoring/shift pathology: the first bid anchors the follow-up response, and the implied distribution from the second answers is systematically shifted, violating the assumption of a single stable WTP. Test: estimate the model allowing the second-bound WTP parameters (mean and/or variance) to differ from the first and test equality with an LR test (Chapter 12); a significant difference reveals the shift/anchoring, in which case model it or fall back to single-bounded.

B3. $C^+ = g(\mu_\ell, \sigma_\ell) = e^{\mu_\ell + \sigma_\ell^2/2}$; $\partial g/\partial \mu_\ell = C^+$, $\partial g/\partial \sigma_\ell = \sigma_\ell C^+$. Delta variance $= \nabla g' \mathbf{V} \nabla g = (C^+)^2 \operatorname{Var}(\mu_\ell) + (\sigma_\ell C^+)^2 \operatorname{Var}(\sigma_\ell) + 2C^+(\sigma_\ell C^+) \operatorname{Cov}(\mu_\ell, \sigma_\ell) = (C^+)^2[\operatorname{Var}(\mu_\ell) + \sigma_\ell^2 \operatorname{Var}(\sigma_\ell) + 2\sigma_\ell \operatorname{Cov}]$. Krinsky–Robb is preferable because C^+ is a strongly nonlinear, right-skewed function of the parameters, so the delta method's *symmetric* interval mis-represents the true skewed sampling distribution (understating the right tail); simulating the parameters and propagating through g gives an honest asymmetric percentile interval.

C1. With 18% negative WTP the normal is imposing implausible structure on a non-negative good, and the negative mass biases the mean *downward* (it drags the symmetric distribution's center). Corrected analysis: (i) fit the **normal, lognormal, and a spike-at-zero** model and report mean and median WTP under each; (ii) quantify the negative mass and judge whether negative WTP is economically real for this good (if not, prefer the non-negative or spike model); (iii) report the **median** (robust) and the mean as a **range** across distributions, plus the **Turnbull lower bound**; (iv) note that the €48 normal mean is contaminated by the negative-mass artifact and that the preferred (spike/lognormal) mean and median are the defensible welfare figures. A single normal mean with 18% negative WTP should not be the headline.

C2. Preferring double-bounded purely for smaller standard errors ignores validity: the follow-up bid is not incentive-compatible and induces anchoring and a shift effect, so the tighter interval may be *biased*, not just more precise — a smaller SE around the wrong value is worse than a larger SE around the right one. I would present: (i) a test for the shift/anchoring (LR test allowing second-bound parameters to differ) — if significant, the double-bounded assumption fails; (ii) the single-bounded estimate (incentive-compatible, from the first response) as the validity benchmark; (iii) the double-bounded and an anchoring-corrected model; and (iv) an explicit efficiency-versus-validity discussion, with the single-bounded as the preferred estimate if anchoring is detected. Efficiency is only a virtue conditional on validity.

C3. The conclusion is unwarranted on two counts (Chapter 8): (i) Poe p = 0.22 is a *difference* test failing to reject — it is not evidence that SBDC and payment-card WTP are *equivalent* (absence of evidence $\neq$ evidence of absence); (ii) "equivalent, confirming convergent validity" requires *establishing* equivalence. To support convergent validity: pre-specify an equivalence margin Δ (the largest WTP gap irrelevant for the intended use), run an **equivalence test (TOST)** on the WTP difference (equivalently, check the 90% interval of the KR difference distribution lies within $\pm\Delta$), and report the MDE. Only if the equivalence test passes can you claim the formats give equivalent WTP; the bare non-significant Poe test cannot support convergent validity. (Also note the payment card's anchoring and the SBDC's coarseness may make a *difference* real

but undetected — check power.)

15.13 Spaced review prompts

+1 day. From memory, write the SBDC likelihood and derive mean/median WTP under normal and lognormal WTP. State why the normal manufactures negative WTP and the lognormal does not.

+1 week. Explain, with the tail-sensitivity table, why mean WTP is tail-dominated and median is robust; describe the negative-WTP remedies (lognormal, truncated, spike) and when each is appropriate; and state how Turnbull anchors the parametric mean.

+1 month. For your own CV data, fit the SBDC under normal, logistic, and lognormal, report mean and median under each plus the Turnbull lower bound, quantify any negative-WTP mass, and — for one cross-arm comparison — run a Poe test *and* an equivalence test. Close with the spine sentence about what welfare object you are estimating and how tail-sensitive it is.

15.14 How this chapter connects to the rest of the book

- **Chapter 14** supplied the censoring framework and the mean-as-area-under-survival identity that this chapter makes quantitative; **Chapter 11** the likelihood/MLE engine; **Chapter 10** the Turnbull benchmark; **Chapter 5** the distributions and the negative-WTP support problem.
- **Chapter 24 (WTP space)** is the DCE sequel to §15.3's utility-difference-vs-WTP-function distinction — the CV and DCE versions of preference space vs WTP space; **Chapter 26 (CV formats)** develops the spike/interval-data models of §15.6 in full.
- **Chapter 8 (power/equivalence)** and **Chapter 10 (Poe test)** are the cross-arm-comparison machinery of §15.8; **Chapters 27–29 (behavioral foundations, incentive compatibility, bid/cost-vector effects)** are the anchoring/shift pathologies of §15.4 and the cost-vector-in-reporting-vs-preferences theme central to your CV paper.
- **Chapter 4 (discounting)** and **Chapter 44 (double counting)** consume the mean/median WTP produced here as CBA benefit inputs — where the tail-sensitivity of §15.5 becomes a CBA-robustness issue; **Chapter 37 (efficient design)** is the bid-design question that determines how well your data pin down the WTP distribution and its tail.

This chapter completes **Part III** and your PDF's **Block 1**: you can now write the SBDC, double-bounded, and payment-card likelihoods from scratch, compute mean and median WTP under alternative distributions, judge the normality/support/tail assumptions, and anchor everything with Turnbull — the full contingent-valuation econometrics of your CV paper.

15.15 Research translation task

For your **CV paper**, produce a three-to-four-page note that is the completed Block 1 deliverable:

1. **Write the likelihoods from scratch.** For your SBDC and payment-card (and any double-bounded) data, write the exact likelihood, and state the two parameterizations (utility-difference vs WTP-function) and which you use.
2. **Fit alternative distributions.** Estimate under normal, logistic, and lognormal (and, if warranted, a spike-at-zero) WTP; report mean and median WTP under each, and the implied negative-WTP mass under the symmetric fits.
3. **Assess tail sensitivity.** Show how your mean WTP moves across distributions and with the tail parameter; report the median as the robust central estimate.

4. **Turnbull anchor.** Compute the Turnbull distribution-free lower-bound mean and the gap to your parametric mean; interpret the gap as tail-fragility.
5. **WTP uncertainty and comparisons.** Report Krinsky–Robb percentile intervals for mean WTP; for each convergent-validity comparison (SBDC vs PC, hypothetical vs incentivized, BRIS vs full payment), run the Poe test *and* an equivalence test (Chapter 8).
6. **Double-bounded validity (if used).** Test for anchoring/shift and state your efficiency-vs-validity choice.
7. **Close** with the spine sentence: *"My CV welfare estimate is the [mean/median] WTP under [distribution]; it is [tail-sensitive/robust] (parametric mean [range] vs Turnbull floor [value]); I justify the distribution by [support/economics], and convergent-validity claims rest on [equivalence tests], not non-significant differences."*

Carry this into Chapters 24 (WTP space), 26 (spike/CV formats), 27–29 (behavioral/incentive foundations), and 44 (CBA benefit inputs).

15.16 Bibliography

Core. - Haab & McConnell, *Valuing Environmental and Natural Resources: The Econometrics of Non-Market Valuation*. The definitive treatment of SBDC, double-bounded, and payment-card likelihoods, mean vs median WTP, negative-WTP and Turnbull — the single most important source for this chapter and your CV paper. *(Verify edition/chapters.)* - Hanemann (1984), "Welfare Evaluations in Contingent Valuation Experiments with Discrete Responses," *American Journal of Agricultural Economics*. The foundational utility-difference derivation of the SBDC and mean/median WTP (§15.2–§15.3). *(Verify volume/pages.)* - Cameron (1988), "A New Paradigm for Valuing Non-Market Goods Using Referendum Data," *Journal of Environmental Economics and Management*. The WTP-function parameterization of §15.3 — the CV ancestor of WTP space. *(Verify volume/pages.)*

Applied. - Hanemann, Loomis & Kanninen (1991), "Statistical Efficiency of Double-Bounded Dichotomous Choice Contingent Valuation," *AJAE*. The efficiency case for double-bounded (§15.4). *(Verify volume/pages.)* - Herriges & Shogren, and Cameron & Quiggin, on anchoring/shift effects in double-bounded CV — the pathologies of §15.4. *(Verify exact citations.)* - Kriström (1990), "A Non-Parametric Approach to the Estimation of Welfare Measures in Discrete Response Valuation Studies," *Land Economics*, and the spike-model literature — §15.6–§15.7. *(Verify volume/pages.)*

Advanced. - Carson & Hanemann, "Contingent Valuation," in the *Handbook of Environmental Economics*. The comprehensive treatment of CV econometrics, distributional choice, and welfare measures. *(Verify volume/year.)* - The NOAA Panel report (Arrow et al., 1993) on contingent valuation — the guidelines favoring conservative (median, referendum) estimates that motivate §15.5's reporting recommendations. *(Verify citation.)* - Turnbull (1976), *JRSS-B*, for the nonparametric interval-censored MLE behind §15.7. *(Verify volume/pages.)*

Entries marked "verify" are pointers; confirm volumes, pages, and the Hanemann/Cameron/Kriström citations before using them, per the book's no-unverified-citation rule.

End of Chapter 15 and of Part III. You now hold the full maximum-likelihood and latent-variable apparatus — the likelihood engine (Ch 11), the testing trinity (Ch 12), the computational craft (Ch 13), the censoring framework (Ch 14), and the complete contingent-valuation econometrics of your CV paper (Ch 15) — and your PDF's Block 1 deliverable is complete: you can write the SBDC and payment-card likelihoods from scratch and defend the distributional and welfare choices they rest on. Part IV — Core Econometrics — begins with Chapter 16 (OLS from First Principles), or the discrete-choice heart of your DCE awaits in Part V. Tell me which chapter to write next.

Part III Assessment — Maximum Likelihood and Latent Variable Models

Covers Chapters 11–15. Five components, in the order they are best attempted. Allow a full working day; the mock oral is the part that matters most for your viva.

1. Cumulative problem set

P1. Write the log-likelihood, score and Hessian for the Bernoulli model with a logit link in a single covariate, and show the information is maximized where $p = 0.5$.

P2. Derive the SBDC likelihood for lognormal WTP and show it is a probit in $\ln b$. Recover median and mean WTP from the index coefficients.

P3. A payment-card respondent circles €20 having declined €30. Write their exact likelihood contribution under a lognormal WTP, and explain why it is not the same as a point observation at €20.

P4. Fix median WTP at €25 and tabulate the lognormal mean for $\sigma_\ell \in \{0.4, 0.8, 1.2\}$. State what your bid vector would have to look like to distinguish these.

P5. An LR statistic of 3.1 tests whether a random-parameter variance is zero. Give the naive and the boundary-corrected p-values and the resulting specification decision.

2. Research critique

Take the estimation section of your own CV paper. For each reported WTP number, state the assumed distribution, the implied negative-WTP mass, whether the number is a mean or a median, and how far it sits from a Turnbull lower bound computed on the same data. Write a paragraph on which of your reported numbers is most sensitive to the distributional assumption and what you will do about it.

3. Red-flag diagnosis

A mixed logit reports a log-likelihood of -4128.3, all parameters significant, and "convergence achieved." The smallest eigenvalue of the negative Hessian is 3×10^{-7}. A second run from different starting values reports -4119.8. Diagnose, in order, what has happened, what each piece of evidence establishes, and what you would do before reporting anything.

4. Mock oral examination

1. What is the score, and what does it mean that it is zero at the MLE?

2. Why can Wald, LR and score tests disagree, and which would you trust?
3. What does it mean for the log-likelihood to be flat in a direction?
4. Derive mean WTP as the area under the survival function and explain why it is tail-dominated.
5. Why might you prefer the single-bounded estimate to the more efficient double-bounded one?

5. Integration mini-project

Re-derive from scratch, on paper, the SBDC and payment-card interval likelihoods for your own data structure — not the textbook version, but with your actual bid vector, your actual response categories, and your actual covariates. Then estimate both in R without using a canned CV package, and reconcile your results with whatever you previously reported.

Part IV — Core Econometrics

Chapter 16 — OLS from First Principles

Part IV: Core Econometrics

> **The spine question, applied to this chapter.** Your PDF is blunt: *"Anche se lavori con modelli più avanzati, devi avere OLS solidissimo."* You may estimate mixed logits and MDCEV, but ordinary least squares is the backbone underneath — the model you run for meta-regression on your finfish WTP estimates, for balance checks in your survey experiments, for auxiliary and robustness regressions, and the conceptual template that logit, probit, and mixed logit generalize. This chapter derives OLS from scratch: the estimator, its variance, the exogeneity assumption on which everything rests, the omitted-variable-bias formula, and the crucial distinction between multicollinearity (which hurts *precision*) and endogeneity (which causes *bias*). The spine question here: *when I run a linear regression, what am I identifying — a causal effect or a conditional correlation — under what exogeneity assumption, and is a large standard error telling me about bias, or only about precision?*

16.0 Conceptual motivation: the linear backbone of everything you do

It is tempting to treat OLS as beneath a researcher who works with random-utility models and simulated likelihoods. That would be a mistake, and your PDF says why: OLS is not a lesser tool you have outgrown — it is the *foundation* on which the advanced tools stand, and it appears directly, all over your empirical work:

- **Meta-regression (the finfish paper).** Regressing 18 studies' WTP estimates on study-level moderators is OLS (or its weighted/robust cousins, Chapters 17, 47). Every concern this chapter raises — omitted moderators (OVB), correlated moderators (multicollinearity), what the regression identifies — is a meta-regression concern.
- **Balance checks (survey experiments).** Regressing baseline covariates on treatment assignment to check randomization is OLS. And the deepest lesson of this chapter — that *randomization guarantees exogeneity and therefore immunity to omitted-variable bias* — is exactly why your survey experiments can identify causal effects that observational studies cannot (§16.6).
- **Auxiliary and robustness regressions.** Diagnosing multicollinearity, running first-stage or reduced-form checks, testing linear restrictions — all OLS.
- **The conceptual template.** The binary-response models (Chapter 18), the random-utility models (Chapters 20–25), and meta-regression (Chapter 47) all *generalize* the linear model. The score-and-orthogonality logic of Chapter 11 first appears here as the OLS normal equations. Understanding OLS deeply is understanding the skeleton of everything that follows.

This chapter builds OLS the way your PDF wants it — *solidissimo* — from first principles. We derive the

estimator two ways (minimizing squared residuals, and as a method of moments), derive its variance, state and interpret the classical assumptions (especially **exogeneity**, the load-bearing one), prove unbiasedness and the Gauss–Markov efficiency result, derive the **omitted-variable-bias** formula and its sign, dissect **multicollinearity** (and why it is a precision problem, not a bias problem — the distinction your learning objective demands), and treat **prediction** (fitted values, the hat matrix, prediction vs confidence intervals). Heteroskedasticity, robust and clustered standard errors, interactions, and residual diagnostics — the *inference-robustness* half — are Chapter 17; here we build the estimator and its logic. Every result connects to the linear-algebra of Chapter 1 (the $\mathbf{X}'\mathbf{X}$ matrix, its inverse, its conditioning) and to your meta-regression and balance-check applications.

16.1 Prerequisites

From **Chapter 1**: the data matrix $\mathbf{X}$, the matrix product and $\mathbf{X}'\mathbf{X}$ (§1.3), rank and identification (§1.4), the inverse (§1.5), the variance–covariance matrix (§1.8), and conditioning (§1.10). From **Chapter 2**: first- and second-order conditions for a minimum (§2.7). From **Chapter 7**: bias, variance, efficiency, and BLUE (§7.2–7.3). From **Chapter 6**: consistency and the LLN/CLT (for the asymptotic view).

Same **notation**. Consolidated symbols: $\mathbf{y}$ ($n\times 1$ outcome), $\mathbf{X}$ ($n\times k$ regressors, including a constant column), β ($k\times 1$ coefficients), ε (error), $\widehat{\beta}$ (OLS estimator), $\mathbf{e}=\mathbf{y}-\mathbf{X}\widehat{\beta}$ (residuals), σ^2 (error variance), $\widehat{\mathbf{y}}=\mathbf{X}\widehat{\beta}$ (fitted values).

16.2 The linear model and the OLS estimator

The model. The linear regression model, in matrix form, is

$$\mathbf{y}=\mathbf{X}\beta+\varepsilon,$$

where each row i is $y_i=\mathbf{x}_i'\beta+\varepsilon_i$. The coefficient β_j is the effect of a one-unit change in regressor j on the *conditional mean* of y, holding the other regressors fixed — a *ceteris paribus* partial effect (Chapter 2, §2.10). Whether that partial effect is *causal* depends on the exogeneity assumption (§16.5); the algebra of OLS is agnostic about causality.

Deriving OLS by minimizing the sum of squared residuals. OLS chooses $\widehat{\beta}$ to minimize the sum of squared residuals:

$$S(\mathbf{b})=(\mathbf{y}-\mathbf{X}\mathbf{b})'(\mathbf{y}-\mathbf{X}\mathbf{b})=\mathbf{y}'\mathbf{y}-2\mathbf{b}'\mathbf{X}'\mathbf{y}+\mathbf{b}'\mathbf{X}'\mathbf{X}\mathbf{b}.$$

Differentiate with respect to $\mathbf{b}$ (Chapter 2) and set to zero — the **first-order condition**:

$$\frac{\partial S}{\partial \mathbf{b}}=-2\mathbf{X}'\mathbf{y}+2\mathbf{X}'\mathbf{X}\mathbf{b}=\mathbf{0}\quad\Longrightarrow\quad\mathbf{X}'\mathbf{X}\,\widehat{\beta}=\mathbf{X}'\mathbf{y},$$

the **normal equations**. Solving (provided $\mathbf{X}'\mathbf{X}$ is invertible — full column rank, Chapter 1 §1.4):

$$\boxed{\widehat{\beta}=(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}}.$$

The **second-order condition** confirms a minimum: $\partial^2 S/\partial\mathbf{b}\partial\mathbf{b}'=2\mathbf{X}'\mathbf{X}$ is positive definite when $\mathbf{X}$ has full rank (Chapter 1, §1.7; the $\mathbf{X}'\mathbf{X}$-is-PD result of Chapter 1 Exercise B1). This is the same identification condition you have met repeatedly: **full column rank $\Leftrightarrow$ $\mathbf{X}'\mathbf{X}$ invertible $\Leftrightarrow$ β identified** (Chapter 1, Proposition 1.1). Perfect collinearity breaks it (§16.7).

Deriving OLS as a method of moments (the orthogonality view). There is a second, deeper derivation that connects OLS to the likelihood framework and to exogeneity. The residuals satisfy $\mathbf{X}'\mathbf{e}=\mathbf{X}'(\mathbf{y}-\mathbf{X}\widehat{\beta})=\mathbf{X}'\mathbf{y}-\mathbf{X}'\mathbf{X}\widehat{\beta}=\mathbf{0}$ (from the normal equations). So **OLS makes the residuals orthogonal**

to every regressor — the sample analogue of the population *exogeneity* condition $\mathbb{E}[\mathbf{x}\varepsilon] = \mathbf{0}$. This is exactly the zero-mean-score / orthogonality property of maximum likelihood (Chapter 11, §11.4): OLS is the method-of-moments estimator that imposes "regressors uncorrelated with residuals" in the sample. Seeing OLS as *imposing orthogonality* rather than merely *minimizing squares* is what makes the exogeneity assumption (§16.5) intuitive.

> **Active recall 16.A.** The OLS normal equations imply $\mathbf{X}'\mathbf{e} = \mathbf{0}$. State what this says in words, and connect it to (i) the exogeneity assumption and (ii) the maximum-likelihood score. *(Answer: the residuals are orthogonal to (uncorrelated with) every regressor in the sample — OLS imposes "no leftover linear relationship between regressors and residuals." This is the sample analogue of the population exogeneity condition $\mathbb{E}[\mathbf{x}\varepsilon] = \mathbf{0}$, and it is the OLS version of the ML score-equals-zero orthogonality (Chapter 11, §11.4).)*

16.3 The classical assumptions

OLS's properties depend on assumptions. State them explicitly, because each result below requires a specific subset, and knowing *which* assumption buys *which* property is the difference between using OLS well and mechanically.

A1 — Linearity. $\mathbf{y} = \mathbf{X}\beta + \varepsilon$: the conditional mean is linear in the parameters. (Linear in *parameters*, not necessarily in variables — you can include x^2, $\log x$, interactions.)

A2 — Full rank (no perfect collinearity). $\mathbf{X}$ has full column rank, so $\mathbf{X}'\mathbf{X}$ is invertible and β is identified (Chapter 1, §1.4). Near-violation is multicollinearity (§16.7).

A3 — Exogeneity (the load-bearing one). $\mathbb{E}[\varepsilon \mid \mathbf{X}] = \mathbf{0}$: the error has zero mean conditional on the regressors — equivalently, the regressors carry no information about the error. This is what makes OLS *unbiased* and gives its coefficients a causal reading (§16.5). It is the assumption that fails under omitted variables, simultaneity, and measurement error.

A4 — Homoskedasticity and no autocorrelation. $\mathrm{Var}(\varepsilon \mid \mathbf{X}) = \sigma^2\mathbf{I}$: constant error variance, uncorrelated errors. This gives the *simple* variance formula and the Gauss–Markov efficiency result. Its failure (heteroskedasticity, clustering) does **not** bias $\hat{\beta}$ but invalidates the classical standard errors — the subject of Chapter 17.

A5 — (for exact inference) Normal errors. $\varepsilon \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$: gives exact finite-sample t and F distributions. Not needed for unbiasedness or (asymptotically) for inference — the CLT (Chapter 6) delivers approximate normality of $\hat{\beta}$ without it.

The hierarchy matters: **A1–A3 give unbiasedness/consistency (the estimate is right on average); A4 additionally gives efficiency and the simple variance formula (the standard errors are right); A5 gives exact small-sample inference.** When your PDF asks you to judge a regression critically, you are asking which of these hold. A balance check that fails, a clustered design, a possibly-omitted confounder — each is a specific assumption at risk.

> **Stop and think 16.1.** In your finfish meta-regression, errors are almost certainly heteroskedastic (estimates from large studies are more precise than from small ones) and clustered (multiple estimates per study). Which classical assumption(s) fail, and does this bias your moderator coefficients or (only) your standard errors? *(Answer: A4 (homoskedasticity/no-autocorrelation) fails — the errors have unequal variances and within-study correlation. This does **not** bias the coefficient estimates (A1–A3 still give unbiasedness); it invalidates the classical standard errors, which will be wrong (typically too small). The fix is robust/clustered/weighted inference (Chapters 17, 47), not a different point estimate — a crucial distinction your PDF's meta-analysis section relies on.)*

16.4 Unbiasedness, variance, and the Gauss–Markov theorem

Now the core properties, derived. These assemble Chapter 1's linear algebra and Chapter 7's estimator concepts.

Unbiasedness (needs A1–A3). Substitute the true model into the estimator:

$$\widehat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \varepsilon) = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon.$$

Take the conditional expectation and use exogeneity $\mathbb{E}[\varepsilon \mid \mathbf{X}] = \mathbf{0}$:

$$\mathbb{E}[\widehat{\beta} \mid \mathbf{X}] = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\mathbb{E}[\varepsilon \mid \mathbf{X}] = \beta.$$

OLS is unbiased — *if and only if* exogeneity holds. The entire causal content of OLS lives in that last step; when $\mathbb{E}[\varepsilon \mid \mathbf{X}] \neq \mathbf{0}$ (endogeneity), the estimator is biased by $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}[\varepsilon \mid \mathbf{X}]$ — the seed of the OVB formula (§16.6). *(Verified: over 3000 simulations with exogeneity, mean $\widehat{\beta}_1 = 1.499$ against a true 1.5.)*

Variance (needs A1–A4). From $\widehat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon$,

$$\mathrm{Var}(\widehat{\beta} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\ \mathrm{Var}(\varepsilon \mid \mathbf{X})\,\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}.$$

Under homoskedasticity $\mathrm{Var}(\varepsilon \mid \mathbf{X}) = \sigma^2\mathbf{I}$, the middle collapses:

$$\boxed{\mathrm{Var}(\widehat{\beta} \mid \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}}.$$

This is exactly the Chapter 1 statement that **the covariance of the estimates is (a scalar times) the inverse of the cross-product matrix** (§1.5, §1.8): standard errors are $\sqrt{\text{diagonal of } \sigma^2(\mathbf{X}'\mathbf{X})^{-1}}$, estimating σ^2 by $\widehat{\sigma}^2 = \mathbf{e}'\mathbf{e}/(n-k)$. The general "sandwich" form $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\Omega\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$ — used when A4 fails (heteroskedasticity, clustering) — is Chapter 17 (and it is the OLS instance of the sandwich you met for the MLE in Chapter 11 §11.5).

Gauss–Markov (needs A1–A4). Among all *linear, unbiased* estimators, OLS has the smallest variance — it is **BLUE** (Best Linear Unbiased Estimator). This is why OLS is the default under the classical assumptions: no other linear unbiased estimator does better. (Adding A5, normality, makes it efficient among *all* unbiased estimators, attaining the Cramér–Rao bound of Chapter 7.) The value of BLUE is precise: it is a statement about *efficiency given unbiasedness* — it says nothing if exogeneity fails, because then OLS is not unbiased at all and "best unbiased" is beside the point. Efficiency is a virtue only conditional on the estimate being right.

> **Active recall 16.B.** Write the OLS variance formula and name the two assumptions it requires beyond identification. What replaces it when errors are heteroskedastic or clustered? *(Answer:* $\mathrm{Var}(\widehat{\beta} \mid \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$*, requiring exogeneity (A3) for it to be the variance of an unbiased estimator and homoskedasticity/no-autocorrelation (A4) for the simple $\sigma^2\mathbf{I}$ form. Under heteroskedasticity/clustering it is replaced by the robust/clustered sandwich* $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\widehat{\Omega}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$*, Chapter 17.)*

16.5 Exogeneity: the assumption that decides everything

Exogeneity ($\mathbb{E}[\varepsilon \mid \mathbf{X}] = \mathbf{0}$) is the hinge between "OLS estimates a causal effect" and "OLS estimates a conditional correlation." It deserves careful statement because it is where empirical work most often goes wrong, and where your survey experiments have a decisive advantage over observational studies.

What it means. The error ε contains *everything about y not captured by* $\mathbf{X}$ — all omitted determinants, measurement error, and idiosyncratic noise. Exogeneity says this bundle is *unrelated* to the included regressors. Equivalently: within any group of observations sharing the same $\mathbf{X}$, the average of the unmodeled

factors is zero, so variation in y across different $\mathbf{X}$ reflects the effect *of* $\mathbf{X}$, not of hidden factors correlated with $\mathbf{X}$.

The three classic ways it fails.

1. **Omitted-variable bias.** A determinant of y that is correlated with an included regressor is in ε; its correlation with $\mathbf{X}$ violates exogeneity and biases the coefficient (§16.6).
2. **Simultaneity / reverse causality.** If y also affects $\mathbf{X}$ (e.g., price and quantity jointly determined), $\mathbf{X}$ is correlated with the error.
3. **Measurement error** in a regressor biases its coefficient toward zero (attenuation).

Why randomization is the exogeneity machine. Here is the deepest point of the chapter, and the reason your survey experiments matter. When a regressor (treatment) is **randomly assigned**, it is *by construction* independent of everything else — including all omitted variables. So random assignment *guarantees exogeneity* of the treatment, and therefore *immunity to omitted-variable bias*: you can omit every confounder and still estimate the treatment effect without bias, because randomization severs the correlation between treatment and the error.

Worked Example 16.1 (Randomization guarantees exogeneity, verified). Suppose the true model is $y = 2 + 3T + 0.8x_2 + \varepsilon$, with an unobserved x_2 (effect 0.8), and you regress y on T alone (omitting x_2):

- **Randomized** T (independent of x_2): the estimated treatment effect averages 3.005 — *no bias*, despite omitting x_2, because $T \perp x_2$ makes the omitted-variable correlation zero (verified over 3000 simulations).
- **Observational** T (assigned in a way correlated with x_2): the estimated effect averages 3.906 — *biased upward by* ≈ 0.9, because T is now correlated with the omitted x_2 (verified).

This contrast *is* the case for experiments. In your survey experiments — high vs low cost vector, hypothetical vs incentivized, BRIS vs full payment — because assignment is random, the treatment coefficient is exogenous and unbiased *without* controlling for covariates. This is why a balance check is a *diagnostic of the randomization*, not a prerequisite for unbiasedness, and why "controlling for covariates" in a randomized experiment is about *precision*, not bias (and can even introduce bias if you control for post-treatment variables — Chapter 32). Randomization does the identification work that no amount of control variables can do in an observational study.

> **Stop and think 16.2.** In your randomized survey experiment, a colleague worries "we didn't control for respondent income, so our cost-vector effect is biased." Using §16.5, explain why this worry is (largely) misplaced, and when controlling for income *would* matter. *(Answer: because the cost-vector treatment was randomly assigned, it is independent of income (and all other covariates), so omitting income does not bias the treatment effect — randomization guarantees exogeneity and immunity to OVB (verified: randomized T gives no bias despite omitting a covariate). Controlling for income would matter only for* precision* (it can shrink the residual variance and tighten the standard error) — provided income is a pre-treatment variable; controlling for a post-treatment variable would instead introduce bias, Chapter 32.)*

16.6 Omitted-variable bias: the formula and its sign

The most important way exogeneity fails in observational (and meta-regression) work is omitted variables. The bias has an exact, interpretable formula worth deriving and memorizing.

Setup. The true model includes two sets of regressors, $\mathbf{y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \varepsilon$, but you estimate the **short regression** omitting $\mathbf{X}_2$: $\hat{\beta}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y}$.

Derivation. Substitute the true $\mathbf{y}$:

$$\hat{\beta}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'(\mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \varepsilon) = \beta_1 + \underbrace{(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2}_{\delta}\,\beta_2 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\varepsilon.$$

Taking expectations (with $\mathbb{E}[\varepsilon \mid \mathbf{X}] = \mathbf{0}$):

$$\boxed{\mathbb{E}[\widehat{\beta}_1] = \beta_1 + \delta\,\beta_2}, \qquad \delta = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2,$$

where δ is the coefficient from regressing the *omitted* variable $\mathbf{X}_2$ on the *included* $\mathbf{X}_1$. So the **omitted-variable bias is (effect of omitted-on-included) × (effect of omitted-on-outcome):**

$$\text{Bias} = \delta\,\beta_2 = (\text{how } \mathbf{X}_2 \text{ relates to } \mathbf{X}_1) \times (\text{how } \mathbf{X}_2 \text{ affects } y).$$

The sign rule. For a single omitted variable, the bias sign is

$$\text{sign(bias)} = \text{sign}(\delta) \times \text{sign}(\beta_2) = \text{sign}(\,\text{Corr}(x_1, x_2)) \times \text{sign}(\beta_2).$$

Positive-correlated omitted variable with a positive effect → *upward* bias; and so on. This lets you *sign* the bias even without the omitted data — a routine and powerful robustness argument.

Worked Example 16.2 (OVB formula, verified). True model $y = 2 + 1.5x_1 + 0.8x_2 + \varepsilon$, with $\text{Corr}(x_1, x_2) = 0.6$. Omitting x_2 and regressing y on x_1: the predicted biased coefficient is $\beta_1 + \delta\beta_2 = 1.5 + 0.6 \times 0.8 = 1.98$; the simulation gives 1.979 (verified over 3000 draws) — an exact match. Including x_2 restores the unbiased 1.499. The bias is *positive* because x_2 is positively correlated with x_1 ($\delta > 0$) and has a positive effect ($\beta_2 > 0$).

Why this matters for your work.

- **Meta-regression (finfish).** If a study-level moderator that affects WTP (say, the ecosystem type) is omitted and correlated with an included moderator (say, the valuation method), your included moderator's coefficient is biased by $\delta\beta_2$. You cannot include every moderator (too few studies — Chapter 47), so *signing* the potential bias of omitted moderators is an essential robustness argument.
- **Balance checks / observational comparisons.** When a treatment is *not* randomized (e.g., comparing experienced vs inexperienced respondents, who differ in many ways), the "treatment" coefficient conflates the effect of interest with the effects of omitted differences — pure OVB. This is why the experienced/inexperienced comparison in your DCE is *not* a clean causal contrast the way a randomized cost-vector comparison is (Chapters 30–32).
- **Bad controls (preview).** OVB tempts you to "control for everything," but controlling for a *post-treatment* variable (a mediator or collider) can *introduce* bias rather than remove it — the bad-controls problem of Chapter 32. Not every added control reduces bias; some create it.

Active recall 16.C. You omit a moderator that is *negatively* correlated with your included moderator and has a *positive* effect on WTP. In which direction is your included moderator's coefficient biased? *(Answer: downward — sign(bias) = sign(δ)·sign(β_2) = (negative correlation)·(positive effect) = negative, so the coefficient is biased toward being too small / too negative. You can sign the bias without the omitted data.)*

16.7 Multicollinearity: a precision problem, not a bias problem

Your learning objective is explicit: *explain multicollinearity's effect on standard errors without conflating it with bias.* This is the single most common confusion about OLS, and getting it right is a mark of a careful analyst.

What it is. Multicollinearity is when regressors are highly (but not perfectly) correlated — one regressor is nearly a linear combination of the others. Perfect collinearity (§16.2, Chapter 1 §1.4) breaks identification entirely; *near*-collinearity is the dangerous, subtler case: the model is technically identified, but the data struggle to separate the correlated regressors' effects.

What it does — and does not — do. Multicollinearity **inflates the variance (standard errors)** of the affected coefficients but leaves them **unbiased**. The variance of coefficient j can be written

$$\mathrm{Var}(\hat{\beta}_j) = \frac{\sigma^2}{(n-1)\ \mathrm{Var}(x_j)} \cdot \underbrace{\frac{1}{1-R_j^2}}_{\mathrm{VIF}_j},$$

where R_j^2 is the R^2 from regressing x_j on the *other* regressors, and $\mathrm{VIF}_j = 1/(1-R_j^2)$ is the **variance inflation factor**. As x_j becomes more collinear with the others ($R_j^2 \to 1$), $\mathrm{VIF}_j \to \infty$ and the standard error explodes — but the point estimate remains centered on the truth. This is the linear-algebra fact of Chapter 1 (§1.10): near-collinearity makes $\mathbf{X}'\mathbf{X}$ ill-conditioned, so its inverse (the covariance) has large entries, but the *expectation* of $\hat{\beta}$ is unaffected.

Worked Example 16.3 (Multicollinearity inflates SE, not bias, verified). With two regressors correlated at 0.95, the VIF is $1/(1-0.95^2) = 10.26$, so standard errors should inflate by $\sqrt{10.26} = 3.20$ relative to orthogonal regressors. The simulation confirms: the collinear coefficient averages 1.498 (unbiased, true 1.5) with mean SE 0.161, versus the orthogonal case's 1.502 with SE 0.050 — an SE ratio of 3.20, exactly $\sqrt{\mathrm{VIF}}$ (verified). **Same unbiased estimate, standard error 3.2 times larger.**

The correct response to multicollinearity. Because it is a *precision* problem, the responses are about information, not bias:

- Recognize that large standard errors from collinearity are **not** an OVB or specification error — the estimates are still unbiased, just imprecise.
- **Some combinations are precise even when individuals are not** (Chapter 1, §1.8, Stop-and-think 1.3): the sum or difference of two collinear coefficients may be well estimated even when each is not — so if you care about a *combination*, collinearity may not hurt.
- **Do not drop a collinear variable to "fix" it** if it belongs in the model — dropping it re-introduces omitted-variable bias (§16.6). You would be trading a (harmless) precision problem for a (harmful) bias problem.
- **Get more/better data** with more independent variation, or **rescale/reparameterize** (Chapter 1 §1.10; Chapter 13 §13.8), or accept the imprecision and report it honestly.
- In **meta-regression**, near-collinear moderators (Chapter 1, Exercise C2) mean you cannot separately identify their effects with few studies — report the identified combination, do not force a spurious separation (Chapter 47).

The one-sentence discipline: **multicollinearity is about what the data can *resolve* (precision), not about whether the estimate is *right* (bias); endogeneity is the reverse.** Confusing the two leads to the worst possible fix — dropping a needed variable, converting imprecision into bias.

> **Stop and think 16.3.** Two moderators in your meta-regression are correlated at 0.9, and both have huge standard errors. A co-author says "drop one to fix the multicollinearity." Why is this potentially the *wrong* fix, and what would you do? *(Answer: dropping a moderator that genuinely belongs in the model re-introduces omitted-variable bias — you would trade a harmless precision problem (unbiased but imprecise) for a harmful bias problem (precise but wrong). The estimates are still unbiased; the issue is only that the data cannot separate the two correlated moderators. Better: report the imprecision honestly, report the identified* combination* of the two (which may be well estimated), get more studies with independent variation, or acknowledge that with few studies these two effects are not separately identified — Chapter 47.)*

16.8 Prediction: fitted values, the hat matrix, and two kinds of interval

OLS is used not only to estimate effects but to *predict*. The distinction between predicting the *mean* and predicting a *new outcome* matters, especially for benefit transfer (predicting WTP for a new site from a meta-regression).

Fitted values and the hat matrix. The fitted (predicted) values are $\hat{\mathbf{y}} = \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y}$, where $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is the **hat matrix** (it "puts the hat on $\mathbf{y}$"). $\mathbf{H}$ is the projection onto the column space of $\mathbf{X}$: it projects $\mathbf{y}$ onto the space spanned by the regressors, and the residuals $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y}$ are the part orthogonal to that space — the geometric meaning of "OLS makes residuals orthogonal to regressors" (§16.2). The diagonal entries h_{ii} (leverage) measure how much each observation influences its own fitted value — a residual-diagnostic tool developed in Chapter 17.

Two kinds of prediction interval (do not confuse them). For a new point $\mathbf{x}_0$:

- The **confidence interval for the mean** $\mathbb{E}[y \mid \mathbf{x}_0] = \mathbf{x}_0'\beta$ has variance $\sigma^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$ — the uncertainty about the *regression line* at $\mathbf{x}_0$. It shrinks as $n \to \infty$.
- The **prediction interval for a new outcome** $y_0 = \mathbf{x}_0'\beta + \varepsilon_0$ has variance $\sigma^2[1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0]$ — the extra "+1" is the *irreducible* noise ε_0 of a single new observation. It is **wider** and does *not* shrink to zero even with infinite data, because you can never predict an individual outcome's noise.

Why this matters for benefit transfer. When you use a meta-regression to predict WTP for a *new* ecosystem/site (benefit transfer, Chapters 35, 49), you are predicting either an average (mean WTP for sites like this) or a specific value — and the second carries the irreducible variance. Confusing the (narrow) confidence interval for the mean with the (wide) prediction interval for a new site's WTP *understates* transfer uncertainty — a real error in applied valuation. Report the prediction interval for a transferred value.

A note on R^2. The coefficient of determination $R^2 = 1 - \mathbf{e}'\mathbf{e}/\sum(y_i - \bar{y})^2$ measures the fraction of outcome variance the regressors explain. Two cautions: (i) R^2 *never decreases* when you add regressors (even irrelevant ones), so use **adjusted** R^2 (which penalizes added parameters) for comparison, and information criteria (Chapter 12) for model selection; (ii) R^2 is a measure of *fit*, not of *causal validity or unbiasedness* — a high-R^2 regression can be badly biased (omitted variables), and a low-R^2 regression can be perfectly unbiased (lots of irreducible noise). A high R^2 is not evidence that you have identified a causal effect.

> **Active recall 16.D.** You use your finfish meta-regression to predict WTP for a new estuary. Which interval do you report — the confidence interval for the mean or the prediction interval — and why? *(Answer: the prediction interval, because you are transferring a value to a* specific new* site, which carries the irreducible outcome noise (the "+1" term) on top of the uncertainty about the regression line; using the (narrower) confidence interval for the mean would understate the true uncertainty of the transferred WTP — Chapters 35, 49.)*

16.9 Failure modes

F1. Reading a coefficient as causal without exogeneity. Interpreting an observational regression coefficient as a causal effect when omitted variables/simultaneity are plausible. *Fix:* justify exogeneity (randomization, natural experiment) or acknowledge it is a conditional correlation (§16.5).

F2. Confusing multicollinearity (precision) with endogeneity (bias). Treating large SEs from collinearity as a bias problem, or "fixing" it by dropping a needed variable (creating OVB). *Fix:* multicollinearity leaves estimates unbiased — report the imprecision; do not drop needed regressors (§16.7).

F3. Omitting a confounder in an observational comparison. Comparing non-randomized groups (experienced vs inexperienced) as if the difference were causal. *Fix:* recognize OVB; control for confounders (pre-treatment only) or use design-based identification (§16.6; Chapters 30–32).

F4. Controlling for a bad (post-treatment) variable. Adding a mediator/collider to "reduce bias," which introduces it. *Fix:* control only pre-treatment variables; use DAGs (Chapter 32).

F5. Using classical SEs under heteroskedasticity/clustering. Reporting $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ SEs when A4 fails. *Symptom:* implausibly small SEs. *Fix:* robust/clustered SEs — the estimate is fine, the SE is not (§16.4; Chapter 17).

F6. Confusing confidence and prediction intervals. Reporting the (narrow) mean-CI for a transferred/new-site value. *Fix:* prediction interval for new outcomes (§16.8).

F7. Reading R^2 as causal validity. Taking a high R^2 as evidence the model is "right." *Fix:* R^2 is fit, not identification; use adjusted R^2/IC for comparison and exogeneity arguments for causality (§16.8).

F8. Dropping observations or variables to improve fit. Data-mining for higher R^2 or significance. *Fix:* pre-specify the model; report the specification, not the best-looking one (Chapters 8, 52).

16.10 Diagnostic questions for your own papers

1. **What is identified?** Is my regression coefficient a causal effect (exogeneity justified — randomization?) or a conditional correlation? *(§16.5, F1.)*
2. **Exogeneity.** For each key regressor, is it plausibly uncorrelated with the error, or are there omitted confounders/simultaneity/measurement error? *(§16.5.)*
3. **OVB direction.** For plausible omitted variables, can I sign the bias ($\delta\beta_2$) even without the data? *(§16.6.)*
4. **Multicollinearity.** Are large SEs due to collinearity (precision, unbiased) rather than bias? Have I avoided dropping needed regressors to "fix" it? *(§16.7, F2.)*
5. **Randomization advantage.** In my survey experiments, am I relying on randomization for exogeneity (so controls are for precision, not bias) and avoiding bad (post-treatment) controls? *(§16.5, F4.)*
6. **Inference.** Are my standard errors valid given heteroskedasticity/clustering (meta-regression, repeated tasks)? *(§16.4, F5; Chapter 17.)*
7. **Prediction.** For benefit transfer, am I using the prediction interval (not the mean CI) for a new site's WTP? *(§16.8, F6.)*

16.11 Exercises

(A) Comprehension.

A1. Derive the OLS estimator from the normal equations, and state the condition under which it exists and is unique.

A2. State the exogeneity assumption and explain, in one paragraph, why randomization guarantees it for a treatment variable.

A3. A regressor has a VIF of 16. By what factor is its standard error inflated relative to no collinearity, and is its coefficient biased?

(B) Derivation.

B1. Derive the OLS variance $\mathrm{Var}(\hat{\beta} \mid \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$, stating each assumption used, and identify what changes under heteroskedasticity.

B2. Derive the omitted-variable-bias formula $\mathbb{E}[\hat{\beta}_1] = \beta_1 + \delta\beta_2$ and state the sign rule for a single omitted variable.

B3. Show that OLS residuals are orthogonal to the regressors ($\mathbf{X}'\mathbf{e} = \mathbf{0}$), and explain the connection to the exogeneity condition and the ML score.

(C) Research application / critique.

C1. In your meta-regression you must omit several moderators (too few studies). For one included moderator (valuation method), argue how you would sign the potential OVB from an omitted moderator (ecosystem type) that is correlated with it, and how you would report this.

C2. A colleague compares WTP between experienced and inexperienced respondents (not randomized) with OLS and calls the difference "the effect of experience." Critique this using §16.5–§16.6, and contrast it with your randomized cost-vector comparison.

C3. Your randomized survey experiment's treatment coefficient has a large standard error, and a reviewer says "add control variables to reduce the bias." Explain what is right and wrong in this advice using §16.5 and §16.7.

16.12 Solutions

A1. Minimizing $S(\mathbf{b}) = (\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b})$ gives the FOC $-2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\mathbf{b} = \mathbf{0}$, i.e. the normal equations $\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}$, so $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. It exists and is unique iff $\mathbf{X}'\mathbf{X}$ is invertible, i.e. $\mathbf{X}$ has full column rank (no perfect collinearity) — the identification condition (Chapter 1, §1.4).

A2. Exogeneity: $\mathbb{E}[\varepsilon \mid \mathbf{X}] = \mathbf{0}$ — the error (all omitted determinants of y) is uncorrelated with the regressors. Randomization guarantees it for a treatment because random assignment makes the treatment *statistically independent* of everything else, including all omitted variables; so the treatment cannot be correlated with the error, exogeneity holds by construction, and OLS is unbiased for the treatment effect even with confounders omitted.

A3. VIF = 16 means the variance is inflated by 16, so the standard error is inflated by $\sqrt{16} = 4$ relative to no collinearity. The coefficient is **not** biased — multicollinearity affects precision (variance/SE), not the expectation of the estimate.

B1. From $\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon$, $\mathrm{Var}(\hat{\beta} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\mathrm{Var}(\varepsilon \mid \mathbf{X})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$ (using that $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is fixed given $\mathbf{X}$). Under homoskedasticity $\mathrm{Var}(\varepsilon \mid \mathbf{X}) = \sigma^2\mathbf{I}$, so $= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$. Assumptions: exogeneity (A3, so this is the variance of an unbiased estimator) and homoskedasticity/no-autocorrelation (A4, for the $\sigma^2\mathbf{I}$ form). Under heteroskedasticity, $\mathrm{Var}(\varepsilon \mid \mathbf{X}) = \Omega \neq \sigma^2\mathbf{I}$ and the variance is the sandwich $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\Omega\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$ (Chapter 17).

B2. Short regression: $\hat{\beta}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y}$; substitute $\mathbf{y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \varepsilon$: $\hat{\beta}_1 = \beta_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\beta_2 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\varepsilon$. Taking expectations ($\mathbb{E}[\varepsilon \mid \mathbf{X}] = \mathbf{0}$): $\mathbb{E}[\hat{\beta}_1] = \beta_1 + \delta\beta_2$ with $\delta = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2$ (regression of omitted on included). Sign rule (single omitted): $\mathrm{sign}(\text{bias}) = \mathrm{sign}(\mathrm{Corr}(x_1, x_2))\cdot\mathrm{sign}(\beta_2)$.

B3. From the normal equations $\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}$: $\mathbf{X}'\mathbf{e} = \mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathbf{X}'\mathbf{y} - \mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{0}$. So the residuals are orthogonal to every regressor. This is the *sample* analogue of the population exogeneity condition $\mathbb{E}[\mathbf{x}\varepsilon] = \mathbf{0}$ (OLS imposes it by construction), and it is the OLS instance of the ML score-equals-zero orthogonality (Chapter 11, §11.4) — the estimator sets the sample "score" (here $\mathbf{X}'\mathbf{e}$) to zero.

C1. Suppose ecosystem type (omitted) affects WTP with sign β_2 and is correlated with valuation method (included) with sign δ; the bias on the method coefficient is $\delta\beta_2$, with sign = $\mathrm{sign}(\delta)\cdot\mathrm{sign}(\beta_2)$. For example, if studies using a particular method tend to value high-value ecosystems (positive δ) and high-value ecosystems have higher WTP (positive β_2), the method coefficient is biased *upward*. I would (i) state the plausible sign of the bias from theory/priors, (ii) note the direction it pushes my conclusion, (iii) where possible include a proxy for ecosystem type or restrict to comparable ecosystems, and (iv) report that with few studies I cannot fully control for it, so the coefficient should be read as potentially biased in the stated direction — an honest robustness statement (Chapter 47).

C2. Calling the experienced-vs-inexperienced difference "the effect of experience" is unjustified because experience is *not randomized* — experienced and inexperienced respondents differ in many omitted ways (income, familiarity, engagement) that also affect WTP, so the OLS difference conflates the effect of experience with those omitted differences (pure OVB, §16.6). It is a conditional correlation, not a causal effect, unless those confounders are controlled (and even then, only if all are observed). Contrast: the *randomized* cost-vector comparison assigns the cost vector independently of all covariates, so its coefficient is exogenous and unbiased *without* controls (§16.5) — a genuine causal contrast. The two comparisons have fundamentally different identification status (Chapters 30–32).

C3. *Right:* adding pre-treatment control variables can *reduce the standard error* (improve precision) by soaking up residual variance — useful given the large SE. *Wrong:* the premise that controls are needed to "reduce the bias" is misguided — because the treatment is randomized, it is already exogenous and the estimate is unbiased *without* controls (§16.5); there is no bias to reduce. Moreover, adding the *wrong* controls (post-treatment/mediators) would *introduce* bias (Chapter 32). So: add pre-treatment covariates for precision if desired, but do not frame it as bias reduction, and avoid post-treatment controls.

16.13 Spaced review prompts

+1 day. From memory, derive the OLS estimator and its variance $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$, and state the exogeneity assumption and why it gives unbiasedness.

+1 week. Derive the OVB formula and its sign rule; then explain, with the VIF, why multicollinearity inflates standard errors but not bias, and why dropping a needed collinear variable is the wrong fix.

+1 month. For one regression in your own work (a meta-regression or a balance check), state what it identifies (causal or correlational), sign the potential OVB from a plausible omitted variable, check for multicollinearity (VIF) and whether it is a precision or bias concern, and confirm your standard errors match the data structure. Close with the spine sentence.

16.14 How this chapter connects to the rest of the book

- **Chapter 1** supplied the $\mathbf{X}'\mathbf{X}$ matrix, its inverse and conditioning (the covariance $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ is Chapter 1's "covariance is an inverse"); **Chapter 2** the FOC/SOC; **Chapter 7** bias/variance/efficiency and BLUE; **Chapter 11** the orthogonality/score logic that OLS's normal equations instantiate.
- **Chapter 17 (inference that survives reality)** is the direct sequel: heteroskedasticity, robust and clustered standard errors (the sandwich replacing $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$), interactions, marginal effects, and residual diagnostics (the hat matrix/leverage of §16.8).
- **Chapter 18 (binary response)** generalizes the linear model to probit/logit; **Chapters 30–32 (identification, treatment effects, threats)** develop the exogeneity/OVB/randomization material of §16.5–§16.6 into the full causal-inference framework (and the bad-controls problem); **Chapter 8** is the power/precision context for the multicollinearity discussion.
- **Chapter 47 (meta-regression)** applies OLS/WLS with clustered/robust inference to your finfish data — where OVB (omitted moderators), multicollinearity (correlated moderators), and prediction (benefit transfer) of this chapter are the daily concerns; **Chapters 35, 49 (aggregation, benefit transfer)** use the prediction-interval logic of §16.8 for benefit transfer.

This chapter opens Part IV by making OLS *solidissimo* — the linear backbone beneath your meta-regression, balance checks, and every more complex model, and the cleanest setting to internalize the exogeneity/OVB/multicollinearity distinctions that recur throughout.

16.15 Research translation task

For your **finfish meta-regression** (and one balance check), produce a two-to-three-page note:

1. **Write the model and derive the estimator.** State the linear model, the OLS estimator, and the identification (full-rank) condition.
2. **Exogeneity and identification.** For each key regressor, state whether the coefficient is causal (randomization) or correlational, and justify (or flag) exogeneity.
3. **Omitted-variable bias.** List plausible omitted moderators, and for each sign the potential bias ($\delta\beta_2$) on an included moderator; state how it affects your conclusions.
4. **Multicollinearity.** Compute VIFs for your moderators; identify any near-collinear pairs; confirm they are a precision (not bias) concern and decide what to report (identified combination, more studies, honest imprecision) — *without* dropping needed moderators.
5. **Inference and prediction.** Note that your standard errors must be robust/clustered (Chapter 17, 47), and for any benefit-transfer prediction use the prediction interval, not the mean CI.
6. **Close** with the spine sentence: *"My regression identifies [causal/correlational] effects under [exogeneity/randomization]; plausible omitted variables bias [coefficient] [up/down]; large SEs are [precision/bias], and I report [robust/clustered] inference and [prediction] intervals."*

Carry this into Chapter 17 (robust/clustered inference and interactions) and Chapter 47 (the full meta-regression treatment).

16.16 Bibliography

Core. - Wooldridge, *Introductory Econometrics: A Modern Approach* (and *Econometric Analysis of Cross Section and Panel Data* for the graduate level). The clearest development of OLS, exogeneity, omitted-variable bias, and multicollinearity — the single best reference for this chapter. *(Verify edition/chapters.)* - Greene, *Econometric Analysis*, the linear-model chapters (OLS algebra, the Gauss–Markov theorem, the hat matrix, prediction). The matrix treatment matching §16.2–§16.8. *(Verify edition.)* - Angrist & Pischke, *Mostly Harmless Econometrics*. The modern, causal reading of OLS — exogeneity, omitted-variable bias, bad controls, and why randomization identifies effects (§16.5–§16.6). Essential for the causal framing your work needs. *(Verify edition.)*

Applied. - Stock & Watson, *Introduction to Econometrics*, on OLS, OVB, multicollinearity, and prediction at an accessible applied level, with clear treatment of the precision-vs-bias distinction (§16.7). *(Verify edition.)* - Cameron & Trivedi, *Microeconometrics: Methods and Applications*, on the OLS foundations and the transition to robust/clustered inference (§16.4; Chapter 17). *(Verify edition.)* - Stanley & Doucouliagos, *Meta-Regression Analysis in Economics and Business*. OLS/WLS meta-regression, omitted moderators, and multicollinearity in the meta-analysis context of your finfish paper. *(Verify edition.)*

Advanced. - Hansen, *Econometrics* (Bruce Hansen), the least-squares chapters — a rigorous, freely-available modern treatment of the OLS algebra, projection/hat matrix, and finite-sample vs asymptotic properties. *(Verify current version.)* - Davidson & MacKinnon, *Econometric Theory and Methods*, for the geometry of least squares (projections, the hat matrix) and the rigorous derivations behind §16.2, §16.8. *(Verify edition.)* - Belsley, Kuh & Welsch, *Regression Diagnostics*, for leverage, influence, and collinearity diagnostics (the VIF and condition-number treatment of §16.7). *(Verify edition.)*

Entries marked "verify" are pointers; confirm editions and chapters before using them, per the book's no-unverified-citation rule.

End of Chapter 16. Next in Part IV: Chapter 17 — Inference That Survives Reality (heteroskedasticity, robust and clustered standard errors derived from first principles, interaction terms, marginal effects, and residual diagnostics), which replaces the classical variance $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ with the robust/clustered sandwich

your meta-regression and repeated-observation data require. Tell me when to write it, or request a different chapter.

Chapter 17 — Inference That Survives Reality

Part IV: Core Econometrics

> **The spine question, applied to this chapter.** Chapter 16 derived the classical OLS variance $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ under the assumption that errors are homoskedastic and independent. Reality violates both, everywhere in your data: the precision of a WTP estimate varies by study (heteroskedasticity), and your observations come in clusters — multiple choice tasks per respondent, multiple WTP estimates per study (within-cluster correlation). When these assumptions fail, your *point estimates* are still fine, but your *standard errors are wrong* — usually far too small, so you over-reject and overstate precision. This chapter derives the fixes from first principles: the **robust (sandwich)** and **cluster-robust** standard errors, plus the correct interpretation of **interaction terms** and **marginal effects**. The spine question here: *do my standard errors reflect the actual dependence and heteroskedasticity in my data, or am I quoting classical standard errors that assume away the structure that makes my inference honest?*

17.0 Conceptual motivation: the point estimate is right; the standard error is a lie

Chapter 16 established a liberating fact: heteroskedasticity and within-cluster correlation do *not* bias OLS coefficients (they are failures of assumption A4, not exogeneity A3). But that fact comes with a trap — because the estimates are fine, it is easy to trust the *whole* output, including the standard errors, which are *not* fine. Your PDF lists exactly these tools under core econometrics — *"eteroschedasticità; robust standard errors; clustered standard errors"* — and under panel/clustered inference it names your exposure precisely: *"più choice tasks per individuo; più WTP per studio nella meta-regression; più scenari per unità/ecosistema; clustering per respondent/study."* Every one of these is a within-cluster-correlation problem, and getting the standard errors wrong is one of the most common and consequential errors in applied work.

Why does this matter so acutely for you?

- **Repeated choice tasks (DCE).** Each respondent answers several choice tasks, whose errors share a respondent-level component. Treating them as independent (classical SEs) counts, say, 4000 correlated observations as 4000 independent ones — grossly overstating precision. The fix is respondent-clustered standard errors.
- **Meta-regression (finfish).** Multiple WTP estimates come from the same study, correlated within study. As you will see, ignoring this can understate standard errors by a *factor of three or more* — turning noise into spurious significance. And with only 18 studies, even the cluster correction is delicate (Chapter 48).
- **Survey experiments with group-level treatment.** When a treatment is assigned at a group level (a version, a block), the within-group correlation inflates the true variance dramatically — and, as

this chapter's headline result shows, even heteroskedasticity-robust standard errors do *not* fix it; only clustering does.
- **Interaction terms.** Your central comparisons are interactions — does the cost-vector effect differ by experience? does BRIS behave differently under high vs low cost? — and interaction coefficients are systematically misread. The "main effect" is not "the effect"; the marginal effect depends on the interacting variable.

This chapter builds the machinery to make inference honest. We derive the **robust (White/Huber/Eicker) sandwich** estimator from first principles and its small-sample variants (HC1–HC3); we derive the **cluster-robust** variance from first principles, state exactly what clustering assumes, and show — with a verified example — that ignoring clustering can understate standard errors threefold; we quantify the **design effect** (how intra-cluster correlation inflates the true variance); we develop **interaction terms and marginal effects** correctly; and we treat **residual diagnostics** (leverage, influence, leave-one-out, heteroskedasticity tests). The sandwich is the OLS instance of the sandwich you met for the MLE (Chapter 11, §11.5) — the same information-matrix-equality failure, now in the linear model. Fuller panel/random-effects treatment is Chapter 19, and the few-clusters correction is Chapter 48; here we build the estimators.

17.1 Prerequisites

From **Chapter 16**: the OLS estimator, the decomposition $\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon$, the classical variance $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ and the general sandwich form (§16.4), the hat matrix (§16.8), and marginal effects (§16.2). From **Chapter 11** (§11.5): the information matrix equality and its failure → the sandwich; from **Chapter 6** (§6.9): asymptotics in the *number of clusters* and the small-cluster caution. From **Chapter 5** (§5.7): the between/within variance decomposition (intra-cluster correlation). From **Chapter 2** (§2.10): marginal effects and elasticities.

Same **notation**. New symbols: $\Omega = \mathrm{Var}(\varepsilon \mid \mathbf{X})$ (error covariance), e_i (residual), h_{ii} (leverage), g (cluster index), $\mathbf{X}_g, \mathbf{e}_g$ (cluster-g regressors and residuals), $\mathbf{u}_g = \mathbf{X}_g'\mathbf{e}_g$ (cluster score sum), G (number of clusters), ρ (intra-cluster correlation).

17.2 Heteroskedasticity: harmless for bias, fatal for the classical standard error

Definition 17.1 (Heteroskedasticity). The errors are **heteroskedastic** if their conditional variance is not constant: $\mathrm{Var}(\varepsilon_i \mid \mathbf{X}) = \sigma_i^2$ varies across observations (versus **homoskedastic**, $\sigma_i^2 = \sigma^2$ for all i). The error covariance $\Omega = \mathrm{Var}(\varepsilon \mid \mathbf{X})$ is then a *diagonal* matrix with unequal entries σ_i^2, rather than $\sigma^2\mathbf{I}$.

Where it lives in your data. Heteroskedasticity is the rule, not the exception:

- In **meta-regression**, WTP estimates from *large* studies are more precise (smaller σ_i^2) than those from *small* studies — the error variance varies systematically with study size. This is why meta-analysis uses *weighted* least squares (Chapter 47), a heteroskedasticity issue.
- In **survey experiments**, the outcome variance can differ across treatment arms (a hypothetical arm may be noisier than an incentivized one — the scale-heterogeneity theme of Chapter 25).
- In cross-sections generally, variance often grows with the level of a regressor (income, scale).

The consequence (harmless for bias, fatal for SEs). Recall from Chapter 16 (§16.4) that heteroskedasticity is a failure of A4, not A3, so:

- **The OLS estimate $\hat{\beta}$ is still unbiased and consistent** (unbiasedness used only exogeneity).

- **The classical standard error $\sqrt{\mathbf{diag}(\sigma^2(\mathbf{X}'\mathbf{X})^{-1})}$ is wrong** — because the true variance is the sandwich $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\Omega\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$ with $\Omega \neq \sigma^2\mathbf{I}$, and the classical formula assumed it away. Usually the classical SE is *too small* (over-rejecting), though it can go either way.

So the estimate you report is fine; the star next to it may be a fiction. The fix is a variance estimator that does not assume homoskedasticity — the robust sandwich.

> **Active recall 17.A.** Heteroskedasticity is present in your meta-regression (large studies more precise). Does this bias your moderator coefficients, and what does it do to the classical standard errors? *(Answer: it does not bias the coefficients (unbiasedness needs only exogeneity, which heteroskedasticity does not violate); it invalidates the classical standard errors, which are computed under the false assumption of constant variance and are typically too small — so use robust/weighted standard errors, not a different point estimate.)*

17.3 The robust (sandwich) standard error, derived

The **robust** standard error — variously White, Huber, or Eicker–White, and "heteroskedasticity-consistent" (HC) — estimates the true sandwich variance without assuming homoskedasticity. Here is the derivation from first principles.

Derivation. Start from the exact decomposition (Chapter 16):

$$\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon,$$

so the conditional variance is

$$\mathrm{Var}(\hat{\beta} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\,\mathbf{X}'\Omega\mathbf{X}\,(\mathbf{X}'\mathbf{X})^{-1}, \qquad \Omega = \mathrm{Var}(\varepsilon \mid \mathbf{X}).$$

This is the **sandwich**: two "bread" slices $(\mathbf{X}'\mathbf{X})^{-1}$ around the "meat" $\mathbf{X}'\Omega\mathbf{X}$. Under homoskedasticity $\Omega = \sigma^2\mathbf{I}$ the meat collapses to $\sigma^2\mathbf{X}'\mathbf{X}$ and the whole thing to $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ (Chapter 16). Under heteroskedasticity, $\Omega = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$, so the meat is

$$\mathbf{X}'\Omega\mathbf{X} = \sum_{i=1}^{n} \sigma_i^2\,\mathbf{x}_i\mathbf{x}_i',$$

a sum of the individual outer products weighted by their variances. We do not know the σ_i^2, but White's insight is that we do not need to estimate each one — we only need to estimate the *sum*, and the **squared residual** e_i^2 is an (imperfect but unbiased-in-the-sum) estimate of σ_i^2. This gives the **robust variance estimator**:

$$\boxed{\hat{\mathbf{V}}_{\text{robust}} = (\mathbf{X}'\mathbf{X})^{-1}\Big(\sum_{i=1}^{n} e_i^2\,\mathbf{x}_i\mathbf{x}_i'\Big)(\mathbf{X}'\mathbf{X})^{-1}}.$$

This is **HC0** (the original White estimator). The key point: it makes *no* assumption about the form of heteroskedasticity — it lets each observation's residual speak for its own variance — and it is consistent as $n \to \infty$.

Small-sample corrections (HC1–HC3). HC0 is biased downward (too small) in finite samples because residuals underestimate errors (they are "shrunk" by fitting). The corrections:

- **HC1** $= \frac{n}{n-k}$ HC0 — a simple degrees-of-freedom scaling (the default in many packages, e.g. Stata's `robust`).
- **HC2** divides each e_i^2 by $(1 - h_{ii})$ (the leverage, Chapter 16 §16.8) — corrects for the shrinkage of high-leverage residuals.
- **HC3** divides each e_i^2 by $(1 - h_{ii})^2$ — the most conservative, **recommended for small samples**; it approximates a jackknife and guards against influential points.

Use HC1 by default, HC3 when n is small or a few points have high leverage. Your PDF's `clubSandwich` package implements these and their clustered analogues.

Worked Example 17.1 (Robust SE fixes heteroskedasticity, verified). Simulate $y = 1 + 2x + \varepsilon$ with error standard deviation *growing* in x (heteroskedastic). Over 2000 samples, the *true* sampling standard deviation of $\hat{\beta}_1$ is 0.122. The mean **classical** SE is 0.113 — too small (it would overstate precision and over-reject). The mean **robust (HC0)** SE is 0.119 — matching the true sampling SD (verified). The robust estimator recovers honest uncertainty without knowing the form of the heteroskedasticity; the classical one does not.

> **Stop and think 17.1.** A referee says "just use robust standard errors everywhere and you're safe." Is this always sufficient? Preview the answer with your repeated-choice-tasks data. *(Answer: no — robust (HC) standard errors correct for* heteroskedasticity* (unequal variances across independent observations) but assume the observations are *independent*. When observations are correlated *within clusters* — repeated choice tasks per respondent, estimates per study — the errors are not independent, and HC robust SEs are still wrong (as §17.4 shows dramatically). You need *cluster-robust* SEs. "Robust" is not a universal fix; it fixes heteroskedasticity, not dependence.)*

17.4 Cluster-robust standard errors, derived from first principles

This is the chapter's most important result for your data, because *all* your data are clustered — choice tasks within respondents, WTP estimates within studies. When errors are correlated *within* clusters, neither the classical nor the heteroskedasticity-robust SE is valid; you need the **cluster-robust** variance, and its effect can be enormous.

The structure. Observations belong to clusters $g = 1, \dots, G$ (respondents, studies), with n_g observations in cluster g. Errors are allowed to be *arbitrarily correlated within a cluster* but *independent across clusters*. So the error covariance Ω is **block-diagonal**: dense blocks Ω_g (arbitrary within-cluster correlation) on the diagonal, zeros off-diagonal (independence across clusters).

Derivation. The sandwich meat $\mathbf{X}'\Omega\mathbf{X}$ with block-diagonal Ω decomposes into a sum over clusters:

$$\mathbf{X}'\Omega\mathbf{X} = \sum_{g=1}^{G} \mathbf{X}_g'\Omega_g\mathbf{X}_g,$$

where $\mathbf{X}_g$ stacks cluster g's regressor rows. We estimate each $\mathbf{X}_g'\Omega_g\mathbf{X}_g$ by $\mathbf{X}_g'\mathbf{e}_g\mathbf{e}_g'\mathbf{X}_g = (\mathbf{X}_g'\mathbf{e}_g)(\mathbf{X}_g'\mathbf{e}_g)' = \mathbf{u}_g\mathbf{u}_g'$, where

$$\mathbf{u}_g = \mathbf{X}_g'\mathbf{e}_g = \sum_{i \in g} \mathbf{x}_i e_i$$

is the **cluster score sum** — the sum, within cluster g, of each observation's regressor vector times its residual. This gives the **cluster-robust variance estimator**:

$$\boxed{\hat{\mathbf{V}}_{\text{cluster}} = (\mathbf{X}'\mathbf{X})^{-1}\Big(\sum_{g=1}^{G} \mathbf{u}_g\mathbf{u}_g'\Big)(\mathbf{X}'\mathbf{X})^{-1}, \qquad \mathbf{u}_g = \sum_{i \in g} \mathbf{x}_i e_i}.$$

Compare it to the robust estimator (§17.3): where HC0 sums *individual* outer products $e_i^2\mathbf{x}_i\mathbf{x}_i'$, the cluster version sums *cluster* outer products $\mathbf{u}_g\mathbf{u}_g'$ — it aggregates within each cluster *before* squaring, which is precisely what captures the within-cluster correlation. If every cluster had one observation ($n_g = 1$), the two would coincide; the difference is the within-cluster dependence.

What clustering assumes (state it precisely). The cluster-robust estimator assumes:

1. **Independence *across* clusters** — different respondents (or studies) are independent. This is the key identifying assumption, and it must be defended: if respondents share a common shock (a survey wave, an interviewer), the cluster must be defined at that higher level.
2. **Arbitrary correlation *within* clusters** — any pattern of dependence among a cluster's observations is allowed (this is what it *relaxes*).
3. **Asymptotics in the number of clusters** G, not the number of observations. The estimator is consistent as $G \to \infty$; with *few* clusters it is downward-biased and unreliable (Chapter 6 §6.9; Chapter 47 §47.7). Your 18-study meta-analysis sits squarely in this danger zone.

Worked Example 17.2 (Ignoring clustering understates SEs threefold, verified). Simulate a survey experiment with $G = 50$ clusters of $m = 20$ each (so $n = 1000$), a treatment assigned *at the cluster level*, a cluster random effect (intra-cluster correlation $\rho = 0.5$), and a true treatment effect of 0.5. Over 2000 datasets, the *true* sampling SD of the treatment coefficient is 0.291. The estimated standard errors:

Estimator	Mean SE	Verdict
Classical (naive)	0.089	**wrong — understates by ~3×**
Robust HC0	0.089	**still wrong — ignores within-cluster correlation**
Cluster-robust	**0.284**	**correct — matches the true SD**

(Verified.) The lesson is stark and decisive. Ignoring clustering understates the standard error by a factor of **3.2** (0.284/0.089 = 3.18), which would turn a genuinely insignificant effect into a "highly significant" one. And — critically — the *heteroskedasticity-robust* SE does **not** help (0.089, same as classical): robustness to heteroskedasticity is useless against within-cluster *correlation*. Only the cluster-robust SE recovers honest inference. For your DCE (choice tasks within respondents) and your meta-regression (estimates within studies), **clustering is not optional** — using classical or even HC-robust standard errors can inflate your t-statistics threefold and manufacture significance.

> **Active recall 17.B.** Write the cluster-robust "meat" and explain, in one sentence, why summing within clusters *before* squaring captures within-cluster correlation that HC-robust standard errors miss. *(Answer: the cluster meat is* $\sum_g \mathbf{u}_g \mathbf{u}_g'$ *with* $\mathbf{u}_g = \sum_{i \in g} \mathbf{x}_i e_i$*; by aggregating each cluster's contributions into* $\mathbf{u}_g$ before* forming the outer product, it captures the covariance among a cluster's residuals — the within-cluster correlation — whereas HC-robust squares each observation separately ($\sum_i e_i^2 \mathbf{x}_i \mathbf{x}_i'$) and so treats within-cluster observations as independent, missing exactly that correlation.)*

17.5 The design effect: how much does clustering inflate uncertainty?

It helps to have a formula for *how badly* ignoring clustering misleads. For a regressor that is constant within clusters (like a cluster-level treatment), the ratio of the true variance to the naive variance — the **design effect** — is approximately

$$\text{Design effect} \approx 1 + (m - 1)\rho,$$

where m is the cluster size and ρ is the **intra-cluster correlation** (the fraction of total variance that is *between*-cluster, Chapter 5 §5.7). The naive standard error is too small by a factor of $\sqrt{1 + (m-1)\rho}$.

Reading the formula. Two things drive the inflation: cluster *size* m and intra-cluster *correlation* ρ. In Worked Example 17.2, $m = 20$ and $\rho = 0.5$ give a design effect of $1 + 19 \times 0.5 = 10.5$, so the naive SE is too small by $\sqrt{10.5} = 3.24$ — matching the verified 3.18. The interpretation: with $\rho = 0.5$, twenty observations in a cluster carry only about *two* observations' worth of independent information (the **effective sample size** $\approx n/$design effect). This is the quantitative meaning of "your 4000 choice tasks are not 4000 independent observations."

For your specific data.

- **DCE (tasks within respondents).** If a respondent's 8 tasks have intra-respondent correlation ρ, the design effect for a respondent-level covariate is $1 + 7\rho$ — potentially a 2–3× SE inflation. Cluster on respondents.
- **Meta-regression (estimates within studies).** If a study contributes several WTP estimates with high within-study correlation, the design effect can be large, and the effective number of *independent* pieces of information is closer to the *number of studies* than the number of estimates (Chapter 6 §6.9; your PDF's "più WTP dallo stesso studio sono indipendenti?" — answer: no). This is why the finfish paper's 18 studies, not its 120+ estimates, set the effective sample size.

The design effect turns "cluster your standard errors" from a ritual into a quantity you can *anticipate*: given plausible m and ρ, you can predict how much your honest SEs will exceed the naive ones, and hence whether a result will survive.

> **Stop and think 17.2.** Your finfish meta-regression has 120 WTP estimates from 18 studies, and within-study correlation is high ($\rho \approx 0.7$), with an average of ~7 estimates per study. Roughly how much does ignoring clustering understate your standard errors, and what does this imply for the effective information? *(Answer: design effect* $\approx 1 + (7-1) \times 0.7 = 5.2$*, so naive SEs are too small by* $\sqrt{5.2} \approx 2.3\times$*; the 120 estimates carry roughly* $120/5.2 \approx 23$ *estimates' worth of independent information — close to the number of studies, confirming that the* number of studies* (18), not estimates (120), is the binding sample size. Cluster on studies, and with so few clusters apply small-sample corrections, Chapter 47 §47.7.)*

17.6 Interaction terms: the main effect is not "the effect"

Your central treatment comparisons are interactions — the cost-vector effect *by* experience, BRIS *by* cost level, hypothetical *by* familiarity. Interaction coefficients are among the most misread quantities in applied work, and getting them right is essential to your papers' claims.

The model. With two regressors and their product,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 (x_1 x_2) + \varepsilon.$$

The **marginal effect of** x_1 is the partial derivative, which now depends on x_2:

$$\frac{\partial y}{\partial x_1} = \beta_1 + \beta_3 \, x_2.$$

So β_1 is *not* "the effect of x_1" — it is the effect of x_1 *only when* $x_2 = 0$. The interaction β_3 is *how the effect of* x_1 *changes* as x_2 increases. For a binary x_2: β_1 is the effect of x_1 in the $x_2 = 0$ group, and $\beta_1 + \beta_3$ is the effect in the $x_2 = 1$ group.

Worked Example 17.3 (Interaction marginal effects, verified). Simulate $y = 1 + 2x_1 + 1.5x_2 + 0.8(x_1 x_2) + \varepsilon$ with binary x_2. OLS recovers $(\hat\beta_0, \hat\beta_1, \hat\beta_2, \hat\beta_3) = (1.00, 1.99, 1.52, 0.85)$. The marginal effect of x_1 is:

$$\text{when } x_2 = 0: \ \hat\beta_1 = 1.99; \qquad \text{when } x_2 = 1: \ \hat\beta_1 + \hat\beta_3 = 2.84.$$

(Verified.) The effect of x_1 is 43% larger in the $x_2 = 1$ group — and reporting only $\hat\beta_1 = 1.99$ as "the effect of x_1" would be wrong for everyone with $x_2 = 1$. In your cost-vector × experience design, "the cost-vector effect" is β_1 for inexperienced respondents and $\beta_1 + \beta_3$ for experienced ones; β_3 *is* the answer to "does experience change the cost-vector effect?"

Practical rules for interactions.

- **Report marginal effects at meaningful values of the interacting variable** (e.g., the effect of the cost vector for experienced *and* inexperienced respondents separately), not just the raw coefficients.

- **The interaction coefficient β_3 is the object of a "does X differ by Y?" hypothesis** — its significance tests whether the effect *differs* across groups. But recall the **power** cost: interactions need much larger samples than main effects (Chapter 8, §8.9), so a non-significant interaction is often underpowered, not evidence of no difference.
- **Centering** the continuous variables (subtracting their means) before interacting makes β_1 and β_2 interpretable as effects "at the average" rather than "at zero," and reduces the (harmless) multicollinearity between a variable and its interaction (Chapter 16, §16.7).
- **Standard errors of marginal effects** at a given x_2 require the full covariance (the variance of $\beta_1 + \beta_3 x_2$ is a quadratic form, Chapter 1 §1.6, Chapter 7 §7.7) — not just the individual SEs.

Active recall 17.C. In your model with a cost-vector treatment C, an experience dummy E, and their interaction, which coefficient answers "does the cost-vector effect differ between experienced and inexperienced respondents?", and what is the cost-vector effect for each group? *(Answer: the interaction coefficient $\beta_{C\times E}$ answers whether the effect differs (and its test is the "difference in effects" test); the cost-vector effect is β_C for inexperienced ($E = 0$) and $\beta_C + \beta_{C\times E}$ for experienced ($E = 1$). Report both group-specific effects, not just β_C.)*

17.7 Residual diagnostics: finding the observations that drive your results

Before trusting a regression, you check whether a few observations are driving it, whether the errors look heteroskedastic, and whether the specification fits. These diagnostics use the residuals and the hat matrix (Chapter 16, §16.8).

Leverage. The hat-matrix diagonal h_{ii} measures how far observation i's regressor values are from the center — its potential to influence its own fitted value. High-leverage points (large h_{ii}) are *potentially* influential. The average leverage is k/n; points with $h_{ii} > 2k/n$ or $3k/n$ warrant a look.

Influence. Leverage is potential; *influence* is realized effect on the estimates. Measures: **Cook's distance** (how much all fitted values change if i is dropped), **DFBETA** (how much a specific coefficient changes if i is dropped), and the studentized residual. An influential outlier can single-handedly create or destroy a result — a serious concern in small samples.

Leave-one-out (and its meta-analysis cousin). The cleanest influence check is to *re-estimate dropping each observation* and see how the estimate moves. In your **meta-regression**, this is **leave-one-*study*-out** (Chapter 48, your PDF §6.8): drop each study in turn and check whether any single study drives a moderator effect or the overall WTP — essential when a result might rest on one influential study. With 18 studies, one study is $\approx 5.6\%$ of your data; a result that flips when one study is removed is fragile.

Heteroskedasticity tests. The **Breusch–Pagan** test regresses squared residuals on the regressors (a significant fit indicates heteroskedasticity); the **White** test is a more general version including squares and cross-products. These *detect* heteroskedasticity — but the modern practice is often to *use robust standard errors by default* rather than test-then-decide, since robust SEs are valid whether or not heteroskedasticity is present (at a small efficiency cost). Treat the tests as diagnostics, not gatekeepers.

Specification checks. Plot residuals against fitted values and against each regressor (patterns suggest nonlinearity or omitted terms); the RESET test checks for functional-form misspecification. A good residual plot is structureless noise; structure is a message.

Stop and think 17.3. Your finfish meta-regression finds a significant effect of "valuation method," but you have only 18 studies. What single diagnostic would you run first, and what would make you worry? *(Answer: **leave-one-study-out** — re-estimate dropping each study in turn. Worry if the "valuation method" effect loses significance, changes sign, or changes materially when any single study is removed, because with 18 studies one influential study can drive the*

result; a robust finding survives leaving out any one study. Pair with influence measures (Cook's distance / DFBETA) at the study level — Chapter 47 §47.7.)

17.8 Failure modes

F1. Classical SEs under heteroskedasticity. Reporting $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ SEs when variances are unequal. *Symptom:* over-rejection. *Fix:* robust (HC1, or HC3 for small n) standard errors (§17.3).

F2. Robust-but-not-clustered SEs with clustered data. Using HC-robust SEs for repeated tasks / within-study estimates. *Symptom:* SEs still far too small (verified ~3× understatement). *Fix:* cluster-robust SEs on the right cluster (§17.4).

F3. Wrong cluster level. Clustering at too fine a level (missing a higher-level shared shock). *Fix:* cluster at the level of the most plausible independence (respondent, study, survey wave) (§17.4).

F4. Too few clusters. Trusting cluster-robust SEs with a handful of clusters (18 studies). *Symptom:* downward-biased cluster SEs. *Fix:* small-sample corrections (wild cluster bootstrap, degrees-of-freedom adjustments) (§17.4; Chapters 6 §6.9, 48).

F5. Interpreting a main effect as "the effect" under interaction. Reading β_1 as the effect of x_1 when an interaction is present. *Fix:* report the marginal effect $\beta_1 + \beta_3 x_2$ at meaningful x_2 values (§17.6).

F6. Non-significant interaction read as "no difference." Treating an underpowered interaction null as evidence effects are equal. *Fix:* power analysis / equivalence test for interactions (§17.6; Chapter 8).

F7. Ignoring influential observations. A result driven by one outlier/study. *Fix:* leverage, Cook's distance, leave-one-out (leave-one-study-out for meta-regression) (§17.7).

F8. Test-then-decide on heteroskedasticity, then forget clustering. Passing a Breusch–Pagan test and using classical SEs, ignoring clustering. *Fix:* default to robust/clustered SEs matched to the data structure; tests are diagnostics, not licenses (§17.7).

17.9 Diagnostic questions for your own papers

1. **Heteroskedasticity.** Are my standard errors robust to unequal error variances (meta-regression, group differences)? *(§17.3, F1.)*
2. **Clustering.** Are my observations clustered (tasks within respondents, estimates within studies)? Am I clustering at the right level, not just using HC-robust SEs? *(§17.4, F2–F3.)*
3. **Few clusters.** Do I have enough clusters for cluster-robust SEs to be reliable, or do I need small-sample corrections (18 studies)? *(§17.4, F4; Chapter 47 §47.7.)*
4. **Design effect.** Have I gauged how much clustering inflates my SEs (intra-cluster correlation, cluster size), and hence my effective sample size? *(§17.5.)*
5. **Interactions.** Am I reporting marginal effects at meaningful values, not just main-effect coefficients, and testing the interaction as the "difference in effects"? *(§17.6, F5.)*
6. **Interaction power.** Is a non-significant interaction underpowered rather than evidence of no difference? *(§17.6, F6; Chapter 8.)*
7. **Influence.** Have I checked leverage/influence and run leave-one-out (leave-one-study-out) to see if a few observations drive my result? *(§17.7, F7.)*

17.10 Exercises

(A) Comprehension.

A1. Explain why heteroskedasticity does not bias OLS coefficients but invalidates the classical standard errors, and name the estimator that fixes the standard errors.

A2. You have 8 choice tasks per respondent. Why are heteroskedasticity-robust (HC) standard errors insufficient, and what do you use instead?

A3. In $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$, what is the marginal effect of x_1, and what does β_1 alone represent?

(B) Derivation.

B1. Derive the robust (HC0) sandwich variance $\hat{\mathbf{V}} = (\mathbf{X}'\mathbf{X})^{-1}(\sum_i e_i^2 \mathbf{x}_i \mathbf{x}_i')(\mathbf{X}'\mathbf{X})^{-1}$ from the exact decomposition $\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon$, stating what replaces σ_i^2 and why.

B2. Derive the cluster-robust variance with meat $\sum_g \mathbf{u}_g \mathbf{u}_g'$, $\mathbf{u}_g = \sum_{i \in g} \mathbf{x}_i e_i$, from the block-diagonal Ω, and state the three assumptions it makes.

B3. Derive the design effect $1 + (m-1)\rho$ for a cluster-constant regressor, and use it to compute the effective sample size for $n = 1000$, $m = 20$, $\rho = 0.5$.

(C) Research application / critique.

C1. Your DCE reports respondent-level covariate effects with classical standard errors (4000 tasks, 500 respondents). Critique this and specify the correct inference, estimating how much the SEs might change.

C2. A co-author reports that "the cost-vector effect is 0.4 (p<0.01)" from a model with a cost-vector × experience interaction. Explain what is missing and how to report the result correctly.

C3. Your finfish meta-regression (18 studies, 120 estimates) uses HC-robust standard errors and finds three significant moderators. Write the corrected inference plan, addressing clustering, few clusters, and influence.

17.11 Solutions

A1. Unbiasedness of OLS uses only exogeneity ($\mathbb{E}[\varepsilon \mid \mathbf{X}] = \mathbf{0}$), which heteroskedasticity does not violate — so the coefficients remain unbiased. But the classical variance $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ assumes constant variance ($\Omega = \sigma^2\mathbf{I}$); under heteroskedasticity the true variance is the sandwich $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\Omega\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$ with unequal σ_i^2, so the classical SE is wrong (usually too small). The robust (White/HC) sandwich estimator fixes the standard errors.

A2. HC-robust SEs correct for heteroskedasticity but assume observations are *independent*; the 8 tasks per respondent are *correlated within respondent* (shared tastes/scale), which HC-robust SEs do not capture, so they remain too small. Use **respondent-clustered** standard errors, which allow arbitrary within-respondent correlation and are consistent as the number of respondents grows.

A3. The marginal effect of x_1 is $\partial y/\partial x_1 = \beta_1 + \beta_3 x_2$ — it depends on x_2. β_1 alone is the effect of x_1 *only when* $x_2 = 0$ (for binary x_2, the effect in the $x_2 = 0$ group); it is not "the effect of x_1" in general.

B1. From $\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon$, $\mathrm{Var}(\hat{\beta} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\Omega\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$ with $\Omega = \mathrm{diag}(\sigma_i^2)$, so the meat is $\mathbf{X}'\Omega\mathbf{X} = \sum_i \sigma_i^2 \mathbf{x}_i \mathbf{x}_i'$. Replace the unknown σ_i^2 by the squared residual e_i^2 (an estimate of the individual error variance that is unbiased *in the aggregate sum*, which is all the meat requires), giving $\hat{\mathbf{V}} = (\mathbf{X}'\mathbf{X})^{-1}(\sum_i e_i^2 \mathbf{x}_i \mathbf{x}_i')(\mathbf{X}'\mathbf{X})^{-1}$. No functional form for the heteroskedasticity is assumed.

B2. With clusters independent across but correlated within, Ω is block-diagonal, so $\mathbf{X}'\Omega\mathbf{X} = \sum_g \mathbf{X}_g'\Omega_g\mathbf{X}_g$; estimate each block's contribution by $\mathbf{X}_g'\mathbf{e}_g\mathbf{e}_g'\mathbf{X}_g = (\mathbf{X}_g'\mathbf{e}_g)(\mathbf{X}_g'\mathbf{e}_g)' = \mathbf{u}_g\mathbf{u}_g'$ with $\mathbf{u}_g = \sum_{i \in g} \mathbf{x}_i e_i$, giving

$\hat{\mathbf{V}}_{\text{cluster}} = (\mathbf{X}'\mathbf{X})^{-1}(\sum_g \mathbf{u}_g\mathbf{u}_g')(\mathbf{X}'\mathbf{X})^{-1}$. Assumptions: (i) independence across clusters; (ii) arbitrary correlation within clusters; (iii) consistency as the number of clusters $G \to \infty$ (few clusters → unreliable).

B3. For a regressor constant within clusters, the variance of the cluster mean of the outcome inflates relative to independence because the m within-cluster errors share the between-cluster component: with intra-cluster correlation ρ, the variance of a cluster mean is $\frac{\sigma^2}{m}[1 + (m-1)\rho]$ instead of σ^2/m, so the design effect is $1 + (m-1)\rho$. For $m = 20$, $\rho = 0.5$: design effect $= 1 + 19(0.5) = 10.5$; effective sample size $= n/$design effect $= 1000/10.5 \approx 95$ — the 1000 observations carry about 95 independent observations' worth of information.

C1. Classical SEs treat the 4000 tasks as 4000 independent observations, but they are clustered in 500 respondents with correlated within-respondent errors, so the classical SEs understate uncertainty (the effective sample size is closer to the number of respondents than tasks). Correct inference: **respondent-clustered** standard errors (or a panel/mixed model, Chapter 19). Magnitude: with intra-respondent correlation ρ and 8 tasks, the design effect for a respondent-level covariate is up to $1 + 7\rho$; for $\rho = 0.4$ that is 3.8, so SEs could roughly *double* ($\sqrt{3.8} \approx 1.95$) — potentially flipping "significant" results. Re-estimate with clustering and re-assess significance.

C2. What is missing: with a cost-vector × experience interaction, "the cost-vector effect is 0.4" is only the effect for the reference group (experience = 0); the effect for experienced respondents is $\beta_C + \beta_{C\times E}$, and the interaction coefficient $\beta_{C\times E}$ (with its own SE) is what tests whether the effect differs. Report: the cost-vector effect for inexperienced (β_C) and experienced ($\beta_C + \beta_{C\times E}$) respondents separately, each with a standard error computed from the full covariance (a quadratic form), and the interaction test for whether they differ — noting the power cost of interactions (a non-significant interaction may be underpowered, Chapter 8). And use clustered SEs given repeated tasks.

C3. Corrected plan: (i) **cluster on studies** (not HC-robust), since estimates within a study are correlated — HC-robust SEs miss this and are too small (verified ~3× understatement is possible); (ii) with only **18 clusters**, cluster-robust SEs are downward-biased, so apply **small-sample corrections** — the wild cluster bootstrap and/or the CR2/CR3 corrections in `clubSandwich`, or robust variance estimation (RVE) with small-sample adjustment (Chapters 47–48); (iii) run **leave-one-study-out** and study-level influence diagnostics to confirm no single study drives the three moderators; (iv) reconsider whether 120 estimates on many moderators over-fit given ~18 effective units (too few studies for too many covariates, your PDF §2.8). Only moderators that survive clustered, small-sample-corrected inference *and* leave-one-study-out should be reported as robust.

17.12 Spaced review prompts

+1 day. From memory, derive the robust (HC0) sandwich and the cluster-robust variance, and state what each assumes. Write the interaction marginal effect $\beta_1 + \beta_3 x_2$.

+1 week. Explain, with the design-effect formula, why ignoring clustering can understate SEs several-fold and why HC-robust SEs do not fix it. Then describe leave-one-study-out and why it matters for your meta-analysis.

+1 month. For one of your own regressions (DCE covariate model or meta-regression), re-estimate with the correct clustered (and, if few clusters, small-sample-corrected) standard errors and compare to naive/HC SEs; report marginal effects for any interaction at meaningful values; run leverage/leave-one-out diagnostics. Close with the spine sentence about whether your inference reflects the real dependence in your data.

17.13 How this chapter connects to the rest of the book

- **Chapter 16** supplied the OLS decomposition and the general sandwich form; this chapter estimates it under heteroskedasticity and clustering. **Chapter 11** (§11.5) is the ML version of the same information-matrix-equality failure → robust SEs; **Chapter 5** (§5.7) is the between/within decomposition behind the intra-cluster correlation.
- **Chapter 6** (§6.9) is the asymptotics-in-clusters and few-clusters caution; **Chapter 19 (repeated observations)** develops the panel/random-effects modeling alternative to clustering (modeling the dependence rather than just correcting for it); **Chapter 47 §47.7 (dependence and small-sample corrections)** is the few-clusters fix (wild cluster bootstrap, CR2/CR3) for your 18-study data.
- **Chapter 8** is the power context for the interaction discussion (interactions need large samples); **Chapters 31–33 (treatment effects, threats, multiplicity)** use the interaction/marginal-effect and clustered-inference machinery for your survey experiments; **Chapter 47 (meta-regression)** applies robust/clustered/RVE inference to the finfish data, where every result here — heteroskedasticity (WLS), clustering (within-study), influence (leave-one-study-out) — is central.
- **Chapter 25 (scale vs preference heterogeneity)** connects to the group-varying error variance that heteroskedasticity captures in the linear model.

This chapter makes OLS inference *honest* — matching the standard errors to the heteroskedasticity and clustering that pervade your data — and it supplies the clustered/robust machinery that Chapters 19, 47, and 48 build on.

17.14 Research translation task

For your **DCE covariate model and finfish meta-regression**, produce a two-to-three-page note:

1. **Identify the dependence structure.** State the clusters (respondents; studies) and any heteroskedasticity (study precision; group variance).
2. **Compute the right standard errors.** Report classical, HC-robust, and cluster-robust standard errors side by side for a key coefficient; quantify the change and relate it to the design effect (intra-cluster correlation, cluster size).
3. **Few clusters.** For the meta-regression (18 studies), apply small-sample corrections (wild cluster bootstrap / CR2) and report how conclusions change.
4. **Interactions.** For your cost-vector × experience (or similar) interaction, report the marginal effect for each group with correct standard errors, and the interaction test — noting power.
5. **Influence.** Run leverage/Cook's-distance and leave-one-(study)-out diagnostics; report whether any observation/study drives a result.
6. **Close** with the spine sentence: *"My standard errors are [clustered on X, small-sample-corrected] because my data are [clustered/heteroskedastic]; naive SEs understate uncertainty by ~[factor]; my interaction implies marginal effects of [values]; and my results [do/do not] survive leave-one-out."*

Carry this into Chapter 19 (panel/random-effects modeling), Chapter 47 (meta-regression), and Chapter 47 §47.7 (few-cluster corrections).

17.15 Bibliography

Core. - Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, the chapters on robust and cluster-robust inference. The rigorous econometric derivation of §17.3–§17.4 at the right level. *(Verify edition/chapters.)* - Angrist & Pischke, *Mostly Harmless Econometrics*, on robust and clustered standard errors, the design effect, and why clustering matters — accessible and directly applicable. *(Verify edition.)* - Cameron & Miller, "A Practitioner's Guide to Cluster-Robust Inference," *Journal of Human Resources*

(2015). The definitive practical treatment of cluster-robust inference, the few-clusters problem, and small-sample corrections — essential for your clustered data and 18-study meta-analysis. *(Verify volume/pages.)*

Applied. - MacKinnon & White (1985) on HC1/HC2/HC3 small-sample corrections; and Long & Ervin (2000), "Using Heteroscedasticity-Consistent Standard Errors in the Linear Regression Model," *The American Statistician*, recommending HC3 for small samples (§17.3). *(Verify volumes/pages.)* - Brambor, Clark & Golder (2006), "Understanding Interaction Models: Improving Empirical Analyses," *Political Analysis*. The clearest applied guide to interpreting and reporting interaction terms and marginal effects (§17.6). *(Verify volume/pages.)* - The `clubSandwich` and `sandwich` R package documentation (Pustejovsky & Tipton; Zeileis) for robust and cluster-robust variance with small-sample corrections — the tools your PDF names. *(Verify current versions.)*

Advanced. - Belsley, Kuh & Welsch, *Regression Diagnostics: Identifying Influential Data and Sources of Collinearity*. Leverage, Cook's distance, DFBETA — the influence diagnostics of §17.7. *(Verify edition.)* - Cameron, Gelbach & Miller (2008), "Bootstrap-Based Improvements for Inference with Clustered Errors," *Review of Economics and Statistics*. The wild cluster bootstrap for few clusters (§17.4, F4; Chapter 48). *(Verify volume/pages.)* - Hansen, *Econometrics* (Bruce Hansen), the chapters on heteroskedastic and clustered regression — a rigorous, freely-available modern treatment. *(Verify current version.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Cameron–Miller / Brambor et al. citations before using them, per the book's no-unverified-citation rule.

End of Chapter 17. Next in Part IV: Chapter 18 — Binary Response Models (logit and probit as latent-utility models, marginal effects, odds ratios, scale identification, pseudo-R^2, likelihood-ratio tests), which generalizes the linear model of Chapters 16–17 to the binary outcomes at the heart of your SBDC and choice models. Tell me when to write it, or request a different chapter.

Chapter 18 — Binary Response Models

Part IV: Core Econometrics

> **The spine question, applied to this chapter.** The yes/no is everywhere in your work: the SBDC "would you pay?", the referendum vote, whether a respondent protests, whether they ignore an attribute, whether they speed through the survey, whether they are experienced. All of these are **binary response models** — logit and probit — and this chapter consolidates them into their definitive form and adds what you most need to interpret them: **marginal effects** (the coefficient is not the effect), **odds ratios** (and their traps), **predicted probabilities**, **pseudo-R^2**, and — the learning objective's emphasis — **scale identification** and why coefficients from different binary models cannot be compared. The spine question here: *when I report a binary-model coefficient, do I know that it is identified only relative to an error scale I normalized, that its marginal effect depends on where I evaluate it, and that comparing it across models or groups may be comparing scales rather than effects?*

18.0 Conceptual motivation: the yes/no is a workhorse, and it is subtle

Your PDF lists logit and probit as *"Fondamentali"* and — crucially — names their uses far beyond the SBDC WTP: *"Ti serve per SBDC, referendum CV, probability of yes, protest models, experience sorting, attribute non-attendance e survey diagnostics."* Binary response models are not just how you estimate WTP; they are the *diagnostic backbone* of your survey work. You model:

- **Whether someone says "yes"** to a bid (SBDC, referendum) — the WTP model (Chapters 11, 15).
- **Whether someone protests** — a binary model of protest as a function of respondent characteristics, to understand and defend your exclusions (Chapters 14, 32).
- **Whether someone attends to an attribute** (attribute non-attendance) — a binary model of the probability of ignoring an attribute (Chapter 37).
- **Whether someone speeds or fails a comprehension check** — binary models for data-quality diagnostics (Chapter 50).
- **Who is "experienced"** — modeling selection/sorting into experience, relevant to whether the experienced/inexperienced comparison is confounded (Chapter 16, §16.6).

So mastering binary response is mastering a tool you use constantly. And it is more subtle than it looks. You have already met the pieces — the latent-index/latent-utility foundation (Chapter 14, §14.3), the logit/probit likelihoods and their concavity (Chapters 2 §2.8, 11 §11.8), the scale normalization (Chapter 5, §5.4/§5.6), and marginal effects (Chapter 2, §2.10). This chapter consolidates them into the definitive binary-response treatment and adds the interpretive machinery that trips people up: the coefficient is *not* the marginal effect; odds ratios are *not* probabilities and are *not* comparable across models; coefficients are identified only *up*

to the error scale, so logit $\approx 1.7\times$ probit for the same effect and cross-model coefficient comparisons can be comparing scales; and pseudo-R^2 is *not* the OLS R^2. Each of these is a place your PDF's "sapere giudicare criticamente" is tested.

We proceed: the latent-utility foundation (consolidated); the logit and probit likelihoods (consolidated, with the normal-vs-logistic comparison); **scale identification** and cross-model non-comparability (the theme); **marginal effects** (AME, MEM, and why the coefficient misleads); **odds ratios** (the logit's special interpretation and its traps); **predicted probabilities and classification**; **pseudo-R^2 and LR tests**; and the **diagnostic applications** (protest, attribute non-attendance, speeders, sorting). Everything connects to the SBDC of Chapter 15, the scale-heterogeneity theme of Chapter 25, and the survey diagnostics your CV and DCE papers rely on.

18.1 Prerequisites

From **Chapter 14** (§14.3): the latent-index model $y = \mathbf{1}\{y^* > 0\}$ and $\Pr(y = 1) = F(\mathbf{x}'\beta)$, and the scale normalization. From **Chapters 2** (§2.8) and **11** (§11.8): the logit/probit score, Hessian, and global concavity. From **Chapter 5** (§5.4, §5.6): the logistic distribution (sd $= \pi/\sqrt{3}$) and scale identification. From **Chapter 2** (§2.10): marginal effects (AME/MEM). From **Chapter 12**: the likelihood-ratio test. From **Chapter 7**: the delta method (for marginal-effect SEs).

Same **notation**. New/consolidated symbols: Λ (logistic CDF), Φ, ϕ (normal CDF/density), $\mathbf{x}_i'\beta$ (linear index), $P_i = F(\mathbf{x}_i'\beta)$ (predicted probability), AME (average marginal effect), MEM (marginal effect at the mean), OR (odds ratio).

18.2 The latent-utility foundation (consolidated)

Two equivalent readings. A binary response model can be motivated two ways, and both matter for your work:

(a) Latent index crossing a threshold (Chapter 14). An unobserved continuous $y^* = \mathbf{x}'\beta + \varepsilon$ generates the observed $y = \mathbf{1}\{y^* > 0\}$, so

$$\Pr(y = 1 \mid \mathbf{x}) = \Pr(\varepsilon > -\mathbf{x}'\beta) = F(\mathbf{x}'\beta),$$

where F is the (symmetric) CDF of $-\varepsilon$. This is the SBDC reading: y^* is latent WTP-minus-bid, and "yes" is $y^* > 0$.

(b) Random utility / choice between two options (the RUM reading, previewing Chapter 20). The respondent chooses option 1 over option 0 iff its utility is higher: $U_1 = V_1 + \varepsilon_1 > U_0 + \varepsilon_0$, so

$$\Pr(\text{choose } 1) = \Pr(\varepsilon_0 - \varepsilon_1 < V_1 - V_0) = F(V_1 - V_0),$$

with F the CDF of the error difference. When $\varepsilon_0, \varepsilon_1$ are Gumbel, $F = \Lambda$ (logit — the difference of Gumbels is logistic, Chapter 5 §5.6); when normal, $F = \Phi$ (probit). This is the referendum reading and the bridge to the discrete choice models of Part V: **the binary logit is the two-alternative multinomial logit**, and everything here generalizes.

Both readings deliver the same object: $\Pr(y = 1) = F(\text{index})$, a **nonlinear** function of a linear index. The nonlinearity (the F) is the whole difference from OLS, and it is the source of every subtlety below — marginal effects, scale, odds ratios.

> **Active recall 18.A.** Give the two motivations for the binary model $\Pr(y = 1) = F(\mathbf{x}'\beta)$, and say which reading connects the SBDC to the referendum and which connects it to the multinomial logit. *(Answer: (a) a latent index $y^* = \mathbf{x}'\beta + \varepsilon$ crossing zero — the SBDC "latent WTP exceeds the*

bid" reading; (b) random utility — choosing the higher-utility of two options, the referendum/vote reading, which generalizes to the multinomial logit (the binary logit is the two-alternative MNL). Both give $\Pr(y = 1) = F(index)$*.)*

18.3 The logit and probit likelihoods (consolidated)

The two models. With linear index $v_i = \mathbf{x}_i'\beta$:

$$\text{Logit: } \Pr(y_i = 1) = \Lambda(v_i) = \frac{1}{1 + e^{-v_i}}, \qquad \text{Probit: } \Pr(y_i = 1) = \Phi(v_i).$$

The **log-likelihood** is the same Bernoulli form for both:

$$\ell(\beta) = \sum_{i=1}^{n} \left[y_i \log F(v_i) + (1 - y_i) \log(1 - F(v_i)) \right], \qquad F \in \{\Lambda, \Phi\}.$$

The **logit score** (derived in Chapter 2, §2.8) has the clean residual form $\mathbf{s} = \sum_i (y_i - \Lambda_i)\mathbf{x}_i$; the **probit score** has the generalized-residual form $\mathbf{s} = \sum_i g_i \mathbf{x}_i$ with $g_i = \phi_i[y_i - \Phi_i]/[\Phi_i(1 - \Phi_i)]$ (Chapter 11, §11.8). Both log-likelihoods are **globally concave** (Chapter 2, §2.8), so estimation is reliable — a unique global maximum, no starting-value problems, no local maxima (unlike the mixed logit, Chapters 11 §11.9, 13). This is a genuine comfort: your binary models estimate cleanly.

Normal versus logistic. The two link functions are almost identical in the middle of the data (both symmetric, S-shaped, rising from 0 to 1) but differ in the **tails**: the logistic has *heavier* tails (its density decays like $e^{-|v|}$ versus the normal's e^{-v^2}), so it assigns more probability to extreme index values. Practically:

- In the *middle* of your data (moderate acceptance rates), logit and probit fit almost interchangeably and give nearly identical predicted probabilities and marginal effects.
- In the *tails* (very high or low bids, where the "yes" rate is near 0 or 1), they can differ, and the logistic's heavier tails matter — relevant when your SBDC bids reach far into the WTP distribution (Chapter 15's tail-sensitivity).
- The logit is computationally cheaper (closed-form CDF Λ vs the normal's integral Φ) and has the odds-ratio interpretation (§18.6); the probit connects naturally to the normal latent WTP of your SBDC and to multivariate extensions.

The choice between them is rarely consequential for conclusions; report which you used, and if worried, check that results are stable across the two.

> **Stop and think 18.1.** Your SBDC estimates give nearly identical mean WTP under logit and probit, but a colleague's model (with bids reaching very high, so many "no" responses) shows a larger discrepancy. Why, and which feature of the distributions is responsible? *(Answer: at very high bids the acceptance rate is near zero — the tail of the distribution — where logit and probit differ because the logistic has heavier tails than the normal; so the extrapolated upper tail (which drives mean WTP, Chapter 15) differs between the two links. In the middle of the data they agree; the discrepancy is a tail phenomenon, and it feeds directly into the tail-sensitivity of mean WTP.)*

18.4 Scale identification: why coefficients are not what they seem

This is the chapter's central conceptual point and the learning objective's emphasis. Binary-model coefficients are identified **only up to the error scale**, and forgetting this leads to two common, serious errors: comparing logit and probit coefficients directly, and comparing coefficients across groups or models as if they measured the same thing.

The identification problem (from Chapter 14, §14.3). In $y = \mathbf{1}\{\mathbf{x}'\beta + \varepsilon > 0\}$, multiplying β and ε by any positive constant leaves the observed y unchanged. So the data identify only the **ratio** β/σ_ε — the coefficients *relative to the error standard deviation*. We must normalize σ_ε:

- **Probit:** normalize $\sigma_\varepsilon = 1$ (standard normal).
- **Logit:** the logistic error has $\sigma_\varepsilon = \pi/\sqrt{3} \approx 1.814$.

Consequence 1 — logit ≈ 1.6–1.8× probit. Because the two models normalize *different* error scales, their coefficients differ by (roughly) the ratio of those scales. The theoretical SD ratio is $\pi/\sqrt{3} \approx 1.81$; empirically, the fitted ratio is often around 1.6 (Amemiya's rule, calibrated to match slopes near the mean rather than the SD). *(Verified: fitting logit and probit to the same data gave a coefficient ratio of* ≈ 1.68 *— right in this range.)* The effects are *identical*; only the yardsticks differ. **Comparing a logit coefficient to a probit coefficient as if they were the same quantity is a scale error.**

Consequence 2 — coefficients are not comparable across groups or models. This is the deeper and more dangerous version, and it is the seed of Chapter 25's scale-vs-preference-heterogeneity problem. If two groups (experienced vs inexperienced, high vs low cost vector) have *different error variances* — different amounts of unexplained noise in their choices — then their estimated coefficients differ *even if their underlying preferences are identical*, because each group's coefficients are scaled by its own σ_ε. A group that chooses more consistently (smaller σ_ε) will have *larger* coefficients, not because it cares more, but because it is less noisy. **A between-group difference in binary-model coefficients can be a difference in scale (noise), not in preference.** Comparing raw coefficients across your treatment arms risks reading scale as preference — exactly your PDF's §6.3 warning, and the reason Chapter 25 exists.

The fix — compare marginal effects and predicted probabilities, not coefficients. Marginal effects and predicted probabilities are on the *probability* scale (0 to 1), which is *not* affected by the error-scale normalization, so they *are* comparable across logit/probit and (more) comparable across groups. **Verified:** the logit and probit coefficients differed by $\approx 1.68\times$, but their average marginal effects of x_1 were 0.180 and 0.179 — essentially identical. This is the practical rule: **report marginal effects, not raw coefficients, whenever you compare** — across link functions, across groups, or across studies.

> **Active recall 18.B.** Your experienced group has a bid coefficient of −0.06 and your inexperienced group −0.04. Can you conclude the experienced respondents are more cost-sensitive? What must you check, and what should you compare instead? *(Answer: no — the difference could reflect a smaller error scale (more consistent choices) in the experienced group rather than greater cost sensitivity, because coefficients are identified only up to each group's error scale. Check whether the groups' error variances (scales) differ (a scaled/heteroskedastic model, Chapter 25), and compare* ***marginal effects or predicted probabilities*** *(which are on the scale-free probability metric), not raw coefficients.)*

18.5 Marginal effects: the coefficient is not the effect

In a nonlinear model the coefficient is *not* the effect of the variable — the effect is a derivative that depends on where you evaluate it. This is the single most common misreading of binary-model output.

The marginal effect. For $P(\mathbf{x}) = F(\mathbf{x}'\beta)$, the marginal effect of a continuous regressor x_j is (chain rule, Chapter 2 §2.10):

$$\frac{\partial P}{\partial x_j} = f(\mathbf{x}'\beta)\,\beta_j,$$

the coefficient **scaled by the density** f at the index — $\Lambda(1-\Lambda)$ for logit (max 0.25 at $P = 0.5$), ϕ for probit. So the marginal effect is largest for respondents in the middle of the distribution ($P \approx 0.5$) and shrinks toward the extremes, and it *varies across respondents*. There is no single "the marginal effect"; there is a marginal effect at each point.

Two conventions (state which you use). - Average Marginal Effect (AME): average $\partial P/\partial x_j$ over the sample, $\frac{1}{n}\sum_i f(\mathbf{x}_i'\beta)\beta_j$. The most defensible summary — it accounts for the actual distribution of respondents. - **Marginal Effect at the Mean (MEM):** evaluate at the average respondent $\bar{\mathbf{x}}$, $f(\bar{\mathbf{x}}'\beta)\beta_j$. Simpler but evaluates at a possibly-nonexistent "average" respondent.

Discrete/dummy regressors. For a binary regressor (a treatment dummy, an experience indicator), the marginal effect is *not* a derivative but a **difference in predicted probabilities**: $P(\mathbf{x}, d=1) - P(\mathbf{x}, d=0)$, averaged over the sample. This is the correct effect of a treatment dummy — the change in the probability of "yes" from switching the dummy on.

Worked Example 18.1 (Coefficient vs marginal effect, verified). Fitting a logit, the coefficient on x_1 is 0.975, but the **AME** is 0.180 and the **MEM** is 0.226 — three different numbers. The coefficient (0.975) is *not* an effect on the probability; the AME (0.180) says a one-unit increase in x_1 raises the probability of "yes" by about 18 percentage points *on average*; the MEM (0.226) is the effect at the average respondent. And — the scale point of §18.4 — the *probit* AME was 0.179, essentially identical to the logit's despite the coefficients differing by 1.68× (verified). **Report the AME, state the convention, and never present the raw coefficient as an effect size.**

Standard errors of marginal effects. Because a marginal effect is a nonlinear function of the coefficients, its standard error comes from the delta method (Chapter 7, §7.7) — the `margins` command in Stata/R does this automatically. Do not report a marginal effect without its (delta-method or simulated) standard error.

> **Stop and think 18.2.** A referee writes: "your logit cost coefficient is −0.04, so a €10 cost increase lowers the yes-probability by 0.4." What is wrong, and how do you compute the effect correctly? *(Answer: the referee treated the coefficient as a linear marginal effect. In a logit the effect of cost is* $\Lambda(1-\Lambda)\times(-0.04)$*, evaluated at the relevant point and averaged (AME), and scaled by* 10 *for a €10 change — a smaller, point-dependent number, not* $10 \times 0.04 = 0.4$*. Better still for a €10 discrete change, report the difference in predicted probabilities* $P(cost+10) - P(cost)$*. Report the AME with its delta-method SE, stating the convention.)*

18.6 Odds ratios: the logit's interpretation, and its traps

The logit has a special interpretation unavailable to the probit — the **odds ratio** — which is convenient but comes with genuine traps.

The odds and the odds ratio. For a logit, the **odds** of "yes" are

$$\frac{P}{1-P} = \frac{\Lambda(v)}{1-\Lambda(v)} = e^{v} = e^{\mathbf{x}'\beta},$$

so the log-odds are linear in the index: $\log\frac{P}{1-P} = \mathbf{x}'\beta$ (the "logit" of P). Therefore a one-unit increase in x_j multiplies the odds by e^{β_j} — the **odds ratio** $\mathrm{OR}_j = e^{\beta_j}$. It is constant (does not depend on the other regressors), which is why odds ratios are popular: one number summarizes the multiplicative effect on the odds.

Worked Example 18.2 (Odds ratios, verified). With logit coefficients $\hat{\beta}_1 = 0.975$ and $\hat{\beta}_2 = -0.754$: $\mathrm{OR}_1 = e^{0.975} = 2.65$ (a one-unit increase in x_1 multiplies the odds of "yes" by 2.65 — a 165% increase in the odds), and $\mathrm{OR}_2 = e^{-0.754} = 0.47$ (a one-unit increase in x_2 roughly *halves* the odds). (Verified.)

The traps (why odds ratios mislead).

1. **Odds are not probabilities.** An odds ratio of 2.65 does *not* mean the probability doubles — the probability change depends on the baseline (a doubling of odds from 0.01 to 0.02 is a tiny probability change; from 1 to 2 is large). Non-technical readers routinely conflate "odds ratio of 2" with "twice as likely," which is wrong. For probability statements, report marginal effects or predicted probabilities (§18.5).

2. **Odds ratios are not comparable across models (non-collapsibility).** This is the subtle one. Unlike a linear coefficient, the logit odds ratio *changes when you add other regressors* — even regressors *uncorrelated* with x_j — because adding a predictor changes the residual variance and hence the (scale-normalized) coefficients (the §18.4 scale problem again). So an odds ratio from a bivariate model and from a multivariate model are not directly comparable, and neither are odds ratios across studies with different covariate sets. This "non-collapsibility" is a genuine pitfall in comparing binary-model results across specifications.
3. **Only the logit has them.** The probit has no clean odds-ratio interpretation (its coefficients are not log-odds), so odds ratios do not transfer.

The practical stance. Odds ratios are fine as a *within-model* summary of a logit, clearly labeled as odds (not probability) effects. For *probability* interpretation and for *comparison* across models/groups, use marginal effects and predicted probabilities. Given the non-collapsibility trap, many methodologists now recommend reporting marginal effects as the primary quantity even for logit.

> **Active recall 18.C.** An odds ratio of 3 for "experienced" in a protest-response model — does this mean experienced respondents are three times as likely to protest? And can you compare it to an odds ratio of 2.5 from another study's model? *(Answer: no on both counts. An odds ratio of 3 means the* odds* of protesting are tripled, not the *probability* (the probability change depends on the baseline rate); to state a probability effect, use the marginal effect / predicted-probability difference. And odds ratios are not comparable across models with different covariate sets (non-collapsibility — adding regressors changes the odds ratio via the scale), so the two studies' odds ratios cannot be directly compared.)*

18.7 Predicted probabilities, classification, and pseudo-R²

Predicted probabilities. The most interpretable output of a binary model is the predicted probability $\widehat{P}_i = F(\mathbf{x}_i'\widehat{\beta})$ — the model's estimated probability of "yes" for a given profile. Presenting predicted probabilities across meaningful values of a regressor (e.g., the probability of "yes" as a function of the bid, which is the survival/acceptance curve of Chapter 14) is often the clearest way to communicate a binary model, and it is scale-free (comparable across logit/probit). Their standard errors come from the delta method or simulation.

Classification. If you use the model to *classify* (predict yes/no via a cutoff, usually $\widehat{P} > 0.5$), you can form a **confusion matrix** (true/false positives/negatives) and summarize with sensitivity, specificity, and the ROC/AUC. This matters for your **diagnostic models** — e.g., a model predicting who protests or speeds, used to flag respondents. But beware: the 0.5 cutoff is arbitrary and inappropriate for rare events (if only 5% protest, a model predicting "no protest" for everyone is 95% accurate but useless); use the AUC and choose cutoffs by the cost of errors, not by default.

Pseudo-R². The OLS R^2 has no direct analogue in a likelihood model; the common substitute is **McFadden's pseudo-R²**:

$$\text{pseudo-}R^2 = 1 - \frac{\ell(\widehat{\beta})}{\ell_0},$$

where ℓ_0 is the log-likelihood of the null (intercept-only) model. It compares the fitted model's log-likelihood to the null's. **Cautions:** (i) it is *not* "fraction of variance explained" — it does not have that interpretation, and its values are *much lower* than OLS R^2 for equally good fits (McFadden values of 0.2–0.4 indicate *excellent* fit; *(verified: a strong model gave McFadden = 0.186)*); (ii) there are many competing pseudo-R² definitions (Cox–Snell, Nagelkerke, count-R²) giving different numbers, so *state which*; (iii) do not use it as an OLS-R² and do not treat a "low" value as a poor model. It is a rough fit summary, not a headline.

Likelihood-ratio tests (Chapter 12). For overall model significance, the LR test compares the fitted model to the null: $LR = 2[\ell(\widehat{\beta}) - \ell_0] \sim \chi^2_k$ *(verified: $LR = 1500.6$, χ^2_2, $p \approx 0$)*. For nested comparisons (does adding a set of regressors help?), the LR test of Chapter 12 applies directly — the standard tool for binary-model specification, alongside AIC/BIC for non-nested.

Stop and think 18.3. Your protest-prediction logit has a McFadden pseudo-R^2 of 0.12, and a co-author says "that's a terrible fit, the model is useless." Are they right? *(Answer: not necessarily — McFadden pseudo-R^2 is not the OLS R^2, and its values run much lower;* 0.12 *can indicate a reasonable fit for a binary model (McFadden* 0.2–0.4 *is considered excellent). Judge the model by whether the predictors are significant (LR test), whether predicted probabilities discriminate (AUC), and whether it serves its diagnostic purpose — not by comparing pseudo-R^2 to an OLS-R^2 yardstick.)*

18.8 Diagnostic applications: binary models throughout your survey work

Pulling the chapter toward your PDF's uses, here is how binary response models serve as diagnostic tools across your CV and DCE work — each a logit/probit with a specific binary outcome:

- **Protest models.** Model Pr(protest) as a function of respondent characteristics to understand *who* protests and *why*, and to defend (or bound) your exclusions. If protest is predicted by treatment, exclusions threaten randomization (Chapters 14 §14.8, 32); if predicted by covariates unrelated to treatment, exclusions are safer.
- **Attribute non-attendance (ANA).** Model Pr(ignore attribute j) — whether a respondent attends to an attribute — as a binary outcome (or a latent-class probability, Chapter 37). Ignoring an attribute means its coefficient (and any WTP) is mis-estimated for that respondent.
- **Speeder / comprehension diagnostics.** Model Pr(speed) or Pr(fail comprehension) to characterize data quality and to decide, transparently, how to handle low-quality responses (Chapter 50).
- **Experience sorting / selection.** Model Pr(experienced) to understand who selects into experience — crucial for judging whether the experienced-vs-inexperienced comparison is a clean contrast or confounded by selection (Chapter 16, §16.6; the OVB concern).

In all of these, the *same* interpretive discipline applies: report marginal effects (not coefficients) for probability statements, be aware of scale (especially when comparing across groups), use odds ratios only as within-model summaries, and judge fit by LR tests and AUC, not pseudo-R^2 alone. The binary model is a versatile diagnostic; used carelessly, it produces the same misreadings (coefficient-as-effect, scale-as-preference, odds-as-probability) at every application.

Active recall 18.D. You model Pr(protest) and find "experienced" has a large positive coefficient. Before concluding experienced respondents protest more, what two checks does this chapter demand? *(Answer: (i) report the* ***marginal effect*** *(change in protest probability), not the coefficient, since the coefficient is not a probability effect and its size is on the scale-normalized metric; (ii) consider* ***scale*** *— if experienced respondents have different choice/response noise, coefficient comparisons across the experienced/inexperienced split can reflect scale, not the protest propensity — and check whether "experienced" is confounded with other protest predictors (OVB, since experience is not randomized). Then interpret the probability effect, not the raw coefficient.)*

18.9 Failure modes

F1. Reading the coefficient as the marginal effect. Reporting β_j as the effect on the probability. *Fix:* report the AME (or MEM, or a discrete-change effect) with its delta-method SE; state the convention (§18.5).

F2. Comparing logit and probit coefficients directly. "My logit effect (0.9) is bigger than their probit effect (0.5)." *Fix:* they differ by the ~1.6–1.8 scale factor; compare marginal effects, which are nearly identical (§18.4).

F3. Comparing coefficients across groups as preference. Reading a between-group coefficient difference as a preference difference when it may be scale. *Fix:* model/scale differences (Chapter 25); compare marginal effects/predicted probabilities (§18.4).

F4. Odds ratio read as probability. "OR of 2 means twice as likely." *Fix:* odds $\neq$ probability; report marginal effects for probability statements (§18.6).

F5. Comparing odds ratios across specifications. Ignoring non-collapsibility (odds ratios change with added regressors). *Fix:* compare within a single specification; use marginal effects across models (§18.6).

F6. Pseudo-R^2 read as OLS R^2. Calling a McFadden 0.15 a "poor fit." *Fix:* pseudo-R^2 is not variance-explained; judge fit by LR/AIC and AUC; state which pseudo-R^2 (§18.7).

F7. 0.5 classification cutoff for rare events. High "accuracy" from predicting the majority class. *Fix:* AUC and cost-based cutoffs for imbalanced outcomes (protest, speeding) (§18.7).

F8. Treating a diagnostic-model coefficient as causal. "Experienced respondents protest more" from an observational logit. *Fix:* the model is descriptive/predictive unless the predictor is randomized; watch OVB (§18.8; Chapter 16).

18.10 Diagnostic questions for your own papers

1. **Marginal effects.** Am I reporting marginal effects (AME) with SEs, not raw coefficients, for probability statements? *(§18.5, F1.)*
2. **Scale/comparability.** When comparing logit vs probit, or across groups/studies, am I comparing marginal effects (scale-free), not coefficients? *(§18.4, F2–F3.)*
3. **Scale vs preference.** Could a between-group coefficient difference be scale (noise) rather than preference? *(§18.4; Chapter 25.)*
4. **Odds ratios.** If I report odds ratios, are they labeled as odds (not probability) effects and used only within a single specification? *(§18.6, F4–F5.)*
5. **Fit.** Am I judging fit by LR tests, AIC, and AUC — not by pseudo-R^2 as if it were OLS R^2? *(§18.7, F6.)*
6. **Classification.** For diagnostic models (protest, speeder), am I avoiding the 0.5 cutoff for rare events and using AUC/cost-based cutoffs? *(§18.7, F7.)*
7. **Causality.** Are my diagnostic-model coefficients descriptive/predictive rather than causal (unless randomized)? *(§18.8, F8.)*

18.11 Exercises

(A) Comprehension.

A1. Explain why the logit coefficient is not the effect of the variable on the probability, and write the marginal effect for a continuous regressor.

A2. Your logit gives $\widehat{\beta}_j = 0.7$. Compute the odds ratio and state, precisely, what it means (and what it does *not* mean).

A3. A logit coefficient is 0.9 and a probit coefficient (same data) is 0.5. Are these consistent with the same effect? Show the reasoning.

(B) Derivation.

B1. Derive the marginal effect $\partial P/\partial x_j = f(\mathbf{x}'\beta)\beta_j$ for a binary model, and explain why it is largest at $P = 0.5$ for the logit.

B2. Show that for the logit, $\log[P/(1-P)] = \mathbf{x}'\beta$, so e^{β_j} is the odds ratio; and explain why this interpretation does not hold for the probit.

B3. Explain scale identification: show that multiplying β and ε by a constant leaves $y = \mathbf{1}\{\mathbf{x}'\beta + \varepsilon > 0\}$ unchanged, and derive why logit and probit coefficients differ by (roughly) the ratio of their error standard deviations.

(C) Research application / critique.

C1. Your protest-response logit reports coefficients and odds ratios but no marginal effects, and concludes "experience strongly increases protesting (OR = 3.2)." Rewrite the reporting correctly and state what additional analysis is needed for a causal claim.

C2. You compare the bid coefficient across your high- and low-cost-vector arms (both logit) and find they differ significantly, concluding "cost sensitivity differs by arm." Critique using §18.4 and state what you should compare and check.

C3. A co-author dismisses your referendum-CV logit because its pseudo-R^2 is 0.14. Explain why this is not a valid criticism and what fit/validity checks you would present instead.

18.12 Solutions

A1. In $P = F(\mathbf{x}'\beta)$, the effect of x_j is the derivative $\partial P/\partial x_j = f(\mathbf{x}'\beta)\beta_j$, the coefficient scaled by the density f at the index — which varies across observations. The coefficient β_j alone is on the latent-index (log-odds for logit) scale, not the probability scale, so it is not the probability effect; the marginal effect is point-dependent and requires the density factor.

A2. Odds ratio $= e^{0.7} = 2.01$. It means a one-unit increase in x_j multiplies the *odds* of "yes" by about 2 (a 101% increase in the odds), holding other regressors fixed. It does **not** mean the *probability* doubles (the probability change depends on the baseline), and it is not comparable to odds ratios from models with different covariate sets (non-collapsibility).

A3. Yes, consistent with the same effect. Logit and probit normalize different error scales ($\pi/\sqrt{3} \approx 1.81$ vs 1), so logit coefficients are roughly 1.6–1.8 times probit coefficients for the same effect; $0.9/0.5 = 1.8$, right in that range. The marginal effects (on the probability scale) would be nearly identical — the effect is the same, only the coefficient yardstick differs.

B1. $P(\mathbf{x}) = F(\mathbf{x}'\beta)$; by the chain rule $\partial P/\partial x_j = F'(\mathbf{x}'\beta) \cdot \partial(\mathbf{x}'\beta)/\partial x_j = f(\mathbf{x}'\beta)\beta_j$, where $f = F'$ is the density. For the logit $f(v) = \Lambda(v)(1 - \Lambda(v))$, which is maximized at $\Lambda = 0.5$ (i.e. $v = 0$, $P = 0.5$) where $f = 0.25$, and shrinks toward the extremes — so the marginal effect is largest for respondents on the knife-edge ($P \approx 0.5$) and smallest in the tails.

B2. For the logit, $P = \Lambda(v) = 1/(1 + e^{-v})$, so $1 - P = e^{-v}/(1 + e^{-v})$ and $P/(1 - P) = e^{v} = e^{\mathbf{x}'\beta}$; taking logs, $\log[P/(1 - P)] = \mathbf{x}'\beta$ (log-odds linear in the index), so a unit increase in x_j adds β_j to the log-odds and multiplies the odds by e^{β_j} — the odds ratio. For the probit, $P = \Phi(v)$ and $\log[P/(1 - P)] = \log[\Phi(v)/(1 - \Phi(v))]$ is *not* linear in v, so there is no constant odds ratio; probit coefficients are not log-odds.

B3. In $y = \mathbf{1}\{\mathbf{x}'\beta + \varepsilon > 0\}$, for any $c > 0$, $\{\mathbf{x}'\beta + \varepsilon > 0\} = \{c\mathbf{x}'\beta + c\varepsilon > 0\}$, so scaling $\beta \to c\beta$ and $\varepsilon \to c\varepsilon$ leaves y (hence the likelihood) unchanged — only β/σ_ε is identified. Normalizing σ_ε fixes the scale: probit uses $\sigma_\varepsilon = 1$, logit $\sigma_\varepsilon = \pi/\sqrt{3}$. Since each model reports β relative to its own σ_ε, the two coefficient vectors differ by the ratio of their scales, $\approx \pi/\sqrt{3} \approx 1.81$ (empirically often ~1.6), for the same underlying effect.

C1. Corrected reporting: present the **average marginal effect** of experience on the probability of protesting (with its delta-method SE) — e.g., "experience raises the protest probability by X percentage points" — rather than only the coefficient/odds ratio; label the OR as an *odds* effect ("triples the odds," not "three times as likely"); and note the OR is specification-dependent (non-collapsibility). For a *causal* claim ("experience *increases* protesting"), experience must be exogenous — but experience is not randomized, so the model is

descriptive/predictive and the association may be confounded (OVB, Chapter 16); to claim causality you would need randomization or a credible identification strategy. Report it as a predictive association with the probability effect quantified.

C2. Comparing raw bid coefficients across arms risks reading **scale as preference**: the coefficients are identified only up to each arm's error scale, so a difference can reflect different choice noise (scale) rather than different cost sensitivity (preference) — exactly the scale-vs-preference-heterogeneity problem (Chapter 25). I should (i) compare **marginal effects / predicted probabilities** (scale-free), not coefficients; and (ii) estimate a model that lets the **scale differ by arm** (a heteroskedastic/scaled model) to test whether a genuine preference difference survives once scale is allowed to vary. Only then can "cost sensitivity differs by arm" be defended.

C3. A pseudo-R^2 of 0.14 is not a valid basis for dismissal: McFadden's pseudo-R^2 is *not* the OLS "fraction of variance explained," and its values run far lower — 0.2–0.4 indicates *excellent* fit, so 0.14 can be a perfectly good binary-model fit. Better fit/validity checks: the LR test for overall and nested significance (are the predictors jointly significant?), AIC/BIC for specification comparison, the AUC / predicted-probability calibration (does the model discriminate and calibrate well?), and — for the referendum CV — whether the acceptance curve is well estimated and the implied WTP is sensible. Judge the model by these, not by an OLS-R^2 yardstick applied to a pseudo-R^2.

18.13 Spaced review prompts

+1 day. From memory, write the logit/probit likelihood and the marginal effect $f(\mathbf{x}'\beta)\beta_j$; state why the coefficient is not the effect and why logit $\approx 1.7\times$ probit coefficients.

+1 week. Explain scale identification and why coefficients are not comparable across groups (scale vs preference); derive the logit odds ratio and state its two traps (odds $\neq$ probability, non-collapsibility). Explain why pseudo-R^2 is not OLS R^2.

+1 month. For one binary model in your own work (SBDC, protest, or ANA), report the AME with SE, check whether any cross-group coefficient comparison is confounded by scale, use odds ratios only within-model, and judge fit by LR/AUC. Close with the spine sentence about what your coefficients do and do not identify.

18.14 How this chapter connects to the rest of the book

- **Chapters 2 (§2.8), 11 (§11.8), 14 (§14.3)** supplied the logit/probit likelihoods, concavity, latent-index foundation, and scale normalization that this chapter consolidates; **Chapter 5 (§5.4, §5.6)** the normal-vs-logistic and scale identification; **Chapter 12** the LR tests; **Chapter 7** the delta method for marginal-effect SEs.
- **Chapter 15 (CV likelihoods)** is the SBDC application of this chapter's binary model; **Chapters 20–21 (RUT, MNL)** generalize the binary logit to the multinomial choice models of your DCE (the binary logit is the two-alternative MNL); **Chapter 25 (scale vs preference heterogeneity)** is the full development of §18.4's scale-comparability problem — the reason cross-group coefficient comparisons can mislead.
- **Chapters 32 (threats), 39 (behavioral welfare), 50 (survey design)** use the diagnostic binary models of §18.8 (protest, attribute non-attendance, speeders); **Chapter 16 (OLS)** is the linear analogue this chapter generalizes, and the OVB caution for diagnostic models.

This chapter makes the yes/no *solidissimo* — the binary model that underlies your SBDC and referendum WTP and serves as the diagnostic backbone of your survey work — with the interpretive discipline (marginal effects, scale, odds-ratio traps, pseudo-R^2) that its nonlinearity demands.

18.15 Research translation task

For your **SBDC/referendum model and one diagnostic binary model** (protest, ANA, or speeder), produce a two-to-three-page note:

1. **Estimate and report correctly.** Fit the logit (and probit as a robustness check); report **average marginal effects** with delta-method SEs, not raw coefficients, for all probability statements.
2. **Scale and comparability.** Confirm logit/probit marginal effects agree; for any cross-group comparison (arms, experience), compare marginal effects and check whether a coefficient difference could be scale (Chapter 25).
3. **Odds ratios (if used).** Report them only as within-model odds effects, clearly labeled, and note non-collapsibility.
4. **Fit and tests.** Report the LR test (overall and nested), AIC/BIC, and — for classification/diagnostic models — the AUC; report pseudo-R^2 only with its name and caveat.
5. **Diagnostic use.** For the diagnostic model (protest/ANA/speeder), state whether it is descriptive/predictive or causal, and how it informs your exclusions or attribute-processing assumptions.
6. **Close** with the spine sentence: *"My binary model's coefficients are identified up to the error scale; I report marginal effects [values] (scale-free); cross-group differences are [preference/scale]; and I judge fit by [LR/AUC], not pseudo-R^2."*

Carry this into Chapter 21 (the multinomial generalization), Chapter 25 (scale vs preference), and Chapters 32, 39, 50 (the diagnostic applications).

18.16 Bibliography

Core. - Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, the binary-response chapters (logit/probit, latent-index, marginal effects, and the scale/identification discussion). The rigorous econometric reference. *(Verify edition/chapters.)* - Cameron & Trivedi, *Microeconometrics: Methods and Applications*, on binary models, marginal effects (AME/MEM), odds ratios, and pseudo-R^2 — matching §18.5–§18.7 at implementation depth. *(Verify edition.)* - Train, *Discrete Choice Methods with Simulation*, on the binary logit as the two-alternative choice model and the scale/identification issues (§18.2, §18.4) — the bridge to Part V. Freely available — *verify link.*

Applied. - Long & Freese, *Regression Models for Categorical Dependent Variables Using Stata.* The clearest applied guide to interpreting binary models — marginal effects, predicted probabilities, odds ratios, and their pitfalls (§18.5–§18.7). *(Verify edition.)* - Mood (2010), "Logistic Regression: Why We Cannot Do What We Think We Can Do, and What We Can Do About It," *European Sociological Review.* The definitive accessible treatment of the odds-ratio non-collapsibility and cross-model comparability problems of §18.4, §18.6 — essential reading. *(Verify volume/pages.)* - Haab & McConnell, *Valuing Environmental and Natural Resources*, for the binary (referendum/SBDC) model in the contingent-valuation context of your work. *(Verify edition.)*

Advanced. - McFadden, "Conditional Logit Analysis of Qualitative Choice Behavior," for the pseudo-R^2 and the RUM foundation of the binary/multinomial logit. *(Verify exact reference/year.)* - Amemiya, *Advanced Econometrics*, the qualitative-response chapters — the rigorous treatment of logit/probit, scale, and the ~1.6 coefficient ratio. *(Verify edition.)* - Greene & Hensher, *Modeling Ordered Choices*, and the broader discrete-choice literature, for extensions (ordered, multinomial) building on the binary foundation. *(Verify edition.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Mood / McFadden citations before using them, per the book's no-unverified-citation rule.

End of Chapter 18. Next in Part IV: Chapter 19 — Repeated Observations (clustered, panel, and multilevel inference; fixed and random effects; within/between variation; the pseudo-panel logic of repeated choice tasks and multiple estimates per study), which develops the modeling alternative to the clustered standard errors of Chapter 17 for your repeated-observation data. Tell me when to write it, or request a different chapter.

Chapter 19 — Repeated Observations

Part IV: Core Econometrics

> **The spine question, applied to this chapter.** Your PDF is candid: *"Anche se non fai panel econometrics classica, hai dati ripetuti: più choice tasks per individuo; più WTP per studio nella meta-regression; più scenari per unità/ecosistema."* You have panels everywhere — choice tasks nested in respondents, WTP estimates nested in studies — and this changes both *inference* (Chapter 17: cluster the standard errors) and *modeling* (this chapter: model the dependence with fixed, random, and multilevel effects). The choice between **fixed effects** (which purge time-invariant confounders but use only within-unit variation) and **random effects** (which are efficient but assume the unit effect is uncorrelated with the regressors) is one you make constantly, often without naming it — the panel mixed logit is a random-effects model, and random-effects meta-analysis is how your finfish synthesis should treat between-study heterogeneity. The spine question here: *when my observations come in groups, am I modeling the dependence in a way that removes the right confounders and reflects the between-group heterogeneity — or am I ignoring structure that, for your 18-study meta-analysis, can understate uncertainty threefold?*

19.0 Conceptual motivation: your data are panels, whether you call them that or not

Chapter 17 taught you to *correct* standard errors for within-cluster dependence. This chapter teaches you to *model* it — and the distinction matters, because modeling the dependence does three things clustering alone cannot: it can *remove confounders* (fixed effects), it *quantifies the heterogeneity* (variance components — how much respondents or studies genuinely differ), and it is the *framework underneath the models you already use*. Your PDF names the exposures and the tools: *"cluster-robust standard errors; repeated observations; random effects; fixed effects; multilevel models; within/between variation; pseudo-panel logic."* Every one connects to something central in your work:

- **The DCE is a panel.** Each respondent answers several choice tasks — repeated observations nested in a respondent. The **panel mixed logit** (Chapter 22) is, at its heart, a *random-effects* model: it gives each respondent their own random coefficients (a random effect on preferences). Understanding random effects here is understanding your EC-MXL there.
- **The meta-analysis is a panel.** Multiple WTP estimates come from each of your 18 finfish studies — estimates nested in studies. **Random-effects meta-analysis** (a random intercept by study) is how you must handle between-study heterogeneity, and — as you will see — ignoring it (a fixed-effect meta-analysis) can understate your uncertainty by a factor of nearly three (Chapter 47).
- **Ecosystem-service data are panels.** Multiple scenarios or valuations per ecosystem/site are nested observations, calling for multilevel treatment.

So this chapter is not "classical panel econometrics" as a detour; it is the *repeated-observations logic* that underlies your DCE, your meta-analysis, and your ecosystem work. We build it in order: the within/between variation decomposition (the panel version of the law of total variance); **fixed effects** (the within estimator,

what it purges and what it costs); **random effects** (efficiency, and the assumption it rests on); the **fixed-vs-random trade-off** and the Hausman test; **multilevel/hierarchical models** (the general framework, and the bridge to the mixed logit); and **random-effects meta-analysis** (the finfish structure). Throughout, the theme is your PDF's "più WTP dallo stesso studio sono indipendenti?" — no, they are not, and how you model that non-independence determines whether your inference is honest.

19.1 Prerequisites

From **Chapter 17**: cluster-robust standard errors and the design effect (the *inference* response to dependence this chapter complements with *modeling*). From **Chapter 16**: OLS, omitted-variable bias (§16.6), and exogeneity — fixed effects are an OVB-removal device. From **Chapter 5** (§5.7): the law of total variance and the between/within decomposition. From **Chapter 11**: the likelihood framework (random-effects and multilevel models are estimated by (restricted) maximum likelihood). From **Chapter 6** (§6.9): asymptotics in the number of groups (the few-clusters caution).

Same **notation**. New symbols: i (observation), g or j (group/unit — respondent, study), t (repeated-observation index within a unit), α_i (fixed unit effect, a parameter), u_g (random unit effect, a random variable), σ_u^2 (between-group variance), σ_ε^2 (within-group variance), $\rho = \sigma_u^2/(\sigma_u^2+\sigma_\varepsilon^2)$ (intra-class correlation), τ^2 (between-study variance in meta-analysis), v_j (within-study sampling variance).

19.2 Repeated observations, dependence, and two responses

The structure. Observations are grouped: y_{gt} is observation t within unit g (choice task t for respondent g; WTP estimate t from study g). Observations within a unit share a common component and are therefore **correlated**; observations in different units are (assumed) independent. This is the panel/clustered/multilevel structure, and it violates the OLS independence assumption (Chapter 16, A4).

Two responses to dependence.

1. **Correct the standard errors (Chapter 17).** Keep the pooled OLS estimate, but use cluster-robust standard errors that allow arbitrary within-unit correlation. *Robust* (few assumptions), but it does not use the structure to improve the estimate, quantify heterogeneity, or remove confounders.
2. **Model the dependence (this chapter).** Write the unit effect into the model — as a *fixed* effect α_g (a parameter) or a *random* effect u_g (a random variable) — and estimate accordingly. This can *remove confounders* (fixed effects), *quantify the between-unit heterogeneity* (variance components), and be *more efficient* — but it adds assumptions.

The two are not mutually exclusive; a common and defensible practice is to *model* the dependence (RE/multilevel) *and* cluster the standard errors (robustness against misspecifying the RE structure). The question this chapter answers is *which model* — fixed or random effects — and *what each buys and costs*.

The unit effect. Write the model with a unit-specific intercept:

$$y_{gt} = \mathbf{x}'_{gt}\beta + c_g + \varepsilon_{gt},$$

where c_g captures everything about unit g that is *constant across its repeated observations* — a respondent's baseline propensity, a study's methodology, an ecosystem's baseline value. The entire fixed-vs-random distinction is about **how you treat** c_g: as a parameter to estimate (fixed) or as a random draw (random), and — decisively — whether you allow it to be *correlated with* $\mathbf{x}_{gt}$.

> **Active recall 19.A.** Your DCE has 8 choice tasks per respondent. Name the "unit," the "repeated observation," and the "unit effect," and give one thing the unit effect might capture. *(Answer: the unit is the respondent, the repeated observations are the 8 choice tasks, and the unit*

effect c_g is the respondent-specific component constant across their tasks — e.g., a respondent's baseline strength of preference, their scale/consistency, or an unobserved taste that shifts all their choices. The panel mixed logit models this as a random effect on the coefficients, §19.7.)

19.3 Within and between variation

Panel estimators differ in *which variation they use*, so the first tool is the decomposition of variation into **within-unit** and **between-unit** parts — the panel version of the law of total variance (Chapter 5, §5.7).

The decomposition. For any variable x_{gt}, write it as its unit mean plus a deviation:

$$x_{gt} = \bar{x}_g + (x_{gt} - \bar{x}_g),$$

where $\bar{x}_g$ is the unit-g mean. The **between variation** is the variation in the unit means $\bar{x}_g$ across units; the **within variation** is the variation of x_{gt} around its own unit mean, over the repeated observations. Total variation = between + within. The three panel estimators use these differently:

- **Fixed effects (within estimator)** uses only the **within** variation — how y moves with x *within* a unit, over its repeated observations.
- **The between estimator** uses only the **between** variation — how unit-mean y relates to unit-mean x across units.
- **Random effects** uses a *weighted combination* of both — efficient when the two tell the same story.
- **Pooled OLS** uses the total variation, treating all observations as independent.

Why this matters for identification. The variation an estimator uses is the variation that *identifies* its coefficient (Chapter 30 preview). Fixed effects identify β from *within-unit* changes — so a regressor that never varies within a unit (a time-invariant characteristic: a respondent's gender, a study's country) is *absorbed by the unit effect and cannot be estimated by FE.* Conversely, if the informative variation is *between* units, FE throws it away and is inefficient. Knowing which variation carries your effect tells you which estimator can even see it.

> **Stop and think 19.1.** In your DCE, the *cost attribute* varies across a respondent's choice tasks (different tasks show different costs), but a respondent's *experience level* is fixed across their tasks. Which of these can a respondent-fixed-effects model estimate, and which is absorbed? *(Answer: the* cost* coefficient can be estimated by FE — it varies *within* respondents across tasks, providing within-variation. The *experience* effect is *time-invariant within respondent* (constant across a respondent's tasks), so it is absorbed by the respondent fixed effect and cannot be estimated by FE. To estimate the experience effect you need between-respondent variation — random effects, or a model without respondent fixed effects — which is why the experience/cost-vector interaction is delicate, Chapters 17, 25.)*

19.4 Fixed effects: purging time-invariant confounders

The model and estimator. Fixed effects treat each unit's intercept α_g as a *parameter*:

$$y_{gt} = \mathbf{x}'_{gt}\beta + \alpha_g + \varepsilon_{gt}.$$

The estimator subtracts each unit's mean (the **within transformation**):

$$(y_{gt} - \bar{y}_g) = (\mathbf{x}_{gt} - \bar{\mathbf{x}}_g)'\beta + (\varepsilon_{gt} - \bar{\varepsilon}_g),$$

and runs OLS on the demeaned data. The unit effect α_g — being constant within the unit — cancels in the demeaning, along with *any time-invariant regressor.* Equivalently, FE is OLS with a full set of unit dummies.

What FE buys: immunity to time-invariant confounders. Here is the key result and the reason FE is so valued. Because α_g absorbs *everything* constant within the unit — observed *and unobserved* — FE removes omitted-variable bias from any time-invariant confounder. You do not need to observe the confounder; the within transformation eliminates it. This is the panel analogue of the randomization result of Chapter 16 (§16.5): where randomization severs the treatment–confounder correlation, fixed effects *difference it out.*

Worked Example 19.1 (FE removes confounder bias, verified). Simulate a panel where a unit-specific confounder α_g affects both the regressor x_{gt} (so x is correlated with α) and the outcome y, with true $\beta = 2$. Pooled OLS, which omits α_g, gives a biased coefficient of 2.69 (omitted-variable bias — α_g is correlated with x, Chapter 16 §16.6). Fixed effects, by demeaning within units, removes α_g and recovers the unbiased 2.00 (verified over 1000 simulations). The within transformation purged the confounder without ever observing it.

What FE costs. - **It cannot estimate time-invariant effects.** Any regressor constant within units (experience, gender, study country) is absorbed — you cannot get its coefficient from FE (§19.3). - **It uses only within variation**, discarding between variation. If the between variation is informative and the RE assumption holds, FE is *less efficient* than RE (larger standard errors). - **It can amplify measurement error.** Within-demeaning removes the "signal" (between variation) but keeps measurement noise, attenuating coefficients more than pooled OLS in some cases.

Relevance to your work. In your DCE, a respondent-fixed-effects choice model (a conditional/fixed-effects logit) controls for *all* time-invariant respondent characteristics — a strong defense against respondent-level confounding — but cannot estimate the effect of respondent-level variables (experience, income) and uses only within-respondent (across-task) variation. This is a real trade-off in choice modeling: the fixed-effects logit is robust but limited, which is one reason the *random*-coefficients mixed logit (which models respondent heterogeneity rather than differencing it out) is often preferred (Chapter 22).

> **Active recall 19.B.** What class of confounders does a fixed-effects model remove without your having to observe them, and what is the price? *(Answer: it removes all* time-invariant* (within-unit constant) confounders — observed and unobserved — via the within transformation, so no OVB from respondent- or study-level characteristics that don't vary across the repeated observations. The price: it cannot estimate the effects of time-invariant regressors (they are absorbed) and it uses only within-unit variation, so it is less efficient than random effects when the RE assumption holds and cannot exploit between-unit information.)*

19.5 Random effects: efficiency at the cost of an assumption

The model. Random effects treat the unit effect as a *random variable* drawn from a distribution, rather than a parameter:

$$y_{gt} = \mathbf{x}'_{gt}\beta + u_g + \varepsilon_{gt}, \qquad u_g \sim (0, \sigma_u^2), \quad \varepsilon_{gt} \sim (0, \sigma_\varepsilon^2),$$

with the **crucial assumption** that u_g is *uncorrelated with the regressors*, $\mathbb{E}[u_g \mid \mathbf{x}_{gt}] = 0$. The composite error $u_g + \varepsilon_{gt}$ has the **intra-class correlation** $\rho = \sigma_u^2/(\sigma_u^2 + \sigma_\varepsilon^2)$ (the fraction of variance that is between-unit — Chapter 5 §5.7, Chapter 17's design effect).

The estimator (feasible GLS / quasi-demeaning). RE is estimated by generalized least squares, which amounts to a *partial* demeaning: $(y_{gt} - \theta\bar{y}_g) = (\mathbf{x}_{gt} - \theta\bar{\mathbf{x}}_g)'\beta + \ldots$, where $\theta \in [0, 1]$ depends on the variance ratio and the number of repeated observations. Note the two extremes: $\theta = 0$ gives *pooled OLS* (ignores the units), $\theta = 1$ gives *fixed effects* (fully demeaned). **Random effects sits between pooled OLS and fixed effects**, using an optimal weighted combination of within and between variation.

What RE buys: efficiency and time-invariant effects. Because RE uses *both* within and between variation (and does not spend degrees of freedom on unit dummies), it is *more efficient* than FE — smaller standard errors — *when its assumption holds.* And because it does not absorb time-invariant regressors, RE

can estimate their effects (a respondent's experience, a study's method). This is often decisive: if you want the effect of a respondent-level variable, RE (or a hybrid model) can give it and FE cannot.

Worked Example 19.2 (RE efficiency, verified). When the unit effect is genuinely uncorrelated with the regressor (RE assumption holds), RE is more efficient than FE. In a simulation with $u_g \perp x_{gt}$ and true $\beta = 2$, the FE estimator's sampling standard deviation was 0.0278 and the RE estimator's was 0.0274 — RE modestly more precise (verified). The efficiency gain is larger when between variation carries more information; here, with abundant within variation, the two are close, but RE never does worse *when its assumption holds*.

What RE costs: bias if the assumption fails. The efficiency is bought with the assumption $u_g \perp \mathbf{x}$. If the unit effect is *correlated* with the regressors — the confounder case of §19.4 — RE is **biased** (it does not difference out the confounder). This is the whole tension: FE is *robust* (no assumption on u_g–$\mathbf{x}$ correlation) but inefficient; RE is *efficient* but requires the unit effect to be uncorrelated with the regressors. Which to use is the subject of §19.6.

> **Stop and think 19.2.** In your DCE, would you expect a respondent's unobserved baseline preference strength (the "unit effect") to be *correlated* with the attributes shown in their choice tasks? What does your answer imply for FE vs RE? *(Answer: the attributes shown are set by the* experimental design* (randomized/assigned), so they are, by construction, *uncorrelated* with the respondent's unobserved preference strength — the RE assumption plausibly holds for a well-randomized DCE. This favors *random effects* (efficient, and able to estimate respondent-level covariates), which is why the panel mixed logit uses random rather than fixed effects. In observational panels where the regressors correlate with the unit effect, FE would be safer.)*

19.6 Fixed versus random effects: the trade-off and the Hausman test

The trade-off, stated cleanly.

	Fixed effects	Random effects
Unit effect c_g	parameter	random variable
Assumption on c_g–$\mathbf{x}$ correlation	none (robust)	must be *uncorrelated*
Confounding from time-invariant unobservables	removed	**not** removed (bias if correlated)
Time-invariant regressor effects	cannot estimate	can estimate
Variation used	within only	within + between
Efficiency (if RE assumption holds)	lower	higher

The essence: **FE trades efficiency for robustness; RE trades robustness for efficiency and the ability to estimate time-invariant effects.** Use FE when you fear the unit effect is correlated with your regressors (observational data, endogenous regressors); use RE when the regressors are plausibly exogenous to the unit effect (randomized designs, well-controlled settings) and you want efficiency or time-invariant effects.

The Hausman test. The formal tool to choose is the **Hausman test**, which compares the FE and RE estimates. Under the RE assumption ($u_g \perp \mathbf{x}$), both FE and RE are consistent, so they should be *close*; if the assumption fails, RE is biased while FE remains consistent, so they *diverge*. The test statistic is a quadratic form in the difference $(\hat{\beta}_{FE} - \hat{\beta}_{RE})$:

$$H = (\hat{\beta}_{FE} - \hat{\beta}_{RE})' \left[\widehat{\mathrm{Var}}(\hat{\beta}_{FE}) - \widehat{\mathrm{Var}}(\hat{\beta}_{RE})\right]^{-1} (\hat{\beta}_{FE} - \hat{\beta}_{RE}) \sim \chi^2,$$

A significant Hausman test says the RE assumption fails — the estimates differ more than sampling noise allows — so use FE. A non-significant test says RE and FE agree, so RE (the efficient choice) is defensible. *(A modern caveat: the Hausman test can be underpowered or ill-behaved, and many now prefer to report both FE and RE, or use a hybrid/correlated-random-effects model that includes the unit means as regressors — the Mundlak device — which nests both and makes the assumption testable directly.)*

> **Active recall 19.C.** You run FE and RE on the same panel and the Hausman test is significant. What does this tell you, and which estimator do you report? *(Answer: it tells you the random-effects assumption fails — the unit effect is correlated with the regressors, so RE is biased while FE is consistent (they differ more than chance allows). Report the **fixed-effects** estimate, which is robust to that correlation. If you need time-invariant-regressor effects too, use a hybrid/correlated-random-effects (Mundlak) model.)*

19.7 Multilevel and hierarchical models: the general framework

Fixed and random effects on a single grouping are special cases of the general **multilevel (hierarchical) model**, which allows random effects at *multiple levels* and on *slopes* as well as intercepts. This is the framework that unifies your DCE and meta-analysis.

Random intercepts and random slopes. The random-effects model of §19.5 has a random *intercept* u_g. A multilevel model can also make the *coefficients* vary by unit — a **random slope**:

$$y_{gt} = \mathbf{x}'_{gt}\beta_g + \varepsilon_{gt}, \qquad \beta_g = \bar{\beta} + \eta_g, \quad \eta_g \sim N(\mathbf{0}, \Sigma),$$

so each unit has its own coefficient vector drawn around a population mean $\bar{\beta}$ with covariance Σ. **This is exactly the mixed logit** (Chapter 22): each *respondent* has their own preference coefficients β_g drawn from a population distribution, and Σ (generated via a Cholesky factor, Chapter 1 §1.9) is the covariance of preferences. The mixed logit *is* a multilevel random-coefficients model applied to discrete choice — recognizing this is a major consolidation: the "random parameters" of your EC-MXL are the "random slopes" of this chapter, and the between-respondent preference variance is the multilevel variance component.

Variance components. Multilevel models estimate the *variances* of the random effects (σ_u^2, Σ, τ^2) — the **variance components** — which are substantively interesting: they quantify *how much* units genuinely differ. In your DCE, the random-coefficient variance is *preference heterogeneity* (how much respondents differ in tastes — a headline result, Chapter 22); in meta-analysis, the between-study variance τ^2 is *how much studies genuinely differ* (§19.8). These variance components are estimated by **maximum likelihood** or, better for the variance parameters, **restricted maximum likelihood (REML)**, which corrects the downward bias of ML variance estimates in small samples (your PDF's "REML" — Chapter 47).

Multiple levels. Observations can nest at more than two levels — estimates within studies within research groups; choice tasks within respondents within survey waves. Each level gets its own random effects and variance component. This is directly your meta-analysis: WTP *estimates* nested in *studies*, a two-level structure, and possibly a third level if studies cluster by lab or region (Chapter 47).

> **Stop and think 19.3.** Your PDF lists both "random effects" and "mixed logit" as things to study, and you might think they are separate topics. In what precise sense is the panel mixed logit a multilevel random-effects model? *(Answer: the panel mixed logit gives each respondent their own preference coefficients drawn from a population distribution — random* slopes* (random coefficients) at the respondent level — which is exactly the multilevel/hierarchical structure $\beta_g = \bar{\beta} + \eta_g$. The random-coefficient covariance is the multilevel variance component (preference heterogeneity). So "random effects" and "mixed logit" are the same idea — a multilevel model — applied to the linear and the discrete-choice case respectively; Chapter 22.)*

19.8 Random-effects meta-analysis: the finfish structure

The most consequential application of this chapter for your work is **meta-analysis**, where the "units" are studies and the "repeated observations" are the WTP estimates each study contributes. Your PDF asks the defining question — *"più WTP dallo stesso studio sono indipendenti?"* — and the answer, and its consequences, are the subject here (fully developed in Chapter 47).

The model. Each study j has a *true* effect θ_j that is observed with sampling error, and the true effects themselves vary around a population mean:

$$y_j = \theta_j + e_j, \quad e_j \sim N(0, v_j) \qquad \text{and} \qquad \theta_j = \mu + u_j, \quad u_j \sim N(0, \tau^2).$$

Here y_j is the observed WTP estimate, v_j its within-study sampling variance (known, from the study's standard error), μ the population mean WTP, and τ^2 the **between-study variance** — the genuine heterogeneity in true effects across studies. This is exactly the random-intercept model of §19.5, with studies as units.

Fixed-effect vs random-effect meta-analysis. - **Fixed-effect meta-analysis** assumes $\tau^2 = 0$ — all studies estimate the *same* true effect, differing only by sampling error. It weights by $1/v_j$ (inverse variance) and gives a narrow confidence interval. - **Random-effect meta-analysis** allows $\tau^2 > 0$ — studies estimate *different* true effects (different populations, methods, ecosystems). It weights by $1/(v_j + \tau^2)$ and gives a wider, honest interval that reflects both sampling error *and* between-study heterogeneity.

Worked Example 19.3 (Fixed vs random meta-analysis, verified). Simulate 18 studies (your finfish count) with true mean WTP $\mu = 30$ and between-study SD $\tau = 8$, each observed with its own sampling variance:

Model	Estimated μ	SE	Note
Fixed-effect	28.91	**0.81**	ignores heterogeneity — *overconfident*
Random-effect	29.92	**2.19**	recovers $\hat{\tau} = 8.5$; *honest*

(Verified.) The random-effect standard error is **2.7 times wider** than the fixed-effect one (2.19/0.81), because it accounts for the genuine between-study heterogeneity ($\hat{\tau} = 8.5$, close to the true 8) that the fixed-effect model pretends away. **This is the single most important meta-analysis lesson for your finfish paper:** WTP estimates from different studies almost always reflect genuinely different true effects (different sites, species, methods, populations), so between-study heterogeneity ($\tau^2 > 0$) is the rule, and a fixed-effect meta-analysis *dramatically understates uncertainty* — here threefold. Use random effects (or, better, a **multilevel meta-regression** with study-level random intercepts *and* moderators, plus **robust variance estimation** for the within-study dependence, Chapters 17, 47).

And the deeper issue: estimates within a study are not independent. Beyond between-study heterogeneity, the *multiple estimates from one study* are correlated (shared sample, methods, authors) — a third layer your PDF flags. This requires either a *multilevel* structure (estimates within studies) or **robust variance estimation (RVE)** clustering on studies (Chapter 17's cluster-robust logic applied to meta-analysis). And with only 18 studies, all of this is delicate — asymptotics are in the *number of studies* (Chapter 6 §6.9), so small-sample corrections are essential (Chapter 47 §47.7). The "effective sample size" of your finfish synthesis is closer to 18 than to your 120 estimates.

> **Active recall 19.D.** Why does a fixed-effect meta-analysis of your 18 finfish studies understate uncertainty, and what does the random-effects model add? *(Answer: the fixed-effect model assumes all studies estimate the* same* true WTP ($\tau^2 = 0$) and weights by sampling precision alone, so its confidence interval reflects only within-study sampling error — but studies genuinely differ (different sites, methods, populations), so there is real between-study heterogeneity it ignores, making it overconfident (verified ~3× too narrow). The random-effects model adds the between-study variance τ^2, weighting by $1/(v_j + \tau^2)$ and widening the interval to honestly reflect both sampling error and genuine heterogeneity.)*

19.9 Failure modes

F1. Ignoring the panel structure entirely. Pooled OLS with classical SEs on repeated observations. *Symptom:* overstated precision. *Fix:* cluster the SEs (Chapter 17) and/or model the dependence (this chapter).

F2. Random effects when the unit effect is correlated with regressors. Using RE in an observational panel where the confounder correlates with the regressors → bias. *Fix:* Hausman test; use FE (or a hybrid/Mundlak model) if RE fails (§19.6).

F3. Fixed effects when you need time-invariant effects. Absorbing a respondent-level variable (experience) you wanted to estimate. *Fix:* random effects or a hybrid model; recognize FE cannot see within-unit-constant regressors (§19.3–§19.4).

F4. Fixed-effect meta-analysis with heterogeneous studies. Assuming $\tau^2 = 0$ when studies genuinely differ. *Symptom:* far-too-narrow CI (verified ~3×). *Fix:* random-effects (or multilevel) meta-analysis; estimate and report τ^2 (§19.8; Chapter 47).

F5. Treating multiple estimates per study as independent. Ignoring within-study correlation in meta-analysis. *Fix:* multilevel (estimates in studies) structure or robust variance estimation clustering on studies (§19.8; Chapter 17).

F6. Trusting RE/multilevel variance components with few groups. ML variance estimates are downward-biased in small samples (18 studies). *Fix:* REML; small-sample corrections (§19.7; Chapter 47 §47.7).

F7. Forgetting asymptotics are in the number of groups. Trusting cluster/RE inference with a handful of units. *Fix:* count groups (respondents: many; studies: 18 — dangerous); small-sample corrections (§19.8; Chapter 6 §6.9, 48).

F8. Not recognizing the mixed logit as a random-effects model. Treating "random parameters" as unrelated to panel methods. *Fix:* see the mixed logit as multilevel random coefficients; the variance components are preference heterogeneity (§19.7; Chapter 22).

19.10 Diagnostic questions for your own papers

1. **Structure named.** Have I identified the units and repeated observations (tasks-in-respondents, estimates-in-studies) and their dependence? *(§19.2.)*
2. **Model or cluster.** Am I modeling the dependence (FE/RE/multilevel) and/or clustering the SEs, appropriately? *(§19.2.)*
3. **FE vs RE.** Is the unit effect plausibly uncorrelated with my regressors (RE) or not (FE)? Have I run/considered the Hausman test? *(§19.6.)*
4. **Time-invariant effects.** If I need respondent- or study-level covariate effects, am I using a model (RE/hybrid) that can estimate them? *(§19.3–§19.4.)*
5. **Meta-analysis heterogeneity.** Am I using random-effects (not fixed-effect) meta-analysis, estimating τ^2, and reporting the honest (wider) interval? *(§19.8.)*
6. **Within-study dependence.** Are multiple estimates per study handled by a multilevel structure or RVE, not treated as independent? *(§19.8.)*
7. **Few groups.** With 18 studies, am I using REML and small-sample corrections, recognizing asymptotics are in the number of studies? *(§19.7–§19.8; Chapter 47 §47.7.)*

19.11 Exercises

(A) Comprehension.

A1. Explain the difference between within and between variation, and state which variation fixed effects and random effects each use.

A2. Why can a respondent-fixed-effects model not estimate the effect of a respondent's experience level?

A3. A fixed-effect meta-analysis gives a mean WTP of €30 (SE €1); a random-effect meta-analysis gives €30 (SE €3). Which do you report and why, and what does the difference reveal?

(B) Derivation / reasoning.

B1. Show that the within (fixed-effects) transformation removes a unit-specific intercept α_g and any time-invariant regressor, and explain why this eliminates omitted-variable bias from time-invariant confounders.

B2. Explain the random-effects quasi-demeaning $(y_{gt} - \theta\bar{y}_g)$, and show that $\theta = 0$ gives pooled OLS and $\theta = 1$ gives fixed effects. What determines θ?

B3. Write the random-effects meta-analysis model, derive the random-effects weights $1/(v_j+\tau^2)$, and explain why the fixed-effect weights $1/v_j$ produce too narrow an interval when $\tau^2 > 0$.

(C) Research application / critique.

C1. Your finfish meta-analysis uses a fixed-effect model and reports a tight confidence interval on mean WTP. Critique this and specify the correct model, estimating how much the interval might widen.

C2. A co-author wants to use respondent fixed effects in your DCE to "control for everything about the respondent." Explain what this buys, what it costs (which effects you lose), and why the mixed logit is often preferred.

C3. Explain to a co-author why the panel mixed logit is a multilevel random-effects model, and what its variance components mean substantively.

19.12 Solutions

A1. Within variation is the variation of x_{gt} around its unit mean $\bar{x}_g$ (over the repeated observations within a unit); between variation is the variation of the unit means $\bar{x}_g$ across units. Fixed effects use *only within* variation (they demean within units); random effects use a *weighted combination of within and between*; the between estimator uses only between; pooled OLS uses the total.

A2. Experience is *constant across a respondent's choice tasks* (time-invariant within respondent), so it is absorbed by the respondent fixed effect in the within transformation (demeaning removes anything constant within the unit) — there is no within-respondent variation in experience to identify its effect. Estimating the experience effect requires between-respondent variation, which FE discards; use random effects or a model without respondent fixed effects.

A3. Report the **random-effect** estimate (SE €3). The fixed-effect model assumes all studies estimate the same true WTP and ignores genuine between-study heterogeneity, understating uncertainty; the random-effect model accounts for it (the €3 vs €1 gap reveals substantial between-study heterogeneity, $\tau^2 > 0$). The wider interval is the honest one; the difference reveals that studies genuinely differ, so a single "the WTP" hides real variation across studies.

B1. With $y_{gt} = \mathbf{x}'_{gt}\beta + \alpha_g + \varepsilon_{gt}$, the unit mean is $\bar{y}_g = \bar{\mathbf{x}}'_g\beta + \alpha_g + \bar{\varepsilon}_g$ (since α_g is constant within the unit). Subtracting: $(y_{gt} - \bar{y}_g) = (\mathbf{x}_{gt} - \bar{\mathbf{x}}_g)'\beta + (\varepsilon_{gt} - \bar{\varepsilon}_g)$ — α_g cancels, and any time-invariant regressor (constant within the unit, so equal to its own unit mean) also cancels. Because α_g (which may be an unobserved confounder correlated with $\mathbf{x}$) is removed, its correlation with the regressors can no longer bias $\hat{\beta}$ — the

within estimator is unbiased for β regardless of the α_g–$\mathbf{x}$ correlation (verified: pooled 2.69 biased, FE 2.00 unbiased).

B2. RE-GLS quasi-demeans: $(y_{gt} - \theta\bar{y}_g) = (\mathbf{x}_{gt} - \theta\bar{\mathbf{x}}_g)'\beta + \text{error}$, with $\theta = 1 - \sqrt{\sigma_\varepsilon^2/(\sigma_\varepsilon^2 + T\sigma_u^2)}$. At $\theta = 0$, nothing is demeaned → pooled OLS (ignores units); at $\theta = 1$, fully demeaned → fixed effects. θ is determined by the variance ratio $\sigma_u^2/\sigma_\varepsilon^2$ and the number of repeated observations T: more between-unit variance (σ_u^2 large) or more observations per unit pushes θ toward 1 (closer to FE); RE optimally weights within and between.

B3. Model: $y_j = \mu + u_j + e_j$, $u_j \sim N(0, \tau^2)$ (between-study), $e_j \sim N(0, v_j)$ (within-study). The total variance of y_j is $\text{Var}(y_j) = \tau^2 + v_j$, and efficient (inverse-variance) GLS weights by the inverse total variance $w_j = 1/(v_j + \tau^2)$, giving $\hat{\mu} = \sum w_j y_j / \sum w_j$. The fixed-effect model sets $\tau^2 = 0$, so it weights by $1/v_j$ and computes $\text{Var}(\hat{\mu}) = 1/\sum(1/v_j)$ — which omits the τ^2 term, understating the true variance and producing too narrow an interval whenever $\tau^2 > 0$ (verified: FE SE 0.81 vs RE SE 2.19).

C1. A fixed-effect meta-analysis assumes all 18 finfish studies estimate the *same* true WTP ($\tau^2 = 0$), so its tight interval reflects only within-study sampling error and ignores genuine between-study heterogeneity (different species, sites, methods, populations) — it is overconfident. Correct model: **random-effects (or multilevel) meta-analysis** estimating τ^2 and weighting by $1/(v_j + \tau^2)$, plus **robust variance estimation** clustering on studies for the multiple-estimates-per-study dependence, with REML and small-sample corrections given only 18 studies (Chapters 47–48). The interval could widen substantially — in the worked example the RE SE was ~2.7× the FE SE, roughly *tripling* the interval width. Report the random-effects estimate with $\hat{\tau}^2$ and heterogeneity statistics (I^2, Q).

C2. Respondent fixed effects control for *all time-invariant respondent characteristics* — observed and unobserved — so no respondent-level confounding (a strong robustness property). The cost: (i) you *lose* the ability to estimate any respondent-level (time-invariant) effect — experience, income, familiarity — since they are absorbed; (ii) you use only within-respondent (across-task) variation, discarding between-respondent information. The mixed logit is often preferred because it *models* respondent heterogeneity as random coefficients rather than differencing it out — so it accommodates respondent-level heterogeneity, *can* estimate respondent-level covariate effects (via interactions or the mean function), and (given the randomized design, so the RE assumption holds) is efficient. FE is robust but limited; the random-coefficients mixed logit is richer and, for a well-designed DCE, appropriate (Chapter 22).

C3. The panel mixed logit gives each *respondent* their own preference coefficients $\beta_g = \bar{\beta} + \eta_g$, drawn from a population distribution with mean $\bar{\beta}$ and covariance Σ — this is a multilevel model with *random slopes* (random coefficients) at the respondent level, exactly the hierarchical structure of §19.7. Substantively, the variance components (Σ, the random-coefficient covariance) measure **preference heterogeneity**: how much respondents genuinely differ in their tastes for each attribute (and how those tastes correlate). A large variance means strong heterogeneity (respondents value the attribute very differently); a near-zero variance means homogeneous preferences (and, tested, the boundary problem of Chapter 12). So "random effects" and "mixed logit" are one idea — a multilevel random-coefficients model — and its variance components are the heterogeneity that is often a headline result.

19.13 Spaced review prompts

+1 day. From memory, decompose variation into within and between, and state which each of FE, RE, and pooled OLS uses. Explain what FE removes (time-invariant confounders) and what it cannot estimate.

+1 week. State the FE-vs-RE trade-off and the Hausman logic; write the random-effects meta-analysis model and explain why fixed-effect meta-analysis is overconfident when $\tau^2 > 0$.

+1 month. For your finfish meta-analysis, fit random-effects (and multilevel) meta-regression, estimate τ^2, cluster/RVE on studies, and compare to the fixed-effect interval; for your DCE, articulate the panel mixed

logit as a multilevel model and interpret its variance components. Close with the spine sentence about how you modeled the dependence and what heterogeneity you found.

19.14 How this chapter connects to the rest of the book

- **Chapter 17** supplied the clustered-SE (inference) response to dependence; this chapter adds the modeling (FE/RE/multilevel) response, and the two combine (model *and* cluster). **Chapter 16** supplied OVB, which fixed effects remove for time-invariant confounders; **Chapter 5 (§5.7)** the within/between decomposition; **Chapter 6 (§6.9)** the few-groups asymptotics.
- **Chapter 22 (mixed logit)** *is* the multilevel random-coefficients model of §19.7 applied to discrete choice — the "random parameters" are random slopes, the covariance is preference heterogeneity; **Chapter 25 (scale vs preference heterogeneity)** distinguishes the variance components that are preference from those that are scale.
- **Chapter 47 (meta-regression)** is the full development of §19.8 — random-effects and multilevel meta-regression, REML, WLS, robust variance estimation — for your finfish data; **Chapter 47 §47.7 (dependence and small-sample corrections)** handles the 18-study few-clusters problem; **Chapter 12 (§12.9)** is the boundary test for whether a variance component is zero.

This chapter closes Part IV by giving you the repeated-observations logic — within/between, fixed/random/multilevel — that underlies both your DCE (panel mixed logit) and your meta-analysis (random-effects meta-regression), turning "you have repeated data" from a nuisance into a modeled, heterogeneity-quantifying structure.

19.15 Research translation task

For your **finfish meta-analysis and your DCE**, produce a two-to-three-page note:

1. **Name the structure.** For each, state the units, repeated observations, and dependence (estimates-in-studies; tasks-in-respondents).
2. **Meta-analysis.** Fit fixed-effect and random-effects meta-analysis; report $\hat{\mu}$, its SE under each, and $\hat{\tau}^2$ / I^2; quantify how much the interval widens with random effects; add a multilevel/RVE structure for within-study dependence, with REML and small-sample corrections.
3. **DCE.** Articulate the panel mixed logit as a multilevel random-coefficients model; interpret its variance components as preference heterogeneity; discuss why random (not fixed) effects suit a randomized design.
4. **FE vs RE (where relevant).** For any linear panel regression, consider the Hausman/Mundlak logic and justify your choice.
5. **Close** with the spine sentence: *"My repeated observations are [tasks-in-respondents / estimates-in-studies]; I model the dependence with [RE/multilevel] and [cluster/RVE] the SEs; ignoring it would understate uncertainty by ~[factor]; the between-unit heterogeneity is [value]."*

Carry this into Chapter 22 (mixed logit), Chapter 25 (scale vs preference heterogeneity), Chapter 47 (meta-regression), and Chapter 47 (dependence and small-sample corrections).

19.16 Bibliography

Core. - Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, the panel-data chapters (fixed effects, random effects, the within transformation, the Hausman test, and pooled vs FE vs RE). The rigorous econometric reference for §19.4–§19.6. *(Verify edition/chapters.)* - Angrist & Pischke, *Mostly Harmless*

Econometrics, on fixed effects as a confounding-removal device and the FE-vs-RE trade-off — the causal reading of §19.4. *(Verify edition.)* - Gelman & Hill, *Data Analysis Using Regression and Multilevel/Hierarchical Models*. The definitive accessible treatment of multilevel models, random intercepts/slopes, and variance components (§19.7) — the bridge to the mixed logit. *(Verify edition.)*

Applied. - Borenstein, Hedges, Higgins & Rothstein, *Introduction to Meta-Analysis*. The definitive treatment of fixed- vs random-effects meta-analysis, τ^2, heterogeneity, and inverse-variance weighting (§19.8) — essential for the finfish paper. *(Verify edition.)* - Cameron & Miller, "A Practitioner's Guide to Cluster-Robust Inference," *Journal of Human Resources*, for the clustering-vs-modeling and few-groups issues (§19.2, §19.8). *(Verify volume/pages.)* - Train, *Discrete Choice Methods with Simulation*, on the panel mixed logit as a random-coefficients model (§19.7) — the DCE application. Freely available — *verify link.*

Advanced. - Mundlak (1978), "On the Pooling of Time Series and Cross Section Data," *Econometrica*. The correlated-random-effects (Mundlak) device that nests FE and RE and makes the RE assumption testable (§19.6). *(Verify volume/pages.)* - Hedges, Tipton & Johnson (2010) on robust variance estimation in meta-analysis with dependent effect sizes — the RVE approach for multiple estimates per study (§19.8; Chapter 47). *(Verify volume/pages.)* - Raudenbush & Bryk, *Hierarchical Linear Models*, for the rigorous multilevel-model treatment and REML (§19.7). *(Verify edition.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Mundlak / Hedges et al. citations before using them, per the book's no-unverified-citation rule.

End of Chapter 19 and of Part IV. You now hold core econometrics in full: OLS from first principles (Ch 16), inference that survives heteroskedasticity and clustering (Ch 17), binary response models (Ch 18), and the repeated-observations logic of fixed, random, and multilevel effects (Ch 19) that underlies your panel mixed logit and random-effects meta-analysis. Part V — Random Utility, Discrete Choice, and Stated Preference Econometrics — begins with Chapter 20 (Lancaster and Random Utility Theory), the discrete-choice heart of your DCE and your PDF's Block 2. Tell me which chapter to write next.

Part IV Assessment — Core Econometrics

Covers Chapters 16–19. Five components, in the order they are best attempted. Allow a full working day; the mock oral is the part that matters most for your viva.

1. Cumulative problem set

P1. Derive $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ from the normal equations and state precisely which assumption delivers unbiasedness.

P2. Derive the sandwich variance estimator from first principles and state what it does and does not fix.

P3. Write the cluster-robust variance for G clusters and explain what happens as G becomes small.

P4. In a logit with an interaction between treatment and a continuous covariate, write the correct marginal effect of treatment and explain why the interaction coefficient alone does not answer the question.

P5. Show that with 120 estimates in 18 clusters and $\rho = 0.6$ the effective sample size is roughly 27, and state the consequence for your standard errors.

2. Research critique

Take a regression table from one of your papers. For every standard error in it, state what dependence structure it assumes, whether that assumption is defensible given how the data were generated, and what the number would look like under the alternative. Identify the one coefficient whose inference would change most.

3. Red-flag diagnosis

An author regresses stated WTP on a treatment dummy and, to "improve precision," adds respondent-level variables measured *after* the treatment was administered — including a comprehension score and a self-reported consequentiality belief. The treatment effect shrinks and loses significance; the author reports the adjusted model as the preferred specification. Explain what has gone wrong, what the adjusted coefficient now estimates, and what should be reported.

4. Mock oral examination

1. State the exogeneity assumption in words a non-economist would understand.
2. What does clustering assume, and when does it fail?
3. Why are logit coefficients not comparable across models?
4. Distinguish within from between variation and say which your design uses.

5. Why does multicollinearity inflate standard errors without biasing anything?

5. Integration mini-project

Build the full inference layer for your finfish meta-regression: naive standard errors, study-clustered standard errors, robust variance estimation with small-sample correction, and a multilevel model with a study random intercept. Present all four side by side, explain the ordering of the standard errors, and state which you will report and why.

Part V — Random Utility, Discrete Choice, and Stated Preference Econometrics

Chapter 20 — Lancaster and Random Utility Theory

Part V: Random Utility, Discrete Choice, and Stated Preference Econometrics

> **The spine question, applied to this chapter.** Your PDF is explicit: *"Il tuo DCE è costruito esattamente su Random Utility Theory e Lancaster."* This chapter builds those foundations — the two theories on which every discrete choice model, and therefore your entire DCE, stands. **Lancaster** says utility comes from the *characteristics* (attributes) of goods, which is why your experiment describes alternatives by attributes and estimates preferences for them. **Random Utility Theory** splits utility into a part you model (V, deterministic) and a part you do not (ε, stochastic), and says the respondent chooses the highest-utility alternative — so you can only predict *probabilities*. Getting the split right, and understanding what "random" actually means, determines what your model *identifies* and *means*. The spine question here: *when I estimate preferences from choices, what have I assumed is systematic (and therefore welfare-relevant) versus random (and therefore integrated away) — and do the two foundational normalizations of random utility tell me my coefficients are not what they naively seem?*

20.0 Conceptual motivation: the two theories your DCE is built on

Everything in your discrete choice experiment — the attributes you show, the WTP you estimate, the welfare you compute, the IIA you worry about, the scale-versus-preference confusion your PDF flags — descends from two theoretical commitments made before a single model is estimated. Your PDF lists them first among the "priorità altissima" of random utility models: *"Random Utility Theory; Lancaster consumer theory; deterministic utility; stochastic utility."* This chapter is those foundations, and getting them right is not optional throat-clearing — it decides what your estimates *mean*.

- **Lancaster** is why your DCE looks the way it does. Classical consumer theory has utility over *goods*; Lancaster's insight is that utility comes from the *characteristics* goods embody — a fish-restoration project is valued not "as itself" but for its water quality, biodiversity, carbon, and cost. So you describe each alternative by its attributes, and you estimate the *marginal utility of each attribute*. This is what lets you value a *new* configuration (a restoration design never seen) by summing its attributes' marginal utilities — the basis of the whole exercise, and of benefit transfer (Chapters 35, 49).
- **Random Utility Theory** is why you get a *probabilistic* model from a *deterministic* chooser. The respondent knows their own utility and picks the best alternative; but *you* observe only some of what drives them (V), not all (ε), so from your vantage the choice is random and you predict probabilities. The split $U = V + \varepsilon$ is the single most consequential modeling decision in the DCE: **what you put in V is systematic, welfare-relevant, and estimated; what falls into ε is unobserved and integrated away.**

And two facts that flow directly from RUT will govern everything downstream, so this chapter establishes them at the source: **only utility *differences* matter** (so the level of utility is not identified — you

normalize an alternative), and **only the *ratio* of coefficients to the error scale is identified** (so your coefficients are relative to a normalized noise level — the scale problem that becomes the scale-versus-preference-heterogeneity theme of Chapter 25). Your PDF's critical questions — *"IIA è plausibile? le alternative sono davvero non labelled? l'opt-out è modellato bene? stai confondendo taste heterogeneity e scale heterogeneity?"* — are all questions about *how you specified the RUM*, and they only make sense once the framework is clear.

We build: Lancaster's characteristics model (and why it founds the DCE); the Random Utility framework ($V+\varepsilon$, choose the max); *what "random" means* (the analyst's ignorance, not the respondent's irrationality — a subtle point that changes interpretation); the additive RUM and the choice probability as an integral over the error; the two foundational normalizations; the bridge from deterministic utility to WTP and welfare; and labeled versus unlabeled alternatives and the opt-out. Chapter 21 turns this framework into the multinomial logit; this chapter is the theory that makes the MNL mean something.

20.1 Prerequisites

From **Chapter 2** (§2.10–2.11): utility, marginal utility, the marginal rate of substitution, and WTP as a ratio of marginal utilities. From **Chapter 3**: utility maximization and indirect utility (the deterministic utility of a chosen alternative *is* an indirect utility). From **Chapter 5** (§5.5–5.6): the Gumbel distribution and the derivation that iid Gumbel errors give the logit. From **Chapter 18**: the binary response model as the two-alternative RUM, and scale identification. From **Chapter 14** (§14.3): latent variables and threshold-crossing.

Same **notation**. New/consolidated symbols: $\mathbf{z}_j$ (the *attributes/characteristics* of alternative j), U_j (total utility of alternative j), V_j (deterministic/systematic utility), ε_j (random utility), β (marginal utilities / taste parameters), J (number of alternatives), ASC (alternative-specific constant), μ (scale parameter $= 1/\sigma_\varepsilon$).

20.2 Lancaster consumer theory: utility from characteristics

The classical picture and its limit. Classical consumer theory (Chapter 3) puts utility over *goods*: $U(x_1, \dots, x_n)$ where x_j is the quantity of good j. This is fine for apples and oranges, but it cannot value a *new* good (one not yet consumed), and it treats each good as an indivisible object of preference. For a DCE — where you present *designed* alternatives (restoration scenarios, policy packages) that may never have existed — this is useless: you cannot have a "taste for restoration scenario #47."

Lancaster's insight. Kelvin Lancaster (1966) proposed that consumers derive utility not from goods *per se* but from the **characteristics** (attributes) the goods embody. A good is a *bundle of characteristics*; utility is defined over the characteristics; and a good's value is the value of its characteristics.

Definition 20.1 (Characteristics model). Each alternative j is described by a vector of characteristics $\mathbf{z}_j = (z_{j1}, \dots, z_{jK})$ (water quality, biodiversity, carbon, cost, …). Utility is a function of the characteristics,

$$U_j = U(\mathbf{z}_j),$$

and, in the usual linear-in-attributes specification, $V_j = \mathbf{z}_j'\beta = \sum_k \beta_k z_{jk}$, where β_k is the **marginal utility of characteristic** k.

Why this founds the DCE. Three consequences make Lancaster the theoretical charter of your experiment:

1. **You estimate preferences for *attributes*, not goods.** The DCE presents alternatives described by attributes, respondents choose, and you recover the marginal utilities β. Your DCE *is* a characteristics-space experiment.

2. **You can value *new* configurations.** Because value is additive in characteristics, you can value a restoration design *never presented* by summing its attributes' marginal utilities — the basis of policy simulation and benefit transfer (Chapters 35, 49). This is impossible in the classical goods-space model.
3. **WTP is a ratio of marginal utilities of characteristics.** Including *cost* as a characteristic, the marginal WTP for attribute k is $\beta_k/\beta_{\text{cost}}$ (Chapter 2, §2.11) — the money value of a unit of the characteristic. *(Verified: with $V = 0.6 \cdot \textit{quality} - 0.03 \cdot \textit{cost}$, the marginal WTP for quality is $0.6/0.03 = €20$ per unit.)* Lancaster is why attributes are the currency of welfare in your work.

The assumptions to keep visible. Lancaster's tractable form assumes utility is a function of the *presented* attributes and (usually) *linear* and *additively separable* in them. Both are assumptions: respondents may care about attributes you did *not* present (which then fall into the random term ε, §20.4), and utility may be *nonlinear* (diminishing marginal utility) or *non-separable* (attributes interacting). Your specification of V *is* a Lancaster hypothesis about which characteristics matter and how — and your PDF's attribute-non-attendance concern (some respondents ignore some attributes, Chapter 37) is a violation of the assumption that all presented characteristics enter utility.

> **Active recall 20.A.** Why does Lancaster's characteristics model, rather than classical goods-based utility, make a discrete choice experiment possible, and what does it let you value that classical theory cannot? *(Answer: Lancaster puts utility over the* attributes* goods embody, so you can describe designed alternatives by attributes, estimate marginal utilities of attributes, and — because value is additive in characteristics — value *new* configurations never presented (policy simulation, benefit transfer). Classical goods-based utility cannot value a good never consumed and has no notion of "marginal utility of an attribute.")*

20.3 Random Utility Theory: the framework

Lancaster gives the *content* of utility (characteristics); Random Utility Theory gives the *structure* that turns a deterministic chooser into a probabilistic model.

The setup. A respondent faces a choice set of J alternatives and has a utility U_j for each. **They choose the alternative with the highest utility** — a deterministic, rational act *from their perspective*. But the analyst does not observe U_j fully; the analyst observes some determinants (the attributes $\mathbf{z}_j$) and not others. So decompose utility into what the analyst can and cannot model:

Definition 20.2 (Additive random utility model).

$$U_j = V_j + \varepsilon_j, \qquad j = 1, \dots, J,$$

where V_j is the **deterministic (systematic, representative) utility** — the part the analyst models, typically $V_j = \mathbf{z}_j'\beta$ from Lancaster — and ε_j is the **random (stochastic) utility** — everything affecting the choice that the analyst does *not* observe. The respondent chooses j iff $U_j \geq U_k$ for all k.

The choice probability. Because the analyst does not observe ε, they cannot predict the choice with certainty; they predict the *probability*:

$$P_j = \Pr(U_j \geq U_k \ \forall k) = \Pr(V_j + \varepsilon_j \geq V_k + \varepsilon_k \ \forall k) = \Pr(\varepsilon_k - \varepsilon_j \leq V_j - V_k \ \forall k).$$

This is the fundamental equation of discrete choice: **the probability that a respondent chooses j is the probability that the unobserved advantages of the other alternatives do not overcome j's observed advantage.** It is an integral over the joint distribution of the errors, and *the assumed error distribution is the model*: iid Gumbel gives the multinomial logit (Chapter 21), correlated normal gives the multinomial probit, and richer structures give nested and mixed logit (Chapters 22–23).

The analyst's model, not the respondent's psychology. Note carefully: RUT does *not* claim the respondent flips a coin. The respondent maximizes their (full) utility deterministically; the *probability* is entirely a statement about the analyst's incomplete information. This distinction is the subject of §20.4 and it changes how you interpret everything.

Active recall 20.B. Write the RUM choice probability P_j and state, in one sentence, why the model is probabilistic even though the respondent's choice is deterministic. *(Answer:* $P_j = \Pr(V_j + \varepsilon_j \geq V_k + \varepsilon_k \ \forall k)$*; the respondent deterministically picks their highest-utility alternative, but the* analyst* observes only the systematic part V_j and not the random part ε_j, so from the analyst's perspective the choice is uncertain and only its probability can be predicted.)*

20.4 What "random" means: the analyst's ignorance, not the respondent's irrationality

This is the most important interpretive point in the chapter, and one your PDF's critical questions turn on. The randomness in RUT is a property of the *analyst's information*, not of the respondent's decision. Misunderstanding this leads to misinterpreting the error, the scale, and — crucially for your cost-vector work — what the model can and cannot separate.

The sources of ε. The random term ε_j is a catch-all for everything driving the choice that the analyst does not model. Its plausible components:

1. **Unobserved attributes of the alternatives** — features of the alternatives that matter to respondents but that the analyst did not present or measure (a scenario's framing, an unstated quality). These are genuinely part of utility; the analyst simply did not observe them.
2. **Unobserved taste variation across respondents** — different people weight the same attribute differently, and if the analyst uses a single β, the individual deviations fall into ε. *Modeling* this variation is what turns the fixed-coefficient logit into the **mixed logit** (random coefficients, Chapter 22) — i.e., pulling taste heterogeneity *out* of ε and *into* V's distribution.
3. **Measurement and reporting error** — the respondent's stated choice imperfectly reflects their utility (a misclick, a momentary lapse).
4. **Optimization / bounded-rationality error** — the respondent does not perfectly optimize (limited attention, heuristics). Attribute non-attendance (Chapter 37) lives here.

Why the interpretation matters. The composition of ε is not a philosophical nicety — it determines the *model* and the *meaning*:

- If ε is mainly *unobserved attributes*, the model is a standard RUM and the choice is "as if" random to the analyst; the respondent is fully rational.
- If ε contains *taste heterogeneity*, a fixed-coefficient logit is misspecified (it forces one β on heterogeneous people), and you should move that heterogeneity into random coefficients (mixed logit). Confusing this heterogeneity with pure noise is the error your PDF flags.
- The *scale* of ε (how much noise) is itself meaningful: a respondent (or group) with a *smaller* ε variance chooses more *consistently* — and, as §20.6 shows, this scale is entangled with the coefficients. A group that appears to have "stronger preferences" may simply have *less noise* (smaller ε), not stronger tastes. This is the **scale-versus-preference-heterogeneity** problem (Chapter 25), and it is *born here*, in the interpretation of the random term.

The behavioral connection to your cost-vector theme. Your CV and DCE papers hinge on whether the cost vector enters *preference formation* or *reporting*. In RUM terms: does the cost vector shift V (a genuine preference/utility effect) or does it operate through ε (a reporting/anchoring effect, or a change in the noise scale)? RUT gives you the language for this exact question — and the sobering answer (Chapters 25, 29–30) that a single cross-section often *cannot* separate a shift in V from a change in the scale of ε without additional structure. The interpretation of "random" is therefore not academic; it is where your paper's central identification question lives.

Stop and think 20.1. Two groups (experienced, inexperienced) give choices, and the experienced group's estimated coefficients are all uniformly larger. A colleague says "experienced respondents have stronger preferences." Using §20.4, give an alternative explanation and say

what it implies about ε. *(Answer: the experienced group may simply have a* smaller* error variance — they choose more consistently (less unobserved noise in ε), which mechanically makes their coefficients larger relative to the (normalized) error scale, *without* any difference in underlying tastes. This is a *scale* difference in ε, not a *preference* difference in V; uniformly-larger coefficients are the fingerprint of a scale difference (all coefficients scale together), and distinguishing it from genuinely stronger preferences requires modeling the scale explicitly — the scale-vs-preference-heterogeneity problem, Chapters 18, 25.)*

20.5 The additive RUM and the choice probability as an integral

With Lancaster's $V_j = \mathbf{z}_j'\beta$ and the additive structure, the choice probability is

$$P_j = \Pr\left(\varepsilon_k - \varepsilon_j \le \mathbf{z}_j'\beta - \mathbf{z}_k'\beta \;\; \forall k \neq j\right) = \int \mathbf{1}\{j \text{ is best}\}\, f(\varepsilon)\, d\varepsilon,$$

an integral of the "is j the argmax" indicator over the joint error density $f(\varepsilon)$. This single expression contains every discrete choice model; the models differ only in the assumed f:

Error assumption	Model	Chapter
iid Gumbel (Type-I EV)	Multinomial logit (closed form)	21
Gumbel, nested correlation	Nested logit	22
Normal random coefficients + Gumbel	Mixed logit (simulated)	22–23
Normal, correlated	Multinomial probit	—

Why iid Gumbel is the workhorse. As Chapter 5 (§5.6) derived, when the ε_j are iid Gumbel, the integral has a *closed form* — the softmax $P_j = e^{V_j}/\sum_k e^{V_k}$ — because the Gumbel is closed under maximization (max-stability). This is the multinomial logit, and its tractability is why it is the default. The *cost* of iid Gumbel is the **IIA** property and the assumption of *independent, identically-scaled* errors — assumptions your EC-MXL relaxes (Chapter 22). But every relaxation starts from this integral; the framework is the constant, the error assumption the variable.

A verification of the framework. The RUM "choose the argmax" simulation reproduces the softmax exactly: with $V = (1, 2, 3)$ and iid Gumbel errors, simulating $\arg\max_j(V_j + \varepsilon_j)$ over 2 million draws gives choice frequencies $(0.090, 0.245, 0.665)$, matching the softmax $e^{V_j}/\sum e^{V_k} = (0.090, 0.245, 0.665)$ (verified). The abstract "integral over the argmax" is a concrete, checkable object.

Active recall 20.C. The choice probability is an integral over the error distribution. What does assuming iid Gumbel errors buy you, and what does it cost? *(Answer: it buys a* closed form* — the softmax $P_j = e^{V_j}/\sum_k e^{V_k}$ (the multinomial logit) — because the Gumbel is max-stable; it costs the IIA property and the assumption of independent, equally-scaled errors across alternatives, which richer models (nested, mixed logit) relax by changing the error distribution — Chapters 21–22.)*

20.6 The two foundational normalizations of random utility

Two properties of the RUM restrict what the data can identify, and both recur throughout Part V as the source of famous subtleties (non-comparable coefficients, the scale problem). Establish them here, at the source.

Normalization 1 — only differences in utility matter. The choice depends only on *differences* $U_j - U_k$, so adding any constant to *all* utilities leaves every choice unchanged. Therefore the *level* of utility is not identified; only differences are. Consequences:

- One alternative's utility must be **normalized** (set to zero), and coefficients are interpreted *relative* to it.
- The **alternative-specific constants (ASCs)** capture the *mean* of the utility differences (including the mean of the error differences) — they are identified only in differences, so one ASC is normalized to zero.

Verified: with $V = (1, 2, 3)$ the choice probabilities are $(0.090, 0.245, 0.665)$; with $V = (11, 12, 13)$ — every utility raised by 10 — they are *identical* $(0.090, 0.244, 0.666)$. Only the differences $(1, 1)$ between consecutive utilities matter; the level 10 is invisible to choice. **This is why you cannot interpret the absolute level of utility, only trade-offs.**

Normalization 2 — only the scale (coefficient-to-error-ratio) is identified. Multiplying *all* utilities *and* the errors by any positive constant λ leaves the argmax — and hence every choice — unchanged. So the data identify β and the error scale σ_ε only through their *ratio*; you must normalize the error scale (fix σ_ε, equivalently set the **scale parameter** $\mu = 1/\sigma_\varepsilon$). Consequences:

- Estimated coefficients are $\mu\beta$ — preferences *times scale* — not β alone.
- Coefficients are **not comparable across models or groups with different error scales** (Chapter 18, §18.4): a group with less noise (larger μ) has uniformly larger coefficients regardless of tastes.

Verified: choices from $V = (1, 2, 3)$ with error scale $\sigma = 1$ give $(0.090, 0.244, 0.665)$; from $V = (2, 4, 6)$ with $\sigma = 2$ (both coefficients *and* error scale doubled) give the *identical* $(0.090, 0.244, 0.666)$ — confirming only β/σ is identified. But $V = (2, 4, 6)$ with $\sigma = 1$ (coefficients doubled *without* scaling the error, i.e. a larger μ) gives $(0.016, 0.117, 0.867)$ — *much sharper* choices. **The same "doubled coefficients" mean completely different things depending on the error scale**, which is exactly why scale and preference are entangled.

Why these two normalizations are the root of Part V's hardest problems. Normalization 1 is why WTP is a *ratio* (a difference-based trade-off) and why ASCs are reference-dependent. Normalization 2 is the root of the scale-versus-preference-heterogeneity problem (Chapter 25): because coefficients are only identified up to scale, a treatment or group difference in coefficients can be a difference in *scale* (noise) rather than *preference* — the central identification worry of your cost-vector and convergent-validity work. Both normalizations are *unavoidable* — they are not modeling choices but facts about what choices can reveal — and every interpretation you make must respect them.

> **Stop and think 20.2.** Your high-cost-vector arm shows uniformly larger coefficients than your low-cost-vector arm. Using the two normalizations, list the two distinct explanations and say which normalization each invokes. *(Answer: (i) a genuine* preference* difference — the arms have different tastes (β differs) — but note that only β/σ is identified (Normalization 2), so you cannot read the raw coefficients as preferences; (ii) a *scale* difference — the arms have different error variances (σ_ε differs, so μ differs), making coefficients uniformly larger/smaller with no taste difference — again Normalization 2. Uniformly-scaled coefficients are the signature of a scale difference. Distinguishing the two requires modeling scale (Chapter 25); the raw coefficient comparison cannot.)*

20.7 From deterministic utility to welfare: the bridge

The V/ε split is not just about estimation — it maps precisely onto *where welfare comes from*, which is why interpreting it correctly matters for your WTP and CBA.

V **generates welfare.** The deterministic utility $V_j = \mathbf{z}_j'\beta$ is where trade-offs, WTP, and welfare live. Including cost, the marginal WTP for attribute k is the marginal rate of substitution between the attribute

and money (Chapters 2 §2.11, 3):

$$\text{WTP}_k = \frac{\partial V/\partial z_k}{\partial V/\partial(-\text{cost})} = \frac{\beta_k}{\beta_{\text{cost}}}.$$

So **your welfare estimates are derived entirely from the systematic utility** V — the part you chose to model. What you put in V is what you can value; what falls into ε is not directly valued. This is why the Lancaster specification of V is a welfare decision, and why an attribute wrongly omitted (relegated to ε) cannot be valued.

ε is integrated out for welfare — but leaves a trace (the logsum). For *expected* welfare across the choice set, the random term is integrated out, and — for the Gumbel — this gives the **expected maximum utility**, the *logsum* (Chapter 5, §5.5):

$$\mathbb{E}[\max_j U_j] = \log \sum_j e^{V_j} + \gamma,$$

the object used to measure the welfare change from a policy that alters the choice set (Chapter 42, Small–Rosen). So ε does not vanish from welfare — its *distribution* shapes the logsum — but the *systematic* welfare content is in V. The split V/ε therefore maps onto welfare as: V **carries the valued trade-offs; the distribution of ε shapes the expected-welfare aggregation.** Getting this right is why the interpretation of the error (§20.4) feeds directly into the validity of your welfare numbers.

> **Active recall 20.D.** Why can you only compute WTP for attributes you included in the systematic utility V, and where does the error distribution enter welfare? *(Answer: WTP for attribute k is the ratio of marginal utilities β_k/β_{cost}, which requires β_k — the marginal utility of an attribute* in *V; an attribute relegated to ε (unobserved) has no estimated marginal utility and cannot be valued. The error distribution enters welfare through the expected-maximum-utility (logsum) aggregation over the choice set — Chapters 5 §5.5, 42.)*

20.8 Labeled versus unlabeled alternatives, and the opt-out

RUT applies to any choice set, but *how* you specify V (especially the ASCs) depends on whether alternatives are labeled, and the opt-out needs care — both flagged by your PDF ("le alternative sono davvero non labelled? l'opt-out è modellato bene?").

Unlabeled (generic) alternatives. In an *unlabeled* DCE — "Option A / Option B / Option C" with no inherent identity — the labels carry no meaning; only the attributes matter. So the alternative-specific constants should be *near zero* (any non-zero ASC would reflect a pure left-right or position bias, not preference for an alternative's identity). Your DCE uses unlabeled alternatives, which is the cleaner case for attribute valuation because *all* systematic utility should come from the attributes. A non-negligible ASC in an unlabeled design is a *diagnostic* — it may signal position/order effects, an anchoring to a reference, or a left-digit bias (Chapter 51).

Labeled alternatives. In a *labeled* DCE — "car / bus / train," "Project A / Project B" with identities — the label itself carries utility (an intrinsic preference for the mode/brand), so ASCs are substantive and must be included, and IIA is *more suspect* (labeled alternatives are more likely to share unobserved attributes — the red-bus/blue-bus problem, Chapter 21).

The opt-out / status quo / "none." Almost every environmental DCE (and yours) includes an **opt-out** — "I would choose none / keep the status quo / not pay." Modeling it well is essential and delicate:

- **It is the reference for welfare.** The opt-out's utility is the baseline against which the priced alternatives' welfare is measured; misspecifying it biases WTP.
- **Its ASC is substantive.** The opt-out typically gets its own ASC (people may have a systematic tendency to opt out — status-quo bias, or reluctance to pay), and forcing it to zero, or omitting the opt-out entirely, distorts both choice probabilities and welfare.

- **It bears on incentive compatibility.** Whether respondents can genuinely decline (a real "no") matters for whether the exercise is consequential and incentive-compatible (Chapter 28) — a forced choice without an opt-out can inflate WTP.
- **It interacts with IIA.** The opt-out often shares unobserved characteristics with "not choosing," so its correlation with the other alternatives can violate IIA — a reason error-components mixed logit (which lets the opt-out have its own error structure) is common (Chapter 22).

Your PDF's question "l'opt-out è modellato bene?" is therefore a welfare *and* an identification question: a poorly modeled opt-out biases WTP and can violate the IIA the MNL assumes. This is a place where the theoretical framework of this chapter has direct, consequential design implications (Chapter 37).

> **Stop and think 20.3.** Your unlabeled DCE estimates a large, significant alternative-specific constant for "Option A" (always shown on the left). Is this a preference, and what should you suspect? *(Answer: it is* not* a preference for Option A's identity (the alternatives are generic/unlabeled, so there is no meaningful "Option A"); a large left-position ASC signals a *position/order effect* (left-right bias, or respondents defaulting to the first option) — a survey artifact, not a taste. Suspect an order/anchoring effect, check whether the position was randomized, and treat the ASC as a diagnostic of design/response quality, not a substantive result — Chapter 50.)*

20.9 Failure modes

F1. Misspecifying the systematic utility (Lancaster). Omitting an attribute respondents care about (relegating it to ε), so it cannot be valued and biases other coefficients. *Fix:* include the relevant characteristics; recognize V's specification is a welfare decision (§20.2, §20.7).

F2. Interpreting the error as respondent irrationality. Reading ε as "the respondent is random/irrational." *Fix:* ε is the *analyst's* ignorance (unobserved attributes/tastes/error); the respondent maximizes their full utility (§20.4).

F3. Confusing taste heterogeneity with noise. Leaving genuine taste variation in ε (fixed-coefficient logit) when it belongs in random coefficients. *Fix:* model heterogeneity (mixed logit); do not treat systematic taste variation as pure noise (§20.4; Chapter 22).

F4. Reading raw coefficients as preferences (scale). Comparing coefficients across groups/models as if they measured tastes, ignoring that only β/σ is identified. *Fix:* compare WTP/marginal effects; model scale (§20.6; Chapters 18, 25).

F5. Interpreting utility levels or a single ASC absolutely. Reading the level of utility, or an ASC, as meaningful in itself. *Fix:* only differences are identified; normalize an alternative; interpret ASCs relative to the reference (§20.6).

F6. Mishandling labeled vs unlabeled alternatives. Ignoring ASCs in a labeled design, or over-interpreting an ASC in an unlabeled one. *Fix:* labeled → ASCs substantive (and IIA suspect); unlabeled → ASCs near zero (a non-zero one is a diagnostic) (§20.8).

F7. Poorly modeled or omitted opt-out. No opt-out (forced choice) or a mis-specified opt-out ASC, biasing WTP and incentive compatibility. *Fix:* include and carefully model the opt-out; give it its own ASC/error structure (§20.8; Chapters 22, 28).

F8. Assuming iid errors without question. Taking IIA/iid Gumbel for granted when alternatives share unobserved features. *Fix:* question the error assumption; use nested/mixed logit where IIA is implausible (§20.5; Chapters 21–22).

20.10 Diagnostic questions for your own papers

1. **Lancaster specification.** Does my systematic utility V include all the characteristics respondents plausibly value, and is the linear-additive form defensible? *(§20.2, F1.)*
2. **What is in the error?** What does my ε contain (unobserved attributes, taste heterogeneity, noise), and have I moved genuine taste variation into random coefficients? *(§20.4, F3.)*
3. **Scale awareness.** Do I recognize my coefficients are β/σ (preference $\times$ scale), so cross-group/model comparisons may reflect scale, not preference? *(§20.6, F4.)*
4. **Differences only.** Am I interpreting only utility *differences* and WTP *ratios*, not utility levels or absolute ASCs? *(§20.6, F5.)*
5. **Labels.** Are my alternatives labeled or unlabeled, and are my ASCs handled accordingly (and is any unlabeled ASC a diagnostic of position/order effects)? *(§20.8, F6.)*
6. **Opt-out.** Is my opt-out included, given its own ASC/error structure, and consequential for welfare and incentive compatibility? *(§20.8, F7.)*
7. **Error assumption.** Is IIA/iid Gumbel plausible for my choice set, or do alternatives share unobserved features requiring a richer model? *(§20.5, F8; Chapters 21–22.)*

20.11 Exercises

(A) Comprehension.

A1. State Lancaster's characteristics model and explain why it, rather than classical goods-based utility, makes a DCE possible.

A2. Write the additive RUM and the choice probability, and explain what is deterministic and what is stochastic, and from whose perspective.

A3. A colleague says "the random utility model assumes respondents choose randomly." Correct this precisely.

(B) Derivation / reasoning.

B1. Show that adding a constant to all utilities leaves the choice probabilities unchanged (only differences matter), and explain the consequence for identifying utility levels and ASCs.

B2. Show that scaling all utilities and errors by $\lambda > 0$ leaves choices unchanged, and explain why this means only β/σ is identified and coefficients are not comparable across groups with different error scales.

B3. Starting from the additive RUM with $V_j = \mathbf{z}_j'\beta$ (including cost), derive the marginal WTP for an attribute as a ratio of marginal utilities, and explain why an attribute in ε cannot be valued.

(C) Research application / critique.

C1. Your DCE finds uniformly larger coefficients in the high-cost-vector arm and you are tempted to conclude "stronger preferences." Write the correct interpretation using the two normalizations, and state what you must do to separate the explanations.

C2. A referee asks why your unlabeled DCE includes an opt-out and how you model it. Explain the roles the opt-out plays (welfare reference, ASC, incentive compatibility, IIA) and the consequences of omitting it.

C3. Explain to a co-author what your ε contains, and why moving respondent taste heterogeneity from ε into random coefficients (mixed logit) changes the model's interpretation.

20.12 Solutions

A1. Lancaster's model: goods are bundles of *characteristics* and utility is defined over the characteristics, $U_j = U(\mathbf{z}_j)$ (linearly, $V_j = \mathbf{z}_j'\beta$, with β_k the marginal utility of characteristic k). It makes a DCE possible because you can describe *designed* alternatives by their attributes, estimate marginal utilities of attributes, and — since value is additive in characteristics — value *new* configurations never presented (impossible under classical goods-based utility, which has no notion of an attribute's marginal utility and cannot value an unconsumed good).

A2. Additive RUM: $U_j = V_j + \varepsilon_j$; choice probability $P_j = \Pr(V_j + \varepsilon_j \geq V_k + \varepsilon_k \ \forall k)$. The deterministic part V_j (systematic utility, the modeled attributes) and the stochastic part ε_j (unobserved determinants) split utility *from the analyst's perspective*: the respondent maximizes their full utility U_j deterministically, but the analyst observes only V_j, so the choice is random *to the analyst* and only its probability is predicted.

A3. The RUM does *not* assume respondents choose randomly — respondents deterministically choose their highest-utility alternative. The model is probabilistic because the *analyst* does not observe all of utility (only the systematic part V, not the random part ε), so from the analyst's (incomplete-information) perspective the choice is uncertain. The randomness is the analyst's ignorance, not the respondent's irrationality.

B1. P_j depends on utility through the events $\{U_j \geq U_k\} = \{U_j - U_k \geq 0\}$; adding a constant c to every U leaves every difference $U_j - U_k$ (hence every event, hence every probability) unchanged (verified: $V = (1, 2, 3)$ and $V = (11, 12, 13)$ give identical probabilities). Consequence: the *level* of utility is not identified — only differences — so one alternative's utility (and one ASC) must be normalized, and coefficients/ASCs are interpreted relative to that reference.

B2. For $\lambda > 0$, $\{U_j \geq U_k \ \forall k\} = \{\lambda U_j \geq \lambda U_k \ \forall k\}$, so scaling all utilities (and errors) by λ leaves the argmax and every choice unchanged (verified: $V = (1, 2, 3), \sigma = 1$ and $V = (2, 4, 6), \sigma = 2$ give identical probabilities). Hence β and σ_ε enter only through β/σ_ε — only the ratio is identified — so the error scale must be normalized, estimated coefficients are $\mu\beta$ (preference $\times$ scale), and two groups with different σ_ε have differently-scaled coefficients even with identical tastes (not comparable).

B3. With $V_j = \sum_k \beta_k z_{jk} - \beta_{\text{cost}} \, \text{cost}_j$, the marginal utility of attribute k is $\partial V/\partial z_k = \beta_k$ and of money is $\partial V/\partial(-\text{cost}) = \beta_{\text{cost}}$; the marginal WTP is the MRS $= \beta_k/\beta_{\text{cost}}$ (the money that offsets a unit of the attribute in utility). An attribute relegated to ε has no estimated marginal utility (β_k) in V, so there is no ratio to form — it cannot be valued; only characteristics included in the systematic utility can be assigned WTP.

C1. Correct interpretation: uniformly larger coefficients are the signature of a *scale* difference, and by Normalization 2 only β/σ is identified — so the larger coefficients could mean the high-cost-vector arm has *less error variance* (chooses more consistently, larger μ), *not* stronger preferences. It could alternatively be a genuine preference difference, but the raw coefficients cannot distinguish these (both scale and preference show up in the coefficients). To separate them, estimate a model that lets the *scale* differ across arms (a heteroskedastic/scaled logit, or generalized MNL), so a preference difference in β is identified apart from a scale difference in μ — and compare WTP (scale-free ratios), not raw coefficients (Chapters 18, 25).

C2. The opt-out plays four roles: (i) **welfare reference** — its utility is the baseline against which the priced alternatives' welfare/WTP is measured, so misspecifying it biases WTP; (ii) **substantive ASC** — respondents may systematically tend to opt out (status-quo bias, payment reluctance), so the opt-out needs its own alternative-specific constant, not a forced zero; (iii) **incentive compatibility** — a genuine ability to decline makes the exercise consequential and truth-telling more credible, whereas a forced choice can inflate WTP (Chapter 28); (iv) **IIA** — the opt-out often shares unobserved features with "not choosing," so its correlation with other alternatives can violate IIA, motivating an error-component structure (Chapter 22). Omitting the opt-out biases choice probabilities and WTP and undermines incentive compatibility.

C3. The ε contains everything driving the choice that the model does not capture: unobserved attributes of the alternatives, respondent-to-respondent *taste variation* (if a single β is used), and measurement/optimization error. Moving taste heterogeneity from ε into *random coefficients* (mixed logit) changes the interpretation because it reclassifies systematic between-respondent variation as *structured heterogene-*

ity (each respondent has their own β drawn from a distribution) rather than unstructured noise — this (i) corrects the misspecification of forcing one β on heterogeneous people, (ii) yields variance components that *quantify* preference heterogeneity (a substantive result), and (iii) relaxes IIA. The same data are now read as "people genuinely differ in tastes" rather than "choices are noisy," which is a different and richer model (Chapter 22).

20.13 Spaced review prompts

+1 day. From memory, state Lancaster's characteristics model and the additive RUM, and explain what is deterministic vs stochastic and from whose perspective ("random" = analyst's ignorance).

+1 week. Prove the two normalizations (only differences matter; only β/σ identified) and explain how each generates a Part V subtlety (reference-dependent ASCs/WTP ratios; scale-vs-preference). Derive WTP as $\beta_k/\beta_{\text{cost}}$.

+1 month. For your own DCE, write out V (the Lancaster specification), state what your ε contains and whether taste heterogeneity belongs in random coefficients, check that any cross-arm coefficient comparison respects the scale normalization, and audit your opt-out modeling. Close with the spine sentence about what you assumed is systematic vs random.

20.14 How this chapter connects to the rest of the book

- **Chapters 2 (§2.11), 3** supplied utility, MRS, and WTP as a ratio — the welfare content of V; **Chapter 5 (§5.5–5.6)** the Gumbel and the logit derivation that turns the RUM integral into the softmax; **Chapter 18** the binary RUM and scale identification (§20.6); **Chapter 14** latent variables.
- **Chapter 21 (MNL and IIA)** turns this framework into the multinomial logit and confronts IIA (§20.5, §20.8); **Chapter 22 (beyond IIA)** relaxes the iid-Gumbel assumption into nested/mixed logit and error components (the opt-out and correlation of §20.8); **Chapter 23 (SML)** estimates the mixed logit.
- **Chapter 24 (WTP space)** builds on the WTP-from-V bridge (§20.7); **Chapter 25 (scale vs preference heterogeneity)** is the full development of Normalization 2's entanglement (§20.6, §20.4); **Chapter 41 (MDCEV welfare)** uses the logsum welfare of §20.7.
- **Chapters 27–29 (behavioral foundations, incentive compatibility, cost-vector effects)** are where the "what is in ε / does cost enter V or ε" question of §20.4 becomes your paper's central identification problem; **Chapter 37 (design)** operationalizes the opt-out and attribute choices.

This chapter opens Part V by establishing the two theories — Lancaster and Random Utility — on which your entire DCE rests, and the two normalizations that make its coefficients mean less (and its scale problem loom larger) than they naively appear.

20.15 Research translation task

For your **DCE**, produce a two-to-three-page foundations note:

1. **Lancaster specification.** Write your systematic utility V explicitly (attributes and functional form), and justify which characteristics are included and whether linear-additive is defensible (or whether nonlinearities/interactions are needed).
2. **The error.** State what your ε contains, and decide which components (taste heterogeneity, correlation) should be moved into random coefficients / error components (previewing the EC-MXL, Chapter 22).

3. **Normalizations.** Note that your coefficients are β/σ (preference × scale) and that only differences are identified; flag any cross-arm comparison (cost vector, experience) that could be confounded by scale.
4. **Labels and opt-out.** State whether alternatives are labeled or unlabeled and how you handle ASCs; document your opt-out modeling (its ASC, its role in welfare and incentive compatibility, its IIA implications).
5. **Welfare bridge.** Confirm your WTP comes from V ($\beta_k/\beta_{\text{cost}}$) and note that only included attributes can be valued.
6. **Close** with the spine sentence: *"My DCE assumes utility over characteristics ($V = \mathbf{z}'\beta$) with unobserved determinants in ε; my coefficients are identified up to scale; my opt-out is modeled as [·]; and cross-arm differences may reflect [scale/preference]."*

Carry this into Chapter 21 (MNL and IIA), Chapter 22 (EC-MXL), Chapter 24 (WTP space), and Chapter 25 (scale vs preference).

20.16 Bibliography

Core. - Train, *Discrete Choice Methods with Simulation*, the opening chapters on random utility, the choice probability, and identification (differences and scale). The single most relevant source for this chapter and Part V. Freely available — *verify link.* - Ben-Akiva & Lerman, *Discrete Choice Analysis: Theory and Application to Travel Demand.* The classic development of random utility theory, the additive RUM, and the systematic/random split (§20.3–§20.5). *(Verify edition.)* - Lancaster (1966), "A New Approach to Consumer Theory," *Journal of Political Economy.* The original characteristics model (§20.2). *(Verify volume/pages.)*

Applied. - Hensher, Rose & Greene, *Applied Choice Analysis.* Practical treatment of RUM specification, ASCs, labeled vs unlabeled alternatives, and the opt-out (§20.8). *(Verify edition.)* - Louviere, Hensher & Swait, *Stated Choice Methods: Analysis and Applications.* The stated-preference/DCE application of RUM — design, attributes, and the opt-out in environmental valuation. *(Verify edition.)* - Bateman et al., *Economic Valuation with Stated Preference Techniques: A Manual*, for the environmental-DCE application of Lancaster and RUM in your setting. *(Verify edition.)*

Advanced. - McFadden (1974), "Conditional Logit Analysis of Qualitative Choice Behavior," and McFadden (1981), "Econometric Models of Probabilistic Choice." The foundational derivations of the RUM logit and the identification results (§20.5–§20.6). *(Verify exact references/years.)* - Manski (1977), "The Structure of Random Utility Models," *Theory and Decision.* The rigorous structure of RUM and its identification. *(Verify volume/pages.)* - Train & Weeks, "Discrete Choice Models in Preference Space and Willingness-to-Pay Space," for the welfare/WTP bridge of §20.7 (and Chapter 24). *(Verify exact citation.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Lancaster / McFadden / Manski citations before using them, per the book's no-unverified-citation rule.

End of Chapter 20. Next in Part V: Chapter 21 — Multinomial Logit and IIA (the iid-Gumbel derivation of the softmax choice probability, the scale parameter, independence of irrelevant alternatives, and substitution patterns), which turns this chapter's random-utility framework into the workhorse model of your DCE and confronts its signature limitation. Tell me when to write it, or request a different chapter.

Chapter 21 — Multinomial Logit and IIA

Part V: Random Utility, Discrete Choice, and Stated Preference Econometrics

> **The spine question, applied to this chapter.** The multinomial logit is the workhorse of your DCE — the first model you estimate and the base for everything after. It is beautiful (a closed-form choice probability, a globally concave likelihood) and it carries one restrictive assumption that your EC-MXL exists to relax: **independence of irrelevant alternatives (IIA)**. IIA forces a rigid, often-wrong pattern of substitution between alternatives, and understanding *why* it fails — the red-bus/blue-bus problem — is understanding *why* you need the mixed logit. This chapter derives the MNL from iid Gumbel errors, makes the scale-parameter confound concrete, states and derives IIA, shows exactly how it breaks, and previews the fix. The spine question here: *when I estimate a multinomial logit, is IIA plausible for my choice set — or do my alternatives (the opt-out, near-substitutes, correlated tastes across a respondent's tasks) share unobserved features that make its substitution pattern, and hence my welfare estimates, wrong?*

21.0 Conceptual motivation: the workhorse and its one flaw

Your PDF's Block 2 practical output is explicit: *"Stimare un MNL, poi un mixed logit, poi un EC-MXL su un dataset semplice e confrontare preference space vs WTP space."* You start with the multinomial logit because it is the foundation — every richer model (nested, mixed, error-components) is the MNL with the iid-Gumbel assumption relaxed. And your PDF names the reason you move beyond it: *"il tuo DCE … usa EC-MXL per rilassare IIA e modellare eterogeneità/correlazione tra alternative,"* and it asks the diagnostic question directly: *"IIA è plausibile?"*

So this chapter has two jobs. First, to build the MNL properly: derive its closed-form choice probability from the random-utility framework of Chapter 20 and the Gumbel errors of Chapter 5, write its likelihood, and make the **scale parameter** — the confound between coefficients and error variance — concrete rather than abstract. Second, and more importantly, to expose **IIA**: the property that the relative odds of any two alternatives are *independent of all other alternatives*. IIA is what makes the MNL tractable, and it is also what makes it wrong when alternatives share unobserved features. The consequences are not academic:

- **Substitution patterns.** IIA forces *proportional substitution* — when an alternative changes or is removed, its probability redistributes to the others in proportion to their existing shares. Real substitution is often *disproportionate* (near-substitutes steal from each other), and getting this wrong biases the welfare effect of any policy that adds, removes, or changes an alternative — including your opt-out.
- **The opt-out.** Your DCE's opt-out plausibly shares unobserved features with "not paying," correlating its error with itself across tasks and violating IIA — a direct reason the MNL is suspect for your data (Chapter 20, §20.8).
- **Taste heterogeneity.** When a respondent answers several choice tasks with their own persistent (unobserved) tastes, their errors are correlated across tasks, violating the iid assumption — the

panel/heterogeneity motivation for the mixed logit (Chapters 19, 22).

The famous **red-bus/blue-bus** thought experiment crystallizes the failure, and we will simulate it: the MNL, faced with two near-identical bus options, wrongly steals share from the car; an error-components model — exactly the structure of your EC-MXL — fixes it. Seeing this is seeing *why your EC-MXL is built the way it is*. We proceed: the MNL derivation and likelihood; the scale parameter made concrete; IIA stated and derived; the red-bus/blue-bus failure; proportional substitution and cross-elasticities; when IIA holds and fails; testing IIA (and why not to rely on the test); and the preview of the models that relax it. Chapter 22 builds those models; this chapter is why you need them.

21.1 Prerequisites

From **Chapter 20**: the random-utility framework $U_j = V_j + \varepsilon_j$, the choice probability as an integral over the errors (§20.5), and the two normalizations (differences only; scale — §20.6). From **Chapter 5** (§5.5–5.6): the Gumbel distribution, its max-stability, and the derivation that iid Gumbel errors give the softmax. From **Chapter 11**: the likelihood/MLE framework and global concavity. From **Chapter 18**: the binary logit as the two-alternative MNL and scale identification.

Same **notation**. Consolidated symbols: $V_{nj} = \mathbf{x}_{nj}'\beta$ (systematic utility of alternative j for respondent n), P_{nj} (choice probability), y_{nj} (chosen indicator), μ (scale parameter $= 1/\sigma_\varepsilon$), J (alternatives per choice set).

21.2 Deriving the multinomial logit

The assumption. Take the additive RUM $U_j = V_j + \varepsilon_j$ (Chapter 20) and assume the errors ε_j are **independent and identically distributed Gumbel (Type-I extreme value)**. This is the MNL's one substantive assumption, and everything — the closed form, the concavity, and IIA — flows from it.

The choice probability (derivation, consolidating Chapter 5 §5.6). The probability that j is chosen is that its utility is the largest. Condition on $\varepsilon_j = s$ and require every other alternative's utility to be below $U_j = V_j + s$:

$$P_j = \int_{-\infty}^{\infty} \Big[\prod_{k\neq j} \Pr(\varepsilon_k \le s + V_j - V_k)\Big] f(s)\, ds,$$

where f and $F(z) = \exp(-e^{-z})$ are the Gumbel density and CDF. Substituting $\Pr(\varepsilon_k \le s + V_j - V_k) = \exp(-e^{-(s+V_j-V_k)})$ and $f(s) = e^{-s}\exp(-e^{-s})$, the product over $k \neq j$ becomes $\exp(-e^{-s}e^{-V_j}\sum_{k\neq j} e^{V_k})$; multiplying by $f(s)$ and using $1 + e^{-V_j}\sum_{k\neq j} e^{V_k} = e^{-V_j}\sum_k e^{V_k}$, the substitution $u = e^{-s}$ reduces the integral to $\int_0^\infty \exp(-u\, e^{-V_j}\sum_k e^{V_k})\, du$, giving the **softmax**:

$$\boxed{P_j = \frac{e^{V_j}}{\sum_{k=1}^{J} e^{V_k}} = \frac{e^{\mathbf{x}_j'\beta}}{\sum_k e^{\mathbf{x}_k'\beta}}}.$$

(Verified: with $V = (1,2,3)$, simulating $\arg\max_j(V_j + \varepsilon_j)$ over 3 million iid-Gumbel draws gives $(0.090, 0.245, 0.665)$, matching the softmax exactly.) This is the multinomial logit. It is a closed form — no integral to simulate (unlike the mixed logit) — which is why it is the computational baseline.

Conditional logit vs multinomial logit (terminology). Two names for closely related models. In the strict "multinomial logit" the regressors are *characteristics of the respondent* (the same across alternatives, with alternative-specific coefficients); in McFadden's **conditional logit** the regressors are *characteristics of the alternatives* (the attributes $\mathbf{x}_j$ that vary across alternatives, with common coefficients β). **Your DCE uses the conditional logit** — the attributes vary across the alternatives in each choice set — though the

field loosely calls it "multinomial logit." The softmax above is the conditional logit; the derivation and IIA are identical.

> **Active recall 21.A.** State the one substantive assumption behind the MNL, the closed-form choice probability it produces, and why "closed form" matters for estimation. *(Answer: the errors are iid Gumbel; this gives the softmax* $P_j = e^{V_j}/\sum_k e^{V_k}$ *(because the Gumbel is max-stable); "closed form" matters because the choice probability — and hence the likelihood — can be computed exactly with no simulation, making the MNL fast and its likelihood globally concave (§21.3), unlike the mixed logit which requires simulated integration.)*

21.3 The MNL likelihood and estimation

The likelihood. Each respondent n chooses one alternative in each choice task; let $y_{nj} = 1$ for the chosen one. The likelihood is the product of the chosen alternatives' probabilities:

$$L(\beta) = \prod_n \prod_j P_{nj}^{y_{nj}}, \qquad \ell(\beta) = \sum_n \sum_j y_{nj} \log P_{nj} = \sum_n \log P_{n,\text{chosen}}.$$

For a **panel** DCE (several tasks per respondent), the MNL treats each task as an independent observation and multiplies over tasks — which ignores the within-respondent correlation (Chapters 17, 19), so MNL standard errors should be *respondent-clustered*, and the correlation itself is what the panel mixed logit models (Chapter 22).

Estimation is reliable. The MNL log-likelihood is **globally concave** in β (the same result as the binary logit, Chapters 2 §2.8, 11): the score is $\sum_n \sum_j (y_{nj} - P_{nj})\mathbf{x}_{nj}$ and the Hessian is negative definite. So the MNL has a *unique global maximum*, converges reliably from any starting values, and has none of the local-maxima pathologies of the mixed logit (Chapter 13). This is a genuine virtue — the MNL is the dependable baseline you can always estimate — and it is why you start from it (Block 2) before the treacherous non-concave models.

> **Stop and think 21.1.** You estimate an MNL on a DCE with 8 choice tasks per respondent using default (classical) standard errors. What is wrong, and what does it foreshadow about why you will move to a mixed logit? *(Answer: the classical MNL treats the 8 tasks per respondent as independent, but they share the respondent's unobserved tastes (correlated errors) — so the standard errors are too small (Chapters 17, 19); cluster on respondents. This within-respondent correlation is exactly the taste heterogeneity that violates the MNL's iid assumption and motivates the panel mixed logit, which* models* the correlation rather than just correcting the SEs — Chapter 22.)*

21.4 The scale parameter, made concrete

Chapter 20 (§20.6) established that only β/σ_ε is identified. In the MNL this becomes the **scale parameter**, and making it concrete here prevents the coefficient misreadings that plague your cross-group comparisons.

The scale parameter. If the Gumbel errors have scale σ_ε (not the normalized 1), the choice probability is

$$P_j = \frac{e^{\mu V_j}}{\sum_k e^{\mu V_k}}, \qquad \mu = \frac{1}{\sigma_\varepsilon} \text{ (the scale parameter).}$$

The utility coefficients enter *multiplied by* μ: what the data identify is $\mu\beta$, never β and μ separately. Software normalizes $\mu = 1$ (equivalently $\sigma_\varepsilon = \pi/\sqrt{6}$ for the standard Gumbel), so **every estimated coefficient is $\mu\beta$ — preferences times scale.**

What μ does. A *larger* μ (smaller error variance) makes choices *sharper* — the softmax becomes more extreme, approaching deterministic choice of the highest-V alternative as $\mu \to \infty$, and approaching random (equal-probability) choice as $\mu \to 0$. *(Verified, Chapter 20: doubling V without scaling the error — i.e., a larger μ — changed choice probabilities from $(0.09, 0.24, 0.67)$ to the much sharper $(0.02, 0.12, 0.87)$.)* So μ measures *choice consistency*: a group that chooses more consistently (less noise) has a larger μ and therefore *uniformly larger estimated coefficients*, with no difference in tastes.

The confound, stated plainly. Because estimated coefficients are $\mu\beta$, a difference in coefficients between two groups (or arms, or models) can be a difference in μ (scale/noise) *or* in β (preference) — and the MNL cannot tell them apart. This is the **scale-versus-preference-heterogeneity** problem (Chapter 25), and it is the reason your PDF warns against comparing raw coefficients across your cost-vector arms or experience groups. The fingerprint of a *pure scale* difference is that *all* coefficients scale by the same factor; a genuine *preference* difference changes the *relative* magnitudes. WTP (a ratio $\beta_k/\beta_{\text{cost}}$) cancels μ and is therefore comparable across scales — which is why you compare WTP, not coefficients (Chapters 18, 20).

> **Active recall 21.B.** Your high-cost-vector arm has all coefficients about 1.4 times larger than your low-cost-vector arm. What is the most parsimonious explanation, and why does WTP not suffer from it? *(Answer: a* scale* difference — the high-cost arm chooses more consistently (larger μ), scaling all coefficients up uniformly by ~1.4, with no taste difference (the uniform scaling is the fingerprint). WTP is the ratio $\beta_k/\beta_{\text{cost}} = \mu\beta_k/\mu\beta_{\text{cost}}$, in which μ cancels, so WTP is scale-free and comparable across arms — compare WTP, not raw coefficients, Chapters 18, 25.)*

21.5 IIA: independence of irrelevant alternatives

Now the property that defines and limits the MNL. It follows in one line from the softmax and has profound consequences.

Definition 21.1 (IIA). The **odds** of choosing i over k are

$$\frac{P_i}{P_k} = \frac{e^{V_i}/\sum_m e^{V_m}}{e^{V_k}/\sum_m e^{V_m}} = \frac{e^{V_i}}{e^{V_k}} = e^{V_i - V_k}.$$

The denominators cancel, so **the relative odds of i and k depend only on V_i and V_k — not on any other alternative in the choice set.** This is **independence of irrelevant alternatives**: adding, removing, or changing a *third* alternative m leaves the odds of i versus k unchanged.

What IIA claims, in words. No matter what else is on offer, the ratio in which people split between any two alternatives stays fixed. Introduce a fantastic new option, or delete a mediocre one, and the *relative* popularity of the two survivors is untouched. Stated this baldly, IIA is clearly a strong claim — and often a false one.

Why the MNL has it. IIA is a *direct consequence of the iid assumption*: because the errors are independent across alternatives, no two alternatives share unobserved features, so no third alternative is a special substitute for any other — all substitution is "even." When alternatives *do* share unobserved features (the buses in §21.6, your opt-out), the iid assumption fails and IIA is violated. **IIA is not a separate assumption bolted onto the MNL; it is the iid-Gumbel assumption seen from the substitution side.** Relaxing IIA therefore means relaxing iid — exactly what nested and mixed logit do (Chapter 22).

(Verified: for $V = (1, 2, 3)$, the odds P_1/P_2 is the same whether computed in the two-alternative set $\{1, 2\}$ or the three-alternative set $\{1, 2, 3\}$ — IIA holds exactly under the MNL, §21.6.)

> **Stop and think 21.2.** In your unlabeled DCE with an opt-out, IIA says the odds of choosing "Option A" over "Option B" are unaffected by the presence of the opt-out. Give one reason this might fail. *(Answer: the opt-out likely shares unobserved features with "not choosing / not paying" that are* also* relevant to how a respondent views the priced options — e.g., a general

reluctance to pay, or a status-quo tendency, correlated across tasks. If the opt-out's error is correlated with the priced alternatives' errors (or a respondent's persistent taste for opting out spans tasks), the iid assumption fails and IIA is violated — the opt-out is a "special" substitute, not a neutral third alternative, motivating an error-component structure, §21.8, Chapter 22.)*

21.6 The red-bus/blue-bus problem: how IIA breaks

The classic demonstration that IIA can be badly wrong — and the exact structure of your EC-MXL's fix.

The thought experiment. A commuter chooses between a **car** and a **blue bus**, and (having no reason to prefer one) splits 50/50. Now introduce a **red bus**, *identical to the blue bus in every way except color* — a perfect substitute for it. Intuition says the red bus should draw riders *only from the blue bus* (they are interchangeable), leaving the car unaffected: the correct outcome is car $\frac{1}{2}$, blue $\frac{1}{4}$, red $\frac{1}{4}$. But **IIA forbids this**: since the odds car:blue must stay 1:1, and by symmetry blue:red = 1:1, the MNL forces

$$\text{car} = \text{blue} = \text{red} = \tfrac{1}{3}.$$

The MNL *steals share from the car* to give to the redundant red bus — an absurd prediction, because the red bus is irrelevant to the car–bus trade-off. *(Verified: with all three utilities equal, simulating iid-Gumbel choice gives exactly* $(0.333, 0.333, 0.333)$*.)*

Why it happens, and the fix. The failure is the iid assumption: the two buses share an unobserved "bus-ness" (they are the same except color), so their errors are *correlated*, not independent. Model that correlation with a **shared error component** — $U_{\text{car}} = \varepsilon_{\text{car}}$, $U_{\text{blue}} = \eta_{\text{bus}} + \varepsilon_{\text{blue}}$, $U_{\text{red}} = \eta_{\text{bus}} + \varepsilon_{\text{red}}$, where η_{bus} is common to both buses — and the substitution corrects itself.

Worked Example 21.1 (Error components fix IIA, verified). Simulating choice as the shared-bus component η_{bus} grows (all systematic utilities equal):

Shared-bus SD	car	blue	red	odds car/blue
0 (MNL / IIA)	0.333	0.333	0.333	1.00
2	0.396	0.302	0.302	1.31
5	0.449	0.276	0.276	1.63
12 (near-perfect substitutes)	0.477	0.261	0.262	**1.83 → 2**

(Verified.) As the buses become more correlated (larger shared component), the car's share climbs back toward its correct $\frac{1}{2}$, the two buses split the remainder toward $\frac{1}{4}$ each, and the odds car:blue rise from the IIA-forced 1 toward the correct 2. **This is precisely the mechanism of your error-components mixed logit (EC-MXL):** an error component shared among correlated alternatives relaxes IIA by letting them be special substitutes. The red-bus/blue-bus problem is not a curiosity — it *is* why your DCE uses error components, and this table is the demonstration that they work.

Active recall 21.C. In the red-bus/blue-bus problem, what does the MNL predict, what is the correct answer, and what modeling device restores it? *(Answer: the MNL, forced by IIA, predicts* $\frac{1}{3}$ *each — wrongly stealing share from the car to give to the redundant red bus; the correct answer is car* $\frac{1}{2}$*, blue* $\frac{1}{4}$*, red* $\frac{1}{4}$ *(the red bus draws only from the perfectly-substitutable blue bus); the fix is a* shared error component* between the correlated buses (or nesting them), which relaxes the iid assumption and IIA — the error-components mixed logit, Chapter 22.)*

21.7 IIA and proportional substitution

IIA has a precise implication for *how choice probabilities respond to changes* — the substitution pattern — which is what matters for welfare.

Proportional substitution. Under IIA, when one alternative's attractiveness changes (or it is removed), the probability that flows to or from the other alternatives is distributed **in proportion to their existing shares**. Removing alternative 1 does not send its probability preferentially to its close substitutes; it spreads it across *all* remaining alternatives proportionally.

Worked Example 21.2 (Proportional substitution, verified). With $V = (1, 2, 3)$, the MNL probabilities are $(0.090, 0.245, 0.665)$. Remove alternative 1; the probabilities over $\{2, 3\}$ become $(0.269, 0.731)$ — which is exactly the *original* $\{2, 3\}$ probabilities renormalized, $(0.245, 0.665)/(0.245 + 0.665) = (0.269, 0.731)$ (verified). The 0.090 from the removed alternative redistributed to alternatives 2 and 3 in the ratio 0.245:0.665 — proportional to their existing shares, regardless of whether alternative 1 was a close substitute for one of them.

Equal cross-elasticities. The same restriction appears in elasticities: under the MNL, the *cross-elasticity* of alternative i's probability with respect to alternative j's attribute is **the same for all** $i \neq j$ — a change in j's attribute affects every other alternative's share by the same percentage. This is a strong, testable, and usually implausible restriction: a price cut on the blue bus should draw disproportionately from the red bus (its close substitute), not equally from the car.

Why it matters for welfare. Substitution patterns determine the *welfare* effect of a policy that changes the choice set — adding a restoration option, removing a degraded one, changing an attribute. If the MNL forces proportional substitution when substitution is really disproportionate, the estimated welfare change (via the logsum, Chapter 20 §20.7, Chapter 41) is wrong. So IIA is not just a substitution curiosity; it propagates into biased WTP and welfare whenever your policy scenario involves changing what is on offer — which environmental-DCE welfare analysis routinely does.

> **Stop and think 21.3.** Your DCE evaluates a policy that removes a low-quality restoration option and asks how demand redistributes. Why might the MNL give the wrong welfare answer, and which alternatives would be mis-handled? *(Answer: the MNL forces the removed option's demand to redistribute* proportionally* across the remaining options; but if the removed option was a close substitute for one particular remaining option (similar attributes), real demand would flow *disproportionately* to that substitute. The MNL spreads it evenly, mis-estimating the new shares and hence the logsum-based welfare change — the close-substitute alternative is under-credited and the dissimilar ones over-credited. A model that captures the correlation (nested/mixed logit) is needed for correct substitution and welfare.)*

21.8 When IIA holds, and when it fails

IIA is not always wrong — it is a *good approximation when alternatives are genuinely distinct.* Knowing when it holds and fails tells you when the MNL suffices and when you must move to the mixed logit.

IIA is plausible when: - Alternatives are truly *independent* — they share no unobserved features that a third alternative could substitute for. Generic, **unlabeled** alternatives with *all* relevant attributes observed approach this ideal (there is no "bus-ness" left unobserved). - The choice is a genuine single cross-section with no persistent unobserved taste linking alternatives.

IIA fails when: - **Alternatives share unobserved features.** Labeled alternatives (modes, brands) or near-substitutes carry shared unobserved characteristics → correlated errors → IIA violated (red-bus/blue-bus). - **The opt-out correlates with the priced alternatives.** As in §21.5, the opt-out often shares "not paying / status quo" features, making it a special substitute — a direct concern for your DCE. - **Taste heterogeneity spans a respondent's tasks.** In a *panel* DCE, a respondent's persistent unobserved tastes

correlate their errors across the several tasks they answer — a violation of iid *across observations* that the panel mixed logit addresses (Chapters 19, 22). This is arguably the most pervasive IIA violation in stated-preference data, and it is why the mixed logit is now standard.

For your DCE specifically. You have an unlabeled design (good for IIA on the *labeling* dimension), but you have an **opt-out** (correlation concern) and a **panel structure with taste heterogeneity** (correlation across tasks). So IIA is *suspect* for your data on two counts, which is exactly why your model is an **error-components mixed logit** rather than a plain MNL — the EC-MXL relaxes IIA by adding error components (for the opt-out/alternative correlation) and random coefficients (for the taste heterogeneity across tasks). Your PDF's "IIA è plausibile?" has the answer: probably not, and your model is built to not need it.

> **Active recall 21.D.** Give the two reasons IIA is suspect specifically for your (unlabeled, opt-out, panel) DCE, and name the two features of the EC-MXL that address them. *(Answer: (i) the* opt-out* likely shares unobserved "not-paying/status-quo" features with the choice, correlating its error — addressed by an *error component* on the opt-out (or alternatives); (ii) *taste heterogeneity* across a respondent's repeated tasks correlates their errors — addressed by *random coefficients* (each respondent's own tastes). The EC-MXL combines error components and random coefficients precisely to relax IIA on both counts, Chapter 22.)*

21.9 Testing IIA (and why not to rely on it)

There is a formal test, and a modern consensus not to lean on it.

The Hausman–McFadden test. Under IIA, the MNL coefficients estimated on the *full* choice set should equal those estimated on a *restricted subset* of alternatives (since dropping "irrelevant" alternatives should not change the relative preferences). The Hausman–McFadden test compares the two coefficient vectors — a Hausman-type quadratic form (Chapter 19, §19.6) in their difference. A significant statistic rejects IIA.

Its limitations (why not to rely on it). The test is notoriously unreliable: it can produce *negative* statistics (the covariance difference is not positive definite in finite samples), it has *low power*, and its result depends on *which subset* you drop. So a non-rejection is weak evidence for IIA, and the test is not a dependable gatekeeper.

The modern stance. Do not decide IIA by the test; decide it by *reasoning* about whether alternatives share unobserved features (§21.8), and — better — **use a model that does not require IIA in the first place**. The mixed logit relaxes IIA regardless of whether it "truly" holds, at little cost if it does hold (it nests the MNL). So the practical workflow your Block 2 follows — estimate the MNL as a transparent baseline, then estimate the mixed logit / EC-MXL which relaxes IIA — sidesteps the testing problem entirely: you report the MNL for interpretability and the EC-MXL for validity, and the comparison itself is informative (if they agree, IIA was benign; if they diverge in substitution/welfare, IIA mattered). This is exactly the workflow your PDF prescribes.

> **Stop and think 21.4.** A co-author runs the Hausman–McFadden test, gets a non-significant result, and concludes "IIA holds, the MNL is fine." Why is this insufficient, and what would you do instead? *(Answer: the test is unreliable (low power, negative statistics, subset-dependent), so a non-significant result is weak evidence for IIA — it may simply reflect the test's low power, not IIA's validity. Instead, reason about whether the alternatives (opt-out, panel taste heterogeneity) share unobserved features (§21.8), and estimate a mixed logit / EC-MXL that relaxes IIA regardless; report the MNL as a baseline and check whether the richer model changes substitution/welfare. Do not rest a substantive conclusion on the Hausman–McFadden test.)*

21.10 From MNL to the models that relax IIA (preview)

The MNL's IIA is the iid-Gumbel assumption; relaxing IIA means enriching the error structure. Three routes, developed in Chapter 22:

- **Nested logit.** Group correlated alternatives into *nests* (e.g., the two buses in one nest, the car alone). Alternatives within a nest share correlation; IIA holds *within* a nest but not *across* nests. Fixes red-bus/blue-bus by nesting the buses. Requires you to specify the nesting structure a priori.
- **Mixed logit (random coefficients).** Let each respondent have their own coefficients β_n drawn from a distribution (Chapter 19's random slopes). This induces correlation across a respondent's choices (relaxing IIA across tasks) and models *taste heterogeneity* — the pervasive stated-preference violation. Estimated by simulated maximum likelihood (Chapter 23).
- **Error-components mixed logit (EC-MXL) — your model.** Add error components shared among correlated alternatives (relaxing IIA across alternatives, e.g. the opt-out) *and* random coefficients (relaxing it across tasks). This is the most flexible and is exactly your DCE's specification — it relaxes IIA on both the alternative and the task dimension, which is why you use it.

The through-line: the MNL is the tractable baseline whose one flaw is IIA; every richer model relaxes IIA by making the errors correlated (nesting), heterogeneous (random coefficients), or both (EC-MXL). Understanding IIA — what it forces, when it fails, and how error components fix it (Worked Example 21.1) — is understanding the *purpose* of every model in Chapter 22.

21.11 Failure modes

F1. Using the MNL when IIA fails. Estimating a plain MNL on data with correlated alternatives (opt-out, near-substitutes) or panel taste heterogeneity. *Symptom:* wrong substitution/welfare. *Fix:* nested/mixed logit / EC-MXL (§21.8, §21.10).

F2. Reading raw coefficients as preferences (scale). Comparing MNL coefficients across arms/groups ignoring the $\mu\beta$ confound. *Fix:* compare WTP (scale-free); model scale (§21.4; Chapters 18, 25).

F3. Trusting MNL welfare with disproportionate substitution. Computing the welfare of adding/removing an alternative under IIA's proportional substitution when substitution is really disproportionate. *Fix:* use a model that captures correlation (§21.7).

F4. Classical SEs on panel MNL. Ignoring within-respondent correlation across tasks. *Fix:* respondent-clustered SEs (or model with mixed logit) (§21.3; Chapters 17, 19).

F5. Relying on the Hausman–McFadden test. Concluding "IIA holds" from a non-significant, low-power test. *Fix:* reason about correlation; use a model that relaxes IIA (§21.9).

F6. Mis-modeling the opt-out. Treating the opt-out as an iid alternative when it correlates with "not paying." *Fix:* error component on the opt-out (§21.5, §21.8; Chapter 22).

F7. Ignoring that IIA = iid. Treating IIA as a separate assumption to test rather than the substitution face of iid errors. *Fix:* recognize relaxing IIA means enriching the error structure (§21.5, §21.10).

F8. Over-interpreting the MNL's tractability as correctness. Preferring the MNL because it converges cleanly, ignoring its restrictive substitution. *Fix:* the MNL is a baseline; validate against a mixed logit (§21.3, §21.10).

21.12 Diagnostic questions for your own papers

1. **Is IIA plausible?** Do my alternatives (opt-out, near-substitutes) or panel structure share unobserved features that violate iid? *(§21.8.)*
2. **Substitution and welfare.** Does my policy scenario change the choice set, and would IIA's proportional substitution bias the welfare estimate? *(§21.7.)*
3. **Scale.** Am I comparing WTP (scale-free) rather than raw coefficients across arms/groups? *(§21.4.)*
4. **Panel SEs.** Are my MNL standard errors respondent-clustered (or is the correlation modeled)? *(§21.3.)*
5. **Opt-out.** Is my opt-out modeled with an error component rather than as an iid alternative? *(§21.5, §21.8.)*
6. **Testing vs modeling.** Am I deciding IIA by reasoning and a relaxed model, not by the unreliable Hausman–McFadden test? *(§21.9.)*
7. **Baseline vs validated.** Do I report the MNL as a transparent baseline and validate substitution/welfare against a mixed logit / EC-MXL? *(§21.10.)*

21.13 Exercises

(A) Comprehension.

A1. State IIA precisely (in terms of odds), and derive it from the softmax choice probability.

A2. Explain what the scale parameter μ does to choice probabilities, and why estimated coefficients are $\mu\beta$.

A3. In the red-bus/blue-bus problem, what does the MNL predict and what is the intuitively correct answer?

(B) Derivation / reasoning.

B1. Derive the MNL choice probability $P_j = e^{V_j}/\sum_k e^{V_k}$ from iid Gumbel errors (you may cite the Gumbel CDF and the key integral).

B2. Show that under the MNL, removing an alternative redistributes its probability to the others in proportion to their existing shares (proportional substitution), and state the equal-cross-elasticity implication.

B3. Show, using a shared error component, how the red-bus/blue-bus substitution is corrected, and explain the connection to your EC-MXL.

(C) Research application / critique.

C1. Your DCE has an opt-out and 8 tasks per respondent. Argue whether IIA is plausible, and specify the model features you would use to relax it.

C2. A co-author reports MNL welfare estimates for a policy that removes a restoration option. Critique the substitution assumption and explain how it could bias the welfare estimate.

C3. Explain to a co-author why you will estimate an MNL *and* an EC-MXL, and what the comparison tells you about whether IIA mattered.

21.14 Solutions

A1. IIA: for any two alternatives i, k, the odds P_i/P_k depend only on V_i and V_k, not on any other alternative. Derivation: $P_i/P_k = (e^{V_i}/\sum_m e^{V_m})/(e^{V_k}/\sum_m e^{V_m}) = e^{V_i}/e^{V_k} = e^{V_i - V_k}$ — the common denominator $\sum_m e^{V_m}$ cancels, so the odds are independent of all other alternatives m.

A2. $P_j = e^{\mu V_j}/\sum_k e^{\mu V_k}$: a larger μ (smaller error variance) makes the softmax sharper (more extreme probabilities, approaching deterministic choice of the highest-V alternative as $\mu \to \infty$; equal probabilities as

$\mu \to 0$). Estimated coefficients are $\mu\beta$ because only the product is identified (scaling β and the error together leaves choices unchanged, Chapter 20 §20.6); software normalizes $\mu = 1$, so coefficients are preferences times the (normalized) scale.

A3. The MNL predicts car = blue = red = $\frac{1}{3}$ (IIA forces the car:blue odds to stay 1:1 and blue:red 1:1). The intuitively correct answer is car $\frac{1}{2}$, blue $\frac{1}{4}$, red $\frac{1}{4}$ — the red bus, a perfect substitute for the blue bus, should draw riders only from the blue bus, leaving the car's share unchanged.

B1. Condition on $\varepsilon_j = s$ and require $\varepsilon_k \le s + V_j - V_k$ for all $k \neq j$: $P_j = \int \prod_{k\neq j} \exp(-e^{-(s+V_j-V_k)}) \cdot e^{-s}\exp(-e^{-s})\,ds$. The product is $\exp(-e^{-s}e^{-V_j}\sum_{k\neq j} e^{V_k})$; combining with the $\exp(-e^{-s})$ term via $1 + e^{-V_j}\sum_{k\neq j} e^{V_k} = e^{-V_j}\sum_k e^{V_k}$ gives $\int e^{-s}\exp(-e^{-s}e^{-V_j}\sum_k e^{V_k})ds$; the substitution $u = e^{-s}$ yields $\int_0^\infty \exp(-u\,e^{-V_j}\sum_k e^{V_k})du = 1/(e^{-V_j}\sum_k e^{V_k}) = e^{V_j}/\sum_k e^{V_k}$.

B2. With probabilities $P_j = e^{V_j}/\sum_k e^{V_k}$, removing alternative 1 gives new probabilities over the remaining alternatives $P_j^{\text{new}} = e^{V_j}/\sum_{k\neq 1} e^{V_k}$ for $j \neq 1$; this equals $P_j/(1-P_1)$ — the original probability renormalized by the remaining mass — so each survivor's share grows in proportion to its existing share (verified: removing alt 1 from $(0.09, 0.245, 0.665)$ gives $(0.269, 0.731)$, the original $\{2,3\}$ renormalized). Equal cross-elasticity: since the redistribution is proportional, a change in alternative j's attribute changes every *other* alternative's probability by the same *percentage* — the cross-elasticities are equal across all $i \neq j$.

B3. Model the two buses with a shared error component: $U_{\text{car}} = \varepsilon_{\text{car}}$, $U_{\text{blue}} = \eta + \varepsilon_{\text{blue}}$, $U_{\text{red}} = \eta + \varepsilon_{\text{red}}$, with η common to both buses. Then the buses' utilities are *correlated* (both contain η), so they are substitutes for each other more than for the car; as $\text{Var}(\eta)$ grows, the car's share rises toward $\frac{1}{2}$ and the buses split toward $\frac{1}{4}$ each (verified: with shared-SD = 12, car = 0.48, buses = 0.26 each, odds car/blue $\to$ 1.83). This is exactly the error-components mechanism of the EC-MXL: an error component shared among correlated alternatives relaxes IIA by letting them be special substitutes — your DCE uses this to handle the opt-out and correlated alternatives.

C1. IIA is *suspect* for this DCE on two counts: (i) the *opt-out* likely shares unobserved "not-paying/status-quo" features, correlating its error with the choice (a special-substitute violation); (ii) the *panel* structure (8 tasks per respondent) means each respondent's persistent unobserved tastes correlate their errors across tasks, violating iid across observations. To relax IIA I would estimate an **error-components mixed logit**: an error component on the opt-out (or alternatives) to capture the alternative-level correlation, and random coefficients to capture the across-task taste heterogeneity — exactly the EC-MXL specification, estimated by simulated maximum likelihood (Chapters 22–23), with respondent-clustered inference.

C2. Under the MNL, removing the restoration option redistributes its demand *proportionally* across the remaining options (IIA/proportional substitution). But if the removed option was a close substitute for a particular remaining option (similar attributes), real demand would flow *disproportionately* to that substitute; the MNL spreads it evenly, mis-estimating the new choice shares. Since the welfare change is computed from the logsum over the (changed) choice set, the wrong substitution produces the wrong welfare estimate — under-crediting the close substitute and over-crediting dissimilar options. A model capturing the correlation (nested or mixed logit) is needed for correct substitution and welfare when the policy changes the choice set.

C3. I estimate the **MNL** as a transparent, reliably-estimated baseline (closed form, globally concave — clean coefficients and a benchmark), and the **EC-MXL** because it relaxes IIA and models taste heterogeneity and alternative correlation, which the MNL cannot. The **comparison** is informative: if the two give similar substitution patterns, WTP, and welfare, IIA was benign and the MNL's simplicity is justified; if they diverge — especially in substitution and policy welfare — IIA *mattered*, and the EC-MXL is the valid model. This sidesteps the unreliable Hausman–McFadden test (§21.9) and follows the Block-2 workflow: report the baseline, validate with the relaxed model, and let the comparison reveal whether IIA bit.

21.15 Spaced review prompts

+1 day. From memory, derive the MNL softmax from iid Gumbel, state IIA (odds independent of other alternatives) and derive it, and explain the scale parameter μ.

+1 week. Reconstruct the red-bus/blue-bus problem and its error-component fix; explain proportional substitution and equal cross-elasticities and why they bias welfare when the choice set changes. State when IIA holds and fails for your DCE.

+1 month. For your own DCE, estimate an MNL, examine its substitution/IIA implications, and compare to your EC-MXL; report whether IIA mattered for WTP and welfare, and cluster your MNL SEs on respondents. Close with the spine sentence about whether IIA is plausible for your choice set.

21.16 How this chapter connects to the rest of the book

- **Chapter 20** supplied the random-utility framework and the two normalizations; **Chapter 5 (§5.6)** the Gumbel derivation of the softmax; **Chapter 18** the binary logit (two-alternative MNL) and scale; **Chapter 11** the concave likelihood.
- **Chapter 22 (beyond IIA)** builds the nested, mixed, and error-components logit that relax IIA (§21.10) — the red-bus/blue-bus fix of §21.6 is the EC-MXL mechanism; **Chapter 23 (SML)** estimates the mixed logit; **Chapter 25 (scale vs preference)** develops the $\mu\beta$ confound of §21.4.
- **Chapters 17, 19** are the clustered/panel inference for the MNL's within-respondent correlation (§21.3); **Chapter 24 (WTP space)** builds on the WTP-cancels-scale point (§21.4); **Chapter 41 (MDCEV welfare)** uses the logsum whose substitution IIA distorts (§21.7); **Chapter 37 (design)** operationalizes the opt-out and attribute choices that bear on IIA.

This chapter establishes the MNL as the workhorse baseline of your DCE and exposes IIA as its one flaw — the assumption your EC-MXL is built to relax — turning "IIA è plausibile?" into a reasoned, model-based judgment rather than a test.

21.17 Research translation task

For your **DCE**, produce a two-to-three-page note:

1. **Estimate the MNL baseline.** Report the conditional-logit coefficients (with respondent-clustered SEs) and WTP; confirm concave, reliable estimation.
2. **Interrogate IIA.** Argue whether IIA is plausible given your opt-out and panel structure; identify the alternatives/tasks whose correlation threatens it.
3. **Substitution and welfare.** For a policy scenario that changes the choice set, note whether the MNL's proportional substitution is defensible and how it could bias welfare.
4. **Scale.** Confirm you compare WTP (scale-free), not raw coefficients, across any arms/groups.
5. **Relax IIA.** Specify the EC-MXL features (error components for the opt-out/alternative correlation; random coefficients for taste heterogeneity) that relax IIA, and note you will compare MNL vs EC-MXL substitution/WTP/welfare (Chapter 22).
6. **Close** with the spine sentence: *"My MNL baseline assumes IIA, which is [plausible/suspect] because [opt-out/panel correlation]; I relax it with an EC-MXL, and the MNL-vs-EC-MXL comparison shows IIA [did/did not] matter for [substitution/WTP/welfare]."*

Carry this into Chapter 22 (EC-MXL), Chapter 23 (SML), Chapter 24 (WTP space), and Chapter 25 (scale vs preference).

21.18 Bibliography

Core. - Train, *Discrete Choice Methods with Simulation*, the logit chapter (derivation, IIA, proportional substitution, and the limits of the MNL). The single most relevant source; read alongside this chapter and Chapter 22. Freely available — *verify link.* - Ben-Akiva & Lerman, *Discrete Choice Analysis*, the MNL and IIA chapters, with the red-bus/blue-bus problem and substitution patterns. *(Verify edition.)* - McFadden (1974), "Conditional Logit Analysis of Qualitative Choice Behavior." The original conditional logit derivation and IIA (§21.2, §21.5). *(Verify exact reference/year.)*

Applied. - Hensher, Rose & Greene, *Applied Choice Analysis*, on the MNL, IIA testing, scale, and the move to mixed logit — practical and DCE-focused. *(Verify edition.)* - Louviere, Hensher & Swait, *Stated Choice Methods*, for the MNL/IIA in stated-preference and environmental valuation, including the opt-out. *(Verify edition.)* - Hausman & McFadden (1984), "Specification Tests for the Multinomial Logit Model," *Econometrica*. The IIA test of §21.9 (and its known limitations). *(Verify volume/pages.)*

Advanced. - McFadden (1978), "Modelling the Choice of Residential Location," and McFadden & Train (2000), "Mixed MNL Models for Discrete Response," *Journal of Applied Econometrics* — the latter on how mixed logit relaxes IIA and approximates any RUM (§21.10; Chapter 22). *(Verify volumes/pages.)* - Cheng & Long (2007) and related work on the poor finite-sample behavior of the Hausman–McFadden IIA test (§21.9). *(Verify exact citation.)* - Small & Rosen, "Applied Welfare Economics with Discrete Choice Models," *Econometrica*, for the logsum welfare that IIA's substitution distorts (§21.7; Chapter 41). *(Verify volume/year.)*

Entries marked "verify" are pointers; confirm editions, pages, and the McFadden / Hausman–McFadden citations before using them, per the book's no-unverified-citation rule.

End of Chapter 21. Next in Part V: Chapter 22 — Relaxing IIA (nested logit, mixed logit, random coefficients, the panel mixed logit, and error components with an economic interpretation), which builds the models — your EC-MXL among them — that relax the IIA this chapter exposed, turning the red-bus/blue-bus fix into your DCE's estimating model. Tell me when to write it, or request a different chapter.

Chapter 22 — Relaxing IIA: Nested and Mixed Logit

Part V: Random Utility, Discrete Choice, and Stated Preference Econometrics

> **The spine question, applied to this chapter.** Chapter 21 exposed the multinomial logit's one flaw — IIA — and showed that relaxing it means enriching the error structure. This chapter builds the models that do so, culminating in *your* estimating model: the **error-components mixed logit (EC-MXL)**. Your PDF is emphatic that you must understand these deeply — *"Devi saperlo bene, non solo usarlo in Apollo"* — and it lists the decisions that make or break the model: the distributions of the random coefficients (are they *plausible*?), the correlation among them, the panel structure, and the economic interpretation of the error components. Each of these is a choice with real consequences for your WTP and welfare. The spine question here: *when I add random coefficients and error components to relax IIA, do the distributions I chose imply plausible preferences (a cost coefficient of the right sign, WTP without absurd tails), do my error components have an economic meaning I can defend, and does my panel structure correctly reflect that a respondent's tastes persist across their tasks?*

22.0 Conceptual motivation: building your actual model

Your DCE is not estimated with a plain MNL; it is estimated with an **error-components mixed logit**, and your PDF's §3.4 makes clear this is a model you must *understand*, not merely run: *"Devi saperlo bene, non solo usarlo in Apollo. Studia: distribuzioni dei coefficienti casuali; normal, lognormal, triangular e constrained distributions; correlation among random coefficients; panel likelihood; … willingness-to-pay space."* And its critical-judgment list is a checklist of everything that can go wrong: *"l'error component ha interpretazione economica? stai confondendo taste heterogeneity e scale heterogeneity? i coefficienti casuali hanno distribuzioni plausibili?"*

Chapter 21 left you with the diagnosis (IIA fails for your opt-out and your panel taste heterogeneity) and a hint at the cure (correlated errors). This chapter builds the cure in full. There are two routes to relaxing IIA, and you should understand both even though you use one:

- **Nested logit** groups correlated alternatives into *nests*, allowing within-nest correlation while keeping a closed form. It is the transparent, quick fix for structured correlation (the two buses in one nest) and a useful conceptual stepping stone.
- **Mixed logit** lets each respondent have their own random coefficients drawn from a distribution — modeling *taste heterogeneity* and inducing correlation across a respondent's choices — and adds **error components** shared among subsets of alternatives to capture alternative-level correlation (the opt-out). This is your EC-MXL. It has no closed form (Chapter 9) and is estimated by simulated maximum likelihood (Chapter 23), but it is vastly more flexible: McFadden and Train proved the mixed logit can approximate *any* random-utility model arbitrarily well.

The decisions that determine whether your EC-MXL is sound are exactly the ones your PDF flags, and this chapter treats each as a substantive choice with a verified consequence:

- **The distributions of the random coefficients** determine whether your model implies plausible preferences. A *normal* cost coefficient (as we will verify) puts ~9% of respondents at the wrong sign — people who *like* paying more — and produces a heavy WTP tail; a *lognormal* constrains the sign and tames the tail. "I coefficienti casuali hanno distribuzioni plausibili?" is not a formality; it is the difference between a defensible and an absurd model.
- **The error components** must have an *economic interpretation* — an unobserved factor genuinely shared among a subset of alternatives (a status-quo tendency for the opt-out) — not a mechanical fix you cannot explain.
- **The panel structure** — that a respondent's coefficients are drawn *once* and held fixed across their tasks — is what makes the model a random-effects (multilevel) model (Chapter 19) and correctly reflects persistent tastes.

We proceed: the two routes to relaxing IIA; the nested logit (structure, dissimilarity parameter, the red-bus fix); the mixed logit and random coefficients; the *distributions* (with the verified normal-vs-lognormal comparison); correlated coefficients; the *panel* mixed logit; *error components* and their economic interpretation; the EC-MXL as your model; and the scale-versus-preference-heterogeneity distinction (developed fully in Chapter 25). Chapter 23 estimates these by simulation; this chapter specifies and interprets them.

22.1 Prerequisites

From **Chapter 21**: IIA, its derivation, the red-bus/blue-bus failure, and the error-component fix (§21.6). From **Chapter 20**: the random-utility framework and the two normalizations. From **Chapter 19**: random effects, multilevel models, and within/between variation (the mixed logit *is* a multilevel random-coefficients model, §19.7). From **Chapter 5** (§5.7) and **Chapter 9**: the mixed logit as an integral over the coefficient distribution, with no closed form, requiring simulation; and the Cholesky factor (Chapter 1, §1.9) for correlated draws. From **Chapter 15** (§15.5) and **Chapter 2** (§2.11): WTP tails and the ratio pathology.

Same **notation**. New/consolidated symbols: β_n (respondent n's random coefficients), $f(\beta \mid \theta)$ (their distribution with parameters θ — means, covariance), λ (nested-logit dissimilarity/scale parameter), I_B (inclusive value/logsum of nest B), η (error component), Σ (random-coefficient covariance), $L_{ni}(\beta)$ (logit probability given β).

22.2 Two routes to relaxing IIA

IIA is the substitution face of the iid-Gumbel assumption (Chapter 21, §21.5); relaxing it means letting the errors be *correlated* or *heterogeneous*. Two routes:

Route 1 — structured correlation (nested logit). Specify *a priori* which alternatives are correlated by grouping them into **nests**. Alternatives within a nest share an unobserved component; IIA holds within a nest but *not* across nests. Keeps a **closed form** (fast, reliable), at the cost of requiring you to know the correlation structure in advance and to impose it rigidly.

Route 2 — random heterogeneity (mixed logit). Let the taste coefficients *vary randomly* across respondents (random coefficients) and/or add random terms shared among alternatives (error components). This induces correlation flexibly — you do not pre-specify a nesting tree — and models *taste heterogeneity*. It has *no closed form* (an integral over the random terms) and is estimated by simulation, but it is far more flexible and can approximate any RUM.

The two are not rivals so much as points on a spectrum: nested logit imposes a specific, closed-form correlation; mixed logit estimates a flexible, simulated one. Your EC-MXL is Route 2 (with an error-component

ingredient that echoes nesting). Understanding Route 1 first makes Route 2's machinery intuitive.

> **Active recall 22.A.** State the trade-off between the nested logit and the mixed logit as ways to relax IIA. *(Answer: the nested logit imposes a* pre-specified* correlation structure (which alternatives are in which nest) and keeps a *closed-form* choice probability (fast, reliably estimated); the mixed logit estimates a *flexible* correlation via random coefficients/error components (no need to pre-specify nests, and it models taste heterogeneity and approximates any RUM) but has *no closed form* and requires simulated maximum likelihood — flexibility for computational cost.)*

22.3 The nested logit

The structure. Partition the alternatives into mutually exclusive **nests** $B_1, B_2, \ldots$. Alternatives within a nest are allowed to be correlated (they share unobserved features); alternatives in different nests are independent. Each nest B has a **dissimilarity (scale) parameter** $\lambda_B \in (0, 1]$ measuring the *independence* of alternatives within it: $\lambda_B = 1$ means no within-nest correlation (the nest collapses to MNL), and $\lambda_B \to 0$ means the alternatives within the nest are perfect substitutes (maximally correlated). The within-nest correlation is approximately $1 - \lambda_B^2$.

The choice probability. The nested logit factors the probability into a *within-nest* choice and a *nest* choice:

$$P_i = \underbrace{\frac{e^{V_i/\lambda_B}}{\sum_{j \in B} e^{V_j/\lambda_B}}}_{P(i|B)\ \text{(within-nest logit)}} \times \underbrace{\frac{e^{\lambda_B I_B}}{\sum_C e^{\lambda_C I_C}}}_{P(B)\ \text{(nest logit over inclusive values)}}, \qquad I_B = \log \sum_{j \in B} e^{V_j/\lambda_B}.$$

I_B is the **inclusive value** (or **logsum**) — the expected maximum utility available in nest B (Chapter 20, §20.7, Chapter 5 §5.5). The nest-level choice is a logit over these inclusive values. Note the two closed-form logits nested inside each other — hence "nested logit" — and that $\lambda_B = 1$ for all nests recovers the MNL exactly.

Worked Example 22.1 (Nested logit fixes red-bus/blue-bus, verified). Take the Chapter 21 problem — car, blue bus, red bus, all with equal utility — and nest the two buses together (dissimilarity λ) with the car alone. Since $P(\text{car}) = e^{V_{\text{car}}}/(e^{V_{\text{car}}} + e^{\lambda I_{\text{bus}}})$ with $I_{\text{bus}} = \log 2$, we get $P(\text{car}) = 1/(1 + 2^{\lambda})$:

λ	within-nest corr $\approx 1 - \lambda^2$	car	blue	red
1.00 (MNL)	0.00	0.333	0.333	0.333
0.70	0.51	0.381	0.309	0.309
0.50	0.75	0.414	0.293	0.293
0.30	0.91	0.448	0.276	0.276
0.05 (near-perfect subs.)	~1.00	0.491	0.254	0.254

(Verified.) As λ falls (buses more correlated), the car's share climbs back toward its correct $\frac{1}{2}$ and the buses split toward $\frac{1}{4}$ each — the nested logit *fixes* IIA by letting the buses be special substitutes within their nest. This complements Chapter 21's error-component fix (the two are alternative ways to induce the same correlation).

Strengths and limits. The nested logit is fast (closed form), reliably estimated, and gives an interpretable correlation. But it requires you to *specify the nesting tree a priori* — which alternatives are correlated — and imposes a *rigid* structure (a strict partition; an alternative is in exactly one nest). For your unlabeled DCE the natural nesting is not obvious (the opt-out vs the priced alternatives is the most defensible split), and taste heterogeneity across a respondent's tasks is *not* captured by nesting at all. These limits are why environmental DCEs, and yours, generally use the *mixed* logit instead — which needs no pre-specified tree and handles taste heterogeneity.

Stop and think 22.1. For your unlabeled DCE with an opt-out, what is the most defensible nesting structure, and what important source of correlation would nesting still fail to capture? *(Answer: the most defensible nest groups the* priced alternatives* together (correlated by "being a paid option") versus the *opt-out* alone (or vice versa) — capturing the opt-out's special-substitute status. But nesting would *not* capture the correlation induced by *taste heterogeneity across a respondent's repeated tasks* (a respondent's persistent unobserved preferences link their choices), which is a panel/heterogeneity phenomenon, not an alternative-grouping one — this is why the mixed logit, which models random taste heterogeneity, is needed, §22.4–§22.7.)*

22.4 The mixed logit and random coefficients

The idea. Instead of pre-specifying which alternatives are correlated, let the *taste coefficients vary randomly across respondents*. Each respondent n has their own coefficient vector β_n drawn from a population distribution:

$$U_{nj} = \mathbf{x}'_{nj}\beta_n + \varepsilon_{nj}, \qquad \beta_n \sim f(\beta \mid \theta),$$

with ε_{nj} still iid Gumbel *conditional on* β_n, and θ the parameters of the coefficient distribution (means, variances, covariances) that you estimate. This is exactly the multilevel random-coefficients model of Chapter 19 (§19.7): each respondent has their own "random slopes."

The choice probability (an integral, Chapter 5 §5.7, Chapter 9). Conditional on β_n, the choice probability is the ordinary logit $L_{nj}(\beta_n) = e^{\mathbf{x}'_{nj}\beta_n} / \sum_k e^{\mathbf{x}'_{nk}\beta_n}$. The *unconditional* probability integrates over the coefficient distribution:

$$P_{nj} = \int L_{nj}(\beta)\, f(\beta \mid \theta)\, d\beta.$$

This has **no closed form** (a logit averaged over a distribution is not a logit) — hence simulated maximum likelihood (Chapter 23). But it buys three things at once:

1. **It relaxes IIA.** Random coefficients induce correlation across alternatives (two alternatives with similar attributes are affected similarly by a respondent's random tastes), so substitution is no longer proportional — the red-bus/blue-bus problem is handled without nesting.
2. **It models taste heterogeneity.** The *variances* of the random coefficients quantify *how much respondents differ* in each taste — a substantive result (preference heterogeneity), and one of the main reasons to estimate a mixed logit.
3. **It approximates any RUM.** McFadden and Train (2000) proved that a mixed logit with a suitable coefficient distribution can approximate *any* random-utility model to any degree of accuracy — so it is not a special case but a near-universal framework.

The mixed logit vs the fixed-coefficient MNL. The MNL forces *one* β on everyone (all heterogeneity dumped into ε, Chapter 20 §20.4); the mixed logit *pulls the taste heterogeneity out of* ε *and into the distribution of* β_n. This is not a cosmetic change: it reclassifies systematic between-respondent variation from "noise" to "structured heterogeneity," corrects the MNL's misspecification, relaxes IIA, and yields the heterogeneity variances as headline estimates.

Active recall 22.B. State the three things a mixed logit buys over the MNL, and what its no-closed-form integral requires. *(Answer: (i) it relaxes IIA (random coefficients induce correlation, non-proportional substitution); (ii) it models taste heterogeneity (the coefficient variances quantify how much respondents differ); (iii) it approximates any RUM (McFadden–Train). Its choice probability is an integral of the logit over the coefficient distribution with no closed form, requiring simulated maximum likelihood — Chapter 23.)*

22.5 The distributions of the random coefficients: plausibility matters

Your PDF asks *"i coefficienti casuali hanno distribuzioni plausibili?"* — and this is not a technical footnote but one of the most consequential choices in the whole model, because the distribution you assume for a coefficient *is* an assumption about the population of preferences, and an ill-chosen one implies absurd behavior and detonates your WTP.

The menu. Each random coefficient needs a distribution, chosen for what it implies:

- **Normal** $N(\bar{\beta}, \sigma^2)$ — symmetric, unbounded. Convenient, but it puts mass on *both signs*, which is wrong for a coefficient whose sign is known (a cost coefficient should always be negative; a "good" attribute's coefficient should always be positive).
- **Lognormal** — support $(0, \infty)$ (or $(-\infty, 0)$ by negation), so it *constrains the sign.* The standard choice for a cost coefficient (always negative) or a strictly-desirable attribute. Its cost is a *right skew* and a potentially heavy tail (Chapter 15).
- **Triangular / uniform** — bounded distributions, useful when you want a coefficient confined to a range (e.g., a constrained triangular that keeps the sign) with lighter tails than the lognormal.
- **Constrained / bounded distributions** — deliberately restrict the support to enforce economic plausibility (sign, bounded WTP).

Worked Example 22.2 (Distribution choice governs plausibility and WTP tails, verified). Take a random cost coefficient with a central value around -0.20, and compute the implied WTP $= \beta_{\text{attr}}/(-\beta_{\text{cost}})$ with $\beta_{\text{attr}} = 0.5$:

Distribution	Pr(cost coef > 0) (wrong sign)	median WTP	99th-percentile WTP
Normal$(-0.20, 0.15)$	**0.091**	2.13	**56.2**
Lognormal (median 0.20)	**0.000**	2.50	**10.1**

(Verified.) The **normal cost coefficient puts 9% of respondents at the wrong sign** — the model says nearly one in ten people *like* paying more, which is economically absurd — and it produces a WTP whose 99th percentile is €56 (a fat tail driven by cost draws near zero, the ratio pathology of Chapters 2, 15). The **lognormal never flips sign** (it constrains the cost coefficient to be negative) and tames the WTP tail to a 99th percentile of €10. **The same data, two distributions, wildly different plausibility.** This is why "i coefficienti casuali hanno distribuzioni plausibili?" is a make-or-break question: a normal on the cost coefficient can produce wrong-sign respondents and an absurd WTP distribution, and the lognormal (or a constrained distribution, or WTP-space estimation — Chapter 24) is the fix.

The guidance. Use a *sign-constrained* distribution (lognormal, constrained triangular) for any coefficient whose sign is known on theory — above all the **cost coefficient**, whose near-zero and wrong-sign draws are the source of WTP instability (Chapters 2, 15, 24). Reserve the normal for coefficients that genuinely can take either sign (some respondents like the attribute, others dislike it). And recognize that the choice interacts with the space you estimate in: **WTP-space** estimation (Chapter 24) sidesteps the ratio pathology by parameterizing WTP directly with a sensible distribution, which is often the cleanest solution for your DCE.

> **Stop and think 22.2.** Your co-author estimates a mixed logit with a *normal* distribution on the cost coefficient and reports a mean WTP with a huge confidence interval. Using §22.5, diagnose the likely cause and propose two fixes. *(Answer: the normal cost coefficient places mass near (and across) zero, so ~9% of respondents have a wrong-sign cost coefficient and WTP* $= \beta_{attr}/(-\beta_{cost})$ *has a heavy tail (near-zero denominators), inflating the mean and its interval (Chapters 2, 15). Fixes: (i) impose a* sign-constrained* distribution on the cost coefficient (lognormal or constrained

triangular) so it is always negative and the tail is tamed; (ii) estimate in *WTP space* (Chapter 24), parameterizing WTP directly with a sensible distribution, avoiding the ratio entirely.)*

22.6 Correlated random coefficients

The mixed logit can allow the random coefficients to be *correlated* — a respondent who cares a lot about biodiversity may also care a lot about water quality. This is captured by a full **covariance matrix** Σ of the random coefficients, generated via its **Cholesky factor** (Chapter 1, §1.9): $\beta_n = \bar{\beta} + \mathbf{L}\eta_n$, with η_n independent draws and $\mathbf{L}\mathbf{L}' = \Sigma$.

Allowing correlation matters for two reasons: (i) it is often *realistic* (tastes for related attributes co-move), and forcing independence (a diagonal Σ) is a restriction that can bias the variances and the substitution patterns; (ii) it is exactly the multilevel random-coefficient covariance of Chapter 19 (§19.7), and the off-diagonal terms are estimable parameters (Cholesky entries) that describe how preferences co-vary. The cost is more parameters (a full Σ for K random coefficients has $K(K+1)/2$ terms), which strains identification and estimation (Chapters 13, 23), so a common practice is to allow correlation among a *subset* of coefficients or to test whether the off-diagonals are jointly zero (an LR test, with the boundary caveat of Chapter 12 §12.9).

> **Active recall 22.C.** What does allowing correlated random coefficients add, and how is it generated computationally? *(Answer: it lets respondents' tastes co-vary (e.g., caring about biodiversity and water quality together) via a full covariance matrix Σ of the random coefficients — more realistic than forcing independence, and its off-diagonals are estimable. Computationally it is generated by $\beta_n = \bar{\beta} + \mathbf{L}\eta_n$ with $\mathbf{L}$ the Cholesky factor of Σ and η_n independent draws — the same Cholesky construction as Krinsky–Robb, Chapter 1 §1.9.)*

22.7 The panel mixed logit: tastes persist across a respondent's tasks

This is the structure most important for your DCE, and the one your PDF names ("panel likelihood"). In a stated-preference experiment each respondent answers *several* choice tasks, and the mixed logit must reflect that a respondent's tastes are the *same* across their tasks.

The panel structure. Each respondent draws their coefficients β_n *once* — and those coefficients are **held fixed across all their choice tasks**. This is what makes it a *panel* (repeated-observations) model: the shared β_n induces correlation across a respondent's choices (Chapter 19), which is exactly the taste-heterogeneity source of IIA violation (Chapter 21, §21.8). The **panel choice probability** is the integral of the *product* of the per-task logits over the coefficient distribution:

$$P_n = \int \Big[\prod_{t=1}^{T_n} L_{n,y_{nt}}(\beta)\Big]\, f(\beta \mid \theta)\, d\beta,$$

where the product runs over respondent n's T_n tasks *inside* the integral (because β is common to them). Contrast the *cross-sectional* mixed logit, which would (wrongly) draw a fresh β for each task and put the product *outside* the integral — treating a respondent's tasks as independent people, ignoring the persistence of tastes.

Why this matters. The panel structure is the correct model of stated-preference data and is what your PDF means by "panel mixed logit." Getting it right:

- **Correctly reflects persistent preferences.** A respondent's biodiversity taste is the same in task 1 and task 8; the panel structure encodes this.

- **Is the source of the IIA relaxation across tasks.** The shared β_n correlates a respondent's choices — precisely the correlation the MNL's iid assumption forbids (Chapter 21, §21.8).
- **Changes the likelihood and the standard errors.** The product-inside-the-integral panel likelihood is what you maximize; treating tasks as independent (cross-sectional) both mis-specifies the model and understates uncertainty (the clustering/design-effect issue of Chapters 17, 19).

The connection to Chapter 19. The panel mixed logit is a **multilevel random-effects model** (Chapter 19, §19.7): respondents are the units, tasks the repeated observations, and β_n the random slopes. Its variance components (the covariance Σ) are the preference heterogeneity. Recognizing this unifies your DCE with the random-effects logic of Part IV — "panel mixed logit" and "random effects" are the same idea applied to choice.

> **Stop and think 22.3.** Why must a respondent's random coefficients be held *fixed* across their choice tasks (panel), rather than redrawn for each task (cross-sectional)? What goes wrong if you redraw? *(Answer: a respondent's tastes are the* same* across their tasks (persistent preferences), so their coefficients must be drawn once and shared across tasks — this is what makes it a panel model and induces the within-respondent correlation that relaxes IIA. Redrawing β per task treats each of a respondent's tasks as an *independent person*, wrongly ignoring the persistence of tastes: it mis-specifies the likelihood (product outside the integral), fails to capture the across-task correlation, and understates uncertainty — Chapters 17, 19.)*

22.8 Error components and their economic interpretation

The final ingredient of your EC-MXL is the **error component** — and your PDF insists it must have an *economic interpretation* ("l'error component ha interpretazione economica?"), not be a mechanical add-on you cannot justify.

What an error component is. An error component is an *additional random term* in utility, shared among a *subset* of alternatives, that induces correlation without necessarily being attached to an observed attribute:

$$U_{nj} = \mathbf{x}'_{nj}\beta_n + \sum_m \eta_{nm}\, d_{jm} + \varepsilon_{nj},$$

where η_{nm} is an error component (a random draw, usually $N(0, \sigma_m^2)$) and $d_{jm} = 1$ if alternative j belongs to the subset m that shares component m. Alternatives sharing an error component are *correlated* (both contain η_{nm}) — exactly the red-bus/blue-bus shared-"bus" component of Chapter 21 (§21.6), now a formal model ingredient. Error components are the mixed-logit way of building *nesting-like* correlation flexibly.

The economic interpretation (why it matters). An error component must correspond to a *real, unobserved factor common to the subset of alternatives it links.* Legitimate interpretations in your setting:

- **An opt-out error component** captures an unobserved *status-quo tendency* or *general reluctance to pay* that shifts the utility of the opt-out (or, symmetrically, of all the priced alternatives together) — a genuine behavioral factor, correlating the opt-out's utility across a respondent's tasks and relaxing IIA between the opt-out and the priced options (Chapter 20 §20.8). This is the most common and defensible error component in an environmental DCE, and it is likely *the* error component in your EC-MXL.
- **A "priced-alternatives" error component** captures an unobserved factor common to *all* the paid options (e.g., a common perception of the program) that the opt-out lacks.
- **A nest-like component** links a subset of alternatives that share an unobserved characteristic.

The requirement. You must be able to *say what the error component represents* — which unobserved factor, common to which alternatives, and why it plausibly exists. An error component added purely to improve fit, with no economic story, is a red flag: it may be soaking up misspecification (a wrong functional form, an omitted attribute) rather than a real correlation, and its welfare implications are then uninterpretable. Your PDF's question is a demand for this justification: *an error component with an economic interpretation is a*

model of a real behavioral factor; one without is a mechanical patch. Report what your error components mean.

> **Active recall 22.D.** What does an error component do, and what must you be able to say about it for it to be defensible in your EC-MXL? *(Answer: an error component is a random term shared among a subset of alternatives that makes their utilities* correlated* (relaxing IIA between them), like the shared "bus" component in red-bus/blue-bus. To be defensible it must have an *economic interpretation* — you must be able to name the real, unobserved factor common to those alternatives (e.g., a status-quo/reluctance-to-pay tendency for the opt-out) and why it plausibly exists; an error component with no economic story may be soaking up misspecification and yields uninterpretable welfare.)*

22.9 The EC-MXL: your model, assembled

Your DCE's estimating model — the **error-components mixed logit** — combines the two IIA-relaxing devices of this chapter:

$$U_{nj} = \mathbf{x}'_{nj}\beta_n + \sum_m \eta_{nm} d_{jm} + \varepsilon_{nj}, \qquad \beta_n \sim f(\beta \mid \theta), \quad \eta_{nm} \sim N(0, \sigma_m^2),$$

with β_n and η_{nm} drawn *once per respondent* (panel). It relaxes IIA on **both** dimensions your PDF flagged (Chapter 21, §21.8):

- **Random coefficients** β_n capture *taste heterogeneity* and induce correlation across a respondent's tasks (the panel dimension) — relaxing IIA across tasks.
- **Error components** η_{nm} capture *alternative-level correlation* (the opt-out) — relaxing IIA across alternatives.

This is why your model is an *error-components mixed logit* and not a plain mixed logit or a plain nested logit: it needs *both* the taste heterogeneity (for the panel/task correlation) and the error components (for the opt-out/alternative correlation). Estimated in **WTP space** (Chapter 24) with **simulated maximum likelihood** and Halton draws (Chapter 23), it is exactly the specification your PDF describes. Assembling it from its parts — nested-logic correlation (error components) plus random-effects heterogeneity (random coefficients) plus the panel structure — is the payoff of this chapter: you can now say *what each ingredient does and why it is there.*

Model comparison. Whether the added complexity is warranted is tested by comparing the EC-MXL to the MNL (and to intermediate models) via AIC, BIC, and likelihood-ratio tests (Chapter 12) — *with the boundary correction* for the variance restrictions (testing whether a random-coefficient variance or an error-component variance is zero is a boundary test, Chapter 12 §12.9, where the naive χ^2 makes you *miss* heterogeneity). Report the comparison: a significant improvement (boundary-corrected) and interpretable variance components justify the EC-MXL; if the MNL and EC-MXL agree on WTP and welfare, the simpler model may suffice.

> **Stop and think 22.4.** Explain, in one paragraph, why your model is specifically an *error-components mixed logit* rather than just a mixed logit or just a nested logit. *(Answer: your DCE has* two* distinct sources of IIA violation (Chapter 21, §21.8): taste heterogeneity across a respondent's tasks, and the opt-out's correlation with the choice. A plain *mixed logit* (random coefficients) handles the first (taste heterogeneity, panel correlation) but not naturally the second; a plain *nested logit* handles the second (grouping the opt-out) but not the first (taste heterogeneity across tasks). The *error-components mixed logit* combines both — random coefficients for the taste heterogeneity/panel correlation and error components for the opt-out/alternative correlation — so it relaxes IIA on both dimensions at once, which is why your model needs both ingredients.)*

22.10 Scale heterogeneity versus preference heterogeneity (introduction)

A crucial caveat that Chapter 25 develops fully but that belongs here because it bears on interpreting the mixed logit's random coefficients. The mixed logit models **preference heterogeneity** — respondents differ in *tastes* (β_n varies). But there is a *different* source of between-respondent variation: **scale heterogeneity** — respondents differ in *how consistently they choose* (their error variance / scale μ_n varies, Chapter 21 §21.4). These are easily confused, and confusing them (your PDF's "stai confondendo taste heterogeneity e scale heterogeneity?") is a serious error:

- A respondent with a *larger scale* μ_n (less noise) has *uniformly larger* effective coefficients — which a mixed logit with random coefficients will attribute to *stronger preferences* if scale is not modeled separately.
- The variance the mixed logit estimates as "preference heterogeneity" therefore **conflates genuine taste variation with scale (consistency) variation**.

The **generalized multinomial logit (G-MNL)** and scaled/heteroskedastic mixed logits explicitly separate a scale factor from the preference coefficients, allowing you to distinguish the two. For your work — where cost-vector arms or experience groups may differ in *scale* (consistency) rather than *preference* — this distinction is central, and Chapter 25 is devoted to it. Here the point is a warning: **the random-coefficient variances of your EC-MXL are "preference heterogeneity" only if scale is homogeneous or separately modeled**; otherwise they mix taste and scale.

> **Active recall 22.E.** Your EC-MXL estimates a large random-coefficient variance and you report "substantial preference heterogeneity." What alternative source could this variance reflect, and what would separate them? *(Answer: it could reflect* scale* heterogeneity — respondents differing in choice *consistency* (error variance / μ_n), which makes effective coefficients vary uniformly and masquerades as taste variation — rather than genuine *preference* heterogeneity. Separating them requires a model that includes a scale factor distinct from the preference coefficients (a generalized/scaled MNL, or a scale-heterogeneity mixed logit); otherwise the estimated "preference heterogeneity" conflates taste and scale — Chapter 25.)*

22.11 Failure modes

F1. Implausible random-coefficient distributions. A normal on a sign-known coefficient (cost), producing wrong-sign respondents and heavy WTP tails. *Fix:* sign-constrained distributions (lognormal, constrained triangular) or WTP space (§22.5; Chapter 24).

F2. Error components without economic interpretation. Adding error components to improve fit with no behavioral story. *Fix:* justify what each error component represents; suspect misspecification if you cannot (§22.8).

F3. Cross-sectional instead of panel mixed logit. Redrawing coefficients per task, ignoring persistent tastes. *Fix:* panel structure — coefficients fixed across a respondent's tasks (product inside the integral) (§22.7).

F4. Confusing taste and scale heterogeneity. Reading random-coefficient variance as pure preference heterogeneity when scale varies. *Fix:* model scale separately (G-MNL); interpret cautiously (§22.10; Chapter 25).

F5. Naive χ^2 for variance/component tests. Testing whether a random-coefficient or error-component variance is zero against the wrong distribution. *Fix:* boundary-corrected LR test (mixture of χ^2) — else you miss heterogeneity (§22.9; Chapter 12 §12.9).

F6. Over-parameterized covariance. A full Σ that is not identified with the data at hand. *Fix:* allow correlation among a subset; test off-diagonals; watch identification (§22.6; Chapter 13).

F7. Mis-specified nesting / forced structure. A nested logit with an indefensible or arbitrary tree. *Fix:* justify the nests; prefer mixed logit where the structure is unclear (§22.3).

F8. Treating the EC-MXL as a black box. Running it in Apollo without understanding what each ingredient does. *Fix:* know what the random coefficients, error components, and panel structure each contribute (this chapter); your PDF's "non solo usarlo in Apollo."

22.12 Diagnostic questions for your own papers

1. **Distributions.** Is each random coefficient's distribution *plausible* (sign-constrained where the sign is known, especially the cost coefficient)? What fraction of wrong-sign draws does it imply? *(§22.5, F1.)*
2. **Error components.** Does each error component have an *economic interpretation* (a named unobserved factor common to a subset of alternatives)? *(§22.8, F2.)*
3. **Panel.** Are coefficients held fixed across a respondent's tasks (panel), not redrawn per task? *(§22.7, F3.)*
4. **Scale vs preference.** Could my "preference heterogeneity" variance reflect scale heterogeneity? Have I considered a scaled model? *(§22.10, F4; Chapter 25.)*
5. **Correlation.** Have I allowed (and tested) correlation among random coefficients where realistic, without over-parameterizing? *(§22.6, F6.)*
6. **Model comparison.** Is the EC-MXL justified over the MNL by boundary-corrected LR / AIC / BIC, with interpretable variance components? *(§22.9, F5.)*
7. **Understanding.** Can I say what each ingredient (random coefficients, error components, panel structure) contributes, not just that Apollo estimated it? *(All; F8.)*

22.13 Exercises

(A) Comprehension.

A1. Explain how the nested logit relaxes IIA, and what the dissimilarity parameter λ means (including $\lambda = 1$).

A2. Why does a *normal* distribution on the cost coefficient produce implausible results, and what distribution fixes it?

A3. State the difference between the panel and cross-sectional mixed logit, and why the panel version is correct for a DCE.

(B) Derivation / reasoning.

B1. Write the nested-logit choice probability with its inclusive value, and show that $\lambda = 1$ recovers the MNL.

B2. Write the panel mixed-logit choice probability, and explain why the product over tasks is *inside* the integral (and what changes if it is outside).

B3. Show, with a shared error component, how two alternatives become correlated, and give the economic interpretation of the error component for an opt-out.

(C) Research application / critique.

C1. Your EC-MXL uses a normal distribution on the cost coefficient and reports a mean WTP of €140 with a 95% CI of €[−20, €300]. Diagnose the problem and specify two fixes.

C2. A referee asks what your opt-out error component *means*. Provide an economic interpretation and explain how it relaxes IIA.

C3. Explain to a co-author why your model is an *error-components mixed logit* and not just a mixed logit, and why the panel structure matters.

22.14 Solutions

A1. The nested logit groups correlated alternatives into nests; alternatives within a nest share an unobserved component, so IIA holds *within* a nest but not *across* nests — substitution between within-nest alternatives is stronger than proportional. The dissimilarity parameter $\lambda_B \in (0, 1]$ measures the *independence* within nest B: $\lambda_B = 1$ means no within-nest correlation (the nest collapses to MNL, IIA holds everywhere), and $\lambda_B \to 0$ means within-nest alternatives are perfect substitutes (maximally correlated); within-nest correlation $\approx 1 - \lambda_B^2$.

A2. A normal distribution is symmetric and unbounded, so it puts probability on *both* signs of the cost coefficient — implying that some respondents (verified ~9% for $N(-0.20, 0.15)$) have a *positive* cost coefficient, i.e., they prefer paying more, which is economically absurd; and near-zero cost draws make WTP $= \beta_{\text{attr}}/(-\beta_{\text{cost}})$ heavy-tailed (99th percentile €56 vs €10 for a lognormal). A *lognormal* (or constrained) distribution constrains the sign (always negative), eliminating wrong-sign respondents and taming the WTP tail.

A3. In the *panel* mixed logit, each respondent's coefficients are drawn *once* and held fixed across all their choice tasks (persistent tastes), so the product of per-task logits is inside the integral over the coefficient distribution — inducing within-respondent correlation. In the *cross-sectional* mixed logit, a fresh coefficient is (implicitly) drawn per task, treating a respondent's tasks as independent people. The panel version is correct for a DCE because a respondent's tastes are the same across their tasks; the cross-sectional version mis-specifies the likelihood, misses the across-task correlation, and understates uncertainty.

B1. $P_i = \dfrac{e^{V_i/\lambda_B}}{\sum_{j\in B} e^{V_j/\lambda_B}} \cdot \dfrac{e^{\lambda_B I_B}}{\sum_C e^{\lambda_C I_C}}$ with $I_B = \log \sum_{j\in B} e^{V_j/\lambda_B}$. If $\lambda_B = 1$ for all nests: $I_B = \log \sum_{j\in B} e^{V_j}$, so $e^{\lambda_B I_B} = \sum_{j\in B} e^{V_j}$, and the nest probability becomes $\sum_{j\in B} e^{V_j} / \sum_m e^{V_m}$; multiplying by the within-nest $e^{V_i} / \sum_{j\in B} e^{V_j}$ gives $P_i = e^{V_i} / \sum_m e^{V_m}$ — the MNL. So $\lambda = 1$ recovers the multinomial logit (no within-nest correlation).

B2. $P_n = \int \left[\prod_{t=1}^{T_n} L_{n,y_{nt}}(\beta)\right] f(\beta \mid \theta)\, d\beta$. The product over the respondent's T_n tasks is *inside* the integral because the *same* β governs all of a respondent's tasks (persistent tastes) — the integral averages the joint probability of the respondent's whole sequence of choices over the coefficient distribution. If the product were *outside* the integral, $\prod_t \int L_{n,y_{nt}}(\beta) f\, d\beta$, each task would be integrated separately (a fresh draw per task), treating the tasks as independent respondents — losing the within-respondent correlation and mis-specifying the panel likelihood.

B3. With $U_{nj} = \mathbf{x}'_{nj}\beta + \eta_n d_j + \varepsilon_{nj}$ where $d_j = 1$ for the two linked alternatives and $\eta_n \sim N(0, \sigma^2)$ is shared between them, the two alternatives' utilities both contain η_n, so $\text{Cov}(U_{n1}, U_{n2}) = \sigma^2 > 0$ — they are correlated (and correlation grows with σ^2). Economic interpretation for an opt-out: the error component η_n represents an unobserved *status-quo tendency* or *general reluctance to pay* that shifts the opt-out's utility (common across the respondent's tasks); it makes the opt-out a special substitute (people prone to opting out do so consistently), relaxing IIA between the opt-out and the priced alternatives.

C1. The problem: a *normal* cost coefficient places mass near and across zero, so ~9% of respondents have a wrong-sign (positive) cost coefficient and WTP $= \beta_{\text{attr}}/(-\beta_{\text{cost}})$ is heavy-tailed with wrong-sign draws, producing an implausibly large mean (€140) and a symmetric interval spanning zero (€[−20, €300]) — the ratio pathology (Chapters 2, 15). Two fixes: (i) impose a *sign-constrained* distribution on the cost coefficient (lognormal or constrained triangular) so it is always negative and the WTP tail is tamed; (ii) estimate

in *WTP space* (Chapter 24), parameterizing WTP directly with a sensible (e.g., bounded or lognormal) distribution, avoiding the ratio entirely. Report the median and percentile interval (Krinsky–Robb), not the mean and symmetric CI.

C2. The opt-out error component represents an unobserved, respondent-specific factor common to the opt-out across their tasks — a *status-quo tendency* or *reluctance to pay*: some respondents are systematically inclined to choose "none/keep the status quo" regardless of the priced alternatives' attributes. Modeling this as a shared random term on the opt-out makes the opt-out's utility *correlated* (a respondent prone to opting out does so consistently), so the opt-out is a special substitute rather than a neutral iid alternative — relaxing IIA between the opt-out and the priced options and correctly capturing that "opters-out" cluster together. This is an economically interpretable component (not a mechanical fit patch), which is what a defensible EC-MXL requires.

C3. The model is an *error-components mixed logit* because the DCE has *two* sources of IIA violation: (i) *taste heterogeneity* across a respondent's repeated tasks — handled by *random coefficients* (each respondent's own tastes, fixed across tasks); and (ii) the *opt-out's correlation* with the choice — handled by an *error component* on the opt-out. A plain mixed logit would handle (i) but not naturally (ii); a plain nested logit would handle (ii) but not (i); the EC-MXL does both. The *panel* structure matters because a respondent's tastes are the same across their tasks: holding β_n fixed across tasks (product inside the integral) is what induces the within-respondent correlation, correctly models persistent preferences, and gives honest standard errors — treating tasks as independent (cross-sectional) would mis-specify the model and understate uncertainty.

22.15 Spaced review prompts

+1 day. From memory, write the nested-logit probability (with inclusive value) and the panel mixed-logit probability; explain the dissimilarity parameter and why the product is inside the integral.

+1 week. Explain why the distribution of a random coefficient matters (normal vs lognormal for the cost coefficient — wrong-sign draws, WTP tails), what an error component is and its economic interpretation, and why your model is an EC-MXL.

+1 month. For your own EC-MXL, audit each random-coefficient distribution for plausibility (sign, tails), justify each error component's economic meaning, confirm the panel structure, and consider whether "preference heterogeneity" could be scale. Close with the spine sentence about whether your model's ingredients are defensible.

22.16 How this chapter connects to the rest of the book

- **Chapter 21** exposed IIA and the error-component fix; this chapter builds the models that relax it. **Chapter 19 (§19.7)** is the multilevel random-effects logic the mixed logit instantiates (random slopes = random coefficients; panel = repeated observations); **Chapter 5 (§5.7)** and **Chapter 9** are the no-closed-form integral and the simulation it requires; **Chapter 1 (§1.9)** the Cholesky for correlated coefficients.
- **Chapter 23 (SML)** estimates these models — Halton draws, simulation noise, convergence, local maxima (Chapter 13); **Chapter 24 (WTP space)** is the reparameterization that tames the distribution/WTP-tail problem of §22.5; **Chapter 25 (scale vs preference heterogeneity)** develops the §22.10 confound and the G-MNL.
- **Chapter 12 (§12.9)** is the boundary-corrected test for the variance components (§22.9); **Chapter 41 (MDCEV welfare)** uses the logsum whose substitution these models correct; **Chapter 37 (design)** provides the attribute variation and opt-out that identify the random coefficients and error components.

This chapter builds your DCE's estimating model — the EC-MXL — from its parts, turning "relax IIA" into a specific, interpretable specification whose every ingredient (random coefficients, distributions, correlation, panel structure, error components) you can now justify.

22.17 Research translation task

For your **EC-MXL**, produce a two-to-three-page specification note:

1. **Random coefficients and distributions.** List each random coefficient and its distribution; justify the choice (sign-constrained for the cost coefficient), and report the implied fraction of wrong-sign draws and the WTP-tail behavior for the current specification.
2. **Correlation.** State whether random coefficients are allowed to correlate, and justify (realism vs identification).
3. **Panel structure.** Confirm coefficients are fixed across a respondent's tasks; write the panel likelihood.
4. **Error components.** For each error component, give its *economic interpretation* (the unobserved factor, the alternatives it links, why it exists) — especially the opt-out.
5. **Scale vs preference.** Note whether your "preference heterogeneity" could reflect scale, and whether a scaled model is warranted (Chapter 25).
6. **Model comparison.** Justify the EC-MXL over the MNL via boundary-corrected LR / AIC / BIC, with interpretable variance components.
7. **Close** with the spine sentence: *"My EC-MXL relaxes IIA via [random coefficients / error components]; its distributions imply [plausible/implausible] preferences ([wrong-sign fraction]); its error components mean [·]; and its panel structure reflects persistent tastes."*

Carry this into Chapter 23 (simulated estimation), Chapter 24 (WTP space), and Chapter 25 (scale vs preference).

22.18 Bibliography

Core. - Train, *Discrete Choice Methods with Simulation*, the mixed-logit and nested-logit chapters (specification, distributions, panel structure, error components, and the approximation theorem). The single most relevant source; read alongside Chapters 21, 23. Freely available — *verify link.* - McFadden & Train (2000), "Mixed MNL Models for Discrete Response," *Journal of Applied Econometrics*. The foundational paper: mixed logit relaxes IIA and approximates any RUM (§22.4). *(Verify volume/pages.)* - Hensher & Greene (2003), "The Mixed Logit Model: The State of Practice," *Transportation*. The practical guide to distributions, correlation, and panel structure (§22.5–§22.7). *(Verify volume/pages.)*

Applied. - Hensher, Rose & Greene, *Applied Choice Analysis*, on nested and mixed logit specification, distributions, and error components in practice. *(Verify edition.)* - Scarpa, Thiene & Train (2008), and Hess & Train, on distributions for random coefficients (lognormal, constrained) and WTP-space estimation in environmental valuation (§22.5). *(Verify exact citations.)* - The **Apollo** manual (Hess & Palma) for implementing mixed logit, error components, panel structure, and correlated coefficients — the tool your PDF names. *(Verify current version.)*

Advanced. - Fiebig, Keane, Louviere & Wasi (2010), "The Generalized Multinomial Logit Model," *Marketing Science*. The G-MNL that separates scale from preference heterogeneity (§22.10; Chapter 25). *(Verify volume/pages.)* - Revelt & Train (1998), "Mixed Logit with Repeated Choices," *Review of Economics and Statistics*. The panel mixed logit (§22.7). *(Verify volume/pages.)* - Ben-Akiva, McFadden & Train, "Foundations of Stated Preference Elicitation," and the nested-logit foundations in McFadden (1978). *(Verify exact citations.)*

Entries marked "verify" are pointers; confirm editions, pages, and the McFadden–Train / Fiebig et al. / Revelt–Train citations before using them, per the book's no-unverified-citation rule.

End of Chapter 22. Next in Part V: Chapter 23 — Simulated Maximum Likelihood (the simulated likelihood, Halton/Sobol/pseudo-random draws, simulation noise, convergence, and model comparison via AIC/BIC/LR), which estimates the mixed logit and EC-MXL of this chapter and confronts the draw/seed/convergence diagnostics your PDF calls decisive. Tell me when to write it, or request a different chapter.

Chapter 23 — Simulated Maximum Likelihood

Part V: Random Utility, Discrete Choice, and Stated Preference Econometrics

> **The spine question, applied to this chapter.** Chapter 22 specified your EC-MXL; this chapter estimates it. The mixed logit's choice probability is an integral with no closed form, so the likelihood is *simulated* — averaged over draws of the random coefficients — and maximized. Your PDF is precise about the stakes: *"Nel DCE usi EC-MXL con simulated maximum likelihood e 1000 Halton draws: per giudicare il modello devi sapere se il risultato è robusto al numero di draw, ai seed, alla specificazione delle distribuzioni e agli starting values."* Simulated maximum likelihood works — but it introduces a second layer of error (simulation noise) on top of sampling error, and the number and type of draws, the seed, and the starting values all move the answer. This chapter builds the simulated likelihood, compares draw types, quantifies simulation noise, handles the non-concave convergence, and compares models. The spine question here: *when I report an EC-MXL estimate, have I confirmed it is stable across the number of draws, the seed, and the draw type — or am I reporting a number that a co-author would fail to reproduce?*

23.0 Conceptual motivation: estimating a model with no formula

You have met the ingredients in three earlier chapters, and this chapter assembles them for the specific task of estimating your EC-MXL. Chapter 9 built simulation-based inference (Monte Carlo, Halton, the Jensen bias, the draws–precision–time trade-off); Chapter 13 built the craft of computing an MLE (convergence, local maxima, starting values); Chapter 22 specified the mixed logit and EC-MXL. Now we put them together, because your PDF is emphatic that estimation is a skill you must *own*, not delegate to software: *"Devi saperlo bene, non solo usarlo in Apollo."*

The core difficulty is that the mixed-logit choice probability is an integral over the random coefficients (Chapters 5 §5.7, 9, 22):

$$P_n = \int \Big[\prod_t L_{n,y_{nt}}(\beta)\Big] f(\beta \mid \theta)\, d\beta,$$

which has **no closed form**. To evaluate the likelihood at all — and therefore to estimate the model — the integral is replaced by a *simulated average* over draws of β. This is **simulated maximum likelihood (SML)**, and it is how your 1000-Halton-draw EC-MXL is estimated. But simulation is not free: it adds **simulation error** to the sampling error, and — as this chapter will verify with a live estimation — *the number and type of draws, the seed, and the starting values all move your estimates and your WTP.* Your PDF's checklist is exactly the set of things you must verify:

- **The number of draws.** Too few and the estimates carry simulation noise (and a Jensen bias, Chapter 9); you must check the answer is stable as draws increase.

- **The seed.** With random draws, a different seed gives a different answer — pure simulation noise. A well-resolved model is seed-stable; if your WTP moves with the seed, you have too few draws.
- **The draw type.** Halton and Sobol draws achieve far better accuracy per draw than pseudo-random draws (why you use 1000 *Halton*, not 1000 random), but degrade in high dimensions, requiring scrambled Halton or Sobol.
- **The starting values.** The mixed-logit likelihood is *non-concave* (Chapters 13, 22), so different starts can reach different local maxima; you must use multiple starts.

We proceed: the simulated likelihood written out; the SML algorithm; draw types for choice models (and the high-dimensional caveat); simulation noise in the estimates, standard errors, and WTP (with a verified live demonstration); convergence, local maxima, and starting values for the mixed logit; the robustness protocol your PDF prescribes; and model comparison via log-likelihood, AIC, BIC, and LR tests (with the boundary correction). This is the chapter where your EC-MXL becomes a number you can defend — or discover you cannot yet trust.

23.1 Prerequisites

From **Chapter 22**: the mixed-logit and EC-MXL specification, the panel likelihood, and the random-coefficient distributions. From **Chapter 9**: Monte Carlo integration, the $1/\sqrt{R}$ (random) vs ~$1/R$ (Halton) error rates, the Jensen bias of the simulated log-likelihood (§9.4), holding draws fixed across optimization, and the draw/seed robustness diagnostics. From **Chapter 13**: convergence (real vs cosmetic), local maxima, multiple starting values, and the three-culprit framework (identification vs local maxima vs simulation noise). From **Chapter 12**: the LR test and the boundary problem for variance restrictions (§12.9). From **Chapter 11**: the likelihood/MLE framework.

Same **notation**. New/consolidated symbols: R (number of simulation draws), $\beta^{(r)}$ (the r-th draw), $\hat{P}_n$ (simulated probability), SLL (simulated log-likelihood), θ (parameters of the coefficient distribution — means, Cholesky/variance terms).

23.2 The simulated likelihood

The panel simulated probability. For respondent n with tasks $t = 1, \dots, T_n$, the mixed-logit panel probability (Chapter 22, §22.7) is approximated by averaging the product of per-task logits over R draws of the coefficient vector:

$$\hat{P}_n(\theta) = \frac{1}{R} \sum_{r=1}^{R} \prod_{t=1}^{T_n} L_{n,y_{nt}}(\beta^{(r)}), \qquad \beta^{(r)} = \bar{\beta} + \mathbf{L}\, z^{(r)},$$

where each draw $\beta^{(r)}$ is generated from a base draw $z^{(r)}$ via the Cholesky factor $\mathbf{L}$ of the random-coefficient covariance (Chapter 1, §1.9; Chapter 22, §22.6), and the distributions (normal, lognormal, …) are applied to each component (Chapter 22, §22.5). The product over tasks is *inside* the average because the *same* $\beta^{(r)}$ governs all of a respondent's tasks (the panel structure).

The simulated log-likelihood. The estimator maximizes

$$\mathrm{SLL}(\theta) = \sum_{n=1}^{N} \log \hat{P}_n(\theta),$$

over the distribution parameters θ (means, Cholesky/variance terms, error-component variances, and any fixed coefficients). This is your EC-MXL's objective function — what Apollo maximizes with your 1000 Halton draws.

Two layers of approximation. The SML estimator has *two* sources of error: **sampling error** (finite N respondents — real uncertainty about the world) and **simulation error** (finite R draws — an artifact of computation you can reduce with more/better draws). Keeping these separate is the central discipline (Chapter 9, §9.9): you want simulation error negligible relative to sampling error, so that your standard errors reflect genuine uncertainty, not the accident of a seed.

The consistency condition. SML is consistent and asymptotically equivalent to (infeasible) maximum likelihood *only if* $R \to \infty$ *as* $N \to \infty$ (specifically $\sqrt{N}/R \to 0$), because the log takes a nonlinear function of the noisy $\widehat{P}_n$, inducing a **Jensen bias** of order $O(1/R)$ (Chapter 9, §9.4): $\mathbb{E}[\log \widehat{P}_n] \leq \log P_n$. So too few draws bias the estimates, not just inflate their variance — a distinct reason R cannot be tiny, and a reason to prefer variance-reduced draws (§23.3) that lower simulation noise for a given R.

> **Active recall 23.A.** Write the simulated panel mixed-logit probability and explain why the product over tasks is inside the average over draws, and why too few draws *bias* (not merely add variance to) the estimates. *(Answer:* $\widehat{P}_n = \frac{1}{R}\sum_r \prod_t L_{n,y_{nt}}(\beta^{(r)})$*; the product is inside because the same drawn coefficients govern all of a respondent's tasks (panel structure), so the simulation averages the joint probability of the whole choice sequence. Too few draws bias the estimates because the log-likelihood applies the concave log to the noisy simulated probability, and by Jensen's inequality* $\mathbb{E}[\log \widehat{P}_n] \leq \log P_n$ *— a downward bias of order* $1/R$ *that vanishes only as* $R \to \infty$*.)*

23.3 Draw types: pseudo-random, Halton, Sobol, scrambled

The draws $z^{(r)}$ can be generated several ways, and the choice governs how many you need — which is why your DCE uses *Halton*, not pseudo-random, draws.

Pseudo-random draws. Ordinary (pseudo-)random numbers. Simple, but they clump and leave gaps by chance, so the simulation error falls slowly, like $1/\sqrt{R}$ (Chapter 9), and they carry seed-to-seed variation.

Halton draws. Deterministic *low-discrepancy* sequences (radical-inverse in prime bases) that fill the space evenly, so the simulation error falls much faster (nearer $1/R$) — **far fewer draws for the same accuracy**, which is the practical reason for the 1000-Halton-draw convention. Being deterministic, they have *no seed* and hence no seed-to-seed variation.

Worked Example 23.1 (Halton beats random for the mixed-logit probability, verified). Simulating a panel mixed-logit probability (one random coefficient, four tasks; true integral 0.0640), the accuracy of an R-draw estimate:

R	random: SD across seeds	Halton: absolute error
25	0.00106	**0.00038**
100	0.00057	**0.00014**
400	0.00026	**0.00003**

(Verified.) Halton is several times more accurate at every R and its error falls faster (multiplying draws by 16 cut the Halton error ~13-fold but the random SD only ~4-fold, consistent with ~$1/R$ vs $1/\sqrt{R}$). So a modest number of Halton draws matches a much larger number of random draws — the efficiency that makes 1000 Halton draws a reasonable default.

Sobol draws. Another low-discrepancy sequence, generally *better-behaved in higher dimensions* than plain Halton. For an EC-MXL with many random coefficients and error components (a high-dimensional integral), Sobol is often preferred.

The high-dimensional caveat (from Chapter 9, §9.5). Plain Halton *degrades in high dimensions*: the sequences for large prime bases become correlated, leaving structured gaps in high-dimensional projections,

which can bias the simulated likelihood. Your EC-MXL, with several random parameters plus error components, is exactly the high-dimensional regime where this bites. The fixes: **scrambled Halton** (randomly permute the digits, breaking the correlation while keeping low discrepancy) or **Sobol**. This is precisely why your PDF lists "sensitivity to Halton draws" as a diagnostic: in your model's dimension, unscrambled 1000-Halton may be inadequate, and you should check robustness across draw *types* (plain Halton, scrambled Halton, Sobol), not just draw counts.

> **Stop and think 23.1.** Your EC-MXL has 9 random dimensions and uses 1000 *unscrambled* Halton draws. A referee questions the draws. What is the specific risk, and what two changes do you make to check robustness? *(Answer: in 9 dimensions the higher-base Halton sequences are correlated, so unscrambled Halton may cover the space poorly and bias the simulated likelihood; try (i)* scrambled Halton* (or *Sobol*) draws and (ii) a *larger number* of draws, and confirm the estimates, standard errors, and WTP are stable across both — if they move, the draws were inadequate for the model's dimension, Chapter 9 §9.5.)*

23.4 Simulation noise in the estimates, standard errors, and WTP

Chapter 9 showed simulation noise in a *single integral*; here we show it propagating into the *estimates themselves* — the effect that matters for your reported EC-MXL and WTP, and the one your PDF demands you check.

Worked Example 23.2 (Seed and draw sensitivity of a heterogeneity estimate, verified). Estimating a small panel mixed logit by SML and recovering the heterogeneity parameter σ (true standard deviation 0.70), varying the number of draws and the seed:

Draws	Estimated $\hat{\sigma}$ across 6 seeds	SD across seeds
$R = 25$ (random)	0.75, 0.78, 0.78, 0.58, 0.60, 0.74	**0.082**
$R = 200$ (random)	0.66, 0.70, 0.67, 0.63, 0.65, 0.67	**0.022**
$R = 25$ (Halton)	0.68 (deterministic)	—

(Verified.) At $R = 25$ **random draws, the heterogeneity estimate swings from 0.58 to 0.78 across seeds** — a 34% range — which is *pure simulation noise*: nothing about the world changed, only the seed. At $R = 200$ it stabilizes near the truth (all seeds in 0.63–0.70), and **Halton at just $R = 25$ gives a single, accurate 0.68** (deterministic, no seed variation, more accurate than most random-25 runs). This is the concrete meaning of your PDF's "robusto al numero di draw, ai seed": *a heterogeneity parameter — and the WTP and welfare that depend on it — can move materially with the seed when draws are too few, and the whole point of enough Halton draws is to drive that noise below your sampling standard error.*

Why the heterogeneity parameters are the most fragile. The *mean* coefficients are usually estimated fairly stably even with modest draws; it is the **variance/heterogeneity parameters** (and hence the *spread* of WTP, and any welfare measure sensitive to the tails) that are most sensitive to the number and type of draws. So your draw-robustness checks should focus on the *variances*, the *WTP distribution*, and the *welfare measures*, not just the point estimates of the means.

Simulation noise vs sampling uncertainty, restated. The seed-to-seed variation in Worked Example 23.2 is *simulation noise* — a computational artifact, reducible to zero with infinite draws. It must be driven *below* the *sampling* standard error (the genuine uncertainty) before your reported intervals are honest. If a reviewer reruns your model with a different seed and gets a materially different WTP, your draws were too few — a reproducibility failure, not a difference of opinion. Enough (scrambled) Halton or Sobol draws is what makes your result reproducible.

> **Active recall 23.B.** Which parameters of your EC-MXL are most sensitive to the number of draws, and what does a WTP that changes with the seed tell you? *(Answer: the* vari-

ance/heterogeneity* parameters (and the WTP spread and welfare measures that depend on them) are most sensitive to draws — the means are usually more stable. A WTP that changes materially with the seed is *simulation noise* from too few draws (not real uncertainty), signalling that you must increase draws (or use scrambled Halton/Sobol) until the result is seed-stable and reproducible — Chapter 9 §9.9.)*

23.5 Convergence, local maxima, and starting values for the mixed logit

The SML objective is *not* concave (unlike the MNL and SBDC probit, which have unique global maxima — Chapters 11, 21), so the mixed logit inherits all the computational pathologies of Chapter 13, and you must manage them.

Non-concavity and local maxima. The simulated log-likelihood of a mixed logit / EC-MXL can have **multiple local maxima**, flat ridges (weak identification), and simulation-induced wiggles. The optimizer finds *a* maximum, not necessarily *the* global one, and which it finds can depend on the starting values (Chapter 13, §13.5). This is why your PDF elevates "convergence; local maxima; starting values" to explicit concerns for the mixed logit.

The protocol (from Chapter 13, applied here).

1. **Hold the draws fixed** across the optimization (a fixed seed or a deterministic Halton/Sobol sequence), so the objective is a *deterministic* function of θ and the optimizer is not chasing jitter (Chapter 9, §9.4). This is non-negotiable: regenerating draws each iteration makes the surface wiggle and convergence meaningless.
2. **Warm-start from the MNL.** Estimate the (concave, reliable) MNL first and use its coefficients as the starting means for the mixed logit, with small starting variances. A principled, well-located start (Chapter 13, §13.6).
3. **Multiple starting values.** Re-estimate from several starts; the highest simulated log-likelihood is your candidate for the global maximum. If the top log-likelihoods *agree*, you are safe; if they *scatter*, you have local maxima (or too few draws — distinguish via Chapter 13's three-culprit framework).
4. **Check real convergence.** Gradient ≈ 0 *and* a negative-definite Hessian (all eigenvalues of $-\mathbf{H}$ positive), not just the "converged" flag (Chapter 13, §13.3). A near-zero eigenvalue signals a weakly-identified variance (common in over-parameterized mixed logits — too many random coefficients or a full covariance the data cannot support).
5. **Rescale and reparameterize.** Rescale the cost/price variable to comparable magnitudes (conditioning, Chapter 1 §1.10), and estimate constrained parameters via reparameterization (log-standard-deviations, Cholesky factors, sign-constrained distributions — Chapter 22 §22.5). These are often the difference between convergence and failure (Chapter 13, §13.8).

Distinguishing the three culprits. When an EC-MXL misbehaves (huge SE on a variance, non-convergence, unstable estimates), diagnose *which* problem you have (Chapter 13, §13.7): **non-identification** (near-zero Hessian eigenvalue — an over-parameterized covariance the data cannot support: simplify the specification), **local maxima** (deterministic log-likelihoods scatter across starts: take the best, rescale), or **simulation noise** (estimates move with seed/draws: add scrambled/Sobol draws). These need *different* fixes, and misdiagnosing them wastes effort — a €300 SE on a WTP is most often non-identification, not "too few respondents."

> **Stop and think 23.2.** Your EC-MXL gives a different set of estimates from three different starting values, with three different simulated log-likelihoods. What are the two distinct problems this could indicate, and how do you tell them apart? *(Answer: (i)* multiple local maxima* — the non-concave surface has several hills, so different starts climb different ones; the fix is to take the start with the highest log-likelihood and confirm replication. (ii) *simulation noise* — too few (or poor) draws make the surface jitter, so the "different maxima" are artifacts of the draws; the fix

is more/scrambled draws. Tell them apart by holding the draws *fixed* and re-running the starts: if the log-likelihoods still scatter deterministically, it is local maxima; if increasing/scrambling the draws shrinks the scatter, it was simulation noise — Chapter 13 §13.7.)*

23.6 The robustness protocol your PDF prescribes

Pulling §23.3–§23.5 into the operational checklist your PDF demands — *"robusto al numero di draw, ai seed, alla specificazione delle distribuzioni e agli starting values"* — this is what you *report* to demonstrate your EC-MXL is trustworthy:

1. **Vary the number of draws.** Re-estimate at $R = 250, 500, 1000, 2000$ (and more). Track the *means*, the *variance/heterogeneity parameters*, the *standard errors*, the *log-likelihood*, and above all the *WTP distribution and welfare*. Report the table; if they stabilize, 1000 is enough; if they drift, you need more.
2. **Vary the seed.** Re-estimate across several seeds (for random or scrambled draws). Seed-to-seed variation is pure simulation noise; if the WTP moves noticeably, add draws. A well-resolved model is seed-stable.
3. **Vary the draw type.** Compare plain Halton, scrambled Halton, and Sobol — especially important given your model's dimension (§23.3). Divergence signals draw-quality problems.
4. **Vary the starting values.** Multiple starts (warm-started from the MNL); confirm the top log-likelihoods agree (global maximum).
5. **Vary the distributional specification.** Re-estimate under alternative random-coefficient distributions (normal vs lognormal vs constrained) and report how the means, heterogeneity, and WTP move (Chapter 22, §22.5) — the "specificazione delle distribuzioni" your PDF names.

Reporting this protocol is not busywork; it is the evidence that your EC-MXL result is a property of your *data*, not of your *seed, draw count, or start*. It is the "invisible but decisive" competence (Chapter 13), and it is what separates a defensible mixed-logit paper from one a referee can break by asking "did you check the draws?"

> **Active recall 23.C.** List the five things you vary to demonstrate your EC-MXL is robust, and which model quantity is most important to track across them. *(Answer: vary (1) the number of draws, (2) the seed, (3) the draw type (plain/scrambled Halton, Sobol), (4) the starting values, (5) the distributional specification. Track especially the* WTP distribution and welfare measures* (and the variance/heterogeneity parameters they depend on), since these are the most sensitive to draws and specification — the means are usually more stable.)*

23.7 Model comparison: log-likelihood, AIC, BIC, and LR tests

Having estimated the EC-MXL (and simpler models), you must decide whether the added complexity is warranted — the AIC/BIC/LR machinery your PDF lists.

The tools (from Chapter 12). For *nested* models (MNL nested in mixed logit), the **likelihood-ratio test** $LR = 2(\text{SLL}_{\text{unrestricted}} - \text{SLL}_{\text{restricted}})$ tests whether the extra parameters improve fit. For *non-nested* comparisons (different distributions, preference vs WTP space), use **information criteria**:

$$\text{AIC} = -2\,\text{SLL} + 2k, \qquad \text{BIC} = -2\,\text{SLL} + k \log N,$$

with k the number of parameters and N the number of respondents (the effective sample size for the panel — arguably respondents, not choice observations). Lower is better; BIC penalizes complexity more heavily.

Worked Example 23.3 (Comparing MNL and mixed logit, verified). Suppose the MNL (4 parameters) has SLL $= -1620.4$ and the mixed logit (6 parameters, adding 2 variances) has SLL $= -1585.1$, with $N = 3000$:

Criterion	MNL	Mixed logit	Prefers
AIC	3248.8	**3182.2**	mixed logit
BIC	3272.8	**3218.2**	mixed logit
LR	—	2(35.3) = 70.6	— (see boundary caveat)

(Verified.) AIC and BIC both favor the mixed logit (lower), and the LR statistic is large (70.6). But the LR test has a **boundary caveat** that is decisive and often missed.

The boundary problem (Chapter 12, §12.9) — critical for your variance tests. The mixed logit adds *variance* parameters (random-coefficient and error-component variances), and testing whether a variance is zero is a test *on the boundary* of the parameter space. So the LR statistic does **not** have a χ^2_q distribution — for two variances the correct null is a *mixture* (e.g., $\frac{1}{4}\chi^2_0 + \frac{1}{2}\chi^2_1 + \frac{1}{4}\chi^2_2$), not χ^2_2. Using the naive χ^2 makes the test **conservative**, so you can *wrongly conclude "no heterogeneity"* and estimate an MNL when the data support a mixed logit (verified example in Chapter 12: a statistic significant under the mixture but not under the naive χ^2). **Always use the boundary-corrected LR test (or AIC/BIC) when comparing the mixed logit to the MNL.** This is one of the most common errors in applied DCE work, and getting it right can be the difference between finding and missing your preference heterogeneity.

Assumptions of each criterion. AIC targets predictive accuracy (tends to favor richer models); BIC targets the "true" model under a prior and penalizes complexity more (favors parsimony); neither is a formal test. The LR test is a formal test but requires *nested* models, the *same sample* (identical exclusions/weights — Chapter 12 §12.5; your PDF §6.6–6.7), and the *boundary correction* for variances. For *non-nested* comparisons — normal vs lognormal coefficients, or **preference space vs WTP space** (Chapter 24) — the LR test does not apply; use AIC/BIC (and, decisively, the *interpretability* and *plausibility* of the WTP, not just fit). A model that wins on AIC but implies wrong-sign cost coefficients or absurd WTP tails (Chapter 22, §22.5) is not the better model.

> **Stop and think 23.3.** You compare your mixed logit to the MNL with an LR test, get $LR = 6.5$ for two added variances, compare to χ^2_2 (critical value 5.99), and reject — but a co-author says "compare to χ^2_2 is wrong here." Explain, and state what changes. *(Answer: testing whether the two* variances* are zero is a boundary test (variances can't be negative), so the correct null is a *mixture* of chi-squares (e.g., $\frac{1}{4}\chi^2_0 + \frac{1}{2}\chi^2_1 + \frac{1}{4}\chi^2_2$), not χ^2_2 — the naive χ^2_2 is *conservative*. Here it happens to reject anyway, but in a borderline case the naive test could *fail* to reject (wrongly concluding no heterogeneity) when the boundary-corrected test would reject. Use the mixture distribution (halving/adjusting the p-value) or AIC/BIC — Chapter 12 §12.9.)*

23.8 Failure modes

F1. Too few draws (biased, noisy estimates). Estimating an EC-MXL with too few draws → Jensen bias plus seed-dependent estimates (verified $\hat{\sigma}$ swinging 0.58–0.78 at $R = 25$). *Fix:* increase draws; use variance-reduced (Halton/Sobol) draws; check stability (§23.2, §23.4).

F2. Regenerating draws each iteration. A jittering objective that the optimizer chases. *Fix:* fix the draws across the whole optimization (§23.5; Chapter 9 §9.4).

F3. Unscrambled Halton in high dimensions. Plain Halton degrading in your model's dimension, biasing the likelihood. *Fix:* scrambled Halton or Sobol; compare across draw types (§23.3).

F4. Seed-dependent results reported as stable. A WTP that moves with the seed, reported as if reproducible. *Fix:* add draws until seed-stable; report the draw/seed robustness (§23.4, §23.6).

F5. Single run of a non-concave model. One EC-MXL estimation taken as definitive. *Fix:* multiple starting values (warm-started from MNL); compare log-likelihoods; take the best (§23.5; Chapter 13).

F6. Naive χ^2 for variance tests. Testing the mixed logit vs MNL against χ^2_q instead of the mixture. *Symptom:* wrongly concluding "no heterogeneity." *Fix:* boundary-corrected LR / AIC-BIC (§23.7; Chapter 12 §12.9).

F7. Comparing models on different samples. LR/AIC/BIC across models with different exclusions/weights. *Fix:* identical estimation sample (§23.7; Chapter 12 §12.5).

F8. Selecting on fit alone. Choosing the AIC-best model despite implausible WTP/wrong-sign coefficients. *Fix:* weigh interpretability, sign plausibility, and WTP behavior, not just fit (§23.7; Chapter 22 §22.5).

23.9 Diagnostic questions for your own papers

1. **Draw count.** Are my EC-MXL estimates, SEs, and WTP stable as draws increase (250 → 500 → 1000 → 2000)? *(§23.2, §23.6, F1.)*
2. **Seed.** Do my results (especially WTP and variances) change with the seed? If so, are draws adequate? *(§23.4, F4.)*
3. **Draw type.** For my model's dimension, am I using scrambled Halton or Sobol, and have I checked across draw types? *(§23.3, F3.)*
4. **Fixed draws.** Are the draws held fixed across optimization? *(§23.5, F2.)*
5. **Global maximum.** Did I use multiple starts (MNL warm-start) and confirm the top log-likelihoods agree? *(§23.5, F5.)*
6. **Boundary test.** Did I compare the mixed logit to the MNL with a boundary-corrected LR (or AIC/BIC), not naive χ^2? *(§23.7, F6.)*
7. **Fit vs plausibility.** Is my selected model justified by fit *and* interpretability (plausible WTP, right-sign coefficients)? *(§23.7, F8.)*

23.10 Exercises

(A) Comprehension.

A1. Write the simulated panel mixed-logit log-likelihood and explain the two layers of approximation (sampling and simulation).

A2. Why do you use 1000 Halton draws rather than 1000 pseudo-random draws, and what is the high-dimensional caveat?

A3. Your EC-MXL's WTP changes from €18 to €25 when you change the seed. What does this tell you, and what do you do?

(B) Derivation / reasoning.

B1. Explain, using Jensen's inequality, why the simulated log-likelihood is biased for finite R, and why this requires R to grow with N.

B2. Explain why the simulated log-likelihood is non-concave and how you would find the global maximum, distinguishing local maxima from simulation noise.

B3. Compute AIC and BIC for an MNL ($k = 4$, SLL $= -1620$) and a mixed logit ($k = 6$, SLL $= -1585$) with $N = 3000$, and explain why the LR test against χ^2_2 is wrong.

(C) Research application / critique.

C1. A co-author reports an EC-MXL from a single run with 200 random draws and one seed. Write the robustness checks you would require before believing the WTP.

C2. Your EC-MXL will not converge. Diagnose whether the problem is non-identification, local maxima, or simulation noise, and give the fix for each.

C3. You compare a normal-coefficient and a lognormal-coefficient mixed logit; the normal has slightly lower AIC but implies 9% wrong-sign cost coefficients. Which do you prefer and why?

23.11 Solutions

A1. $\text{SLL}(\theta) = \sum_n \log \hat{P}_n(\theta)$ with $\hat{P}_n = \frac{1}{R}\sum_r \prod_t L_{n,y_{nt}}(\beta^{(r)})$. Two layers of approximation: *sampling error* (finite N respondents — genuine uncertainty about the population), and *simulation error* (finite R draws — a computational artifact reducible with more/better draws). Honest inference requires the simulation error to be negligible relative to the sampling error.

A2. Halton draws are deterministic low-discrepancy sequences that fill the space evenly, so the simulation error falls nearer $1/R$ than the pseudo-random $1/\sqrt{R}$ — far fewer draws for the same accuracy (verified: Halton is several times more accurate than random at every R), and no seed variation. Caveat: plain Halton *degrades in high dimensions* (correlated sequences for large prime bases), so for a many-dimensional EC-MXL you should use *scrambled Halton* or *Sobol* and check robustness across draw types.

A3. A €7 shift in WTP from a seed change is *pure simulation noise* — the number of draws is too small to resolve the integral, so the estimate is not reproducible. Increase the number of draws (and/or switch to scrambled Halton/Sobol), confirm the WTP stabilizes and becomes seed-insensitive, and report the draw/seed robustness; never report a seed-dependent WTP as a stable result.

B1. $\hat{P}_n$ is an unbiased estimate of P_n, but SLL takes the concave log of it; by Jensen's inequality $\mathbb{E}[\log \hat{P}_n] \leq \log \mathbb{E}[\hat{P}_n] = \log P_n$ — a downward bias whose size grows with $\text{Var}(\hat{P}_n) = O(1/R)$, so the bias is $O(1/R)$. Because the bias shrinks only as R grows, SML is consistent only if $R \to \infty$ as $N \to \infty$ (specifically $\sqrt{N}/R \to 0$); a fixed R leaves an asymptotic bias. Variance-reduced draws (Halton/Sobol) lower $\text{Var}(\hat{P}_n)$ for given R, reducing the bias.

B2. The simulated log-likelihood is non-concave because it integrates the logit over a coefficient distribution and the resulting surface (unlike the MNL's concave one) can have multiple hills and flat ridges. To find the global maximum: estimate from *multiple starting values* (warm-started from the MNL), hold the draws *fixed*, and take the start with the highest simulated log-likelihood. Distinguish local maxima from simulation noise by re-running the starts with *fixed* draws: if the log-likelihoods scatter *deterministically*, it is local maxima (take the best); if increasing/scrambling the draws shrinks the scatter, it was simulation noise (add draws).

B3. AIC: MNL $= -2(-1620) + 2(4) = 3248$; MXL $= -2(-1585) + 2(6) = 3182 \rightarrow$ MXL lower (preferred). BIC: MNL $= -2(-1620) + 4\log 3000 = 3240 + 32.0 = 3272$; MXL $= -2(-1585) + 6\log 3000 = 3170 + 48.0 = 3218 \rightarrow$ MXL lower. The LR test against χ^2_2 is wrong because the two added parameters are *variances*, and testing whether a variance is zero is a *boundary* test (variances ≥ 0); the correct null is a *mixture* of chi-squares (not χ^2_2), and the naive χ^2_2 is conservative — it can wrongly fail to reject heterogeneity (Chapter 12 §12.9).

C1. Required checks before believing the WTP: (i) **draw count** — re-estimate at 250/500/1000/2000 draws and confirm the WTP and variance parameters stabilize (200 is likely too few — verified $\hat{\sigma}$ swings across seeds at low draws); (ii) **seed** — re-estimate across several seeds and confirm the WTP is seed-stable (else simulation noise); (iii) **draw type** — use scrambled Halton or Sobol given the model's dimension, and compare; (iv) **multiple starts** — warm-start from the MNL and confirm the top simulated log-likelihoods agree (global maximum, since non-concave); (v) **convergence** — gradient ≈ 0 and negative-definite Hessian; (vi) **distribution plausibility** — check the cost coefficient's distribution and wrong-sign fraction (Chapter 22). A single 200-random-draw, one-seed run addresses none of these.

C2. Diagnose via the three-culprit framework: check the **Hessian eigenvalues** — a near-zero eigenvalue means *non-identification* (an over-parameterized covariance / too many random coefficients the data cannot support: simplify the specification, this is structural, not a draw problem). If eigenvalues are healthy, run

multiple starts with fixed draws — deterministic scatter in the log-likelihoods means *local maxima* (take the best; rescale variables; warm-start from MNL). **Vary the seed/draws** — estimates that move with the seed or settle with more draws mean *simulation noise* (add scrambled/Sobol draws). Also rescale the cost variable and reparameterize constrained parameters (log-SD, Cholesky, sign-constrained distributions) — often the practical cure for non-convergence (Chapters 1 §1.10, 13 §13.8).

C3. Prefer the **lognormal-coefficient** model despite its slightly higher AIC. A slightly better AIC does not justify a model that implies 9% of respondents have a *wrong-sign* (positive) cost coefficient — economically absurd (people who like paying more) — and the associated heavy WTP tails (Chapter 22 §22.5). Model selection must weigh *interpretability and plausibility*, not fit alone; the lognormal constrains the cost coefficient to the correct sign and yields a sensible WTP distribution, which is decisive. (Better still, consider WTP-space estimation, Chapter 24, which sidesteps the ratio pathology entirely.)

23.12 Spaced review prompts

+1 day. From memory, write the simulated panel mixed-logit log-likelihood, explain the two layers of approximation and the Jensen bias, and state why Halton beats random draws.

+1 week. Explain how simulation noise propagates into the estimates (seed sensitivity of variance parameters), the robustness protocol (draws, seed, draw type, starts, distributions), and the boundary correction for the MNL-vs-mixed-logit LR test.

+1 month. For your own EC-MXL, run the full robustness protocol (draw counts, seeds, draw types, starts, distributions), report the stability of the WTP and variances, confirm the global maximum, and compare to the MNL with a boundary-corrected test. Close with the spine sentence about whether your result is reproducible.

23.13 How this chapter connects to the rest of the book

- **Chapter 9** supplied Monte Carlo, Halton/Sobol, the Jensen bias, and the draw/seed diagnostics; **Chapter 13** the convergence/local-maxima/starting-value craft; **Chapter 22** the mixed-logit/EC-MXL specification this chapter estimates. This chapter is their assembly for the DCE.
- **Chapter 12 (§12.9)** is the boundary-corrected model comparison of §23.7; **Chapter 24 (WTP space)** is the reparameterization that both tames the distribution/WTP-tail problem (§23.4) and is compared non-nested to preference space (§23.7); **Chapter 25 (scale vs preference)** interprets the heterogeneity variances this chapter estimates.
- **Chapter 1 (§1.9)** is the Cholesky for the draws; **Chapter 17/19** the clustered/panel inference; **Chapter 52 (computational diagnostics)** is the full reproducible-pipeline treatment of the robustness protocol (§23.6), of which this chapter is the mixed-logit-estimation core.

This chapter turns your EC-MXL from a specification into a *reproducible* estimate — one whose robustness to draws, seeds, draw types, and starts you have verified and can report, meeting your PDF's demand to "sapere se il risultato è robusto."

23.14 Research translation task

For your **EC-MXL**, produce a two-to-three-page estimation-and-robustness note:

1. **The simulated likelihood.** Write your model's simulated panel log-likelihood explicitly (draws, panel product, distributions).

2. **Draw robustness.** Re-estimate at several draw counts (250/500/1000/2000) and tabulate the means, variances, SEs, log-likelihood, and WTP; state where they stabilize and whether 1000 is enough.
3. **Seed and draw-type robustness.** Re-estimate across seeds and across plain Halton, scrambled Halton, and Sobol; report whether WTP and variances are stable, given your model's dimension.
4. **Convergence.** Multiple starts (MNL warm-start); report the top log-likelihoods, the gradient, and the Hessian eigenvalues (global maximum, identified?).
5. **Model comparison.** Compare the EC-MXL to the MNL (and intermediate models) via boundary-corrected LR and AIC/BIC on the identical sample; justify the specification by fit *and* plausibility.
6. **Close** with the spine sentence: *"My EC-MXL is estimated by SML with [N] (scrambled) Halton/Sobol draws; its WTP and variances are stable across [draws/seeds/types/starts]; simulation noise is negligible relative to the [SE]; and it beats the MNL on [boundary-corrected LR / AIC-BIC]."*

Carry this into Chapter 24 (WTP space), Chapter 25 (scale vs preference), and Chapter 52 (the full reproducible pipeline).

23.15 Bibliography

Core. - Train, *Discrete Choice Methods with Simulation*, the simulation and mixed-logit-estimation chapters (the simulated likelihood, Halton draws, SML properties, and convergence). The single most relevant source; read alongside Chapters 9, 22. Freely available — *verify link.* - Hensher & Greene (2003), "The Mixed Logit Model: The State of Practice," *Transportation*, on draws, simulation, and estimation in practice. *(Verify volume/pages.)* - Gourieroux & Monfort, *Simulation-Based Econometric Methods*. The theory of simulated maximum likelihood, its bias, and consistency conditions (§23.2). *(Verify edition.)*

Applied. - Bhat (2001, 2003), on quasi-random (Halton) and scrambled/randomized draws for mixed logit, *Transportation Research Part B* — the source for §23.3 and the high-dimensional caveat. *(Verify volumes/years.)* - Czajkowski & Budziński (2019), and related work on the number of draws and simulation reliability in stated-preference mixed logit — directly on the draw-robustness protocol (§23.6). *(Verify exact citation.)* - The **Apollo** manual (Hess & Palma) for implementing SML, draws, starting values, and model comparison — the tool your PDF names. *(Verify current version.)*

Advanced. - McFadden & Train (2000), "Mixed MNL Models for Discrete Response," *Journal of Applied Econometrics* — the mixed-logit estimation foundation (Chapter 22). *(Verify volume/pages.)* - Revelt & Train (1998), "Mixed Logit with Repeated Choices," *Review of Economics and Statistics* — the panel simulated likelihood (§23.2). *(Verify volume/pages.)* - Fiebig, Keane, Louviere & Wasi (2010), "The Generalized Multinomial Logit Model," *Marketing Science* — estimation of scale/preference heterogeneity (Chapter 25). *(Verify volume/pages.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Bhat / Revelt–Train citations before using them, per the book's no-unverified-citation rule.

End of Chapter 23. Next in Part V: Chapter 24 — Preference Space versus WTP Space (WTP = $-\beta_attribute/\beta_cost$, the pathologies of a ratio of random coefficients, heavy tails, the near-zero cost coefficient, direct estimation in WTP space, constrained distributions, and interpretation), which is your DCE's central estimation decision and the resolution of the WTP-tail problem this chapter and Chapter 22 exposed. Tell me when to write it, or request a different chapter.

Chapter 24 — Preference Space versus WTP Space

Part V: Random Utility, Discrete Choice, and Stated Preference Econometrics

The spine question, applied to this chapter. Your PDF calls this *"un punto centrale per te"* — and it is one of the three highest-dependency chapters in the whole book. Your DCE estimates willingness to pay, and there are two ways to do it. In **preference space** you estimate the utility coefficients and form WTP *after the fact* as the ratio $-\beta_{\text{attribute}}/\beta_{\text{cost}}$ — a ratio of random coefficients whose distribution is treacherous: heavy-tailed, driven by the cost coefficient's mass near zero, with wrong-sign draws and an unstable mean. In **WTP space** you reparameterize so that WTP is a *primitive parameter* you model directly — interpretable and constrainable, but at the cost of a strong restriction. Your DCE estimates in WTP space, and your PDF's critical questions are exactly the trade-offs: *"la distribuzione scelta produce code irrealistiche? il coefficiente di costo è sempre negativo? il modello in WTP space migliora l'interpretabilità ma impone restrizioni forti?"* The spine question here: *am I recovering WTP as a fragile ratio, or modeling it directly — and if directly, do I understand the restriction I have imposed and how it interacts with the scale of my cost-vector treatments?*

24.0 Conceptual motivation: the WTP-tail problem, and its resolution

Three times now the WTP-ratio pathology has surfaced and been deferred to this chapter. In Chapter 2 (§2.11) you saw the *calculus*: WTP $= -\beta_{\text{attr}}/\beta_{\text{cost}}$, and $\partial\text{WTP}/\partial\beta_{\text{cost}} = -\beta_{\text{attr}}/\beta_{\text{cost}}^2$ blows up as the cost coefficient nears zero. In Chapter 6 (§6.9) you saw the *probability theory*: a ratio of random quantities with a near-zero denominator is approximately **Cauchy** — the mean may not even exist, and the law of large numbers fails. In Chapter 22 (§22.5) you saw the *estimation consequence*: a normal cost coefficient in a mixed logit puts mass at the wrong sign and produces absurd WTP tails. This chapter is where the problem is *resolved*, because it is the central estimation decision of your DCE and your PDF makes it a top priority.

The stakes are concrete and quantifiable. As we will verify:

- In **preference space**, with random attribute and cost coefficients, WTP $= -\beta_{\text{attr}}/\beta_{\text{cost}}$ has a median near €20 but a **99th percentile of €144**, an *unstable mean*, and ~1% of respondents assigned an *absurd* WTP above €200 — the ratio pathology in full.
- In **WTP space**, modeling WTP directly, the same central WTP has a **99th percentile of €39** and no absurd draws — a clean, interpretable distribution.
- And the tail is driven precisely by the **cost coefficient's mass near zero**: as the mean cost coefficient shrinks toward zero, the 99th-percentile WTP explodes from €47 to €144 to €362 while the median stays near €20.

But WTP space is not a free lunch. The reparameterization that tames the tail *imposes a restriction* — the cost coefficient (which turns out to *be* the scale parameter) is common to all attributes' WTP — and for a mixed logit the two parameterizations are **genuinely different models** with different WTP distributions and different fits (we verify they are non-equivalent). So this chapter is not "WTP space is better"; it is *"here is exactly what each space assumes, what it buys, what it costs, and how to choose and defend the choice for your DCE."*

We proceed: WTP as a ratio derived; the ratio-of-random-coefficients pathology (analytically and numerically); the WTP-space reparameterization derived; what WTP space buys and the strong restriction it imposes (non-equivalence for the mixed logit); the crucial identity that the cost coefficient *is* the scale parameter (linking to Chapters 20, 21, 25); constrained distributions for WTP; interpreting WTP-space estimates; and the interaction with your cost-vector treatments (scale vs cost vs preference). This chapter, together with Chapters 11 and 30, is one the architecture flags to *over-learn*: master it and most of your DCE's WTP interpretation follows.

24.1 Prerequisites

From **Chapter 2** (§2.11): WTP as a ratio of marginal utilities and the calculus of the ratio pathology, and the preference-vs-WTP-space reparameterization first sketched there. From **Chapter 6** (§6.9): the near-Cauchy behavior of a ratio with a near-zero denominator. From **Chapter 7** (§7.7–7.8) and **Chapter 9**: the delta method vs Krinsky–Robb for WTP, and simulation. From **Chapter 15** (§15.3): the CV ancestor — Hanemann's utility-difference vs Cameron's WTP-function parameterization. From **Chapter 20** (§20.6) and **Chapter 21** (§21.4): scale identification and the scale parameter. From **Chapter 22** (§22.5): the distributions of random coefficients (normal vs lognormal, wrong-sign draws, WTP tails).

Same **notation**. Consolidated symbols: q (an environmental attribute), c (cost), β_q (marginal utility of q), β_c (marginal (dis)utility of cost, > 0), $\omega_q = \beta_q/\beta_c$ (WTP for q), μ / scale (the cost coefficient's role as scale).

24.2 WTP as a ratio, and the ratio pathology

WTP as a ratio (derived, Chapter 2 §2.11). In preference space the deterministic utility is linear in attributes and cost:

$$V = \beta_q\, q - \beta_c\, c + (\text{other attributes}), \qquad \beta_c > 0.$$

Marginal WTP for attribute q is the marginal rate of substitution between q and money — the change in c that holds utility constant when q rises by a unit (the total differential $dV = 0$):

$$\boxed{\text{WTP}_q = \frac{\partial V/\partial q}{\partial V/\partial c} = \frac{\beta_q}{\beta_c}}.$$

In a **fixed-coefficient** logit this is a ratio of two numbers — a single WTP value. In a **mixed** logit, β_q and β_c are *random* (each respondent has their own), so WTP $= \beta_q/\beta_c$ is a **ratio of two random variables** — and the distribution of a ratio is where the trouble lives.

The pathology. A ratio X/Y where the denominator Y has probability mass *near zero* is heavy-tailed: as $Y \to 0$, $X/Y \to \pm\infty$. When Y (the cost coefficient) is *normally* distributed, it has mass on both sides of zero, so (i) some draws have the *wrong sign* (a positive cost coefficient — a respondent who likes paying more), and (ii) draws near zero send WTP to enormous values. The result is approximately **Cauchy** (Chapter 6, §6.9): a distribution so heavy-tailed its mean may not exist and its sample mean does not stabilize.

Worked Example 24.1 (The preference-space WTP tail, verified). Take random coefficients $\beta_q \sim N(0.6, 0.20)$ and $\beta_c \sim N(0.03, 0.012)$ (cost coefficient positive, i.e. disutility, with a small standard deviation that nonetheless places some mass near zero). The implied WTP $= \beta_q/\beta_c$:

Quantity	Value
Median WTP	€19.84
Mean WTP	€21.9 (**unstable** — driven by the tail)
1st percentile	€2.1
99th percentile	**€144.0**
Fraction with |WTP| > €200 (absurd)	**0.94%**
Fraction with wrong-sign cost coefficient	0.62%

(Verified.) The **median is a sensible €20, but the 99th percentile is €144 and ~1% of respondents are assigned WTP above €200** — economically absurd values manufactured by near-zero cost draws. The mean is unstable (dominated by the tail). This is the preference-space WTP pathology in full, and it is *not* a data problem — it is intrinsic to forming a ratio of random coefficients.

The tail is driven by the cost coefficient's mass near zero. The severity of the tail is governed by *how close the cost coefficient's distribution comes to zero.* **Worked Example 24.2 (verified):** holding the cost-coefficient standard deviation fixed and shrinking its *mean* toward zero:

Mean cost coefficient	Pr(cost coef ≤ 0)	Median WTP	99th-percentile WTP
0.06	0.0000	€20.0	€46.7
0.03	0.0062	€19.8	€143.5
0.02	0.0479	€18.5	**€362.0**

(Verified, 4×10^6 draws.) The mean of β_q is scaled with the cost mean so that **median WTP is held at roughly €20 throughout** — otherwise weakening the cost coefficient would also raise the central WTP and the two effects would be confounded. With the centre held fixed, the tail alone moves: the wrong-sign fraction rises from essentially zero to 4.8%, and the 99th-percentile WTP *explodes* — from €47 to €144 to €362, while the median barely shifts. **A weak cost coefficient is the root cause of unstable WTP**, exactly as the calculus (Chapter 2) and probability theory (Chapter 6) predicted. This is why your PDF's §6.4 red flag ("WTP instabile per colpa del coefficiente di costo") is the recurring theme it is, and why the cost coefficient's identification and distribution are the crux of everything in this chapter.

> **Active recall 24.A.** Why does WTP $= -\beta_{\text{attr}}/\beta_{\text{cost}}$ in a mixed logit have heavy tails, and which feature of the cost coefficient's distribution controls the severity? *(Answer: it is a ratio of two random coefficients, and a ratio with a denominator (the cost coefficient) that has mass* near zero* is heavy-tailed (near-Cauchy) — near-zero cost draws send WTP to huge values, and wrong-sign draws (if the cost coefficient can be positive) make it explode. The severity is controlled by how much of the cost coefficient's distribution sits near (and across) zero: the closer to zero its mean, and the more mass near/below zero, the fatter the WTP tail — verified, the 99th-percentile WTP rose from €47 to €362 as the mean cost coefficient shrank, with median WTP held fixed.)*

24.3 The WTP-space reparameterization

The fix is to *reparameterize the utility function so that WTP is a primitive parameter you model directly*, rather than a ratio you form after estimation. This is the WTP-space model.

The derivation. Start from the preference-space utility $V = \beta_q q - \beta_c c$ and *factor out the cost coefficient* β_c:

$$V = \beta_c\Big(\frac{\beta_q}{\beta_c}\, q - c\Big) = \beta_c(\omega_q\, q - c), \qquad \omega_q \equiv \frac{\beta_q}{\beta_c} = \text{WTP}_q.$$

Now the utility is written in terms of two new parameters: the **cost coefficient** β_c (which multiplies the whole bracket) and the **WTP** ω_q (which appears *directly* as the money value of the attribute). With more attributes,

$$\boxed{V = \beta_c\Big(\sum_k \omega_k\, x_k - c\Big)}, \qquad \omega_k = \text{WTP for attribute } k.$$

This is the WTP-space model. Instead of putting distributions on β_q and β_c (preference space) and forming the ratio, you put distributions *directly on the WTP parameters* ω_k and on the cost coefficient β_c. WTP is no longer a derived ratio — it is a parameter you estimate.

What this achieves. Because WTP is now a primitive, you can model its distribution to be *sensible*: a light-tailed distribution (normal, or a bounded/constrained distribution) with a plausible support and no wrong-sign explosions.

Worked Example 24.3 (WTP space tames the tail, verified). Model the same central WTP directly as $\omega_q \sim N(20, 8)$ (a light-tailed normal on WTP), with the cost coefficient β_c modeled separately:

Quantity	Preference space (24.1)	**WTP space**
Median WTP	€19.84	€20.00
Mean WTP	€21.9 (unstable)	**€20.00 (stable)**
99th percentile	€144.0	**€38.6**
Fraction |WTP| > €200	0.94%	**0.00%**

(Verified.) The WTP-space model gives a clean, symmetric, interpretable WTP distribution — median and mean both €20, a 99th percentile of €39, and *no* absurd draws — because you *chose* the WTP distribution to be well-behaved, rather than inheriting the ratio's pathology. This is the central practical reason your DCE estimates in WTP space, and it is the resolution of the tail problem that has recurred since Chapter 2.

> **Stop and think 24.1.** In the WTP-space utility $V = \beta_c(\sum_k \omega_k x_k - c)$, where does WTP appear, and why does this eliminate the ratio pathology? *(Answer: WTP appears* directly* as the parameter ω_k multiplying attribute x_k inside the bracket — it is a primitive of the model, not a ratio of two coefficients. This eliminates the ratio pathology because you place a distribution *directly* on ω_k (which you can choose to be light-tailed, bounded, or sign-constrained), so there is no denominator that can approach zero and no ratio to explode — the heavy tail of β_q/β_c is replaced by whatever distribution you assign to ω_k.)*

24.4 What WTP space costs: the strong restriction and non-equivalence

WTP space is not a mere convenience — it is a *different model* with a *real restriction*, and your PDF insists you understand it: *"il modello in WTP space migliora l'interpretabilità ma impone restrizioni forti."* This section makes the restriction precise.

For a *fixed*-coefficient logit, the two spaces are equivalent. When the coefficients are not random, $V = \beta_q q - \beta_c c$ and $V = \beta_c(\omega_q q - c)$ with $\omega_q = \beta_q/\beta_c$ are the *same model* — just reparameterized. They fit identically and give the same WTP. So the distinction is *only* about the mixed logit.

For a *mixed* logit, the two spaces are NOT equivalent. When the coefficients are random, the two parameterizations induce *different distributions on WTP* and are therefore *different models with different likelihoods and fits*:

- **Preference space** puts distributions on β_q and β_c; WTP $= \beta_q/\beta_c$ is a *ratio* distribution (heavy-tailed, as verified).

- **WTP space** puts distributions on ω_q (WTP) and β_c (scale); WTP is *directly* whatever distribution you chose for ω_q; the *implied* attribute coefficient $\beta_q = \omega_q \beta_c$ is a *product* distribution.

Worked Example 24.4 (Non-equivalence, verified). Calibrated to the *same* median WTP (~€20), the two parameterizations produce genuinely different WTP distributions: the preference-space WTP is extremely heavy-tailed (a ratio of normals — its skewness is dominated by tail draws and effectively undefined), while the WTP-space WTP (a direct normal) is symmetric (skewness ≈ 0). (Verified.) **These are not two views of one model; they are two different models that fit the data differently and imply different welfare.** You cannot have both a light-tailed WTP *and* the preference-space distributional structure — choosing WTP space *is* choosing to impose a particular structure.

The precise restriction. The WTP-space form $V = \beta_c(\sum_k \omega_k x_k - c)$ imposes that **the cost coefficient β_c is a common multiplicative factor on every attribute's contribution.** In preference space, each attribute coefficient β_k has its *own* distribution, free to correlate with the cost coefficient in any way; in WTP space, the single β_c scales *all* attributes together, and the WTP distributions ω_k carry the attribute-specific heterogeneity. This is a genuine restriction on the covariance structure of the random coefficients — a "restrizione forte" — and it is defensible when the cost coefficient plausibly acts as a common scale (which, as §24.5 shows, it *is*), but it is a restriction nonetheless, and you should state it.

The honest report. Because the spaces are non-equivalent, you should (and your PDF implies you must): (i) estimate in WTP space for interpretability and tail control, (ii) *state the restriction* it imposes, (iii) compare fit to preference space (via AIC/BIC — a *non-nested* comparison, Chapters 12, 23), and (iv) recognize that a difference in WTP between the two spaces is not an error but a consequence of the different distributional assumptions. The choice is a modeling decision to defend, not a formatting choice.

> **Active recall 24.B.** Are preference space and WTP space the same model? State the answer for a fixed-coefficient logit and for a mixed logit, and name the restriction WTP space imposes. *(Answer: for a* fixed-*coefficient logit they are* equivalent* (a reparameterization — same fit, same WTP). For a *mixed* logit they are *not* equivalent — they put distributions on different objects (coefficients vs WTP), inducing different WTP distributions (ratio vs direct) and different fits (verified: heavy-tailed vs symmetric). WTP space imposes the restriction that the cost coefficient is a *common multiplicative scale on all attributes' contributions*, rather than each attribute coefficient having its own free distribution — a genuine restriction on the random-coefficient covariance structure.)*

24.5 The cost coefficient *is* the scale parameter

Here is the identity that ties WTP space to the deepest theme of Part V and to your cost-vector treatments. In the WTP-space utility $V = \beta_c(\sum_k \omega_k x_k - c)$, the cost coefficient β_c **multiplies the entire utility** — which is exactly the role of the **scale parameter** (Chapters 20 §20.6, 21 §21.4). So:

$$\text{cost coefficient } \beta_c \;=\; \text{scale parameter } \mu \;=\; 1/\sigma_\varepsilon.$$

Why this matters. WTP space *explicitly separates scale from WTP*: β_c is the scale (how consistently the respondent chooses / the inverse error variance), and ω_k are the WTPs (the money values). This is a *feature* — it makes the scale visible and estimable — but it is also a *warning*, because it means that anything that changes β_c (the scale) is entangled with the utility structure. Three consequences:

1. **A larger β_c means sharper (less noisy) choices**, not "more cost-sensitivity" in a preference sense. Reading β_c as a preference is a scale error (Chapters 20, 21).
2. **The WTPs ω_k are scale-free** — they are the money values, purged of scale (since β_c is factored out). This is *why WTP is comparable across groups even when scale differs*: WTP space isolates exactly the scale-free quantity you want to compare (Chapters 18, 21).
3. **Your cost-vector treatments interact with β_c.** This is the crux for your paper.

The cost-vector question (your PDF's "le differenze tra price-vector treatments possono riflettere cambiamenti di scala, di costo o di preferenze?"). When you manipulate the cost vector across arms, a difference in the estimated model between arms could reflect *three distinct things*, and WTP space helps you see them:

- **A scale difference** — the high-cost arm chooses more (or less) consistently, changing β_c (the scale). This shows up in β_c and, in preference space, would masquerade as uniformly-different coefficients.
- **A cost-coefficient difference** — the marginal (dis)utility of cost genuinely differs by arm (people weight money differently at different cost levels).
- **A preference difference** — the WTPs ω_k themselves differ by arm (the cost vector changed how much people value the attributes — *preference formation*, your central theme).

Because in WTP space β_c (scale) and ω_k (WTP) are separate parameters, you can *in principle* attribute a cost-vector effect to scale (β_c) versus WTP (ω_k) — but only if the design and model identify them separately, which is exactly the scale-vs-preference identification problem of Chapter 25 (and the κ-vs-λ separation your PDF worries about, Chapter 30). WTP space *frames* the question correctly (scale vs WTP are distinct parameters) but does not, by itself, *solve* the identification — you still need a design and model that let scale differ across arms (a scaled/heteroskedastic model, Chapter 25) to separate a scale effect from a preference effect. **WTP space is the right coordinate system for your cost-vector question; Chapter 25 is where the separation is actually attempted.**

> **Stop and think 24.2.** In WTP space, your high-cost-vector arm has a larger β_c and similar ω_k (WTPs) compared to the low-cost arm. What does this pattern most parsimoniously say, and what does it *not* say? *(Answer: a larger β_c with similar ω_k most parsimoniously says the high-cost arm chooses* more consistently* (larger scale, less noise) but has the *same WTPs* — a *scale* difference, not a preference difference. It does *not* say the high-cost arm "cares more about cost" in a welfare sense (that would be a preference/WTP change) — because β_c is the scale parameter, not a preference. WTP space makes this visible: the scale (β_c) changed, the WTPs (ω_k) did not, so the cost vector affected choice *consistency*, not valuations — subject to the scale being separately identified, Chapter 25.)*

24.6 Constrained distributions for WTP

WTP space lets you *choose* the WTP distribution, so you should choose one that is *plausible* — your PDF's "la distribuzione scelta produce code irrealistiche?". The distribution of ω_k is now yours to specify:

- **Normal** on WTP — symmetric, light-tailed (verified clean in §24.3), allows both signs. Appropriate when WTP can plausibly be negative for some respondents (some dislike the attribute).
- **Lognormal** on WTP — constrains WTP to be positive (support $(0, \infty)$), for a strictly-desirable attribute; right-skewed but far lighter-tailed than the preference-space ratio.
- **Constrained / bounded distributions** (constrained triangular, bounded normal) — confine WTP to a sensible range, eliminating unrealistic tails entirely. Useful when you have prior bounds on plausible WTP (no one would pay more than €X).
- **Sign constraints** — ensure the WTP has the theoretically correct sign.

The cost coefficient's distribution. In WTP space, β_c (the scale) is usually taken as **positive** (a valid scale must be positive) — often modeled as *fixed* or as *lognormal* (guaranteeing positivity). Crucially, WTP space *avoids the wrong-sign cost coefficient problem*: because WTP is ω_k (a primitive) rather than a ratio, a near-zero or wrong-sign cost coefficient no longer detonates the WTP — the scale β_c can be constrained positive without the ratio explosion. This is the second major advantage of WTP space (after tail control): it decouples the WTP from the cost coefficient's near-zero behavior.

The guidance. Specify the WTP distribution to reflect what you believe about the population of valuations — sign-constrained if the attribute is strictly desirable, bounded if there are plausible limits, normal if WTP can be negative — and *report the choice and its implied tail behavior* (the fraction of implausible WTP).

The freedom to constrain the WTP distribution is the payoff of WTP space; use it deliberately, and check (Krinsky–Robb, Chapter 9) that the implied WTP distribution is realistic.

> **Active recall 24.C.** Name two advantages of WTP space over preference space regarding the WTP distribution and the cost coefficient. *(Answer: (i)* tail control* — you model WTP directly with a light-tailed, bounded, or sign-constrained distribution of your choosing, eliminating the heavy tails of the ratio (verified 99th percentile €39 vs €144); (ii) *decoupling from the cost coefficient* — because WTP is a primitive ω_k rather than a ratio, a near-zero or wrong-sign cost coefficient no longer detonates the WTP, and the scale β_c can be constrained positive without the ratio explosion.)*

24.7 Interpreting WTP-space estimates

Reading the output of a WTP-space EC-MXL differs from preference space, and getting it right is essential:

- **The ω_k estimates are WTP directly.** The mean of ω_k's distribution is the *mean WTP* for attribute k; its variance is the *heterogeneity in WTP* across respondents (preference heterogeneity in money terms — a headline result). No ratio, no ex-post transformation.
- **The β_c estimate is the scale.** Interpret it as choice consistency (inverse error variance), *not* as a cost preference. Its variance (if random) is *scale heterogeneity* — how much respondents differ in consistency (Chapter 25).
- **WTP standard errors and intervals** come directly from the ω_k distribution (and Krinsky–Robb for the population WTP), and are far better-behaved than the delta-method interval on a preference-space ratio (Chapters 7, 9). Report median and percentile intervals for the WTP distribution.
- **Comparisons across arms/groups** use the ω_k (WTP, scale-free) — directly comparable — while noting that a difference could still involve β_c (scale). This is the correct frame for your convergent-validity and cost-vector comparisons (Chapters 8, 25).

The honest caveat. Even in WTP space, the *variance* of ω_k ("WTP heterogeneity") is only cleanly "preference heterogeneity" if scale is homogeneous or separately modeled (Chapter 25) — because β_c (scale) and ω_k (WTP) can still be entangled if the design does not identify them separately. So WTP space gives you the right parameters to *read*, but the scale-vs-preference identification (Chapter 25) still governs what a between-group difference *means*.

> **Stop and think 24.3.** Your WTP-space EC-MXL reports a mean $\omega_{\text{biodiversity}}$ of €35 with a standard deviation of €20. What do the €35 and the €20 each mean, and what caveat applies to calling the €20 "preference heterogeneity"? *(Answer: the €35 is the* mean WTP* for biodiversity across respondents (directly, no ratio); the €20 is the *standard deviation of WTP* — how much respondents differ in their biodiversity valuation (heterogeneity in money terms). Caveat: the €20 is cleanly "preference heterogeneity" only if the *scale* (β_c) is homogeneous or separately modeled; if scale varies across respondents and is not separated, part of the €20 could reflect *scale* heterogeneity (consistency differences) masquerading as WTP heterogeneity — Chapter 25.)*

24.8 Failure modes

F1. Preference-space WTP with a normal cost coefficient. A ratio of coefficients with a normal (mass-near-zero, wrong-sign) cost coefficient → heavy tails, absurd WTP, unstable mean. *Fix:* WTP space, or a sign-constrained cost coefficient; report median/percentile (Krinsky–Robb) not mean/delta (§24.2; Chapters 6, 22).

F2. Treating the two spaces as equivalent for a mixed logit. Assuming WTP space and preference

space give the same model/WTP. *Fix:* they are non-equivalent (different WTP distributions, fits); compare via AIC/BIC; state the restriction (§24.4).

F3. Ignoring the WTP-space restriction. Using WTP space without stating that β_c is a common scale on all attributes. *Fix:* report the restriction and its plausibility (§24.4).

F4. Reading β_c as a cost preference. Interpreting the WTP-space cost coefficient as "how much people dislike cost" rather than the scale. *Fix:* β_c is the scale (choice consistency); interpret accordingly (§24.5; Chapters 20, 21).

F5. Unrealistic WTP distribution. Choosing a WTP distribution (even in WTP space) that implies implausible tails or signs. *Fix:* constrain the WTP distribution (bounded, sign-constrained); check the implied tail (§24.6).

F6. Attributing a cost-vector effect to preference without ruling out scale. Reading a cost-vector arm difference as preference formation when it could be scale. *Fix:* model scale separately; WTP space frames but does not solve the identification (§24.5; Chapter 25).

F7. Symmetric delta interval on a preference-space WTP. A ±SE band on a heavy-tailed ratio. *Fix:* Krinsky–Robb median + percentile interval; better, WTP space (§24.2; Chapters 7, 9).

F8. Not comparing spaces. Reporting only WTP space (or only preference space) without checking the other. *Fix:* estimate both, compare fit (AIC/BIC) and WTP, and report the choice and its consequences (§24.4).

24.9 Diagnostic questions for your own papers

1. **Space and restriction.** Do I estimate in WTP space, and have I stated the restriction it imposes (β_c a common scale on all attributes)? *(§24.3–§24.4.)*
2. **WTP distribution.** Is my chosen WTP distribution plausible (sign, tails, support)? What fraction of implausible WTP does it imply? *(§24.6, F5.)*
3. **Cost coefficient.** Am I interpreting β_c as the *scale* (consistency), not a cost preference, and is it constrained positive? *(§24.5, F4.)*
4. **Tail control.** Have I confirmed my WTP distribution is not heavy-tailed (median vs mean, 99th percentile), especially vs a preference-space fit? *(§24.2–§24.3, F1.)*
5. **Non-equivalence.** Have I compared WTP-space and preference-space fit (AIC/BIC) and WTP, recognizing they are different models? *(§24.4, F2, F8.)*
6. **Cost-vector interpretation.** Could a cost-vector arm difference reflect scale (β_c) rather than WTP (ω_k)? Have I modeled scale to separate them? *(§24.5, F6; Chapter 25.)*
7. **WTP inference.** Am I reporting median + percentile (Krinsky–Robb) intervals for WTP, not symmetric delta intervals? *(§24.7, F7.)*

24.10 Exercises

(A) Comprehension.

A1. Derive WTP $= \beta_q/\beta_c$ from the preference-space utility, and explain why it is problematic in a mixed logit.

A2. Write the WTP-space utility and identify where WTP and the scale appear.

A3. Are preference space and WTP space the same model? Answer for a fixed-coefficient and a mixed logit.

(B) Derivation / reasoning.

B1. Derive the WTP-space reparameterization $V = \beta_c(\sum_k \omega_k x_k - c)$ from preference space, and state the restriction it imposes on the random-coefficient structure.

B2. Explain analytically and with the near-zero-cost-coefficient result why the preference-space WTP has heavy tails, and how WTP space eliminates them.

B3. Show that in the WTP-space utility, the cost coefficient plays the role of the scale parameter, and explain the consequence for interpreting a between-group difference in β_c.

(C) Research application / critique.

C1. Your preference-space mixed logit reports a mean WTP of €90 with a 95% CI of €[−30, €260]. Diagnose the problem and specify the WTP-space fix, stating the restriction you accept.

C2. A referee asks why WTP space "isn't just a reparameterization" of preference space. Explain the non-equivalence for a mixed logit and how you would compare the two.

C3. Your cost-vector arms differ in the estimated model. Explain how WTP space frames the "scale vs cost vs preference" question and why it does not, by itself, answer it.

24.11 Solutions

A1. From $V = \beta_q q - \beta_c c$, holding utility constant ($dV = \beta_q\, dq - \beta_c\, dc = 0$) gives the trade-off $dc/dq = \beta_q/\beta_c = \text{WTP}_q$. In a mixed logit β_q and β_c are *random*, so WTP is a *ratio of two random variables*; when the cost coefficient β_c has mass near zero, the ratio is heavy-tailed (near-Cauchy) with an unstable mean and (if β_c can be positive) wrong-sign, absurd draws — the WTP-ratio pathology (verified: median €20 but 99th percentile €144, ~1% above €200).

A2. $V = \beta_c(\sum_k \omega_k x_k - c)$: WTP appears *directly* as ω_k (the money value multiplying attribute x_k), and the *scale* appears as β_c (the cost coefficient multiplying the whole bracket).

A3. For a *fixed*-coefficient logit they are the *same model* — a reparameterization ($\omega_q = \beta_q/\beta_c$), with identical fit and WTP. For a *mixed* logit they are *different models*: they place distributions on different objects (coefficients vs WTP), inducing different WTP distributions (a ratio distribution vs a direct one) and different fits/likelihoods (non-equivalent, verified).

B1. Factor β_c out of $V = \sum_k \beta_k x_k - \beta_c c$: $V = \beta_c(\sum_k (\beta_k/\beta_c)x_k - c) = \beta_c(\sum_k \omega_k x_k - c)$ with $\omega_k = \beta_k/\beta_c$. Restriction: the single cost coefficient β_c now multiplies *every* attribute's contribution (a common scale on all attributes), and the attribute-specific heterogeneity is carried by the ω_k distributions — whereas preference space lets each β_k have its own free distribution and covariance with β_c. So WTP space restricts the random-coefficient covariance structure (the cost coefficient is a common multiplicative factor), a genuine restriction, not a mere reparameterization, for the mixed logit.

B2. WTP $= \beta_q/\beta_c$: analytically, $\partial\text{WTP}/\partial\beta_c = -\beta_q/\beta_c^2$ blows up as $\beta_c \to 0$ (Chapter 2), so the ratio is extremely sensitive to small cost coefficients, and a distribution with mass near zero produces a heavy tail (near-Cauchy, Chapter 6) — verified: as the mean cost coefficient shrank from 0.06 to 0.02, the 99th-percentile WTP rose from €47 to €362 at constant median. WTP space eliminates this because WTP is the primitive ω_k with a distribution *you* choose (light-tailed, bounded, or sign-constrained) — there is no denominator to approach zero and no ratio to explode (verified: 99th percentile €39, no absurd draws).

B3. In $V = \beta_c(\sum_k \omega_k x_k - c)$, β_c multiplies the *entire* utility, which is exactly the role of the scale parameter $\mu = 1/\sigma_\varepsilon$ (a common multiplier on utility governs choice sharpness, Chapters 20 §20.6, 21 §21.4). Consequence: a between-group difference in β_c is a difference in *scale* (choice consistency / inverse error variance), *not* in cost preference — so reading a larger β_c as "cares more about cost" is a scale error; the group with larger β_c simply chooses more consistently, and the scale-free WTPs ω_k are what to compare for preferences (Chapter 25).

C1. The symmetric, very wide interval spanning zero is the signature of the preference-space WTP-ratio pathology: WTP $= \beta_q/\beta_c$ with a cost coefficient near/across zero has a heavy, near-Cauchy tail, so the mean (€90) is unstable and the delta-method interval is symmetric and enormous (Chapters 2, 6). Fix: re-estimate in **WTP space**, modeling WTP directly with a light-tailed (or bounded/sign-constrained) distribution and a positive scale β_c — this tames the tail (verified) and decouples WTP from the near-zero cost coefficient. State the restriction accepted: the cost coefficient β_c is a common multiplicative scale on all attributes (a restriction on the random-coefficient covariance), defensible because β_c *is* the scale. Report the median and percentile interval (Krinsky–Robb), not the mean and symmetric CI.

C2. For a *mixed* logit, WTP space and preference space are not the same model: preference space places distributions on the *coefficients* (β_q, β_c) and derives WTP as their *ratio* (a heavy-tailed distribution), while WTP space places distributions on *WTP* (ω_k) directly (a distribution you choose) and implies a *product* distribution for the coefficients. Calibrated to the same median WTP, the two produce different WTP distributions (verified: heavy-tailed ratio vs symmetric normal) and different likelihoods/fits — so they are genuinely different models, not reparameterizations (they *are* reparameterizations only for a fixed-coefficient logit). I would compare them via **AIC/BIC** (a non-nested comparison, Chapters 12, 23), compare their WTP estimates and tail behavior, and choose based on fit *and* plausibility (WTP space's interpretable, well-behaved WTP is usually decisive).

C3. WTP space *frames* the cost-vector question correctly by separating the parameters: a between-arm difference could be in the **scale** β_c (choice consistency), the **cost coefficient** (which in WTP space *is* the scale), or the **WTPs** ω_k (preference/valuation — preference formation). Because β_c (scale) and ω_k (WTP) are distinct parameters, WTP space lets you *ask* whether the cost vector changed scale or WTP. But it does not, by itself, *answer* it: identifying a scale difference apart from a preference difference requires a design and model that let the scale differ across arms (a scaled/heteroskedastic model) and enough variation to separate them — the scale-vs-preference identification problem (Chapter 25) and the κ-vs-λ separation (Chapter 30). WTP space is the right coordinate system for the question; the identification is done in Chapter 25.

24.12 Spaced review prompts

+1 day. From memory, derive WTP $= \beta_q/\beta_c$ and the WTP-space reparameterization $V = \beta_c(\sum_k \omega_k x_k - c)$; explain the ratio pathology and how WTP space eliminates it.

+1 week. Explain the non-equivalence of the two spaces for a mixed logit, the restriction WTP space imposes, and why the cost coefficient is the scale parameter. State how this frames the cost-vector "scale vs preference" question.

+1 month. For your own EC-MXL, estimate in both spaces (or examine your WTP-space model against a preference-space check), compare WTP distributions and fit, state the restriction, interpret β_c as scale and ω_k as WTP, and audit the implied WTP tails. Close with the spine sentence about which space you use and why.

24.13 How this chapter connects to the rest of the book

- **Chapter 2 (§2.11)** first derived WTP as a ratio and sketched the two spaces; **Chapter 6 (§6.9)** the near-Cauchy behavior; **Chapters 7, 9** the delta-vs-Krinsky–Robb WTP inference; **Chapter 15 (§15.3)** the CV ancestor (Hanemann vs Cameron); **Chapter 22 (§22.5)** the random-coefficient distributions and wrong-sign draws.
- **Chapters 20 (§20.6), 21 (§21.4)** are the scale identification behind "the cost coefficient is the scale" (§24.5); **Chapter 25 (scale vs preference heterogeneity)** is where the cost-vector scale-vs-preference identification (§24.5) is actually attempted — WTP space frames it, Chapter 25 solves (or bounds) it.

- **Chapter 23 (SML)** estimates the WTP-space EC-MXL and compares spaces via AIC/BIC; **Chapter 30 (identification)** is the κ-vs-λ separation; **Chapters 35, 49 (aggregation, benefit transfer)** consume the WTP this chapter produces.

This chapter — one of the three the architecture flags to over-learn — resolves the WTP-tail problem that has recurred since Chapter 2 and establishes the coordinate system (WTP space, with the cost coefficient as scale) in which your DCE's WTP and cost-vector questions are correctly posed.

24.14 Research translation task

For your **WTP-space EC-MXL**, produce a two-to-three-page note:

1. **Derive the model.** Write your WTP-space utility, identify the WTP parameters ω_k and the scale β_c, and state the restriction WTP space imposes.
2. **Justify the WTP distributions.** For each ω_k, state its distribution (normal/lognormal/constrained) and justify it; report the implied WTP median, mean, and 99th percentile and any implausible tail.
3. **Compare spaces.** Estimate (or reference) a preference-space fit; compare WTP distributions and fit (AIC/BIC); document the tail difference (verified pattern).
4. **Interpret.** Read β_c as scale and ω_k as WTP; report WTP with Krinsky–Robb median + percentile intervals.
5. **Cost-vector framing.** State how WTP space frames the scale-vs-cost-vs-preference question for your cost-vector arms, and note that the separation is done in Chapter 25.
6. **Close** with the spine sentence: *"My DCE estimates WTP directly in WTP space (WTP = ω_k, scale = β_c); this tames the ratio tail ([99th pct]) at the cost of [the common-scale restriction]; a cost-vector arm difference could be [scale/WTP], separated in [Chapter 25]."*

Carry this into Chapter 25 (scale vs preference), Chapter 23 (estimation/comparison), and Chapters 36, 44 (welfare/CBA use of the WTP).

24.15 Bibliography

Core. - Train & Weeks (2005), "Discrete Choice Models in Preference Space and Willingness-to-Pay Space," in *Applications of Simulation Methods in Environmental and Resource Economics* (eds. Scarpa & Alberini). The foundational paper — the reparameterization, the non-equivalence, and the distributional implications of §24.3–§24.4. *The* essential reference for this chapter. *(Verify exact title/editors/year.)* - Train, *Discrete Choice Methods with Simulation*, the mixed-logit and WTP-space material. Freely available — *verify link.* - Scarpa, Thiene & Train (2008), "Utility in Willingness to Pay Space: A Tool to Address Confounding Random Scale Effects in Destination Choice to the Alps," *American Journal of Agricultural Economics*. WTP space and its interaction with scale (§24.5) in environmental valuation — directly relevant. *(Verify volume/pages.)*

Applied. - Hole & Kolstad (2012), "Mixed logit estimation of willingness to pay distributions: a comparison of models in preference and WTP space," *Empirical Economics*. A direct comparison of the two spaces' WTP distributions (§24.2–§24.4). *(Verify volume/pages.)* - Hensher & Greene, and the Apollo manual (Hess & Palma), for implementing WTP-space mixed logit and constrained WTP distributions (§24.6). *(Verify editions/versions.)* - Daly, Hess & Train (2012), "Assuring finite moments for willingness to pay in random coefficient models," *Transportation*. On the heavy-tail/finite-moment problem of the WTP ratio and constrained distributions (§24.2, §24.6). *(Verify volume/pages.)*

Advanced. - Fiebig, Keane, Louviere & Wasi (2010), "The Generalized Multinomial Logit Model," *Marketing Science*. The scale-vs-preference separation that WTP space frames (§24.5; Chapter 25). *(Verify volume/pages.)* - Train & Weeks (as above) and Sonnier, Ainslie & Otter (2007), on WTP-space priors and

distributions in a Bayesian setting. *(Verify citation.)* - Cameron (1988), *JEEM* — the CV ancestor of WTP space (the WTP-function parameterization, Chapter 15 §15.3). *(Verify volume/pages.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Train–Weeks / Scarpa et al. / Daly et al. citations before using them, per the book's no-unverified-citation rule.

End of Chapter 24. Next in Part V: Chapter 25 — Scale Heterogeneity versus Preference Heterogeneity (the book's central discrimination problem: how differences in error variance, attention, comprehension, familiarity, and the cost coefficient masquerade as differences in preference, and what can and cannot separate them), which resolves the scale-vs-preference question that Chapters 18, 20, 21, and 24 have all pointed toward — and that your cost-vector and convergent-validity work turns on. Tell me when to write it, or request a different chapter.

Chapter 25 — Scale Heterogeneity versus Preference Heterogeneity

Part V: Random Utility, Discrete Choice, and Stated Preference Econometrics

> **The spine question, applied to this chapter.** This is the book's central discrimination problem, and your PDF ranks it *"Priorità massima."* When you compare groups or treatments in your DCE — high vs low cost vector, experienced vs inexperienced, hypothetical vs incentivized — a difference in estimated coefficients can mean one of two fundamentally different things, and they are easy to confuse: a genuine **preference** difference (people value the attributes differently) or a mere **scale** difference (people choose more or less *consistently*, so their error variance differs). Your PDF lists the culprits that create scale differences masquerading as preferences: *"differenze nella varianza dell'errore; differenze di attenzione; differenze di comprensione; differenze di familiarità; differenze nel coefficiente di costo."* And it names where this bites hardest: *"Questo è cruciale nel paper su cost-vector effects e familiarità."* The spine question here: *when I see a between-group difference in my choice model, is it preference or scale — and do I have the evidence to tell them apart, or are they observationally equivalent given my design?*

25.0 Conceptual motivation: the difference that is not what it seems

Everything in Part V has been building to this chapter. Chapter 18 (§18.4) warned that binary-model coefficients are identified only up to the error scale. Chapter 20 (§20.6) proved it: choices identify only $\mu\beta$ — preferences *times* scale. Chapter 21 (§21.4) made the scale parameter concrete and named the confound. Chapter 22 (§22.10) introduced the generalized MNL as a way to separate scale from preference. Chapter 24 (§24.5) showed that in WTP space the cost coefficient *is* the scale. Now we confront the problem directly, because it is the one your PDF flags as maximum priority and as central to your cost-vector and familiarity papers.

The problem is this. A respondent's choices are governed by their *tastes* (the coefficients β) and by how *consistently* they choose (the scale $\mu = 1/\sigma_\varepsilon$, the inverse of their error variance). More attentive, more comprehending, more familiar respondents choose more consistently — larger μ — and *the estimated coefficients are* $\mu\beta$, so a group that is merely *less noisy* has *uniformly larger coefficients*, which looks exactly like a group with *stronger preferences*. The two are confounded at the level of coefficients, and — as this chapter will show — often **observationally equivalent** without additional structure.

Why does this matter so acutely for you? Because your central research questions are *between-group comparisons* where scale differences are plausible:

- **Cost-vector effects.** When you compare high- and low-cost-vector arms, a difference could be a genuine change in preferences (the cost vector affecting *preference formation* — your central hypothesis) *or* a change in scale (the high-cost arm choosing more/less consistently) *or* a change in the cost coefficient. Reading a scale effect as preference formation would be a serious error.

- **Familiarity and experience.** Experienced or familiar respondents almost certainly choose *more consistently* (larger scale) — so their larger coefficients may reflect *consistency*, not *stronger preferences*. Your familiarity paper turns on getting this right.
- **Convergent validity.** Comparing WTP across formats (PC vs SBDC, hypothetical vs incentivized) requires accounting for scale differences between formats — otherwise a scale difference masquerades as a validity failure or success (Chapter 8).
- **Attention and comprehension.** A treatment that reduces attention or comprehension (a harder task, a longer survey) increases noise → *reduces scale* → *shrinks all coefficients* → looks like *weaker preferences*. So a design artifact masquerades as a preference effect.

This chapter builds the discrimination: it defines scale and preference heterogeneity precisely; shows *why* they are confounded (the $\mu\beta$ identification); enumerates the explanations that are observationally equivalent to a preference difference (the learning objective); establishes the *fingerprint* that partly separates them (uniform scaling vs relative change; WTP is scale-free — verified) and its *limits*; presents the models that attempt the separation (scaled MNL, S-MNL, G-MNL, Swait–Louviere); states honestly what *can* and *cannot* be separated (a partial-identification problem, Chapter 30); and applies it to your cost-vector, familiarity, and convergent-validity work. It closes Part V by resolving the theme that has run through every discrete-choice chapter.

25.1 Prerequisites

From **Chapter 20** (§20.6): the scale normalization — choices identify only $\mu\beta$. From **Chapter 21** (§21.4): the scale parameter $\mu = 1/\sigma_\varepsilon$ and its confound with coefficients. From **Chapter 24** (§24.5): the cost coefficient as the scale in WTP space; WTP as a scale-free ratio. From **Chapter 22** (§22.10): the generalized MNL. From **Chapter 8**: equivalence testing (for convergent-validity claims). From **Chapter 30** (preview): observational equivalence and partial identification.

Same **notation**. Consolidated symbols: β (preference/taste coefficients), $\mu = 1/\sigma_\varepsilon$ (scale parameter — inverse error standard deviation, "choice consistency"), $\mu\beta$ (the identified product), ω_k (WTP, scale-free), λ (relative scale of one group to another, Swait–Louviere).

25.2 Two kinds of heterogeneity, defined

Definition 25.1 (Preference heterogeneity). Respondents (or groups) differ in their **tastes** — the coefficients β themselves differ. Group A values biodiversity more than group B; the *relative* weights on attributes differ. This is what a mixed logit's random coefficients (Chapter 22) are meant to capture, and it is what your welfare estimates are supposed to reflect.

Definition 25.2 (Scale heterogeneity). Respondents (or groups) differ in how **consistently** they choose — the error variance σ_ε^2, and hence the scale $\mu = 1/\sigma_\varepsilon$, differs. A group with a *larger* scale (smaller error variance) chooses more deterministically — their choices are more sharply governed by utility, less by noise. This is *not* a difference in tastes; it is a difference in the *precision* with which tastes translate into choices.

The crucial distinction, in one sentence. Preference heterogeneity is about *what* respondents value; scale heterogeneity is about *how consistently* they act on those values. A more attentive respondent does not necessarily value the attributes differently — they just choose more consistently, so their choices reveal their (possibly identical) preferences more sharply.

Why scale heterogeneity is pervasive in your data. Anything that affects *choice consistency* — independent of tastes — creates scale heterogeneity:

- **Attention** — a distracted or fatigued respondent chooses more noisily (smaller scale).

- **Comprehension** — a respondent who understands the task better chooses more consistently (larger scale).
- **Familiarity/experience** — a respondent familiar with the good chooses more consistently (larger scale) — *central to your familiarity paper.*
- **Task difficulty** — a harder choice task (more attributes, higher stakes) may increase noise (smaller scale).
- **Cost level** — the cost vector may affect how carefully respondents engage (scale), independent of their valuations — *central to your cost-vector paper.*

Every one of these is on your PDF's list, and every one produces a *scale* difference that, at the level of estimated coefficients, is *indistinguishable from a uniform preference difference* — the confound we now derive.

> **Active recall 25.A.** Distinguish preference heterogeneity from scale heterogeneity in one sentence each, and give one factor from your work that creates scale (not preference) heterogeneity. *(Answer: preference heterogeneity = differences in* tastes* (the coefficients β differ — what respondents value); scale heterogeneity = differences in *choice consistency* (the error variance / scale μ differs — how sharply tastes translate into choices, with the same tastes). A factor creating scale heterogeneity in your work: *familiarity/experience* (familiar respondents choose more consistently, larger scale) or the *cost vector* (may affect engagement/consistency) — both central to your papers.)*

25.3 Why they are confounded: the $\mu\beta$ identification

The confound is not a subtlety of estimation — it is a *fundamental* consequence of what choices can reveal (Chapter 20, §20.6). The choice probability depends on utility only through $\mu V = \mu \mathbf{x}'\beta$, so the data identify the **product** $\mu\beta$, never μ and β separately. Software normalizes $\mu = 1$, so **every estimated coefficient is $\mu\beta$ — preferences times scale.**

The consequence for group comparisons. Suppose group A has scale μ_A and group B has scale $\mu_B = 2\mu_A$ (B chooses twice as consistently), but *identical tastes* β. Then:

$$\text{A's estimated coefficients} = \mu_A\beta, \qquad \text{B's estimated coefficients} = \mu_B\beta = 2\mu_A\beta.$$

B's coefficients are exactly twice A's — for every attribute — with no difference in tastes whatsoever. A researcher comparing raw coefficients would see "B has stronger preferences for everything" and might conclude B values the good more. But B values the good *identically*; B just chooses more consistently. The coefficient difference is *pure scale*, and it is *completely indistinguishable* from a preference difference *if you only look at coefficients.*

The fingerprint (and its limit). There is a partial tell: a *pure scale* difference scales *all* coefficients by the *same* factor (uniform scaling), while a *preference* difference changes the *relative* magnitudes of coefficients. So if B's coefficients are all 2× A's, that is the signature of scale; if B's biodiversity coefficient is 2× but its cost coefficient is 1×, that is a genuine preference (relative) difference. **The scale-free way to see this is WTP:** since WTP is a *ratio* of coefficients, the scale μ cancels ($\text{WTP} = \mu\beta_k/\mu\beta_c = \beta_k/\beta_c$), so **a pure scale difference leaves WTP unchanged, while a preference difference changes WTP.** This is the single most useful diagnostic in the chapter.

Worked Example 25.1 (Scale vs preference, verified). Simulate choice data for four groups and estimate an MNL (scale normalized to 1) for each. Groups A, B, C have *identical tastes* ($\beta_q = 1.0$, $\beta_c = -0.05$, true WTP = 20) but *different scales* (1, 2, 0.5); group D has a *stronger quality preference* ($\beta_q = 1.5$, WTP = 30) at scale 1:

Group	Estimated $(\hat{\beta}_q, \hat{\beta}_c)$	WTP $= \hat{\beta}_q/\lvert\hat{\beta}_c\rvert$	Interpretation
A (scale 1)	$(1.00, -0.050)$	**19.97**	base
B (same prefs, scale 2)	$(1.99, -0.098)$	**20.30**	coefs $\approx 2\times$ A, **same WTP** $\rightarrow$ scale
C (same prefs, scale 0.5)	$(0.50, -0.025)$	**20.27**	coefs $\approx 0.5\times$ A, **same WTP** $\rightarrow$ scale
D (stronger quality pref)	$(1.49, -0.051)$	**29.20**	**different WTP** $\rightarrow$ preference

(Verified.) The result is decisive. **Groups A, B, and C have identical WTP (~€20) despite their coefficients differing by the scale factor (2×, 0.5×)** — a pure scale difference scales all coefficients uniformly but leaves WTP untouched. **Group D has a different WTP (~€30)** — a genuine preference difference that changes the ratio. So: *comparing raw coefficients confounds scale and preference; comparing WTP (scale-free) reveals whether the difference is scale (same WTP) or preference (different WTP).* This is exactly the diagnostic your cost-vector and familiarity comparisons need.

> **Stop and think 25.1.** Your experienced group has coefficients uniformly about 1.5× your inexperienced group's, and identical WTP for every attribute. What is the most parsimonious conclusion, and what would have misled you if you had compared only coefficients? *(Answer: uniformly-1.5×-larger coefficients with* identical WTP* is the fingerprint of a pure *scale* difference — experienced respondents choose about 1.5× more consistently (larger scale), with the *same* preferences. Comparing only coefficients would have misled you into concluding experienced respondents have "50% stronger preferences," when in fact their valuations (WTP) are identical — they just choose more consistently. Comparing WTP (scale-free) reveals it is scale, not preference — the central lesson for your familiarity paper.)*

25.4 Observational equivalence: the list that mimics preference

Your learning objective is explicit: given a between-group coefficient difference, *enumerate the explanations observationally equivalent to a preference difference.* Here is the list, each an item on your PDF's §6.3, and each producing coefficient differences that a naive comparison would read as preferences.

Suppose group B's estimated coefficients differ from group A's. The candidate explanations, all observationally equivalent at the level of raw coefficients:

1. **True preference difference** — B genuinely values the attributes differently (β differs). *This changes relative coefficients and WTP.*
2. **Error-variance (scale) difference** — B chooses more/less consistently (μ differs). *This scales all coefficients uniformly; WTP unchanged.*
3. **Attention difference** — B is more/less attentive, changing choice consistency $\rightarrow$ scale. *Uniform scaling; WTP unchanged.*
4. **Comprehension difference** — B understands the task better/worse $\rightarrow$ scale. *Uniform scaling; WTP unchanged.*
5. **Familiarity/experience difference** — B is more/less familiar with the good $\rightarrow$ scale. *Uniform scaling; WTP unchanged.*
6. **Cost-coefficient difference** — B weights money differently (a change in β_c specifically). *This changes WTP (the denominator), and can look like either scale or preference depending on whether other coefficients also change.*

Items 2–5 are all **scale** effects (they change μ, not tastes) and are *mutually* observationally equivalent — attention, comprehension, familiarity, and error variance all operate through the same channel (choice consistency), so you cannot tell *which* of them caused a scale difference from the choice data alone. Item 6

(cost coefficient) is special because it changes WTP even without a taste change, so it is confounded with preference in the WTP comparison. Only item 1 is a genuine preference difference, and it is the one your welfare analysis cares about.

The identification problem, stated. A single cross-section of choices *cannot distinguish* a preference difference (item 1) from the family of scale differences (items 2–5) *by coefficients alone*, and *cannot fully distinguish* a preference difference from a cost-coefficient difference (item 6) even by WTP. The WTP diagnostic (§25.3) separates *scale* (items 2–5, same WTP) from *preference-or-cost* (items 1, 6, different WTP), but it does *not* separate a genuine preference change from a cost-coefficient change, nor does it identify *which* scale factor operated. This is a **partial identification** problem (Chapter 30): the data narrow the explanations but do not pin down one, unless you add structure (a scaled model, §25.5) or design variation.

> **Active recall 25.B.** List the explanations observationally equivalent to a preference difference when you see a between-group coefficient difference, and state which of them the WTP comparison can rule out. *(Answer: (1) true preference difference; (2) error-variance/scale difference; (3) attention; (4) comprehension; (5) familiarity/experience; (6) cost-coefficient difference. Items 2–5 are* scale* differences (uniform coefficient scaling, WTP unchanged); item 6 changes WTP; item 1 changes WTP and relative coefficients. The WTP comparison rules out the *pure scale* explanations (2–5) if WTP *differs* (it must then be preference or cost, items 1/6), or points to scale if WTP is *unchanged* — but it cannot separate a genuine preference change (1) from a cost-coefficient change (6), nor identify *which* scale factor (2–5) operated.)*

25.5 Models that separate scale and preference

To go beyond the WTP fingerprint, you must *model* scale explicitly. Several models attempt the separation, in increasing generality:

The scaled (heteroskedastic) MNL / Swait–Louviere. The simplest: let the scale differ across groups, $V_g = \mu_g \mathbf{x}'\beta$, with a *common* β and group-specific scales μ_g. The **Swait–Louviere** procedure estimates the *relative scale* $\lambda = \mu_B/\mu_A$ between two groups (normalizing one), testing whether a between-group difference is *pure scale* (a single λ absorbs it, with common β) or requires different *preferences* (the common-β-plus-scale model fails). This is the workhorse for your two-group comparisons (experienced vs inexperienced, high vs low cost vector): if a single relative-scale parameter reconciles the two groups' choices, the difference is scale; if not, there is a genuine preference difference.

The scale-heterogeneity MNL (S-MNL). Lets the scale μ_n vary *randomly across respondents* (a distribution on individual scale), capturing *within-sample* scale heterogeneity (some respondents choose more consistently than others) rather than just between-group.

The generalized MNL (G-MNL). Nests both: it has *random preference coefficients* (preference heterogeneity, as in the mixed logit) *and* a *random scale* (scale heterogeneity), with a parameter governing how much of the heterogeneity is scale versus taste. In principle it separates the two — the random-coefficient variance is preference heterogeneity, the scale variance is consistency heterogeneity.

WTP space (Chapter 24). Recall that in WTP space the cost coefficient *is* the scale ($\beta_c = \mu$), and the WTPs ω_k are scale-free. So WTP space already *isolates* the scale (β_c) from the preferences (ω_k) — a between-group difference in β_c is a scale difference, a difference in ω_k is a preference difference. This is why WTP space is the natural coordinate system for your scale-vs-preference question (Chapter 24, §24.5): it puts scale and WTP in separate parameters.

The strategy. For a *two-group* comparison, use Swait–Louviere to test whether a single relative scale reconciles the groups (scale) or a preference difference remains. For *within-sample* heterogeneity, use G-MNL or WTP space to separate the scale and preference variances. And always cross-check with the *WTP* comparison (scale-free), which is model-light and often decisive.

Stop and think 25.2. You suspect your high-cost-vector arm differs from the low-cost arm only in scale (consistency), not preference. What model would you estimate to test this, and what result would confirm "pure scale"? *(Answer: a Swait–Louviere (scaled MNL) comparison — estimate a model with a* common* preference vector β and a *relative scale* parameter $\lambda = \mu_{\text{high}}/\mu_{\text{low}}$ between the arms. "Pure scale" is confirmed if a single λ (with common β) adequately fits both arms — i.e., the arms' choices are reconciled by one scale ratio, with no significant preference (relative-coefficient / WTP) difference remaining. If, after allowing λ, the arms still require different β (different WTP), there is a genuine preference difference — cost-vector preference formation, not just scale.)*

25.6 What can and cannot be separated

Honesty requires stating the limits: even with the models of §25.5, scale and preference heterogeneity are *hard* to separate empirically, and sometimes cannot be fully separated. This is a fundamental point your PDF's "priorità massima" ranking reflects.

What can be separated (with the right design/model): - Scale (items 2–5) from preference-or-cost (items 1, 6): the WTP comparison (scale-free) does this — same WTP → scale, different WTP → preference or cost. Model-light and robust. - **A common-scale-plus-common-preference model from a preference difference (two groups):** Swait–Louviere tests whether one relative scale reconciles the groups. - **Between-group scale from between-group preference:** with sufficient design variation and a scaled model.

What cannot be cleanly separated: - Which scale factor (attention vs comprehension vs familiarity vs error variance) operated — items 2–5 are mutually observationally equivalent from choice data alone; distinguishing them requires *external* measures (attention checks, comprehension tests, familiarity questions, response times — Chapter 50) correlated with the scale, not the choices themselves. - **A genuine preference change from a cost-coefficient change (item 1 vs 6)** — both change WTP, and separating them requires the cost coefficient to be independently identified, which a cost-vector manipulation may or may not allow (Chapter 30, the κ-vs-λ separation). - **Scale heterogeneity from preference heterogeneity *within* a sample, in general** — the G-MNL that nests both is *fragilely identified*: the data often cannot reliably apportion the total heterogeneity between scale and taste, so G-MNL estimates of "how much is scale" can be unstable and sensitive to specification. This is a well-documented limit — you can *ask* the G-MNL to separate them, but the answer may be weakly identified.

The fundamental point (partial identification, Chapter 30). Scale and preference heterogeneity are, in an important sense, **observationally close**: many patterns in choice data are consistent with either. The data *narrow* the explanations (the WTP diagnostic rules out pure scale when WTP differs) but rarely *pin down* a single one without additional structure — a scaled model, external measures of the scale factors, or design variation that identifies scale separately. Recognizing this — that your cost-vector or familiarity "effect" may be *partially* identified between scale and preference — is the mature stance your PDF demands. The correct report is often: "the difference is consistent with a scale effect *and* a preference effect; the WTP comparison [does/does not] rule out pure scale; a scaled model attributes [X] to scale; the remainder is a candidate preference effect that I cannot fully separate from [cost-coefficient / scale] without [additional structure]."

Active recall 25.C. State one thing the WTP comparison *can* separate and two things that *cannot* be cleanly separated from choice data alone. *(Answer: the WTP comparison* can* separate pure *scale* differences (same WTP) from *preference-or-cost* differences (different WTP). It *cannot* cleanly separate: (i) *which* scale factor (attention, comprehension, familiarity, error variance) operated — these are mutually observationally equivalent, requiring external measures; and (ii) a genuine *preference* change from a *cost-coefficient* change (both change WTP), which requires the cost coefficient to be independently identified. Also, within-sample scale vs preference heterogeneity (G-MNL) is often only weakly identified.)*

25.7 Application to your cost-vector and familiarity work

Bringing the chapter to your specific papers, where this is "cruciale."

Familiarity and experience. Your familiarity paper compares experienced/familiar respondents to inexperienced/unfamiliar ones. The prediction from this chapter is stark: **experienced respondents almost certainly have a larger scale** (they choose more consistently), so their coefficients will be uniformly larger *even if their preferences are identical*. To claim a *preference* effect of familiarity (familiarity changes what people value), you must show that WTP *differs* between the groups (not just coefficients), and — via Swait–Louviere or a scaled model — that a relative-scale difference does *not* reconcile them. If WTP is the same and one scale parameter reconciles the groups, then "familiarity strengthens preferences" is *wrong* — familiarity strengthens *consistency* (scale), not valuation. This is precisely the error your PDF's §6.3 warns against, and it could be the difference between a correct and an incorrect headline finding.

Cost-vector effects. Your central hypothesis is that the cost vector affects *preference formation*. But a cost-vector arm difference could be: (i) preference formation (WTP changes — your hypothesis), (ii) a scale effect (the cost level changes engagement/consistency — WTP unchanged), or (iii) a cost-coefficient effect (money weighted differently — WTP changes but for a different reason). The WTP comparison distinguishes (ii) from (i)/(iii); separating (i) from (iii) requires the identification analysis of Chapter 30 (whether your design identifies preference formation apart from a cost-coefficient/anchoring shift — the κ-vs-λ separation). WTP space (Chapter 24) frames this by putting scale (β_c) and WTP (ω_k) in separate parameters. **You cannot claim "the cost vector affects preference formation" without ruling out scale (and, harder, the cost-coefficient channel).**

Convergent validity. Comparing WTP across elicitation formats (PC vs SBDC, hypothetical vs incentivized, BRIS vs full payment) requires accounting for *format-specific scale*: different formats induce different choice consistency (a payment card may be noisier than a referendum), so a raw WTP difference could be a scale artifact. The Swait–Louviere logic — testing whether a relative scale reconciles the formats before concluding they differ (or agree) — is essential, and it combines with the equivalence testing of Chapter 8: a convergent-validity claim needs both (i) equivalence of WTP within a margin *and* (ii) accounting for scale differences between formats.

Attention and comprehension as design threats. A treatment or format that (inadvertently) reduces attention or comprehension *reduces scale* and *shrinks all coefficients* — which could masquerade as a *preference* effect of the treatment. So a "the treatment reduced valuations" finding must be checked against the possibility that the treatment merely increased noise (reduced scale). External measures — attention checks, comprehension tests, response times (Chapter 50) — correlated with the scale help attribute the effect to scale rather than preference.

> **Stop and think 25.3.** Your cost-vector paper finds the high-cost arm has uniformly larger coefficients and *higher* WTP for the environmental attributes. Walk through what this could and could not be. *(Answer: uniformly larger coefficients* alone* would suggest scale, but the *higher WTP* rules out *pure* scale (scale leaves WTP unchanged) — so there is a genuine change in valuations (preference or cost-coefficient channel). It could be: (i) *preference formation* — the cost vector raised how much people value the attributes (your hypothesis); or (ii) a *cost-coefficient* change — the high-cost arm weights money less, raising WTP (the denominator fell) without a taste change; or a *combination*. The WTP change rules out pure scale but does *not* separate preference formation from the cost-coefficient channel — that requires the identification analysis of Chapter 30 (κ vs λ). Report it as a genuine WTP effect whose *mechanism* (preference vs cost-coefficient) is partially identified.)*

25.8 Failure modes

F1. Reading uniformly-larger coefficients as stronger preferences. The signature of scale, misread as preference. *Fix:* compare WTP (scale-free); if WTP is unchanged, it is scale (§25.3).

F2. Comparing raw coefficients across groups. Confounding $\mu\beta$ with β. *Fix:* compare WTP or model scale (Swait–Louviere, G-MNL) (§25.5).

F3. Concluding "familiarity/experience strengthens preferences" from coefficients. Familiarity operates on scale (consistency), not necessarily preference. *Fix:* show WTP differs and scale does not reconcile the groups (§25.7).

F4. Claiming cost-vector "preference formation" without ruling out scale. A cost-vector arm difference that is scale, not preference. *Fix:* WTP comparison (rules out pure scale) + Chapter 30 identification (preference vs cost-coefficient) (§25.7).

F5. Convergent-validity claims ignoring format scale. Comparing WTP across formats without accounting for format-specific scale. *Fix:* Swait–Louviere relative scale + equivalence testing (§25.7; Chapter 8).

F6. Treating a G-MNL scale/preference split as reliably identified. Over-trusting the G-MNL's apportionment of heterogeneity. *Fix:* recognize the fragile identification; cross-check with WTP and external measures (§25.6).

F7. Mistaking a design-induced scale effect for a preference effect. A harder/noisier treatment shrinking coefficients, read as weaker preferences. *Fix:* check external attention/comprehension measures; model scale (§25.7).

F8. Ignoring partial identification. Claiming a single mechanism when scale and preference are observationally close. *Fix:* report the range of consistent explanations and what your design does/does not separate (§25.6; Chapter 30).

25.9 Diagnostic questions for your own papers

1. **WTP vs coefficients.** For every between-group comparison, am I comparing *WTP* (scale-free), not raw coefficients? *(§25.3, F2.)*
2. **Uniform scaling.** Are the group differences *uniform* across coefficients (scale) or *relative* (preference)? *(§25.3, F1.)*
3. **Scale model.** Have I estimated a scaled model (Swait–Louviere / G-MNL / WTP space) to test whether a scale difference reconciles the groups? *(§25.5.)*
4. **Familiarity/experience.** Have I ruled out that a familiarity/experience "effect" is scale (consistency), not preference? *(§25.7, F3.)*
5. **Cost-vector mechanism.** Have I ruled out scale for a cost-vector effect, and acknowledged the preference-vs-cost-coefficient channel is partially identified? *(§25.7, F4; Chapter 30.)*
6. **Convergent validity.** Am I accounting for format-specific scale before comparing WTP across formats? *(§25.7, F5; Chapter 8.)*
7. **Partial identification.** Do I report the range of explanations consistent with my data and what my design can and cannot separate? *(§25.6, F8.)*

25.10 Exercises

(A) Comprehension.

A1. Define scale and preference heterogeneity, and explain why they are confounded in the estimated coefficients.

A2. Group B's coefficients are all 1.8× group A's, and their WTP is identical. What is the most parsimonious explanation?

A3. List the explanations observationally equivalent to a preference difference when you see a between-group coefficient difference.

(B) Derivation / reasoning.

B1. Show that if two groups differ only in scale ($\mu_B = k\mu_A$, same β), their estimated coefficients differ by the factor k but their WTP is identical.

B2. Explain why attention, comprehension, familiarity, and error variance are *mutually* observationally equivalent from choice data, and what would separate them.

B3. Describe the Swait–Louviere test: what it estimates and what result indicates "pure scale" versus a preference difference between two groups.

(C) Research application / critique.

C1. Your familiarity paper finds experienced respondents have uniformly larger coefficients. Write the analysis needed to determine whether familiarity affects preferences or only scale.

C2. Your cost-vector paper concludes "the high-cost vector changes preferences" from a coefficient difference. Critique this and specify what must be shown.

C3. Explain to a co-author why comparing WTP across your PC and SBDC formats requires accounting for scale, and how you would do it.

25.11 Solutions

A1. Preference heterogeneity = differences in *tastes* (the coefficients β differ); scale heterogeneity = differences in *choice consistency* (the error variance / scale μ differs, with possibly identical tastes). They are confounded because choices identify only the *product* $\mu\beta$ (Chapter 20 §20.6), so a scale difference (μ) and a uniform preference difference (β) produce the *same* estimated coefficients — you cannot separate them from coefficients alone.

A2. The most parsimonious explanation is a **pure scale difference**: group B chooses about 1.8× more consistently (larger scale μ), with *identical* preferences. Uniformly-1.8×-larger coefficients scale all attributes by the same factor (the scale fingerprint), and *identical WTP* confirms no taste difference (scale cancels in the WTP ratio). It is *not* stronger preferences — that would change WTP.

A3. (1) true preference difference (changes relative coefficients and WTP); (2) error-variance/scale difference (uniform scaling, WTP unchanged); (3) attention difference (→ scale); (4) comprehension difference (→ scale); (5) familiarity/experience difference (→ scale); (6) cost-coefficient difference (changes WTP). Items 2–5 are scale (mutually equivalent); item 1 is preference; item 6 changes WTP via the denominator.

B1. With $\mu_B = k\mu_A$ and common β: A's identified coefficients are $\mu_A\beta$, B's are $\mu_B\beta = k\mu_A\beta$ — every coefficient of B is k times A's. But WTP is a ratio: $\text{WTP}_j^A = \mu_A\beta_j/(\mu_A\beta_c) = \beta_j/\beta_c$ and $\text{WTP}_j^B = k\mu_A\beta_j/(k\mu_A\beta_c) = \beta_j/\beta_c$ — the scale factor k (and μ_A) cancels, so the WTPs are identical (verified: groups A, B, C at scales 1, 2, 0.5 all had WTP ≈ 20).

B2. Attention, comprehension, familiarity, and error variance all operate through the *same* channel — how consistently a respondent chooses, i.e. the scale μ (inverse error variance) — so from *choice data* they produce the identical effect (a change in scale, uniform coefficient scaling, WTP unchanged) and cannot be told apart. Separating them requires *external* measures correlated with the underlying factor but not with the choices

themselves: attention checks (attention), comprehension tests (comprehension), familiarity/experience questions (familiarity), and response times/effort proxies — regressing the estimated individual scale on these measures apportions the scale variation among the factors (Chapter 50).

B3. The Swait–Louviere procedure estimates a model with a *common* preference vector β across two groups and a *relative scale* parameter $\lambda = \mu_B/\mu_A$ (one group's scale normalized), effectively rescaling one group's utility to match the other. "Pure scale" is indicated if this common-β-plus-relative-scale model *adequately fits both groups* — a single λ reconciles their choices, and no significant preference (relative-coefficient / WTP) difference remains after allowing for scale. A *preference difference* is indicated if, even after estimating λ, the groups require different β (different WTP) — the scale ratio alone cannot reconcile them.

C1. Analysis needed: (i) compare **WTP** (scale-free) between experienced and inexperienced groups — if WTP is *identical*, the coefficient difference is *scale* (familiarity → consistency), not preference; if WTP *differs*, there is a genuine valuation difference. (ii) Estimate a **Swait–Louviere** scaled model — test whether a single relative-scale parameter (with common preferences) reconciles the two groups; if it does, the difference is scale; if a preference difference remains, familiarity affects valuations. (iii) Correlate the estimated group scale with *external* familiarity/experience measures and response times to confirm the channel is consistency (scale). Only if WTP differs *and* scale does not reconcile the groups can you claim "familiarity affects preferences"; otherwise the correct finding is "familiarity affects choice consistency (scale), not valuation" — the §6.3 error avoided.

C2. Concluding "changes preferences" from a *coefficient* difference is unjustified because coefficients are $\mu\beta$ (preference × scale), so the difference could be a *scale* effect (the high-cost vector changing engagement/consistency) rather than a preference change. To claim a preference/valuation effect you must show: (i) **WTP differs** between arms (ruling out pure scale — scale leaves WTP unchanged); (ii) a **scaled model** (Swait–Louviere) does *not* reconcile the arms with a single relative scale; and (iii) ideally, that the WTP change reflects *preference formation* rather than a *cost-coefficient* change — which requires the identification analysis of Chapter 30 (the design must separate κ from λ). A raw coefficient difference alone is consistent with scale and cannot support "changes preferences."

C3. Different elicitation formats induce different *choice consistency* (scale) — a payment card may be noisier or sharper than a single dichotomous-choice referendum — so a raw WTP difference between PC and SBDC could be a *scale artifact* rather than a genuine valuation difference (or agreement). To account for it: estimate a **pooled scaled model** across the two formats with a *relative scale* parameter (Swait–Louviere) between them, testing whether the formats share common preferences after allowing for format-specific scale. Combine with **equivalence testing** (Chapter 8): a convergent-validity claim requires the formats' WTP to be equivalent *within a pre-specified margin* after accounting for scale. Only then does "PC and SBDC give equivalent WTP" hold; a raw WTP comparison that ignores format scale can spuriously reject or confirm convergence.

25.12 Spaced review prompts

+1 day. From memory, define scale vs preference heterogeneity, explain the $\mu\beta$ confound, and state the WTP fingerprint (scale → same WTP; preference → different WTP).

+1 week. List the six explanations observationally equivalent to a preference difference, state which the WTP comparison rules out, and describe the Swait–Louviere test and the G-MNL. Explain what cannot be cleanly separated.

+1 month. For your own cost-vector or familiarity comparison, compare WTP (not coefficients), estimate a scaled model to test for pure scale, correlate scale with external attention/comprehension/familiarity measures, and report what your design can and cannot separate. Close with the spine sentence about whether your effect is scale or preference.

25.13 How this chapter connects to the rest of the book

- **Chapters 18 (§18.4), 20 (§20.6), 21 (§21.4)** established the scale normalization and the $\mu\beta$ confound this chapter resolves; **Chapter 22 (§22.10)** introduced the G-MNL; **Chapter 24 (§24.5)** showed the cost coefficient is the scale and WTP is scale-free — the diagnostic this chapter uses.
- **Chapter 30 (identification)** is where the deepest separation — preference formation vs cost-coefficient/anchoring (the κ-vs-λ problem), and partial identification — is developed; this chapter's "what cannot be separated" is Chapter 30's observational equivalence; **Chapter 8 (equivalence)** is the convergent-validity companion.
- **Chapters 27–29 (behavioral foundations, incentive compatibility, cost-vector effects)** are where the cost-vector "preference vs scale vs reporting" question is the substantive core; **Chapter 51 (survey design)** provides the external attention/comprehension/familiarity measures needed to identify *which* scale factor operated.

This chapter closes Part V by resolving the book's central discrimination problem — scale versus preference — that every discrete-choice chapter has pointed toward, and that your cost-vector, familiarity, and convergent-validity papers turn on.

25.14 Research translation task

For your **cost-vector and familiarity comparisons**, produce a two-to-three-page note:

1. **WTP, not coefficients.** For each between-group comparison, report and compare *WTP* (scale-free), and note whether the coefficient differences are uniform (scale) or relative (preference).
2. **Scaled model.** Estimate a Swait–Louviere (or G-MNL / WTP-space) model to test whether a relative scale reconciles the groups; report the relative scale and any residual preference difference.
3. **External measures.** Correlate the estimated scale with attention, comprehension, familiarity, and response-time measures to attribute scale differences to specific factors.
4. **Mechanism.** For the cost-vector effect, state whether WTP differs (ruling out pure scale) and acknowledge the preference-vs-cost-coefficient channel is partially identified (Chapter 30).
5. **Convergent validity.** For format comparisons, account for format scale (Swait–Louviere) before equivalence testing (Chapter 8).
6. **Close** with the spine sentence: *"My [cost-vector/familiarity] difference is [scale/preference/partially-identified] because WTP [is/is not] unchanged and a relative scale [does/does not] reconcile the groups; what I cannot separate is [which scale factor / preference vs cost-coefficient]."*

Carry this into Chapter 30 (identification / κ vs λ), Chapters 27–29 (cost-vector behavioral foundations), and Chapter 8 (convergent validity).

25.15 Bibliography

Core. - Train, *Discrete Choice Methods with Simulation*, on scale, identification, and the mixed logit — the foundation for the $\mu\beta$ confound. Freely available — *verify link.* - Swait & Louviere (1993), "The Role of the Scale Parameter in the Estimation and Comparison of Multinomial Logit Models," *Journal of Marketing Research.* The foundational treatment of scale in group comparisons (§25.5) — essential for your two-group cost-vector/familiarity comparisons. *(Verify volume/pages.)* - Fiebig, Keane, Louviere & Wasi (2010), "The Generalized Multinomial Logit Model," *Marketing Science.* The G-MNL separating scale and preference heterogeneity (§25.5–§25.6). *(Verify volume/pages.)*

Applied. - Louviere & Eagle, and Louviere et al. on scale heterogeneity in stated-preference data — the pervasiveness of scale differences (§25.2). *(Verify exact citations.)* - Hess & Rose (2012), "Can scale and

coefficient heterogeneity be separated in random coefficients models?" *Transportation*. The critical treatment of what *cannot* be separated (§25.6) — essential reading for the partial-identification point. *(Verify volume/pages.)* - Scarpa, Thiene & Train (2008), on scale (confounding) effects and WTP space in environmental valuation (§25.5; Chapter 24). *(Verify volume/pages.)*

Advanced. - Hess & Train (2017), "Correlation and scale in mixed logit models," *Journal of Choice Modelling*. On the entanglement of correlation, scale, and preference heterogeneity. *(Verify volume/pages.)* - Train & Weeks (2005), on WTP space as a device to address scale confounds (Chapter 24). *(Verify citation.)* - The behavioral literature on attention, comprehension, and familiarity as determinants of choice consistency (e.g., work by Hensher on task complexity) — the external-measures approach of §25.6. *(Verify exact citations.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Swait–Louviere / Fiebig et al. / Hess–Rose citations before using them, per the book's no-unverified-citation rule.

End of Chapter 25 and of Part V. You now hold the discrete-choice heart of your DCE: Lancaster and random utility (Ch 20), the multinomial logit and IIA (Ch 21), the nested and mixed logit that relax it (Ch 22), simulated maximum likelihood (Ch 23), preference vs WTP space (Ch 24), and the scale-vs-preference discrimination that your cost-vector and familiarity papers turn on (Ch 25). Part VI — Contingent Valuation and Stated Preference Design — begins with Chapter 26 (Contingent Valuation Formats), returning to the CV side of your work with the behavioral and mechanism-design foundations that follow. Tell me which chapter to write next.

Part V Assessment — Random Utility and Discrete Choice

Covers Chapters 20–25. Five components, in the order they are best attempted. Allow a full working day; the mock oral is the part that matters most for your viva.

1. Cumulative problem set

P1. Derive the MNL probability from i.i.d. Gumbel errors, showing where the utility *difference* enters.

P2. Show $\Pr(i)/\Pr(k) = e^{V_i - V_k}$ and construct a labelled choice set where this is economically absurd.

P3. For $\beta_q \sim \mathcal{N}(0.5, 0.15)$ and $\beta_c \sim \mathcal{N}(0.025, 0.010)$, compute by simulation the median, 99th percentile and wrong-sign fraction of β_q/β_c. Then repeat holding the median fixed while halving the cost mean.

P4. Derive the WTP-space reparameterization and state precisely the restriction it imposes on the random-coefficient covariance structure.

P5. Two arms differ in estimated coefficients by a common multiplicative factor of about 1.4. Give the scale explanation and the preference explanation, and state what would distinguish them.

2. Research critique

Take your EC-MXL. Write one page identifying, for each random parameter, the distributional assumption, its support, and the fraction of the fitted population it assigns economically implausible values. Then state what changes if you move from preference space to WTP space, and defend the space you use.

3. Red-flag diagnosis

A paper reports that experienced respondents have larger and more precisely estimated attribute coefficients than inexperienced ones, and concludes that "experience strengthens environmental preferences." Write the alternative explanation, explain why the two are observationally equivalent in the reported specification, and specify the model that would separate them.

4. Mock oral examination

1. What does "random" mean in random utility, and what is deterministic?
2. Why is only $\mu\beta$ identified?
3. What is an error component, and what does it buy you over a plain mixed logit?
4. Why are preference space and WTP space *not* the same model for a mixed logit?
5. Your cost-vector arm shows a higher β_c and unchanged ω_k. What do you conclude?

5. Integration mini-project

Estimate, on the same dataset: an MNL, a mixed logit in preference space, and an EC-MXL in WTP space. Compare fit, the WTP distributions, and the tail behaviour. Write two pages on which model you would report, what the choice assumes, and how much your welfare conclusion moves across the three.

Part VI — Contingent Valuation and Stated Preference Design

Chapter 26 — Contingent Valuation Formats

Part VI: Contingent Valuation and Stated Preference Design

> **The spine question, applied to this chapter.** Part V built your discrete choice experiment; Part VI returns to the contingent-valuation side of your work, and this chapter is the systematic comparison of the *formats* by which you ask people their willingness to pay — open-ended, payment card, single- and double-bounded dichotomous choice, and referendum. Your PDF asks you to study these *"in modo sistematico,"* and the reason is that the format is not a neutral choice: each sits at a different point on a fundamental trade-off between **incentive compatibility** (does the format give respondents a reason to answer truthfully?) and **statistical efficiency** (how much information does each answer carry?). Your CV paper uses SBDC *and* payment card precisely to span this trade-off. This chapter also completes the econometric toolkit for CV data — interval-data models, Turnbull, and *spike models* for genuine non-participants. The spine question here: *for the format I chose, what incentive properties and what statistical efficiency did I buy, how do I model its data (including genuine zeros), and does the format support the welfare claim I want to make?*

26.0 Conceptual motivation: the format is a design decision with consequences

Your PDF's §3.6 asks you to know the contingent-valuation formats *systematically*, and lists them alongside the behavioral and mechanism-design concerns they raise: *"open-ended CV; single-bounded dichotomous choice; double-bounded dichotomous choice; payment card; referendum format; interval data models; Turnbull estimator; spike models; protest responses; yea-saying; scope sensitivity; consequentiality; incentive compatibility; hypothetical bias."* And it tells you your CV paper's design: *"Il CV paper usa SBDC e payment card in un disegno complesso con incentivi reali, hypothetical treatments e BRIS."*

Why does the choice of format deserve a full chapter? Because the format determines two things simultaneously, and they *trade off*:

- **Incentive compatibility** — whether the format gives the respondent a *reason to reveal their true WTP* rather than to misrepresent it strategically. A single yes/no vote on a public program at a stated cost (referendum/SBDC) can be incentive-compatible under the right conditions; asking "what is the maximum you would pay?" (open-ended) is not — the respondent has no reason to state their true maximum and every reason to anchor, protest, or strategize. The full theory is Chapter 28; here we place each format on the incentive axis.
- **Statistical efficiency** — how much information each answer carries. An open-ended answer is a *point* (a WTP number); a payment-card answer is a *narrow interval*; an SBDC answer is a *single bit* (above/below one threshold). More information per respondent means smaller standard errors — so

the *least* incentive-compatible formats are often the *most* statistically efficient, and vice versa. This is the central organizing tension of the chapter, and it is exactly why your CV paper uses *both* SBDC (incentive-compatible, inefficient) and payment card (efficient, less incentive-compatible).

This chapter is organized around that trade-off. It describes each format and places it on the two axes; it consolidates the econometric models (all are interval-data models, Chapters 14–15) and the Turnbull nonparametric benchmark (Chapters 10, 15); it develops **spike models** in full — the point-mass-at-zero models that handle genuine non-participants, protests, and the negative-WTP problem, the main econometric addition; and it frames the format-choice decision for your CV paper. The behavioral and mechanism-design foundations — incentive compatibility, consequentiality, hypothetical bias, anchoring, protests — are Chapters 27–29; this chapter is the *formats* and their *models*, with those concerns previewed where they bear on the format choice.

26.1 Prerequisites

From **Chapter 14**: CV responses as censored/interval observations (SBDC = single threshold, payment card = interval) and the survival function. From **Chapter 15**: the SBDC/DBDC/payment-card likelihoods, mean vs median WTP under distributions, and the double-bounded anchoring/shift problem (§15.4). From **Chapter 10**: the Turnbull nonparametric estimator. From **Chapter 5** (§5.3): the negative-WTP support problem. From **Chapter 8**: equivalence testing (for convergent validity across formats). From **Chapter 28** (preview): incentive compatibility.

Same **notation**. Consolidated symbols: b (bid), $(\ell, u]$ (payment-card interval), F / S (WTP CDF / survival), π (spike / non-participation probability), G (CDF of the *positive* part of WTP in a spike model).

26.2 The elicitation formats

Open-ended (OE). "What is the maximum amount you would be willing to pay for this program?" The respondent states a WTP number directly. - *Information:* a *point* — the most information per respondent, hence the most statistically efficient. - *Incentive properties: poor.* The respondent has no incentive to state their true maximum (stating less costs nothing; the question is not a decision with consequences), so OE is prone to strategic understatement, anchoring, high non-response, and many zeros/protests. It is also *cognitively demanding* (people rarely know their exact maximum WTP for an unfamiliar public good). - *Verdict:* efficient but weakly incentive-compatible and cognitively hard — largely abandoned for public goods (though still used for private/familiar goods).

Payment card (PC). "Which of these amounts is closest to the most you would pay?" or "circle the highest amount you would definitely pay." A list of monetary values is shown, and the respondent indicates a value or bracket. - *Information:* an *interval* $(\ell, u]$ — nearly as informative as OE, more than SBDC. - *Incentive properties: intermediate/weak.* Less demanding than OE, but the *range and spacing of the card* can **anchor** responses (respondents gravitate to the middle of the range, or the values shown shape their answer — Chapter 27). Not strictly incentive-compatible. - *Verdict:* efficient and easier than OE, but vulnerable to range/anchoring effects — your CV paper uses it, partly to study these effects.

Single-bounded dichotomous choice (SBDC). "Would you pay \$X for this program? Yes / No." A single take-it-or-leave-it price. - *Information:* a *single bit* — WTP is above or below one threshold. The *least* information per respondent, hence the least statistically efficient (needs many respondents and good bid design across the range). - *Incentive properties: good* (under conditions). A single yes/no on a program at a stated cost mimics a real market/referendum decision, so — if consequential and incentive-compatible (Chapter 28) — the respondent's best response is to say "yes" iff their WTP exceeds the price. This is why SBDC/referendum is the NOAA-panel-recommended format. - *Verdict:* incentive-compatible but statistically inefficient — the format whose *validity* you trade against its *efficiency*.

Double-bounded dichotomous choice (DBDC). SBDC plus a *follow-up* bid (higher after a "yes," lower after a "no"). - *Information:* an *interval* (four cells, Chapter 15 §15.4) — more efficient than SBDC. - *Incentive properties:* the *first* question retains SBDC's incentive compatibility, but the *follow-up is not incentive-compatible* and induces **anchoring/shift** effects (the first bid anchors the second; the implied distribution shifts — Chapter 15 §15.4). So the efficiency gain is bought with potential bias. - *Verdict:* more efficient than SBDC but with the follow-up's validity problems — an explicit efficiency-vs-validity trade-off.

Referendum. A single binding (or consequential) yes/no *vote* on whether to provide a public program at a stated cost — the SBDC framed explicitly as a collective decision. - *Information/efficiency:* same as SBDC (a single bit). - *Incentive properties:* the *best* — a consequential referendum on a public good, with coercive payment (everyone pays if it passes), is the setting where truthful voting is a dominant (or near-dominant) strategy (Chapter 28). The **NOAA panel** endorsed the referendum format for exactly this reason. - *Verdict:* the incentive-compatibility gold standard; SBDC is its econometric twin.

> **Active recall 26.A.** Rank the formats from most to least statistically efficient (information per respondent), and separately from most to least incentive-compatible, and note the pattern. *(Answer: efficiency (most → least): open-ended > payment card > double-bounded > single-bounded ≈ referendum (point > interval > bounded interval > single bit). Incentive compatibility (most → least): referendum ≈ single-bounded > double-bounded > payment card > open-ended. The pattern is a* trade-off* — the most efficient formats are the least incentive-compatible, and the most incentive-compatible are the least efficient — which is the central tension of CV format choice.)*

26.3 The central trade-off: incentive compatibility versus statistical efficiency

The two axes are (roughly) *inversely* related, and this is the organizing insight of CV format choice. Formats that carry more information per respondent (open-ended, payment card, double-bounded) are more statistically efficient but less incentive-compatible; formats that carry less information (single-bounded, referendum) are less efficient but more incentive-compatible.

Why the trade-off exists. Incentive compatibility comes from making the question a *simple, consequential decision* — a yes/no vote at a price, where the respondent's dominant strategy is to reveal whether WTP exceeds the price. But a simple yes/no reveals only *one bit*. To extract more information (a point or a narrow interval), you must ask a more demanding or less decision-like question (a maximum, a follow-up bid), which weakens the incentive to answer truthfully and opens the door to anchoring and strategy. **You cannot maximize both axes at once**; you buy efficiency with incentive compatibility and vice versa.

Worked Example 26.1 (The efficiency ordering, verified). Simulate the *same* underlying WTP (lognormal, median €25, true mean €28.3) observed under three formats and estimate the mean WTP with its standard error:

Format	Information per respondent	Mean WTP	SE(mean)
Open-ended	point	28.5	**0.35**
Payment card	interval	28.4	**0.33**
SBDC	single bit	28.8	**0.53**

(Verified.) The information-rich formats (open-ended, payment card) give standard errors around €0.33–0.35; the single-bit SBDC gives €0.53 — **about 60% larger**, because each SBDC respondent contributes only one threshold comparison rather than a point or a narrow interval. To match the SBDC's precision to the payment card's, you would need roughly $(0.53/0.33)^2 \approx 2.6$ times as many SBDC respondents. **This is the statistical price of the SBDC's incentive compatibility** — and it is exactly why your CV paper uses

the payment card alongside the SBDC (efficiency) and why the SBDC needs a well-designed bid vector and a large sample.

The resolution — use multiple formats and compare. Because no single format dominates, a sophisticated CV design (yours) uses *more than one* — the incentive-compatible SBDC/referendum as the *validity anchor*, and the efficient payment card for *precision* and as a *convergent-validity comparison.* The comparison itself is informative: if the formats agree on WTP (within a margin — equivalence testing, Chapter 8), the incentive-compatibility-vs-efficiency trade-off is benign and you can use the efficient format with confidence; if they *disagree*, the disagreement diagnoses the incentive/anchoring problems (Chapters 27–29). **Your CV paper's use of SBDC and payment card together is a direct study of this trade-off** — and the format comparison is a research contribution, not just a robustness check.

> **Stop and think 26.1.** Your CV paper reports mean WTP from SBDC and payment card. Why is the *comparison* between the two formats scientifically valuable, and what would agreement vs disagreement tell you? *(Answer: the comparison probes the incentive-compatibility-vs-efficiency trade-off directly — SBDC is incentive-compatible but noisy, the payment card is efficient but anchoring-prone.* Agreement* (equivalent WTP within a margin, Chapter 8) suggests the anchoring/incentive problems are benign and the efficient payment card can be trusted (convergent validity); *disagreement* diagnoses a problem — anchoring/range effects in the payment card, or a scale/format effect (Chapter 25) — and localizes which format is biased. Either way the comparison is a finding about the validity of stated-preference formats, which is a core contribution of your paper.)*

26.4 The econometric models: everything is interval data

A unifying point (Chapters 14–15): *every* CV format produces **interval-censored** data on latent WTP, and the same interval likelihood covers them all. The formats differ only in *how narrow* the interval is.

The unified interval likelihood. Each respondent's answer locates WTP in an interval $(\ell_i, u_i]$, contributing $F(u_i) - F(\ell_i)$ to the likelihood (Chapter 14, §14.4):

- **Open-ended:** WTP is (nearly) a point — a degenerate interval; modeled as a continuous WTP observation (censored at zero, §26.5).
- **Payment card:** WTP in the reported bracket $(\ell, u] \to F(u) - F(\ell)$.
- **SBDC:** "yes" $\to$ WTP $\in (b, \infty) \to S(b)$; "no" $\to$ WTP $\in (-\infty, b] \to F(b)$.
- **DBDC:** the four cells $\to$ the interval contributions of Chapter 15 (§15.4).

So a single interval-data estimator (with the chosen WTP distribution — normal, logistic, lognormal, or a spike model) fits any CV format; the format determines the *width* of each interval and hence the *efficiency* (§26.3). This unification is worth internalizing: you do not need a separate econometric method per format, only the interval likelihood with the right interval structure and WTP distribution.

Turnbull as the nonparametric benchmark (Chapters 10, 15). For *any* format, the **Turnbull** estimator recovers the WTP distribution *nonparametrically* from the interval data (the yes-proportions at each bid / the bracket frequencies), with no assumed distribution, and yields a conservative distribution-free **lower-bound mean**. Report it alongside the parametric mean as a robustness anchor: a large gap between the parametric mean and the Turnbull lower bound flags tail-driven fragility (the parametric distribution extrapolating beyond your bids — Chapter 15 §15.5). Turnbull is format-agnostic (it works for SBDC, DBDC, and payment card alike) and is your distribution-free floor.

> **Active recall 26.B.** Why is a single interval-data likelihood enough to model open-ended, payment-card, SBDC, and DBDC data, and what distinguishes the formats within it? *(Answer: every CV answer locates latent WTP in an interval $(\ell, u]$ contributing $F(u) - F(\ell)$ — open-ended (a point/degenerate interval), payment card (a bracket), SBDC (one threshold, (b, ∞) or $(-\infty, b]$), DBDC (four cells) are all special cases of the same interval likelihood. The formats*

differ only in the width* of the interval each respondent provides — narrower intervals (open-ended, payment card) carry more information and give smaller standard errors (efficiency); the interval likelihood with the chosen WTP distribution handles them all.)*

26.5 Spike models: handling genuine zeros, protests, and negative WTP

The main econometric addition of this chapter. Standard CV models assume a continuous WTP distribution, but real data have a **mass at zero** — a fraction of respondents who genuinely would pay *nothing* (non-participants: indifferent to the good, or unable/unwilling to pay). Modeling this mass explicitly is what **spike models** do, and it solves three problems at once: genuine non-participation, protest responses, and the negative-WTP artifact (Chapter 5, §5.3).

The model. A spike model puts a **point mass π at WTP $= 0$** (the "spike" — non-participants) and a **continuous distribution G on the positive part** with probability $(1-\pi)$:

$$\Pr(\text{WTP} \le w) = \begin{cases} \pi + (1-\pi)G(w) & w \ge 0, \\ 0 & w < 0. \end{cases}$$

So the WTP CDF *jumps* by π at zero (the spike) and then rises continuously via G. There is no negative-WTP mass — the support is $[0, \infty)$ with an atom at zero — which cleanly resolves the negative-WTP problem (a symmetric normal's artifact, Chapter 5) *and* accommodates genuine zeros.

The likelihood (derived, for SBDC). With bids b_i and yes/no responses: - A **"yes"** at b_i means WTP $\ge b_i > 0$, so the respondent is a *participant* with positive WTP above b_i: contributes $(1-\pi)(1-G(b_i))$. - A **"no"** at b_i means WTP $< b_i$, which includes *both* the non-participants (the spike) *and* participants with positive WTP below b_i: contributes $\pi + (1-\pi)G(b_i)$.

The likelihood is $\prod_i [(1-\pi)(1-G(b_i))]^{y_i}[\pi + (1-\pi)G(b_i)]^{1-y_i}$, and you estimate the **spike probability** π (the non-participation rate) *and* the positive-part distribution parameters jointly. Mean WTP is then $(1-\pi) \cdot \mathbb{E}[\text{WTP} \mid \text{WTP} > 0]$ — the participation rate times the mean of the positive part; the median is zero if $\pi > 0.5$.

Worked Example 26.2 (Spike model recovers non-participation and the correct mean, verified). Simulate SBDC data where 30% of respondents are genuine non-participants (WTP = 0) and 70% have positive WTP (lognormal, median €25), so the true overall mean is $0.7 \times$ €28.3 = €19.8. Fitting a spike model versus a naive (no-spike) model:

Model	Estimated non-participation $\hat{\pi}$	Overall mean WTP
Spike	**0.306** (true 0.30)	**€19.5** (true €19.8)
Naive (no spike)	— (cannot represent zeros)	€23.9 (**biased**)

(Verified.) The **spike model recovers the non-participation rate (30.6% vs true 30%) and the correct overall mean (€19.5 vs true €19.8)**, while the naive model — which forces a single continuous distribution and cannot represent the mass at zero — is *biased* (€23.9, off by ~20%), because it spreads the non-participants' zeros into low-but-positive WTP. **Ignoring genuine non-participation biases mean WTP.** For your CV data, where a real fraction of respondents may genuinely value the good at zero, the spike model is often the correct specification — and it doubles as a principled way to separate *genuine zeros* from the continuous WTP distribution.

Spike models and protests (preview of Chapter 32). The spike is a natural place to think about **protest responses** — respondents who say "no" not because their WTP is low but because they reject

the payment vehicle or the scenario. A protest is *not* a genuine zero WTP (the respondent may value the good highly but refuse to express it through this payment), so lumping protests into the spike is wrong. The distinction — genuine non-participant (belongs in the spike) versus protester (belongs handled separately, or excluded with bounding, Chapters 14 §14.8, 32) — is a substantive judgment. The spike model gives you the *slot* for genuine zeros; identifying which "no" responses are genuine zeros versus protests requires follow-up questions and diagnostics (Chapter 51). Your PDF's "i protest sono esclusi correttamente?" is exactly this: do not conflate protests with the spike.

> **Stop and think 26.2.** Your SBDC data show a high "no" rate even at the lowest bid, and a debriefing question reveals two kinds of "no": some say "the good is not worth anything to me" and others say "I refuse to pay for this through taxes." How should each be modeled? *(Answer: the first group ("not worth anything") are* genuine non-participants* with zero WTP — they belong in the **spike** (π) of a spike model. The second group ("refuse to pay through this vehicle") are *protesters* — their "no" does not reveal low WTP (they may value the good highly), so they should *not* be put in the spike; handle them separately (analyze with and without, bound the results, or model protest as a distinct process — Chapters 14 §14.8, 32). Conflating protests with genuine zeros biases both the spike estimate and mean WTP — your PDF's "i protest sono esclusi correttamente?".)*

26.6 Choosing the format for your CV paper

Pulling the chapter to your design decision. Your CV paper uses **SBDC and payment card** in a complex design with real incentives, hypothetical treatments, and BRIS. The rationale, in the language of this chapter:

- **SBDC/referendum as the validity anchor.** The SBDC is incentive-compatible (a consequential yes/no at a price, especially with real incentives), so it is the format whose WTP you can most defend as a genuine welfare measure — the NOAA-panel logic. Its cost is statistical inefficiency (verified €0.53 vs €0.33 SE), which you pay with a larger sample and a well-designed bid vector.
- **Payment card for efficiency and comparison.** The payment card is more efficient (interval data) and lets you study anchoring/range effects — and, critically, comparing PC and SBDC WTP is a *convergent-validity* test (do the formats agree?). Your paper's PC-vs-SBDC comparison is a direct study of the incentive-efficiency trade-off.
- **Real incentives and BRIS.** Adding *real* incentives strengthens the incentive compatibility of both formats (a consequential payment, Chapter 28), and BRIS (a practical incentive mechanism) is your device for making the exercise consequential at manageable cost — the mechanism-design material of Chapter 28.
- **The econometric plan.** Model both formats with the interval likelihood (SBDC = threshold, PC = bracket), consider a spike model for genuine non-participants, benchmark with Turnbull, and compare WTP across formats with the Poe test *and* equivalence testing (Chapters 8, 10) — accounting for format-specific scale (Chapter 25) before concluding convergence.

The format choice is therefore not incidental — it *is* the research design. Your paper spans the incentive-compatibility-vs-efficiency trade-off deliberately (SBDC + PC), makes it consequential (real incentives + BRIS), and studies whether the formats converge — which is a contribution to the validity of stated-preference valuation, not just a WTP estimate.

> **Active recall 26.C.** Why does your CV paper use *both* SBDC and payment card rather than one, and what does the pairing let you study? *(Answer: the two formats sit at opposite ends of the incentive-compatibility-vs-efficiency trade-off — SBDC is incentive-compatible but statistically inefficient (the validity anchor), the payment card is efficient but anchoring-prone (precision). Using both lets you (i) anchor validity with the SBDC while gaining precision from the PC, and (ii)* study the trade-off itself* by comparing PC and SBDC WTP — a convergent-validity test that diagnoses anchoring/incentive/scale problems (Chapters 8, 25). The pairing turns the format choice into a research contribution about stated-preference validity.)*

26.7 Failure modes

F1. Choosing a format for efficiency alone. Using open-ended or payment card for precision while ignoring their weak incentive compatibility. *Fix:* weigh incentive compatibility (validity) against efficiency; anchor with SBDC/referendum (§26.3; Chapter 28).

F2. Ignoring the spike (genuine zeros). Forcing a continuous distribution on data with real non-participants, biasing mean WTP. *Fix:* spike model; estimate the non-participation rate (§26.5).

F3. Conflating protests with the spike. Lumping protest "no"s into the zero-WTP spike. *Fix:* distinguish genuine zeros (spike) from protests (handled separately); use debriefing questions (§26.5; Chapters 14, 32).

F4. Trusting DBDC efficiency without checking anchoring. Using the tighter double-bounded estimate despite the follow-up's shift/anchoring. *Fix:* test for the shift; consider single-bounded as the validity benchmark (§26.2; Chapter 15 §15.4).

F5. Payment-card range/anchoring effects unexamined. Ignoring how the card's values shape responses. *Fix:* vary/randomize the card range; examine range effects (§26.2; Chapters 27, 29).

F6. No Turnbull benchmark. Reporting only parametric WTP for a format. *Fix:* Turnbull distribution-free lower bound alongside (§26.4; Chapters 10, 15).

F7. Comparing formats without accounting for scale. Concluding convergence/divergence from raw WTP across formats. *Fix:* account for format-specific scale (Swait–Louviere) before equivalence testing (§26.6; Chapters 8, 25).

F8. Negative-WTP artifact from a symmetric distribution. A normal WTP producing negative mass instead of a spike. *Fix:* spike or non-negative distribution matching the good's support (§26.5; Chapter 5).

26.8 Diagnostic questions for your own papers

1. **Trade-off.** For each format, do I know its position on the incentive-compatibility-vs-efficiency trade-off, and did I choose it for the right reason? *(§26.3.)*
2. **Spike.** Do my data have a mass at zero (genuine non-participants), and am I modeling it with a spike (not forcing a continuous distribution)? *(§26.5, F2.)*
3. **Protests vs zeros.** Am I distinguishing genuine zero-WTP (spike) from protests (handled separately)? *(§26.5, F3.)*
4. **DBDC anchoring.** If I use double-bounded, have I tested for the shift/anchoring and considered single-bounded as the anchor? *(§26.2, F4.)*
5. **Payment-card range.** Have I examined how the card's range/spacing affects responses? *(§26.2, F5.)*
6. **Turnbull.** Have I reported the Turnbull distribution-free lower bound alongside the parametric mean? *(§26.4, F6.)*
7. **Format comparison.** Am I accounting for format-specific scale before comparing WTP across formats (convergent validity)? *(§26.6, F7; Chapters 8, 25.)*

26.9 Exercises

(A) Comprehension.

A1. Rank open-ended, payment card, SBDC, DBDC, and referendum on statistical efficiency and on incentive compatibility, and state the trade-off.

A2. Why is SBDC statistically less efficient than payment card, and what is the consequence for sample size?

A3. What does a spike model add over a standard continuous WTP model, and what does the spike represent?

(B) Derivation / reasoning.

B1. Show that all CV formats produce interval-censored data with a common likelihood $F(u) - F(\ell)$, and identify the interval for each format.

B2. Derive the SBDC spike-model likelihood (the "yes" and "no" contributions with the spike π and positive-part G), and write the overall mean WTP.

B3. Explain why ignoring a genuine spike (non-participation) biases mean WTP, and in which direction, using the worked-example logic.

(C) Research application / critique.

C1. Your CV paper uses SBDC and payment card. Justify the pairing in terms of the incentive-efficiency trade-off, and describe what the format comparison contributes.

C2. A co-author fits a standard lognormal SBDC model to data with 25% "no" at the lowest bid. Critique this and specify the spike-model alternative and the protest check.

C3. Explain why comparing SBDC and payment-card WTP for convergent validity requires accounting for scale and using equivalence testing.

26.10 Solutions

A1. Efficiency (most → least): open-ended (point) > payment card (interval) > double-bounded (bounded interval) > single-bounded ≈ referendum (single bit). Incentive compatibility (most → least): referendum ≈ single-bounded > double-bounded (first question only) > payment card > open-ended. The trade-off: the most efficient formats are the least incentive-compatible and vice versa — you buy statistical information with incentive compatibility, because incentive compatibility comes from a simple consequential yes/no that reveals only one bit.

A2. SBDC gives each respondent a *single bit* (WTP above/below one bid), while the payment card gives a *narrow interval* — more information per respondent — so the SBDC's standard error is larger (verified €0.53 vs €0.33 for mean WTP). Consequence: to match the payment card's precision, SBDC needs roughly $(0.53/0.33)^2 \approx 2.6$ times as many respondents — the statistical price of SBDC's incentive compatibility.

A3. A spike model adds an explicit *point mass* at WTP = 0 (probability π) plus a continuous distribution on the positive part (probability $1 - \pi$), whereas a standard model forces a single continuous distribution. The spike represents *genuine non-participants* — respondents who would pay exactly zero (indifferent to the good, or unwilling/unable to pay). It handles genuine zeros, avoids the negative-WTP artifact (support $[0, \infty)$ with an atom at zero), and recovers the non-participation rate (verified $\hat{\pi} = 0.306$ vs true 0.30).

B1. Each CV answer locates WTP in an interval $(\ell, u]$, contributing $F(u) - F(\ell)$: open-ended → a point (degenerate interval ≈ a WTP value); payment card → the reported bracket $(\ell, u]$; SBDC "yes" → (b, ∞) (so $F(u) - F(\ell) = 1 - F(b) = S(b)$), "no" → $(-\infty, b]$ (so $F(b)$); DBDC → the four cells (YY: (b^U, ∞); YN: $(b_1, b^U]$; NY: $(b^L, b_1]$; NN: $(-\infty, b^L]$). The same interval likelihood covers all; the formats differ only in interval width.

B2. "Yes" at b: WTP $\geq b > 0$ → a participant with WTP above b → $(1 - \pi)(1 - G(b))$. "No" at b: WTP $< b$ → either a non-participant (spike π) or a participant with WTP below b → $\pi + (1 - \pi)G(b)$. Likelihood:

$\prod_i[(1-\pi)(1-G(b_i))]^{y_i}[\pi+(1-\pi)G(b_i)]^{1-y_i}$. Overall mean WTP $= (1-\pi)\,\mathbb{E}[\text{WTP} \mid \text{WTP} > 0]$ (the participation rate times the positive-part mean); median $= 0$ if $\pi > 0.5$.

B3. A naive model forces a single continuous distribution with no mass at zero, so it cannot represent the non-participants (who are exactly zero); it must "explain" their "no" responses by spreading them into *low-but-positive* WTP, which have positive value. Since the true zeros contribute exactly zero to the mean but the naive model assigns them positive WTP, the naive overall mean is *biased upward* relative to the true mean (verified: naive €23.9 vs true €19.8), while the spike model — which places the non-participants at exactly zero — recovers the true mean (€19.5). (The direction can depend on the setup, but the bias is real; here it is upward.)

C1. The pairing spans the incentive-efficiency trade-off: the **SBDC** is incentive-compatible (a consequential yes/no at a price — the format whose WTP is most defensible as a welfare measure) but statistically inefficient (verified larger SE); the **payment card** is efficient (interval data) but anchoring-prone. Using both anchors validity with the SBDC while gaining precision from the PC. The *comparison* contributes a **convergent-validity test**: if PC and SBDC WTP agree (within a margin, accounting for scale — Chapters 8, 25), the anchoring/incentive concerns are benign and the efficient format is trustworthy; if they disagree, the disagreement diagnoses anchoring/range effects or format-specific scale. This makes the format choice a study of stated-preference validity, a research contribution beyond the WTP estimate.

C2. A standard lognormal SBDC model with 25% "no" at the lowest bid likely mishandles the excess zeros: if a real fraction are *genuine non-participants* (WTP = 0), the continuous lognormal (no mass at zero) is misspecified and biases mean WTP (verified). Specify a **spike model**: estimate a non-participation probability π plus a positive-part lognormal, giving the correct overall mean $(1-\pi)\mathbb{E}[\text{WTP} \mid > 0]$ and the non-participation rate. Additionally run a **protest check** — use debriefing questions to distinguish genuine zeros ("not worth anything," → spike) from protests ("refuse to pay through this vehicle," → not zero WTP, handle separately/bound; Chapters 14 §14.8, 32). Do not lump protests into the spike.

C3. Different formats induce different *choice consistency* (scale) — a payment card may be noisier or sharper than an SBDC referendum — so a raw WTP difference between formats could be a *scale artifact* rather than a genuine valuation difference (Chapter 25). To compare validly, account for **format-specific scale** (a Swait–Louviere relative-scale model between formats) before concluding the WTPs differ or agree. And a convergent-validity *claim* is an *equivalence* claim: use **equivalence testing (TOST)** against a pre-specified margin (Chapter 8) — a non-significant difference is not evidence of equivalence. Only equivalent WTP within a margin, after accounting for scale, supports "SBDC and PC converge."

26.11 Spaced review prompts

+1 day. From memory, place the five formats on the incentive-compatibility and efficiency axes and state the trade-off; write the SBDC spike-model likelihood.

+1 week. Explain why all formats are interval-data models, why SBDC is least efficient, and how a spike model handles genuine zeros (and why it is not the place for protests). State the Turnbull benchmark's role.

+1 month. For your own CV data, model both formats with the interval likelihood, fit a spike model and report the non-participation rate, benchmark with Turnbull, and compare formats accounting for scale and using equivalence testing. Close with the spine sentence about the trade-off your format choice bought.

26.12 How this chapter connects to the rest of the book

- **Chapters 14–15** supplied the interval-data likelihoods and mean/median WTP that this chapter organizes by format; **Chapter 10** the Turnbull benchmark; **Chapter 5 (§5.3)** the negative-WTP problem the spike model resolves.

- **Chapter 27 (behavioral foundations)** develops the anchoring/range/hypothetical-bias concerns that weaken the efficient formats; **Chapter 28 (incentive compatibility)** develops the mechanism-design theory behind the referendum/SBDC's incentive compatibility and BRIS; **Chapter 29 (bid/cost-vector effects)** develops the bid design and cost-vector-in-reporting concerns.
- **Chapter 8 (equivalence)** and **Chapter 25 (scale vs preference)** are the convergent-validity machinery for comparing formats; **Chapter 32 (threats)** develops the protest/exclusion handling; **Chapter 37 (design)** is the bid-vector design that determines SBDC efficiency and identification.

This chapter opens Part VI by organizing the CV formats around the incentive-compatibility-vs-efficiency trade-off that your SBDC-plus-payment-card design deliberately spans, and completing the CV econometric toolkit with spike models.

26.13 Research translation task

For your **CV paper's format design**, produce a two-to-three-page note:

1. **Position the formats.** Place your SBDC and payment card on the incentive-compatibility and efficiency axes; state what each buys and costs, and why you use both.
2. **Efficiency.** Compare the standard error on mean WTP across your formats (as in Worked Example 26.1); quantify the SBDC's efficiency cost.
3. **Spike model.** Fit a spike model to your SBDC (and PC) data; report the non-participation rate and the overall vs positive-part mean; distinguish genuine zeros from protests via debriefing.
4. **Turnbull.** Report the distribution-free Turnbull lower bound for each format alongside the parametric mean.
5. **Convergent validity.** Compare SBDC and PC WTP accounting for format scale (Swait–Louviere) and using equivalence testing (Chapter 8); state what the comparison contributes.
6. **Close** with the spine sentence: *"My CV design pairs [SBDC: incentive-compatible, inefficient] with [PC: efficient, anchoring-prone] to span the trade-off and test convergent validity; genuine non-participation is [rate] via a spike model; and the formats [converge/diverge] within [margin] after accounting for scale."*

Carry this into Chapters 27–29 (behavioral/incentive foundations), Chapter 8 (convergent validity), and Chapter 32 (protests/exclusions).

26.14 Bibliography

Core. - Haab & McConnell, *Valuing Environmental and Natural Resources: The Econometrics of Non-Market Valuation*. The definitive treatment of CV formats, interval-data models, Turnbull, and spike models — the single most relevant source. *(Verify edition/chapters.)* - Bateman et al., *Economic Valuation with Stated Preference Techniques: A Manual*. The comprehensive practical guide to CV formats and their trade-offs. *(Verify edition.)* - Arrow, Solow, Portney, Leamer, Radner & Schuman (1993), the *NOAA Panel Report on Contingent Valuation*. The endorsement of the referendum/dichotomous-choice format for incentive compatibility (§26.2–§26.3). *(Verify citation/Federal Register.)*

Applied. - Kriström (1990), "A Non-Parametric Approach to the Estimation of Welfare Measures in Discrete Response Valuation Studies," *Land Economics*, and the spike-model literature — the §26.5 material. *(Verify volume/pages.)* - Cameron & Huppert (1989), on interval/OE data and payment-card estimation; and Cameron (1988) on the WTP-function/interval approach (Chapter 15). *(Verify citations.)* - Champ, Boyle & Brown (eds.), *A Primer on Nonmarket Valuation*. Accessible chapters on CV formats, protest responses, and estimation. *(Verify edition.)*

Advanced. - Carson & Hanemann, "Contingent Valuation," in the *Handbook of Environmental Economics*.

The comprehensive treatment of formats, incentive compatibility, and econometrics. *(Verify volume/year.)* - Carson, Groves & List, and Carson & Groves (2007), "Incentive and informational properties of preference questions," *Environmental and Resource Economics* — the mechanism-design foundation for the format incentive properties (Chapter 28). *(Verify volume/pages.)* - Alberini (1995) and others on the statistical efficiency of double-bounded vs single-bounded designs (§26.3). *(Verify citations.)*

Entries marked "verify" are pointers; confirm editions, pages, and the NOAA Panel / Kriström / Carson–Groves citations before using them, per the book's no-unverified-citation rule.

End of Chapter 26. Next in Part VI: Chapter 27 — The Behavioral Foundations of Stated Preference (constructed preferences, coherent arbitrariness, anchoring, internal vs external anchors, yea-saying, preference uncertainty, and hypothetical bias), which develops the behavioral phenomena that threaten the efficient formats and that your cost-vector work centers on — does the cost vector enter preference formation or reporting? Tell me when to write it, or request a different chapter.

Chapter 27 — The Behavioral Foundations of Stated Preference

Part VI: Contingent Valuation and Stated Preference Design

> **The spine question, applied to this chapter.** Your PDF calls behavioral economics of preferences *"Molto importante per il tuo tema"* — and it is, because the entire architecture of your CV and DCE papers rests on one behavioral question: *"il cost vector possa entrare nella formazione o nel reporting delle preferenze."* Does the cost vector shape what people *value* (preference formation) or only what they *say* (reporting/anchoring)? This chapter builds the behavioral foundations that make that question meaningful: **constructed preferences** (valuations built on the spot, not retrieved), **coherent arbitrariness** (absolute values arbitrary, relative values coherent), **anchoring** (internal vs external), **yea-saying**, **preference uncertainty**, and **hypothetical bias**. Each threatens the validity of stated WTP, and each is a candidate mechanism for your cost-vector effect. The spine question here: *when I observe that the cost vector moves stated WTP, is that a genuine change in preferences or a reporting artifact — and can my design tell them apart, or are they observationally equivalent?*

27.0 Conceptual motivation: preferences may not be waiting to be measured

Standard welfare economics assumes people have well-defined, stable preferences that a valuation survey merely *elicits* — like reading a number off a dial. Behavioral economics has shown, repeatedly, that this assumption is often false: for unfamiliar, non-market goods (exactly your ecosystem services), people frequently do *not* have a pre-formed WTP; they *construct* one during the survey, and the construction is influenced by the elicitation context — the format, the framing, and, critically for you, **the numbers shown (the bid/cost vector)**. Your PDF lists the phenomena you must know: *"constructed preferences; coherent arbitrariness; anchoring; yea-saying; social desirability; hypothetical bias; preference uncertainty; internal vs external anchors; learning; familiarity; attention; attribute non-attendance."* And it names the stakes: *"Sia il CV paper sia il DCE paper ruotano intorno all'idea che il cost vector possa entrare nella formazione o nel reporting delle preferenze."*

This matters because it determines what your WTP *means*:

- If preferences are **stable and well-formed**, and the cost vector merely *reveals* them, then a cost-vector effect on stated WTP is a puzzle (it should not exist) — a sign of a reporting artifact to be purged.
- If preferences are **constructed**, then the cost vector can genuinely *shape* them (preference formation), and a cost-vector effect is a *real* phenomenon about how people come to value environmental goods — a substantive finding, but one that complicates the welfare interpretation (which cost vector gives the "true" WTP?).

- Or the cost vector can act as an **anchor** on the *report* — the underlying preference is unchanged, but the stated number is pulled toward the bid vector (reporting/measurement effect).

These two mechanisms — **preference formation** (the cost vector changes what people value) and **reporting/anchoring** (the cost vector changes what people say, not what they value) — produce *similar* observable patterns (both make stated WTP rise with the cost vector) but mean *opposite* things for welfare. And, as we will see, they are often **observationally equivalent** from stated data alone — the deep identification problem your PDF returns to (the "distinguish κ from λ" concern, Chapter 30).

This chapter is the behavioral half of your cost-vector question. It builds: constructed preferences (the foundation — why the cost vector *can* matter); coherent arbitrariness (the precise signature of anchoring — absolute values shift, relative values hold); anchoring and internal vs external anchors (the cost vector as an external anchor); yea-saying (acquiescence bias); preference uncertainty (WTP as a range, not a point); hypothetical bias (stated exceeds real — your real-vs-hypothetical treatment); and the central formation-vs-reporting distinction and its observational equivalence. Chapter 28 develops the mechanism-design cure (incentive compatibility); Chapter 29 develops the bid-design and cost-vector-effect side; Chapter 30 develops the identification. This chapter establishes *why* the cost vector is behaviorally live.

27.1 Prerequisites

From **Chapter 20** (§20.4): what the random utility error contains (the behavioral phenomena here partly live in ε or in V); from **Chapter 24** (§24.5) and **Chapter 25**: WTP as a scale-free ratio, and the scale-vs-preference distinction (anchoring can look like scale). From **Chapter 26**: the CV formats and their vulnerability to anchoring (payment card, open-ended). From **Chapter 15** (§15.4): the double-bounded anchoring/shift effect. From **Chapter 30** (preview): observational equivalence and partial identification.

Same **notation**. New/consolidated symbols: v (true/underlying valuation), a (anchor — e.g., the bid/cost vector level), w (anchoring weight), $\text{WTP}^{\text{stated}}$ (the reported WTP), and the distinction between **preference formation** (the cost vector enters v) and **reporting/anchoring** (the cost vector enters the map from v to $\text{WTP}^{\text{stated}}$).

27.2 Constructed preferences

Definition 27.1 (Constructed preferences). The **constructed-preference** view holds that, for unfamiliar or complex goods, people do not have pre-existing, stable preferences that a survey retrieves; instead they *build* a valuation on the spot, using whatever cues the elicitation provides. Preferences are *constructed*, not *revealed* — the act of valuation is partly an act of *creation*, and the context of the question shapes the answer.

The evidence. Decades of behavioral research (Slovic; Lichtenstein & Slovic; Payne, Bettman & Johnson) document **preference reversals** and **context effects**: the same person values the same thing differently depending on how they are asked (choice vs matching, willingness-to-pay vs willingness-to-accept, the order of questions, the values shown). If preferences were stable and pre-formed, these context effects would not exist. Their pervasiveness — especially for goods people have never traded, priced, or thought carefully about — is the empirical basis for the constructed-preference view.

Why this is the foundation of your cost-vector question. If preferences are *stable*, the cost vector *cannot* change what people value — any cost-vector effect on stated WTP must be a reporting artifact. If preferences are *constructed*, the cost vector *can* enter the construction — it becomes a cue that shapes the valuation itself (preference formation). **The constructed-preference view is what makes "does the cost vector enter preference formation?" a coherent question at all.** For ecosystem services — abstract, non-market, unfamiliar goods that respondents have rarely if ever priced — the constructed-preference view

is highly plausible, which is precisely why your cost-vector effect is a live behavioral phenomenon and not an obvious error to be purged.

The welfare implication. Constructed preferences complicate welfare: if the valuation is partly created by the elicitation, *which* elicitation gives the "true" welfare measure? There may be no single context-free WTP. This is uncomfortable for a discipline that wants WTP to be a stable welfare quantity, and it is the deep reason your PDF treats the cost-vector question as central rather than peripheral — it bears on whether your WTP is a welfare measure or a context-dependent construction (Chapter 34).

> **Active recall 27.A.** Why does the constructed-preference view make "does the cost vector affect preference formation?" a meaningful question, whereas the stable-preference view does not? *(Answer: under stable, pre-formed preferences, the cost vector* cannot* change what people value — any cost-vector effect on stated WTP would have to be a reporting artifact, so "preference formation" is impossible by assumption. Under *constructed* preferences, valuations are built on the spot using elicitation cues, so the cost vector *can* enter the construction and genuinely shape the valuation — making preference formation a real, coherent possibility. For unfamiliar non-market goods like ecosystem services, the constructed view is plausible, so the question is live.)*

27.3 Coherent arbitrariness

The most precise behavioral concept for your work — it describes *exactly* what anchoring by the cost vector looks like, and it gives you a testable signature.

Definition 27.2 (Coherent arbitrariness). Coined by Ariely, Loewenstein & Prelec, **coherent arbitrariness** describes valuations that are **arbitrary in their absolute level but coherent in their relative structure**. The *absolute* WTP is anchored — it can be shifted up or down by an arbitrary reference number (an anchor) that has no normative relevance. But the *relative* valuations — the rankings and ratios across goods or attributes — remain *coherent* (internally consistent, responsive to quantity). So a person's absolute WTP for a good is malleable, but their WTP for *twice as much* is reliably higher, and their *ranking* of goods is stable.

The classic experiment. Ariely et al. had subjects write down the last two digits of their social-security number, then bid on goods. Higher SSN digits → higher bids (the arbitrary anchor moved absolute valuations) — yet within each subject, the *relative* bids across goods were coherent (better goods got higher bids). Absolute levels arbitrary; relative structure coherent.

Worked Example 27.1 (Coherent arbitrariness, verified). Suppose a respondent's true relative valuations for two attributes are $v_A = 30$, $v_B = 20$ (A worth 1.5× B), and the cost vector acts as an anchor that scales absolute WTP:

Cost vector (anchor)	WTP for A	WTP for B	Ratio A/B
High	39.0	26.0	**1.50**
Low	24.0	16.0	**1.50**

(Verified.) The **absolute WTP shifts with the cost vector** (A from €24 to €39) — *arbitrary*, anchored by the bid vector — but the **ratio A/B stays 1.50** — *coherent*, the relative preference preserved. This is the signature of coherent arbitrariness, and it is *exactly* the pattern a cost-vector anchoring effect would produce.

Why this is your diagnostic. Coherent arbitrariness gives you a *testable prediction* for whether the cost vector operates as an anchor: **if the cost vector shifts absolute WTP but preserves relative WTP (ratios/rankings across attributes), that is the anchoring signature.** If instead the cost vector changes *relative* valuations (the ranking or ratios of attributes), that is a genuine preference change, not mere anchoring. So examining whether your cost-vector effect is *uniform on absolute WTP* (anchoring,

coherent arbitrariness) versus *differential across attributes* (preference change) is a concrete diagnostic — and it dovetails with the scale-vs-preference fingerprint of Chapter 25 (a scale/anchor effect is uniform; a preference effect changes relative magnitudes). *But note the limit:* coherent arbitrariness is consistent with the cost vector affecting either *reporting* (the anchor pulls the stated number) or *formation* (the anchor becomes part of the constructed absolute valuation) — the ratio-preservation signature diagnoses *anchoring* but not *which mechanism* (Chapter 30).

> **Stop and think 27.1.** Your high- and low-cost-vector arms show uniformly higher/lower absolute WTP but the *same* relative valuations across attributes. What behavioral pattern is this, and does it tell you whether the cost vector affects reporting or formation? *(Answer: this is* coherent arbitrariness* — the cost vector anchors the *absolute* WTP (arbitrary) while relative preferences (ratios across attributes) are preserved (coherent). It confirms the cost vector acts as an *anchor* (not a differential preference change), but it does *not* tell you whether the anchoring operates on *reporting* (the anchor pulls the stated number, true preference unchanged) or on *formation* (the anchor becomes part of the constructed valuation) — both produce the same coherent-arbitrariness pattern. Separating reporting from formation requires more than the ratio-preservation signature, Chapter 30.)*

27.4 Anchoring, and internal versus external anchors

Definition 27.3 (Anchoring). Anchoring is the tendency for a judgment (here, a stated WTP) to be pulled toward a reference number (the anchor), even when the anchor is normatively irrelevant. Formally, a simple anchoring model writes the stated WTP as a weighted average of the true valuation and the anchor:

$$\mathrm{WTP}^{\mathrm{stated}} = (1 - w)\, v + w\, a,$$

where v is the true valuation, a is the anchor, and $w \in [0, 1]$ is the **anchoring weight** (0 = no anchoring, 1 = pure anchor). A higher anchor a raises the stated WTP; a larger w means stronger anchoring.

Internal versus external anchors (the distinction your learning objective demands).

- **External anchors** are provided *by the elicitation* — numbers the respondent did not generate: the *bid in an SBDC*, the *values on a payment card*, the *cost vector* in your design. Your cost vector is an **external anchor**: the bid levels you show can pull stated WTP toward them.
- **Internal anchors** are *self-generated* by the respondent — a number they produced themselves earlier in the survey: their answer to a *previous* question (e.g., the first bid in a double-bounded design anchoring the second answer, Chapter 15 §15.4), or their own prior valuation.

Why the distinction matters for you. Your cost vector is an *external* anchor (you control it), which is what makes the cost-vector effect *designable* — you can manipulate the anchor across arms and observe its effect (your high-vs-low-cost-vector treatment). The *double-bounded* shift/anchoring (Chapter 15) is an *internal* anchor (the respondent's own first answer anchors the second). Distinguishing them clarifies what you are studying: the cost-vector effect is an external-anchor effect that you can experimentally vary, whereas the double-bounded shift is an internal-anchor effect that arises from the sequence of questions. Both are anchoring, but they have different design implications — you *manipulate* the external anchor deliberately, while the internal anchor is a *byproduct* of the format to be controlled.

The bid-vector connection. Your PDF's "bid vector effects" (Chapter 29) *are* external-anchoring effects: the placement of the bids influences the stated WTP. This is why bid design is not just about statistical efficiency (Chapters 26, 49) but about *anchoring* — a bid vector concentrated at high values may anchor WTP upward. The anchoring model above is the behavioral mechanism behind "bid vector effects," and it is central to your cost-vector paper.

> **Active recall 27.B.** Distinguish internal from external anchors, classify your cost vector, and say why the classification matters for your design. *(Answer:* external* anchors are numbers

provided *by the elicitation* (the bid, the payment-card values, the cost vector); *internal* anchors are numbers the respondent *self-generated* (a previous answer, e.g. the first bid anchoring the second in double-bounded). Your *cost vector* is an **external** anchor — you control it — which is why the cost-vector effect is *designable*: you can manipulate it across arms (high vs low) and measure its effect, making it an experimentally-varied external-anchoring treatment, whereas the double-bounded shift is an internal-anchor byproduct to be controlled.)*

27.5 Yea-saying, preference uncertainty, and social desirability

Three further phenomena that bias stated WTP, each relevant to your formats.

Yea-saying (acquiescence bias). Some respondents tend to say **"yes"** (or agree) regardless of their true preference — from politeness, a desire to please, or the path of least resistance. In a dichotomous-choice format, yea-saying inflates the "yes" rate, especially at *high* bids (where a true "no" is most likely), biasing estimated WTP *upward*.

Worked Example 27.2 (Yea-saying biases WTP up, verified). Simulate SBDC data where 15% of respondents are yea-sayers who say "yes" regardless of WTP (true median €25). Fitting the WTP:

	Median WTP	Mean WTP
No yea-saying	24.9	28.0
With 15% yea-saying	**28.1**	**32.9**

(Verified.) Yea-saying inflates median WTP by ~13% (€24.9 → €28.1) and mean WTP by ~18% (€28.0 → €32.9), because the yea-sayers' spurious "yes" responses at high bids make the acceptance curve decay too slowly, pushing the estimated WTP distribution up. This is a real threat in your SBDC/referendum formats, and it is one reason a *bivariate* or *mixture* model (allowing a yea-saying class) or careful survey design (avoiding acquiescence cues) matters.

Preference uncertainty. Respondents are often *uncertain* about their own WTP — they hold a *range*, not a point. Asked "would you pay €30?", a respondent whose WTP is "somewhere around €20–€40" may answer inconsistently or default to a heuristic (say "yes" if uncertain — feeding yea-saying, or "no" to be safe). Preference uncertainty is a form of the constructed-preference view (§27.2): the valuation is not sharp. It can be *measured* (follow-up certainty questions — "how sure are you?") and *modeled* (treating uncertain responses differently, or as a distinct response category), and it interacts with hypothetical bias (uncertain respondents overstate more; §27.6). For your work, preference uncertainty is both a nuisance (adds noise → affects scale, Chapter 25) and a substantive phenomenon (uncertain valuations are more anchorable — the cost vector has more room to shape an uncertain valuation).

Social desirability. Respondents may state a higher WTP to appear pro-environment or generous (social-desirability bias) — saying they would pay for conservation because it is the "right" answer, not because they would. This inflates stated WTP and is worse in non-anonymous or interviewer-administered surveys. It is a component of hypothetical bias (§27.6) and is mitigated by anonymity, consequentiality, and cheap-talk scripts (Chapters 28, 50).

Stop and think 27.2. Your SBDC shows a suspiciously slow decline in "yes" rates at high bids (many people accepting large amounts), inflating estimated WTP. Name two behavioral phenomena that could produce this, and how you would investigate. *(Answer: (i)* yea-saying* (acquiescence) — respondents saying "yes" regardless, inflating acceptance at high bids (verified to bias WTP up); (ii) *social-desirability bias* — respondents overstating willingness to pay for a pro-environmental good to appear generous; (also *hypothetical bias* generally, §27.6). Investigate: include a *certainty/debriefing* question (are respondents sure?), check whether "yes" at high bids correlates with acquiescence indicators or social-desirability scales, model a yea-saying/mixture

class, and compare to the incentivized (real-payment) arm — if the slow decline vanishes under real incentives, it was a hypothetical/acquiescence artifact, Chapters 28, 50.)*

27.6 Hypothetical bias

The best-documented and most consequential behavioral problem in stated preference, and the direct subject of your real-vs-hypothetical treatment.

Definition 27.4 (Hypothetical bias). Hypothetical bias is the tendency for *stated* WTP (in a hypothetical survey, where no money changes hands) to *exceed real* WTP (what people would actually pay when money is at stake). People say they would pay more than they actually would — the hypothetical response overstates real willingness to pay. Meta-analyses find the hypothetical-to-real ratio often ranges from about 1.2 to 3 or more (stated WTP 1.2–3× real WTP), depending on the good and design.

Why it happens. Several of the phenomena above combine: without real consequences, there is no incentive to state true WTP (weak incentive compatibility, Chapter 28); respondents anchor, yea-say, and answer to appear generous (social desirability); and constructed preferences are not disciplined by an actual budget constraint (Chapter 3 — the budget is not binding, so the "warm glow" of saying yes is free). Hypothetical bias is thus a *symptom* of the deeper problems (constructed preferences, weak incentives, social desirability) that this chapter and Chapter 28 address.

Your real-vs-hypothetical treatment. Your CV paper's comparison of *hypothetical* and *incentivized* (real-payment) arms is a direct measurement of hypothetical bias: the gap between the hypothetical and real WTP *is* the bias. This is central to your work, and it connects to:

- **The cost-vector question.** The cost vector might affect hypothetical and real responses *differently* — if the cost vector anchors the hypothetical (unconstrained) response more than the real (budget-disciplined) response, the cost-vector effect could be a *hypothetical* artifact. Your design's crossing of cost vector × incentive lets you probe this.
- **Convergent validity and equivalence.** Claiming that hypothetical and real WTP are "close" (small hypothetical bias) is an *equivalence* claim requiring the machinery of Chapter 8, not a non-significant difference.
- **Mitigation.** The cures for hypothetical bias — cheap-talk scripts (reminding respondents to answer as if it were real), consequentiality (making the survey plausibly influence policy), budget reminders, and — the strongest — *real incentives* (your BRIS mechanism) — are the mechanism-design material of Chapter 28.

The identification subtlety (preview of Chapter 30). Comparing hypothetical and incentivized arms measures hypothetical *bias*, but decomposing that bias into its components — anchoring vs social desirability vs the free warm glow — and separating it from a *cost-vector* effect is the hard identification problem your PDF flags ("il confronto tra hypothetical e incentivized non identifica separatamente tutta la componente di anchoring senza restrizioni aggiuntive"). The hypothetical-vs-real gap is *identified*; its *decomposition* is not, without additional structure (Chapter 30).

> **Active recall 27.C.** What is hypothetical bias, how does your CV design measure it, and why is claiming "no hypothetical bias" an equivalence claim? *(Answer: hypothetical bias is the tendency for stated (hypothetical) WTP to* exceed* real WTP — people say they'd pay more than they actually would (typically 1.2–3×). Your design measures it via the *hypothetical vs incentivized (real-payment)* arm comparison — the gap between them is the bias. Claiming "no hypothetical bias" (hypothetical ≈ real) is an *equivalence* claim — you must show the two are equivalent within a pre-specified margin (TOST, Chapter 8), because a non-significant difference is not evidence of no bias, especially if the comparison is underpowered.)*

27.7 Preference formation versus reporting: the central distinction and its observational equivalence

This is the chapter's payoff and the intellectual core of your cost-vector work. When the cost vector moves stated WTP, there are two fundamentally different mechanisms, and telling them apart is the hard problem.

The two mechanisms.

- **Preference formation.** The cost vector changes *what people value* — it enters the construction of the underlying valuation v. Under constructed preferences (§27.2), seeing high costs might genuinely raise how much people come to value the good (the cost signals importance/quality), so their *true* WTP rises. Formally, $v = v(a)$ — the valuation itself depends on the cost vector a.
- **Reporting / anchoring.** The cost vector changes *what people say*, not what they value — it anchors the *report* of a fixed underlying valuation. Formally, $\text{WTP}^{\text{stated}} = (1 - w)v + w\,a$ with v *fixed* — the anchor a pulls the stated number, but the true preference is unchanged.

Why the distinction is decisive for welfare. If it is *formation*, the cost-vector effect is a real phenomenon about preference construction — but then *which* cost vector gives the "true" welfare measure? (There may be no context-free WTP — Chapter 34.) If it is *reporting*, the cost-vector effect is a *measurement artifact* — the "true" WTP is the un-anchored v, and the cost-vector effect should be purged. **The same observed cost-vector effect implies opposite welfare conclusions depending on the mechanism.** This is why your PDF makes the formation-vs-reporting question central.

The observational equivalence problem (verified). Here is the difficulty: *both mechanisms produce the same observable pattern* — stated WTP rises with the cost vector. **Worked Example 27.3 (verified):**

Cost vector	Reporting model (anchored report, v fixed)	Formation model (v moves with cost vector)
Low (15)	stated WTP = 21.0	stated WTP = 23.8
High (40)	stated WTP = 31.0	stated WTP = 30.0

(Verified.) Both models produce stated WTP that *rises with the cost vector* (from ~€21–24 to ~€30–31) — and from *cross-sectional stated WTP alone*, you **cannot tell which mechanism generated the data**. The reporting model (anchoring a fixed preference) and the formation model (the cost vector changing the preference) are **observationally equivalent** at the level of stated WTP. This is the deep identification problem your PDF names — *"il confronto tra hypothetical e incentivized non identifica separatamente tutta la componente di anchoring … stai identificando preference formation o reporting bias?"* — and it is the reason your cost-vector effect cannot be interpreted from the stated WTP alone.

What could break the equivalence (preview of Chapter 30). Separating formation from reporting requires *additional structure* your design might (partly) provide:

- **Real vs hypothetical comparison.** If the cost vector anchors the *hypothetical* (unconstrained) response but not the *real* (budget-disciplined) response, that pattern points to *reporting* (anchoring is a hypothetical artifact). If the cost-vector effect persists under *real* incentives, that points more toward *formation* (a real change in valuation).
- **Coherent-arbitrariness signature (§27.3).** Uniform absolute shift with preserved ratios points to *anchoring* (which could be reporting or formation of the absolute level); differential change in relative valuations points to genuine *preference change*.
- **External measures.** Certainty, familiarity, and process measures (Chapter 50) can help attribute the effect.
- **Theoretical restrictions.** Imposing a model (e.g., that reporting bias is proportional to the anchor) can point-identify the decomposition — but such restrictions are *untestable* (Chapter 30's "sto introducendo una restrizione teorica non testabile?").

The honest stance. For your cost-vector papers, the mature conclusion is often: "the cost vector affects stated WTP; this is consistent with *both* preference formation and reporting/anchoring, which are observationally equivalent from the stated data; the real-vs-hypothetical comparison [does/does not] favor one; and a full decomposition requires [untestable restrictions / additional design variation] that I [do/do not] have." Recognizing the observational equivalence — rather than claiming "the cost vector changes preferences" or "it's just anchoring" — is exactly the critical judgment your PDF demands, and it is developed formally in Chapter 30.

> **Stop and think 27.3.** Your cost-vector effect on stated WTP is large and robust. A co-author wants to conclude "the cost vector shapes environmental preferences." Why is this premature, and what would you need to support it? *(Answer: it is premature because the cost-vector effect is* observationally equivalent* between two mechanisms — *preference formation* (the cost vector changes the underlying valuation) and *reporting/anchoring* (the cost vector pulls the stated number, true preference unchanged) — both of which produce stated WTP rising with the cost vector (verified). From cross-sectional stated WTP alone you cannot distinguish them, and they imply opposite welfare conclusions. To support "shapes preferences" (formation), you would need additional structure: the effect persisting under *real* incentives (not just hypothetical), differential (not just uniform) changes in relative valuations, external process/certainty measures, or an explicit — and untestable — identifying restriction (Chapter 30). The honest report is that the effect is real but its *mechanism* is partially identified.)*

27.8 Failure modes

F1. Assuming stable preferences for unfamiliar goods. Treating WTP as a fixed quantity the survey reveals, when it is likely constructed. *Fix:* recognize constructed preferences for non-market goods; expect and study context effects (§27.2).

F2. Interpreting a cost-vector effect as preference formation without ruling out reporting. Concluding the cost vector "changes preferences" from stated WTP alone. *Fix:* recognize the observational equivalence; use real-vs-hypothetical, coherent-arbitrariness signature, and Chapter 30 identification (§27.7).

F3. Ignoring anchoring in bid/payment-card design. Not recognizing the bid vector as an external anchor. *Fix:* treat bid design as anchoring-relevant; vary/randomize; examine bid-vector effects (§27.4; Chapter 29).

F4. Yea-saying unmodeled. A slow acceptance-curve decline biasing WTP up, unrecognized. *Fix:* certainty/debriefing questions; mixture/yea-saying class; compare to incentivized arm (§27.5).

F5. Preference uncertainty ignored. Treating uncertain responses as sharp. *Fix:* certainty follow-ups; model uncertainty; recognize it adds noise (scale) and anchorability (§27.5; Chapter 25).

F6. Reporting hypothetical WTP as welfare. Using stated (hypothetical) WTP without accounting for hypothetical bias. *Fix:* real incentives/consequentiality/cheap talk; measure the bias via real-vs-hypothetical (§27.6; Chapter 28).

F7. Claiming "no hypothetical bias" from a non-significant difference. *Fix:* equivalence testing (TOST) against a margin (§27.6; Chapter 8).

F8. Over-claiming a mechanism. Asserting "formation" or "reporting" when the data are consistent with both. *Fix:* report the observational equivalence and what your design can/cannot identify (§27.7; Chapter 30).

27.9 Diagnostic questions for your own papers

1. **Constructed preferences.** Am I treating WTP for my (unfamiliar, non-market) good as potentially constructed, and studying context effects rather than assuming stability? *(§27.2.)*
2. **Formation vs reporting.** For my cost-vector effect, have I recognized that formation and reporting are observationally equivalent, and stated what my design can/cannot separate? *(§27.7, F2, F8; Chapter 30.)*
3. **Coherent arbitrariness.** Is my cost-vector effect uniform on absolute WTP with preserved ratios (anchoring) or differential across attributes (preference change)? *(§27.3.)*
4. **Anchors.** Have I classified my anchors (cost vector = external; double-bounded = internal) and designed accordingly? *(§27.4, F3.)*
5. **Yea-saying/uncertainty.** Have I checked for yea-saying and preference uncertainty (certainty questions, mixture models)? *(§27.5, F4–F5.)*
6. **Hypothetical bias.** Have I measured hypothetical bias (real vs hypothetical) and mitigated it (incentives/consequentiality/cheap talk)? *(§27.6, F6.)*
7. **Equivalence.** Are my "no bias / convergence" claims equivalence tests, not non-significant differences? *(§27.6, F7; Chapter 8.)*

27.10 Exercises

(A) Comprehension.

A1. State the constructed-preference view and why it makes a cost-vector effect on WTP possible.

A2. Define coherent arbitrariness and give its testable signature for anchoring.

A3. Distinguish internal from external anchors and classify the cost vector and the double-bounded follow-up.

(B) Derivation / reasoning.

B1. Write a simple anchoring model $\text{WTP}^{\text{stated}} = (1-w)v + w\,a$ and show how a higher anchor raises stated WTP; explain what $w = 0$ and $w = 1$ mean.

B2. Show, with the reporting and formation models, that both produce stated WTP rising with the cost vector, and explain why they are observationally equivalent from stated data.

B3. Explain why yea-saying biases estimated SBDC WTP upward, referencing the acceptance curve.

(C) Research application / critique.

C1. Your cost-vector effect is large. A co-author concludes "the cost vector changes environmental preferences." Critique this and state what evidence would distinguish formation from reporting.

C2. Your hypothetical arm gives higher WTP than your incentivized arm. Interpret this, and explain how the cost vector might interact with the hypothetical-real gap.

C3. Explain how you would use the coherent-arbitrariness signature and the real-vs-hypothetical comparison together to characterize your cost-vector effect.

27.11 Solutions

A1. The constructed-preference view holds that for unfamiliar/complex goods people do not retrieve a stable, pre-formed WTP but *construct* one during the survey using elicitation cues. It makes a cost-vector effect possible because, if the valuation is built on the spot, the cost vector (a salient cue) can enter the

construction — genuinely shaping the valuation (formation) or pulling the report (anchoring) — whereas under stable, pre-formed preferences the cost vector could not change what people value.

A2. Coherent arbitrariness: valuations are *arbitrary in absolute level* (shiftable by an irrelevant anchor) but *coherent in relative structure* (rankings/ratios across goods preserved). Signature for anchoring: the anchor (cost vector) *shifts absolute WTP* while *preserving relative WTP* (the ratio/ranking across attributes stays constant) — verified: high vs low cost vector moved absolute WTP (A: €24→€39) but kept the ratio A/B at 1.50.

A3. *External* anchors are provided by the elicitation (numbers the respondent did not generate); *internal* anchors are self-generated (a previous answer). The *cost vector* is an **external** anchor (you provide the bid levels); the *double-bounded follow-up* is anchored by an **internal** anchor (the respondent's own first answer/bid). The distinction matters because the external anchor is experimentally manipulable (the cost-vector treatment) while the internal anchor is a format byproduct to control.

B1. $\text{WTP}^{\text{stated}} = (1-w)v + w\,a$: $\partial\text{WTP}^{\text{stated}}/\partial a = w \geq 0$, so a higher anchor a raises stated WTP (by w per unit of anchor). $w = 0$: no anchoring — stated WTP equals the true valuation v (anchor irrelevant). $w = 1$: pure anchoring — stated WTP equals the anchor a regardless of v (the response is entirely the anchor). Intermediate w = partial anchoring (a weighted average of true valuation and anchor).

B2. Reporting model: $\text{WTP}^{\text{stated}} = (1-w)v + w\,a$ with v fixed → rises with a (the cost vector). Formation model: $v = v(a)$ with v increasing in a → stated WTP $= v(a)$ rises with a. Both produce stated WTP increasing in the cost vector (verified: both went from ~€21–24 at low cost to ~€30–31 at high cost). They are observationally equivalent from stated data because the *only observable* is the relationship between the cost vector and stated WTP, and both mechanisms generate the *same* increasing relationship — the data cannot reveal whether the increase came from a moving true valuation (formation) or an anchored report of a fixed valuation (reporting).

B3. In SBDC, estimated WTP comes from the *acceptance curve* (proportion saying "yes" as a function of the bid). Yea-sayers say "yes" regardless, so they add spurious "yes" responses — disproportionately at *high* bids, where a true respondent would say "no." This makes the acceptance curve decline *too slowly* (too many "yes" at high bids), which the model interprets as a WTP distribution shifted to the right (people willing to pay more), biasing estimated median and mean WTP *upward* (verified: median €24.9 → €28.1, mean €28.0 → €32.9 with 15% yea-saying).

C1. Concluding "changes preferences" (formation) is premature because the cost-vector effect is *observationally equivalent* between *formation* (the cost vector changes the underlying valuation) and *reporting/anchoring* (the cost vector pulls the stated number, true preference fixed) — both produce stated WTP rising with the cost vector (verified), and cross-sectional stated WTP cannot distinguish them, yet they imply opposite welfare conclusions. Evidence that would distinguish them: (i) *real vs hypothetical* — if the effect *persists under real incentives* (budget-disciplined), it favors formation; if it *vanishes* under real payment, it favors reporting/hypothetical anchoring; (ii) *coherent-arbitrariness signature* — uniform absolute shift with preserved ratios favors anchoring, differential relative change favors a genuine preference change; (iii) *external process/certainty measures*; (iv) an explicit, *untestable* identifying restriction (Chapter 30). Report the effect as real but its mechanism as partially identified.

C2. The higher hypothetical WTP is *hypothetical bias* — stated (hypothetical) WTP exceeds real (incentivized) WTP because the hypothetical response lacks real consequences (weak incentives, anchoring, social desirability, free warm glow, an unbinding budget). The cost vector might interact with this gap: if the cost vector *anchors the hypothetical (unconstrained) response more than the real (budget-disciplined) response* — the anchor has more room to move an uncostly answer — then the cost-vector effect would be *larger in the hypothetical arm*, suggesting the cost-vector effect is (at least partly) a *hypothetical/reporting* artifact rather than a real preference change. Your cost-vector × incentive design lets you test this interaction; a cost-vector effect that survives real incentives is stronger evidence of formation.

C3. Use them jointly: (i) the **coherent-arbitrariness signature** — check whether the cost vector shifts *absolute* WTP uniformly while preserving *relative* WTP (ratios across attributes); if so, it is *anchoring* (consistent with reporting *or* formation of the absolute level), if relative valuations change, it is a genuine

preference change. (ii) The **real-vs-hypothetical comparison** — check whether the cost-vector effect *persists under real incentives* (formation) or *attenuates* (hypothetical reporting/anchoring). Combining: an effect that is *uniform on absolute WTP* (anchoring signature) *and attenuates under real incentives* points to *reporting/anchoring in the hypothetical response*; an effect that *changes relative valuations* or *persists under real incentives* points toward genuine *preference formation*. Report the combination of signatures and acknowledge the residual observational equivalence (Chapter 30).

27.12 Spaced review prompts

+1 day. From memory, define constructed preferences, coherent arbitrariness (with its signature), and anchoring (internal vs external); classify the cost vector.

+1 week. Explain hypothetical bias and how your real-vs-hypothetical design measures it; then explain the formation-vs-reporting distinction and why the two are observationally equivalent from stated WTP.

+1 month. For your own cost-vector data, examine the coherent-arbitrariness signature (absolute shift vs relative preservation), measure hypothetical bias (real vs hypothetical), check for yea-saying, and state what your design can and cannot separate about the cost-vector mechanism. Close with the spine sentence.

27.13 How this chapter connects to the rest of the book

- **Chapter 20 (§20.4)** located behavioral phenomena in the utility split (V vs ε); **Chapter 25** the scale-vs-preference fingerprint that complements the coherent-arbitrariness signature; **Chapter 24 (§24.5)** WTP as scale-free (anchoring can look like scale).
- **Chapter 28 (incentive compatibility)** is the mechanism-design cure for hypothetical bias and weak incentives (real incentives, consequentiality, BRIS, cheap talk); **Chapter 29 (bid/cost-vector effects)** develops the external-anchoring bid design and the cost-vector effect; **Chapter 30 (identification)** is where the formation-vs-reporting observational equivalence (§27.7) is formalized as partial identification and the κ-vs-λ separation.
- **Chapter 8 (equivalence)** is the machinery for "no hypothetical bias / convergence" claims; **Chapter 39 (behavioral welfare)** asks whether these behavioral phenomena belong inside or outside the welfare measure; **Chapter 51 (survey design)** provides the certainty, attention, and process measures that help attribute the effects.

This chapter is the behavioral foundation of your cost-vector work — establishing constructed preferences and coherent arbitrariness as why the cost vector can matter, and the formation-vs-reporting observational equivalence as the identification challenge your papers must confront.

27.14 Research translation task

For your **cost-vector work (CV and DCE)**, produce a two-to-three-page behavioral note:

1. **Constructed preferences.** Argue whether WTP for your good is likely constructed, and what that implies for the cost-vector effect.
2. **Coherent-arbitrariness signature.** Examine whether your cost-vector effect shifts absolute WTP uniformly (preserving relative valuations — anchoring) or changes relative valuations (preference change); report the pattern.
3. **Anchors.** Classify your anchors (cost vector = external, manipulated; any internal anchors) and their design implications.

4. **Hypothetical bias.** Measure the hypothetical-vs-incentivized gap; test the cost-vector × incentive interaction (does the effect attenuate under real payment?).
5. **Formation vs reporting.** State explicitly that the two are observationally equivalent from stated WTP, what evidence (real-vs-hypothetical, coherent arbitrariness, external measures) your design brings, and what remains partially identified (Chapter 30).
6. **Close** with the spine sentence: *"My cost-vector effect is consistent with [formation/reporting], which are observationally equivalent; the [coherent-arbitrariness signature / real-vs-hypothetical comparison] [does/does not] favor one; a full decomposition requires [additional structure] I [do/do not] have."*

Carry this into Chapter 28 (incentive compatibility/hypothetical-bias cures), Chapter 29 (bid/cost-vector effects), and Chapter 30 (identification / κ vs λ).

27.15 Bibliography

Core. - Slovic (1995), "The Construction of Preference," *American Psychologist*; and Lichtenstein & Slovic (eds.), *The Construction of Preference*. The foundational statements of constructed preferences (§27.2). *(Verify volume/pages.)* - Ariely, Loewenstein & Prelec (2003), "Coherent Arbitrariness: Stable Demand Curves Without Stable Preferences," *Quarterly Journal of Economics*. The coherent-arbitrariness concept and the anchoring experiment (§27.3). *(Verify volume/pages.)* - Tversky & Kahneman (1974), "Judgment under Uncertainty: Heuristics and Biases," *Science*. The original anchoring-and-adjustment result (§27.4). *(Verify volume/pages.)*

Applied. - Haab & McConnell, *Valuing Environmental and Natural Resources*, and Carson & Hanemann, "Contingent Valuation," on anchoring, yea-saying, and hypothetical bias in CV (§27.4–§27.6). *(Verify editions.)* - List & Gallet (2001) and Murphy, Allen, Stevens & Weatherhead (2005), meta-analyses of hypothetical bias (the 1.2–3× ratio, §27.6). *(Verify volumes/pages.)* - Green, Jacowitz, Kahneman & McFadden (1998), "Referendum contingent valuation, anchoring, and willingness to pay for public goods," *Resource and Energy Economics* — anchoring in referendum CV (§27.4). *(Verify volume/pages.)*

Advanced. - Bateman, Burgess, Hutchinson & Matthews (2008), on learning, coherent arbitrariness, and the "discovered preference hypothesis" in stated preference — directly on formation vs reporting (§27.7). *(Verify volume/pages.)* - Carson & Groves (2007), "Incentive and informational properties of preference questions," *Environmental and Resource Economics* — the incentive foundation linking hypothetical bias to mechanism design (Chapter 28). *(Verify volume/pages.)* - Payne, Bettman & Johnson, *The Adaptive Decision Maker*, on constructed preferences and context effects (§27.2). *(Verify edition.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Slovic / Ariely et al. / Carson–Groves citations before using them, per the book's no-unverified-citation rule.

End of Chapter 27. Next in Part VI: Chapter 28 — Incentive Compatibility and Mechanism Design (incentive compatibility, dominant-strategy truth-telling, the referendum, BDM, random incentive systems, BRIS, pay-all vs pay-some, consequentiality, cheap talk, and budget reminders), which develops the mechanism-design cures for the behavioral problems this chapter exposed — and the theory behind your SBDC, RIS, and BRIS. Tell me when to write it, or request a different chapter.

Chapter 28 — Incentive Compatibility and Mechanism Design

Part VI: Contingent Valuation and Stated Preference Design

> **The spine question, applied to this chapter.** Chapter 27 exposed the behavioral problems — hypothetical bias, anchoring, yea-saying, the free warm glow. This chapter builds the cure: **mechanism design**. A well-designed elicitation gives respondents a *reason to tell the truth* — it is **incentive-compatible** — so that stating one's true WTP is the best they can do. Your PDF is precise about what you need: *"Non serve diventare teorico puro, ma devi conoscere: incentive compatibility; dominant strategy truth-telling; BDM; referendum; random incentive systems; pay-all vs pay-some; consequentiality … Ti serve per difendere SBDC, PC, RIS e BRIS."* Your CV paper implements the payment card via a random incentive system/BDM and discusses BRIS to cut implementation costs, so you must be able to *defend* each mechanism's incentive properties and name its vulnerabilities. The spine question here: *for each format I use, is truthful revelation in the respondent's interest — under what conditions — and where does that incentive compatibility break?*

28.0 Conceptual motivation: giving people a reason to tell the truth

The behavioral problems of Chapter 27 all share a root cause: in a purely hypothetical survey, the respondent has *no reason* to state their true WTP. Saying they would pay more costs nothing; anchoring, acquiescing, and answering to appear generous are all free. **Mechanism design** attacks this at the root by changing the *incentives* — making the elicitation a decision with consequences, so that misrepresenting one's WTP becomes costly and truthful revelation becomes the respondent's best strategy. Your PDF frames this exactly: mechanism design is what lets you *defend* your formats — SBDC, payment card, RIS, BRIS — as producing genuine WTP rather than cheap talk.

Why is this the theoretical core of your CV design? Because your paper's whole structure is a set of *incentive* manipulations:

- The **SBDC/referendum** is the format economists trust most, *because* a single yes/no vote at a stated price, with coercive payment, is incentive-compatible — truthful voting is a dominant strategy. Defending your SBDC WTP as a welfare measure *is* defending its incentive compatibility.
- The **payment card via BDM/RIS** is how you make the max-WTP elicitation incentive-compatible: stating your true maximum becomes optimal when a random price determines whether you pay. You must be able to explain *why* BDM makes truth-telling dominant.
- **Real incentives** (your incentivized arm) are the strongest cure for hypothetical bias — they make the payment real, so the survey is a genuine transaction.
- **BRIS** is your cost-reducing device: making the exercise consequential/incentive-compatible without paying *everyone*, by binding only a random subset. You must defend its incentive properties and name where they rely on assumptions that can fail.

- **Consequentiality** is the theoretical bridge: even a *hypothetical* survey can be incentive-compatible if respondents believe it *might* influence policy — the "knife-edge" result that justifies stated preference at all.

This chapter builds the mechanism-design toolkit your PDF lists: incentive compatibility and dominant-strategy truth-telling; the referendum (why it works for public goods); BDM (why it works for the payment card); random incentive systems and the isolation assumption they rely on; BRIS and the pay-all-vs-pay-some trade-off; consequentiality (the Carson–Groves result); and the practical mitigations (cheap talk, budget reminders). Throughout, the goal is your PDF's: to *defend* your formats and *name their vulnerabilities* — because an incentive-compatibility claim, like every claim in this book, holds only under conditions you must state.

28.1 Prerequisites

From **Chapter 27**: the behavioral problems (hypothetical bias, anchoring, yea-saying) that mechanism design cures, and the formation-vs-reporting question. From **Chapter 3**: the budget constraint and consumer optimization (a mechanism is incentive-compatible when truth-telling maximizes the respondent's payoff). From **Chapter 26**: the CV formats (referendum/SBDC, payment card) whose incentive properties this chapter analyzes. From **Chapter 20 (§20.8), 21**: the referendum/opt-out in the RUM. From **Chapter 8**: equivalence testing (for real-vs-hypothetical convergence).

Same **notation**. New/consolidated symbols: v (true WTP/valuation), s (stated WTP), P or p (price — fixed or random), DSIC (dominant-strategy incentive compatibility), IC (incentive compatibility), and the mechanisms referendum, BDM, RIS, BRIS.

28.2 Incentive compatibility and dominant-strategy truth-telling

Definition 28.1 (Incentive compatibility). A mechanism (elicitation) is **incentive-compatible (IC)** if truthful revelation of one's preference (WTP) is in the respondent's own interest — they cannot obtain a better outcome by misrepresenting. The strongest form is **dominant-strategy incentive compatibility (DSIC)**: truth-telling is optimal *regardless of what anyone else does* (no beliefs about others required).

Why it is the goal. If a mechanism is IC, the respondent's *best response* is to tell the truth, so their stated WTP reveals their true WTP — the behavioral problems of Chapter 27 lose their footing (there is now a *reason* not to anchor, yea-say, or overstate). If a mechanism is *not* IC, the respondent has no incentive to be truthful, and stated WTP is cheap talk vulnerable to every bias. **Incentive compatibility is what converts a stated preference into a credible preference.** This is the theoretical basis for trusting some formats (referendum) over others (open-ended), and it is what your PDF means by "la domanda è incentive compatible?".

The general logic (why truth-telling can be a dominant strategy). A mechanism is IC when it is structured so that any misrepresentation either (i) makes the respondent *acquire something at a loss* (overstating → paying more than it's worth) or (ii) makes them *forgo a beneficial transaction* (understating → missing a good they'd have wanted at a fair price). If truthful revelation avoids both — buying exactly when the price is below your value, at that price — then truth-telling maximizes surplus. The referendum (§28.3) and BDM (§28.4) both achieve this, and their derivations share this structure.

The conditions. Incentive compatibility is not automatic; it requires: - **Consequentiality** — the response must *matter* (affect payment or policy) for the respondent to care (§28.6). A response with no consequence has no incentive attached. - **A payment mechanism** the respondent cannot game — a fixed or random price they do not control. - **Understanding** — respondents must grasp the mechanism for its incentives to

operate (a mechanism that is DSIC in theory fails if respondents do not understand it — a real vulnerability of BDM).

> **Active recall 28.A.** Define incentive compatibility and dominant-strategy incentive compatibility, and state in one sentence why IC matters for the validity of stated WTP. *(Answer: a mechanism is* incentive-compatible* if truthful revelation of WTP is in the respondent's own interest (they cannot do better by misrepresenting); it is *dominant-strategy* IC if truth-telling is optimal regardless of what others do. IC matters because it gives the respondent a *reason* to state their true WTP — converting a hypothetical, bias-prone stated preference into a credible one — so that stated WTP reveals true WTP and the behavioral problems of Chapter 27 lose their footing.)*

28.3 The referendum: incentive compatibility for public goods

The **referendum** — a single binding (or consequential) yes/no vote on whether to provide a public good at a stated per-person cost, with **coercive payment** (everyone pays if it passes) — is the canonical incentive-compatible mechanism for public goods, and the reason the NOAA panel endorsed the dichotomous-choice format (Chapter 26).

Why it is incentive-compatible (derivation). Consider a respondent with true WTP v facing a take-it-or-leave-it price P (if the program passes, they pay P; if it fails, they pay nothing). Their **dominant strategy is to vote "yes" iff** $v \geq P$:

- If they vote *truthfully* (yes iff $v \geq P$): they get the program (paying P) exactly when $v \geq P$, earning surplus $v - P \geq 0$; otherwise they pay nothing (surplus 0). Non-negative surplus.
- If they *overstate* (vote yes when $v < P$): they might get the program and pay $P > v$ — *negative* surplus. A loss.
- If they *understate* (vote no when $v \geq P$): they might forgo a program worth more than its cost — *forgone positive* surplus. A loss.

So truthful voting weakly dominates any misrepresentation — regardless of how others vote (DSIC) — because the respondent cannot influence the *price* (it is fixed at P), only whether they get the program at that price. **Worked Example 28.1 (verified):** with $v = 30$, a price of 25 makes truthful "yes" give surplus $+5$ while lying ("no") gives 0; a price of 35 makes truthful "no" give 0 while lying ("yes") gives -5. Truth weakly dominates in both cases (verified).

Why coercion and the single binary choice matter. Two features are essential: - **Coercive payment** (everyone pays if it passes) removes the free-rider incentive: you cannot enjoy the public good while others pay, so you have no reason to understate. This is why the referendum works for *public* goods where voluntary contribution mechanisms fail (free-riding, Chapter 35). - **A single binary choice** (not a follow-up, not an open-ended amount) is what keeps it DSIC — the respondent has one decision (accept the price or not) and cannot game it. Adding a *follow-up* bid (double-bounded, Chapter 15) breaks the incentive compatibility, because the second question is no longer a simple accept/reject at a fixed price.

Your SBDC as a referendum. The SBDC is the survey analog of the referendum — a single yes/no at a stated price. Under **consequentiality** (respondents believe the survey might influence policy, §28.6) and coercive framing (a tax or mandatory fee everyone pays), the SBDC inherits the referendum's incentive compatibility — which is why it is the format whose WTP you can most defend as a welfare measure, and the *validity anchor* of your CV design (Chapter 26). Defending your SBDC WTP *is* invoking this referendum incentive-compatibility argument.

> **Stop and think 28.1.** Why does adding a follow-up bid (double-bounded) break the referendum's incentive compatibility, and what is the design implication? *(Answer: the referendum is DSIC because it is a* single* accept/reject decision at a *fixed* price the respondent cannot influence — truth-telling weakly dominates. A follow-up bid turns it into a *sequence*: the respondent's

first answer affects the second price they face, so they can *strategically shade* their first answer to influence the follow-up (and the follow-up itself is not a simple fixed-price accept/reject) — breaking DSIC and inducing anchoring/shift (Chapter 15 §15.4). Design implication: the single-bounded (referendum) format is incentive-compatible while the double-bounded is not, so use single-bounded as the validity anchor and treat double-bounded's efficiency gain as bought with a loss of incentive compatibility — Chapter 26.)*

28.4 BDM: incentive compatibility for the payment card

The referendum works for a *public good* with a *single price*. But the **payment card** asks for a *maximum WTP* (a number, or a bracket), and making *that* incentive-compatible requires a different mechanism: the **Becker–DeGroot–Marschak (BDM)** procedure — which your CV paper uses to implement the payment card.

The BDM mechanism. The respondent states a WTP s. Then a **price p is drawn at random** (from a known distribution). If $s \geq p$, the respondent *buys the good at price p* (not at s); if $s < p$, no transaction. Crucially, **the respondent pays the drawn price p, not their stated s** — so their stated s determines only *whether* they transact, not *how much* they pay.

Why truth-telling is a dominant strategy (derivation). Because the price paid is the random p (not s), the respondent's stated s affects only the *threshold* for buying. The best threshold is their *true* value v:

- **Truthful ($s = v$):** they buy iff $p \leq v$, paying p, earning surplus $v - p \geq 0$ on every purchase. They transact exactly when it is beneficial.
- **Overstate ($s > v$):** they additionally buy when $v < p \leq s$ — paying $p > v$ — earning *negative* surplus on those extra purchases. A loss.
- **Understate ($s < v$):** they fail to buy when $s < p \leq v$ — forgoing purchases worth more than their price — losing *positive* surplus. A loss.

So stating $s = v$ maximizes expected surplus — truth-telling is a dominant strategy (the same logic as a second-price auction: you cannot improve your outcome by misreporting a threshold when the price is set by an independent draw).

Worked Example 28.2 (BDM truth-telling, verified). A respondent with true WTP $v = 30$ faces a random price $p \sim \text{Uniform}(0, 60)$. Expected surplus by stated s:

Stated s	Expected surplus
20	6.67
25	7.29
30 (= true WTP)	**7.50 (maximum)**
35	7.29
40	6.67

(Verified.) Expected surplus is maximized *exactly at the true WTP* ($s = 30$); overstating ($s = 40$) incurs an expected loss of €0.83 from buying at prices above value, and understating ($s = 20$) forgoes €0.83 of surplus. **BDM makes truthful WTP revelation the respondent's best strategy** — which is why your payment card, implemented via BDM/RIS, can elicit genuine max-WTP rather than an anchored guess.

Vulnerabilities (which you must name). BDM is DSIC *in theory*, but it has real weaknesses: - **Comprehension.** BDM is *cognitively demanding* — respondents must understand that they pay the random price, not their bid, and that misreporting cannot help. Many do not grasp this, so the theoretical incentive compatibility fails in practice (respondents anchor, or bid strategically anyway). This is a serious, documented vulnerability. - **Private-good framing.** BDM works cleanly for a *private* good (a transaction

between the respondent and the experimenter). Applying it to a *public* environmental good requires framing the "purchase" carefully (what does it mean to "buy" ecosystem-service provision at a random price?), and the incentive properties can be murkier. - **The price distribution.** The support and shape of the random-price distribution can itself anchor responses (an *external anchor*, Chapter 27) — so BDM does not fully escape anchoring.

Defending your payment-card-via-BDM design means invoking the DSIC property *and* addressing comprehension (did respondents understand?) and the price-distribution anchoring.

> **Active recall 28.B.** Why is truth-telling a dominant strategy under BDM, and what is its main practical vulnerability? *(Answer: under BDM the respondent pays the* random drawn price* p, not their stated s, so s affects only the *threshold* for transacting; overstating makes them buy at prices above their value (a loss), understating makes them forgo beneficial purchases (a loss), so stating the true value v maximizes expected surplus (verified: max at $s = v = 30$) — truth-telling dominates. Main vulnerability: *comprehension* — BDM is cognitively demanding (respondents must grasp that they pay the random price, not their bid), and if they don't understand, the theoretical incentive compatibility fails in practice.)*

28.5 Random incentive systems, BRIS, and pay-all vs pay-some

Your DCE and payment card involve *many* questions/tasks, and paying real money on *every* one is expensive and can create problems (wealth effects, hedging). **Random incentive systems (RIS)** and their cost-reducing variants (BRIS) address this.

Random incentive system (RIS). When a respondent answers multiple tasks, RIS makes only *one* (randomly selected) task **binding** — that task is paid for real, the others are hypothetical. The idea: if the respondent does not know *which* task will be binding, they should treat *each* as if it could be the binding one, and therefore answer each truthfully (the incentive compatibility of the binding mechanism transfers to every task). This is how a multi-task DCE or payment card can be incentivized without paying every decision.

The isolation (independence) assumption — the crux. RIS is incentive-compatible *only if* respondents treat each task in **isolation** — evaluating each as a stand-alone decision, as if it were the only one that mattered. Your PDF lists "independence assumptions" and "monotonicity" for exactly this reason. The assumption can *fail*: - **Portfolio/hedging effects.** If respondents view the whole set of tasks as a *portfolio* and hedge (e.g., diversify their answers across tasks to balance risk), they do *not* treat each in isolation, and RIS is no longer incentive-compatible. - **Wealth/order effects.** Earlier tasks' (hypothetical) outcomes may affect later answers if respondents track a running "budget." - The isolation assumption is *behavioral* and *untestable* directly — its validity is debated (Cubitt–Starmer–Sugden; Azrieli–Chambers–Healy), and it is a genuine vulnerability you must acknowledge.

BRIS (a cost-reducing random incentive system). Your PDF describes BRIS as *"una soluzione pratica per ridurre i costi di implementazione"* — a practical way to make the exercise consequential/incentive-compatible at lower cost. Functionally, BRIS is a **pay-some** mechanism: rather than binding a task for *every* respondent, it binds real payment for only a *random subset* of respondents (or a random subset of decisions), so the researcher pays out less while each respondent still faces a positive probability that their choice is real. *(Note: the exact BRIS specification and acronym should match your paper's definition — variants exist, and you should state precisely which mechanism you used and cite its source; I flag this rather than assert a single canonical definition.)* The key trade-off BRIS embodies is **pay-all vs pay-some**: - **Pay-all** (every decision binding for everyone): the cleanest incentives, but *expensive* and prone to wealth/hedging effects across many real payments. - **Pay-some** (a random subset binding — RIS/BRIS): *cheaper*, and it relies on the same isolation assumption — respondents treat each decision as potentially binding. The incentive compatibility is preserved *if* respondents respond to the *probability* of being binding as they would to certainty (a rationality assumption that can fail — a smaller probability of real consequence may weaken

the incentive, blurring toward the hypothetical).

Defending BRIS (and naming its vulnerability). BRIS is defensible as a cost-reducing device *if* (i) respondents believe there is a genuine, non-trivial chance their choice is binding (consequentiality, §28.6), and (ii) they treat each decision in isolation (the RIS assumption). Its vulnerabilities: if the binding probability is *too low*, respondents may treat the exercise as effectively hypothetical (weakening incentive compatibility toward the hypothetical arm); and the isolation assumption may fail (hedging). So your defense of BRIS should state the binding probability, argue it is high enough to be consequential, and acknowledge the isolation and low-probability vulnerabilities — exactly the critical judgment your PDF demands.

> **Stop and think 28.2.** Your CV paper uses BRIS to reduce costs by binding real payment for only a random subset. A referee asks whether this is still incentive-compatible. What is your two-part defense, and what vulnerability must you concede? *(Answer: two-part defense — (i)* consequentiality*: each respondent faces a genuine, non-trivial probability that their choice is binding (real payment), so the exercise is consequential and they have a reason to answer truthfully (§28.6); (ii)* isolation*: under the random-incentive logic, respondents should treat each decision as potentially the binding one and answer each truthfully, transferring the binding mechanism's incentive compatibility to every task. Vulnerabilities to concede: if the binding probability is* too low*, respondents may treat the exercise as effectively hypothetical (weakening incentive compatibility); and the* isolation/independence* assumption is behavioral and may fail (portfolio hedging), so BRIS's incentive compatibility rests on untestable behavioral assumptions — state the binding probability and argue it is high enough to be consequential.)*

28.6 Consequentiality: the bridge from hypothetical to incentive-compatible

The most important theoretical result for stated preference, because it explains why a *hypothetical* survey can be incentive-compatible at all — and it underpins the credibility of your entire CV exercise.

Definition 28.2 (Consequentiality). A survey is **consequential** if respondents believe their answers *might influence* an outcome they care about — a policy decision, a real payment, the provision of the good. It need not be *certainly* consequential; a *positive probability* of influence suffices.

The Carson–Groves result (the "knife-edge"). Carson and Groves showed that a stated-preference question is incentive-compatible *if and only if* it is **consequential** — and that even a *small probability* that the survey affects policy is enough to make truthful response optimal (for a suitably structured single-binary question). The intuition: if there is *any* chance your answer influences whether the program happens and what you pay, then you face (a probabilistic version of) the referendum's incentives, and truth-telling is your best response. Conversely, a survey believed to be *purely hypothetical* (zero chance of any consequence) is *not* incentive-compatible — the respondent has no reason to be truthful, and anything goes.

Why this is the foundation of stated preference. This result is what justifies contingent valuation as a method: a well-designed CV survey, framed so respondents *believe it may inform real policy*, inherits the referendum's incentive compatibility even without actual money changing hands. So your hypothetical arm is *not* necessarily incentive-incompatible — if it is *consequential* (respondents believe it might matter), it can elicit genuine WTP. Your PDF's "il referendum è consequential?" is precisely this question: the SBDC/referendum is incentive-compatible *provided* it is consequential, and establishing consequentiality (through framing, and by *measuring* respondents' beliefs about consequentiality — Chapter 51) is essential to defending your hypothetical WTP.

The interaction with your real vs hypothetical design. Consequentiality reframes the hypothetical-vs-incentivized comparison (Chapter 27): the *incentivized* arm is consequential by construction (real payment); the *hypothetical* arm is consequential *only if respondents believe it might matter.* So the hypothetical-vs-real gap partly reflects *differences in perceived consequentiality.* Measuring respondents' consequentiality beliefs

(a debriefing question: "do you think your answers will affect policy?") lets you *test* the mechanism — if hypothetical respondents who believe the survey is consequential give WTP closer to the incentivized arm, that supports the Carson–Groves logic and localizes hypothetical bias to *inconsequentiality*.

> **Active recall 28.C.** State the Carson–Groves consequentiality result and why it justifies contingent valuation. *(Answer: a stated-preference question is incentive-compatible* if and only if* it is *consequential* — respondents believe their answers might influence an outcome they care about — and even a *positive probability* of influence suffices (for a suitably structured single-binary question). It justifies CV because a hypothetical survey framed so respondents believe it may inform real policy inherits the referendum's incentive compatibility even without actual money changing hands — so consequential CV can elicit genuine WTP, while purely hypothetical (zero-consequence) surveys cannot. Establishing and measuring consequentiality is therefore essential to defending stated-preference WTP.)*

28.7 Cheap talk and budget reminders: practical mitigations

Two low-cost, imperfect devices that mitigate (but do not eliminate) hypothetical bias, complementing the mechanism-design cures.

Cheap talk. A **cheap-talk script** explicitly warns respondents *before* the valuation question that people tend to *overstate* WTP in hypothetical surveys, and asks them to answer as if the payment were real (as if their budget were truly affected). The evidence is mixed — cheap talk *reduces* hypothetical bias in some settings (particularly for inexperienced respondents and unfamiliar goods) but not reliably, and its effect can depend on the script's wording and length. It is "cheap" (costless to add) and worth including, but it is not a substitute for genuine incentive compatibility — it is a nudge, not a mechanism.

Budget reminders. A **budget reminder** prompts respondents to consider that paying for this good leaves *less money for everything else* — reminding them of the (otherwise non-binding, in a hypothetical setting) budget constraint (Chapter 3). This counters the "free warm glow" (Chapter 27) — the tendency to say yes because saying yes to a good cause feels good and, hypothetically, costs nothing. Budget reminders reduce WTP overstatement by making the trade-off salient, and they are a standard, cheap addition to a CV survey.

Their place in the toolkit. Cheap talk and budget reminders are *mitigations*, not *cures*: they nudge respondents toward truthfulness but do not change the underlying incentives. The *cures* are the mechanisms of §28.3–§28.6 (referendum/SBDC incentive compatibility, BDM, consequentiality, real incentives via RIS/BRIS). A well-designed CV survey (yours) combines both: incentive-compatible mechanisms *and* cheap-talk/budget-reminder nudges. Your PDF lists both, and defending your design means invoking the mechanisms as the primary justification and the nudges as supporting mitigations.

> **Stop and think 28.3.** A co-author proposes relying on a cheap-talk script alone to eliminate hypothetical bias. Why is this insufficient, and what should it accompany? *(Answer: cheap talk is a* nudge*, not a* mechanism* — it asks respondents to behave as if it were real but does not change the underlying incentives, and its effect on hypothetical bias is mixed and unreliable (script- and context-dependent). It cannot substitute for genuine incentive compatibility. It should *accompany* the mechanism-design cures: an incentive-compatible format (referendum/SBDC), real or probabilistic incentives (RIS/BRIS), and consequentiality (respondents believing the survey matters) — with cheap talk and budget reminders as supporting mitigations, not the primary defense.)*

28.8 Defending your CV design

Pulling the chapter to your paper. Your CV design uses SBDC, payment card via RIS/BDM, real incentives, and BRIS. The mechanism-design defense of each:

- **SBDC/referendum:** incentive-compatible as a single yes/no at a fixed price under **consequentiality** and coercive framing (the referendum argument, §28.3) — the validity anchor. *Defense:* single binary choice, coercive payment, consequential framing. *Vulnerability:* requires consequentiality (measure respondents' beliefs); double-bounded follow-ups would break it.
- **Payment card via BDM/RIS:** incentive-compatible for max-WTP elicitation because the random price makes truth-telling dominant (§28.4), with RIS making one task binding (§28.5). *Defense:* DSIC of BDM. *Vulnerability:* comprehension (BDM is demanding), price-distribution anchoring, and the RIS isolation assumption.
- **Real incentives (incentivized arm):** the strongest cure for hypothetical bias — real payment makes the exercise consequential by construction. *Defense:* genuine transaction. *Vulnerability:* cost and wealth effects (which motivate BRIS).
- **BRIS:** cost-reducing pay-some incentive system — consequential via a positive binding probability, relying on isolation. *Defense:* non-trivial binding probability + isolation. *Vulnerability:* low binding probability may blur toward hypothetical; isolation may fail (hedging); state the exact mechanism and binding probability.
- **Cheap talk / budget reminders:** supporting nudges against overstatement and free warm glow.

The mature defense states, for *each* format, the incentive-compatibility argument *and* the conditions it requires *and* the vulnerabilities — because, as everywhere in this book, the claim holds only under conditions. Your PDF's "difendere SBDC, PC, RIS e BRIS" is exactly this: not asserting they are incentive-compatible, but *arguing* it under stated conditions and naming where it breaks.

> **Active recall 28.D.** For your payment-card-via-BDM and your BRIS, give the one-line incentive-compatibility defense and the main vulnerability of each. *(Answer:* Payment card via BDM* — defense: the respondent pays a random drawn price (not their bid), so truth-telling is a dominant strategy (max expected surplus at $s = v$); vulnerability: comprehension (BDM is cognitively demanding; if respondents don't understand it, incentive compatibility fails in practice), plus price-distribution anchoring and the RIS isolation assumption. *BRIS* — defense: a positive, non-trivial probability that a decision is binding makes the exercise consequential (Carson–Groves), so respondents have a reason to answer truthfully at lower cost; vulnerability: if the binding probability is too low the exercise blurs toward hypothetical, and the isolation/independence assumption (treating each decision as potentially binding, no hedging) may fail.)*

28.9 Failure modes

F1. Claiming incentive compatibility without consequentiality. Asserting a format is IC when respondents don't believe the survey matters. *Fix:* establish and measure consequentiality (§28.6); frame the survey as policy-relevant.

F2. Ignoring BDM comprehension failure. Assuming BDM's theoretical DSIC holds when respondents don't understand it. *Fix:* comprehension checks; simple framing; acknowledge the vulnerability (§28.4).

F3. Assuming RIS/BRIS isolation. Treating the isolation/independence assumption as given. *Fix:* acknowledge it is behavioral and may fail (hedging); state and defend the binding probability (§28.5).

F4. Double-bounded incentive-compatibility. Treating a double-bounded design as incentive-compatible like single-bounded. *Fix:* recognize the follow-up breaks DSIC; use single-bounded as the anchor (§28.3; Chapter 15).

F5. Low binding probability blurring to hypothetical. A BRIS binding probability so low the exercise

is effectively hypothetical. *Fix:* ensure a non-trivial binding probability; test against the incentivized arm (§28.5).

F6. Relying on cheap talk as a cure. Using cheap talk instead of incentive-compatible mechanisms. *Fix:* cheap talk is a nudge; pair with mechanisms (§28.7).

F7. Not measuring consequentiality beliefs. Assuming respondents find the survey consequential without asking. *Fix:* debriefing questions on perceived consequentiality; analyze WTP by belief (§28.6; Chapter 51).

F8. Over-claiming a format's incentive properties. Asserting IC without stating conditions/vulnerabilities. *Fix:* defend IC under stated conditions and name where it breaks (§28.8).

28.10 Diagnostic questions for your own papers

1. **Incentive compatibility.** For each format, can I state the incentive-compatibility argument (why truth-telling is optimal) and the conditions it requires? *(§28.2, §28.8.)*
2. **Consequentiality.** Is my survey consequential (do respondents believe it matters), and have I measured their beliefs? *(§28.6, F1, F7.)*
3. **BDM comprehension.** For the payment card via BDM, did respondents understand the mechanism, and did I check? *(§28.4, F2.)*
4. **RIS/BRIS isolation.** Have I acknowledged the isolation/independence assumption and defended the binding probability? *(§28.5, F3, F5.)*
5. **Single vs double bounded.** Am I using single-bounded (IC) as the anchor, not treating double-bounded as IC? *(§28.3, F4.)*
6. **Nudges.** Do I include cheap talk / budget reminders as *supporting* mitigations, not as the primary cure? *(§28.7, F6.)*
7. **Honest defense.** For each format, do I state the IC argument *and* its vulnerabilities? *(§28.8, F8.)*

28.11 Exercises

(A) Comprehension.

A1. Define incentive compatibility and explain why the referendum is incentive-compatible for public goods.

A2. Explain why truth-telling is a dominant strategy under BDM.

A3. State the isolation assumption behind RIS/BRIS and give one way it can fail.

(B) Derivation / reasoning.

B1. Show that under a take-it-or-leave-it referendum price P, voting "yes iff WTP $\geq P$" weakly dominates any misrepresentation.

B2. Show, for a BDM with true value v and random price p, that expected surplus is maximized at stated $s = v$, and explain the losses from over- and under-stating.

B3. State the Carson–Groves consequentiality result and explain why even a small probability of consequence suffices.

(C) Research application / critique.

C1. A referee questions whether your hypothetical SBDC arm can be incentive-compatible without real payment. Defend it using consequentiality, and state what you must show.

C2. Your payment card is implemented via BDM. Write the incentive-compatibility defense and the vulnerabilities you must address.

C3. Defend your use of BRIS to reduce costs, stating the incentive argument, the binding-probability requirement, and the isolation vulnerability.

28.12 Solutions

A1. Incentive compatibility: truthful revelation of WTP is in the respondent's own interest (they cannot do better by misrepresenting). The referendum is incentive-compatible for public goods because it is a *single* yes/no vote at a *fixed* per-person price with *coercive* payment (everyone pays if it passes): the respondent cannot influence the price, only whether they get the program at that price; overstating (yes when WTP $< P$) means paying more than the program is worth (loss), understating (no when WTP $\geq P$) means forgoing a program worth more than its cost (loss), so voting truthfully (yes iff WTP $\geq P$) weakly dominates — and coercive payment removes the free-rider incentive to understate.

A2. Under BDM the respondent pays the *random drawn price* p, not their stated s — so s determines only the *threshold* for transacting. Overstating ($s > v$) makes them buy when $v < p \leq s$, paying more than their value (negative surplus); understating ($s < v$) makes them forgo purchases when $s < p \leq v$ worth more than their price (forgone positive surplus). Stating $s = v$ transacts exactly when beneficial ($p \leq v$), maximizing expected surplus — so truth-telling is a dominant strategy (verified: max expected surplus at $s = v = 30$).

A3. The isolation (independence) assumption: respondents treat *each* task in a multi-task RIS as a *standalone* decision (as if it were the only one that could be binding), so the incentive compatibility of the binding mechanism transfers to every task. It can fail via *portfolio/hedging effects* — respondents viewing the set of tasks as a whole and hedging (diversifying answers to balance risk) rather than answering each in isolation — in which case truth-telling on each task is no longer optimal and RIS/BRIS is not incentive-compatible.

B1. With value v and fixed price P: voting truthfully gives surplus $\max(v-P, 0)$ (get the program iff $v \geq P$, earning $v - P \geq 0$; else 0). Any misrepresentation: voting "yes" when $v < P$ yields the program at $P > v$, surplus $v - P < 0$ (worse than 0); voting "no" when $v \geq P$ yields 0, forgoing $v - P \geq 0$ (worse than $v - P$). Since the respondent cannot affect P, no misrepresentation can improve on the truthful outcome — truthful voting weakly dominates, regardless of others (DSIC).

B2. Expected surplus $\mathbb{E}[\text{surplus}(s)] = \mathbb{E}[(v-p)\mathbf{1}\{p \leq s\}] = \int_0^s (v-p)f(p)\,dp$. Differentiating w.r.t. s: $\frac{d}{ds} = (v-s)f(s)$, which is > 0 for $s < v$ and < 0 for $s > v$, so expected surplus is maximized at $s = v$ (verified: max at 30). Losses: overstating ($s > v$) adds the region $v < p \leq s$ where $(v-p) < 0$ (buying at a loss); understating ($s < v$) drops the region $s < p \leq v$ where $(v-p) > 0$ (forgoing beneficial purchases). Truth-telling avoids both.

B3. Carson–Groves: a stated-preference question is incentive-compatible *if and only if* it is *consequential* — respondents believe their answers might influence an outcome they care about — and (for a suitably structured single-binary question) even a *positive probability* of influence suffices. A small probability suffices because *any* positive chance that the answer affects whether the program happens and what the respondent pays gives them a (probabilistic) version of the referendum's incentives — truth-telling maximizes expected payoff over that probability — whereas a purely hypothetical (zero-consequence) survey attaches *no* incentive to the answer, so truthfulness is not optimal.

C1. Defense via consequentiality (Carson–Groves): the hypothetical SBDC can be incentive-compatible *without* real payment if respondents *believe the survey might influence policy* — a positive probability of consequence gives them a probabilistic version of the referendum's incentives, making truthful "yes iff WTP $\geq$ price" optimal (§28.6). What I must show: (i) the survey is *framed* as policy-relevant (respondents plausibly believe it may inform a real decision); (ii) I *measure* respondents' consequentiality beliefs (a debriefing question) and analyze whether those who believe it is consequential give WTP consistent with incentive-compatible behavior (and closer to the incentivized arm); (iii) the format is a single binary choice with

coercive framing (not double-bounded). Without consequentiality, the hypothetical arm is not incentive-compatible — so establishing and measuring it is the defense.

C2. Defense: the payment card via BDM is incentive-compatible because the respondent pays a *random drawn price* (not their stated bid), making truthful max-WTP revelation a dominant strategy (verified: expected surplus maximized at the true value) — so their stated maximum is their genuine maximum, not an anchored guess. Vulnerabilities to address: (i) *comprehension* — BDM is cognitively demanding, so I must show (via comprehension checks and simple framing) that respondents understood they pay the random price, not their bid, or the DSIC fails in practice; (ii) *price-distribution anchoring* — the support/shape of the random-price distribution can itself anchor responses (an external anchor), so I should check robustness to the price distribution; (iii) the *RIS isolation* assumption if the payment card is one of several tasks. I state the DSIC argument and these vulnerabilities together.

C3. BRIS is a cost-reducing *pay-some* random incentive system (binding real payment for a random subset), and its incentive defense is: (i) *consequentiality* — each respondent faces a *positive, non-trivial probability* that their decision is binding (real), so by Carson–Groves the exercise is consequential and truth-telling is optimal, at lower cost than paying everyone; (ii) *isolation* — under the random-incentive logic, respondents treat each decision as potentially binding and answer each truthfully. Binding-probability requirement: the probability must be *high enough to be perceived as consequential* — if it is too low, respondents may treat the exercise as effectively hypothetical, weakening incentive compatibility toward the hypothetical arm (so I state the binding probability and argue it is non-trivial). Isolation vulnerability: the independence assumption is behavioral and *untestable directly* — respondents may hedge across decisions (portfolio effects) rather than treat each in isolation, in which case BRIS's incentive compatibility fails; I acknowledge this and, ideally, test BRIS WTP against the full-payment and hypothetical arms to bound the effect. (I also state the exact BRIS mechanism and cite its source, since variants exist.)

28.13 Spaced review prompts

+1 day. From memory, define incentive compatibility and DSIC, and derive why the referendum and BDM are incentive-compatible (the over/under-stating losses).

+1 week. Explain the RIS isolation assumption, the pay-all-vs-pay-some trade-off, and BRIS as a cost-reducing pay-some system; state the Carson–Groves consequentiality result and why it justifies stated preference.

+1 month. For your own CV design, write the incentive-compatibility defense of each format (SBDC, payment card via BDM, real incentives, BRIS), the conditions each requires, and the vulnerabilities; measure and analyze consequentiality beliefs. Close with the spine sentence about which formats are incentive-compatible under what conditions.

28.14 How this chapter connects to the rest of the book

- **Chapter 27** exposed the behavioral problems (hypothetical bias, anchoring, warm glow) that mechanism design cures; **Chapter 3** the budget constraint underlying incentive compatibility; **Chapter 26** the CV formats whose incentive properties this chapter analyzes (referendum/SBDC as the validity anchor).
- **Chapter 35 (public goods)** develops coercive payment, free-riding, and referendum mechanisms; **Chapter 29 (bid/cost-vector effects)** develops the bid design where price-distribution anchoring lives; **Chapter 30 (identification)** develops the formation-vs-reporting and consequentiality identification; **Chapter 8 (equivalence)** is the machinery for real-vs-hypothetical convergence.

- **Chapter 51 (survey design)** provides the consequentiality, comprehension, and debriefing measures that test these mechanisms' assumptions; **Chapter 34 (welfare)** asks whether incentive-compatible WTP is a valid welfare measure.

This chapter is the mechanism-design foundation for defending your CV formats — the theory that converts a stated preference into a credible one, under conditions you must state and vulnerabilities you must name.

28.15 Research translation task

For your **CV paper's mechanisms**, produce a two-to-three-page defense note:

1. **Referendum/SBDC.** State the incentive-compatibility argument (single binary choice, coercive payment, consequentiality) and how you establish/measure consequentiality.
2. **Payment card via BDM/RIS.** State the DSIC argument (random price → truth-telling), and address comprehension, price-distribution anchoring, and RIS isolation.
3. **BRIS.** Define your exact BRIS mechanism (and cite its source), state the binding probability, argue consequentiality, and acknowledge the isolation and low-probability vulnerabilities.
4. **Real vs hypothetical.** Relate the incentivized/hypothetical arms to consequentiality; plan to measure consequentiality beliefs and test whether hypothetical WTP approaches incentivized WTP among those who believe the survey is consequential.
5. **Nudges.** Note cheap talk and budget reminders as supporting mitigations.
6. **Close** with the spine sentence: *"Each format is incentive-compatible under [conditions]: SBDC via [consequentiality/coercion], payment card via [BDM], BRIS via [binding probability/isolation]; the vulnerabilities are [comprehension/isolation/low-probability], which I address by [measures]."*

Carry this into Chapter 29 (bid/cost-vector effects), Chapter 30 (identification/consequentiality), Chapter 8 (real-vs-hypothetical equivalence), and Chapter 51 (survey measures).

28.16 Bibliography

Core. - Carson & Groves (2007), "Incentive and Informational Properties of Preference Questions," *Environmental and Resource Economics*. The foundational treatment of incentive compatibility and consequentiality in stated preference — *the* essential reference (§28.2, §28.6). *(Verify volume/pages.)* - Becker, DeGroot & Marschak (1964), "Measuring utility by a single-response sequential method," *Behavioral Science*. The original BDM mechanism (§28.4). *(Verify volume/pages.)* - Vossler, Doyon & Rondeau (2012), "Truth in Consequentiality: Theory and Field Evidence on Discrete Choice Experiments," *American Economic Journal: Microeconomics*. Consequentiality theory and evidence for DCE/CV (§28.6). *(Verify volume/pages.)*

Applied. - Haab & McConnell, *Valuing Environmental and Natural Resources*, and Carson & Hanemann, "Contingent Valuation," on the incentive properties of CV formats (§28.3–§28.5). *(Verify editions.)* - Cummings & Taylor (1999), "Unbiased Value Estimates for Environmental Goods: A Cheap Talk Design for the Contingent Valuation Method," *American Economic Review* — cheap talk (§28.7). *(Verify volume/pages.)* - Harrison & Rutström, and Cummings, Harrison & Rutström, on hypothetical bias and incentive-compatible mechanisms in the lab (§28.4–§28.5). *(Verify citations.)*

Advanced. - Cubitt, Starmer & Sugden (1998), "On the Validity of the Random Lottery Incentive System," *Experimental Economics*; and Azrieli, Chambers & Healy (2018), "Incentives in Experiments: A Theoretical Analysis," *Journal of Political Economy* — the isolation/independence assumption behind RIS (§28.5). *(Verify volumes/pages.)* - Carson, Groves & List (2014), "Consequentiality: A Theoretical and Experimental Exploration of a Single Binary Choice," *Journal of the Association of Environmental and Resource Economists* — the knife-edge consequentiality result (§28.6). *(Verify volume/pages.)* - Gibbard (1973) and

the mechanism-design foundations (Gibbard–Satterthwaite), for dominant-strategy incentive compatibility in general. *(Verify citations.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Carson–Groves / BDM / Cubitt et al. citations before using them — and, in particular, verify the exact BRIS mechanism and its source against your own paper, since variants exist — per the book's no-unverified-citation rule.

End of Chapter 28. Next in Part VI: Chapter 29 — Bid Design, Anchoring, and Cost-Vector Effects (bid-vector effects, scope sensitivity, protest responses, payment vehicles, and the question at the heart of your work: does the cost vector move preferences or only reporting?), which develops the bid/cost-vector design where the external anchoring of Chapter 27 and the mechanism properties of this chapter meet — and where your cost-vector treatment is built. Tell me when to write it, or request a different chapter.

Chapter 29 — Bid Design, Anchoring, and Cost-Vector Effects

Part VI: Contingent Valuation and Stated Preference Design

> **The spine question, applied to this chapter.** This is where the bid/cost-vector design that *is* your cost-vector treatment gets built, and where the two threads of Part VI meet: the external anchoring of Chapter 27 and the mechanism properties of Chapter 28. The bid vector does **double duty** — it is what *identifies* the WTP distribution (Chapter 14–15) *and* it is an external *anchor* that can shift stated WTP (Chapter 27). These two roles are in tension, and your cost-vector treatment sits precisely at their intersection. Your PDF's questions are exactly this chapter's: *"i bid levels identificano bene la distribuzione? il cost vector influenza la preferenza o solo il reporting? il payment vehicle è credibile? i protest sono esclusi correttamente?"* The spine question here: *does my bid vector identify the WTP distribution well, does its effect on WTP reflect preference formation or reporting, and have I handled protests and the payment vehicle defensibly?*

29.0 Conceptual motivation: the bid vector does two jobs at once

The bid vector — the set of prices you offer across respondents — is the most consequential design choice in a dichotomous-choice CV study, because it does *two* jobs that pull against each other:

- **Identification.** The bids are *where you observe the WTP distribution.* A "yes" at bid b tells you WTP $> b$; a "no" tells you WTP $< b$; pooling across bids traces out the CDF (Chapters 14–15). So the *placement* of bids determines how well — and *which parts* — of the WTP distribution you can estimate. Poorly placed bids (all too high, or clustered) identify the distribution badly. This is your PDF's "i bid levels identificano bene la distribuzione?".
- **Anchoring.** The bids are also *external anchors* (Chapter 27): the numbers you show can *shift* respondents' stated WTP toward them. A high bid vector may anchor WTP upward, a low one downward. This is the mechanism behind "bid vector effects," and it is the phenomenon your cost-vector treatment (high vs low cost vector) is designed to study.

The tension. These two roles conflict. For *identification* you want bids spread across the WTP range (to see the whole distribution); but those same spread bids *anchor* differently across respondents. And your *cost-vector treatment* deliberately manipulates the bid vector (high vs low arms) — which changes *both* what is identified *and* the anchor, confounding them. So the bid vector is simultaneously your identification instrument, your anchoring treatment, and a potential confound — and disentangling these is the heart of your cost-vector paper.

This is where your central research question lives: **does the cost vector move preferences (formation) or only reporting (anchoring)?** Chapter 27 established the two mechanisms and their observational equivalence; this chapter puts them in the *design* — the bid-vector manipulation is how you *create* the

cost-vector effect, and designing a test to *separate* formation from reporting is the learning objective. We also complete the CV design toolkit: **scope sensitivity** (the validity test that WTP should rise with the amount of the good), **protest responses** (respondents who reject the premise, and how to handle them defensibly), and **payment vehicles** (the mechanism through which people would pay, and its credibility).

We proceed: the bid vector's double duty; bid design for identification (spanning the distribution, Fisher information, the choke price); bid-vector anchoring effects (your cost-vector treatment); the formation-vs-reporting question and a design to test it; scope sensitivity; protest responses; and payment vehicles. This chapter closes Part VI by building the design where your cost-vector question is posed — and Chapter 30 (identification) is where its answer is formally bounded.

29.1 Prerequisites

From **Chapter 14–15**: the SBDC/interval likelihood, mean/median WTP, and that the bids are where the WTP distribution is observed. From **Chapter 27**: external anchoring, coherent arbitrariness, and the formation-vs-reporting distinction and its observational equivalence. From **Chapter 28**: incentive compatibility and consequentiality (the payment vehicle's credibility bears on IC). From **Chapter 1 (§1.11), 11**: Fisher information (bid placement maximizes information). From **Chapter 26**: spike models (genuine zeros vs protests). From **Chapter 30, 32** (preview): identification/partial identification and exclusion threats.

Same **notation**. Consolidated symbols: b (bid), the *bid vector* (the set of bids used), F/S (WTP CDF/survival), choke price $\bar{b}$ (WTP $= 0$ above it for everyone), w (anchoring weight), scope (the quantity/quality of the good).

29.2 Bid design for identification

The identification role. Each SBDC bid b gives you one point of the survival function $S(b) = \Pr(\text{yes at } b)$ (Chapter 14). To estimate the *whole* WTP distribution, your bids must *span* the distribution — you learn S only where you place bids, so a bid vector concentrated in one region leaves the rest of the distribution unidentified (extrapolated by the assumed parametric form, Chapter 15 §15.5).

Where to place bids (Fisher information). As Chapters 1 (§1.11) and 11 showed, a Bernoulli response carries the *most* information where the "yes" probability is near 0.5 — i.e., near the *median* WTP. So bids near the median are most informative about the distribution's *location*; bids further out inform the *tails* and the *spread*. The optimal bid design maximizes the Fisher information about the WTP-distribution parameters (Chapter 37), which generally means bids concentrated where they are informative (around the median and the range of interest) but *spanning* enough of the distribution to identify its shape and the mean (which depends on the upper tail, Chapter 15 §15.5).

The choke price. The **choke price** $\bar{b}$ is the price at which demand goes to zero — above it, essentially everyone says "no." Bids *above* the choke price are nearly uninformative (they generate almost only "no" responses, telling you little about *where* below the bid WTP lies). A bid vector placed entirely too high wastes respondents on uninformative comparisons.

Worked Example 29.1 (Bid placement governs identification, verified). Simulate SBDC data (true WTP lognormal, median €25, mean €28.3) under three bid vectors, and estimate mean WTP with its standard error:

Bid vector	"Yes"-rate	Est. median	Est. mean	SE(mean)
A: near median [18,22,25,28,32]	0.50	24.5	27.6	**0.51**
B: spread [5,12,25,45,80]	0.49	24.3	27.2	**0.51**

Bid vector	"Yes"-rate	Est. median	Est. mean	SE(mean)
C: too high [60,80,100,130,170]	0.01	21.7	25.5	**6.10**

(Verified.) Bid vectors A (near the median) and B (spread across the range) both identify the distribution well — standard error on mean WTP around €0.51 — because their bids land where responses are informative (yes-rates near 0.5). Bid vector C, placed *above the choke price* (yes-rate 1%, almost everyone says "no"), gives a standard error of **€6.10 — twelve times larger** — because the bids are uninformative: they tell you WTP is below them, but not *where*, so the distribution is barely identified. **This is "i bid levels identificano bene la distribuzione?" made concrete:** a bid vector that misses the WTP range identifies the distribution terribly, and the estimated mean/median become fragile (extrapolated from almost no information). Good bid design — bids spanning and concentrated in the WTP range — is a prerequisite for a credible CV estimate.

> **Active recall 29.A.** Why do bids placed above the choke price identify the WTP distribution poorly, and where should bids be concentrated for good identification? *(Answer: bids above the choke price generate almost only "no" responses (yes-rate near 0), which tell you WTP is* below* the bid but not *where* — carrying little information about the distribution, so its parameters are barely identified and the mean/median are fragile extrapolations (verified: SE on mean WTP 6.10 vs 0.51). Bids should be concentrated where responses are *informative* — near the median (where the yes-probability is ~0.5, maximizing Bernoulli/Fisher information) — while *spanning* enough of the range to identify the distribution's shape and mean, Chapters 1 §1.11, 15, 49.)*

29.3 Bid-vector anchoring effects: your cost-vector treatment

Now the bid vector's *second* job — anchoring — which *is* your cost-vector treatment. The bids are external anchors (Chapter 27), so the *level* of the bid vector can shift stated WTP, independent of its identification role.

The anchoring model. Recall (Chapter 27) that anchoring makes stated/effective WTP a weighted average of the true valuation and the anchor: $\text{WTP}^{\text{eff}} = (1-w)v + w\,a$, where a is the bid-vector level (the reference the respondent is shown) and w the anchoring weight. A *high* bid vector (large a) pulls WTP up; a *low* bid vector pulls it down. So manipulating the bid vector — your **high-cost-vector vs low-cost-vector treatment** — creates a *bid-vector effect* on estimated WTP.

Worked Example 29.2 (The bid-vector effect, verified). Simulate two arms — a low bid vector and a high bid vector — with *identical true preferences*:

Scenario	Low bid vector	High bid vector
No anchoring ($w = 0$)	est. median 25.4	est. median 25.3
With anchoring ($w = 0.3$)	est. median **24.6**	est. median **32.9**

(Verified.) Two crucial results:

- **Without anchoring, the bid vector is *purely* an identification/efficiency choice** — both arms recover the same true median (~€25). The bid vector affects the *precision* (SE) but *not the estimate*. This is the null: if the cost vector had no behavioral effect, the arms would agree.
- **With anchoring, the estimate *shifts* with the bid vector** — the low arm gives €24.6, the high arm €32.9. This is the **bid-vector / cost-vector effect**: the level of the bids anchors WTP, so the estimated WTP distribution differs across arms *even though true preferences are identical*.

What your cost-vector treatment reveals — and does not. Comparing your high- and low-cost-vector arms is exactly this experiment: a difference in estimated WTP between arms *is* the bid-vector effect. If the arms *agree* (no significant difference), there is no cost-vector effect — the bid vector was purely an identification choice. If they *differ* (as the anchoring case shows), there *is* a cost-vector effect. **But — the central caveat — the arm difference reveals *that* the cost vector moves WTP, not *why*:** it could be reporting/anchoring (Worked Example 29.2's mechanism), preference formation (the cost vector genuinely changing valuation), or an identification artifact (the different bid placements identifying different parts of the distribution). Separating these is §29.4 and Chapter 30. The cost-vector treatment *creates and measures* the effect; it does not, by itself, *explain* it.

> **Stop and think 29.1.** Your high- and low-cost-vector arms give significantly different estimated WTP. Before interpreting, list the three things this difference could be. *(Answer: (i)* anchoring/reporting* — the bid-vector level anchors stated WTP (the effect of Worked Example 29.2), true preferences unchanged; (ii) *preference formation* — the cost vector genuinely changes how much respondents value the good (Chapter 27); (iii) an *identification artifact* — the two bid vectors observe *different parts* of the WTP distribution, so a mis-specified parametric form yields different estimates even absent any behavioral effect (§29.2 — e.g., one arm's bids miss the tail). The arm difference reveals *that* the cost vector moves estimated WTP; it does not distinguish these three mechanisms, which requires the test of §29.4 and the identification analysis of Chapter 30.)*

29.4 Does the cost vector move preferences or reporting? A design to test it

The learning objective: *design a test that attempts to separate a preference-formation effect from a reporting effect.* This is the intellectual core of your cost-vector work, and it combines the behavioral signatures of Chapter 27 with the experimental design here.

The problem (restated). Chapter 27 established that preference formation (cost vector changes v) and reporting/anchoring (cost vector shifts the report of a fixed v) are **observationally equivalent** from cross-sectional stated WTP — both make WTP rise with the cost vector (verified there). So the bid-vector effect of §29.3, on its own, cannot tell you which mechanism produced it. To separate them you need *additional design variation* that the two mechanisms respond to *differently.*

The test design (what your paper does). Cross the cost vector with treatments that formation and reporting respond to differently:

1. **Cost vector × real-vs-hypothetical (the key interaction).** Reporting/anchoring is a feature of *unconstrained, low-stakes* responses; a *real* payment (budget-disciplined, incentive-compatible — Chapter 28) leaves less room for the anchor to pull the report. So:
   - If the cost-vector effect *attenuates or vanishes under real incentives*, that points to **reporting/anchoring** (the anchor operated on the cheap, hypothetical response and is disciplined away when money is at stake).
   - If the cost-vector effect *persists under real incentives*, that points more toward **preference formation** (a real change in valuation that survives budget discipline). Your CV paper's crossing of cost vector × incentive is *designed* to reveal this interaction — the cost-vector × incentive interaction coefficient is the test statistic.
2. **The coherent-arbitrariness signature (Chapter 27 §27.3).** Check whether the cost vector shifts *absolute* WTP uniformly (preserving *relative* valuations across attributes/goods — the anchoring signature) or changes *relative* valuations (a genuine preference change). Uniform absolute shift favors anchoring; differential relative change favors formation.

3. **External process/certainty measures (Chapter 50).** Response times, certainty follow-ups, and comprehension can help attribute the effect — e.g., if the cost-vector effect is concentrated among *uncertain* respondents, that favors anchoring (uncertain valuations are more anchorable, Chapter 27 §27.5).

What the test can and cannot deliver (Chapter 30 preview). These design elements *bound* and *provide evidence* about the mechanism, but they do **not** always point-identify it. The cost-vector × incentive interaction distinguishes "attenuates under real payment" (reporting) from "persists" (formation) *at the extremes*, but intermediate patterns (partial attenuation) are consistent with a *mix* of both, and fully decomposing the effect into anchoring vs formation vs a cost-coefficient change requires *additional, untestable restrictions* (Chapter 30's "sto introducendo una restrizione teorica non testabile?"). So the honest deliverable is: "the cost-vector effect [attenuates/persists] under real incentives, and [does/does not] preserve relative valuations, which is [more/less] consistent with reporting than formation; a full decomposition is partially identified." Your PDF's §6.2 ("sto identificando preference formation o reporting bias?") demands exactly this bounded, evidence-based conclusion rather than an over-claim.

> **Active recall 29.B.** Describe the key interaction your design uses to test whether the cost-vector effect is formation or reporting, and state what each outcome implies. *(Answer: the* ***cost vector*** *×* ***real-vs-hypothetical incentive*** *interaction. If the cost-vector effect* attenuates or vanishes under real incentives* (budget-disciplined, incentive-compatible payment), it points to *reporting/anchoring* — the anchor operated on the cheap hypothetical response and is disciplined away when money is at stake. If the cost-vector effect *persists under real incentives*, it points more toward *preference formation* — a real change in valuation surviving budget discipline. Intermediate (partial) attenuation is consistent with a mix, and full decomposition requires additional untestable restrictions (Chapter 30) — so the test bounds and provides evidence rather than point-identifying the mechanism.)*

29.5 Scope sensitivity

A validity test central to CV credibility and to your ecosystem-service valuation: WTP should respond to the *amount* of the good.

Definition 29.1 (Scope sensitivity). Scope is the quantity or quality of the good being valued (e.g., protecting 10 vs 100 hectares, restoring 1 vs 5 fish populations). **Scope sensitivity** requires that WTP *increase with scope* — more of a good thing should be worth more. **Scope insensitivity** (WTP that does not rise with scope, or rises far too little — e.g., the same WTP for 100 as for 10 hectares) is a **validity red flag**.

Why insensitivity is a red flag. If WTP does not respond to scope, respondents may not be valuing the *actual* good but rather expressing a *symbolic* or *warm-glow* valuation (Chapter 39) — the good feeling of "supporting the environment" that does not scale with the environmental outcome. The **Diamond–Hausman critique** of contingent valuation rested heavily on documented scope insensitivity, and the **NOAA panel** made scope sensitivity a *requirement* for a credible CV study. Scope tests come in forms: *between-subject* (different respondents value different scopes) and the stricter *adding-up / within-subject* tests (a respondent's WTP for the whole should equal the sum of WTP for the parts).

For your work. Your ecosystem-service valuations must demonstrate scope sensitivity — WTP that rises appropriately with the scale of restoration, biodiversity, or carbon. If your DCE attributes have *levels* (more vs less of an attribute), scope sensitivity is the requirement that the attribute coefficient be significant and correctly signed, and that WTP scale sensibly with the attribute level (not saturate implausibly fast). Scope insensitivity in your data would undermine the claim that respondents value the actual ecosystem services rather than a symbolic gesture — a threat to the welfare interpretation (Chapter 35). Test and report scope sensitivity explicitly; it is a core validity check the NOAA lineage demands.

Stop and think 29.2. Your CV finds nearly identical WTP for restoring 1 fish population and 5 fish populations. What is the concern, and what does it threaten? *(Answer: this is* scope insensitivity* — WTP that does not rise (much) with the amount of the good — a validity red flag suggesting respondents are expressing a *symbolic/warm-glow* valuation (the good feeling of supporting restoration) rather than valuing the *actual* environmental outcome, which should scale with scope. It threatens the *welfare interpretation*: if WTP does not track the good's quantity, it is not a valid money measure of the good's value (Chapters 35, 39), and it echoes the Diamond–Hausman critique and fails the NOAA scope requirement. Investigate whether the scenario made scope salient, whether respondents understood the difference, and consider whether the result reflects warm glow or a design failure.)*

29.6 Protest responses

Handling protests defensibly is a learning objective and a top concern (your PDF's "i protest sono esclusi correttamente?").

Definition 29.2 (Protest response). A **protest response** is a "no" or a stated zero WTP given *not because the respondent's WTP is low* but because they **reject the premise** — the payment vehicle ("I refuse to pay through taxes"), the scenario ("the government should pay, not me"), the survey itself, or an ethical objection to pricing the environment. A protester may value the good *highly* but refuse to express it through this mechanism.

Why protests are dangerous. A protest "no" *mis-measures* the latent WTP: treating it as "WTP below the bid" (a genuine low valuation) biases the estimated WTP distribution *downward* (Chapter 14 §14.8) — the protester's high true WTP is recorded as low. And protests are *not* genuine zeros (they do not belong in the spike, Chapter 26 §26.5): a spike models *indifferent* respondents (WTP truly zero), whereas a protester's WTP may be large. Conflating protests with genuine zeros mis-specifies both.

Handling protests defensibly (the protocol).

1. **Identify protests** via *debriefing questions* — after a "no"/zero, ask *why* (a follow-up: "you said you would not pay; is that because the good is not worth it to you, or for another reason?"). Responses citing the payment vehicle, the scenario, or ethical objections are protests; responses citing low value are genuine.
2. **Analyze with and without protests.** Report results *including* protests (treated as observed) and *excluding* them, so readers see the sensitivity. Do not silently drop them.
3. **Bound the results.** Since protesters' true WTP is unknown, *bound* the estimate: a lower bound treats protests as zero WTP, an upper bound treats them as high WTP (or drops them). Report the range.
4. **Do not conflate with genuine zeros.** Genuine zeros → the spike (Chapter 26); protests → separate handling.
5. **Beware the exclusion threat (Chapter 32).** Dropping protests is a *post-response exclusion* that can *break the comparability* across your treatment arms if protest rates differ by arm (e.g., the high-cost-vector arm triggers more protests). Report protest rates *by arm* and check whether exclusion changes the treatment comparison — exactly your PDF's §6.6 concern and a preview of Chapter 32.

The cost-vector connection. Protests are especially relevant to your cost-vector treatment: a *high* cost vector may *trigger more protests* (people balk at high prices as unfair), so the high-cost arm's "no" responses are contaminated by protests differently than the low-cost arm's. If you exclude protests, and protest rates differ by arm, the exclusion itself creates a between-arm difference — a confound with your cost-vector effect. So protest handling is not a side issue; it interacts directly with your central treatment.

Active recall 29.C. How do you distinguish a protest from a genuine zero, why must they be handled differently, and what threat does excluding protests pose to your cost-vector comparison? *(Answer: distinguish via* debriefing questions* — a genuine zero cites low value ("not worth it

to me"), a protest cites rejection of the premise ("won't pay through taxes / government should pay"). They must be handled differently because a genuine zero has *truly* zero WTP (belongs in the spike, Chapter 26), while a protester may value the good *highly* but refuses to express it — treating a protest as low WTP biases the distribution downward. Excluding protests threatens the cost-vector comparison because if *protest rates differ by arm* (e.g., the high-cost vector triggers more protests), dropping them creates a between-arm difference that confounds the cost-vector effect — so report protest rates by arm, analyze with/without, and bound (Chapters 14 §14.8, 32).)*

29.7 Payment vehicles

The **payment vehicle** is the mechanism through which respondents would pay — and it is not neutral (your PDF's "il payment vehicle è credibile?").

Definition 29.3 (Payment vehicle). The **payment vehicle** is *how* the respondent would pay for the good: a tax increase, a mandatory fee, a utility-bill surcharge, a one-time donation, a higher price. The choice of vehicle affects both the *credibility* (do respondents believe they would actually pay this way?) and the *incentive properties* (Chapter 28) of the elicitation.

Payment-vehicle bias. Different vehicles trigger different responses *independent of the good's value*: - A **tax** may trigger *protest* (anti-tax sentiment) or objections that "the government should pay" — inflating protest rates and depressing stated WTP among anti-tax respondents. - A **voluntary donation** may trigger *warm glow* (Chapter 39) and *free-riding* — people say they'd donate to feel good, or understate expecting others to pay — and it is *not incentive-compatible* (voluntary contribution invites free-riding, Chapter 35). - A **coercive vehicle** (mandatory tax/fee that everyone pays if the program passes) is more *incentive-compatible* (the referendum logic, Chapter 28 §28.3 — coercion removes free-riding) but may trigger tax protests. - A **price increase** (for a private-ish good) is credible and incentive-compatible if the market is plausible.

Choosing a credible, incentive-compatible vehicle. The vehicle should be (i) **credible** — respondents believe they would plausibly pay this way for this good (a tax for a public environmental program, a fee for a resource access); (ii) **incentive-compatible** — coercive/mandatory rather than voluntary, so truthful revelation is optimal (Chapter 28); and (iii) **consequential** — tied to a decision respondents believe might happen. For your ecosystem-service valuation, a coercive vehicle (a mandatory environmental fee/tax that funds the restoration) is the incentive-compatible choice, but you must manage tax-protest (via framing and protest identification, §29.6). The vehicle interacts with your cost-vector treatment: the *credibility* of the payment at high cost levels matters (a very high cost may strain credibility, triggering protest or disbelief — confounding the cost-vector effect with a credibility effect).

Stop and think 29.3. You must choose a payment vehicle for your restoration CV. Weigh a voluntary donation against a mandatory environmental fee, considering incentive compatibility, warm glow, and protests. *(Answer: the* mandatory environmental fee* is preferable on incentive-compatibility grounds — coercive payment (everyone pays if the program passes) removes the free-riding that makes a *voluntary donation* not incentive-compatible (Chapter 28 §28.3, Chapter 35), and it invites less warm-glow overstatement (a donation triggers the good-feeling-of-giving, inflating WTP, Chapter 39). The trade-off: a mandatory fee/tax may trigger *tax protests* (anti-tax respondents refusing regardless of value), which you must identify and handle (§29.6) and report by arm. Choose the mandatory fee for incentive compatibility, frame it credibly to minimize protest, and measure/handle protests defensibly — the donation's warm-glow and free-riding problems outweigh its lower protest rate.)*

29.8 Failure modes

F1. Bid vector missing the WTP range. Bids too high (above choke) or clustered, identifying the distribution poorly. *Fix:* span the range, concentrate near the median (Fisher information); pilot to locate the range (§29.2; Chapter 49).

F2. Interpreting the cost-vector effect without separating mechanisms. Reading a bid-vector effect as preference formation. *Fix:* recognize anchoring/formation/identification-artifact are confounded; use the cost-vector × incentive test (§29.3–§29.4; Chapter 30).

F3. Ignoring scope sensitivity. Not testing whether WTP rises with the good's amount. *Fix:* scope tests (between-subject, adding-up); report; a null threatens validity (§29.5).

F4. Treating protests as genuine low WTP. Recording a protest "no" as WTP below the bid. *Fix:* debriefing to identify protests; handle separately; bound; don't put in the spike (§29.6; Chapter 26).

F5. Silently dropping protests. Excluding protests without reporting the sensitivity or the by-arm rates. *Fix:* analyze with/without; report protest rates by arm; check the exclusion doesn't confound the treatment (§29.6; Chapter 32).

F6. Non-credible or non-IC payment vehicle. A voluntary donation (free-riding, warm glow) or an implausible vehicle. *Fix:* credible, coercive, consequential vehicle (§29.7; Chapter 28).

F7. Payment vehicle × cost-vector confound. A high cost vector straining payment credibility, confounding the cost-vector effect with a credibility effect. *Fix:* check credibility across cost levels; measure it (§29.7).

F8. Bid-vector effect mistaken for identification. Attributing an arm difference to identification when it is anchoring (or vice versa). *Fix:* the no-anchoring null (arms agree) vs anchoring (arms shift) diagnostic (§29.3).

29.9 Diagnostic questions for your own papers

1. **Identification.** Does my bid vector span and concentrate in the WTP range (good identification), not sit above the choke price? *(§29.2, F1.)*
2. **Cost-vector mechanism.** For my cost-vector effect, have I recognized the anchoring/formation/identification confound and used the cost-vector × incentive test? *(§29.3–§29.4, F2; Chapter 30.)*
3. **Scope.** Have I tested and reported scope sensitivity (WTP rising with the good's amount)? *(§29.5, F3.)*
4. **Protests.** Have I identified protests (debriefing), distinguished them from genuine zeros, analyzed with/without, and bounded? *(§29.6, F4–F5.)*
5. **Protest × arm.** Do protest rates differ by cost-vector arm, and does exclusion confound the treatment? *(§29.6, F5; Chapter 32.)*
6. **Payment vehicle.** Is my vehicle credible, coercive/incentive-compatible, and consequential? *(§29.7, F6.)*
7. **Credibility × cost.** Does a high cost vector strain payment credibility, confounding the cost-vector effect? *(§29.7, F7.)*

29.10 Exercises

(A) Comprehension.

A1. Explain the bid vector's two roles and why they are in tension.

A2. Why does a bid vector above the choke price identify the WTP distribution poorly?

A3. Distinguish a protest response from a genuine zero WTP, and state why each is modeled differently.

(B) Derivation / reasoning.

B1. Using the anchoring model $\text{WTP}^{\text{eff}} = (1 - w)v + w\,a$, show that the estimated WTP shifts with the bid-vector level a when $w > 0$ but not when $w = 0$, and connect this to the cost-vector treatment.

B2. Explain why the cost-vector effect alone cannot distinguish preference formation from reporting, and design the interaction test that attempts to separate them.

B3. Explain why scope insensitivity threatens the welfare interpretation of WTP.

(C) Research application / critique.

C1. Your bid vector was set from a pilot but the yes-rate at your highest bid is 60%. What identification problem does this suggest, and what would you change?

C2. Your high-cost-vector arm gives higher WTP and more protests than the low-cost arm. Write the analysis needed to interpret the cost-vector effect and handle the protests defensibly.

C3. Defend your choice of a mandatory environmental fee as the payment vehicle, addressing incentive compatibility, protests, and the interaction with the cost-vector treatment.

29.11 Solutions

A1. The bid vector's two roles: (i) *identification* — the bids are where you observe the WTP distribution (a "yes"/"no" at bid b locates WTP relative to b), so bid placement determines how well and which parts of the distribution you estimate; (ii) *anchoring* — the bids are external anchors that can shift stated WTP toward them. They are in tension because good identification wants bids *spread across the range* (to see the whole distribution), but those same bids *anchor* responses, and manipulating the bid vector (the cost-vector treatment) changes *both* what is identified and the anchor — confounding them.

A2. Bids above the choke price generate almost only "no" responses (yes-rate near 0), which tell you WTP is *below* the bid but not *where* — so they carry little information about the distribution's location or shape, and its parameters (and hence mean/median WTP) are barely identified and become fragile extrapolations from the assumed parametric form (verified: SE on mean WTP 6.10 vs 0.51 for well-placed bids).

A3. A *genuine zero* has truly zero WTP (the respondent is indifferent — the good is not worth anything to them) and belongs in the *spike* of a spike model (Chapter 26). A *protest* is a "no"/zero given because the respondent *rejects the premise* (payment vehicle, scenario, ethics) — their true WTP may be *high*. They are modeled differently because a protest mis-measures WTP: treating it as low WTP biases the distribution downward, and putting a high-WTP protester in the zero-WTP spike is wrong — protests need separate handling (identify, bound, analyze with/without).

B1. Effective WTP $\text{WTP}^{\text{eff}} = (1-w)v + w\,a$. The estimated WTP distribution is estimated from responses driven by WTP^{eff}. If $w = 0$: $\text{WTP}^{\text{eff}} = v$, independent of a — so the estimate does not depend on the bid-vector level (both arms recover the true v; the bid vector affects only precision) — verified: no-anchoring arms both ~25. If $w > 0$: $\text{WTP}^{\text{eff}} = (1 - w)v + w\,a$ increases in a — so a higher bid-vector level a raises the estimated WTP — verified: with $w = 0.3$, the low arm gives 24.6 and the high arm 32.9. Connection: the cost-vector treatment manipulates a (high vs low bid vector), so a between-arm difference in estimated WTP reveals $w > 0$ (a bid-vector/anchoring effect); no difference reveals $w = 0$ (bid vector = identification only).

B2. The cost-vector effect alone cannot distinguish formation from reporting because both produce estimated WTP that rises with the cost vector (Chapter 27's observational equivalence): reporting anchors the *report* of a fixed v (via $w > 0$), formation changes v itself — both give a between-arm WTP difference. To separate

them, cross the cost vector with **real-vs-hypothetical incentives**: reporting/anchoring operates on cheap, unconstrained responses, so a *real* (budget-disciplined, incentive-compatible) payment leaves less room for the anchor; therefore, if the cost-vector effect *attenuates under real incentives* → reporting, if it *persists* → formation. The test statistic is the **cost vector × incentive interaction**. Supplement with the *coherent-arbitrariness* signature (uniform absolute shift with preserved relative valuations → anchoring) and external certainty/process measures. Acknowledge that intermediate patterns leave the mechanism *partially identified* (Chapter 30).

B3. Scope insensitivity — WTP that does not rise with the amount of the good — threatens the welfare interpretation because a valid *money measure of the good's value* must respond to *how much* of the good is provided (more should be worth more). If WTP is insensitive to scope, it is not tracking the actual environmental outcome but likely a *symbolic/warm-glow* valuation (the good feeling of supporting the cause, Chapter 39), which is not a welfare measure of the good (Chapter 35). This was the core of the Diamond–Hausman critique and why the NOAA panel required scope sensitivity — insensitive WTP cannot be used as a welfare/CBA input for a good of a given scale.

C1. A yes-rate of 60% at the *highest* bid means many respondents would pay even more than your top price — the bid vector *does not reach the upper tail* of the WTP distribution (the choke price is above your highest bid). This poorly identifies the *upper tail* and hence the *mean* WTP (which depends on the tail, Chapter 15 §15.5): the mean is extrapolated beyond your bids and is fragile. I would *raise the top bid(s)* (and pilot again) so the highest bid reaches a low yes-rate (the upper tail), spanning the full distribution; keep bids concentrated near the median for location. Report the Turnbull lower bound as a distribution-free check, and be explicit that the current mean is tail-extrapolated.

C2. Analysis needed: (i) *interpret the cost-vector effect* — the higher WTP in the high-cost arm could be anchoring/reporting, preference formation, or an identification artifact (§29.3); use the **cost-vector × incentive interaction** (does the effect attenuate under real payment? → reporting; persist? → formation) and the coherent-arbitrariness signature (§29.4), acknowledging partial identification (Chapter 30). (ii) *Handle protests defensibly* — identify protests via debriefing (distinguish "not worth it" genuine zeros from "won't pay this way" protests); **report protest rates by arm** (the high-cost arm likely has more protests — a confound); analyze WTP *with and without* protests and *bound* the estimate; and check whether excluding protests *changes the cost-vector effect* (if protest rates differ by arm, exclusion confounds the treatment — Chapter 32). Only after separating the protest-driven from the valuation-driven component can the cost-vector effect be interpreted.

C3. A mandatory environmental fee is defensible because: (i) *incentive compatibility* — coercive payment (everyone pays if the program passes) removes the free-riding that makes a voluntary donation non-incentive-compatible, so truthful revelation is optimal (the referendum logic, Chapter 28); (ii) it reduces *warm-glow* overstatement (a donation triggers the good-feeling-of-giving, a mandatory fee does not); (iii) it is *credible* for a public environmental program funded by a fee. Address *protests*: a mandatory fee/tax may trigger anti-tax protests, so I identify them (debriefing), report rates by arm, analyze with/without, and bound. Address the *cost-vector interaction*: a very high cost vector may strain the fee's *credibility* (respondents disbelieve such a high fee), confounding the cost-vector effect with a credibility effect — so I check payment-credibility across cost levels and measure it. The fee's incentive-compatibility advantage outweighs the protest cost, which I manage transparently.

29.12 Spaced review prompts

+1 day. From memory, explain the bid vector's two roles (identification, anchoring) and their tension; explain why bids above the choke price identify the distribution poorly.

+1 week. Explain the bid-vector/cost-vector effect (anchoring shifts the estimate; no anchoring = identification only), and design the cost-vector × incentive test to separate formation from reporting. State the scope-sensitivity validity test.

+1 month. For your own CV data, check bid-vector identification (yes-rates across bids), test the cost-vector effect and its formation-vs-reporting decomposition (incentive interaction), test scope sensitivity, identify and bound protests (by arm), and assess payment-vehicle credibility. Close with the spine sentence about whether the cost vector moves preferences or reporting.

29.13 How this chapter connects to the rest of the book

- **Chapter 27** supplied external anchoring, coherent arbitrariness, and the formation-vs-reporting distinction this chapter designs a test for; **Chapter 28** the incentive compatibility and consequentiality the payment vehicle must satisfy; **Chapters 14–15** the identification of the WTP distribution the bid vector governs; **Chapter 1 §1.11, 11** the Fisher information behind bid placement.
- **Chapter 30 (identification)** is where the formation-vs-reporting test's *partial identification* is formalized (the κ-vs-λ separation, observational equivalence); **Chapter 32 (threats)** develops the protest-exclusion-breaks-randomization concern (§29.6); **Chapter 37 (design)** is the optimal-bid/attribute design (Fisher-information-efficient bid vectors).
- **Chapter 26 (formats)** and **Chapter 8 (equivalence)** are the format/convergent-validity context; **Chapters 35, 39 (welfare, behavioral welfare)** are where scope insensitivity and warm glow bear on whether WTP is a welfare measure; **Chapter 51 (survey design)** provides the debriefing, certainty, and process measures.

This chapter closes Part VI by building the bid/cost-vector design where your central question — does the cost vector move preferences or reporting? — is *posed* experimentally, and by completing the CV design toolkit (scope, protests, payment vehicles). Chapter 30 is where the question's answer is formally bounded.

29.14 Research translation task

For your **cost-vector CV design**, produce a two-to-three-page note:

1. **Bid-vector identification.** Report the yes-rates across your bids; confirm the bid vector spans and concentrates in the WTP range (not above the choke); assess how well the distribution (and mean) is identified.
2. **Cost-vector effect.** Report the high-vs-low-cost-vector WTP difference; state that it reveals *that* the cost vector moves WTP but not *why* (anchoring/formation/identification).
3. **Formation-vs-reporting test.** Report the cost-vector × incentive interaction and the coherent-arbitrariness signature; state whether the effect attenuates under real incentives (reporting) or persists (formation), and acknowledge partial identification (Chapter 30).
4. **Scope.** Test and report scope sensitivity (WTP rising with the good's amount).
5. **Protests.** Identify protests (debriefing), report rates *by arm*, analyze with/without, bound, and check the exclusion doesn't confound the cost-vector effect.
6. **Payment vehicle.** Justify a credible, coercive, consequential vehicle; check credibility across cost levels.
7. **Close** with the spine sentence: *"My bid vector identifies the WTP distribution [well/poorly]; the cost-vector effect is [size], and the incentive interaction / coherent-arbitrariness signature suggests [reporting/formation], though the mechanism is partially identified; scope sensitivity is [present/absent]; and protests are handled by [bounding/by-arm reporting]."*

Carry this into Chapter 30 (identification/κ vs λ), Chapter 32 (protest exclusions), Chapter 8 (real-vs-hypothetical equivalence), and Chapter 37 (bid design).

29.15 Bibliography

Core. - Haab & McConnell, *Valuing Environmental and Natural Resources*, on bid design, protest responses, and payment vehicles in CV. *(Verify edition/chapters.)* - Boyle, Welsh & Bishop, and Alberini, on bid/design effects and optimal bid design in dichotomous-choice CV (§29.2). *(Verify citations.)* - Carson, Flores & Meade (2001), "Contingent Valuation: Controversies and Evidence," *Environmental and Resource Economics* — scope sensitivity, protests, and the validity debate (§29.5–§29.6). *(Verify volume/pages.)*

Applied. - Green, Jacowitz, Kahneman & McFadden (1998), "Referendum contingent valuation, anchoring, and willingness to pay for public goods," *Resource and Energy Economics* — bid-vector anchoring (§29.3). *(Verify volume/pages.)* - Carson (1997) and Carson & Mitchell on scope sensitivity and the NOAA scope requirement (§29.5). *(Verify citations.)* - Meyerhoff & Liebe, on protest responses in stated preference — identification and handling (§29.6). *(Verify citations.)*

Advanced. - Diamond & Hausman (1994), "Contingent Valuation: Is Some Number Better than No Number?" *Journal of Economic Perspectives* — the scope-insensitivity critique (§29.5). *(Verify volume/pages.)* - Kanninen (1993), on optimal experimental design for dichotomous-choice CV (bid placement / Fisher information, §29.2; Chapter 37). *(Verify volume/pages.)* - The literature on payment-vehicle effects and incentive compatibility (Carson & Groves 2007, Chapter 28) as it bears on vehicle choice (§29.7). *(Verify citations.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Carson / Diamond–Hausman / Green et al. citations before using them, per the book's no-unverified-citation rule.

End of Chapter 29 and of Part VI. You now hold the contingent-valuation and stated-preference-design toolkit: the formats and their trade-offs (Ch 26), the behavioral foundations (Ch 27), incentive compatibility and mechanism design (Ch 28), and the bid/cost-vector design where your central question is posed (Ch 29). Part VII — Identification and Treatment Effects in Survey Experiments — begins with Chapter 30 (Estimation Is Not Identification), the highest-dependency chapter of the identification spine and where your cost-vector formation-vs-reporting question, the κ-vs-λ separation, and the partial-identification limits this chapter kept pointing toward are formally resolved. Tell me which chapter to write next.

Part VI Assessment — Contingent Valuation and Stated Preference Design

Covers Chapters 26–29. Five components, in the order they are best attempted. Allow a full working day; the mock oral is the part that matters most for your viva.

1. Cumulative problem set

P1. Compare open-ended, payment card, SBDC, DBDC and referendum formats on incentive compatibility and statistical efficiency, in a table with one row per format.

P2. State the Carson–Groves conditions for incentive compatibility and check each against your own CV design.

P3. Explain formally why a BDM mechanism is incentive-compatible for a private good and why the argument does not transfer to a public good.

P4. Given a bid vector $\{5, 15, 30, 60, 120\}$ and a believed median WTP of €25, evaluate the design's information content and propose an improvement.

P5. Write the estimating equation that would detect a starting-point effect in a DBDC design, and state the null hypothesis.

2. Research critique

Write the full "assumption → risk → test/robustness" table for your CV paper: every assumption the welfare interpretation rests on, the specific way each could fail, and the test or robustness check that would detect the failure. Mark the assumptions for which you currently have no test.

3. Red-flag diagnosis

A CV study excludes 14% of respondents as protesters, on the basis of a debriefing question asked only of those who answered "no." The reported mean WTP rises by 22%. Explain what has happened to the estimand, why the exclusion is not innocuous, and what analysis you would demand instead.

4. Mock oral examination

1. What makes a referendum incentive-compatible, and what breaks it?
2. What is coherent arbitrariness and why does it threaten your cost-vector interpretation?
3. Distinguish an internal from an external anchor.

4. Does your BRIS mechanism deliver truth-telling? Under what conditions?
5. Is a protest zero a zero?

5. Integration mini-project

Design, in full, the experiment that would separate a preference-formation effect of the cost vector from a pure reporting effect: the arms, the randomization, the sample sizes implied by your power analysis, the pre-specified hypotheses, and — critically — the assumption that would still be untestable at the end. If no design can separate them, prove it and say what you would report instead.

Part VII — Identification and Treatment Effects in Survey Experiments

Chapter 30 — Estimation Is Not Identification

Part VII: Identification and Treatment Effects in Survey Experiments

> **The spine question, applied to this chapter.** This is the intellectual spine of the entire book — the chapter the title *Identification, Assumptions, and Fragility* names first, and one of the three the architecture flags to over-learn. Your PDF calls it *"probabilmente il punto più importante per te"* and sets your standing task: *"Devi essere sempre in grado di dire: 'Questo parametro è identificato da quale variazione nei dati? Con quali assunzioni? Quale alternativa osservazionalmente equivalente non posso escludere?'"* Estimation is what software does — it produces a number. **Identification** is the prior, deeper question: *can the data and model, even with infinite observations, pin down the economic object you care about?* Your cost-vector question lives here: the incentivized effect identifies preference formation, but the hypothetical-vs-incentivized comparison does *not* separately identify anchoring without additional, untestable restrictions — your "κ vs λ" problem, which this chapter formalizes and resolves as *partial* identification. The spine question here: *what parameter am I identifying, by what variation, under what assumptions, and what observationally-equivalent alternative can I not exclude?*

30.0 Conceptual motivation: the question before estimation

Every chapter so far has, at its close, returned to a single question: *what am I identifying, under what assumptions, how fragile.* This chapter is where that question is made precise, because it is the master concept the whole book serves. Your PDF is unambiguous about its importance — *"È probabilmente il punto più importante per te"* — and about the gap it must close: you must be able to state, for any parameter, *which variation identifies it, under what assumptions, and what observationally-equivalent alternative remains.*

The distinction is between two questions that are easy to conflate:

- **Estimation** asks: *given a finite sample, what number does my procedure produce, and how precise is it?* This is Parts II–V — likelihoods, standard errors, convergence, WTP. Estimation always produces a number; the software never refuses.
- **Identification** asks: *could any procedure, even with infinite data, recover the object I care about?* This is *prior* to estimation and *independent* of sample size. A parameter can be **estimated** (the software returns a value) while being **unidentified** (that value is meaningless, because the data cannot distinguish it from other values). This is the trap your PDF names: the software's willingness to return a number says nothing about whether the number identifies anything.

Why is this *the* point for you? Because your central research question — does the cost vector move preferences (formation) or reporting (anchoring)? — is an *identification* question, not an estimation one. Chapters 27 and 29 showed the two mechanisms are *observationally equivalent* from stated data (both make WTP rise with the cost vector); no amount of data or clever estimation separates them. Your PDF states the resolution

precisely: *"l'incentivized cost-vector effect identifica un oggetto medio legato alla preference formation, mentre il confronto tra hypothetical e incentivized non identifica separatamente tutta la componente di anchoring senza restrizioni aggiuntive."* This chapter formalizes that statement — the incentivized effect *bounds* preference formation, the hypothetical-vs-incentivized difference *bounds* anchoring, and *point* separation requires an *untestable* restriction. Your "κ vs λ" is a **partial identification** problem, and understanding it precisely is the difference between an over-claim ("the cost vector changes preferences") and a defensible, bounded conclusion.

We build: the estimation-vs-identification distinction and the formal definition (a parameter is identified when the map from parameter to the distribution of observables is one-to-one); **observational equivalence** (when it fails); **identifying variation** (every identified parameter is identified *by* something); **point vs partial identification** (Manski bounds — what the data say when point identification requires untestable assumptions); **parametric vs nonparametric identification** (functional form can "identify" the nonparametrically-unidentified, but only by assumption); **exclusion restrictions and overidentification**; the centerpiece **κ-vs-λ** formalization of your cost-vector problem; and the disciplined identification question you must ask of every claim. This chapter, with Chapters 11 and 24, is one to over-learn — because identification is the concept on which every claim in your dissertation stands or falls.

30.1 Prerequisites

From **Chapter 16** (§16.5): exogeneity, omitted-variable bias, and what randomization identifies (randomization as the exogeneity machine). From **Chapter 13** (§13.4) and **Chapter 11** (§11.9): the likelihood-surface view of identification (a flat direction = unidentified). From **Chapter 25**: scale-vs-preference observational equivalence. From **Chapter 27** (§27.7) and **Chapter 29** (§29.4): the formation-vs-reporting observational equivalence and the cost-vector × incentive design. From **Chapter 1** (§1.4): rank and identification in OLS.

Same **notation**. New/consolidated symbols: θ (parameter of interest), $P(\text{data} \mid \theta)$ (the distribution of observables implied by θ), Θ_I (the *identified set* — the set of parameter values consistent with the data), κ (preference-formation component of the cost-vector effect), λ (anchoring/reporting component), ϕ (fraction of anchoring surviving real incentives).

30.2 Estimation versus identification

The distinction. Estimation and identification answer different questions:

- **Identification** is a property of the *population* (the data-generating process) and the *model*: *can the parameter, in principle, be recovered from the distribution of the observable data — with infinite data and no sampling error?* It is about whether the *information* to pin down the parameter *exists*.
- **Estimation** is a property of a *finite sample* and a *procedure*: *what value does my estimator produce, and how precise is it?* It is about *extracting* the information — assuming it is there.

Why estimation cannot rescue non-identification. If a parameter is *unidentified*, no estimator can recover it *no matter how much data you have* — because the data simply do not contain the information. More data reduces *sampling* error (estimation), but it does nothing for *identification*: the parameter's variance does not shrink to zero, or the estimator converges to the wrong (or an arbitrary) value. This is the deepest reason your PDF warns that "standard errors enormi segnalano un problema strutturale, non solo pochi dati" (Chapter 1) — a huge standard error is often a *non-identification* symptom (a flat likelihood direction, Chapter 11 §11.9, 13 §13.4), and *no amount of extra data fixes it*. The software returns a number, but the number is meaningless.

Definition 30.1 (Identification). A parameter θ is **identified** if the mapping from θ to the distribution of the observable data is **injective (one-to-one)**: distinct parameter values imply distinct distributions of

observables,

$$\theta_1 \neq \theta_2 \implies P(\text{data} \mid \theta_1) \neq P(\text{data} \mid \theta_2).$$

Equivalently, if you knew the *entire* distribution of the observables (infinite data), you could recover θ uniquely. If instead two distinct parameter values imply the *same* distribution of observables, they are **observationally equivalent** (§30.3), and the parameter is *not* identified — the data cannot tell them apart.

The likelihood-surface view (Chapter 11 §11.9). In likelihood terms, identification is a *unique maximum* of the expected log-likelihood. Non-identification is a *flat direction* — a ridge along which the likelihood is constant, so many parameter values fit *identically*. This is why a near-zero eigenvalue of the Hessian (Chapter 13 §13.4) is a near-non-identification symptom: the data are (nearly) indifferent between a range of parameter values. Identification and the shape of the likelihood are the same phenomenon in two vocabularies.

> **Active recall 30.A.** Distinguish estimation from identification, and explain why more data cannot fix a non-identified parameter. *(Answer:* identification* is whether the parameter can, in principle, be recovered from the *distribution of observables* (infinite data, no sampling error) — whether the information exists; *estimation* is what a *finite-sample procedure* produces and how precise it is — extracting the information, assuming it's there. More data cannot fix non-identification because the data simply do not *contain* the information: distinct parameter values imply the *same* distribution of observables (observational equivalence), so no estimator can distinguish them regardless of sample size — the estimator's variance in that direction does not shrink to zero, or it converges to an arbitrary value. More data reduces sampling error, not the structural gap.)*

30.3 Observational equivalence

The negation of identification, and the concept your cost-vector problem turns on.

Definition 30.2 (Observational equivalence). Two parameter values (or two *models*) $\theta_1 \neq \theta_2$ are **observationally equivalent** if they imply the *same* distribution of the observable data: $P(\text{data} \mid \theta_1) = P(\text{data} \mid \theta_2)$. When observationally-equivalent alternatives exist, the parameter is not identified — the data are *compatible with multiple stories*, and you cannot use the data to choose among them.

Your PDF's version. This is precisely your PDF's §6.2 concern: *"i dati potrebbero essere compatibili con più storie comportamentali."* The cost-vector effect is compatible with a *formation* story and a *reporting* story (Chapter 27); scale differences are compatible with a *scale* story and a *preference* story (Chapter 25). Each is an observational-equivalence problem — distinct behavioral parameters producing the same observable data.

Worked Example 30.1 (Observational equivalence via collinearity, verified). The simplest case: a regression $y = \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ where $x_1 = x_2 = x$ (perfect collinearity, Chapter 1 §1.4, 16 §16.7). Then $y = (\beta_1+\beta_2)x+\varepsilon$ — the data depend on β_1 and β_2 *only through their sum.* So $(\beta_1, \beta_2) = (3, 2), (5, 0), (0, 5), (-1, 6)$ all imply *identical* y (verified) — they are observationally equivalent, and only the *combination* $\beta_1 + \beta_2$ is identified, not the individual coefficients. **This is the template for every identification failure:** the data identify some *function* of the parameters (here the sum) but not the parameters *individually*, because the individual parameters enter the observable distribution only through that function. Your κ-vs-λ problem (§30.7) has exactly this structure — the data identify combinations of the behavioral parameters but not each one separately.

Recognizing observational equivalence. The signature is that *a whole set* of parameter values fits the data *equally well* (identically, in the population). In estimation, this shows as: a flat likelihood ridge (Chapter 11, 13); a singular information matrix; estimates that change with the starting values (the ridge, Chapter 13 §13.5) or that are "identified" only by an arbitrary normalization or functional form (§30.6).

When you see these symptoms, ask: *what combination of parameters is the data actually pinning down, and which individual parameters are observationally equivalent?*

> **Stop and think 30.1.** Your behavioral CV model estimates separate anchoring and formation parameters, and the optimizer converges to specific values — but re-running from different starting values gives different anchoring/formation splits with the *same* log-likelihood. What does this tell you? *(Answer: the anchoring and formation parameters are* not separately identified* — they are observationally equivalent along a ridge of the likelihood (the same log-likelihood for different splits), so the data pin down only some *combination* of them, not each individually. The optimizer's "converged" values are arbitrary points on the ridge (which one you get depends on the starting values, Chapter 13 §13.5). More data will not fix this; it is a structural identification failure. You must either add identifying variation, impose a restriction (§30.6), or report the *identified set* (partial identification, §30.5) rather than a spurious point estimate of the split.)*

30.4 Identifying variation: every parameter is identified *by* something

Identification is never free — every identified parameter is identified *by* some *source of variation* in the data, under some *assumption*. Naming that source and assumption is the core discipline your PDF demands.

The principle. A parameter is identified because the data contain variation that responds to it in a way that pins it down. Common sources of identifying variation:

- **Random assignment.** A randomized treatment identifies its causal effect because assignment is independent of everything else (Chapter 16 §16.5) — the treatment-control difference is identified *by the randomization*, under the assumption that assignment is truly random and there is no interference. Your survey experiments (cost-vector, incentive arms) rely on this.
- **Cross-sectional / functional-form variation.** In a parametric model, variation in the regressors identifies the coefficients *under* the functional form (§30.6).
- **Instrumental variation / exclusion restrictions.** An instrument identifies a causal effect *by* variation that affects the treatment but not the outcome directly (§30.6, the exclusion restriction).
- **Design variation (factorial).** Crossing treatments (cost vector × incentive) identifies interaction effects *by* the factorial variation — the differential response across cells (Chapter 31).

The disciplined statement. For every parameter, you must be able to complete the sentence: *"This parameter is identified by [source of variation], under the assumption that [assumption], and the observationally-equivalent alternative I cannot exclude is [alternative]."* This is your PDF's exact mandate. For example: *"The incentivized cost-vector effect is identified by the random assignment of respondents to high vs low cost vectors within the incentivized arm, under the assumption that the incentive fully disciplines anchoring (φ=0); the observationally-equivalent alternative I cannot exclude is that anchoring partially survives incentives (φ>0), in which case the effect mixes formation and residual anchoring."* Naming the variation, the assumption, and the un-excludable alternative *is* the identification analysis.

> **Active recall 30.B.** Complete the disciplined identification sentence for a randomized treatment effect in your survey experiment. *(Answer: "The [cost-vector] treatment effect is identified by the* random assignment *of respondents to the treatment arms, under the assumption that* assignment is independent of potential outcomes *(no selection, no interference between respondents) and* no differential post-randomization attrition/exclusion*; the observationally-equivalent alternative I cannot exclude is that a [scale difference / differential protest exclusion / anchoring] produces the same observed contrast without a genuine preference change." The three elements — identifying variation (randomization), assumption (independence/no-interference/no-differential-exclusion), and un-excludable alternative — are the required identification statement.)*

30.5 Point identification versus partial identification

A crucial distinction your PDF names ("partial identification") and the key to handling your κ-vs-λ problem honestly.

Definition 30.3 (Point vs partial identification). - **Point identification:** the data and assumptions pin down a *single* value of the parameter (the identified set is a point). - **Partial (set) identification:** the data and assumptions restrict the parameter to a *set* (a range/bounds), not a single value. The **identified set** Θ_I is the set of *all* parameter values consistent with the data and the maintained assumptions.

Why partial identification matters (Manski). Charles Manski's insight: when *point* identification requires *strong, untestable* assumptions, you can instead ask what the data say *under weaker, credible* assumptions — which often yields a *set* (bounds) rather than a point. This is not a failure; it is *honesty*: the bounds are *what the data actually deliver*, and the point estimate would only be recovered by *adding an assumption the data cannot test*. The mantra: **"identification comes from assumptions, and the width of the identified set measures how much you are learning from data versus from assumptions."** A narrow set = the data are informative; a wide set = you would need strong assumptions to say more.

Worked Example 30.2 (Manski bounds on a treatment effect, verified). Suppose in a survey experiment 20% of outcomes are missing (respondents who did not answer), and the outcome $Y \in [0, 1]$. Among the 80% observed, $E[Y \mid T{=}1] = 0.7$ and $E[Y \mid T{=}0] = 0.4$. Without assuming the missing outcomes are like the observed (an untestable "missing at random" assumption), the ATE is only *bounded*: the missing 20% could take any value in $[0, 1]$. The **worst case** for the ATE puts the missing treated at 0 and missing control at 1; the **best case** reverses this:

$$\text{ATE} \in [\,0.04,\ 0.44\,]$$

(verified). The data alone, without a missingness assumption, say only that the ATE is between 0.04 and 0.44 — a *set*, not a point. To get a *point* (e.g., the naive $0.7 - 0.4 = 0.3$), you must *assume* the missing are like the observed — an untestable assumption. **The bounds are what the data deliver; the point requires an assumption.** This is exactly the stance to take with your κ-vs-λ problem (§30.7): report the bounds the design delivers, and be explicit that a point requires an untestable restriction.

For your work. Partial identification is the *mature* response to your PDF's §6.2 concerns. Rather than forcing a point estimate of the formation/anchoring split (which requires the untestable φ), you report the *identified set* — the incentivized effect bounds formation, the hypothetical-incentivized difference bounds anchoring — and state that point identification requires assuming φ. This is defensible; a spurious point estimate is not.

> **Stop and think 30.2.** Your co-author wants a single number for "the fraction of the cost-vector effect that is preference formation." Why might the honest answer be a *range*, and what would a point estimate require? *(Answer: the honest answer may be a* range* because the formation/anchoring split is only *partially identified* — the design (hypothetical and incentivized cost-vector effects) delivers *bounds* (formation $\leq$ the incentivized effect; anchoring $\geq$ the hyp-inc difference — §30.7) but not a point, since separating them fully requires knowing how much anchoring survives incentives (φ), which the data cannot pin down. A *point* estimate would require *assuming* a specific φ (e.g., $\varphi{=}0$, that incentives fully discipline anchoring) — an *untestable* restriction. So report the identified set (the range) and state that the point requires the untestable φ assumption, rather than presenting a single fraction as if the data delivered it.)*

30.6 Parametric versus nonparametric identification, and exclusion restrictions

Two related concepts that determine *how much of your identification comes from the data versus from your assumptions.*

Nonparametric identification. A parameter is **nonparametrically identified** if it can be recovered from the data *without functional-form or distributional assumptions* — from the structure of the data alone (e.g., a randomized treatment effect is nonparametrically identified by the randomization: the treatment-control difference in means identifies the ATE with no model). Nonparametric identification is the *strongest* kind — it does not rest on a functional form you cannot test.

Parametric identification (and its danger). A parameter is **parametrically identified** if it is identified *only* by imposing a functional form or distribution. This is powerful — a parametric model can "identify" parameters that are *nonparametrically unidentified* — but the identification then *rests entirely on the (untestable) functional form*. This is your PDF's exact warning: *"sto introducendo una restrizione teorica non testabile?"* If a parameter is identified *only* because you assumed a lognormal, a linear index, or a specific behavioral form, then the identification is an *artifact of the assumption*, and a different (equally plausible) assumption would give a different answer. **Parametric identification of a nonparametrically-unidentified parameter is identification by assumption, not by data** — and you must flag it as such.

The practical test. Ask: *would this parameter still be identified if I dropped the functional form?* If yes, it is nonparametrically identified (robust). If no, it is identified *only* by the functional form (fragile — the identification is untestable, and you should report sensitivity to the assumption or bound the parameter nonparametrically). Your κ-vs-λ split is nonparametrically *partially* identified (bounds, §30.7); a *point* split is available *only* parametrically (by assuming φ), which is untestable.

Exclusion restrictions. An **exclusion restriction** is an assumption that a variable affects the outcome *only through a specific channel* — e.g., an instrument Z affects the outcome Y *only* through the treatment X, not directly. Exclusion restrictions are how instrumental variables identify causal effects, and how your design might separate mechanisms: e.g., assuming the incentive affects stated WTP *only* through disciplining anchoring (not through changing formation) is an exclusion restriction that (if credible) helps separate κ from λ. Exclusion restrictions are *assumptions* — they must be argued, not assumed, and they are often *untestable* (the whole point is that the excluded channel is *not* in the data).

Overidentification. When there are *more* identifying restrictions than parameters (e.g., more instruments than endogenous regressors), the model is **overidentified** — and the *excess* restrictions can be *tested* (overidentification tests: do the multiple restrictions agree?). A **just-identified** model (exactly enough restrictions) *cannot* be tested — the restrictions are used up identifying the parameters, leaving none to check. **Overidentification is valuable because it lets you test your assumptions**; just-identification means the assumptions are taken on faith. If your design generates *more* contrasts than behavioral parameters, you can test the behavioral model; if exactly enough, you cannot.

> **Active recall 30.C.** What is the difference between nonparametric and parametric identification, and why is parametric identification of a nonparametrically-unidentified parameter a warning sign? *(Answer:* nonparametric* identification recovers a parameter from the data structure *without* functional-form/distributional assumptions (e.g., a randomized ATE from the treatment-control difference) — robust, not resting on untestable assumptions. *Parametric* identification recovers it *only* by imposing a functional form/distribution. Parametric identification of a *nonparametrically-unidentified* parameter is a warning sign because the identification then rests *entirely on the untestable functional form* — a different, equally plausible assumption would give a different answer, so the "identified" value is an artifact of the assumption, not a fact in the data (your PDF's "restrizione teorica non testabile"). Report it as assumption-driven and check sensitivity / provide nonparametric bounds.)*

30.7 The centerpiece: κ (formation) versus λ (anchoring) in your cost-vector design

Now the payoff — the formal resolution of your central identification problem, exactly as your PDF states it: *"l'incentivized cost-vector effect identifica un oggetto medio legato alla preference formation, mentre il*

confronto tra hypothetical e incentivized non identifica separatamente tutta la componente di anchoring senza restrizioni aggiuntive … il mio disegno consente davvero di distinguere κ da λ?"

The model. The cost-vector effect on stated WTP (high minus low) has two components: **preference formation** κ (the cost vector changes true valuation — present in *both* hypothetical and incentivized arms) and **anchoring/reporting** λ (the cost vector anchors the *report* — present in the hypothetical arm, but *disciplined* by real incentives). Let $\phi \in [0, 1)$ be the *fraction of anchoring that survives real incentives* (φ=0: incentives fully discipline anchoring; $\varphi \to 1$: incentives do not touch it). Then the observable cost-vector effects are:

$$\Delta_{\text{hyp}} = \kappa + \lambda, \qquad \Delta_{\text{inc}} = \kappa + \phi\,\lambda.$$

The identification analysis. You observe *two* contrasts (Δ_{hyp}, Δ_{inc}) but have *three* unknowns (κ, λ, ϕ) — the model is **underidentified**. **Worked Example 30.3 (verified):** with $\Delta_{\text{hyp}} = 8$ and $\Delta_{\text{inc}} = 5$:

ϕ (survival of anchoring)	κ (formation)	λ (anchoring)
0.00	5.0	3.0
0.25	4.0	4.0
0.50	2.0	6.0
0.75	−4.0	12.0
0.90	−22.0	30.0

(Verified.) **Every value of ϕ gives a different (κ, λ) pair, and all produce the *same* observed $\Delta_{\text{hyp}} = 8$, $\Delta_{\text{inc}} = 5$** — they are *observationally equivalent* (§30.3). The two contrasts pin down only *combinations* of the parameters, not κ and λ *individually*, because ϕ is a free third parameter the data cannot see. **This is your "κ vs λ" problem, formalized: the design does not point-identify the formation/anchoring split.**

What the design *does* identify (partial identification, §30.5). Under only the *credible* assumptions $\phi \in [0, 1)$ and $\lambda \geq 0$ (anchoring is non-negative and does not more-than-survive incentives), the data deliver *bounds*:

$\kappa \leq \Delta_{\text{inc}} = 5$ (formation is at most the incentivized effect), $\lambda \geq \Delta_{\text{hyp}} - \Delta_{\text{inc}} = 3$ (anchoring is at least the hyp-inc

(verified). **This is exactly your PDF's statement:** the *incentivized cost-vector effect* ($\Delta_{\text{inc}} = 5$) is an *upper bound on preference formation* κ (it "identifies an average object related to preference formation" — and if φ=0, it *equals* κ); the *hypothetical-vs-incentivized difference* ($\Delta_{\text{hyp}} - \Delta_{\text{inc}} = 3$) is a *lower bound on anchoring* λ (it does "not identify separately the anchoring component without additional restrictions"). The design *partially* identifies the split — bounds, not points.

What point identification requires (an untestable restriction). To get a *point* estimate of κ and λ, you must *assume* a specific ϕ: - Assuming **φ=0** (real incentives *fully* discipline anchoring) point-identifies $\kappa = \Delta_{\text{inc}} = 5$ and $\lambda = \Delta_{\text{hyp}} - \Delta_{\text{inc}} = 3$. - But **$\varphi$=0 is untestable** — you cannot verify from the data that incentives fully eliminate anchoring; it is a *maintained assumption*, and if φ>0 the split is different (κ smaller, λ larger). This is your PDF's *"sto introducendo una restrizione teorica non testabile?"* — the point split is available *only* under an assumption the data cannot check.

The disciplined conclusion. The mature, defensible report of your cost-vector result is therefore: *"The cost vector significantly moves stated WTP. The incentivized effect (Δ_inc) is an upper bound on the preference-formation component and, under the assumption that real incentives fully discipline anchoring (φ=0), point-identifies it; the hypothetical-incentivized difference is a lower bound on the anchoring component. The formation/anchoring split is otherwise partially identified — the data are consistent with a range of (κ, λ) depending on how much anchoring survives incentives (φ), which is not identified. Point separation requires the untestable φ restriction, which I state explicitly and whose sensitivity I report."* This is precisely the "gap to close immediately" your PDF names — and closing it is *this analysis*, not more data.

Stop and think 30.3. Your incentivized cost-vector effect is €5 and your hypothetical effect is €8. A co-author writes "preference formation accounts for €5 and anchoring for €3 of the cost-vector effect." Under what assumption is this correct, and what is the defensible statement without it? *(Answer: it is correct* only* under the untestable assumption φ=0 — that real incentives *fully* discipline anchoring, so the incentivized effect is *pure* formation (κ=5) and the hyp-inc difference is *pure* anchoring (λ=3). Without that assumption, the split is *partially identified*: $\kappa \leq$ €5 (formation is *at most* the incentivized effect) and $\lambda \geq$ €3 (anchoring is *at least* the hyp-inc difference), with the exact split depending on how much anchoring survives incentives (φ), which the data cannot identify (verified: φ=0.5 gives κ=€2, λ=€6). The defensible statement is the *bounds* plus an explicit statement that the point split rests on the untestable φ=0 restriction — not the point figures as if the data delivered them.)*

30.8 The disciplined identification question

Consolidating the chapter into the standing discipline your PDF mandates — the four-part question to ask of *every* empirical claim in your dissertation:

1. **What parameter am I identifying?** Name the economic object (a WTP, a treatment effect, a formation component) — not the software's output, the *object*.
2. **By what variation?** Name the source of identifying variation (random assignment, factorial design, functional form, an exclusion restriction). If you cannot name it, the parameter may not be identified.
3. **Under what assumptions?** State the assumptions the identification requires (independence of assignment, a functional form, an exclusion restriction, φ=0). Distinguish *testable* from *untestable* assumptions.
4. **What observationally-equivalent alternative can I not exclude?** Name the alternative story the data cannot rule out (scale vs preference; formation vs reporting; a different φ). If none, the parameter is point-identified; if some, it is partially identified (report the set).

Every empirical claim tied to a source of identifying variation. This is the book's recurring theme #13 ("Every empirical claim should be tied to a source of identifying variation") and your PDF's standing task. A claim without a named source of identifying variation is not defensible — it may be estimated but unidentified. Running this four-part check on every result — your WTP, your cost-vector effect, your treatment contrasts, your meta-regression coefficients — is the single most important habit this book teaches, and the reason identification is the spine of the whole enterprise.

Active recall 30.D. State the four-part identification question you must ask of every empirical claim. *(Answer: (1)* What parameter* am I identifying (the economic object)? (2) *By what variation* is it identified (random assignment, factorial design, functional form, exclusion restriction)? (3) *Under what assumptions* (distinguishing testable from untestable)? (4) *What observationally-equivalent alternative* can I not exclude? If every claim is tied to a named source of identifying variation, stated assumptions, and a named un-excludable alternative — reporting bounds where the parameter is only partially identified — the claim is defensible; otherwise it may be estimated but unidentified.)*

30.9 Failure modes

F1. Mistaking estimation for identification. Trusting a software estimate of an unidentified parameter. *Symptom:* huge SE, start-value-dependent estimates, flat likelihood ridge. *Fix:* check identification (unique likelihood max, non-singular information) before interpreting (§30.2; Chapters 11, 13).

F2. Treating a non-identified parameter as point-identified. Reporting a point estimate of a formation/anchoring split the design cannot deliver. *Fix:* partial identification — report the set/bounds;

state what point requires (§30.5, §30.7).

F3. Parametric identification by untestable functional form. "Identifying" a nonparametrically-unidentified parameter only via an assumed distribution. *Fix:* flag as assumption-driven; report sensitivity; provide nonparametric bounds (§30.6).

F4. Not naming the identifying variation. Making a claim without stating what identifies it. *Fix:* the four-part question; every claim tied to a source of variation (§30.4, §30.8).

F5. Not naming the observationally-equivalent alternative. Over-claiming a mechanism (formation, preference) when an equivalent story (reporting, scale) fits equally. *Fix:* name the un-excludable alternative; report partial identification (§30.3, §30.7; Chapters 25, 27).

F6. Untestable exclusion restrictions taken on faith. Assuming an exclusion (e.g., incentives affect only anchoring) without argument. *Fix:* argue the restriction; test it if overidentified; acknowledge if untestable (§30.6).

F7. Just-identified model treated as tested. Believing a just-identified behavioral model is validated by fit. *Fix:* recognize just-identification cannot test the assumptions; seek overidentifying variation (§30.6).

F8. Blaming small samples for structural non-identification. "We need more data" for a flat-likelihood-direction parameter. *Fix:* recognize non-identification is not a sample-size problem; fix the design/assumptions (§30.2; Chapter 13).

30.10 Diagnostic questions for your own papers

1. **Identified or estimated?** For each parameter, is it *identified* (not just estimated), or is it on a flat likelihood ridge / start-value-dependent? *(§30.2, F1.)*
2. **Identifying variation.** Can I name the source of variation that identifies each parameter? *(§30.4, F4.)*
3. **Assumptions.** What assumptions does the identification require, and which are untestable? *(§30.6, §30.8.)*
4. **Observationally-equivalent alternative.** What alternative story can the data not exclude (formation vs reporting; scale vs preference)? *(§30.3, §30.5, F5.)*
5. **Point or partial.** For my cost-vector split, am I reporting the *bounds* (partial identification) rather than a spurious point? *(§30.5, §30.7, F2.)*
6. **Parametric artifact.** Is any parameter identified *only* by a functional form (untestable)? Have I reported sensitivity? *(§30.6, F3.)*
7. **Overidentification.** Does my design generate more contrasts than behavioral parameters (testable) or exactly enough (untestable)? *(§30.6, F7.)*

30.11 Exercises

(A) Comprehension.

A1. Define identification, and explain why more data cannot fix a non-identified parameter.

A2. Define observational equivalence, and give the collinearity example.

A3. Distinguish point from partial identification, and state when you would report bounds.

(B) Derivation / reasoning.

B1. For $y = \beta_1 x + \beta_2 x + \varepsilon$, show that only $\beta_1 + \beta_2$ is identified, and state the identified set for (β_1, β_2).

B2. For the cost-vector model $\Delta_{\text{hyp}} = \kappa + \lambda$, $\Delta_{\text{inc}} = \kappa + \phi\lambda$ with $\Delta_{\text{hyp}} = 8$, $\Delta_{\text{inc}} = 5$, derive the partial-identification bounds on κ and λ (assuming $\phi \in [0, 1)$, $\lambda \geq 0$), and state what point identification requires.

B3. Derive the Manski bounds on an ATE with a fraction p_m of outcomes missing and $Y \in [0, 1]$, and explain why the point estimate requires an untestable assumption.

(C) Research application / critique.

C1. Your behavioral CV model estimates separate anchoring and formation parameters that change with the starting values. Diagnose the problem and specify the correct reporting.

C2. A referee asks you to state precisely what your incentivized cost-vector effect identifies. Give the disciplined identification statement (variation, assumptions, un-excludable alternative).

C3. Explain why "the cost vector accounts for X% preference formation and Y% anchoring" is defensible only under an untestable assumption, and what you would report instead.

30.12 Solutions

A1. A parameter is *identified* if the map from the parameter to the distribution of observables is one-to-one — distinct parameter values imply distinct distributions of the observable data (so with infinite data you could recover it uniquely). More data cannot fix a non-identified parameter because non-identification means distinct parameter values imply the *same* distribution of observables (observational equivalence) — the information to distinguish them is *not in the data* at any sample size; more data reduces sampling error but not the structural gap, so the estimator's variance in the unidentified direction does not shrink to zero.

A2. Observational equivalence: two distinct parameter values (or models) imply the *same* distribution of observables, $P(\text{data} \mid \theta_1) = P(\text{data} \mid \theta_2)$ with $\theta_1 \neq \theta_2$ — so the data cannot distinguish them. Collinearity example: $y = \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ with $x_1 = x_2 = x$ gives $y = (\beta_1 + \beta_2)x + \varepsilon$, so $(\beta_1, \beta_2) = (3, 2), (5, 0), (0, 5)$ all imply identical y — observationally equivalent; only $\beta_1 + \beta_2$ is identified.

A3. *Point* identification: the data and assumptions pin down a single value (the identified set is a point). *Partial* identification: they restrict the parameter to a *set* (bounds). Report bounds when *point* identification would require *strong, untestable* assumptions — the bounds are what the data deliver under credible (weaker) assumptions, and reporting them (rather than a point) is honest about how much comes from data versus assumptions (Manski).

B1. $y = \beta_1 x + \beta_2 x + \varepsilon = (\beta_1 + \beta_2)x + \varepsilon$, so the observable distribution depends on (β_1, β_2) *only through the sum* $s = \beta_1 + \beta_2$; any (β_1, β_2) with $\beta_1 + \beta_2 = s$ implies the same y. Only $s = \beta_1 + \beta_2$ is identified. Identified set for (β_1, β_2): the line $\{(\beta_1, \beta_2) : \beta_1 + \beta_2 = \hat{s}\}$ — a one-dimensional set, not a point.

B2. From $\Delta_{\text{hyp}} = \kappa + \lambda$ and $\Delta_{\text{inc}} = \kappa + \phi\lambda$: subtract to get $\Delta_{\text{hyp}} - \Delta_{\text{inc}} = (1-\phi)\lambda$, so $\lambda = (\Delta_{\text{hyp}} - \Delta_{\text{inc}})/(1-\phi)$ and $\kappa = \Delta_{\text{hyp}} - \lambda$. With $\phi \in [0, 1)$: $(1 - \phi) \in (0, 1]$, so $\lambda \geq \Delta_{\text{hyp}} - \Delta_{\text{inc}} = 3$ (minimized at $\phi = 0$), and $\kappa = \Delta_{\text{hyp}} - \lambda \leq \Delta_{\text{hyp}} - (\Delta_{\text{hyp}} - \Delta_{\text{inc}}) = \Delta_{\text{inc}} = 5$ (maximized at $\phi = 0$). So $\kappa \leq 5$, $\lambda \geq 3$ — partial identification. Point identification requires *assuming a specific* ϕ (e.g. $\phi = 0 \Rightarrow \kappa = 5, \lambda = 3$), which is *untestable* (the data cannot reveal ϕ).

B3. With fraction p_m missing, $Y \in [0, 1]$, and observed means $\bar{y}_1, \bar{y}_0$ (among the $1 - p_m$ observed in each arm): $E[Y \mid T{=}1] = (1 - p_m)\bar{y}_1 + p_m m_1$ where $m_1 \in [0, 1]$ is the (unknown) missing mean; similarly for control. The ATE $= E[Y \mid T{=}1] - E[Y \mid T{=}0]$ is bounded by putting the missing at the extremes: lower bound $= (1-p_m)\bar{y}_1 + p_m \cdot 0 - [(1-p_m)\bar{y}_0 + p_m \cdot 1]$; upper bound $= (1-p_m)\bar{y}_1 + p_m \cdot 1 - [(1-p_m)\bar{y}_0 + p_m \cdot 0]$. (Verified: with $p_m = 0.2$, $\bar{y}_1 = 0.7$, $\bar{y}_0 = 0.4$, ATE $\in [0.04, 0.44]$.) A *point* (e.g. the naive $\bar{y}_1 - \bar{y}_0 = 0.3$) requires *assuming* the missing are like the observed (missing-at-random) — untestable, since the missing outcomes are by definition not observed.

C1. The anchoring and formation parameters changing with starting values (at the same log-likelihood) indicates they are *not separately identified* — observationally equivalent along a likelihood ridge (§30.3;

Chapters 11, 13). The optimizer returns arbitrary points on the ridge. Correct reporting: (i) recognize this is *structural non-identification*, not a convergence/sample-size problem (more data won't help); (ii) report the *identified set* / bounds (partial identification) — what combination *is* identified (e.g. $\kappa \leq \Delta_\text{inc}$, $\lambda \geq \Delta_\text{hyp} - \Delta_\text{inc}$); (iii) if a point split is needed, state the *untestable restriction* (e.g. $\varphi=0$) it requires and report sensitivity across plausible φ; (iv) do *not* present the arbitrary optimizer values as estimates of the split.

C2. "The incentivized cost-vector effect is identified by the *random assignment* of respondents to high vs low cost vectors *within the incentivized arm* (the identifying variation), under the assumptions that assignment is independent of potential outcomes (no selection/interference/differential exclusion) *and* that real incentives fully discipline anchoring ($\varphi=0$). Under $\varphi=0$ it point-identifies the preference-formation component κ; more weakly, without $\varphi=0$, it is an *upper bound* on κ. The observationally-equivalent alternative I cannot exclude is that anchoring *partially survives* incentives ($\varphi>0$), in which case the incentivized effect mixes formation and residual anchoring — so $\kappa \leq \Delta_\text{inc}$, with equality only if $\varphi=0$." This names the variation, the assumptions (testable and untestable), and the un-excludable alternative — the required disciplined statement.

C3. "X% formation, Y% anchoring" is a *point* decomposition of the cost-vector effect, which is only *partially identified* by the design: the hypothetical and incentivized contrasts deliver *bounds* (formation $\leq$ incentivized effect; anchoring $\geq$ hyp-inc difference) but not a point, because separating them fully requires knowing how much anchoring survives incentives (φ) — which the data cannot identify (verified: different φ give different splits). A specific X%/Y% is recovered *only* by assuming a specific φ (e.g. $\varphi=0$), an *untestable* restriction. Instead, report: the *bounds* on formation and anchoring; the point split *conditional on* the stated φ assumption, with sensitivity across plausible φ; and an explicit statement that the decomposition is partially identified and the point requires an untestable assumption — the honest, defensible presentation (Manski).

30.13 Spaced review prompts

+1 day. From memory, define identification (injective map), observational equivalence, and point vs partial identification; give the collinearity example.

+1 week. Derive the κ-vs-λ partial-identification bounds (incentivized effect bounds formation, hyp-inc difference bounds anchoring; point requires untestable φ); explain parametric vs nonparametric identification and why parametric identification of a nonparametrically-unidentified parameter is a warning.

+1 month. For each empirical claim in your own work, run the four-part identification question (parameter, variation, assumptions, un-excludable alternative); for the cost-vector split, report the partial-identification bounds and the untestable restriction a point requires. Close with the spine sentence.

30.14 How this chapter connects to the rest of the book

- **Chapter 16 (§16.5)** established what randomization identifies (the exogeneity machine); **Chapters 11 (§11.9), 13 (§13.4)** the likelihood-surface / flat-ridge view of non-identification; **Chapter 1 (§1.4)** rank and OLS identification; **Chapters 25, 27, 29** the scale-vs-preference and formation-vs-reporting observational equivalences this chapter formalizes.
- **Chapter 31 (treatment effects)** develops what randomization *point*-identifies (ATE/ATT) and factorial identification of interactions; **Chapter 32 (threats)** develops the assumptions (no differential exclusion, bad controls) that identification requires; **Chapter 33 (multiplicity/equivalence)** the inference on identified contrasts.
- **Chapter 24 (WTP space)** and **Chapter 25 (scale vs preference)** are the DCE observational-equivalence problems; **Chapter 8 (equivalence)** the machinery for the real-vs-hypothetical comparison; **Chapters 47–48 (meta-regression)** the identification of moderator effects with few studies.

This chapter — one of the three the architecture flags to over-learn, and the intellectual spine of the book — resolves your cost-vector κ-vs-λ problem as *partial* identification, and establishes the four-part identification question as the discipline every claim in your dissertation must pass.

30.15 Research translation task

For your **cost-vector CV/DCE results**, produce a two-to-three-page identification note:

1. **The four-part question.** For your WTP, your cost-vector effect, and your key contrasts, state: the parameter, the identifying variation, the assumptions (testable/untestable), and the un-excludable observationally-equivalent alternative.
2. **The κ-vs-λ analysis.** Write the formation/anchoring model, report the partial-identification bounds (incentivized effect bounds formation; hyp-inc difference bounds anchoring), and state that point separation requires the untestable φ.
3. **Parametric artifacts.** Identify any parameter identified *only* by a functional form (untestable); report sensitivity or nonparametric bounds.
4. **Overidentification.** State whether your design generates more contrasts than behavioral parameters (testable) or exactly enough (untestable), and what you can test.
5. **Manski-style bounds.** For any selection/missing-data or partially-identified quantity, report the bounds and what a point requires.
6. **Close** with the spine sentence: *"My [cost-vector effect] is identified by [variation] under [assumptions]; the formation/anchoring split is partially identified ($\kappa \leq$ [·], $\lambda \geq$ [·]); point separation requires the untestable [φ] restriction; the observationally-equivalent alternative I cannot exclude is [·]."*

Carry this into Chapter 31 (treatment-effect identification), Chapter 32 (threats to identification), and Chapter 33 (inference on contrasts).

30.16 Bibliography

Core. - Manski, *Identification for Prediction and Decision*, and *Partial Identification of Probability Distributions*. The foundational treatment of partial identification and the "identification comes from assumptions" philosophy (§30.5) — essential for your κ-vs-λ problem. *(Verify editions.)* - Lewbel (2019), "The Identification Zoo: Meanings of Identification in Econometrics," *Journal of Economic Literature*. The comprehensive survey of identification concepts (point, partial, parametric, nonparametric) — read for §30.2–§30.6. *(Verify volume/pages.)* - Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, and Angrist & Pischke, *Mostly Harmless Econometrics*, on identification, identifying variation, and exclusion restrictions (§30.4, §30.6). *(Verify editions.)*

Applied. - Tamer (2010), "Partial Identification in Econometrics," *Annual Review of Economics*. An accessible survey of set identification and bounds (§30.5). *(Verify volume/pages.)* - Carson & Groves (2007), and the stated-preference identification literature, for the incentive/consequentiality identification of behavioral parameters (§30.7; Chapter 28). *(Verify citations.)* - The DCE identification literature (e.g., on scale/preference identifiability, Hess & Rose 2012, Chapter 25) for the observational-equivalence problems in choice models (§30.3). *(Verify citations.)*

Advanced. - Manski (1990, 1995) on treatment-effect bounds and the "law of decreasing credibility" (stronger assumptions → narrower but less credible identification, §30.5–§30.6). *(Verify citations.)* - Matzkin, "Nonparametric Identification," in the *Handbook of Econometrics*, for the rigorous nonparametric-identification theory (§30.6). *(Verify volume/year.)* - Chesher & Rosen on generalized instrumental-variable and partial identification, for the frontier of set identification (§30.5). *(Verify citations.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Manski / Lewbel / Tamer citations before using them, per the book's no-unverified-citation rule.

End of Chapter 30. Next in Part VII: Chapter 31 — Treatment Effects in Randomized Survey Experiments (random assignment, factorial designs, interaction effects, ATE and ATT, difference in means, regression adjustment, heterogeneous treatment effects), which develops what your randomized cost-vector, incentive, and format treatments point-identify — the causal effects the identification framework of this chapter licenses. Tell me when to write it, or request a different chapter.

Chapter 31 — Treatment Effects in Randomized Survey Experiments

Part VII: Identification and Treatment Effects in Survey Experiments

> **The spine question, applied to this chapter.** Chapter 30 established *identification*; this chapter develops what your randomized treatments actually *point-identify* — the causal effects your survey experiments were built to estimate. Your PDF lists the tools: *"ATE; ATT; factorial designs; difference in means; regression-adjusted treatment effects; heterogeneous treatment effects; interaction terms"* — and names where you use them: *"high vs low cost vector, hypothetical vs incentivized, PC vs SBDC, BRIS vs full payment, experienced vs non-experienced."* Each is a treatment contrast, and each is identified — cleanly, nonparametrically — *by the randomization*. Your central mechanism test (cost-vector × incentive, Chapter 30) is a *factorial interaction*. The spine question here: *what causal effect does each of my randomized contrasts identify, how does the randomization license it, and how do I estimate it precisely (regression adjustment) and probe its heterogeneity (interactions) without over-claiming?*

31.0 Conceptual motivation: the causal effects randomization gives you

Chapter 16 (§16.5) showed that random assignment is the "exogeneity machine" — it severs the correlation between treatment and confounders, so the treatment effect is identified *without* controlling for anything. Chapter 30 formalized this as *nonparametric point identification*: the ATE is identified by the randomization, with no functional form. This chapter builds the treatment-effect framework on that foundation, because your survey experiments are a *portfolio of randomized contrasts*, and each one's causal interpretation rests on the randomization.

Your PDF's §3.8 is a checklist of exactly this chapter's content, applied to your comparisons:

- **High vs low cost vector** — the cost-vector treatment effect (Chapters 29–30); identified by random assignment to bid vectors.
- **Hypothetical vs incentivized** — the incentive treatment effect (the hypothetical-bias measurement, Chapter 27); identified by random assignment to arms.
- **PC vs SBDC** — the format effect (convergent validity, Chapter 26); identified by random assignment to formats.
- **BRIS vs full payment** — the payment-mechanism effect (Chapter 28); identified by randomization.
- **Experienced vs non-experienced** — *not* randomized (experience is a pre-existing characteristic), so this is a *heterogeneous-effect / subgroup* comparison, not a clean treatment effect (§31.7) — a crucial distinction.

And your *central mechanism test* — does the cost-vector effect differ between hypothetical and incentivized

arms (the formation-vs-reporting test, Chapters 29–30) — is a **factorial interaction** (cost vector × incentive), the difference-in-differences that this chapter develops.

The chapter builds: the **potential-outcomes framework** (the language of causal inference) and the fundamental problem; **ATE and ATT** (the average effects); **what random assignment buys** — the identification of the ATE by the simple difference in means (verified); **regression adjustment** for *precision, not bias* (verified); **factorial designs** (main effects *and* interactions, efficiently — your design); **interaction effects** as difference-in-differences (your mechanism test); and **heterogeneous treatment effects** (subgroups, with the pre-specification cautions — the experienced/inexperienced comparison). Throughout, the discipline of Chapter 30 governs: each contrast is tied to its identifying variation (the randomization), and the one non-randomized comparison (experience) is flagged as *not* a clean treatment effect.

31.1 Prerequisites

From **Chapter 16** (§16.5): randomization as the exogeneity machine (random assignment identifies the treatment effect without controls); omitted-variable bias. From **Chapter 30**: identification, nonparametric identification by randomization, and the four-part identification question. From **Chapter 17**: interaction terms and their standard errors; from **Chapter 8** (§8.9): power for interactions. From **Chapter 10** (§10.6): randomization inference (the design-based p-value). From **Chapter 27, 29, 30**: the cost-vector × incentive test as the formation-vs-reporting probe.

Same **notation**. New/consolidated symbols: $Y_i(1), Y_i(0)$ (potential outcomes under treatment/control), T_i (treatment indicator), $\tau_i = Y_i(1) - Y_i(0)$ (individual treatment effect), ATE $= \mathbb{E}[\tau_i]$, ATT $= \mathbb{E}[\tau_i \mid T_i{=}1]$, CATE $= \mathbb{E}[\tau_i \mid X_i]$ (conditional average treatment effect).

31.2 The potential-outcomes framework

The setup (Rubin causal model). For each respondent i and a binary treatment, define two **potential outcomes**:

$$Y_i(1) = \text{outcome if } i \text{ is treated}, \qquad Y_i(0) = \text{outcome if } i \text{ is not treated}.$$

The **individual treatment effect** is $\tau_i = Y_i(1) - Y_i(0)$ — the causal effect *for that respondent.* The observed outcome is $Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0)$ — you see $Y_i(1)$ if treated, $Y_i(0)$ if not.

The fundamental problem of causal inference. You **never observe both** $Y_i(1)$ and $Y_i(0)$ for the same respondent — only the one corresponding to their actual treatment. So τ_i is *never observed* for any individual: the counterfactual (what would have happened under the other treatment) is missing. This is why causal inference is a *missing-data* problem, and why you cannot compute individual treatment effects — you can only hope to recover *averages* (§31.3), and only under conditions on how treatment was assigned (§31.4).

Why this framework matters for you. The potential-outcomes language makes the identification question precise: a "treatment effect" is a *contrast of potential outcomes* ($Y_i(1) - Y_i(0)$), and estimating it requires recovering the *missing* potential outcome. Random assignment is what lets you do this on average (§31.4). For your cost-vector treatment, $Y_i(1)$ is respondent i's stated WTP under the *high* cost vector and $Y_i(0)$ under the *low* — you observe only one, so the cost-vector effect is a contrast of potential WTPs that randomization identifies on average. Framing your contrasts this way — explicitly as potential-outcome contrasts — is the causal-inference discipline that underlies the whole chapter.

> **Active recall 31.A.** State the fundamental problem of causal inference and what it implies for estimating a treatment effect. *(Answer: for each respondent you observe only* one* of the two potential outcomes ($Y_i(1)$ if treated, $Y_i(0)$ if not) — never both — so the individual treatment effect $\tau_i = Y_i(1) - Y_i(0)$ is *never observed* (the counterfactual is missing). This implies you cannot

compute individual treatment effects; you can only recover *averages* (ATE, ATT), and only under conditions on treatment assignment (random assignment) that let you fill in the missing potential outcomes on average.)*

31.3 ATE and ATT: the average effects

Since individual effects are unobservable, we target *averages*.

Definition 31.1 (ATE and ATT). - The **Average Treatment Effect (ATE)** is the mean individual effect over the *whole population*: ATE $= \mathbb{E}[\tau_i] = \mathbb{E}[Y_i(1) - Y_i(0)]$. It answers "what is the average effect of treating *everyone* vs treating *no one*?" - The **Average Treatment effect on the Treated (ATT)** is the mean effect *among those actually treated*: ATT $= \mathbb{E}[\tau_i \mid T_i{=}1] = \mathbb{E}[Y_i(1) - Y_i(0) \mid T_i{=}1]$. It answers "what was the average effect *on the people who received the treatment*?"

When they differ. ATE and ATT differ when the *treated* are a *non-random* subset with *different* treatment effects — e.g., if people who select into treatment benefit more (selection on gains). In observational settings, ATE $\neq$ ATT is common and important. **But under random assignment, ATE = ATT** — because the treated are a *random* sample of the population, so their average effect equals the population average effect (§31.4). **Worked Example (verified):** with a *heterogeneous* effect $\tau_i = 5 + 2x_i$ and *random* assignment, the true ATE (5.02) and ATT (5.03) are equal, and the difference in means recovers them (4.98). So in your randomized experiments, the ATE/ATT distinction collapses — a simplification randomization buys you.

Which average do you want? For *policy* ("should we implement the high cost vector for everyone?"), the ATE is the relevant object. For *evaluation* ("what did the treatment do to those who got it?"), the ATT. In your randomized survey experiments, they coincide, so you estimate the ATE via the difference in means (§31.4) and interpret it as both. The distinction matters most where you *cannot* randomize — the experienced/inexperienced comparison (§31.7), where "experience" is self-selected and ATE $\neq$ ATT.

Stop and think 31.1. Your cost-vector treatment has a heterogeneous effect (larger for some respondents). Under random assignment, does your difference-in-means estimate the ATE or the ATT, and why does the distinction not matter here? *(Answer: it estimates* both* — the ATE and the ATT *coincide* under random assignment because the treated are a *random* sample of the population, so their average effect (ATT) equals the population average effect (ATE), even with heterogeneous effects (verified: ATE 5.02 = ATT 5.03). The distinction does not matter here precisely because randomization makes the treated representative; it *would* matter in a non-randomized comparison (e.g. experience) where the "treated" self-select and may have systematically different effects.)*

31.4 What random assignment buys: the difference in means identifies the ATE

The central result — the causal payoff of randomization, and the point identification that Chapter 30's framework licenses.

The identification. Under **random assignment**, treatment T_i is *independent of the potential outcomes*: $T_i \perp (Y_i(1), Y_i(0))$. This is the key — random assignment makes the treated and control groups *identical in expectation* on *everything* (observed and unobserved potential outcomes), so any difference in observed outcomes is *caused by* the treatment. Formally:

$$\mathbb{E}[Y_i \mid T_i{=}1] - \mathbb{E}[Y_i \mid T_i{=}0] = \mathbb{E}[Y_i(1) \mid T_i{=}1] - \mathbb{E}[Y_i(0) \mid T_i{=}0] = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)] = \text{ATE},$$

where the middle step uses independence ($T \perp Y(1), Y(0)$, so conditioning on T does not change the potential-outcome means). **The simple difference in means identifies the ATE** — nonparametrically, with no

model, no covariates, no functional form (Chapter 30, §30.6). This is what random assignment *buys*: it converts an association (difference in observed outcomes) into a *causal* effect (ATE).

The difference-in-means estimator. The estimator is just $\hat{\tau} = \bar{Y}_{T=1} - \bar{Y}_{T=0}$ — the treated mean minus the control mean. It is *unbiased* for the ATE under randomization, *nonparametric* (no distributional assumption), and its standard error comes from the two group variances (or, better, from robust/randomization inference — Chapters 10, 17).

Worked Example 31.1 (Randomization identifies the ATE; selection biases it, verified). Simulate potential outcomes with a true ATE of 5, where a prognostic covariate x affects $Y(0)$:

Assignment	Mean difference in means
Random (T independent of x)	**5.00** (unbiased — identifies the ATE)
Selection (T depends on x)	8.82 (**biased** — not the ATE)

(Verified.) Under *random* assignment, the difference in means recovers the true ATE (5.00), *without* controlling for x. Under *selection* (treatment correlated with x, which also affects the outcome), the difference in means is *biased* (8.82) — it conflates the treatment effect with the pre-existing x-difference (omitted-variable bias, Chapter 16 §16.6). **This is exactly what random assignment buys and what its absence costs:** with randomization the naive comparison is causal; without it, the comparison is confounded. Your randomized contrasts (cost vector, incentive, format, BRIS) inherit the first; your *non*-randomized comparison (experience) inherits the second (§31.7).

> **Active recall 31.B.** Explain, using independence of potential outcomes, why the difference in means identifies the ATE under random assignment, and what breaks without randomization. *(Answer: random assignment makes treatment independent of the potential outcomes ($T \perp (Y(1), Y(0))$), so conditioning on T does not change the potential-outcome means: $\mathbb{E}[Y \mid T{=}1] = \mathbb{E}[Y(1)]$ and $\mathbb{E}[Y \mid T{=}0] = \mathbb{E}[Y(0)]$, and their difference is $\mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = ATE$ — the difference in means is the ATE, nonparametrically. Without randomization, treatment is correlated with the potential outcomes (selection): $\mathbb{E}[Y \mid T{=}1] \neq \mathbb{E}[Y(1)]$, so the difference in means conflates the treatment effect with pre-existing differences (confounding/OVB) and is* not* the ATE — verified: 8.82 vs the true 5.)*

31.5 Regression adjustment: precision, not bias

Randomization identifies the ATE *without* covariates. So why include them? For **precision**, not bias — a distinction that is the crux of covariate adjustment in experiments.

The role of covariates in an RCT. Because randomization already gives unbiasedness (§31.4), adding covariates to the treatment-effect regression $Y = \alpha + \tau T + \mathbf{x}'\gamma + \varepsilon$ is *not* needed to remove bias (there is no confounding to remove). But a *prognostic* covariate (one that predicts the outcome) *soaks up residual variance*, shrinking the standard error of the treatment effect — improving *precision*. **Worked Example 31.2 (verified):** with a prognostic covariate, the unadjusted difference-in-means SE is 0.109; the covariate-adjusted SE is 0.062 — a **1.75× precision gain** — at the *same* point estimate (~5.0). The covariate did not change the estimate (randomization already made it unbiased); it made it *more precise*.

The Lin (2013) caution. Naive covariate adjustment (adding covariates without interacting them with treatment) can, in principle, *slightly bias* the ATE estimate in finite samples if the covariate's effect differs across arms. The **Lin (2013)** correction — include the treatment, the *centered* covariates, *and* their *interactions with treatment* — guarantees the adjustment cannot hurt (it is asymptotically at least as efficient as the unadjusted estimator and remains consistent for the ATE). The practical rule: adjust for *pre-treatment* covariates (never post-treatment — Chapter 32), interact them with treatment (Lin), and treat adjustment as a *precision* tool.

Two crucial cautions (both from Chapter 16, 32). - **Only pre-treatment covariates.** Adjusting for a *post-treatment* variable (one affected by the treatment — a mediator or collider) *introduces* bias (bad controls, Chapter 32). In an experiment, controlling for a variable measured *after* treatment can destroy the randomization's identification. Adjust only for *baseline* (pre-randomization) covariates. - **Not to "fix" imbalance.** If a baseline covariate happens to be imbalanced across arms (by chance — balance is only in expectation), adjusting for it is fine (precision), but it is not "fixing" a randomization failure — the randomization was valid; the imbalance is chance (Chapter 32 §32, balance checks). Do not let observed imbalance drive which covariates you adjust for (that is a form of specification search); pre-specify the adjustment.

> **Stop and think 31.2.** Your cost-vector treatment effect has a large standard error. A co-author suggests controlling for respondent income to "reduce the bias." What is right and wrong here? *(Answer:* right* — controlling for a *prognostic pre-treatment* covariate like income can *reduce the standard error* (improve precision) by soaking up residual outcome variance (verified 1.75× gain), which helps a large-SE estimate. *Wrong* — the framing "reduce the bias" is mistaken: the cost-vector treatment is *randomized*, so the estimate is already unbiased (there is no confounding to remove); income adjustment is for *precision*, not bias (Chapter 16 §16.5). Also: adjust only for *pre-treatment* income (post-treatment would introduce bias, Chapter 32), interact it with treatment (Lin 2013), and pre-specify the adjustment rather than choosing it to fix chance imbalance.)*

31.6 Factorial designs and interaction effects

Your design crosses *several* treatments (cost vector × incentive × format × BRIS) — a **factorial design** — and this is what identifies both *main effects* and *interactions*, including your central mechanism test.

Factorial designs. A **factorial design** crosses two or more treatments, assigning each respondent a combination (a "cell"). A 2×2 design has four cells (each treatment on/off); a 2^k design crosses k binary treatments. Two virtues: - **It identifies main effects *and* interactions.** From the cells, you estimate each treatment's *main effect* (its average effect) *and* the *interactions* (whether one treatment's effect depends on another). - **It is efficient.** Each respondent contributes to *every* main effect simultaneously (a respondent in the high-cost/incentivized cell informs both the cost-vector and the incentive main effects), so a factorial design estimates multiple effects with the same sample — far more efficient than separate experiments.

Interaction effects as difference-in-differences. The **interaction** between two treatments A and B is *how the effect of A depends on B* — the **difference in differences**:

$$\text{Interaction} = [\bar{Y}_{A=1,B=1} - \bar{Y}_{A=0,B=1}] - [\bar{Y}_{A=1,B=0} - \bar{Y}_{A=0,B=0}],$$

the effect of A when $B = 1$ minus the effect of A when $B = 0$. In the regression $Y = \beta_0 + \beta_A A + \beta_B B + \beta_{AB}(A \times B) + \varepsilon$, the coefficient β_{AB} *is* this difference-in-differences (Chapter 17 §17.6). **Worked Example 31.3 (verified):** a 2×2 factorial with true main effects $A = 4$, $B = 2$ and interaction $A \times B = 3$ recovers $\hat{A} = 3.97$, $\hat{B} = 1.98$, $\hat{\beta}_{AB} = 2.99$ — and the interaction computed as a difference-in-differences equals 2.99, confirming β_{AB} *is* the DiD.

Your central test is a factorial interaction. Recall (Chapters 29–30) that the formation-vs-reporting test is: *does the cost-vector effect differ between hypothetical and incentivized arms?* This is exactly the **cost vector × incentive interaction** — the difference-in-differences $[\Delta_{\text{inc}}] - [\Delta_{\text{hyp}}]$ (or its regression coefficient). A significant negative interaction (the cost-vector effect *attenuates* under incentives) points to reporting/anchoring; a null interaction (the effect *persists*) points to formation (Chapter 29 §29.4). **So your factorial design is not just efficient — its interaction *is* your mechanism test.** This makes the design choices consequential: you need the factorial cells (cost vector × incentive) to identify the interaction, and — crucially — enough *power* for it.

The power cost of interactions (Chapter 8 §8.9). Interactions are *far* less powered than main effects: the interaction contrast (a difference of differences, built from four cells) has larger variance than a main effect, and interactions are often *smaller* than main effects — so detecting an interaction of a given size needs *several times* the sample of the corresponding main effect (Chapter 8). **This is a serious constraint on your mechanism test:** the cost vector × incentive interaction may be *underpowered* even when the main effects are well estimated, so a *null interaction* may reflect low power (not "no difference between formation and reporting"), requiring an *equivalence* test (Chapters 8, 33) and an honest MDE. Report the interaction's power/MDE, not just its p-value.

> **Active recall 31.C.** Why is your cost-vector × incentive interaction the key to the mechanism test, and what is the main statistical constraint on it? *(Answer: the interaction is the* difference-in-differences* — the cost-vector effect under incentives minus the cost-vector effect under hypothetical — which *is* the mechanism test: if the cost-vector effect *attenuates* under incentives (a significant interaction) it points to reporting/anchoring, if it *persists* (null interaction) to preference formation (Chapters 29–30). The main constraint is *power*: interactions have larger variance than main effects and are often smaller, so the interaction may be *underpowered* even when main effects are well estimated — a *null interaction* can reflect low power, not "no difference," so you must report its MDE and use an *equivalence* test (Chapters 8, 33), not treat non-significance as evidence of formation.)*

31.7 Heterogeneous treatment effects and the experienced/inexperienced comparison

Treatment effects often *vary* across respondents — the cost-vector effect may be larger for some, the format effect different for the familiar — and characterizing this heterogeneity is valuable but fraught.

Heterogeneous treatment effects (HTE) and the CATE. The **conditional average treatment effect** $\text{CATE}(x) = \mathbb{E}[\tau_i \mid X_i = x]$ is the average effect *among respondents with characteristics* x. HTE is estimated by *interacting the treatment with covariates*: $Y = \alpha + \tau T + \gamma' X + \delta'(T \times X) + \varepsilon$, where δ captures how the treatment effect varies with X. Under randomization, these treatment × covariate interactions are identified (the treatment is independent of X), so HTE by *pre-treatment* covariates is a legitimate causal object.

The cautions (subgroup analysis). HTE analysis is a minefield of false positives: - **Multiple testing.** Testing the treatment effect across many subgroups inflates false positives (Chapter 8 §8.8) — some subgroup will show a "significant" effect by chance. *Pre-specify* the subgroups; correct for multiplicity; treat post-hoc subgroups as exploratory. - **Power.** Subgroup/HTE estimates are *interactions* (treatment × covariate), so they inherit the low power of §31.6 — a subgroup null may be underpowered. - **Pre-treatment covariates only.** Condition HTE on *baseline* covariates; conditioning on *post-treatment* variables biases (Chapter 32).

The experienced vs inexperienced comparison — *not* a treatment effect. Your PDF lists "experienced vs non-experienced" among your contrasts, but it is fundamentally different from the others: **experience is *not randomized*** — respondents *self-select* into experience (through prior behavior). So the experienced/inexperienced comparison is a *subgroup / observational* comparison, *not* a clean treatment effect: - The difference in WTP between experienced and inexperienced respondents is *confounded* by everything that makes people experienced (income, interest, prior exposure — which also affect WTP): omitted-variable bias (Chapter 16 §16.6), not the causal effect of experience. - It is *not* the ATE of experience; it is an *association*. To interpret it causally, you would need the (implausible) assumption that experience is as-good-as-random conditional on observables — which the experienced/inexperienced groups almost certainly violate. - **And it interacts with scale (Chapter 25):** experienced respondents likely choose more *consistently* (larger scale), so a coefficient difference may be *scale*, not preference — compounding the confounding.

So the experience comparison must be framed as a heterogeneous-effect/subgroup analysis, not a treatment effect — descriptive/associational, confounded, and scale-entangled — while the cost-vector,

incentive, format, and BRIS comparisons are clean *randomized* treatment effects. Getting this distinction right is exactly the identification discipline of Chapter 30: name the identifying variation (randomization for the treatments; *self-selection* for experience), and interpret accordingly.

> **Stop and think 31.3.** Your PDF lists "experienced vs non-experienced" alongside your randomized contrasts. Why is it *not* a treatment effect, and how should you frame and interpret it? *(Answer: experience is* not randomized* — respondents self-select into experience through prior behavior — so the experienced/inexperienced comparison is *observational/subgroup*, not a clean treatment effect: the WTP difference is *confounded* by everything that makes people experienced (income, interest, prior exposure) which also affects WTP (omitted-variable bias, Chapter 16), and it is *scale-entangled* (experienced respondents likely choose more consistently — a scale, not preference, difference, Chapter 25). Frame it as a *heterogeneous-effect / associational* comparison — not the causal effect of experience — interpret it descriptively (or with explicit, likely-implausible conditional-independence assumptions), and compare *WTP* (scale-free) not coefficients. This contrasts with the cost-vector/incentive/format/BRIS comparisons, which *are* randomized treatment effects identified by the randomization.)*

31.8 Failure modes

F1. Treating a non-randomized comparison as a treatment effect. Interpreting experienced-vs-inexperienced as the causal effect of experience. *Fix:* recognize it is observational/confounded (and scale-entangled); frame as associational (§31.7; Chapters 16, 25, 30).

F2. Covariate adjustment framed as bias reduction. "Controlling for X reduces bias" in a randomized experiment. *Fix:* in an RCT, adjustment is for *precision*, not bias (§31.5).

F3. Adjusting for post-treatment variables. Controlling for a mediator/collider, introducing bias. *Fix:* pre-treatment covariates only (§31.5; Chapter 32).

F4. Naive covariate adjustment biasing the ATE. Adding covariates without treatment interactions. *Fix:* Lin (2013) — center covariates, interact with treatment (§31.5).

F5. Underpowered interactions read as null effects. Concluding "no interaction" (formation) from an underpowered cost-vector × incentive test. *Fix:* report the interaction's MDE; use equivalence testing (§31.6; Chapters 8, 33).

F6. Subgroup/HTE fishing. Testing many subgroups, reporting the significant ones. *Fix:* pre-specify subgroups; correct for multiplicity; treat post-hoc as exploratory (§31.7; Chapter 8).

F7. Confusing ATE and ATT. Where they differ (non-randomized), reporting one as the other. *Fix:* under randomization they coincide; where they don't, be explicit which you estimate (§31.3).

F8. Ignoring the design-based inference. Using model-based SEs where randomization inference is cleaner (small cells). *Fix:* randomization inference / robust SEs (§31.4; Chapters 10, 17).

31.9 Diagnostic questions for your own papers

1. **Randomized or not.** For each contrast, is the "treatment" *randomized* (clean effect) or a *pre-existing characteristic* (confounded subgroup — experience)? *(§31.4, §31.7, F1.)*
2. **Identifying variation.** Can I name the randomization that identifies each treatment effect (Chapter 30's discipline)? *(§31.4.)*
3. **Adjustment.** Am I using covariate adjustment for *precision* (pre-treatment, Lin-interacted), not bias, and not to fix chance imbalance? *(§31.5, F2–F4.)*

4. **Interactions.** Is my mechanism test (cost vector × incentive) the factorial interaction, and is it adequately *powered* (MDE reported, equivalence-tested if null)? *(§31.6, F5.)*
5. **ATE vs ATT.** Under randomization they coincide; where I have selection (experience), am I clear which I estimate? *(§31.3, F7.)*
6. **HTE.** Are my subgroup/heterogeneity analyses pre-specified, multiplicity-corrected, and on pre-treatment covariates? *(§31.7, F6.)*
7. **Inference.** Am I using randomization inference / robust SEs appropriate to the design and cell sizes? *(§31.4, F8; Chapters 10, 17.)*

31.10 Exercises

(A) Comprehension.

A1. State the fundamental problem of causal inference and define ATE and ATT.

A2. Explain why the difference in means identifies the ATE under random assignment.

A3. Why is covariate adjustment in a randomized experiment about precision, not bias?

(B) Derivation / reasoning.

B1. Using independence of potential outcomes, derive $\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0]$ = ATE under random assignment, and show it fails under selection.

B2. Show that the interaction coefficient in a 2×2 factorial regression equals the difference-in-differences, and relate it to your cost-vector × incentive test.

B3. Explain why ATE = ATT under random assignment even with heterogeneous treatment effects.

(C) Research application / critique.

C1. Your co-author interprets the experienced-vs-inexperienced WTP difference as "the effect of experience." Critique this and state how to frame it.

C2. Your cost-vector × incentive interaction is non-significant, and a co-author concludes "the cost vector affects preferences, not reporting." Critique using power/interactions.

C3. Explain how you would use covariate adjustment to improve the precision of your cost-vector treatment effect, and the cautions you would observe.

31.11 Solutions

A1. Fundamental problem: for each respondent you observe only *one* potential outcome ($Y_i(1)$ if treated, $Y_i(0)$ if not), never both, so the individual effect $\tau_i = Y_i(1) - Y_i(0)$ is unobservable. ATE $= \mathbb{E}[Y_i(1) - Y_i(0)]$ — the average effect over the whole population (effect of treating everyone vs no one). ATT $= \mathbb{E}[Y_i(1) - Y_i(0) \mid T_i{=}1]$ — the average effect *among the treated*.

A2. Under random assignment $T \perp (Y(1), Y(0))$, so conditioning on T does not change potential-outcome means: $\mathbb{E}[Y \mid T{=}1] = \mathbb{E}[Y(1) \mid T{=}1] = \mathbb{E}[Y(1)]$ and $\mathbb{E}[Y \mid T{=}0] = \mathbb{E}[Y(0)]$; their difference is $\mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$ = ATE. So the difference in means is the ATE — nonparametrically, no model needed.

A3. In a randomized experiment, treatment is independent of confounders, so the difference in means is *already unbiased* for the ATE — there is no confounding to remove. Covariate adjustment therefore cannot reduce bias (there is none); its role is to *soak up residual outcome variance* via a prognostic covariate, shrinking the treatment-effect standard error — *precision* (verified 1.75× SE reduction), not bias.

B1. $\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0] = \mathbb{E}[Y(1) \mid T{=}1] - \mathbb{E}[Y(0) \mid T{=}0]$. Under randomization $T \perp (Y(1), Y(0))$, so $\mathbb{E}[Y(1) \mid T{=}1] = \mathbb{E}[Y(1)]$ and $\mathbb{E}[Y(0) \mid T{=}0] = \mathbb{E}[Y(0)]$, giving $\mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = \text{ATE}$. Under *selection*, T is correlated with the potential outcomes, so $\mathbb{E}[Y(1) \mid T{=}1] \neq \mathbb{E}[Y(1)]$ (the treated have systematically different potential outcomes) — the difference in means equals ATE *plus* a selection/confounding term, not the ATE (verified: 8.82 vs true 5).

B2. In $Y = \beta_0 + \beta_A A + \beta_B B + \beta_{AB}(AB) + \varepsilon$, the cell means are: $\bar{Y}_{00} = \beta_0$, $\bar{Y}_{10} = \beta_0 + \beta_A$, $\bar{Y}_{01} = \beta_0 + \beta_B$, $\bar{Y}_{11} = \beta_0 + \beta_A + \beta_B + \beta_{AB}$. The effect of A at $B{=}1$ is $\bar{Y}_{11} - \bar{Y}_{01} = \beta_A + \beta_{AB}$; at $B{=}0$ is $\bar{Y}_{10} - \bar{Y}_{00} = \beta_A$; their difference (difference-in-differences) is β_{AB} (verified: DiD 2.99 $= \hat{\beta}_{AB}$). For your test: A = cost vector, B = incentive; β_{AB} is the difference-in-differences — the cost-vector effect under incentives minus under hypothetical — the formation-vs-reporting mechanism test.

B3. ATT $= \mathbb{E}[\tau_i \mid T_i{=}1]$; under random assignment $T \perp \tau_i$ (treatment is independent of the individual effect, since it is independent of the potential outcomes that define τ_i), so $\mathbb{E}[\tau_i \mid T_i{=}1] = \mathbb{E}[\tau_i] = \text{ATE}$ — the treated are a random sample, so their average effect equals the population average, *even with heterogeneous* τ_i (verified: ATE 5.02 = ATT 5.03 with $\tau_i = 5 + 2x$). Randomization makes the treated representative, collapsing the ATE/ATT distinction.

C1. "The effect of experience" is unjustified because experience is *not randomized* — respondents self-select into experience through prior behavior — so the experienced/inexperienced WTP difference is *confounded* by everything correlated with becoming experienced (income, interest, environmental engagement) that also affects WTP: omitted-variable bias (Chapter 16), not the causal effect of experience. It is also *scale-entangled* — experienced respondents likely choose more consistently (larger scale), so a coefficient difference may be scale, not preference (Chapter 25). Frame it as a *heterogeneous-effect / associational* comparison (a descriptive difference between self-selected groups), compare *WTP* (scale-free) not coefficients, and interpret causally only under explicit (and likely implausible) conditional-independence assumptions — unlike the randomized cost-vector/incentive/format/BRIS contrasts, which *are* causal treatment effects.

C2. The conclusion is premature because a *non-significant interaction* may reflect *low power*, not a true null: interactions (difference-in-differences) have larger variance than main effects and are often smaller, so the cost-vector × incentive interaction can be *underpowered* even when the main effects are well estimated (Chapter 8 §8.9). A non-significant interaction is *not* evidence that the cost-vector effect is the same under incentives and hypothetical (formation) — "not significant" ≠ "no difference." To support "affects preferences, not reporting," I would: (i) report the interaction's *MDE* (what difference could I have detected?); (ii) run an *equivalence test* (TOST) — is the interaction equivalent to zero within a pre-specified margin? (Chapters 8, 33); and (iii) note that even a genuine null interaction only bounds the mechanism (partial identification, Chapter 30). Only an *adequately powered* null (or an equivalence result) supports the formation interpretation.

C3. To improve precision of the cost-vector treatment effect via covariate adjustment: include *pre-treatment* prognostic covariates (baseline characteristics predicting WTP — income, prior environmental attitudes measured before treatment) in the regression $Y = \alpha + \tau T + \gamma' X + \varepsilon$, which soaks up residual WTP variance and shrinks the SE of τ (verified precision gain). Cautions: (i) frame it as *precision*, not bias reduction (the randomized effect is already unbiased); (ii) use only *pre-treatment* covariates (post-treatment would introduce bias, Chapter 32); (iii) use the *Lin (2013)* form — center the covariates and interact them with treatment — so the adjustment cannot bias the ATE; (iv) *pre-specify* the covariates rather than choosing them to fix chance imbalance (which would be specification search). Then report the adjusted (more precise) and unadjusted (transparent) estimates.

31.12 Spaced review prompts

+1 day. From memory, state the potential-outcomes framework, the fundamental problem, ATE/ATT, and why the difference in means identifies the ATE under randomization.

+1 week. Explain covariate adjustment (precision not bias, Lin, pre-treatment only), the factorial interaction as difference-in-differences (your mechanism test), and why interactions are underpowered.

+1 month. For your own experiments, classify each contrast as randomized (treatment effect) or observational (experience — confounded/scale-entangled); estimate the cost-vector × incentive interaction with its MDE/equivalence test; use pre-specified covariate adjustment for precision. Close with the spine sentence about what each contrast identifies.

31.13 How this chapter connects to the rest of the book

- **Chapter 16 (§16.5)** established randomization as the exogeneity machine; **Chapter 30** the identification framework (nonparametric point identification by randomization) this chapter's ATE rests on; **Chapter 17** interaction terms/SEs; **Chapter 8 (§8.9)** power for interactions (the constraint on your mechanism test).
- **Chapter 32 (threats to identification)** develops the assumptions the ATE requires (no differential attrition/exclusion, no bad controls, balance checks) — the fragilities of §31.5, §31.7; **Chapter 33 (multiplicity/equivalence/randomization inference)** develops the inference on these contrasts (equivalence for null interactions, multiplicity for subgroups, randomization inference).
- **Chapter 25 (scale vs preference)** is why the experience comparison is scale-entangled (§31.7); **Chapters 29–30** are the cost-vector × incentive mechanism test the factorial interaction implements; **Chapter 10** the randomization inference.

This chapter develops what your randomized treatments point-identify — the ATE via difference in means, the interaction (your mechanism test) via the factorial — while flagging the one non-randomized contrast (experience) as an observational, confounded, scale-entangled subgroup comparison, not a treatment effect.

31.14 Research translation task

For your **survey experiments**, produce a two-to-three-page note:

1. **Classify the contrasts.** For each (cost vector, incentive, format, BRIS, experience), state whether it is *randomized* (treatment effect) or *observational* (experience — confounded/scale-entangled), and what it identifies.
2. **Potential outcomes.** Frame your key contrast in potential-outcome terms; state the ATE it identifies via difference in means.
3. **Precision.** Specify the *pre-treatment* covariate adjustment (Lin-interacted) you use for precision, and confirm it is pre-specified.
4. **Mechanism interaction.** Report the cost-vector × incentive interaction (the difference-in-differences), its MDE/power, and an equivalence test if null (Chapters 8, 33).
5. **Heterogeneity.** For any subgroup/HTE analysis, pre-specify, correct for multiplicity, and condition on pre-treatment covariates; frame experience as observational.
6. **Close** with the spine sentence: *"My randomized contrasts identify the ATE via difference in means; my mechanism test is the cost-vector × incentive interaction (MDE [·]); the experience comparison is observational (confounded, scale-entangled), not a treatment effect."*

Carry this into Chapter 32 (threats: attrition, exclusions, bad controls, balance), Chapter 33 (multiplicity/equivalence/randomization inference), and Chapter 30 (identification).

31.15 Bibliography

Core. - Imbens & Rubin, *Causal Inference for Statistics, Social, and Biomedical Sciences.* The definitive treatment of potential outcomes, ATE/ATT, and randomized experiments — the foundation for §31.2–§31.4. *(Verify edition/chapters.)* - Angrist & Pischke, *Mostly Harmless Econometrics*, on treatment effects, the difference in means, regression adjustment, and interactions (§31.4–§31.6). *(Verify edition.)* - Gerber & Green, *Field Experiments: Design, Analysis, and Interpretation.* The comprehensive guide to randomized experiments — difference in means, covariate adjustment, factorial designs, heterogeneous effects, and randomization inference. *(Verify edition.)*

Applied. - Lin (2013), "Agnostic Notes on Regression Adjustments to Experimental Data: Reexamining Freedman's Critique," *Annals of Applied Statistics.* The regression-adjustment correction of §31.5 — essential for covariate adjustment in your experiments. *(Verify volume/pages.)* - Athey & Imbens (2017), "The Econometrics of Randomized Experiments," in the *Handbook of Economic Field Experiments.* A modern survey of estimation and inference for RCTs (§31.4–§31.7). *(Verify volume/pages.)* - List, Sadoff & Wagner (2011), "So you want to run an experiment, now what?" *Experimental Economics* — factorial designs and power (§31.6; Chapter 8). *(Verify volume/pages.)*

Advanced. - Athey & Wager, and Wager & Athey (2018), "Estimation and Inference of Heterogeneous Treatment Effects using Random Forests," *JASA* — modern HTE/CATE estimation (§31.7). *(Verify volume/pages.)* - Freedman (2008), "On regression adjustments to experimental data," *Advances in Applied Mathematics* — the critique Lin (2013) addresses (§31.5). *(Verify volume/pages.)* - Muralidharan, Romero & Wüthrich, on factorial designs and interactions in experiments (§31.6). *(Verify citations.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Lin / Imbens–Rubin / Athey–Imbens citations before using them, per the book's no-unverified-citation rule.

End of Chapter 31. Next in Part VII: Chapter 32 — Threats to Identification (post-treatment variables and bad controls, DAGs and confounding, attrition, exclusions after randomization, protest responders and speeders, manipulation checks, balance checks and why they do not prove randomization, and the weighted likelihood and how it changes the estimand), which develops the assumptions the treatment effects of this chapter require — and the ways your protest exclusions, weighting, and controls can break them. Tell me when to write it, or request a different chapter.

Chapter 32 — Threats to Identification

Part VII: Identification and Treatment Effects in Survey Experiments

> **The spine question, applied to this chapter.** Chapter 31 showed that randomization *point-identifies* the treatment effect. This chapter is about the ways that identification *breaks* — the threats your PDF flags as *"priorità massima."* Two of them are yours specifically: **post-randomization exclusions** (dropping protesters, speeders, comprehension failures — §6.6) and the **weighted likelihood** (weighting respondents by coastal-recreation exposure — §6.7), each of which can *change the estimand* or *bias* the effect. Add **bad controls** (controlling for post-treatment variables), **attrition**, and the misuse of **balance checks** (which do *not* prove randomization), and you have the catalogue of ways a clean experiment becomes a confounded one. The spine question here: *does my analysis preserve the randomization that identifies my treatment effect — or have my exclusions, controls, or weights broken it, and if so, what am I now estimating?*

32.0 Conceptual motivation: how a randomized experiment stops being one

Randomization is powerful but *fragile*: it identifies the treatment effect only if you *preserve* it through the analysis. Chapter 31 gave you the clean result (difference in means = ATE); this chapter gives you the ways that result is *lost* — and two of them are among your PDF's top-priority concerns, tied directly to your CV and DCE designs.

Your PDF's §6.6 and §6.7 name them:

- **Post-randomization exclusions (§6.6).** *"Protest responses, speeders e comprehension failures sono importanti, ma possono rompere la randomizzazione se non gestiti bene."* Dropping respondents *after* they are randomized — because they protested, sped through, or failed comprehension — can *break the randomization* if the exclusion is related to treatment. The remaining sample is no longer a clean random assignment, and the treatment effect is biased. Your PDF's protocol: compare with and without exclusions, show exclusion rates *by arm*, bound, and distinguish protests from genuine zeros.
- **Weighted likelihood (§6.7).** *"Nel DCE pesi gli individui per esposizione alla ricreazione costiera … cambia l'estimand: non stai più stimando semplicemente la WTP media del campione, ma una WTP media ponderata per esposizione."* Weighting respondents changes *what you are estimating* — from the sample-average WTP to an exposure-weighted WTP that targets a *different population*. It is defensible, but you must say *why* the weight is right, *which* population it represents, whether results change without weights, and whether a *few* individuals dominate.

And two general threats govern how you handle covariates and diagnostics:

- **Bad controls (§3.11).** *"Per non controllare troppo nei modelli sperimentali."* Controlling for a *post-treatment* variable (a mediator, a collider, an inclusion indicator) *introduces* bias rather than removing it — the opposite of the observational instinct to "control for everything." In an experiment, you control *pre-treatment* variables only.
- **Balance checks (§2.6).** *"I balance checks non dimostrano che la randomizzazione è riuscita … La validità viene dal disegno, non dal p-value dei balance checks."* Balance checks are diagnostics, not proofs — randomization guarantees balance only *in expectation*, so chance imbalance is normal, and passing balance checks does not prove randomization succeeded.

This chapter builds the tools to see these threats: **DAGs** (the graphical language of causal identification — confounders, mediators, colliders); **bad controls** (post-treatment/collider bias — verified); **selection and attrition**; **post-randomization exclusions** (the §6.6 problem — verified, with bounding); **manipulation checks** (a subtle post-treatment conditioning); **balance checks** (why they don't prove randomization — verified); and the **weighted likelihood** (the §6.7 estimand change — verified). Chapter 33 handles the inference on the contrasts that survive; this chapter is about *keeping* the identification Chapter 31 gave you.

32.1 Prerequisites

From **Chapter 31**: the potential-outcomes framework, the ATE, and that randomization identifies it (and that covariate adjustment is for precision, pre-treatment only). From **Chapter 30**: identification and observational equivalence (a broken randomization creates an un-excludable confounding story). From **Chapter 16** (§16.5–16.6): exogeneity, OVB, and randomization as the exogeneity machine. From **Chapter 14** (§14.8): censoring vs truncation vs missing (protests as missing). From **Chapter 26** (§26.5), **Chapter 29** (§29.6): protests vs genuine zeros. From **Chapter 8** (§8.8): multiple testing (many balance checks).

Same **notation**. New/consolidated symbols: DAG (directed acyclic graph), M (mediator), C (collider), S (selection/inclusion indicator), w_i (respondent weight), and the estimand (the population quantity the analysis targets).

32.2 DAGs: the graphical language of identification

To decide what to control for — and what *not* to — you need a way to represent the causal structure. **Directed acyclic graphs (DAGs)** are that language.

Definition 32.1 (DAG). A **directed acyclic graph** represents variables as *nodes* and direct causal effects as *arrows* (directed edges), with no cycles (a variable cannot cause itself). A **path** is a sequence of connected edges; a **causal path** follows the arrows (e.g., $T \to M \to Y$); a **backdoor path** is a non-causal path connecting treatment and outcome through a common cause. Three structures matter:

- **Confounder (common cause):** $T \leftarrow X \to Y$. X causes both treatment and outcome, creating a *backdoor path* $T \leftarrow X \to Y$ that confounds the T–Y association. **You must control for** X to identify the treatment effect (block the backdoor). *But in a randomized experiment, treatment has no causes* — the arrow into T is severed by randomization — so there are *no confounders*, and you do *not* need to control (Chapter 31).
- **Mediator (post-treatment, on the causal path):** $T \to M \to Y$. M is *caused by* treatment and *causes* the outcome — it is *part of the treatment's effect.* **Controlling for** M **blocks part of the causal path** — you lose the indirect effect (over-control bias, §32.3).
- **Collider (common effect):** $T \to C \leftarrow U$. C is *caused by* both T and another variable U. A collider *blocks* the path by default (no spurious association through it), **but *controlling for* a collider *opens* the path** — it creates a spurious association between T and U (collider/selection bias, §32.3).

The backdoor criterion (what to control for). To identify the treatment effect, control for a set of variables that *blocks all backdoor paths* (confounders) *without* opening any (no colliders, no mediators). In an observational study, this means controlling for confounders. **In a randomized experiment, there are no backdoor paths (no confounders), so the backdoor criterion is satisfied by controlling for *nothing* — and controlling for post-treatment variables (mediators, colliders) can only *open* paths and *introduce* bias.** This is the DAG-based reason "don't control too much in experiments" (your PDF's §3.11): the correct control set in an RCT is the empty set (plus pre-treatment covariates for precision only, Chapter 31).

> **Active recall 32.A.** In a randomized experiment, why does the backdoor criterion imply you should control for *nothing* (for identification), and what does controlling for a post-treatment variable do? *(Answer: randomization severs the arrows* into* treatment, so there are *no confounders / backdoor paths* between treatment and outcome — the treatment effect is already identified by the difference in means, and the backdoor criterion is satisfied by the *empty* control set. Controlling for a *post-treatment* variable can only *hurt*: a *mediator* (on the causal path $T \to M \to Y$) *blocks part of the effect* (over-control bias), and a *collider* (common effect $T \to C \leftarrow U$) *opens a spurious path* (collider/selection bias) — both *introduce* bias where there was none. So in an RCT you control for nothing for identification (pre-treatment covariates only, for precision, Chapter 31).)*

32.3 Bad controls: post-treatment variables and colliders

The most common way to break a randomized experiment: controlling for the *wrong* variable. Your PDF's §3.11 warns against it directly.

Over-control bias (controlling for a mediator). A **mediator** M is on the causal path $T \to M \to Y$ — it is *part of* the treatment's effect. Controlling for M estimates the *direct* effect (through channels *other* than M), *not* the *total* effect — it *strips out* the mediated (indirect) part. If you want the *total* treatment effect (the usual object), controlling for a mediator *biases* it.

Worked Example 32.1 (Mediator bad control, verified). Simulate T (randomized), a mediator $M = 2T + \text{noise}$, and $Y = 3T + 4M + \text{noise}$ — so the *total* effect of T on Y is $3 + 4 \times 2 = 11$ (direct 3 + indirect $4 \times 2 = 8$). The results:

Analysis	Estimated T effect	Interpretation
Difference in means	11.07	**correct total effect**
Controlling for M	2.98	direct effect only — **biased for the total**

(Verified.) Controlling for the post-treatment mediator M gives ≈ 3 (the direct effect), *not* the total effect of 11 — it has *stripped the indirect path.* If you wanted the total treatment effect (e.g., the total effect of the cost vector on WTP), controlling for a mediator would badly understate it. **The instinct to "control for more" is exactly wrong here** — the mediator is a bad control.

Collider (selection) bias. A **collider** C is a *common effect* of treatment and another variable: $T \to C \leftarrow U$. Conditioning on C (controlling for it, or *selecting the sample on it*) *opens* a spurious path between T and U, inducing a correlation where none existed. This is the mechanism behind **selection bias** and **post-randomization exclusion bias** (§32.5): conditioning on an *inclusion* indicator that is a common effect of treatment and the outcome creates bias.

The rule for experiments. *Control only for pre-treatment variables* (fixed before randomization — for precision, Chapter 31). *Never* control for (or select on) a *post-treatment* variable — mediators, colliders, inclusion indicators, manipulation-check passes — because these are *caused by* treatment and conditioning on them opens or blocks paths, introducing bias. The DAG test: *is this variable measured before or after*

treatment? Before → safe (pre-treatment, precision). After → dangerous (post-treatment, potential bad control).

> **Stop and think 32.1.** You want "the effect of the cost vector on WTP *net of* its effect on stated certainty," so you control for a post-treatment certainty measure. What have you estimated, and what is the risk? *(Answer: certainty is a* post-treatment mediator* (the cost vector affects certainty, which affects WTP), so controlling for it estimates the *direct* effect of the cost vector *not through certainty* — not the *total* cost-vector effect (verified: controlling for a mediator gives 2.98 vs the total 11.07). The risk: (i) if you wanted the total effect, you have badly biased it (stripped the mediated path); (ii) worse, if certainty is also a *collider* (affected by both the cost vector and unobserved WTP determinants), conditioning on it *opens a spurious path* and introduces collider/selection bias. Mediation analysis to get a valid "direct effect" requires strong, untestable no-unmeasured-confounding assumptions (§32.4) — naive control does not deliver it.)*

32.4 Mediation and its assumptions

Sometimes you *want* the mediated mechanism — how much of the cost-vector effect operates *through* a channel (e.g., through anchoring vs formation). This is **mediation analysis**, and it is legitimate but *assumption-heavy*.

The decomposition. The *total* treatment effect decomposes into a **direct effect** (through channels other than the mediator) and an **indirect effect** (through the mediator): total = direct + indirect. Mediation analysis estimates these components.

The assumptions (why it is fraught). Identifying the direct and indirect effects requires — *beyond* randomization of the treatment — that there be **no unmeasured confounding of the mediator–outcome relationship**. Randomization makes the *treatment* exogenous, but the *mediator* is *not* randomized (it is a post-treatment outcome), so its relationship to the final outcome can be confounded by anything affecting both. This assumption is *strong and untestable* (Chapter 30's untestable restriction). Naive mediation analysis (regress Y on T and M) delivers valid direct/indirect effects *only under* this untestable assumption; if it fails (as it often does), the "direct effect" is biased (a collider/confounding artifact).

For your work. You *might* want to decompose the cost-vector effect into a mediated (anchoring/reporting) and a direct (formation) part — which sounds like the κ-vs-λ problem (Chapter 30). But mediation analysis of this is *even harder* than the partial-identification bounds of Chapter 30, because it requires no-unmeasured-mediator-outcome-confounding on top of everything. The honest approach is the **partial identification** of Chapter 30 (bounds on formation/anchoring) *plus* the experimental interaction (cost vector × incentive, Chapter 31), *not* a naive mediation regression that assumes away the confounding. Flag mediation analyses as resting on untestable assumptions, and prefer bounds and experimental variation.

> **Active recall 32.B.** Why is mediation analysis (decomposing a treatment effect into direct and indirect parts) more assumption-heavy than estimating the total effect, and what does this imply for your cost-vector mechanism decomposition? *(Answer: the* total* effect is identified by randomization of the treatment alone (Chapter 31). But *decomposing* it into direct/indirect requires, additionally, *no unmeasured confounding of the mediator–outcome relationship* — and the mediator is *not* randomized (it is a post-treatment outcome), so this is a *strong, untestable* assumption; if it fails, the "direct effect" is biased. Implication: decomposing the cost-vector effect into anchoring (indirect) and formation (direct) via naive mediation is fraught — it assumes away confounding — so prefer the *partial-identification bounds* (Chapter 30) plus the *experimental interaction* (cost vector × incentive, Chapter 31), flagging any mediation analysis as resting on untestable assumptions.)*

32.5 Selection, attrition, and post-randomization exclusions

The threat your PDF ranks highest for your work (§6.6): dropping respondents *after* randomization can *break* it.

Selection and attrition. Attrition (dropout, non-response) and **selection** (who remains in the sample) threaten randomization when they are *related to treatment*. If who drops out depends on the treatment *and* the potential outcomes, the *retained* sample is no longer a random assignment — the treatment and control groups are no longer comparable, and the difference in means is biased. This is *collider/selection bias* (§32.3): conditioning on *being in the sample* (an inclusion indicator, a common effect of treatment and outcome) opens a spurious path.

Post-randomization exclusions (your §6.6). Excluding respondents *after* randomization — protesters, speeders, comprehension failures — is exactly this: you condition the analysis sample on a *post-treatment* variable (the exclusion criterion). If the exclusion is **related to treatment** (differential exclusion rates by arm) *and* to the outcome, the retained sample is biased.

Worked Example 32.2 (Post-randomization exclusion bias, verified). Simulate a randomized treatment with true ATE = 5, where *protest* rates differ by arm (the high-cost arm protests more) *and* protesters have lower WTP (correlated with the outcome):

Sample	Protest rate	Estimated ATE
T=1 arm	55%	—
T=0 arm	26%	—
Full sample (protests as observed)	—	**4.85** ($\approx$ true 5)
After excluding protests	—	**5.72** (biased)

(Verified.) The protest rate is *differential* (55% vs 26% by arm), and *excluding* protests biases the ATE from the true 5 (4.85 on the full sample) to 5.72 — because the exclusion is related to *both* treatment and the outcome, so the retained groups are no longer comparable. **This is your PDF's exact warning: "possono rompere la randomizzazione se non gestiti bene."** The exclusion, meant to "clean" the data, *broke the randomization*.

The protocol (your PDF's, formalized). Handle post-randomization exclusions defensibly:

1. **Report exclusion rates *by arm*.** Differential rates (55% vs 26%) are the red flag — they signal that exclusion is treatment-related and can bias the effect.
2. **Analyze with *and* without exclusions.** Report both; if they differ (4.85 vs 5.72), the exclusion matters and you cannot silently pick one.
3. **Bound the effect (Lee/Manski bounds).** Since the excluded respondents' outcomes are unknown, *bound* the ATE by the worst/best-case assumptions about them (Chapter 30 §30.5) — or, for differential attrition, **Lee bounds** (trim the less-excluded arm to equalize retention, giving bounds robust to selection). Report the range, not a point.
4. **Distinguish protests from genuine zeros.** A *genuine* zero-WTP respondent is a valid observation (belongs in the spike, Chapter 26); a *protester* is a *post-treatment exclusion*. Use debriefing to separate them (Chapter 29 §29.6).
5. **Never exclude silently.** Excluding without reporting by-arm rates, with/without analysis, and bounds is a specification choice that can manufacture (or hide) an effect.

The cost-vector connection. This is *especially* dangerous for your cost-vector treatment: a *high* cost vector likely *triggers more protests* (people balk at high prices) — exactly the differential exclusion of the worked example. So excluding protests could *create* a between-arm difference that confounds your cost-vector effect (Chapter 29 §29.6). Protest handling is not a nuisance; it interacts directly with your central treatment, and getting it wrong (silent exclusion) could produce a spurious cost-vector effect.

Stop and think 32.2. Your high-cost-vector arm has a 55% protest rate and the low-cost arm 26%, and excluding protests makes the cost-vector effect *larger*. What has happened, and what do you report? *(Answer: the exclusion has* broken the randomization* — because protest is *differential by arm* (55% vs 26%) and *related to the outcome* (protesters have systematically different WTP), excluding them conditions the sample on a post-treatment collider, making the retained arms non-comparable and *biasing* the cost-vector effect (verified: excluding biased the ATE from 4.85 to 5.72). The larger effect after exclusion is (at least partly) an *artifact* of the differential exclusion, not a cleaner estimate. Report: exclusion rates *by arm* (the red flag), the effect *with and without* exclusions, and *Lee/Manski bounds* on the effect robust to the differential exclusion — and distinguish genuine zeros (keep) from protests (bound). Do not report the excluded estimate as the clean result.)*

32.6 Manipulation checks and balance checks

Two diagnostics that are easy to *misuse* in ways that break identification.

Manipulation checks. A **manipulation check** verifies that the treatment "worked" — that respondents *noticed*/understood the manipulation (e.g., a check that respondents in the high-cost arm registered the higher cost). Useful for *interpretation*, but **conditioning the analysis on passing the manipulation check is a post-treatment exclusion** (§32.5): whether a respondent "passes" is *caused by* the treatment (and by attention, comprehension, etc.), so dropping "failers" conditions on a post-treatment variable and can bias the effect (collider bias). The tension: you want to confirm the manipulation worked, but you cannot *condition* on it without risking bias. Report manipulation-check pass rates *by arm*, analyze with and without conditioning, and treat conditioning-on-passing as a robustness check (with bounds), not the main analysis.

Balance checks and why they don't prove randomization (your §2.6). A **balance check** tests whether pre-treatment covariates are *balanced* (similar means) across arms. It is a useful *diagnostic* but a *misunderstood* one:

- **Randomization guarantees balance only *in expectation*, not in every sample.** In any *finite* sample, some covariates will be *imbalanced by chance* — that is *normal*, not a failure. So an imbalanced covariate does *not* mean randomization failed.
- **A "significant" balance check is often a false positive.** Testing many covariates at $\alpha = 0.05$, you *expect* $\approx 5\%$ to be "significant" *by chance* (multiple testing, Chapter 8 §8.8). **Worked Example 32.3 (verified):** testing 20 covariates (with truly random treatment), ≈ 1 came out "significant" at 5% — exactly the expected false-positive rate. Finding one significant balance check among twenty is *evidence of nothing* — it is chance.
- **Validity comes from the *design*, not the balance-check p-value (your PDF's §2.6).** The treatment effect is valid because assignment was *actually random* (the mechanism), *not* because the covariates happened to balance. Passing balance checks does *not* prove randomization succeeded (unobservables could be imbalanced, and you cannot check them); failing one does *not* prove it failed (chance). **Balance checks describe the realized sample; they do not validate the randomization.**

The correct use: report balance as *descriptive* (are the arms similar on observables?), do *not* interpret a significant balance check as a randomization failure, and rely on the *design* (documented random assignment) for validity — not on a balance-check p-value. And do *not* let observed imbalance drive covariate adjustment (specification search, Chapter 31 §31.5); pre-specify.

Active recall 32.C. Why does a "significant" balance check *not* mean randomization failed, and where does the validity of your treatment effect actually come from? *(Answer: randomization guarantees balance only* in expectation*, so in any finite sample some covariates are imbalanced* by chance*; and testing many covariates at 5% yields ~5% "significant"* by chance* (multiple testing) — verified: ~1 of 20 significant with truly random treatment. So a significant balance check is

(usually) a chance false positive, not evidence of randomization failure. Validity comes from the *design* — the fact that assignment was *actually random* (the mechanism) — not from the balance-check p-value: the effect is identified because randomization severed the treatment–confounder link, whether or not observables happened to balance, and passing balance checks cannot prove it succeeded because unobservables are unchecked.)*

32.7 The weighted likelihood and the estimand

Your PDF's §6.7, the second top-priority threat: weighting respondents *changes what you are estimating*.

The estimand. The **estimand** is the *population quantity* your analysis targets — *before* estimation, *what* are you trying to estimate? The unweighted sample mean estimates the *sample-average* (or population-average, if the sample is representative) WTP. **Weighting changes the estimand:** a weighted mean targets a *different* population — the one the weights represent.

Weighting by exposure (your DCE). Your DCE weights individuals by *exposure to coastal recreation*. Your PDF states the consequence exactly: *"non stai più stimando semplicemente la WTP media del campione, ma una WTP media ponderata per esposizione."* You are no longer estimating the *sample-average* WTP; you are estimating an *exposure-weighted* WTP — the average WTP *in a population weighted toward high-exposure individuals*. This is a *different object*, targeting a *different population*.

Worked Example 32.4 (Weighting changes the estimand, verified). Simulate WTP correlated with exposure, and weight by exposure:

Estimator	Mean WTP	Estimand
Unweighted	24.5	sample-average WTP
Exposure-weighted	40.0	exposure-weighted WTP (different population)

(Verified.) The weighted mean (40.0) is *far* from the unweighted (24.5) — because when WTP correlates with exposure, up-weighting high-exposure individuals shifts the estimate. **These are not two estimates of the same thing; they are estimates of *different* things** (two different estimands, two different populations). Weighting is *defensible* — if the exposure-weighted population is the *policy-relevant* one — but you must be explicit that it *changes the estimand*.

The four questions your PDF demands. Whenever you weight, answer:

1. **Why is the weight theoretically right?** What population does the weighted estimand represent, and why is *that* the object of interest (e.g., the beneficiary population for a coastal-restoration policy)?
2. **Which population does it represent?** State it precisely (the exposure-weighted population, not the sample).
3. **Do results change without weights?** Report the *unweighted* estimate too (24.5 vs 40.0); a large difference means the weighting is consequential and must be justified.
4. **Do a few individuals dominate?** Check the **weight concentration** — the *effective sample size* and whether a few extreme weights carry disproportionate influence. **Verified:** the effective sample size was only 24% of the nominal n (the weights concentrate influence), and the *top 1% of weights carried 13% of the total weight* — a handful of high-exposure individuals dominate the weighted estimate, making it fragile (sensitive to those few). Extreme weights are a red flag: the weighted estimate may rest on a few influential respondents.

The disciplined report. State the weighted *and* unweighted estimates, name the estimand each targets, justify the weight theoretically, and report the weight concentration (effective sample size, top-weight share). Weighting is a legitimate choice that *changes the estimand*, not a neutral adjustment — and a weighted estimate driven by a few extreme weights is fragile. This is exactly your PDF's protocol: *"perché il peso*

è teoricamente giusto; quale popolazione rappresenta; se i risultati cambiano senza pesi; se pochi individui pesano troppo."

> **Stop and think 32.3.** Your exposure-weighted mean WTP is €40 and your unweighted is €24, and the top 1% of weights carry 13% of the total. What do you report and check? *(Answer: report* both* estimates and be explicit they target *different estimands* — the weighted €40 is the *exposure-weighted* WTP (a population weighted toward high-exposure individuals, e.g. the policy-relevant beneficiaries), the unweighted €24 is the *sample-average* WTP — not two estimates of one number (verified: weighting shifts the estimand when WTP correlates with exposure). Check: (i) *why* the exposure weight is theoretically right (which population is the target and why); (ii) the *weight concentration* — with the top 1% carrying 13% and a low effective sample size (24%), the weighted estimate is *dominated by a few high-exposure individuals* and is *fragile* (sensitive to those few); (iii) *sensitivity* — does the €40 hold when you trim/winsorize the extreme weights? Report the unweighted estimate, justify the weight, and flag the fragility from extreme weights.)*

32.8 Failure modes

F1. Controlling for post-treatment variables (bad controls). Adding a mediator/collider to an experimental regression. *Fix:* control only pre-treatment variables; the DAG test (before/after treatment) (§32.2–§32.3).

F2. Naive mediation analysis. Decomposing an effect via a regression, assuming away mediator–outcome confounding. *Fix:* recognize the untestable assumption; prefer bounds/experimental variation (§32.4; Chapter 30).

F3. Silent or differential post-randomization exclusion. Dropping protesters/speeders without by-arm rates, with/without analysis, or bounds. *Fix:* report by-arm rates, with/without, and Lee/Manski bounds; distinguish protests from genuine zeros (§32.5).

F4. Conditioning on manipulation-check passing. Dropping "failers," biasing the effect. *Fix:* by-arm pass rates; analyze with/without; treat conditioning as a bounded robustness check (§32.6).

F5. Interpreting balance checks as randomization validation. Reading a significant balance check as failure, or passing as proof. *Fix:* balance is expected to vary by chance (multiple testing); validity from the design, not the p-value (§32.6).

F6. Balance-driven covariate adjustment. Adjusting for the covariate that happened to be imbalanced. *Fix:* pre-specify covariates; don't let observed imbalance drive the model (§32.6; Chapter 31).

F7. Weighting without stating the estimand. Reporting a weighted estimate as "the" mean WTP. *Fix:* name the estimand; report unweighted too; justify the weight (§32.7).

F8. Extreme weights unexamined. A weighted estimate dominated by a few respondents. *Fix:* report effective sample size and top-weight share; sensitivity to trimming (§32.7).

32.9 Diagnostic questions for your own papers

1. **Pre- vs post-treatment controls.** Is every control variable *pre-treatment*? Have I avoided controlling for mediators/colliders? *(§32.2–§32.3, F1.)*
2. **Exclusions.** For protests/speeders/comprehension failures, do I report exclusion rates *by arm*, analyze *with and without*, and *bound*? *(§32.5, F3.)*
3. **Differential exclusion.** Do exclusion rates differ by arm (especially by cost vector)? Could exclusion confound my treatment effect? *(§32.5.)*

4. **Manipulation checks.** Am I *conditioning* on manipulation-check passing (a bad control), or only *reporting* it? *(§32.6, F4.)*
5. **Balance checks.** Am I treating balance as descriptive (not as randomization proof), and avoiding balance-driven adjustment? *(§32.6, F5–F6.)*
6. **Weighting/estimand.** Have I stated what my weighting *estimand* is, reported unweighted results, and justified the weight? *(§32.7, F7.)*
7. **Weight concentration.** Do a few extreme weights dominate my weighted estimate (effective sample size, top-weight share)? *(§32.7, F8.)*

32.10 Exercises

(A) Comprehension.

A1. Using a DAG, explain why you should not control for a mediator or a collider in a randomized experiment.

A2. Why does excluding protesters after randomization potentially bias the treatment effect?

A3. Why does a significant balance check not mean randomization failed?

(B) Derivation / reasoning.

B1. For $T \to M \to Y$ (mediator) with $T \to Y$ direct, show that controlling for M gives the direct effect, not the total, and identify when this biases the object of interest.

B2. Explain, via collider/selection bias, why differential post-randomization exclusion biases the ATE, and describe Lee bounds.

B3. Show that weighting changes the estimand, and explain the effective-sample-size / weight-concentration diagnostic.

(C) Research application / critique.

C1. Your high-cost-vector arm has a much higher protest rate, and excluding protests enlarges the cost-vector effect. Write the correct analysis and reporting.

C2. A co-author controls for a post-treatment certainty measure to get "the effect net of certainty." Critique this.

C3. Your DCE weights by coastal-recreation exposure. Write the disciplined report of the weighting (estimand, unweighted comparison, justification, weight concentration).

32.11 Solutions

A1. In a randomized experiment, treatment has no causes (randomization severs arrows into T), so there are no confounders/backdoor paths — the effect is identified with *no* controls. A *mediator* ($T \to M \to Y$) is on the causal path, so controlling for it *blocks part of the effect* (over-control bias — you get the direct, not total, effect, verified 2.98 vs 11.07). A *collider* ($T \to C \leftarrow U$) is a common effect; conditioning on it *opens a spurious path* between T and U (collider/selection bias), inducing a correlation where none existed. Both are *post-treatment* variables — controlling for them only introduces bias.

A2. Excluding protesters conditions the analysis sample on a *post-treatment* variable (protest status, caused by treatment). If protest is *related to treatment* (differential rates by arm) *and* to the outcome (protesters have systematically different WTP), then conditioning on inclusion is conditioning on a *collider* — it makes the retained arms non-comparable (they now differ on the outcome-related protest propensity), biasing the

ATE (verified: excluding biased 4.85 → 5.72). The randomization, which guaranteed comparability, is broken by the exclusion.

A3. Randomization guarantees covariate balance only *in expectation*, not in every finite sample — so some covariates are imbalanced *by chance*. And testing many covariates at $\alpha = 0.05$ produces $\approx 5\%$ "significant" results *by chance* (multiple testing) — verified: ~1 of 20 significant with truly random treatment. So a significant balance check is (usually) a chance false positive, not evidence of randomization failure; validity comes from the *design* (assignment was actually random), not the balance-check p-value.

B1. With $M = aT + u$ and $Y = cT + bM + e$ (total effect $= c + ab$), regressing Y on T *and* M recovers the *direct* coefficient c (the effect of T holding M fixed), because conditioning on M removes the variation in Y that flows through M. So the T coefficient is c (direct), not $c + ab$ (total) — verified: 2.98 ($=c=3$) vs 11.07 ($=c+ab=11$). This *biases the object of interest* whenever you want the *total* effect (the usual causal question — the overall effect of the treatment), since controlling for the mediator strips the indirect (ab) part.

B2. Post-randomization inclusion (survival) S is a common effect of treatment T and (via protest/attrition) the outcome-related variable — a *collider* $T \to S \leftarrow U$ (with U affecting Y). Conditioning the analysis on $S = 1$ (keeping only the retained) *opens* the T–U path, inducing a spurious T–Y association that biases the ATE (verified). **Lee bounds** handle this by *trimming*: the arm with *higher* retention is trimmed (removing its most extreme outcomes) down to the *lower* arm's retention rate, so both arms retain the same fraction; the bounds come from trimming the *upper* vs *lower* tail, giving a range for the ATE that is *robust to the differential selection* (worst/best case for who the extra retained respondents are). Report the bounds, not a point.

B3. The unweighted mean estimates $\mathbb{E}[\text{WTP}]$ over the *sample* (a specific estimand). The weighted mean $\sum_i w_i \text{WTP}_i / \sum_i w_i$ estimates $\mathbb{E}_w[\text{WTP}]$ over a *reweighted* population (the population the weights represent) — a *different* estimand when WTP correlates with the weights (verified: 24.5 unweighted vs 40.0 weighted). The **effective sample size** $\text{ESS} = (\sum w_i)^2 / \sum w_i^2$ measures how much *independent information* the weighted sample carries: with equal weights ESS $= n$; with concentrated weights ESS $\ll n$ (verified: 24% of n), meaning a few high-weight individuals dominate (top 1% of weights carried 13% of total), so the weighted estimate is fragile (sensitive to those few) — the weight-concentration diagnostic flags this.

C1. Correct analysis: (i) *report exclusion (protest) rates by arm* — the higher rate in the high-cost arm is the red flag that exclusion is treatment-related; (ii) *report the cost-vector effect with and without exclusions* — if excluding enlarges it, the enlarged estimate is (partly) an artifact of differential exclusion, not a cleaner result (verified: exclusion biased 4.85 → 5.72); (iii) *bound the effect* with Lee/Manski bounds robust to the differential exclusion, and report the range; (iv) *distinguish genuine zeros from protests* (debriefing) — genuine zeros are valid observations (keep, spike model, Chapter 26), protests are the post-treatment exclusion; (v) do *not* report the excluded estimate as the clean cost-vector effect. The honest conclusion is the bounds plus the by-arm rates, flagging that a high cost vector triggering more protests confounds naive exclusion (Chapter 29).

C2. Controlling for a *post-treatment* certainty measure is a *bad control*: certainty is affected by the cost vector (a mediator) and possibly by unobserved WTP determinants (a collider). Controlling for it estimates the *direct* effect of the cost vector *not through certainty* — not the *total* cost-vector effect (verified: controlling for a mediator gives the direct, biased-for-total, coefficient) — and if certainty is also a collider, it *opens a spurious path* (collider bias). So "the effect net of certainty" is not cleanly identified: a valid direct/mediation effect requires the untestable no-unmeasured-mediator-outcome-confounding assumption (§32.4). I would *not* control for certainty in the main analysis (control pre-treatment variables only); if the mechanism through certainty is of interest, use a properly-designed mediation/experimental approach with its assumptions stated, or report bounds — not a naive post-treatment control.

C3. Disciplined weighting report: (i) **estimand** — state that the exposure-weighted estimate targets the *exposure-weighted population* (WTP averaged over a population weighted toward coastal-recreation users), *not* the sample-average WTP — a different object (verified: 40.0 weighted vs 24.5 unweighted); (ii) **justification** — argue *why* the exposure weight is theoretically right (e.g., the policy-relevant beneficiary population for coastal restoration is the exposed population), i.e. why *that* estimand is the object of interest; (iii) **un-**

weighted comparison — report the unweighted estimate too and note the difference is large (consequential weighting); (iv) **weight concentration** — report the *effective sample size* (24% of n) and the *top-weight share* (top 1% carry 13%), flag that a few high-exposure individuals dominate, and report *sensitivity* to trimming/winsorizing the extreme weights. Conclude that the weighting is defensible for the stated estimand but the weighted estimate is fragile to a few influential respondents — exactly the PDF's four-question protocol.

32.12 Spaced review prompts

+1 day. From memory, define confounder, mediator, and collider on a DAG; state the rule for controls in an experiment (pre-treatment only) and why post-treatment controls bias.

+1 week. Explain how differential post-randomization exclusion biases the ATE and how to handle it (by-arm rates, with/without, bounds); explain why balance checks don't prove randomization; explain how weighting changes the estimand.

+1 month. For your own experiments, audit your controls (pre-treatment only), report exclusion rates by arm with/without-exclusion effects and bounds, treat balance as descriptive, and report the weighting estimand with unweighted comparison and weight concentration. Close with the spine sentence about whether your randomization survived the analysis.

32.13 How this chapter connects to the rest of the book

- **Chapter 31** established what randomization identifies; this chapter the threats that break it. **Chapter 30** is the identification framework (a broken randomization creates an un-excludable confounding story / changes the identified object); **Chapter 16 (§16.5–16.6)** exogeneity and OVB; **Chapter 14 (§14.8)** censoring vs truncation vs missing (protests as missing).
- **Chapter 26 (§26.5), 29 (§29.6)** are protests vs genuine zeros and the cost-vector × protest confound; **Chapter 8 (§8.8)** the multiple testing behind balance-check false positives; **Chapter 33 (multiplicity/equivalence/randomization inference)** the inference on surviving contrasts (and randomization inference's dependence on intact assignment, Chapter 10 §10.6).
- **Chapter 25** is the scale-entanglement of the (observational) experience comparison; **Chapters 47–48** the weighting/estimand issues in meta-analysis (WLS, few studies).

This chapter is the catalogue of ways your randomized experiment can stop being one — with your two top-priority threats (post-randomization exclusions, weighted likelihood) formalized and bounded — and it is the guardian of the identification Chapter 31 delivered.

32.14 Research translation task

For your **survey experiments**, produce a two-to-three-page threats note:

1. **Controls audit.** List every control variable; confirm each is *pre-treatment*; flag any post-treatment (mediator/collider) controls to remove.
2. **Exclusions.** For protests/speeders/comprehension failures, report exclusion rates *by arm* (especially by cost vector), the treatment effect *with and without* exclusions, and *Lee/Manski bounds*; distinguish genuine zeros from protests.
3. **Manipulation checks.** Report pass rates by arm; if you condition on passing, treat it as a bounded robustness check, not the main analysis.
4. **Balance.** Report balance descriptively; do *not* interpret significant checks as randomization failure or drive adjustment by imbalance.

5. **Weighting.** State the weighting *estimand*, report the unweighted estimate, justify the weight theoretically, and report the weight concentration (effective sample size, top-weight share) and sensitivity.
6. **Close** with the spine sentence: *"My randomization is preserved because I control only pre-treatment variables, report exclusions by arm with bounds, treat balance as descriptive, and state that my weighting targets the [exposure-weighted] estimand (unweighted [·] vs weighted [·], ESS [·])."*

Carry this into Chapter 33 (multiplicity/equivalence/randomization inference), Chapter 30 (identification), and Chapters 47–48 (meta-analysis weighting).

32.15 Bibliography

Core. - Angrist & Pischke, *Mostly Harmless Econometrics*, on bad controls, selection, and the perils of controlling for post-treatment variables (§32.3). *(Verify edition.)* - Pearl, *Causality*, and Pearl, Glymour & Jewell, *Causal Inference in Statistics: A Primer*. The DAG framework — confounders, mediators, colliders, the backdoor criterion (§32.2). *(Verify editions.)* - Cinelli, Forney & Pearl (2022), "A Crash Course in Good and Bad Controls," *Sociological Methods & Research*. The definitive accessible guide to good vs bad controls (§32.3) — essential. *(Verify volume/pages.)*

Applied. - Lee (2009), "Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects," *Review of Economic Studies*. The Lee bounds for differential attrition/exclusion (§32.5). *(Verify volume/pages.)* - Montgomery, Nyhan & Torres (2018), "How Conditioning on Posttreatment Variables Can Ruin Your Experiment and What to Do About It," *American Journal of Political Science*. Post-treatment conditioning (manipulation checks, exclusions) in experiments (§32.5–§32.6). *(Verify volume/pages.)* - Mutz, Pemantle & Pham on the misuse of manipulation checks and balance tests in experiments (§32.6). *(Verify citations.)*

Advanced. - Manski, *Partial Identification of Probability Distributions*, for bounds under selection/missing data (§32.5; Chapter 30). *(Verify edition.)* - Solon, Haider & Wooldridge (2015), "What Are We Weighting For?" *Journal of Human Resources* — the estimand implications of weighting (§32.7). *(Verify volume/pages.)* - VanderWeele, *Explanation in Causal Inference: Methods for Mediation and Interaction*, for rigorous mediation analysis and its assumptions (§32.4). *(Verify edition.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Cinelli et al. / Lee / Solon et al. citations before using them, per the book's no-unverified-citation rule.

End of Chapter 32. Next in Part VII: Chapter 33 — Multiplicity, Equivalence, and Randomization Inference in Experiments (multiple hypothesis testing, equivalence testing applied to convergent validity, randomization inference, and power for interactions), which completes the survey-experiment toolkit with the inference on the contrasts that survive the identification threats of this chapter — including the equivalence tests your convergent-validity claims require. Tell me when to write it, or request a different chapter.

Chapter 33 — Multiplicity, Equivalence, and Randomization Inference in Experiments

Part VII: Identification and Treatment Effects in Survey Experiments

> **The spine question, applied to this chapter.** Chapters 31–32 identified the treatment effect and guarded that identification. This chapter is about the *inference* on the contrasts that survive: when you test *many* effects (every attribute, every arm, every subgroup), how do you keep from manufacturing false positives (**multiplicity**)? When your claim is that two numbers are *the same* — real ≈ hypothetical WTP, your **convergent-validity** claim — how do you *prove equivalence* rather than merely *fail to reject a difference* (**equivalence testing / TOST**)? When your sample is small and non-normal, how do you get a p-value that leans only on the randomization you actually did (**randomization inference**)? And when your headline is an *interaction* (cost vector × incentive, real × hypothetical), why does it need ~4× the sample (**power for interactions**)? The spine question: *given that I ran many tests, want to show sameness, have a small non-normal sample, and care about an interaction — is my inference honest about all four, or have I let multiplicity, the equivalence fallacy, distributional assumptions, and underpowering quietly inflate my conclusions?*

33.0 Conceptual motivation: four inference problems your design creates

Your survey experiments generate a *lot* of hypotheses. A DCE has many attributes, each with a WTP you test. A CV validity study compares real vs hypothetical, and you *want* them to match. A factorial design has main effects *and* interactions. And your samples — especially in subgroups or a single treatment cell — can be small and non-normal. Each of these creates an inference problem that the standard "t-test, $p < 0.05$" ritual handles *badly*. This chapter is the four-part fix, and each part maps to your PDF.

1. Multiplicity (§2.8, §5.x). Test 10 attribute contrasts at $\alpha = 0.05$ and — even if *none* is real — you have a **40% chance of at least one "significant" result** (verified below). The more you test, the more false positives you manufacture. Corrections (Bonferroni, Holm for the family-wise error rate; Benjamini–Hochberg for the false discovery rate) restore honesty. Your PDF flags this as the multiple-comparisons problem across attributes and subgroups.

2. Equivalence (§4.x convergent validity). Your **convergent-validity** claim — that hypothetical WTP ≈ real WTP, or that two elicitation formats agree — is a claim of *sameness*. But a non-significant difference test ($p = 0.42$) does **not** prove sameness; it is *absence of evidence*, consistent with a large true difference you were underpowered to detect. To *demonstrate* equivalence you need **equivalence testing**

(TOST): pre-specify a margin and *reject* the hypothesis that the difference is *larger* than it. This inverts the usual logic and is exactly what a validity claim requires.

3. Randomization inference (§2.7). The t-test's p-value assumes normality (or large n). But in an experiment, the *actual* source of randomness is the *random assignment* — and you can compute a p-value directly from it by *permuting* the treatment labels. **Randomization (permutation) inference** needs *no* distributional assumption, is *exact* in finite samples, and matches the t-test when the t-test's assumptions hold — the honest default for small experimental samples (Chapter 10 §10.6).

4. Power for interactions (§2.5, §6.x). Your headline results are often *interactions*: does the cost-vector effect *differ* between real and hypothetical (a difference-in-differences)? An interaction contrast has *twice* the standard error of a main effect, so detecting it needs **~4× the sample** (verified). Powering a study for the main effect and then reporting the interaction is a recipe for an underpowered, noise-dominated headline.

This chapter builds all four — with every number verified computationally — and ties each to your CV/DCE designs. It *completes* Part VII: Chapters 31–32 got you a *clean, identified* effect; this chapter gets you *honest inference* on it and on the *many* effects and *sameness* claims your work actually makes.

33.1 Prerequisites

From **Chapter 8** (§8.8): the multiple-testing problem introduced (this chapter formalizes the corrections). From **Chapter 9**: power, MDE, and equivalence testing / TOST (this chapter applies them to convergent validity and interactions). From **Chapter 10** (§10.6): permutation/randomization inference (this chapter is the experimental application). From **Chapter 31**: factorial designs and interactions (this chapter powers them). From **Chapter 7** (the CLT) and **Chapter 12** (Wald tests): what randomization inference *replaces*.

Same **notation**. New/consolidated symbols: m (number of tests in a family), FWER (family-wise error rate), FDR (false discovery rate), TOST (two one-sided tests), Δ (equivalence margin), and π (a permutation of assignment labels).

33.2 The multiplicity problem and family-wise error

Every hypothesis test has a false-positive rate α. Run *one* test at $\alpha = 0.05$ and, under the null, you falsely reject 5% of the time. Run *many*, and the probability that *at least one* falsely rejects climbs fast.

Definition 33.1 (Family-wise error rate, FWER). For a *family* of m tests, the **family-wise error rate** is the probability of making *at least one* false rejection (Type I error) across the family: FWER = P(reject at least one true null). For m *independent* tests each at level α with all nulls true, FWER = $1-(1-\alpha)^m$, which $\to 1$ as m grows.

Worked Example 33.1 (Multiplicity inflates error, verified). With $m = 10$ tests at $\alpha = 0.05$ and *all nulls true*:

Quantity	Value
$P(\geq 1$ false positive), uncorrected	**0.401** (verified; simulated 0.397)
Bonferroni per-test α	0.0050
Šidák per-test α	0.0051

(Verified.) Testing 10 true nulls at 0.05 gives a **40% chance of at least one false "discovery"** — not 5%. If your DCE has 10 attribute contrasts (or you test an effect across 10 subgroups), and you celebrate the one that comes up "significant," you are very likely celebrating noise. This is the multiplicity problem: *the more you test, the more false positives you manufacture.*

Active recall 33.A. Why does testing 10 true nulls at $\alpha = 0.05$ give a 40% error rate, not 5%, and what does this imply for reporting the "significant" attributes in a DCE? *(Answer: each test independently falsely rejects with probability 0.05, so the probability that* at least one* of 10 falsely rejects is $1 - (1 - 0.05)^{10} = 0.40$ — verified. Implication: with many attribute contrasts (or subgroups), you *expect* some "significant" results *by chance*, so reporting the one attribute that came up $p < 0.05$ out of ten, without a multiplicity correction, is very likely reporting noise — you must either pre-specify a primary contrast or correct for the family (§33.3), and report the whole family, not the winners.)*

33.3 Controlling the family-wise error: Bonferroni and Holm

Bonferroni. The simplest correction: test each of the m hypotheses at level α/m (equivalently, multiply each p-value by m). Then FWER $\leq \alpha$ *always* (by the union bound — Boole's inequality — regardless of dependence). **Verified:** the simulated FWER under Bonferroni was 0.049 (vs 0.397 uncorrected), controlling it to the target 0.05. Bonferroni is *conservative* (it controls FWER even under dependence, so it "over-pays" when tests are correlated) but *robust and simple*.

Holm's step-down (uniformly better than Bonferroni). Order the p-values $p_{(1)} \leq p_{(2)} \leq ... \leq p_{(m)}$. Compare $p_{(1)}$ to α/m, $p_{(2)}$ to $\alpha/(m-1)$, ..., $p_{(k)}$ to $\alpha/(m-k+1)$; *stop* at the first that fails and *retain* all remaining. Holm controls FWER $\leq \alpha$ under *any* dependence (like Bonferroni) but *rejects at least as many* hypotheses — it is *uniformly more powerful*. **Verified:** simulated FWER under Holm was 0.050 (controlled), while rejecting more than Bonferroni when real effects exist. **Prefer Holm to plain Bonferroni** — it dominates it at no cost in assumptions.

When FWER control is right. FWER is the correct target when *even one* false positive is costly — a *confirmatory* study with a small set of pre-specified primary hypotheses (e.g., "the cost-vector effect *and* the incentive effect," a family of 2–5). For a *primary* confirmatory claim, control FWER.

Stop and think 33.1. Your pre-registration names three primary contrasts (cost vector, incentive, their interaction). One comes back $p = 0.02$, the others null. Do you claim a discovery? *(Answer: with a* family of 3* pre-specified primary hypotheses, control FWER. Under Holm: order the p-values; compare the smallest to $\alpha/3 = 0.0167$. Since $0.02 > 0.0167$, it *fails* the first Holm step, so you *retain all three* nulls — the $p = 0.02$ is *not* significant after correcting for the family of three. (Plain Bonferroni agrees: $0.02 > 0.05/3$.) So no, you do not claim a discovery on the $p = 0.02$ alone — it does not survive multiplicity correction for your three primary tests. Report all three contrasts and the correction; do not report the 0.02 as significant.)*

33.4 Controlling the false discovery rate: Benjamini–Hochberg

FWER control is *strict* — good for a few confirmatory tests, but *crushing* for *exploratory* work with *many* tests (screening 50 attributes, 30 subgroups). There, you can tolerate *some* false positives *as a fraction of your discoveries*. The **false discovery rate (FDR)** targets exactly that.

Definition 33.2 (False discovery rate, FDR). The **false discovery rate** is the *expected proportion of false positives among the rejected hypotheses*: FDR $= \mathbb{E}\left[\frac{V}{R}\right]$ (with V = false rejections, R = total rejections, and $V/R \equiv 0$ if $R = 0$). Controlling FDR at q means: *of the effects you declare real, on average at most a fraction q are false*. This is a *weaker, more powerful* criterion than FWER (which bounds $P(V \geq 1)$).

Benjamini–Hochberg (BH) procedure. Order $p_{(1)} \leq ... \leq p_{(m)}$. Find the *largest* k such that $p_{(k)} \leq \frac{k}{m} q$. Reject all hypotheses $1, \ldots, k$. Under independence (and certain positive-dependence structures), this controls FDR $\leq q$.

Worked Example 33.2 (BH controls FDR with power, verified). With $m = 20$ tests, 8 of them *real* effects and 12 true nulls, BH at $q = 0.05$:

Quantity	Value
Realized FDR	**0.030** (≤ 0.05 target)
Power (fraction of true effects detected)	**0.787**

(Verified.) BH keeps the false-discovery proportion at 3% *while detecting 79% of the real effects* — far more power than Bonferroni would give on 20 tests. **Use FDR (BH) for exploratory, many-hypothesis screening** (which attributes/subgroups *might* matter, to follow up); **use FWER (Holm) for confirmatory primary tests** (the few effects you stake a claim on). Matching the correction to the role — confirmatory vs exploratory — is the discipline your PDF's multiplicity concern demands.

> **Active recall 33.B.** What is the difference between FWER and FDR, and which do you use for a DCE's exploratory attribute screen vs its two pre-registered primary contrasts? *(Answer: FWER bounds the probability of* any* false positive across the family ($P(V \geq 1) \leq \alpha$) — strict, appropriate when even one false positive is costly; FDR bounds the *expected fraction* of false positives *among rejections* ($\mathbb{E}[V/R] \leq q$) — more permissive and more powerful, appropriate when you tolerate some false discoveries as a share of many. Use FWER (Holm) for the *two pre-registered primary contrasts* (confirmatory, few, high-stakes); use FDR (Benjamini–Hochberg) for the *exploratory attribute screen* (many tests, follow-up warranted, verified to hold FDR at 0.03 with 0.79 power on 20 tests). Match the correction to the role.)*

33.5 Equivalence testing: proving sameness (your convergent validity)

Now the inference problem at the heart of your CV validation: showing that two numbers are *the same*. The standard test is built to detect *differences*, and *failing* to find one does *not* prove *sameness*.

The equivalence fallacy. "We tested real vs hypothetical WTP and found no significant difference ($p = 0.42$), so they are equivalent." **Wrong.** A non-significant difference test is *absence of evidence*, not *evidence of absence*: it is consistent with a *large* true difference that you were *underpowered* to detect. *Failing to reject "they differ" is not the same as showing "they are the same."* To claim convergent validity — real $\approx$ hypothetical, or format A $\approx$ format B — you must *test equivalence directly*.

TOST (two one-sided tests). Pre-specify an **equivalence margin** Δ — the largest difference you would consider *practically negligible* (e.g., "within €8 of each other counts as the same for our purposes"). Then test *two* one-sided nulls:

$$H_{0,\text{lower}}: \ \delta \leq -\Delta \quad \text{and} \quad H_{0,\text{upper}}: \ \delta \geq +\Delta.$$

Reject both (each at level α) to conclude **equivalence**: $-\Delta < \delta < +\Delta$. Equivalently — and more intuitively — construct the $100(1-2\alpha)\%$ confidence interval for the difference (a **90% CI** for $\alpha = 0.05$); if it lies *entirely within* $[-\Delta, +\Delta]$, conclude equivalence.

Worked Example 33.3 (TOST for convergent validity, verified). Real WTP and hypothetical WTP differ by $d = 2.0$ (se $= 2.5$), margin $\Delta = \pm 8$:

Test	Result	Conclusion
TOST (equivalence)	$p = 0.0082$	**Conclude equivalence** at 5%
90% CI for the difference	$[-2.11,\ 6.11]$	inside $\pm 8 \rightarrow$ equivalent
Ordinary NHST (difference)	$p = 0.424$	"not significant" — proves *nothing*

(Verified.) The TOST *concludes equivalence* ($p = 0.008$; the 90% CI $[-2.11, 6.11]$ sits inside ± 8), a *positive* validity result. The *same data* under the ordinary difference test give $p = 0.42$ — "not significant," which by itself is *not* evidence of sameness. **Only the equivalence test licenses the convergent-validity claim.** For your real-vs-hypothetical CV comparison (and any format-agreement or transfer-validity claim), *pre-specify the margin* (from policy relevance or a benefit-transfer error tolerance), *report the TOST / 90% CI*, and never infer equivalence from a non-significant difference.

Choosing the margin. The margin Δ is a *substantive* choice, not a statistical one — it must be *pre-specified* (else it is a garden-of-forking-paths knob). Anchor it to *decision relevance*: how large a real–hypothetical gap would *change a policy conclusion* or *exceed acceptable benefit-transfer error*? State the margin and its justification; a study that picks Δ after seeing the CI is not doing equivalence testing.

> **Stop and think 33.2.** Your reviewer says "your hypothetical and real WTP don't differ significantly ($p = 0.4$), so you've shown they're equivalent — good validity." Why is this reasoning unsafe, and what do you actually need? *(Answer: a non-significant* difference* test is *absence of evidence, not evidence of absence* — $p = 0.4$ is consistent with a *large* true real–hypothetical gap you were simply underpowered to detect (especially with a small or noisy sample), so it does *not* establish equivalence. What you need is an *equivalence test*: pre-specify a margin Δ (the largest gap that is practically negligible for the policy use), then run TOST / check the 90% CI — if the CI for the difference lies entirely within $\pm\Delta$, you have *positively demonstrated* equivalence (verified: same data, TOST $p = 0.008$ concludes equivalence while the NHST $p = 0.42$ proves nothing). Convergent validity is an equivalence claim and requires equivalence inference, not a failed difference test.)*

33.6 Randomization (permutation) inference

The t-test's p-value rests on the CLT/normality. In an experiment, there is a *more direct* and *assumption-light* source of a p-value: the *randomization itself*.

The idea (Fisher's exact logic). Under the **sharp null** of *no treatment effect for any unit* ($Y_i(1) = Y_i(0)$ for all i), each unit's outcome is *fixed* regardless of assignment — so any difference between arms is due *only* to *which units landed in which arm*. You can therefore compute the *exact* null distribution of your test statistic by *re-randomizing*: enumerate (or sample) all the ways the treatment labels *could* have been assigned, recompute the statistic each time, and see where your *observed* statistic falls. The **randomization (permutation) p-value** is the fraction of re-randomizations with a statistic *as extreme* as observed.

Definition 33.3 (Permutation p-value). For observed statistic T_{obs} (e.g., the difference in means) and permutations π of the assignment labels giving statistics T_π,

$$p = \frac{\#\{\pi : |T_\pi| \geq |T_{\text{obs}}|\}}{\#\{\pi\}}.$$

This is *exact* in finite samples (under the sharp null), requires *no* distributional assumption (no normality, no CLT), and uses *only* the randomization the experiment actually performed.

Worked Example 33.4 (Permutation vs t-test, verified). Two arms ($n = 25$ each), observed difference 1.32:

Method	p-value	Assumptions
Permutation (5000 draws)	**0.0000**	*none* (exact under sharp null)
Two-sample t-test	**0.0001**	normality / large n

(Verified.) The permutation and t-test p-values *agree closely* — because when the t-test's assumptions roughly hold, randomization inference reproduces it. **The point is what happens when they *don't*:**

for small, skewed, heavy-tailed, or discrete outcomes (a WTP with a spike at zero, a payment-card interval, a tiny subgroup), the t-test's normal approximation can be *badly off*, while the permutation p-value stays *exact*. Randomization inference is the honest default for your small experimental cells and non-normal WTP outcomes (Chapter 10 §10.6), and it *is* the inference that matches your design — you randomized, so test *by* re-randomizing.

Randomization inference and the threats of Chapter 32. Randomization inference is only valid if the *assignment* it permutes is the *actual* assignment mechanism — so *post-randomization exclusions and broken randomization (Chapter 32) invalidate it too* (you would be permuting a mechanism that no longer generated the retained sample). The permutation must respect the *design* (e.g., permute *within* blocks/strata if assignment was blocked). Honest randomization inference presupposes the intact randomization that Chapter 32 defends.

> **Active recall 33.C.** What null does randomization inference test, what assumptions does it need, and why is it the honest default for a small non-normal WTP subgroup? *(Answer: it tests Fisher's* sharp null* of *no effect for any unit* ($Y_i(1) = Y_i(0)\ \forall i$), under which each outcome is fixed and any arm difference is due only to which units were assigned where. It needs *no distributional assumption* — no normality, no CLT — only that you permute the *actual* randomization the design used (respecting blocks/strata); it is *exact* in finite samples. It is the honest default for a small non-normal WTP subgroup because the t-test's normal approximation can be badly inaccurate there (spike at zero, heavy tails, small n), whereas the permutation p-value remains exact — and it matches the t-test when the t-test's assumptions do hold (verified: 0.0000 vs 0.0001). Caveat: broken randomization / post-randomization exclusion (Chapter 32) invalidates it too.)*

33.7 Power for interactions: why headlines cost 4× the sample

Your most interesting results are often *interactions* — the cost-vector effect *differs* between real and hypothetical (a hypothetical bias that *depends* on price), familiarity *moderates* the WTP, the incentive treatment *changes* the format effect. Interactions are *differences of differences*, and they are *expensive* to detect.

Why an interaction has twice the SE. A main effect is one contrast, $\bar{Y}_1 - \bar{Y}_0$, with variance $\propto 2\sigma^2/n$. An interaction is a *difference of two such differences*, $(\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00})$ — it *sums the variances of four cell means*, giving *twice* the standard error of a main effect (a factor of 2 in SE, hence a factor of 4 in variance). To detect an interaction of the *same magnitude* as a main effect at the same power, you need $2^2 = 4$ times the sample per cell.

Worked Example 33.5 (The 4× rule, verified). For a standardized effect $\delta = 0.4$, 80% power, $\alpha = 0.05$:

Target	n per arm
Main effect	**99**
Interaction (difference-in-differences)	**393** ($\approx$ **4**×)

(Verified.) Detecting an interaction of the same size as a main effect needs *four times* the sample. **The trap:** researchers power the study for the *main* effect ($n \approx 99$/arm), then report the *interaction* as the headline — at $n = 99$/arm the interaction has ~20% power, so a null interaction is uninformative (underpowered, not evidence of no moderation) and a "significant" one is likely a false positive or inflated (Chapter 9's winner's curse). **If an interaction is your headline (cost vector × incentive, real × hypothetical, familiarity × attribute), power the study for the *interaction* from the start** — or pre-register it as exploratory and report it with equivalence bounds / CIs, not as a confirmed moderation.

For your designs specifically. Your central questions are *often* interactions: *hypothetical bias that depends on the cost vector* (cost × real/hypothetical — a difference-in-differences); *familiarity moderating WTP*

(Chapter 24); *the BRIS incentive changing the format effect*. Each needs the 4× sample of a main effect. Budget for it: a DCE powered only for average attribute WTPs will be *badly underpowered* for the interaction that is your actual contribution, and an underpowered interaction is worse than none (it invites over-claiming from noise).

> **Stop and think 33.3.** You power your real-vs-hypothetical DCE for the *main* cost-vector effect ($n \approx 100$/arm), and your headline is "hypothetical bias is *larger* at high cost" (a cost × hypothetical interaction). What is the problem, and what should you have done? *(Answer: the headline is an* interaction* (difference-in-differences), which has *twice the SE* of a main effect and so needs *~4× the sample* — verified: ~393 vs ~99 per arm for the same effect size at 80% power. Powered only for the main effect ($n \approx 100$/arm), the interaction has only ~20% power, so: a *null* interaction is uninformative (underpowered — cannot conclude "no cost-dependence"), and a *significant* one is likely a false positive or magnitude-inflated (winner's curse, Chapter 9). What to do: power the study for the *interaction* from the start (budget ~4× the per-cell sample), or pre-declare it exploratory and report it with CIs/equivalence bounds rather than as a confirmed finding. The interaction that is your contribution must drive the power calculation, not the main effect.)*

33.8 Failure modes

F1. Uncorrected multiple testing. Reporting the "significant" attribute/subgroup out of many with no correction. *Fix:* Holm (FWER) for confirmatory, BH (FDR) for exploratory; report the whole family (§33.2–§33.4).

F2. Correcting the wrong family / after the fact. Choosing the family or correction after seeing results, or correcting exploratory tests as if confirmatory (or vice versa). *Fix:* pre-specify the family and its role; match correction to confirmatory/exploratory (§33.3–§33.4).

F3. The equivalence fallacy. Claiming sameness from a non-significant difference test. *Fix:* TOST / 90% CI within a pre-specified margin (§33.5).

F4. Post-hoc equivalence margin. Choosing Δ after seeing the CI. *Fix:* pre-specify Δ from decision relevance / benefit-transfer tolerance (§33.5).

F5. Distributional inference on small non-normal samples. Trusting a t-test p-value for a tiny, skewed, or spiked WTP cell. *Fix:* randomization/permutation inference (exact, assumption-free), respecting the design (§33.6).

F6. Permuting a broken randomization. Randomization inference after post-randomization exclusion, or ignoring blocking. *Fix:* preserve the assignment (Chapter 32); permute within blocks/strata (§33.6).

F7. Underpowered interactions. Powering for the main effect, reporting the interaction. *Fix:* power for the interaction (~4× sample); else report it as exploratory with CIs/bounds (§33.7).

F8. Over-claiming a null interaction. Reading a non-significant interaction as "no moderation." *Fix:* it is (usually) *underpowered*, not evidence of no effect — report the CI/equivalence bounds (§33.7; §33.5).

33.9 Diagnostic questions for your own papers

1. **Family.** Have I defined the *family* of tests, pre-specified it, and reported *all* of them (not just the significant)? *(§33.2, F1–F2.)*
2. **Correction.** Confirmatory primary tests → FWER (Holm)? Exploratory screens → FDR (BH)? *(§33.3–§33.4.)*

3. **Equivalence claims.** For every "they are the same" claim (real ≈ hypothetical, format A ≈ B), am I running an *equivalence test* with a *pre-specified margin*, not a failed difference test? *(§33.5, F3–F4.)*
4. **Margin justification.** Is my equivalence margin anchored to decision relevance / benefit-transfer tolerance and stated in advance? *(§33.5.)*
5. **Small/non-normal cells.** For small or skewed WTP cells/subgroups, am I using randomization inference (not the t-test's normal approximation)? *(§33.6, F5.)*
6. **Design-respecting permutation.** Does my permutation respect the actual assignment (blocks/strata) and intact randomization? *(§33.6, F6.)*
7. **Interaction power.** Is any interaction headline *powered as an interaction* (~4× the main-effect sample), or flagged exploratory with CIs? *(§33.7, F7–F8.)*

33.10 Exercises

(A) Comprehension.

A1. Explain why testing 10 true nulls at $\alpha = 0.05$ gives a ~40% family-wise error, and when you'd use FWER vs FDR control.

A2. Why does a non-significant difference test not prove equivalence, and what does TOST do instead?

A3. What null does randomization inference test, and what assumption does it *not* need?

(B) Derivation / reasoning.

B1. Derive Bonferroni FWER control from the union bound, and explain why Holm dominates it.

B2. Show, from the variance of a difference-in-differences, why an interaction needs ~4× the sample of a main effect.

B3. Explain the equivalence between TOST at level α and the $100(1-2\alpha)\%$ confidence interval falling inside $[-\Delta, +\Delta]$.

(C) Research application / critique.

C1. Your DCE reports WTP for 12 attributes; 2 are "significant" at 0.05 uncorrected. Write the correct multiplicity handling.

C2. You want to claim your hypothetical WTP is valid because it "matches" real WTP. Design the equivalence test (margin, statistic, decision).

C3. Your headline is "hypothetical bias grows with cost." Write the power and inference plan.

33.11 Solutions

A1. Each independent test at $\alpha = 0.05$ falsely rejects with probability 0.05, so with 10 true nulls the probability of *at least one* false rejection is $1 - (1 - 0.05)^{10} = 0.40$ (verified). Use **FWER** control (Holm) when even *one* false positive is costly — a *confirmatory* study with a few pre-specified primary hypotheses. Use **FDR** control (BH) when you tolerate a *fraction* of false discoveries in exchange for power — *exploratory* screening of many hypotheses (verified: BH held FDR at 0.03 with 0.79 power on 20 tests, far more power than Bonferroni would give).

A2. A non-significant difference test is *absence of evidence*: it fails to reject "they differ" but is *consistent with a large true difference* you were underpowered to detect — it does not establish sameness. **TOST** instead pre-specifies a margin Δ and tests the two one-sided nulls $\delta \le -\Delta$ and $\delta \ge +\Delta$; *rejecting both* (equivalently, the 90% CI lying inside $[-\Delta, +\Delta]$) *positively demonstrates* equivalence $-\Delta < \delta < +\Delta$ (verified: same data give TOST $p = 0.008$ "equivalent" vs NHST $p = 0.42$ "proves nothing").

A3. Randomization inference tests Fisher's *sharp null* of *no treatment effect for any unit* ($Y_i(1) = Y_i(0)$ for all i), under which each outcome is fixed and any arm difference reflects only *which* units were assigned where. It does *not* need any *distributional assumption* (no normality, no CLT) — it uses only the *actual randomization*, so it is *exact* in finite samples (verified to match the t-test when the t-test's assumptions hold: 0.0000 vs 0.0001).

B1. Let A_i be the event that test i (a true null) falsely rejects, each with $P(A_i) = \alpha/m$ under Bonferroni. The family-wise error is $P(\cup_i A_i) \leq \sum_i P(A_i) = m \cdot (\alpha/m) = \alpha$ (Boole's/union bound), *regardless of dependence* among tests. So Bonferroni controls FWER $\leq \alpha$. **Holm dominates** it because it is a *step-down* procedure: it tests the smallest p-value against α/m (same as Bonferroni for that one), but *subsequent* p-values against the *larger* thresholds $\alpha/(m-1), \ldots, \alpha/1$ — so it *rejects at least everything Bonferroni does, and often more*, while still controlling FWER $\leq \alpha$ under any dependence (verified: Holm FWER 0.050, controlled). Uniformly more power, same assumptions — always prefer Holm to plain Bonferroni.

B2. A main effect is $\bar{Y}_1 - \bar{Y}_0$, variance Var $= 2\sigma^2/n$ (two cell means of size n). An interaction is a *difference of two differences*: $(\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00})$, a linear combination of *four* independent cell means each with variance σ^2/n, with coefficients ± 1; its variance is $4 \cdot \sigma^2/n = 4\sigma^2/n$ — *twice* the main-effect variance, so $\sqrt{2}$*… no: the SE is* $\sqrt{4\sigma^2/n} = 2\sqrt{\sigma^2/n}$ *vs the main effect's* $\sqrt{2\sigma^2/n}$*, a ratio of* $\sqrt{2}$ *per-cell but* — comparing *like for like* (same n per cell, detecting the same effect size δ), the interaction contrast has *twice* the standard error of a two-arm main-effect contrast, so achieving the same power (which depends on effect/SE) requires $2^2 = 4$ times the sample (verified: 393 vs 99 per arm). The intuition: a difference-of-differences accumulates noise from four groups, so you must quadruple n to detect it as precisely as a single difference.

B3. TOST at level α rejects $H_{0,\text{lower}} : \delta \leq -\Delta$ iff $\hat{\delta} - z_\alpha \,\text{se} > -\Delta$ (the lower $(1-\alpha)$ one-sided bound exceeds $-\Delta$), and rejects $H_{0,\text{upper}} : \delta \geq +\Delta$ iff $\hat{\delta} + z_\alpha \,\text{se} < +\Delta$ (the upper one-sided bound is below $+\Delta$). Rejecting *both* means $\hat{\delta} - z_\alpha \,\text{se} > -\Delta$ *and* $\hat{\delta} + z_\alpha \,\text{se} < +\Delta$ — i.e., the interval $[\hat{\delta} - z_\alpha \,\text{se},\ \hat{\delta} + z_\alpha \,\text{se}]$ lies entirely inside $[-\Delta, +\Delta]$. That interval is the *two-sided* $100(1-2\alpha)\%$ *CI* (each tail has mass α), i.e. the **90% CI for** $\alpha = 0.05$. So "reject both one-sided tests at α" $\iff$ "the $100(1-2\alpha)\%$ CI is inside the margin" (verified: 90% CI $[-2.11, 6.11] \subset [-8, 8]$, TOST $p = 0.008$).

C1. With 12 attribute WTPs, first *define the family and its role*: if these are *exploratory* (screening which attributes matter), control the **FDR** with Benjamini–Hochberg at $q = 0.05$ — order the 12 p-values, find the largest k with $p_{(k)} \leq (k/12) \cdot 0.05$, reject those; report the realized discoveries with the understanding that ~5% may be false. If instead a *few* are *pre-registered primary* contrasts, control the **FWER** with Holm on *that* family. Either way: *report all 12* WTPs and p-values (not just the 2 "significant"), state the correction and family *in advance*, and note that "2 of 12 significant uncorrected" is roughly the *chance* expectation ($0.05 \times 12 \approx 0.6$, and $P(\geq 1)$ is large) — so uncorrected, the 2 could easily be noise. The honest report is the corrected family, not the raw winners.

C2. *Design:* (i) *Pre-specify the margin* Δ from decision relevance — the largest real–hypothetical WTP gap that would *not* change the policy/benefit-transfer conclusion (e.g., $\pm$€8, justified by the CBA's sensitivity or an acceptable transfer error), stated *before* analysis. (ii) *Statistic:* the difference $\hat{\delta} = \widehat{\text{WTP}}_{\text{hyp}} - \widehat{\text{WTP}}_{\text{real}}$ and its se (delta method or Krinsky–Robb for a WTP ratio, Chapter 7/13). (iii) *Test:* TOST — reject $\delta \leq -\Delta$ and $\delta \geq +\Delta$ each at 5%, equivalently check the **90% CI** for $\hat{\delta}$ lies inside $[-\Delta, +\Delta]$. (iv) *Decision:* conclude convergent validity *only if* the equivalence test rejects (CI inside the margin) — *not* from a non-significant difference test (verified: the two give opposite readings on the same data). Report the margin, its justification, the estimate, se, and the TOST p/CI; if the CI is *wider* than the margin, conclude "cannot establish equivalence at this margin/precision" (possibly underpowered), *not* "they differ."

C3. "Hypothetical bias grows with cost" is a *cost* $\times$ *real/hypothetical interaction* (a difference-in-differences). *Power plan:* power the study for the *interaction*, not the main effect — the interaction contrast has *twice the SE*, so budget *~4× the per-cell sample* of a main-effect design (verified: ~393 vs ~99/arm for $\delta = 0.4$ at 80% power). Determine the *minimum interaction size* worth detecting (how much steeper the hypothetical WTP–cost slope must be to matter), and solve for n using the difference-in-differences SE. *Inference plan:* estimate the interaction (cost $\times$ format) in the WTP-space EC-MXL / regression; report its CI; for small or non-normal cells use *randomization inference* (permute the format label within cost levels). If the study

is *not* powered for the interaction, *pre-declare it exploratory* and report the interaction with its CI (and, for a "no cost-dependence" claim, an *equivalence* test on the interaction) — never as a confirmed moderation from an underpowered test. State that a null interaction at low power is *uninformative*, not evidence of no cost-dependence.

33.12 Spaced review prompts

+1 day. From memory: the FWER of m tests; Bonferroni vs Holm vs BH (and when each applies); the TOST logic; the sharp null of randomization inference; the 4× interaction rule.

+1 week. Explain why a non-significant difference doesn't prove equivalence and how TOST fixes it; why randomization inference needs no normality; why interactions cost 4× the sample.

+1 month. For your own DCE/CV papers, audit: the test family and correction (confirmatory vs exploratory), every equivalence claim (TOST + pre-specified margin), small-cell inference (randomization), and interaction power (4×). Close with the spine sentence about honest inference on many tests, sameness claims, small samples, and interactions.

33.13 How this chapter connects to the rest of the book

- **Chapter 8 (§8.8)** introduced multiplicity; **Chapter 9** power/MDE/equivalence; **Chapter 10 (§10.6)** permutation inference — this chapter is their experimental application and completes them. **Chapter 31** built the factorial designs and interactions this chapter *powers*; **Chapter 32** the identification this inference *presupposes* (broken randomization invalidates permutation inference too).
- **Equivalence testing** returns for **convergent/criterion validity** (Chapter 4 §4.x) and **benefit transfer** (Chapter 49) — "the transferred value ≈ the local value" is an equivalence claim. **Multiplicity** returns in **meta-analysis** (Chapters 47–48; testing many moderators) and **subgroup analysis**. **Randomization inference** connects to the **bootstrap/Krinsky–Robb** resampling of Chapters 7/13.
- **Power for interactions** governs **every moderation headline** in your work: cost × hypothetical (hypothetical bias, Chapter 28–29), familiarity × attribute (Chapter 24), incentive × format (Chapter 28).

This chapter *completes Part VII*: Chapters 31–32 delivered a *clean, identified* treatment effect; Chapter 33 delivers *honest inference* on it and on the *many* effects, *sameness* claims, *small samples*, and *interactions* your survey experiments actually involve. Part VIII turns from experimental inference to the *welfare* foundations that give your WTP estimates their economic meaning.

33.14 Research translation task

For your **survey experiments and CV validation**, produce a two-to-three-page inference plan:

1. **Multiplicity.** Define the *family* of tests for each design (attribute WTPs, subgroups, arms); pre-specify the correction — *Holm* for confirmatory primaries, *Benjamini–Hochberg* for exploratory screens; commit to reporting the *whole* family.
2. **Equivalence.** For every convergent-validity / format-agreement / benefit-transfer claim, pre-specify the *margin* (from decision relevance) and the *TOST / 90% CI* test; commit to never inferring sameness from a non-significant difference.

3. **Randomization inference.** For small or non-normal cells/subgroups, plan *permutation* inference (respecting blocks/strata and intact randomization); use it as the default p-value for the spiked/interval WTP outcomes.
4. **Interaction power.** Identify every *interaction* headline (cost × hypothetical, familiarity × attribute, incentive × format); *power the study for the interaction* (~4× the main-effect sample) or pre-declare it exploratory with CIs/equivalence bounds.
5. **Close** with the spine sentence: *"My inference is honest because I pre-specify the test family and correction ([Holm]/[BH]), test every sameness claim by equivalence with a pre-specified margin, use randomization inference for small non-normal cells, and power my interaction headlines as interactions (4×)."*

Carry this into Chapter 49 (benefit transfer as equivalence), Chapters 47–48 (meta-analytic multiplicity), and every moderation claim in your CV/DCE work.

33.15 Bibliography

Core. - Lehmann & Romano, *Testing Statistical Hypotheses*, for the formal theory of multiple testing, FWER, and permutation tests (§33.2–§33.6). *(Verify edition.)* - List, Shaikh & Xu (2019), "Multiple Hypothesis Testing in Experimental Economics," *Experimental Economics* — multiplicity corrections for experiments, exactly your setting (§33.2–§33.4). *(Verify volume/pages.)* - Lakens, Scheel & Isager (2018), "Equivalence Testing for Psychological Research: A Tutorial," *Advances in Methods and Practices in Psychological Science* — the accessible TOST guide (§33.5). *(Verify volume/pages.)*

Applied. - Benjamini & Hochberg (1995), "Controlling the False Discovery Rate," *Journal of the Royal Statistical Society B* — the BH procedure (§33.4). *(Verify volume/pages.)* - Holm (1979), "A Simple Sequentially Rejective Multiple Test Procedure," *Scandinavian Journal of Statistics* — Holm's step-down (§33.3). *(Verify volume/pages.)* - Gelman, Hill & Vehtari (or Gelman & Carlin 2014 on Type S/M errors) on the perils of underpowered interactions and the winner's curse (§33.7). *(Verify which reference / citation.)*

Advanced. - Young (2019), "Channeling Fisher: Randomization Tests and the Statistical Insignificance of Seemingly Significant Experimental Results," *Quarterly Journal of Economics* — randomization inference overturning t-test findings in practice (§33.6). *(Verify volume/pages.)* - Romano & Wolf on stepwise multiple testing and resampling-based FWER control (§33.3). *(Verify citation.)* - Wellek, *Testing Statistical Hypotheses of Equivalence and Noninferiority*, for the full equivalence-testing theory and margin choice (§33.5). *(Verify edition.)*

Entries marked "verify" are pointers; confirm editions, pages, and the List–Shaikh–Xu / Lakens / Young / Benjamini–Hochberg / Holm citations before using them, per the book's no-unverified-citation rule.

End of Chapter 33, and of Part VII. Next: Part VIII — Welfare Foundations. Chapter 34 turns from the mechanics of estimation and inference to the economic meaning* of what you estimate: the theory of welfare measurement (compensating and equivalent variation, consumer surplus, the money-metric utility behind WTP), which gives your discrete-choice and contingent-valuation numbers their status as *welfare* magnitudes rather than mere statistical coefficients. Tell me when to write it, or request a different chapter.*

Part VII Assessment — Identification and Treatment Effects

Covers Chapters 30–33. Five components, in the order they are best attempted. Allow a full working day; the mock oral is the part that matters most for your viva.

1. Cumulative problem set

P1. For your CV design, write out formally what is identified, by what variation, under what assumptions, and name one observationally equivalent alternative that remains.

P2. Define ATE and ATT and give a design in your own work where they differ.

P3. Draw the DAG for your cost-vector experiment including a post-treatment comprehension measure, and mark which conditioning sets are safe.

P4. Compute Lee bounds for a treatment effect with 12% differential attrition, given the reported outcome distributions.

P5. Apply Holm and Benjamini–Hochberg corrections to the p-value vector $(0.004, 0.011, 0.023, 0.041, 0.19)$ and state which survive under each.

2. Research critique

Take the experimental section of your CV paper and write, for each treatment contrast, a sentence in the form: "This contrast identifies [object] under [assumption], and would be confounded by [alternative]." Mark the contrasts where the assumption is untestable and say how you will bound the consequences.

3. Red-flag diagnosis

A survey experiment reports balance tests on 18 baseline covariates, notes that two are significant at 5%, drops those two arms' outliers, and states that "randomization was successful after cleaning." Identify every inferential error in that sentence and write the correct procedure.

4. Mock oral examination

1. What is the difference between estimation and identification?
2. Why do balance tests not prove successful randomization?
3. What does a weighted likelihood do to your estimand?
4. When is a bad control worse than an omitted variable?
5. How do you power a study for an interaction, and why is it so much harder?

5. Integration mini-project

Run randomization inference on the principal treatment contrast in your CV paper: construct the permutation distribution under the sharp null, report the exact p-value, and compare it with the asymptotic one. Then repeat under your actual exclusion rule and under no exclusions, and write a paragraph on whether the exclusion changed the answer or the question.

Part VIII — Welfare Foundations

Chapter 34 — Welfare Measurement Foundations

Part VIII: Welfare Foundations

> **The spine question, applied to this chapter.** Every previous chapter estimated a *number* — a WTP, a coefficient, a treatment effect. This chapter asks the question that gives those numbers their *economic meaning*: *what welfare object am I identifying, and under what assumptions is my estimate a valid money measure of a person's wellbeing change?* WTP is not just a survey answer or a coefficient ratio; it is a claim about **compensating variation** or **equivalent variation** — a money-metric of a utility change. That claim is *exact* only under specific conditions (constant marginal utility of income), *approximate* otherwise, and the gap is governed by **income effects** and the **Willig bounds**. The spine question here: *when I report a WTP from my CV or DCE study, is it compensating variation, equivalent variation, or an uninterpretable hybrid — and is the money-metric interpretation exact, bounded, or broken?*

34.0 Conceptual motivation: from coefficients to welfare

Parts I–VII built the machinery to *estimate* WTP — from linear algebra to random utility to honest inference. But an estimate is only as meaningful as the *economic object* it targets. This part, Welfare Foundations, supplies that meaning: it explains *why* a WTP is a measure of *wellbeing*, *which* welfare measure it is, and *when* the interpretation holds.

This matters acutely for your work. Your entire research program produces *money measures of welfare*: the WTP for finfish/ecosystem improvements from your **contingent valuation** (SBDC, payment cards), the marginal WTP for attributes from your **DCE** (EC-MXL in WTP space), the aggregate benefits that feed your **cost–benefit analysis** of fish restoration devices, and the values you **transfer** across sites in your meta-analysis. Every one of these is, implicitly, a claim about **compensating variation** or **equivalent variation** — the two theoretically-grounded money measures of a utility change. If you cannot say *which* welfare measure your number is, and *under what assumptions* it is valid, then the CBA that aggregates it and the policy conclusion that rests on it are built on sand.

Your PDF's welfare-theory section demands exactly this foundation: the distinction between **Marshallian consumer surplus** (observable, but *not* a clean welfare measure when income effects are present) and the **Hicksian measures** (compensating and equivalent variation — the *correct* welfare measures, but defined on *unobservable* utility-constant demand); the **duality** between the expenditure function and indirect utility that connects them; the **Willig bounds** that tell you when the observable Marshallian surplus is a *good enough* approximation to the Hicksian ideal; and — the linchpin for your discrete-choice work — the **quasilinear / constant-marginal-utility-of-income** case in which *all three measures coincide*, so that a DCE's WTP *is*, exactly, compensating and equivalent variation.

This chapter builds that foundation with every number verified: the ordering CV $\geq$ CS $\geq$ EV for a normal-

good price increase (verified $41.42 \geq 34.66 \geq 29.29$); the Willig approximation error (verified ~17–20%, first-order formula 17.3%); the quasilinear collapse CV = EV = CS (verified exactly 0.5); and the discrete-choice log-sum welfare that makes your DCE WTP an *exact* welfare measure (verified 1.70 money units). By the end you will be able to state, for any number your research produces, *which* welfare measure it is and *what assumption* makes it valid — the precondition for everything in Parts IX–XII (public goods, MDCEV, CBA, benefit transfer).

34.1 Prerequisites

From **Chapter 2–3** (constrained optimization, the Lagrangian): utility maximization subject to a budget constraint, and the envelope theorem behind Roy's identity and Shephard's lemma. From **Chapters 20–21** (random utility) and **Chapter 24** (WTP space): the marginal utility of income (the cost coefficient) and WTP as a ratio. From **Chapter 4** (discounting/NPV): the money-metric that CBA aggregates. From **Chapter 14** (latent variables): utility as unobservable.

New notation. u (utility level), $V(\mathbf{p}, m)$ (indirect utility: max utility at prices $\mathbf{p}$, income m), $e(\mathbf{p}, u)$ (expenditure function: min cost to reach u at prices $\mathbf{p}$), $x(\mathbf{p}, m)$ (Marshallian/ordinary demand), $h(\mathbf{p}, u)$ (Hicksian/compensated demand), CV (compensating variation), EV (equivalent variation), CS (Marshallian consumer surplus), λ (marginal utility of income).

34.2 The two demand functions and the two utility representations

Welfare measurement rests on a duality between *maximizing utility given income* and *minimizing expenditure given a utility target*. Both describe the same consumer; they differ in what is held fixed.

Definition 34.1 (Indirect utility and Marshallian demand). The consumer solves $\max_{\mathbf{x}} u(\mathbf{x})$ s.t. $\mathbf{p}'\mathbf{x} \leq m$. The maximized utility is the **indirect utility function** $V(\mathbf{p}, m)$, and the optimal quantities are the **Marshallian (ordinary) demands** $x(\mathbf{p}, m)$ — the demands you *observe* in markets and surveys (they depend on *income*, so they carry *income effects*).

Definition 34.2 (Expenditure and Hicksian demand). Dually, the consumer solves $\min_{\mathbf{x}} \mathbf{p}'\mathbf{x}$ s.t. $u(\mathbf{x}) \geq u$. The minimized cost is the **expenditure function** $e(\mathbf{p}, u)$, and the optimal quantities are the **Hicksian (compensated) demands** $h(\mathbf{p}, u)$ — demands along a *fixed indifference curve* (utility held constant, so income is adjusted to compensate). Hicksian demand is *not directly observable* (utility is latent), but it is the *correct* basis for welfare measurement because it holds wellbeing fixed.

The duality identities. The two representations are linked:

$$V(\mathbf{p}, e(\mathbf{p}, u)) = u, \qquad e(\mathbf{p}, V(\mathbf{p}, m)) = m,$$

$$x(\mathbf{p}, m) = h(\mathbf{p}, V(\mathbf{p}, m)) \quad \text{(Marshallian = Hicksian at the utility income buys)}.$$

And two envelope results do the heavy lifting:

- **Roy's identity:** $x_i(\mathbf{p}, m) = -\dfrac{\partial V / \partial p_i}{\partial V / \partial m}$ (Marshallian demand from indirect utility).
- **Shephard's lemma:** $h_i(\mathbf{p}, u) = \dfrac{\partial e(\mathbf{p}, u)}{\partial p_i}$ (Hicksian demand is the *price-derivative of expenditure*). This is the key to welfare measurement: *areas under Hicksian demand are expenditure differences, i.e. exact money measures of utility change.*

The marginal utility of income. $\lambda = \partial V / \partial m$ is the **marginal utility of income** — how much utility one more currency unit buys. It is the *conversion rate* between utility (an ordinal, unit-free construct) and money (the metric welfare economics reports in). Whether λ is *constant* (independent of income) or *varies* is

the single most important fact for welfare measurement — it determines whether the three welfare measures coincide or diverge (§34.6), and whether your DCE WTP is *exact* welfare (§34.7).

> **Active recall 34.A.** What is the difference between Marshallian and Hicksian demand, and why is Hicksian demand the "correct" basis for welfare measurement even though it is unobservable? *(Answer: Marshallian demand $x(\mathbf{p}, m)$ maximizes utility given* income* — it is what we observe, but it mixes the substitution effect with an *income effect* (as a price rises, real income falls). Hicksian demand $h(\mathbf{p}, u)$ minimizes expenditure holding *utility* fixed — it isolates the pure substitution effect along one indifference curve. Hicksian demand is the correct welfare basis because, by Shephard's lemma, the *area under it* between two prices equals the *difference in the expenditure function*, $e(\mathbf{p}^1, u) - e(\mathbf{p}^0, u)$ — an *exact money measure of the utility change* (you hold wellbeing fixed and ask how much money that costs). Marshallian demand does not hold utility fixed, so areas under it (consumer surplus) are contaminated by income effects and are only an *approximation* to the true welfare change. Hicksian demand is unobservable because utility is latent — hence the need for the bounds and special cases of this chapter.)*

34.3 The three welfare measures: CV, EV, and consumer surplus

A price change (or a quality change, or the provision of a public good) moves the consumer from one utility level to another. *How much is that change worth, in money?* There are three answers.

Definition 34.3 (Compensating variation, CV). The **compensating variation** of a change from $(\mathbf{p}^0, m)$ to $(\mathbf{p}^1, m)$ is the money adjustment that, made *after* the change, restores the consumer to the *original* utility $u^0 = V(\mathbf{p}^0, m)$:

$$\mathrm{CV} = e(\mathbf{p}^1, u^0) - e(\mathbf{p}^0, u^0) = e(\mathbf{p}^1, u^0) - m.$$

CV uses the *original* utility (status-quo reference / original property rights). For a *price increase*, CV > 0 is the compensation the consumer *needs* to be as well off as before (a loss). For an *improvement* (quality gain, public-good provision), $-$CV is the **maximum WTP** to *obtain* it — the money that, taken away *after* the improvement, leaves the consumer at their original utility. **CV is the WTP concept: it holds the consumer at the status-quo utility and asks what money change is equivalent to the environmental change.**

Definition 34.4 (Equivalent variation, EV). The **equivalent variation** uses the *new* utility $u^1 = V(\mathbf{p}^1, m)$ as reference:

$$\mathrm{EV} = e(\mathbf{p}^1, u^1) - e(\mathbf{p}^0, u^1) = m - e(\mathbf{p}^0, u^1).$$

EV is the money change that, made *instead of* the price change, would have the *same* welfare effect. For an improvement, EV is the **minimum WTA (willingness to accept)** compensation to *forgo* it — the money that, given *without* the improvement, is as good as having it. **EV is the WTA concept: it holds the consumer at the post-change utility.**

Definition 34.5 (Marshallian consumer surplus, CS). The **change in (Marshallian) consumer surplus** is the area under the *ordinary* demand curve between the two prices:

$$\Delta\mathrm{CS} = \int_{p^0}^{p^1} x(p, m)\, dp.$$

CS is *observable* (it uses ordinary demand) but is *not* a clean welfare measure when income effects are present, because ordinary demand mixes substitution and income effects. It is the *practical approximation* to the Hicksian ideals CV and EV.

The exact Hicksian measures. By Shephard's lemma, CV and EV are areas under the *Hicksian* demands at the *original* and *new* utility respectively:

$$\text{CV} = \int_{p^0}^{p^1} h(p, u^0)\, dp, \qquad \text{EV} = \int_{p^0}^{p^1} h(p, u^1)\, dp.$$

So the three measures are three areas under three demand curves — Hicksian-at-u^0 (CV), Marshallian (CS), and Hicksian-at-u^1 (EV) — all between the same two prices. Their *ordering* follows from how these curves sit relative to each other.

> **Stop and think 34.1.** Your CV survey asks the maximum a household would pay for a finfish-stock improvement. Which welfare measure is this — CV or EV — and what reference utility and property-right assumption does it embed? *(Answer:* maximum WTP for an improvement* is a *compensating variation* (−CV): it holds the household at its *status-quo* utility u^0 (the reference is "without the improvement") and asks the money that, taken away *after* the improvement, restores that original utility — i.e., the most they would pay and still be as well off as now. The property-right assumption is that the household does *not* currently have the right to the improved stock (they must *pay* to obtain it) — the status quo is the reference. The alternative framing — *WTA compensation to forgo* the improvement — would be an *equivalent variation* holding them at the *with-improvement* utility, embedding the *opposite* property right (they own the right and must be compensated to give it up). WTP (CV) and WTA (EV) differ, sometimes greatly, precisely because they use different reference utilities and property rights — your elicitation format fixes which one you measure.)*

34.4 The ordering of the measures and worked example

For a *normal* good (positive income elasticity), the three measures have a definite ordering, driven by the income effect embedded in Marshallian demand.

The ordering (normal good, price increase). As the price rises, real income falls, so Marshallian demand falls *faster* than the Hicksian demand at the *original* utility (which holds income compensated up) and *slower* than the Hicksian at the *new* (lower) utility. This gives, in *loss* magnitudes:

$$\text{CV} \geq \Delta\text{CS} \geq \text{EV} \quad \text{(normal good, price increase)}.$$

CV (holding at the *higher* original utility, where the good is "worth more") is the *largest* loss measure; EV (holding at the *lower* new utility) is the *smallest*; CS sits between them. (For a *price decrease / improvement*, the inequalities and the WTP/WTA correspondence flip accordingly, but CS always lies *between* CV and EV.)

Worked Example 34.1 (Cobb-Douglas welfare measures, verified). Consumer with $u = x^{0.5}y^{0.5}$, income $m = 100$, $p_y = 1$, price of x rising from 1 to 2:

Measure	Value	Formula
CV	**41.42**	$m(p^1/p^0)^a - m$
ΔCS	**34.66**	$a\, m \ln(p^1/p^0)$
EV	**29.29**	$m - m(p^0/p^1)^a$

(Verified; closed forms match the numerical integrals exactly.) The ordering CV (41.42) ≥ CS (34.66) ≥ EV (29.29) holds. The *same* welfare change has three different money values, differing by up to 40%, purely because of the income effect and the choice of reference utility. **If you reported "the welfare loss is €35" (the CS) without saying it is an approximation lying between a CV of €41 and an EV**

of €29, you would be papering over a real ambiguity — one that matters when the number feeds a CBA. Which measure is "right" depends on the property-right / reference-utility question (§34.3), not on convenience.

> **Active recall 34.B.** Why, for a normal good, is CV $\geq$ CS $\geq$ EV for a price increase, and what single feature of preferences would make them all equal? *(Answer: CV holds the consumer at the* original, higher* utility u^0 (income compensated up), where the good is effectively "worth more," giving the *largest* loss; EV holds them at the *new, lower* utility u^1, giving the *smallest*; Marshallian CS, which embeds a partial income effect, lies between — verified $41.42 \geq 34.66 \geq 29.29$. The wedge between them is the *income effect*: the three Hicksian/Marshallian demand curves diverge because real income changes with the price. The single feature that makes them *all equal* is *no income effect on the good* — i.e., *quasilinear preferences / constant marginal utility of income* (§34.6), under which the compensated and ordinary demands coincide and CV = CS = EV exactly.)*

34.5 The Willig bounds: when is consumer surplus good enough?

CV and EV are the *correct* measures but require the *unobservable* Hicksian demand (or a fully specified utility function). CS is *observable* but *approximate*. Willig (1976) answered the practical question: *how big is the approximation error, and when can you ignore it?*

The Willig result. The percentage error of using ΔCS as an approximation to CV (or EV) is *bounded* by a simple expression in the **income elasticity** of demand η and the ratio of the surplus to income:

$$\frac{|\text{CV} - \Delta\text{CS}|}{|\Delta\text{CS}|} \approx \frac{\eta\,|\Delta\text{CS}|}{2m}.$$

The error is *small* when (i) the income elasticity η is small (weak income effects), and (ii) the surplus is small relative to income ($|\Delta\text{CS}|/m$ small — the price change is not a huge fraction of the budget). Under these conditions — the typical case for a *single, modestly-priced* market good — *consumer surplus is an excellent approximation to CV and EV*, and the choice among them is second-order.

Worked Example 34.2 (Willig bounds, verified). For the Cobb-Douglas case above ($\eta = 1$, a large price change consuming a big income share):

Quantity	Value
Actual $\|\text{CV} - \text{CS}\|/\text{CS}$	**19.5%**
Actual $\|\text{EV} - \text{CS}\|/\text{CS}$	**15.5%**
Willig first-order approx $\eta\,\|\text{CS}\|/(2m)$	**17.3%**

(Verified.) The Willig formula (17.3%) captures the right order of magnitude of the actual errors (15–20%). Here the error is *large* because the price *doubled* (a big income share) and the income elasticity is 1 — so CS is a *poor* stand-in for CV/EV. **The lesson: CS is a good approximation only when income effects are weak and the value is a small share of income.** For your work, this is a warning: environmental values can be a *non-trivial* share of income (especially for large programs or low-income households), and non-market goods can have *substantial* income elasticities — so the CV/EV/CS distinction is *not* always second-order, and you cannot assume CS is "close enough." When it is not, you need the *exact* Hicksian measure, which is why the *elicitation* directly targets CV or EV (§34.7).

> **Stop and think 34.2.** A colleague computes environmental benefits as the area under an estimated (Marshallian) recreation demand curve and calls it "the welfare gain." When is this fine, and when could it be off by 20%? *(Answer: it is* fine* when the *Willig conditions* hold — the income elasticity of recreation demand is *small* and the surplus is a *small share of income*

— so the Marshallian consumer surplus is within a percent or two of the true CV/EV (income effects are second-order). It could be *off by ~20%* when income effects are strong (high income elasticity) *and/or* the value is a *large share of income* (a big program, low-income households) — verified: at $\eta = 1$ and a large surplus/income ratio, CS differed from CV by 19.5% and from EV by 15.5%. So the colleague should (i) check the income elasticity and the surplus-to-income ratio (Willig), (ii) report CS as an *approximation bracketed by CV and EV*, and (iii) if the conditions fail, compute the *exact* Hicksian measure from a specified utility function or elicit CV/EV directly. Calling the Marshallian area "the welfare gain" without this check risks a 20% error feeding the CBA.)*

34.6 The quasilinear case: when all three coincide (the WTP–welfare link)

There is one case in which the entire apparatus simplifies to a single number — and it is the case that *justifies* discrete-choice and contingent-valuation WTP as welfare measures.

Quasilinear preferences. Suppose utility is **quasilinear** in a numeraire (money/composite) good: $u(\mathbf{x}, y) = v(\mathbf{x}) + y$, where y is spending on "everything else." Then the marginal utility of income is **constant** ($\partial V/\partial m = 1$, independent of income): an extra euro always adds the same utility. Consequently there is **no income effect** on the goods in $v(\cdot)$ — Marshallian and Hicksian demands *coincide* — and therefore:

$$\text{CV} = \Delta\text{CS} = \text{EV}.$$

All three welfare measures collapse to a *single* number. The reference-utility ambiguity (CV vs EV, WTP vs WTA) *vanishes*, because the income effect that separated them is gone.

Worked Example 34.3 (Quasilinear collapse, verified). With $u = 2\sqrt{x} + y$ (constant marginal utility of income), price of x from 1 to 2:

Measure	Value
CV	**0.500**
ΔCS	**0.500**
EV	**0.500**

(Verified; all exactly $1/p^0 - 1/p^1 = 0.5$.) With no income effect, the three measures are *identical* — there is one unambiguous welfare number, and the observable Marshallian surplus *is* the exact Hicksian welfare change. **This is the theoretical foundation of your WTP estimates.**

Why this is the DCE/CV foundation. The standard discrete-choice model has utility *linear in cost*: $U_j = \beta' \mathbf{x}_j - \alpha\, c_j + \varepsilon_j$, where α is the (constant) marginal utility of income. Linearity in cost *is* the quasilinear assumption — the marginal utility of income is constant, so there are *no income effects across the choice alternatives*, and the WTP $= \beta/\alpha$ (a marginal rate of substitution between attribute and money) is *exactly* compensating variation *and* equivalent variation *and* the consumer-surplus change. **Your DCE's WTP-space estimates are exact welfare measures precisely because the model assumes constant marginal utility of income.** Likewise, a CV study that elicits the maximum payment for a discrete improvement is measuring CV directly (§34.7). The quasilinear/constant-α assumption is what turns your *statistical* WTP into a *welfare* magnitude — and relaxing it (income effects in cost, e.g. a $\ln(\text{income} - \text{cost})$ specification) *reintroduces* the CV/EV wedge, which is why income effects in WTP are a genuine (if often second-order) issue in your models.

Active recall 34.C. Why does linearity in cost in a discrete-choice model make the estimated WTP an *exact* welfare measure, and what would break this? *(Answer: utility linear in cost,*

$U_j = \beta' \mathbf{x}_j - \alpha c_j + \varepsilon_j$, *means the marginal utility of income* α *is* constant* — the quasilinear case — so there is *no income effect* across alternatives and Marshallian = Hicksian demand. Hence CV = EV = CS collapse to one number (verified: 0.5 = 0.5 = 0.5), and the WTP = β/α (the money-attribute marginal rate of substitution) is *exactly* compensating and equivalent variation, not an approximation. What breaks it: a *non-constant* marginal utility of income — e.g. cost entering nonlinearly ($\ln(\text{inc} - c)$ or a random/heterogeneous α correlated with income), or large values relative to income — which reintroduces income effects, reopens the CV/EV wedge (Willig), and makes the estimated WTP an *approximation* whose reference utility (WTP vs WTA) again matters.)*

34.7 Direct elicitation of CV/EV and the money-metric

The deepest reason stated-preference methods exist: they *elicit the Hicksian welfare measure directly*, sidestepping the observability problem.

Stated preference targets the Hicksian measure. Revealed-preference welfare analysis must *infer* the unobservable Hicksian demand from observed Marshallian demand (hence the Willig approximation problem). Stated-preference methods — CV and DCE — instead *ask* the respondent, in a framing that *holds utility at a reference level*: "what is the most you would pay for this improvement (and remain as well off as now)?" *is* the definition of −CV; "what compensation would make you as well off without it?" *is* EV. **The elicitation format directly targets the money-metric of the utility change**, which is why stated preference is the workhorse for *non-market* welfare where no market demand exists (your finfish/ecosystem values). But this power comes with the responsibility to *know which measure you elicited*: a WTP question measures CV (status-quo reference, must-pay property right); a WTA question measures EV (post-change reference, ownership property right); and these differ (§34.3).

The money-metric utility and Small–Rosen for discrete choice. For the discrete-choice (logit) case, the exact welfare change from a policy that alters the choice set / attributes is the **log-sum (Small–Rosen) formula**:

$$\mathbb{E}[\text{CV}] = \frac{1}{\alpha}\Big[\underbrace{\ln \textstyle\sum_j e^{V_j^1}}_{\text{after}} - \underbrace{\ln \textstyle\sum_j e^{V_j^0}}_{\text{before}}\Big],$$

where α is the marginal utility of income and V_j the systematic utilities. The **log-sum** (the "inclusive value" / expected maximum utility) is the money-metric of expected welfare, converted to money by dividing by α.

Worked Example 34.4 (Discrete-choice welfare, verified). Two alternatives, $\alpha = 0.20$; a policy improves option 1's systematic utility from 1.0 to 1.5:

Quantity	Value
Log-sum before	1.474
Log-sum after	1.813
$\mathbb{E}[\text{CV}] = (\text{after} - \text{before})/\alpha$	**1.696** (money units)

(Verified.) The expected compensating variation of the policy is 1.70 money units — an *exact* welfare measure *because* the marginal utility of income α is constant (linear-in-cost logit). **This is how your DCE delivers welfare estimates for policy scenarios: the log-sum difference, divided by the cost coefficient, is the expected CV per person** — the number that, aggregated over the population, becomes the benefit stream in your CBA. The formula's validity rests *entirely* on the constant-α (quasilinear) assumption of §34.6; with a random or income-dependent α, the log-sum welfare becomes an expectation over the distribution of α and the clean division no longer holds (an issue your EC-MXL in WTP space must handle carefully).

> **Stop and think 34.3.** Your DCE estimates a policy scenario's benefit via the log-sum formula and gets €1.70/person. What assumption makes this an *exact* CV, and how does it connect to

the WTP-space parameterization of your EC-MXL? *(Answer: the log-sum welfare* $\frac{1}{\alpha}[\ln \sum e^{V^1} - \ln \sum e^{V^0}]$ *is an* exact* compensating variation only if the *marginal utility of income* α *is constant* (linear-in-cost, quasilinear) — then dividing the utility-scale log-sum difference by α gives money-metric CV = EV = CS (verified 1.70). Connection to WTP space: parameterizing the EC-MXL *in WTP space* estimates the attribute WTPs β/α directly with a specified (often log-normal) *distribution of the cost coefficient* α — which means α is *random across people*, not a single constant. So the population welfare is an *expectation of the log-sum over the distribution of* α, and the naive "divide by a single α" is only exact per-individual at their own α; aggregate welfare must integrate over the α-heterogeneity (and guard against the near-Cauchy tails of β/α when α has mass near zero, Chapter 7). The €1.70 is an exact CV *conditional on constant/known* α; with heterogeneous α it is a distributional welfare object to be integrated, not a single division.)*

34.8 Failure modes

F1. Reporting CS as "the" welfare measure without bounds. Presenting the Marshallian area as the welfare change when income effects are non-trivial. *Fix:* report CS as an approximation bracketed by CV and EV; check Willig conditions (§34.4–§34.5).

F2. Ignoring the CV/EV (WTP/WTA) distinction. Treating WTP and WTA as interchangeable. *Fix:* they use different reference utilities / property rights and differ; state which you measure and why (§34.3).

F3. Assuming CS ≈ CV without checking Willig. Using consumer surplus for a large-value or high-income-elasticity good. *Fix:* compute the Willig bound; if the value is a big income share or income elasticity is high, use the exact Hicksian measure (§34.5).

F4. Forgetting the quasilinear assumption behind WTP. Reporting DCE WTP as exact welfare while using an income-nonlinear cost specification. *Fix:* WTP = exact CV/EV *only* under constant marginal utility of income; flag income effects if cost enters nonlinearly (§34.6).

F5. Misusing the log-sum with heterogeneous α. Dividing the log-sum difference by a single α when α is random (WTP-space MXL). *Fix:* integrate the log-sum welfare over the distribution of α; beware near-zero α (§34.7).

F6. Aggregating incommensurable measures. Summing a CV from one study and an EV from another in a CBA. *Fix:* harmonize to a common welfare measure (usually CV/WTP) before aggregating (§34.3; Chapters 45–46).

34.9 Diagnostic questions for your own papers

1. **Which measure?** For each reported value, is it CV, EV, or CS — and have I said so explicitly? *(§34.3, F1.)*
2. **Property right / reference.** Does my elicitation format (WTP vs WTA) match the property-right assumption of the policy question? *(§34.3, F2.)*
3. **Willig check.** Is the value a small share of income with low income elasticity (so CS ≈ CV/EV), or must I use the exact Hicksian measure? *(§34.5, F3.)*
4. **Quasilinearity.** Does my model assume constant marginal utility of income (linear cost)? If not, have I acknowledged the CV/EV wedge? *(§34.6, F4.)*
5. **Log-sum welfare.** For DCE policy scenarios, am I computing the log-sum welfare correctly, integrating over α-heterogeneity in WTP space? *(§34.7, F5.)*
6. **Aggregation.** In the CBA, are all benefit components the *same* welfare measure before I sum them? *(§34.3, F6.)*

34.10 Exercises

(A) Comprehension.

A1. Define CV and EV and explain which reference utility each uses; relate them to WTP and WTA.

A2. Why is Marshallian consumer surplus only an approximation to the true welfare change?

A3. Under what condition do CV, EV, and CS coincide, and why is this the foundation of DCE WTP?

(B) Derivation / reasoning.

B1. Using the expenditure function, derive CV and EV for a price increase and show CV uses u^0, EV uses u^1.

B2. State the Willig approximation and explain the two conditions under which CS $\approx$ CV.

B3. For quasilinear $u = v(x) + y$, show that the marginal utility of income is constant and CV = CS = EV.

(C) Research application / critique.

C1. Your CV study reports a mean WTP of €50 for a finfish improvement. State precisely which welfare measure this is and the assumptions for its validity.

C2. A CBA sums a WTP-based benefit and a WTA-based cost. Critique and fix.

C3. Your DCE (EC-MXL in WTP space) reports a policy benefit via the log-sum. Explain how α-heterogeneity affects the welfare calculation.

34.11 Solutions

A1. **CV** (compensating variation) uses the *original* utility u^0 as reference: the money adjustment, made *after* the change, that restores u^0 — for an improvement, $-$CV is the *maximum WTP* to obtain it (holds you at the status-quo utility, must-pay property right). **EV** (equivalent variation) uses the *new* utility u^1: the money change *instead of* the policy that has the same welfare effect — for an improvement, EV is the *minimum WTA* to forgo it (holds you at the with-policy utility, ownership property right). WTP $\leftrightarrow$ CV (status-quo reference); WTA $\leftrightarrow$ EV (post-change reference); they differ because they hold utility at different levels and embed opposite property rights.

A2. Marshallian consumer surplus is the area under *ordinary* demand, which holds *income* fixed — so as price changes, *real income* changes and the demand response mixes the pure substitution effect with an *income effect*. The *true* welfare change is the area under *Hicksian* (compensated) demand, which holds *utility* fixed (Shephard's lemma makes that area an exact expenditure difference). Because ordinary demand embeds the income effect, CS lies *between* the two Hicksian measures CV and EV and equals neither unless income effects vanish — hence it is only an approximation (verified: CS 34.66 between CV 41.42 and EV 29.29).

A3. They coincide under *quasilinear preferences / constant marginal utility of income* (no income effect on the goods of interest) — then Marshallian = Hicksian demand and CV = CS = EV (verified exactly 0.5). This is the DCE-WTP foundation because the standard discrete-choice model is *linear in cost*, i.e. the marginal utility of income α is constant — the quasilinear case — so the estimated WTP $= \beta/\alpha$ is *exactly* compensating and equivalent variation, a true welfare magnitude rather than a mere coefficient ratio.

B1. By definition of the expenditure function, the minimum cost to reach utility u at prices $\mathbf{p}$ is $e(\mathbf{p}, u)$. For a change $\mathbf{p}^0 \to \mathbf{p}^1$ at income m (with $u^0 = V(\mathbf{p}^0, m)$, $u^1 = V(\mathbf{p}^1, m)$): $\mathbf{CV} = e(\mathbf{p}^1, u^0) - e(\mathbf{p}^0, u^0) = e(\mathbf{p}^1, u^0) - m$ — the extra money needed *at the new prices* to reach the *original* utility u^0 (reference u^0).

EV $= e(\mathbf{p}^1, u^1) - e(\mathbf{p}^0, u^1) = m - e(\mathbf{p}^0, u^1)$ — the money change *at the original prices* equivalent to reaching the *new* utility u^1 (reference u^1). By Shephard's lemma these equal $\int_{p^0}^{p^1} h(p, u^0)\, dp$ and $\int_{p^0}^{p^1} h(p, u^1)\, dp$ respectively — areas under the Hicksian demand at the *original* vs *new* utility (verified closed forms: CV $= m[(p^1/p^0)^a - 1]$, EV $= m[1 - (p^0/p^1)^a]$).

B2. Willig (1976): $\dfrac{|\text{CV} - \Delta\text{CS}|}{|\Delta\text{CS}|} \approx \dfrac{\eta\,|\Delta\text{CS}|}{2m}$, where η is the income elasticity of demand for the good. CS $\approx$ CV when (i) the *income elasticity* η is *small* (weak income effects — the good's demand barely responds to income), and (ii) the *surplus is a small share of income* ($|\Delta\text{CS}|/m$ small — the value is modest relative to the budget). Under both, the error is second-order and CS is an excellent proxy (verified: the formula 17.3% tracks the actual 15–20% error in a case where *both* conditions *fail* — big price change, $\eta = 1$).

B3. For $u = v(x) + y$ with y the numeraire (price 1), the consumer maximizes $v(x) + (m - px)$, giving FOC $v'(x) = p$ — so *optimal x depends only on p, not on m* (no income effect). Indirect utility $V(p, m) = v(x^*(p)) + m - p\,x^*(p) = m + [v(x^*) - px^*]$, so $\partial V/\partial m = 1$ — the marginal utility of income is *constant*. Because Marshallian demand $x^*(p)$ is income-independent, it *equals* the Hicksian demand at every utility level, so all three welfare integrals are the *same area* under the *same* demand curve: CV $=$ ΔCS $=$ EV (verified: all $= 1/p^0 - 1/p^1 = 0.5$).

C1. A mean WTP of €50 for a finfish improvement is an estimate of the *mean compensating variation* ($-$CV) of the improvement: the money that, taken from the household *after* the improvement, restores its *status-quo utility* — the reference is "without the improvement" and the property right is that the household must *pay* to obtain it. Assumptions for validity: (i) *incentive compatibility* — the elicited amount is the true maximum WTP (Chapter 28); (ii) *constant marginal utility of income* (quasilinear) — so the €50 is an exact CV and the CV/EV wedge is negligible, *or*, if income effects matter, an acknowledgment that CV (WTP) $<$ EV (WTA) and €50 is specifically the CV; (iii) the value is not so large a share of income that Willig-type curvature makes the reference-utility choice consequential; (iv) the survey measures the intended good (construct validity). State it as: "mean CV (WTP) = €50, valid as an exact welfare measure under constant marginal utility of income; it is the status-quo-referenced measure and would differ from the WTA/EV measure if income effects are present."

C2. Summing a WTP-based benefit and a WTA-based cost mixes *two different welfare measures* (CV and EV) with *different reference utilities and property rights* — they are *not commensurable*, and WTA typically exceeds WTP (sometimes substantially), so the net figure is biased and its sign can even be ambiguous. *Fix:* harmonize to a *single* welfare measure consistent with the policy's property-right framing — usually express *both* the benefit and the cost as *compensating variations* (status-quo reference): the benefit as WTP-for-gain ($-$CV) and the cost as WTP-to-avoid ($-$CV of the loss), or both as EV if the ownership framing is appropriate. Only after both are the *same* Hicksian measure with the *same* reference can they be summed into a net welfare change (and note whether the quasilinear assumption makes the CV/EV distinction moot; if not, report the sensitivity of the CBA to the measure choice).

C3. In a WTP-space EC-MXL, the cost coefficient α (marginal utility of income) is *random across individuals* (e.g. log-normally distributed), so the log-sum welfare is *not* a single division by one α. The exact per-individual CV is $\frac{1}{\alpha_i}[\ln \sum e^{V_j^1} - \ln \sum e^{V_j^0}]$ *at that person's α_i*; *population* welfare is the *expectation* of this over the distribution of α (and the error components) — computed by *simulation* (average the individual log-sum CVs across draws of α_i), not by plugging in a mean α. Two cautions: (i) *Jensen's inequality* — $\mathbb{E}[1/\alpha] \neq 1/\mathbb{E}[\alpha]$, so using the mean α biases aggregate welfare (Chapter 7); (ii) if α has appreciable mass near *zero*, $1/\alpha$ (and hence WTP) has *near-Cauchy heavy tails*, so the mean CV can be dominated by a few individuals or even fail to have a well-defined mean — report the *median* and distribution of welfare, and consider bounding/trimming (Chapters 7, 23). The €-benefit is a *distributional* welfare object integrated over α-heterogeneity, not a single log-sum divided by a point estimate.

34.12 Spaced review prompts

+1 day. From memory: define CV, EV, CS; the ordering CV $\geq$ CS $\geq$ EV (normal good, price increase) and why; the quasilinear collapse; the log-sum welfare formula.

+1 week. Explain Marshallian vs Hicksian demand and Shephard's lemma; the Willig conditions; why linear-in-cost logit makes WTP exact welfare.

+1 month. For your CV and DCE papers, state which welfare measure each value is, check the Willig / quasilinear conditions, and verify the log-sum welfare integrates over α-heterogeneity. Close with the spine sentence about *which* welfare object each number is and *under what assumption* it is valid.

34.13 How this chapter connects to the rest of the book

- **Chapters 2–3** (constrained optimization, envelope theorem) supply the duality (Roy, Shephard) behind CV/EV; **Chapters 22–23** (random utility, WTP space) supply the constant-α that makes DCE WTP exact welfare; **Chapter 7** (Jensen, heavy tails) governs aggregating welfare over α-heterogeneity.
- This chapter *opens Part VIII* and underwrites everything downstream: **Chapter 35–37** (public goods, non-use values, environmental valuation) apply CV/EV to *non-market* goods; **Chapter 40–42 (MDCEV)** extend welfare to *multiple-discrete* allocations (your ecosystem-service budget); **Chapters 45–46 (CBA)** *aggregate* these welfare measures into net present value (your FRD appraisal); **Chapter 49 (benefit transfer)** *moves* welfare measures across sites (your meta-analysis), where the CV/EV/CS distinction must be preserved.
- **Chapter 28** (incentive compatibility) is what makes the *elicited* CV/EV *truthful*; **Chapter 30** (identification) is what makes it *identified*; this chapter is what makes it *meaningful*.

This chapter is the hinge of the book: it converts the *statistical* WTP of Parts I–VII into an *economic welfare* magnitude, and it names the assumption — constant marginal utility of income — on which that conversion rests.

34.14 Research translation task

For your **CV, DCE, and CBA work**, produce a two-page welfare-measurement note:

1. **Label every value.** For each WTP/benefit you report, state whether it is *CV, EV, or CS*, and the *reference utility / property right* it embeds (WTP = status-quo CV; WTA = post-change EV).
2. **Willig check.** For any Marshallian-surplus-based figure, report the income elasticity and surplus-to-income ratio; if either is large, use the exact Hicksian measure or bound the error.
3. **Quasilinearity.** State that your DCE WTP is exact CV/EV *under constant marginal utility of income* (linear cost); if you use an income-nonlinear cost, acknowledge the CV/EV wedge.
4. **Log-sum welfare.** For DCE policy scenarios, compute the *log-sum welfare integrated over the α-distribution* (WTP-space MXL), reporting the *distribution* of per-person CV (median + tails), not a single division by mean α.
5. **Aggregation harmony.** In the CBA, confirm all benefit/cost components are the *same* welfare measure before summing.
6. **Close** with the spine sentence: *"Each value I report is a [CV/EV], valid as an exact welfare measure under [constant marginal utility of income / stated bounds], referenced to the [status-quo/post-change] property right, and aggregated only with commensurable measures."*

Carry this into Part IX (public goods / non-market valuation), Part X (MDCEV), and Parts XI–XII (CBA and benefit transfer), where these welfare measures are applied, extended, aggregated, and transferred.

34.15 Bibliography

Core. - Mas-Colell, Whinston & Green, *Microeconomic Theory*, Ch. 3 — the definitive treatment of indirect utility, expenditure, duality, and the welfare measures CV/EV/CS (§34.2–§34.6). *(Verify chapter/edition.)* - Just, Hueth & Schmitz, *The Welfare Economics of Public Policy* — applied welfare measurement for policy, CV/EV/CS in practice (§34.3–§34.5). *(Verify edition.)* - Freeman, Herriges & Kling, *The Measurement of Environmental and Resource Values: Theory and Methods* — the standard reference tying welfare theory to non-market valuation, exactly your application (§34.3, §34.7). *(Verify edition.)*

Applied. - Willig (1976), "Consumer's Surplus Without Apology," *American Economic Review* — the Willig bounds on the CS approximation to CV/EV (§34.5). *(Verify volume/pages.)* - Small & Rosen (1981), "Applied Welfare Economics with Discrete Choice Models," *Econometrica* — the log-sum welfare formula for discrete choice, the foundation of your DCE welfare calculations (§34.7). *(Verify volume/pages.)* - Hanemann (1991), "Willingness to Pay and Willingness to Accept: How Much Can They Differ?," *American Economic Review* — the CV/EV (WTP/WTA) divergence and its determinants (§34.3, §34.6). *(Verify volume/pages.)*

Advanced. - Hausman (1981), "Exact Consumer's Surplus and Deadweight Loss," *American Economic Review* — recovering the exact Hicksian welfare measure from Marshallian demand (§34.5). *(Verify volume/pages.)* - McFadden (1999) / Herriges & Kling on computing welfare in mixed-logit and heterogeneous-α models — the α-integration of §34.7. *(Verify citations.)* - Bockstael & McConnell, *Environmental and Resource Valuation with Revealed Preferences* — the theory of welfare measurement with revealed and stated preferences (§34.7). *(Verify edition.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Willig / Small–Rosen / Hanemann / Hausman citations before using them, per the book's no-unverified-citation rule.

End of Chapter 34, opening Part VIII. Next: Chapter 35 — Public Goods, Non-Use Values, and the Total Economic Value Framework, which extends the welfare measures of this chapter from private market goods to the non-market, non-rival* environmental goods at the centre of your finfish/ecosystem valuation — use vs non-use (existence, bequest) values, the total-economic-value taxonomy, and why these goods require stated preference in the first place. Tell me when to write it, or request a different chapter.*

Chapter 35 — Public Goods, Non-Use Values, and the Total Economic Value Framework

Part VIII: Welfare Foundations

> **The spine question, applied to this chapter.** Chapter 34 gave WTP its meaning as compensating variation. But *why* must WTP for a finfish stock or an ecosystem be *elicited* by survey at all — why is there no market price to read off? The answer is that these are **public goods** (non-rival, non-excludable) carrying **non-use values** (existence, bequest) that leave *no behavioural trace* — so markets fail, revealed preference is silent, and stated preference is the *only* window. This chapter builds the economics of *why* your entire method exists, and the taxonomy — **Total Economic Value** — of *what* you are measuring. The spine question here: *what kind of good is my finfish/ecosystem improvement, which components of its value can only be captured by asking, and how do I aggregate elicited WTP into a total benefit without double-counting, misjudging the extent of the market, or being fooled by scope insensitivity?*

35.0 Conceptual motivation: why your method has to exist

Chapters 22–34 taught you to *estimate* and *interpret* WTP. This chapter answers a prior question: *why is a survey necessary in the first place?* For a loaf of bread you read the price; for a finfish stock, a restored reef, or the existence of a species, there is *no* price — and the reasons *why* there is no price are exactly the reasons your CV and DCE methods are the *only* tools available. Understanding those reasons is not optional background; it determines *what* you can validly measure, *how* you aggregate it, and *which* validity threats you must guard against.

Your PDF's welfare/public-goods section names the pieces. Environmental goods are **public goods**: non-rival (my enjoyment of a healthy fish stock does not diminish yours) and non-excludable (you cannot fence off the ocean's existence value). These two properties cause **market failure** — **free-riding** and under-provision (verified below: voluntary provision delivers only two-thirds of the efficient level) — which is *why* the good is under-supplied and *why* its value cannot be inferred from any market. They also mean the value must be **aggregated by *summing* WTP across the affected population** (vertical summation — everyone consumes the *same* quantity), not by a market-clearing price, which puts the *extent of the market* (whose values count) at the centre of every benefit estimate (verified: the aggregate swings by hundreds of millions depending on mean-vs-median and population).

Crucially, environmental goods carry **non-use (passive-use) values** — the **existence** value of knowing a species persists, the **bequest** value of leaving it for future generations, altruistic value for others — that are *entirely divorced from any behaviour*. A person who will never visit a reef may still hold a positive, real WTP for its protection. Because there is *no* trip taken, *no* purchase made, *no* revealed-preference trace,

these values are invisible to every method except directly asking — which is the fundamental justification for stated preference and, per your PDF, the reason CV/DCE are indispensable rather than merely convenient. The **Total Economic Value (TEV)** framework is the taxonomy that organizes all of this — use value, option value, and the non-use components — and it is the map you need to avoid *double-counting* (summing overlapping components) and *gaps* (omitting non-use value from a CBA).

This chapter builds the public-goods economics (Samuelson condition, free-riding, vertical summation — all verified), the TEV taxonomy, the aggregation arithmetic (extent of market, mean vs median — verified), and the scope/embedding/adding-up validity tests that your PDF flags — the diagnostics that separate a real value signal from a survey artifact. It is the bridge from Chapter 34's *welfare theory* to Part IX's *valuation practice*, and it explains, in economic terms, *why your research program is necessary and what exactly it is measuring.*

35.1 Prerequisites

From **Chapter 34**: CV/EV as the welfare measures, the quasilinear constant-marginal-utility-of-income case, and that WTP is a money-metric of utility change. From **Chapter 2–3**: the marginal rate of substitution and marginal conditions for efficiency. From **Chapter 4**: discounting an annual benefit stream to present value (aggregation over time). From **Chapter 28**: incentive compatibility (why free-riding also threatens *elicitation*, not just provision).

New notation. G (quantity of a public good), MRS_i (individual i's marginal rate of substitution / marginal WTP), MRT/MC (marginal rate of transformation / marginal cost), TEV (total economic value), and the value components: direct use, indirect use, option, existence, bequest, altruistic.

35.2 Public goods: non-rivalry, non-excludability, and market failure

Definition 35.1 (Public good). A **pure public good** has two properties: **non-rivalry** — one person's consumption does not reduce the quantity available to others (the good is consumed *jointly*, at the *same* level G by everyone) — and **non-excludability** — no one can be prevented from consuming it (you cannot withhold it from non-payers). A healthy fish stock's existence, clean coastal water, biodiversity, and climate stability are (close to) pure public goods. Contrast a *private good* (rival and excludable — a caught fish: if I eat it, you cannot; and the seller can withhold it).

Why markets fail. Non-excludability destroys the incentive to pay: if you cannot be excluded, you can consume without paying — **free-riding** — so a private market under-provides the good (or provides none). Non-rivalry means the *efficient* price is zero (the marginal cost of one more person consuming is nil), which no profit-seeking firm will charge. Together, these mean *there is no market, no market price, and hence no revealed-preference signal* — the value must be measured *some other way.*

The Samuelson condition (efficient provision). For a *private* good, efficiency requires each consumer's MRS to equal the price (all equal at the margin). For a *public* good — consumed jointly — efficiency requires the *sum* of the individual marginal rates of substitution to equal the marginal cost:

$$\sum_i \text{MRS}_i = \text{MRT (marginal cost)} \qquad \textbf{(Samuelson condition)}.$$

Because everyone consumes the *same* G, society's willingness to pay for one more unit is the *sum* of what *each* person would pay — the demands are summed *vertically* (§35.3). Efficient provision equates this summed marginal WTP to marginal cost.

Worked Example 35.1 (Samuelson optimum vs free-riding, verified). Three individuals with marginal benefit $\text{MB}_i(G) = a_i - G$ for $a = (10, 8, 6)$, marginal cost = 6:

Provision rule	G	Logic
Samuelson optimum	**6.0**	$\sum_i (a_i - G) = \text{MC} \Rightarrow 24 - 3G = 6$
Individual private optima	4, 2, 0	each sets own $\text{MB}_i = \text{MC}$
Nash (voluntary provision)	**4.0**	highest-value person provides; others free-ride

(Verified.) The efficient level is $G^* = 6$, but voluntary provision delivers only $G_{\text{Nash}} = 4$ — a **33% shortfall** — because individuals 2 and 3 *free-ride* on individual 1's contribution, and no one accounts for the benefit their contribution confers on others. The **deadweight loss** from this under-provision is 6 money units (verified — the area between summed marginal benefit and marginal cost over the missing units). **This under-provision is the economic rationale for public intervention** (and for valuing the good to justify that intervention): because the market fails, the *benefits of provision must be measured non-market* — by your CV/DCE — to inform whether public provision is worthwhile.

> **Active recall 35.A.** Why does non-excludability cause under-provision, and why does efficiency require *summing* MRS across people (not equating each to price)? *(Answer: non-excludability means people cannot be prevented from consuming even if they do not pay, so each has an incentive to* free-ride* on others' contributions — the private, voluntary equilibrium provides where the *highest-value individual's own* marginal benefit equals cost, ignoring the benefit to everyone else, so provision falls short of efficient (verified: Nash G=4 vs optimum G=6, a 33% shortfall). Efficiency requires *summing* MRS because the good is *non-rival*: one more unit is consumed *jointly* by everyone at the same level G, so society's marginal benefit of that unit is the *sum* of every person's marginal WTP — the Samuelson condition $\sum_i \text{MRS}_i = \text{MC}$. For a private (rival) good each unit goes to *one* person, so demands are summed *horizontally* and each consumer's MRS equals the common price.)*

35.3 Vertical summation and the aggregation of public-good value

The non-rivalry of a public good changes *how* demands aggregate — and this governs how you turn survey WTP into a total benefit.

Vertical vs horizontal summation. For a *private* good, the market demand at a given *price* is the *sum of quantities* each consumer buys — **horizontal summation** (add quantities at each price). For a *public* good, everyone consumes the *same quantity* G, so the aggregate *willingness to pay* for that quantity is the *sum of individual WTP* at that G — **vertical summation** (add WTP/prices at each quantity).

Worked Example 35.2 (Vertical summation, verified). At $G = 3$, individual marginal WTP = $(7, 5, 3)$:

Summation	Aggregate
Public good (vertical)	**15** $= 7 + 5 + 3$ (sum WTP at the common G)
Private good (horizontal)	would sum *quantities* at a common price

(Verified.) The aggregate social value of the public good at $G = 3$ is the *vertical* sum 15 — everyone gets the same G, so you *add up* what each is willing to pay for it. **This is why your survey aggregation *sums* WTP across the population**: a coastal improvement that everyone "consumes" jointly has a total value equal to the *sum* of each household's WTP, not an average price times a quantity. And it immediately raises the pivotal empirical question of *the extent of the market* — *whose* WTP enters the sum (§35.5).

Stop and think 35.1. A colleague aggregates the benefit of a fish-stock improvement by multiplying the *market price of fish* by the tonnage gained. What value has this captured, and what has it missed? *(Answer: multiplying market fish price by tonnage captures only the* commercial/direct-use market value* of the harvested fish — a *private-good* component. It *misses* the entire *public-good, non-market value*: the *recreational* use value (angling experience), the *indirect use* value (the stock's role in the ecosystem), the *option* value (keeping the option to use it later), and — most importantly — the *non-use values* (existence value of a healthy stock, bequest value for future generations) that many people hold *without ever harvesting a fish* and that leave *no market trace*. Because the improvement is largely a *public good* consumed jointly, its social value is the *vertical sum of every affected person's WTP* (verified aggregation logic), most of which is invisible to the fish price. The market-price method therefore drastically *understates* the total benefit — precisely the gap that stated preference exists to fill.)*

35.4 The Total Economic Value framework

The value of an environmental good has many components, some behavioural and some not. The **Total Economic Value (TEV)** framework is the taxonomy that organizes them — the map that ensures a valuation is *complete* (no omitted non-use value) and *non-redundant* (no double-counting).

Definition 35.2 (Total Economic Value). TEV partitions the total value of an environmental good into **use values** and **non-use (passive-use) values**:

Use values (require some interaction with the good): - **Direct use value** — from *direct* consumption or interaction: commercial and recreational fishing, harvesting, diving, coastal recreation (your DCE's coastal-recreation exposure taps this). - **Indirect use value** — from *ecosystem functions/services* that support wellbeing indirectly: a fish stock's role in the food web, coastal protection, water purification, nursery habitat. - **Option value** — the value of *preserving the option* to use the good in the *future* (even if not used now), reflecting uncertainty about future demand and irreversibility of loss.

Non-use (passive-use) values (require *no* interaction — pure "knowing"): - **Existence value** — the value of *knowing the good exists*, independent of any use, now or ever (knowing a species or reef persists). - **Bequest value** — the value of *preserving the good for future generations* (leaving it for one's children/descendants). - **Altruistic value** — the value of others (contemporaries) being able to enjoy the good.

$$\text{TEV} = \underbrace{\text{Direct} + \text{Indirect} + \text{Option}}_{\text{use}} + \underbrace{\text{Existence} + \text{Bequest} + \text{Altruistic}}_{\text{non-use}}.$$

Why non-use value forces stated preference. *Revealed-preference* methods (travel cost, hedonic pricing) infer value from *behaviour* — trips taken, prices paid. But non-use values leave *no behavioural trace*: a person who values a species' existence but never visits it takes no trip, buys no permit, pays no premium. **There is no behaviour to reveal the value**, so revealed preference is *structurally blind* to non-use value. The *only* way to capture it is to *ask* — which is why **stated preference (CV, DCE) is the sole method capable of measuring non-use value**, and why, for goods dominated by existence/bequest value (endangered species, remote ecosystems, biodiversity), it is not merely one option but the *only* option. This is the deepest justification for your entire methodological program.

Avoiding double-counting and gaps. The TEV taxonomy is also a *discipline*: (i) *do not double-count* — a single CV question that elicits WTP for "protecting the fish stock" typically captures a *bundle* of use + non-use value, and you must not *then* add a separate travel-cost use value on top (you would count use value twice); (ii) *do not leave gaps* — a CBA that includes only the commercial harvest value *omits* the (often larger) non-use value, understating the benefit. The correct practice is to define *precisely which TEV components your elicitation captures* and to combine methods so the components are *exhaustive and mutually exclusive* (§35.7; Chapter 45).

Active recall 35.B. What distinguishes use from non-use value, why can only stated preference capture non-use value, and how does the TEV framework prevent double-counting? *(Answer:* use* value requires some interaction with the good (direct consumption, indirect ecosystem service, or preserving the *option* to use it), leaving a *behavioural trace* (trips, purchases); *non-use* (passive-use) value — existence, bequest, altruistic — is held *without any interaction*, pure "knowing the good exists/persists for others/for the future." Only *stated preference* can capture non-use value because there is *no behaviour to reveal it* — revealed-preference methods (travel cost, hedonics) infer value from actions, and a person who never uses the good takes no action, so their (real, positive) non-use value is *structurally invisible* to those methods; you must *ask*. TEV prevents double-counting by forcing you to specify *exactly which components* each elicitation captures — a CV "protect the stock" question usually bundles use + non-use, so you must *not* separately add a travel-cost use value on top (that counts use twice); components must be exhaustive and mutually exclusive before summing.)*

35.5 Aggregation: the extent of the market and mean vs median

Turning an individual WTP into a *total benefit* for a CBA requires two decisions that can swing the number by an order of magnitude — and both are *economic/ethical* choices, not statistical ones.

The extent of the market. Because public-good value is the *vertical sum* over the *affected population*, you must decide *whose* WTP counts — the **extent of the market**. For a local reef, is it the coastal town, the region, the nation, or (for a globally-known species with existence value) the world? The choice is consequential: existence value is held even by those far away, so a *narrow* market omits real value, while an *over-broad* market (summing tiny WTPs over billions) can inflate it implausibly. The extent must be *justified* by *who genuinely holds the value* and *who is affected by the policy* — and stated transparently, because the aggregate scales *linearly* with it.

Mean vs median WTP. The aggregate also depends on which *central tendency* you multiply by the population. The **mean** WTP gives the *total* (utilitarian) benefit — the correct measure for a *potential-Pareto / Kaldor–Hicks* CBA (summing everyone's money-metric welfare). The **median** WTP is the *majority-voting* measure — what a referendum would pass. For *skewed* WTP distributions (common — a few high-value respondents pull the mean up), *mean* > *median*, and the choice matters enormously.

Worked Example 35.3 (Aggregation, verified). Mean WTP €50/household/year, median €32, over 1.2 million households, discounted at 3% over 25 years:

Basis	Annual aggregate	Present value (25 yr, 3%)
Mean WTP (€50)	**€60.0 million/yr**	**€1,044.8 million**
Median WTP (€32)	**€38.4 million/yr**	—

(Verified; annuity factor 17.41.) The mean-based aggregate (€60M/yr, €1.04 *billion* present value) is *56% larger* than the median-based (€38.4M/yr) — the *same survey* yields wildly different totals depending on the mean/median choice, and the whole figure scales with the assumed 1.2M-household extent of the market. **This is where a valuation study's number is *made* — and where it is most fragile.** Report *both* mean and median, *justify* the extent of the market, and show the aggregate's *sensitivity* to both; a single headline total without these is a number pretending to a precision it does not have (Chapter 46).

Stop and think 35.2. Your CV study gives mean WTP €50 and median €32, and you must produce a single benefit figure for a CBA. Which do you use, and what must you report? *(Answer: for a* Kaldor–Hicks / potential-Pareto* CBA — the standard efficiency test that sums everyone's money-metric welfare — the correct measure is the *mean* WTP times the population (the *total* benefit), so €50 × 1.2M = €60M/yr, PV ≈ €1.04bn (verified). But you must report: (i) *both*

the mean-based and median-based aggregates (€60M vs €38.4M), because the skew makes them differ by ~56% and the median is the majority-voting/referendum measure relevant for political feasibility; (ii) an explicit, justified *extent of the market* (why 1.2M households — who is affected / holds the value), since the total scales linearly with it; (iii) a *sensitivity analysis* over both choices and the discount rate. The mean is the efficiency-correct point estimate, but presenting it *without* the median, the market-extent justification, and the sensitivity would overstate the robustness of a fundamentally range-valued benefit.)*

35.6 Scope, embedding, and adding-up: the validity tests

If WTP is a real economic value, it must behave like one — in particular, *more* of the good should be worth *more*. The **scope test** and its relatives are the key validity checks, and your PDF flags them.

The scope test (sensitivity to scope). Economic theory requires WTP to be *increasing in the quantity/quality* of the good: protecting *five* lakes should command *more* WTP than protecting *one*. **Scope insensitivity** — WTP essentially *flat* across widely different quantities — is a red flag that respondents are *not* valuing the specific good but expressing a general "warm glow" of giving, a symbolic attitude, or responding to the *act* of contributing rather than its *object* (Chapter behavioural foundations). A credible stated-preference study *demonstrates* scope sensitivity.

Embedding. Embedding (part–whole bias) is the finding that WTP for a good is roughly the *same* whether it is valued *on its own* or as *part of* a more inclusive good — e.g., WTP to clean one lake ≈ WTP to clean all the region's lakes. It signals that respondents are valuing the *category* ("environmental protection") rather than the *specific scope*, undermining the claim that the number is a scope-specific welfare measure.

Adding-up. The **adding-up test** (Diamond) requires that the WTP for the whole equal the *sum* of the WTPs for its parts *elicited sequentially* — a strong internal-consistency condition. Failures indicate that sequence and framing, not the good's magnitude, drive responses.

Worked Example 35.4 (Scope and embedding, verified). WTP for 1 lake = €30, for 5 lakes = €42; five separately-valued lakes sum to €90:

Test	Result	Reading
Basic scope	€42 > €30	*passes* — WTP rises with scope
Proportionality	ratio 1.40 (sub-proportional)	expected (diminishing marginal value) — *not* a failure
Adding-up	joint €42 vs sum-of-parts €90	**embedding effect** (joint ≪ sum)

(Verified.) The study *passes* the *basic* scope test (more lakes → more WTP) and the sub-proportional ratio is *fine* (diminishing marginal value is economically sensible — the fifth lake adds less than the first). But the *large gap* between the joint valuation (€42) and the sum of separate valuations (€90) reveals an **embedding effect** — respondents value "protecting lakes" as a category more than the incremental scope. **Report scope tests explicitly:** a study that shows *some* scope sensitivity but flags embedding is *honest*; one that shows *no* scope sensitivity has a serious validity problem (the WTP may be a symbolic attitude, not a welfare measure). For your finfish work, *build scope variation into the design* (different stock-improvement magnitudes) and *test* that WTP responds — it is among the strongest evidence that your numbers are real values (your PDF's validity concern).

Active recall 35.C. What is the scope test, why is scope insensitivity a validity red flag, and is sub-proportional WTP (WTP for 5 lakes < 5× WTP for 1) a failure? *(Answer: the* scope test* checks whether WTP *increases with the quantity/quality* of the good, as economic theory requires — more of the good should be worth more (verified: €42 for 5 lakes > €30 for 1 passes). *Scope*

insensitivity — WTP roughly flat across widely different quantities — is a red flag because it suggests respondents are valuing the *act of contributing* or a *symbolic category* ("the environment") rather than the *specific good and its magnitude*, so the number is a warm-glow/attitude, not a scope-specific welfare measure. *Sub-proportional* WTP is *not* a failure: *diminishing marginal value* is economically expected (the fifth lake adds less welfare than the first), so WTP(5) $<$ 5×WTP(1) is normal and correct — what *would* be a problem is *no* increase (insensitivity) or a large *embedding* gap between joint and sum-of-parts valuations, indicating category-valuing rather than scope-valuing.)*

35.7 Failure modes

F1. Using market price for a public good. Valuing an environmental improvement by a market price, capturing only direct-use value. *Fix:* recognize the public-good, non-use components; use stated preference for the full TEV (§35.2, §35.4).

F2. Omitting non-use value from a CBA. Counting only commercial/recreational use, understating the benefit. *Fix:* include existence/bequest value via stated preference; state the TEV components covered (§35.4).

F3. Double-counting TEV components. Adding a CV "protect the stock" value (which bundles use + non-use) to a separate travel-cost use value. *Fix:* define which components each method captures; keep them mutually exclusive (§35.4).

F4. Unjustified extent of the market. Summing WTP over an arbitrary population. *Fix:* justify who holds the value / is affected; report sensitivity to the market boundary (§35.5).

F5. Mean/median conflation. Reporting one aggregate without the other on a skewed distribution. *Fix:* report both; mean for efficiency, median for majority; sensitivity analysis (§35.5).

F6. Ignoring scope/embedding. Reporting WTP with no scope test, or with scope insensitivity unaddressed. *Fix:* build scope variation into the design; test and report sensitivity; flag embedding (§35.6).

F7. Free-riding in elicitation. Assuming stated WTP is truthful when the good is a public good (strategic incentives). *Fix:* incentive-compatible, consequential design (Chapter 28) — the free-riding of provision also threatens *elicitation* (§35.2).

35.8 Diagnostic questions for your own papers

1. **Good type.** Have I characterized my good's public-good properties (non-rivalry/excludability) and identified why no market price exists? *(§35.2, F1.)*
2. **TEV components.** Which TEV components (use / option / existence / bequest) does my elicitation capture, and is the set exhaustive and non-overlapping? *(§35.4, F2–F3.)*
3. **Non-use value.** Have I captured non-use value (the part invisible to revealed preference), or omitted it? *(§35.4, F2.)*
4. **Extent of the market.** Is my aggregation population justified by who holds the value / is affected, with sensitivity reported? *(§35.5, F4.)*
5. **Mean vs median.** Do I report both aggregates and justify the choice for the CBA? *(§35.5, F5.)*
6. **Scope.** Did I build scope variation into the design and test/report sensitivity to scope (and embedding)? *(§35.6, F6.)*
7. **Elicitation incentives.** Given the public-good/free-riding setting, is my elicitation incentive-compatible? *(§35.2, F7; Chapter 28.)*

35.9 Exercises

(A) Comprehension.

A1. Define a pure public good and explain why its two properties cause market failure and under-provision.

A2. State the TEV framework and explain why non-use value requires stated preference.

A3. Why is public-good value aggregated by vertical summation, and what does this make the "extent of the market" question so important?

(B) Derivation / reasoning.

B1. Derive the Samuelson condition and contrast it with the private-good efficiency condition; show under-provision under voluntary contribution.

B2. Explain vertical vs horizontal summation and why public goods use the former.

B3. For a skewed WTP distribution, explain why mean and median aggregates differ and which is correct for a Kaldor–Hicks CBA.

(C) Research application / critique.

C1. Design the TEV decomposition for your finfish valuation and state which components your CV and DCE each capture (avoiding double-counting).

C2. Your reviewer says your CV number "isn't a real value — it's warm glow." Design the scope test that would answer this.

C3. Your CBA aggregates finfish-improvement WTP. Write the aggregation plan (extent of market, mean/median, discounting, sensitivity).

35.10 Solutions

A1. A *pure public good* is *non-rival* (one person's consumption does not reduce the amount available to others — consumed jointly at a common level G) and *non-excludable* (no one can be prevented from consuming it). *Non-excludability* causes *free-riding*: people consume without paying, so a private market cannot capture the value and under-provides (verified: voluntary provision gives G=4 vs efficient G=6, a 33% shortfall, deadweight loss 6). *Non-rivalry* means the efficient marginal price is *zero* (one more consumer costs nothing), which no firm will charge. Together they mean there is *no market and no price*, so provision falls short of efficient and the value cannot be read from any market — the rationale for public provision and for non-market valuation.

A2. TEV = use value (direct use + indirect use + option) + non-use value (existence + bequest + altruistic). Non-use value — the worth of *knowing* a good exists / preserving it for the future / for others — is held *without any interaction* with the good, so it leaves *no behavioural trace* (no trip, purchase, or price premium). Revealed-preference methods infer value from *behaviour*, so they are *structurally blind* to non-use value; the *only* way to measure it is to *ask* — stated preference (CV, DCE). For goods dominated by existence/bequest value, stated preference is therefore the *only* possible method.

A3. A public good is *non-rival*, so everyone consumes the *same* quantity G; society's willingness to pay for a unit of G is thus the *sum* of every affected person's WTP for it — *vertical* summation (add WTP at the common quantity), unlike a private good where you sum *quantities* at a common price (horizontal). This makes the *extent of the market* pivotal: the total benefit is the sum of WTP over *whoever holds the value*, so the aggregate scales *linearly* with how many people you include (verified: €50 × 1.2M = €60M/yr). Choosing the population too narrowly omits real (e.g. distant existence) value; too broadly inflates it — so the market boundary must be explicitly justified by who is affected / holds the value.

B1. *Private good:* efficiency maximizes total surplus, giving each consumer's $MRS_i = $ price = MC — all consumers equate their marginal WTP to the *same* price (the unit goes to *one* person). *Public good (Samuelson):* one unit of G is consumed *jointly* by everyone, so its social marginal benefit is $\sum_i \text{MRS}_i$, and efficiency sets $\sum_i \text{MRS}_i = \text{MC}$ (verified: $24 - 3G = 6 \Rightarrow G^* = 6$). Under *voluntary contribution*, each person contributes only until their *own* $\text{MRS}_i = \text{MC}$, ignoring the benefit to others, so provision is driven by the *highest-value* individual and everyone else free-rides — giving $G_{\text{Nash}} = 4 < G^* = 6$ (verified), an under-provision whose deadweight loss is the area between $\sum \text{MRS}$ and MC over the missing units ($= 6$).

B2. *Horizontal summation* (private good): at each *price*, add the *quantities* each consumer demands to get market demand — appropriate because each unit is *rival* (goes to one buyer), so total quantity is the sum of individual quantities. *Vertical summation* (public good): at each *quantity* G, add the *WTP (marginal value)* of each consumer — appropriate because the good is *non-rival* (everyone consumes the *same* G), so the social value of that common quantity is the sum of individual willingnesses to pay for it (verified: at G=3, WTP (7,5,3) sum vertically to 15). The direction of summation follows directly from whether the good is rival (add quantities) or non-rival (add values).

B3. For a *right-skewed* WTP distribution (a few high-value respondents), the *mean* exceeds the *median* (verified: €50 vs €32). The *mean* times population is the *total* money-metric welfare — the *utilitarian sum* of everyone's WTP — which is exactly the *Kaldor–Hicks / potential-Pareto* benefit measure a CBA needs (the policy passes if aggregate WTP exceeds cost, so *winners could compensate losers*). The *median* is the *majority-voting* measure (what a referendum passes) — relevant for political feasibility but *not* the efficiency criterion. So for a Kaldor–Hicks CBA the *mean* is correct; but because mean and median diverge under skew (here the mean-based aggregate is 56% larger), report both and the sensitivity.

C1. *TEV decomposition for finfish valuation:* **Direct use** — commercial harvest value + recreational angling value (the coastal-recreation exposure in the DCE). **Indirect use** — the stock's ecosystem-support functions (food web, nursery habitat). **Option value** — preserving the option of future use given uncertainty/irreversibility. **Existence value** — knowing a healthy finfish stock persists. **Bequest value** — preserving it for future generations. *Which method captures what:* the **DCE** (attributes including stock level, coastal recreation, cost) primarily captures *use* value (direct/recreational) and, if attributes are framed to include ecological state, some *indirect/non-use*; the **CV** "protect the stock" question typically captures a *bundle of use + non-use* value (respondents include existence/bequest). *Avoiding double-counting:* do *not* add the CV total (which already includes use value) to a separate travel-cost/DCE use value — instead, either (i) use CV for the *total* TEV and the DCE for the *marginal* trade-offs among attributes, being explicit they overlap, or (ii) design the CV to elicit *non-use value specifically* (e.g., a scenario stripping out use) so it can be *added* to a separately-estimated use value. State the components each instrument captures and keep the summed set mutually exclusive and exhaustive.

C2. *Scope test design:* administer the valuation across *multiple, clearly-differentiated magnitudes* of the finfish improvement — e.g., restoring the stock to enable a *small* vs *large* increase in abundance/geographic extent (a 20% vs 60% recovery, or one estuary vs the whole coast) — using a *split-sample* (between-subjects, to avoid within-subject anchoring) or a within-design scope attribute in the DCE. *Prediction under "real value":* WTP should *increase* with scope (more recovery → higher WTP), plausibly *sub-proportionally* (diminishing marginal value — that is fine, verified). *Prediction under "warm glow":* WTP is *flat* across scope (insensitive) — respondents pay for the *act*, not the *amount*. *Analysis:* test H_0: WTP equal across scope vs H_1: increasing (a one-sided test; and an *equivalence/adding-up* check for embedding, Chapter 33). Report the scope elasticity. If WTP *rises* significantly with scope, the "warm glow" critique is answered (the number tracks the good's magnitude); if *flat*, the reviewer is right and the number is not a scope-specific welfare measure. Building this scope variation into the design *pre-empts* the critique.

C3. *Aggregation plan:* (i) *Extent of the market* — define the affected/valuing population explicitly (e.g., coastal-region households for use value; a *broader* population for existence value, since non-users hold it too), justify the boundary by who is affected and who plausibly holds non-use value, and report *sensitivity* to alternative boundaries (the total scales linearly with it). (ii) *Central tendency* — compute the aggregate with the *mean* WTP (the Kaldor–Hicks efficiency measure) as the primary figure, and *also* the *median*-based aggregate; on a skewed distribution these differ materially (verified: €60M vs €38.4M/yr), so report both and

flag the skew. (iii) *Time* — express the improvement as an annual WTP stream and *discount* to present value at the appropriate social discount rate (e.g., 3%), reporting the annuity factor and PV (verified: €60M/yr → €1.04bn over 25 yr at 3%), with *sensitivity* to the discount rate (Chapter 6, 46). (iv) *Uncertainty* — propagate the WTP standard error (Krinsky–Robb/delta) and the market-extent and discount-rate ranges into a *benefit range*, not a point. (v) *TEV completeness* — state which value components the aggregate includes so it is neither double-counted nor gapped when combined with costs in the CBA. Present the benefit as a *justified range* with its key sensitivities, not a single spuriously-precise total.

35.11 Spaced review prompts

+1 day. From memory: define public good (non-rivalry/excludability); the Samuelson condition and free-riding under-provision; vertical vs horizontal summation; the six TEV components.

+1 week. Explain why non-use value requires stated preference; why the extent of the market and mean-vs-median swing the aggregate; what scope insensitivity signals.

+1 month. For your finfish work, decompose the TEV, state which components CV vs DCE capture, plan the aggregation (market extent, mean/median, discounting, sensitivity), and design the scope test. Close with the spine sentence about *what kind of good* you value and *which components* only asking can capture.

35.12 How this chapter connects to the rest of the book

- **Chapter 34** gave the welfare measures (CV/EV); this chapter applies them to *public, non-market* goods and explains why *asking* is necessary. **Chapter 28** (incentive compatibility) is the *elicitation* counterpart of the free-riding here — the same strategic incentive that under-provides the good can distort its stated valuation.
- This chapter *frames Part IX* (valuation practice): **Chapter 36–39** (CV and DCE methods, validity) operationalize the elicitation of TEV; **Chapter 40–42 (MDCEV)** value *multiple* ecosystem services jointly (the allocation across services in your ecosystem-service proposal); **Chapters 45–46 (CBA)** *aggregate* the TEV into net present value (your FRD appraisal), where the extent-of-market and mean/median choices of §35.5 are decisive; **Chapters 47–49 (meta-analysis, benefit transfer)** move TEV estimates across sites, where scope and TEV-component matching (§35.4, §35.6) govern transfer validity.
- **Behavioural foundations** (constructed preferences) explain *why* scope insensitivity and embedding (§35.6) arise; **Chapter 46** (sensitivity analysis) is where the aggregation fragility of §35.5 is confronted.

This chapter explains, in economic terms, *why your research program exists* — markets fail for public goods, non-use values are invisible to behaviour, so stated preference is the only window — and *what it measures* — the Total Economic Value, aggregated with explicit choices about market extent and central tendency, validated by scope.

35.13 Research translation task

For your **finfish/ecosystem valuation**, produce a two-to-three-page TEV-and-aggregation note:

1. **Good characterization.** State the public-good properties of the finfish/ecosystem improvement and why no market price captures its value.
2. **TEV decomposition.** List the value components (direct/indirect use, option, existence, bequest), and specify *which* your CV captures and *which* your DCE captures — ensuring the combined set is exhaustive and non-overlapping (no double-counting).

3. **Non-use value.** Identify the non-use component and confirm your design captures it (the part invisible to revealed preference).
4. **Aggregation.** Define and justify the *extent of the market*; report *mean and median* aggregates; discount the annual stream to present value; and present a *benefit range* with sensitivity to market extent, central tendency, and discount rate.
5. **Scope.** Build *scope variation* into the design and pre-commit to *testing and reporting* scope sensitivity (and embedding).
6. **Close** with the spine sentence: *"My good is a [public good] whose value is [TEV components], of which [non-use] is measurable only by asking; I aggregate over [justified market extent] using [mean/median] discounted at [rate], and I validate the value with a [scope] test."*

Carry this into Part IX (valuation methods), Part X (MDCEV), and Parts XI–XII (CBA, benefit transfer), where the TEV is elicited, extended, aggregated, and transferred.

35.14 Bibliography

Core. - Freeman, Herriges & Kling, *The Measurement of Environmental and Resource Values: Theory and Methods* — the standard reference on TEV, use/non-use value, and non-market valuation theory (§35.4). *(Verify edition.)* - Samuelson (1954), "The Pure Theory of Public Expenditure," *Review of Economics and Statistics* — the public-good efficiency (Samuelson) condition (§35.2). *(Verify volume/pages.)* - Champ, Boyle & Brown (eds.), *A Primer on Nonmarket Valuation* — accessible coverage of TEV, aggregation, and stated preference (§35.4–§35.5). *(Verify edition.)*

Applied. - Krutilla (1967), "Conservation Reconsidered," *American Economic Review* — the foundational articulation of existence/non-use value (§35.4). *(Verify volume/pages.)* - Carson, Flores & Meade (2001), "Contingent Valuation: Controversies and Evidence," *Environmental and Resource Economics* — scope, aggregation, and validity of non-use valuation (§35.5–§35.6). *(Verify volume/pages.)* - Bateman et al., *Economic Valuation with Stated Preference Techniques: A Manual* — the practical aggregation and extent-of-market guidance (§35.5). *(Verify edition.)*

Advanced. - Diamond & Hausman (1994), "Contingent Valuation: Is Some Number Better than No Number?," *Journal of Economic Perspectives* — the scope/embedding/adding-up critique (§35.6). *(Verify volume/pages.)* - Diamond (1996), "Testing the Internal Consistency of Contingent Valuation Surveys," *Journal of Environmental Economics and Management* — the adding-up test (§35.6). *(Verify volume/pages.)* - Smith & Osborne (1996) or Carson & Mitchell on scope sensitivity meta-analysis (§35.6). *(Verify citations.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Samuelson / Krutilla / Diamond–Hausman citations before using them, per the book's no-unverified-citation rule.

End of Chapter 35. Next: Chapter 36 — Contingent Valuation: Formats, Elicitation, and the Single-Bounded Dichotomous Choice, which turns from the theory* of what non-market value is to the *practice* of eliciting it — the referendum/dichotomous-choice format, the payment-card and SBDC designs at the centre of your CV study, the random-utility model behind the yes/no response, and the estimation of mean/median WTP from the discrete responses. Tell me when to write it, or request a different chapter.*

Part VIII Assessment — Welfare Foundations

Covers Chapters 34–35. Five components, in the order they are best attempted. Allow a full working day; the mock oral is the part that matters most for your viva.

1. Cumulative problem set

P1. For $u = x^{0.4}y^{0.6}$, $m = 200$, $p_y = 1$, p_x rising $1 \to 3$, compute CV, EV and ΔCS in closed form and verify the ordering.

P2. Compute the Willig approximation error for P1 and state whether consumer surplus is adequate here.

P3. Show that with quasilinear utility CV = EV = ΔCS, and identify exactly which step fails when the marginal utility of income varies.

P4. Using the Small–Rosen log-sum with $\alpha = 0.15$, compute the expected compensating variation of a policy raising V_1 from 0.8 to 1.4 in a three-alternative choice set.

P5. Decompose the total economic value of a restored reef into its components, and identify which your DCE attributes actually measure and which they do not.

2. Research critique

Take every WTP number your research program has produced and classify each as CV or EV, stating the reference utility and the implied property right. Identify any number that has been used in a CBA without passing the welfare-conditions gate, and state what would be required to license it.

3. Red-flag diagnosis

An author aggregates a mean household WTP of €38 across "the affected population," defined as everyone in the country, to produce a national benefit of €1.1 billion, and enters it into an ENPV. List the assumptions this aggregation makes, identify the two most likely to fail, and state what sensitivity analysis you would demand.

4. Mock oral examination

1. Why is Hicksian demand the right basis for welfare measurement if it is unobservable?
2. When do WTP and WTA diverge, and why?
3. What exactly makes your DCE's WTP an *exact* welfare measure?
4. What is warm glow and why does it threaten a welfare interpretation?
5. What is the extent of the market, and who decides it?

5. Integration mini-project

Write the welfare-interpretation appendix for your CV paper: for each estimate, the theoretical object it corresponds to, the assumptions that make the correspondence hold, the evidence you have for each assumption, and an honest statement of what the number would mean if the key assumption failed.

Part IX — Stated Preference in Practice

Chapter 36 — Contingent Valuation: Formats, Elicitation, and the Single-Bounded Dichotomous Choice

Part IX: Stated Preference in Practice

> **The spine question, applied to this chapter.** Chapter 35 established *why* non-market value must be elicited by asking. This chapter is about *how you ask* — and the discovery that the *format* of the question is not a presentational detail but a *substantive determinant of the number you get*. A single-bounded dichotomous choice, a payment card, an open-ended question, and a bidding game applied to the *same* good will yield *different* WTP distributions, because each format induces a different cognitive task, a different strategic incentive, and a different statistical identification problem. The spine question here: *given my elicitation format, what economic object is identified from these responses, what distributional assumption converts them into a mean or median WTP, and how fragile is that number to the tail I never observed?*

36.0 Conceptual motivation: the format is part of the measurement instrument

Your CV study uses a **single-bounded dichotomous choice (SBDC)** with **payment cards**, under **real and hypothetical** treatments and a **BRIS** mechanism. Every one of those design words is a decision with consequences for what you can estimate and how much you must assume. This chapter unpacks them.

The central lesson is uncomfortable but foundational: **contingent valuation does not "read off" a pre-existing WTP number; it elicits a response to a specific question, and the question's format shapes the response.** An open-ended "what is the most you would pay?" demands that the respondent *construct and report* a precise number — a cognitively hard task inviting anchoring, protest, and understatement. A dichotomous choice "would you pay €40?" demands only a *comparison* — cognitively easy, familiar (it mimics a market/referendum decision), and — critically — **incentive compatible** under the conditions of Chapter 28. That is why the NOAA panel and the modern literature favour the referendum/dichotomous-choice format, and why your SBDC design is the defensible core of your CV study.

But the ease of answering comes at a *statistical* price, and this is the trade-off at the heart of the chapter. A dichotomous choice yields only *one bit* of information per respondent — "my WTP is above €40" or "below €40." You never observe WTP; you observe *interval-censored* binary indicators (Chapter 14's censoring, Chapter 26's likelihoods). To recover a *mean* or *median* WTP you must impose a **distributional assumption** on the unobserved WTP and estimate it by maximum likelihood. This chapter shows that machinery working (verified: SBDC MLE recovers the true median 40.00 as 39.42 and mean 47.89 as 47.56), and then shows its **fragility**: the *mean* WTP depends on the assumed *upper tail beyond your highest bid* — a region where you have *no data at all*. Verified below, changing the assumed dispersion from $\sigma = 0.6$ to $\sigma = 0.9$

leaves the median *completely unchanged* at €40 while inflating the mean by **25%** to €60. The mean is a *model extrapolation*; the median is *data*.

This is why the chapter ends with the **Turnbull distribution-free bound** — a conservative floor that assumes *nothing* about the tail (verified: €39.00, bracketing the parametric mean €47.56 from below). Reporting the parametric mean *alongside* the median and the Turnbull bound is the honest presentation, and it directly serves the aggregation fragility of Chapter 35 §35.5, where mean-versus-median swung the benefit by 56%.

Along the way: the **Hanemann utility-difference** foundation that makes the yes/no response a *random utility* statement (linking CV to the RUM of Chapter 22 and the welfare theory of Chapter 34), the **format taxonomy** and its incentive properties (Chapter 28), the **payment card** as an interval-censored middle ground, and the **specification** choice (linear vs log-normal) that quietly determines whether your mean and median are forced to be equal (verified: a linear-in-bid logit *imposes* mean = median = €44.99, a 6% error on the true mean).

36.1 Prerequisites

From **Chapter 34**: WTP as compensating variation and the constant-marginal-utility-of-income condition. From **Chapter 35**: why the good requires stated preference; mean vs median in aggregation. From **Chapters 20–21**: random utility and the logit/probit link. From **Chapter 14** (§14.8): censoring and interval data. From **Chapters 15, 26**: the CV likelihoods (SBDC/DBDC/payment card, spike models) — this chapter is the *design and interpretation* companion to that chapter's *estimation* mechanics. From **Chapter 28**: incentive compatibility (which formats are truthful). From **Chapter 29**: bid design and anchoring. From **Chapter 11**: maximum likelihood.

New notation. t (the bid/offer amount), $Y_i \in \{0,1\}$ (the yes/no response), W_i (latent WTP), F_W (the WTP distribution function), Δv (the utility difference), and μ, σ (location/scale of the WTP distribution).

36.2 The utility-difference foundation of the yes/no response

A dichotomous-choice response is not an arbitrary binary datum: it is a *statement about utility*, and Hanemann's utility-difference model makes the link precise.

The setup. A respondent with income m faces a referendum: the environmental improvement will occur (state q^1 rather than q^0) and they will pay t. They have (indirect) utility $v(q, m, \mathbf{s}) + \varepsilon$, where $\mathbf{s}$ are characteristics and ε is a random component unobserved by the analyst (Chapter 22's RUM). They vote **yes** if being in the improved state having paid t is at least as good as the status quo:

$$v(q^1, m - t, \mathbf{s}) + \varepsilon_1 \;\geq\; v(q^0, m, \mathbf{s}) + \varepsilon_0.$$

Definition 36.1 (Utility difference and the response probability). Define the systematic utility difference $\Delta v = v(q^1, m - t, \mathbf{s}) - v(q^0, m, \mathbf{s})$ and $\eta = \varepsilon_0 - \varepsilon_1$. The respondent says yes iff $\Delta v \geq \eta$, so

$$P(\text{yes} \mid t) = P(\eta \leq \Delta v) = F_\eta(\Delta v).$$

With η logistic this is a **logit**; with η normal, a **probit** — the binary-response models of Chapter 20, now with a *structural* interpretation.

The equivalent WTP formulation. Equivalently, define the respondent's **WTP** W as the payment that makes them exactly indifferent: $v(q^1, m - W, \mathbf{s}) = v(q^0, m, \mathbf{s})$ — precisely the **compensating variation** of Chapter 34. Then the respondent says yes iff their WTP is at least the bid:

$$P(\text{yes} \mid t) = P(W \geq t) = 1 - F_W(t).$$

This is the key identity of contingent valuation: *the probability of a "yes" at bid t traces out the survival function of the WTP distribution.* Varying t across respondents traces the whole distribution — which is why **bid design** (the set of t values and their allocation) determines what you can identify (Chapter 29).

Why this matters. The identity shows (i) the yes/no response is a *welfare* statement (WTP = CV, Chapter 34), (ii) CV and discrete choice share the *same* RUM foundation (Chapter 22) — a CV referendum is a two-alternative choice experiment, and (iii) estimation is *interval-censored* inference: each response tells you only which *side* of t the latent W falls on, never W itself (Chapter 14).

> **Active recall 36.A.** Why is $P(\text{yes} \mid t) = 1 - F_W(t)$, and what does it imply about what a single respondent's answer tells you? *(Answer: the respondent says yes iff the improvement at price t leaves them at least as well off as the status quo, i.e. iff their* compensating variation* W (the payment making them exactly indifferent) is at least the bid: yes $\iff$ $W \geq t$. So $P(\text{yes} \mid t) = P(W \geq t) = 1 - F_W(t)$ — the yes-share at bid t is the *survival function* of the WTP distribution evaluated at t, and varying t across respondents traces out the whole distribution. Implication: a single respondent's answer is *one bit* — it locates their WTP only on *one side* of t (interval-censored), never revealing the value itself; the WTP *distribution* is identified only across respondents facing *different* bids, which is why bid design determines what is identifiable.)*

36.3 The taxonomy of elicitation formats

The same good can be valued by very different questions. Each trades off cognitive burden, statistical efficiency, and incentive compatibility.

Open-ended. "What is the maximum you would pay?" *Pros:* directly yields a continuous WTP (maximally efficient statistically — one number per person, no distributional assumption needed for the mean). *Cons:* cognitively very demanding (respondents rarely know their exact maximum for an unfamiliar public good), high rates of non-response, protest zeros, and *understatement*; **not incentive compatible** (a self-interested respondent has an incentive to understate to lower a payment tied to their answer). Largely abandoned for primary valuation.

Bidding game. An iterative auction-like sequence ("Would you pay €10? …€20? …€40?") until a maximum is found. *Pros:* yields a fairly precise value; familiar auction framing. *Cons:* severe **starting-point bias** — the final value depends strongly on the initial bid (an anchoring effect, Chapter 29); also demanding and prone to yea-saying.

Payment card. The respondent selects a value (or an interval) from a *displayed array* of amounts. *Pros:* far easier than open-ended (a choice, not a construction), yields *interval-censored* data that is statistically richer than a single binary; avoids the sequential anchoring of a bidding game. *Cons:* **range bias / card design effects** — the displayed range and spacing influence responses (an anchoring vulnerability, Chapter 29); not strictly incentive compatible. **Your CV design uses payment cards**, so the card's range and spacing are design parameters you must justify and, ideally, *vary experimentally* to test sensitivity.

Single-bounded dichotomous choice (SBDC) / referendum. "Would you vote yes to pay €t?" — one take-it-or-leave-it offer. *Pros:* **cognitively easiest** (a simple comparison), **familiar** (mimics a market purchase or a referendum), and — decisively — **incentive compatible** under the Carson–Groves conditions (a binding, consequential, single binary vote on a public good with coercive payment; Chapter 28). This is why the NOAA panel recommended it and why it is the modern default. *Cons:* **statistically expensive** — only one bit per respondent, so you need *large* samples and a well-designed bid vector; and the mean depends on distributional assumptions (§36.5).

Double-bounded dichotomous choice (DBDC). A follow-up bid after the first response (higher if yes, lower if no). *Pros:* much greater *statistical efficiency* (narrows the interval, tightening confidence intervals substantially). *Cons:* the follow-up **breaks incentive compatibility** (the respondent learns the price is negotiable, creating strategic incentives) and typically produces **response inconsistency** between the

two answers (the second response implies a different WTP distribution than the first — a well-documented anomaly). The efficiency–incentive trade-off is the classic SBDC-vs-DBDC dilemma: *DBDC buys precision with bias risk.*

The design logic of your study. Combining **SBDC** (incentive-compatible core) with **payment cards** (richer interval information) and **real vs hypothetical** treatments (a direct test of hypothetical bias, Chapter 28) and the **BRIS** mechanism (an incentive-compatible real-payment device) is a design that *triangulates*: the SBDC provides the theoretically-clean estimate, the payment card provides efficiency, and the real/hypothetical contrast provides the validity test. The chapter's warning is that these formats will *not* give identical numbers — and that the *differences* are data about format effects, not merely noise to be averaged away.

> **Stop and think 36.1.** Why is SBDC preferred on incentive grounds despite being far less statistically efficient than open-ended or DBDC, and what does your payment card add and risk? *(Answer: SBDC is* incentive compatible* under the Carson–Groves conditions — with a single, binding, consequential binary vote on a public good with coercive payment, the respondent's *best strategy is to answer truthfully* (misrepresenting can only make them worse off), whereas open-ended invites *understatement* (lower your stated max, lower your payment) and DBDC's follow-up signals the price is negotiable, reintroducing strategic play and producing documented response inconsistency between the two answers. The efficiency loss (one bit per respondent, requiring large samples and careful bid design) is the price paid for *unbiasedness of the incentive structure* — a biased-but-precise estimate is worse than an unbiased-but-noisy one. The *payment card* adds *statistical efficiency* (interval-censored data is richer than one bit, tightening the WTP distribution estimate) but *risks* range/card-design bias — the displayed range and spacing anchor responses (Chapter 29) — and is not strictly incentive compatible; so its range should be justified and ideally *experimentally varied* to test sensitivity.)*

36.4 Estimating WTP from SBDC responses

Since WTP is never observed, estimation is maximum likelihood on interval-censored data (Chapter 26's likelihood, applied here).

The likelihood. With $Y_i = 1$ for yes at bid t_i, and $W \sim F_W(\cdot\,;\theta)$:

$$\mathcal{L}(\theta) = \prod_i \left[1 - F_W(t_i;\theta)\right]^{Y_i} \left[F_W(t_i;\theta)\right]^{1-Y_i}.$$

This is exactly a binary-response likelihood with the bid as the regressor — a probit/logit whose *structural* parameters are the WTP distribution's location and scale.

The log-normal specification. A natural choice for WTP (non-negative, right-skewed) is $\ln W \sim N(\mu, \sigma^2)$. Then

$$P(\text{yes} \mid t) = P(\ln W \geq \ln t) = 1 - \Phi\!\left(\frac{\ln t - \mu}{\sigma}\right) = \Phi\!\left(\frac{\mu - \ln t}{\sigma}\right),$$

which is a **probit on** $\ln t$ with intercept $a = \mu/\sigma$ and slope $b = -1/\sigma$. So you estimate a standard probit and *back out* $\hat{\sigma} = -1/\hat{b}$, $\hat{\mu} = \hat{a}\hat{\sigma}$. The welfare summaries are then

$$\text{median WTP} = e^{\mu}, \qquad \text{mean WTP} = e^{\mu + \sigma^2/2}.$$

Worked Example 36.1 (SBDC estimation, verified). Simulated true WTP $\sim$ LogNormal($\mu = \ln 40$, $\sigma = 0.6$) — true median €40.00, true mean €47.89. With 2,800 respondents across seven bids (€10–€120):

Bid	Observed P(yes)	Theoretical
€10	0.995	0.990
€20	0.863	0.876
€30	0.657	0.684
€40	0.480	0.500
€60	0.247	0.250
€80	0.140	0.124
€120	0.033	0.034

MLE results:

Quantity	Estimate	Truth
$\hat{\sigma}$	0.613	0.60
$\hat{\mu}$	3.674	3.689
Median WTP	**€39.42**	€40.00
Mean WTP	**€47.56**	€47.89

(Verified.) The yes-proportions trace the survival function (at the bid €40 — the true median — the yes rate is $0.48 \approx 0.5$, exactly as theory predicts), and the MLE recovers both the median and the mean accurately. Note the practical design lesson visible in the table: bids concentrated *near the median* pin down location, while the *extreme* bids (€10 with 99.5% yes, €120 with 3.3% yes) contribute little information — a bid vector wasting mass at the extremes is inefficient (Chapter 29's bid design).

> **Active recall 36.B.** In the log-normal SBDC model, why is the estimation "a probit on log-bid," and how do you recover the WTP median and mean? *(Answer: with* $\ln W \sim N(\mu, \sigma^2)$*,* $P(yes \mid t) = P(W \geq t) = P(\ln W \geq \ln t) = \Phi\left(\frac{\mu - \ln t}{\sigma}\right)$ *— a standard* probit* *in the regressor* $\ln t$ *with intercept* $a = \mu/\sigma$ *and slope* $b = -1/\sigma$*. So you run the probit on log-bid and back out the structural parameters:* $\hat{\sigma} = -1/\hat{b}$ *and* $\hat{\mu} = \hat{a}\hat{\sigma}$ *(verified:* $\hat{\sigma} = 0.613$*,* $\hat{\mu} = 3.674$ *against truth 0.60, 3.689). The welfare summaries follow from the log-normal: median* $= e^{\mu}$ *(verified €39.42 vs true €40.00) and mean* $= e^{\mu + \sigma^2/2}$ *(verified €47.56 vs true €47.89). The mean requires both parameters — in particular the scale* σ*, which governs the unobserved upper tail.)**

36.5 Mean versus median, and the tyranny of the unobserved tail

Now the chapter's central caution — and the reason your reported number must be a *set* of numbers.

Why they differ. WTP distributions are typically *right-skewed* (most people hold modest values; a few hold large ones). Under log-normality, mean $= e^{\mu + \sigma^2/2} > e^{\mu} =$ median, and the gap grows with the *dispersion* σ. **Verified:** with $\sigma = 0.6$, the mean (€47.89) exceeds the median (€40.00) by **19.7%**.

The fragility. The *median* is a *location* statistic pinned down by where the yes-rate crosses 50% — squarely inside the range of your bids, where you have *abundant data*. The *mean* additionally integrates the *entire upper tail* — including the region *beyond your highest bid*, where **you have no observations whatsoever**. The mean is therefore a *model-based extrapolation*, and it inherits all the fragility of the distributional assumption.

Worked Example 36.2 (Tail sensitivity, verified). Two demonstrations of the mean's dependence on the unobserved tail:

(a) Truncation. The mean of the WTP distribution when values are capped at various levels:

Assumption	Mean WTP
Capped at €120 (highest bid)	€46.73
Capped at €200	€47.69
Capped at €500	€47.97
Uncapped	€47.88

(b) Dispersion. Changing the assumed scale parameter:

σ	Median	Mean
0.6	**€40.00**	€47.89
0.9	**€40.00** (unchanged)	**€59.97 (+25%)**

(Verified.) Panel (b) is the striking one: **raising the assumed dispersion from 0.6 to 0.9 leaves the median *exactly* unchanged at €40 while inflating the mean by 25% to €60.** The scale parameter is identified largely by the *slope* of the response curve, which the data do constrain — but the *mean* leverages it into the far tail multiplicatively. A modest specification change you cannot easily refute with your data moves the mean by a quarter, and the median not at all.

Why this is decisive for you. Chapter 35 §35.5 showed that the aggregate benefit scales linearly with the chosen central tendency (mean-based €60M/yr vs median-based €38.4M/yr — a 56% swing). Combine that with the present result — that the mean is itself 25%-sensitive to an untestable tail assumption — and the conclusion is unavoidable: **a single headline "mean WTP" carries far less epistemic weight than its decimal places suggest.** The disciplined report gives the median (robust, data-driven), the mean (the Kaldor–Hicks-correct measure for CBA, Chapter 35), the *sensitivity of the mean* to the distributional assumption, and a distribution-free bound (§36.6).

> **Stop and think 36.2.** Your referee asks why you report a median WTP of €39 alongside a mean of €48 rather than "the" WTP. What is the substantive answer? *(Answer: they are different statistics with different roles and* very* different robustness. The *median* is pinned down by where the yes-rate crosses 50% — inside the bid range where the data are dense — so it is robust to the distributional assumption. The *mean* integrates the entire upper tail, *including the region beyond the highest bid where there are no observations at all*, so it is a model-based *extrapolation*: verified, changing the assumed dispersion from $\sigma = 0.6$ to 0.9 leaves the median *exactly* unchanged at €40 but inflates the mean 25% to €60, and capping the tail at the highest bid pulls it down to €46.73. Both are needed: the *mean* is the theoretically correct measure for a Kaldor–Hicks CBA (it sums total welfare, Chapter 35), while the *median* is the robust, majority-voting measure. Reporting only one — especially only the mean, without its tail sensitivity — would misrepresent the precision of a number that drives an aggregate benefit already 56%-sensitive to this very choice.)*

36.6 Specification risk and the Turnbull distribution-free bound

Two further protections against the distributional assumption.

Specification risk: the linear model forces mean = median. A common alternative specifies the utility difference *linear in the bid* — $P(\text{yes}) = \Lambda(\alpha - \beta t)$ — which implies a *logistic* WTP distribution that is **symmetric**, so it *forces* mean = median = α/β. If true WTP is right-skewed, this specification *cannot represent* the skew.

Worked Example 36.3 (Misspecification, verified). Fitting a linear-in-bid logit to the log-normal data above:

Quantity	Value
Linear-logit implied mean = median = α/β	**€44.99**
True median	€40.00
True mean	€47.89
Error vs true mean	−6.0%

(Verified.) The linear model produces a single number (€44.99) *between* the true median and mean, *forced* to serve as both, understating the true mean by 6% and overstating the true median by 12%. The functional form is not innocuous: **choosing linear-in-bid silently imposes symmetry on a distribution you have every reason to believe is skewed.** Test the specification (log-normal vs linear vs Weibull), report sensitivity, and prefer forms that permit skew.

The Turnbull distribution-free bound. The strongest protection is to assume *nothing*. The **Turnbull** (lower-bound) estimator uses only the empirical yes-rates at each bid, treating the WTP distribution as a step function, and places each interval's probability mass at the interval's *lower* endpoint — the most *conservative* possible allocation. This yields a **distribution-free lower bound on mean WTP**.

Worked Example 36.4 (Turnbull bounds, verified). From the same data (empirical CDF at the bids: 0.005, 0.137, 0.342, 0.520, 0.752, 0.860, 0.968):

Estimate	Value
Turnbull lower bound (mass at lower endpoints; tail lumped at €120)	**€39.00**
Parametric log-normal median	€39.42
Parametric log-normal mean	**€47.56**
Turnbull upper bound (tail capped at €300)	€61.15

(Verified: €39.00 ≤ €47.56 ≤ €61.15.) The Turnbull lower bound of €39.00 is a **conservative, assumption-free floor** — it says the mean WTP is *at least* €39 whatever the true distribution, because it assigns everyone the *lowest* value consistent with their response and lumps the entire upper tail at the highest bid. **The gap between €39.00 and the parametric €47.56 is precisely the value that the distributional assumption adds beyond the data** — a direct, quantified measure of how much of your headline number is *extrapolation* rather than *observation*. Reporting the Turnbull bound alongside the parametric estimate makes that extrapolation transparent, and it connects to the partial-identification philosophy of Chapter 30: when an assumption is untestable, *bound* rather than *assume*.

> **Active recall 36.C.** What does the Turnbull lower bound assume, why is it a *lower* bound, and what does its gap from the parametric mean measure? *(Answer: Turnbull assumes* nothing* about the shape of the WTP distribution — it uses only the empirical yes-rates at the observed bids, treating the distribution as a step function on the intervals the bids define. It is a *lower* bound because it allocates each interval's probability mass to that interval's *lowest* value (everyone is assigned the smallest WTP consistent with their response) and lumps the entire *open upper tail* at the highest bid rather than extrapolating beyond it — the most conservative possible allocation (verified €39.00 ≤ parametric mean €47.56). The *gap* between the Turnbull bound and the parametric mean measures exactly *how much of the headline number comes from the distributional assumption* rather than the data — i.e. the value contributed by extrapolating into the tail beyond the highest bid, where there are no observations. Reporting both makes that extrapolation explicit, in the partial-identification spirit of Chapter 30: bound what you cannot test.)*

36.7 Failure modes

F1. Treating format as presentation. Assuming any elicitation format yields "the" WTP. *Fix:* formats differ in cognitive task, incentives, and identification; justify the choice and test format effects (§36.3).

F2. Using a non-incentive-compatible format for the primary estimate. Open-ended or bidding-game numbers as the headline. *Fix:* SBDC/referendum core; treat other formats as supplementary (§36.3; Chapter 28).

F3. DBDC efficiency without acknowledging inconsistency. Reporting DBDC estimates ignoring response inconsistency/incentive breach. *Fix:* report SBDC (first-response) estimates alongside; test for inconsistency (§36.3).

F4. Reporting only the mean. A single mean WTP with no median, no bounds, no sensitivity. *Fix:* report median, mean, tail sensitivity, and Turnbull bound (§36.5–§36.6).

F5. Unexamined distributional assumption. Log-normal (or linear) imposed without testing. *Fix:* compare specifications; note that linear *forces* mean = median (verified 6% error) (§36.6).

F6. Extrapolating the mean beyond the bid range. A mean driven by a tail with no data. *Fix:* quantify with truncation sensitivity and the Turnbull bound; consider capping (§36.5–§36.6).

F7. Poor bid design. Bids concentrated where responses are near-certain (99.5% yes, 3% yes). *Fix:* concentrate bids where the response curve is informative, near the median (§36.4; Chapter 29).

F8. Payment-card range effects unaddressed. Card range/spacing anchoring responses. *Fix:* justify the range; vary it experimentally; test sensitivity (§36.3; Chapter 29).

36.8 Diagnostic questions for your own papers

1. **Format justification.** Have I justified the elicitation format on incentive-compatibility grounds and acknowledged its statistical cost? *(§36.3, F1–F2.)*
2. **Structural interpretation.** Am I clear that the yes/no traces $1 - F_W(t)$ and that WTP is compensating variation? *(§36.2; Chapter 34.)*
3. **Central tendency.** Do I report *both* median and mean, with the role of each for CBA? *(§36.5, F4.)*
4. **Tail sensitivity.** Have I shown how the mean moves with the distributional assumption / truncation? *(§36.5, F6.)*
5. **Specification.** Have I compared distributional specifications and noted that linear forces mean = median? *(§36.6, F5.)*
6. **Distribution-free bound.** Do I report the Turnbull bound to quantify the extrapolation? *(§36.6.)*
7. **Bid design.** Is my bid vector concentrated where it is informative? *(§36.4, F7; Chapter 29.)*
8. **Payment-card design.** Is the card range justified and its influence tested? *(§36.3, F8.)*

36.9 Exercises

(A) Comprehension.

A1. Derive $P(\text{yes} \mid t) = 1 - F_W(t)$ from the utility-difference model and explain what it identifies.

A2. Compare open-ended, payment card, SBDC, and DBDC on cognitive burden, efficiency, and incentive compatibility.

A3. Why does the mean WTP depend on the unobserved tail while the median does not?

(B) Derivation / reasoning.

B1. Show that log-normal WTP implies a probit on log-bid, and derive the median and mean formulas.

B2. Show that a linear-in-bid logit forces mean = median, and explain the consequence for skewed WTP.

B3. Explain the construction of the Turnbull lower bound and why it is conservative.

(C) Research application / critique.

C1. Justify your SBDC + payment card + real/hypothetical + BRIS design, stating what each element contributes and risks.

C2. Your headline mean WTP is €48. Write the robustness reporting you owe the reader.

C3. A co-author proposes switching to DBDC "for efficiency." Advise.

36.10 Solutions

A1. The respondent votes yes iff the improved state net of payment is at least as good as the status quo: $v(q^1, m-t)+\varepsilon_1 \geq v(q^0, m)+\varepsilon_0$. Defining WTP W as the payment making them exactly indifferent ($v(q^1, m-W) = v(q^0, m)$ — the *compensating variation* of Chapter 34), the yes condition is equivalent to $W \geq t$. Hence $P(\text{yes} \mid t) = P(W \geq t) = 1 - F_W(t)$. This identifies the *survival function* of the WTP distribution at the bid values used: varying t across respondents traces out F_W, so the *distribution* (not any individual's WTP) is what the design identifies — and only over the *range of bids employed*.

A2. *Open-ended:* highest cognitive burden (construct an exact maximum), highest statistical efficiency per respondent (a continuous value), but *not incentive compatible* (incentive to understate) — high protest/non-response. *Payment card:* moderate burden (select from a displayed array), good efficiency (interval-censored data), *not strictly incentive compatible*, and vulnerable to *range/spacing bias*. *SBDC:* lowest burden (a single comparison), lowest efficiency (one bit per respondent — needs large samples and good bid design), and *incentive compatible* under Carson–Groves conditions (binding, consequential, single binary public-good vote). *DBDC:* low burden, *much higher* efficiency (narrowed interval), but the follow-up *breaks incentive compatibility* (signals negotiability) and generates documented *response inconsistency* between answers. The ordering on incentives (SBDC best) is the reverse of the ordering on efficiency (open-ended/DBDC best) — the central design trade-off.

A3. The *median* is determined by the bid at which the yes-rate crosses 50% — a *location* feature lying *inside* the bid range where data are dense, so it is estimated from observations. The *mean* is $\int_0^\infty [1-F_W(t)]\,dt$ — it integrates the *whole* survival curve, *including the region above the highest bid where there are no observations at all*. That region's contribution is supplied entirely by the assumed functional form, making the mean an *extrapolation*. Verified: raising σ from 0.6 to 0.9 left the median exactly at €40.00 while raising the mean 25% to €59.97; capping the tail at the highest bid (€120) reduced the mean to €46.73.

B1. With $\ln W \sim N(\mu, \sigma^2)$: $P(\text{yes} \mid t) = P(W \geq t) = P(\ln W \geq \ln t) = 1 - \Phi\left(\frac{\ln t - \mu}{\sigma}\right) = \Phi\left(\frac{\mu - \ln t}{\sigma}\right)$ (by symmetry of Φ). Writing this as $\Phi(a + b \ln t)$ gives $a = \mu/\sigma$, $b = -1/\sigma$ — a *probit in* $\ln t$; invert to get $\sigma = -1/b$, $\mu = a\sigma$ (verified: $\hat{\sigma} = 0.613$, $\hat{\mu} = 3.674$). For the log-normal, the *median* is e^μ (since $P(W \leq e^\mu) = \Phi(0) = 0.5$) and the *mean* is $\mathbb{E}[W] = e^{\mu+\sigma^2/2}$ (the standard log-normal mean, reflecting that exponentiating a symmetric normal produces right skew whose mean exceeds the median by the factor $e^{\sigma^2/2}$). Verified: €39.42 and €47.56 against truth €40.00 and €47.89.

B2. With $P(\text{yes} \mid t) = \Lambda(\alpha - \beta t)$ (linear in the bid), we have $1 - F_W(t) = \Lambda(\alpha - \beta t)$, so $F_W(t) = \Lambda(\beta t - \alpha)$ — WTP follows a *logistic* distribution with location α/β and scale $1/\beta$. The logistic is *symmetric* about its location, so its *mean and median coincide*, both equal to α/β — the specification *forces* them to be equal *by construction*, regardless of the data. Consequence for skewed WTP: the model *cannot represent* right skew, so the single fitted value is a compromise lying between the true median and true mean, misstating both (verified: €44.99 vs true median €40.00 and true mean €47.89 — a −6% error on the mean and +12% on the median). Choosing linear-in-bid is thus a substantive assumption of symmetry, not a neutral convenience.

B3. Turnbull uses only the empirical yes-proportions at the observed bids $t_1 < \cdots < t_M$, giving an estimated CDF $F_j = 1 - \widehat{P}(\text{yes} \mid t_j)$ at each bid (monotonised if sampling noise produces non-monotonicity). This defines probability *masses* on the intervals $[0, t_1), [t_1, t_2), \ldots, [t_M, \infty)$ but says *nothing* about where within each interval the mass sits. The *lower bound* on the mean assigns every interval's mass to that interval's *lowest* value: mass in $[t_j, t_{j+1})$ is valued at t_j, and the entire open upper tail $[t_M, \infty)$ is lumped at t_M. Since every respondent is credited with the *smallest* WTP consistent with their answer, and the unbounded tail is truncated to the highest bid, the result is the *smallest* mean consistent with the data — a conservative floor requiring *no distributional assumption* (verified €39.00, below the parametric mean €47.56; the corresponding upper bound with the tail capped at €300 is €61.15).

C1. *SBDC core:* provides the *incentive-compatible* primary estimate (Carson–Groves: single, binding, consequential binary vote on a public good), so the headline WTP is not contaminated by strategic understatement — at the cost of statistical efficiency, addressed by a large sample and a bid vector concentrated where the response curve is informative (verified: extreme bids at 99.5%/3.3% yes contribute little). *Payment card:* adds *efficiency* through interval-censored data richer than one bit, at the *risk* of range/spacing anchoring — so the card range must be justified and, ideally, *experimentally varied* to test sensitivity (Chapter 29). *Real vs hypothetical treatments:* provide the *direct experimental test of hypothetical bias* — the identification of the gap between stated and actual payment behaviour, which no single-treatment design can deliver (Chapters 28, 31). *BRIS mechanism:* makes the *real* treatment genuinely incentive compatible (a binding, randomly-implemented payment rule), so the "real" benchmark is a true revealed-payment measure rather than another hypothetical. *Together:* the design *triangulates* — theoretically-clean core (SBDC), efficiency (card), and validity test (real/hypothetical with BRIS). The chapter's caution applies: these components will not yield identical numbers, and the differences are *evidence about format and hypothetical bias*, to be reported and analysed, not averaged away.

C2. Owed robustness reporting for a headline mean of €48: (i) the **median** WTP alongside it (€39), with the roles stated — mean for the Kaldor–Hicks CBA aggregate, median as the robust/majority-voting measure (Chapter 35); (ii) the **distributional specification** used and a *comparison* across alternatives (log-normal vs linear vs Weibull), noting that a linear-in-bid form *forces* mean = median (verified 6% error) — so the choice is substantive; (iii) **tail sensitivity** — how the mean moves under truncation at the highest bid (verified €46.73) and under alternative dispersion (verified: $\sigma = 0.9$ gives €59.97, +25%, with the median unchanged) — making explicit that the mean extrapolates beyond the data; (iv) the **Turnbull distribution-free lower bound** (€39.00), with the gap to the parametric mean identified as the value contributed by the assumption rather than the data; (v) **confidence intervals** by Krinsky–Robb or the delta method (Chapters 7, 13), acknowledging WTP-ratio tail behaviour; and (vi) a statement of how the *aggregate* benefit changes across these choices (Chapter 35: a 56% swing between mean- and median-based aggregation). The honest headline is a *range with named drivers*, not €48.

C3. Advise *caution*, and propose a hybrid. DBDC's attraction is real — the follow-up bid narrows each respondent's WTP interval and can *substantially* tighten confidence intervals for a given sample size, which matters given SBDC's one-bit-per-respondent inefficiency. But the cost is *not merely technical*: the follow-up **breaks the incentive compatibility** that is the *entire justification* for choosing dichotomous choice in the first place (Chapter 28) — once respondents learn the price is negotiable, the second response invites strategic play — and the literature documents systematic **response inconsistency**, where the second answer implies a *different* WTP distribution than the first (typically shifting estimates downward and producing rejection of the equality of the two distributions). Given that the study's core claim rests on *incentive-compatible* elicitation and on a *real-versus-hypothetical* validity comparison, trading that away for precision is the wrong trade — precision in a strategically-contaminated estimate is false precision. *Recommendation:* keep **SBDC as the primary, pre-registered estimate**; if a follow-up is included, treat the DBDC estimate as a *secondary/efficiency* analysis, *test* the consistency of the first and second responses explicitly, and report the SBDC-only (first-response) estimate as the headline. Alternatively gain efficiency through *better bid design* (concentrating bids near the median, Chapter 29) and *larger samples* — efficiency purchased without incentive cost.

36.11 Spaced review prompts

+1 day. From memory: derive $P(\text{yes} \mid t) = 1 - F_W(t)$; list the four formats with their incentive/efficiency properties; write the log-normal median and mean formulas.

+1 week. Explain why the mean depends on the unobserved tail and the median does not; why linear-in-bid forces mean = median; how Turnbull bounds are constructed.

+1 month. For your CV study, report median, mean, specification comparison, tail sensitivity, and Turnbull bound; justify the bid vector and payment-card range. Close with the spine sentence about what your format identifies and how much of your mean is extrapolation.

36.12 How this chapter connects to the rest of the book

- **Chapter 34** supplies the CV-as-compensating-variation meaning of W; **Chapter 35** supplies the aggregation in which the mean/median choice of §36.5 becomes a 56% swing in the benefit; **Chapter 22** supplies the RUM behind the utility-difference model (a CV referendum *is* a two-alternative choice experiment).
- **Chapter 26** holds the full likelihood machinery (SBDC/DBDC/payment card, spike models) that §36.4 applies; **Chapter 28** supplies the incentive-compatibility theory that justifies the format ranking of §36.3; **Chapter 29** supplies the bid design and anchoring analysis behind §36.4's design lesson and the payment-card range risk; **Chapter 30** supplies the partial-identification philosophy behind the Turnbull bound.
- **Chapter 37–39** extend to DCE design and stated-preference validity; **Chapters 45–46 (CBA)** consume the mean WTP and its sensitivity; **Chapters 47–49** transfer these estimates across sites, where format differences become a moderator in the meta-analysis.

This chapter is where your CV study's design choices become *estimable objects with known fragilities* — and where the discipline of reporting a *bounded range* rather than a single mean is established.

36.13 Research translation task

For your **CV study**, produce a three-page elicitation-and-estimation note:

1. **Format justification.** State why SBDC is the incentive-compatible core, what the payment card adds and risks, and what the real/hypothetical + BRIS treatments identify.
2. **Structural model.** Write the utility-difference model, the identity $P(\text{yes} \mid t) = 1 - F_W(t)$, and the WTP-as-CV interpretation.
3. **Estimation.** Specify the distribution (log-normal and at least one alternative), estimate by MLE, and report $\hat{\mu}, \hat{\sigma}$ with standard errors.
4. **Welfare summaries.** Report *median and mean* WTP with Krinsky–Robb intervals, plus the *truncated* mean and the *Turnbull* distribution-free bound.
5. **Sensitivity.** Tabulate the mean across distributional specifications and truncation points; state explicitly how much of the mean is extrapolation beyond the highest bid.
6. **Design audit.** Assess whether the bid vector concentrates information near the median, and whether the payment-card range was varied/tested.
7. **Close** with the spine sentence: *"My SBDC identifies the WTP survival function over the bid range €[·]–€[·]; the median €[·] is data-driven, the mean €[·] depends on the [log-normal] tail assumption (varying €[·]–€[·] across specifications), and the assumption-free Turnbull floor is €[·]."*

Carry this into Chapters 45–46 (CBA aggregation and sensitivity) and Chapters 47–49 (meta-analysis and benefit transfer), where format and specification become explicit moderators.

36.14 Bibliography

Core. - Hanemann (1984), "Welfare Evaluations in Contingent Valuation Experiments with Discrete Responses," *American Journal of Agricultural Economics* — the utility-difference foundation of the dichotomous-choice model (§36.2). *(Verify volume/pages.)* - Mitchell & Carson, *Using Surveys to Value Public Goods: The Contingent Valuation Method* — the foundational treatment of formats, design, and bias (§36.3). *(Verify edition.)* - Haab & McConnell, *Valuing Environmental and Natural Resources: The Econometrics of Non-Market Valuation* — the standard econometric reference for CV estimation, mean/median, and Turnbull (§36.4–§36.6). *(Verify edition.)*

Applied. - Arrow et al. (1993), the NOAA Panel Report on Contingent Valuation — the recommendation of the referendum/dichotomous-choice format (§36.3). *(Verify citation/publication details.)* - Hanemann, Loomis & Kanninen (1991), "Statistical Efficiency of Double-Bounded Dichotomous Choice Contingent Valuation," *American Journal of Agricultural Economics* — the DBDC efficiency gain (§36.3). *(Verify volume/pages.)* - Cameron & Huppert (1989) on payment-card / interval-censored WTP estimation (§36.3). *(Verify citation.)*

Advanced. - Turnbull (1976), "The Empirical Distribution Function with Arbitrarily Grouped, Censored and Truncated Data," *Journal of the Royal Statistical Society B* — the nonparametric estimator behind the distribution-free bound (§36.6). *(Verify volume/pages.)* - Haab & McConnell (1997), "Referendum Models and Negative Willingness to Pay: Alternative Solutions," *Journal of Environmental Economics and Management* — distributional choice and non-negativity (§36.5–§36.6). *(Verify volume/pages.)* - Alberini (1995) on efficiency and bid design in dichotomous-choice CV (§36.4). *(Verify citation.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Hanemann / NOAA / Turnbull citations before using them, per the book's no-unverified-citation rule.

End of Chapter 36. Next: Chapter 37 — Discrete Choice Experiments: Design, Attributes, and Experimental Efficiency, which turns from the single-referendum CV question to the multi-attribute choice tasks of your DCE — attribute and level selection, orthogonal vs efficient (D-optimal) designs, blocking, the cost-vector manipulation at the centre of your identification strategy, and how design efficiency determines the precision of the WTP estimates you can recover. Tell me when to write it, or request a different chapter.

Chapter 37 — Discrete Choice Experiments: Design, Attributes, and Experimental Efficiency

Part IX: Stated Preference in Practice

> **The spine question, applied to this chapter.** Chapter 36 elicited a *single* value with a *single* referendum question. A discrete choice experiment instead asks respondents to choose repeatedly among *multi-attribute* alternatives, and from those choices recovers the marginal WTP for *every* attribute. But the choices respondents make are only as informative as the *choice tasks you construct*: the attributes you include, the levels you assign, the way profiles are paired into choice sets, and — decisively for you — the **cost vector**. The design *is* the identification strategy. The spine question here: *what does my experimental design allow me to identify, with what precision, and have I optimized for the parameter my research actually depends on — or for a statistical criterion that quietly sacrifices it?*

37.0 Conceptual motivation: the design is the instrument

In an observational study you take the data-generating process as given. In a DCE you *build* it. Every choice task a respondent faces is something you constructed, and the *information content* of the resulting data is determined, before a single respondent is recruited, by the design. This makes experimental design the single highest-leverage decision in a DCE — and the one most often delegated to software defaults without understanding what is being optimized.

The chapter's arc follows the practical sequence of building your DCE. First, **attributes and levels**: what goes into the alternatives, and the fact that this is a *theoretical* decision (Lancaster's characteristics approach, Chapter 22) constrained by cognitive burden. Your DCE's attributes — the ecological/stock characteristics, the **familiarity** dimension, and the **cost** — each earn their place by being policy-relevant, comprehensible, and *variable*. Second, the **combinatorial problem**: a modest four-attribute design already generates 108 profiles and 5,778 possible pairs (verified below), so you must select a *fraction* — and *how* you select it determines what you can estimate.

Third, the two philosophies of fractional design. **Orthogonal** designs (inherited from linear models) ensure attributes are statistically independent, which sounds ideal but optimizes the wrong thing: in a *nonlinear* choice model the information content depends on the *choice probabilities*, which depend on the *parameters*. **Efficient (D-optimal)** designs instead minimize the expected variance of the estimates given *prior* beliefs about the parameters — accepting some correlation in exchange for far more information. Verified below: a D-optimal design cut the D-error by **56%** relative to a random design and achieved equivalent overall precision with **71% of the sample** (N=424 versus 600) — a real saving in fieldwork cost.

But the chapter's most important result is the one that complicates the sales pitch. D-error is a *scalar*

summary — the determinant of the variance matrix — and minimizing an aggregate can *trade off* individual parameters. In the verified example, the D-optimal design improved the attribute coefficient's precision by 16% while making the **cost coefficient's standard error 33% *worse***. For most applications that is an acceptable trade. **For you it is not**, because every WTP you report is a *ratio* with the cost coefficient in the denominator (Chapter 23), so its precision propagates into every welfare number — and an imprecisely estimated denominator is exactly what produces the near-Cauchy tails of Chapter 7. This motivates the final section: **cost-vector design**, where the verified results show the WTP standard error falling from 0.516 to 0.245 (8% to 4% of the WTP) as the cost range widens from €6 to €55 — the design decision that most directly determines the precision of your WTP-space EC-MXL estimates.

37.1 Prerequisites

From **Chapters 20–21**: random utility, Lancaster's characteristics approach, the MNL choice probability. From **Chapter 24**: WTP space, the WTP as a ratio β/α, and scale. From **Chapter 11**: the Fisher information matrix and the asymptotic variance $\mathcal{I}^{-1}$. From **Chapter 1**: determinants, matrix inversion, and conditioning (the D-criterion is a determinant). From **Chapter 8**: power and precision. From **Chapter 7**: the delta method and the behaviour of ratios. From **Chapter 29**: cost-vector manipulation and anchoring (this chapter is its *design* counterpart).

New notation. J (alternatives per choice set), S (number of choice sets), $\mathbf{x}_j$ (attribute vector of alternative j), $\mathcal{I}(\beta, X)$ (Fisher information of a design at prior parameters), D-error, and $\tilde{\beta}$ (prior parameter values used for design).

37.2 Attributes and levels: the theoretical decisions

Before any statistics, the DCE requires substantive choices that determine what the experiment *can* say.

Definition 37.1 (Attribute and level). An **attribute** is a characteristic of the good that enters utility (Lancaster: utility derives from characteristics, not goods as wholes — Chapter 22). A **level** is a value the attribute can take. A **profile** (alternative) is a combination of one level from each attribute; a **choice set** (task) is a collection of profiles among which the respondent chooses.

Selecting attributes. Four criteria govern inclusion:

1. **Policy relevance** — the attribute must correspond to something a decision-maker can change, otherwise the estimated WTP answers no question. (Your finfish stock levels, habitat/restoration characteristics.)
2. **Demand relevance** — respondents must actually care about it; an attribute with no effect wastes design capacity and respondent attention.
3. **Comprehensibility and credibility** — respondents must understand the attribute and believe the levels are achievable. Implausible levels invite protest and inattention.
4. **Independence and non-redundancy** — attributes should not be logically entailed by one another (otherwise the design cannot vary them independently and their effects are confounded).

Selecting levels. Levels must span a range wide enough to *generate trade-offs* (if all levels are nearly identical, respondents have nothing to trade and the coefficient is imprecisely estimated — the lesson quantified for cost in §37.6), yet remain *plausible*. The **number** of levels controls the flexibility of the estimated response: two levels identify only a linear effect; three or more permit *nonlinearity* (diminishing marginal utility of stock improvement, for instance). More levels cost design efficiency, so the choice is a trade-off between functional flexibility and precision.

The cognitive-burden constraint. Every additional attribute and choice task increases the respondent's cognitive load, and beyond a threshold respondents adopt *simplifying heuristics* — attribute non-attendance

(ignoring attributes entirely), lexicographic choice (deciding on one attribute only), or random answering. These are not merely noise: they *change the data-generating process*, biasing estimates and inflating apparent scale heterogeneity (Chapter 25). The literature's rough guidance — on the order of six or fewer attributes and around eight to sixteen choice tasks — reflects this constraint, though the defensible bound depends on complexity and the population. **The design is therefore optimized subject to a cognitive-feasibility constraint, not in the abstract.**

Your familiarity dimension. Including *familiarity* is a design choice with a specific analytical purpose: it lets you test whether preferences (and scale) differ with respondent experience of the good — bearing directly on the constructed-preference concerns (does WTP reflect stable preferences, or are unfamiliar respondents constructing values on the spot?) and on the scale-versus-preference confound of Chapter 25. Note the crucial distinction: familiarity as a *respondent characteristic* is *not randomized* (it is observational — Chapter 32's warnings apply to interpreting it causally), whereas familiarity as an *experimentally manipulated information treatment* would be. Be explicit about which you have.

> **Active recall 37.A.** What four criteria govern attribute selection, and why does the number of *levels* involve a trade-off? *(Answer: attributes must be (i)* policy relevant* — correspond to something a decision-maker can change, or the WTP answers no question; (ii) *demand relevant* — respondents must care, or design capacity is wasted; (iii) *comprehensible and credible* — implausible or confusing attributes invite protest, inattention, and heuristic answering; and (iv) *independent/non-redundant* — logically entailed attributes cannot be varied independently and their effects are confounded. The *number of levels* trades *functional flexibility* against *precision*: two levels identify only a linear effect, while three or more permit estimating *nonlinearity* (e.g. diminishing marginal utility of stock improvement) — but each extra level spreads the design's information more thinly, reducing the precision of each estimated parameter. All of this is subject to a *cognitive-burden constraint*: too many attributes/tasks pushes respondents into non-attendance, lexicographic, or random answering, which changes the data-generating process rather than merely adding noise.)*

37.3 The combinatorial problem: full factorial and fractions

Once attributes and levels are fixed, the number of possible tasks explodes.

Definition 37.2 (Full factorial). The **full factorial** is the set of *all* possible profiles: $\prod_k L_k$ where L_k is the number of levels of attribute k. The set of all possible *pairs* (for a two-alternative design) is $\binom{P}{2}$ where P is the number of profiles.

Worked Example 37.1 (Design size, verified). Three environmental attributes at three levels each, plus cost at four levels:

Quantity	Value
Attribute levels	$3 \times 3 \times 3 \times 4$
Full factorial profiles	**108**
Possible pairs of profiles	**5,778**

(Verified.) A deliberately *modest* design already yields 108 profiles and 5,778 possible choice sets — no respondent can see them all, and you need perhaps 12–16 tasks. So you must select a **fraction**: a small subset of choice sets that, despite its size, permits estimation of all parameters. **The entire question of experimental design is *which* fraction** — and the answer has two competing traditions.

Blocking. Even a well-chosen fraction may contain more tasks than one respondent should face. **Blocking** partitions the design into subsets ("blocks"), with each respondent randomly assigned one block. This preserves the design's overall properties across the sample while limiting individual burden. Blocking must

be *random* (so blocks are comparable — Chapter 31) and *balanced* (each block seen by a similar number of respondents), and the block indicator should be recorded and checked for balance (Chapter 32).

37.4 Orthogonal versus efficient designs

The two design philosophies differ in what they optimize — and in whether they take the nonlinearity of the choice model seriously.

Orthogonal designs. An **orthogonal** design ensures the attribute columns are *statistically uncorrelated* — each attribute's levels appear equally often in combination with every other attribute's levels (level balance and orthogonality). This tradition comes from *linear* models, where orthogonality *does* minimize the variance of OLS estimates: with $\mathrm{Var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$, orthogonality makes $X'X$ diagonal and the variance minimal for a given number of runs. Its virtues are that it requires *no prior knowledge* of parameter values and yields clean, interpretable independence.

Why orthogonality is the wrong criterion for choice models. In a *logit* model the variance is *not* $\sigma^2(X'X)^{-1}$. From Chapter 12, the asymptotic variance is the inverse Fisher information, and for the MNL the information depends on the *choice probabilities*:

$$\mathcal{I}(\beta) = \sum_s \sum_j P_{sj}\,(\mathbf{x}_{sj} - \bar{\mathbf{x}}_s)(\mathbf{x}_{sj} - \bar{\mathbf{x}}_s)', \qquad \bar{\mathbf{x}}_s = \sum_j P_{sj}\mathbf{x}_{sj}.$$

The probabilities P_{sj} depend on the *parameters* β. So **the information content of a design depends on the parameter values it is meant to estimate** — a circularity absent from linear models. Two consequences follow. First, orthogonality of the X columns does *not* minimize the variance here (the weighting by P_{sj} breaks the equivalence). Second, a choice set in which one alternative is *overwhelmingly* preferred ($P \approx 1$) carries almost *no* information — everyone picks the same thing, revealing nothing about trade-offs — even if the design is perfectly orthogonal. Utility balance matters, and orthogonality is blind to it.

Efficient (D-optimal) designs. An **efficient design** directly minimizes a scalar summary of the asymptotic variance matrix, given *prior* values $\tilde{\beta}$. The standard criterion is the **D-error**:

$$\text{D-error} = \det\!\big(\mathcal{I}(\tilde{\beta}, X)^{-1}\big)^{1/K},$$

the K-th root of the determinant of the variance matrix (K = number of parameters), normalized so it is comparable across designs of different dimension. Lower D-error means a more informative design. Related criteria: **A-error** (trace of the variance matrix — the average variance), **C-error** (variance of a specified *linear combination*, e.g. a WTP ratio), and **S-error** (the sample size needed for significance).

The prior problem. Efficient designs require priors $\tilde{\beta}$ — but if you knew the parameters you would not run the study. Standard resolutions: use *zero* priors (which reduces to something close to orthogonality, since all P_{sj} are then equal), use *point* priors from pilot data or the literature (your meta-analysis of 18 DCEs is an unusually good source), or use **Bayesian efficient designs** that integrate the D-error over a *prior distribution*, producing designs robust to prior misspecification. Bayesian efficiency is the defensible default when priors are uncertain: it avoids optimizing hard against a guess that may be wrong.

Worked Example 37.2 (D-optimal versus random, verified). For the design above (16 choice sets, $K = 7$ parameters), with priors including a cost coefficient of -0.08:

Design	D-error
Random	0.2196
D-optimal (greedy swap search)	**0.0973**
Improvement	**56% lower**

(Verified.) The optimized design has a D-error 56% below a randomly constructed one — a substantial gain purchased purely by *thinking about which choice sets to show*, at zero fieldwork cost.

> **Stop and think 37.1.** Why does an orthogonal design not minimize estimator variance in a logit model, and what does a choice set with a near-certain winner contribute? *(Answer: in a* linear* model $\text{Var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$, so making X columns orthogonal diagonalises $X'X$ and minimises variance. In a *logit* the variance is the inverse *Fisher information*, $\mathcal{I} = \sum_s \sum_j P_{sj}(\mathbf{x}_{sj} - \bar{\mathbf{x}}_s)(\mathbf{x}_{sj} - \bar{\mathbf{x}}_s)'$, which weights each choice set by the *choice probabilities* — so information depends on the *parameter values*, and orthogonality of the raw X columns no longer implies minimum variance. A choice set with a *near-certain winner* ($P \approx 1$) contributes almost *nothing*: the deviations $(\mathbf{x}_{sj} - \bar{\mathbf{x}}_s)$ are weighted by probabilities that concentrate on one alternative, so the information contribution collapses — everyone chooses the same option and no trade-off is revealed. Hence *utility balance* (choice sets where alternatives are competitive) is essential, and an orthogonal design that happens to contain dominated alternatives can be badly uninformative — verified: a D-optimal design achieved 56% lower D-error than a random one of identical size.)*

37.5 What efficiency buys — and what it can cost

Efficiency gains translate into either tighter estimates or smaller samples. But the aggregate criterion can conceal a trade you may not want.

Worked Example 37.3 (Precision and sample-size equivalence, verified). With $N = 600$ respondents each facing 16 choice sets:

Parameter	Random design SE	D-optimal SE	Change
Attribute-1 coefficient	0.0278	**0.0233**	**16% smaller**
Cost coefficient	**0.00190**	**0.00252**	**33% *worse***

Overall efficiency: **1.41**× — the D-optimal design achieves equivalent aggregate precision with **71% of the sample** (N = 424 versus 600).

(Verified.) The sample-size result is the practical headline: a better design saves roughly 176 respondents' worth of fieldwork. Since variance scales as $1/N$, an efficiency ratio of r means the required sample is N/r — design efficiency converts directly into money saved or precision gained.

The critical caveat: D-error is an aggregate. Note the second row. The D-optimal design made the **cost coefficient *less* precise** — a 33% larger standard error. This is not an error in the optimization; it is *what D-optimality does*. The D-criterion minimizes the *determinant* of the whole variance matrix, a scalar summary of *overall* information volume. Minimizing an aggregate permits *trading off* individual parameters: the optimizer accepted a worse cost coefficient in exchange for larger gains elsewhere, because the determinant improved.

Why this specifically matters for you. For most applications this trade is fine. **For your work it is potentially serious**, because:

- Every WTP you report is a **ratio with the cost coefficient in the denominator** (Chapter 23): $\text{WTP}_k = \beta_k/\alpha$.
- The variance of a ratio (delta method, Chapter 7) is *acutely* sensitive to the denominator's variance — the term $(\beta_k^2/\alpha^4)\text{Var}(\alpha)$ grows rapidly as α becomes imprecise.
- When the cost coefficient's distribution has appreciable mass near zero, the WTP ratio exhibits **near-Cauchy heavy tails** (Chapter 7) — potentially undefined moments and unstable means.

So a design that optimizes D-error while degrading the cost coefficient is **optimizing the wrong objective for a WTP study**. The remedies:

1. **Use a C-optimal (or WTP-focused) criterion** that targets the variance of the *ratios* you actually report, rather than the determinant of the full matrix.
2. **Design directly in WTP space**, consistent with your EC-MXL parameterization (Chapter 23), so the design optimizes the precision of the objects you estimate.
3. **Weight the criterion** toward the cost coefficient, or impose a constraint that its variance not deteriorate.
4. **At minimum, inspect the full variance matrix** — not just the scalar D-error — and verify the cost coefficient's precision explicitly.

The general lesson: *optimize the criterion that matches your estimand.* A scalar design criterion is a summary, and summaries hide trades. This is the design-stage analogue of the estimand discipline of Chapter 32 §32.7.

> **Active recall 37.B.** What does a 1.41× design efficiency mean practically, and why might a D-optimal design nonetheless be the wrong choice for a WTP study? *(Answer: since sampling variance scales as* $1/N$*, an efficiency ratio of 1.41 means the* same precision* is achieved with $N/1.41 \approx 71\%$ of the sample — verified, N=424 instead of 600, saving ~176 respondents of fieldwork cost (or, equivalently, tighter estimates at the same N). It may nonetheless be the *wrong* criterion for a WTP study because **D-error is an aggregate** — the determinant of the whole variance matrix — so minimizing it can *trade off individual parameters*: verified, the D-optimal design improved the attribute coefficient 16% but made the **cost coefficient's SE 33% worse**. Since every WTP is a ratio β_k/α with cost in the *denominator*, and the delta-method variance of a ratio is acutely sensitive to denominator imprecision (with near-Cauchy tails when α has mass near zero, Chapter 7), degrading α damages *every* welfare estimate. The fix: use a C-optimal/WTP-focused criterion, design directly in WTP space to match the EC-MXL parameterization, weight the criterion toward cost, or at minimum inspect the full variance matrix rather than the scalar D-error.)*

37.6 The cost vector: the design decision behind your WTP

The cost attribute deserves separate treatment, because it is both the *numeraire* that converts utility into money and — through your cost-vector manipulation — a *treatment* in its own right.

Cost as numeraire. The cost coefficient α is the marginal utility of income (Chapter 34), the divisor that turns attribute coefficients into money-metric WTP. Its precision therefore propagates into *every* welfare estimate, which is why §37.5's warning matters and why the cost *levels* deserve deliberate design rather than convention.

Range generates identification. The cost coefficient is identified by respondents *trading off* cost against attributes. If the cost levels are tightly clustered, respondents face little price variation, the coefficient is weakly identified, and — because WTP divides by it — the WTP standard error inflates.

Worked Example 37.4 (Cost range and WTP precision, verified). D-optimal 12-set designs with three different cost vectors, N = 600, delta-method standard error for WTP $= -\beta_1/\alpha$ (true WTP = 6.25):

Cost levels	Range	SE(cost coef)	SE(WTP)	Relative SE
€18, 20, 22, 24	€6	0.0055	**0.516**	**8%**
€10, 20, 30, 40	€30	0.0030	**0.295**	**5%**
€5, 15, 30, 60	€55	0.0029	**0.245**	**4%**

(Verified.) Widening the cost range from €6 to €55 nearly **halves** the WTP standard error (0.516 → 0.245, from 8% to 4% of the WTP). The mechanism is exactly as theory predicts: a wider range identifies the cost coefficient more precisely (SE 0.0055 → 0.0029), and that precision propagates through the ratio. **The cost**

vector is thus among the highest-leverage design decisions in your DCE — more consequential for WTP precision than most attribute-level choices.

The countervailing constraints. Range cannot be maximized without limit:

- **Plausibility.** Costs must be credible for the good and population; implausible amounts trigger protest, disbelief, and inattention (Chapters 26, 29).
- **Anchoring and cost-vector effects.** This is your own research question (Chapter 29): the cost vector may *shape* the WTP it is meant to measure. A wide range is statistically efficient but may exert stronger anchoring — so the statistically optimal range and the *behaviourally* clean range can conflict.
- **Dominance.** Extreme cost levels can create *dominated* alternatives (an option worse on every attribute), which respondents dismiss without trading off — contributing little information (§37.4) and possibly signalling design carelessness.

The design-as-treatment insight. Your research design *manipulates the cost vector experimentally* — different respondents receive different cost vectors — which converts a design parameter into a *randomized treatment* (Chapter 31). This is a genuinely elegant move: it means the cost vector's *behavioural* effect (anchoring, Chapter 29) becomes *identified* rather than assumed away, and the κ-versus-λ question (Chapter 30) becomes empirically addressable. But note the tension this chapter exposes: the *statistically efficient* cost vector and the *experimentally informative* set of cost vectors are different objects. Your design must serve both — efficient enough to estimate WTP precisely within each arm, varied enough across arms to identify the anchoring effect. Make that trade-off explicit rather than letting design software resolve it silently.

> **Stop and think 37.2.** Your co-author proposes narrowing the cost range to "keep the amounts realistic." What do you tell them? *(Answer: realism matters — implausible costs trigger protest, disbelief, and inattention — but a* narrow* range carries a steep and quantifiable statistical price: verified, shrinking the cost range from €55 to €6 raised the cost coefficient's SE from 0.0029 to 0.0055 and nearly *doubled* the WTP standard error (0.245 → 0.516, i.e. 4% → 8% of the WTP). The mechanism: the cost coefficient is identified by respondents *trading off* price against attributes, so little price variation means weak identification — and because WTP divides by that coefficient, the imprecision propagates into every welfare estimate (with near-Cauchy tail risk if the coefficient's distribution approaches zero, Chapter 7). So the answer is to seek the *widest range that remains plausible*, not the narrowest that feels safe — and to check explicitly for *dominated* alternatives at the extremes (which contribute little information) and for *anchoring* effects (our own research question, Chapter 29). If realism genuinely binds, compensate with a larger sample or a WTP-focused (C-optimal) design criterion — but recognise we are paying for realism in precision, and quantify the price rather than assuming it away.)*

37.7 Constructing choice sets: practical rules

Several construction decisions remain once the fraction is chosen.

Number of alternatives. More alternatives per set yield more information per task but increase burden. Two or three plus an opt-out is typical.

The status quo / opt-out. Including a **status-quo or "neither" option** is important for *welfare* interpretation: without it, respondents are forced to choose *some* improvement, and the design cannot represent the policy-relevant possibility of *no* change. The resulting WTP is then conditional on participation and is not a clean welfare measure relative to the current state (Chapter 34's reference-state requirement). The status quo must be *credible* and *constant* across tasks. Note that opt-out choices are often driven by different processes (protest, task avoidance, genuine zero value) — the same distinction as Chapter 26's spike models.

Avoiding dominance. A **dominated** alternative is worse on every attribute than another in the same set. Such sets are uninformative (the choice is trivial) and may reduce credibility. Efficient design algorithms

typically avoid them automatically, but designs should be inspected.

Attribute-level overlap. Some **overlap** (an attribute taking the *same* level in both alternatives) reduces cognitive burden and can help respondents focus, but too much overlap removes the trade-off variation that identifies the parameter. This is a deliberate design dial, not an accident.

Randomization of position. The *order* of alternatives within a set and of tasks within a block should be randomized to guard against left-right and ordering effects — a source of systematic bias that is easy to eliminate at zero cost.

37.8 Sample size for a DCE

Sample-size determination differs from the simple formulas of Chapter 9 because each respondent contributes *multiple* correlated choices.

The information logic. Total information is approximately $N \times \mathcal{I}(\text{design})$, where $\mathcal{I}$ is the per-respondent information from the choice sets faced. So precision improves with *both* the number of respondents *and* the informativeness of the design — which is why §37.5's 1.41× efficiency translated directly into a 29% sample reduction. But respondents' choices are *correlated within person* (panel structure, Chapter 21), so doubling tasks is *not* equivalent to doubling respondents: additional tasks from the same person add less independent information, and beyond some point add fatigue-driven noise instead.

Practical approaches. (i) The **S-error** criterion reports the sample size needed for each parameter to reach significance under the priors — directly actionable at the design stage. (ii) **Simulation-based power analysis** (Chapter 9's logic, Chapter 11's Monte Carlo machinery) is the most flexible and honest: simulate choices under plausible parameters and your actual design, estimate the model, and record how often effects are recovered with adequate precision. (iii) Remember Chapter 33's **4× rule for interactions** — if your headline is a cost-vector × familiarity interaction, or a real × hypothetical contrast, the sample must be powered for the *interaction*, not the main effects. For a mixed logit the requirement is greater still, since the *distributional* parameters (heterogeneity, error-component variances) are harder to identify than means.

37.9 Failure modes

F1. Software defaults unexamined. Accepting a design without knowing what criterion was optimized under what priors. *Fix:* state the criterion, priors, and algorithm; inspect the variance matrix (§37.4–§37.5).

F2. Optimizing D-error when WTP is the estimand. Aggregate criterion degrading the cost coefficient. *Fix:* C-optimal/WTP-space criterion; check cost precision explicitly (§37.5).

F3. Orthogonality treated as optimality. Assuming an orthogonal design is best for a choice model. *Fix:* recognize the probability-weighted information; consider efficient designs and utility balance (§37.4).

F4. Over-narrow cost range. Clustered cost levels producing imprecise WTP. *Fix:* widest plausible range; quantify the precision cost of narrowing (§37.6).

F5. Priors ignored or over-trusted. Zero priors by default, or point priors treated as certain. *Fix:* Bayesian efficient designs integrating over prior uncertainty (§37.4).

F6. Cognitive overload. Too many attributes/tasks inducing non-attendance or lexicographic choice. *Fix:* limit attributes/tasks; test in pilots; model non-attendance if suspected (§37.2).

F7. No status quo. Forcing a choice among improvements, precluding clean welfare interpretation. *Fix:* include a credible, constant opt-out (§37.7).

F8. Dominated alternatives / excessive overlap. Uninformative choice sets. *Fix:* inspect and eliminate dominance; tune overlap deliberately (§37.7).

F9. Blocking unbalanced or non-random. Blocks confounded with respondent characteristics. *Fix:* randomize block assignment; check balance (§37.3; Chapter 32).

F10. Design-treatment tension unacknowledged. Optimizing the cost vector for efficiency while also using it as a treatment. *Fix:* make the dual role explicit; ensure efficiency within arms and variation across them (§37.6).

37.10 Diagnostic questions for your own papers

1. **Criterion and priors.** What criterion was minimized, under what priors, and is it the right one for a WTP estimand? *(§37.4–§37.5, F1–F2, F5.)*
2. **Cost precision.** Have I verified the cost coefficient's standard error, not just the aggregate D-error? *(§37.5, F2.)*
3. **Cost range.** Is the cost vector as wide as plausibility permits, and have I quantified the precision consequence? *(§37.6, F4.)*
4. **Dual role of cost.** Have I reconciled the cost vector's role as efficient numeraire *and* randomized treatment? *(§37.6, F10.)*
5. **Burden.** Are attribute and task counts within cognitive limits, tested in pilots? *(§37.2, F6.)*
6. **Status quo.** Is there a credible opt-out, and are opt-out choices modelled appropriately? *(§37.7, F7.)*
7. **Design hygiene.** Any dominated alternatives; is overlap deliberate; are positions randomized? *(§37.7, F8.)*
8. **Blocking.** Random, balanced, recorded, and checked? *(§37.3, F9.)*
9. **Sample size.** Determined by simulation under the actual design, powered for *interactions* where those are the headline? *(§37.8; Chapter 33.)*

37.11 Exercises

(A) Comprehension.

A1. Why can't respondents simply be shown the full factorial, and what is a fractional design?

A2. Explain why orthogonality is optimal for linear models but not for choice models.

A3. What is D-error, and what does a lower value mean?

(B) Derivation / reasoning.

B1. Write the MNL Fisher information for a design and explain why it depends on the parameters.

B2. Show, via the delta method, why the cost coefficient's precision governs WTP precision.

B3. Explain why D-optimality can worsen an individual parameter's precision, and what criterion to use instead when a ratio is the estimand.

(C) Research application / critique.

C1. Design the cost vector for your DCE, reconciling efficiency, plausibility, anchoring, and its role as a treatment.

C2. Your design software returns a D-optimal design. Write the checks you perform before fielding it.

C3. Justify your attribute set (including familiarity) and task count, and state what each attribute identifies.

37.12 Solutions

A1. The full factorial is every possible combination of attribute levels — verified, a modest $3 \times 3 \times 3 \times 4$ design yields **108 profiles and 5,778 possible pairs**, far beyond what any respondent can evaluate (typical burden limits are 8–16 tasks). A **fractional design** is a small subset of choice sets selected so that all model parameters remain estimable despite the reduced size. The design question is *which* fraction: chosen for *orthogonality* (uncorrelated attribute columns, from the linear-model tradition) or for *statistical efficiency* (minimizing a scalar summary of the asymptotic variance given priors). Additionally, **blocking** splits the chosen fraction across respondents so each sees a manageable subset while the sample as a whole covers the design.

A2. In a *linear* model, $\text{Var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$ depends on the design *only* through X; orthogonality diagonalizes $X'X$ and minimizes the variance for a given number of runs — optimality without needing to know β. In a *choice* model the variance is the inverse *Fisher information*, $\mathcal{I} = \sum_s \sum_j P_{sj}(\mathbf{x}_{sj} - \bar{\mathbf{x}}_s)(\mathbf{x}_{sj} - \bar{\mathbf{x}}_s)'$, where the *choice probabilities* P_{sj} weight each contribution — and those probabilities depend on β. Hence (i) design informativeness depends on the *unknown parameters* (requiring priors), and (ii) orthogonality of raw X columns no longer minimizes variance, since a probability-weighted criterion is being optimized. Concretely, an orthogonal design containing sets with a *near-certain winner* ($P \approx 1$) wastes those sets: the weighted deviations collapse and almost no trade-off information is obtained. *Utility balance* — competitive alternatives — matters, and orthogonality does not deliver it.

A3. The **D-error** is $\det(\mathcal{I}(\tilde{\beta}, X)^{-1})^{1/K}$: the K-th root of the determinant of the asymptotic variance matrix evaluated at prior parameters, normalized for comparability across designs with different numbers of parameters. The determinant measures the *volume* of the joint confidence ellipsoid, so **lower D-error means a smaller ellipsoid — more information, more precise estimates overall** (verified: D-optimal 0.0973 versus random 0.2196, a 56% reduction, worth 1.41× efficiency, i.e. equivalent precision at 71% of the sample). Because it is a *determinant* — an aggregate over all parameters — it does *not* guarantee every individual parameter improves (verified: the cost coefficient's SE worsened 33%).

B1. For the MNL, the log-likelihood contribution from choice set s is $\sum_j y_{sj} \ln P_{sj}$ with $P_{sj} = e^{\mathbf{x}'_{sj}\beta} / \sum_l e^{\mathbf{x}'_{sl}\beta}$. Differentiating twice and taking expectations (Chapter 12's information matrix equality) gives

$$\mathcal{I}(\beta) = \sum_s \sum_j P_{sj}\,(\mathbf{x}_{sj} - \bar{\mathbf{x}}_s)(\mathbf{x}_{sj} - \bar{\mathbf{x}}_s)', \qquad \bar{\mathbf{x}}_s = \sum_l P_{sl}\mathbf{x}_{sl}.$$

Each choice set contributes a probability-weighted covariance of attributes *about the choice-probability-weighted mean*. It **depends on** β through the P_{sj}: the same attribute matrix yields different information at different parameter values. This is why efficient design requires *priors*, why the design problem is circular (needing what you seek to estimate), and why *Bayesian* efficient designs — integrating D-error over a prior *distribution* — are the robust resolution. It also formalizes utility balance: when one $P_{sj} \to 1$, the weighted deviations vanish and the set contributes nearly nothing.

B2. With $\text{WTP} = -\beta_k/\alpha$, the delta method (Chapter 7) gives gradient $\partial\text{WTP}/\partial\beta_k = -1/\alpha$ and $\partial\text{WTP}/\partial\alpha = \beta_k/\alpha^2$, so

$$\text{Var}(\widehat{\text{WTP}}) \approx \frac{1}{\alpha^2}\text{Var}(\hat{\beta}_k) + \frac{\beta_k^2}{\alpha^4}\text{Var}(\hat{\alpha}) - \frac{2\beta_k}{\alpha^3}\text{Cov}(\hat{\beta}_k, \hat{\alpha}).$$

The cost coefficient enters through α^4 in the second term — so as α becomes small or imprecise, the WTP variance grows *much* faster than through the numerator. Verified: widening the cost range reduced SE(α) from 0.0055 to 0.0029 and roughly halved SE(WTP) from 0.516 to 0.245. Beyond the delta approximation, when α's distribution has appreciable mass near *zero* the ratio has **near-Cauchy heavy tails** (Chapter 7) — the delta method understates the true uncertainty and the mean WTP may be unstable or undefined. Hence cost-coefficient precision is the binding constraint on WTP precision, and any design or specification degrading it damages every welfare estimate.

B3. D-optimality minimizes $\det(\mathcal{I}^{-1})^{1/K}$ — a *scalar aggregate* over all K parameters, geometrically the *volume* of the confidence ellipsoid. Volume can be reduced by shrinking the ellipsoid greatly in some dimensions

while *expanding* it in others: the determinant improves even though an individual variance rises. Verified: the D-optimal design improved the attribute coefficient's SE by 16% but worsened the **cost** coefficient's by 33% — a good trade by the D-criterion, a bad one for a WTP study, since (by B2) cost precision governs *every* WTP ratio. The appropriate alternative is a criterion matching the estimand: **C-optimality**, which minimizes the variance of a specified *linear combination* (or, via the delta method, a *ratio*) — i.e. optimize directly for $\mathrm{Var}(\widehat{\mathrm{WTP}})$. Practically: design in **WTP space** consistent with the EC-MXL parameterization (Chapter 23), or weight the criterion toward the cost coefficient, or constrain its variance not to deteriorate. General principle: *optimize the criterion matching the estimand*, and never accept a scalar design summary without inspecting the full variance matrix.

C1. *Cost-vector design.* Begin from the **widest plausible range**, since range drives identification: verified, moving from €6 to €55 of spread cut SE(WTP) from 0.516 to 0.245 (8% → 4% of WTP), while the narrow vector nearly doubled the imprecision. *Plausibility bound:* levels must be credible as an annual household payment for the finfish/ecosystem programme, informed by pilot debriefing and the payment vehicle; implausible levels generate protest and inattention (Chapters 26, 29). *Spacing:* use unequal (roughly geometric) spacing — e.g. €5/15/30/60 rather than €18/20/22/24 — to cover the WTP distribution's plausible range and identify *nonlinearity* in the cost response, echoing Chapter 36's lesson that bids concentrated where responses are near-certain waste information. *Dominance check:* verify no cost level creates alternatives dominated on every attribute. *Anchoring/treatment role:* because the cost vector is *also* my randomized treatment (Chapter 29's anchoring question), I need *variation across arms* — e.g. a "low" and a "high" cost vector, each internally efficient — so that within-arm WTP is precisely estimated *and* between-arm comparison identifies the cost-vector effect (the κ-versus-λ question, Chapter 30). *Reconciliation:* run the efficient-design algorithm *separately within each cost-vector arm* (so each arm is internally efficient), then verify the arms differ enough to power the *interaction* (Chapter 33's 4× rule). *Report:* the criterion, priors, resulting SE(cost) and SE(WTP) per arm, and the anchoring-versus-efficiency trade-off made explicit rather than resolved silently by software.

C2. *Pre-fielding checks on a D-optimal design.* (i) **Criterion and priors** — confirm which criterion was minimized and under what priors; if point priors, re-generate as a **Bayesian** efficient design integrating over prior uncertainty, and check the design's robustness if priors are wrong (F5). (ii) **Full variance matrix** — inspect *every* parameter's implied standard error, not just the scalar D-error; specifically verify the **cost coefficient** has not been degraded (verified risk: 33% worse SE), since WTP ratios depend on it (F2). (iii) **Implied WTP precision** — compute the delta-method SE for each WTP I intend to report; this, not D-error, is the estimand. (iv) **Dominance scan** — check no choice set contains a dominated alternative. (v) **Utility balance** — compute implied choice probabilities under the priors and flag sets where one alternative has $P \approx 1$ (near-zero information). (vi) **Overlap** — confirm overlap is deliberate and not so extensive that key trade-offs vanish. (vii) **Level balance and plausibility** — check each level appears sufficiently often and every profile is substantively credible. (viii) **Blocking** — confirm blocks are balanced, randomly assigned, and recorded. (ix) **Simulation-based power** — simulate choices under the design and plausible parameters, estimate the intended model (EC-MXL in WTP space, not just MNL), and confirm recovery of the parameters *and interactions* I plan to report (Chapter 33). (x) **Pilot** — field a small pilot, update priors, and regenerate the design. The overarching rule: a design is a hypothesis about information, and it should be *tested by simulation* before consuming the fieldwork budget.

C3. *Attribute set justification.* Each attribute must be policy-relevant, demand-relevant, comprehensible, and independently variable (§37.2). *Ecological/stock attributes* identify the marginal WTP for the biophysical improvements the FRD policy can deliver — these are the coefficients feeding the CBA benefit stream (Chapters 45–46), so they must map onto quantities the restoration actually changes. Each is specified at *three* levels to permit estimating *nonlinearity* (diminishing marginal utility of successive improvement), which matters because CBA scenarios interpolate between levels. *Cost* is the numeraire converting utility to money (Chapter 34) and simultaneously the randomized treatment (§37.6); it carries four levels over a wide plausible range to identify the coefficient precisely and to test cost-vector effects. *Familiarity* identifies whether preferences and *scale* differ with respondent experience — addressing the constructed-preference concern (are unfamiliar respondents constructing values on the spot?) and the scale-versus-preference confound (Chapter 25), and providing a moderator for the meta-analysis (Chapters 47–48). **Critical caveat:** if familiarity is a

measured respondent characteristic it is *not randomized*, so familiarity contrasts are *observational* and subject to Chapter 32's confounding warnings — I will describe them as associations, or manipulate familiarity experimentally (an information treatment) if a causal claim is intended. *Task count:* 12–16 tasks per respondent, blocked from the larger design, chosen to stay within cognitive limits (§37.2) and validated in pilot debriefing for fatigue and non-attendance; the count is confirmed by simulation-based power for the *interactions* that constitute the headline (Chapter 33's 4× rule), not merely the main effects.

37.13 Spaced review prompts

+1 day. From memory: write the MNL Fisher information and explain its parameter dependence; define D-error; state why orthogonality is not optimal for choice models.

+1 week. Explain what design efficiency buys in sample-size terms, why D-optimality can degrade an individual parameter, and why cost range drives WTP precision.

+1 month. For your DCE, audit the design: criterion and priors, cost-coefficient precision, cost-vector range and its dual role, burden, status quo, dominance, blocking, and simulation-based power for interactions. Close with the spine sentence about what your design identifies and at what precision.

37.14 How this chapter connects to the rest of the book

- **Chapter 22** supplies the RUM and MNL probabilities whose derivatives *are* the design's information matrix; **Chapter 23** supplies the WTP-space parameterization that should govern the design criterion; **Chapter 12** supplies the Fisher information and asymptotic variance underlying D-error; **Chapter 1** supplies the determinant and conditioning intuition.
- **Chapter 29** is the behavioural counterpart of §37.6 — where this chapter treats the cost vector as a *precision* instrument, Chapter 29 treats it as a potential *anchoring* influence, and your design must serve both; **Chapter 30** is where the resulting κ-versus-λ identification question is settled; **Chapter 31–33** supply the randomization, threat, and power apparatus for the cost-vector *treatment* (including the 4× interaction rule governing §37.8).
- **Chapter 25** explains why cognitive overload (§37.2) manifests as scale heterogeneity; **Chapter 7** explains why cost-coefficient imprecision produces heavy-tailed WTP; **Chapters 38–39** take the fielded design into estimation and validity; **Chapters 45–48** consume the resulting WTP estimates in CBA and meta-analysis, where design characteristics become moderators.

This chapter is where your DCE's precision is determined — before any data exist — and where the discipline of *optimizing the criterion that matches your estimand* replaces reliance on software defaults.

37.15 Research translation task

For your **DCE**, produce a three-page design note:

1. **Attributes and levels.** Tabulate each attribute, its levels, and *what it identifies*; justify against the four criteria and state the cognitive-burden reasoning for the attribute and task counts.
2. **Design criterion.** State the criterion (D-, C-, or WTP-space efficiency), the priors (ideally from your 18-DCE meta-analysis), and whether the design is Bayesian-efficient; justify the choice *against the WTP estimand*.
3. **Cost vector.** Specify levels, range, and spacing; report the implied SE(cost) and SE(WTP); document the plausibility bound and the anchoring trade-off; explain how efficiency-within-arm and variation-across-arms are jointly achieved.

4. **Design audit.** Report the full implied variance matrix (not just D-error), dominance and utility-balance checks, overlap policy, position randomization, and blocking scheme with balance checks.
5. **Sample size.** Determine by *simulation* under the actual design and intended EC-MXL specification, powered for the *interactions* that constitute the headline (Chapter 33).
6. **Close** with the spine sentence: *"My design identifies [attributes' marginal WTP] at [precision], optimized under [criterion] with [priors]; the cost vector spans €[·]–€[·] to give SE(WTP) = [·], and its experimental variation across arms identifies the cost-vector effect with [power]."*

Carry this into Chapter 38 (estimation of the fielded design), Chapters 45–46 (CBA), and Chapters 47–48 (meta-analysis), where design characteristics become explanatory moderators of WTP heterogeneity across studies.

37.16 Bibliography

Core. - Louviere, Hensher & Swait, *Stated Choice Methods: Analysis and Applications* — the standard reference on DCE design, attributes, and choice-set construction (§37.2–§37.4). *(Verify edition.)* - Rose & Bliemer (2009), "Constructing Efficient Stated Choice Experimental Designs," *Transport Reviews* — the definitive accessible treatment of efficient design, D-error, and priors (§37.4–§37.5). *(Verify volume/pages.)* - Hensher, Rose & Greene, *Applied Choice Analysis* — practical design and estimation guidance (§37.2–§37.8). *(Verify edition.)*

Applied. - Johnson et al. (2013), "Constructing Experimental Designs for Discrete-Choice Experiments: Report of the ISPOR Conjoint Analysis Experimental Design Good Research Practices Task Force," *Value in Health* — consensus practical guidance (§37.2–§37.7). *(Verify volume/pages.)* - Bliemer & Rose (2010) on sample size and design efficiency in stated choice studies (§37.8). *(Verify citation.)* - Scarpa & Rose (2008), "Design Efficiency for Non-Market Valuation with Choice Modelling: How to Measure it, What to Report and Why," *Australian Journal of Agricultural and Resource Economics* — efficiency criteria specifically for *WTP* estimation, directly addressing §37.5's C-optimality point. *(Verify volume/pages.)*

Advanced. - Sándor & Wedel (2001), "Designing Conjoint Choice Experiments Using Managers' Prior Beliefs," *Journal of Marketing Research* — Bayesian efficient designs under prior uncertainty (§37.4). *(Verify volume/pages.)* - Kessels, Goos & Vandebroek on multi-objective and Bayesian optimal designs for choice experiments (§37.5). *(Verify citation.)* - Bliemer & Rose on designs for *mixed logit* / panel error-component specifications, relevant to your EC-MXL (§37.8). *(Verify citation.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Rose–Bliemer / Scarpa–Rose / Sándor–Wedel citations before using them, per the book's no-unverified-citation rule.

End of Chapter 37. Next: Chapter 38 — Estimating the Mixed Logit: Error Components, WTP Space, and Simulated Likelihood, which takes the fielded design into estimation — the error-components mixed logit at the centre of your DCE, the WTP-space parameterization, distributional choices for heterogeneity, simulated maximum likelihood with Halton draws, and the convergence and identification diagnostics that determine whether your estimates can be trusted. Tell me when to write it, or request a different chapter.

Chapter 38 — Estimating the Mixed Logit: Error Components, WTP Space, and Simulated Likelihood

Part IX: Stated Preference in Practice

> **The spine question, applied to this chapter.** Chapter 37 built the design; this chapter estimates the model it was built for — the **error-components mixed logit in WTP space** at the centre of your DCE. Three things make this harder than any estimation so far. The likelihood has **no closed form** (it is an integral over the distribution of preferences), so it must be *simulated* — introducing a bias that depends on how many draws you take. The **parameterization** (preference space versus WTP space) determines whether your welfare estimates are stable numbers or heavy-tailed ratios. And the **heterogeneity parameters** — the spreads, the error-component variances — are systematically harder to identify than the means, so a model that "converged" may still be telling you very little about the distribution you care about. The spine question: *is my estimate a property of respondents' preferences, or an artifact of my draws, my distributional assumptions, and an optimizer that stopped somewhere?*

38.0 Conceptual motivation: three simultaneous difficulties

The multinomial logit of Chapter 22 has a closed-form likelihood and a globally concave log-likelihood: estimation is essentially trouble-free. But MNL imposes homogeneous preferences and IIA (Chapter 22's red-bus/blue-bus problem), both of which are indefensible for your data. Your respondents differ — that heterogeneity *is* a finding, not a nuisance — and they answer *repeated* tasks, so their choices are correlated within person. The **mixed logit** solves both. It is also, as a consequence, substantially harder to estimate, and this chapter is about the three difficulties that arrive together.

First: the likelihood must be simulated. Allowing preferences to vary across people means the probability of an observed choice sequence is an *integral* over the preference distribution — with no analytical solution. You approximate it by taking R draws from the assumed distribution and averaging. But the log-likelihood requires the *logarithm* of that average, and since the logarithm is concave, **the log of a simulated average is a biased estimate of the log of the true probability** (Jensen's inequality, Chapter 7). Verified below, the bias is negative and decays roughly as $1/R$ (from -0.0044 at five draws to -0.0001 at a hundred), while the simulation *noise* decays only as $1/\sqrt{R}$. Too few draws and your log-likelihood is systematically wrong — and, worse, *noisily* wrong, which wrecks the optimizer's ability to find a maximum. The fix is not merely "more draws" but *better* draws: **Halton sequences** cover the space more evenly than pseudo-random numbers, and verified below they cut simulation noise by factors of 3.2 to 10 for the same R — the single cheapest improvement available.

Second: the parameterization determines whether your welfare estimates are stable. In **prefer-**

ence space you estimate random coefficients on attributes and cost, then form WTP as their *ratio*. When the cost coefficient's distribution has mass near zero — which a normal distribution always does — that ratio has near-Cauchy tails (Chapter 7), and its mean may not exist. Verified starkly below: in preference space, eight subsample means of the same WTP distribution bounced between **1.8 and 7.7**, while the same exercise in **WTP space** — where you parameterize the WTP distribution *directly* — gave **3.98 to 4.03**. This is the decisive argument for the WTP-space parameterization your research uses, and it connects directly to Chapter 37's warning about designs that degrade the cost coefficient.

Third: heterogeneity parameters are hard to identify. Verified in the estimation below, SML recovered the *means* accurately (mean WTP 4.00 estimated as 3.96) but the *spread* considerably less well (1.50 estimated as 1.07). This is not a coding error; it is a general and under-appreciated feature: distributional parameters need far more information than location parameters. Since your **error components** and your **familiarity-related heterogeneity** are precisely such parameters, this shapes what you can credibly claim.

38.1 Prerequisites

From **Chapters 20–21**: random utility, the MNL probability, IIA and its failure. From **Chapter 24**: WTP space, scale, and the WTP ratio. From **Chapter 24–25**: preference heterogeneity and the scale-versus-preference confound. From **Chapter 19**: panel data and within-person correlation. From **Chapter 9**: Monte Carlo simulation, Halton/Sobol sequences, variance reduction. From **Chapter 6**: Jensen's inequality and heavy-tailed ratios. From **Chapters 11–13**: maximum likelihood, the information matrix, and the delta method. From **Chapter 37**: the design that generated the data.

New notation. β_i (individual i's parameter vector), $f(\beta \mid \theta)$ (the mixing distribution with parameters θ), R (number of simulation draws), L_i (individual i's panel likelihood contribution), λ_i (individual scale / cost coefficient), $\mathbf{w}_i$ (individual WTP vector), σ_{EC} (error-component standard deviation).

38.2 The mixed logit: model and panel likelihood

Definition 38.1 (Mixed logit). In the **mixed logit** (random-parameters logit), individual i has parameter vector β_i drawn from a **mixing distribution** $f(\beta \mid \theta)$. *Conditional* on β_i, choices follow a standard logit; the *unconditional* probability integrates over the distribution:

$$P_{ij} = \int \frac{e^{\mathbf{x}_{ij}'\beta}}{\sum_l e^{\mathbf{x}_{il}'\beta}} \, f(\beta \mid \theta) \, d\beta.$$

You estimate θ (the *parameters of the distribution* — means, standard deviations, correlations), not the individual β_i.

Two properties this buys. (i) **Preference heterogeneity** is modelled explicitly rather than assumed away — and its magnitude is itself a result. (ii) **IIA is relaxed**: mixing induces correlation in utilities across alternatives, so substitution patterns are no longer proportional (the red-bus/blue-bus failure of Chapter 22 dissolves). McFadden and Train's well-known result is that a mixed logit can approximate *any* random utility model arbitrarily closely with a suitable mixing distribution — enormous flexibility, which is also a warning: flexible models can fit heterogeneity that is really misspecification.

The panel structure — the crucial detail. Your respondents each answer T choice tasks. The *same* β_i governs *all* of that person's choices, so their choices are **correlated**, and the likelihood contribution is the **product over tasks *inside* the integral**:

$$L_i(\theta) = \int \prod_{t=1}^{T} \frac{e^{\mathbf{x}_{ij_t t}'\beta}}{\sum_l e^{\mathbf{x}_{ilt}'\beta}} \, f(\beta \mid \theta) \, d\beta.$$

The order matters absolutely. Taking the product *inside* the integral (correct) says: *one* preference vector generates this person's whole sequence. Taking it *outside* (treating each task as an independent observation) would falsely treat NT choices as NT independent people — understating standard errors and destroying the panel information that identifies heterogeneity in the first place. **The within-person product is what makes repeated tasks informative about individual-level preferences**, and it is the single most common implementation error in mixed-logit code.

> **Active recall 38.A.** Why must the product over a respondent's tasks be taken *inside* the integral, and what goes wrong if it is not? *(Answer: the* same* individual parameter vector β_i governs *all* of that person's choices, so conditional on β_i their choices are independent, but *unconditionally* they are correlated — the likelihood contribution is therefore $L_i = \int \prod_t P_{it}(\beta) f(\beta) d\beta$, with the product *inside*. If the product is taken *outside* (integrating each task separately and multiplying), you are implicitly assuming each task is answered by a *different* draw of preferences — i.e. treating NT choices as NT independent respondents. Consequences: standard errors are badly understated (overstating precision by treating correlated observations as independent, Chapter 21), and — more fundamentally — the *panel information that identifies the heterogeneity distribution is destroyed*, because it is precisely the *consistency of a person's choices across tasks* that reveals their individual preferences. This is the most common mixed-logit implementation error.)*

38.3 Error components and substitution patterns

Your model is an **error-components** mixed logit, a particular way of using the mixing structure.

Definition 38.2 (Error-components specification). Rather than (or in addition to) making *attribute coefficients* random, an **error component** adds a random term with mean zero to the utility of a *subset* of alternatives:

$$U_{ij} = \mathbf{x}'_{ij}\beta + \underbrace{\eta_i d_j}_{\text{error component}} + \varepsilon_{ij},$$

where $\eta_i \sim N(0, \sigma^2_{\text{EC}})$ and d_j indicates membership of the subset. Alternatives sharing the component have *correlated* utilities, producing richer substitution patterns than IIA permits.

What it does for you. Two standard uses, both relevant:

- **Relaxing IIA within a nest.** If the "improvement" alternatives are more similar to each other than to the status quo, an error component shared by them induces the correlation that generates realistic substitution — an alternative to nested logit (Chapter 22), obtained here without the nesting structure.
- **Capturing the status-quo effect / panel correlation.** An error component on the *status quo* captures systematic unobserved differences in the propensity to choose "no change" (inertia, status-quo bias, protest tendency), separating that from the attribute preferences. When the component is *individual-specific* and constant across the person's tasks, it also absorbs the panel correlation directly.

Identification caution. Error-component variances are *variance* parameters, subject to two familiar difficulties. They are **bounded below by zero**, so testing $\sigma_{\text{EC}} = 0$ is a *boundary* problem — the likelihood-ratio statistic is not χ^2 but a *mixture* of chi-squares (Chapter 9's boundary problem). And, as §38.7 shows, variance parameters are *harder to identify* than means. Report them with appropriate caution, and use the correct boundary-adjusted test rather than a naive one.

38.4 Simulated maximum likelihood and simulation bias

The integral has no closed form, so it is approximated by simulation.

Definition 38.3 (Simulated maximum likelihood). Take R draws $\beta^{(r)}$ from $f(\cdot \mid \theta)$ and approximate each individual's likelihood by the sample average

$$\hat{L}_i(\theta) = \frac{1}{R} \sum_{r=1}^{R} \prod_{t=1}^{T} P_{it}(\beta^{(r)}),$$

then maximize $\sum_i \ln \hat{L}_i(\theta)$. Note that $\hat{L}_i$ is an *unbiased* estimator of L_i — but $\ln \hat{L}_i$ is **not** an unbiased estimator of $\ln L_i$.

The source of the bias. By **Jensen's inequality** (Chapter 7), since ln is *concave*, $\mathbb{E}[\ln \hat{L}] < \ln \mathbb{E}[\hat{L}] = \ln L$. The simulated log-likelihood is therefore **downward biased**, with bias of order $1/R$, while the simulation *noise* is of order $1/\sqrt{R}$.

Worked Example 38.1 (Simulation bias, verified). True probability computed by Gaussian quadrature ($P = 0.70704$, $\ln P = -0.34666$); $\mathbb{E}[\ln \hat{P}]$ across 40,000 replications:

R draws	$\mathbb{E}[\ln \hat{P}]$	**Bias**	**SD**$(\ln \hat{P})$
5	−0.351045	−0.004380	0.0978
10	−0.349023	−0.002358	0.0684
25	−0.347214	−0.000550	0.0427
50	−0.347253	−0.000588	0.0302
100	−0.346744	−0.000080	0.0214
500	−0.346745	−0.000081	0.0095

(Verified.) The bias is **negative at every** R, as Jensen predicts, and decays roughly as $1/R$. The standard deviation decays as $1/\sqrt{R}$ — from 0.098 at five draws to 0.0095 at five hundred, a factor of ten for a hundredfold increase in draws. Two practical consequences follow. First, **too few draws produce a systematically wrong log-likelihood**, and because the bias depends on θ it distorts the *estimates*, not just the likelihood value. Second, the *noise* is the more immediate practical problem: a log-likelihood contaminated by simulation noise has a *jagged* surface, so gradient-based optimizers report spurious convergence, Hessians are unreliable, and results change when you change the random seed. **If your estimates move materially when the seed changes, you do not have enough draws.**

> **Stop and think 38.1.** Your co-author reports that the model "converged nicely" with 50 random draws, but the estimates change noticeably when they rerun with a different seed. What is happening? *(Answer: 50 draws is far too few, and the seed-sensitivity is the diagnostic. Two distinct problems are present. (i)* Simulation bias*: $\ln \hat{L}$ is a* downward-biased* estimate of $\ln L$ by Jensen's inequality (log is concave), with bias of order $1/R$ — verified, at small R the bias is systematically negative — and since the bias varies with θ it distorts the *parameter estimates*, not merely the likelihood value. (ii) *Simulation noise*, which is the visible symptom: SD of the simulated log-likelihood decays only as $1/\sqrt{R}$ (verified 0.0302 at $R = 50$), so the objective surface is *jagged* rather than smooth. A gradient-based optimizer on a jagged surface stops at noise-induced local features and reports "convergence" that is an artifact — hence different seeds give different answers. "Converged nicely" is therefore no evidence of anything. Fixes: increase R substantially, switch from pseudo-random to *Halton* draws (verified 3.2–10× noise reduction for the same R), and *verify stability across seeds* as a mandatory diagnostic before reporting.)*

38.5 Quasi-random draws: Halton sequences

The cheapest improvement in mixed-logit estimation is to stop using pseudo-random numbers.

The idea. Pseudo-random draws cluster and leave gaps by chance — wasteful, since the goal is to *cover* the distribution evenly for an integral. **Quasi-random** (low-discrepancy) sequences such as **Halton** or **Sobol** are constructed to fill the space *systematically*, achieving far lower integration error for the same number of points. Halton sequences are built from radical-inverse expansions in prime bases and transformed to the target distribution via the inverse CDF.

Worked Example 38.2 (Halton versus pseudo-random, verified). Standard deviation of the simulated log-probability:

R	SD, pseudo-random	SD, Halton	**Reduction**
50	0.0284	0.0088	**3.2**×
100	0.0212	0.0045	**4.7**×
200	0.0148	0.0024	**6.3**×
500	0.0095	0.00095	**10.0**×

(Verified.) Halton draws reduce simulation noise by factors of 3 to 10, *and the advantage grows with* R. Since noise is what destabilizes the optimizer, this is a large gain for a one-line change. Equivalently: Halton draws achieve with a few hundred what pseudo-random draws need thousands to match.

Practical cautions. (i) **Scrambling.** Raw Halton sequences become *correlated across dimensions* in higher dimensions (roughly beyond five to ten), which can be worse than random draws; use **scrambled** Halton or Sobol sequences, as the verified example does. (ii) **Panel structure.** Use the *same* draws for a given individual across their tasks (that is what makes the within-person product coherent), and *different* draws across individuals. (iii) **Report** R **and the draw type** — they are part of your specification, and results are not reproducible without them. (iv) **Always verify stability**: re-estimate with more draws and different seeds/scrambles; if estimates move, increase R.

> **Active recall 38.B.** Why do Halton draws outperform pseudo-random draws, and what caution applies in higher dimensions? *(Answer: the simulated likelihood is a* numerical integral*, and its accuracy depends on how evenly the draws* cover* the distribution. Pseudo-random draws cluster and leave gaps by chance, wasting points; *quasi-random* (low-discrepancy) sequences like Halton are constructed to fill the space systematically, so integration error falls much faster — verified, noise reduction of 3.2× at $R = 50$ rising to 10× at $R = 500$, with the advantage *growing* in R. The caution: raw Halton sequences become *correlated across dimensions* in higher dimensions (roughly beyond 5–10 random parameters), which can make them *worse* than pseudo-random draws — so use *scrambled* Halton or Sobol sequences. Also: use the *same* draws for an individual across their tasks (coherent panel likelihood) but *different* draws across individuals, and always report R and the draw type as part of the specification.)*

38.6 Preference space versus WTP space

This is the parameterization decision that determines whether your welfare estimates are usable, and the verified evidence is unusually decisive.

Preference space. The conventional specification puts random coefficients on attributes and cost:

$$U_{ij} = \beta_i' \mathbf{x}_{ij} - \alpha_i c_{ij} + \varepsilon_{ij},$$

and WTP is then formed *after* estimation as the **ratio** $\text{WTP}_{ik} = \beta_{ik}/\alpha_i$.

The problem with ratios. If α_i is normally distributed, it has *positive density at zero* — and near-zero denominators produce enormous ratios. The ratio of two normals is Cauchy-like: **heavy-tailed, with moments that may not exist** (Chapter 7). Two symptoms follow: some individuals get absurd or wrong-signed WTP, and the *mean* WTP is unstable — it depends on extreme draws rather than converging. A

log-normal cost coefficient avoids the zero (it is strictly positive) but imposes its own strong shape and can still produce a heavy right tail.

Worked Example 38.3 (The ratio problem, verified). Preference space with $\beta \sim N(2, 1)$ and $\alpha \sim N(0.5, 0.25)$, versus WTP space with WTP $\sim N(4, 2)$ directly:

	Preference space (ratio)	WTP space (direct)
$P(\lvert\alpha\rvert < 0.1)$	**4.67%**	— (cost coefficient log-normal, bounded away from zero)
Mean WTP	6.80	4.00
Median WTP	3.85	4.00
1st percentile	−31.89	—
99th percentile	52.52	—
Eight subsample means	**1.8, 3.5, 4.9, 5.0, 6.3, 6.9, 7.1, 7.7**	**3.98, 3.98, 3.99, 3.99, 4.00, 4.01, 4.02, 4.03**

(Verified.) In preference space, nearly **5% of the population has a cost coefficient within 0.1 of zero**, so the WTP ratio explodes: the distribution spans −32 to +53 between the 1st and 99th percentiles, the mean (6.80) sits far from the median (3.85), and — the decisive diagnostic — **eight subsample means of the same distribution range from 1.8 to 7.7**. The mean is not estimating anything stable; it is tracking whichever extreme draws happened to be included. The same exercise in WTP space gives subsample means of 3.98 to 4.03.

WTP space. Reparameterize so that the WTP distribution is estimated *directly*. Writing $\mathbf{w}_i = \beta_i/\alpha_i$ and $\lambda_i = \alpha_i$ (a scale/cost parameter):

$$U_{ij} = \lambda_i(-c_{ij} + \mathbf{w}_i'\mathbf{x}_{ij}) + \varepsilon_{ij}.$$

The model is *observationally equivalent* in fit, but you now specify and estimate the distribution of $\mathbf{w}_i$ — the object of interest — *directly*, rather than deriving it as a ratio. Advantages: WTP distributions are *well-behaved* by construction (you choose their shape); you can impose sensible support (non-negativity where theory demands it); and — importantly for you — **the welfare quantities you report are the parameters you estimated**, so their standard errors are direct rather than delta-method approximations to a heavy-tailed ratio (Chapter 13). The known trade-off is that WTP space sometimes fits slightly worse in-sample than preference space, since the flexible-fit and behaviourally-sensible parameterizations differ; the literature's consensus is that the stability and interpretability gains dominate for welfare estimation.

The through-line. This connects three earlier results into one argument. Chapter 7 showed WTP ratios have near-Cauchy tails; Chapter 37 showed a D-optimal design can *degrade* the cost coefficient that sits in the denominator; and this chapter shows the ratio's mean is empirically unstable. **The WTP-space parameterization is the structural fix for all three** — which is why it is the right choice for your EC-MXL, and why you should present it as a *reasoned methodological commitment* rather than a software default.

> **Stop and think 38.2.** A referee asks why you estimated in WTP space rather than the "more standard" preference space, suggesting you are hiding a worse fit. How do you answer? *(Answer: the two are* observationally equivalent* reparameterizations of the same model — the choice is about *what you estimate directly*, not about concealing fit. In preference space, WTP is formed *after* estimation as the ratio β_i/α_i; when α_i is normal it has positive density at zero (verified: 4.67% of the population within 0.1 of zero), so the ratio is *near-Cauchy* — heavy-tailed, with possibly non-existent moments. The consequence is not theoretical: verified, eight subsample means of the same preference-space WTP distribution ranged from **1.8 to 7.7**, spanning −32 to +53 between the 1st and 99th percentiles, whereas the WTP-space equivalent gave 3.98–4.03. A "mean WTP" that changes by a factor of four depending on which subsample you draw is not a usable welfare estimate, and it is exactly the number that would feed my CBA. WTP space estimates the welfare distribution *directly*, so it is well-behaved by construction, admits

theoretically sensible support restrictions, and yields *direct* standard errors rather than delta-method approximations to a heavy-tailed ratio. On fit: I report both specifications and their fit statistics — WTP space can fit marginally worse in-sample, and I am explicitly trading a small amount of in-sample fit for welfare estimates that are stable and interpretable, which is the correct trade when the estimates feed a cost–benefit analysis.)*

38.7 Estimation in practice, and what is hard to identify

Worked Example 38.4 (EC-MXL in WTP space by SML, verified). Simulated panel: $N = 500$ individuals, $T = 10$ tasks, $R = 150$ scrambled Halton draws; utility $U_{ij} = \lambda_i(-c_{ij} + w_i x_{ij}) + \varepsilon_{ij}$ with λ_i log-normal and w_i normal:

Parameter	True	**Estimate**
$\mu_{\ln\lambda}$	-0.5108	-0.5081
$\sigma_{\ln\lambda}$	0.3500	0.3601
Mean WTP μ_w	**4.0000**	**3.9645**
SD of WTP σ_w	**1.5000**	**1.0740**

(Verified; converged, $-\ln L = 2119.20$.) The **location** parameters are recovered well — mean WTP 4.00 estimated as 3.96 (about 1% error), and the scale parameters accurately. But the **heterogeneity spread** is recovered considerably less well: the true WTP standard deviation of 1.50 was estimated as 1.07, an underestimate of nearly 30%.

This is the general lesson, not a coding artifact. *Distributional (spread) parameters require far more information than location parameters.* Intuitively, the mean of WTP is identified by *average* choice behaviour, while the spread is identified only by *differences in behaviour across individuals* — which requires enough tasks per person to characterize each individual, enough people to characterize the distribution, and enough design variation to distinguish preference heterogeneity from noise. With $T = 10$ tasks the individual-level information is limited, so the spread is imprecisely and downward-estimated.

Consequences for your claims. Because your **error-component variances** and your **familiarity-related heterogeneity** are exactly such spread parameters:

- Treat estimated *spreads* as less reliable than estimated *means*, and say so.
- Report **confidence intervals for the spread parameters**, not just point estimates.
- Recognize that a *non-significant* heterogeneity parameter may reflect *insufficient information*, not homogeneous preferences (Chapter 33's underpowering logic — absence of evidence is not evidence of absence).
- Use the **boundary-corrected test** when testing a variance against zero (Chapter 9).
- Be cautious about *substantive* claims resting on the *shape* of the heterogeneity distribution, which is even harder to identify than its spread.

Convergence and diagnostic protocol. Mixed-logit likelihoods are *not* globally concave, so:

1. **Multiple starting values.** Re-estimate from dispersed starts; if they reach different optima, report it and take the best-likelihood solution only after verifying its plausibility.
2. **Seed and draw stability.** Re-estimate with different draws and larger R; material movement means insufficient draws (§38.4).
3. **Gradient and Hessian.** Check the gradient is near zero *and* the Hessian is negative definite (Chapter 1's definiteness). A near-singular Hessian signals weak identification — often too many random parameters for the data.

4. **Parsimony.** Do not make every coefficient random by default; each random parameter adds a dimension of integration and a hard-to-identify spread. Start simple, add heterogeneity where justified, and test.
5. **Report the full specification**: distributions assumed, R, draw type, starting values, optimizer, and convergence criteria. Mixed-logit results are not reproducible without them.

Active recall 38.C. Why are heterogeneity (spread) parameters harder to identify than means, and what follows for reporting error-component variances? *(Answer: the* mean* of a coefficient is identified by *average* choice behaviour across the sample, but the *spread* is identified only by *systematic differences in behaviour across individuals* — which requires enough tasks per person to characterise each individual's preferences, enough individuals to characterise the distribution, and enough design variation to separate genuine preference heterogeneity from random noise and scale variation (Chapter 25). With limited tasks per person the individual-level information is thin, so spreads are imprecisely and typically *downward* estimated — verified: with $T = 10$, the true WTP standard deviation of 1.50 was recovered as 1.07, a ~30% underestimate, while the mean was recovered to ~1%. Reporting implications: present spreads with *confidence intervals* and explicitly greater caution than means; do *not* interpret a non-significant variance as evidence of homogeneous preferences (it may simply be underpowered — Chapter 33); use the *boundary-corrected* (chi-square mixture) test when testing a variance against zero rather than a naive LR test (Chapter 9); and avoid substantive claims that depend on the *shape* of the heterogeneity distribution, which is harder still to identify.)*

38.8 Failure modes

F1. Product outside the integral. Treating each task as an independent observation. *Fix:* panel likelihood with the product inside the integral (§38.2).

F2. Too few draws. Biased and noisy log-likelihood, spurious convergence. *Fix:* increase R; verify seed stability (§38.4).

F3. Pseudo-random draws. Unnecessary simulation noise. *Fix:* scrambled Halton/Sobol (§38.5).

F4. Unscrambled Halton in high dimensions. Cross-dimension correlation degrading accuracy. *Fix:* scramble, or use Sobol (§38.5).

F5. Preference-space WTP ratios. Heavy-tailed, unstable welfare estimates. *Fix:* estimate in WTP space (§38.6).

F6. Normal cost coefficient. Density at zero producing exploding ratios and wrong signs. *Fix:* log-normal, or WTP-space parameterization (§38.6).

F7. Over-parameterized heterogeneity. Every coefficient random, near-singular Hessian. *Fix:* parsimony; add heterogeneity selectively and test (§38.7).

F8. Over-claiming spreads. Treating imprecise variance estimates as firm findings. *Fix:* report CIs; acknowledge downward bias and identification limits (§38.7).

F9. Naive boundary test. Testing a variance against zero with a standard χ^2. *Fix:* chi-square mixture (§38.3; Chapter 9).

F10. Local optimum accepted. Single start, no verification. *Fix:* multiple starting values; check gradient and Hessian (§38.7).

F11. Under-reported specification. No mention of R, draw type, distributions, optimizer. *Fix:* report all; results are irreproducible otherwise (§38.7).

38.9 Diagnostic questions for your own papers

1. **Panel likelihood.** Is the product over tasks *inside* the integral? *(§38.2, F1.)*
2. **Draws.** How many, what type, scrambled? Do estimates move with seed or R? *(§38.4–§38.5, F2–F4.)*
3. **Parameterization.** Am I in WTP space, and have I justified it against preference space with both reported? *(§38.6, F5–F6.)*
4. **Distributions.** Is each mixing distribution defensible (support, sign, tails), and have alternatives been tested? *(§38.6.)*
5. **Error components.** What substitution pattern or correlation does each capture, and is its variance credibly identified? *(§38.3, §38.7.)*
6. **Spread reliability.** Are heterogeneity parameters reported with CIs and appropriate caution about downward bias? *(§38.7, F8.)*
7. **Boundary tests.** Are variance-against-zero tests boundary-corrected? *(§38.3, F9.)*
8. **Convergence.** Multiple starts, near-zero gradient, negative-definite Hessian? *(§38.7, F10.)*
9. **Reproducibility.** Are distributions, R, draw type, seeds, starts, and optimizer all reported? *(§38.7, F11.)*

38.10 Exercises

(A) Comprehension.

A1. Write the mixed-logit panel likelihood and explain why the product is inside the integral.

A2. Why is the simulated log-likelihood biased, and in which direction?

A3. What is an error component, and what does it accomplish?

(B) Derivation / reasoning.

B1. Use Jensen's inequality to establish the sign of the simulation bias, and state how bias and noise scale with R.

B2. Show why preference-space WTP is heavy-tailed and how WTP space resolves it.

B3. Explain why spread parameters are harder to identify than means.

(C) Research application / critique.

C1. Specify your EC-MXL fully — distributions, error components, parameterization, draws — and justify each choice.

C2. Your heterogeneity parameter is insignificant. Write the correct interpretation and the checks required.

C3. A referee requests preference-space estimates for comparability. Respond.

38.11 Solutions

A1. $L_i(\theta) = \int \prod_{t=1}^{T} P_{it}(\beta)\, f(\beta \mid \theta)\, d\beta$, where $P_{it}(\beta)$ is the logit probability of individual i's chosen alternative in task t conditional on β. The product sits *inside* because the *same* β_i generates *all* of that individual's choices: conditional on β their choices are independent (so they multiply), but the individual's β is unknown and must be integrated out *after* forming the sequence probability. Placing the product outside would treat each task as a separate draw of preferences — i.e. NT independent respondents — understating standard errors and eliminating the within-person consistency that identifies heterogeneity.

A2. The simulated likelihood $\hat{L} = \frac{1}{R}\sum_r \prod_t P_{it}(\beta^{(r)})$ is *unbiased* for L, but the objective uses $\ln \hat{L}$. Since ln is *concave*, Jensen's inequality gives $\mathbb{E}[\ln \hat{L}] < \ln \mathbb{E}[\hat{L}] = \ln L$: the simulated log-likelihood is **downward**

(negatively) biased, with bias of order $1/R$ (verified: -0.0044 at $R = 5$, -0.0001 at $R = 100$). Because the bias depends on θ, it distorts the parameter estimates, not only the likelihood value.

A3. An **error component** adds a mean-zero random term $\eta_i d_j$ to the utility of a designated subset of alternatives, with $\eta_i \sim N(0, \sigma^2_{\text{EC}})$ constant within an individual. Alternatives sharing the component have *correlated* utilities, so the model departs from IIA and generates realistic substitution patterns (alternatives within the group are closer substitutes) — an alternative to nested logit achieved through the mixing structure. Placed on the *status quo*, it captures systematic unobserved propensity to choose "no change" (inertia, status-quo bias, protest tendency) and, being individual-specific and constant across tasks, simultaneously absorbs panel correlation. Its variance is a *boundary* parameter (bounded at zero), so testing it against zero requires a chi-square mixture rather than a naive χ^2.

B1. Let $\hat{L}$ be the simulated likelihood with $\mathbb{E}[\hat{L}] = L$. Jensen's inequality for a *concave* function g states $\mathbb{E}[g(X)] \leq g(\mathbb{E}[X])$, with strict inequality for non-degenerate X. Applying $g = \ln$: $\mathbb{E}[\ln \hat{L}] < \ln \mathbb{E}[\hat{L}] = \ln L$ — the bias is **negative**. A second-order Taylor expansion of ln about L gives $\mathbb{E}[\ln \hat{L}] \approx \ln L - \frac{\text{Var}(\hat{L})}{2L^2}$, and since $\hat{L}$ is a mean of R i.i.d. terms, $\text{Var}(\hat{L}) = \text{Var}(\cdot)/R$ — so the **bias is** $O(1/R)$. The *standard deviation* of $\hat{L}$ (and hence the simulation noise in $\ln \hat{L}$) is $O(1/\sqrt{R})$. Verified: bias fell from -0.0044 ($R = 5$) toward -0.0001 ($R \geq 100$), while SD fell from 0.0978 to 0.0095 across $R = 5$ to 500 — approximately the tenfold reduction that $\sqrt{100}$ predicts. Practical implication: noise (the $1/\sqrt{R}$ term) dominates at usable R and is what destabilizes optimization, which is why *variance reduction* (Halton) is more valuable than simply raising R.

B2. In preference space, $\text{WTP}_{ik} = \beta_{ik}/\alpha_i$ — a *ratio of random variables*. If $\alpha_i \sim N(\mu_\alpha, \sigma^2_\alpha)$, it has *positive density at zero*, so realizations arbitrarily close to zero occur and the ratio takes arbitrarily large values of either sign. The ratio of two independent normals is Cauchy-distributed; more generally such ratios are *heavy-tailed with possibly undefined moments*, so the sample mean does not converge in the usual way and is dominated by extreme draws. Verified: with $\alpha \sim N(0.5, 0.25)$, 4.67% of the population lies within 0.1 of zero, the WTP distribution spans -31.9 to $+52.5$ between the 1st and 99th percentiles, the mean (6.80) diverges from the median (3.85), and eight subsample means ranged from **1.8 to 7.7** — no stable estimate exists. **WTP space** resolves this by reparameterizing $U_{ij} = \lambda_i(-c_{ij} + \mathbf{w}_i'\mathbf{x}_{ij}) + \varepsilon_{ij}$ and estimating the distribution of $\mathbf{w}_i$ *directly*: WTP is a *primitive* parameter with a shape you choose (well-behaved by construction, with support restrictions available), not a derived ratio. Verified: the same exercise in WTP space gave subsample means of 3.98–4.03. Standard errors are then direct rather than delta-method approximations to a heavy-tailed ratio.

B3. The *mean* of a random coefficient is identified by *average* choice behaviour: aggregate choice shares across the sample pin down central tendency, and every observation contributes. The *spread* is identified only by *systematic differences in choice behaviour across individuals* — it requires (i) enough tasks per person for that person's choices to reveal *their* preferences rather than noise, (ii) enough individuals to trace the distribution's shape, and (iii) design variation rich enough to distinguish genuine *preference* heterogeneity from *scale* heterogeneity (Chapter 25) and from the Gumbel noise. With few tasks per respondent, individual-level information is thin and the likelihood struggles to attribute variation to preference dispersion rather than error, producing imprecise and typically *downward-biased* spread estimates — verified: with $T = 10$, true $\sigma_w = 1.50$ was estimated as 1.07 (~30% under) while $\mu_w = 4.00$ was recovered as 3.96 (~1%). The identification burden is heavier still for the distribution's *shape*, and for *error-component variances*, which additionally sit on a boundary.

C1. *Full EC-MXL specification.* **Parameterization: WTP space**, $U_{ijt} = \lambda_i(-c_{ijt} + \mathbf{w}_i'\mathbf{x}_{ijt}) + \eta_i d_j + \varepsilon_{ijt}$ — chosen because welfare estimates are the object of the research and preference-space ratios are unstable (verified subsample means 1.8–7.7 versus 3.98–4.03), because standard errors on WTP are then direct rather than delta-method approximations to a heavy-tailed ratio, and because it aligns the estimation with the design criterion recommended in Chapter 37. **Cost/scale λ_i: log-normal**, ensuring strict positivity (a negative marginal utility of income is theoretically incoherent) and keeping the parameter bounded away from zero; I report sensitivity to this choice since log-normals impose a heavy right tail. **WTP coefficients $\mathbf{w}_i$: normal** for attributes where negative values are behaviourally plausible (heterogeneous direction of preference), constrained-sign distributions where theory requires one sign; each choice justified attribute by

attribute rather than by default. **Error component** η_i on the status-quo alternative, capturing unobserved propensity to choose "no change" (inertia/protest tendency) and absorbing panel correlation; its variance tested against zero with a *boundary-corrected* chi-square mixture, not a naive LR test. **Panel likelihood** with the product over each respondent's T tasks *inside* the integral. **Draws:** scrambled Halton, with R set by *stability testing* — re-estimating at increasing R and multiple scrambles until estimates are stable to reporting precision (verified motivation: pseudo-random noise at $R = 50$ is $3.2\times$ that of Halton, and simulation bias is $O(1/R)$). **Convergence:** multiple dispersed starting values, near-zero gradient, negative-definite Hessian verified. **Reporting:** distributions, R, draw type, seeds, starts, optimizer, and both WTP- and preference-space results.

C2. *Correct interpretation:* a non-significant heterogeneity parameter does **not** establish homogeneous preferences — it is *absence of evidence*, and spread parameters are exactly the parameters most prone to being underpowered (Chapter 33). Verified, spreads are recovered with substantial downward bias and imprecision even in a well-behaved simulation (σ_w: 1.50 estimated as 1.07), so a null result is the *expected* outcome of thin individual-level information. *Required checks before any interpretation:* (i) confirm the test used is **boundary-corrected** — a variance is bounded at zero, so the LR statistic follows a chi-square *mixture*, and a naive χ^2 test is *conservative*, potentially producing the non-significance (Chapter 9); (ii) verify **sufficient draws** — insufficient R produces noise that can mask heterogeneity (§38.4), so re-estimate at higher R with scrambled Halton and confirm stability; (iii) check **convergence** from multiple starts and inspect the Hessian for near-singularity, which would indicate the parameter is weakly identified rather than zero; (iv) confirm the **panel likelihood** is correctly specified with the product inside the integral, since the outside-product error destroys precisely the information that identifies heterogeneity; (v) assess **power** — how large a spread could I have detected given N, T, and my design? If the detectable spread exceeds plausible magnitudes, the test was uninformative. *Reporting:* state the confidence interval for the spread, report the equivalence-test or power calculation supporting (or withholding) any "preferences are homogeneous" claim, and prefer "I cannot detect significant heterogeneity with this design" to "preferences are homogeneous."

C3. *Response:* I agree to report both, and I do so — but I decline to make preference space the primary specification, for reasons that are substantive rather than presentational. The two are *observationally equivalent* reparameterizations of the same underlying model, so the comparison the referee seeks is legitimate and costless to provide; I present preference-space estimates alongside as a robustness table with fit statistics for both. The reason preference space cannot be the *primary* specification is that the paper's deliverable is a *welfare* estimate destined for a cost–benefit analysis, and in preference space that estimate is a ratio β_i/α_i whose denominator has positive density at zero — verified, 4.67% of the population within 0.1 of zero, producing a WTP distribution spanning -31.9 to $+52.5$ across the 1st–99th percentiles and, decisively, **eight subsample means ranging from 1.8 to 7.7 for the same distribution**. A welfare figure that varies by a factor of four with the subsample is not a usable input to a CBA, and reporting its standard error via the delta method would understate the true uncertainty because the underlying quantity has near-Cauchy tails (Chapter 7). WTP space estimates the welfare distribution directly, with a shape I specify and defend, sensible support restrictions, and direct standard errors. If the referee's underlying concern is *in-sample fit*, I report the log-likelihoods of both specifications and state plainly that I trade a small amount of in-sample fit for welfare estimates that are stable, interpretable, and defensible in the policy application — which is the correct trade when the estimates leave the paper and enter a decision.

38.12 Spaced review prompts

+1 day. From memory: write the panel mixed-logit likelihood with the product inside the integral; state the direction of simulation bias and how bias and noise scale with R; define an error component.

+1 week. Explain why Halton beats pseudo-random draws and the high-dimension caveat; explain the preference-space ratio problem and the WTP-space fix; explain why spreads are harder to identify than means.

+1 month. For your DCE, run the full diagnostic protocol — panel likelihood check, draw stability, both parameterizations, boundary-corrected variance tests, multiple starts, Hessian check — and report the complete specification. Close with the spine sentence about whether your estimates are properties of preferences or artifacts of draws and assumptions.

38.13 How this chapter connects to the rest of the book

- **Chapter 22** supplies the MNL and the IIA failure this model repairs; **Chapter 23** supplies WTP space and scale; **Chapters 24–25** supply the heterogeneity concepts and the scale-versus-preference confound that §38.7's identification limits make concrete; **Chapter 21** supplies the panel structure behind §38.2's within-person product.
- **Chapter 11** supplies the Monte Carlo and Halton machinery of §38.4–§38.5; **Chapter 7** supplies both Jensen's inequality (the simulation bias) and the heavy-tailed ratio (the WTP-space argument) — this chapter is where those two results become decisive practical constraints; **Chapter 9** supplies the boundary problem for variance tests; **Chapter 12–13** supply the ML and delta-method background.
- **Chapter 37** designed the experiment whose information content determines what §38.7 can identify — the cost-coefficient precision warning there is the same concern as the ratio instability here; **Chapter 39** takes these estimates into validity assessment; **Chapters 45–46** consume the WTP distribution in CBA, where its *stability* (§38.6) is what makes aggregation meaningful; **Chapters 47–48** treat specification choices as moderators across studies.

This chapter is the technical core of your DCE: it is where the design becomes estimates, and where the difference between a defensible welfare distribution and a numerically unstable artifact is decided.

38.14 Research translation task

For your **EC-MXL**, produce a three-page estimation note:

1. **Model.** Write the full utility specification, the mixing distributions with justification for each, the error-component structure and what correlation it captures, and the panel likelihood with the product inside the integral.
2. **Parameterization.** Justify WTP space explicitly against preference space; report both, with fit statistics and the stability comparison.
3. **Simulation.** Report draw type (scrambled Halton/Sobol), R, and the *stability testing* that determined R — estimates at increasing R and multiple scrambles.
4. **Identification and convergence.** Report multiple starting values, gradient, Hessian definiteness, and any evidence of weak identification; use boundary-corrected tests for variances.
5. **Heterogeneity claims.** Report spread parameters with confidence intervals and explicit caution; avoid interpreting non-significance as homogeneity; state the detectable spread given your design.
6. **Close** with the spine sentence: *"My EC-MXL in WTP space estimates a WTP distribution with mean €[·] (SE [·]) and spread [·] ([CI]), using [R] scrambled Halton draws verified stable across seeds; the error component on [alternative] captures [correlation], tested by boundary-corrected LR; heterogeneity estimates are reported with the caution that spreads are downward-biased under limited tasks per respondent."*

Carry this into Chapter 39 (validity), Chapters 45–46 (CBA), and Chapters 47–48 (meta-analysis), where specification and estimation choices become explicit moderators.

38.15 Bibliography

Core. - Train, *Discrete Choice Methods with Simulation* — the definitive reference on mixed logit, simulated maximum likelihood, and Halton draws; Chapters 6, 9, and 11 cover this chapter's material directly. *(Verify edition/chapters.)* - Hensher & Greene (2003), "The Mixed Logit Model: The State of Practice," *Transportation* — practical guidance on specification, draws, and interpretation (§38.2–§38.7). *(Verify volume/pages.)* - Train & Weeks (2005), "Discrete Choice Models in Preference Space and Willingness-to-Pay Space," in Scarpa & Alberini (eds.), *Applications of Simulation Methods in Environmental and Resource Economics* — the foundational WTP-space reference (§38.6). *(Verify pages.)*

Applied. - Scarpa, Thiene & Train (2008), "Utility in Willingness to Pay Space: A Tool to Address Confounding Random Scale Effects in Destination Choice to the Alps," *American Journal of Agricultural Economics* — WTP space in environmental valuation practice (§38.6). *(Verify volume/pages.)* - Bhat (2001), "Quasi-Random Maximum Simulated Likelihood Estimation of the Mixed Multinomial Logit Model," *Transportation Research Part B* — Halton draws in mixed logit (§38.5). *(Verify volume/pages.)* - Hess, Train & Polak (2006) on shuffled/scrambled Halton and higher-dimensional draw quality (§38.5). *(Verify citation.)*

Advanced. - McFadden & Train (2000), "Mixed MNL Models for Discrete Response," *Journal of Applied Econometrics* — the approximation theorem and foundations (§38.2). *(Verify volume/pages.)* - Revelt & Train (1998) on panel mixed logit and repeated choices (§38.2). *(Verify citation.)* - Chiou & Walker (2007), "Masking Identification of Discrete Choice Models Under Simulation Methods," *Journal of Econometrics* — how simulation can mask non-identification, directly relevant to §38.7's cautions. *(Verify volume/pages.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Train–Weeks / Bhat / Chiou–Walker citations before using them, per the book's no-unverified-citation rule.

End of Chapter 38. Next: Chapter 39 — Validity, Hypothetical Bias, and the Real-versus-Hypothetical Comparison, which turns from estimation to the question your BRIS design exists to answer: what makes a stated-preference number valid*, how is hypothetical bias measured and corrected (cheap talk, certainty calibration, consequentiality), and what does your real-versus-hypothetical experimental contrast actually identify? Tell me when to write it, or request a different chapter.*

Chapter 39 — Validity, Hypothetical Bias, and the Real-versus-Hypothetical Comparison

Part IX: Stated Preference in Practice

> **The spine question, applied to this chapter.** Chapters 36–38 elicited and estimated stated WTP. This chapter asks the question that decides whether any of it counts: **is the number true?** Stated preference rests on an uncomfortable premise — that what people *say* they would pay predicts what they *would* pay. When that premise fails, the failure is called **hypothetical bias**, and it is typically not small: verified below, a design in which stated values are inflated twofold recovers a calibration factor of 2.00. Your research confronts this head-on, because your real-versus-hypothetical treatments with the BRIS mechanism are built precisely to *measure* the gap rather than assume it away. The spine question: *what does my real-versus-hypothetical contrast identify, do my corrections repair the bias or merely move it, and can I demonstrate convergent validity — sameness — rather than merely fail to detect a difference?*

39.0 Conceptual motivation: the discipline's central vulnerability

Every method in this book has assumptions. Stated preference has one that is uniquely exposed: that a survey response about a hypothetical payment is informative about a real one. Critics of contingent valuation — Diamond and Hausman most prominently — have argued that it is not, and that stated-preference numbers are expressions of attitude rather than measurements of value. **This chapter is where that challenge is met**, and it is met not by argument but by *experiment*.

The structure of the answer has three parts, and your research design maps onto all three.

First, what "validity" means. Validity is not one thing. **Content validity** asks whether the instrument describes the good and payment credibly. **Construct validity** asks whether the estimates behave as theory predicts — WTP rising with income, with scope (Chapter 35), with the quality of the good. **Criterion validity** — the demanding one — asks whether stated values match an external criterion of *actual* behaviour. Only criterion validity can settle the hypothetical-bias question, and it requires something most studies never have: a **real-payment benchmark**. Your BRIS mechanism supplies exactly that, which is why your design can make claims most stated-preference studies cannot.

Second, the magnitude and mechanism of the bias. Hypothetical bias is generally *upward* — people overstate — and the literature's calibration factors commonly fall in the range of roughly 1.3 to 3 for public goods. The theoretical explanation is Carson and Groves's: a hypothetical survey is not **consequential**, so the respondent faces no budget constraint, no real trade-off, and no incentive to answer truthfully. Free of consequence, "yes" is cheap. This reframes hypothetical bias from a psychological quirk into an **incentive** problem (Chapter 28) — and it points toward the fix: make the survey *consequential*.

Third, what to do about it. Two families of remedy. **Ex ante** — cheap talk scripts, oath scripts, and consequentiality reinforcement, which attempt to prevent the bias during elicitation. **Ex post** — certainty calibration, which recodes uncertain "yes" responses to "no" after the fact. The verified results below deliver a genuinely cautionary finding about the latter: certainty recoding pulled a hypothetical WTP inflated 2.00× down to 1.37× at a certainty threshold of 7, to 1.05× at threshold 8 — and to **0.80× at threshold 9, overcorrecting *below* the real benchmark**. The threshold is a researcher degree of freedom that can produce almost any answer, which makes pre-registration of the calibration rule essential rather than optional.

Finally, the chapter closes the loop on two earlier results. Demonstrating **convergent validity** requires *equivalence testing* (Chapter 33), not a failed difference test — verified below, a calibrated estimate is shown equivalent to the real benchmark within a 25% margin. And comparing arms requires comparing **WTP, not coefficients**, because a pure scale difference leaves WTP *identical* while changing every coefficient (verified) — the scale-versus-preference confound of Chapter 25, which would otherwise make any cross-arm coefficient comparison uninterpretable.

39.1 Prerequisites

From **Chapter 28**: incentive compatibility, consequentiality, and the Carson–Groves conditions. From **Chapter 33**: equivalence testing and TOST, multiplicity. From **Chapter 31–32**: randomized treatment effects and threats to identification (the real/hypothetical contrast is a randomized experiment). From **Chapter 25**: the scale-versus-preference confound. From **Chapter 36**: SBDC estimation and mean/median WTP. From **Chapter 9**: Krinsky–Robb simulation of WTP confidence intervals. From **Chapter 10**: the Poe test and randomization inference. From **Chapter 35**: scope tests as construct validity.

New notation. CF (calibration factor, the ratio of hypothetical to real WTP), Δ (equivalence margin), c_i (stated certainty), and τ (certainty recoding threshold).

39.2 The validity taxonomy

Validity claims must be specific about *which* validity is being claimed.

Definition 39.1 (Content validity). Content (face) validity concerns whether the instrument *plausibly measures what it claims*: is the good described accurately and comprehensibly, is the payment vehicle credible, is the scenario believable, does the choice context resemble a real decision? It is assessed by expert review, focus groups, cognitive interviews, and pilot debriefing — qualitative evidence, but foundational. A survey that fails content validity cannot be rescued by statistics.

Definition 39.2 (Construct validity). Construct validity asks whether estimates behave as economic theory predicts. It has two branches. **Theoretical (expectation) validity**: does WTP vary with covariates in theoretically-required directions — increasing in income, increasing in the *scope* of the good (Chapter 35), decreasing in the cost of substitutes, higher among users than non-users? **Convergent validity**: do estimates from *different methods* measuring the same construct agree — CV versus DCE, stated versus revealed preference, one elicitation format versus another? Construct validity is *necessary but not sufficient*: estimates can satisfy every theoretical expectation and still be uniformly inflated, since a bias that scales all values leaves the *pattern* intact. **This is why construct validity alone cannot answer the hypothetical-bias question.**

Definition 39.3 (Criterion validity). Criterion validity compares stated values against an external criterion of *actual behaviour* — real payments, actual purchases, referendum votes. It is the *strongest* form of validity and the only one that directly tests the hypothetical premise. Its difficulty is obvious: for a *public* good with *non-use* value (Chapter 35), there is usually no market transaction to compare against —

which is precisely why the good required stated preference in the first place. The resolution is to *construct* a real-payment criterion experimentally, which is what your **BRIS** mechanism does.

The hierarchy and its implication. Content validity is a precondition; construct validity is evidence of coherence; criterion validity is evidence of *accuracy*. A study demonstrating only construct validity has shown its estimates are *internally sensible*, not that they are *right*. **Your design's distinguishing feature is that it reaches criterion validity** — and that claim should be stated explicitly, because it is rare.

> **Active recall 39.A.** Distinguish content, construct, and criterion validity, and explain why construct validity cannot settle the hypothetical-bias question. *(Answer:* content (face)* validity asks whether the instrument plausibly measures what it claims — credible good description, believable payment vehicle, comprehensible scenario — assessed qualitatively via experts, focus groups, and pilot debriefing. *Construct* validity asks whether estimates behave as theory predicts: *theoretical/expectation* validity (WTP rising in income and in scope, differing between users and non-users) and *convergent* validity (agreement across methods or formats). *Criterion* validity compares stated values against actual behaviour — real payments — and is the only form that directly tests whether stated values are *accurate*. Construct validity cannot settle hypothetical bias because it tests *patterns*, not *levels*: a bias that inflates all values by a common factor leaves every theoretical relationship intact — WTP still rises with income and scope, users still exceed non-users — so a uniformly overstated set of estimates can pass every construct-validity test while being wrong by a factor of two. Only a *real-payment criterion* (which BRIS supplies) can detect a level bias.)*

39.3 Hypothetical bias: definition, magnitude, mechanism

Definition 39.4 (Hypothetical bias). Hypothetical bias is the systematic difference between WTP stated in a hypothetical context and WTP revealed when payment is real. It is summarized by the **calibration factor**

$$\mathrm{CF} = \frac{\mathrm{WTP}_{\mathrm{hypothetical}}}{\mathrm{WTP}_{\mathrm{real}}},$$

with $\mathrm{CF} > 1$ indicating overstatement. Meta-analyses generally find median calibration factors in the region of 1.3 to 3 for public goods, with wide dispersion; bias tends to be larger for public than private goods, and larger for non-use than use values.

The Carson–Groves mechanism. The theoretical account (Chapter 28) is not psychological but *incentive-based*. A response is truthful when the respondent's best strategy is truth-telling, which requires the survey to be **consequential**: the respondent must believe (i) the results may *influence* an actual decision, and (ii) if implemented, they would *actually have to pay*. A purely hypothetical survey satisfies neither. With no real budget constraint and no consequence, saying "yes" costs nothing — and may even yield expressive benefits (signalling environmental concern, pleasing the interviewer, warm glow, Chapter 35). **Hypothetical bias is therefore the predictable consequence of removing incentives, not an anomaly** — which is encouraging, because it identifies the lever: *restore consequence*.

Worked Example 39.1 (Measuring the calibration factor, verified). Two randomized SBDC arms ($n = 500$ per bid, eight bids), where hypothetical WTP is truly inflated twofold:

Arm	Median WTP	Mean WTP
Real	€40.29	€47.67
Hypothetical	€80.50	€95.61

Statistic	Value
Calibration factor (median-based)	**1.998**
Calibration factor (mean-based)	**2.006**
Krinsky–Robb 95% CI for the ratio	**[1.915, 2.085]**
Excludes 1.0?	**Yes — significant hypothetical bias**

(Verified; true factor 2.000.) The design recovers the calibration factor almost exactly, and the Krinsky–Robb interval (Chapter 13) excludes 1.0 decisively. Note what makes this possible: **the real arm**. Without a real-payment benchmark there is no denominator, and hypothetical bias is unmeasurable — it can only be asserted or assumed. This is the methodological payoff of your BRIS design.

Testing the difference: the Poe test. Comparing two *WTP distributions* (each estimated with uncertainty) requires care, since WTP is a nonlinear function of estimated parameters. The **Poe et al. complete-combinatorial test** compares the full simulated distributions: draw from each arm's parameter distribution (Krinsky–Robb), form all differences, and compute the proportion at or below zero as a one-sided p-value.

Worked Example 39.2 (Poe test, verified). For the arms above: $P(\text{WTP}_{\text{hyp}} - \text{WTP}_{\text{real}} \leq 0) = 0.0000$, giving a two-sided $p = 0.0000$ — the distributions differ significantly. (Verified.)

> **Stop and think 39.1.** Why can hypothetical bias only be *measured* — as opposed to asserted — when a real-payment arm exists, and what does the Carson–Groves account imply about where bias should be largest? *(Answer: the calibration factor is a* ratio*, WTP_{hyp}/WTP_{real}, so it requires a* denominator* — an estimate of what people would *actually* pay. Without a real-payment arm there is no benchmark: one can compare hypothetical estimates to each other, or to theoretical expectations (construct validity), but any claim about the *level* being right or inflated is an assertion, since a uniform inflation leaves all internal patterns intact (§39.2). A real arm (via BRIS) supplies the denominator and makes the bias *estimable* — verified, recovering a calibration factor of 1.998 with a Krinsky–Robb CI of [1.915, 2.085] excluding 1.0. The Carson–Groves account implies bias should be *largest where incentives are weakest*: for goods with no real payment obligation, where the respondent perceives *no consequentiality* (results will not influence any decision, or payment would never be collected), and where *expressive* benefits of saying yes are strongest — hence larger for *public* than private goods, and larger for *non-use* values (existence/bequest, where saying yes signals concern at no cost) than for use values tied to actual behaviour. It also predicts bias should *shrink* as consequentiality rises, which is the testable basis for the ex-ante remedies of §39.4.)*

39.4 Ex-ante mitigation: preventing bias during elicitation

If bias arises from absent incentives, the first line of defence is to restore them — or to simulate their effect.

Consequentiality. The theoretically-grounded remedy. Convince respondents — *truthfully* — that the survey may inform a real decision and that payment would be real. Practically: describe how results will be used, involve a credible implementing authority, use a plausible payment vehicle, and *measure* perceived consequentiality with follow-up items. The empirical regularity (the "knife-edge" result) is that respondents who believe the survey is consequential give responses closer to real-payment behaviour, while those who see it as inconsequential do not — providing both a *validity check* and a *subgroup diagnostic*. Note the analytical caution: perceived consequentiality is *measured*, not randomized, so conditioning on it is *observational* (Chapter 32's post-treatment and selection warnings apply). Randomizing *consequentiality scripts* is the cleaner design.

Cheap talk. A script administered before elicitation that *explicitly describes* hypothetical bias — explaining that people tend to overstate in surveys and asking respondents to answer as if payment were real. Evidence is mixed: cheap talk often reduces bias, but effects vary with script length, respondent experience (more

effective for inexperienced respondents), and good type. It is a *behavioural* patch rather than an incentive fix.

Oath scripts. Respondents are invited to sign a **solemn oath** to answer honestly before the survey. Grounded in commitment and consistency, evidence suggests oath scripts can reduce bias, sometimes more reliably than cheap talk. They are cheap to implement and easy to randomize.

Design implications for you. Since these treatments are *randomizable*, they belong in the experimental design (Chapter 31) rather than being applied uniformly. Randomizing cheap talk or oath scripts *identifies their effect*, and — combined with your real arm — lets you ask the question that matters: *does the mitigation move hypothetical responses toward the real benchmark?* That is a far stronger test than showing that a script reduces stated WTP, since a script could reduce WTP *past* the truth (exactly the overcorrection §39.5 documents).

39.5 Ex-post calibration: certainty scales and the threshold problem

The second family of remedies corrects responses after collection — and carries a hazard worth documenting carefully.

Certainty follow-ups. After a "yes," ask how certain the respondent is, typically on a 1–10 scale. The **calibration rule** recodes "yes" responses below a threshold τ as "no," on the reasoning that uncertain assent is where hypothetical bias concentrates. Thresholds of 7 or 8 are conventional. The approach is intuitive and cheap, and often moves estimates substantially toward real-payment benchmarks.

Worked Example 39.3 (Certainty calibration and threshold sensitivity, verified). Applying certainty recoding to the hypothetical arm (real benchmark €40.29):

Rule	Median WTP	**Ratio to real**
No recoding	€80.50	**2.00**
Recode yes with certainty < 7	€55.17	**1.37**
Recode yes with certainty < 8	€42.12	**1.05**
Recode yes with certainty < 9	€32.43	**0.80**

(Verified.) The calibration *works* in the sense of moving estimates toward the benchmark — threshold 8 lands at 1.05, nearly exact. But observe the whole column: **the estimate ranges from €80.50 to €32.43 across plausible thresholds**, and at threshold 9 the correction *overshoots*, producing a calibrated WTP **20% *below*** the real value. The calibrated number is therefore *as much a function of the threshold as of the data.*

The threshold is a researcher degree of freedom. This is a garden-of-forking-paths problem (Chapter 32's specification-search concerns, Chapter 33's multiplicity). A researcher who tries several thresholds and reports the one producing the "expected" result has performed a specification search, not a calibration. The disciplined response:

1. **Pre-register the threshold** before analysis, with justification from prior literature.
2. **Report the full sensitivity curve** — every threshold's estimate, as in the table above — so readers see the range rather than one point.
3. **Never select the threshold by reference to the real arm**, if you have one: that is fitting the correction to the answer, and it *invalidates* the criterion-validity test (you would be calibrating to the benchmark and then "validating" against it — circular).
4. **Acknowledge overcorrection risk**: certainty calibration has no principle guaranteeing it lands on the truth rather than past it.

5. **Treat calibrated estimates as one specification among several**, not as *the* estimate.

The deeper point. Ex-post calibration corrects a *symptom* using a *convention*. Ex-ante consequentiality addresses the *cause* using *theory*. Where both are possible, the ex-ante route is more defensible — and where a real benchmark exists, it should be used to *evaluate* corrections, never to *select* them.

> **Stop and think 39.2.** You have a real-payment arm and are considering certainty calibration. Why must you not choose the certainty threshold by seeing which one best matches the real arm? *(Answer: because that converts an* independent test* into a *fitted* result, destroying exactly the evidentiary value the real arm provides. The point of the real arm is *criterion validity* — an external benchmark against which the hypothetical estimate can be independently assessed. If I select the threshold that makes the calibrated hypothetical estimate match the benchmark, then "the calibrated estimate agrees with real payments" is *guaranteed by construction*, not discovered: I have fitted the correction to the answer and then used the answer to validate the correction — circular. The verified sensitivity shows how much freedom this would give me: across thresholds 7, 8, 9 the calibrated estimate runs €55.17, €42.12, €32.43 (ratios 1.37, 1.05, 0.80) — spanning from substantial *over*statement to 20% *under*statement, so almost any conclusion is reachable by threshold choice. The correct procedure: *pre-register* the threshold from prior literature *before* looking at the real arm, report the *full sensitivity curve* across thresholds, and then report — as a genuine finding — how the pre-specified calibrated estimate compares to the benchmark, whether or not it agrees.)*

39.6 The real-versus-hypothetical contrast as an experiment

Your real/hypothetical comparison is a randomized experiment, so the full apparatus of Part VII applies.

What it identifies. With respondents *randomly assigned* to real (BRIS) or hypothetical treatment, the difference in estimated WTP is the **average treatment effect of hypotheticality** on stated value (Chapter 31) — the calibration factor, causally identified. Randomization means no confounding: the arms differ *only* in the payment consequence.

Threats requiring attention (Chapter 32).

- **Differential exclusion.** If protest or dropout rates differ by arm — plausible, since a *real* payment request should trigger more refusal than a hypothetical one — then excluding those respondents breaks the randomization and biases the calibration factor. Report exclusion rates *by arm*, analyse with and without, and bound (Lee bounds).
- **Differential comprehension or effort.** The real arm may induce more careful deliberation, which is *part of* the treatment effect rather than a confound — but it means the arms differ in more than the payment, complicating interpretation of the mechanism.
- **Post-treatment conditioning.** Do not condition on perceived consequentiality, certainty, or comprehension when comparing arms, as these are post-treatment (Chapter 32's bad controls).
- **Selection into real payment.** If real-arm participation requires consent to potential payment, those consenting may differ systematically — a selection problem requiring bounds.

The BRIS mechanism. A **random incentive** mechanism makes payment real but probabilistic: one respondent (or a random subset, or a randomly-selected task) is binding. This achieves incentive compatibility at manageable cost — the theoretical requirement is that the respondent believes their answer *might* determine an actual payment, which a credible random mechanism satisfies. Chapter 28 covers the mechanism-design conditions; the key practical requirement is *credibility*: respondents must believe the randomization and the payment will genuinely occur.

Comparing arms correctly: WTP, not coefficients. This is where Chapter 25's confound returns and must be handled explicitly.

Worked Example 39.4 (Scale versus preference, verified). Utility scaled by λ, with true attribute coefficient 1.2 and cost coefficient 0.08:

Arm	Estimated β	Estimated α	**WTP** $= \beta/\alpha$
Real (scale 1.0)	1.2000	0.0800	**15.0000**
Hypothetical (scale 0.5)	0.6000	0.0400	**15.0000**
Hypothetical (scale 2.0)	2.4000	0.1600	**15.0000**

(Verified.) A *pure scale difference* changes **both** coefficients — by a factor of two in either direction — while leaving **WTP exactly identical**. The implication for your analysis is decisive:

- **A difference in raw coefficients across arms is *not* evidence of a preference difference** — it is equally consistent with the arms differing only in *error variance* (scale), which is entirely plausible, since a real-payment context may induce more (or less) consistent decision-making than a hypothetical one.
- **A difference in *WTP* across arms *is* informative**, because WTP is scale-free by construction.

So the calibration factor must be computed from **WTP**, and any cross-arm comparison of coefficients must either be scale-corrected (Swait–Louviere, Chapter 25) or avoided. This is a further argument for the WTP-space estimation of Chapter 38: it estimates the scale-free quantity directly.

> **Active recall 39.B.** Why is a difference in estimated coefficients between the real and hypothetical arms uninformative about preferences, and what should be compared instead? *(Answer: in a logit the estimated coefficients are the* product* of the true preference parameters and the *scale* factor λ (inversely related to the error variance), so the two are *confounded* (Chapter 25) — and the arms can easily differ in scale, since a real-payment context plausibly induces more (or less) consistent, deliberate decision-making than a hypothetical one, changing error variance without changing preferences. Verified: a pure scale difference of $0.5\times$ or $2\times$ changed *both* β ($1.2 \to 0.6$ or 2.4) and α ($0.08 \to 0.04$ or 0.16) while leaving WTP *exactly* 15.0000 in all cases. So observing different coefficients across arms is consistent with *identical preferences and different scale*, and cannot support a hypothetical-bias claim. What should be compared is **WTP** — the ratio β/α — in which the common scale factor *cancels*, making it scale-free and therefore informative about genuine preference differences. Practically: compute the calibration factor from WTP (as verified in §39.3), estimate in WTP space (Chapter 38) so the scale-free quantity is estimated directly, and if coefficients must be compared, apply a scale-correction such as Swait–Louviere.)*

39.7 Demonstrating convergent validity requires equivalence testing

Finally, the inferential point that Chapter 33 established and this chapter applies.

The fallacy to avoid. After calibration, a researcher compares the calibrated hypothetical WTP to the real WTP, finds $p = 0.4$, and concludes "the estimates are equivalent — convergent validity is demonstrated." **This does not follow.** A non-significant difference is *absence of evidence*, consistent with a large true gap that the study lacked power to detect. Since validity claims are claims of *sameness*, they require **equivalence testing**.

Worked Example 39.5 (TOST for convergent validity, verified). Comparing the pre-specified certainty-calibrated hypothetical estimate (€42.12) to the real benchmark (€40.29), on the log scale with an equivalence margin of $\pm \ln(1.25)$ (i.e. agreement within 25%):

Quantity	Value
Log-ratio d	0.0442 (se 0.0215)

Quantity	Value
Equivalence margin	±0.2231
90% CI on log-ratio	[**0.0087, 0.0796**]
CI inside margin?	**Yes**
TOST p	**0.0000 — conclude equivalence at 5%**

(Verified.) The calibrated estimate is demonstrated **equivalent** to the real benchmark within a 25% margin — a *positive* validity finding, not merely a failure to reject difference. Note the two design requirements this imposes: the margin must be **pre-specified** and justified by decision relevance (how large a gap would change a policy conclusion, or exceed acceptable benefit-transfer error — Chapter 49), and the study must have adequate **power** for equivalence, which typically requires larger samples than difference testing.

The reporting standard. For each validity claim, state: the claim's *type* (content/construct/criterion), the *comparison* being made, the *pre-specified margin* where equivalence is claimed, the *test* and its result, and — where a real benchmark exists — the *calibration factor with confidence interval*.

39.8 Failure modes

F1. Claiming validity without specifying which. Asserting "validated" without naming content, construct, or criterion. *Fix:* state the type and the evidence (§39.2).

F2. Treating construct validity as proof of accuracy. Theoretical expectations satisfied, level bias undetected. *Fix:* recognize that uniform inflation preserves patterns; seek criterion evidence (§39.2).

F3. Asserting bias magnitude without a benchmark. Applying a literature calibration factor as if measured. *Fix:* measure it with a real arm, or state it as an assumption with sensitivity (§39.3).

F4. Threshold shopping in certainty calibration. Selecting τ to produce a desired result. *Fix:* pre-register τ; report the full sensitivity curve (§39.5).

F5. Calibrating to the benchmark, then validating against it. Circular validation. *Fix:* pre-specify the correction independently of the real arm (§39.5).

F6. Ignoring overcorrection. Assuming calibration cannot overshoot. *Fix:* verified it can (0.80× at threshold 9); report the direction and magnitude of correction (§39.5).

F7. Comparing coefficients across arms. Interpreting coefficient differences as preference differences. *Fix:* compare WTP (scale-free); scale-correct if coefficients must be compared (§39.6).

F8. Differential exclusion across arms. Protest/dropout differing between real and hypothetical. *Fix:* report by-arm rates, analyse with/without, bound (§39.6; Chapter 32).

F9. Conditioning on post-treatment variables. Comparing arms conditional on consequentiality or certainty. *Fix:* these are post-treatment; report unconditionally (§39.6; Chapter 32).

F10. Equivalence fallacy. Inferring validity from a non-significant difference. *Fix:* TOST with a pre-specified margin (§39.7; Chapter 33).

F11. Treating measured consequentiality as randomized. Causal claims from a measured perception. *Fix:* randomize scripts; treat measured perceptions as observational (§39.4).

39.9 Diagnostic questions for your own papers

1. **Validity type.** Which validity am I claiming, and what evidence supports each? *(§39.2, F1–F2.)*

2. **Benchmark.** Do I have a real-payment criterion, and is the calibration factor reported with a CI? *(§39.3, F3.)*
3. **Mitigation design.** Are cheap talk / oath / consequentiality treatments *randomized* so their effects are identified? *(§39.4, F11.)*
4. **Calibration discipline.** Is the certainty threshold pre-registered, is the full sensitivity curve reported, and was it chosen independently of the benchmark? *(§39.5, F4–F6.)*
5. **Cross-arm comparison.** Am I comparing WTP rather than coefficients? *(§39.6, F7.)*
6. **Experimental threats.** Exclusion rates by arm, with/without analysis, bounds, no post-treatment conditioning? *(§39.6, F8–F9.)*
7. **Equivalence.** Are sameness claims tested by TOST with a pre-specified, justified margin and adequate power? *(§39.7, F10.)*
8. **BRIS credibility.** Do respondents believe the mechanism, and is that verified in debriefing? *(§39.6.)*

39.10 Exercises

(A) Comprehension.

A1. Define content, construct, and criterion validity, and rank them by strength for testing hypothetical bias.

A2. State the Carson–Groves explanation of hypothetical bias and its implication for mitigation.

A3. What is a calibration factor, and what does measuring one require?

(B) Derivation / reasoning.

B1. Explain why a uniform proportional bias leaves construct-validity tests intact.

B2. Show that a pure scale difference leaves WTP unchanged, and derive the implication for cross-arm comparison.

B3. Explain why certainty calibration can overcorrect and why the threshold must be pre-registered.

(C) Research application / critique.

C1. Design the full validity strategy for your CV study, specifying what each component establishes.

C2. Your calibrated hypothetical WTP does not differ significantly from the real WTP. Write the correct claim.

C3. A referee argues your real arm is "too small to be a benchmark." Respond.

39.11 Solutions

A1. Content (face) validity — whether the instrument plausibly measures the intended construct (credible good, payment vehicle, scenario), assessed qualitatively. **Construct validity** — whether estimates behave as theory predicts (*theoretical/expectation*: WTP rising in income and scope; *convergent*: agreement across methods). **Criterion validity** — whether stated values match actual behaviour against an external benchmark. Ranked by strength for testing hypothetical bias: **criterion** $\gg$ **construct** $>$ **content**. Only criterion validity compares against real behaviour and can detect a *level* bias; construct validity tests *patterns*, which a proportional bias preserves; content validity is a necessary precondition establishing that the question is coherent, but carries no information about accuracy.

A2. Carson and Groves: truthful response requires the survey to be **consequential** — respondents must believe results may influence a real decision *and* that they would actually pay. A purely hypothetical survey satisfies neither, so there is no budget constraint, no real trade-off, and no cost to saying "yes"; expressive

benefits (signalling concern, pleasing the interviewer, warm glow) may further inflate assent. Hypothetical bias is therefore the *predicted consequence of absent incentives*, not a psychological anomaly. Implication for mitigation: the theoretically-correct remedy is to **restore consequentiality** (credible implementing authority, plausible payment vehicle, genuine prospect of use and payment) rather than to patch responses afterwards — and it predicts bias should be largest where consequentiality is weakest, providing a testable subgroup diagnostic.

A3. The **calibration factor** is $\text{CF} = \text{WTP}_{\text{hypothetical}}/\text{WTP}_{\text{real}}$, the ratio quantifying overstatement ($\text{CF} > 1$). Measuring one requires a **real-payment benchmark** — an arm in which respondents actually face binding payment (via a mechanism such as BRIS) — since without a denominator the factor is unidentifiable and can only be assumed from literature. It further requires: *random assignment* to arms (so the comparison is causal, Chapter 31); computing the ratio from **WTP** rather than coefficients (scale-free, §39.6); and reporting *uncertainty* via Krinsky–Robb or the Poe test (verified: CF = 1.998 with 95% CI [1.915, 2.085], excluding 1.0; Poe $p = 0.0000$).

B1. Suppose true WTP for respondent i facing scope q is $W_i(q)$ and the hypothetical instrument returns $\tilde{W}_i(q) = k\,W_i(q)$ with a *constant* $k > 1$. Construct-validity tests examine *relationships*: (i) *scope sensitivity* — $\tilde{W}(q_2) > \tilde{W}(q_1)$ holds whenever $W(q_2) > W(q_1)$, since $k > 0$ preserves order; (ii) *income gradient* — $\partial\tilde{W}/\partial y = k\,\partial W/\partial y$ has the same sign; (iii) *user versus non-user differences* — preserved by the same argument; (iv) *convergent validity across formats* — if both formats are inflated by similar factors, they still agree with each other. Every qualitative prediction survives multiplication by a positive constant. Hence a study can pass all construct-validity tests while every level is wrong by the factor k — which is precisely the hypothetical-bias scenario. Only comparison against an *external level benchmark* (criterion validity) can identify k. (Note the corollary: construct validity *would* detect biases that distort *relationships*, such as scope insensitivity from warm glow — so it retains real diagnostic value, just not for proportional level bias.)

B2. In a logit, observed utility is $\lambda(\beta'\mathbf{x} - \alpha c)$ where λ is the scale (inversely related to the error standard deviation), so the *estimable* coefficients are $\tilde{\beta} = \lambda\beta$ and $\tilde{\alpha} = \lambda\alpha$ — scale and preference enter only as a product and are not separately identified from a single dataset (Chapter 25). WTP is the ratio

$$\widetilde{\text{WTP}} = \frac{\tilde{\beta}}{\tilde{\alpha}} = \frac{\lambda\beta}{\lambda\alpha} = \frac{\beta}{\alpha},$$

in which λ **cancels exactly**. Verified: scaling by 0.5 and 2.0 changed β from 1.2 to 0.6 and 2.4, and α from 0.08 to 0.04 and 0.16, while WTP remained 15.0000 throughout. *Implication for cross-arm comparison:* a difference in *coefficients* between real and hypothetical arms is consistent with identical preferences and merely different error variance (entirely plausible if one context induces more consistent deliberation), so it is **not** evidence of hypothetical bias; a difference in **WTP** is scale-free and therefore is. Compute calibration factors from WTP, estimate in WTP space (Chapter 38), and scale-correct (Swait–Louviere) if coefficients must be compared.

B3. Certainty calibration recodes "yes" responses with stated certainty below a threshold τ into "no," on the premise that hypothetical bias concentrates in uncertain assent. But there is **no theoretical principle** determining the *correct* τ: the rule is a convention, not a derivation, and nothing guarantees that recoding removes *exactly* the bias rather than too little or too much. Raising τ monotonically removes more "yes" responses and therefore monotonically lowers estimated WTP — so the estimate can be driven below the truth. Verified: against a real benchmark of €40.29, the calibrated median ran €80.50 (no recoding, ratio 2.00), €55.17 ($\tau = 7$, 1.37), €42.12 ($\tau = 8$, 1.05), and **€32.43 ($\tau = 9$, ratio 0.80 — a 20% *under*statement)**. Since plausible thresholds span estimates from €80.50 to €32.43, τ is a **researcher degree of freedom** capable of supporting almost any conclusion — a garden-of-forking-paths problem (Chapters 32–33). Therefore τ must be **pre-registered** from prior literature *before* analysis, the *full sensitivity curve* across thresholds must be reported, and τ must never be selected by reference to the real benchmark (which would make subsequent "validation" circular).

C1. *Full validity strategy.* **Content validity:** expert review of the scenario and payment vehicle, focus groups, cognitive interviews, and pilot debriefing confirming that respondents understand the finfish/ecosystem good, find the payment vehicle credible, and interpret the scenario as intended — reported

qualitatively as a precondition. **Construct validity:** (i) *theoretical/expectation* — test that WTP rises with income, differs between users and non-users, and responds to *scope* (built into the design as a deliberate scope treatment, Chapter 35), reporting each predicted sign; (ii) *convergent* — compare CV and DCE estimates of the same construct, tested by **equivalence** with a pre-specified margin (§39.7), not by a failed difference test. **Criterion validity:** the core contribution — randomize respondents to *real* (BRIS) versus *hypothetical* payment, estimate WTP in each arm, and report the **calibration factor with a Krinsky–Robb CI** and a **Poe test** (verified machinery: CF 1.998, CI [1.915, 2.085], Poe $p = 0.0000$). **Mitigation experiments:** randomize cheap talk and/or oath scripts as *additional arms*, so their effects are causally identified, and evaluate them against the real benchmark — asking not merely "does the script lower stated WTP?" but "does it move stated WTP *toward the truth* without overshooting?" **Ex-post calibration:** pre-register the certainty threshold, report the full sensitivity curve, and never select τ using the real arm. **Comparisons:** always on **WTP**, not coefficients (§39.6), estimated in WTP space (Chapter 38). **Experimental hygiene:** report exclusion rates *by arm* (expecting higher refusal in the real arm), analyse with and without exclusions, bound (Lee), and avoid post-treatment conditioning on consequentiality or certainty (Chapter 32). *What each establishes:* content — the question is coherent; construct — the estimates are internally sensible; criterion — the estimates are *accurate*, which is the claim most stated-preference studies cannot make and which the BRIS design uniquely supports.

C2. The correct claim depends entirely on whether an *equivalence* test was conducted, and a non-significant difference alone supports **no validity claim**. *Incorrect:* "the calibrated and real estimates do not differ significantly ($p = 0.4$), so convergent validity is established." That is *absence of evidence*: it is consistent with a substantial true gap that the study was underpowered to detect, and with small samples a non-significant result is close to *guaranteed* regardless of the truth. *Correct procedure:* pre-specify an **equivalence margin** Δ justified by decision relevance (how large a discrepancy would change a policy conclusion or exceed acceptable benefit-transfer error — e.g. $\pm 25\%$, $\Delta = \ln 1.25$), then run **TOST**: reject both $H_0 : \delta \leq -\Delta$ and $H_0 : \delta \geq +\Delta$, equivalently check that the **90% CI** for the log-ratio lies entirely inside $[-\Delta, +\Delta]$. *Correct claim if it passes* (verified example): "the calibrated hypothetical estimate (€42.12) is statistically **equivalent** to the real-payment benchmark (€40.29) within a pre-specified $\pm 25\%$ margin (log-ratio 0.0442, 90% CI [0.0087, 0.0796], TOST $p < 0.001$), supporting criterion validity at that tolerance." *Correct claim if it fails:* "I cannot establish equivalence at the $\pm 25\%$ margin; the CI admits differences up to $[\cdot]\%$, so the data are inconclusive at this precision" — explicitly *not* "the estimates agree." I would also report the achieved power for equivalence, since an inconclusive result may reflect sample size rather than disagreement.

C3. *Response.* The concern is legitimate in form — real-payment arms are expensive and therefore usually small, and a small benchmark yields a wide confidence interval on the *denominator* of the calibration factor, propagating into the factor itself. But it is answerable with evidence rather than assertion, in four steps. **(i) Report the precision I actually have.** The calibration factor's uncertainty is quantified by Krinsky–Robb simulation propagating *both* arms' parameter uncertainty (verified machinery: CI [1.915, 2.085] in the illustrative design), and by the Poe complete-combinatorial test comparing the full distributions. If that interval is tight enough to exclude 1.0 and to bound the factor usefully, the arm *is* adequate for the claim being made — and I state the claim at the precision the interval supports, not beyond. **(ii) Distinguish the claim from the precision.** Even a modest real arm supports the qualitative criterion-validity claim (bias exists, direction, order of magnitude) while supporting only a *range* for the factor. I will state which claims the arm can and cannot bear, rather than over-reading it. **(iii) Power and design.** I report the *a priori* power calculation for the arm, and note that the real arm's efficiency was maximized by concentrating bids where the response curve is informative (Chapter 36) rather than spreading them thinly — the same efficiency logic as Chapter 37. **(iv) Equivalence framing.** Where I claim *agreement* after calibration, I use TOST with a pre-specified margin (§39.7) and report the achieved equivalence power, so a small arm cannot masquerade as evidence of sameness through mere non-significance — precisely the fallacy the referee should be worried about. If, after this, the interval is genuinely too wide for the claims made, the correct response is to *weaken the claims* (report the factor as a range, treat criterion validity as suggestive) rather than to abandon the benchmark — since a wide but honest estimate of hypothetical bias is far more informative than the alternative, which is *assuming* a literature calibration factor with no benchmark at all (F3).

39.12 Spaced review prompts

+1 day. From memory: the three validity types and their strength ordering; the Carson–Groves mechanism; the definition of the calibration factor; why WTP not coefficients.

+1 week. Explain why construct validity cannot detect proportional bias; why certainty calibration can overcorrect and must be pre-registered; why convergent validity needs TOST.

+1 month. For your CV study, execute the full validity strategy: report the calibration factor with CI and Poe test, the pre-registered calibration with full sensitivity curve, exclusion rates by arm with bounds, and equivalence tests with justified margins. Close with the spine sentence about what your real-versus-hypothetical contrast identifies.

39.13 How this chapter connects to the rest of the book

- **Chapter 28** supplies the incentive-compatibility theory that *explains* hypothetical bias and motivates BRIS; **Chapter 33** supplies the equivalence testing that validity claims require; **Chapters 31–32** supply the experimental identification and threat analysis for the real/hypothetical contrast; **Chapter 25** supplies the scale confound that forces WTP-based comparison; **Chapter 38** supplies the WTP-space estimation that delivers the scale-free quantity directly.
- **Chapter 35** supplies the scope tests that constitute construct validity; **Chapter 36** supplies the SBDC estimation applied in both arms; **Chapter 13** supplies Krinsky–Robb; **Chapter 10** supplies the Poe test's resampling logic.
- Downstream: **Chapters 45–46 (CBA)** consume calibrated WTP, where the calibration factor's uncertainty becomes a first-order sensitivity parameter; **Chapters 47–48 (meta-analysis)** treat hypothetical bias, elicitation format, and consequentiality as *moderators* explaining variation across your 18 DCEs; **Chapter 49 (benefit transfer)** inherits the validity question in transferred form.

This chapter is where your research makes its most distinctive contribution: most stated-preference studies *assume* validity or borrow a calibration factor; yours *measures* it.

39.14 Research translation task

For your **CV study**, produce a three-page validity note:

1. **Validity map.** State what content, construct, and criterion evidence you have, and what each establishes and does not.
2. **Calibration factor.** Report WTP by arm, the calibration factor, its Krinsky–Robb CI, and the Poe test — computed from **WTP**, not coefficients.
3. **Mitigation.** Report randomized cheap talk / oath / consequentiality treatments and evaluate each *against the real benchmark* (does it move estimates toward truth without overshooting?).
4. **Ex-post calibration.** State the pre-registered threshold, report the *full sensitivity curve* across thresholds, and confirm the threshold was chosen independently of the real arm.
5. **Experimental hygiene.** Exclusion rates by arm, with/without analysis, Lee bounds, no post-treatment conditioning.
6. **Equivalence.** For each sameness claim, report the pre-specified margin, the TOST result, and the achieved power.
7. **Close** with the spine sentence: *"My randomized real-versus-hypothetical contrast identifies a calibration factor of [·] (95% CI [·, ·]) computed from scale-free WTP; the pre-registered certainty calibration at $\tau = [\cdot]$ yields [·], shown equivalent to the real benchmark within a pre-specified $\pm[\cdot]$% margin (TOST $p = [\cdot]$)."*

Carry this into Chapters 45–46 (CBA sensitivity to the calibration factor) and Chapters 47–48 (hypothetical bias and consequentiality as meta-analytic moderators).

39.15 Bibliography

Core. - Carson & Groves (2007), "Incentive and Informational Properties of Preference Questions," *Environmental and Resource Economics* — the incentive-theoretic account of hypothetical bias and consequentiality (§39.3–§39.4). *(Verify volume/pages.)* - Murphy, Allen, Stevens & Weatherhead (2005), "A Meta-Analysis of Hypothetical Bias in Stated Preference Valuation," *Environmental and Resource Economics* — the standard reference for calibration-factor magnitudes (§39.3). *(Verify volume/pages.)* - List & Gallet (2001), "What Experimental Protocol Influence Disparities Between Actual and Hypothetical Stated Values?," *Environmental and Resource Economics* — meta-analytic evidence on the hypothetical–real gap (§39.3). *(Verify volume/pages.)*

Applied. - Cummings & Taylor (1999), "Unbiased Value Estimates for Environmental Goods: A Cheap Talk Design for the Contingent Valuation Method," *American Economic Review* — the cheap talk script (§39.4). *(Verify volume/pages.)* - Jacquemet, Joule, Luchini & Shogren (2013), "Preference Elicitation Under Oath," *Journal of Environmental Economics and Management* — oath scripts (§39.4). *(Verify volume/pages.)* - Champ, Bishop, Brown & McCollum (1997) on certainty calibration of contingent values (§39.5). *(Verify citation.)* - Poe, Giraud & Loomis (2005), "Computational Methods for Measuring the Difference of Empirical Distributions," *American Journal of Agricultural Economics* — the complete-combinatorial test (§39.3). *(Verify volume/pages.)*

Advanced. - Vossler, Doyon & Rondeau (2012), "Truth in Consequentiality: Theory and Field Evidence on Discrete Choice Experiments," *American Economic Journal: Microeconomics* — consequentiality theory and evidence for DCEs (§39.4). *(Verify volume/pages.)* - Diamond & Hausman (1994), "Contingent Valuation: Is Some Number Better than No Number?," *Journal of Economic Perspectives* — the foundational validity critique this chapter answers (§39.2). *(Verify volume/pages.)* - Haghani et al. (2021) systematic reviews of hypothetical bias and mitigation methods in stated choice (§39.4–§39.5). *(Verify citation.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Carson–Groves / Murphy et al. / Vossler et al. / Poe et al. citations before using them, per the book's no-unverified-citation rule.

End of Chapter 39, and of Part IX. Next: Part X — Multiple Discreteness. Chapter 40 introduces the MDCEV framework, which extends discrete choice from "which one do you choose?" to "how much of each do you consume?" — the multiple-discrete-continuous structure needed for your ecosystem-services allocation proposal, where respondents divide a budget across several services rather than selecting a single alternative. Tell me when to write it, or request a different chapter.

Part IX Assessment — Stated Preference in Practice

Covers Chapters 36–39. Five components, in the order they are best attempted. Allow a full working day; the mock oral is the part that matters most for your viva.

1. Cumulative problem set

P1. Derive the SBDC as a referendum in the RUM, and show where the scale parameter goes.

P2. Given four attributes at three levels each, construct a D-efficient design of 12 choice tasks under a zero prior, and compute its D-error.

P3. Explain how a Bayesian-efficient design differs, and what happens if your prior sign is wrong.

P4. Write the simulated log-likelihood for an EC-MXL in WTP space and state where the Halton draws enter.

P5. Given real and hypothetical treatment means of €19 and €31, compute the hypothetical-bias ratio, and design the equivalence test that would support a calibration claim.

2. Research critique

Take your DCE design and evaluate it against the criteria of Chapter 37: attribute independence, level balance, overlap, dominance, choke prices, opt-out framing and cognitive burden. Identify the one design choice that most constrains what you can identify, and state what you would change.

3. Red-flag diagnosis

A paper calibrates hypothetical WTP by multiplying by a factor of 0.6 taken from a published meta-analysis of a different good, and reports the calibrated value with the original confidence interval. Identify every problem, and state what the reported uncertainty should have been.

4. Mock oral examination

1. What does the cost-vector manipulation identify, and what does it not?
2. Why is D-efficiency not the same as identification?
3. What is hypothetical bias — a bias in what, exactly?
4. Does a cheap-talk script solve it? What is the evidence?
5. Should attribute non-attendance be inside or outside your welfare measure? Defend your answer.

5. Integration mini-project

Estimate your EC-MXL in WTP space, then produce the full validity dossier: the real-versus-hypothetical contrast with an equivalence test against a pre-specified margin, the scope test, the non-attendance diagnostics, and a paragraph on which validity claim your design can and cannot support.

Part X — Multiple Discreteness

Chapter 40 — The MDCEV Framework: Multiple Discreteness and Allocation

Part X: Multiple Discreteness

> **The spine question, applied to this chapter.** Every choice model so far has asked *which one?* — the respondent picks a single alternative and the analyst recovers the marginal WTP for its attributes. But your ecosystem-services proposal asks a different question: *given a budget, how much would you allocate to each of several services?* Respondents choose **several** services (not one), in **continuous** amounts (not just yes/no), and typically **not all** of them (some receive nothing). No model in Parts I–IX can represent that. The **MDCEV** framework can, by replacing the linear utility that forces single choice with a *satiating* utility that makes variety optimal. The spine question: *what does the satiation structure I impose actually identify — and are my satiation parameters recovering preferences, or absorbing a functional form I chose and cannot test?*

40.0 Conceptual motivation: why standard discrete choice cannot do this

Consider the survey task your ecosystem-services proposal contemplates. A respondent is given a conservation budget and asked to divide it among several services — fish-stock recovery, water quality, habitat restoration, coastal protection, biodiversity. The observed response is a *vector*: perhaps €40 to one service, €25 to another, €35 to a third, and nothing to the remaining two. Three features of that response defeat the models built so far.

It is multiple. The respondent chose *three* services, not one. A multinomial logit assigns probability to *exactly one* alternative being chosen; it has no way to represent choosing several simultaneously.

It is continuous. The respondent chose *how much*, not merely *whether*. The quantities carry information — €40 versus €5 to the same service means something quite different — and a discrete-choice model discards it entirely.

It has corners. Two services received *exactly zero*. This is not missing data or refusal; it is an optimal choice of zero, a **corner solution**. A standard continuous-demand model (a system of linear expenditure equations, say) predicts interior solutions with everything receiving a positive share, so it cannot generate the zeros that dominate real allocation data. Verified below, respondents in a simulated five-service allocation consumed on average **3.95 of 5** services — never reliably all, essentially never just one. **MNL forces exactly one; a linear allocation model forces all; the data do neither.**

The economic source of the problem is the linearity of utility. In a standard random utility model, utility is linear in attributes, so marginal utility is constant: if service A is better than B at the first euro, it

is better at every euro, and the optimal allocation is a *corner* — spend everything on the single best option. Linear utility *logically entails* single discrete choice. To get variety, you need **diminishing marginal utility**: the value of an additional euro to a service *falls* as that service receives more, so at some point another service becomes the better use of the next euro. Variety-seeking is not a behavioural add-on — it is what satiation implies.

Bhat's **MDCEV** (multiple discrete-continuous extreme value) model builds exactly this: an additively separable utility with a satiating (non-linear) sub-utility per good, plus extreme-value randomness. The KKT conditions of the constrained optimization then deliver, remarkably, a *closed-form* likelihood — the same analytical tractability that makes logit attractive, extended to allocation. This chapter builds that model, derives its corner-solution logic, estimates it (verified recovery below), and confronts its two identification hazards: the **location normalization** (baseline utilities are identified only as *differences* from the outside good — verified, all estimates shifted by a common -0.749) and the α–γ **ridge** (two parameters generating nearly indistinguishable satiation, verified correlations of 0.974 to 0.997), which is why the literature fixes one of them by convention rather than estimating both.

40.1 Prerequisites

From **Chapter 2–3**: constrained optimization, the Lagrangian, and **Karush–Kuhn–Tucker conditions with inequality constraints** — the mathematical core of this chapter. From **Chapters 5, 20–21**: random utility, the Gumbel distribution, and the logit closed form. From **Chapter 34**: the budget constraint and the marginal utility of income. From **Chapter 14**: corner solutions as censoring. From **Chapter 11**: maximum likelihood. From **Chapter 38**: simulation and identification diagnostics.

New notation. x_k (quantity/expenditure allocated to good k), x_1 or x_{out} (the **outside good** / numeraire), ψ_k (**baseline marginal utility** of good k), γ_k (**translation** parameter), α_k (**satiation** exponent), E (budget), p_k (price), λ (Lagrange multiplier on the budget), M (number of goods consumed).

40.2 The MDCEV utility function

Definition 40.1 (MDCEV utility). Bhat's specification is **additively separable** across goods, with each good contributing a *satiating* sub-utility:

$$U(\mathbf{x}) = \sum_k \frac{\gamma_k}{\alpha_k}\,\psi_k \left[\left(\frac{x_k}{\gamma_k}+1\right)^{\alpha_k} - 1\right].$$

Three parameters per good, each with a distinct economic role:

- ψ_k — **the baseline marginal utility.** The marginal utility of good k *evaluated at zero consumption*: $\partial U/\partial x_k|_{x_k=0} = \psi_k$. It governs the **extensive margin** — whether the good is consumed at all — and is where observed attributes and randomness enter: $\psi_k = \exp(\beta' \mathbf{z}_k + \varepsilon_k)$ with ε_k i.i.d. Gumbel. The exponential guarantees positivity (marginal utility at zero must be positive for the good to be desirable).
- γ_k — **the translation parameter.** It "translates" the utility function away from the origin, which is what permits **corner solutions**: with $\gamma_k > 0$, marginal utility at zero is *finite* ($= \psi_k$), so a good can optimally receive nothing. (Contrast a logarithmic utility $\ln x_k$, whose marginal utility $\to \infty$ as $x_k \to 0$, forcing every good to be consumed — an interior solution always.) γ_k also controls the *rate* of satiation, governing the **intensive margin**.
- α_k — **the satiation exponent.** With $\alpha_k < 1$, marginal utility *diminishes* in x_k. As $\alpha_k \to 1$ utility becomes linear (no satiation, single discrete choice returns); as $\alpha_k \to -\infty$ satiation is immediate. The limiting case $\alpha_k \to 0$ gives the widely-used logarithmic form $\gamma_k \psi_k \ln(x_k/\gamma_k + 1)$.

Marginal utility. Differentiating,

$$\frac{\partial U}{\partial x_k} = \psi_k \left(\frac{x_k}{\gamma_k} + 1 \right)^{\alpha_k - 1},$$

which equals ψ_k at $x_k = 0$ and declines in x_k whenever $\alpha_k < 1$. **This is the whole model in one line:** a finite, good-specific marginal utility at zero (permitting corners) that decreases with consumption (permitting variety).

The outside good. One alternative — the numeraire, "money spent on everything else" — is typically assumed *always consumed*, with $\gamma_{\text{out}} = 1$ and often $\alpha_{\text{out}} \to 0$, giving $\psi_{\text{out}} \ln(x_{\text{out}})$. Its role is essential: it absorbs unallocated budget and provides the reference against which all other baseline utilities are measured (§40.6). In your application, the outside good is the part of the conservation budget *not* allocated to any listed service — or, in a household framing, income retained for other purposes.

> **Active recall 40.A.** What economic role does each of ψ_k, γ_k, α_k play, and why does linear utility make multiple discreteness impossible? *(Answer: ψ_k is the **baseline marginal utility** — marginal utility evaluated at* zero* consumption — governing the *extensive* margin (whether the good is consumed at all); observed attributes and the Gumbel error enter here via $\psi_k = \exp(\beta' \mathbf{z}_k + \varepsilon_k)$. γ_k is the **translation** parameter, which shifts the utility function off the origin so that marginal utility at zero is *finite* rather than infinite — this is what makes *corner solutions* (optimal zeros) possible — and it also controls the *rate* of satiation, driving the *intensive* margin (how much). α_k is the **satiation exponent**: $\alpha_k < 1$ gives diminishing marginal utility, with $\alpha_k \to 1$ removing satiation entirely and $\alpha_k \to 0$ giving the logarithmic form. **Linear utility makes multiple discreteness impossible** because it implies *constant* marginal utility: if good A yields more utility per euro than B at the first euro, it does so at *every* euro, so the optimum is to spend the entire budget on the single best good — a corner. Variety therefore requires *diminishing* marginal utility, so that after enough spending on A the next euro is better spent on B. Multiple discreteness is a direct consequence of satiation, not a separate behavioural assumption.)*

40.3 The KKT conditions and corner solutions

The respondent maximizes $U(\mathbf{x})$ subject to the budget constraint $\sum_k p_k x_k = E$ and non-negativity $x_k \geq 0$. Because the non-negativity constraints may *bind*, this is a Karush–Kuhn–Tucker problem (Chapter 3), not a plain Lagrangian.

The Lagrangian and KKT conditions. With multiplier λ on the budget:

$$\mathcal{L} = U(\mathbf{x}) - \lambda \left(\sum_k p_k x_k - E \right),$$

the KKT conditions are

$$\frac{\partial U}{\partial x_k} - \lambda p_k = 0 \;\text{ if } x_k > 0, \qquad \frac{\partial U}{\partial x_k} - \lambda p_k \leq 0 \;\text{ if } x_k = 0.$$

Substituting the marginal utility:

$$\psi_k \left(\frac{x_k}{\gamma_k} + 1 \right)^{\alpha_k - 1} = \lambda p_k \quad (x_k > 0), \qquad \psi_k \leq \lambda p_k \quad (x_k = 0).$$

Reading the conditions. The interpretation is transparent and worth stating plainly:

- **The consumption rule.** Good k is consumed **if and only if its baseline marginal utility per euro exceeds the shadow value of the budget**: $\psi_k / p_k > \lambda$. Goods whose *very first* euro is worth

less than the budget's shadow value optimally receive **nothing**. This is the corner-solution mechanism, and it is entirely intuitive: you fund a service only if its first euro beats the best alternative use of that euro.

- **The quantity rule.** For consumed goods, allocate until marginal utility per euro is *equalized* across all consumed goods at λ — the classical equimarginal principle. Satiation is what makes this stop short of spending everything on one good.
- λ **is endogenous.** The shadow value depends on which goods are consumed, which depends on λ — so the solution requires an algorithm, not a formula (§40.4).

Worked Example 40.1 (Corner solutions are the norm, verified). Five inside goods with baseline utilities $V = (1.2, 0.8, 0.4, 1.0, 0.2)$, unit prices, budget $E = 100$, $\gamma = (2, 4, 6, 3, 5)$, Gumbel scale 1:

Good	Baseline V_k	P(consumed)	Mean $x_k \mid x_k > 0$
1	1.20	0.870	19.76
2	0.80	0.818	23.46
3	0.40	0.746	24.11
4	1.00	0.837	21.88
5	0.20	0.680	20.01

Number of inside goods consumed: mean **3.95**; distribution over $0, \ldots, 5$: (0.003, 0.056, 0.077, 0.148, 0.282, 0.434).

(Verified.) Consumption probabilities rise with baseline utility (0.68 for the least attractive service, 0.87 for the most), and the *number* of services funded varies across respondents, concentrated at four and five but with real mass at two and three. **This pattern — most but not all goods, varying by respondent — is precisely what neither MNL (exactly one) nor a linear allocation model (all) can produce.** It is the empirical signature of multiple discreteness.

> **Stop and think 40.1.** Your respondent allocates nothing to "coastal protection." Under MDCEV, what does that zero mean, and how does it differ from a zero in a standard demand model or a protest zero in contingent valuation? *(Answer: under MDCEV the zero is an* ***optimal corner solution*** *— it means the KKT inequality binds, $\psi_{coastal}/p_{coastal} \leq \lambda$: the service's* baseline* marginal utility per euro (its value at the very first euro) does not exceed the shadow value of the budget, so the respondent optimally funds other services instead. It is a *genuine economic choice* carrying information about relative preferences, not an absence of data. This differs from a standard *continuous demand* model, which (with marginal utility unbounded at zero, as under log utility) predicts interior solutions where everything receives a positive amount — such a model cannot generate the zero at all and would have to treat it as censoring or measurement error. It also differs from a **protest zero** in CV (Chapters 26, 29), which reflects rejection of the payment vehicle or scenario rather than a genuine valuation of zero; a protest zero is *not* an optimal allocation and must be modelled separately or bounded. The practical implication: in MDCEV zeros are *modelled*, not dropped — they are a large share of the information (verified, respondents left roughly one of five services unfunded on average) — but I must still distinguish an economic corner from a protest or task-avoidance zero using debriefing, exactly as in Chapter 26.)*

40.4 Solving the allocation: the forecasting algorithm

Because λ and the consumed set are jointly determined, computing the optimal allocation requires an iterative procedure — needed both for simulation and for policy forecasting.

The algorithm (Bhat / Pinjari–Bhat). For the γ-profile ($\alpha_k = 0$) with an always-consumed outside good:

1. **Sort** inside goods by baseline marginal utility per euro, ψ_k/p_k, in descending order.
2. **Initialize** with only the outside good consumed: $\lambda = \psi_{\text{out}}/E$.
3. **Add** the highest-ranked remaining good if $\psi_k/p_k > \lambda$; otherwise **stop**.
4. **Recompute** the multiplier over the current consumed set S using the budget constraint:

$$\lambda = \frac{\psi_{\text{out}} + \sum_{k\in S} \gamma_k \psi_k}{E + \sum_{k\in S} \gamma_k p_k}.$$

5. **Repeat** from step 3.
6. **Allocate**: $x_k = \gamma_k \left(\dfrac{\psi_k}{\lambda p_k} - 1 \right)$ for $k \in S$, and $x_{\text{out}} = \psi_{\text{out}}/\lambda$.

The algorithm terminates in at most K steps and is exact — the sorting guarantees that once a good fails the condition, all lower-ranked goods fail it too.

Worked Example 40.2 (What γ does, verified). Varying γ_3 while holding everything else fixed:

γ_3	P(good 3 consumed)	Mean $x_3 \mid x_3 > 0$
0.5	0.738	**4.68**
2.0	0.729	**12.82**
6.0	0.746	**26.25**
12.0	0.718	**33.68**

(Verified.) A twenty-four-fold increase in γ_3 leaves the *probability of consumption* essentially unchanged (0.72–0.75, no trend) while the *amount consumed* rises more than sevenfold (4.68 to 33.68). **This cleanly separates the two margins: ψ (through V) governs *whether*; γ governs *how much*.** The intuition is that γ determines how quickly marginal utility falls — a large γ means slow satiation, so once a good is worth funding it absorbs a large share before the equimarginal condition is met. For your application this is exactly the distinction between *which* ecosystem services people care about at all and *how much* of the budget a service can absorb before people prefer to spread it elsewhere — two substantively different questions your data can answer separately.

40.5 The likelihood

The remarkable feature of MDCEV — and the reason it is usable — is that the KKT conditions plus i.i.d. Gumbel errors yield a **closed-form** likelihood, with no simulation required.

Derivation sketch. Take logarithms of the KKT equality for consumed goods:

$$\underbrace{V_k + (\alpha_k - 1)\ln\left(\frac{x_k}{\gamma_k} + 1\right) - \ln p_k}_{W_k} + \varepsilon_k = \ln\lambda,$$

so for every consumed good the quantity $W_k + \varepsilon_k$ takes the *same* value $\ln\lambda$, while for every non-consumed good $W_k + \varepsilon_k \le \ln\lambda$. The observed allocation therefore corresponds to an event about the ε's: equality for the consumed set, inequality for the rest. With ε i.i.d. Gumbel, this probability has an analytical form. Changing variables from ε to the observed $\mathbf{x}$ contributes the Jacobian, which produces the c_k terms below.

Definition 40.2 (MDCEV likelihood). With M goods consumed (including the outside good), scale σ, and

$$W_k = V_k + (\alpha_k - 1)\ln\left(\frac{x_k}{\gamma_k} + 1\right) - \ln p_k, \qquad c_k = \frac{1 - \alpha_k}{x_k + \gamma_k},$$

the probability of the observed allocation is

$$P(\mathbf{x}) = \frac{1}{\sigma^{M-1}} \left(\prod_{k \in S} c_k \right) \left(\sum_{k \in S} \frac{1}{c_k} \right) \frac{\prod_{k \in S} e^{W_k/\sigma}}{\left(\sum_{j=1}^{K} e^{W_j/\sigma} \right)^M} (M-1)!$$

Structurally this is a logit kernel (the exponential ratio) multiplied by Jacobian terms handling the continuous quantities. When $M = 1$ it collapses to the standard MNL probability — **MDCEV nests multinomial logit as the special case in which exactly one good is chosen**, which is a useful sanity check and clarifies the relationship between the frameworks.

Worked Example 40.3 (Maximum likelihood recovery, verified). Estimating on 2,500 simulated allocations (γ-profile, $\alpha = 0$):

Parameter	True	Estimate
γ_1	2.000	**2.219**
γ_2	4.000	**4.225**
γ_3	6.000	**5.722**
γ_4	3.000	**3.318**
γ_5	5.000	**5.027**
σ	1.000	**0.998**

(Verified; converged.) The satiation parameters and the scale are recovered accurately. The baseline utilities require a separate discussion — which is the subject of the next section, because their apparent "failure" to recover is in fact a structural feature.

40.6 Identification I: the location normalization

The finding. In the same estimation, every baseline utility estimate was *shifted downward by a common amount*:

Parameter	True	Estimate	Shift
V_{out}	1.500	0.777	−0.723
V_1	1.200	0.440	−0.760
V_2	0.800	0.027	−0.773
V_3	0.400	−0.314	−0.714
V_4	1.000	0.173	−0.827
V_5	0.200	−0.500	−0.700

Mean shift −0.749, **standard deviation only 0.043** — a *common* shift, not idiosyncratic estimation error. And the *differences* are recovered accurately:

Good	True $V_k - V_{\text{out}}$	Estimated	Error
1	−0.300	−0.336	−0.036
2	−0.700	−0.750	−0.050
3	−1.100	−1.091	+0.009
4	−0.500	−0.604	−0.104
5	−1.300	−1.276	+0.024

(Verified.)

The explanation. Inspect the KKT conditions: the consumption rule compares ψ_k/p_k against λ, and λ is itself determined by the ψ's through the budget constraint. Multiplying *every* ψ_k by a common constant (equivalently, adding a constant to every V_k) scales λ by the same constant, leaving every comparison — and hence every allocation — **unchanged**. The *level* of baseline utility is therefore not identified; only *differences* are. This is exactly the location normalization of multinomial logit (Chapter 22), where only utility differences matter, appearing here in allocation form.

The practical rule. Normalize $V_{\text{out}} = 0$ and report all baseline utilities *relative to the outside good*. Then V_k reads as "the baseline attractiveness of service k relative to retaining the money" — which is, conveniently, the substantively meaningful comparison for your application. Reporting un-normalized levels invites the mistake of interpreting a number that carries no information.

> **Active recall 40.B.** Why are MDCEV baseline utilities identified only up to a common shift, and what is the reporting convention? *(Answer: the KKT conditions determine consumption by comparing each good's baseline marginal utility per euro, ψ_k/p_k, against the budget's shadow value λ — and λ is itself a function of the ψ's via the budget constraint (verified formula: $\lambda = (\psi_{out} + \sum_S \gamma_k\psi_k)/(E + \sum_S \gamma_k p_k)$). Multiplying every ψ_k by a common constant scales λ by that same constant, so every comparison ψ_k/p_k versus λ — and therefore the entire optimal allocation — is* unchanged*. The data thus contain no information about the* level* of baseline utility, only about *differences*: this is the same location normalization as in MNL, where only utility differences enter choice probabilities. Verified empirically: all six baseline estimates were shifted by a common −0.749 (sd only 0.043, confirming a systematic rather than random shift), while the differences from the outside good were recovered to within about 0.05. Convention: **normalize $V_{\text{out}} = 0$** and report every baseline relative to the outside good — which reads naturally as the attractiveness of funding a service versus retaining the money.)*

40.7 Identification II: the α–γ ridge

The second identification issue is subtler and, unlike the first, cannot be resolved by a normalization alone.

The problem. Both α_k and γ_k generate *satiation*:

$$\gamma\text{-profile } (\alpha = 0): \quad \frac{\partial U}{\partial x} = \frac{\psi}{x/\gamma + 1}; \qquad \alpha\text{-profile } (\gamma = 1): \quad \frac{\partial U}{\partial x} = \psi\,(x+1)^{\alpha-1}.$$

Both are decreasing in x; both flatten as their parameter grows. Since ψ is a free baseline, the empirical question is whether the two *shapes* are distinguishable.

Worked Example 40.4 (The identification ridge, verified). Fitting an α-profile (with free baseline scale) to mimic a γ-profile:

Target γ	Best-fitting α	Baseline scale	RMS gap	**Correlation**
2.0	0.1997	1.130	9.0%	**0.9966**
4.0	0.3672	1.235	12.6%	**0.9873**
8.0	0.5156	1.293	12.6%	**0.9744**

(Verified.) Correlations between the two satiation curves run from 0.974 to 0.997 — highly similar though not identical, with residual RMS gaps of 9–13%. The parameters are therefore **weakly separately identified**: the likelihood exhibits a near-flat *ridge* along which increases in α trade off against decreases in γ with little change in fit. In finite samples this manifests as a near-singular Hessian, slow or failed convergence, estimates highly sensitive to starting values, and results that do not replicate.

The standard resolution. Fix one parameter by convention:

- **γ-profile** ($\alpha_k = 0$ for all inside goods): satiation is captured entirely by γ_k, giving the logarithmic sub-utility $\gamma_k \psi_k \ln(x_k/\gamma_k + 1)$. This is the most common choice, and the one used in the verified estimation above (where γ recovered accurately precisely *because* α was fixed).
- **α-profile** ($\gamma_k = 1$): satiation captured by α_k.
- Occasionally a hybrid: γ free for some goods, α for others, guided by theory.

The honest statement. The choice of profile is an *assumption*, not a finding, and it is largely untestable from the data — which is precisely Chapter 30's untestable-restriction situation. The consequences for your reporting are concrete:

1. **State the profile explicitly** and justify it (convention, comparability with prior literature, convergence behaviour).
2. **Report sensitivity**: estimate under the alternative profile and confirm that substantive conclusions — which services are valued, forecast allocations, welfare rankings — are unchanged. If conclusions *do* change, that is a first-order finding requiring disclosure.
3. **Do not over-interpret the satiation parameter** as a deep behavioural constant; it is a shape parameter of a functional form you selected.
4. **Never estimate both freely** without reporting Hessian conditioning and multiple starting values — the ridge will produce numbers, but unstable ones.

Stop and think 40.2. A co-author proposes estimating α_k and γ_k freely for every service "to let the data decide." What do you say? *(Answer: the data largely* cannot* decide, because the two parameters generate nearly the same satiation curve. Verified: an α-profile fitted to mimic a γ-profile achieves correlations of 0.974–0.997 between the marginal-utility curves, with residual gaps of only 9–13% — so the likelihood has a near-flat *ridge* along which α and γ trade off with almost no change in fit. Estimating both freely will therefore produce a near-singular Hessian, estimates that swing with starting values, standard errors that are unreliable (the information matrix is nearly non-invertible), and results that will not replicate — the model will return numbers, but they will be arbitrary points on the ridge rather than estimates. The disciplined approach: *fix* one by convention — $\alpha = 0$ (the γ-profile, most common, and verified above to recover γ accurately once α is fixed) or $\gamma = 1$ (the α-profile) — state the choice explicitly as an assumption rather than a finding, and then *report sensitivity* by re-estimating under the alternative profile to confirm that the substantive conclusions (which services are valued, forecast allocations, welfare rankings) are unaffected. If those conclusions do change across profiles, that is itself an important finding to disclose, not a nuisance to suppress. This is Chapter 30's untestable-restriction logic: where an assumption cannot be tested, impose it transparently and bound its consequences.)*

40.8 Application to your ecosystem-services proposal

Mapping the framework onto your design:

The goods are the ecosystem services among which respondents allocate — fish-stock recovery, water quality, habitat restoration, coastal protection, biodiversity. The **outside good** is the unallocated portion of the budget (money retained, or allocated to no listed service) — and it must be *explicitly offered*, since without it the model forces the full budget onto the listed services and cannot distinguish "I value these highly" from "I had no alternative."

The budget must be credible, clearly stated, and — for welfare interpretation — connected to a real constraint. A hypothetical allocation of an unspecified budget invites the consequentiality problems of Chapter 39.

What the parameters tell you. ψ_k (via V_k, relative to the outside good) identifies *which* services command support at the margin — the extensive margin, and the natural input to a priority ranking. γ_k identifies *satiation*: how much a service can absorb before respondents prefer to diversify. Verified above, these are separately identified and answer genuinely different policy questions — a service with high V but low γ is

one everyone wants funded *a little*, whereas high V and high γ marks a service that can absorb substantial resources.

Why this is more informative than a ranking exercise. Asking respondents to *rank* services yields ordinal information only; asking them to *allocate* yields both the extensive margin (which services) and the intensive margin (how much), and — because the budget constraint forces genuine trade-offs — it is closer to the actual policy problem. This is the substantive case for MDCEV over simpler elicitation.

Cautions carried forward. *Cognitive burden* (Chapter 37) is significant: allocating a budget across several services is harder than choosing between two profiles, so the number of services must be limited and the task carefully piloted. *Zeros must be interpreted* (§40.3) — an economic corner is not a protest, and debriefing must distinguish them (Chapter 26). *Hypothetical bias* (Chapter 39) applies with full force and arguably more, since allocating a hypothetical budget is even less consequential than a hypothetical payment. And *welfare analysis in MDCEV is harder* than in MNL: there is no simple log-sum formula, and welfare measures require simulation-based computation of the compensating variation using the forecasting algorithm of §40.4 — a topic for the next chapter.

40.9 Failure modes

F1. Using MNL for allocation data. Forcing a single choice on a multiple-discrete response. *Fix:* MDCEV (§40.0).

F2. Using a continuous demand system. Predicting interior solutions, unable to generate zeros. *Fix:* translated utility permitting corners (§40.2).

F3. Dropping zeros. Treating unfunded services as missing. *Fix:* zeros are optimal choices carrying information; model them (§40.3).

F4. Conflating corners with protests. Treating a protest zero as an economic corner. *Fix:* debrief and distinguish; model or bound protests separately (§40.3; Chapter 26).

F5. No outside good. Forcing the entire budget onto listed alternatives. *Fix:* include and explicitly offer an outside option (§40.2, §40.8).

F6. Interpreting un-normalized baselines. Reporting V_k levels as meaningful. *Fix:* normalize $V_{\text{out}} = 0$; report differences (§40.6).

F7. Estimating α and γ freely. Near-singular Hessian, unstable estimates. *Fix:* fix one profile; report sensitivity to the other (§40.7).

F8. Over-interpreting satiation parameters. Treating γ as a behavioural constant rather than a shape parameter of a chosen form. *Fix:* interpret cautiously; test alternative profiles (§40.7).

F9. Ignoring cognitive burden. Too many services for a reliable allocation task. *Fix:* limit the number; pilot extensively (§40.8; Chapter 37).

F10. Assuming MNL-style welfare formulas. Applying a log-sum to MDCEV. *Fix:* simulation-based welfare via the forecasting algorithm (§40.8).

40.10 Diagnostic questions for your own papers

1. **Model match.** Does my response format genuinely involve multiple discreteness, justifying MDCEV over MNL? *(§40.0, F1.)*
2. **Outside good.** Is an outside option explicitly offered, and is the budget credible? *(§40.2, §40.8, F5.)*
3. **Zeros.** Are zeros modelled as corners, and have I distinguished them from protests? *(§40.3, F3–F4.)*

4. **Normalization.** Is $V_{\text{out}} = 0$ imposed and are baselines reported as differences? *(§40.6, F6.)*
5. **Profile choice.** Which profile ($\alpha = 0$ or $\gamma = 1$) did I impose, is it stated as an assumption, and is sensitivity reported? *(§40.7, F7–F8.)*
6. **Convergence.** Multiple starting values, Hessian conditioning checked? *(§40.7.)*
7. **Margins.** Do I report and interpret the extensive (ψ) and intensive (γ) margins separately? *(§40.4, §40.8.)*
8. **Burden and validity.** Is the task cognitively feasible, piloted, and is hypothetical bias addressed? *(§40.8, F9; Chapter 39.)*

40.11 Exercises

(A) Comprehension.

A1. Explain why linear utility implies single discrete choice and what MDCEV changes.

A2. State the KKT consumption rule and interpret it economically.

A3. What are the roles of ψ, γ, and α, and which margins do they govern?

(B) Derivation / reasoning.

B1. Derive the marginal utility of the MDCEV sub-utility and show it is finite at zero and diminishing.

B2. Derive the KKT conditions and show that a common scaling of all ψ_k leaves the allocation unchanged.

B3. Explain the α–γ identification problem and the standard resolution.

(C) Research application / critique.

C1. Design the MDCEV allocation task for your ecosystem-services study, specifying goods, outside good, budget, and what each parameter will identify.

C2. Your estimation with both α and γ free fails to converge. Diagnose and fix.

C3. A referee asks why you did not simply ask respondents to rank the services. Respond.

40.12 Solutions

A1. With linear utility $U = \sum_k \beta_k x_k$, marginal utility β_k is *constant* in x_k. Maximizing subject to a budget therefore allocates everything to the good with the highest β_k/p_k: if a good is the best use of the first euro, it remains the best use of every euro, so the optimum is a corner with a *single* good consumed. Linear utility thus *logically entails* single discrete choice, which is why MNL represents exactly that. MDCEV changes this by making the sub-utility **non-linear and satiating** — $\partial U/\partial x_k = \psi_k(x_k/\gamma_k + 1)^{\alpha_k - 1}$ *decreases* in x_k for $\alpha_k < 1$ — so as a good receives more, its marginal utility per euro falls until another good becomes the better use of the next euro. Variety then emerges as the optimum. The translation parameter γ_k simultaneously keeps marginal utility *finite* at zero, so goods can still optimally receive nothing — giving *multiple discreteness*: several goods consumed, some at zero.

A2. The KKT conditions give: consume good k (i.e. $x_k > 0$) **if and only if** $\psi_k/p_k > \lambda$, where ψ_k is baseline marginal utility (at zero consumption), p_k the price, and λ the Lagrange multiplier on the budget. For consumed goods, allocate until $\psi_k(x_k/\gamma_k + 1)^{\alpha_k - 1}/p_k = \lambda$ for all of them. *Economic reading:* λ is the **shadow value of the budget** — the utility yielded by the best marginal use of one more euro. A service is funded only if its *very first* euro delivers more utility per euro than that benchmark; services failing this test optimally receive nothing (a corner). Among funded services, spending continues until marginal utility per euro is *equalized* across them at λ — the equimarginal principle — with satiation ensuring this happens before the budget is exhausted on any one good.

A3. $\psi_k = \exp(\beta' \mathbf{z}_k + \varepsilon_k)$ is the **baseline marginal utility** at zero consumption, governing the **extensive margin** (whether a good is consumed at all), since the KKT rule compares ψ_k/p_k to λ; attributes and randomness enter here. γ_k is the **translation** parameter: it makes marginal utility at zero *finite* (enabling corner solutions) and controls the *rate* of satiation, governing the **intensive margin** (how much is consumed given consumption) — verified, a twenty-four-fold rise in γ_3 raised mean consumption from 4.68 to 33.68 while leaving P(consumed) flat at 0.72–0.75. α_k is the **satiation exponent**, also generating diminishing marginal utility ($\alpha_k < 1$), with $\alpha_k \to 1$ giving linearity (no satiation) and $\alpha_k \to 0$ the logarithmic form. Because γ and α both produce satiation, they are only weakly separately identified (§40.7).

B1. From $U_k = \frac{\gamma_k}{\alpha_k}\psi_k[(x_k/\gamma_k + 1)^{\alpha_k} - 1]$, differentiate with respect to x_k:

$$\frac{\partial U_k}{\partial x_k} = \frac{\gamma_k}{\alpha_k}\psi_k \cdot \alpha_k \left(\frac{x_k}{\gamma_k} + 1\right)^{\alpha_k - 1} \cdot \frac{1}{\gamma_k} = \psi_k \left(\frac{x_k}{\gamma_k} + 1\right)^{\alpha_k - 1}.$$

Finite at zero: setting $x_k = 0$ gives $\partial U_k/\partial x_k = \psi_k(1)^{\alpha_k - 1} = \psi_k < \infty$ — finite, which is what permits corner solutions (a good with low enough ψ_k fails the KKT inequality and receives nothing). Contrast $\ln(x_k)$, whose derivative $1/x_k \to \infty$ as $x_k \to 0$, forcing strictly positive consumption of every good. *Diminishing:* differentiating again,

$$\frac{\partial^2 U_k}{\partial x_k^2} = \frac{\psi_k(\alpha_k - 1)}{\gamma_k}\left(\frac{x_k}{\gamma_k} + 1\right)^{\alpha_k - 2} < 0 \quad \text{whenever } \alpha_k < 1,$$

since $\psi_k > 0$, $\gamma_k > 0$, and the bracketed term is positive. So marginal utility declines in consumption — satiation — which is what generates variety.

B2. *KKT derivation.* Maximize $U(\mathbf{x})$ subject to $\sum_k p_k x_k = E$ and $x_k \geq 0$. Form $\mathcal{L} = U(\mathbf{x}) - \lambda(\sum_k p_k x_k - E) + \sum_k \mu_k x_k$ with $\mu_k \geq 0$, $\mu_k x_k = 0$ (complementary slackness). Stationarity gives $\partial U/\partial x_k - \lambda p_k + \mu_k = 0$. For $x_k > 0$: $\mu_k = 0$, so $\partial U/\partial x_k = \lambda p_k$. For $x_k = 0$: $\mu_k \geq 0$, so $\partial U/\partial x_k = \lambda p_k - \mu_k \leq \lambda p_k$, i.e. $\psi_k \leq \lambda p_k$. *Invariance to common scaling.* Suppose every baseline is scaled: $\psi_k \mapsto c\,\psi_k$ for all k including the outside good, $c > 0$. The budget-constraint solution for the multiplier (verified form) is

$$\lambda = \frac{\psi_{\text{out}} + \sum_{k \in S} \gamma_k \psi_k}{E + \sum_{k \in S} \gamma_k p_k} \;\mapsto\; \frac{c\left(\psi_{\text{out}} + \sum_{k \in S} \gamma_k \psi_k\right)}{E + \sum_{k \in S} \gamma_k p_k} = c\,\lambda,$$

so λ scales by the same factor c. Every KKT comparison then becomes $c\psi_k/p_k$ versus $c\lambda$ — identical to ψ_k/p_k versus λ — so the consumed set is unchanged; and the quantities $x_k = \gamma_k(c\psi_k/(c\lambda p_k) - 1) = \gamma_k(\psi_k/(\lambda p_k) - 1)$ are unchanged too. Hence the allocation is *invariant*, the level of ψ (equivalently the additive level of V) is **not identified**, and only differences are — verified: all baseline estimates shifted by a common -0.749 (sd 0.043) while differences recovered to within ~0.05. Normalize $V_{\text{out}} = 0$.

B3. *The problem.* Both parameters generate diminishing marginal utility: the γ-profile ($\alpha = 0$) gives MU $= \psi/(x/\gamma + 1)$ and the α-profile ($\gamma = 1$) gives MU $= \psi(x + 1)^{\alpha - 1}$. Since ψ is a free baseline, only the *shapes* need be distinguished — and they are nearly the same. Verified: fitting an α-profile (with free baseline scale) to a γ-profile achieved correlations of 0.9966, 0.9873, and 0.9744 for $\gamma = 2, 4, 8$ respectively, with residual RMS gaps of only 9–13%. The likelihood therefore has a near-flat **ridge** in (α, γ): many combinations fit almost equally well, so the parameters are only *weakly separately identified*. Symptoms in practice: near-singular Hessian, unreliable standard errors, convergence failure or dependence on starting values, and non-replicable estimates. *Resolution.* Fix one by convention — the **γ-profile** ($\alpha_k = 0$, the common default, yielding logarithmic sub-utility and verified accurate recovery of γ once α is fixed) or the **α-profile** ($\gamma_k = 1$) — and treat the choice as an explicit, largely untestable *assumption* (Chapter 30). Then report *sensitivity*: re-estimate under the alternative profile and confirm substantive conclusions are unchanged; disclose it as a finding if they are not.

C1. *Task design.* **Goods:** four to six ecosystem services — fish-stock recovery, water quality, habitat restoration, coastal protection, biodiversity — each described concretely, at a comprehensible and policy-relevant level, and *independently* fundable (not logically entailed by one another), with the number kept low

given the substantial cognitive burden of allocation tasks (Chapter 37). **Outside good:** an explicitly offered "leave unallocated / retain for other purposes" option — *essential*, since without it the design forces the full budget onto listed services and cannot distinguish genuine enthusiasm from a constrained task (F5); it also provides the normalization reference. **Budget:** a credible, clearly-stated amount tied to a plausible payment vehicle and, ideally, made *consequential* (Chapter 39), since a hypothetical allocation of an unspecified budget is even less incentive-compatible than a hypothetical payment. **What each parameter identifies:** V_k (normalized to $V_{\text{out}} = 0$) gives the **extensive margin** — which services command support relative to retaining the money, the natural input to a priority ranking; γ_k gives the **intensive margin** — how much a service absorbs before respondents diversify, verified to be separately identified from V_k. The policy contrast is substantive: high V with low γ marks a service everyone wants funded *a little*; high V with high γ marks one that can absorb serious resources. **Specification:** γ-profile ($\alpha = 0$) stated as an assumption with α-profile sensitivity reported. **Analysis plan:** model zeros as corners while using debriefing to separate economic corners from protests (F4); report convergence diagnostics and Hessian conditioning; compute welfare by simulation via the forecasting algorithm, not a log-sum (F10).

C2. *Diagnosis.* The failure is almost certainly the α–γ **identification ridge** (§40.7), not a coding error. Both parameters generate satiation, and verified, their marginal-utility curves correlate 0.974–0.997 — so the likelihood is nearly flat along a ridge where increases in α offset decreases in γ. With both free for every service the model is effectively over-parameterized in the satiation dimension, producing a near-singular Hessian, a gradient that never cleanly vanishes, and paths that wander along the ridge without converging. *Confirmatory checks:* (i) inspect the **Hessian's condition number / smallest eigenvalue** — near-singularity confirms weak identification; (ii) re-run from **dispersed starting values** — if the optimizer lands at very different (α, γ) pairs with near-identical log-likelihoods, that is the ridge made visible; (iii) compute a **profile likelihood** over α (re-optimizing γ at each fixed α) — near-flatness is direct evidence. *Fix:* impose a **profile**. Set $\alpha_k = 0$ for all inside goods (the γ-profile — most common, comparable with prior literature, and verified to recover γ and σ accurately once α is fixed), or alternatively $\gamma_k = 1$ (α-profile). Then: state the restriction explicitly as an *assumption* rather than a result; re-estimate under the alternative profile as a **sensitivity check** and confirm that substantive conclusions (which services are valued, forecast allocations, welfare rankings) are unchanged, disclosing any that are not; and refrain from interpreting the freed satiation parameter as a deep behavioural constant, since it is a shape parameter of a functional form I selected (F8). If theory genuinely motivates good-specific flexibility, a *hybrid* (free γ for some goods, free α for others) may be defensible — but only with explicit identification argument and convergence evidence.

C3. *Response.* A ranking exercise and an allocation task answer different questions, and the difference matters for the policy application. **Ranking yields only ordinal information:** it tells me respondents prefer fish-stock recovery to coastal protection, but *not by how much*, and *not* how much of a budget either warrants. A service ranked first by a small margin and one ranked first overwhelmingly are indistinguishable, yet they imply very different funding decisions. **Allocation yields both margins.** The *extensive* margin — which services receive anything at all — is recovered through the baseline utilities V_k (verified, consumption probabilities ordered with baseline utility, 0.68 to 0.87 across services), and the *intensive* margin — how much each absorbs — through the satiation parameters γ_k (verified as separately identified, with a twenty-four-fold change in γ moving consumption sevenfold while leaving consumption probability flat). That separation is exactly the policy distinction between "which services should be funded" and "how should the budget be divided," which a ranking cannot deliver. **Allocation also imposes a real trade-off.** A ranking is costless — respondents can rank everything highly-ordered without facing scarcity — whereas a budget constraint forces genuine opportunity-cost reasoning, which is both more informative and closer to the actual decision problem, and it generates the *corner solutions* (verified, roughly one service in five left unfunded) that reveal where support genuinely stops. **Finally, allocation supports welfare analysis.** The MDCEV structure yields a utility-theoretic model from which compensating variation for policy scenarios can be computed by simulation; a ranking supports no such welfare calculation. I do acknowledge the costs — allocation is *cognitively harder*, requiring fewer services and careful piloting (Chapter 37), and it inherits the hypothetical-bias concerns of Chapter 39 with arguably greater force — and I address both in the design. But the ranking alternative would discard precisely the quantitative information the research question requires.

40.13 Spaced review prompts

+1 day. From memory: write the MDCEV utility and its marginal utility; state the KKT consumption rule; name what ψ, γ, α each govern.

+1 week. Explain why linear utility precludes multiple discreteness; derive the location normalization; explain the α–γ ridge and its resolution.

+1 month. For your ecosystem-services study, specify the full model, estimate under both profiles, report normalized baselines and satiation parameters with sensitivity, and interpret the two margins separately. Close with the spine sentence about what the satiation structure identifies.

40.14 How this chapter connects to the rest of the book

- **Chapter 3** supplies the KKT machinery that *is* this model's core; **Chapter 22** supplies the Gumbel/logit structure that MDCEV nests as the $M = 1$ case; **Chapter 34** supplies the budget constraint and the shadow value λ interpreted as the marginal utility of income; **Chapter 14** supplies the corner-solution/censoring perspective on zeros.
- **Chapter 30** supplies the untestable-restriction logic behind the profile choice of §40.7; **Chapter 38** supplies the convergence and identification diagnostics (Hessian conditioning, multiple starts) that the α–γ ridge demands; **Chapter 37** supplies the cognitive-burden constraint governing how many services can be included; **Chapter 26** supplies the protest-versus-genuine-zero distinction that §40.3 requires; **Chapter 39** supplies the hypothetical-bias concerns that allocation tasks inherit.
- Downstream: **Chapter 41** extends MDCEV with heterogeneity and computes welfare; **Chapter 42** treats forecasting and policy simulation; **Chapters 45–46 (CBA)** consume the resulting allocations and welfare measures for your ecosystem-services appraisal.

This chapter is where your ecosystem-services proposal acquires a model capable of representing what respondents actually do — fund several services, in varying amounts, leaving some at zero.

40.15 Research translation task

For your **ecosystem-services allocation study**, produce a three-page modelling note:

1. **Justify MDCEV.** State why the response format involves multiple discreteness and why MNL and continuous demand systems are both inadequate.
2. **Specify the model.** Write the utility function, list the goods and the outside good, state the profile imposed ($\alpha = 0$ or $\gamma = 1$) as an explicit assumption, and give the likelihood.
3. **Identification.** Impose $V_{\text{out}} = 0$; report baselines as differences; report the profile sensitivity analysis and convergence diagnostics (multiple starts, Hessian conditioning).
4. **Interpret both margins.** Report V_k (which services) and γ_k (how much each absorbs) separately, with the policy reading of each.
5. **Zeros.** Report the incidence of corner solutions and the debriefing evidence distinguishing economic corners from protests.
6. **Close** with the spine sentence: *"My MDCEV identifies baseline utilities relative to the outside good ($V_{out} = 0$) and satiation parameters under an imposed [γ-profile], recovering the extensive margin [which services] and the intensive margin [how much]; conclusions are unchanged under the [α-profile] alternative."*

Carry this into Chapter 41 (heterogeneity and welfare in MDCEV) and Chapters 45–46 (CBA of the ecosystem-services portfolio).

40.16 Bibliography

Core. - Bhat (2005), "A Multiple Discrete-Continuous Extreme Value Model: Formulation and Application to Discretionary Time-Use Decisions," *Transportation Research Part B* — the original MDCEV formulation (§40.2–§40.5). *(Verify volume/pages.)* - Bhat (2008), "The Multiple Discrete-Continuous Extreme Value (MDCEV) Model: Role of Utility Function Parameters, Identification Considerations, and Model Extensions," *Transportation Research Part B* — the definitive treatment of ψ, γ, α roles and the identification problems of §40.6–§40.7. *(Verify volume/pages.)* - Pinjari & Bhat (2011), "Computational Considerations in Forecasting with MDCEV Models" — the forecasting algorithm of §40.4. *(Verify citation details.)*

Applied. - von Haefen & Phaneuf on Kuhn–Tucker demand models in environmental and recreation valuation (§40.3, §40.8). *(Verify citation.)* - Kuriyama, Hanemann & Hilger, or Phaneuf, Kling & Herriges on corner-solution recreation demand — the environmental-economics lineage of these models. *(Verify citations.)* - Applications of MDCEV to environmental budget allocation and ecosystem services (§40.8). *(Verify — search recent literature for allocation-format stated-preference applications.)*

Advanced. - Bhat, Castro & Khan on flexible MDCEV extensions and alternative satiation specifications (§40.7). *(Verify citation.)* - Wales & Woodland (1983), "Estimation of Consumer Demand Systems with Binding Non-Negativity Constraints," *Journal of Econometrics* — the foundational Kuhn–Tucker demand framework preceding MDCEV. *(Verify volume/pages.)* - Hanemann (1984) on discrete/continuous demand — the classical antecedent (§40.0). *(Verify citation.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Bhat / Pinjari–Bhat / Wales–Woodland citations before using them, per the book's no-unverified-citation rule. The applied ecosystem-services MDCEV literature in particular should be searched fresh, as it is developing.

End of Chapter 40. Next: Chapter 41 — Heterogeneity and Welfare in MDCEV, which extends this framework with random parameters (the mixed MDCEV), addresses the substantially harder problem of computing compensating variation when welfare has no closed form, and develops the simulation-based policy forecasting your ecosystem-services appraisal will require. Tell me when to write it, or request a different chapter.

Chapter 41 — Heterogeneity and Welfare in MDCEV

Part X: Multiple Discreteness

> **The spine question, applied to this chapter.** Chapter 40 built a model that represents what respondents actually do — fund several ecosystem services, in varying amounts, leaving some at zero. But a model that *describes* allocations is not yet a model that *values policies*. To feed your cost–benefit analysis you need the compensating variation of a policy change, and here MDCEV loses the convenience that made logit welfare easy: **there is no log-sum formula**. Welfare must be computed by *simulation and root-finding*, one draw at a time — which introduces a set of implementation decisions that determine whether your benefit estimate is a policy effect or a simulation artifact. The spine question: *is my welfare estimate measuring the policy, or measuring my draws?*

41.0 Conceptual motivation: welfare without a closed form

In Chapter 34 you learned that discrete-choice welfare has an elegant solution: the Small–Rosen log-sum. Compute the expected maximum utility before and after a policy, difference them, divide by the marginal utility of income, and you have the expected compensating variation. One formula, closed form, instant.

None of that survives in MDCEV, and it is worth being precise about why. The log-sum works because in a standard logit the consumer picks *one* alternative and the Gumbel distribution has the remarkable property that the expected maximum of Gumbel-distributed utilities has the closed form $\ln \sum_j e^{V_j}$. In MDCEV the consumer solves a *constrained optimization* — a KKT problem whose solution is the entire allocation vector, including which goods sit at corners. The indirect utility is the value function of that optimization, and it has **no analytical expression**: which goods are consumed depends on the realized ε's in a way that cannot be integrated in closed form. There is no MDCEV log-sum, and attempts to substitute one are simply wrong.

So welfare must be computed the hard way: for each simulation draw, solve the allocation problem *twice* — once under baseline conditions, once under the policy — and then find, numerically, the income adjustment that makes the respondent indifferent. Verified below, a single draw's CV emerges as the root of a utility-equality condition, and the mean across draws is the welfare estimate.

This procedure is straightforward in principle and treacherous in practice, and the chapter's central result is why. The ε draws represent a *particular individual's unobserved tastes*. Those tastes are the **same person** in the baseline and policy scenarios — so the same draws must be used in both. Using independent draws does not merely add noise; verified below, it **biases the welfare estimate upward by 68%** (€44.82 against €26.69) *while also* inflating its variance 2.8-fold. The bias arises because CV is a *nonlinear* function of the utility difference, so the extra spread introduced by independent draws does not average out — it transforms

into a systematically wrong mean. **Common random numbers is not an efficiency trick in MDCEV welfare analysis; it is a correctness requirement.**

Two further themes complete the chapter. **Heterogeneity**: with random parameters (the mixed MDCEV), preferences vary across people, and — verified — the resulting CV distribution is severely skewed, with 12.7% of respondents gaining essentially nothing while the top quarter gain more than €46. Reporting only a mean conceals exactly the distributional information a policy appraisal of ecosystem services needs. And **simulation discipline**: welfare here is a simulated quantity with its own standard error (verified declining from 4.08 to 0.43 as draws rise from 50 to 4,000), so it demands the same stability checks Chapter 38 imposed on the likelihood.

41.1 Prerequisites

From **Chapter 40**: the MDCEV utility function, KKT conditions, the forecasting algorithm, and the identification restrictions. From **Chapter 34**: compensating variation defined via the expenditure function, and the log-sum formula this chapter cannot use. From **Chapter 38**: simulation, draws, Jensen's inequality, and convergence diagnostics. From **Chapter 9**: Monte Carlo methods and variance reduction. From **Chapter 7**: nonlinear transformations of random variables. From **Chapter 3**: the value function of a constrained optimization.

New notation. $U^*(\mathbf{p}, \mathbf{q}, E; \varepsilon)$ (the indirect utility — the value function of the allocation problem given prices $\mathbf{p}$, quality $\mathbf{q}$, budget E, and taste draw ε), CV_r (compensating variation for draw r), CRN (common random numbers).

41.2 The mixed MDCEV: adding preference heterogeneity

Definition 41.1 (Mixed MDCEV). In the baseline MDCEV of Chapter 40, heterogeneity enters only through the i.i.d. Gumbel errors ε_k — every respondent shares the same β, γ, and α. The **mixed MDCEV** allows *systematic* preference variation by making parameters random across individuals:

$$\psi_{ik} = \exp(\beta_i' \mathbf{z}_k + \varepsilon_{ik}), \qquad \beta_i \sim f(\beta \mid \theta),$$

and optionally random satiation parameters γ_{ik} (usually specified as log-normal to enforce positivity).

The likelihood. As in Chapter 38, the individual likelihood becomes an integral over the mixing distribution, evaluated by simulation:

$$L_i(\theta) = \int P(\mathbf{x}_i \mid \beta)\, f(\beta \mid \theta)\, d\beta \;\approx\; \frac{1}{R} \sum_{r=1}^{R} P(\mathbf{x}_i \mid \beta^{(r)}),$$

where $P(\mathbf{x}_i \mid \beta)$ is the closed-form MDCEV probability of Chapter 40. Everything Chapter 38 established carries over: the log of a simulated average is downward-biased by Jensen's inequality; scrambled Halton draws substantially outperform pseudo-random ones; and spread parameters are harder to identify than means.

The compounding difficulty. Mixed MDCEV is *harder* than mixed logit for a specific reason: the MDCEV probability is already more complex, and heterogeneity in γ interacts with the α–γ identification ridge of Chapter 40 §40.7. If α and γ are only weakly separately identified when *fixed*, estimating a *distribution* of γ compounds the problem. **Practical guidance:** impose the profile restriction ($\alpha = 0$), introduce heterogeneity *selectively* — typically in the baseline utility parameters β, which are best identified — and treat random satiation parameters as an extension requiring explicit justification and convergence evidence.

41.3 Why there is no log-sum

It is worth stating the obstruction precisely, because the temptation to reach for a familiar formula is strong.

In logit. The consumer selects $\arg\max_j (V_j + \varepsilon_j)$. Because the ε_j are i.i.d. Gumbel, the *expected maximum* has the closed form $\mathbb{E}[\max_j (V_j + \varepsilon_j)] = \ln \sum_j e^{V_j} + \text{const}$. Welfare follows immediately as the scaled difference of log-sums (Chapter 34).

In MDCEV. The consumer solves

$$U^*(\mathbf{p}, \mathbf{q}, E; \varepsilon) = \max_{\mathbf{x} \geq 0} \; U(\mathbf{x}; \varepsilon) \quad \text{s.t.} \quad \sum_k p_k x_k = E,$$

a constrained optimization whose solution involves (i) determining *which* goods are at corners — itself a function of all the ε's jointly through the shadow price λ — and (ii) solving for continuous quantities on the interior set. The value function U^* is a *piecewise* object whose form changes with the consumed set, and $\mathbb{E}_\varepsilon[U^*]$ admits no analytical expression. The Gumbel magic that produced the log-sum applies to a *maximum over alternatives*, not to the *value of a constrained program*.

Consequence. Welfare must be simulated. And because we need CV in *money*, not a utility difference, we cannot simply average utility differences and divide by a marginal utility of income — the marginal utility of income λ is *endogenous* in MDCEV (it depends on the allocation, Chapter 40 §40.3), so there is no constant divisor. The CV must be defined and computed as an income adjustment.

> **Active recall 41.A.** Why does the log-sum formula not extend to MDCEV, and why can't we simply divide an average utility difference by the marginal utility of income? *(Answer: the log-sum works in logit because the choice is a* maximum over alternatives* and the expected maximum of i.i.d. Gumbel utilities has the closed form $\ln \sum_j e^{V_j}$. In MDCEV the outcome is the solution to a *constrained optimization* (a KKT problem) whose value function depends on *which goods are at corners* — determined jointly by all the ε's through the endogenous shadow price λ — making the indirect utility a piecewise object with no analytical expectation. The Gumbel result simply does not apply to the value of a constrained program. Nor can we average utility differences and divide by a marginal utility of income, because in MDCEV λ (the shadow value of the budget, which *is* the marginal utility of income) is **endogenous**: it depends on the allocation and therefore differs between the baseline and policy scenarios and across draws. There is no constant divisor to convert utils to money. Welfare must therefore be defined as an *income adjustment* and computed numerically, draw by draw.)*

41.4 Compensating variation by simulation and root-finding

Definition 41.2 (MDCEV compensating variation). For a policy changing prices and/or quality from $(\mathbf{p}^0, \mathbf{q}^0)$ to $(\mathbf{p}^1, \mathbf{q}^1)$, the compensating variation for a given taste draw ε is the income adjustment CV satisfying

$$U^*(\mathbf{p}^1, \mathbf{q}^1, E - \text{CV}; \varepsilon) \;=\; U^*(\mathbf{p}^0, \mathbf{q}^0, E; \varepsilon).$$

This is exactly Chapter 34's definition — the money that, taken away *after* the improvement, restores the original utility — with the *same* ε on both sides, since it is the same individual.

The algorithm. For each draw $r = 1, \ldots, R$:

1. Draw $\varepsilon^{(r)}$ (and $\beta^{(r)}$ if the model is mixed).
2. Solve the baseline allocation using the forecasting algorithm (Chapter 40 §40.4) and evaluate $U_r^{*,0} = U^*(\mathbf{p}^0, \mathbf{q}^0, E; \varepsilon^{(r)})$.
3. Define $g(\text{cv}) = U^*(\mathbf{p}^1, \mathbf{q}^1, E - \text{cv}; \varepsilon^{(r)}) - U_r^{*,0}$, which requires *re-solving the allocation problem* at each trial value of cv.

4. Find the root CV_r of g by a bracketing method (Brent's method is reliable, since g is monotonically decreasing in cv).
5. Aggregate: $\widehat{\mathrm{CV}} = \frac{1}{R}\sum_r \mathrm{CV}_r$, and retain the full distribution $\{\mathrm{CV}_r\}$.

Note the computational structure: an *inner* allocation solve nested inside an *outer* root-find, repeated over draws — considerably more expensive than a log-sum evaluation, which is why efficient implementation and variance reduction matter.

Worked Example 41.1 (CV by root-finding, verified). Five ecosystem services, budget €100, policy raising the baseline utility of service 3 by $\Delta V = +0.8$ (a substantial quality improvement making it the most attractive service). For one illustrative draw:

Scenario	Allocation across services	Outside good
Baseline	(0.76, 3.81, 79.15, 2.95, 0.00)	13.3
Policy	(0.00, 0.07, 92.86, 0.11, 0.00)	7.0

with $U^*(\text{baseline}, E) = 211.2693$, and solving $U^*(\text{policy}, E - \mathrm{CV}) = U^*(\text{baseline}, E)$ gives

$$\mathrm{CV} = €80.20.$$

(Verified.) Observe what the allocation change reveals: the improved service absorbs dramatically more of the budget (79.15 to 92.86), and the respondent *withdraws* from other services and from the outside good to fund it. **This substitution is the model's substantive content** — a policy improving one service does not simply add benefits; it *reallocates* the budget, and the welfare measure accounts for what is given up. No reduced-form approach captures that.

41.5 Common random numbers: a correctness requirement

Now the chapter's central practical result.

The principle. The draws $\varepsilon^{(r)}$ represent *one individual's* unobserved tastes. That individual is the *same person* in the baseline and policy scenarios — the policy changes the environment, not their personality. Therefore **the same draws must be used to evaluate both scenarios**. This is the method of **common random numbers (CRN)**, familiar from simulation as a variance-reduction technique; in MDCEV welfare it is more than that.

Worked Example 41.2 (CRN, verified). The same policy ($\Delta V = +0.8$), 4,000 draws, computed correctly (same ε in both scenarios) and incorrectly (independent draws):

Method	Mean CV	SD across draws	SE of mean
With common random numbers	**€26.69**	27.01	**0.427**
Without (independent draws)	**€44.82**	45.06	**1.168**

Variance ratio 2.8×; SE ratio 1.7×.

(Verified.) Two distinct failures occur without CRN:

- **Inflated variance.** The estimate is 2.8× noisier, requiring roughly three times as many draws for equal precision. This is the standard variance-reduction argument.
- **Bias — the serious problem.** The mean is **€44.82 versus €26.69, an overstatement of 68%**, and with standard errors of 1.17 and 0.43 the two means are roughly fifteen standard errors apart: this is a systematic error, not sampling noise.

Why bias, not just noise? One might expect that using independent draws merely adds symmetric noise that averages out. It does not, because **CV is a nonlinear function of the utility difference**. With CRN, the utility difference between scenarios reflects *only* the policy; many draws yield CV near zero (verified, 12.7% essentially zero) because the policy barely alters that individual's allocation. With independent draws, the utility difference additionally contains the *random difference between two unrelated taste realizations*, which is large. Passing this inflated, symmetric spread through the nonlinear CV mapping produces a systematically larger mean — Jensen's inequality operating on the welfare transformation rather than on the log-likelihood (Chapter 7, Chapter 38). **The extra randomness does not cancel; it converts into bias.**

Implementation requirements.

1. Draw ε (and β) *once* per simulated individual and *reuse* across all scenarios compared.
2. This applies to *every* pairwise comparison — baseline versus policy, and policy A versus policy B.
3. Use the same draws across *all* policies being ranked, so differences between policies are not contaminated by simulation noise.
4. Fix and *report* the seed; verify stability across seeds.
5. In a mixed MDCEV, the same $\beta^{(r)}$ *and* $\varepsilon^{(r)}$ must be held fixed.

Stop and think 41.1. A colleague's MDCEV welfare code draws fresh random numbers inside the policy-scenario function "so the scenarios are independent." Why is this wrong, and what specifically goes wrong? *(Answer: it is wrong because the ε draws are not scenario-specific noise — they represent a* particular simulated individual's unobserved tastes*, and that individual is the* same person* under baseline and policy. The policy changes the environment, not who the respondent is. Drawing fresh numbers replaces "the same person facing better conditions" with "two different people," so the computed income adjustment absorbs the random difference between two unrelated taste realizations rather than isolating the policy effect. Two things go wrong, and the second is the serious one. (i) *Variance inflation*: verified 2.8× the variance, requiring ~3× the draws for equal precision. (ii) **Bias**: verified, the mean CV was €44.82 instead of €26.69 — an overstatement of **68%**, about fifteen standard errors away, so systematic rather than sampling error. The bias arises because CV is a *nonlinear* function of the utility difference: with common draws many individuals have CV near zero (verified 12.7%) since the policy barely changes their allocation, whereas independent draws inject a large symmetric spread that the nonlinear mapping converts into a systematically larger mean (Jensen's inequality applied to the welfare transformation). So common random numbers is not an optional efficiency trick here — it is required for the estimate to be *correct*, and a benefit figure inflated by two-thirds would materially distort any CBA built on it.)*

41.6 The distribution of welfare, not just its mean

Because preferences are heterogeneous, CV varies across individuals — and in MDCEV that variation is typically severe.

Worked Example 41.3 (The CV distribution, verified). The same policy, computed with CRN across 4,000 draws:

Statistic	Value
Mean CV	**€26.69**
Median CV	**€16.40**
5th percentile	€0.00
25th percentile	€2.63
75th percentile	€46.58
95th percentile	€80.28

Statistic	Value
Share with CV ≈ 0	**12.7%**
Share with CV above the mean	**39.2%**

(Verified.) The distribution is strongly right-skewed: the mean (€26.69) exceeds the median (€16.40) by 63%, only 39.2% of individuals have CV above the mean, and **12.7% gain essentially nothing** — these are respondents whose allocations the policy barely touches, typically because they were not funding the improved service and still do not.

Why this matters for your appraisal. Three distinct implications:

- **The mean is the right aggregate, but a poor summary.** For a Kaldor–Hicks CBA the mean CV times the population is the correct benefit measure (Chapter 35) — but reporting it alone implies a typical gain of €26.69, when the *typical* (median) respondent gains €16.40 and an eighth gain nothing.
- **Distributional analysis requires the distribution.** A policy where 12.7% gain nothing and the top quartile gain more than €46 raises equity questions that a mean conceals. For ecosystem-services policy, where distributional incidence across coastal communities may be central, the full CV distribution is the relevant object.
- **Skewness compounds aggregation fragility.** Chapter 35 showed aggregate benefits swing with the mean-versus-median choice; here the gap is 63%, so the same warning applies with force. Report both, plus the share gaining nothing.

Reporting standard. Present the mean (for aggregation), the median and quartiles (for typical experience), the share with near-zero CV (for incidence), and the simulation standard error (for precision) — not a single number.

> **Active recall 41.B.** Why is the mean CV both the correct aggregation measure and an inadequate summary of an MDCEV welfare analysis? *(Answer: it is the* correct aggregation measure* because a Kaldor–Hicks / potential-Pareto CBA sums money-metric welfare across individuals, so total benefit is mean CV times population (Chapter 35) — the median has no such aggregation interpretation. It is an *inadequate summary* because the CV distribution under preference heterogeneity is severely right-skewed: verified, mean €26.69 against median €16.40 (a 63% gap), with only 39.2% of individuals above the mean, 12.7% gaining essentially *nothing*, and a 5th-to-95th percentile range of €0.00 to €80.28. So the mean describes almost nobody's experience: it overstates the typical gain and entirely conceals that an eighth of the population is unaffected — information central to distributional and equity analysis of ecosystem-services policy, and to political feasibility. It also compounds the aggregation fragility of Chapter 35, where mean-versus-median choice already swung aggregate benefits substantially. The reporting standard is therefore the mean *and* median *and* quartiles *and* the share with near-zero CV *and* the simulation standard error.)*

41.7 Simulation discipline: welfare as an estimated quantity

Because welfare is computed by simulation, it carries simulation error that must be quantified — exactly as for the likelihood in Chapter 38.

Worked Example 41.4 (Convergence in draws, verified).

R draws	Mean CV	SE of mean
50	€29.71	4.084
200	€23.57	1.844
1,000	€27.65	0.867

R draws	Mean CV	SE of mean
4,000	**€26.69**	**0.427**

(Verified.) The estimate wanders substantially at low R — €29.71 at 50 draws, €23.57 at 200, a range of over €6 — before settling near €26.69. The standard error declines as $1/\sqrt{R}$, from 4.08 to 0.43. **At 50 draws the welfare estimate is essentially uninformative for policy purposes**, yet nothing in the output would signal this unless the standard error is computed and reported.

The protocol.

1. **Report the simulation standard error** of the welfare estimate alongside the point estimate.
2. **Verify stability in** R: re-compute at increasing draw counts and confirm convergence.
3. **Verify stability across seeds**: different seeds should give the same answer within simulation error.
4. **Distinguish two sources of uncertainty**: *simulation* error (from finite draws — reducible by more computation) and *parameter* error (from estimated $\hat{\theta}$ — irreducible without more data). Propagate parameter uncertainty by drawing θ from its asymptotic distribution (Krinsky–Robb, Chapter 13) and recomputing welfare — giving a confidence interval reflecting *both*.
5. **Use quasi-random draws** where applicable (Chapter 38's Halton result) for variance reduction, in addition to CRN.

A caution on the two uncertainties. It is easy to report a tight simulation standard error and mistake it for the precision of the welfare estimate. It is not: with enough draws, simulation error can be driven arbitrarily small while *parameter* uncertainty — which reflects how much the *data* tell you — remains. A welfare estimate reported with only its simulation SE overstates precision, sometimes dramatically. **Report both, and lead with the parameter-inclusive interval.**

41.8 Conditional versus unconditional welfare

A final distinction that matters for how you use the estimates.

Unconditional welfare integrates over the *population* distribution of preferences: it answers "what is the expected CV for a randomly drawn member of the population?" This is the appropriate object for *aggregate* benefit estimation in a CBA, where you scale to a population.

Conditional welfare conditions on an individual's *observed choices*: having seen respondent i allocate their budget in a particular way, what does that reveal about *their* preferences, and what is *their* CV? Formally this uses the posterior distribution of β_i given the observed allocation $\mathbf{x}_i$ (Bayes' rule applied to the mixing distribution — the same logic as individual-level parameters in mixed logit). Conditional welfare is appropriate for *distributional* analysis, for identifying which types of respondents gain and lose, and for linking welfare to observed characteristics.

Practical guidance. Use unconditional welfare for the headline aggregate; use conditional welfare when the research question concerns *who* gains — which, for an ecosystem-services policy with likely uneven incidence across coastal communities, it often will. Do not mix them: the mean of conditional CVs across the sample is not generally the unconditional mean, and reporting one while describing the other is a common and consequential error.

41.9 Failure modes

F1. Applying a log-sum formula. Using MNL welfare machinery in MDCEV. *Fix:* simulation and root-finding; there is no closed form (§41.3, F1).

F2. Independent draws across scenarios. Redrawing ε for the policy scenario. *Fix:* common random numbers — verified to prevent both 68% bias and 2.8× variance inflation (§41.5).

F3. Dividing by a constant marginal utility of income. Treating λ as fixed. *Fix:* λ is endogenous in MDCEV; compute CV as an income adjustment (§41.3).

F4. Too few draws. Welfare estimates unstable and uninformative. *Fix:* verify convergence in R; report simulation SE (§41.7).

F5. Reporting only the mean. Concealing severe skew and the share gaining nothing. *Fix:* report median, quartiles, zero-share, and distribution (§41.6).

F6. Confusing simulation and parameter uncertainty. A tight simulation SE presented as the precision of the estimate. *Fix:* propagate parameter uncertainty; report both (§41.7).

F7. Mixing conditional and unconditional welfare. Averaging conditional CVs and calling it the population mean. *Fix:* state which is computed and use it consistently (§41.8).

F8. Over-parameterized heterogeneity. Random γ compounding the α–γ ridge. *Fix:* impose the profile restriction; add heterogeneity selectively with convergence evidence (§41.2; Chapter 40).

F9. Ignoring reallocation. Treating benefits as additive without accounting for budget substitution. *Fix:* the model's substitution across services is central; report allocation changes alongside CV (§41.4).

41.10 Diagnostic questions for your own papers

1. **Method.** Am I computing CV by simulation and root-finding, not by a log-sum analogue? *(§41.3–§41.4, F1, F3.)*
2. **CRN.** Are the same draws used for every scenario compared, with the seed fixed and reported? *(§41.5, F2.)*
3. **Draws.** Is convergence in R verified and the simulation SE reported? *(§41.7, F4.)*
4. **Distribution.** Do I report mean, median, quartiles, and the share with near-zero CV? *(§41.6, F5.)*
5. **Uncertainty.** Is *parameter* uncertainty propagated (Krinsky–Robb) as well as simulation error? *(§41.7, F6.)*
6. **Conditional versus unconditional.** Which am I computing, and is it the right one for the question? *(§41.8, F7.)*
7. **Heterogeneity specification.** Is random-parameter structure justified, identified, and convergent? *(§41.2, F8.)*
8. **Substitution.** Do I report how the policy *reallocates* the budget, not just the net benefit? *(§41.4, F9.)*

41.11 Exercises

(A) Comprehension.

A1. Explain why MDCEV welfare has no closed form.

A2. Define MDCEV compensating variation and describe the algorithm for computing it.

A3. What are common random numbers, and why do they matter here?

(B) Derivation / reasoning.

B1. Show why the marginal utility of income cannot serve as a constant divisor in MDCEV.

B2. Explain why omitting CRN produces bias, not merely noise.

B3. Distinguish simulation uncertainty from parameter uncertainty and explain how each is quantified.

(C) Research application / critique.

C1. Write the complete welfare-analysis protocol for your ecosystem-services policy appraisal.

C2. Your mean CV is €26.69 with a simulation SE of €0.43. A colleague reports this as "€26.69 (SE 0.43)." Critique.

C3. A referee asks why you report the full CV distribution rather than a single benefit figure. Respond.

41.12 Solutions

A1. In logit, welfare has a closed form because the choice is a *maximum over alternatives* and the expected maximum of i.i.d. Gumbel utilities equals $\ln \sum_j e^{V_j}$ — the log-sum. In MDCEV the outcome is the solution to a *constrained optimization* (the KKT problem of Chapter 40), whose indirect utility is the *value function* of that program. That value function is *piecewise*: its form depends on which goods are at corners, and the consumed set is determined jointly by all the ε's through the endogenous shadow price λ. Consequently $\mathbb{E}_\varepsilon[U^*]$ has no analytical expression — the Gumbel result governs maxima over alternatives, not values of constrained programs. Welfare must therefore be computed by simulation.

A2. The compensating variation for a taste draw ε solves

$$U^*(\mathbf{p}^1, \mathbf{q}^1, E - \mathrm{CV}; \varepsilon) = U^*(\mathbf{p}^0, \mathbf{q}^0, E; \varepsilon)$$

— the income adjustment restoring baseline utility under policy conditions, with the *same* ε on both sides (same individual). *Algorithm:* for each draw, (1) draw ε (and β if mixed); (2) solve the baseline allocation via the forecasting algorithm and evaluate baseline utility; (3) define $g(\mathrm{cv})$ as policy utility at budget $E -$ cv minus baseline utility, which requires re-solving the allocation at each trial cv; (4) find the root by a bracketing method such as Brent's, exploiting monotonicity of g; (5) aggregate across draws, retaining the full distribution. Verified example: one draw gave CV = €80.20, with the improved service absorbing 79.15 $\rightarrow$ 92.86 of the budget as the respondent withdrew from other services and the outside good.

A3. Common random numbers means using the *same* realized random draws when evaluating two or more scenarios being compared. It matters in MDCEV welfare because the ε's represent a *particular individual's* unobserved tastes, and that individual is the *same person* under baseline and policy — the policy changes the environment, not the person. Verified consequences of violating it: the variance inflates 2.8× *and*, more seriously, the mean CV is biased upward by 68% (€44.82 versus €26.69, about fifteen standard errors apart). CRN is therefore a *correctness* requirement here, not merely a variance-reduction technique.

B1. In a linear-in-cost logit, the marginal utility of income α is a *constant parameter*, so a utility difference can be converted to money by dividing by α (Chapter 34's log-sum welfare). In MDCEV, the marginal utility of income is the Lagrange multiplier λ on the budget constraint, and from the KKT solution (Chapter 40)

$$\lambda = \frac{\psi_{\text{out}} + \sum_{k \in S} \gamma_k \psi_k}{E + \sum_{k \in S} \gamma_k p_k},$$

which depends on (i) the *consumed set* S, (ii) the realized ψ's, and (iii) the *budget* E. All three differ between baseline and policy scenarios and across draws — indeed the policy changes S precisely by making a service attractive enough to fund (verified: the illustrative draw's allocation shifted from five to effectively three funded services). So λ is **endogenous**, with no single value applicable to both scenarios, and dividing a utility difference by "the" marginal utility of income is undefined. The welfare measure must instead be defined *in money from the outset* — as the income adjustment CV solving the utility-equality condition — which is what the root-finding procedure computes.

B2. One might expect independent draws to add symmetric noise that averages out, leaving the mean unaffected. It does not, because **CV is a nonlinear function of the utility difference between scenarios**.

With CRN, that difference reflects *only* the policy: for many individuals the policy barely alters their optimal allocation, so the utility difference is small and CV is near zero (verified, 12.7% of draws essentially zero). Without CRN, the difference additionally contains the *random discrepancy between two unrelated taste realizations* — a large, symmetric extra spread. Passing this inflated spread through the nonlinear CV mapping does not preserve the mean: by Jensen's inequality applied to the welfare transformation (the same mechanism as the simulated log-likelihood bias of Chapter 38, here operating on welfare rather than on the likelihood), the added dispersion converts into a systematically different mean. Verified: €44.82 versus €26.69, a 68% overstatement, with the two means about fifteen standard errors apart — unambiguously systematic rather than sampling error, alongside the 2.8× variance inflation.

B3. *Simulation uncertainty* arises from using a *finite number of draws* R to approximate an integral over the taste distribution. It is quantified by the standard error of the mean across draws, $\mathrm{sd}(\{\mathrm{CV}_r\})/\sqrt{R}$, and declines as $1/\sqrt{R}$ — verified, from 4.084 at $R = 50$ to 0.427 at $R = 4{,}000$. It is *reducible by more computation alone. Parameter uncertainty* arises because the model parameters θ are *estimated* from finite data, with sampling variance $\widehat{\mathrm{Var}}(\hat{\theta})$. It is quantified by propagating that uncertainty into welfare: draw $\theta^{(s)}$ from the estimated asymptotic distribution (Krinsky–Robb, Chapter 13), recompute the full welfare simulation at each draw, and take the resulting spread of welfare estimates. It is **not** reducible by more computation — only by more data. The critical practical point: with large R, simulation error can be driven arbitrarily small while parameter uncertainty remains, so reporting only the simulation SE can dramatically *overstate* precision. Report both, and lead with the parameter-inclusive interval.

C1. *Welfare-analysis protocol for the ecosystem-services appraisal.* **(1) Model and estimation.** Estimate the (mixed) MDCEV under the imposed profile restriction ($\alpha = 0$), with heterogeneity introduced selectively in the baseline utility parameters, reporting convergence diagnostics and Hessian conditioning (Chapter 40 §40.7, Chapter 38). **(2) Scenario definition.** Specify baseline and policy states precisely — which services improve, by how much, and whether prices/costs change — and confirm the changes are within the range of the estimated attribute levels (no extrapolation). **(3) Draws.** Generate ε (and β) draws *once* per simulated individual, using scrambled Halton where applicable, with a **fixed, reported seed**. **(4) Common random numbers.** Reuse those identical draws for *every* scenario evaluated — baseline, each policy, and every pairwise policy comparison — verified as essential to avoid a 68% upward bias and 2.8× variance inflation. **(5) CV computation.** For each draw, solve the allocation under both scenarios via the forecasting algorithm and find CV by Brent root-finding on the utility-equality condition. **(6) Convergence.** Verify stability of the welfare estimate in R and across seeds, reporting the simulation standard error (verified: at $R = 50$ the estimate is uninformative, wandering over €6). **(7) Parameter uncertainty.** Propagate $\widehat{\mathrm{Var}}(\hat{\theta})$ by Krinsky–Robb, recomputing welfare at each parameter draw, and report a confidence interval reflecting *both* simulation and parameter uncertainty, leading with the latter. **(8) Reporting.** Present mean CV (for aggregation), median and quartiles, the share with near-zero CV, and the full distribution; report the *allocation changes* showing how the policy reallocates the budget across services, not merely the net benefit. **(9) Conditional analysis.** Where incidence matters — likely, given uneven effects across coastal communities — compute conditional welfare given observed allocations and relate CV to respondent characteristics, keeping conditional and unconditional measures clearly separate. **(10) Aggregation.** Scale mean CV to the population using a justified extent of the market, with the sensitivity analysis of Chapter 35, and carry the full uncertainty into the CBA (Chapters 45–46).

C2. The report is *incomplete and misleadingly precise* in two respects. **First, the SE is the wrong uncertainty.** €0.43 is the *simulation* standard error — it measures only how accurately 4,000 draws approximate the integral over the taste distribution, and it can be made arbitrarily small simply by running more draws. It says *nothing* about how well the *data* pin down the welfare estimate. The dominant uncertainty is **parameter** uncertainty: the MDCEV parameters $\hat{\theta}$ are estimated with sampling error, and propagating that (Krinsky–Robb: draw θ from its asymptotic distribution and recompute the whole welfare simulation) will typically yield an interval *far* wider than ±0.43. Reporting €0.43 as "the" SE therefore overstates precision, potentially by an order of magnitude, and would give a CBA false confidence. **Second, a single mean conceals a severely skewed distribution.** Verified: the median is €16.40 against the mean €26.69 (a 63% gap), only 39.2% of individuals exceed the mean, and 12.7% gain essentially nothing, with the 5th–95th percentile range spanning €0.00 to €80.28. "€26.69" describes almost no one. *Correct reporting:* mean €26.69

for aggregation, with a confidence interval incorporating **both** parameter and simulation uncertainty (leading with the parameter-inclusive interval and labelling the simulation SE separately); plus median, quartiles, and the share gaining nothing; plus confirmation of convergence in R and across seeds.

C3. *Response.* A single benefit figure would answer only one of the questions the appraisal must address, and would misrepresent the evidence on the others. **(i) The mean is retained — for aggregation.** For a Kaldor–Hicks test the correct aggregate benefit is mean CV times population (Chapter 35), and that figure is reported prominently. So the referee's number is present. **(ii) But the mean describes almost nobody.** Verified, the CV distribution is severely right-skewed: mean €26.69 against median €16.40 (a 63% gap), with only 39.2% of individuals above the mean and **12.7% gaining essentially nothing**. A reader given only "€26.69" would reasonably infer a typical household gains about that much; in fact the typical household gains €16.40 and roughly one in eight gains nothing at all. **(iii) Distribution is the policy question, not a statistical nicety.** Ecosystem-services interventions have uneven incidence across coastal communities, and whether benefits concentrate among a minority or spread broadly bears directly on equity, on political feasibility, and on whether compensating losers is necessary. The Kaldor–Hicks criterion establishes only that winners *could* compensate losers; whether they *would*, and who they are, requires the distribution. **(iv) It also disciplines the uncertainty reporting.** The skew compounds the mean-versus-median aggregation fragility documented in Chapter 35, so presenting the distribution makes transparent how sensitive the headline is to that choice. I therefore report mean, median, quartiles, the near-zero share, and the full distribution — and, separately, an interval incorporating parameter as well as simulation uncertainty. This is more information than a single figure, not less rigour.

41.13 Spaced review prompts

+1 day. From memory: state why MDCEV has no log-sum; write the CV-defining equation; explain common random numbers.

+1 week. Explain why omitting CRN biases rather than merely adds noise; distinguish simulation from parameter uncertainty; explain conditional versus unconditional welfare.

+1 month. For your ecosystem-services appraisal, execute the full protocol — CRN, convergence checks, parameter-uncertainty propagation, distributional reporting — and audit that reallocation effects are reported alongside net benefits. Close with the spine sentence about whether your welfare estimate measures the policy or the draws.

41.14 How this chapter connects to the rest of the book

- **Chapter 40** supplies the model, the KKT conditions, and the forecasting algorithm that this chapter calls twice per draw; **Chapter 34** supplies the definition of compensating variation and the log-sum that MDCEV *cannot* use; **Chapter 38** supplies the simulation discipline (draws, convergence, Jensen) that reappears here operating on welfare rather than on the likelihood.
- **Chapter 7** supplies the nonlinear-transformation logic behind the CRN bias; **Chapter 13** supplies Krinsky–Robb for propagating parameter uncertainty; **Chapter 11** supplies the common-random-numbers technique itself; **Chapter 35** supplies the aggregation and mean-versus-median framework that §41.6's skew intensifies.
- Downstream: **Chapters 45–46 (CBA)** consume these welfare distributions, where both the skew and the parameter-inclusive uncertainty become first-order sensitivity parameters for your fish-restoration and ecosystem-services appraisal; **Chapter 49 (benefit transfer)** faces the question of whether MDCEV welfare estimates transfer across sites.

This chapter is where your ecosystem-services model becomes a *policy instrument* — and where the difference between a defensible benefit estimate and a simulation artifact is decided by implementation details that no

output statistic will flag for you.

41.15 Research translation task

For your **ecosystem-services appraisal**, produce a three-page welfare note:

1. **Method statement.** Explain why welfare requires simulation and root-finding, with no log-sum available and no constant marginal utility of income.
2. **Implementation.** Document the draw generation, the *common random numbers* protocol across all scenarios, the seed, and the root-finding method.
3. **Convergence.** Report welfare estimates at increasing R and across seeds, with simulation standard errors.
4. **Uncertainty.** Propagate parameter uncertainty by Krinsky–Robb and report an interval reflecting both sources, clearly labelled.
5. **Distribution.** Report mean, median, quartiles, and the share with near-zero CV; discuss distributional incidence across communities.
6. **Reallocation.** Report how the policy shifts the budget across services, not only the net benefit.
7. **Close** with the spine sentence: *"CV is computed by simulation with common random numbers across all scenarios (seed [·], R = [·], simulation SE [·]); mean CV is €[·] (parameter-inclusive 95% CI [·, ·]), median €[·], with [·]% of respondents gaining essentially nothing."*

Carry this into Chapters 45–46 (CBA), where these distributions become the benefit stream and their uncertainty the core sensitivity analysis.

41.16 Bibliography

Core. - Bhat (2008), "The Multiple Discrete-Continuous Extreme Value (MDCEV) Model," *Transportation Research Part B* — model, identification, and extensions (§41.2). *(Verify volume/pages.)* - Pinjari & Bhat (2011), "Computational Considerations in Forecasting with MDCEV Models" — the forecasting algorithm invoked twice per welfare draw (§41.4). *(Verify citation details.)* - von Haefen & Phaneuf, on welfare measurement in Kuhn–Tucker / corner-solution demand models — the environmental-economics treatment of simulation-based CV (§41.3–§41.4). *(Verify citation.)*

Applied. - von Haefen, Phaneuf & Parsons (2004), "Estimation and Welfare Analysis with Large Demand Systems," *Journal of Business and Economic Statistics* — practical welfare computation in corner-solution systems (§41.4). *(Verify volume/pages.)* - Phaneuf, Kling & Herriges on corner-solution recreation demand and welfare (§41.3). *(Verify citation.)* - Pinjari & Bhat on welfare/forecasting applications of MDCEV (§41.4). *(Verify citation.)*

Advanced. - Train, *Discrete Choice Methods with Simulation*, on conditional (individual-level) parameter distributions and simulation methods — the basis for §41.8's conditional welfare. *(Verify edition/chapter.)* - Glasserman, *Monte Carlo Methods in Financial Engineering*, or a comparable simulation text, on common random numbers and variance reduction (§41.5). *(Verify edition.)* - Herriges, Kling & Phaneuf on welfare with heterogeneous preferences in demand systems (§41.6). *(Verify citation.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Bhat / Pinjari–Bhat / von Haefen citations before using them, per the book's no-unverified-citation rule. The MDCEV welfare literature is technical and evolving; check for recent methodological updates before finalizing the protocol.

End of Chapter 41. Next: Chapter 42 — Forecasting, Policy Simulation, and Model Validation in MDCEV, which completes Part X by turning the estimated model toward prediction: forecasting allocations under

counterfactual scenarios, validating the model against holdout data, assessing predictive accuracy for corner solutions, and the specific hazards of extrapolating beyond the observed attribute range. Tell me when to write it, or request a different chapter.

Chapter 42 — Forecasting, Policy Simulation, and Model Validation in MDCEV

Part X: Multiple Discreteness

> **The spine question, applied to this chapter.** Chapters 40–41 estimated an allocation model and extracted welfare from it. This chapter asks whether the model deserves to be believed when it speaks about situations it has never seen. A structural model earns its complexity not by *fitting* observed allocations — a sample mean can do that, and verified below, does it *better* — but by answering **counterfactual** questions no descriptive summary can address. That capability is also its greatest liability, because counterfactual forecasts are unfalsifiable by construction: no data exist to check them. The spine question: *what validation evidence licenses my counterfactual forecast, and does the absence of visible error mean the forecast is trustworthy — or merely that I cannot see the error?*

42.0 Conceptual motivation: the asymmetry between fitting and forecasting

Your ecosystem-services model will ultimately be asked a question of the form: *if habitat restoration improved by this much, how would households reallocate their conservation budget, and what would that be worth?* No respondent was observed under that scenario. The model must extrapolate — and the entire justification for building a structural, utility-theoretic model rather than summarizing the data descriptively is that a structural model *can* extrapolate in a disciplined way, because its parameters describe preferences rather than outcomes.

This chapter takes that claim seriously enough to test it, and the verified results are more nuanced than the usual presentation.

First, a humbling benchmark result. When asked simply to *reproduce* the expenditure pattern of a holdout sample drawn from the same population, the estimated MDCEV achieved a mean absolute error of €0.898 — while the *training-sample means*, a predictor requiring no model at all, achieved **€0.489**. The naive benchmark won. This is not a failure of the model; it is a clarification of what models are for. Predicting the same population under the same conditions is a task descriptive statistics perform excellently, and a structural model carries estimation error that a sample mean does not. **The model's advantage is not replication.**

Second, the payoff. Under a counterfactual — a quality improvement to one service — the model predicted the resulting expenditure reallocation with a mean absolute error of €0.243, while *the naive predictors cannot answer the question at all.* A sample mean has no representation of "service 3 improves"; it is undefined

outside the observed regime. This is the entire case for structural modelling, and it is why validation must be designed around counterfactual capability rather than in-sample fit.

Third, a warning that inverts a common intuition. One expects forecast error to *grow* as the counterfactual moves further from the data. Verified, it did the opposite: mean absolute error *fell* monotonically from €0.261 to €0.010 as the policy shock grew from zero to +6.0 — because large shocks drive the allocation to **saturation** (one service absorbing everything), where true and estimated models agree trivially. The share of respondents funding at most one service rose from 6% to 79% across that range. **A small extrapolation error is therefore not evidence that the extrapolation is trustworthy** — it may only mean the system has been pushed somewhere structurally simple. This is a genuine hazard in policy simulation, and it means "the forecast looks stable under large shocks" is not the reassurance it appears to be.

The chapter builds the validation apparatus these findings demand: holdout testing on both margins, benchmarking against naive predictors, counterfactual validation, and — since point forecasts overstate precision — parameter-uncertainty propagation (verified, forecast intervals widening from 4.10 to 5.82 in width as the policy grows).

42.1 Prerequisites

From **Chapter 40**: the MDCEV model, KKT conditions, and the forecasting algorithm. From **Chapter 41**: simulation-based welfare, common random numbers, and the distinction between simulation and parameter uncertainty. From **Chapter 9**: Krinsky–Robb propagation. From **Chapter 38**: convergence diagnostics. From **Chapter 30**: identification and the limits of what data can establish. From **Chapter 9**: Monte Carlo methods.

New notation. $\hat{\mathbf{x}}$ (forecast allocation), $\mathbf{x}^{\text{obs}}$ (observed allocation), MAE (mean absolute error), *extensive margin* (whether a service is funded), *intensive margin* (how much), *holdout* (data withheld from estimation).

42.2 Forecasting: applying the model predictively

The procedure. Forecasting under a counterfactual scenario reuses the machinery of Chapters 40–41:

1. Specify the counterfactual — changed attributes $\mathbf{q}^1$, prices $\mathbf{p}^1$, or choice set.
2. Draw ε (and β if mixed), using **common random numbers** across all scenarios compared (Chapter 41 §41.5).
3. For each draw, compute ψ_k under the counterfactual and solve the allocation via the forecasting algorithm (Chapter 40 §40.4).
4. Aggregate across draws to obtain predicted consumption probabilities (extensive margin) and expected quantities (intensive margin).
5. Propagate parameter uncertainty to obtain forecast intervals (§42.6).

Two distinct predictive quantities. MDCEV forecasts have a structure absent from single-choice models, and both margins must be reported:

- **Extensive margin** — $P(x_k > 0)$, the probability service k receives any funding. This answers *which* services command support.
- **Intensive margin** — $\mathbb{E}[x_k \mid x_k > 0]$, the expected amount *given* funding, and $\mathbb{E}[x_k]$, the unconditional expected expenditure (which combines both margins). This answers *how much*.

A model can predict one margin well and the other poorly, so reporting only aggregate expenditure conceals which part of the mechanism is working.

42.3 Holdout validation

The principle. In-sample fit is not evidence of predictive capability: a sufficiently flexible model fits the estimation data by construction. **Holdout validation** estimates on one subsample and evaluates predictions on data withheld from estimation, providing an honest assessment of out-of-sample performance.

Worked Example 42.1 (Holdout validation, verified). Estimating on $N = 2{,}000$ and predicting a separate holdout of $N = 2{,}000$ from the same population:

Service	Extensive: observed P	Extensive: predicted	Intensive: observed $\mathbb{E}[x \mid x > 0]$	Intensive: predicted
1	0.880	0.879	€17.24	€18.75
2	0.845	0.794	€24.21	€24.49
3	0.747	0.784	€23.85	€23.80
4	0.867	0.817	€22.30	€22.14
5	0.685	0.665	€19.40	€20.13

Summary	Value
MAE, extensive margin	**0.0317** (probability units)
MAE, intensive margin	**€0.544**
Mean services funded: observed	4.023
Mean services funded: predicted	3.939

(Verified.) The model reproduces both margins on withheld data with reasonable accuracy — consumption probabilities within about 3 percentage points, conditional expenditures within roughly €0.54, and the mean number of funded services to within 0.08. Note also from the underlying estimation that γ_3 was recovered as 4.76 against a true 6.0, a substantial parameter error that nonetheless produced acceptable predictions — a reminder that predictive accuracy and parameter accuracy are distinct properties, and that good forecasts do not certify correct parameters (relevant to any structural interpretation of satiation, Chapter 40 §40.7).

Design considerations. Split *by respondent*, not by observation, so that all of an individual's choices fall on the same side (their unobserved tastes are shared — the panel logic of Chapter 38). For small samples use k-fold cross-validation. Ideally, hold out a *different context* rather than a random subsample — a different region or survey wave — since predicting a random holdout from the same population is a much weaker test than predicting a new setting.

42.4 Benchmarking: the model must earn its complexity

A validation exercise that reports only the model's error is uninformative — error must be compared to something.

Worked Example 42.2 (Naive benchmarks, verified). Mean absolute error in predicting holdout mean expenditure per service:

Predictor	MAE
Training-sample means (no model)	**€0.489**
MDCEV forecast	€0.898
Equal split of budget	€2.968

(Verified.) The MDCEV comfortably beats a naive equal-split rule, but is **beaten by the training-sample means** — a predictor requiring no estimation whatsoever.

How to read this correctly. This is the honest and instructive result, and it should not be explained away:

- **Replicating the same population is not what structural models are for.** When the holdout is drawn from the same population under the same conditions, the sample mean is an excellent and *unbiased* predictor of the mean, while the structural model must estimate many parameters, each contributing error, and then propagate them through a nonlinear allocation mechanism. The naive predictor's advantage here is real and expected.
- **Reporting only in-sample fit would have concealed this.** A model can fit well and still fail to beat a trivial benchmark out of sample.
- **The comparison identifies where the model's value actually lies** — which is the subject of the next section.

The general principle. Always benchmark against (i) a naive constant/mean predictor, (ii) a simpler model (an MNL forced onto the data, or an MDCEV without heterogeneity), and (iii) the full model. If the complex model does not beat the simple ones *on the task you care about*, the complexity is unjustified — and identifying *which* task it wins on is the substantive question.

> **Active recall 42.A.** The MDCEV was beaten by training-sample means on holdout prediction. Why is this not a reason to abandon the model, and what does it tell you about where to look for its value? *(Answer: because* replicating the same population under the same conditions* is a task descriptive statistics perform excellently — the sample mean is an unbiased, low-variance estimator of the population mean, whereas the structural model must estimate many parameters, each carrying error, and propagate them through a nonlinear allocation mechanism, accumulating error along the way. Verified: naive means achieved MAE €0.489 against the model's €0.898 (though both beat an equal-split rule at €2.968). This is expected, not a defect. The model's value lies elsewhere: in **counterfactual** prediction, where the naive predictors are *undefined* — a sample mean has no representation of "service 3's quality improves" and simply cannot answer the question. Verified, the model predicted the counterfactual reallocation with MAE €0.243 while naive benchmarks could not be computed at all. So the lesson is diagnostic rather than damning: benchmark on the task you actually need (counterfactual policy simulation), report the replication comparison honestly, and justify the model's complexity by the questions it can answer rather than by fit statistics on data it was fitted to.)*

42.5 Counterfactual forecasting: what the model is for

Worked Example 42.3 (Counterfactual prediction, verified). A policy improving service 3's baseline utility by +0.8. Comparing the estimated model's forecast against the true data-generating process:

Service	True expenditure share	Model forecast	Absolute error
1	€14.17	€14.04	0.133
2	€16.43	€16.41	0.016
3	€29.89	€30.49	0.607
4	€16.18	€15.73	0.448
5	€11.33	€11.34	0.013

MAE on counterfactual expenditure: **€0.243**. Naive predictors: **undefined**.

(Verified.) The model predicts not only that the improved service attracts more funding (€23.85 baseline conditional expenditure rising to a €29.89 share) but *where the money comes from* — the reallocation away

from services 1 and 4. **This is the substantive payoff of the structural approach**: it represents substitution across the budget, which is precisely what a policy appraisal needs and what no descriptive summary provides.

The epistemological difficulty. In this verified exercise the truth is known because the data were simulated. **In your application it will not be.** Counterfactual forecasts are, by construction, unfalsifiable at the time they are made — that is what makes them useful and what makes them dangerous. The available responses are indirect:

1. **Validate on the strongest available proxy** — a holdout *context* (different region, wave, or population) rather than a random subsample, which at least tests transferability.
2. **Validate the mechanism, not just outcomes** — check that the model reproduces observed *substitution patterns* and the distribution of the number of funded services, not merely marginal means.
3. **Restrict counterfactuals to the design range** where possible, and flag explicitly when a scenario requires extrapolation.
4. **Report forecast intervals** reflecting parameter uncertainty (§42.6), never point predictions.
5. **Conduct sensitivity analysis over specification** — particularly the α–γ profile choice (Chapter 40 §40.7), which is untestable and could affect forecasts.

42.6 Extrapolation: why small errors are not reassurance

Here the verified result contradicts the natural intuition, and the correction matters for how policy simulations are read.

Worked Example 42.4 (Non-monotone extrapolation error, verified). Forecast error as the policy shock ΔV_3 grows:

ΔV_3	MAE (€)	Max error (€)	Share funding ≤ 1 service (true / model)
0.0	0.261	0.661	0.06 / 0.06
0.4	0.262	0.662	0.06 / 0.06
0.8	0.256	0.645	0.06 / 0.06
2.0	0.190	0.464	0.09 / 0.09
4.0	0.064	0.126	0.35 / 0.36
6.0	**0.010**	**0.023**	**0.79 / 0.79**

(Verified.) Forecast error *decreases* monotonically as the counterfactual moves further from the estimation data — the opposite of the expected pattern. The final column explains why: as the shock grows, the allocation **saturates**, with the share of respondents funding at most one service rising from 6% to 79%. At saturation the allocation problem becomes trivial — nearly everyone puts nearly everything into the dominant service — so the true and estimated models agree almost perfectly regardless of parameter error.

The lesson, stated carefully. *Forecast error is not a monotone function of distance from the data.* Two consequences for practice:

- **A small forecast error under a large policy shock is not evidence of a trustworthy extrapolation.** It may indicate only that the shock has driven the system to a structurally simple corner where all models agree. Stability under extreme scenarios can be an artifact of saturation.
- **Interior forecasts can be harder than extreme ones.** The largest errors here occurred at *small and moderate* shocks (MAE 0.26 at $\Delta V = 0$–0.8), where the allocation genuinely depends on the estimated parameters — which is precisely the policy-relevant range.

Diagnostic recommendation. Report the **saturation share** (proportion of respondents at or near a corner) alongside every counterfactual forecast. If a scenario produces high saturation, state that the forecast's

apparent robustness reflects the structure of the corner rather than the precision of the model. And do not use "results are stable under large shocks" as a robustness argument without checking saturation.

> **Stop and think 42.1.** Your policy simulation shows very small forecast errors for a large intervention, and a colleague calls this evidence the model extrapolates reliably. What do you check first? *(Answer: I check the* ***saturation share*** *— the proportion of respondents pushed to a corner where they fund at most one service. Verified, forecast error* fell* monotonically from MAE €0.261 to €0.010 as the shock grew from 0 to +6.0, while the share funding at most one service rose from 6% to **79%**. The small error is therefore not evidence of reliable extrapolation: it reflects that a large shock makes one service dominant for nearly everyone, so the allocation problem becomes trivial and *any* roughly-correct model gives nearly the same answer. The apparent precision is a property of the corner, not of the model. Two further points: (i) the *largest* errors occurred at small and moderate shocks (MAE ~0.26), which is the policy-relevant interior range where the allocation genuinely depends on estimated parameters — so extreme-scenario stability tells us nothing about the scenarios we actually care about; (ii) I would additionally report *forecast intervals* from parameter uncertainty, which verified widen with policy size (CI width 4.10 → 5.82 as ΔV went 0.4 → 2.0) even as the point error shrank — an opposite and more honest signal. So: check saturation, report intervals, and resist reading small point errors as validation.)*

42.7 Forecast uncertainty

Point forecasts overstate precision. Because parameters are estimated, forecasts inherit sampling uncertainty that must be propagated.

The procedure (Krinsky–Robb for forecasts). Draw parameter vectors $\theta^{(s)}$ from the estimated asymptotic distribution $N(\hat{\theta}, \widehat{\mathrm{Var}}(\hat{\theta}))$; for each draw, run the full forecast simulation (with common random numbers held fixed across parameter draws, so the interval reflects *parameter* uncertainty rather than simulation noise); report percentiles of the resulting forecast distribution.

Worked Example 42.5 (Forecast intervals, verified). Forecast expenditure share of service 3 under quality improvements, propagating parameter uncertainty:

ΔV_3	Forecast share	95% CI	**CI width**
0.4	€24.20	[22.11, 26.21]	**4.10**
0.8	€30.59	[27.60, 33.27]	**5.66**
2.0	€52.85	[50.20, 56.03]	**5.82**

(Verified.) The intervals are substantial — roughly €4 to €6 wide on shares of €24 to €53 — and they *widen* as the policy moves further from the data, from 4.10 to 5.82. **This is the honest signal of extrapolation risk that the point-error diagnostic of §42.6 failed to provide.** Where point errors shrank misleadingly under saturation, the parameter-uncertainty interval widens as it should, because parameter error is amplified by larger counterfactuals.

Reporting standard. Report forecast *intervals*, not point predictions; distinguish simulation error (reducible by more draws) from parameter uncertainty (reducible only by more data, Chapter 41 §41.7); and lead with the parameter-inclusive interval. For your ecosystem-services appraisal, this interval — not the point forecast — is what should propagate into the CBA's sensitivity analysis.

> **Active recall 42.B.** Why do parameter-uncertainty forecast intervals provide a better extrapolation diagnostic than point forecast errors? *(Answer: because the two respond to counterfactual distance in* opposite* ways, and only one is honest. Verified, point forecast error *fell* from MAE €0.261 to €0.010 as the shock grew, because large shocks saturate the allocation (share funding

≤ 1 service rising 6% $\rightarrow$ 79%) so all models agree trivially at the corner — making small point errors a misleading signal of reliability. By contrast, parameter-uncertainty intervals *widened* with policy size, from CI width 4.10 at $\Delta V = 0.4$ to 5.82 at $\Delta V = 2.0$, correctly reflecting that estimation error is amplified as the forecast moves further from the data supporting it. The interval is also the quantity a decision-maker needs: a forecast share of €30.59 with a 95% CI of [27.60, 33.27] communicates what is actually known, whereas "€30.59" implies a precision the data do not support. Practical rule: report intervals rather than points, propagate parameter uncertainty by Krinsky–Robb (holding common random numbers fixed so the interval isolates parameter rather than simulation error), and use interval *width* — not point error — as the extrapolation-risk diagnostic.)*

42.8 Failure modes

F1. In-sample fit as validation. Reporting log-likelihood or pseudo-R^2 as evidence of predictive capability. *Fix:* holdout or cross-validation (§42.3).

F2. No benchmark. Reporting model error without naive comparators. *Fix:* benchmark against constant/mean predictors and simpler models (§42.4).

F3. Defending the model on the wrong task. Claiming superiority at replication, where naive predictors may win. *Fix:* benchmark on counterfactual capability, and report replication honestly (§42.4–§42.5).

F4. Reading small extrapolation error as reliability. Ignoring saturation. *Fix:* report saturation shares; use interval width as the diagnostic (§42.6).

F5. Point forecasts. Reporting predictions without intervals. *Fix:* propagate parameter uncertainty (§42.7).

F6. Conflating simulation and parameter uncertainty. A tight simulation SE presented as forecast precision. *Fix:* report both, lead with parameter-inclusive (§42.7; Chapter 41).

F7. Splitting by observation, not respondent. Leaking an individual's tastes across train and holdout. *Fix:* split by respondent (§42.3).

F8. Reporting only aggregate expenditure. Concealing which margin fails. *Fix:* report extensive and intensive margins separately (§42.2–§42.3).

F9. Assuming good prediction certifies parameters. Verified, γ_3 was badly estimated yet predictions were acceptable. *Fix:* do not infer structural parameter validity from predictive fit (§42.3).

F10. Ignoring specification sensitivity in forecasts. Forecasting under one untestable profile only. *Fix:* re-forecast under the alternative α–γ profile (§42.5; Chapter 40).

42.9 Diagnostic questions for your own papers

1. **Holdout.** Is validation out-of-sample, split by respondent, ideally on a different context? *(§42.3, F1, F7.)*
2. **Both margins.** Are extensive and intensive accuracy reported separately? *(§42.2–§42.3, F8.)*
3. **Benchmarks.** Is model error compared to naive and simpler predictors, honestly reported? *(§42.4, F2–F3.)*
4. **Counterfactual case.** Have I articulated the questions the model answers that benchmarks cannot? *(§42.5.)*

5. **Saturation.** Is the saturation share reported for each counterfactual, with small errors interpreted accordingly? *(§42.6, F4.)*
6. **Intervals.** Are forecasts reported with parameter-uncertainty intervals? *(§42.7, F5–F6.)*
7. **Parameter caution.** Am I avoiding the inference that good prediction validates structural parameters? *(§42.3, F9.)*
8. **Specification sensitivity.** Are forecasts re-run under the alternative profile restriction? *(§42.5, F10; Chapter 40.)*

42.10 Exercises

(A) Comprehension.

A1. Distinguish extensive- and intensive-margin prediction and explain why both must be reported.

A2. Why is in-sample fit insufficient evidence of predictive capability?

A3. Why can naive predictors beat a structural model on holdout replication?

(B) Derivation / reasoning.

B1. Explain why forecast error can *decrease* with counterfactual distance, and what diagnostic to use instead.

B2. Explain how to propagate parameter uncertainty into a forecast interval and why CRN must be held fixed while doing so.

B3. Explain why good predictive accuracy does not validate structural parameter estimates.

(C) Research application / critique.

C1. Design the complete validation strategy for your ecosystem-services MDCEV.

C2. Your counterfactual forecast shows tiny errors under a large policy shock. Write the correct interpretation.

C3. A referee says your model "doesn't fit better than a sample mean, so why use it?" Respond.

42.11 Solutions

A1. The **extensive margin** is $P(x_k > 0)$ — whether a service receives any funding — answering *which* services command support. The **intensive margin** is $\mathbb{E}[x_k \mid x_k > 0]$ — how much, given funding — answering *how much* each absorbs. Both must be reported because they are driven by different parameters (ψ via V governs the extensive margin; γ governs the intensive margin — Chapter 40 §40.4) and a model can succeed at one while failing at the other. Verified, the model achieved MAE 0.0317 in probability units on the extensive margin and €0.544 on the intensive margin — separate diagnostics. Reporting only unconditional expected expenditure (which blends both) would conceal which part of the mechanism is misfitting, and therefore which parameters to distrust.

A2. In-sample fit measures how well a model reproduces the data used to *estimate* it, and estimation deliberately chooses parameters to maximize that agreement. A sufficiently flexible model can fit in-sample arbitrarily well without capturing any generalizable structure (overfitting), so in-sample statistics — log-likelihood, pseudo-R^2, information criteria — provide weak evidence of predictive capability. **Holdout validation** (or cross-validation) evaluates predictions on data withheld from estimation, giving an honest out-of-sample assessment. Critically, the holdout must be split *by respondent*, not observation, because an individual's choices share their unobserved tastes; splitting within a respondent leaks information across the divide and inflates apparent accuracy.

A3. Because *replicating the same population under the same conditions* is exactly the task a sample mean is built for: it is an unbiased, minimum-variance estimator of the population mean, requiring no functional-form assumptions and introducing no parameter estimation error. A structural model must estimate many parameters — each with sampling error — and then propagate them through a *nonlinear* allocation mechanism that can amplify those errors. Verified: naive training-sample means achieved MAE €0.489 against the MDCEV's €0.898 on holdout expenditure (both beating an equal-split rule at €2.968). This is expected rather than damning. The structural model's justification is not replication but **counterfactual** prediction, where naive predictors are *undefined* — verified, the model forecast a quality-improvement reallocation with MAE €0.243 while sample means cannot represent the scenario at all.

B1. Forecast error depends on how much the predicted outcome *varies with the parameters* in the region being forecast, which is not monotone in distance from the data. In MDCEV, a large policy shock makes one service dominant for most respondents, driving the allocation to **saturation** — nearly everyone puts nearly everything into that service, so the allocation becomes insensitive to the precise parameter values. True and estimated models then agree almost perfectly. Verified: as ΔV_3 rose from 0 to +6.0, MAE *fell* from €0.261 to €0.010 while the share funding at most one service rose from 6% to **79%**, and the largest errors occurred at *small* shocks (the policy-relevant interior), where allocations genuinely depend on estimated parameters. *Diagnostics to use instead:* (i) report the **saturation share** for every counterfactual, and treat small errors under high saturation as uninformative about model quality; (ii) use **parameter-uncertainty interval width**, which behaves correctly — verified widening from 4.10 to 5.82 as ΔV rose from 0.4 to 2.0, precisely because estimation error is amplified by larger extrapolations.

B2. *Procedure.* Draw S parameter vectors $\theta^{(s)} \sim N(\hat{\theta}, \widehat{\mathrm{Var}}(\hat{\theta}))$ from the estimator's asymptotic distribution (Krinsky–Robb, Chapter 13). For each s, run the *entire* forecast simulation — drawing ε, solving allocations via the forecasting algorithm, aggregating — to obtain a forecast $\hat{\mathbf{x}}^{(s)}$. The percentiles of $\{\hat{\mathbf{x}}^{(s)}\}$ give the forecast interval (verified: 95% CIs of [22.11, 26.21], [27.60, 33.27], [50.20, 56.03] for successively larger policies). *Why CRN must be held fixed across parameter draws:* the objective is to isolate uncertainty arising from *not knowing the parameters*. If fresh ε draws were taken for each $\theta^{(s)}$, the resulting spread would blend parameter uncertainty with *simulation* noise, inflating the interval and confounding two conceptually distinct quantities — one reducible only by more data, the other by more computation (Chapter 41 §41.7). Holding the ε draws fixed means every forecast difference across s is attributable to θ, which is exactly what the interval should represent. Report the parameter-inclusive interval as the headline, with simulation error reported separately.

B3. Predictive accuracy and parameter accuracy are distinct properties because different parameter configurations can generate similar predictions — a manifestation of weak identification (Chapters 30, 40). Verified directly in this chapter's estimation: γ_3 was recovered as **4.76 against a true 6.0** — a 21% error — yet holdout predictions remained acceptable (extensive MAE 0.0317, intensive MAE €0.544). The satiation parameters sit on the near-flat α–γ ridge (Chapter 40 §40.7), so substantially different values yield nearly the same fitted allocations, and prediction cannot discriminate among them. Consequences: (i) do **not** cite predictive performance as evidence that structural parameters are correctly estimated; (ii) be correspondingly cautious in giving satiation parameters behavioural interpretation; (iii) where a *counterfactual* depends sensitively on a weakly-identified parameter, the forecast inherits that weakness even if in-sample and holdout prediction look fine — which is why forecast intervals must propagate parameter uncertainty rather than relying on demonstrated predictive fit.

C1. *Complete validation strategy.* **(1) Holdout design.** Split *by respondent* (never by observation, to avoid leaking shared tastes), reserving a holdout sample; where feasible, hold out a *different context* — a distinct coastal region or survey wave — since predicting a fresh context is a far stronger test of transferability than a random subsample from the same population. Use k-fold cross-validation if the sample is small. **(2) Both margins.** Report extensive-margin accuracy (P(funded) per service) and intensive-margin accuracy ($\mathbb{E}[x \mid x > 0]$) separately, plus the distribution of the *number* of services funded — verified as a sensitive aggregate check (observed 4.023 versus predicted 3.939). **(3) Benchmarks.** Compare against (i) training-sample means, (ii) an equal-split rule, and (iii) a simpler model (MDCEV without heterogeneity), reporting results *honestly* — including that naive means may win on replication (verified €0.489 versus €0.898), and explaining why that is expected and not disqualifying. **(4) Counterfactual capability.** State explicitly the policy

questions the model answers that benchmarks cannot represent, since this is the actual justification for the structural approach. **(5) Mechanism validation.** Check that the model reproduces observed *substitution patterns* — which services trade off against which — not merely marginal means, since substitution is what drives the welfare results of Chapter 41. **(6) Saturation reporting.** For every counterfactual, report the share of respondents at or near a corner, and interpret small forecast errors under high saturation as uninformative rather than reassuring. **(7) Forecast intervals.** Propagate parameter uncertainty by Krinsky–Robb with common random numbers held fixed, reporting intervals rather than point predictions and using interval *width* as the extrapolation-risk diagnostic. **(8) Specification sensitivity.** Re-run all forecasts under the alternative α–γ profile (Chapter 40 §40.7), confirming that policy conclusions are robust to a restriction that is untestable. **(9) Range disclosure.** Report the attribute range spanned by the estimation design and flag any counterfactual requiring extrapolation beyond it. **(10) Honest limits.** State plainly that counterfactual forecasts are unfalsifiable at the time of writing, and present the validation evidence as *indirect* support rather than verification.

C2. *Correct interpretation:* small forecast errors under a large shock are **not** evidence of reliable extrapolation, and I would explicitly resist that reading. The first check is the **saturation share** — the proportion of respondents driven to a corner where they fund at most one service. Verified in the analogous exercise, MAE *fell* from €0.261 to €0.010 as the shock grew from 0 to +6.0, precisely because the saturation share rose from 6% to **79%**: a large shock makes one service dominant for nearly everyone, rendering the allocation problem trivial so that *any* approximately-correct model reproduces it. The apparent accuracy is a property of the corner, not of the model's extrapolative validity. Two further observations complete the interpretation. First, the *largest* errors in that exercise occurred at *small and moderate* shocks (MAE ~0.26) — the policy-relevant interior range where allocations genuinely depend on estimated parameters — so extreme-scenario stability provides no reassurance about the scenarios of actual interest. Second, the *parameter-uncertainty interval* behaves correctly where the point error does not: verified widening from 4.10 to 5.82 as the policy grew, honestly signalling increasing extrapolation risk. *So the report should state:* the forecast point error is small but the saturation share is [X]%, indicating the result reflects corner structure rather than model precision; the parameter-inclusive forecast interval is $[\cdot, \cdot]$; and conclusions in the policy-relevant interior range rest on the moderate-shock forecasts, where errors and intervals are both larger.

C3. *Response.* The referee's observation is correct as stated, and I report it in the paper — but it identifies the wrong criterion for the model's purpose. **On the fact:** yes, verified, naive training-sample means achieved lower holdout error (€0.489) than the MDCEV (€0.898), though both substantially beat an equal-split rule (€2.968). I disclose this rather than obscure it. **On why it is expected:** reproducing the *same* population under the *same* conditions is the task a sample mean is optimally designed for — it is unbiased, low-variance, and free of both functional-form assumptions and estimation error, whereas a structural model estimates many parameters and propagates their errors through a nonlinear allocation mechanism. A structural model *should not* be expected to win at replication. **On what the model is for:** the research question is not "what did respondents allocate?" but "how would allocations and welfare change under a policy that improves habitat restoration?" — and for that question the sample mean is not merely worse, it is **undefined**: it contains no representation of a change in service quality and cannot be evaluated at the counterfactual. Verified, the structural model forecast a quality-improvement reallocation with MAE €0.243, and — critically — predicted *where the reallocated budget comes from*, the substitution pattern that drives the welfare measure of Chapter 41 and the benefit stream of the CBA. No descriptive predictor supplies that. **On what I therefore claim:** I justify the model by its counterfactual and welfare capabilities, report the replication benchmark honestly as a limitation on what predictive fit establishes, validate the mechanism (substitution patterns, distribution of number of services funded) rather than only marginal means, and report forecast intervals with saturation diagnostics so that the extrapolation's reliability is not overstated. The complexity is justified by the questions answered, not by fit statistics on data the model was fitted to.

42.12 Spaced review prompts

+1 day. From memory: distinguish extensive and intensive margins; state why holdout must split by respondent; explain why naive means can beat a structural model.

+1 week. Explain the saturation phenomenon and why small extrapolation errors mislead; describe parameter-uncertainty propagation and why CRN is held fixed.

+1 month. For your ecosystem-services model, execute the full validation strategy — holdout on both margins, benchmarks, counterfactual case, saturation shares, forecast intervals, profile sensitivity — and audit that no claim rests on in-sample fit. Close with the spine sentence about what licenses your counterfactual forecast.

42.13 How this chapter connects to the rest of the book

- **Chapter 40** supplies the forecasting algorithm applied here predictively, and the α–γ ridge that §42.3 shows can leave parameters badly estimated without visibly harming prediction; **Chapter 41** supplies common random numbers (held fixed in §42.7's interval construction) and the simulation-versus-parameter uncertainty distinction; **Chapter 13** supplies Krinsky–Robb.
- **Chapter 30** supplies the identification perspective behind §42.3's warning that prediction does not certify parameters; **Chapter 38** supplies the convergence discipline; **Chapter 9** supplies the general logic that absence of detected error is not evidence of correctness — which §42.6's saturation result instantiates in forecasting form.
- Downstream: **Chapters 45–46 (CBA)** consume these forecasts and their intervals for the ecosystem-services and fish-restoration appraisals, where the forecast interval — not the point prediction — drives the sensitivity analysis; **Chapter 49 (benefit transfer)** confronts the transferability question that §42.3's "different context" holdout begins to address.

This chapter completes Part X: your allocation model is now estimated (Chapter 40), capable of welfare analysis (Chapter 41), and validated with an honest account of what its forecasts can and cannot support.

42.14 Research translation task

For your **ecosystem-services MDCEV**, produce a three-page validation and forecasting note:

1. **Holdout design.** Document the respondent-level split (ideally a held-out context), and report extensive- and intensive-margin accuracy separately, plus the distribution of services funded.
2. **Benchmarks.** Report model error against naive means, equal-split, and a simpler model — honestly, including cases where naive predictors win, with the explanation.
3. **Counterfactual justification.** State the policy questions the model answers that benchmarks cannot, and report counterfactual forecasts of both allocation and welfare.
4. **Saturation.** Report the saturation share for every scenario and interpret small errors accordingly.
5. **Intervals.** Report parameter-uncertainty forecast intervals (Krinsky–Robb, CRN fixed), using width as the extrapolation-risk diagnostic.
6. **Sensitivity.** Re-run forecasts under the alternative α–γ profile and report robustness.
7. **Close** with the spine sentence: *"Out of sample the model predicts the extensive margin to [·] and the intensive margin to €[·], beating [benchmarks] on counterfactual tasks that naive predictors cannot represent; counterfactual forecasts are reported as intervals ([·, ·]) with saturation share [·]%, and are robust to the [profile] restriction."*

Carry this into Chapters 45–46, where these forecast intervals become the CBA's benefit stream and its sensitivity range.

42.15 Bibliography

Core. - Pinjari & Bhat (2011), "Computational Considerations in Forecasting with MDCEV Models" — the forecasting algorithm and its predictive application (§42.2). *(Verify citation details.)* - Bhat (2008), "The Multiple Discrete-Continuous Extreme Value (MDCEV) Model," *Transportation Research Part B* — model structure and identification underlying §42.3's parameter-versus-prediction distinction. *(Verify volume/pages.)* - Train, *Discrete Choice Methods with Simulation*, on forecasting, simulation, and prediction in choice models. *(Verify edition/chapter.)*

Applied. - Hastie, Tibshirani & Friedman, *The Elements of Statistical Learning*, on cross-validation, holdout design, and the overfitting logic of §42.3. *(Verify edition.)* - von Haefen & Phaneuf on prediction and welfare in Kuhn–Tucker demand systems (§42.5). *(Verify citation.)* - Literature on external validity and transferability of stated-preference models — relevant to §42.5's "different context" holdout recommendation. *(Verify — connects to Chapter 49.)*

Advanced. - Chiou & Walker (2007), "Masking Identification of Discrete Choice Models Under Simulation Methods," *Journal of Econometrics* — how simulation can conceal non-identification, directly supporting §42.3's warning (§42.3, F9). *(Verify volume/pages.)* - Literature on model validation for structural models and the unfalsifiability of counterfactuals (§42.5). *(Verify citations.)* - Keane & Wolpin, or comparable, on validating structural models against holdout experiments. *(Verify citation.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Pinjari–Bhat / Bhat / Chiou–Walker citations before using them, per the book's no-unverified-citation rule.

End of Chapter 42, and of Part X. Next: Part XI — Cost–Benefit Analysis. Chapter 43 turns from estimating and forecasting values to the appraisal framework that consumes them: the foundations of CBA, the Kaldor–Hicks criterion and its ethical content, the structure of a benefit–cost account, and the standing question of whose costs and benefits count — the framework into which your fish-restoration device and ecosystem-services estimates will feed. Tell me when to write it, or request a different chapter.

Part X Assessment — Multiple Discreteness

Covers Chapters 40–42. Five components, in the order they are best attempted. Allow a full working day; the mock oral is the part that matters most for your viva.

1. Cumulative problem set

P1. Derive the MDCEV Kuhn–Tucker conditions and state the consumption rule in words.

P2. Show that the multiplier is $\lambda = (\psi_0 + \sum_S \gamma_k \psi_k)/(E + \sum_S \gamma_k p_k)$ and use it to prove that baseline utilities are identified only up to a common shift.

P3. Show that the MDCEV likelihood collapses to the MNL probability when $M = 1$.

P4. Demonstrate numerically that an α-profile can mimic a γ-profile, and report the correlation between the two marginal-utility curves.

P5. Compute the exact compensating variation of a policy that raises one service's baseline utility by 0.5, using the forecasting algorithm.

2. Research critique

Take your MDCEV proposal and write one page on the zeros: what fraction you expect, what economic mechanism generates them, what survey artifacts could generate the same pattern, and exactly what debriefing evidence would distinguish the two.

3. Red-flag diagnosis

An MDCEV paper estimates α_k and γ_k freely for all seven goods, reports convergence, and interprets the estimated satiation parameters as behavioural constants that differ meaningfully across ecosystem services. State what is wrong, what the estimates actually are, and what should have been done.

4. Mock oral examination

1. Why does linear utility make multiple discreteness impossible?
2. What does the translation parameter do that a log utility cannot?
3. What is a corner solution, economically?
4. Why can you not estimate α and γ freely?
5. Is your outside good a genuine numeraire? What follows if it is not?

5. Integration mini-project

Derive the compensating variation for your MDCEV proposal analytically as far as it goes, then implement the simulation-based computation. Report which parameters the welfare number actually depends on, and run the sensitivity analysis across the γ- and α-profiles to show whether your substantive conclusions survive the profile choice.

Part XI — Cost–Benefit Analysis

Chapter 43 — Foundations of Cost–Benefit Analysis

Part XI: Cost–Benefit Analysis

> **The spine question, applied to this chapter.** Everything in Parts I–X produced *numbers*: a WTP, a welfare distribution, a forecast allocation. This part asks what to *do* with them — and the answer is not arithmetic but a normative framework with a specific ethical structure that must be understood before it can be used honestly. Cost–benefit analysis answers a precise question: *does this project increase aggregate welfare, measured as the sum of money-metric willingness to pay across everyone with standing?* Each element of that sentence — aggregate, sum, money-metric, standing — encodes a value judgement. The spine question: *what exactly does my CBA verdict claim, whose welfare does it count and at what weight, and would a defensible alternative specification reverse it?*

43.0 Conceptual motivation: an ethical framework wearing arithmetic clothes

Your fish-restoration device appraisal will produce a number — a net present value — and that number will be used to argue that the project should or should not proceed, potentially with blue-finance funding attached. It is tempting to treat the NPV as a technical output, the objective part of the analysis, with the value judgements confined to some separate section on ethics. **That is precisely backwards.** The NPV *embeds* the value judgements; they are in the aggregation rule, not adjacent to it.

Consider what the verified results in this chapter demonstrate. A project with winners gaining €275m and losers losing €200m **passes** the standard CBA test with an aggregate of +€75m — while the commercial trawlers who bear €140m of that loss receive *nothing*, because the criterion requires only that compensation be *possible*, not that it be *paid*. Apply an equity weight of 2.2 to the affected low-income group — a weight within the range some appraisal guidance recommends — and the same project's verdict **flips to −€93m and fails**. Nothing about the underlying facts changed. The reversal came entirely from an ethical parameter that the unweighted analysis had silently set to 1.

This is the chapter's central claim: **unweighted CBA is not neutral.** Weighting every euro equally regardless of who holds it is itself a substantive ethical position — arguably a defensible one, but a position nonetheless, and one that should be stated rather than smuggled in as the absence of a choice.

Three further foundations follow. **The counterfactual**: benefits are measured *with-project versus without-project*, never *before versus after*. Verified below, for a declining fish stock, the before/after comparison understated the true benefit by **83%** — because it credited the project only with improvement over the starting point, ignoring the decline that would otherwise have occurred. For your FRD appraisal, where the relevant baseline is a stock trajectory under continued pressure, this is not a technicality but the difference between a project that appears marginal and one that is clearly worthwhile.

The decision rule: NPV, benefit–cost ratio, and internal rate of return can rank projects differently, and verified below they do — NPV selecting the full rollout while BCR and IRR both select the small pilot. NPV is the correct rule for mutually exclusive alternatives because it maximizes total welfare rather than welfare per euro, and the BCR is additionally *manipulable* by accounting convention (verified, shifting from 0.907 to 0.922 purely by whether an operating cost is netted from benefits or added to costs, while NPV is invariant).

The counting rules: transfers are not benefits, and secondary "multiplier" effects are usually not additional welfare. Verified, counting a subsidy alongside its financing tax and adding a jobs multiplier inflated a €14.5m NPV to €21.5m — a 48% overstatement — which matters directly when blue-finance instruments make subsidies and their fiscal counterparts highly visible in the project account.

43.1 Prerequisites

From **Chapter 34**: compensating variation as the money-metric of welfare — the quantity CBA aggregates. From **Chapter 35**: aggregation across the population, the extent of the market, and mean-versus-median. From **Chapter 4**: discounting, present value, and the NPV/IRR mechanics. From **Chapters 41–42**: the welfare distributions and forecasts that feed the benefit stream. From **Chapter 39**: the calibration factor whose uncertainty propagates into benefits.

New notation. NPV (net present value), BCR (benefit–cost ratio), IRR (internal rate of return), B_t, C_t (benefits and costs in period t), r (social discount rate), *standing* (whose welfare counts), w_g (distributional weight on group g).

43.2 The welfare-economic foundation: Pareto and Kaldor–Hicks

Definition 43.1 (Pareto improvement). A change is a **Pareto improvement** if at least one person is made better off and **no one** is made worse off. The criterion is compelling — it requires no interpersonal comparison and commands near-universal assent — and almost useless in practice, because virtually every real policy harms someone. An FRD programme that restricts trawling improves angling and harms trawlers; it is not a Pareto improvement, and neither is essentially any environmental policy.

Definition 43.2 (Kaldor–Hicks / potential Pareto improvement). A change satisfies the **Kaldor–Hicks criterion** if the winners' gains *exceed* the losers' losses — so that winners *could in principle* compensate losers and still be better off, converting the change into a Pareto improvement. Equivalently: the *sum* of compensating variations across all affected parties is positive.

$$\sum_i \mathrm{CV}_i > 0.$$

This is the criterion CBA implements. When an appraisal reports NPV > 0, it is reporting that the Kaldor–Hicks test passes.

Worked Example 43.1 (The compensation test, verified). An FRD programme's distributional incidence:

Group	Net gain (€m)
Recreational anglers	+180.0
Coastal tourism	+95.0
Commercial trawlers	−140.0
Taxpayers (capital cost)	−60.0

Group	Net gain (€m)
Aggregate	+75.0 — **Kaldor–Hicks PASSES**

Winners gain €275.0m; losers lose €200.0m (of which trawlers bear €140.0m); the surplus of €75.0m is what winners could pay over and above full compensation.

(Verified.) The test passes. But note carefully what has and has not been established: the winners *could* compensate the losers. **Whether they do is a separate question that the criterion does not address.** If no compensation mechanism exists, the project proceeds, trawlers are €140m worse off, and the appraisal records a success.

The three ethical commitments. Kaldor–Hicks embeds value judgements that should be stated explicitly:

1. **Compensation is hypothetical.** The test asks whether compensation is *possible*, not whether it occurs. A project can pass while inflicting large uncompensated losses on identifiable groups.
2. **A euro is a euro.** Summing CVs weights everyone's money equally, regardless of income. Since willingness to pay is constrained by ability to pay, this systematically privileges the preferences of the wealthy — a poor household's intense preference registers as a smaller number than a rich household's mild one.
3. **Welfarism and money-metric measurement.** Only welfare counts, and welfare is measured by willingness to pay — excluding rights-based, procedural, or non-welfarist considerations (Chapter 44 develops this).

The Scitovsky reversal. A technical difficulty compounds the ethical ones. The Kaldor test asks whether winners could compensate losers *in the new state*; the Hicks test asks whether losers could bribe winners to prevent the change *in the old state*. Because relative prices and endowments differ between states, both tests can pass simultaneously — the **Scitovsky paradox** — implying that a move from A to B *and* the reverse move from B to A both "improve" welfare. This is incoherent, and it arises because the criterion is not a complete ordering of social states. The **Scitovsky double criterion** (require the Kaldor test to pass *and* the Hicks test to fail) resolves the paradox but is rarely applied in practice. For most appraisals the issue is second-order; it matters when a project substantially alters relative prices or the income distribution — which large-scale marine spatial reallocation potentially does.

> **Active recall 43.A.** What exactly does a positive NPV establish under the Kaldor–Hicks criterion, and what does it *not* establish? *(Answer: a positive NPV establishes that the* ***sum of compensating variations across everyone with standing is positive*** *— that the winners' gains exceed the losers' losses, so winners* could in principle* fully compensate losers and still be better off, converting the change into a potential Pareto improvement. Verified example: aggregate +€75m from winners gaining €275m against losers losing €200m. What it does **not** establish: (i) that compensation will *actually be paid* — the test requires only that it be *possible*, so the project can proceed with trawlers bearing an uncompensated €140m loss while the appraisal records success; (ii) that the distribution of gains and losses is acceptable — summing CVs weights every euro equally regardless of who holds it, which systematically privileges those with greater ability to pay, and verified, applying an equity weight of 2.2 to the affected low-income group flips the verdict to −€93m; (iii) that the change is unambiguously an improvement — the Scitovsky paradox means the criterion can endorse both a move and its reverse when relative prices shift; (iv) anything about non-welfarist considerations such as rights or procedural fairness, which the money-metric framework excludes by construction.)*

43.3 Distributional weights: the choice that is always made

Since unweighted aggregation is itself a weighting scheme, the question is not *whether* to weight but *which* weights to use — and to be explicit about it.

Worked Example 43.2 (Weights reverse the verdict, verified). The same FRD programme, applying an equity weight of 2.2 to commercial trawlers (a group with lower average income):

Group	Weight	Weighted gain (€m)
Recreational anglers	1.0	+180.0
Coastal tourism	1.0	+95.0
Commercial trawlers	**2.2**	−308.0
Taxpayers	1.0	−60.0
Weighted total		−93.0 — **FAILS**

(Verified.) The verdict reverses: +€75m becomes −€93m. **The project's desirability is not a fact about the world independent of the ethical parameter.**

Where weights come from. The standard justification is *diminishing marginal utility of income*: an additional euro is worth more to a poorer person. With an iso-elastic utility function, the weight on group g is

$$w_g = \left(\frac{\bar{y}}{y_g}\right)^{\eta},$$

where y_g is group income, $\bar{y}$ average income, and η the elasticity of marginal utility of income. Setting $\eta = 0$ gives the unweighted case; $\eta = 1$ (a common value in official guidance) means a group with half the average income receives twice the weight; higher η implies stronger redistributive preference. Several national appraisal guidelines specify distributional weighting, and η also appears in the Ramsey discounting formula (Chapter 6), where it governs intertemporal rather than interpersonal aggregation — a connection worth noting, since consistency across the two uses is often violated.

The disciplined practice. Rather than choosing one weighting scheme, report the *sensitivity of the verdict to the weight*:

1. Report the **unweighted** NPV, stating explicitly that it embeds $\eta = 0$.
2. Report **weighted** NPVs across a range of η (say 0 to 2).
3. Identify the **switching value** — the weight at which the verdict reverses — and assess whether it lies within a defensible range. If a project passes only at $\eta = 0$ and fails at $\eta = 0.5$, that is decision-relevant information.
4. Report the **distributional incidence** separately (who gains and loses, and by how much), so that decision-makers can apply their own weights.

For your FRD appraisal specifically. The distributional structure is politically salient: if benefits accrue substantially to recreational anglers and tourism operators while costs fall on commercial fishers — often a lower-income group with fewer alternative livelihoods — then the unweighted verdict and the weighted verdict may genuinely differ, and presenting only the former would be advocacy rather than analysis. This connects to the welfare *distribution* of Chapter 41 (verified there: 12.7% gaining essentially nothing) — the incidence data are available from the model and should be used.

> **Stop and think 43.1.** Your FRD appraisal shows NPV = +€75m unweighted. A stakeholder objects that this ignores the impact on fishing communities. What is the analytically correct response? *(Answer: not to defend the unweighted figure as "the objective result," because unweighted aggregation is* not* neutral — it applies a weight of exactly 1.0 to every euro regardless of who holds it, which is a substantive ethical choice (formally $\eta = 0$ in an iso-elastic weighting scheme), and one that systematically privileges those with greater ability to pay since WTP is bounded by income. The correct response has four parts. (i) *Acknowledge the embedded judgement*: state that +€75m assumes a euro to a trawler and a euro to a tourism operator count identically. (ii) *Report the incidence*: winners gain €275m, losers lose €200m, of which commercial trawlers bear €140m — and note that Kaldor–Hicks requires only that compensation be *possible*, so absent an actual transfer mechanism those losses stand uncompensated. (iii) *Show the sensitivity*: verified,

an equity weight of 2.2 on the affected group reverses the verdict to −€93m, so I report weighted NPVs across a range of η and identify the **switching value** at which the decision flips, letting the decision-maker judge whether that weight is defensible. (iv) *Consider compensation design*: since the surplus is €75m, an actual compensation scheme — transition support, licence buy-backs, blue-finance-funded diversification — could convert the potential Pareto improvement into an actual one, which is the substantive answer to the objection rather than a rhetorical one.)*

43.4 Standing: whose welfare counts

Before aggregating, one must decide *over whom* — and the boundary is a choice, not a discovery.

Definition 43.3 (Standing). Standing is the set of parties whose costs and benefits enter the account. The main boundaries:

- **Geographic.** Local, regional, national, or global? Since non-use values are held by distant populations (Chapter 35), a narrow boundary can omit substantial genuine benefit, while an expansive one may aggregate negligible per-capita values over vast populations to implausible totals. For a marine restoration project with existence value, this choice can dominate the result.
- **Temporal.** Future generations count — but at what discount rate? This is the subject of Chapter 44, and it is the single most consequential parameter in long-horizon environmental appraisal.
- **Jurisdictional.** Do benefits to non-citizens count? A national appraisal typically restricts standing to residents, which for transboundary marine ecosystems is analytically awkward: fish stocks and water quality do not respect borders.
- **Legal and moral status.** Do illegal activities have standing (the gains to illegal fishers)? Do non-human entities? Standard practice excludes both, but the exclusion is a normative position rather than a technical necessity.

The practical rule. State the standing boundary *explicitly*, justify it by reference to who is genuinely affected and to the decision-maker's remit, and — where the boundary is contestable — report sensitivity to alternatives. For your FRD appraisal, the relevant questions are whether non-local existence value counts (it is real, per Chapter 35, but its aggregation is fragile per §35.5) and whether the appraisal is national or regional in scope, since blue-finance instruments may have international funders with a broader implicit boundary than a national appraisal would use.

43.5 The counterfactual: with/without, not before/after

The principle. Benefits and costs are *incremental* — the difference between the world **with** the project and the world **without** it, over the same period. The without-project baseline is a *counterfactual projection*, not the present state.

Worked Example 43.3 (The before/after error, verified). A fish stock declining 3% annually without intervention; the project halts the decline and produces 0.5% annual growth:

Year	Without project	With project	Before/after measure	With/without measure
0	1000.0	1000.0	0.0	0.0
5	858.7	1025.3	25.3	**166.5**
10	737.4	1051.1	51.1	**313.7**
19	560.6	1099.4	99.4	**538.8**

Total over 20 years	Value
Before/after	979
With/without	**5,772**
Understatement	**4,793 units (83%)**

(Verified.) The before/after comparison credits the project only with the improvement over the *starting* stock, ignoring the decline that would have occurred anyway — understating the true benefit by **83%**. The gap widens over time as the counterfactual deteriorates.

Why this matters acutely for your FRD appraisal. Restoration projects are typically undertaken *because* a resource is declining. The entire value of "halting decline" is invisible to a before/after comparison, which sees only the difference between the ending and starting states. A stock that is merely *stabilized* shows *zero* before/after benefit while delivering large genuine benefits. Conservation projects are therefore the class of intervention most severely misvalued by this error.

Constructing the counterfactual. The without-project baseline requires: (i) an explicit trajectory (stock dynamics, pressure trends, existing regulation); (ii) justification from biological and management evidence, not assumption; (iii) sensitivity analysis over baseline assumptions, since — as the table shows — the benefit estimate is *highly* sensitive to the assumed decline rate; and (iv) honesty that the counterfactual is itself a forecast with uncertainty (Chapter 42's cautions apply). A steeper assumed decline mechanically produces larger benefits, which creates an obvious incentive problem that transparent sensitivity analysis is designed to police.

> **Active recall 43.B.** Why does the before/after comparison understate the benefits of a conservation project, and by how much can this matter? *(Answer: because it compares the* ending* state to the *starting* state, crediting the project only with improvement over the status quo ante — while the correct measure is the difference between the world *with* the project and the *counterfactual* world without it. Conservation projects are typically undertaken precisely *because* a resource is declining, so the without-project trajectory falls over time; the project's real value includes all the decline it *prevents*, which the before/after comparison cannot see. Verified: a stock declining 3% annually without intervention and growing 0.5% with it showed a before/after benefit of 979 units against a true with/without benefit of 5,772 — an **83% understatement**, with the gap widening over time (25.3 versus 166.5 at year 5; 99.4 versus 538.8 at year 19). In the limiting case, a project that merely *stabilizes* a declining stock registers *zero* before/after benefit while delivering substantial genuine value. The corollary caution: because the counterfactual decline rate drives the benefit estimate mechanically, it must be justified from biological and management evidence and subjected to explicit sensitivity analysis, not assumed.)*

43.6 Decision rules: NPV, BCR, and IRR

Definitions. With benefits B_t, costs C_t, and social discount rate r over horizon T:

$$\text{NPV} = \sum_{t=0}^{T} \frac{B_t - C_t}{(1+r)^t}, \qquad \text{BCR} = \frac{\sum_t B_t/(1+r)^t}{\sum_t C_t/(1+r)^t}, \qquad \text{IRR: } \sum_t \frac{B_t - C_t}{(1+\text{IRR})^t} = 0.$$

Worked Example 43.4 (Conflicting rankings, verified). Two mutually exclusive FRD options at $r = 4\%$ — a small pilot (-100 then $+40$ for four years) and a full rollout (-800 then $+240$ for four years):

Option	NPV	BCR	IRR
A: small pilot	€45.20	**1.452**	**21.9%**

Option	NPV	BCR	IRR
B: full rollout	**€71.17**	1.089	7.7%

NPV selects B; BCR and IRR both select A.

(Verified.) The rules disagree, and the disagreement is systematic rather than incidental.

Why NPV is correct for mutually exclusive choices. NPV measures *total* welfare gain; BCR and IRR measure welfare *per euro invested*. When choosing between mutually exclusive alternatives — as with project scale — the objective is to maximize total welfare, so NPV is the correct criterion. The pilot is more *efficient* per euro but delivers less total benefit; if both are affordable and only one can be chosen, B is the welfare-maximizing choice. BCR and IRR become relevant only under a **binding budget constraint** across independent projects, where ranking by BCR approximates optimal portfolio selection.

Two further defects.

The BCR is manipulable. Its value depends on the arbitrary convention of whether a recurring cost is netted from benefits or added to costs. **Verified:** for an illustrative project with operating costs, netting them from benefits gives BCR = 0.907; adding them to costs gives BCR = 0.922 — while **NPV is identical either way**. A ratio whose value depends on accounting presentation cannot be a reliable decision criterion, and the manipulability is exploitable when a threshold (BCR > 1) determines funding.

The IRR is unreliable. It can have multiple solutions when the cash-flow sign changes more than once — common in environmental projects with decommissioning or restoration costs at the end — and none when no root exists. It also implicitly assumes reinvestment at the IRR itself, and ignores scale entirely (as the verified example shows).

Recommendation. Report NPV as the decision criterion. Report BCR and IRR as supplementary information if convention requires, with explicit statement that NPV governs where they conflict. Under a binding budget constraint across independent projects, rank by BCR while noting the netting convention used.

43.7 Counting rules: transfers, secondary effects, and double counting

Definition 43.4 (Transfer versus real resource cost). A **real resource cost** consumes resources that could have produced value elsewhere — labour, materials, capital, foregone ecosystem services. A **transfer** moves purchasing power between parties without consuming resources — taxes, subsidies, grants. **Transfers net to zero in the social account**: the recipient's gain exactly offsets the payer's loss. They matter for *distributional* analysis but not for the *efficiency* verdict.

Secondary effects. "Multiplier" or "induced" effects — jobs created in supplying industries, local spending by project workers — are generally *not* additional welfare in an economy near full employment, because the resources would have been employed elsewhere producing comparable value. They represent *redistribution* of economic activity, not creation. Counting them is among the most common ways appraisals are inflated, particularly when a project is being advocated.

Worked Example 43.5 (Double counting, verified). An FRD project account:

Item	Value (€m)	Type
FRD capital and installation	−12.0	REAL
Annual maintenance (PV)	−4.5	REAL
WTP for stock improvement	+31.0	REAL

Item	Value (€m)	Type
Subsidy received from fund	+9.0	TRANSFER
Tax paid to finance subsidy	−9.0	TRANSFER
Local jobs "multiplier"	+7.0	SECONDARY

Total	Value
Naive (everything counted)	+€21.5m
Correct economic NPV (real only)	+€14.5**m**
Overstatement	**€7.0m (48%)**

(Verified.) Counting the subsidy without its financing tax, and adding the jobs multiplier, inflates the NPV by 48%. Note that the transfers happen to net to zero here *if both are counted* — the more common error is counting only the *received* subsidy while omitting the tax that funds it, which is a pure inflation of the account.

Relevance to blue finance. Your FRD appraisal is linked to blue-finance instruments, which makes this discipline directly consequential. Blue bonds, concessional loans, and grants are *financing* arrangements — they determine *who bears* the cost, not *whether* resources are consumed. A project does not become economically worthwhile because it is grant-funded; the grant is a transfer from the funder to the project. The economic appraisal must be **separated from the financial appraisal**:

- The **economic (social) appraisal** asks whether the project increases aggregate welfare, counting real resource costs and benefits regardless of who pays.
- The **financial appraisal** asks whether the project is bankable for a particular party, counting cash flows including transfers.

Both are legitimate and both are needed — a project can be economically worthwhile but financially unviable (justifying subsidy) or financially attractive but economically wasteful (a warning sign). Conflating them, by counting concessional finance as an economic benefit, produces appraisals that recommend projects because they are funded rather than because they are worthwhile.

> **Stop and think 43.2.** A colleague argues the FRD project's NPV should include the €9m blue-finance grant as a benefit and the local employment multiplier, since "these are real gains to the region." How do you respond? *(Answer: both belong in a* financial* or *regional* account, not in the *economic* NPV, and including them overstates the social verdict — verified, doing so inflated a €14.5m NPV to €21.5m, a **48% overstatement**. On the *grant*: it is a **transfer**, not a resource. It moves purchasing power from the funder to the project without consuming or creating real resources, so in the social account the recipient's €9m gain is exactly offset by the funder's €9m outlay and it nets to zero. Counting the grant while omitting its source is the classic inflation error. Blue finance determines *who bears* the cost, not *whether* the project uses real resources — a project is not made economically worthwhile by being grant-funded. On the *multiplier*: in an economy near full employment, jobs in supplying industries represent activity *relocated* from elsewhere rather than created, so secondary effects are redistribution rather than additional welfare; counting them double-counts the resources already valued at their opportunity cost. The constructive response: keep two separate accounts — an *economic appraisal* (real resource costs and benefits, ignoring transfers, answering "does this increase aggregate welfare?") and a *financial appraisal* (all cash flows including grants, answering "is this bankable for this party?"). Both are legitimate and the colleague's concerns belong squarely in the second; and if regional employment effects are genuinely a policy objective, they should be reported as a *separate distributional impact*, not folded into the efficiency number.)*

43.8 Failure modes

F1. Treating the NPV as ethically neutral. Presenting unweighted aggregation as objective. *Fix:* state that it embeds $\eta = 0$; report weighted sensitivity (§43.3).

F2. Ignoring uncompensated losers. Reporting KH pass without incidence. *Fix:* report distributional incidence; consider actual compensation design (§43.2–§43.3).

F3. Before/after instead of with/without. Understating conservation benefits. *Fix:* explicit counterfactual trajectory with sensitivity (§43.5).

F4. Unjustified counterfactual. Assuming a decline rate that mechanically drives benefits. *Fix:* evidence-based baseline; sensitivity analysis (§43.5).

F5. Ranking by BCR or IRR. Selecting the wrong mutually exclusive option. *Fix:* NPV governs; BCR only under budget constraint (§43.6).

F6. Reporting a manipulable BCR. Not disclosing the netting convention. *Fix:* state the convention; note NPV invariance (§43.6).

F7. IRR with sign changes. Multiple or undefined roots. *Fix:* use NPV; check sign pattern (§43.6).

F8. Counting transfers as benefits. Grants and subsidies in the economic account. *Fix:* separate economic from financial appraisal (§43.7).

F9. Counting multipliers. Secondary effects as additional welfare. *Fix:* exclude in full-employment settings; report separately if relevant (§43.7).

F10. Unstated standing boundary. Aggregating over an undefined population. *Fix:* state and justify standing; report sensitivity (§43.4).

43.9 Diagnostic questions for your own papers

1. **Criterion.** Have I stated that NPV implements Kaldor–Hicks, and what that does and does not establish? *(§43.2, F1.)*
2. **Incidence.** Do I report who gains and loses, and whether compensation is actual or hypothetical? *(§43.2–§43.3, F2.)*
3. **Weights.** Do I report weighted sensitivity and the switching value? *(§43.3, F1.)*
4. **Standing.** Is the boundary explicit, justified, and tested for sensitivity? *(§43.4, F10.)*
5. **Counterfactual.** Is the baseline a with/without projection, evidence-based, with sensitivity? *(§43.5, F3–F4.)*
6. **Decision rule.** Is NPV the criterion, with BCR/IRR conflicts disclosed? *(§43.6, F5–F7.)*
7. **Transfers.** Are blue-finance flows excluded from the economic account and handled in a separate financial appraisal? *(§43.7, F8.)*
8. **Secondary effects.** Are multipliers excluded from the efficiency verdict? *(§43.7, F9.)*

43.10 Exercises

(A) Comprehension.

A1. Distinguish Pareto and Kaldor–Hicks improvements and explain why CBA uses the latter.

A2. Explain why before/after differs from with/without and when the gap is largest.

A3. Why is NPV preferred to BCR and IRR for mutually exclusive projects?

(B) Derivation / reasoning.

B1. Show that unweighted CBA is equivalent to distributional weighting with $\eta = 0$, and derive the iso-elastic weight.

B2. Explain the Scitovsky paradox and the double criterion.

B3. Show why transfers net to zero in the social account but matter distributionally.

(C) Research application / critique.

C1. Write the complete appraisal framework for your FRD project, specifying criterion, standing, counterfactual, and accounts.

C2. Your FRD NPV is +€75m unweighted but −€93m at η implying weight 2.2 on fishers. Write the reporting.

C3. A funder asks you to include the blue-finance grant and regional employment effects in the NPV. Respond.

43.11 Solutions

A1. A **Pareto improvement** makes at least one person better off and *no one* worse off — requiring no interpersonal comparison, but almost never satisfied by real policy, since virtually every intervention harms someone (an FRD programme restricting trawling harms trawlers). A **Kaldor–Hicks (potential Pareto) improvement** requires only that winners' gains *exceed* losers' losses, so that winners *could* compensate losers and still gain — equivalently, $\sum_i \mathrm{CV}_i > 0$. CBA uses Kaldor–Hicks because Pareto is too restrictive to evaluate any realistic policy, and because summing money-metric welfare produces a workable single criterion. The price is three ethical commitments: compensation is *hypothetical* (verified: a project passes at +€75m while trawlers bear an uncompensated €140m loss), every euro counts equally regardless of who holds it, and only welfare measured by WTP counts.

A2. Before/after compares the ending state to the *starting* state; **with/without** compares the world with the project to the *counterfactual* world without it over the same period. They differ whenever the without-project baseline is *not constant* — i.e. whenever conditions would change anyway. The gap is **largest for conservation projects**, because these are undertaken precisely when a resource is *declining*: the project's real value includes all the decline it prevents, which before/after cannot see. Verified: a stock declining 3% annually without intervention and growing 0.5% with it gave a before/after benefit of 979 against a true 5,772 — an **83% understatement** — with the gap widening over time. In the limiting case, a project that merely *stabilizes* a declining resource shows *zero* before/after benefit despite delivering substantial value.

A3. NPV measures the *total* welfare gain; BCR and IRR measure welfare *per euro invested*. For **mutually exclusive** alternatives the objective is to maximize *total* welfare, so NPV is correct. Verified: a small pilot had BCR 1.452 and IRR 21.9% against a full rollout's 1.089 and 7.7%, yet the rollout's NPV (€71.17) exceeded the pilot's (€45.20) — BCR and IRR select the more *efficient* option, NPV the more *valuable* one. Two additional defects: the **BCR is manipulable** by accounting convention (verified, 0.907 versus 0.922 depending on whether an operating cost is netted from benefits or added to costs, while NPV is identical), and the **IRR** can have multiple or no roots when cash-flow signs change repeatedly — common in environmental projects with end-of-life restoration costs — and implicitly assumes reinvestment at the IRR. BCR is appropriate only for ranking *independent* projects under a binding budget constraint.

B1. Weighted aggregate welfare is $\sum_g w_g \sum_{i \in g} \mathrm{CV}_i$. Setting $w_g = 1$ for all g gives $\sum_i \mathrm{CV}_i$ — the unweighted CBA sum. So unweighted CBA is *not* the absence of weighting but the special case in which all weights equal one. To derive the iso-elastic weight, suppose utility of income is $u(y) = y^{1-\eta}/(1-\eta)$ for $\eta \neq 1$, so marginal utility is $u'(y) = y^{-\eta}$. The social value of an extra euro to group g relative to a group at average

income $\bar{y}$ is

$$w_g = \frac{u'(y_g)}{u'(\bar{y})} = \frac{y_g^{-\eta}}{\bar{y}^{-\eta}} = \left(\frac{\bar{y}}{y_g}\right)^{\eta}.$$

Hence $\eta = 0$ gives $w_g = 1$ for all groups (the unweighted case, confirming the equivalence); $\eta = 1$ gives weights inversely proportional to income (a group at half average income receives weight 2); larger η implies stronger redistributive preference. Verified consequence: a weight of 2.2 on the affected group turned +€75m into −€93m. Note η also appears in the Ramsey rule for *intertemporal* aggregation (Chapter 6), and consistency between its interpersonal and intertemporal uses is frequently violated in practice.

B2. The **Kaldor** test asks, evaluating at the *new* state's prices and endowments, whether winners could compensate losers for moving A→B. The **Hicks** test asks, at the *old* state's prices, whether losers could bribe winners *not* to move. Because relative prices and endowments differ between states, the two tests use different valuation bases, and **both can pass simultaneously** — the **Scitovsky paradox** — implying that A→B *and* B→A are both improvements. This is incoherent: the criterion fails to generate a consistent ordering of social states. The **Scitovsky double criterion** requires that the Kaldor test pass *and* the Hicks test fail, ruling out the reversal. It is rarely applied because the paradox is usually second-order, arising only when a project substantially alters relative prices or the income distribution — though large-scale marine spatial reallocation is exactly the sort of intervention where it could bite, and where the standard single test should be treated with corresponding caution.

B3. A **transfer** moves purchasing power between parties without consuming resources: for a subsidy of amount S, the recipient gains $+S$ and the funding party (taxpayers, donors) loses $-S$, so the contribution to $\sum_i \mathrm{CV}_i$ is $+S - S = 0$. A **real resource cost** consumes inputs with alternative productive uses, so it enters the sum once, at opportunity cost, with no offsetting gain. Verified: an account containing a +€9m subsidy and its −€9m financing tax contributes zero to the economic NPV, and the correct real-only NPV was €14.5m against a naive €21.5m — a 48% overstatement once a jobs multiplier was also included. Transfers nonetheless matter **distributionally**: they change *who* bears costs and receives benefits, which is precisely the incidence information needed for the weighted analysis of §43.3 and for compensation design. Hence the practical rule: exclude transfers from the *efficiency* verdict, but record them in the *distributional* account and in the separate *financial* appraisal.

C1. *Appraisal framework for the FRD project.* **(1) Criterion.** NPV implementing Kaldor–Hicks, stated explicitly — with the accompanying statement that a positive NPV establishes that winners *could* compensate losers, not that they will. **(2) Standing.** Define the boundary explicitly: primary appraisal at national scale covering resident households; report sensitivity to (i) a regional boundary and (ii) a boundary including non-resident existence value, since non-use values are held by distant populations (Chapter 35) and blue-finance funders may operate with an international remit. Justify the primary choice by the decision-maker's jurisdiction. **(3) Counterfactual.** Construct an evidence-based without-project stock trajectory from biological and management data (not assumption), model with-project dynamics, and measure benefits as **with/without** — verified as essential, since before/after understated a conservation benefit by 83%. Run sensitivity over the assumed decline rate, disclosing that steeper assumed decline mechanically increases benefits. **(4) Two separate accounts.** An *economic appraisal* counting real resource costs (capital, installation, maintenance, foregone commercial catch at opportunity cost) and real benefits (WTP for stock improvement from the CV/DCE work, recreational and non-use value), **excluding** blue-finance grants, subsidies, and employment multipliers; and a distinct *financial appraisal* covering all cash flows including concessional finance, answering bankability. **(5) Benefit stream.** Use the calibrated WTP estimates (Chapter 39's calibration factor with its confidence interval) and the MDCEV welfare distributions (Chapter 41), aggregated per Chapter 35 with the extent-of-market and mean-versus-median sensitivities carried through. **(6) Decision rule.** NPV as criterion; report BCR and IRR supplementarily with the netting convention stated and any conflict disclosed. **(7) Distribution.** Report incidence by group; report weighted NPVs across $\eta \in [0, 2]$ and the switching value; assess compensation mechanisms (licence buy-backs, transition support) that could convert potential into actual Pareto improvement. **(8) Uncertainty.** Carry forward the WTP confidence intervals, the forecast intervals of Chapter 42, and discount-rate sensitivity (Chapter 44) into a full sensitivity analysis rather than a point NPV.

C2. *Reporting.* I would present the verdict as *conditional on an ethical parameter*, not as a single number. **Headline:** "The project's economic NPV is +€75m under equal weighting of all affected parties ($\eta = 0$) — the conventional Kaldor–Hicks criterion — and −€93m when commercial fishers receive an equity weight of 2.2 reflecting lower group income. The verdict is therefore not robust to the distributional weight." **Supporting elements:** (i) *Incidence table* — recreational anglers +€180m, coastal tourism +€95m, commercial trawlers −€140m, taxpayers −€60m, showing winners gain €275m and losers lose €200m. (ii) *Explicit statement of the embedded judgement* — that the unweighted figure assigns weight 1.0 to every euro regardless of holder, which is a substantive ethical position, not neutrality, and one that privileges those with greater ability to pay. (iii) *Weight sensitivity schedule* — NPV across $\eta \in [0, 2]$, with the **switching value** identified: the weight at which the verdict flips, and an assessment of whether that value lies within the range used in national appraisal guidance. (iv) *Compensation analysis* — since the unweighted surplus is €75m, an actual compensation scheme (licence buy-backs, transition support, blue-finance-funded diversification) could in principle deliver a genuine Pareto improvement; I would cost such a scheme and report the NPV net of it, since this converts a contested distributional question into a design question. (v) *Explicit recommendation framing* — the analysis does not determine the decision; it establishes that the project is welfare-improving under equal weighting and welfare-reducing under plausible equity weighting, so the choice turns on a value judgement properly belonging to the decision-maker. Presenting only +€75m would be advocacy rather than analysis.

C3. *Response.* I would decline to include either in the economic NPV, while offering the funder the analysis they actually need in the right place. **On the grant.** A blue-finance grant is a **transfer**, not a resource: it changes *who pays* for the project without changing *whether* real resources are consumed. In the social account the project's €9m receipt is exactly offset by the funder's €9m outlay, netting to zero — verified, and counting it (with an employment multiplier) inflated a €14.5m NPV to €21.5m, a **48% overstatement**. Including it would mean the project appears more economically worthwhile *because it is funded*, which inverts the logic: the appraisal exists to determine whether funding is warranted, so using the funding as evidence of merit is circular. **On the employment multiplier.** In an economy near full employment, jobs in supplying industries and induced local spending represent activity *relocated* from elsewhere rather than created; the labour and capital would have produced comparable value in alternative uses, which is already captured by valuing project inputs at opportunity cost. Counting the multiplier therefore double-counts. **What I offer instead.** (i) A separate **financial appraisal** presenting all cash flows including the grant, answering the funder's legitimate question of bankability and the required subsidy level — noting that a project with positive economic NPV but negative financial NPV is precisely the case where concessional finance is *justified*, which strengthens rather than weakens the funding argument. (ii) A separate **distributional and regional impact statement** documenting employment and local economic effects as *incidence*, which is genuine policy-relevant information, reported alongside rather than folded into the efficiency verdict. (iii) A note that this separation *protects* the funder: an appraisal that counts its own subsidy as a benefit is vulnerable to exactly the criticism that blue-finance instruments are most exposed to — that they fund projects because money is available rather than because the projects are worthwhile.

43.12 Spaced review prompts

+1 day. From memory: define Pareto and Kaldor–Hicks; state the three ethical commitments; write the iso-elastic weight formula.

+1 week. Explain the before/after error and why conservation projects are worst affected; explain why NPV beats BCR and IRR; explain why transfers net to zero.

+1 month. For your FRD appraisal, construct the full framework — standing, counterfactual, separate economic and financial accounts, NPV with weighted sensitivity and switching value, incidence table. Close with the spine sentence about what the verdict claims and under what ethical parameter.

43.13 How this chapter connects to the rest of the book

- **Chapter 34** supplies compensating variation, the quantity summed here; **Chapter 35** supplies population aggregation, the extent of the market, and the mean-versus-median choice whose fragility now propagates into the NPV; **Chapter 6** supplies discounting mechanics; **Chapters 39, 41, 42** supply the calibrated WTP, welfare distributions, and forecast intervals that constitute the benefit stream.
- **Chapter 41's** distributional CV results (verified there: 12.7% gaining essentially nothing) are the raw material for §43.3's incidence analysis — the model already provides what the weighted appraisal requires.
- Downstream: **Chapter 44** takes up discounting and intergenerational equity, the parameter that dominates long-horizon environmental appraisal; **Chapters 45–46** develop risk, uncertainty, and sensitivity analysis, converting the point NPV of this chapter into a defensible range; **Chapter 49 (benefit transfer)** addresses appraisals relying on values estimated elsewhere.

This chapter is where your research output becomes a policy verdict — and where the discipline of stating the verdict's ethical content, rather than presenting it as arithmetic, is established.

43.14 Research translation task

For your **FRD appraisal**, produce a four-page framework note:

1. **Criterion.** State that NPV implements Kaldor–Hicks and what that does and does not establish.
2. **Standing.** Define and justify the boundary; report sensitivity to regional/national/international alternatives and to inclusion of non-resident existence value.
3. **Counterfactual.** Document the evidence-based without-project stock trajectory and its sensitivity; measure all benefits with/without.
4. **Accounts.** Present separate economic and financial appraisals; list every item classified as real, transfer, or secondary, with blue-finance flows excluded from the economic account.
5. **Decision rule.** Report NPV as criterion, with BCR/IRR supplementary and netting conventions disclosed.
6. **Distribution.** Report the incidence table, weighted NPVs across $\eta \in [0, 2]$, the switching value, and a costed compensation option.
7. **Close** with the spine sentence: *"The FRD project has economic NPV €[·] under equal weighting ($\eta = 0$), measured with/without against a [·]% annual decline baseline, over [standing boundary], excluding €[·] of blue-finance transfers; the verdict switches sign at an equity weight of [·] on [group], and losers bear €[·] of uncompensated loss absent a compensation scheme."*

Carry this into Chapter 44 (discounting), Chapters 45–46 (uncertainty and sensitivity), where the point NPV becomes a range.

43.15 Bibliography

Core. - Boardman, Greenberg, Vining & Weimer, *Cost–Benefit Analysis: Concepts and Practice* — the standard textbook treatment of criteria, standing, counterfactuals, and counting rules (§43.2–§43.7). *(Verify edition.)* - Pearce, Atkinson & Mourato, *Cost–Benefit Analysis and the Environment: Recent Developments* (OECD) — CBA specifically for environmental appraisal, including distributional weighting. *(Verify edition/year.)* - HM Treasury, *The Green Book: Appraisal and Evaluation in Central Government* — an influential official guide specifying distributional weighting and counterfactual construction (§43.3, §43.5). *(Verify current edition — updated periodically.)*

Applied. - Kaldor (1939) and Hicks (1939) — the original compensation-criterion papers (§43.2). *(Verify journal details.)* - Scitovsky (1941), "A Note on Welfare Propositions in Economics," *Review of Economic*

Studies — the reversal paradox and double criterion (§43.2). *(Verify volume/pages.)* - Drèze & Stern (1987), "The Theory of Cost–Benefit Analysis," in *Handbook of Public Economics* — shadow prices and the formal structure of social appraisal. *(Verify volume/pages.)*

Advanced. - Adler & Posner, *New Foundations of Cost–Benefit Analysis* — the philosophical critique and defence of the welfarist framework (§43.2). *(Verify edition.)* - Johansson & Kriström, *Cost–Benefit Analysis for Project Appraisal* — rigorous welfare-theoretic treatment. *(Verify edition.)* - Literature on blue finance and marine project appraisal — relevant to §43.7's economic/financial separation. *(Verify — search current sources, as this field is developing rapidly.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Kaldor / Hicks / Scitovsky / Drèze–Stern citations before using them, per the book's no-unverified-citation rule. Official appraisal guidance (Green Book and equivalents) is revised periodically — always cite the current edition and check whether your jurisdiction's guidance specifies distributional weights.

End of Chapter 43. Next: Chapter 44 — Discounting, Time, and Intergenerational Equity, which takes up the parameter that dominates long-horizon environmental appraisal: how future benefits are weighed against present costs, the Ramsey framework and its ethical components, the descriptive-versus-prescriptive debate, declining discount rates, and why a marine restoration project delivering benefits over a century is more sensitive to the discount rate than to almost any other assumption. Tell me when to write it, or request a different chapter.

Chapter 44 — Discounting, Time, and Intergenerational Equity

Part XI: Cost–Benefit Analysis

> **The spine question, applied to this chapter.** Chapter 43 established that CBA's aggregation rule carries ethical content. This chapter examines the parameter where that content is most concentrated and most consequential. Marine restoration delivers benefits over decades to centuries; the discount rate determines what those benefits are worth today, and — verified below — moving from 1% to 7% shrinks the present value of a benefit a century hence by a factor of **321**. No other assumption in your appraisal has comparable leverage. Worse, the rate is not an empirical quantity awaiting better measurement: it decomposes into components that are partly observable and partly *ethical*, and reasonable people using the same framework reach rates differing fourfold. The spine question: *what does my discount rate assert about the standing of future generations, and does my project's verdict survive the range of defensible alternatives?*

44.0 Conceptual motivation: the parameter that decides

Your fish-restoration device appraisal has a structural feature shared by most environmental interventions: **costs now, benefits later**. Capital is spent immediately; ecosystem recovery accrues over decades, with stock rebuilding, habitat maturation, and non-use values persisting far beyond the installation's lifetime. The discount rate governs the exchange between those two — and it does so with a severity that is easy to underestimate.

Verified below: a benefit of €1,000,000 arriving in year 100 has a present value of **€369,711 at 1%** and **€1,152 at 7%**. At the higher rate, a million euros of future benefit is worth roughly a tenth of a percent of its face value — effectively nothing. **The rate, not the ecology, decides whether long-horizon benefits register at all.** Push to year 200 and at 7% the present value falls to €1. A project delivering enormous but distant benefits can be made to pass or fail almost at will by a defensible-sounding choice of rate.

This would be tolerable if the correct rate were an empirical fact. It is not. The **Ramsey formula** decomposes the social discount rate into three components — pure time preference, the elasticity of marginal utility of consumption, and the growth rate — of which the first is straightforwardly *ethical*: how much weight do we give to a person's welfare simply because they live later? Verified below, Stern's parameters ($\delta = 0.1\%$, $\eta = 1$, $g = 1.3\%$) give $r = 1.4\%$ while Nordhaus's ($\delta = 1.5\%$, $\eta = 2$, $g = 2\%$) give $r = 5.5\%$ — and the present value of a century-distant benefit differs by a factor of **53** between them. The disagreement is not about arithmetic or data; it is about whether to *describe* observed market behaviour or *prescribe* impartiality between generations.

Two further results shape practice. First, **uncertainty about the rate is not neutral**: because expectations must be taken over discount *factors* rather than rates, an uncertain rate implies a **declining**

certainty-equivalent rate — verified falling from 3.6% at year 1 to 1.4% at year 300 when rates of 1%, 3%, and 7% are equally likely. Second, and importantly for how this is reported, declining rates matter *only where benefits are back-loaded*: verified, the uplift over a constant 3.5% rate is **+1%** for benefits in years 1–50 but **+504%** for a lump sum at year 200. Quoting a single "declining rates raise the NPV by X%" figure without stating the benefit timing is meaningless.

The chapter closes with the practical discipline this demands: report the **switching value** — verified at 5.61% for an illustrative FRD project, which passes below that rate and fails above it — so that the verdict's dependence on a contested ethical parameter is visible rather than buried.

44.1 Prerequisites

From **Chapter 4**: present value, annuities, and NPV mechanics. From **Chapter 43**: the Kaldor–Hicks criterion, standing (including *temporal* standing), and distributional weights — the parameter η reappears here governing intertemporal rather than interpersonal aggregation. From **Chapter 34**: welfare measurement. From **Chapter 3**: intertemporal optimization.

New notation. r (social discount rate), δ (pure rate of time preference / utility discount rate), η (elasticity of marginal utility of consumption), g (growth rate of per-capita consumption), DF_t (discount factor at horizon t), CER (certainty-equivalent rate), STPR (social time preference rate).

44.2 Why discount at all? Three distinct rationales

Discounting is often defended by a single argument, but three separate rationales operate, and they justify different things.

1. Pure time preference (δ). People appear to value present consumption above future consumption simply because it is sooner — impatience. Applied *within* a lifetime this is uncontroversial (individuals may legitimately be impatient about their own consumption). Applied *across generations* it becomes the claim that a future person's welfare counts less *because* they are born later, which many philosophers and economists — Ramsey, Pigou, Sen, Parfit among them — have regarded as ethically indefensible. Ramsey himself called discounting future utility "a practice which is ethically indefensible and arises merely from the weakness of the imagination."

2. Diminishing marginal utility of growing consumption ($\eta \cdot g$). If future generations will be *richer* (growth $g > 0$), then an extra euro is worth *less* to them than to us, by the elasticity η of marginal utility of consumption. This rationale is *not* about impatience or the moral status of future people — it is a straightforward application of diminishing marginal utility, entirely compatible with treating all generations impartially. Note its conditionality: **if growth is zero or negative, this component vanishes or reverses.** For catastrophic environmental scenarios where future generations may be *worse* off, this term can argue for a *negative* contribution.

3. Opportunity cost of capital. Resources invested elsewhere earn a return, so a project must beat that alternative. This justifies discounting at the marginal productivity of capital — but the two rationales (time preference and opportunity cost) coincide only in idealized frictionless markets. In practice they diverge, and the divergence underlies the *descriptive* versus *prescriptive* dispute of §44.4.

Worked Example 44.1 (The annihilation of distant benefits, verified). Present value of €1,000,000 received in year T:

Year	$r = 1\%$	$r = 3\%$	$r = 5\%$	$r = 7\%$
10	€905,287	€744,094	€613,913	€508,349

Year	$r = 1\%$	$r = 3\%$	$r = 5\%$	$r = 7\%$
25	€779,768	€477,606	€295,303	€184,249
50	€608,039	€228,107	€87,204	€33,948
100	**€369,711**	€52,033	€7,604	**€1,152**
200	€136,686	€2,707	€58	**€1**

(Verified.) The ratio of present values at 1% versus 7% for a year-100 benefit is **321×**. At 7%, benefits beyond about a century are numerically invisible regardless of magnitude — a consequence not of any judgement about their importance but of exponential arithmetic. **For your FRD appraisal, where ecosystem recovery and non-use values persist for generations, this is the single most consequential modelling choice you will make.**

> **Active recall 44.A.** What are the three rationales for discounting, and which of them survives the objection that future generations should count equally? *(Answer: (1)* ***Pure time preference*** *(δ) — valuing present consumption more simply because it is sooner. Applied* across generations* this asserts that a future person's welfare matters less *because* they are born later, which is the component most vulnerable to the impartiality objection and which Ramsey called "ethically indefensible." (2) **Diminishing marginal utility of growing consumption** (ηg) — if future generations are richer, an extra euro is worth less to them. This **survives** the impartiality objection entirely: it does not devalue future *people*, only future *euros*, and is fully compatible with treating all generations equally. Note its conditionality — if growth is zero or negative (plausible under environmental catastrophe), this component vanishes or reverses. (3) **Opportunity cost of capital** — resources could earn a return elsewhere, so projects must beat that alternative; this justifies discounting on productivity grounds rather than ethical ones, and coincides with the time-preference rationale only in frictionless markets. So the impartiality objection targets δ specifically, which is why the Stern Review set δ near zero (0.1%, reflecting only extinction risk) while retaining ηg — verified giving $r = 1.4\%$ against Nordhaus's 5.5%, a 53-fold difference in the present value of a century-distant benefit.)*

44.3 The Ramsey formula

Definition 44.1 (Ramsey rule). The social discount rate decomposes as

$$r = \delta + \eta\, g,$$

where δ is the **pure rate of time preference** (the utility discount rate), η is the **elasticity of marginal utility of consumption**, and g is the growth rate of per-capita consumption. The first term is the ethical discount on future *welfare*; the second is the adjustment for future *wealth*.

The components and their status.

- δ — **ethical.** Not observable. Prescriptive analysts set it near zero, retaining only a small allowance for the probability of civilizational extinction (the argument being that we discount future welfare only insofar as the future may not exist). Descriptive analysts infer it from observed savings behaviour, yielding higher values.
- η — **partly empirical, partly ethical.** It governs *both* intertemporal substitution (how much we care about smoothing consumption over time) *and*, as Chapter 43 showed, *interpersonal* distributional weights. Estimates range from about 1 to 3. **Note the consistency requirement:** the same η should govern both uses. An appraisal applying $\eta = 1$ for intergenerational discounting while implicitly using $\eta = 0$ (unweighted aggregation) for contemporaneous distribution is internally inconsistent — a common and rarely noticed error.

- g — **empirical but uncertain.** Historical per-capita growth of roughly 1.3–2% is the usual anchor, but projecting it over a century is heroic, and environmental degradation may itself reduce g — an endogeneity the standard formula ignores.

Worked Example 44.2 (Parameter choices and their consequences, verified).

Stance	δ	η	g	$r = \delta + \eta g$	PV of €1m at year 100
Stern (prescriptive)	0.001	1.0	0.013	**1.40%**	**€249,003**
Nordhaus (descriptive)	0.015	2.0	0.020	**5.50%**	**€4,729**
UK Green Book (STPR)	0.005	1.0	0.020	2.50%	€84,647
High-η variant	0.010	3.0	0.020	7.00%	€1,152

(Verified.) Stern's rate is 1.4%, Nordhaus's 5.5% — and the present value of a century-distant benefit differs by a factor of **53**. Both are internally coherent applications of the same formula. **The disagreement is about δ and η, not about mathematics or data.**

44.4 Descriptive versus prescriptive

The Stern–Nordhaus exchange, though conducted over climate policy, is the defining methodological dispute for any long-horizon environmental appraisal, including yours.

The descriptive position. Use rates *revealed* by actual behaviour — market interest rates, observed savings, the return on capital. The arguments: (i) using rates inconsistent with market behaviour implies the analyst knows better than society what society's preferences are; (ii) if a project's return is below the market rate, the resources would generate more future welfare invested elsewhere — including in ways that benefit future generations; (iii) consistency with how other public investments are appraised.

The prescriptive position. Derive rates from *ethical principles* about intergenerational equity. The arguments: (i) market rates reflect the preferences of *currently living* people trading among themselves, and future generations are unrepresented in those markets — there is no mechanism by which they express preferences; (ii) individual impatience about one's own consumption does not license *social* impatience about other people's welfare; (iii) observed rates embed market failures, myopia, and risk premia inappropriate to a social planner's intergenerational problem.

Assessment. Neither position is obviously correct, and the dispute is genuine rather than a matter of one side's error. Two observations help.

The descriptive argument is strongest for the ηg component and weakest for δ. Market rates plausibly inform the productivity and growth-adjustment terms; they are far less informative about the ethical weight on future welfare, since no market prices intergenerational impartiality.

The prescriptive argument has a practical vulnerability. If a very low rate is applied, almost any project with permanent benefits passes, and appraisal loses discriminating power — the "tyranny of the low rate" mirroring the tyranny of the high one. Very low rates also imply savings behaviour wildly at odds with observation.

What to do. For your FRD appraisal, do not adjudicate the dispute — *report it.* Present the NPV under both descriptive and prescriptive parameterizations, identify the switching value (§44.7), and state which parameter drives the difference. This is more useful to a decision-maker than a single number defended by a position they may not share.

> **Stop and think 44.1.** A funder insists you use the market rate because "that's what the money could earn elsewhere." What is the strongest version of their argument, and where does it fail? *(Answer: the strongest version is a genuine* ***opportunity-cost*** *argument: if capital*

invested in the FRD returns less than it would elsewhere, then diverting it destroys value — and, crucially, the alternative investment would also generate returns that benefit future* generations, potentially more than the FRD does. So using a below-market rate can favour projects that make the future *worse* off than the alternative allocation would. This is a serious argument and should not be dismissed. Where it fails, or at least weakens: (i) market rates are set by trades among *currently living* people — future generations are structurally unrepresented, and no market prices intergenerational impartiality, so observed rates carry little information about the ethical component δ; (ii) the argument conflates *individual impatience* about one's own consumption, which is legitimate, with *social impatience* about other people's welfare, which requires separate justification; (iii) market rates embed risk premia and myopia inappropriate to a social planner's problem, and the relevant environmental benefits are often uncorrelated with market returns; (iv) it assumes the alternative returns are actually reinvested for future benefit rather than consumed. My response: report both. The descriptive rate is defensible for the ηg component and for the opportunity-cost logic; the prescriptive case bears on δ. Verified, the two stances give $r = 5.5\%$ and 1.4%, differing 53-fold in the present value of a century-distant benefit — a range the decision-maker is entitled to see rather than have resolved silently.)*

44.5 Uncertain rates imply declining rates

A result that is technical in origin but consequential in practice: uncertainty about the discount rate does not average out to a constant rate.

The Weitzman–Gollier logic. Suppose the appropriate rate is uncertain across scenarios. The correct procedure is to take the expectation over **discount factors** — the quantities that actually convert future to present value — not over rates:

$$\overline{\mathrm{DF}}_t = \mathbb{E}\left[\frac{1}{(1+r)^t}\right] \neq \frac{1}{(1+\mathbb{E}[r])^t}.$$

Because the discount factor is *convex* in r, Jensen's inequality gives $\mathbb{E}[\mathrm{DF}_t] > \mathrm{DF}_t(\mathbb{E}[r])$, and the gap widens with t. The implied **certainty-equivalent rate**, $\mathrm{CER}_t = \overline{\mathrm{DF}}_t^{-1/t} - 1$, therefore *declines* with horizon, tending toward the lowest possible rate — because at long horizons the low-rate scenarios dominate the expected discount factor (a high-rate scenario contributes essentially zero, so it cannot offset).

Worked Example 44.3 (Declining certainty-equivalent rate, verified). Rates of 1%, 3%, and 7%, each with probability 1/3 (simple average 3.67%):

Year	$\mathbb{E}$[discount factor]	**Certainty-equivalent rate**	Naive average-rate DF
1	0.965184	**3.61%**	0.964630
25	0.480541	**2.97%**	0.406464
50	0.290031	**2.51%**	0.165213
100	0.140966	**1.98%**	0.027295
200	0.046465	**1.55%**	0.000745
300	0.016892	**1.37%**	0.000020

(Verified.) The certainty-equivalent rate falls from 3.61% to 1.37%, approaching the lowest scenario rate. Compare the naive calculation using the average rate: at year 200 it gives a discount factor of 0.000745 against the correct 0.046465 — understating the present value of distant benefits by a factor of **62**. **Uncertainty about the discount rate is therefore an argument for declining rates, not a wash.**

Practical implementation. Several national guidelines specify declining schedules — for instance, stepping from 3.5% for years 0–30 down through 3.0%, 2.5%, 2.0% to 1.5% beyond 300 years. Check your jurisdiction's

current guidance, and note that using a declining schedule creates **time inconsistency**: a project appraised today uses different rates for a given calendar year than an appraisal conducted in twenty years will. This is a known and generally accepted cost of the approach.

> **Active recall 44.B.** Why does uncertainty about the discount rate imply a *declining* certainty-equivalent rate rather than simply using the expected rate? *(Answer: because present value depends on the discount* factor* $1/(1+r)^t$, not on the rate directly, so the expectation must be taken over factors: $\overline{\mathrm{DF}}_t = \mathbb{E}[1/(1+r)^t]$. Since the discount factor is a *convex* function of r, Jensen's inequality gives $\mathbb{E}[\mathrm{DF}_t] > \mathrm{DF}_t(\mathbb{E}[r])$ — the expected factor exceeds the factor at the expected rate — and the gap widens with horizon. The mechanism: at long horizons a high-rate scenario contributes essentially *nothing* to the expected factor, so it cannot offset the low-rate scenario, which comes to dominate. The implied certainty-equivalent rate $\mathrm{CER}_t = \overline{\mathrm{DF}}_t^{-1/t} - 1$ therefore *falls* toward the lowest possible rate. Verified with rates of 1%, 3%, 7% each at probability 1/3 (simple average 3.67%): the CER declined from 3.61% at year 1 to 1.98% at year 100 and 1.37% at year 300, and the naive average-rate calculation understated the year-200 discount factor by a factor of **62** (0.000745 against 0.046465). Practical consequence: honest acknowledgement of rate uncertainty *raises* the weight on distant benefits, and several national guidelines now specify declining schedules — at the cost of time inconsistency, since a future appraisal will assign different rates to the same calendar year.)*

44.6 Where declining rates actually matter

A result that disciplines how the previous section is reported, and which is easy to get wrong.

Worked Example 44.4 (Benefit timing determines the effect, verified). Present value under a constant 3.5% rate versus a declining schedule (3.5% to year 30, then 3.0%, 2.5%, 2.0%):

Benefit profile	PV at constant 3.5%	PV declining	**Uplift**
Uniform 1/yr, years 1–200	28.54	30.86	**+8%**
Uniform 1/yr, years 1–50 only	23.46	23.69	**+1%**
Uniform 1/yr, years 100–200	0.92	2.05	**+123%**
Single lump at year 150	0.57	1.67	**+191%**
Single lump at year 200	0.10	0.62	**+504%**

(Verified; the declining schedule's implied average annual rate over 200 years is 2.574%.)

The lesson. Declining rates raise the present value of a *uniform* 200-year stream by only 8% — because early years dominate the annuity and the schedule barely differs from constant in the first three decades. The effect is **negligible for near-term benefits (+1% over years 1–50)** and **enormous for distant ones (+504% for a year-200 lump sum)**.

Two implications for practice:

1. **Quoting a single "declining rates increase the NPV by X%" figure is meaningless without stating the benefit timing.** The same schedule produces uplifts from 1% to 504% depending on where benefits fall.
2. **The choice matters precisely for back-loaded projects** — ecosystem recovery, carbon sequestration, biodiversity, avoided irreversible loss — and hardly at all for projects paying off within a generation. **For your FRD appraisal the relevant question is therefore empirical: how back-loaded is the benefit stream?** If stock recovery and non-use values accrue mainly in the first few decades, the discounting debate is less decisive than it appears; if substantial value persists beyond a century, it dominates everything else in the appraisal.

44.7 Switching values and reporting discipline

Given that the rate is contested and consequential, reporting a single NPV at a single rate is indefensible.

Definition 44.2 (Switching value). The **switching value** of a parameter is the value at which the decision reverses — here, the discount rate at which NPV = 0.

Worked Example 44.5 (FRD switching value, verified). Capital €12m, net annual benefit €0.70m over 60 years:

Rate	NPV (€m)	Verdict
1.0%	19.47	PASS
2.0%	12.33	PASS
3.0%	7.37	PASS
3.5%	5.46	PASS
5.0%	1.25	PASS
7.0%	−2.17	**FAIL**

Switching value: NPV $= 0$ at $r = 5.61\%$.

(Verified.) The project passes below 5.61% and fails above it. Note that this switching value sits *between* the descriptive (5.5%) and prescriptive (1.4%) parameterizations of §44.3 — so the verdict genuinely depends on which stance is adopted, and reporting either alone would be advocacy.

The reporting standard.

1. **Report NPV across a range of rates**, not one.
2. **Report the switching value** explicitly, and assess whether it falls within the range of defensible rates. A project passing only below 1% is fragile; one passing up to 8% is robust.
3. **State the Ramsey decomposition** underlying any headline rate, so that δ (ethical) is separated from ηg (partly empirical).
4. **Report both descriptive and prescriptive parameterizations.**
5. **Report the declining-rate result with the benefit timing**, per §44.6.
6. **Check η consistency** between intertemporal discounting and any interpersonal distributional weighting (Chapter 43 §43.3).
7. **State the horizon and any terminal value** — see below.

Horizon and terminal values. Truncating the analysis at an arbitrary horizon implicitly assigns zero value beyond it. For persistent ecosystem benefits this can be substantial, so either extend the horizon (the discount factor will handle the diminishing weight) or add an explicit **terminal value** with stated assumptions. Do not let an unstated horizon choice do the work that should be done transparently by the discount rate.

> **Stop and think 44.2.** Your FRD project has a switching value of 5.61%. How do you present the verdict? *(Answer: as **conditional and bounded**, not as a single number. The presentation would be: "The project has positive NPV at any social discount rate below **5.61%** and negative NPV above it. Under the prescriptive (Stern-type) parameterization, $r = 1.4\%$, giving NPV of approximately €19m; under the descriptive (Nordhaus-type) parameterization, $r = 5.5\%$, giving NPV of approximately €0.4m — marginally positive. Under a UK Green Book STPR of 3.5%, NPV is €5.46m." Then four supporting elements. (i)* Assessment of robustness*: the switching value sits **between** the two leading parameterizations, so the verdict is genuinely contested rather than robust — this is the honest headline, and a project passing only below 1% or comfortably above 8% would warrant very different language. (ii)* Ramsey decomposition*: state which component drives the difference — here δ, the ethical weight on future welfare, since ηg is comparatively agreed — so the decision-maker sees that the disagreement is ethical rather than empirical. (iii)* Declining-rate treatment*: report the NPV under the applicable declining schedule **together with***

***the benefit timing**, since verified the uplift ranges from +1% to +504% depending on how back-loaded benefits are; if FRD benefits accrue mainly within 50 years, the declining schedule barely moves the verdict and I should say so. (iv)* Horizon disclosure*: state the 60-year horizon and whether any terminal value is assigned, since truncation implicitly values everything beyond at zero. The decision-maker is entitled to see that the verdict turns on an ethical parameter, and to apply their own.)*

44.8 Further issues: hyperbolic discounting and dual rates

Hyperbolic discounting and time inconsistency. Empirically, individuals discount *hyperbolically* — implying declining rates and generating preference reversals (preferring a larger-later reward when both are distant, but flipping when the sooner one becomes immediate). This is descriptively well-established but poses a problem for *social* appraisal: time-inconsistent social preferences mean a planner's optimal plan today will be abandoned tomorrow, undermining commitment. Note that hyperbolic discounting and the Weitzman declining rate of §44.5 produce similar *schedules* from entirely different reasoning — behavioural inconsistency in one case, rational treatment of uncertainty in the other. **Only the latter is a normative justification for declining rates**; citing hyperbolic discounting as support conflates description with prescription.

Dual discounting. Should environmental benefits be discounted at the same rate as monetary flows? The argument for a *lower* environmental rate: if environmental goods become increasingly *scarce* relative to manufactured goods, their *relative price* rises over time, partially offsetting discounting. Formally, if environmental quality grows at $g_E < g$ (or declines), the effective rate for environmental services is lower — and with sufficient relative scarcity it can approach zero or turn negative. This is theoretically well-grounded (it is simply relative-price change, not special pleading) and increasingly discussed in guidance. The argument against: it introduces an additional contested parameter and can be used to manufacture favourable results.

Relevance to your appraisal. Marine ecosystem services are plausibly becoming scarcer relative to produced goods, which is exactly the condition under which dual discounting applies. If you invoke it, do so *transparently* — state the assumed relative-scarcity trend, justify it from evidence on ecosystem decline, and report results both with and without. It should be a sensitivity case, not the headline.

44.9 Failure modes

F1. Single-rate NPV. Reporting one number at one rate. *Fix:* range plus switching value (§44.7).

F2. Treating the rate as empirical. Presenting r as measured rather than partly chosen. *Fix:* Ramsey decomposition separating δ from ηg (§44.3).

F3. Adjudicating descriptive versus prescriptive silently. Choosing a stance without disclosure. *Fix:* report both (§44.4).

F4. Averaging rates under uncertainty. Using $\mathbb{E}[r]$ rather than $\mathbb{E}[\text{DF}]$. *Fix:* expectation over factors; declining CER (§44.5).

F5. Quoting declining-rate uplift without timing. A single percentage that ranges from 1% to 504%. *Fix:* state benefit timing (§44.6).

F6. η inconsistency. Using $\eta > 0$ intertemporally and $\eta = 0$ interpersonally. *Fix:* consistent η across both (§44.3; Chapter 43).

F7. Arbitrary horizon truncation. Implicitly valuing post-horizon benefits at zero. *Fix:* extend horizon or state terminal value (§44.7).

F8. Citing hyperbolic discounting as normative support. Conflating descriptive behaviour with prescriptive justification. *Fix:* use the Weitzman uncertainty argument instead (§44.8).

F9. Opportunistic dual discounting. Invoking relative scarcity only when it helps. *Fix:* transparent, evidence-based, reported as sensitivity (§44.8).

F10. Ignoring growth endogeneity. Assuming g independent of environmental degradation. *Fix:* note the assumption; consider low-growth sensitivity (§44.3).

44.10 Diagnostic questions for your own papers

1. **Range.** Do I report NPV across rates and the switching value? *(§44.7, F1.)*
2. **Decomposition.** Is the Ramsey decomposition stated, separating ethical from empirical components? *(§44.3, F2.)*
3. **Both stances.** Are descriptive and prescriptive parameterizations both reported? *(§44.4, F3.)*
4. **Uncertainty.** Is rate uncertainty handled via expected discount factors / declining schedule? *(§44.5, F4.)*
5. **Timing.** Is the benefit stream's timing profile reported alongside any declining-rate result? *(§44.6, F5.)*
6. **Consistency.** Is η the same for intertemporal and interpersonal aggregation? *(§44.3, F6.)*
7. **Horizon.** Is the horizon stated and any terminal value justified? *(§44.7, F7.)*
8. **Dual rates.** If used, is the relative-scarcity assumption evidence-based and reported as sensitivity? *(§44.8, F9.)*

44.11 Exercises

(A) Comprehension.

A1. State the Ramsey formula and classify each component as ethical or empirical.

A2. Explain the descriptive versus prescriptive dispute.

A3. Why does discount-rate uncertainty imply declining rates?

(B) Derivation / reasoning.

B1. Derive the Ramsey rule from an intertemporal social welfare function.

B2. Show, using Jensen's inequality, why $\mathbb{E}[\mathrm{DF}] > \mathrm{DF}(\mathbb{E}[r])$ and why the CER declines.

B3. Explain why declining rates matter for back-loaded but not front-loaded benefits.

(C) Research application / critique.

C1. Write the complete discounting protocol for your FRD appraisal.

C2. Your FRD passes at 3.5% but fails at 7%, switching at 5.61%. Write the reporting.

C3. A referee says using anything below the market rate is "advocacy dressed as analysis." Respond.

44.12 Solutions

A1. $r = \delta + \eta g$. δ (pure rate of time preference) is **ethical** — it is the weight placed on future welfare simply because it is future, unobservable in principle, and the component the impartiality objection targets. η (elas-

ticity of marginal utility of consumption) is **partly empirical, partly ethical** — estimable from behaviour (roughly 1 to 3) but also encoding inequality aversion, and it governs *both* intertemporal discounting and interpersonal distributional weights (Chapter 43), requiring consistency across the two uses. g (per-capita consumption growth) is **empirical but deeply uncertain** over century horizons, and potentially *endogenous* to the environmental degradation being appraised. Verified consequence of parameter choice: Stern ($\delta = 0.001$, $\eta = 1$, $g = 0.013$) gives $r = 1.40\%$; Nordhaus ($\delta = 0.015$, $\eta = 2$, $g = 0.020$) gives $r = 5.50\%$ — a 53-fold difference in the present value of a century-distant benefit.

A2. The **descriptive** position holds that the social discount rate should be *inferred from observed behaviour* — market interest rates, savings, returns on capital — on the grounds that (i) the analyst should not override society's revealed preferences, (ii) resources returning less than the market rate would generate more future welfare invested elsewhere, and (iii) consistency with other public appraisal. The **prescriptive** position holds that the rate should be *derived from ethical principles* about intergenerational equity, on the grounds that (i) market rates reflect only trades among *currently living* people, with future generations structurally unrepresented, (ii) individual impatience about one's own consumption does not license social impatience about others' welfare, and (iii) observed rates embed myopia and risk premia inappropriate to a social planner. Assessment: the descriptive argument is strongest for the ηg terms and for opportunity cost, weakest for δ, since no market prices intergenerational impartiality; the prescriptive approach risks losing discriminating power if rates are set very low. The practical resolution is to report both and let the switching value reveal whether the dispute is decisive.

A3. Because present value depends on the discount *factor*, which is *convex* in the rate. Taking expectations over factors rather than rates, Jensen's inequality gives $\mathbb{E}[1/(1+r)^t] > 1/(1+\mathbb{E}[r])^t$, with the gap widening in t. Intuitively, at long horizons a high-rate scenario contributes essentially nothing to the expected factor, so it cannot offset a low-rate scenario, which comes to dominate — and the implied certainty-equivalent rate therefore falls toward the lowest possible rate. Verified with rates 1%, 3%, 7% at equal probability: CER fell from 3.61% (year 1) to 1.98% (year 100) to 1.37% (year 300), while the naive average-rate calculation understated the year-200 discount factor by a factor of 62.

B1. Consider a social planner maximizing $W = \int_0^\infty e^{-\delta t}\, u(c_t)\, N_t\, dt$ where u is utility of per-capita consumption and δ the utility discount rate. The **consumption discount rate** is the rate at which the social value of a marginal unit of consumption declines over time — i.e. the proportional rate of decline of the discounted marginal utility $e^{-\delta t}u'(c_t)$:

$$r_t = -\frac{d}{dt}\ln\left[e^{-\delta t}u'(c_t)\right] = \delta - \frac{d\ln u'(c_t)}{dt}.$$

With iso-elastic utility $u(c) = c^{1-\eta}/(1-\eta)$, marginal utility is $u'(c) = c^{-\eta}$, so $\ln u'(c_t) = -\eta \ln c_t$ and

$$\frac{d\ln u'(c_t)}{dt} = -\eta\,\frac{\dot{c}_t}{c_t} = -\eta g.$$

Substituting gives

$$r = \delta + \eta g.$$

Interpretation: the first term discounts future *welfare* (ethical), the second adjusts for future generations being *richer* so that marginal utility of consumption is lower (a diminishing-marginal-utility adjustment compatible with full impartiality). Note that if $g = 0$ the second term vanishes and $r = \delta$; if $g < 0$ — plausible under severe environmental degradation — the second term is *negative*, potentially implying a rate below δ or even negative.

B2. Let $\phi(r) = (1+r)^{-t}$ for fixed $t > 0$. Then $\phi'(r) = -t(1+r)^{-t-1}$ and $\phi''(r) = t(t+1)(1+r)^{-t-2} > 0$, so ϕ is **strictly convex** in r. Jensen's inequality for a convex function gives

$$\mathbb{E}[\phi(r)] > \phi(\mathbb{E}[r]) \quad\Longleftrightarrow\quad \mathbb{E}\left[\frac{1}{(1+r)^t}\right] > \frac{1}{(1+\mathbb{E}[r])^t}.$$

The curvature ϕ'' grows with t (the factor $t(t+1)$), so the inequality gap widens with horizon. Defining the certainty-equivalent rate by $\overline{\mathrm{DF}}_t = (1+\mathrm{CER}_t)^{-t}$, we get $\mathrm{CER}_t = \overline{\mathrm{DF}}_t^{-1/t} - 1$. As $t \to \infty$, the expected

factor is dominated by the smallest rate in the support, $r_{\min}$: since $\mathbb{E}[\mathrm{DF}_t] \geq p_{\min}(1+r_{\min})^{-t}$ and all other terms decay faster, $\overline{\mathrm{DF}}_t^{-1/t} \to (1+r_{\min})$, so $\mathrm{CER}_t \to r_{\min}$. Verified: with rates 1%, 3%, 7% equally likely, CER declined 3.61% → 2.97% → 2.51% → 1.98% → 1.55% → 1.37% at years 1, 25, 50, 100, 200, 300, monotonically approaching the 1% floor.

B3. Because a declining schedule and a constant rate *differ only at long horizons* — typically the schedule is identical or nearly so for the first few decades, diverging only later. The present value of a benefit at time t depends on the *cumulative* discounting up to t, so benefits arriving before the divergence are essentially unaffected, while benefits arriving well after it accumulate the full compounded difference. Formally, the ratio of declining to constant discount factors is $\prod_{s\leq t}(1+r_c)/(1+r_s)$, which equals 1 while $r_s = r_c$ and grows multiplicatively thereafter. Verified: the uplift was **+1%** for a uniform stream over years 1–50 (entirely within the constant-rate portion of the schedule), **+8%** for a uniform stream over years 1–200 (diluted because early years dominate the annuity), **+123%** for a stream over years 100–200, and **+504%** for a single lump sum at year 200. Practical implication: quoting a single declining-rate uplift without stating benefit timing is meaningless, and the choice of schedule is decisive only for *back-loaded* projects — ecosystem recovery, carbon, biodiversity — and nearly irrelevant for projects paying off within a generation.

C1. *Discounting protocol for the FRD appraisal.* **(1) Characterize the benefit timing profile first**, since verified this determines whether the discounting debate is decisive: report the share of undiscounted benefits accruing in years 0–30, 30–60, 60–100, and beyond. If benefits are front-loaded the rate matters far less than it appears; if substantially back-loaded (ecosystem recovery, persistent non-use value), discounting dominates the appraisal. **(2) Report the Ramsey decomposition** for every rate used, separating δ (ethical) from ηg (partly empirical), so the decision-maker sees which component drives disagreement. **(3) Report NPV under both stances**: descriptive (Nordhaus-type, $r \approx 5.5\%$) and prescriptive (Stern-type, $r \approx 1.4\%$), plus the applicable national guidance rate. **(4) Report the switching value** — verified 5.61% for the illustrative parameters — and assess robustness by whether it lies inside or outside the defensible range; state plainly if, as here, it falls *between* the two leading parameterizations, meaning the verdict is genuinely contested. **(5) Handle rate uncertainty correctly** by taking expectations over *discount factors*, and apply the jurisdiction's declining schedule; report the resulting NPV **together with the benefit timing**, avoiding a bare uplift percentage. **(6) Ensure η consistency** between the intertemporal discount rate and any interpersonal distributional weights used in the Chapter 43 analysis — a frequently violated requirement. **(7) State the horizon explicitly** and justify any terminal value, since truncation implicitly assigns zero to everything beyond. **(8) Consider dual discounting** as a *sensitivity case only*, with the relative-scarcity assumption for marine ecosystem services stated and evidence-based, reported both with and without. **(9) Low-growth sensitivity**: test g near zero, noting that environmental degradation may reduce growth endogenously, which the standard formula ignores. **(10) Present the headline as conditional**: the project's verdict as a function of the rate, not a single number.

C2. *Reporting.* **Headline:** "The FRD project has positive NPV at any social discount rate below **5.61%** and negative NPV above it. At the UK Green Book rate of 3.5% the NPV is €5.46m; under a prescriptive (Stern-type) rate of 1.4% it is approximately €19m; under a descriptive (Nordhaus-type) rate of 5.5% it is marginally positive at about €0.4m; at 7% it is −€2.17m." **Assessment of robustness:** the switching value falls *between* the two leading parameterizations, so the verdict is **genuinely contested rather than robust** — this is the honest characterization, and I would say so explicitly rather than leading with the favourable figure. A project passing only below 1%, or comfortably above 8%, would warrant very different language. **Supporting elements:** (i) *Ramsey decomposition* showing that the difference between the two stances is driven principally by δ — the ethical weight on future welfare — rather than by disputed empirics, so the decision-maker understands the disagreement is normative; (ii) *benefit timing profile*, since verified the declining-rate treatment produces uplifts ranging from +1% to +504% depending on how back-loaded benefits are, and I must state where the FRD stream falls before any declining-rate figure is meaningful; (iii) *declining-schedule NPV* under the applicable guidance, reported with that timing context; (iv) *horizon disclosure* — the 60-year horizon and whether benefits beyond it are assigned a terminal value, since truncation implicitly values them at zero, and for persistent ecosystem services this could be material; (v) *low-growth sensitivity*, since g near zero lowers r and may be the relevant scenario precisely when environmental degradation is severe. **Framing:** the analysis establishes the *conditions* under which the project is

worthwhile, not an unconditional verdict; the choice among defensible rates is a value judgement belonging to the decision-maker.

C3. *Response.* The charge is worth taking seriously, since a low rate can indeed manufacture favourable verdicts, but it misidentifies where the discretion lies — and the symmetric charge applies to the market rate. **First, the market rate is not assumption-free.** Using it embeds specific and contestable values for δ and η — in effect asserting that the ethical weight on future welfare should be inferred from trades among *currently living* people in markets where future generations are structurally unrepresented and have no mechanism to express preferences. That is a substantive normative position, not the absence of one. Verified, the descriptive and prescriptive parameterizations give $r = 5.5\%$ and 1.4%, differing **53-fold** in the present value of a century-distant benefit; whichever is chosen, a large ethical commitment is being made, and defending only one as "analysis" while labelling the other "advocacy" is not sustainable. **Second, the concern cuts both ways.** At 7%, €1m arriving in year 100 is worth €1,152 — about 0.1% of face value — and at year 200 it is worth €1. A high rate does not neutrally decline to take a position; it decides, in advance and almost regardless of the ecological evidence, that long-horizon benefits do not count. That is as much a thumb on the scale as a low rate. **Third, the referee's strongest point should be conceded and addressed.** The genuine opportunity-cost argument — that capital returning less than the market rate might generate more *future* welfare invested elsewhere — is serious, and I address it by reporting the descriptive parameterization prominently rather than dismissing it. **Fourth, my actual practice forecloses the advocacy charge.** I do not select a rate and report a single NPV. I report NPV across the full defensible range, state the **switching value** (5.61%), give the Ramsey decomposition separating the ethical from the empirical components, and present both stances — verified, the switching value falls *between* them, which I report as evidence that the verdict is contested rather than as support for my preferred conclusion. Advocacy would be choosing the rate that delivers the desired answer and reporting only that; disclosing the range and the switching point is the opposite.

44.13 Spaced review prompts

+1 day. From memory: write the Ramsey formula and classify its components; state why a 7% rate annihilates century-distant benefits.

+1 week. Explain the descriptive–prescriptive dispute; derive why uncertain rates imply declining rates; explain why declining rates matter only for back-loaded benefits.

+1 month. For your FRD appraisal, report the benefit timing profile, NPV across rates, the switching value, both parameterizations, and the η consistency check. Close with the spine sentence about what your rate asserts and whether the verdict survives the defensible range.

44.14 How this chapter connects to the rest of the book

- **Chapter 6** supplies the present-value mechanics; **Chapter 43** supplies the Kaldor–Hicks framework, temporal standing, and the parameter η whose interpersonal use must be consistent with its intertemporal use here; **Chapter 7** supplies the Jensen's-inequality argument that drives the declining-rate result of §44.5.
- **Chapters 41–42** supply the long-horizon benefit streams and forecast intervals to which these rates apply — and §44.6's timing result means the *shape* of those streams, not merely their magnitude, determines how much discounting matters.
- Downstream: **Chapters 45–46** integrate discount-rate uncertainty with the other uncertainties (WTP calibration, forecast error, aggregation) into a combined sensitivity analysis, where the switching value of this chapter becomes one dimension of a multi-parameter robustness assessment; **Chapter 49** addresses transferred values whose original appraisals may have used different rates.

This chapter is where your appraisal confronts the parameter with the greatest leverage over its conclusion — and where the discipline of reporting a conditional verdict, rather than a number, is most necessary.

44.15 Research translation task

For your **FRD appraisal**, produce a three-page discounting note:

1. **Benefit timing profile.** Report the share of undiscounted benefits by period (0–30, 30–60, 60–100, 100+ years) — this determines how decisive discounting is.
2. **Ramsey decomposition.** State δ, η, g for every rate used, classifying each component.
3. **Both stances.** Report NPV under descriptive and prescriptive parameterizations plus national guidance.
4. **Switching value.** Report the rate at which the verdict reverses and assess whether it lies within the defensible range.
5. **Uncertainty.** Apply expected discount factors / the declining schedule, reporting results with benefit timing stated.
6. **Consistency.** Verify η matches any interpersonal weighting used in the Chapter 43 distributional analysis.
7. **Horizon.** State the horizon and justify any terminal value.
8. **Close** with the spine sentence: *"The FRD project's NPV is positive for $r <$ [switching value]; under prescriptive parameters ($\delta =$ [·], $\eta =$ [·], $g =$ [·]) NPV is €[·], under descriptive parameters €[·]; [·]% of undiscounted benefits accrue beyond year 60, so the verdict [is/is not] dominated by the discount-rate choice."*

Carry this into Chapters 45–46, where the discount rate joins WTP calibration and forecast uncertainty in a combined sensitivity analysis.

44.16 Bibliography

Core. - Ramsey (1928), "A Mathematical Theory of Saving," *Economic Journal* — the origin of the Ramsey rule and the "ethically indefensible" remark on discounting future utility (§44.3). *(Verify volume/pages.)* - Stern, *The Economics of Climate Change: The Stern Review* — the prescriptive parameterization and its defence (§44.4). *(Verify edition/year.)* - Nordhaus (2007), "A Review of the *Stern Review on the Economics of Global Warming*," *Journal of Economic Literature* — the descriptive critique (§44.4). *(Verify volume/pages.)*

Applied. - Weitzman (1998, 2001), on gamma discounting and declining rates under uncertainty (§44.5). *(Verify which papers and volumes/pages.)* - Gollier, *Pricing the Planet's Future: The Economics of Discounting in an Uncertain World* — the definitive treatment of declining rates (§44.5). *(Verify edition.)* - HM Treasury, *The Green Book* — declining discount-rate schedules in official guidance (§44.5–§44.6). *(Verify current edition; schedules are revised.)*

Advanced. - Arrow et al. (2013), "Determining Benefits and Costs for Future Generations," *Science* — expert synthesis on intergenerational discounting. *(Verify volume/pages.)* - Dasgupta on η, inequality aversion, and the ethics of discounting (§44.3). *(Verify specific work.)* - Hoel & Sterner, or Sterner & Persson, on dual discounting and relative prices of environmental goods (§44.8). *(Verify citations.)* - Drupp, Freeman, Groom & Nesje on expert survey evidence regarding social discount rates. *(Verify citation.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Ramsey / Weitzman / Nordhaus / Arrow et al. citations before using them, per the book's no-unverified-citation rule. Official declining-rate schedules are revised periodically — always cite the current edition of your jurisdiction's guidance.

End of Chapter 44. Next: Chapter 45 — Risk, Uncertainty, and Sensitivity Analysis in Appraisal, which integrates the uncertainties accumulated across this book — WTP calibration factors, welfare distributions, forecast intervals, aggregation choices, and now the discount rate — into a combined assessment, covering expected values under risk, the distinction between risk and deep uncertainty, Monte Carlo appraisal, switching values, and the option value of delay under irreversibility. Tell me when to write it, or request a different chapter.

Chapter 45 — Risk, Uncertainty, and Sensitivity Analysis in Appraisal

Part XI: Cost–Benefit Analysis

> **The spine question, applied to this chapter.** Every previous chapter added uncertainty to your appraisal: the WTP has a confidence interval (Chapter 36), the calibration factor is estimated (Chapter 39), the welfare distribution is simulated (Chapter 41), the forecast is an interval (Chapter 42), the aggregation depends on contested choices (Chapter 43), and the discount rate spans a fourfold range (Chapter 44). This chapter refuses to let those uncertainties collapse into a single NPV. It propagates them jointly — and the result is transformative: verified below, a project reporting a comfortable point NPV of **+€11.35m "PASS"** in fact carries a **49.9% probability of destroying welfare**. The spine question: *is my appraisal reporting a number, or a risk profile — and does the decision I recommend account for the chance that I am wrong, the value of resolving that uncertainty, and the option to wait?*

45.0 Conceptual motivation: a point NPV is a claim about a world that does not exist

The appraisal machinery of Chapters 43–44 produces a net present value. It is enormously tempting to treat that number as *the* answer — to plug in the best estimate of each input, compute NPV, and report "PASS" or "FAIL." This chapter exists to explain why that practice is not merely incomplete but actively misleading, and to build the alternative.

The problem is that a point NPV computed from best-guess inputs answers a question no decision-maker actually faces: *what is the value of the project in the single hypothetical world where every uncertain quantity happens to equal its central estimate?* No such world exists. The real decision is made under a *distribution* of possible worlds, and the object of interest is the *risk profile* of the project across that distribution — its expected value, its variance, and above all its probability of failure.

The verified results in this chapter make the gap concrete. Consider a marginal FRD project — capital large relative to benefits, as many real conservation projects are. Its point-estimate NPV is **+€11.35m**, which a naive appraisal reports as a pass. But propagating the uncertainty in willingness to pay, the calibration factor, the extent of the market, the discount rate, capital cost, and maintenance through a Monte Carlo simulation reveals the truth: the *mean* NPV is €11.8m but the *median* is €0.1m, the standard deviation is €64.8m, and — the number that matters — **the probability of a negative NPV is 49.9%**. The project is, to a first approximation, a coin flip. The point estimate concealed this entirely.

Four further results structure the chapter. First, **plugging in means is not the expected value**: because NPV is nonlinear in its inputs, $\mathbb{E}[\text{NPV}] \neq \text{NPV}(\mathbb{E}[\text{inputs}])$ — a Jensen gap (verified at 4% here, larger when nonlinearity is stronger). Second, **not all uncertainty is equal**: a tornado analysis (verified) ranks the extent of the market, the calibration factor, and the discount rate as the dominant drivers, while capital and

maintenance barely move the verdict — telling you where to concentrate both analysis and data collection. Third, **uncertainty has value**: the expected value of perfect information (verified €18.18m) is the ceiling on what further research or a pilot is worth, and when it is large relative to study costs, *collecting more data before deciding is itself the optimal decision.* Fourth, and most subtly, **irreversibility changes the rule**: when an action cannot be undone and uncertainty will resolve over time, the *option to wait* has value — verified at €29.85m, large enough to reverse a "build now" recommendation even when expected NPV is positive.

Underlying all of this is a distinction that Chapter 44 touched and this chapter formalizes: between **risk** (known probabilities), **Knightian uncertainty** (unknown probabilities), and **deep uncertainty** (unknown outcomes and models) — because the tools differ radically across the three, and misapplying risk tools to deep uncertainty produces false precision of the most dangerous kind.

45.1 Prerequisites

From **Chapter 43**: the NPV criterion and its ethical content. From **Chapter 44**: discount-rate uncertainty and switching values. From **Chapters 36, 39, 41, 42**: the WTP intervals, calibration factor, welfare distributions, and forecast intervals that are the uncertain inputs here. From **Chapter 9**: Monte Carlo simulation. From **Chapter 6**: Jensen's inequality and functions of random variables. From **Chapter 9**: propagating parameter uncertainty (Krinsky–Robb).

New notation. $P(NPV < 0)$ (probability of a negative verdict), EVPI (expected value of perfect information), QOV (quasi-option value), *risk* / *Knightian uncertainty* / *deep uncertainty* (the three regimes), *tornado* (one-way sensitivity ranking).

45.2 Three regimes: risk, uncertainty, and deep uncertainty

The appropriate analytical tool depends on *what kind* of not-knowing you face, and conflating the three is a foundational error.

Definition 45.1 (Risk). Risk describes situations where the possible outcomes and their *probabilities* are known (or credibly estimable). A fair die, an insurance actuarial table, and — approximately — a WTP with a well-estimated sampling distribution are risk. Under risk, expected values, variances, and probability statements are meaningful, and the tools of this chapter (Monte Carlo, probability of failure, EVPI) apply directly.

Definition 45.2 (Knightian uncertainty). Following Knight, **uncertainty** (proper) describes situations where outcomes are known but *probabilities are not.* One can list what might happen but cannot assign credible probabilities. Here, probability-weighted expectations are not straightforwardly available; the tools are *scenario analysis*, *robustness* (how does the decision fare across the range?), and criteria such as *maximin* (maximize the worst case) or *minimax regret* (minimize the largest possible regret).

Definition 45.3 (Deep uncertainty). Deep uncertainty describes situations where even the *set of possible outcomes* — or the correct *model* — is unknown or contested. Long-horizon ecological dynamics, tipping points, and novel interventions often qualify. Here, prediction itself is unreliable, and the appropriate responses are *adaptive management* (decisions revisited as knowledge accrues), *precaution*, *robustness across models*, and preserving *flexibility* rather than optimizing against a specific probability distribution.

Why the distinction is decisive. Applying risk tools to deep uncertainty produces **false precision** — a Monte Carlo simulation with invented probability distributions yields a confident $P(NPV < 0)$ that is an artefact of the assumed distributions, not a property of the world. For your appraisal, the discipline is to *classify each uncertain input by regime*:

- The **WTP sampling distribution** is genuine *risk* — you have an estimated distribution.
- The **calibration factor** is closer to *Knightian uncertainty* — you have an estimate, but transferring it across contexts (Chapter 39) involves unknown probabilities.
- The **long-horizon ecological trajectory** and **century-scale non-use values** are *deep uncertainty* — the models themselves are contested.

Mixing these in a single Monte Carlo without acknowledging the regime differences treats deep uncertainty as if it were risk, which is exactly the false-precision trap. The honest appraisal uses probabilistic tools where probabilities are credible and *scenario/robustness/precautionary* tools where they are not — and says which is which.

> **Active recall 45.A.** Distinguish risk, Knightian uncertainty, and deep uncertainty, and explain why using Monte Carlo simulation on all three produces false precision. *(Answer:* risk* = outcomes and their *probabilities* known/estimable (a well-estimated WTP sampling distribution) — expected values and probability statements are meaningful. *Knightian uncertainty* = outcomes known but *probabilities* unknown (transferring a calibration factor to a new context) — the tools are scenario analysis, robustness, maximin, minimax regret. *Deep uncertainty* = the *set of outcomes* or the correct *model* itself unknown or contested (long-horizon ecological dynamics, tipping points) — the tools are adaptive management, precaution, robustness across models, and preserving flexibility. Monte Carlo produces *false precision* when applied to Knightian or deep uncertainty because it *requires* input probability distributions: if those distributions are invented rather than estimated, the resulting P(NPV<0) is a confident-looking artefact of the analyst's assumptions, not a property of the world — and its spurious precision can be more dangerous than honest acknowledgement of ignorance, because it invites decision-makers to trust a number that encodes guesses. The discipline is to classify each input by regime and use probabilistic tools only where probabilities are credible, scenario/robustness/precautionary tools elsewhere, and to state which is which.)*

45.3 From point estimate to expected value: why plugging in means is wrong

Even in the pure-risk case, the standard practice of computing NPV at the mean of each input is *incorrect* as an estimate of the expected NPV.

The Jensen problem. NPV is a *nonlinear* function of its inputs — it involves the discount factor $1/(1+r)^t$ (convex in r), products of uncertain quantities (WTP × households × calibration), and ratios. For a nonlinear function f, Jensen's inequality (Chapter 7) gives $\mathbb{E}[f(X)] \neq f(\mathbb{E}[X])$ in general. So the NPV computed at mean inputs is *not* the expected NPV.

Worked Example 45.1 (Point estimate versus expected value, verified). A marginal FRD project with uncertain WTP, calibration factor, extent of market, discount rate, capital, and maintenance:

Quantity	Value
Point-estimate NPV (all inputs at means)	**€11.35m**
Monte Carlo $\mathbb{E}$[NPV] (300,000 draws)	**€11.80m**
Jensen gap	€0.45m (4% of point)

(Verified.) The gap is 4% here — modest, but arising purely from the annuity convexity in r and the products of inputs, and it can be much larger when the nonlinearity is stronger (deep in-the-money option-like payoffs, strong skew, or interactions). The point is directional and general: **the point estimate is a specific, usually-biased functional of the input distribution, not their expected outcome.** The correct

expected NPV requires propagating the input distributions through the NPV function — which is what Monte Carlo does.

The deeper point. Even the *expected* NPV is only a summary. The full object is the *distribution* of NPV, and it is the distribution — not any single moment — that a risk-aware appraisal reports.

45.4 Monte Carlo appraisal and the risk profile

The procedure. (i) Represent each uncertain input by a probability distribution reflecting genuine risk (its sampling distribution, or an elicited distribution). (ii) Draw a joint sample, respecting correlations (§45.6). (iii) Compute NPV for each draw. (iv) Report the resulting *distribution* — mean, median, spread, percentiles, and the probability of a negative verdict.

Worked Example 45.2 (The risk profile, verified). The same marginal FRD project, 300,000 draws:

Statistic	Value
Mean NPV	€11.8m
Median NPV	**€0.1m**
Standard deviation	€64.8m
5th / 25th / 75th / 95th percentiles	−€69.7m / −€33.6m / €44.1m / €132.9m
P(NPV < 0)	**49.9%**

(Verified.) The transformation of understanding is complete. The point NPV of €11.35m read as a comfortable pass; the risk profile reveals a project whose median outcome is essentially zero, whose 5th percentile is a €70m loss, and which has an **almost exactly even chance of destroying welfare**. The mean is positive only because the right tail (large gains when benefits are high and the discount rate low) is fatter than the left — the distribution is right-skewed, so mean exceeds median by a wide margin, exactly as with the WTP aggregation of Chapter 35 and the CV distribution of Chapter 41.

What to report. The risk profile, not a number:

1. **The full distribution** (histogram or fan chart), or at minimum mean, median, standard deviation, and key percentiles.
2. **P(NPV < 0)** — the probability the project fails the efficiency test — as a headline result. A 49.9% failure probability is decision-critical information that no point estimate conveys.
3. **The asymmetry** — for a right-skewed profile, report both mean (relevant for a risk-neutral social planner aggregating across many projects) and median (the typical outcome).

Risk attitude. A risk-neutral decision-maker ranks by expected NPV. But a decision-maker facing a *single* large irreversible project may reasonably be *risk-averse*, weighting the downside more heavily — in which case a 49.9% chance of a €70m loss is not offset by a symmetric chance of gain. Whether to incorporate risk aversion (via a utility function over outcomes, or a required risk premium) is a policy choice; the analyst's job is to *reveal the risk profile* so that choice can be made informedly, not to bury it in an expected value.

> **Stop and think 45.1.** Your FRD appraisal reports a point NPV of +€11.35m. The Monte Carlo shows mean €11.8m but P(NPV < 0) = 49.9%. How does this change what you tell the decision-maker? *(Answer: fundamentally — it converts a recommendation into a risk disclosure. The point estimate said "the project passes"; the risk profile says "the project's expected value is positive but it is essentially a coin flip, with a* ***49.9% chance of destroying welfare*** *and a 5th-percentile outcome of a €70m loss." What I tell the decision-maker: (i) the* headline is the failure probability*, not the mean — a near-50% chance of a negative outcome is the decision-critical fact; (ii) the distribution is* right-skewed* (mean €11.8m but median only €0.1m), so the positive mean is driven by a fat upper tail (high-WTP, low-discount-rate worlds) and the *typical* outcome

is roughly break-even; (iii) for a *single, large, irreversible* project a risk-neutral expected-value ranking may be inappropriate — a risk-averse decision-maker would weight the €70m downside heavily; (iv) this level of uncertainty makes the *value of further information* large (verified EVPI €18.18m, §45.7) and the *option to wait* valuable (verified QOV €29.85m, §45.8), so "resolve some uncertainty first — a pilot, better WTP data, a firmer ecological baseline" may dominate "build now." The point NPV would have hidden all of this and recommended proceeding on what is actually a coin flip.)*

45.5 Sensitivity analysis: which uncertainty matters?

Not every uncertain input matters equally. Sensitivity analysis identifies the drivers, focusing both analytical effort and data collection where they change the verdict.

One-way sensitivity and the tornado diagram. Vary each input across its plausible range while holding others at base values; record the *swing* in NPV. Ranking inputs by swing produces a **tornado diagram** (widest bars at top).

Worked Example 45.3 (Tornado, verified). Base NPV €11.35m; each input varied low-to-high:

Input	NPV at low	NPV at high	**Swing**
Extent of market (N_{hh})	−€35.3m	€58.0m	**€93.3m**
Calibration factor	−€30.5m	€57.4m	**€87.9m**
Discount rate	−€28.6m	€54.0m	**€82.6m**
WTP	−€24.5m	€56.2m	**€80.7m**
Capital cost	−€8.7m	€28.3m	€37.0m
Maintenance	€2.8m	€19.9m	€17.1m

(Verified.) The verdict is driven by the **extent of the market, the calibration factor, the discount rate, and WTP** — each capable of flipping the sign single-handedly — while capital and maintenance move it far less. This is directly actionable: it says the appraisal's credibility rests on the aggregation boundary (Chapter 35), the hypothetical-bias correction (Chapter 39), the discount rate (Chapter 44), and the WTP estimate (Chapter 36), and that refining the capital-cost estimate — however tempting as an engineering exercise — would barely change the verdict.

Switching values. For each key input, the *switching value* (the level at which NPV = 0; Chapter 44) tells you how far the input must move to reverse the verdict, and whether that lies within the plausible range. An input whose switching value is far outside its plausible range is not a threat; one whose switching value is *inside* the range is where the decision genuinely hangs.

Scenario analysis. Rather than varying inputs one at a time, construct coherent *scenarios* (optimistic, central, pessimistic) in which correlated inputs move together — a "low benefit" scenario combining low WTP, low extent of market, and high discount rate. This captures the joint movements that one-way analysis misses, and is the appropriate tool under Knightian uncertainty where probabilities cannot be assigned.

The hierarchy of methods. One-way sensitivity identifies drivers (cheap, but ignores interactions); scenario analysis captures coherent joint movements (good for Knightian uncertainty); probabilistic (Monte Carlo) analysis gives the full risk profile (best under genuine risk, requires credible distributions). Use them in combination: the tornado to *find* the drivers, then Monte Carlo over *those* drivers with credible distributions, and scenarios for the deeply-uncertain inputs that resist probabilistic treatment.

Active recall 45.B. What does a tornado diagram tell you, and how should it direct your data-collection effort? *(Answer: a tornado diagram ranks the uncertain inputs by how much each* swings the NPV* when varied across its plausible range (holding others fixed), with the

widest bars — the most influential inputs — at the top. Verified for the marginal FRD project, the extent of the market (swing €93.3m), calibration factor (€87.9m), discount rate (€82.6m), and WTP (€80.7m) each dominated, every one capable of flipping the sign alone, while capital cost (€37.0m) and maintenance (€17.1m) mattered far less. It directs data-collection effort by *identifying where uncertainty actually changes the verdict*: resources should go to tightening the dominant inputs — a firmer aggregation boundary and extent-of-market estimate (Chapter 35), a better-identified calibration factor (Chapter 39), justified discount-rate bounds (Chapter 44), and a more precise WTP (Chapter 36) — and *not* to refining inputs like capital cost that barely move the result, however tractable those might be to measure. The tornado is thus the bridge between the appraisal and the research agenda: it tells you which of the book's earlier estimation problems most needs to be gotten right for *this* decision, and it is the qualitative precursor to the quantitative value-of-information analysis of §45.7.)*

45.6 Correlated inputs

Inputs are rarely independent, and assuming independence misstates the risk profile — usually understating the downside.

The issue. If two benefit drivers are *positively* correlated — for instance, WTP and the extent of the market are both higher in wealthier regions — then good and bad cases *compound*: high-WTP worlds also tend to be high-population worlds, and low-WTP worlds also low-population, fattening both tails of the NPV distribution relative to the independent case.

Worked Example 45.4 (Correlation fattens the tails, verified). Inducing a positive correlation ($\rho = 0.6$) between WTP and the extent of the market:

Quantity	Independent	Correlated
Standard deviation of NPV	298.6	**365.8 (+22% wider)**

(Verified.) Positive correlation between benefit drivers widens the NPV distribution by 22% — both the upside and the downside become more extreme. **Assuming independence understates the risk**, including the probability of a large loss. The mechanism is general: correlated inputs do not average out, they reinforce.

Practical guidance. (i) Identify plausible correlations — benefit drivers moving together (income effects), cost drivers moving together (a project that is expensive tends to be expensive on every line), and benefit–cost correlations. (ii) Represent them in the Monte Carlo via a correlation matrix (Cholesky decomposition on the standardized inputs, Chapter 1) or a copula. (iii) Report the risk profile *with* correlations, and note the direction of the bias from ignoring them. (iv) Where correlations are themselves uncertain, treat the correlation as a scenario variable. For your appraisal, the WTP–population correlation is real (both track regional income), the calibration factor may correlate with WTP (both reflect elicitation quality), and these should not be assumed away.

45.7 The value of information

If uncertainty is large and consequential, *resolving it before deciding* may be worth more than deciding now. This is quantifiable.

Definition 45.4 (Expected value of perfect information, EVPI). EVPI is the expected gain from knowing the true state of the world *before* deciding, relative to deciding under current uncertainty:

$$\text{EVPI} = \mathbb{E}[\max(0, \text{NPV})] - \max(0, \mathbb{E}[\text{NPV}]).$$

The first term is the expected value of a decision-maker who observes the outcome and proceeds only when NPV > 0; the second is the value of the best decision under current uncertainty. EVPI is the **ceiling on what any research, pilot, or delay-to-learn is worth** — no information can be worth more than perfect information.

Worked Example 45.5 (EVPI, verified). For the marginal FRD project:

Quantity	Value
$\mathbb{E}[\text{NPV}]$	€11.8m
$\mathbb{E}[\max(0, \text{NPV})]$	€30.0m
EVPI	**€18.18m**

(Verified.) With a ~50% chance of loss, resolving the uncertainty before committing is worth up to €18.18m. **This is decision-critical:** if a pilot study, a better WTP survey, or a firmer ecological baseline can be conducted for substantially less than €18m and would meaningfully resolve the uncertainty, then *conducting it first is the optimal decision* — not proceeding, and not rejecting, but *learning*. Conversely, when EVPI is small (a robust project), further study is not justified and delay merely postpones benefits.

Refinements. EVPI assumes *perfect* information, which is unattainable; the **expected value of sample information (EVSI)** or **expected value of partial perfect information (EVPPI)** quantify the worth of *realistic* studies that reduce but do not eliminate uncertainty about *specific* inputs. Combined with the tornado (§45.5), EVPPI tells you not just that information is valuable but *which input* it is most valuable to learn about — directing the research budget precisely. For your work, this closes a loop back through the whole book: the appraisal identifies which of the WTP, calibration, forecast, and discount-rate uncertainties most deserves further estimation effort.

> **Stop and think 45.2.** Your FRD appraisal shows P(NPV < 0) = 49.9% and EVPI = €18.18m. A €2m pilot study would substantially resolve the ecological-benefit uncertainty. What is the analytically grounded recommendation? *(Answer:* do the pilot before committing to the full project* — this is very likely the optimal decision, not "build" and not "reject." The reasoning: with a near-50% chance of destroying welfare, deciding now is close to a coin flip on a large irreversible commitment. EVPI of €18.18m is the *ceiling* on the value of resolving the uncertainty, and a €2m pilot costs far less than that ceiling. The relevant comparison is not EVPI directly (that assumes *perfect* information) but the *expected value of the sample information* the pilot would actually provide — but since the pilot addresses one of the dominant tornado drivers (the ecological/benefit uncertainty feeding WTP and extent-of-market) and costs an order of magnitude less than EVPI, it is very likely worth funding: the expected reduction in the chance of a €70m mistake exceeds €2m. The recommendation therefore is a *sequenced* decision — fund the pilot, resolve the dominant uncertainty, then re-appraise — which is precisely the adaptive-management response appropriate when uncertainty is large, resolvable, and the stakes are high. This also connects to the option value of §45.8: waiting to learn is valuable when the action is irreversible, and the pilot is the mechanism for learning. Reporting only the point NPV of €11.35m would have foreclosed this option and recommended immediate commitment on a coin flip.)*

45.8 Irreversibility and the option value of waiting

The final and most subtle tool. When an action is *irreversible* and uncertainty will *resolve over time*, the standard "invest if expected NPV > 0" rule is *wrong* — it omits the value of preserving the option to not

invest.

Definition 45.5 (Quasi-option value). The **quasi-option value** (Arrow–Fisher–Henry) is the value of *delaying an irreversible decision* to await information. Formally, it is the difference between the expected value of a strategy that *waits and learns* (then acts optimally) and the expected value of *committing now*. When positive, it implies that a project with positive expected NPV should nonetheless be *delayed*.

Worked Example 45.6 (Quasi-option value, verified). An irreversible investment; uncertainty resolves next period (good state, probability 0.55, NPV +€60m; bad state, probability 0.45, NPV −€70m):

Strategy	Expected NPV
Invest now	€1.5m
Wait, learn the state, invest only if good	€31.35m
Quasi-option value of waiting	**€29.85m**

(Verified.) Investing now has positive expected NPV (€1.5m), so the naive rule says "build." But *waiting* — and building only if the good state is revealed — has an expected value of €31.35m, because it avoids the €70m loss in the bad state. The **quasi-option value of €29.85m** is the cost of committing prematurely, and it *exceeds* the gain from acting now. **The correct decision is to wait**, despite the positive expected NPV of immediate action.

Why this matters for marine restoration. Two features make quasi-option value central to your appraisal:

- **Irreversibility.** Many marine interventions — and the ecosystem changes they induce or fail to prevent — are difficult or impossible to reverse. Habitat alteration, species loss, and large capital installations all have this character.
- **Resolvable uncertainty.** Ecological monitoring, pilot deployments, and improved valuation *will* reduce uncertainty over time.

Together these mean the *option to wait and learn* has genuine value, and an appraisal that reports only expected NPV — recommending "build" whenever it is positive — systematically *undervalues flexibility* and *overvalues immediate commitment*.

The countervailing consideration. Irreversibility can also cut the *other* way. If the *no-action* baseline involves irreversible *loss* — a fish stock collapsing beyond recovery, a tipping point crossed — then *delay itself is an irreversible commitment* to that loss, and the option value can favour *acting now* (the "precautionary" logic). The quasi-option framework accommodates both: the question is *which* irreversibility dominates — the irreversibility of the intervention or the irreversibility of inaction. For an FRD intended to *halt* an irreversible decline, this tension is central and must be analyzed explicitly rather than assumed in either direction.

> **Active recall 45.C.** What is quasi-option value, why can it make a positive-expected-NPV project one that should be delayed, and how can irreversibility also argue for acting now? *(Answer: quasi-option value (Arrow–Fisher–Henry) is the value of* delaying an irreversible decision to await information* — the difference between the expected value of waiting-and-learning (then acting optimally) and committing immediately. It can make a positive-expected-NPV project one that should be *delayed* because committing now forecloses the option to *avoid* the bad state: verified, investing now had expected NPV €1.5m, but waiting and building only if the good state is revealed had expected value €31.35m, since it avoids the €70m loss — a quasi-option value of €29.85m that exceeds the gain from acting now, so the correct decision is to wait despite the positive expected NPV. The conditions are *irreversibility* (the action cannot be undone) plus *resolvable uncertainty* (waiting yields information). But irreversibility can equally argue for *acting now* when *inaction* is the irreversible commitment: if delay means a fish stock collapses beyond recovery or a tipping point is crossed, then waiting irreversibly forecloses the good outcomes, and the option value favours precautionary action. The framework is symmetric — it directs

attention to *which* irreversibility dominates: that of the intervention (favouring delay) or that of the no-action baseline (favouring action). For an FRD designed to halt an irreversible decline, both are present and the analysis must weigh them explicitly rather than defaulting to either.)*

45.9 Failure modes

F1. Point-estimate appraisal. Reporting NPV at mean inputs as the answer. *Fix:* propagate distributions; report the risk profile (§45.3–§45.4).

F2. Plugging in means for the expectation. Treating NPV(mean) as $\mathbb{E}$[NPV]. *Fix:* Monte Carlo; acknowledge the Jensen gap (§45.3).

F3. Omitting P(NPV < 0). Reporting only the mean. *Fix:* headline the failure probability (§45.4).

F4. Ignoring skew. Reporting mean without median for a skewed profile. *Fix:* report both (§45.4).

F5. False precision under deep uncertainty. Monte Carlo with invented distributions. *Fix:* classify by regime; use scenario/robust/precautionary tools where probabilities are not credible (§45.2).

F6. Assuming independence. Uncorrelated inputs understating the tails. *Fix:* model correlations; note the direction of bias (§45.6).

F7. Uniform sensitivity effort. Refining inputs that do not move the verdict. *Fix:* tornado first; concentrate on drivers (§45.5).

F8. Ignoring the value of information. Deciding now when learning is cheap and uncertainty large. *Fix:* compute EVPI/EVPPI; consider a pilot (§45.7).

F9. Naive expected-NPV rule under irreversibility. "Build if positive" ignoring option value. *Fix:* quasi-option value; weigh both irreversibilities (§45.8).

F10. Ignoring risk aversion for a single large project. Ranking by expected value alone. *Fix:* reveal the risk profile; let the decision-maker apply risk attitude (§45.4).

45.10 Diagnostic questions for your own papers

1. **Regime.** Have I classified each uncertain input as risk, Knightian, or deep uncertainty, and used matching tools? *(§45.2, F5.)*
2. **Risk profile.** Do I report the NPV distribution, median, and P(NPV < 0), not just a point? *(§45.4, F1–F4.)*
3. **Expectation.** Is $\mathbb{E}$[NPV] computed by simulation, not at mean inputs? *(§45.3, F2.)*
4. **Drivers.** Have I produced a tornado and directed effort to the drivers? *(§45.5, F7.)*
5. **Correlation.** Are input correlations modelled, with the independence-bias direction noted? *(§45.6, F6.)*
6. **Value of information.** Have I computed EVPI/EVPPI and considered a pilot before committing? *(§45.7, F8.)*
7. **Irreversibility.** Have I assessed quasi-option value and weighed intervention versus inaction irreversibility? *(§45.8, F9.)*
8. **Risk attitude.** Have I revealed the profile so risk aversion can be applied? *(§45.4, F10.)*

45.11 Exercises

(A) Comprehension.

A1. Distinguish risk, Knightian uncertainty, and deep uncertainty with an environmental example of each.

A2. Why is NPV at mean inputs not the expected NPV?

A3. What does P(NPV < 0) add beyond the expected NPV?

(B) Derivation / reasoning.

B1. Use Jensen's inequality to explain the sign of the point-estimate bias for a convex NPV.

B2. Derive the EVPI formula and interpret each term.

B3. Construct a two-state example where positive expected NPV implies waiting, and explain the role of irreversibility.

(C) Research application / critique.

C1. Design the complete uncertainty analysis for your FRD appraisal, integrating all prior chapters' uncertainties.

C2. Your point NPV is +€11m but P(NPV < 0) = 50%. Write the reporting and recommendation.

C3. A funder wants an immediate build decision. Using option value and EVPI, argue for a pilot.

45.12 Solutions

A1. *Risk* — outcomes and probabilities known: the sampling distribution of an estimated WTP, from which P(NPV < 0) can be credibly computed. *Knightian uncertainty* — outcomes known, probabilities not: whether a calibration factor estimated in one context transfers to your site (you can list the possibilities but cannot credibly probability-weight them). *Deep uncertainty* — outcomes or models themselves unknown/contested: the century-scale ecological response of the marine system to restoration, including possible tipping points, where the *model* is uncertain. The tools differ: expected values and Monte Carlo for risk; scenario analysis, maximin, and minimax regret for Knightian uncertainty; adaptive management, precaution, and robustness across models for deep uncertainty.

A2. Because NPV is a *nonlinear* function of its inputs — convex in the discount rate (via $1/(1+r)^t$), and involving products and ratios of uncertain quantities. For a nonlinear f, Jensen's inequality gives $\mathbb{E}[f(X)] \neq f(\mathbb{E}[X])$: the expected value of the function is not the function of the expected values. Computing NPV at mean inputs therefore yields a specific, generally *biased*, functional of the input distribution rather than the expected NPV. Verified: NPV at mean inputs was €11.35m against a Monte Carlo $\mathbb{E}$[NPV] of €11.80m — a 4% Jensen gap here, larger when nonlinearity is stronger. The correct expected NPV requires propagating the full input distributions through the NPV function.

A3. The expected NPV is a single moment that can be positive even when the project is highly likely to fail, because a fat right tail (large gains in favourable states) pulls the mean up. P(NPV < 0) reports the *probability the project destroys welfare* — the decision-critical fact that a risk-averse or single-project decision-maker most needs. Verified: a point/mean NPV of ~€11m corresponded to P(NPV < 0) = 49.9% — the project is essentially a coin flip, with a median of €0.1m and a 5th percentile of −€70m, none of which is visible in the positive mean. P(NPV < 0) thus converts a "pass" into an honest statement of risk, and is the appropriate headline for a marginal project.

B1. Let NPV $= f(X)$ where X collects the uncertain inputs and f is *convex* in the relevant argument (e.g. the discount factor $1/(1+r)^t$ is convex in r). Jensen's inequality for a convex function states $\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$, with strict inequality for non-degenerate X. So when NPV is convex in an uncertain input, computing it at the *mean* of that input *understates* the expected NPV: the point estimate is a *lower bound*. Verified: the

discount-factor convexity contributed a positive Jensen gap ($\mathbb{E}$[NPV] €11.80m > point €11.35m). The sign can reverse for inputs where NPV is *concave*; the general lesson is that the point estimate is biased in a direction determined by the curvature of NPV in each uncertain input, and only simulation recovers the true expectation.

B2. EVPI = $\mathbb{E}[\max(0, \text{NPV})] - \max(0, \mathbb{E}[\text{NPV}])$. *First term* $\mathbb{E}[\max(0, \text{NPV})]$: the expected payoff to a decision-maker who *learns the true state before deciding* and proceeds only when NPV > 0 (taking zero — the no-project value — when NPV ≤ 0). This is the value of perfectly-informed decision-making. *Second term* $\max(0, \mathbb{E}[\text{NPV}])$: the value of the *best decision under current uncertainty* — proceed if expected NPV is positive (getting $\mathbb{E}$[NPV]), otherwise decline (getting 0). *EVPI* is their difference: the expected gain from resolving all uncertainty before deciding, which equals the expected value of the losses *avoided* by not proceeding in bad states that current information cannot distinguish. It is the *ceiling* on the worth of any research, pilot, or delay-to-learn, since no information can beat perfect information. Verified: $\mathbb{E}[\max(0, \text{NPV})]$ €30.0m minus $\max(0, €11.8\text{m})$ €11.8m gives EVPI €18.18m — so up to €18m of study or delay is justified to resolve this ~50%-failure-probability decision.

B3. Two states next period: good (probability p, NPV $+g$) and bad (probability $1-p$, NPV $-b$), with $b, g > 0$; the investment is *irreversible*. *Invest now:* expected NPV = $pg - (1-p)b$. *Wait, learn, invest only if good:* expected value = $\text{disc} \cdot [p \cdot g + (1-p) \cdot 0] = \text{disc} \cdot pg$ (build in the good state, decline in the bad, avoiding the loss). Waiting dominates when $\text{disc} \cdot pg > pg - (1-p)b$, i.e. when the avoided expected loss $(1-p)b$ exceeds the cost of delaying the good-state gain $(1-\text{disc})pg$. Verified with $p = 0.55$, $g = 60$, $b = 70$, disc = 0.95: invest-now expected NPV = $0.55(60) - 0.45(70) = 33 - 31.5 =$ €1.5m (positive, so the naive rule says build); wait-and-learn = $0.95 \times 0.55 \times 60 =$ €31.35m; quasi-option value = €29.85m. *Role of irreversibility:* it is essential — if the investment were *reversible*, one could build now and undo it in the bad state, capturing the good-state gain without bearing the bad-state loss, so there would be no value to waiting. Irreversibility is what makes premature commitment costly and the option to wait valuable. (Symmetrically, if *inaction* were the irreversible commitment — e.g. delay lets a stock collapse — the option value would favour acting now.)

C1. *Complete uncertainty analysis for the FRD appraisal.* **(1) Classify each input by regime.** WTP sampling distribution and its Krinsky–Robb interval (Chapter 36) — *risk*; calibration factor (Chapter 39) and its transferability — *Knightian*; long-horizon ecological trajectory and century-scale non-use values (Chapters 41–42, 44) — *deep uncertainty*. Use probabilistic tools for the first, scenario/robustness for the second, adaptive-management framing for the third, and state which is which to avoid false precision. **(2) Assemble the input distributions** from the book's earlier estimates: WTP distribution (Chapter 36), calibration factor with CI (Chapter 39), MDCEV welfare distribution (Chapter 41) and forecast intervals (Chapter 42), extent-of-market and mean/median aggregation range (Chapter 35), and the discount-rate range with switching value (Chapter 44). **(3) Model correlations** — WTP with extent of market (both track regional income), calibration with WTP (elicitation quality) — via a correlation matrix, since verified independence understates the spread by ~22%. **(4) Monte Carlo** the NPV, reporting the *full distribution*: mean, median, standard deviation, percentiles, and — as the headline — **P(NPV < 0)**, acknowledging the Jensen gap between the point estimate and $\mathbb{E}$[NPV]. **(5) Tornado** to rank drivers; verified these are extent of market, calibration, discount rate, and WTP — concentrate reporting and any further data collection there. **(6) Switching values** for each driver, assessing whether they lie within plausible ranges (Chapter 44's discount-rate switching value feeds directly in). **(7) EVPI/EVPPI** to determine whether a pilot or further survey is justified, and *which input* it is most valuable to resolve. **(8) Quasi-option value** — assess the irreversibility of the intervention *versus* the irreversibility of continued stock decline, since the FRD is designed to halt an irreversible loss, and determine whether the option value favours waiting-to-learn or precautionary action. **(9) Risk attitude** — present the risk profile so the decision-maker can apply risk aversion appropriate to a single large irreversible project, rather than defaulting to expected-value neutrality. **(10) Integrate with the distributional and discounting analyses** of Chapters 43–44, so the final output is a *conditional, probabilistic verdict* — the project's risk profile as a function of the ethical parameters (weights, discount rate) and the resolvable uncertainties — not a single number.

C2. *Reporting and recommendation.* **Headline:** "The project's expected NPV is positive (€11.8m), but this masks substantial risk: the probability of a negative NPV — of the project *destroying* welfare — is **49.9%**.

The median outcome is approximately break-even (€0.1m), the 5th-percentile outcome is a €70m loss, and the positive mean is driven by a fat upper tail in favourable (high-WTP, low-discount-rate) states." **Supporting elements:** (i) the *full risk profile* (fan chart / percentiles), making clear the distribution is right-skewed so mean exceeds median; (ii) the *tornado*, identifying extent of market, calibration factor, discount rate, and WTP as the inputs that determine the verdict, each able to flip the sign alone; (iii) the *EVPI of €18.18m*, indicating that resolving the uncertainty before committing is worth up to that amount; (iv) the *quasi-option value*, if the intervention is irreversible. **Recommendation:** *do not present this as a pass.* Given a near-50% failure probability on a large irreversible commitment, the analytically grounded recommendation is a **sequenced decision** — fund a pilot or targeted study addressing the dominant uncertainties (ecological benefit, WTP, calibration) for a cost well below EVPI, resolve the uncertainty, then re-appraise. If sequencing is impossible and an immediate decision is forced, the choice depends on the decision-maker's *risk attitude*: a risk-neutral planner aggregating many independent projects might proceed on the positive expected value; a risk-averse decision-maker facing this *single* project should weight the 49.9% chance of a €70m loss heavily and likely decline. The analyst's role is to make the risk explicit; recommending "build" on a point NPV of €11m would misrepresent a coin flip as a sure thing.

C3. *Argument for a pilot.* I would make the case in the funder's own decision-theoretic terms rather than by appealing to caution. **First, the immediate-build decision is close to a coin flip.** The point NPV of €11m looks like a clear pass, but propagating the genuine uncertainties gives $P(\text{NPV} < 0) = 49.9\%$, a median of €0.1m, and a 5th-percentile loss of €70m. Committing now stakes a large irreversible investment on what is, in expectation, barely better than even odds. **Second, the uncertainty is worth resolving.** The expected value of perfect information is €18.18m — the ceiling on what learning the true state before deciding is worth. A pilot need not deliver perfect information to be justified; it need only reduce the dominant uncertainties (the tornado identifies these as ecological benefit, WTP, calibration, and the discount rate) at a cost below the *expected value of the sample information* it would provide. A pilot costing, say, €2m against an €18m information ceiling and a €70m downside is very likely to clear that bar. **Third, irreversibility makes waiting valuable in its own right.** Because the investment is irreversible and the uncertainty *will* resolve with monitoring, there is a *quasi-option value* to waiting — verified in an analogous case at €29.85m, exceeding the €1.5m gain from acting immediately — precisely because waiting lets us avoid committing in the bad state. **The one caveat I would flag honestly:** if the *no-action* baseline is itself irreversible — if delay means the stock collapses beyond recovery — then the option value can favour *acting now*, and the pilot must be designed to resolve the uncertainty *fast enough* not to forfeit the recoverable window. So my recommendation is a *time-bounded pilot*: resolve the dominant uncertainties on a schedule compatible with the ecological deadline, then commit or decline on far better information. This maximizes the funder's expected return and protects them from a 50% chance of a €70m write-off — which is a stronger basis for a blue-finance investment than a point NPV that conceals the risk.

45.13 Spaced review prompts

+1 day. From memory: distinguish the three uncertainty regimes; explain why NPV at means is not $\mathbb{E}[\text{NPV}]$; write the EVPI formula.

+1 week. Explain what $P(\text{NPV} < 0)$ adds; why correlation fattens tails; why irreversibility plus resolvable uncertainty creates option value.

+1 month. For your FRD appraisal, run the full analysis — regime classification, Monte Carlo risk profile, tornado, correlations, EVPI, quasi-option value — and produce a conditional probabilistic verdict. Close with the spine sentence about whether you are reporting a number or a risk profile.

45.14 How this chapter connects to the rest of the book

- This chapter *integrates* the uncertainties of the entire book: the WTP distribution (Chapter 36), calibration factor (Chapter 39), MDCEV welfare distribution and its parameter uncertainty (Chapter 41), forecast intervals (Chapter 42), aggregation and mean/median choices (Chapter 35), and the discount-rate range and switching value (Chapter 44) all enter as inputs to the Monte Carlo here.
- **Chapter 7** supplies Jensen's inequality (the point-estimate bias) and functions of random variables; **Chapter 11** supplies the Monte Carlo and correlation machinery; **Chapter 13** supplies parameter-uncertainty propagation; **Chapter 43** supplies the NPV criterion whose distribution is now the object; **Chapter 44** supplies the discount-rate uncertainty that the tornado (verified) ranks among the top drivers.
- Downstream: **Chapter 46** turns to reporting and reproducibility — presenting this risk profile transparently and making the whole appraisal auditable; **Chapters 47–49 (meta-analysis, benefit transfer)** supply values whose transfer uncertainty becomes another input to analyses like this one.

This chapter is where your appraisal stops pretending to certainty. The honest output is not "the NPV is €11m" but "the project is a near-coin-flip whose verdict turns on four identifiable uncertainties, worth up to €18m to resolve, with a valuable option to wait" — a far more useful thing to tell a decision-maker.

45.15 Research translation task

For your **FRD and ecosystem-services appraisal**, produce a four-page uncertainty note:

1. **Regime classification.** Tabulate every uncertain input as risk, Knightian, or deep uncertainty, with the tool used for each.
2. **Risk profile.** Report the Monte Carlo NPV distribution — mean, median, spread, percentiles, and **P(NPV < 0)** as the headline — with the Jensen gap from the point estimate noted.
3. **Drivers.** Present a tornado and the switching values of the top drivers.
4. **Correlations.** State the modelled correlations and the direction of bias from ignoring them.
5. **Value of information.** Report EVPI/EVPPI and assess whether a pilot or further study is justified, and for which input.
6. **Irreversibility.** Assess quasi-option value, weighing intervention against inaction irreversibility, and state whether it favours delay or precautionary action.
7. **Close** with the spine sentence: *"The FRD project has expected NPV €[·] but P(NPV < 0) = [·]%, driven by [top drivers]; resolving these is worth up to €[EVPI], and the option to wait is worth €[QOV], so the recommendation is [proceed / decline / pilot-then-reappraise]."*

Carry this into Chapter 46 (transparent reporting) and Chapters 47–49, where transferred values add further uncertainty to appraisals of this kind.

45.16 Bibliography

Core. - Knight, *Risk, Uncertainty, and Profit* — the original risk-versus-uncertainty distinction (§45.2). *(Verify edition.)* - Boardman, Greenberg, Vining & Weimer, *Cost–Benefit Analysis: Concepts and Practice* — Monte Carlo appraisal, sensitivity analysis, and expected value (§45.3–§45.5). *(Verify edition.)* - Arrow & Fisher (1974), "Environmental Preservation, Uncertainty, and Irreversibility," *Quarterly Journal of Economics* — the foundational quasi-option-value paper (§45.8). *(Verify volume/pages.)*

Applied. - Dixit & Pindyck, *Investment Under Uncertainty* — the definitive treatment of irreversibility and the option value of waiting (§45.8). *(Verify edition.)* - HM Treasury, *The Green Book* and its supplementary guidance on risk, optimism bias, and Monte Carlo appraisal (§45.4–§45.5). *(Verify current edition.)* -

Raiffa & Schlaifer, or a modern decision-analysis text, on the expected value of information (§45.7). *(Verify citation.)*

Advanced. - Weitzman (2009), "On Modeling and Interpreting the Economics of Catastrophic Climate Change," *Review of Economics and Statistics* — the "dismal theorem" and fat-tailed deep uncertainty (§45.2). *(Verify volume/pages.)* - Lempert, Popper & Bankes on robust decision-making under deep uncertainty (§45.2). *(Verify citation.)* - Fisher & Hanemann, or Traeger, on quasi-option value and the value of information in environmental economics (§45.7–§45.8). *(Verify citations.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Knight / Arrow–Fisher / Dixit–Pindyck / Weitzman citations before using them, per the book's no-unverified-citation rule.

End of Chapter 45. Next: Chapter 46 — Transparent Reporting, Reproducibility, and the Appraisal Document, which completes Part XI by turning the analysis into a defensible, auditable, reproducible deliverable: the structure of an appraisal report, the reproducible-research practices (version control, literate programming, open data and code) your PDF emphasizes, pre-registration and its limits for observational work, and how to present a conditional, uncertain, ethically-loaded verdict honestly to a decision-maker. Tell me when to write it, or request a different chapter.

Chapter 46 — Transparent Reporting, Reproducibility, and the Appraisal Document

Part XI: Cost–Benefit Analysis

> **The spine question, applied to this chapter.** The entire book has been an accumulation of *choices* — which exclusions, which distribution, which calibration threshold, which extent of market, which discount rate — each defensible, each affecting the number. This chapter confronts what that means for credibility. Verified below, 324 individually-defensible analytic pipelines applied to the *same* FRD data produce NPVs ranging from **−€91m to +€313m**, with only 35% passing. A motivated analyst can report almost any verdict honestly. The remedy is not better judgement but *transparency*: pre-specifying the primary analysis, reporting the full range of specifications, and making the entire pipeline computationally reproducible. The spine question: *could an independent analyst, given only my report and materials, reproduce my number exactly — and could they see every choice that could have changed it?*

46.0 Conceptual motivation: the analysis is not the result; the pipeline is

Your study plan places reproducible research among its explicit commitments, and your working practice reflects it — you render your documents through RMarkdown and pandoc to PDF, a literate-programming workflow that binds code, output, and prose into a single reproducible artefact. This chapter explains *why* that discipline is not administrative box-ticking but the load-bearing foundation of the whole enterprise, and it does so by confronting the uncomfortable fact that has haunted every preceding chapter.

That fact is this: **the number you report is a function of choices, not just data.** Every chapter has flagged a fork in the road. Chapter 32: exclude protesters or bound them? Chapter 36: log-normal or linear, mean or median? Chapter 39: which certainty threshold? Chapter 35: which extent of the market? Chapter 44: which discount rate? Each choice is *defensible* — you could justify any of them in a seminar — and each moves the answer. The disturbing consequence, made concrete below, is that the *combination* of defensible choices spans a range so wide that the data barely constrain the verdict. Verified: across 324 defensible specifications the FRD NPV ranges from −€91m (a clear fail) to +€313m (an overwhelming pass), and 65% of the specifications fail. An analyst who wanted the project approved could find a route to +€313m; one who wanted it rejected could find −€91m; both would be telling the truth about their own pipeline.

This is the **garden of forking paths** (Gelman and Loken), and it is more corrosive than outright fraud because it operates through *honest* judgement. No individual choice is wrong. The problem is that the *freedom to choose* — the researcher's degrees of freedom — is not disclosed, so the reader cannot tell

whether the reported number is representative or cherry-picked, and often the *analyst* cannot tell either, having arrived at their pipeline through a sequence of locally-reasonable decisions.

The response has three components, and this chapter builds each. **Transparency about the multiverse**: pre-specify the primary analysis and report the *specification curve* — the distribution of results across defensible choices — so the reader sees the range, not a point drawn from it. **Computational reproducibility**: fix and report the seed, the draw count, the software environment, and the code, so that "the same analysis" yields "the same number" — verified below, a Monte Carlo result varies across seeds by ± 4 percentage points at 200 draws and ± 0.1 at 200,000, so *both* the seed and an adequate draw count are part of the result. **Independent verification**: re-implement key numbers to catch errors — verified below, a checksum-style re-implementation catches an off-by-one bug that shifts an annuity by 3.5%, which is exactly the verify-before-you-write discipline this book has applied to every worked example.

The chapter ends with the appraisal document itself: how to structure a report so that a decision-maker receives an honest, conditional, uncertain, ethically-explicit verdict rather than a false point of certainty — the culmination of everything Parts VIII–XI have built.

46.1 Prerequisites

From **Chapters 32–33**: researcher degrees of freedom, specification search, pre-registration, and multiplicity. From **Chapter 45**: the Monte Carlo risk profile whose reproducibility is now at issue. From **Chapters 36, 39, 44**: the specific defensible choices (distribution, calibration threshold, discount rate) that populate the multiverse. From **Chapter 9**: pseudo-random number generation and seeds. Throughout: the "verify every number computationally before writing" practice this book has modelled.

New notation. *Computational reproducibility* (same code + data $\rightarrow$ same result), *replication* (new data $\rightarrow$ same conclusion), *multiverse / specification curve* (results across all defensible pipelines), *researcher degrees of freedom*, *literate programming*, *provenance* (the traceable chain from raw data to reported number).

46.2 What reproducibility means: three distinct standards

The word "reproducible" conflates several distinct standards, and precision matters because they demand different things.

Definition 46.1 (Computational reproducibility). A result is **computationally reproducible** if an independent analyst, given the *same data and the same code*, obtains the *same result*. This is the *minimum* standard — it does not ask whether the analysis is *correct*, only whether it is *repeatable*. Astonishingly, much published research fails even this: code is unavailable, environments are undocumented, random seeds are unfixed, and re-running the authors' own materials produces different numbers or errors. Computational reproducibility is *necessary but not sufficient* for credibility.

Definition 46.2 (Replicability). A result is **replicable** if an independent analyst, collecting *new data* from the same population and applying the same method, reaches the *same conclusion*. This is a stronger and more scientifically meaningful standard — it tests whether the finding is a property of the world rather than of the particular sample. Replication is what the "reproducibility crisis" in psychology and other fields found wanting.

Definition 46.3 (Robustness / analytical reproducibility). A result is **robust** if it survives *reasonable variations in the analytic choices* — different defensible specifications applied to the *same* data. This is the standard the multiverse analysis (§46.3) directly tests, and the one your appraisal most needs, because — verified — the analytic choices span a range wide enough to reverse the verdict.

The hierarchy and what it requires of you. Computational reproducibility requires *engineering discipline* (code, data, seeds, environment). Replicability requires *new studies* (largely outside a single appraisal's control, though your meta-analysis of 18 DCEs is precisely a replication-aggregation exercise, Chapters 47–48). Robustness requires *transparency about choices* (the specification curve). An appraisal should aim for all three: computationally reproducible by construction, robust via specification reporting, and honest about the replication evidence available.

> **Active recall 46.A.** Distinguish computational reproducibility, replicability, and robustness, and explain why the first is necessary but not sufficient for a credible appraisal. *(Answer:* computational reproducibility* = same data + same code → same result (repeatability of the specific computation); *replicability* = new data from the same population + same method → same conclusion (the finding is a property of the world, not the sample); *robustness/analytical reproducibility* = the result survives reasonable variation in defensible analytic choices applied to the same data. Computational reproducibility is *necessary* because a result that cannot even be re-derived from its own code and data cannot be checked, audited, or corrected — and much research fails this minimum through missing code, undocumented environments, and unfixed seeds. But it is *not sufficient* because it certifies only that the computation *repeats*, not that it is *correct* or *representative*: a computationally-reproducible pipeline can still be one cherry-picked point from a multiverse of 324 defensible specifications spanning −€91m to +€313m (verified), or can faithfully reproduce a coding error every time. Credibility therefore additionally requires *robustness* (transparency about the specification range, so the reader sees whether the number is representative) and, ideally, *replication* evidence (new data confirming the finding). An appraisal should be computationally reproducible by construction, robust via specification reporting, and honest about available replication.)*

46.3 The multiverse and the specification curve

The deepest threat to a CBA's credibility is not error but *undisclosed flexibility*. The multiverse analysis makes that flexibility visible.

The garden of forking paths. At each analytic decision the researcher chooses among defensible options. With even a handful of decisions, the number of possible pipelines explodes combinatorially, and the results scatter across a wide range. Because the researcher typically reports *one* pipeline — often the one arrived at through a sequence of locally-reasonable choices that happened to lead somewhere — the reader sees a point with no sense of the cloud it was drawn from. This is *specification search* (Chapter 32) whether or not the analyst intends it, because the choices are correlated with the results in ways the analyst cannot fully police from the inside.

Definition 46.4 (Multiverse / specification-curve analysis). A **multiverse analysis** (Steegen et al.) computes the result under *every combination* of defensible analytic choices; the **specification curve** (Simonsohn et al.) displays the sorted results with the choices that produced each, revealing the distribution of outcomes and which choices drive them. It converts hidden flexibility into an explicit, reportable object.

Worked Example 46.1 (The FRD multiverse, verified). Six analytic decisions, each flagged in earlier chapters, each with defensible options:

Decision	Options	Chapter
Protest handling	keep / drop uncertain / Lee-bound	32
WTP central tendency	mean / median	35, 36
WTP distribution	log-normal / linear	36
Calibration threshold	$\tau = 7$ / 8 / 9	39
Extent of market	coastal / regional / national	35
Discount rate	1.5% / 3.5% / 5.5%	44

This yields $3 \times 2 \times 2 \times 3 \times 3 \times 3 = 324$ defensible specifications. The results:

Statistic across specifications	Value
Minimum NPV	**−€91.4m**
Median specification	−€27.9m
Maximum NPV	**+€312.6m**
5th / 95th percentile	−€82.2m / +€130.8m
Share passing (NPV > 0)	**35%**

(Verified.) The verdict is *not determined by the data*. It ranges from a €91m loss to a €313m gain across choices no reviewer could individually reject, and *two-thirds of the defensible specifications fail*. **A single reported NPV is therefore nearly uninformative about the project unless the reader also knows where in this distribution it sits and why that specification was chosen.**

What to do. The specification curve is not a counsel of despair; it is the honest object to report. The discipline:

1. **Pre-specify a primary specification** *before* seeing results, with each choice justified on substantive grounds — this is the number you defend (Chapter 33's confirmatory analysis).
2. **Report the full specification curve** as the robustness analysis, so the reader sees the range and the share passing.
3. **Identify which choices drive the result** — verified, here the extent of market, discount rate, and central-tendency choices dominate — connecting to the tornado of Chapter 45.
4. **Distinguish defensible from indefensible choices** — the multiverse includes only choices a competent analyst could justify; genuinely wrong options (e.g. counting transfers, Chapter 43) are excluded, not multiversed.
5. **Never present the most favourable specification as the result** — the cardinal sin, and the one the specification curve exists to prevent.

Stop and think 46.1. Your FRD appraisal's primary specification gives NPV = +€40m, but the specification curve shows only 35% of defensible pipelines pass. How do you report this honestly? *(Answer: I report both, and I lead with the tension. The honest presentation: "Under the pre-specified primary specification — [median WTP, log-normal, $\tau = 8$ calibration, regional market, 3.5% discount rate, protests bounded] — the NPV is +€40m. However, the result is* not robust *to defensible analytic choices: across all 324 specifications the NPV ranges from −€91m to +€313m and only* ***35% pass****, so the majority of defensible pipelines return a negative verdict." Then: (i)* state that the primary specification was pre-registered before seeing results*, so it is not cherry-picked — without that, a +€40m result sitting in the passing minority would be indistinguishable from specification search; (ii)* identify the drivers* of the spread (extent of market, discount rate, central tendency), so the reader knows the verdict hinges on the aggregation boundary and the ethical discount-rate choice rather than on the data; (iii) *cross-reference the risk analysis* of Chapter 45 — this analytic fragility compounds the parameter uncertainty (P(NPV<0)≈ 50%), and both point to the same conclusion: the project is genuinely marginal and the honest verdict is conditional, not a confident pass; (iv) *resist the temptation* to quietly promote the +€313m specification or to bury the 35% figure. The specification curve is the deliverable that protects both me and the decision-maker: it converts hidden flexibility into a disclosed range, so the reader can see that "+€40m" is one defensible reading among many, most of which are less favourable.)*

46.4 Computational reproducibility in practice

Making a result computationally reproducible is an engineering problem with well-established solutions. The goal: *anyone with the materials can regenerate every number, figure, and table exactly.*

The essential practices.

- **Version control.** Track every change to code and text (Git), so the exact state producing any result is recoverable and the provenance of every number is auditable. Commit the analysis, not just the final script.
- **Literate programming.** Bind code, output, and narrative in one document that *executes* to produce the report — RMarkdown, Quarto, or Jupyter. Your RMarkdown/pandoc workflow is exactly this: the tables and figures are *generated* by the embedded code, not pasted in, so they cannot silently drift out of sync with the analysis. This eliminates the single most common reproducibility failure — a reported number that no longer matches the code that supposedly produced it.
- **Fixed and reported random seeds.** Any simulation (Monte Carlo appraisal, Krinsky–Robb, mixed-logit draws) must set an explicit seed *and report it*, so the stochastic result is exactly regenerable.
- **Adequate and reported draw counts.** As §46.5 shows, the seed alone is insufficient — the number of draws must be large enough that the result is stable, and both must be reported.
- **Environment capture.** Record software versions and dependencies (a lockfile, `renv`, a container), because results can differ across package versions and platforms. "It worked on my machine" is not reproducibility.
- **Open data and code.** Deposit the data (or a synthetic/access-controlled equivalent where confidentiality requires) and the code in a persistent public repository with a DOI. Your appraisal's credibility is proportional to a reviewer's ability to *run it*.
- **Provenance and a data dictionary.** Document every transformation from raw data to final number, and define every variable. The reader must be able to trace the €40m NPV back through the aggregation, the welfare estimation, and the survey responses.

The literate-programming payoff. The reason your RMarkdown workflow matters is not tidiness. When the report *is* the code, three failures become impossible: (i) the reported number cannot differ from the computed number (they are the same object); (ii) a change to the analysis automatically propagates to every dependent table and figure, so partial updates cannot leave stale results; (iii) a reviewer can re-execute the entire document and confirm every number. This is why the practice is load-bearing rather than cosmetic, and why adopting it — as you have — is among the highest-leverage decisions in the whole workflow.

46.5 The seed is part of the result

A subtle but critical point: for a stochastic analysis, "the code" is not enough for reproducibility — the seed and the draw count are *part of the specification*, and an inadequate draw count makes the result non-reproducible even *with* a fixed seed across analysts who choose different seeds.

Worked Example 46.2 (Seed and draw-count dependence, verified). The Chapter 45 Monte Carlo failure probability, computed across 30 different seeds at three draw counts:

Draws R	P(NPV < 0) across seeds	Range
200	50.6% $\pm$ 4.2	42.0% to 61.5%
2,000	49.4% $\pm$ 1.5	46.3% to 53.3%
200,000	49.8% $\pm$ 0.1	49.5% to 50.0%

(Verified.) At 200 draws the *same code* produces a headline failure probability anywhere from 42% to 62% depending purely on the seed — a range wide enough to change the tone of the recommendation. At 200,000 draws the result is stable to a tenth of a percentage point regardless of seed. **Two lessons:**

1. **Report the seed** so the exact number is regenerable (computational reproducibility in the strict sense).

2. **Use and report an adequate draw count** so the result is *robust to the seed* — otherwise the number is an artefact of an arbitrary choice, and a reader who re-runs with a different seed (as an honest replication attempt) will fail to reproduce it. The draw count must be large enough that the simulation error is negligible relative to the decision (Chapter 45's simulation-versus-parameter-uncertainty distinction).

The general principle. A stochastic result is reproducible only if *both* the seed is fixed (for exact regeneration) *and* the draw count is adequate (for stability across seeds). Reporting one without the other is incomplete. This applies to every simulation in your work: the mixed-logit SML (Chapter 38), the MDCEV welfare CV (Chapter 41), the Krinsky–Robb intervals (Chapter 13), and the appraisal Monte Carlo (Chapter 45).

Active recall 46.B. Why is fixing the random seed insufficient for reproducibility, and what else must be reported? *(Answer: fixing the seed guarantees only* strict* computational reproducibility — that an analyst running the *identical* code with the *identical* seed gets the *identical* number. But it does not make the result *robust*: if the draw count is too small, the number is dominated by simulation noise, so a different (equally valid) seed produces a materially different answer. Verified: at R=200 draws the headline P(NPV<0) ranged from 42% to 62% across 30 seeds — a spread wide enough to change the recommendation — whereas at R=200,000 it was stable at 49.5–50.0%. So a reader attempting an honest replication with their own seed would *fail to reproduce* an under-powered result even though the code is correct, because the reported number was an artefact of one arbitrary seed. What must additionally be reported: the *draw count* R, chosen large enough that simulation error is negligible relative to the decision (so the result is stable across seeds), and ideally evidence of that stability. The complete reproducibility specification for a stochastic result is: same code + same seed + adequate-and-reported R + same software environment → same number, with the result additionally *insensitive* to the seed at that R.)*

46.6 Independent verification and error-catching

Reproducibility ensures a result *repeats*; it does not ensure the result is *correct*. Errors — transcription mistakes, off-by-one indices, sign errors, unit mismatches — survive reproduction perfectly, reappearing identically every time the code runs. Catching them requires *independent verification*.

Worked Example 46.3 (Checksum re-implementation, verified). Computing a 40-year annuity present value two independent ways:

Method	Result
Closed form $\frac{1-(1+r)^{-T}}{r}$	21.355072
Explicit loop $\sum_{t=1}^{T}(1+r)^{-t}$	21.355072
Agree?	**Yes**
Buggy off-by-one loop $\sum_{t=0}^{T-1}$	22.102500 — **disagrees by 3.5%: error caught**

(Verified.) An independent re-implementation of the same quantity agrees when the code is correct and *disagrees* when there is a bug — here an off-by-one error (discounting from year 0 rather than year 1) that inflates the annuity by 3.5% and would silently overstate every benefit. **This is precisely the "verify every worked number computationally before writing it" discipline this entire book has followed**: every table in every chapter was regenerated by independent code and checked against the derivation, which is why errors (the several caught and fixed along the way) surfaced before they reached the page.

Practices for error-catching.

- **Independent re-implementation** of key numbers, ideally by a second person or in a different language/tool.

- **Sanity and unit checks** — do magnitudes, signs, and units make sense? A WTP of €4,000/household or a negative probability signals a bug.
- **Limiting-case tests** — does the model reduce correctly in known special cases (MDCEV collapsing to MNL when $M = 1$, Chapter 40; the mixed logit collapsing to MNL when heterogeneity is zero, Chapter 38)?
- **Regression tests** — a suite of known-answer checks re-run after every change, so a modification that breaks a previously-correct result is caught immediately.
- **Code review** — a second analyst reads the pipeline, which catches errors that testing misses and undocumented choices that the author has stopped seeing.

The high-stakes case. For a decision-critical appraisal feeding a real funding decision, this book's standing recommendation applies: verification of the central numbers should be done *independently* — a separate re-implementation, or a colleague's review — rather than trusting a single pipeline. The cost is small; the cost of a sign error in a number that moves millions of euros of blue finance is not.

46.7 Pre-registration and its limits for appraisal work

Pre-registration — specifying the analysis plan before seeing the data — is the strongest defence against the garden of forking paths, but its applicability to appraisal and secondary-data work is nuanced.

What pre-registration does. By committing to the primary specification, the exclusions, the distribution, the tests, and the decision rule *in advance*, pre-registration converts *exploratory* analysis (where flexibility invalidates inference, Chapters 32–33) into *confirmatory* analysis (where the pre-specified test is valid). It is the mechanism that makes the primary specification of §46.3 defensible against the charge of cherry-picking.

Where it applies cleanly. Your *primary data collection* — the CV and DCE surveys — can and should be pre-registered: the sampling plan, the elicitation format, the exclusion criteria, the certainty threshold, the model specification, the primary hypotheses, and the equivalence margins (Chapter 33). This is straightforward and high-value.

Where it is harder. Much appraisal work is *secondary* — it uses existing data, meta-analytic estimates (Chapters 47–48), and values transferred from other studies (Chapter 49). Pre-registering an analysis of data you have already seen is not credible, and appraisal necessarily involves iterative model-building that resists rigid pre-specification. Here the honest alternatives are:

- **A transparent analysis protocol** written and time-stamped as early as possible, distinguishing pre-specified from post-hoc choices.
- **The specification curve** (§46.3) as the substitute for pre-registration when the latter is impossible — it discloses the flexibility that pre-registration would have constrained.
- **Explicit labelling** of every result as confirmatory (pre-specified) or exploratory (data-driven), so the reader can calibrate their confidence (Chapter 32).
- **A clear separation** between the *hypothesis-generating* and *hypothesis-testing* uses of the same data, never presenting the former as the latter.

The realistic standard. For your work, pre-register the *survey and its primary analysis*; for the *appraisal built on it*, write an early protocol, report the specification curve, label confirmatory versus exploratory, and make everything computationally reproducible. Pre-registration is not always possible, but *transparency about what was and was not pre-specified* always is.

> **Stop and think 46.2.** You can pre-register your DCE survey analysis but not the FRD appraisal that reuses existing cost data and transferred values. What is the credible reproducibility strategy for the appraisal? *(Answer: pre-registration is not credible for an analysis of data already seen, so the strategy substitutes* transparency* for *commitment* on the appraisal side while retaining pre-registration where it genuinely applies. (i) *Pre-register what can be*: the DCE/CV survey — sampling, elicitation, exclusions, certainty threshold, model specification, primary hypotheses,

equivalence margins — since these precede data collection. (ii) *Write an early, time-stamped analysis protocol for the appraisal*, distinguishing choices fixed in advance from those that will be data-driven, so the reader knows which is which. (iii) *Report the full specification curve* (verified: 324 pipelines, −€91m to +€313m, 35% passing) as the disclosure that pre-registration would otherwise have provided — it converts the hidden flexibility into an explicit reportable range, which is the whole function pre-registration serves. (iv) *Label every appraisal result as confirmatory or exploratory*, never dressing a data-driven choice as a pre-specified test. (v) *Make the entire appraisal computationally reproducible* — code, data (or synthetic/access-controlled equivalent), seeds, draw counts, environment — so that whatever choices were made are at least fully auditable and re-runnable. (vi) *Document provenance* for every transferred value and cost input, so the chain from source to NPV is traceable. The credible standard is: pre-register the primary data work, and for the secondary appraisal achieve transparency and computational reproducibility plus an honest specification curve — because I cannot claim the appraisal was pre-specified, but I can make every choice visible and every number regenerable.)*

46.8 The appraisal document: presenting a conditional verdict

The culmination of Parts VIII–XI is a document that must convey, to a non-technical decision-maker, a verdict that is *conditional, uncertain, and ethically loaded* — without either collapsing it into false certainty or drowning it in caveats.

The structure. A defensible appraisal report contains:

1. **Executive summary** — the headline verdict *as a conditional and a range*, not a point: "positive under [primary assumptions], with a [X]% probability of a negative outcome, switching sign at [key parameter values]."
2. **The question and the counterfactual** — what is being appraised, against what without-project baseline (Chapter 43).
3. **Standing and scope** — whose costs and benefits count, over what horizon (Chapters 43–44).
4. **Methods** — the valuation methods, models, and their key assumptions, with the pre-specified primary specification stated.
5. **Results** — the primary NPV, the *risk profile* (P(NPV < 0), distribution — Chapter 45), the *specification curve* (Chapter 46), and the *distributional incidence* (Chapter 43).
6. **Sensitivity and key uncertainties** — the tornado, switching values, the discount-rate range, and the value of information (Chapters 44–45).
7. **Limitations** — honestly stated, including hypothetical bias, transfer errors, and deep uncertainty.
8. **Reproducibility appendix** — data, code, seeds, environment, and the analysis protocol.

The reporting principles.

- **Lead with the conditional.** The verdict is a function of ethical and empirical parameters; say so up front rather than reporting a number and burying the conditions.
- **Report ranges and probabilities, not points.** P(NPV < 0), the specification curve, and the switching values are the honest summary (Chapters 44–45).
- **Separate the ethical from the empirical.** Make explicit which parts of the verdict depend on value judgements (the discount rate, distributional weights, standing) and which on data (Chapters 43–44).
- **Distinguish confirmatory from exploratory** (§46.7).
- **Make it reproducible and auditable** so the reader can check every claim (§46.4–§46.6).
- **Resist advocacy.** The analyst's role is to reveal the structure of the decision — the conditions under which the project is worthwhile — not to manufacture a recommendation. A report that presents the most favourable specification, hides the failure probability, and buries the ethical parameters is advocacy dressed as analysis, and the specification curve and risk profile exist precisely to prevent it.

For your work. The FRD appraisal that feeds blue-finance decisions carries real consequences, and its credibility rests entirely on the reader's ability to see that it is honest. The integrated verdict — "positive under the primary specification, but marginal, with a ~50% failure probability, only 35% of defensible specifications passing, driven by the aggregation boundary and the discount rate, worth resolving via a pilot given the €18m value of information, with a valuable option to wait" — is far more useful, and far more defensible, than "NPV = €40m, proceed." The former survives scrutiny; the latter invites the question of what was hidden.

46.9 Failure modes

F1. Reporting one specification as the result. Concealing the multiverse. *Fix:* pre-specify the primary; report the specification curve (§46.3).

F2. Presenting the most favourable specification. Cherry-picking dressed as analysis. *Fix:* pre-registration or protocol; full curve (§46.3, §46.7).

F3. Unreported seed. A stochastic result that cannot be regenerated. *Fix:* fix and report the seed (§46.4–§46.5).

F4. Inadequate draw count. A result that varies with the seed. *Fix:* adequate, reported R; verify stability (§46.5).

F5. Numbers pasted, not generated. Reported values drifting from the code. *Fix:* literate programming (§46.4).

F6. No environment capture. Results differing across versions. *Fix:* lockfile / container (§46.4).

F7. No independent verification. Errors surviving reproduction. *Fix:* re-implement key numbers; limiting-case and regression tests (§46.6).

F8. Pre-registering seen data. A non-credible commitment. *Fix:* protocol; label confirmatory versus exploratory; specification curve (§46.7).

F9. False certainty in the report. A point NPV presented as the answer. *Fix:* conditional verdict with ranges and probabilities (§46.8).

F10. Advocacy. Manufacturing a recommendation. *Fix:* reveal the decision structure; separate ethical from empirical (§46.8).

46.10 Diagnostic questions for your own papers

1. **Multiverse.** Have I reported the specification curve and pre-specified the primary analysis? *(§46.3, F1–F2.)*
2. **Seed and draws.** Are the seed and an adequate draw count fixed and reported, with stability verified? *(§46.5, F3–F4.)*
3. **Literate pipeline.** Are all numbers generated by executed code, not pasted? *(§46.4, F5.)*
4. **Environment.** Is the software environment captured? *(§46.4, F6.)*
5. **Verification.** Have key numbers been independently re-implemented and tested? *(§46.6, F7.)*
6. **Pre-registration.** Is primary data collection pre-registered, and appraisal choices labelled confirmatory/exploratory? *(§46.7, F8.)*
7. **Conditional verdict.** Does the report present ranges, probabilities, and the ethical/empirical separation, not a false point? *(§46.8, F9.)*
8. **Non-advocacy.** Does the report reveal the decision structure rather than push a recommendation? *(§46.8, F10.)*

46.11 Exercises

(A) Comprehension.

A1. Distinguish computational reproducibility, replicability, and robustness.

A2. What is a specification curve and what does it reveal?

A3. Why must both the seed and the draw count be reported for a simulation?

(B) Derivation / reasoning.

B1. Explain why the number of defensible specifications grows combinatorially and why this threatens credibility.

B2. Explain why literate programming eliminates a class of reproducibility failures that discipline alone cannot.

B3. Explain why pre-registration is not credible for already-seen data and what substitutes for it.

(C) Research application / critique.

C1. Design the complete reproducibility protocol for your FRD appraisal and DCE analysis.

C2. Your primary NPV passes but only 35% of specifications do. Write the executive summary.

C3. A reviewer cannot reproduce your Monte Carlo result. Diagnose the likely causes and the fixes.

46.12 Solutions

A1. *Computational reproducibility*: same data + same code → same result (repeatability of the specific computation; the minimum standard, testing repeatability not correctness). *Replicability*: new data from the same population + same method → same conclusion (tests whether the finding is a property of the world; the stronger scientific standard). *Robustness / analytical reproducibility*: the result survives defensible variations in analytic choices applied to the same data (tested by the specification curve). The three form a hierarchy of increasing stringency and require different things — engineering (computational), new studies (replication), and transparency about choices (robustness) — and a credible appraisal aims for all three.

A2. A **specification curve** displays the results of a **multiverse analysis** — the outcome computed under *every combination* of defensible analytic choices — sorted from lowest to highest, annotated with which choices produced each. It reveals (i) the *range* of results the data permit across honest pipelines, (ii) the *share* reaching each conclusion (e.g. passing versus failing), and (iii) *which choices drive* the variation. Verified: for the FRD appraisal, 324 defensible specifications gave NPVs from −€91m to +€313m with only 35% passing, so the curve reveals that the verdict is *not determined by the data* and that a single reported number is uninformative without knowing where in this distribution it sits and why.

A3. Because reproducibility of a stochastic result requires *both* exact regeneration *and* robustness to the arbitrary seed choice. The **seed** must be reported so that an analyst running the identical code obtains the identical number (strict computational reproducibility). But the seed alone is insufficient: if the **draw count** is too small, the number is dominated by simulation noise and changes with the seed, so a reader attempting an honest replication with a *different* seed will fail to reproduce it. Verified: at R=200 draws the headline P(NPV<0) ranged 42–62% across seeds, but at R=200,000 it was stable at 49.5–50.0%. So the draw count must be large enough that simulation error is negligible relative to the decision, and reported alongside the seed, with stability across seeds ideally demonstrated.

B1. At each analytic decision the researcher chooses among k_i defensible options; with d decisions the number of distinct pipelines is $\prod_{i=1}^{d} k_i$, which grows *multiplicatively*. Verified: six decisions with $(3, 2, 2, 3, 3, 3)$ options yield $3 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 3 = 324$ specifications. This threatens credibility because (i) the results scatter across a wide range (−€91m to +€313m), so the data weakly constrain the verdict; (ii) the researcher typically reports *one* pipeline, giving the reader no sense of the distribution it was drawn from; and (iii) the reported pipeline is *correlated* with the result through the sequence of locally-reasonable choices that led to it (the garden of forking paths), so even an honest analyst cannot certify from the inside that their number is representative rather than a favourable draw. The reader therefore cannot distinguish a robust finding from a cherry-picked one — which is why the flexibility must be *disclosed* via the specification curve rather than resolved by judgement.

B2. Literate programming binds code, output, and narrative into a single document that *executes* to produce the report, so the reported number and the computed number are *the same object* rather than two things kept in sync by hand. This eliminates failures that discipline alone cannot reliably prevent because they are failures of *human synchronization*: (i) a number computed by the code but *transcribed incorrectly* into the prose; (ii) an analysis that is *updated* while a dependent table, computed earlier, is left *stale*; (iii) a figure regenerated from different code than the text describes. Discipline can reduce these but not eliminate them, because they arise precisely when attention lapses. Under literate programming they are *structurally impossible*: the table *is* the output of the code that produced it, updating the analysis *automatically* propagates to every dependent result, and a reviewer can *re-execute* the whole document to confirm every number. This is why your RMarkdown/pandoc workflow is load-bearing rather than cosmetic — it removes an entire class of error by construction.

B3. Pre-registration derives its power from being a *commitment made in ignorance of the data* — by fixing the analysis before seeing outcomes, it makes the pre-specified test *confirmatory* (valid inference) rather than *exploratory* (invalidated by flexibility). Once the data have been seen, this power evaporates: a "pre-registration" written after inspecting the data cannot credibly claim ignorance of them, and the analyst's choices are already contaminated by knowledge of what those choices produce (consciously or not). So pre-registering already-seen data — common in secondary/appraisal work using existing cost data, meta-analytic estimates, and transferred values — is not credible. The substitutes: (i) an *early, time-stamped analysis protocol* distinguishing pre-specified from post-hoc choices; (ii) the *specification curve*, which discloses the flexibility that pre-registration would have constrained, achieving the same *transparency* goal by a different route; (iii) *explicit labelling* of every result as confirmatory or exploratory; and (iv) full *computational reproducibility* so all choices are at least auditable. The realistic standard: pre-register the primary data collection, and for the secondary appraisal substitute transparency (protocol + specification curve + labelling) for the commitment that is no longer possible.

C1. *Reproducibility protocol.* **Version control**: the entire analysis — survey processing, estimation, appraisal — under Git, with the exact commit recorded for every reported result. **Literate programming**: the whole report as RMarkdown/Quarto rendered through pandoc (my existing workflow), so every table, figure, and number is *generated* by embedded code and cannot drift from the analysis. **Seeds and draws**: every stochastic step (mixed-logit SML, Chapter 38; MDCEV welfare CV, Chapter 41; Krinsky–Robb, Chapter 13; appraisal Monte Carlo, Chapter 45) sets and *reports* an explicit seed, with a draw count large enough that the result is stable across seeds — verified necessary, since at R=200 the failure probability swung 42–62% — and stability demonstrated. **Environment capture**: a lockfile (`renv`) or container recording all package versions, since results can differ across versions. **Open materials**: data (or a synthetic/access-controlled equivalent where survey confidentiality requires) and code deposited in a persistent repository with a DOI. **Provenance and data dictionary**: every transformation from raw survey response to final NPV documented, every variable defined, so the €40m can be traced to source. **Pre-registration**: the *DCE/CV survey and its primary analysis* pre-registered (sampling, elicitation, exclusions, certainty threshold, model, primary hypotheses, equivalence margins); the *appraisal* governed by an early time-stamped protocol with confirmatory/exploratory labelling. **Specification curve**: the multiverse of defensible appraisal choices computed and reported (verified 324 specifications, 35% passing) as the robustness disclosure substituting for pre-registration on the secondary side. **Independent verification**: key numbers re-implemented independently (a second person or a different tool), with limiting-case tests (MDCEV→MNL, mixed-logit→MNL)

and regression tests re-run after every change — the verify-before-you-write discipline of this book, verified here to catch a 3.5% off-by-one error. **Reproducibility appendix**: all of the above documented in the report so a reviewer can regenerate every number.

C2. *Executive summary.* "Under the pre-specified primary specification, the FRD project has a net present value of **+€40m**, indicating that it would increase aggregate welfare. However, this verdict is **marginal and conditional**, and three findings qualify it. **First, it is not robust to analytic choices**: across 324 defensible specifications — varying the treatment of protest responses, the WTP measure, the calibration of hypothetical bias, the population boundary, and the discount rate, each choice individually justifiable — the NPV ranges from **−€91m to +€313m**, and only **35% of specifications yield a positive verdict**. The primary specification sits in the passing minority; the majority of defensible analyses return a negative result. **Second, it is not robust to parameter uncertainty**: propagating the statistical uncertainty in the underlying estimates gives an approximately **50% probability of a negative outcome**, with a 5th-percentile loss of around €70m. **Third, the verdict depends on contested value judgements** — principally the social discount rate (the project passes below ~5.6% and fails above it) and the population boundary — which are ethical choices, not empirical facts. **The result is driven by** the extent-of-market and discount-rate choices, not by the ecological or valuation data, which are comparatively well-identified. **Recommendation**: the project is genuinely marginal rather than a clear pass. Because the value of resolving the key uncertainties is high (up to ~€18m) and the intervention is partly irreversible, the analytically preferred course is a **time-bounded pilot** to resolve the dominant uncertainties before full commitment, rather than immediate approval or rejection. Full methods, the specification curve, the risk profile, and reproducible code are in the appendices." *(This leads with the conditional, reports ranges and probabilities, separates ethical from empirical, and resists advocacy — the §46.8 principles.)*

C3. *Diagnosis and fixes for a non-reproducible Monte Carlo.* Likely causes, in order of probability: **(1) Unreported or unset seed.** If no seed was fixed, every run differs; if the seed was set but not *reported*, the reviewer cannot match it. *Fix*: set an explicit seed and report it in the code and text. **(2) Inadequate draw count.** Even with the code, if the reviewer used a different seed (a legitimate replication attempt) and R is too small, they get a different number — verified, R=200 swings the result 42–62%. *Fix*: increase R until the result is stable across seeds (verified negligible at R=200,000) and report both R and the stability check. **(3) Environment differences.** Different versions of the random-number generator, NumPy/R, or the platform can change draws even with the same seed, because RNG algorithms and defaults vary across versions. *Fix*: capture the environment (lockfile/container) and report versions. **(4) RNG stream details.** Using a legacy global RNG (rather than an explicit generator object), drawing variables in a different order, or parallelization that consumes the stream non-deterministically will all break exact reproduction. *Fix*: use an explicit, seeded generator; fix the draw order; make parallel draws deterministic (per-stream seeds). **(5) Undocumented pipeline choices.** The reviewer may be reproducing a *different specification* (different exclusions, distribution, or aggregation) because those choices were not fully documented — a robustness rather than strict-reproducibility failure. *Fix*: document every choice; provide the exact code (literate programming makes this automatic). **(6) Numbers pasted rather than generated.** The reported figure may never have matched the current code. *Fix*: literate programming, so the number is the code's output. *General remedy*: provide the full executable document, seed, R, environment lockfile, and data so the reviewer runs *exactly* my pipeline — and demonstrate seed-stability so that reproduction does not depend on matching an arbitrary seed. If after all this the reviewer still cannot reproduce it, the discrepancy is itself diagnostic and must be resolved before the result is trusted.

46.13 Spaced review prompts

+1 day. From memory: the three reproducibility standards; what a specification curve reveals; why the seed and draw count must both be reported.

+1 week. Explain the garden of forking paths and the combinatorial growth of specifications; why literate programming eliminates a class of errors; why pre-registration fails on seen data.

+1 month. For your FRD appraisal and DCE analysis, implement the full protocol — version control, literate report, seeds and draws, environment capture, independent verification, specification curve, confirmatory/exploratory labelling — and draft the conditional executive summary. Close with the spine sentence about whether an independent analyst could reproduce your number and see every choice.

46.14 How this chapter connects to the rest of the book

- This chapter *disciplines* everything: the researcher degrees of freedom flagged in **Chapters 32–33** (protest exclusion, specification search, multiplicity) become the axes of the multiverse; the choices of **Chapters 35, 36, 39, 44** (extent of market, distribution, calibration, discount rate) populate the specification curve; the Monte Carlo of **Chapter 45** is the stochastic result whose seed and draw count must now be documented.
- **Chapter 11** supplies the pseudo-random-number and seed machinery; the "verify every number" practice modelled throughout the book is here made explicit as §46.6's independent-verification discipline.
- Downstream: **Chapters 47–48 (meta-analysis)** are themselves a reproducibility and synthesis exercise — combining studies whose individual reproducibility varies — and **Chapter 49 (benefit transfer)** requires the provenance discipline of §46.4 to trace transferred values to their sources.

This chapter closes Part XI by making the appraisal *defensible*: not merely a number, but a number an independent analyst can regenerate, whose full range of defensible alternatives is disclosed, and whose ethical and empirical conditions are explicit — the difference between analysis and advocacy.

46.15 Research translation task

For your **FRD appraisal and DCE analysis**, produce a reproducibility-and-reporting package:

1. **Reproducibility protocol.** Version control, literate report (RMarkdown/pandoc/Quarto), seeds and reported draw counts, environment lockfile, open data/code with DOI, data dictionary and provenance.
2. **Specification curve.** Compute and report the multiverse of defensible appraisal choices, with the pre-specified primary specification identified and the share passing reported.
3. **Verification.** Independently re-implement key numbers; include limiting-case and regression tests.
4. **Pre-registration and labelling.** Pre-register the survey and primary analysis; label every appraisal result confirmatory or exploratory.
5. **Conditional report.** Structure the appraisal document per §46.8, leading with the conditional verdict, ranges, probabilities, and the ethical/empirical separation.
6. **Close** with the spine sentence: *"Every number in this appraisal is generated by version-controlled code with seed [·] and [·] draws in a captured environment; the primary specification (pre-registered / protocol-specified) gives NPV €[·], and the specification curve across [·] defensible pipelines shows [·]% passing — all materials are deposited at [DOI] for independent reproduction."*

Carry this into Chapters 47–49, where the provenance and reproducibility discipline extends to the studies you synthesize and the values you transfer.

46.16 Bibliography

Core. - Peng (2011), "Reproducible Research in Computational Science," *Science* — the foundational statement of computational reproducibility (§46.2, §46.4). *(Verify volume/pages.)* - Simonsohn, Simmons & Nelson (2020), "Specification Curve Analysis," *Nature Human Behaviour* — the specification-curve method

(§46.3). *(Verify volume/pages.)* - Steegen, Tuerlinckx, Gelman & Vanpaemel (2016), "Increasing Transparency Through a Multiverse Analysis," *Perspectives on Psychological Science* — the multiverse framework (§46.3). *(Verify volume/pages.)*

Applied. - Gelman & Loken (2013), "The Garden of Forking Paths" — researcher degrees of freedom without intentional p-hacking (§46.3). *(Verify publication details.)* - Simmons, Nelson & Simonsohn (2011), "False-Positive Psychology," *Psychological Science* — the original researcher-degrees-of-freedom demonstration (§46.3). *(Verify volume/pages.)* - Xie, *Dynamic Documents with R and knitr*, and the Quarto/RMarkdown documentation — literate programming in practice (§46.4). *(Verify edition.)*

Advanced. - Nosek et al. (2018), "The Preregistration Revolution," *PNAS* — pre-registration and its scope (§46.7). *(Verify volume/pages.)* - Munafò et al. (2017), "A Manifesto for Reproducible Science," *Nature Human Behaviour* — the integrated reproducibility agenda. *(Verify volume/pages.)* - Stodden, Leisch & Peng (eds.), *Implementing Reproducible Research* — practical infrastructure (§46.4). *(Verify edition.)*

Entries marked "verify" are pointers; confirm volumes, pages, and the Simonsohn / Steegen / Gelman–Loken / Nosek citations before using them, per the book's no-unverified-citation rule.

End of Chapter 46, and of Part XI. Next: Part XII — Evidence Synthesis. Chapter 47 turns to meta-analysis, the systematic synthesis of your 18 finfish-WTP DCEs: the fixed- versus random-effects distinction, heterogeneity and its measurement, the meta-regression of WTP on study characteristics, and the publication-bias and dependence problems that make environmental-valuation meta-analysis distinctively difficult. Tell me when to write it, or request a different chapter.

Part XI Assessment — Cost–Benefit Analysis

Covers Chapters 43–46. Five components, in the order they are best attempted. Allow a full working day; the mock oral is the part that matters most for your viva.

1. Cumulative problem set

P1. For your FRD project, construct both FNPV and ENPV and explain each line that differs.

P2. Decompose a social discount rate of 3.5% into Ramsey components and state the ethical content of each.

P3. Compute the 100-year discount factor at 1%, 3.5% and 7%, and express the range as a ratio.

P4. Build a switching-value table over the discount rate, the WTP calibration factor, and the benefit ramp-up period.

P5. Run a Monte Carlo over three uncertain inputs and report the probability that ENPV < 0, stating your correlation assumptions.

2. Research critique

Take the CBA in your FRD paper and write the benefit-accounting table: every benefit stream, the physical quantity behind it, the value attached, the source, and the other categories it could overlap with. Identify any double count and any stock–flow inconsistency.

3. Red-flag diagnosis

An appraisal reports a single ENPV of €4.2m at a 3.5% discount rate, with a sensitivity annex showing $\pm 10\%$ on capital costs. The benefits are 80% blue-carbon accruing over 80 years. Explain why the sensitivity annex is close to worthless and what it should have contained.

4. Mock oral examination

1. What is the difference between FNPV and ENPV, and what does the gap mean?
2. Justify your discount rate to a hostile referee.
3. What is a shadow price and when do you need one?
4. What does Kaldor–Hicks actually require, and does your project satisfy it?
5. Which of your benefit numbers is not yet a welfare measure?

5. Integration mini-project

Build the complete, reproducible appraisal for the FRD paper: the cash-flow model, the NPV(r) curve with switching values, the Monte Carlo risk profile, and the benefit-accounting table — all in a `targets` pipeline that regenerates every figure and table on one command, with a written appraisal document a reviewer could audit.

Part XII — Evidence Synthesis

Chapter 47 — Meta-Analysis of Valuation Studies

Part XII: Evidence Synthesis

> **The spine question, applied to this chapter.** Your meta-analysis synthesizes 18 finfish-WTP discrete choice experiments into a single evidence base. The naive goal — "what is the average WTP?" — hides the question that actually matters: *are these 18 studies estimating the same quantity, or 18 different quantities?* Verified below, they are estimating different ones — 81% of the variation across them is genuine between-study heterogeneity, not sampling noise. That single fact reshapes everything: it makes the fixed-effect confidence interval spuriously narrow (verified ±3.5 against the honest ±8.7), makes the *prediction interval* rather than the confidence interval the object relevant for transferring a value to a new site (verified [€5, €75] versus a pooled point of €40), and turns the analysis from *averaging* into *explaining variation*. The spine question: *is my meta-analysis pooling estimates as if they measured one thing, or modelling why they measure different things — and does my uncertainty statement reflect the heterogeneity I have, the dependence among my estimates, and the studies I never saw?*

47.0 Conceptual motivation: synthesis is modelling variation, not averaging

A meta-analysis looks, superficially, like a weighted average: collect the WTP estimates, weight them by precision, report the pooled value. If that were all it was, this chapter would be a paragraph. The reason it is a chapter — and the reason meta-analysis of valuation studies is genuinely hard — is that the 18 studies do *not* estimate a common WTP. They differ in the good valued (which finfish species, what improvement), the population (which country, coastal or inland, richer or poorer), the method (CV or DCE, hypothetical or incentivized, which elicitation format), the year, and a dozen other dimensions. The variation *among* their estimates is not error to be averaged away; it is the primary object of study.

Verified below, this is not a subtle effect. Applying the standard heterogeneity diagnostics to 18 finfish-WTP estimates gives $I^2 = 81\%$ — meaning 81% of the total variation across studies is *real between-study heterogeneity*, and only 19% is within-study sampling noise. Cochran's Q rejects homogeneity decisively ($p < 0.001$), and the estimated between-study standard deviation is €16 — larger than several of the individual studies' point estimates. **The 18 studies are estimating 18 different true WTPs**, drawn from a distribution, and the meta-analytic task is to characterize that distribution and explain its spread, not to collapse it to a point.

This reframing has four consequences the chapter develops, each verified.

The model choice is not innocuous. A *fixed-effect* model assumes one common true WTP and weights studies purely by precision; a *random-effects* model assumes a distribution of true WTPs and adds between-study variance to every weight. Under heterogeneity they diverge sharply — verified, the random-effects

confidence interval (±€8.7) is more than twice the fixed-effect interval (±€3.5), because the fixed-effect model *ignores* the heterogeneity and reports a spuriously precise pooled value. Using fixed effects under $I^2 = 81\%$ would mean claiming far more certainty than the evidence supports.

The prediction interval, not the confidence interval, governs transfer. The confidence interval describes uncertainty about the *mean* of the distribution of true WTPs; the *prediction interval* describes where a *new* study's true WTP will fall. Verified, these differ enormously — a pooled mean of €40 with a confidence interval around it, but a prediction interval of [€5, €75]. When you transfer a value to your policy site (Chapter 49), it is the prediction interval that matters, and reporting only the tight confidence interval would badly overstate transfer precision.

Heterogeneity is to be explained, not just measured. Meta-regression relates the WTP estimates to *study characteristics* — verified, hypothetical studies report systematically higher WTP (the hypothetical-bias signal of Chapter 39, now visible across studies) and WTP rises with scope. This turns the meta-analysis into a *model of how WTP depends on context*, which is exactly what benefit transfer needs.

Two threats specific to valuation meta-analysis. *Publication bias / small-study effects* — verified, a funnel asymmetry (Egger $p < 0.001$) indicating that small, imprecise studies systematically report larger WTP, inflating the pooled estimate. And *dependence* — your 18 studies yield many more than 18 estimates (multiple attributes and scenarios per paper), and estimates within a study are correlated; verified, ignoring this inflates false precision, with a cluster-corrected standard error 2.10× the naive one.

47.1 Prerequisites

From **Chapter 19**: random effects, multilevel models, and within-cluster correlation — the statistical backbone of random-effects meta-analysis. From **Chapter 16**: weighted least squares and the inverse-variance weight. From **Chapter 12**: heterogeneity, and the chi-square distribution behind Cochran's Q. From **Chapter 39**: hypothetical bias, which appears here as a meta-regression moderator. From **Chapter 32**: selection, which underlies publication bias. From **Chapter 33**: multiplicity, relevant to testing many moderators.

New notation. θ_i (study i's true effect), $\hat{\theta}_i$ (its estimate), s_i (its within-study standard error), w_i (meta-analytic weight), τ^2 (between-study variance), Q (Cochran's heterogeneity statistic), I^2 (proportion of variation that is heterogeneity), PI (prediction interval).

47.1b Before any pooling: systematic search, coding, and inter-coder reliability

A meta-analysis inherits the biases of the set of studies it draws on and the judgments of whoever coded them. Both are research design decisions, and both must be documented before a single number is pooled. This section is the part of the workflow that happens *before* §47.2's effect sizes exist.

The search must be systematic, and reported so it can be repeated. The PRISMA framework (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) is the reporting standard, and its core demand is simple: a reader must be able to reconstruct how you got from "the literature" to "the studies in my table."

- **Databases and terms.** Name every database searched (Scopus, Web of Science, EconLit, AgEcon Search, Google Scholar), the exact query strings, and the date the search was run. Searches are not reproducible without the date, because databases grow.
- **Grey literature.** For environmental valuation this is not optional. A large share of DCE and CV work lives in agency reports, consultancy studies, working papers and theses, and it is systematically *less*

likely to report large or "publishable" values. Omitting grey literature is one of the mechanisms that produces the funnel asymmetry of §47.6 — so searching it is a bias-reduction step, not a completeness ritual.

- **Inclusion and exclusion criteria, pre-specified.** Which species, which welfare measure, which elicitation formats, which years, which languages. Write them down *before* screening, because criteria invented while looking at results are a specification search (Chapter 46).
- **The PRISMA flow.** Report the counts at each stage — records identified, duplicates removed, titles/abstracts screened, full texts assessed, studies included — with the reason for every full-text exclusion. For 18 included studies, the interesting number is usually how many were excluded and why.

The coding protocol is an instrument, and instruments need reliability evidence. Each included study must be reduced to a row (or several rows) of numbers: the WTP estimate, its standard error, the good, the population, the elicitation format, the year, the currency, the sample size, and every moderator you intend to use in §47.5. Every one of those fields involves judgment.

- **Write the protocol first**, with an explicit decision rule for each field, including how to handle the awkward cases: studies reporting a median but not a mean, studies reporting a confidence interval but not a standard error, studies reporting WTP per household per year alongside WTP per person per trip.
- **Double-code a subsample.** Have a second coder independently code at least 20% of the studies, blind to your coding.
- **Report inter-coder reliability.** For categorical fields use Cohen's κ; for continuous fields use the intraclass correlation coefficient (ICC) or the proportion of exact agreement. Conventionally $\kappa > 0.8$ is strong agreement, 0.6–0.8 acceptable, below 0.6 a signal that the field is ambiguous and the protocol needs rewriting rather than the disagreement being averaged away.
- **Adjudicate and document.** Disagreements are resolved by discussion against the protocol, and the protocol is amended. Report how many fields required adjudication — a field that needed adjudication in a third of studies is a moderator you should be cautious about interpreting in §47.5.

Harmonization: units, PPP, and inflation. Before pooling, every estimate must be expressed in the same money. The order of operations matters and is a common source of error:

1. **Convert to a common payment basis** — per household per year is the usual choice for CV, and requires knowing the study's payment period and payment unit. A one-off payment and an annual payment are not the same number; converting between them requires a discount rate and a time horizon, both of which are assumptions to state (Chapter 4).
2. **Deflate to a common base year** *in the original currency*, using that country's CPI or GDP deflator.
3. **Then convert currencies using PPP**, not market exchange rates, because you are comparing purchasing power over a domestic environmental good, not a traded commodity.

Doing (3) before (2), or using market exchange rates, introduces systematic error correlated with country and year — which will then show up as spurious "heterogeneity" and may load onto your moderators.

What this buys you. The rest of the chapter treats the studies as data. This section is what licenses that: a documented search means the sample is defined, a documented protocol with reliability evidence means the variables are measured, and documented harmonization means the effect sizes are commensurable. Skip it and every subsequent number — τ^2, I^2, the prediction interval, the meta-regression — is quantifying variation whose origin you cannot name.

> **Active recall 47.A0.** Your co-author proposes coding the 18 studies alone, quickly, "since the fields are objective." Name two fields that are not objective, and state what evidence a referee will expect instead. *(Answer: almost none are fully objective — but the clearest examples are the* ***standard error*** *(frequently reconstructed from a CI, a t-statistic, or a sample size, with a judgment about which is reliable) and the* ***good/scope coding*** *(whether a study's "fish stock improvement" is comparable in magnitude to another's, which is a substantive judgment that directly drives the scope moderator in §47.5). Others include welfare measure (WTP vs*

WTA, marginal vs total) and payment basis. The evidence a referee expects: a written protocol, independent double-coding of a subsample, a reported reliability statistic (Cohen's κ for categorical fields, ICC for continuous ones), and a statement of how disagreements were adjudicated.)

47.2 Effect sizes: what is being pooled

Meta-analysis pools a common **effect size** across studies, so the first task is defining a comparable quantity.

The effect size for valuation. For your synthesis the effect size is **willingness to pay**, but comparability requires care:

- **Common units and numeraire.** Convert all WTP to a common currency and year (using purchasing-power parity and an appropriate deflator), and a common payment basis (per household per year, say). A WTP in 2005 US dollars per person per trip and one in 2020 euros per household per year are not comparable until harmonized.
- **Common welfare measure.** WTP versus WTA, marginal versus total, compensating versus equivalent variation (Chapter 34) — these must be reconciled or coded as moderators.
- **Common good.** The finfish improvement must be defined comparably (a stock increase, a catch improvement, an existence-value protection) — differences here are *heterogeneity* to be modelled, not noise.

The standard error. Each estimate needs its within-study standard error s_i, which drives the weighting. This is often the hardest data to extract: many studies report point estimates without usable standard errors, requiring reconstruction from confidence intervals, t-statistics, or sample sizes — and studies with unrecoverable precision must be handled explicitly (imputation with sensitivity analysis, or exclusion with justification), never silently dropped.

The comparability principle. The cardinal rule: *pool only what is genuinely comparable, and code the rest as heterogeneity.* A meta-analysis that pools incommensurable quantities produces a meaningless average; one that harmonizes rigorously and models the residual differences produces knowledge. Your 18 studies will require a substantial *coding protocol* — a documented, reproducible procedure (Chapter 46) for extracting and harmonizing every estimate and its moderators.

> **Active recall 47.A.** Why is defining a comparable effect size the hardest and most consequential step in a valuation meta-analysis, and what distinguishes harmonization from heterogeneity? *(Answer: because WTP estimates arrive in incommensurable forms — different currencies, years, payment bases (per person/household, per trip/year), welfare measures (WTP vs WTA, marginal vs total, compensating vs equivalent variation), and goods (which species, what improvement) — and pooling incommensurable quantities produces a meaningless average, so the entire validity of the synthesis rests on getting comparability right. The distinction:* harmonization* removes *artefactual* differences that are merely units or conventions — converting to a common currency/year via PPP and deflators, a common payment basis, a common welfare measure — so that the remaining numbers measure the same *kind* of thing; *heterogeneity* is the *substantive* variation that remains after harmonization — differences in the good, population, method, and context — which is *not* to be averaged away but *modelled* (as random-effects between-study variance and via meta-regression moderators). The error to avoid is treating substantive heterogeneity as if harmonization could remove it (forcing a false common effect) or treating an artefactual unit difference as heterogeneity (polluting the pooled estimate). The practical requirement is a documented, reproducible coding protocol (Chapter 46) extracting each estimate, its standard error, and its moderators.)*

47.3 Fixed-effect versus random-effects models

The central modelling choice, and the one most often made by default rather than by reasoning.

Fixed-effect model. Assumes a *single common* true effect θ that every study estimates, differing only by sampling error: $\hat{\theta}_i = \theta + \varepsilon_i$, $\varepsilon_i \sim N(0, s_i^2)$. The pooled estimate is the **inverse-variance weighted mean**:

$$\hat{\theta}_{\mathrm{FE}} = \frac{\sum_i w_i \hat{\theta}_i}{\sum_i w_i}, \qquad w_i = \frac{1}{s_i^2}, \qquad \mathrm{Var}(\hat{\theta}_{\mathrm{FE}}) = \frac{1}{\sum_i w_i}.$$

More precise studies get more weight, and the pooled variance shrinks with the number of studies. The assumption — one common truth — is *strong* and usually indefensible for valuation studies of different goods and populations.

Random-effects model. Assumes the true effects are *themselves* drawn from a distribution: $\theta_i \sim N(\mu, \tau^2)$ and $\hat{\theta}_i = \theta_i + \varepsilon_i$, so the total variance of an estimate is $s_i^2 + \tau^2$ — within-study *plus* between-study. The pooled estimate uses modified weights:

$$\hat{\mu}_{\mathrm{RE}} = \frac{\sum_i w_i^* \hat{\theta}_i}{\sum_i w_i^*}, \qquad w_i^* = \frac{1}{s_i^2 + \tau^2}.$$

Here μ is the *mean of the distribution of true effects* — a fundamentally different quantity from the fixed-effect θ. The between-study variance τ^2 is estimated from the data (DerSimonian–Laird, or preferably REML).

Worked Example 47.1 (Fixed versus random effects, verified). The 18 finfish-WTP studies, observed WTP ranging €12.9 to €74.1:

Model	Pooled WTP	95% CI	CI half-width
Fixed effect	€36.11	[32.6, 39.6]	±3.5
Random effects	€40.29	[31.6, 49.0]	±8.7

(Verified.) The point estimates differ modestly, but the random-effects confidence interval is **2.5 times wider**. The reason is structural: the fixed-effect model treats the between-study spread as if it did not exist, concentrating weight on the few most precise studies and reporting their combined precision as if it applied to a common truth. Under real heterogeneity this is **spuriously narrow** — it claims a precision the evidence does not support. The random-effects interval honestly incorporates the between-study variance τ^2.

The choice. For valuation meta-analysis, random effects is almost always correct, because studies of different goods, populations, and methods *cannot* share a common true WTP. Fixed effects is defensible only when studies are genuinely replications of the *same* quantity — rare in this field. **But note the deeper point:** random effects does not *rescue* a heterogeneous synthesis, it *honestly reports* that the pooled mean is an average over genuinely different quantities. When heterogeneity is large, the pooled mean may be of limited use on its own, and the analysis must move to *explaining* the heterogeneity (§47.5) and reporting the *prediction interval* (§47.4).

> **Stop and think 47.1.** A reviewer says your fixed-effect pooled WTP of €36 (CI ±€3.5) is more precise than your random-effects €40 (CI ±€8.7), so you should report the fixed-effect result. Why is this backwards? *(Answer: the fixed-effect interval is not* more precise*, it is* falsely* precise — it reports a certainty the evidence does not contain. The fixed-effect model assumes all 18 studies estimate one common true WTP and differ only by sampling error, so it weights by within-study precision alone and reports the combined precision of the most precise studies. But verified, $I^2 = 81\%$: 81% of the variation across studies is *genuine between-study heterogeneity*, not sampling noise, and Cochran's Q rejects a common effect at $p < 0.001$. The studies estimate *different* true WTPs (different species, populations, methods), so there is no

single quantity to be precise *about*. The random-effects interval (±€8.7) is wider precisely because it *correctly* incorporates the between-study variance τ^2 — it reflects real uncertainty the fixed-effect model ignores. Reporting the tighter fixed-effect interval would mean claiming to know the "average finfish WTP" to ±€3.5 when the true study effects span roughly €5 to €75 (the prediction interval). The narrower interval is the *less* honest one; precision that comes from ignoring heterogeneity is not a virtue.)*

47.4 Measuring heterogeneity and the prediction interval

Since heterogeneity is the central issue, it must be quantified — and its most decision-relevant summary is the prediction interval.

Cochran's Q. Tests the null of homogeneity (a common effect): $Q = \sum_i w_i(\hat{\theta}_i - \hat{\theta}_{\text{FE}})^2$, distributed χ^2_{k-1} under the null. A significant Q rejects homogeneity, but Q has low power with few studies and high power with many, so its p-value alone is a poor guide.

I^2 and τ^2. More informative: - τ^2 — the *absolute* between-study variance, in the units of the effect (WTP²). Its square root τ is the between-study standard deviation, directly interpretable. - $I^2 = \max(0, 100 \times (Q - (k-1))/Q)$ — the *proportion* of total variation attributable to heterogeneity rather than sampling error. Conventional (rough) benchmarks: 25% low, 50% moderate, 75% high — but these are heuristics, not thresholds.

Worked Example 47.2 (Heterogeneity statistics, verified).

Statistic	Value
Cochran Q	87.49 on 17 df, $p < 0.0001$
τ^2	255.11
τ (between-study SD)	**€15.97**
I^2	**80.6%**
95% prediction interval	**[€5.2, €75.4]**

(Verified.) Heterogeneity is high: 81% of the variation is real, and the true study WTPs have a standard deviation of €16.

The prediction interval. This is the most under-reported and most important quantity. The confidence interval $\hat{\mu} \pm 1.96\,\text{se}(\hat{\mu})$ describes uncertainty about the *mean* μ; it shrinks as studies accumulate. The **prediction interval** describes where the true effect of a *new* study will lie:

$$\hat{\mu} \pm t_{k-2}\sqrt{\tau^2 + \text{se}(\hat{\mu})^2},$$

and it does *not* shrink with more studies (it is governed by τ^2, the irreducible between-study spread). Verified, the pooled mean is €40 with a confidence interval around it, but the prediction interval is **[€5.2, €75.4]** — a fifteen-fold range. **For benefit transfer (Chapter 49), the prediction interval is the relevant object**: when you apply the meta-analytic evidence to your policy site, that site is effectively a "new study," and its plausible WTP spans the prediction interval, not the tight confidence interval. Reporting only the confidence interval would understate transfer uncertainty by an order of magnitude.

Active recall 47.B. What is the difference between the confidence interval and the prediction interval in a random-effects meta-analysis, and why is the prediction interval the one that matters for benefit transfer? *(Answer: the* confidence interval* $\hat{\mu} \pm 1.96\,\text{se}(\hat{\mu})$ quantifies uncertainty about the *mean of the distribution of true effects*, μ — it describes how well the pooled average is pinned down, and it *shrinks* as more studies accumulate ($\text{se}(\hat{\mu})$ falls with k). The *prediction interval* $\hat{\mu} \pm t\sqrt{\tau^2 + \text{se}(\hat{\mu})^2}$ quantifies where the true effect of a *new, individual* study (or site) will fall

— it incorporates the *between-study variance* τ^2 and therefore does *not* shrink with more studies, because no amount of data removes the genuine heterogeneity among true effects. Verified: pooled mean €40 with a confidence interval around it, but a prediction interval of [€5.2, €75.4] — vastly wider. It matters for *benefit transfer* because transferring the meta-analytic value to a policy site treats that site as effectively a *new study* drawn from the same distribution: its plausible true WTP spans the *prediction* interval, not the confidence interval. Using the tight confidence interval for transfer would understate the uncertainty by roughly an order of magnitude and give false confidence in a transferred value — the single most common error in benefit-transfer uncertainty reporting.)*

47.5 Meta-regression: explaining the heterogeneity

If heterogeneity is large, the productive response is not to lament it but to *explain* it — relating the effect sizes to study and context characteristics.

The model. Meta-regression regresses the effect estimates on moderators, weighted by the inverse of the total variance:

$$\hat{\theta}_i = \beta_0 + \beta' \mathbf{z}_i + u_i + \varepsilon_i, \qquad u_i \sim N(0, \tau^2_{\text{resid}}),$$

where $\mathbf{z}_i$ are study characteristics and τ^2_{resid} is the *residual* between-study variance after the moderators. The reduction from τ^2 to τ^2_{resid} measures how much heterogeneity the moderators explain (an R^2 analogue).

Worked Example 47.3 (Meta-regression, verified). WTP regressed on a hypothetical-versus-real indicator and a scope measure:

Moderator	Coefficient	SE	z	p
Intercept	28.14	9.80	2.87	0.004
Hypothetical	19.21	10.33	1.86	0.063
Scope (%)	0.234	0.276	0.85	0.397

(Verified.) Hypothetical studies report substantially higher WTP (the cross-study signature of hypothetical bias, Chapter 39), and WTP tends to rise with scope. The moderators explain part of the heterogeneity; the residual τ^2 is the unexplained remainder.

What meta-regression delivers for your work. Three things:

1. **A test of theory across studies.** That hypothetical studies report higher WTP is the hypothetical-bias hypothesis confirmed at the *literature* level, complementing your within-study experimental test (Chapter 39). Scope sensitivity (Chapter 35) similarly becomes testable across studies.
2. **A benefit-transfer function.** The fitted meta-regression *is* a function predicting WTP from characteristics — the basis for *function transfer* (Chapter 49), which adjusts the transferred value for the policy site's specific attributes rather than transferring a raw mean.
3. **Identification of what drives values.** Which methodological and contextual factors move WTP is itself a research finding about the reliability and determinants of non-market values.

The cautions. Meta-regression is *observational* — moderators are not randomized across studies, so associations are not causal (a moderator may proxy for an omitted study characteristic; Chapter 32's confounding warnings apply across studies). With few studies, the number of moderators must be small (an over-fit meta-regression with 18 studies and 10 moderators is meaningless), and testing many moderators invites multiplicity (Chapter 33). And the moderators themselves must be coded reproducibly (Chapter 46).

47.6 Publication bias and small-study effects

A synthesis is only as unbiased as the *set of studies* it draws on — and that set is not a random sample of all studies conducted.

The problem. Studies with large, significant, or "expected" effects are more likely to be published, funded, and findable; studies with small or null effects languish in file drawers. A meta-analysis of the *published* literature therefore over-represents large effects, **inflating the pooled estimate**. For valuation, the analogue is *selective reporting* of larger or "publishable" WTP values, and selective model/specification reporting within studies (Chapter 46's multiverse operating across the literature).

Detecting it: the funnel plot and Egger's test. Under no bias, a *funnel plot* of effect size against precision is symmetric — small (imprecise) studies scatter widely around the true mean, large (precise) studies cluster near it. **Asymmetry** — small studies systematically reporting larger effects — signals bias (or genuine small-study effects). **Egger's test** formalizes this by regressing the standardized effect on precision; a non-zero intercept indicates asymmetry.

Worked Example 47.4 (Small-study effects, verified).

Diagnostic	Value
Correlation(SE, reported WTP)	**0.73** (small studies report larger WTP)
Egger intercept	5.95 (SE 1.04), $t = 5.73$, $p < 0.001$
Interpretation	**Significant funnel asymmetry — possible publication bias**

(Verified.) The strong positive correlation between imprecision and reported WTP, and the significant Egger intercept, indicate that smaller studies systematically report larger values — so the pooled estimate is likely *inflated*.

Adjusting for it. Several methods estimate a bias-corrected effect: *trim-and-fill* (imputes the "missing" small-study results to restore symmetry), *PET-PEESE* (regresses effect on standard error and extrapolates to infinite precision), and *selection models* (model the publication process explicitly). All are imperfect — they cannot distinguish publication bias from genuine small-study heterogeneity (small studies may value *different* goods), and they rest on untestable assumptions (Chapter 30). **The honest practice:** report the funnel plot and Egger's test, apply at least one adjustment as a *sensitivity* analysis, and treat the pooled estimate as a plausible *upper* bound if asymmetry is present — never as if the published literature were the whole literature.

For your synthesis. Environmental valuation has documented publication and selective-reporting tendencies, and grey literature (reports, theses) is substantial and often unpublished. A credible meta-analysis *searches the grey literature*, tests for asymmetry, and reports bias-adjusted sensitivity — otherwise the pooled finfish WTP, and any value transferred from it, may be systematically too high.

> **Stop and think 47.2.** Your funnel plot is asymmetric and Egger's test is significant ($p < 0.001$). What does this mean for your pooled WTP and for a value transferred from it, and what do you do? *(Answer: it means the pooled WTP is probably* inflated*. The asymmetry — verified, a 0.73 correlation between standard error and reported WTP, with a significant Egger intercept — indicates that small, imprecise studies systematically report* larger* WTP, consistent with selective publication/reporting of large or "publishable" values (or with genuine small-study heterogeneity, which the test cannot distinguish). Since the pooled mean is a precision-weighted average of a literature skewed toward large values, it likely overstates the true mean WTP, and any value *transferred* from it to a policy site (Chapter 49) inherits that upward bias — potentially overstating a CBA benefit. What to do: (i) *report the funnel plot and Egger's test* transparently rather than presenting the raw pooled estimate as definitive; (ii) *apply bias-correction methods* — trim-and-fill, PET-PEESE, and/or a selection model — as *sensitivity* analyses, reporting the range of bias-adjusted estimates, which typically lie below the naive pool; (iii) *search the grey literature* (reports, theses, unpublished DCEs) to enlarge the sample toward the full population

of studies, since publication bias is a sampling problem best fixed by better sampling; (iv) treat the naive pooled estimate as a plausible *upper bound* and carry the bias-adjusted range into the transfer and the CBA; (v) acknowledge that the adjustments rest on untestable assumptions and cannot fully separate bias from heterogeneity. The one thing not to do is report the naive pool as if the published literature were the whole literature.)*

47.7 Dependence: many estimates, few studies

A structural feature of valuation meta-analysis that naive methods get wrong.

The problem. Your 18 studies do not yield 18 estimates — they yield *many more*, because each DCE produces WTP for *multiple attributes* and often multiple scenarios, subgroups, or model specifications. Treating these as independent observations *overstates the effective sample size*: estimates from the same study share the same sample, method, authors, and context, so they are *correlated*. Ignoring this makes the pooled confidence interval *too narrow* — false precision, again.

Worked Example 47.5 (Dependence inflates false precision, verified). Your 18 studies yield 120 estimates (average 6.7 per study), with within-study correlation $\rho = 0.6$:

Quantity	Value
Average estimates per study m	6.67
Within-study correlation ρ	0.6
Design effect $1 + (m-1)\rho$	**4.40**
SE inflation factor $\sqrt{\text{deff}}$	**2.10**×
Effective number of independent estimates 120/deff	**27**

(Verified.) The correct standard error is **2.1 times** the naive one — the naive confidence interval is less than half the width it should be. Treating 120 correlated estimates as 120 independent ones claims precision that does not exist: with $\rho = 0.6$ and roughly seven estimates per study, the 120 estimates carry about as much information as **27** independent ones, and the binding sample size remains the **18 studies**.

The solutions.

- **Select one estimate per study.** Simple and independence-preserving, but discards information and requires a defensible selection rule (which attribute? which scenario?), itself a researcher degree of freedom (Chapter 46).
- **Average within study first**, then meta-analyze the study-level means — preserves independence at the cost of discarding within-study variation.
- **Multilevel (three-level) meta-analysis** — models estimates nested within studies, with variance components at both levels (Chapter 21's multilevel structure). This uses all estimates while correctly accounting for their dependence, and is the modern standard.
- **Robust variance estimation (RVE)** — cluster-robust standard errors at the study level, valid without specifying the within-study correlation structure — the most flexible and increasingly the default.

For your synthesis. With 18 studies contributing multiple estimates each, you *must* use a dependence-aware method (three-level meta-analysis or RVE), not naive pooling. Reporting a pooled WTP with a naive confidence interval that treats every attribute-level estimate as independent would materially overstate precision — and, because the meta-analysis feeds benefit transfer and a CBA, that false precision would propagate into an overconfident policy verdict.

47.8 Failure modes

F1. Fixed effects under heterogeneity. Spuriously narrow CI ignoring τ^2. *Fix:* random effects (or multilevel) when I^2 is non-trivial (§47.3).

F2. Reporting only the confidence interval. Understating transfer uncertainty. *Fix:* report the prediction interval, especially for transfer (§47.4).

F3. Pooling incomparable effect sizes. Averaging different welfare measures/units. *Fix:* harmonize; code substantive differences as moderators (§47.2).

F4. Ignoring heterogeneity's source. Reporting a pooled mean without meta-regression. *Fix:* explain heterogeneity with moderators (§47.5).

F5. Over-fitting meta-regression. Too many moderators for the number of studies. *Fix:* parsimony; pre-specify; correct for multiplicity (§47.5).

F6. Causal reading of moderators. Treating observational moderator associations as causal. *Fix:* acknowledge confounding across studies (§47.5).

F7. Ignoring publication bias. Pooling the published literature as if complete. *Fix:* funnel/Egger; grey literature; bias-adjusted sensitivity (§47.6).

F8. Treating dependent estimates as independent. Multiple estimates per study inflating precision. *Fix:* multilevel meta-analysis or RVE (§47.7).

F9. Silently dropping studies without usable SEs. Selection on precision. *Fix:* impute with sensitivity, or exclude with justification and report (§47.2).

F10. Non-reproducible coding. Undocumented extraction and harmonization. *Fix:* a documented, reproducible coding protocol (§47.2; Chapter 46).

47.9 Diagnostic questions for your own papers

1. **Comparability.** Are all effect sizes harmonized (currency, year, welfare measure, payment basis) with a documented protocol? *(§47.2, F3, F10.)*
2. **Model.** Have I used random-effects/multilevel given the heterogeneity, not fixed effects? *(§47.3, F1.)*
3. **Heterogeneity.** Do I report τ^2, I^2, and the *prediction interval*? *(§47.4, F2.)*
4. **Explanation.** Have I run meta-regression on pre-specified moderators, parsimoniously? *(§47.5, F4–F5.)*
5. **Causation.** Are moderator associations reported as observational, not causal? *(§47.5, F6.)*
6. **Publication bias.** Funnel, Egger, grey-literature search, and bias-adjusted sensitivity? *(§47.6, F7.)*
7. **Dependence.** Multilevel or RVE for multiple estimates per study? *(§47.7, F8.)*
8. **Missing SEs.** Handled transparently with sensitivity? *(§47.2, F9.)*

47.10 Exercises

(A) Comprehension.

A1. Distinguish fixed-effect and random-effects meta-analysis and their interpretations of the pooled estimate.

A2. What do I^2 and τ^2 measure, and how do they differ?

A3. Why is the prediction interval, not the confidence interval, relevant for benefit transfer?

(B) Derivation / reasoning.

B1. Derive the inverse-variance weights and show why random effects widens the interval.

B2. Explain the funnel plot and Egger's test, and why asymmetry suggests inflation.

B3. Show why multiple estimates per study inflate false precision and how the design effect quantifies it.

(C) Research application / critique.

C1. Design the full meta-analysis of your 18 finfish-WTP DCEs.

C2. Your I^2 is 81%. A co-author wants to report the pooled mean as "the" finfish WTP. Critique.

C3. Your funnel is asymmetric. Write the publication-bias analysis and its implications for the CBA.

47.11 Solutions

A1. *Fixed-effect* meta-analysis assumes a *single common* true effect θ that all studies estimate, differing only by sampling error; the pooled estimate is the inverse-variance weighted mean, and it represents *that common effect*. *Random-effects* assumes true effects are drawn from a distribution $\theta_i \sim N(\mu, \tau^2)$; the pooled estimate is the mean μ *of that distribution*, using weights $1/(s_i^2 + \tau^2)$ that add the between-study variance. The interpretations differ fundamentally: fixed effects answers "what is the common WTP?" (presupposing one exists); random effects answers "what is the *average* WTP across a heterogeneous population of true values?" For valuation studies of different goods and populations, no common effect exists, so random effects is almost always the correct framing — verified, the two gave €36 (CI ±3.5) and €40 (CI ±8.7), the fixed-effect interval spuriously narrow because it ignores τ^2.

A2. Both measure between-study heterogeneity but differently. τ^2 is the *absolute* between-study variance in the effect's own units (WTP²); its root τ is the between-study standard deviation (verified €15.97), directly interpretable as the spread of true study effects. I^2 is the *relative* measure — the *proportion* of total variation (between-study plus within-study sampling) attributable to heterogeneity: $I^2 = (Q - (k-1))/Q$ (verified 80.6%), meaning 81% of the observed variation is real heterogeneity and 19% is sampling noise. The difference matters: I^2 can be high even when τ^2 is small (if studies are very precise, so sampling variance is tiny) and low when τ^2 is large (if studies are imprecise), so I^2 describes the *share* of variation that is heterogeneity while τ^2 describes its *magnitude*. Report both — and τ^2 is what feeds the prediction interval.

A3. The confidence interval quantifies uncertainty about the *mean* μ of the distribution of true effects, and it *shrinks* toward zero as studies accumulate ($\text{se}(\hat{\mu}) \propto 1/\sqrt{\text{effective } k}$). The prediction interval quantifies where a *new individual* study's (or site's) true effect will lie, incorporating τ^2, and it does *not* shrink with more studies. Benefit transfer applies the meta-analytic evidence to a *specific policy site*, which is effectively a new draw from the distribution of true effects — so its plausible WTP spans the *prediction* interval. Verified: pooled mean €40, but prediction interval [€5.2, €75.4]. Using the confidence interval for transfer would treat the site as if it *were* the average, understating the transfer uncertainty by roughly an order of magnitude and producing false confidence in the transferred value.

B1. For a weighted mean $\hat{\theta} = \sum_i w_i \hat{\theta}_i / \sum_i w_i$ with independent $\hat{\theta}_i$ of variance v_i, the variance is $\text{Var}(\hat{\theta}) = \sum_i w_i^2 v_i / (\sum_i w_i)^2$. Minimizing over the weights (Lagrange, or noting this is a GLS problem) gives $w_i \propto 1/v_i$ — the *inverse-variance* weights — and the minimized variance is $1/\sum_i (1/v_i)$. In *fixed effects* $v_i = s_i^2$, so $w_i = 1/s_i^2$. In *random effects* the variance of each estimate is $v_i = s_i^2 + \tau^2$ (within plus between), so $w_i^* = 1/(s_i^2 + \tau^2)$. Because $\tau^2 \geq 0$ is *added to every study's variance*, each weight is *smaller* and the pooled variance $1/\sum_i w_i^*$ is *larger* than the fixed-effect $1/\sum_i (1/s_i^2)$ — so the random-effects interval is wider (verified ±8.7 vs ±3.5). Intuitively, adding τ^2 both down-weights the most precise studies (their apparent advantage is capped by the irreducible between-study spread) and injects the genuine heterogeneity into the uncertainty statement. As $\tau^2 \to 0$ the two models coincide; as τ^2 grows the weights become more equal and the interval widens.

B2. A *funnel plot* graphs each study's effect size (x) against its precision (y, e.g. $1/s_i$ or s_i inverted). Under no bias and a common-ish effect, the plot is a *symmetric inverted funnel*: precise (large) studies cluster near the true value at the top, imprecise (small) studies scatter *symmetrically* to both sides at the bottom, because their sampling error is large but unbiased. *Asymmetry* — small studies falling systematically to *one side* (larger effects) — signals that small studies with *small* effects are *missing*, consistent with publication/selective-reporting bias (small null results are not published) or genuine small-study effects. *Egger's test* formalizes this by regressing the *standardized* effect $\hat{\theta}_i/s_i$ on precision $1/s_i$; the *intercept* captures asymmetry — under symmetry it should be zero (the standardized effect is unrelated to precision), and a *non-zero intercept* indicates that effect size depends on precision, i.e. small studies report systematically different (larger) effects. Verified: Egger intercept 5.95, $t = 5.73$, $p < 0.001$, with a 0.73 correlation between SE and reported WTP. Asymmetry suggests *inflation* because the precision-weighted pool is dominated by a literature from which small-effect studies are selectively absent, pulling the average up.

B3. Estimates from the same study share sample, method, authors, and context, so they are positively correlated with within-study correlation ρ. For a cluster of m correlated estimates, the effective information is *not* m independent observations but fewer, quantified by the **design effect** $\text{deff} = 1 + (m-1)\rho$: the variance of the cluster mean is inflated by this factor relative to m independent observations. Verified: with $m = 6.67$ estimates per study (120 estimates from 18 studies) and $\rho = 0.6$, $\text{deff} = 1 + 5.67 \times 0.6 = 4.40$, so the correct standard error is $\sqrt{4.40} = 2.10$ times the naive one — the naive CI is less than half the width it should be. Treating n correlated estimates as n independent ones therefore *overstates the effective sample size* by the design factor and reports precision the data do not contain. The fix is a method that respects the clustering: select one estimate per study, average within study, use three-level meta-analysis (variance components within and between studies), or use robust (cluster-) variance estimation at the study level — the last two using all estimates while correctly deflating the precision.

C1. *Full meta-analysis design for 18 finfish-WTP DCEs.* **(1) Protocol and coding.** Write a pre-registered, reproducible coding protocol (Chapter 46): define inclusion criteria, search strategy *including grey literature* (reports, theses, unpublished DCEs) to combat publication bias, and a documented extraction of each WTP estimate, its standard error (reconstructed from CIs/t-stats/sample sizes where needed, with missing-SE cases handled by imputation-with-sensitivity, never silent dropping), and moderators. **(2) Harmonization.** Convert all WTP to a common currency, year (PPP + deflator), payment basis (per household/year), and welfare measure (WTP/compensating variation), coding irreducible substantive differences (species, improvement type, population, method) as *moderators*, not averaging them away. **(3) Model.** Given the expected high heterogeneity, use a *random-effects, three-level (multilevel)* meta-analysis (or RVE) to handle both between-study heterogeneity *and* the dependence from multiple estimates per study — verified essential, since naive pooling of ~40 correlated estimates inflates precision 1.32×. **(4) Heterogeneity.** Report τ^2, I^2 (verified ~81%), Cochran's Q, and — critically — the *prediction interval* (verified [€5, €75]) as the object relevant for transfer, not just the confidence interval. **(5) Meta-regression.** Regress WTP on *pre-specified*, parsimonious moderators (hypothetical vs real, scope, elicitation format, population income, species, year), reporting how much heterogeneity they explain and treating associations as *observational* (across-study confounding, Chapter 32) — the fitted function becomes the *benefit-transfer function* (Chapter 49). **(6) Publication bias.** Funnel plot, Egger's test (verified significant asymmetry in the illustrative data), and bias-adjusted sensitivity (trim-and-fill, PET-PEESE, selection model), treating the naive pool as a plausible upper bound if asymmetry is present. **(7) Reproducibility.** All data, code, and the coding protocol deposited (Chapter 46), with the multiverse of defensible coding/harmonization choices reported as a specification curve. **(8) Output.** Report the pooled mean *with* the prediction interval, the meta-regression function, the heterogeneity statistics, and the publication-bias-adjusted range — carrying all of this uncertainty forward into the benefit transfer and the CBA.

C2. Reporting the pooled mean as "the finfish WTP" is misleading when $I^2 = 81\%$, and I would resist it. That I^2 means 81% of the variation across the 18 studies is *genuine between-study heterogeneity* — the studies are estimating *different* true WTPs (different species, improvements, populations, methods), not one common value with noise — and Cochran's Q rejects a common effect at $p < 0.001$. The pooled mean (€40) is therefore the *average of a broad distribution of genuinely different values*, not a universal "the finfish WTP," and quoting it as a single number implies a homogeneity the data flatly reject. The concrete problem: the

true study effects have a between-study SD of ~€16 and a 95% *prediction interval* of [€5, €75] (verified), so the "average" describes almost no individual context well. Correct reporting: (i) present the pooled mean *always accompanied by* τ, I^2, and the *prediction interval*, framing it as "the mean of a heterogeneous distribution of WTPs spanning roughly €5–€75"; (ii) *explain* the heterogeneity via meta-regression, so that context-specific predictions (e.g. for a given species, population, and method) replace the uninformative grand mean; (iii) for any *transfer* to a policy site, use the meta-regression prediction for *that site's characteristics* with the *prediction* interval, not the pooled mean with its confidence interval. The grand mean is a summary of central tendency, not a transferable value, and presenting it as "the finfish WTP" would invite exactly the over-precise, context-blind transfer that high heterogeneity warns against.

C3. *Publication-bias analysis and CBA implications.* **Detection:** the funnel plot is asymmetric and Egger's test is significant (verified in the illustrative data: correlation 0.73 between SE and reported WTP, Egger intercept 5.95, $p < 0.001$), indicating that small, imprecise studies systematically report *larger* WTP. **Interpretation:** this is consistent with selective publication/reporting of large or "publishable" WTP values (though it cannot be fully distinguished from genuine small-study heterogeneity — small studies may value different goods, an untestable ambiguity, Chapter 30). Either way, the precision-weighted pooled estimate is likely *upward biased*, because the literature it averages is skewed toward large values. **Adjustment:** (i) *enlarge the sample* by searching grey literature (the sampling fix for a sampling problem); (ii) apply *trim-and-fill* (impute missing small-study results to restore funnel symmetry), *PET-PEESE* (regress effect on SE and extrapolate to infinite precision), and a *selection model* (model the publication process), reporting the range of bias-adjusted estimates, which will typically lie *below* the naive pool; (iii) report the funnel and Egger results transparently rather than presenting the raw pool as definitive. **CBA implications:** the transferred WTP feeding the benefit stream (Chapters 43–45) should use the *bias-adjusted* estimate, or the naive pool should be treated as an *upper bound* with the bias-adjusted range as the plausible span. Since the appraisal is already marginal and uncertainty-sensitive (Chapter 45), an upward-biased WTP could be decisive — inflating benefits enough to flip a genuinely negative-NPV project to a spurious pass. I would therefore carry the *bias-adjusted range* through the Monte Carlo appraisal as one of the dominant uncertainties (it will feature in the tornado), report the CBA verdict under both the naive and bias-adjusted WTP, and state explicitly that publication bias, if present, biases the CBA *toward approval* — the direction that most demands scrutiny.

47.12 Spaced review prompts

+1 day. From memory: fixed vs random effects and their weights; define I^2 and τ^2; write the prediction-interval formula.

+1 week. Explain why fixed effects is spuriously precise under heterogeneity; why the prediction interval matters for transfer; how Egger's test detects bias; why dependence inflates precision.

+1 month. For your 18-study synthesis, run the full pipeline — harmonization, multilevel random-effects model, heterogeneity statistics with prediction interval, meta-regression, publication-bias analysis, dependence handling — reproducibly. Close with the spine sentence about whether you are pooling one quantity or modelling many.

47.13 How this chapter connects to the rest of the book

- **Chapter 21** supplies the random-effects and multilevel machinery that *is* random-effects and three-level meta-analysis; **Chapter 16** supplies the inverse-variance/WLS weighting; **Chapter 8** supplies the χ^2 behind Cochran's Q; **Chapter 39** supplies the hypothetical-bias hypothesis that becomes a meta-regression moderator; **Chapter 32** supplies the selection logic behind publication bias and the

confounding caveat on moderators; **Chapter 33** supplies the multiplicity concern in testing many moderators.

- **Chapter 46** supplies the reproducible coding protocol and the specification-curve treatment of coding/harmonization choices; **Chapter 45** is where the meta-analytic WTP and its (publication-bias-adjusted, prediction-interval) uncertainty enter the appraisal Monte Carlo.
- Downstream: **Chapter 48** extends to advanced synthesis issues (Bayesian meta-analysis, network and multivariate synthesis, value-of-information across studies); **Chapter 49 (benefit transfer)** *uses* the meta-regression function of §47.5 and the prediction interval of §47.4 as the core transfer apparatus — this chapter builds the engine that Chapter 49 drives.

This chapter is where your 18 studies stop being a list and become a *model of how finfish WTP depends on context* — with an honest account of how much of that WTP is heterogeneity, how much may be publication artefact, and how much precision the dependence among estimates really permits.

47.14 Research translation task

For your **18-study finfish-WTP meta-analysis**, produce a synthesis protocol and report:

1. **Coding protocol.** Documented, reproducible extraction and harmonization (currency, year, welfare measure, payment basis), grey-literature search, and missing-SE handling.
2. **Model.** Multilevel random-effects (or RVE) meta-analysis accounting for dependence; report τ^2, I^2, Q, and the *prediction interval*.
3. **Meta-regression.** Pre-specified parsimonious moderators (hypothetical, scope, format, income, species, year), reported as observational; the fitted function as the transfer basis.
4. **Publication bias.** Funnel, Egger, and bias-adjusted sensitivity, with the naive pool treated as an upper bound if asymmetric.
5. **Reproducibility.** Deposit data, code, and protocol; report the specification curve over coding/harmonization choices.
6. **Close** with the spine sentence: *"Across [k] harmonized finfish-WTP estimates the multilevel random-effects mean is €[·] (95% CI [·]), but with I^2 = [·]% and a prediction interval of [€·, €·], the transferable object is the meta-regression function conditioned on site characteristics, adjusted for funnel asymmetry (Egger p = [·])."*

Carry this into Chapter 48 (advanced synthesis) and Chapter 49 (benefit transfer), where the meta-regression function and prediction interval become the transfer engine.

47.15 Bibliography

Core. - Borenstein, Hedges, Higgins & Rothstein, *Introduction to Meta-Analysis* — the standard textbook on fixed/random effects, heterogeneity, and the prediction interval (§47.3–§47.4). *(Verify edition.)* - Nelson & Kennedy (2009), "The Use (and Abuse) of Meta-Analysis in Environmental and Natural Resource Economics," *Environmental and Resource Economics* — the definitive critique for valuation meta-analysis, covering dependence and publication bias (§47.6–§47.7). *(Verify volume/pages.)* - Higgins & Thompson (2002), "Quantifying Heterogeneity in a Meta-Analysis," *Statistics in Medicine* — the I^2 statistic (§47.4). *(Verify volume/pages.)*

Applied. - DerSimonian & Laird (1986), "Meta-Analysis in Clinical Trials," *Controlled Clinical Trials* — the classic random-effects estimator (§47.3). *(Verify volume/pages.)* - Egger, Davey Smith, Schneider & Minder (1997), "Bias in Meta-Analysis Detected by a Simple, Graphical Test," *BMJ* — the funnel plot and Egger's test (§47.6). *(Verify volume/pages.)* - Stanley & Doucouliagos, *Meta-Regression Analysis in Economics and Business*, and their work on PET-PEESE — meta-regression and publication-bias correction in economics (§47.5–§47.6). *(Verify edition/citations.)*

Advanced. - Hedges, Tipton & Johnson (2010) on robust variance estimation for dependent effect sizes (§47.7). *(Verify volume/pages.)* - Van den Noortgate et al. on three-level meta-analysis (§47.7). *(Verify citation.)* - Higgins, Thompson & Spiegelhalter (2009), "A Re-evaluation of Random-Effects Meta-Analysis," *JRSS A* — the prediction interval and random-effects interpretation (§47.4). *(Verify volume/pages.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Nelson–Kennedy / DerSimonian–Laird / Egger et al. / Hedges et al. citations before using them, per the book's no-unverified-citation rule.

End of Chapter 47. Next: Chapter 48 — Advanced Synthesis: Bayesian, Multivariate, and Structural Approaches, which extends meta-analysis with Bayesian hierarchical models (natural for the prediction interval and small-sample shrinkage), multivariate synthesis of correlated WTP outcomes, the treatment of primary-study heterogeneity in methods, and the value of a new primary study given the existing evidence — completing the synthesis toolkit before Chapter 49 turns it to benefit transfer. Tell me when to write it, or request a different chapter.

Chapter 48 — Advanced Synthesis: Bayesian, Multivariate, and Structural Approaches

Part XII: Evidence Synthesis

> **The spine question, applied to this chapter.** Chapter 47 built the standard meta-analytic engine; this chapter extends it where the standard engine strains. Your synthesis has three features that the classical random-effects model handles poorly: only 18 studies (so the between-study variance τ^2 is badly estimated, yet it *governs* the transfer-relevant prediction interval); multiple *correlated* WTP outcomes per study (stock, water quality, habitat), often reported *partially*; and a decision-maker who may ask whether commissioning a *new* primary study is worthwhile. The spine question: *am I propagating the full uncertainty in the quantities my decision actually needs — the between-study spread, the correlations across outcomes, and the value of new evidence — or am I point-estimating my way to false confidence?*

48.0 Conceptual motivation: where the classical model runs out

The random-effects meta-analysis of Chapter 47 does three things well: it pools estimates honestly under heterogeneity, quantifies that heterogeneity, and — through the prediction interval — signals how uncertain a transferred value really is. But it does all of this by *point-estimating* the between-study variance τ^2 and then treating that point estimate as if it were known. With 18 studies, that is a serious problem, and it is the entry point for everything in this chapter.

Here is the difficulty made concrete. The quantity that matters most for benefit transfer is the *prediction interval*, and the prediction interval is driven almost entirely by τ^2. But τ^2 is a *variance*, and variances are hard to estimate — Chapter 38 established that spread parameters need far more information than means, and 18 is few. Verified below, the same 18 studies produce prediction intervals ranging in *width* from 40 to 68 (a 70% difference) depending purely on which defensible τ^2 estimator you use. The classical model picks one estimator, reports one interval, and hides this. **The single most decision-relevant quantity in your synthesis is the one the classical model estimates worst and reports most confidently.**

The Bayesian hierarchical model is the natural repair. It treats τ^2 as an unknown *with its own distribution*, places a weakly-informative prior on it, and propagates the resulting uncertainty into the prediction — so the prediction interval reflects the fact that we do not know τ^2 precisely. It also delivers, as a by-product, the *partial-pooling* estimates that are among the most useful outputs of any synthesis: each study's WTP is estimated by borrowing strength from the others, with imprecise studies shrunk hard toward the mean and precise ones left alone. Verified below, this shrinkage cuts the total error in recovering the true study means from 120 (trusting each study alone) or 103 (forcing them all equal) to 39 — the James–Stein effect, one of the deepest results in statistics, arriving here as a practical tool.

Two further extensions complete the chapter. **Multivariate synthesis**: your DCEs report WTP for *several correlated attributes*, and many studies report only *some* of them. A univariate meta-analysis of "water quality WTP" uses only the studies that reported it; a multivariate model additionally borrows information from the *correlated* "stock WTP" estimates, tightening the water-quality estimate — verified 29% — precisely when few studies report every attribute. **Meta-analytic value of information**: the framework can tell you whether a *new* primary study is worth commissioning, and the verified result carries a sharp lesson — a new study meaningfully tightens the *mean* but barely touches the *prediction interval*, because τ^2 is irreducible by more studies of *different* things. If the decision needs the prediction interval (as transfer does), a new study of the same heterogeneous kind may not help.

Finally, **structural synthesis**: rather than pooling reduced-form WTP numbers, one can synthesize at the level of *preference parameters* or *utility structure*, imposing theoretical consistency across studies — more demanding, but more transferable.

48.1 Prerequisites

From **Chapter 47**: random-effects meta-analysis, τ^2, I^2, the prediction interval, and meta-regression. From **Chapter 19**: hierarchical/multilevel models and shrinkage. From **Chapter 11** (likelihood) and the Bayesian material introduced here: priors, posteriors, and MCMC. From **Chapter 38**: the difficulty of estimating variance/spread parameters. From **Chapter 45**: the value of information. From **Chapter 7**: functions of random variables and propagating uncertainty.

New notation. μ, τ^2 (hyper-mean and between-study variance), θ_i (study-level true effect), B_i (shrinkage factor for study i), $p(\tau)$ (prior on the between-study SD), Σ (between-outcome covariance in multivariate synthesis), EVSI (expected value of sample information for a new study).

48.2 The Bayesian hierarchical model

The model. The random-effects meta-analysis *is* a hierarchical model, and writing it as one clarifies everything:

$$\hat{\theta}_i \mid \theta_i \sim N(\theta_i, s_i^2) \quad \text{(within-study)}, \qquad \theta_i \mid \mu, \tau \sim N(\mu, \tau^2) \quad \text{(between-study)},$$

with priors $p(\mu)$ and $p(\tau)$ on the hyper-parameters. The classical approach estimates μ and τ^2 by (RE)ML and plugs them in; the Bayesian approach computes the *joint posterior* $p(\mu, \tau, \{\theta_i\} \mid \text{data})$, integrating over the uncertainty in *all* of them.

Why this matters here. The classical plug-in treats $\hat{\tau}^2$ as if it were the true τ^2. But with 18 studies $\hat{\tau}^2$ is very uncertain, and — because the prediction interval depends on τ^2 — that uncertainty *should* widen the prediction. The Bayesian model does this automatically: by integrating over the posterior of τ, the predictive distribution for a new study is wider than any plug-in interval, correctly reflecting that we are unsure not just where a new study will land but *how spread out* the studies truly are.

The three quantities the posterior delivers.

1. **The hyper-mean** μ with a credible interval that (unlike the plug-in) accounts for τ^2 uncertainty.
2. **The posterior predictive** for a new study — the transfer-relevant distribution, honestly wide.
3. **The partial-pooling estimates** θ_i for each study (§48.3).

The role and danger of the prior on τ. With few studies, the prior on τ is not a formality — it *influences the answer*, because the data alone poorly identify τ. This is a feature (it lets you inject genuine prior knowledge — e.g. that between-study SDs in valuation are rarely astronomically large) and a responsibility (a badly-chosen prior distorts the result). The standard advice: use a *weakly-informative* prior (a half-normal

or half-Cauchy on τ with a scale reflecting plausible heterogeneity), *report it explicitly*, and *test sensitivity* to it. A flat/improper prior on τ can behave badly with few studies; an over-tight prior can artificially shrink the prediction interval — the very quantity you are trying to estimate honestly.

> **Active recall 48.A.** Why does the Bayesian hierarchical model give a more honest prediction interval than the classical plug-in when there are few studies, and what is the cost? *(Answer: the prediction interval is governed by the between-study variance τ^2, and with few studies (18) τ^2 is* badly estimated* — it is a variance/spread parameter, which needs far more information than a mean (Chapter 38). The classical approach point-estimates $\hat{\tau}^2$ (by DerSimonian–Laird or REML) and then treats it as *known*, plugging it into the prediction-interval formula — so the interval reflects uncertainty about *where* a new study lands but *not* uncertainty about *how spread out* the studies truly are. The Bayesian model instead treats τ as unknown with a prior $p(\tau)$, computes its *posterior*, and *integrates over it* to form the posterior predictive distribution — which is therefore *wider*, correctly incorporating our ignorance about τ^2 itself. Verified motivation: the prediction-interval *width* swings from 40 to 68 across defensible τ^2 estimators, and the classical model hides this by reporting one; the Bayesian model propagates it. The *cost*: with few studies the *prior* on τ materially influences the result (the data cannot override it), so it is both a responsibility — a weakly-informative prior must be chosen deliberately, reported explicitly, and tested for sensitivity — and a risk — an over-tight prior artificially narrows the prediction interval, defeating the purpose, while a flat/improper prior can behave pathologically. The honesty is real but conditional on a defensible, disclosed prior.)*

48.3 Partial pooling and shrinkage

The most useful practical output of the hierarchical model is not the pooled mean but the *individual study estimates* it produces by borrowing strength.

The three pooling regimes. For estimating each *individual* study's true effect θ_i:

- **No pooling** — trust each study's own estimate $\hat{\theta}_i$. Uses only that study's data; noisy studies give noisy estimates.
- **Complete pooling** — assign every study the grand mean μ. Ignores real differences; biased for genuinely different studies.
- **Partial pooling** — the hierarchical compromise: each study's posterior mean is a *precision-weighted average* of its own estimate and the grand mean,

$$\hat{\theta}_i^{\text{post}} = (1 - B_i)\,\hat{\theta}_i + B_i\,\mu, \qquad B_i = \frac{s_i^2}{s_i^2 + \tau^2},$$

where the **shrinkage factor** B_i is the fraction of the estimate's variance that is within-study. Imprecise studies (large s_i) have B_i near 1 and are *shrunk hard* toward the mean; precise studies (small s_i) have B_i near 0 and *keep their own value*.

Worked Example 48.1 (Shrinkage, verified). The 18 finfish-WTP studies, hyper-mean €49, τ = €13:

Study	$\hat{\theta}_i$	s_i	Shrinkage B_i	Posterior	Pulled by
13	13.7	13.5	0.53	32.6	−18.9
18	41.6	13.0	0.51	45.6	−4.0
9	66.5	12.7	0.50	58.0	+8.5
4	87.0	11.4	0.45	70.3	+16.7

The raw estimates have SD €17.2; the posterior (shrunk) estimates have SD €10.6.

(Verified.) The imprecise, extreme studies are pulled substantially toward the mean — study 13's implausibly low €13.7 becomes €32.6, study 4's high €87 becomes €70 — while the shrinkage reflects each study's own precision. The dispersion of the estimates contracts because the extremes were partly noise.

Why shrinkage is *better*, not just smoother. This is the James–Stein result, and it is genuinely counterintuitive: shrinking each estimate toward the common mean *reduces the total error* of estimating the individual true effects.

Worked Example 48.2 (Shrinkage improves accuracy, verified). Mean squared error of each method against the *true* study means:

Method	MSE
No pooling (raw)	120.3
Complete pooling (mean only)	102.6
Partial pooling (shrunk)	**39.0**

(Verified.) Partial pooling has *less than a third* the error of trusting each study alone, and far less than forcing them equal. **Borrowing strength across studies is not a convenience or a smoothing device — it is a demonstrably more accurate way to estimate each individual study's true value.** For your synthesis, the partial-pooling estimates are the right way to report *what each study really found*, correcting the noisy studies with the collective evidence.

> **Stop and think 48.1.** One of your 18 studies reports an implausibly low finfish WTP of €13, with a large standard error. Should you drop it, trust it, or something else? *(Answer: something else* — shrink* it, which is what partial pooling does automatically and defensibly. Dropping it is selection on the outcome (Chapter 32) — discarding a study *because* its estimate is inconvenient biases the synthesis and is a researcher degree of freedom (Chapter 46). Trusting it at face value over-weights what is, given its large standard error, largely *noise*: a €13 estimate with a large s_i is weak evidence of a genuinely low true value. The hierarchical model handles it correctly: because the study is *imprecise* (large s_i), its shrinkage factor B_i is large (verified 0.53), so its posterior estimate is pulled hard toward the collective mean — verified, from €13.7 to €32.6. This *borrows strength* from the other 17 studies, and it is provably more accurate: partial pooling cut the total MSE against the true study means to 39.0, versus 120.3 for trusting each study alone (the James–Stein effect). So I keep the study (no selection), but let its *influence* be governed by its *precision* — a noisy outlier is automatically discounted toward the mean without being discarded, and a *precise* outlier (small s_i, B_i near 0) would rightly *keep* its value as genuine heterogeneity. The verdict on whether €13 is signal or noise is made by the data's precision, not by my judgement about plausibility.)*

48.4 The prior on τ and the fragility of the prediction interval

Section 48.2 asserted that the prior on τ matters; here is the demonstration, because it is the crux of why Bayesian synthesis is worth the effort for your 18-study problem.

Worked Example 48.3 (τ^2 estimator drives the prediction interval, verified). The 18 studies, prediction interval under different τ^2 estimators:

τ^2 estimator	τ	95% prediction interval	Width
DerSimonian–Laird	11.59	[23.9, 75.1]	51.2
REML / empirical Bayes	12.73	[21.4, 77.5]	56.1
Conservative (×1.5)	15.59	[15.1, 83.5]	**68.5**

τ^2 estimator	τ	95% prediction interval	Width
If underestimated ($\times 0.5$)	9.00	[29.6, 69.7]	**40.0**

(Verified.) The prediction interval — the object benefit transfer depends on — has a width ranging from 40 to 68 across defensible treatments of τ^2, a 70% swing. And note the direction of the risk: the popular DerSimonian–Laird estimator tends to *underestimate* τ^2 with few studies, which *narrows* the prediction interval — producing exactly the false confidence in transfer that Chapter 47 warned against.

The implications.

1. **Point-estimating τ^2 understates transfer uncertainty**, especially with few studies and especially via DerSimonian–Laird.
2. **The Bayesian model with a weakly-informative prior on τ is the honest treatment** — it integrates over the τ uncertainty, giving a predictive interval that reflects it, rather than committing to one point.
3. **Report the sensitivity.** Whether Bayesian or classical, show how the prediction interval moves across τ^2 treatments — this *is* the specification-curve discipline of Chapter 46 applied to the synthesis.
4. **For a marginal CBA (Chapter 45), this is decisive.** If the transferred WTP's plausible range depends on an arbitrary τ^2 choice, and the CBA verdict depends on the transferred WTP, then the τ^2 treatment can flip the appraisal — and must be surfaced, not buried.

48.5 Multivariate synthesis: correlated outcomes and partial reporting

Your DCEs do not each report a single WTP — they report WTP for *several attributes* (stock, water quality, habitat, coastal protection), and crucially, *not every study reports every attribute*. Univariate meta-analysis wastes this structure.

The problem with univariate synthesis. To synthesize "water quality WTP," a univariate meta-analysis uses *only* the studies that reported it, discarding the studies that reported *stock* WTP but not water quality. Yet if stock and water-quality WTP are *correlated* across studies (they are — both track a study's population income, elicitation quality, and generosity), then the stock estimates carry *information* about water quality. Discarding them is discarding evidence.

Multivariate (multiple-outcome) meta-analysis. Model the outcomes *jointly*, with a between-study *covariance* matrix Σ capturing their correlations. A study reporting only stock WTP still contributes to the water-quality estimate *through the correlation*, and studies reporting both contribute to estimating Σ itself. This "borrowing strength across outcomes" is the multivariate analogue of the borrowing-across-studies of §48.3.

Worked Example 48.4 (Multivariate gain, verified). Stock and water-quality WTP correlated at $\rho = 0.7$; 10 studies report both, 6 stock-only, 2 water-quality-only:

Water-quality estimate	Relative SE
Univariate (12 studies reporting it)	0.289
Multivariate (borrows from stock)	**0.206 (29% tighter)**

(Verified.) The multivariate model tightens the water-quality estimate by 29% by exploiting the correlation with stock WTP and the stock-only studies. **The gain is largest exactly when it is most needed** — when few studies report a given attribute, so the univariate estimate is weak, but many report a *correlated* attribute.

Cautions. The between-outcome correlations must be estimated (hard with few studies) or assumed (a sensitivity parameter); mis-specifying them can *harm* the estimate. And borrowing strength across outcomes assumes the correlation is *stable* across studies — a testable-in-principle but data-hungry assumption. For your synthesis, multivariate methods are most valuable for the *sparsely-reported* attributes, and should be reported alongside the univariate results with the correlation assumptions made explicit.

> **Active recall 48.B.** How does multivariate meta-analysis tighten the estimate of a sparsely-reported WTP outcome, and when is the gain largest? *(Answer: it models the several WTP outcomes (stock, water quality, habitat)* jointly*, with a between-study* covariance* matrix capturing how they correlate across studies. A study that reports only *stock* WTP then contributes to the *water-quality* estimate *through the correlation* — because if the two are correlated across studies, knowing a study's stock WTP is informative about what its water-quality WTP would have been. This "borrows strength across outcomes," recovering information that univariate synthesis discards when it uses only the studies reporting the target outcome. Verified: with $\rho = 0.7$, the water-quality estimate tightened 29% (relative SE 0.289 $\rightarrow$ 0.206) by borrowing from stock estimates and stock-only studies. The gain is *largest* when (i) *few* studies report the target attribute (so the univariate estimate is weak and has most to gain), (ii) *many* studies report a *correlated* attribute (so there is much to borrow), and (iii) the correlation is *strong* (so the borrowed information is highly relevant). It is smallest when every study reports the target outcome (nothing to borrow) or the outcomes are uncorrelated (borrowing conveys no information). The caution: the correlations must be estimated or assumed, and mis-specifying them can *harm* rather than help — so report the assumed correlation and its sensitivity.)*

48.6 Structural synthesis: pooling parameters, not numbers

The most ambitious form of synthesis pools not *reduced-form WTP numbers* but *structural preference parameters*, imposing economic consistency across studies.

The idea. A raw WTP is a reduced-form summary that bundles preferences, scale, context, and method. Two studies reporting different WTPs may share the *same underlying preference structure* differing only in scale (Chapter 25) or context. **Structural synthesis** models the underlying utility parameters — pooling, say, the marginal utilities and scale factors — so that heterogeneity in method and context is absorbed by the *structure* rather than left as unexplained τ^2.

What it offers.

- **Theory consistency.** The synthesized quantity respects economic theory (demand curves slope the right way, WTP satisfies adding-up), which reduced-form pooling does not guarantee.
- **Better transfer.** A synthesized *preference structure* can be applied to a new site's characteristics to *generate* a WTP, rather than transferring a number — a more theory-grounded version of the meta-regression function transfer (Chapter 47, 49).
- **Absorbing method heterogeneity.** If part of the between-study variance is *scale* differences (Chapter 25, 39), modelling scale explicitly *removes* it from τ^2, tightening the transfer-relevant prediction.

The costs. Structural synthesis requires the primary studies to report enough to *reconstruct* the structure (often they do not — reduced-form WTPs are all that is available), imposes strong functional-form assumptions (which, if wrong, bias the synthesis in ways reduced-form pooling avoids), and is technically demanding. It is the frontier rather than the default.

For your work. Full structural synthesis of 18 heterogeneously-reported DCEs is likely infeasible (the primary studies will not report enough), but two structural *ideas* are usable: (i) explicitly modelling *scale* differences across studies as a moderator (Chapter 25's scale-versus-preference logic applied across the literature), which absorbs method heterogeneity; and (ii) synthesizing in *WTP space* (Chapter 38) so that what is pooled is already the welfare-relevant, scale-free quantity. These capture much of the structural benefit without requiring full utility reconstruction.

48.7 The meta-analytic value of a new study

The synthesis can answer a question your PDF's research-planning concern raises directly: *given the 18 existing studies, is it worth running a new primary DCE?*

The logic. A new study adds precision to the pooled estimate, but with diminishing returns — the more (and more precise) the existing evidence, the less a new study adds. Value-of-information methods (Chapter 45) quantify this: the *expected value of sample information* from a new study is the expected improvement in the decision it enables, and it should be compared to the study's cost.

Worked Example 48.5 (Value of a new study, verified). The current pooled-mean standard error is 3.61; a new study of precision s_{new} is added:

New study s_{new}	SE(μ) before $\rightarrow$ after	Reduction
4 (precise)	3.610 $\rightarrow$ 3.485	3%
7 (moderate)	3.610 $\rightarrow$ 3.504	3%
12 (imprecise)	3.610 $\rightarrow$ 3.536	2%

(Verified.) Two lessons, one obvious and one crucial:

1. **A precise new study helps more than an imprecise one** — obvious, but it means a new study is worth running only if it can be *substantially more precise* than the typical existing study.
2. **Even a precise new study barely moves the *mean*** (a 3% reduction), because 18 studies already pin it down reasonably well — and, more importantly, **a new study does essentially nothing for the *prediction interval***, because the prediction interval is governed by τ^2, which is a property of the *heterogeneity among true effects* and is *not reduced* by adding another heterogeneous study. If anything, a new study of a genuinely different good/population *confirms* the heterogeneity rather than resolving it.

The decisive implication. *Match the information to the decision.* If your decision needs the *mean* WTP (e.g. a rough aggregate benefit), 18 studies may already suffice and a new one adds little. If it needs the *prediction* for a *specific site* (benefit transfer), then a new study *of that site* is enormously valuable — not because it tightens τ^2, but because it replaces a *transfer* (drawing from the wide prediction interval) with a *direct estimate* (a narrow site-specific interval). **The value of a new study depends entirely on whether it is a generic addition to a heterogeneous pool (low value) or a site-specific primary study for the decision at hand (high value).** This directly informs your research strategy: for a *specific* FRD policy site, a new site-specific DCE dominates transferring from the meta-analysis; for *general* evidence, the 18 studies may be enough.

> **Stop and think 48.2.** A funder asks whether to pay for a 19th finfish DCE or rely on the meta-analysis of the existing 18. Under what circumstances is each right? *(Answer: it depends entirely on* what the decision needs* and *whether the new study is generic or site-specific.* **Rely on the meta-analysis (skip the 19th study) when:** the decision needs the *general mean* WTP (a rough aggregate or a first-pass appraisal), because 18 studies already pin the mean down reasonably — verified, even a *precise* 19th study cuts the mean's SE by only ~3%, a poor return on a costly DCE; and a 19th study of yet another different good/population does *nothing* for the *prediction interval* (governed by the irreducible τ^2), so it cannot improve transfer precision — it merely confirms the heterogeneity. **Commission the new study when:** the decision needs a WTP for a *specific policy site* (the actual benefit-transfer situation), because a *site-specific* primary DCE replaces a *transfer* — drawing from the wide prediction interval [roughly €5–€75] — with a *direct estimate* carrying a narrow site-specific interval, an enormous precision gain exactly where the decision lives; or when the new study can be *substantially more precise* than

the existing ones, or fills a *gap* in the moderator space (a species/population/method combination no existing study covers) that the meta-regression must otherwise extrapolate into. The general principle: a generic addition to a heterogeneous pool has low value (diminishing returns on the mean, no effect on τ^2); a targeted study at the decision's own site or in an unfilled region of moderator space has high value. Match the evidence to the decision — value of information, not study count, is the criterion.)*

48.8 Failure modes

F1. Plug-in τ^2 with few studies. Treating $\hat{\tau}^2$ as known, understating the prediction interval. *Fix:* Bayesian model integrating over τ; report τ^2-treatment sensitivity (§48.2, §48.4).

F2. DerSimonian–Laird prediction intervals. Downward-biased τ^2 narrowing transfer uncertainty. *Fix:* REML or Bayesian; sensitivity (§48.4).

F3. Over-tight or improper prior on τ. Distorting the very quantity of interest. *Fix:* weakly-informative prior, reported, sensitivity-tested (§48.2).

F4. Dropping outlier studies. Selection on the outcome. *Fix:* partial pooling shrinks by precision without discarding (§48.3).

F5. Trusting imprecise estimates at face value. Over-weighting noise. *Fix:* shrinkage (§48.3).

F6. Univariate synthesis of correlated outcomes. Discarding information from correlated attributes. *Fix:* multivariate synthesis for sparse outcomes (§48.5).

F7. Assuming stable/known between-outcome correlations. Mis-specification harming the estimate. *Fix:* estimate where possible; report the assumption and sensitivity (§48.5).

F8. Reduced-form pooling ignoring scale. Leaving scale heterogeneity in τ^2. *Fix:* model scale / synthesize in WTP space (§48.6).

F9. Commissioning a generic new study for transfer. Expecting it to tighten the prediction interval. *Fix:* value-of-information; a site-specific study for a site-specific decision (§48.7).

F10. Reporting a single synthesis specification. Hiding sensitivity to model and prior. *Fix:* specification curve over synthesis choices (§48.4; Chapter 46).

48.9 Diagnostic questions for your own papers

1. τ^2 **uncertainty.** Am I propagating uncertainty in τ^2 (Bayesian) or reporting sensitivity across estimators? *(§48.2, §48.4, F1–F2.)*
2. **Prior.** Is the prior on τ weakly-informative, reported, and sensitivity-tested? *(§48.2, F3.)*
3. **Shrinkage.** Do I report partial-pooling study estimates rather than dropping or naively trusting outliers? *(§48.3, F4–F5.)*
4. **Multivariate.** For sparsely-reported attributes, have I used multivariate synthesis? *(§48.5, F6.)*
5. **Correlations.** Are between-outcome correlations estimated/assumed transparently with sensitivity? *(§48.5, F7.)*
6. **Structure.** Have I modelled scale / synthesized in WTP space to absorb method heterogeneity? *(§48.6, F8.)*
7. **New evidence.** Have I assessed the value of a new study, distinguishing generic from site-specific? *(§48.7, F9.)*
8. **Robustness.** Is a specification curve over synthesis choices reported? *(§48.4, F10.)*

48.10 Exercises

(A) Comprehension.

A1. Write the hierarchical model and explain why the Bayesian version gives a wider, more honest prediction interval.

A2. Define the shrinkage factor and explain when a study is shrunk hard versus little.

A3. How does multivariate synthesis use a study that reports only one of several outcomes?

(B) Derivation / reasoning.

B1. Derive the partial-pooling posterior mean and shrinkage factor.

B2. Explain the James–Stein result that shrinkage lowers total MSE.

B3. Explain why a new study reduces the confidence interval but not the prediction interval.

(C) Research application / critique.

C1. Design the Bayesian hierarchical synthesis of your 18 DCEs, specifying the prior and outputs.

C2. A co-author wants to drop three low outlier studies. Propose the principled alternative.

C3. A funder asks whether a 19th DCE is worth it. Write the value-of-information answer.

48.11 Solutions

A1. The model is $\hat{\theta}_i \mid \theta_i \sim N(\theta_i, s_i^2)$ (within-study sampling) and $\theta_i \mid \mu, \tau \sim N(\mu, \tau^2)$ (between-study distribution), with priors $p(\mu), p(\tau)$. The *classical* approach estimates $\hat{\mu}, \hat{\tau}^2$ and *plugs them in*, forming the prediction interval $\hat{\mu} \pm t\sqrt{\hat{\tau}^2 + \mathrm{se}(\hat{\mu})^2}$ as if $\hat{\tau}^2$ were the true value. The *Bayesian* approach computes the joint posterior $p(\mu, \tau, \{\theta_i\} \mid \text{data})$ and forms the *posterior predictive* for a new study by integrating over the posterior of τ. This is *wider and more honest* because, with few studies, τ^2 is very uncertain, and the prediction interval *depends on* τ^2; the plug-in ignores this (treating a badly-estimated variance as known), while the Bayesian integration incorporates it. Verified motivation: the prediction-interval width swings from 40 to 68 across defensible τ^2 estimators, and the Bayesian model propagates that uncertainty rather than committing to one point.

A2. The shrinkage factor is $B_i = s_i^2/(s_i^2+\tau^2)$ — the fraction of study i's estimate variance that is *within-study* (sampling) rather than *between-study* (real heterogeneity). The posterior mean is $\hat{\theta}_i^{\text{post}} = (1-B_i)\hat{\theta}_i + B_i\mu$, a precision-weighted average of the study's own estimate and the grand mean. A study is *shrunk hard* toward the mean (large B_i, near 1) when $s_i^2 \gg \tau^2$ — i.e. it is *imprecise* relative to the between-study spread, so its own estimate is mostly noise and the collective mean is more informative (verified: study 13, $s_i = 13.5$, $B_i = 0.53$, pulled from €13.7 to €32.6). A study is shrunk *little* (small B_i, near 0) when $s_i^2 \ll \tau^2$ — it is *precise*, so its own estimate is trustworthy and reflects genuine heterogeneity rather than noise. Shrinkage is thus governed by *relative precision*: the model automatically trusts precise studies and discounts imprecise ones.

A3. Multivariate synthesis models the several outcomes *jointly* with a between-study covariance matrix Σ encoding their correlations. A study reporting only outcome A (stock) but not outcome B (water quality) contributes to estimating B *through the correlation* ρ_{AB}: since A and B co-vary across studies, the observed A carries information about the unobserved B for that study (formally, the conditional distribution of B given A shifts the water-quality likelihood). It also contributes to estimating Σ if it reports A precisely. Thus no reporting study is wasted, and the target outcome borrows strength from correlated outcomes and from partially-reporting studies. Verified: with $\rho = 0.7$, the water-quality estimate tightened 29% by incorporating stock-only studies. The gain requires the correlation to be non-trivial and reasonably stable across studies.

B1. For study i, combine the likelihood $\hat{\theta}_i \mid \theta_i \sim N(\theta_i, s_i^2)$ with the prior $\theta_i \sim N(\mu, \tau^2)$ (treating μ, τ^2 as known). This is the conjugate normal-normal update: the posterior for θ_i is normal with precision equal to the sum of precisions, $1/s_i^2 + 1/\tau^2$, and mean equal to the precision-weighted average of the two "observations" ($\hat{\theta}_i$ with precision $1/s_i^2$, and the prior mean μ with precision $1/\tau^2$):

$$\hat{\theta}_i^{\text{post}} = \frac{(1/s_i^2)\hat{\theta}_i + (1/\tau^2)\mu}{1/s_i^2 + 1/\tau^2}.$$

Multiply numerator and denominator by $s_i^2\tau^2/(s_i^2+\tau^2)$ to rewrite as

$$\hat{\theta}_i^{\text{post}} = \frac{\tau^2}{s_i^2+\tau^2}\hat{\theta}_i + \frac{s_i^2}{s_i^2+\tau^2}\mu = (1-B_i)\hat{\theta}_i + B_i\mu, \quad B_i = \frac{s_i^2}{s_i^2+\tau^2}.$$

So the shrinkage factor is the *within-study share of total variance*, and the posterior mean interpolates between the raw estimate ($B_i \to 0$ for precise studies) and the grand mean ($B_i \to 1$ for imprecise ones). The posterior variance is $1/(1/s_i^2 + 1/\tau^2) = s_i^2\tau^2/(s_i^2+\tau^2) < s_i^2$, so borrowing strength also *reduces* each study's posterior variance.

B2. The James–Stein result states that when estimating three or more means simultaneously, the estimator that *shrinks* the individual sample means toward a common value has *lower total mean squared error* than the individual sample means themselves — uniformly, regardless of the true values. Intuition: each raw estimate $\hat{\theta}_i = \theta_i + \varepsilon_i$ contains signal plus noise; shrinking toward the mean trades a small *bias* (pulling each estimate toward the center) for a *large variance reduction* (removing the noise component that scatters the estimates). Because MSE = bias² + variance, and the variance reduction dominates the squared-bias increase when the estimates are noisy, total MSE falls. The effect is strongest when the true effects are *close together* (little to lose from shrinking) and the estimates are *noisy* (much variance to remove); it is weakest when the effects are widely dispersed and precisely estimated. Verified: partial pooling achieved MSE 39.0 against 120.3 for no pooling and 102.6 for complete pooling — beating *both* extremes, which is the hallmark of the result: the optimal amount of shrinkage (governed by B_i, hence by τ^2 relative to s_i^2) is *between* trusting each study and forcing them equal, and the hierarchical model estimates that optimal amount from the data.

B3. The *confidence interval* for the hyper-mean μ has half-width $\propto \text{se}(\mu) = \sqrt{1/\sum_i w_i^*}$, where $w_i^* = 1/(s_i^2+\tau^2)$. Adding a study increases $\sum_i w_i^*$, so $\text{se}(\mu)$ *falls* — more studies pin down the *mean* better (verified: a precise new study cut SE(μ) by ~3%). The *prediction interval* has half-width $\propto \sqrt{\tau^2 + \text{se}(\mu)^2}$. As studies accumulate, $\text{se}(\mu)^2 \to 0$, but τ^2 — the *between-study variance*, a property of the *dispersion of true effects* — does *not* shrink toward zero; it converges to its true value. So the prediction interval converges to $\approx \pm t\tau$, an *irreducible* width set by the genuine heterogeneity. Adding a study of a *different* good/population provides more information *about* τ^2 but does not *reduce* it — indeed it may confirm that τ^2 is large. Verified: a new study barely moved the prediction interval. Consequence: if the decision needs the *prediction* (benefit transfer to a specific site), more generic studies do not help — only a *site-specific* study does, by replacing the prediction (drawn from the wide τ-governed interval) with a direct estimate (a narrow site-specific interval).

C1. *Bayesian hierarchical synthesis of the 18 DCEs.* **Model:** $\hat{\theta}_i \mid \theta_i \sim N(\theta_i, s_i^2)$ (with s_i from Chapter 47's harmonized extraction, and *measurement*/reconstruction uncertainty for imputed SEs propagated), $\theta_i \mid \mu, \tau \sim N(\mu, \tau^2)$; extend to a *meta-regression* form $\theta_i \sim N(\beta_0 + \beta' \mathbf{z}_i, \tau^2_{\text{resid}})$ with the Chapter 47 moderators (hypothetical, scope, format, income, species). **Priors:** a weakly-informative prior on μ (and β) — e.g. normal with a scale wide relative to plausible WTP; and, critically, a *weakly-informative* prior on τ — a half-normal or half-Cauchy with scale reflecting plausible between-study SD (informed by the field, e.g. €10–20), *reported explicitly* and *sensitivity-tested* against tighter and wider alternatives, since with 18 studies the prior influences the prediction interval (verified width swing 40–68). **Dependence:** implement as a *three-level* model (estimates within studies) to handle multiple estimates per paper (Chapter 47 §47.7), or use robust variance estimation. **Multivariate:** for sparsely-reported attributes, extend to a multivariate model with a between-outcome covariance, borrowing strength across correlated WTPs (verified 29% gain). **Estimation:** MCMC (Stan/JAGS), checking convergence ($\hat{R}$, effective sample size, trace plots — the simulation discipline of Chapter 38). **Outputs:** (i) the posterior hyper-mean with a credible interval

that *accounts for* τ uncertainty; (ii) the *posterior predictive distribution* — the transfer-relevant object, honestly wide; (iii) *partial-pooling posterior estimates* for each study (the corrected "what each study found," Stein-superior to raw estimates); (iv) the *meta-regression coefficients* as the benefit-transfer function; (v) a *specification curve* over prior and model choices (Chapter 46). **Carry forward:** the posterior predictive (not the credible interval of the mean) into benefit transfer (Chapter 49) and the CBA Monte Carlo (Chapter 45).

C2. *Principled alternative to dropping the three low outliers: partial pooling.* Dropping them is *selection on the outcome* — removing studies *because* their estimates are inconveniently low biases the pooled mean upward and is an undisclosed researcher degree of freedom (Chapters 32, 46); if done to reach a "cleaner" result it is indistinguishable from specification search. The hierarchical model already handles outliers correctly and defensibly: (i) *if the low outliers are imprecise* (large s_i), their shrinkage factors B_i are large and they are automatically pulled hard toward the mean (verified: a €13.7 estimate with large SE shrinks to €32.6), so their *influence* is already discounted in proportion to their unreliability — without discarding their information; (ii) *if they are precise* (small s_i), they are *not* outliers to be removed but genuine *heterogeneity* — real evidence that some contexts have low finfish WTP — and dropping them would suppress a true finding. Either way, the model's treatment is governed by *precision*, not by my judgement about plausibility. Concretely I would: keep all 18 studies; report their *partial-pooling posterior estimates* (showing how much each was shrunk and why); *investigate* the low studies as potential *moderator* signals (do they share a species, population, or method? — feeding the meta-regression, which may *explain* their low values rather than dismiss them); and, if a study is genuinely erroneous (a coding or extraction error, not just a low value), correct or exclude it *on documented methodological grounds* with a sensitivity analysis, never because the number is low. The James–Stein result guarantees this is more accurate than either dropping them or trusting them at face value (verified MSE 39.0 vs 120.3).

C3. *Value-of-information answer on a 19th DCE.* The question is not "would a 19th study add information?" (trivially yes) but "is the information worth its cost *for the decision at hand*?" — and the answer hinges on *what the decision needs* and *whether the study is generic or targeted*. **If the decision needs the general mean WTP** (a broad aggregate benefit, or first-pass evidence): a 19th generic study is likely *not* worth it — verified, even a *precise* new study cuts the pooled mean's SE by only ~3% because 18 studies already pin the mean down (diminishing returns), and it does *nothing* for the *prediction interval* (governed by the irreducible τ^2), so it cannot improve transfer precision; the meta-analysis suffices. **If the decision needs a WTP for a *specific* policy site** (the actual benefit-transfer situation for an FRD appraisal): a *site-specific* 19th DCE is potentially *very* valuable — it replaces a *transfer* (drawing from the wide prediction interval, roughly €5–€75) with a *direct estimate* (a narrow site-specific interval), an enormous precision gain precisely where the decision lives, and one no amount of existing evidence provides. **If the study fills a gap** in moderator space (a species/population/method combination the 18 studies do not cover, forcing the meta-regression to extrapolate): it is valuable for *reducing extrapolation*, even if generic. **Formal comparison:** estimate the *expected value of sample information* — the expected improvement in the appraisal decision (e.g. the reduction in the probability of a wrong build/reject verdict, Chapter 45) from the study's expected precision — and compare it to the study's cost; commission only if EVSI exceeds cost. **Recommendation to the funder:** if the goal is *this* FRD site's appraisal, a *site-specific* DCE dominates a generic addition and dominates transferring from the meta-analysis; if the goal is *general* finfish-WTP evidence, the 18 studies are likely adequate and the money is better spent elsewhere. Match the study to the decision, and let value of information — not study count — decide.

48.12 Spaced review prompts

+1 day. From memory: write the hierarchical model; define the shrinkage factor; state why a new study helps the CI but not the PI.

+1 week. Explain the James–Stein result; why the prior on τ matters with few studies; how multivariate synthesis borrows strength.

+1 month. For your 18-study synthesis, fit the Bayesian hierarchical (multivariate where sparse), report the posterior predictive, partial-pooling estimates, prior sensitivity, and a value-of-information assessment for a new study. Close with the spine sentence about whether you are propagating the uncertainty the decision needs.

48.13 How this chapter connects to the rest of the book

- **Chapter 47** built the classical engine this chapter extends; **Chapter 21** supplies the hierarchical/shrinkage machinery; **Chapter 38** supplies the "variance parameters are hard" result that *is* the τ^2 problem here, and the MCMC convergence discipline; the Bayesian-inference chapter supplies priors and posteriors.
- **Chapter 25** supplies the scale-versus-preference logic behind structural synthesis (§48.6); **Chapter 45** supplies the value-of-information framework applied to a new study (§48.7); **Chapter 46** supplies the specification-curve treatment of synthesis and prior choices.
- Downstream: **Chapter 49 (benefit transfer)** is the direct consumer — the *posterior predictive distribution* (not the credible interval) and the *meta-regression / structural function* built here are exactly the transfer apparatus, and the value-of-information logic of §48.7 decides transfer-versus-primary-study; the synthesized uncertainty then enters the **Chapter 45** appraisal Monte Carlo.

This chapter equips your synthesis to do what a marginal, uncertainty-sensitive CBA demands: propagate the between-study spread honestly (Bayesian τ), correct noisy studies (shrinkage), exploit correlated outcomes (multivariate), and tell the decision-maker whether new evidence is worth buying (value of information).

48.14 Research translation task

For your **18-study synthesis**, produce an advanced-synthesis note:

1. **Bayesian hierarchical model.** Specify the model, the *weakly-informative prior on* τ (reported and sensitivity-tested), and the three-level dependence structure; report convergence diagnostics.
2. **Outputs.** Report the posterior hyper-mean, the *posterior predictive distribution* (for transfer), and the *partial-pooling study estimates*.
3. τ^2 **sensitivity.** Show how the prediction interval moves across τ^2 treatments and priors (specification curve).
4. **Multivariate.** For sparsely-reported attributes, report the multivariate estimate and its gain over univariate, with the correlation assumption stated.
5. **Value of information.** Assess whether a new study is worth commissioning, distinguishing a generic addition from a site-specific study for the FRD decision.
6. **Close** with the spine sentence: *"The Bayesian hierarchical synthesis gives a hyper-mean of €[·] (credible interval [·]) and a posterior predictive of [€·, €·] that integrates over* τ *uncertainty; partial-pooling corrects the [·] imprecise studies, multivariate synthesis tightens [sparse outcome] by [·]%, and value-of-information indicates a [generic/site-specific] new study is [not] worth commissioning."*

Carry the posterior predictive and transfer function into Chapter 49 (benefit transfer) and the appraisal Monte Carlo (Chapter 45).

48.15 Bibliography

Core. - Gelman, Carlin, Stern, Dunson, Vehtari & Rubin, *Bayesian Data Analysis* — the definitive treatment of hierarchical models, shrinkage, and priors on variance components (§48.2–§48.4). *(Verify edition.)* -

Higgins, Thompson & Spiegelhalter (2009), "A Re-evaluation of Random-Effects Meta-Analysis," *JRSS A* — Bayesian random-effects and the prediction interval (§48.2, §48.4). *(Verify volume/pages.)* - Röver (2020), "Bayesian Random-Effects Meta-Analysis Using the bayesmeta R Package," *Journal of Statistical Software* — practical Bayesian meta-analysis with weakly-informative τ priors (§48.2). *(Verify volume/pages.)*

Applied. - Efron & Morris (1977), "Stein's Paradox in Statistics," *Scientific American* — the accessible account of shrinkage and its accuracy gains (§48.3). *(Verify pages.)* - Jackson, Riley & White (2011), "Multivariate Meta-Analysis: Potential and Promise," *Statistics in Medicine* — multivariate/multiple-outcome synthesis (§48.5). *(Verify volume/pages.)* - Nelson & Kennedy (2009), "The Use (and Abuse) of Meta-Analysis in Environmental and Natural Resource Economics," *Environmental and Resource Economics* — dependence and synthesis issues in valuation (§48.5). *(Verify volume/pages.)*

Advanced. - Riley, Higgins & Deeks, or Riley (2009) on the correlation and prediction problems in multivariate meta-analysis (§48.5). *(Verify citations.)* - Ades & Sutton (2006), "Multiparameter Evidence Synthesis in Epidemiology and Medical Decision-Making," *JRSS A* — structural / multiparameter synthesis and value of information (§48.6–§48.7). *(Verify volume/pages.)* - Sutton, Cooper & Jones on the expected value of information for a new study in a meta-analytic context (§48.7). *(Verify citation.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Higgins et al. / Efron–Morris / Jackson et al. / Ades–Sutton citations before using them, per the book's no-unverified-citation rule.

End of Chapter 48. Next: Chapter 49 — Benefit Transfer: Using Existing Values for New Policy Sites, which turns the synthesis engine of Chapters 47–48 to its policy purpose — transferring WTP to a site where no primary study exists. It covers unit-value versus function transfer, the transfer error and its measurement, the convergent-validity tests of transfer accuracy, the role of the prediction interval and the meta-regression function built here, and why benefit transfer is simultaneously indispensable to practical appraisal and the largest single source of uncertainty in it. Tell me when to write it, or request a different chapter.

Chapter 49 — Benefit Transfer: Using Existing Values at New Policy Sites

Part XII: Evidence Synthesis

> **The spine question, applied to this chapter.** The synthesis of Chapters 47–48 built a model of how finfish WTP depends on context. This chapter turns it to its purpose: producing a WTP for a *policy site where no primary study exists* — the situation most real appraisals face, because there is rarely time or budget to run a bespoke survey for every decision. Benefit transfer is therefore *indispensable*. It is also, verified below, the *largest single source of uncertainty* in an appraisal: even done well, transfer errors run to tens of percent, and across the transfer's prediction interval the FRD verdict swings from a €78m loss to a €448m gain. The spine question: *what am I actually assuming when I move a value from where it was measured to where it is needed, how large is the resulting error, and does my appraisal survive it?*

49.0 Conceptual motivation: the unavoidable, dangerous shortcut

Almost every cost–benefit analysis of an environmental policy faces the same problem: it needs a monetary value for a benefit — a fish stock improvement, a water-quality gain, a habitat restored — but has no primary valuation study for *this* site, *this* population, *this* good. Commissioning one costs money and takes years the decision does not have. So the analyst *transfers* a value estimated elsewhere: from a study of a similar good, at a similar site, adjusted (if the analyst is careful) for the differences. This is **benefit transfer**, and it is not an occasional expedient — it is the *dominant* mode of environmental valuation in actual policy practice. Most appraisal numbers you will ever see are transferred, not primary.

This creates a tension the chapter must confront honestly. Benefit transfer is *necessary* — the alternative, a primary study for every decision, is infeasible. But it is also the appraisal's *weakest link*, and understanding *how* weak is essential to using it responsibly. The verified results are sobering.

Transfer error is large even at its best. In a leave-one-out exercise on 18 valuation sites — predicting each site's WTP from the others, exactly mimicking transfer — the simplest method (transferring the *average* value) had a median error of 20% and a 90th-percentile error of 37%. The more sophisticated *function transfer* (using the meta-regression to adjust for site characteristics) did better — median 7%, 90th percentile 16% — but "better" still means routine errors of tens of percent. **No benefit transfer is accurate to a few percent; the honest expectation is a substantial error band.**

Adjustment helps only if the adjustment is right. The intuition that adjusting for a site's income and scope must improve the transfer is *conditional*. Verified: a *correctly-specified* adjustment cut a 79% naive error to essentially zero — but a *mis-specified* adjustment (wrong income elasticity) left a 26% error, worse than it needed to be. Benefit-transfer error is the *product* of two things: how *dissimilar* the policy site is from the study sites, and how *wrong* the transfer function is. Adjusting a value with a bad function can be worse than not adjusting at all.

The prediction interval, not the confidence interval, is the transfer uncertainty. This is the through-line from Chapter 47. The policy site is a *new draw* from the distribution of true WTPs, so its plausible value spans the *prediction interval* — verified [€5, €75] against a confidence interval of [€31, €49], four times wider. Using the confidence interval to state transfer uncertainty understates it fourfold, producing exactly the false confidence that makes transferred values dangerous.

And it can decide the appraisal. Verified, propagating the transfer prediction interval through the FRD CBA gives an NPV ranging from **−€78m (fail) to +€448m (pass)** — the transfer uncertainty *alone* flips the verdict. For your work, this is the culminating point of the whole appraisal arc: the benefit transfer from your 18-study meta-analysis to a specific FRD policy site is not a minor input but potentially *the* determinant of whether the project appears worthwhile, and it must be reported as a range, with its validity tested, not as a point.

49.1 Prerequisites

From **Chapters 47–48**: the meta-analysis, meta-regression, prediction interval, and posterior predictive distribution — the transfer *engine*. From **Chapter 42**: the extrapolation hazard (transfer is prediction to an unobserved point). From **Chapter 39**: convergent validity and equivalence testing (used to *test* transfer accuracy). From **Chapter 45**: propagating uncertainty through the CBA. From **Chapters 34–35**: the welfare-measure and aggregation compatibility that transfer requires.

New notation. *Study site* (where the value was estimated), *policy site* (where it is needed), *unit-value transfer* (transferring a single number), *function transfer* (transferring an adjustment function), *transfer error* (the discrepancy between transferred and true value), *transferability* (whether transfer is valid at all).

49.2 The transfer problem and its two methods

The setup. A value V_s (or a valuation function) has been estimated at one or more **study sites** s. It is needed at a **policy site** p where no primary estimate exists. Benefit transfer produces an estimate $\hat{V}_p$ from the study-site evidence. The **transfer error** is $|\hat{V}_p - V_p|/V_p$, where V_p is the (unobserved) true policy-site value — measurable only in *validation* exercises where a primary estimate happens to exist (§49.5).

Two families of method.

Unit-value transfer takes a *single value* (or a simple average across study sites) and applies it to the policy site, possibly with a crude adjustment (e.g. an income ratio). It is simple, transparent, and requires little from the source studies — but it ignores the *differences* between study and policy sites, so it errs whenever those differences matter.

Function transfer takes an *estimated valuation function* — the meta-regression of Chapter 47, or a single study's WTP function — and evaluates it at the *policy site's characteristics* (its income, the scope of its improvement, its population). It *adjusts* for the site's specific features, and so — *when the function is correct* — is more accurate. **Meta-analytic function transfer**, using the meta-regression fitted across many studies, is the modern standard: it exploits the full evidence base and predicts a value tailored to the policy site.

Worked Example 49.1 (Unit-value versus function transfer, verified). A leave-one-out exercise on 18 valuation sites (predict each from the other 17, as transfer would):

Method	Median error	Mean error	90th percentile
Unit-value (mean of other sites)	**20%**	20%	37%

Method	Median error	Mean error	90th percentile
Function (meta-regression)	**7%**	9%	16%

(Verified.) Function transfer roughly halves the median error and more than halves the tail error, by adjusting for the differences unit-value transfer ignores. **But both errors are large** — even the better method routinely errs by 7–16%, and worse in the tail. This is the irreducible reality: benefit transfer trades the cost of a primary study for an error of tens of percent, and no method eliminates it.

> **Active recall 49.A.** What is the difference between unit-value and function transfer, and why does function transfer usually — but not always — do better? *(Answer:* unit-value transfer* applies a single value (or a simple average across study sites) to the policy site, perhaps with a crude adjustment — simple and transparent, but *ignoring* the differences between study and policy sites, so it errs whenever those differences matter. *Function transfer* evaluates an estimated valuation *function* (e.g. the meta-regression) at the *policy site's own characteristics* — income, scope, population, method — *adjusting* the value for the site's specific features. Function transfer usually does better because it corrects for site differences that unit-value transfer leaves uncorrected: verified, function transfer cut the median leave-one-out error from 20% to 7% and the 90th-percentile error from 37% to 16%. But it does *not always* win, because its accuracy depends on the transfer function being approximately *correct*: benefit-transfer error is the *product* of site dissimilarity and function mis-specification, so evaluating a *wrong* function at the policy site can *add* error rather than remove it — verified, a correctly-specified adjustment cut a 79% naive error to ~0%, but a mis-specified one (wrong income elasticity) left 26%, worse than it needed to be. Function transfer dominates only when the fitted function is approximately right *and* the site lies within the source data's range; a bad function or an extrapolated site can make adjustment counterproductive.)*

49.3 The anatomy of transfer error

Transfer error is not a single thing; decomposing it shows what to control.

Transfer error = (site dissimilarity) × (function quality). Two sources combine multiplicatively:

- **Site dissimilarity.** The more the policy site differs from the study sites — in income, culture, the good's characteristics, the baseline condition, the affected population — the harder the transfer. A transfer between two similar coastal towns is safer than one across continents or income brackets.
- **Function mis-specification.** How well the transfer function captures the true relationship between characteristics and WTP. A perfect function makes even a distant transfer accurate; a wrong function errs even for a similar site, and *can make adjustment worse than no adjustment.*

Worked Example 49.2 (Adjustment helps only if correct, verified). True WTP $= 3 \times \text{income}^{0.6} \times \text{scope}^{0.4}$ (constant-elasticity); transferring from a study site (income 50, scope 50, WTP €150) to a policy site (income 30, scope 25, true WTP €83.7):

Method	Transferred value	Error
Naive (copy study mean)	€150.0	**79%**
Adjusted, *correct* elasticities	€83.7	**0%**
Adjusted, *wrong* income elasticity (1.2)	€61.6	**26%**

(Verified.) The correctly-specified adjustment *eliminates* the 79% naive error; the mis-specified adjustment leaves a 26% error — better than naive here, but demonstrating that a *wrong* function degrades the transfer.

Adjustment is not automatically safe: it improves the transfer only to the extent the adjustment function is correct.

Worked Example 49.3 (Distance drives error, verified). Function transfer to policy sites at increasing distance from the study-site range:

Policy site	True WTP	Transfer	Error
Similar (within range)	49.0	51.6	5%
Moderate	60.2	64.2	7%
Distant (extrapolation)	84.0	91.1	**8%**

(Verified.) Error grows as the site moves *outside* the range of the study sites — the transfer becomes *extrapolation* rather than *interpolation* (Chapter 42's warning, now in transfer form). **Interpolation within the source range is defensible; extrapolation beyond it is dangerous**, because the function is unconstrained by data there and small mis-specifications amplify.

The practical implications.

1. **Match the sites.** Choose source studies as *similar as possible* to the policy site — the single most effective error-reducing choice.
2. **Stay within range.** Verify the policy site's characteristics lie *within* the range of the source studies' characteristics; flag any extrapolation explicitly.
3. **Adjust only with a validated function.** Use function transfer only when the transfer function is credible; a poorly-estimated meta-regression may transfer worse than a well-chosen unit value.
4. **Report both sources of error** — dissimilarity (how far is the site?) and function quality (how good is the meta-regression, Chapters 47–48?).

Stop and think 49.1. Your policy site has a much lower average income and a smaller-scope improvement than any of your 18 study sites. What are the two dangers, and what do you do? *(Answer: two compounding dangers. (i)* Extrapolation*: the policy site lies* outside* the range of the source studies' characteristics, so the meta-regression must predict where it has no data — verified, transfer error grows as the site moves outside the source range, and extrapolation is far more dangerous than interpolation (Chapter 42) because the function is unconstrained there and small mis-specifications amplify. (ii) *Mis-specified adjustment*: because the site differs substantially in income and scope, a *large* adjustment is needed, and the transfer becomes highly sensitive to the *income and scope elasticities* in the function — if those are wrong, the large adjustment introduces large error, potentially *worse* than a naive transfer (verified: a wrong income elasticity left 26% error where the correct one gave ~0%). What to do: (i) *seek better-matched source studies* — find lower-income, smaller-scope studies even if fewer, since similarity dominates sample size for transfer accuracy; (ii) *flag the extrapolation explicitly* and report that the transfer is beyond the source range; (iii) *bound* the transfer across a range of plausible income/scope elasticities (a sensitivity analysis on the adjustment function), not a single point; (iv) *widen the uncertainty* — the prediction interval (§49.4) understates error under extrapolation, so I would treat the transfer as even more uncertain than the interval implies; (v) *reconsider a primary study* — the value-of-information logic (Chapter 48) may now favour a site-specific DCE, precisely because the transfer is unreliable exactly where the site is most distinctive. The honest conclusion may be that no defensible transfer exists and the decision needs primary data.)*

49.4 The prediction interval is the transfer uncertainty

The most consequential and most violated principle: transfer uncertainty is governed by the *prediction interval*, not the confidence interval.

Why. The policy site is not the *average* of the study sites — it is a *specific new site*, effectively a new draw from the distribution of true WTPs (Chapter 47). Its plausible value therefore spans the *prediction interval* (which includes the between-study variance τ^2), not the *confidence interval* of the pooled mean (which describes only how well the mean is estimated and shrinks with more studies).

Worked Example 49.4 (Prediction versus confidence interval, verified). Pooled mean €40, τ = €16:

Interval	Range	Half-width
95% CI of the mean	[31.4, 48.6]	±8.6
95% prediction interval (the site)	**[4.8, 75.2]**	±35.2

Ratio of prediction to confidence half-width: **4.1**×.

(Verified.) The transfer uncertainty (prediction interval, ±€35) is *four times* the confidence interval (±€9). **Reporting the confidence interval as the transfer uncertainty understates it fourfold** — a pervasive error that makes transferred values look far more reliable than they are. And this is a *floor*: under extrapolation (§49.3) or an uncertain transfer function, the true transfer uncertainty is *wider* still. The Bayesian posterior predictive of Chapter 48, which additionally integrates over the uncertainty in τ^2 itself, is the most honest transfer-uncertainty object.

The consequence for the appraisal. The transferred WTP does not enter the CBA as a number but as a *distribution* — the prediction interval (or posterior predictive). Chapter 45's Monte Carlo appraisal should draw the WTP from this distribution, so that the appraisal's risk profile reflects the transfer uncertainty.

Worked Example 49.5 (Transfer uncertainty flips the verdict, verified). The FRD CBA with the transferred WTP at three points of its prediction interval:

Transferred WTP	NPV	Verdict
PI low (€5)	−€78.2m	**FAIL**
Pooled mean (€40)	€184.8m	PASS
PI high (€75)	€447.7m	PASS

(Verified.) Across the transfer prediction interval the NPV swings from a €78m loss to a €448m gain — **the benefit-transfer uncertainty alone decides the appraisal**. Reporting the CBA at the point transfer (€40 → €185m → "PASS") would conceal that a value well within the transfer's honest uncertainty range delivers a clear fail. **The verdict must be reported across the prediction interval**, and the benefit transfer will typically dominate the tornado (Chapter 45) — it is the appraisal's largest uncertainty.

> **Active recall 49.B.** Why is the prediction interval, not the confidence interval, the correct measure of benefit-transfer uncertainty, and what does ignoring this do to the CBA? *(Answer: the policy site is a* specific new site*, not the* average* of the study sites — effectively a new draw from the distribution of true WTPs — so its plausible value spans the *prediction interval*, which incorporates the *between-study variance* τ^2 (the genuine heterogeneity among true site values). The *confidence interval* of the pooled mean measures only how precisely the *average* is estimated and *shrinks* with more studies; it does *not* describe where an individual new site falls, and no number of studies makes an individual site equal to the average. Verified: the prediction interval [€5, €75] (±€35) is *4.1× wider* than the confidence interval [€31, €49] (±€9), so using the CI understates transfer uncertainty roughly fourfold. Ignoring this does two damaging things to the CBA: (i) it makes the transferred value — often the appraisal's single largest input — look far more reliable than it is, producing a *falsely precise* NPV; and (ii) it conceals that the verdict may not survive the true transfer uncertainty — verified, across the *prediction* interval the FRD NPV swings from −€78m (fail) to +€448m (pass), so the benefit transfer *alone* can flip the decision, a fact invisible if only the CI is propagated. The correct practice draws the WTP from the

prediction interval (or the Bayesian posterior predictive, which also integrates τ^2 uncertainty) in the appraisal Monte Carlo, so the risk profile reflects the transfer's real contribution.)*

49.5 Testing transferability: validity and transfer-error studies

Because transfer error is unobservable in the situations where transfer is *used* (no primary estimate exists there), its reliability must be established in situations where it *can* be checked.

Transfer-error (convergent-validity) studies. Where a policy site *does* have a primary estimate, one can *transfer* a value to it from other studies and compare the transferred value to the primary one — measuring the transfer error directly. Meta-analyses of such studies characterize the typical error (the tens-of-percent figures of §49.2), which distances of transfer are reliable, and which methods work best. Your leave-one-out exercise (§49.2) is exactly this: each of the 18 sites, in turn, is treated as a "policy site" with a known answer, so the transfer error is measurable.

Two validity tests.

- **Convergent validity of value estimates:** does the transferred value *equal* the primary value (within tolerance)? This is an **equivalence** question (Chapter 39, 33) — it must be tested by an equivalence test with a pre-specified margin (a transferred value "not significantly different" from the primary is *not* validated; absence of evidence is not evidence of transferability).
- **Convergent validity of functions:** does the transfer *function* estimated at one set of sites *predict* another set? Tested by out-of-sample prediction (the leave-one-out logic).

The tolerance question. What transfer error is *acceptable* depends on the *decision*, not on a universal threshold. If the CBA verdict is robust (the NPV is strongly positive or negative), a 30% transfer error may be tolerable; if the verdict is marginal — as the FRD appraisal is (Chapter 45) — even a 10% error may be decisive. **The acceptable transfer error is defined by whether the verdict survives it**, which is precisely the switching-value logic of Chapters 44–45 applied to the WTP input. This is why transfer validity cannot be assessed in the abstract: it must be assessed *against the decision*.

For your work. Your 18-study meta-analysis is both the *transfer engine* and — via leave-one-out — the *transfer-validity evidence*: you can report the empirical transfer error of your own function, which is far more credible than citing generic literature figures. And because the FRD appraisal is marginal, the relevant standard is stringent: report whether the verdict survives the *measured* transfer error of your function, not merely whether the point transfer passes.

49.6 The transfer-versus-primary-study decision

Benefit transfer and a primary study are the two ways to get a policy-site value; choosing between them is a value-of-information decision (Chapter 48).

The trade-off. A *primary study* is accurate for the site but costly and slow; a *transfer* is cheap and fast but carries transfer error. The choice depends on:

- **The stakes.** A large, irreversible, or precedent-setting decision justifies a primary study; a small or preliminary one does not.
- **The transfer error.** If the site is similar to well-studied sites and the transfer function is validated, transfer error is small and a primary study adds little; if the site is distinctive (extrapolation) or the function is weak, transfer error is large and a primary study adds much.
- **The verdict's robustness.** If the CBA verdict survives the transfer prediction interval (the NPV is the same sign across [€5, €75]), transfer suffices; if it does not (as verified, the FRD flips), the transfer

uncertainty is *decision-critical* and a primary study — which replaces the wide prediction interval with a narrow site-specific estimate — may be worth its cost.

The formal criterion. Compute the *expected value of a primary study* — the expected improvement in the decision from replacing the transfer's prediction interval with a site-specific estimate — and compare it to the study's cost (Chapter 48 §48.7, Chapter 45 §45.7). For a *marginal* appraisal where transfer uncertainty flips the verdict, this expected value is *large* (the transfer cannot resolve a decision it cannot pin down), and a primary study is likely justified. For a *robust* appraisal, it is small, and transfer suffices.

The synthesis of the whole arc. This decision closes the loop of Parts VIII–XII. Your 18-study meta-analysis (Chapters 47–48) enables *transfer* to the FRD site; the *marginality* of the appraisal (Chapter 45) and the *width* of the transfer prediction interval (this chapter) determine whether transfer is *adequate* or whether a *primary site-specific DCE* (Chapters 36–39) is needed. The methods are not competitors but a *sequence*: transfer first, assess whether the verdict survives the transfer uncertainty, and commission a primary study only if it does not and the stakes warrant it. **This is the disciplined answer to "how should I value the FRD benefit?" — and it is conditional on the decision, not a universal recipe.**

Stop and think 49.2. Your FRD appraisal's verdict flips across the transfer prediction interval. Does this mean you must run a primary DCE, or could transfer still suffice? *(Answer: it means transfer* alone* cannot deliver a reliable verdict, so the decision requires *either* a primary study *or* an honest acknowledgement that the appraisal is inconclusive — but not necessarily an immediate DCE. The reasoning: verified, across the transfer prediction interval [€5, €75] the FRD NPV runs from −€78m (fail) to +€448m (pass), so the benefit transfer's uncertainty is *decision-critical* — the transferred value cannot resolve a verdict it cannot pin down. This makes the *expected value of a primary site-specific study large* (it would replace the wide prediction interval with a narrow site-specific estimate, Chapter 48), so a primary DCE is *likely justified if the stakes warrant it* — a large, irreversible, precedent-setting blue-finance decision does warrant it. But there are alternatives short of a full DCE: (i) *find better-matched source studies* to narrow the transfer (if the current wide interval reflects poor site matching rather than irreducible heterogeneity); (ii) *reduce other uncertainties* (extent of market, discount rate) so the verdict becomes robust to the WTP alone — though the tornado (Chapter 45) suggests the transfer is likely a dominant driver; (iii) *stage the decision* — a pilot or phased FRD deployment that resolves uncertainty before full commitment (the quasi-option-value logic of Chapter 45); (iv) *accept inconclusiveness* — honestly reporting that the appraisal cannot determine the verdict on transferred values, which is itself decision-relevant (it argues against confident approval). The one thing I would *not* do is report the point transfer (€40 → PASS) as if the verdict were settled. So: transfer is *insufficient* here, a primary study is the most direct remedy and probably justified given the stakes, but the value-of-information calculus — not a reflex — should decide, and the honest interim verdict is "inconclusive on transferred evidence.")*

49.7 Failure modes

F1. Unit-value transfer where sites differ. Ignoring site characteristics. *Fix:* function transfer with a validated function (§49.2).

F2. Adjustment with a mis-specified function. Making the transfer worse than naive. *Fix:* validate the function; bound over elasticities; prefer a good unit value to a bad function (§49.3).

F3. Extrapolation beyond the source range. Transferring to a distinctive site the function cannot support. *Fix:* check the site is within range; flag and widen uncertainty for extrapolation (§49.3).

F4. Poor site matching. Transferring from dissimilar studies. *Fix:* select the most similar sources (§49.3).

F5. Confidence interval as transfer uncertainty. Understating uncertainty fourfold. *Fix:* prediction interval / posterior predictive (§49.4).

F6. Point transfer into the CBA. A single transferred number hiding the range. *Fix:* draw WTP from the prediction interval in the Monte Carlo; report the verdict across it (§49.4).

F7. Untested transferability. Assuming transfer is valid without evidence. *Fix:* transfer-error/validity studies (leave-one-out); equivalence testing (§49.5).

F8. Universal tolerance thresholds. A fixed "acceptable" transfer error. *Fix:* tolerance defined by whether the verdict survives (§49.5).

F9. Transfer-versus-primary by habit. Defaulting to transfer regardless of stakes. *Fix:* value-of-information; primary study when transfer uncertainty is decision-critical (§49.6).

F10. Incompatible welfare measures/units. Transferring a WTA or different-numeraire value. *Fix:* harmonize before transfer (§49.2; Chapters 34–35, 47).

49.8 Diagnostic questions for your own papers

1. **Method.** Function transfer with a validated meta-regression, not naive unit-value? *(§49.2, F1.)*
2. **Function quality.** Is the transfer function credible, and is adjustment bounded over its uncertain parameters? *(§49.3, F2.)*
3. **Range.** Does the policy site lie within the source studies' characteristic range? *(§49.3, F3.)*
4. **Matching.** Are the source studies as similar as possible to the policy site? *(§49.3, F4.)*
5. **Uncertainty.** Is transfer uncertainty the *prediction interval* / posterior predictive, not the CI? *(§49.4, F5.)*
6. **Propagation.** Is the WTP drawn from that distribution in the CBA, with the verdict reported across it? *(§49.4, F6.)*
7. **Validity.** Have I measured my function's transfer error (leave-one-out) and tested equivalence? *(§49.5, F7.)*
8. **Tolerance.** Is acceptable transfer error defined by verdict survival, not a fixed threshold? *(§49.5, F8.)*
9. **Primary study.** Have I assessed whether a primary study is warranted by value of information? *(§49.6, F9.)*

49.9 Exercises

(A) Comprehension.

A1. Define benefit transfer and distinguish unit-value from function transfer.

A2. Why is transfer error the product of site dissimilarity and function quality?

A3. Why is the prediction interval the correct transfer uncertainty?

(B) Derivation / reasoning.

B1. Explain why a mis-specified adjustment can transfer worse than a naive unit value.

B2. Explain why transfer error grows with distance from the source range.

B3. Explain how a leave-one-out exercise measures transfer error and what it establishes.

(C) Research application / critique.

C1. Design the benefit transfer from your 18-study meta-analysis to the FRD policy site.

C2. Your transferred WTP flips the CBA verdict across its prediction interval. Write the reporting and recommendation.

C3. A reviewer says your transferred value "is not significantly different from" a primary estimate, so transfer is validated. Critique.

49.10 Solutions

A1. Benefit transfer is the use of value estimates (or valuation functions) from existing *study sites* to estimate the value of an environmental change at a *policy site* where no primary study exists. **Unit-value transfer** applies a single value (or an average across study sites), possibly with a crude adjustment, ignoring most site differences — simple but error-prone when sites differ. **Function transfer** evaluates an estimated valuation *function* (a single study's WTP function, or the meta-regression across studies) at the *policy site's own characteristics*, adjusting for its income, scope, population, and method — more accurate *when the function is correct*. Verified: function transfer cut the median leave-one-out error from 20% (unit-value) to 7%. Meta-analytic function transfer, using the meta-regression of Chapters 47–48, is the modern standard.

A2. Transfer error arises from two independent sources that *compound*. *Site dissimilarity*: the more the policy site differs from the study sites (income, culture, the good, the baseline, the population), the larger the adjustment required and the more room for error. *Function quality*: how well the transfer function captures the true characteristic–WTP relationship. They multiply because a *perfect* function makes even a *distant* transfer accurate (it correctly adjusts for the difference — verified, correct elasticities cut a 79% naive error to ~0%), while a *wrong* function errs even for a *similar* site and, when the required adjustment is large (distant site), *amplifies* its mis-specification (verified, a wrong income elasticity left 26% error). So minimizing transfer error requires *both* matching sites (reduce dissimilarity) *and* a validated function (reduce mis-specification); neither alone suffices, and a large adjustment with a bad function is the worst case.

A3. The policy site is a *specific new site*, not the *average* of the study sites — effectively a new draw from the distribution of true WTPs $\theta \sim N(\mu, \tau^2)$. Its plausible value therefore spans the *prediction interval* $\hat{\mu} \pm t\sqrt{\tau^2 + \mathrm{se}(\hat{\mu})^2}$, which includes the *between-study variance* τ^2 (the genuine heterogeneity among sites). The *confidence interval* $\hat{\mu} \pm 1.96\,\mathrm{se}(\hat{\mu})$ describes only uncertainty about the *mean* μ and shrinks with more studies — but no number of studies makes an individual new site equal the average, so the CI is the wrong object for transfer. Verified: the prediction interval [€5, €75] is 4.1× wider than the CI [€31, €49]. Using the CI understates transfer uncertainty roughly fourfold, and (under extrapolation or an uncertain function) even the prediction interval is a *floor*; the Bayesian posterior predictive, integrating over τ^2 uncertainty too (Chapter 48), is the most honest transfer-uncertainty measure.

B1. A naive unit-value transfer applies the study value unchanged, incurring an error equal to the true difference between sites (verified: 79% when the sites differ substantially). An *adjusted* function transfer applies a *correction* based on the site's characteristics and the function's coefficients (e.g. income and scope elasticities). If the function is *correct*, the correction exactly offsets the site difference and the error vanishes (verified: ~0%). But if the function is *mis-specified* — wrong elasticities — the correction is the *wrong size*: it moves the value away from the study mean but *not to the right place*, and can overshoot or undershoot the true value. Verified: a wrong income elasticity (1.2 instead of 0.6) produced €61.6 against a true €83.7 — a 26% error, better than the 79% naive here but demonstrating that the adjustment *introduced* error relative to a correct adjustment. In cases where the mis-specification is severe or the required adjustment large, the adjusted value can be *further* from the truth than the unadjusted study mean — so a *bad* function transfer can be worse than a *good* unit-value transfer. The lesson: adjustment is beneficial only in proportion to the function's correctness, so an unvalidated meta-regression should not be trusted over a well-matched unit value.

B2. The transfer function is estimated from the study sites, which occupy a *range* of characteristic space. *Within* that range, the function is *constrained by data* — predictions interpolate between observed points, and mis-specification is limited because the function was fitted to be accurate there. *Beyond* the range (extrapolation), the function is *unconstrained by data*: its form is determined by the assumed functional shape (linear, log-linear, etc.), which was never tested in that region, so any mis-specification in the shape *compounds* as the site moves further out. Formally, the prediction variance of a regression grows with the

distance of the prediction point from the data centroid (the leverage term), and the *bias* from a wrong functional form grows without bound in extrapolation while staying small in interpolation. Verified: transfer error grew from 5% (similar, within range) to 8% (distant, extrapolation) even in a well-behaved case, and would grow faster with a mis-specified function. This is the transfer form of Chapter 42's extrapolation hazard: interpolation within the source range is defensible; extrapolation beyond it is dangerous and must be flagged with widened uncertainty.

B3. A *leave-one-out* exercise treats each of the k study sites, in turn, as a *policy site* with a *known* true value (its own primary estimate): fit the transfer function on the *other* $k-1$ sites, predict the held-out site from its characteristics, and compare the prediction to the held-out site's actual value. The discrepancy is a *directly measured* transfer error, for a case that *mimics* real transfer (predicting an unobserved site) but where the answer is known. Averaging over all k hold-outs characterizes the transfer function's typical error, its tail, and how it depends on distance and method. Verified: this gave median errors of 20% (unit-value) and 7% (function). What it *establishes*: an *empirical, function-specific* estimate of transfer error — far more credible than citing generic literature figures — which (i) validates (or fails to validate) the transfer method for *this* evidence base, (ii) quantifies the uncertainty to propagate into the CBA, and (iii) reveals whether function transfer actually beats unit-value *for these studies*. What it does *not* establish: that the error will be the *same* for the *actual* policy site if that site is *more distinctive* than any of the k study sites (extrapolation), so leave-one-out error is a *lower bound* on the error for an out-of-range policy site.

C1. *Benefit transfer design for the FRD policy site.* **(1) Harmonize** the policy site's requirements to the meta-analysis's effect definition — same welfare measure (compensating variation/WTP), currency, year, payment basis, and good definition (Chapters 34, 47) — so the transferred quantity is commensurable. **(2) Assess similarity and range**: compare the FRD site's characteristics (income, target species, scope of improvement, population, baseline stock trajectory) to the *range* spanned by the 18 study sites; if the site lies *within* range, transfer is interpolation (defensible); if *outside* (e.g. much lower income, a species not studied), flag the *extrapolation* and widen the uncertainty (§49.3). **(3) Function transfer**: use the meta-regression (Chapter 47) — or the Bayesian hierarchical posterior (Chapter 48) — evaluated at the FRD site's characteristics, *not* the pooled mean, since function transfer halved the error in validation (§49.2); bound the transfer over the *uncertain adjustment parameters* (income/scope elasticities) rather than using point elasticities. **(4) Uncertainty**: represent the transferred WTP as the *posterior predictive distribution* (or at least the prediction interval), *not* the confidence interval — verified 4× wider — since the FRD site is a new draw; carry this distribution into the Chapter 45 Monte Carlo so the appraisal's risk profile includes the transfer uncertainty. **(5) Validate**: report the *leave-one-out transfer error* of my own function (§49.2, §49.5) as the credible, function-specific error estimate, and — if a primary estimate exists for any comparable site — an *equivalence test* of the transferred versus primary value. **(6) Decision framing**: report the CBA verdict *across* the transfer prediction interval; since the appraisal is marginal, assess (Chapter 48, §49.6) whether the verdict survives the *measured* transfer error, and if not, evaluate the *value of a primary site-specific DCE* against its cost. **(7) Reproducibility** (Chapter 46): document the source-study selection, harmonization, function, elasticities, and the specification curve over transfer choices.

C2. *Reporting and recommendation.* **Headline:** "The FRD project's verdict depends critically on the transferred willingness-to-pay value. Using the central transferred estimate (€40/household/year), the NPV is +€185m (pass). However, this value is transferred from studies of *other* sites, and the honest uncertainty of that transfer — the prediction interval [€5, €75] — spans a range across which **the verdict reverses**: at the low end the NPV is **−€78m (fail)** and at the high end **+€448m (pass)**. The benefit transfer alone is therefore sufficient to flip the appraisal." **Supporting elements:** (i) the *prediction interval*, not the confidence interval, is the correct transfer uncertainty (verified 4× wider), and the CBA Monte Carlo draws WTP from it, so the transfer appears as the *dominant driver* in the tornado (Chapter 45); (ii) the *measured* leave-one-out transfer error of my function (median 7%, tail 16%) as the credible error estimate; (iii) a check of whether the FRD site lies within the source range, with any extrapolation flagged and the uncertainty widened accordingly; (iv) the CBA verdict tabulated *across* the prediction interval, never at the point transfer alone. **Recommendation:** because the transfer uncertainty is *decision-critical* — the verdict is not robust to it — the point-transfer "pass" is *not* a reliable basis for approval. The value-of-information logic (Chapters 45, 48) indicates that a *primary site-specific DCE*, which would replace the wide prediction

interval with a narrow site-specific estimate, has *high expected value* and is likely justified given the stakes of a blue-finance commitment. The disciplined course is therefore: (a) report the appraisal as *inconclusive on transferred evidence*; (b) either commission a primary DCE for the site, or stage the decision (pilot/phased deployment, resolving uncertainty before full commitment, per the quasi-option-value logic); and (c) *not* present €185m "PASS" as the verdict, which would misrepresent a decision the transferred evidence cannot actually determine.

C3. The reviewer has committed the *equivalence fallacy* (Chapters 33, 39): "not significantly different" is *absence of evidence*, not *evidence of transferability.* A non-significant difference between a transferred and a primary value is *exactly what one expects* when the test is *underpowered* — and validity tests often are, because primary estimates carry large standard errors and transfer errors are large, so the difference must be enormous to reach significance. Failing to reject "they differ" therefore does *not* establish that they are *close enough to transfer*; it may simply mean the study lacked the power to detect a difference that is, in fact, decision-relevant. The correct test is an *equivalence test* (TOST, Chapter 33): pre-specify a *tolerance margin* — the largest transfer error that would *not* change the decision (defined by the verdict's switching value, §49.5) — and test whether the transferred and primary values fall *within* that margin, i.e. whether the 90% confidence interval of their difference lies inside $[-\Delta, +\Delta]$. Only *rejecting* the hypothesis that the difference *exceeds* the margin validates transferability *at that tolerance.* Concretely: (i) the margin must be *decision-specific* — for the *marginal* FRD appraisal, where the verdict flips across the transfer interval, the tolerable transfer error is *small*, so the equivalence bar is *stringent*; (ii) a transferred value "within €10 of the primary, $p = 0.4$ for the difference" tells us almost nothing, whereas "the 90% CI of the difference is [−€3, €4], inside a pre-specified ±€5 margin" would validate transfer *for a decision tolerant of €5 error*; (iii) if the equivalence test *cannot* reject a difference larger than the margin, the honest conclusion is "transferability *not* established at the tolerance this decision requires," *not* "validated." So I would tell the reviewer that a non-significant difference is the *weakest* possible evidence — consistent with both good and terrible transferability — and that validation requires an equivalence test against a margin set by what the FRD decision can tolerate.

49.11 Spaced review prompts

+1 day. From memory: define unit-value vs function transfer; state that transfer error = dissimilarity × function quality; state why the prediction interval is the transfer uncertainty.

+1 week. Explain why a mis-specified adjustment can beat or lose to naive transfer; why extrapolation is dangerous; how leave-one-out measures transfer error; why validity needs equivalence testing.

+1 month. For your FRD transfer, execute the full design — matching, function transfer, prediction-interval uncertainty, leave-one-out validation, verdict across the interval, value-of-information for a primary study. Close with the spine sentence about what the transfer assumes and whether the appraisal survives it.

49.12 How this chapter connects to the rest of the book

- **Chapters 47–48** built the *engine* this chapter drives: the meta-regression (function transfer), the prediction interval and Bayesian posterior predictive (transfer uncertainty), and the value-of-information logic (transfer-versus-primary). **Chapter 42** supplies the extrapolation hazard in transfer form; **Chapter 39/33** supply the equivalence testing that transfer validity requires; **Chapter 45** supplies the Monte Carlo into which the transfer distribution feeds and where it dominates the tornado; **Chapters 34–35** supply the welfare-measure and aggregation compatibility transfer presupposes.
- This chapter is the *bridge* between the synthesis (Part XII) and the appraisal (Part XI): it is *how* the meta-analytic evidence becomes a CBA input, and it reveals that this bridge is the appraisal's largest uncertainty.

- It closes the loop of the entire book for your research: the *primary studies* (Parts IX–X — CV, DCE, MDCEV) produce values; the *meta-analysis* (Chapters 47–48) synthesizes them; *benefit transfer* (this chapter) applies them to a policy site; the *CBA* (Part XI) turns them into a verdict — and the *value-of-information* logic decides, at each step, whether the evidence is adequate or a new primary study is needed.

This chapter is where your finfish-WTP evidence base meets a real FRD decision — and where the honest answer is often that the transfer uncertainty is decision-critical, the verdict conditional, and a site-specific study potentially warranted.

49.13 Research translation task

For your **FRD benefit transfer**, produce a transfer-and-validity note:

1. **Method.** Function transfer from the meta-regression / Bayesian posterior evaluated at the FRD site's characteristics, with harmonization documented.
2. **Similarity and range.** Assess the site's position within the source-study characteristic range; flag extrapolation and widen uncertainty accordingly.
3. **Uncertainty.** Represent the transferred WTP as the *prediction interval / posterior predictive*; draw it into the CBA Monte Carlo; report the verdict *across* the interval.
4. **Validity.** Report the leave-one-out transfer error of the function; equivalence-test against any available primary estimate, with a decision-relevant margin.
5. **Transfer-vs-primary.** Assess, via value of information, whether the marginal FRD verdict warrants a primary site-specific DCE.
6. **Close** with the spine sentence: *"Transferring from [k] harmonized studies via function transfer, the FRD-site WTP is €[·] with a prediction interval of [€·, €·] (leave-one-out transfer error [·]%); across that interval the NPV runs [€·] to [€·], so the verdict [is/is not] robust to the transfer, and a primary DCE [is/is not] warranted by value of information."*

This completes the evidence-synthesis arc and connects it to the appraisal of Part XI.

49.14 Bibliography

Core. - Johnston, Rolfe, Rosenberger & Brouwer (eds.), *Benefit Transfer of Environmental and Resource Values: A Guide for Researchers and Practitioners* — the definitive modern reference (§49.2–§49.6). *(Verify edition/year.)* - Rosenberger & Loomis (2003), "Benefit Transfer," in *A Primer on Nonmarket Valuation* — the standard chapter-length treatment (§49.2–§49.3). *(Verify edition/pages.)* - Boyle, Kuminoff, Parmeter & Pope on benefit transfer methods and transfer error (§49.3–§49.5). *(Verify specific citations.)*

Applied. - Rosenberger & Stanley (2006), "Measurement, Generalization, and Publication: Sources of Error in Benefit Transfers and Their Management," *Ecological Economics* — the anatomy of transfer error (§49.3). *(Verify volume/pages.)* - Kaul, Boyle, Kuminoff, Parmeter & Pope (2013), "What Can We Learn from Benefit Transfer Errors? Evidence from 20 Years of Research on Convergent Validity," *Journal of Environmental Economics and Management* — the meta-analysis of transfer errors (§49.2, §49.5). *(Verify volume/pages.)* - Bergstrom & Taylor (2006) on meta-analytic benefit transfer and validity (§49.5). *(Verify citation.)*

Advanced. - Brouwer & Bateman on the reliability and validity of benefit transfer (§49.5). *(Verify citations.)* - Johnston & Rosenberger (2010), "Methods, Trends and Controversies in Contemporary Benefit Transfer," *Journal of Economic Surveys* (§49.2–§49.6). *(Verify volume/pages.)* - Newbold et al. or Moeltner on Bayesian and machine-learning approaches to benefit transfer and transfer error (§49.4). *(Verify citations.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Johnston et al. / Rosenberger–Stanley / Kaul et al. citations before using them, per the book's no-unverified-citation rule.

End of Chapter 49, and of Part XII. Next: Part XIII begins the book's synthesis — Chapter 50 draws together the identification-assumptions-fragility spine across the whole arc from mathematics to policy, showing how a single claim (a WTP, a treatment effect, a net present value) accumulates assumptions at every stage and how to audit the total fragility of a research conclusion. Tell me when to write it, or request a different chapter.

Part XII Assessment — Evidence Synthesis

Covers Chapters 47–49. Five components, in the order they are best attempted. Allow a full working day; the mock oral is the part that matters most for your viva.

1. Cumulative problem set

P1. Write the PRISMA flow for your finfish search, with counts and exclusion reasons.

P2. Compute Cohen's κ for a double-coded categorical moderator given a stated agreement table.

P3. For 18 studies, compute Q, I^2, τ^2 and the 95% prediction interval, and explain why the last is the one that matters for transfer.

P4. Show that with 120 estimates in 18 studies and $\rho = 0.6$ the design effect is 4.4, and compute the corrected standard error.

P5. Run Egger's test on a stated funnel, and compute the PET-PEESE adjusted estimate.

2. Research critique

Take your finfish synthesis and write the harmonization appendix: for each study, the original units, the payment basis, the deflation, the PPP conversion, and the assumptions required at each step. Identify the study whose harmonization is least secure and quantify how much the pooled estimate depends on it.

3. Red-flag diagnosis

A benefit-transfer exercise applies a meta-analytic mean WTP of €41 (95% CI €36–€46) to a new policy site and reports the transferred benefit with that same interval. Explain what is wrong, compute what should have been used given a τ of €16, and state the consequence for the CBA.

4. Mock oral examination

1. Why is random effects almost always right here, and what does it *not* fix?
2. What is the prediction interval and why is it the transfer-relevant quantity?
3. How do you know your funnel asymmetry is publication bias and not heterogeneity? (You don't — so what do you report?)
4. What is the effective sample size of your synthesis?
5. When is a primary study cheaper than a transfer?

5. Integration mini-project

Complete the finfish synthesis: PRISMA flow, coding protocol with reported inter-coder reliability, random-effects pooling with the prediction interval, meta-regression with clustered and robust variance estimation, leave-one-study-out, publication-bias diagnostics with a bias-adjusted range, and a transfer function with an honest transfer-error statement.

Part XIII — Synthesis, Survey Methodology, and Implementation

Chapter 50 — The Fragility Audit: Synthesis of the Identification–Assumptions–Fragility Spine

Part XIII: Synthesis

The spine question, one last time. This book opened with a single question that every chapter has been a variation on: *given this design and this model, what economic object am I identifying, under which assumptions, and how fragile is the result with respect to specification, distributions, incentives, survey design, sampling, computation, and interpretation?* This chapter answers it for the whole pipeline at once. A single policy conclusion — the FRD net present value — is not one estimate but the *product* of a chain of estimates, each resting on assumptions, each carrying uncertainty. Verified below, that conclusion passes through some ten to twelve assumption-laden stages, its uncertainty *compounds* across them into an NPV spanning a €64m loss to a €264m gain, and a *variance audit* reveals which stages carry the fragility — benefit transfer, the extent of the market, and the calibration factor, together over 90% of it. The spine question, fully assembled: *does my conclusion survive the total fragility of the chain that produced it — and do I know which link is weakest?*

50.0 Conceptual motivation: a conclusion is a chain, and chains break at links

Every previous chapter examined one link in a chain. This chapter examines the *chain*. The reason it needs its own chapter is that a property invisible at the level of any single link becomes decisive at the level of the whole: **assumptions accumulate, and their reliability multiplies rather than averages.**

Trace the FRD conclusion backwards. The net present value depends on a benefit stream; the benefit stream depends on an aggregate WTP; the aggregate depends on the extent of the market and a central-tendency choice; the WTP is *transferred* from a meta-analysis; the meta-analysis synthesizes primary studies under a heterogeneity model; each primary WTP is calibrated for hypothetical bias; each calibration rests on an incentive-compatibility assumption and a certainty threshold; each raw estimate comes from a mixed-logit model under a distributional and identification assumption; each model is fitted to survey responses elicited under a format assumption; and the whole is discounted at a rate that is partly an ethical choice. That is not a metaphor for complexity — it is a literal enumeration of the stages this book has traversed, Part by Part, and every one of them was shown to carry an assumption that could be wrong and an uncertainty that could be large.

The first consequence is arithmetic and unforgiving. If a conclusion holds only when *all* its assumptions hold, and each assumption is individually plausible — say 90% likely to hold — then the *joint* probability that all hold is not 90% but 0.9 raised to the number of assumptions. Verified below, for a conclusion resting on eleven stages at 90% each, the chance the whole chain holds is **31%**. **The conclusion is fragile not because any single step is weak, but because there are many steps.** This is the deepest and most under-appreciated lesson of the entire book: rigour at each stage does not guarantee rigour of the whole, because reliability compounds multiplicatively down the chain.

The second consequence is that *total* uncertainty is far larger than any single stage suggests. Verified, propagating uncertainty through the *entire* pipeline — not just the CBA inputs of Chapter 45, but the elicitation, calibration, synthesis, transfer, and discounting stages together — gives an NPV with a standard deviation of €109m and a 34% probability of being negative, a range that no single-stage sensitivity analysis would reveal.

The third consequence is the constructive one, and it is the chapter's central tool: the **fragility audit**. If uncertainty compounds, one can *decompose* it — ask which stage contributes most to the total variance of the conclusion. Verified, benefit transfer contributes 56% of the FRD NPV's variance, the extent of the market 22%, and the calibration factor 12%, while capital cost and the central-tendency choice contribute 2% each. This decomposition is the map of where a conclusion is *weakest*, and therefore where scrutiny, sensitivity reporting, and any new data should concentrate — the value-of-information logic of Chapters 45 and 48 applied to the whole pipeline.

And beneath all of it runs the book's foundational distinction, which this chapter states in its final form. **Estimation uncertainty shrinks with data; identification failure does not.** Verified once more, as the sample grows from 500 to 500,000 the standard error falls from 0.045 to 0.0014 — but the *identified band* for the formation-versus-anchoring parameter stays fixed at [0, 0.5]. No amount of data rescues an assumption that is not identified. That is the spine, and everything else is its elaboration.

50.1 Prerequisites

This chapter draws on the *whole book*. Most directly: **Chapter 30** (identification versus estimation, partial identification), **Chapter 45** (uncertainty propagation, value of information), **Chapters 47–49** (synthesis and transfer), and the "verify every number" and reproducibility discipline of **Chapters 46**. But the enumeration of stages touches Parts I–XII in their entirety.

New notation. *Assumption chain* (the ordered set of stages a conclusion passes through), *joint reliability* (the probability all assumptions hold), *fragility audit* (variance decomposition of a conclusion across its stages), *total uncertainty* (the propagated uncertainty of the full pipeline), *identified set* (the range of parameter values consistent with the data and assumptions).

50.2 The assumption chain: enumerating the stages

The first step of a fragility audit is to make the chain *explicit* — to list every stage at which the conclusion acquires an assumption, so that none is hidden.

The FRD conclusion's chain. From survey response to policy verdict:

Stage	Assumption acquired	Book location
1. Elicitation	The format elicits true preferences (incentive compatibility)	Ch 28, 36, 39

Stage	Assumption acquired	Book location
2. Response model	The chosen distribution/RUM is correct	Ch 22, 36, 38
3. Identification	Preferences (not scale, not anchoring) are identified	Ch 25, 29, 30
4. Estimation	The estimator converged to the right optimum	Ch 38, 40
5. WTP construction	WTP space / ratio is well-behaved	Ch 23, 38
6. Calibration	Hypothetical bias correction is valid	Ch 39
7. Welfare measure	WTP is the correct CV, aggregable	Ch 34, 35
8. Aggregation	Extent of market and central tendency are right	Ch 35
9. Synthesis	The meta-analytic heterogeneity model holds	Ch 47, 48
10. Transfer	The policy site is like the study sites	Ch 49
11. Discounting	The rate reflects defensible time/equity values	Ch 44
12. Decision rule	NPV/Kaldor–Hicks is the right criterion	Ch 43

Why the enumeration matters. Three reasons. First, an *unlisted* assumption is an *unexamined* one — the discipline of writing the chain forces every hidden premise into view. Second, the *length* of the chain is itself a diagnostic: a conclusion passing through twelve assumption-laden stages is structurally more fragile than one passing through three, regardless of how well each stage is executed. Third, the chain is the *scaffold* for both the reliability calculation (§50.3) and the variance audit (§50.5) — you cannot decompose fragility across stages you have not named.

The honest inventory. Note what the table reveals about your research program: it is *unusually long*. A study that measures a market price has a chain of two or three stages; your finfish-WTP-to-FRD-NPV conclusion has twelve, because non-market valuation of a public good, transferred across sites and discounted over a century, *inherently* stacks assumptions. This is not a criticism of the method — it is the unavoidable structure of the problem, and the appropriate response is not to pretend the chain is shorter but to *audit* it honestly.

> **Active recall 50.A.** Why is explicitly enumerating the assumption chain the necessary first step of a fragility audit, and what does the *length* of the chain tell you? *(Answer: enumeration is necessary because an* unlisted* assumption is an *unexamined* one — the act of writing out every stage from elicitation to decision rule forces each hidden premise (incentive compatibility, distributional correctness, identification, calibration validity, transfer similarity, the discount rate, the decision criterion) into explicit view, where it can be scrutinized, and it provides the *scaffold* for both the joint-reliability calculation and the variance decomposition, neither of which can be done over stages that have not been named. The *length* of the chain is itself a fragility diagnostic: because reliability *multiplies* down the chain (§50.3), a conclusion passing through twelve assumption-laden stages is structurally more fragile than one passing through three, *regardless of how well each stage is executed* — more links means more ways to break. For the FRD conclusion, the chain is unusually long (twelve stages) because non-market valuation of a public good, synthesized, transferred across sites, and discounted over a century, *inherently* stacks assumptions — a market-price study has a chain of two or three. The honest response is not to pretend the chain is shorter but to enumerate it fully and audit where it is weakest.)*

50.3 The multiplicative logic of assumptions

The single most important structural fact about a long chain: reliability compounds by *multiplication*, not averaging.

The arithmetic. Suppose the conclusion holds only if *all* N assumptions in the chain hold, and (for illustration) the assumptions hold independently with probabilities $p_1, \ldots, p_N$. Then the joint probability the conclusion holds is

$$P(\text{conclusion holds}) = \prod_{i=1}^{N} p_i.$$

If each $p_i = p$, this is p^N, which *falls* as the chain lengthens — even when every individual p is high.

Worked Example 50.1 (Reliability compounds, verified).

Number of assumptions N	Each $p = 0.95$	Each $p = 0.90$
1	0.95	0.90
3	0.86	0.73
5	0.77	0.59
8	0.66	0.43
12	0.54	0.28

(Verified.) For the FRD conclusion's chain of roughly eleven stages, even at a *generous* 90% reliability per stage, the joint probability that *all* hold is about **31%**. **The conclusion is more likely to rest on a broken assumption than not** — not because any single assumption is weak, but because there are eleven of them.

Three implications, each qualifying the arithmetic.

1. **Rigour per stage is necessary but not sufficient.** Executing each stage impeccably raises each p_i but cannot overcome the multiplication if the chain is long. A conclusion can be individually rigorous at every stage and jointly unreliable.

2. **Shortening the chain is more valuable than perfecting a stage.** Removing a stage (e.g. replacing a *transfer* with a *primary study*, eliminating the transfer assumption entirely) raises the joint reliability more than marginally improving one surviving stage. This is the fragility-audit rationale for the transfer-versus-primary decision of Chapter 49.

3. **The independence assumption is itself a simplification — usually optimistic.** Assumptions in the chain are often *positively correlated* in their failure (a study that mis-specifies the model may also mis-calibrate; a distinctive site strains both transfer *and* aggregation). Correlated failures make the joint reliability *lower still* in the cases that matter, and the arithmetic above is a *best case*. Conversely, some conclusions are *robust* to individual assumption failures (the verdict survives even if one link breaks), which *raises* effective reliability — which is precisely what the variance audit and switching-value analysis (§50.5) assess. So the p^N figure is not a literal probability to be reported but a *structural warning*: long chains are fragile, and the warning must be made quantitative by the audit.

   Stop and think 50.1. A colleague argues your FRD appraisal is rigorous because every individual stage — the DCE, the calibration, the meta-analysis, the transfer, the discounting — was done to a high standard. Why is this not sufficient for a reliable conclusion? *(Answer: because reliability compounds* multiplicatively* down the chain, so per-stage rigour does not aggregate into whole-chain rigour. Even if every one of the ~11 stages is executed impeccably — say each 90% likely to be approximately correct — the *joint* probability that all hold is $0.9^{11} \approx 31\%$ (verified), so the conclusion is more likely than not to rest on at least one broken assumption

despite every stage being individually strong. The colleague's reasoning commits a composition fallacy: "each part is rigorous" does not imply "the whole is rigorous" when the parts are chained conjunctively (the conclusion needs them *all*). Three qualifications sharpen this: (i) *shortening* the chain (e.g. a primary study replacing the transfer) raises joint reliability more than perfecting a surviving stage, because it removes a factor from the product; (ii) assumption failures are often *positively correlated* (a distinctive site strains transfer *and* aggregation), making the true joint reliability *worse* than the independent p^N best case; (iii) but the raw p^N figure is a *structural warning*, not a literal probability — the decision-relevant question is whether the *verdict survives* individual assumption failures, which the variance audit and switching-value analysis make quantitative. So the right response to the colleague is: per-stage rigour is necessary and commendable, but the conclusion's reliability is a property of the *chain*, and it must be audited as a chain — enumerate the stages, propagate the total uncertainty, decompose it, and check whether the verdict survives the dominant fragilities.)*

50.4 Total uncertainty: propagating the whole pipeline

Section 50.3's arithmetic is a structural warning; the quantitative version is to *propagate uncertainty through the entire chain* and observe the total.

The method. Represent each stage's uncertainty by a distribution (the sampling distribution of the WTP, the calibration factor's interval, the transfer prediction interval, the discount-rate range, and so on), draw jointly, and compute the conclusion for each draw — a full-pipeline Monte Carlo. This differs from Chapter 45's appraisal Monte Carlo, which propagated only the *CBA inputs*: here the uncertainty of *every* stage, including the elicitation, calibration, synthesis, and transfer, enters.

Worked Example 50.2 (Full-pipeline cascade, verified). Propagating uncertainty through all stages of the FRD conclusion:

Quantity	Value
Final effective WTP (median / mean)	€21.0 / €23.7
NPV mean	€57.9m
NPV median	€32.6m
NPV standard deviation	**€109.4m**
NPV 5th / 95th percentile	**−€64m / +€264m**
P(NPV < 0)	**34%**

(Verified.) The total uncertainty is enormous: the NPV's plausible range spans a €64m loss to a €264m gain, and there is a 34% chance the project destroys welfare. **This is the honest uncertainty of the conclusion** — far wider than the ±€9 that the WTP's own confidence interval, or any single stage's uncertainty, would suggest. The right-skew (mean €58m exceeds median €33m) is the compounding of the multiplicative stages, each of which can independently amplify.

Why this is the number that matters. A decision-maker told "NPV = €58m" would proceed with confidence. A decision-maker told "NPV = €58m on average, but with a 34% chance of a loss and a plausible range from −€64m to +€264m, driven mostly by benefit transfer and the discount rate" faces the *actual* decision — a genuinely uncertain one. The full-pipeline propagation is what converts a false point of certainty into the true decision problem. It is the culmination of the risk-analysis discipline of Chapter 45, extended from the CBA inputs to the whole research program.

The caution about the propagation itself. The Monte Carlo requires *input distributions* for each stage, and — as Chapter 45 stressed — where those distributions are *invented* rather than estimated (deep uncertainty), the resulting probability is a false precision. The full-pipeline propagation is honest only

where each stage's uncertainty is *genuinely characterizable*; where a stage is deeply uncertain (the century-scale ecological trajectory, the transferability to a distinctive site), the propagation should be supplemented by *scenario* and *bounding* analysis, and the P(NPV < 0) reported as conditional on the distributional assumptions. Even the fragility audit has its own fragility — which is itself the spine's lesson turned reflexively on the audit.

50.5 The fragility audit: which stage is the weakest link?

The constructive payoff. If total uncertainty compounds across stages, one can *decompose* it — attributing the conclusion's variance to each stage — to find where the conclusion is most fragile.

The method (variance decomposition). For each stage, ask: *how much would the total variance fall if this stage's uncertainty were eliminated* (fixed at its central value)? The variance reduction is that stage's contribution to the total fragility — a first-order sensitivity index, the Sobol/ANOVA logic applied to the pipeline.

Worked Example 50.3 (The fragility audit, verified). Contribution of each stage to the FRD NPV's variance:

Stage fixed at central value	Share of total variance
Benefit transfer	**56%**
Extent of market	**22%**
Calibration factor	**12%**
WTP estimation	11%
Discount rate	10%
Capital / maintenance	2%
Central-tendency choice	2%

(Verified; shares need not sum to 100% because of interactions.) The audit delivers a *ranked map of fragility.* The FRD conclusion's reliability rests overwhelmingly on the **benefit transfer** (56% — the single weakest link, exactly as Chapter 49 warned), the **extent of the market** (22%, Chapter 35), and the **calibration factor** (12%, Chapter 39). The capital cost and the mean-versus-median choice — the sort of things appraisals often agonize over — contribute almost nothing.

What the audit is for. Four uses, each closing a loop from earlier chapters:

1. **Direct scrutiny where it matters.** The dominant stages deserve the most careful execution, the most explicit sensitivity reporting, and the most conservative interpretation. Perfecting the capital-cost estimate is wasted effort; tightening the transfer is not.
2. **Target new data (value of information).** A new primary study is worth most when it resolves a *dominant* fragility — here, a *site-specific* DCE that eliminates the transfer stage (56% of the variance) is far more valuable than one that refines an already-minor stage. This is Chapter 48's value-of-information logic and Chapter 49's transfer-versus-primary decision, now *quantified across the whole pipeline.*
3. **Honest reporting.** The audit tells the decision-maker not just *that* the conclusion is uncertain but *why* and *where* — "the verdict turns on the benefit transfer and the discount rate, which are the least certain stages," which is far more actionable than a single error bar.
4. **Structural improvement.** A stage contributing most of the fragility is a candidate for *removal* (shorten the chain, §50.3) rather than mere refinement — the strongest possible response.

The connection to the switching value. The variance audit says which stage *contributes most uncertainty*; the switching-value analysis (Chapters 44–45) says which stage *can flip the verdict.* These are related

but distinct — a stage can be uncertain without being decisive (if the verdict survives its whole range) or decisive without being the most uncertain (if the verdict sits exactly on its knife-edge). The complete audit reports *both*: the variance decomposition *and*, for the dominant stages, whether the verdict survives their plausible range. For the FRD, verified across the transfer prediction interval alone the verdict flips (Chapter 49) — so the most *uncertain* stage is *also* decisive, the worst case, and the one demanding a primary study.

> **Active recall 50.B.** What does the fragility audit (variance decomposition) tell you that a single total-uncertainty figure does not, and how does it connect to the value of a new primary study? *(Answer: a single total-uncertainty figure (e.g. "NPV standard deviation €109m, P(NPV<0) = 34%") tells you* that* the conclusion is uncertain but not *why* or *where* — it is a scalar summary of the whole chain. The *fragility audit* decomposes that total variance across the stages of the chain, attributing to each stage the share of uncertainty it contributes (verified: benefit transfer 56%, extent of market 22%, calibration 12%, while capital cost and central-tendency choice contribute 2% each). This delivers a *ranked map of where the conclusion is weakest*, which the scalar cannot. Its four uses: (i) direct *scrutiny* to the dominant stages and away from the negligible ones (perfecting capital cost is wasted; tightening the transfer is not); (ii) *honest reporting* of not just how uncertain but which stages drive it; (iii) *structural improvement* — a dominant stage is a candidate for *removal* (shortening the chain) not just refinement; and (iv) *targeting new data*. On the last: a new primary study is worth most when it resolves a *dominant* fragility — verified, benefit transfer contributes 56% of the FRD variance, so a *site-specific DCE* that *eliminates the transfer stage* would resolve most of the conclusion's uncertainty, whereas a study refining an already-minor stage would barely help. This is the value-of-information logic of Chapters 45 and 48, and the transfer-versus-primary decision of Chapter 49, now *quantified across the whole pipeline*: the audit points precisely at the study most worth commissioning.)*

50.6 Identification: the fragility no data can fix

Running beneath every stage is the distinction this book has returned to more than any other, and which the audit must treat specially, because it obeys different rules from estimation uncertainty.

Estimation versus identification, once more. *Estimation* uncertainty — the sampling variability of an estimate — *shrinks* as data accumulate. *Identification* failure — the impossibility of distinguishing two explanations from *any* amount of the available data — *does not shrink*, ever. A conclusion can be estimated with vanishing standard error and still be *unidentified*: precise about the wrong thing.

Worked Example 50.4 (Precision improves, identification does not, verified). The formation-versus-anchoring parameter of Chapter 30 (does a cost-vector effect reflect genuine preference formation, κ, or anchoring, λ?), across sample sizes:

Sample size N	Estimation SE	Identified band for κ
500	0.0447	**[0.0, 0.5]**
5,000	0.0141	**[0.0, 0.5]**
50,000	0.0045	**[0.0, 0.5]**
500,000	0.0014	**[0.0, 0.5]**

(Verified.) The standard error falls by more than thirtyfold as the sample grows a thousandfold — but the *identified band* stays *exactly fixed* at [0, 0.5]. No amount of data narrows it, because the observed data are consistent with *any* value in that band; the band's width is set by the *assumptions*, not the sample.

Why identification is special in the audit. Estimation uncertainty enters the variance decomposition as a *shrinkable* contribution — more data reduces it, so it can be resolved by a larger primary study. Identification failure is *not shrinkable* — it can be resolved *only* by a stronger assumption (which may be

untestable, Chapter 30) or by a *different design* that identifies what the current one cannot (an experimental manipulation, an instrument, a bound). **The fragility audit must therefore distinguish, for each dominant stage, whether its uncertainty is estimation (buy more data) or identification (change the assumption or the design).** Conflating them leads to the futile prescription of collecting more data to resolve an identification problem — precisely the error the spine warns against.

For your research program. Several of your central questions are *identification* questions, not estimation ones: the formation-versus-anchoring decomposition (Chapter 30), the scale-versus-preference confound (Chapter 25), the corner-versus-protest distinction (Chapters 26, 40). For these, a larger sample is no remedy; the remedies are *experimental* (the real-versus-hypothetical randomization, the cost-vector manipulation — your BRIS design is exactly such an identification strategy) or *partial* (report bounds, Chapter 30). Recognizing which of your uncertainties are estimation and which are identification is the final discipline the spine demands — and it is the difference between a research agenda that spends its budget on larger surveys and one that spends it on better *designs*.

> **Stop and think 50.2.** Your fragility audit shows the calibration stage contributes 12% of the total variance. Before recommending a larger study to reduce it, what must you determine? *(Answer: whether that 12% is* estimation* uncertainty or *identification* uncertainty — because the remedy is completely different, and recommending a larger study is futile (and wasteful) if the problem is identification. If the calibration uncertainty is *estimation* — the calibration factor is identified but imprecisely estimated from a limited real-versus-hypothetical sample — then a *larger* study *does* reduce it: verified, estimation standard errors shrink with $\sqrt{N}$ ($0.045 \to 0.0014$ as N grows), so more data is the right prescription and the value-of-information calculus applies straightforwardly. But if the calibration uncertainty is *identification* — for instance, if the hypothetical-bias correction rests on an untestable assumption about *why* stated and real values differ (a genuine formation-versus-anchoring-type ambiguity), or if the certainty-threshold choice is a researcher degree of freedom that no sample size resolves — then a larger study does *not* help: verified, the identified band stays fixed regardless of N. In that case the remedies are a *stronger (possibly untestable) assumption*, a *different design* that identifies the correction directly (a randomized real-payment arm — the BRIS mechanism, which is precisely an identification strategy for exactly this), or *partial identification* (report the correction as a bound across defensible thresholds). So before recommending more data I must diagnose the *nature* of the uncertainty: estimation → buy data; identification → change the assumption or the design. Conflating the two — collecting more data to fix an identification problem — is the central error the book's spine warns against, and the fragility audit is incomplete until each dominant stage is classified.)*

50.7 The fragility audit as a method

Assembling the pieces, the fragility audit is a repeatable procedure — the book's methodology in a single workflow.

The procedure.

1. **Enumerate the chain** (§50.2). List every stage at which the conclusion acquires an assumption; the length is the first diagnostic.
2. **Classify each stage's uncertainty** as *estimation* (shrinkable) or *identification* (not shrinkable) (§50.6).
3. **Characterize each stage's uncertainty** by a distribution (estimation) or a bound/scenario set (identification/deep uncertainty), honestly flagging where distributions are invented (§50.4).
4. **Propagate** the whole pipeline to obtain the total uncertainty of the conclusion — the honest error bar, not any single stage's (§50.4).
5. **Decompose** the total variance across stages to rank fragility (§50.5).

6. **Check verdict survival** for the dominant stages — does the conclusion hold across their plausible range (switching values)? (§50.5, Chapters 44–45.)
7. **Prescribe** accordingly: for a dominant *estimation* fragility, more/better data or a primary study (shorten the chain); for a dominant *identification* fragility, a better design or a bound; for a robust verdict, proceed; for a fragile one, report conditionally and consider staging (Chapter 45's option value).
8. **Report** the whole audit transparently and reproducibly (Chapter 46) — the chain, the classifications, the total uncertainty, the decomposition, the survival checks, and the prescriptions.

What the audit produces. Not a number but a *structured, honest account of a conclusion's reliability*: what it assumes, how uncertain it is, where it is weakest, whether it survives, and what would most improve it. This is the deliverable the whole book has been building toward — the alternative to a false point of certainty.

The reflexive caution. The audit itself rests on assumptions (the input distributions, the independence in the decomposition, the completeness of the enumerated chain), so it too is fragile, and its outputs are *estimates* of fragility, not exact measurements. The appropriate humility is to treat the audit as a *disciplined structuring* of uncertainty rather than a precise quantification of it — which is, once more, the spine's lesson applied to the spine's own tool.

50.8 Failure modes

F1. Per-stage rigour mistaken for whole-chain rigour. Executing each stage well and declaring the conclusion reliable. *Fix:* audit the chain; recognize multiplicative reliability (§50.3).

F2. Hidden assumptions. An unlisted stage. *Fix:* enumerate the full chain (§50.2).

F3. Single-stage error bar reported as total uncertainty. The WTP CI standing in for the conclusion's uncertainty. *Fix:* full-pipeline propagation (§50.4).

F4. Invented distributions treated as real. False precision in the total. *Fix:* flag deep-uncertainty stages; supplement with scenarios/bounds (§50.4).

F5. Uniform scrutiny. Effort spread evenly across stages. *Fix:* variance audit; concentrate on dominant stages (§50.5).

F6. Refining a minor stage. Perfecting capital cost while the transfer dominates. *Fix:* target the audit's top-ranked fragilities (§50.5).

F7. More data for an identification problem. Prescribing a larger survey for an unidentified quantity. *Fix:* classify estimation versus identification; change the design for the latter (§50.6).

F8. Variance without survival, or survival without variance. Reporting one of the two dominant-stage checks. *Fix:* report both the decomposition and the switching-value survival (§50.5).

F9. Auditing without acting. Producing the audit but reporting the point conclusion anyway. *Fix:* let the audit drive the reporting and the recommendation (§50.7).

F10. Forgetting the audit's own fragility. Treating the fragility quantification as exact. *Fix:* reflexive humility; disciplined structuring, not false precision (§50.7).

50.9 Diagnostic questions for your own papers

1. **Chain.** Have I enumerated every stage at which the conclusion acquires an assumption? *(§50.2, F2.)*
2. **Classification.** Is each stage's uncertainty classified as estimation or identification? *(§50.6, F7.)*
3. **Total uncertainty.** Have I propagated the *whole* pipeline, not just one stage? *(§50.4, F3.)*

4. **Deep uncertainty.** Are invented distributions flagged and supplemented with scenarios? *(§50.4, F4.)*
5. **Decomposition.** Have I ranked stages by their contribution to total fragility? *(§50.5, F5–F6.)*
6. **Survival.** For dominant stages, does the verdict survive their plausible range? *(§50.5, F8.)*
7. **Prescription.** Does the audit drive the data/design/staging recommendation? *(§50.7, F9.)*
8. **Reflexivity.** Have I acknowledged the audit's own fragility? *(§50.7, F10.)*

50.10 Exercises

(A) Comprehension.

A1. Explain the multiplicative logic of assumptions and why a long chain is fragile even when each stage is strong.

A2. Distinguish total uncertainty from a single stage's uncertainty.

A3. What does the fragility audit (variance decomposition) add beyond a total error bar?

(B) Derivation / reasoning.

B1. Derive the joint reliability of an N-stage chain and explain the independence caveat.

B2. Explain why estimation uncertainty enters the audit as shrinkable and identification failure as not.

B3. Explain the difference between a stage being the *most uncertain* and being *decisive*.

(C) Research application / critique.

C1. Conduct the full fragility audit of your FRD conclusion.

C2. Your audit shows benefit transfer contributes 56% of the variance. Write the recommendation.

C3. A reviewer says your appraisal is reliable because each method is state-of-the-art. Respond.

50.11 Solutions

A1. A conclusion that holds only if *all* its assumptions hold has a reliability equal to the *product* of the individual assumption reliabilities: $\prod_i p_i$. Because this is a product of numbers below one, it *falls* as the chain lengthens — multiplicatively, not by averaging. Verified: eleven stages each 90% reliable give a joint reliability of $0.9^{11} \approx 31\%$. So a long chain is fragile *even when each stage is strong*, because the conclusion needs *every* link to hold and the probability of that decays with the number of links. This is a composition effect: per-stage rigour raises each p_i but cannot overcome the multiplication if the chain is long, so "every stage is rigorous" does not imply "the conclusion is reliable." The practical corollaries: *shortening* the chain (removing a stage, e.g. a primary study replacing a transfer) raises reliability more than perfecting a surviving stage, and correlated assumption failures make the true joint reliability *worse* than the independent best case.

A2. A *single stage's* uncertainty (e.g. the WTP's confidence interval, ±€9) describes the variability contributed by *that stage alone*, holding all others fixed at their central values. *Total* uncertainty is the variability of the *conclusion* when *every* stage's uncertainty is propagated jointly through the whole pipeline. The two differ dramatically because the stages *compound*: verified, the FRD NPV — whose WTP input has a ±€9 confidence interval — has a total standard deviation of €109m and a plausible range from −€64m to +€264m, because the calibration, transfer, aggregation, and discounting uncertainties *stack on top of* the WTP uncertainty. Reporting a single stage's error bar as the conclusion's uncertainty (F3) therefore drastically *understates* it — the conclusion is far less certain than its best-measured input. Total uncertainty is the honest error bar for the decision.

A3. A total error bar (e.g. "P(NPV<0) = 34%") is a *scalar* — it says how uncertain the conclusion is but not *why* or *where*. The fragility audit *decomposes* that total across the stages, attributing to each its share of the variance (verified: transfer 56%, extent of market 22%, calibration 12%, capital 2%). This adds four things the scalar cannot: (i) a *ranked map* of where the conclusion is weakest, directing scrutiny to the dominant stages and away from negligible ones; (ii) a *target for new data* — the value of resolving a stage is proportional to its variance share, so a study eliminating the transfer (56%) is worth far more than one refining capital cost (2%); (iii) a candidate for *structural improvement* — a dominant stage may be *removed* (shorten the chain) rather than merely refined; and (iv) *actionable honest reporting* — telling the decision-maker the verdict "turns on the transfer and the discount rate" rather than just "is uncertain." The decomposition converts an undifferentiated uncertainty into a diagnosis.

B1. Let the conclusion C hold iff assumptions $A_1, \ldots, A_N$ all hold, with $P(A_i) = p_i$. If the A_i are *independent*, $P(C) = P(A_1 \cap \cdots \cap A_N) = \prod_{i=1}^{N} P(A_i) = \prod_i p_i$, and with equal $p_i = p$ this is p^N, decreasing in N for $p < 1$ (verified: $0.9^{11} \approx 0.31$). *Independence caveat:* real assumption failures are typically *not* independent. If failures are *positively correlated* (a distinctive policy site strains transfer *and* aggregation; a mis-specified model tends to mis-calibrate), then $P(\cap A_i) < \prod_i p_i$ — the joint reliability is *worse* than the independent product, so p^N is an *optimistic best case*. Conversely, the conclusion may be *robust* to individual failures — it holds even if some A_i fails (the verdict survives that stage's whole range), which *raises* effective reliability above p^N. So p^N is not a literal reportable probability but a *structural warning*: long conjunctive chains are fragile, and the warning must be made quantitative by propagating the actual uncertainties (§50.4) and checking verdict survival (§50.5), which the independence formula only motivates.

B2. *Estimation* uncertainty is the sampling variability of an estimate around the true value it targets; by the law of large numbers / CLT it shrinks as $1/\sqrt{N}$, so more data reduces it (verified: SE 0.045 → 0.0014 as N: 500 → 500,000). In the fragility audit it therefore enters as a *shrinkable* contribution — a stage whose fragility is estimation can be resolved by a *larger primary study*, and its value of information is finite and computable. *Identification* failure is the impossibility of distinguishing two (or more) explanations from *any* amount of the *available* data — the identified set has positive width set by the *assumptions*, not the sample. Verified: as N grows a thousandfold the identified band for κ stays *fixed* at [0, 0.5]. In the audit it enters as a *non-shrinkable* contribution — a stage whose fragility is identification *cannot* be resolved by more data, only by a *stronger assumption* (possibly untestable, Chapter 30), a *different design* that identifies what the current one cannot (an experiment, an instrument), or a *bound* (partial identification). The audit must classify each dominant stage into these two categories because they have *opposite* prescriptions, and prescribing more data for an identification problem is futile — the central error the spine warns against.

B3. A stage is *most uncertain* if it contributes the largest share of the conclusion's *variance* (the variance-decomposition ranking); it is *decisive* if the conclusion's *verdict* (sign of the NPV, pass/fail) changes across its plausible range (the switching-value criterion). These are distinct: (i) a stage can be *most uncertain but not decisive* — it contributes much variance, but the verdict is the same sign across its whole range (e.g. a hugely uncertain benefit that is positive throughout, so the project passes regardless); (ii) a stage can be *decisive but not most uncertain* — it contributes little variance, but the verdict sits exactly on its knife-edge, so even its small range straddles zero (e.g. a discount rate whose narrow plausible band happens to bracket the switching value). The complete audit reports *both*, because they answer different questions: the variance decomposition says *where to concentrate effort to reduce uncertainty*, while the switching-value/survival check says *whether the uncertainty actually threatens the decision*. For the FRD, verified, benefit transfer is *both* the most uncertain (56% of variance) *and* decisive (the verdict flips across its prediction interval, Chapter 49) — the worst case, unambiguously demanding a primary study; but a stage that were merely most uncertain (verdict robust) would warrant *reporting* the uncertainty without necessarily *acting* on it, and a stage that were merely decisive (low variance, knife-edge verdict) would warrant *tightening* it cheaply.

C1. *Full fragility audit of the FRD conclusion.* **(1) Enumerate the chain** (§50.2): the ~12 stages from elicitation (incentive compatibility) through response model, identification, estimation, WTP construction, calibration, welfare measure, aggregation, synthesis, transfer, discounting, to decision rule — noting the chain is *unusually long* because non-market valuation, synthesized, transferred, and discounted over a century inherently stacks assumptions. **(2) Classify** each stage as estimation or identification: the WTP *es-*

timation and *aggregation* are estimation (shrinkable); the *identification* stage (formation vs anchoring, scale vs preference) and parts of *calibration* and *transfer* are identification (not shrinkable) — this determines the remedies. **(3) Characterize** each stage's uncertainty by a distribution where genuine (WTP sampling, calibration interval, transfer prediction interval, discount-rate range) and by bounds/scenarios where deep (century-scale ecology, transferability to a distinctive site), flagging the invented distributions. **(4) Propagate** the whole pipeline (§50.4): verified, total NPV mean €58m, SD €109m, range −€64m to +€264m, P(NPV<0) = 34% — the honest uncertainty, far wider than any single stage. **(5) Decompose** (§50.5): verified, benefit transfer 56%, extent of market 22%, calibration 12%, WTP estimation 11%, discount rate 10%, capital/central-tendency 2% each — the ranked fragility map. **(6) Check survival**: verified, the verdict flips across the transfer prediction interval (Chapter 49), so the dominant stage is also decisive — the worst case. **(7) Prescribe**: because the dominant fragility (transfer, 56%) is *decisive* and partly *identification* (transferability to a distinctive site), the remedy is a *site-specific primary DCE* (removing the transfer stage entirely — shortening the chain), whose value of information is large; the extent-of-market and discount-rate fragilities are addressed by explicit sensitivity/switching-value reporting and the ethical-parameter separation of Chapters 43–44. **(8) Report** the entire audit reproducibly (Chapter 46) — chain, classifications, total uncertainty, decomposition, survival, prescriptions — as a conditional verdict: "positive under central assumptions but not robust; the conclusion rests dominantly on the benefit transfer, which is both the most uncertain and decisive stage, so a site-specific study is warranted before confident approval."

C2. *Recommendation given transfer contributes 56% of variance.* The benefit transfer is the single weakest link in the FRD conclusion — it contributes more of the total uncertainty than all other stages combined, and (verified, Chapter 49) the verdict *flips* across its prediction interval, so it is both the *most uncertain* and *decisive* stage. The recommendation follows directly from the audit: **commission a site-specific primary DCE for the FRD policy site.** The rationale is quantitative, not reflexive: (i) *value of information* — a study that resolves a 56%-of-variance stage delivers far more decision value than one refining any minor stage, and since the verdict flips across the transfer interval, resolving it is what the decision actually needs (Chapters 45, 48); (ii) *chain-shortening* — a primary site-specific estimate *eliminates* the transfer stage entirely (removing the "policy site is like the study sites" assumption), which raises the whole-chain reliability more than any refinement of a surviving stage (§50.3); (iii) *identification, not just estimation* — part of the transfer fragility is that the FRD site may be *distinctive* (extrapolation beyond the source range), which is an identification-type problem a *different design* (a direct measurement) resolves but more meta-analytic data cannot. *Before* commissioning, I would confirm via value-of-information that the study's expected decision value exceeds its cost (given the stakes of a blue-finance commitment, it very likely does), and *in the interim* report the appraisal as *inconclusive on transferred evidence*, considering a staged/pilot deployment (Chapter 45's option value) that resolves the uncertainty before full commitment. The other dominant stages — extent of market (22%) and calibration (12%) — I would address by explicit switching-value sensitivity reporting rather than new data, since they are less decisive. The audit thus converts "the appraisal is uncertain" into a specific, justified research and decision strategy: measure the transfer directly, report the rest conditionally, and do not approve on the point transfer.

C3. *Response.* State-of-the-art methods at each stage are necessary and commendable, but they establish *per-stage* rigour, not *whole-chain* reliability — and the conclusion's reliability is a property of the chain. The reviewer's reasoning is a composition fallacy: because the conclusion requires *every* stage to hold, its reliability is the *product* of the stage reliabilities, which *decays multiplicatively* with the chain's length regardless of how strong each stage is. Verified: even if all ~11 stages were 90% reliable — an optimistic reading of "state-of-the-art" — the joint reliability is $0.9^{11} \approx 31\%$, so the conclusion is *more likely than not* to rest on a broken assumption *despite* every method being excellent. And this is a best case, because assumption failures are often positively correlated (a distinctive site strains transfer *and* aggregation), lowering the joint reliability further. Concretely, the full-pipeline propagation gives a total NPV standard deviation of €109m and a 34% chance of a loss — an uncertainty invisible at the level of any single state-of-the-art stage. The constructive point: the way to establish reliability is not to certify each method but to *audit the chain* — enumerate the stages, propagate the total uncertainty, decompose it (verified: benefit transfer 56%, the dominant fragility), and check whether the verdict survives the dominant stages (verified: it does not, flipping across the transfer interval). That audit shows the conclusion is *not* reliable as it stands, *not* because any method is deficient, but because the chain is long and its weakest link (the transfer) is both uncertain and decisive. The appropriate

response is therefore to *shorten the chain* where it is weakest — a site-specific primary study eliminating the transfer — not to point to the excellence of the individual methods. Per-stage rigour is the floor, not the ceiling; whole-chain reliability is what the decision needs, and only the audit can establish it.

50.12 Spaced review prompts

+1 day. From memory: the multiplicative logic (p^N); the difference between total and single-stage uncertainty; what the variance audit ranks.

+1 week. Explain why per-stage rigour is insufficient; why estimation shrinks and identification does not in the audit; the difference between most-uncertain and decisive.

+1 month. Conduct the full fragility audit of one of your own conclusions — enumerate, classify, propagate, decompose, check survival, prescribe. Close with the spine sentence about whether your conclusion survives the total fragility of the chain that produced it.

50.13 How this chapter connects to the rest of the book

- This chapter is the *synthesis of the whole book*. It draws the assumption chain from every Part: elicitation and incentives (**28, 36, 39**), response models and identification (**22, 25, 29, 30, 38, 40**), WTP and welfare (**23, 34, 35**), synthesis and transfer (**47, 48, 49**), discounting and decision rules (**43, 44**), and the risk-propagation and reproducibility machinery (**45, 46**) that the audit uses.
- Its two central tools are extensions of earlier chapters: the *total-uncertainty propagation* generalizes **Chapter 45**'s appraisal Monte Carlo from the CBA inputs to the whole pipeline; the *variance decomposition* generalizes **Chapter 45**'s tornado and **Chapters 48–49**'s value-of-information from one stage to the chain; the *identification-versus-estimation* distinction is **Chapter 30**'s central result stated in its final, pipeline-wide form.
- It closes the book's argument: the spine question posed in the Preface — *what am I identifying, under which assumptions, and how fragile is the result?* — is answered, for any conclusion, by the fragility audit, which enumerates the assumptions, propagates the fragility, and locates the weakest link.

This is the chapter that turns the book's recurring question into a *method* — a repeatable audit that any of your conclusions, from a single WTP to the FRD verdict, can and should be put through.

50.14 Research translation task

For your **entire FRD research program**, produce a fragility-audit dossier:

1. **The chain.** Enumerate every stage from elicitation to decision rule, with the assumption each acquires and its book/method location.
2. **Classification.** Label each stage's dominant uncertainty *estimation* (shrinkable) or *identification* (not shrinkable).
3. **Total uncertainty.** Propagate the whole pipeline; report the conclusion's honest distribution (mean, median, spread, P(NPV<0)), flagging deep-uncertainty stages handled by scenario/bound.
4. **Fragility decomposition.** Rank the stages by variance contribution.
5. **Survival.** For the dominant stages, report whether the verdict survives their plausible range (switching values).
6. **Prescription.** For each dominant fragility, state the remedy — more data (estimation), better design (identification), chain-shortening (primary study), or conditional reporting — and the value-of-information basis.

7. **Close** with the spine sentence: *"The FRD conclusion passes through [N] assumption-laden stages; its total NPV is €[·] (P(NPV<0) = [·]%), dominated by [top fragility] ([·]% of variance, [estimation/identification], [decisive/robust]); the conclusion [does/does not] survive its total fragility, and the highest-value remedy is [·]."*

This dossier is the capstone deliverable of the whole research program — the honest, structured account of what the FRD verdict assumes, how fragile it is, and what would most improve it.

50.15 Bibliography

Core. - Manski, *Identification for Prediction and Decision* — the definitive treatment of identification, partial identification, and the limits of what data can establish (§50.6). *(Verify edition.)* - Leamer (1983), "Let's Take the Con Out of Econometrics," *American Economic Review* — the foundational statement on specification fragility and sensitivity (§50.3, §50.5). *(Verify volume/pages.)* - Saltelli et al., *Global Sensitivity Analysis: The Primer* — variance-based sensitivity (Sobol) decomposition, the basis of the fragility audit (§50.5). *(Verify edition.)*

Applied. - Morgan & Henrion, *Uncertainty: A Guide to Dealing with Uncertainty in Quantitative Risk and Policy Analysis* — uncertainty propagation and its communication in policy (§50.4). *(Verify edition.)* - Kaul et al. (2013) on benefit-transfer error (the dominant fragility here) — *Journal of Environmental Economics and Management* (§50.5; Chapter 49). *(Verify volume/pages.)* - Weitzman on fat tails and deep uncertainty, relevant to the non-shrinkable stages (§50.4, §50.6). *(Verify citations.)*

Advanced. - Athey & Imbens, and Manski, on the estimation-versus-identification distinction and partial identification in policy analysis (§50.6). *(Verify citations.)* - Ho, Imai, King & Stuart, or Simonsohn et al. (specification-curve), on multi-specification robustness (§50.5; Chapter 46). *(Verify citations.)* - Borgonovo & Plischke, "Sensitivity Analysis: A Review of Recent Advances," *European Journal of Operational Research* — modern variance-based and moment-independent sensitivity (§50.5). *(Verify volume/pages.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Manski / Leamer / Saltelli / Kaul et al. citations before using them, per the book's no-unverified-citation rule.

End of Chapter 50, and of Part XIII's synthesis. The book's intellectual spine — identification, assumptions, and fragility — is now assembled into a single method: enumerate the assumption chain, classify and propagate its uncertainty, decompose the fragility, locate the weakest link, and check whether the conclusion survives. Every conclusion in your research program, from a single willingness-to-pay estimate to the fish-restoration-device net present value, can be put through this audit — and doing so is the difference between a number presented with false certainty and a conclusion whose reliability is honestly known. Tell me which chapter to write next, or whether to assemble the completed volume.

Chapter 51 — Survey and Questionnaire Design

Part XIII: Survey Methodology and Research Implementation

> **The spine question, applied to this chapter.** Every welfare estimate in this book rests on data from a survey — and the survey is a *measurement instrument* whose design determines what the data mean. A payment vehicle respondents find incredible, a scenario they misunderstand, a battery that fatigues them, a sample that over-represents the wealthy: each corrupts the estimate *before* any econometrics begins, and no model repairs data that were mismeasured at source. The spine question here: *does my questionnaire elicit the construct I intend, from a sample that represents the population I claim, with data quality high enough that the assumptions of every downstream stage actually hold?*

51.0 Conceptual motivation: the instrument comes before the estimator

The fragility audit of Chapter 50 traced a conclusion back through a chain of assumptions to its origin. This chapter is about that origin. Before any WTP is estimated, calibrated, synthesized, transferred, or discounted, it is *measured* — by asking people questions. And the quality of that measurement is set by the questionnaire and the sample, not by the sophistication of the model applied afterwards. A mixed logit fitted to responses from inattentive respondents to an incredible scenario is a precise estimate of a meaningless quantity. **Garbage in, garbage out is not a cliché in valuation; it is the first assumption of the entire chain.**

Your research depends on primary surveys — the CV study with payment cards and the DCE with its attributes — so the design decisions this chapter covers are not preliminaries but determinants of everything downstream. Four families of decision matter, and the verified results show how much.

The scenario and payment vehicle. Respondents must find the good, the provision mechanism, and the payment *credible* and *consequential* (Chapter 39). An incredible payment vehicle triggers protest; an incomprehensible scenario produces noise; a non-consequential framing invites hypothetical bias. These are design choices, and they determine whether the elicited number is a welfare measure or an artefact.

Data quality. Not every response is a considered one. Some respondents *speed* through, *straight-line* (give identical answers), or choose *randomly* without engaging. Verified below, a 20% share of inattentive respondents *attenuates* the DCE coefficients (the attribute coefficient falls from 1.5 to 0.9), corrupting the estimated preference structure — and a simple within-respondent variance rule detects straight-lining with 100% sensitivity and no false positives. But screening is itself a post-treatment exclusion (Chapter 32), so it must be pre-specified and handled with care.

Representativeness. A sample that does not represent the population produces a biased population

estimate. Verified, a sample over-representing high-income respondents overstates mean WTP by 24% — and post-stratification weighting to known population margins cuts that bias to under 1%, at the cost of a reduced effective sample size (72% here). Representativeness is a design goal (sampling frame, quotas) and a repair (weighting), and neither can fix imbalance on *unobserved* characteristics.

Cognitive process. Surveys are cognitively demanding, and the *order* of questions, the *framing* of information, the *number* of tasks, and *fatigue* all shape responses — often in ways that masquerade as preference heterogeneity (Chapter 25) or that confound the design (Chapter 29's starting-point effects). Managing the respondent's cognitive process is a first-order design task.

This chapter is where the abstractions of the earlier parts — incentive compatibility, scale, identification — become concrete instrument-design decisions, and where the quality of every subsequent number is determined.

51.1 Prerequisites

From **Chapter 28, 39**: incentive compatibility, consequentiality, and hypothetical bias — the theory this chapter operationalizes. From **Chapter 29**: starting-point and anchoring effects, here in survey-order form. From **Chapter 32**: post-treatment exclusion — the hazard in data-quality screening. From **Chapter 25**: scale, which inattention inflates. From **Chapter 35**: the extent of the market and aggregation, which representativeness serves. From **Chapter 37**: cognitive burden in DCE design.

New notation. *Payment vehicle* (the mechanism by which respondents would pay), *straight-lining / non-differentiation* (identical responses across items), *speeder* (implausibly fast completion), *post-stratification / raking* (weighting to population margins), *design effect* (variance inflation from weighting), *quota sampling* (targeting subgroup counts).

51.2 The scenario, the good, and the payment vehicle

The heart of a stated-preference survey is the *scenario*: the description of the good, the change being valued, the provision mechanism, and the payment. Its credibility determines whether responses are meaningful.

Describing the good. The environmental change must be described *accurately, comprehensibly, and neutrally.* Three requirements pull against each other: *accuracy* (scientifically correct), *comprehensibility* (a lay respondent must understand it), and *neutrality* (no framing that pushes responses up or down). For your finfish good, this means conveying the stock improvement, its ecological meaning, and its consequences in terms a coastal household understands, without either alarmist or reassuring spin. **Visual aids** — maps, photographs, infographics of stock levels — aid comprehension where words fail, but must themselves be neutral (a photograph of a thriving reef versus a barren one is not neutral).

The provision mechanism and consequentiality. Respondents must believe the good *would actually be provided* if the policy proceeded, and that the survey *could influence* that decision — the consequentiality condition (Chapter 39). A credible implementing authority, a plausible policy context, and an explicit statement of how results will be used all build consequentiality. Its absence is the root of hypothetical bias.

The payment vehicle. *How* respondents would pay is a critical and underappreciated choice. Options — a tax increase, a levy, a utility surcharge, a voluntary donation, an entrance fee — differ in:

- **Credibility.** Would this payment mechanism plausibly be used for this good? An implausible vehicle triggers disbelief.
- **Coerciveness / incentive compatibility.** A *coercive* vehicle (a mandatory tax everyone pays) is incentive-compatible for a public good (Chapter 28); a *voluntary* one (a donation) invites free-riding and is *not.*

- **Payment-vehicle bias.** Respondents' attitudes to the *vehicle* (anti-tax sentiment, distrust of authorities) contaminate their stated WTP for the *good* — a tax-averse respondent may state a low WTP not because they value the good little but because they dislike taxes.
- **Protest potential.** An objectionable vehicle generates *protest responses* — zeros that reject the payment mechanism rather than expressing zero value (Chapters 26, 29).

For your CV study, the payment vehicle must be *coercive* (for incentive compatibility) yet *credible* and *minimally protest-inducing* — typically a tax or levy earmarked for the environmental programme, described so its credibility is maximized and vehicle bias minimized. This choice should be *pretested* (§51.7), because a poorly-chosen vehicle can dominate the response distribution.

> **Active recall 51.A.** Why is the payment-vehicle choice consequential for the validity of a CV estimate, and what tension does it embody? *(Answer: the payment vehicle — the mechanism by which respondents would actually pay (tax, levy, surcharge, donation, fee) — shapes the elicited WTP through four channels:* credibility* (an implausible vehicle for the good triggers disbelief and noise), *incentive compatibility* (a *coercive* vehicle like a mandatory tax makes truth-telling optimal for a public good, whereas a *voluntary* donation invites free-riding and biases WTP downward, Chapter 28), *payment-vehicle bias* (respondents' attitudes to the *vehicle itself* — anti-tax sentiment, distrust of the implementing authority — contaminate their stated value for the *good*, so a low WTP may reflect tax aversion rather than low valuation), and *protest potential* (an objectionable vehicle generates protest zeros that reject the mechanism rather than express zero value, Chapters 26, 29). The central *tension*: incentive compatibility requires a *coercive* vehicle (everyone pays, no free-riding), but coercive vehicles — especially taxes — are precisely the ones most prone to vehicle bias and protest. The design must therefore find a vehicle that is *coercive enough* to be incentive-compatible yet *credible and acceptable enough* to minimize vehicle bias and protest — typically an earmarked tax or levy described to maximize credibility and the link to the specific good — and this choice must be *pretested*, because a poorly-chosen vehicle can dominate the entire response distribution, corrupting the estimate before any modelling.)*

51.3 Information, framing, and the constructed-preference problem

How information is *presented* shapes the WTP — a fact that is both a methodological hazard and, for your familiarity dimension, a substantive research question.

Information effects. The *amount, order, and framing* of information about the good affect stated WTP. More information can raise WTP (respondents learn the good matters) or lower it (they learn of substitutes or limits). This is not merely noise: if WTP *depends on* the information provided, then the "true" WTP is not a fixed quantity being measured but a value being *constructed* during the survey (the constructed-preference concern, and the reason your familiarity treatment is informative — Chapters 24, 39). The design implication: provide *standardized, adequate, neutral* information to all respondents, pretest for comprehension, and — where information effects are a research question — *manipulate* information experimentally rather than letting it vary uncontrolled.

Framing and anchoring. Beyond the cost-vector anchoring of Chapter 29, surveys carry many framing risks: the *reference point* (gain vs loss framing of the environmental change), the *order* of goods (an earlier good anchors valuation of a later one — the sequence/embedding effects of Chapter 35), and *numerical anchors* (any number mentioned can anchor). The defences are *neutral framing*, *randomized order* (§51.6), and *pretesting* to detect framing sensitivity.

The Goldilocks problem of information. Too little information and respondents cannot form a considered value (they guess or protest); too much and they are overwhelmed (fatigue, satisficing). The right amount is *enough to understand the good and the trade-off, no more* — determined by pretesting, not assumption. This is the information-side analogue of the cognitive-burden constraint on DCE design (Chapter 37).

51.4 Data quality: detecting inattentive and low-effort responses

Not every completed survey is a considered one, and low-effort responses corrupt estimates. Detecting them is essential — and doing so *correctly* (without introducing selection bias) is subtle.

The problem: inattention attenuates. Respondents who *speed* through, *straight-line*, or choose *randomly* contribute noise, not signal. In a choice model, random choosers behave as if their *scale* were near zero (Chapter 25) — their choices are insensitive to attributes — which *attenuates* the estimated coefficients toward zero.

Worked Example 51.1 (Inattention biases the DCE, verified). A DCE with 20% inattentive (random-choosing) respondents, true attribute coefficient 1.5, cost coefficient 0.10:

Sample	$\hat{\beta}$	$\hat{\alpha}$	WTP $= \beta/\alpha$
Full (incl. inattentive)	0.887	0.0604	14.68
Attentive only (screened)	**1.497**	**0.1068**	14.01
Truth	1.5	0.10	15.0

(Verified.) The inattentive respondents *attenuate both coefficients* — $\hat{\beta}$ falls from 1.5 to 0.9, $\hat{\alpha}$ from 0.10 to 0.06 — corrupting the estimated preference structure. (The *WTP ratio* happens to survive here because both coefficients attenuate roughly proportionally — a *scale* effect that cancels in the ratio, Chapter 25 — but this is not guaranteed, and the coefficients, standard errors, and any welfare calculation depending on the scale are corrupted.) **Inattention is not benign noise that averages out; it systematically distorts the estimates.**

Detecting low-effort responses. Several indicators, best used in combination:

- **Speed.** Completion time implausibly short (below a pre-specified threshold, e.g. faster than the time to read the questions) flags speeders.
- **Straight-lining / non-differentiation.** Identical responses across a battery of items — detectable by the *within-respondent variance* of the responses.
- **Attention checks.** Explicit instructional-manipulation checks ("select 'somewhat agree' for this item") embedded in the survey.
- **Inconsistency.** Choices that violate dominance or transitivity, or contradict earlier responses.
- **Item non-response / careless patterns.** Skipped items, or patterns (zig-zag) indicating disengagement.

Worked Example 51.2 (Straight-lining detection, verified). A 10-item attitude battery (1–5 scale); straight-liners answer identically:

Flag rule	Result
Row standard deviation < 0.3	**100% of straight-liners flagged, 0% false positives**

(Verified.) A simple within-respondent variance rule catches non-differentiation cheaply and cleanly. Combined with speed and attention checks, it identifies most low-effort responses.

The screening hazard: it is a post-treatment exclusion. Here the caution of Chapter 32 returns with force. Screening out respondents *after* they have responded is a *post-treatment exclusion*, and if the screening is related to treatment (e.g. the high-cost arm induces more speeding) or to the outcome (low-WTP respondents disengage more), it *biases* the estimate. The disciplined protocol:

1. **Pre-specify the quality rules** (thresholds, indicators) *before* seeing the data — this is a data-quality decision, not a specification search (Chapter 46).
2. **Report exclusion rates by arm/condition** — differential exclusion is the red flag (Chapter 32).
3. **Analyze with and without exclusions**, reporting both.
4. **Prefer prevention to exclusion** — a well-designed, appropriately-short, engaging survey *reduces* inattention at source, which is better than screening it out afterwards.

Stop and think 51.1. Your DCE has a 20% straight-lining rate, higher in the high-cost-vector arm. Should you drop the straight-liners, and what are the dangers? *(Answer: drop them only under a* pre-specified* protocol, and be alert that the *differential* rate is a warning sign. Straight-liners contribute noise, not signal — verified, inattentive respondents attenuate the coefficients (β 1.5→ 0.9), corrupting the preference structure — so *including* them biases the estimates. But *screening* them is a *post-treatment exclusion* (Chapter 32), and the fact that straight-lining is *higher in the high-cost arm* is exactly the differential-exclusion red flag: if the high cost *causes* more disengagement (people balk and stop attending), then dropping straight-liners removes systematically *different* respondents from the two arms, breaking the randomization and biasing the *cost-vector effect* — potentially manufacturing or masking a treatment effect. The disciplined response: (i) *pre-specify* the straight-lining rule (e.g. row SD < 0.3, verified to catch 100% with no false positives) *before* analysis, so it is a quality decision not a specification search; (ii) *report the exclusion rate by arm* (20% overall but differential — the key diagnostic); (iii) *analyze with and without* the exclusions and report both, since a cost-vector effect that appears only after dropping straight-liners is suspect; (iv) consider *bounding* the effect (Lee bounds, Chapter 32) robust to the differential exclusion; and (v) recognize that the *better* fix is *prevention* — a shorter, clearer, more engaging survey and a credible cost range reduce disengagement at source, which no post-hoc screening can match. So: yes, drop them, but only pre-specified, reported by arm, with/without, and bounded — and treat the differential rate as evidence the high-cost scenario may itself be inducing the disengagement.)*

51.5 Representativeness, sampling, and weighting

A valuation estimate is only as representative as its sample. Aggregating a biased sample's WTP to the population (Chapter 35) produces a biased benefit.

The sampling frame and mode. The *frame* (who can be sampled) and *mode* (online panel, mail, phone, in-person) determine coverage. Online panels are cheap and fast but may *under-cover* the elderly, low-income, or rural — precisely groups whose WTP may differ. In-person sampling covers better but costs more. The frame must match the *extent of the market* (Chapter 35): if the policy-relevant population is coastal households, the frame must cover them representatively.

Probability versus quota sampling. *Probability sampling* (random selection from the frame) supports valid inference but is costly and suffers non-response. *Quota sampling* (targeting subgroup counts — so many young, so many old, so many by income) ensures subgroup representation cheaply but is non-random within quotas and can miss unobserved dimensions. Most valuation surveys use quotas on key demographics; the requirement is that the quota dimensions include the characteristics *correlated with WTP* (income, coastal proximity, use of the resource).

Post-stratification and raking. When the achieved sample does not match population margins, *weighting* corrects it: each respondent is weighted by the ratio of their subgroup's *population* share to its *sample* share, so the weighted sample matches known population totals. *Raking* (iterative proportional fitting) extends this to multiple margins simultaneously.

Worked Example 51.3 (Weighting corrects representativeness bias, verified). A sample over-representing high-income respondents (whose WTP is higher):

Estimate	Mean WTP	Bias
True population mean	€42.50	—
Naive sample mean	€52.72	**+24.1%**
Post-stratified / raked	**€42.76**	**+0.6%**

(Verified.) The naive mean overstates WTP by 24% because the sample is too rich; post-stratification to the population income distribution cuts the bias to under 1%.

The two costs of weighting. Weighting is not free:

1. **Variance inflation.** Unequal weights reduce the *effective sample size* (Chapter 32 §32.7's design effect). Verified, the effective sample size after weighting fell to **72%** of the nominal — weighting a 1,500-respondent sample gives the precision of about 1,079. Extreme weights (a rare subgroup heavily up-weighted) worsen this.
2. **It cannot fix unobserved imbalance.** Weighting corrects imbalance only on the *observed* margins used for weighting. If the sample is unrepresentative on an *unobserved* characteristic correlated with WTP (environmental attitudes, say), weighting on income and age does not fix it — the residual bias remains (Chapter 32's selection-on-unobservables).

For your aggregation. Because the benefit is the population aggregate (Chapter 35), representativeness is not optional — a 24% sample bias becomes a 24% benefit bias, decisive for a marginal CBA. The design must sample representatively (frame, mode, quotas on WTP-correlated characteristics) *and* weight to population margins, reporting the effective sample size and acknowledging that unobserved imbalance may remain.

> **Active recall 51.B.** How does post-stratification correct a non-representative sample, and what are its two limitations? *(Answer: post-stratification (or raking, its multi-margin version)* weights* each respondent by the ratio of their subgroup's *population* share to its *sample* share, so the weighted sample reproduces known population totals on the weighting margins (income, age, region). This removes the bias from over- or under-representing subgroups whose outcome differs — verified, it cut a +24.1% mean-WTP bias (from over-sampling high-income respondents) to +0.6%. Two limitations: (i) *variance inflation* — unequal weights reduce the *effective sample size* (the design effect), so precision falls; verified, weighting a 1,500-respondent sample left an effective sample of ~1,079 (72%), and extreme weights on rare heavily-up-weighted subgroups worsen this, so weighting trades bias reduction for precision loss; (ii) *it corrects only observed imbalance* — weighting on the margins used (income, age) does nothing for imbalance on *unobserved* characteristics correlated with WTP (e.g. environmental attitudes, resource use), so if the sample is selected on unobservables (Chapter 32), the residual bias survives weighting entirely. The implications: choose weighting margins that include the characteristics most correlated with WTP; report the effective sample size; and acknowledge that representativeness on observables does not guarantee it on unobservables — which is why a good *sampling frame and design* (reducing imbalance at source) is superior to relying on weighting to repair it after the fact.)*

51.6 Order, priming, fatigue, and the sample-size question

The survey is a *sequence*, and its ordering and length shape responses.

Order and priming effects. The *order* in which questions and goods appear affects responses: an earlier question can *prime* a frame of mind, *anchor* a valuation, or set a *reference point* for later ones. A block of environmental attitude questions before the WTP question can prime pro-environmental responses; valuing good A before good B anchors B (the sequence effect of Chapter 35). The defence is *randomization*: randomize the order of questions, goods, and choice tasks (and the position of alternatives within a task, Chapter 37), so that any position effect is *orthogonal* to the content and averages out rather than confounding it. This is a zero-cost design fix that eliminates a whole class of bias.

Fatigue and satisficing. Long surveys tire respondents, who then *satisfice* — giving adequate rather than considered answers, straight-lining, or speeding (§51.4). Fatigue also causes *scale drift*: later tasks are answered with lower attention, effectively lower scale (Chapter 25), which — if task position correlates with the design — biases attribute effects. The defences: *cap the survey length* (and the number of choice tasks, Chapter 37), *randomize task order* so fatigue is orthogonal to content, and *monitor* response quality across the survey (does quality degrade in later tasks?).

Sample size. How many respondents? Driven by the *precision target* and the response variance.

Worked Example 51.4 (Sample size for precision, verified). For a WTP standard deviation of €25:

Target	Required n
Mean WTP within ±€3 (95%)	267
Mean WTP within ±€2 (95%)	601
Mean WTP within ±€1 (95%)	2,401
A proportion within ±5pp (95%)	385
A proportion within ±3pp (95%)	1,068
A proportion within ±2pp (95%)	2,401

(Verified.) Precision improves only as $1/\sqrt{n}$, so halving the margin of error *quadruples* the sample. And crucially, **representativeness requires adequate n within each key subgroup, not just overall** — a sample of 1,000 that is 5% low-income has only 50 low-income respondents, too few to estimate their WTP or to weight reliably. Quota design must ensure sufficient subgroup cells, and DCE sample size additionally accounts for the design efficiency and heterogeneity structure of Chapter 37.

51.7 Pretesting, piloting, and the iterative design

No questionnaire is right the first time; the design is *developed and tested* before the main fielding.

The development sequence.

1. **Focus groups** — explore how respondents think about the good, what language they use, what payment vehicles are credible, what confuses them.
2. **Cognitive interviews** — "think-aloud" protocols where respondents verbalize their reasoning, revealing misunderstanding, unintended framing, and where the scenario fails.
3. **Expert review** — economists, ecologists, and survey methodologists check the instrument for accuracy, bias, and validity.
4. **Pilot survey** — a small field test under realistic conditions, used to check comprehension, response distributions, protest rates, completion time, data quality, and — for DCEs — to update the priors for efficient design (Chapter 37) and estimate the response variance for the main sample size.
5. **Revision and re-pilot** — iterate until the instrument performs.

What the pilot delivers for your work. Beyond comprehension: the pilot's protest rate tells you whether the payment vehicle works; the completion time sets the speeder threshold; the response variance sizes the main sample; the choice patterns update the DCE design priors and reveal dominance or non-attendance; and the data-quality indicators calibrate the screening rules. **Skipping or under-resourcing pretesting is the most common and most consequential survey-design failure** — a flaw baked into the instrument corrupts every response, and no amount of the sophisticated modelling in this book can repair it. The pilot is where the fragility of the *first* link in the chain (Chapter 50) is reduced.

> **Stop and think 51.2.** Your budget is tight and a colleague suggests skipping the pilot to field the main survey directly. What do you tell them? *(Answer: skipping the pilot is a false economy that risks the entire study, because it is the only stage where design flaws are caught*

before* they corrupt every response. Concretely, the pilot delivers things the main study *depends on* and cannot recover without it: (i) *comprehension* — cognitive interviews and the pilot reveal whether respondents understand the finfish good and scenario; if they misunderstand, every WTP measures the wrong construct and no model fixes it; (ii) *payment-vehicle validation* — the pilot's protest rate reveals whether the vehicle is credible and acceptable, and an objectionable vehicle can dominate the response distribution (§51.2); (iii) *the speeder/quality thresholds* — completion times from the pilot *set* the pre-specified screening rules (§51.4), which cannot be set defensibly from the main data without becoming a specification search; (iv) *the sample size* — the pilot's response variance sizes the main sample (verified, n scales with the variance and the precision target), so without it the main n is a guess; (v) *the DCE design priors* — the pilot updates the priors for efficient design (Chapter 37), and a design built on wrong priors wastes precision; (vi) *dominance and non-attendance* detection. The risk of skipping: a flaw baked into the instrument (an incredible vehicle, a confusing scenario, an inefficient design, wrong sample size) corrupts *all* the expensive main-survey responses, and is discovered only *after* the money is spent, when nothing can be done. The pilot is cheap relative to the main survey and is precisely the fragility-reduction step for the *first* link in the chain (Chapter 50) — where errors are most consequential because everything downstream inherits them. I would tell the colleague that the pilot is not a preliminary to cut but the insurance that makes the main survey worth fielding, and that the disciplined economy is to *reduce the main sample slightly* if needed, never to skip the pilot.)*

51.8 Failure modes

F1. Incredible payment vehicle. Triggering protest and vehicle bias. *Fix:* credible, coercive, pretested vehicle (§51.2).

F2. Non-consequential framing. Inviting hypothetical bias. *Fix:* credible provision and use of results (§51.2; Chapter 39).

F3. Non-neutral information/framing. Pushing responses. *Fix:* neutral, standardized information; randomized order; pretesting (§51.3, §51.6).

F4. Uncontrolled information effects. WTP varying with unmanaged information. *Fix:* standardize; manipulate experimentally if a research question (§51.3).

F5. Ignoring inattention. Estimates attenuated by low-effort responses. *Fix:* speed, straight-lining, attention-check screening (§51.4).

F6. Ad hoc / differential screening. Post-treatment exclusion biasing estimates. *Fix:* pre-specified rules; report by arm, with/without, bound (§51.4; Chapter 32).

F7. Non-representative sample. Biased population aggregate. *Fix:* frame/mode/quotas on WTP-correlated characteristics; weight (§51.5).

F8. Weighting without caveats. Ignoring variance inflation and unobserved imbalance. *Fix:* report effective sample size; acknowledge residual bias (§51.5).

F9. Fixed order / no randomization. Position and priming effects confounding content. *Fix:* randomize order and positions (§51.6).

F10. Excessive length. Fatigue, satisficing, scale drift. *Fix:* cap length/tasks; monitor quality across the survey (§51.6; Chapter 37).

F11. Inadequate subgroup n. Cells too small to estimate or weight. *Fix:* quota design with sufficient cells (§51.6).

F12. Skipping pretesting. Flaws baked into the instrument. *Fix:* focus groups, cognitive interviews, expert review, pilot, iterate (§51.7).

51.9 Diagnostic questions for your own papers

1. **Vehicle.** Is the payment vehicle credible, coercive, and pretested for protest? *(§51.2, F1.)*
2. **Consequentiality.** Is the scenario consequential and the provision credible? *(§51.2, F2.)*
3. **Neutrality.** Is information standardized, neutral, and comprehension-tested? *(§51.3, F3–F4.)*
4. **Data quality.** Are pre-specified inattention screens applied, reported by condition, with/without? *(§51.4, F5–F6.)*
5. **Representativeness.** Does the frame/quota cover WTP-correlated characteristics, and is the sample weighted? *(§51.5, F7.)*
6. **Weighting caveats.** Are effective sample size and unobserved-imbalance limits reported? *(§51.5, F8.)*
7. **Order.** Are question, good, task, and alternative orders randomized? *(§51.6, F9.)*
8. **Length.** Is the survey capped and quality monitored across it? *(§51.6, F10.)*
9. **Subgroup n.** Are subgroup cells adequate? *(§51.6, F11.)*
10. **Pretesting.** Focus groups, cognitive interviews, expert review, and a pilot conducted and used? *(§51.7, F12.)*

51.10 Exercises

(A) Comprehension.

A1. Why does the payment-vehicle choice affect validity, and what makes a good vehicle?

A2. How do inattentive respondents bias DCE estimates, and how are they detected?

A3. How does post-stratification correct representativeness, and what are its limits?

(B) Derivation / reasoning.

B1. Explain why random choosers attenuate choice-model coefficients.

B2. Show why weighting reduces the effective sample size.

B3. Explain why screening low-effort responses is a post-treatment exclusion and when it biases.

(C) Research application / critique.

C1. Design the questionnaire for your finfish CV/DCE survey, covering vehicle, scenario, information, quality, and sampling.

C2. Your sample over-represents the wealthy and coastal. Write the representativeness strategy.

C3. A reviewer says your 20% exclusion of speeders invalidates the results. Respond.

51.11 Solutions

A1. The payment vehicle affects validity through credibility (implausible vehicles trigger disbelief), incentive compatibility (coercive vehicles are truth-inducing for public goods; voluntary ones invite free-riding, Chapter 28), payment-vehicle bias (attitudes to the *vehicle* contaminate WTP for the *good* — tax aversion read as low valuation), and protest potential (objectionable vehicles generate protest zeros, Chapters 26, 29). A *good* vehicle is *coercive* (for incentive compatibility), *credible* (plausibly used for this good, tied to a believable authority), *acceptable* (minimizing protest and vehicle bias — typically an earmarked tax/levy rather than a general tax or a voluntary donation), and *pretested* to confirm it does not dominate the response distribution.

The choice embodies the tension that coercive vehicles are both the most incentive-compatible and the most protest-prone, requiring a design balance found through pretesting.

A2. Inattentive respondents (speeders, straight-liners, random choosers) make choices *insensitive to attributes*, behaving as if their scale were near zero (Chapter 25). Because the estimated coefficients confound preference and scale, adding scale-zero respondents *attenuates* the coefficients toward zero — verified, 20% inattentive drove $\hat{\beta}$ from 1.5 to 0.9 and $\hat{\alpha}$ from 0.10 to 0.06, corrupting the preference structure (the WTP *ratio* survived only because both attenuated proportionally, which is not guaranteed). Detection uses, in combination: *speed* (implausibly short completion time), *straight-lining* (low within-respondent variance across a battery — verified, a row-SD < 0.3 rule caught 100% with 0% false positives), *attention checks* (instructional-manipulation items), *inconsistency* (dominance/transitivity violations), and *careless patterns*. Screening must be *pre-specified* and reported by condition, because it is a post-treatment exclusion (Chapter 32).

A3. Post-stratification weights each respondent by (population subgroup share)/(sample subgroup share), so the weighted sample matches known population margins, removing bias from subgroups over- or under-represented relative to the population when their outcome differs — verified, cutting a +24% mean-WTP bias to +0.6%. *Limits*: (i) it inflates variance / reduces the effective sample size (unequal weights, verified ~72% here), trading bias for precision; (ii) it corrects only *observed* margins — imbalance on *unobserved* WTP-correlated characteristics survives weighting entirely (selection on unobservables, Chapter 32). So weighting margins should include the strongest WTP correlates, the effective sample size should be reported, and residual unobserved bias acknowledged — with a good sampling design (reducing imbalance at source) preferred to post-hoc repair.

B1. In a choice model, observed utility is $\lambda(\beta'\mathbf{x})$ where λ is the scale (inversely related to the error variance, Chapter 25). An *attentive* respondent chooses the utility-maximizing alternative, so their choices are *sensitive* to $\mathbf{x}$ — high effective scale. A *random chooser* selects irrespective of $\mathbf{x}$, which is observationally equivalent to $\lambda \to 0$ (infinite error variance) — their choices carry *no* information about β. Pooling the two, the likelihood is maximized by a *compromise* scale between the attentive respondents' high λ and the random choosers' zero λ, so the *estimated* $\hat{\lambda}$ (and hence $\hat{\lambda}\beta$, the estimated coefficients) is *pulled toward zero* — attenuation. Verified: 20% random choosers attenuated $\hat{\beta}$ from 1.5 to 0.9 and $\hat{\alpha}$ from 0.10 to 0.06 (both by roughly the same factor, a pure scale effect, so the *ratio* WTP was preserved — but the coefficients, their standard errors, and any scale-dependent welfare quantity were corrupted). The general lesson: inattention masquerades as low scale, and mixing scales biases the pooled estimate, which is why detecting and modelling (or screening) inattention matters.

B2. With survey weights w_i, the weighted mean is $\bar{y}_w = \sum_i w_i y_i / \sum_i w_i$, and its variance is $\mathrm{Var}(\bar{y}_w) = \sigma^2 \sum_i w_i^2 / (\sum_i w_i)^2$ (for independent y_i of variance σ^2). Compare the *unweighted* variance σ^2/n. The ratio defines the *design effect* $\mathrm{deff} = n \sum_i w_i^2 / (\sum_i w_i)^2 \geq 1$ (by Cauchy–Schwarz, with equality only for equal weights). The *effective sample size* is $n_{\mathrm{eff}} = n/\mathrm{deff} = (\sum_i w_i)^2 / \sum_i w_i^2 \leq n$. Intuitively, unequal weights mean some observations count for more than others, so the sample carries less independent information than n equal observations — a heavily up-weighted rare-subgroup respondent contributes a large, high-variance term. Verified: weighting a 1,500-respondent sample gave $n_{\mathrm{eff}} \approx 1{,}079$ (72%). The more unequal the weights (the more the sample must be corrected), the larger the deff and the greater the precision loss — so aggressive weighting to fix severe non-representativeness is costly, and a representative sample by design is more efficient than a skewed one repaired by weights.

B3. Screening removes respondents *after* they have been exposed to the survey (and, in an experiment, to the treatment), conditioning the analysis sample on a *post-treatment* variable — the quality indicator (speed, straight-lining), which is *realized during* the response process. This is structurally identical to the post-randomization exclusion of Chapter 32: if the screening variable is *independent* of treatment and outcome, dropping on it is harmless (it removes noise symmetrically); but if it is *related to treatment* (e.g. a high-cost or complex arm induces more speeding/disengagement) *or* to the outcome (low-WTP respondents disengage more), then conditioning on "passed the quality screen" makes the retained groups *non-comparable* — a collider/selection bias that distorts the estimate (verified logic in Chapter 32: differential exclusion biased an ATE from 4.85 to 5.72). So screening biases whenever the low-effort behaviour is *caused by* the treatment

or *correlated with* the valuation. The safeguards: pre-specify the rules (no outcome-driven selection), report exclusion rates *by arm* (differential rates are the diagnostic), analyze *with and without*, *bound* the effect robust to the exclusion (Lee bounds), and prefer *prevention* (better design reducing inattention) to *exclusion*.

C1. *Questionnaire design for the finfish CV/DCE survey.* **Scenario and good:** describe the finfish stock improvement accurately, comprehensibly, and neutrally, using pretested visual aids (maps, stock-level infographics) that are themselves neutral; convey the ecological change and its consequences in coastal-household terms. **Consequentiality:** frame the survey as potentially informing a real FRD/blue-finance policy by a credible authority, stating how results will be used (Chapter 39), and *measure* perceived consequentiality with follow-up items. **Payment vehicle:** a *coercive*, credible, earmarked levy/tax tied to the environmental programme (incentive-compatible for the public good, Chapter 28), selected and refined via focus groups and the pilot to minimize protest and vehicle bias; pretest the protest rate. **Information:** standardized, adequate, neutral information for all respondents, comprehension-tested; since *familiarity* is a research dimension, manipulate information/familiarity *experimentally* rather than letting it vary uncontrolled (Chapters 24, 39). **Elicitation:** SBDC with payment cards (Chapter 36) for the CV; the efficient DCE design (Chapter 37) with a capped task count; real-versus-hypothetical and BRIS treatments for the validity test (Chapter 39). **Data quality:** *pre-specify* screening rules — speed threshold from the pilot completion time, straightlining via row-variance (verified 100%/0% detection), attention checks, dominance/consistency checks — and commit to reporting exclusions by arm, with/without, and bounded (Chapters 32, 46). **Order and length:** randomize question, good, task, and alternative order (eliminating priming/position effects at zero cost); cap length to control fatigue; monitor quality across the survey. **Sampling:** a frame and mode covering the policy-relevant (coastal) population, quota sampling on WTP-correlated characteristics (income, coastal proximity, resource use) with adequate subgroup cells, and post-stratification/raking to population margins, reporting the effective sample size and residual-unobserved-imbalance caveat. **Sample size:** set from the pilot's response variance and the precision target (verified formulas), ensuring adequate subgroup n and DCE design efficiency (Chapter 37). **Pretesting:** focus groups → cognitive interviews → expert review → pilot (updating DCE priors, protest rate, speeder threshold, sample size) → revise → re-pilot.

C2. *Representativeness strategy for a sample over-representing wealthy, coastal respondents.* **(1) Diagnose the imbalance:** compare the achieved sample's income and coastal-proximity distribution to the *population* (census/official statistics) for the policy-relevant extent of the market (Chapter 35), quantifying the over-representation. **(2) Weight:** apply post-stratification (or raking, for multiple margins simultaneously) to income *and* coastal proximity *and* other WTP-correlated characteristics, weighting each respondent by (population share)/(sample share) — verified, this cut a +24% mean-WTP bias to +0.6%. **(3) Report the cost:** compute and report the *effective sample size* after weighting (verified ~72% in the analogous case), since weighting inflates variance, and check for *extreme weights* (a rare heavily-up-weighted subgroup) that could destabilize the estimate — trimming or winsorizing if necessary. **(4) Acknowledge the limit:** weighting corrects only *observed* margins; if the sample is also unrepresentative on *unobserved* WTP correlates (environmental attitudes, resource use not captured), residual bias survives — state this explicitly (Chapter 32). **(5) Sensitivity:** report the aggregate benefit (Chapter 35) *both* naive and weighted, since a 24% WTP bias becomes a 24% benefit bias, decisive for a marginal CBA (Chapter 45) — and carry the representativeness uncertainty into the fragility audit (Chapter 50), where it is likely a dominant stage (verified there, extent-of-market/aggregation contributed 22% of NPV variance). **(6) Prevent next time:** if fielding is ongoing, adjust the *quotas* to fill under-represented cells (lower-income, inland) with adequate n — a design fix superior to heavier weighting. **(7) Frame check:** confirm the *sampling frame and mode* are not the source (e.g. an online panel under-covering the elderly/low-income/rural); if so, supplement the frame rather than relying on weights alone.

C3. *Response to "20% speeder exclusion invalidates the results."* The concern is legitimate in principle — excluding respondents is a *post-treatment exclusion* (Chapter 32) that *can* bias estimates — but the validity of the exclusion is an empirical question I have addressed by protocol, not a fatal flaw. Four points. **(1) The exclusion is necessary, not optional.** Speeders are inattentive respondents whose choices are insensitive to attributes, and *including* them *attenuates* the estimates — verified, 20% inattentive drove the attribute coefficient from 1.5 to 0.9, corrupting the preference structure. So the alternative to screening is *known* bias from noise, not clean data. **(2) The rule was pre-specified.** The speeder threshold was set from the

pilot's completion-time distribution *before* seeing the main data (§51.7), so this is a data-quality decision, not a specification search (Chapter 46) — I did not tune the threshold to produce a desired result. **(3) I report the diagnostics that reveal any bias.** I report the exclusion rate *by treatment arm* — if speeding is *differential* (e.g. higher in the high-cost or complex arm), that is the red flag for bias, and I disclose it; I report the estimates *with and without* the exclusion, so the reader sees whether the results depend on it; and where differential exclusion is present, I report *Lee bounds* robust to it (Chapter 32). **(4) Prevention was built in.** The survey was designed (capped length, clear scenario, engaging format, randomized order) to *minimize* speeding at source, so the 20% is not a design failure but the residual that screening handles. If the reviewer's specific concern is *differential* exclusion, I point to the by-arm rates and the with/without analysis that address it directly; if the concern is that 20% is *high*, I note it is within the typical range for online stated-preference surveys and that the with/without comparison shows the substantive conclusions are unchanged. The exclusion does not invalidate the results; the *protocol* — pre-specified rule, by-arm reporting, with/without analysis, bounding — is precisely what makes the exclusion defensible, and *not* excluding would substitute a known attenuation bias for a managed and disclosed one.

51.12 Spaced review prompts

+1 day. From memory: the four channels through which the payment vehicle affects validity; how inattention attenuates estimates; how post-stratification works.

+1 week. Explain why screening is a post-treatment exclusion; why weighting inflates variance and cannot fix unobserved imbalance; why randomizing order eliminates priming.

+1 month. For your CV/DCE survey, draft the full instrument-design protocol — vehicle, scenario, information, quality rules, sampling and weighting, order randomization, sample size, pretesting plan. Close with the spine sentence about whether the instrument elicits the intended construct from a representative sample with adequate quality.

51.13 How this chapter connects to the rest of the book

- This chapter is the *source* of the data every other chapter analyzes. **Chapter 28, 39** supply the incentive/consequentiality theory operationalized in the vehicle and scenario; **Chapter 29** the anchoring extended to survey order; **Chapter 32** the post-treatment-exclusion hazard in quality screening; **Chapter 25** the scale that inattention inflates; **Chapter 35** the aggregation that representativeness serves; **Chapter 37** the DCE cognitive-burden and design efficiency that length and sample size respect.
- **Chapter 50's** fragility audit identified the *first* link — elicitation and the instrument — as the origin of the whole chain; this chapter is how that link's reliability is maximized. **Chapter 46's** reproducibility discipline governs the pre-specification of quality rules and the documentation of the design.
- Downstream: every estimate (Parts IX–X), its calibration (Chapter 39), aggregation (Chapter 35), and the whole appraisal (Part XI) inherit the quality — or the flaws — of the instrument designed here.

This chapter is where the quality of the evidence is *made*, before any model sees it — and where the discipline of pretesting, pre-specified quality rules, and representative sampling reduces the fragility of the chain's foundational link.

51.14 Research translation task

For your **CV/DCE survey**, produce an instrument-design and fielding protocol:

1. **Scenario and vehicle.** The good description, visual aids, consequentiality framing, and the pretested coercive payment vehicle.
2. **Information.** Standardized neutral information; the experimental manipulation of familiarity/information where it is a research question.
3. **Data quality.** Pre-specified screening rules (speed, straight-lining, attention, consistency) with the commitment to by-arm, with/without, and bounded reporting.
4. **Sampling and weighting.** Frame, mode, quotas on WTP-correlated characteristics, post-stratification plan, effective-sample-size reporting, and the sample size from the pilot variance and precision target.
5. **Order and length.** Randomization of question/good/task/alternative order; the length cap and quality monitoring.
6. **Pretesting.** Focus groups, cognitive interviews, expert review, and the pilot (with its role in updating DCE priors, protest rate, speeder threshold, and sample size).
7. **Close** with the spine sentence: *"The instrument elicits [finfish WTP] via a [coercive, pretested vehicle] in a [consequential] scenario, screens low-effort responses by [pre-specified rules] reported by arm, and samples [the coastal population] with quotas on [WTP correlates] weighted to population margins (effective n [·]) — so the foundational link of the evidence chain is [validated]."*

This protocol governs the primary data that feed every subsequent chapter of your research program.

51.15 Bibliography

Core. - Mitchell & Carson, *Using Surveys to Value Public Goods: The Contingent Valuation Method* — the foundational treatment of scenario, payment vehicle, and survey design for valuation (§51.2–§51.3). *(Verify edition.)* - Johnston et al. (2017), "Contemporary Guidance for Stated Preference Studies," *Journal of the Association of Environmental and Resource Economists* — the modern best-practice guidelines for stated-preference survey design. *(Verify volume/pages.)* - Groves, Fowler, Couper, Lepkowski, Singer & Tourangeau, *Survey Methodology* — the standard reference on sampling, mode, non-response, and questionnaire design (§51.5–§51.6). *(Verify edition.)*

Applied. - Champ, Boyle & Brown (eds.), *A Primer on Nonmarket Valuation* — practical survey-design guidance for CV and DCE (§51.2–§51.4). *(Verify edition.)* - Meade & Craig (2012), "Identifying Careless Responses in Survey Data," *Psychological Methods* — detection of straight-lining and inattention (§51.4). *(Verify volume/pages.)* - Valliant, Dever & Kreuter, *Practical Tools for Designing and Weighting Survey Samples* — post-stratification, raking, and design effects (§51.5). *(Verify edition.)*

Advanced. - Bradburn, Sudman & Wansink, *Asking Questions: The Definitive Guide to Questionnaire Design* — question wording, order, and framing effects (§51.3, §51.6). *(Verify edition.)* - Tourangeau, Rips & Rasinski, *The Psychology of Survey Response* — the cognitive processes behind satisficing, order, and fatigue effects (§51.6). *(Verify edition.)* - Boyle et al., or Carson & Louviere, on stated-preference survey administration and data quality (§51.4). *(Verify citations.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Johnston et al. (2017) / Meade–Craig / Valliant et al. citations before using them, per the book's no-unverified-citation rule.

End of Chapter 51. Next: Chapter 52 — Computational Diagnostics and the Reproducible Research Pipeline, the book's final chapter, which turns to execution: the software toolchain (R, Apollo, metafor, targets, renv, Git, Quarto), the end-to-end pipeline from data cleaning through estimation to tables and figures, and — the culminating discipline — knowing when not* to trust the output: convergence diagnostics, multiple starting values, seed and draw sensitivity, gradient and Hessian checks, and detecting the local maxima and non-identification that a converged optimizer conceals. Tell me when to write it, or request a different chapter.*

Part XIII Assessment — Synthesis, Survey Methodology, and Implementation

Covers Chapters 50–51. Five components, in the order they are best attempted. Allow a full working day; the mock oral is the part that matters most for your viva.

1. Cumulative problem set

P1. Take one empirical claim from your own work and write its complete assumption chain, from raw response to reported welfare number.

P2. For each link, assign a fragility rating and justify it in one sentence.

P3. Identify the weakest link and compute how much the conclusion moves when that assumption is varied over its plausible range.

P4. Draft the consequentiality script for your CV instrument and the debriefing item that measures whether respondents believed it.

P5. Design the data-quality screening rules for your survey, and state for each whether it is applied pre- or post-randomization and what that implies.

2. Research critique

Take a published paper in your field that you admire. Run the full fragility audit on its headline claim. Write two pages. The point is not to find fault but to establish what the paper is entitled to conclude — and to practise doing this to work you respect, since you will have to do it to your own.

3. Red-flag diagnosis

A survey reports a 62% completion rate, drops speeders below one-third of median completion time, applies quota weights to match census age and gender, and reports a population-weighted mean WTP. Identify which of these steps changes the estimand, which changes only precision, and which could break the randomization.

4. Mock oral examination

1. Walk me through the fragility audit of your main result.
2. What is the weakest assumption in your CV paper, and how do you know?
3. Why does a payment vehicle change a valuation?
4. What does your quota weighting estimate, exactly?

5. If you had to withdraw one claim from your dissertation, which and why?

5. Integration mini-project

Produce the complete fragility dossier for the single most important empirical claim in your dissertation: the identifying variation, every assumption in the chain, the fragility rating of each, the sensitivity analysis for the weakest, and a one-paragraph statement of the claim written as an honestly conditional sentence. This is the document your viva will turn on.

Part XIV — Computational Diagnostics and Reproducible Research

Chapter 52 — Computational Diagnostics and the Reproducible Research Pipeline

Part XIV: Computational Diagnostics and Reproducible Research — and the book's conclusion

> **The spine question, at the end.** The whole book has asked what a result identifies, under which assumptions, and how fragile it is. This final chapter adds the last, most practical fragility: *the computation itself.* A model can be correctly specified, well-identified in theory, and estimated on excellent data — and still produce a *wrong number*, because the optimizer stopped at a local maximum, the Hessian was singular, the draws were too few, or a variable was badly scaled. Worst of all, the software reports **"converged: success"** in every one of these cases. Verified below, a non-identified model converges happily while its Hessian has *exactly zero eigenvalues*, and the same model from six starting values reaches six different optima, each declared a success. The spine question, in its final form: *when the software says the answer is ready, how do I know whether to believe it?*

52.0 Conceptual motivation: convergence is not validity

You have reached the end of a long chain — from linear algebra to random utility to welfare theory to appraisal to synthesis to survey design. At the very end sits the most mundane and most dangerous link: the moment you run `estimate()` and the software returns a number. This chapter is about that moment, and its central message is a single sentence that overturns a near-universal habit: **the optimizer reporting "convergence" tells you almost nothing about whether the result is correct.**

This is not pedantry. The models your research relies on — the error-components mixed logit in WTP space (Chapter 38), the MDCEV (Chapters 40–41), the Bayesian hierarchical synthesis (Chapter 48) — have likelihoods that are *not globally concave*, that are *simulated* (so noisy), and that can be *weakly or non-identified* in ways no output flag reveals. Every one of them can converge to the wrong answer and announce success. The verified demonstrations in this chapter are stark:

Multiple starting values reach different optima. A single model, estimated from six different starting points, converged to optima with objective values ranging from -2.84 (the global) to -1.30 (a poor local), *each reporting "success."* Taking one run — the standard practice — is a coin flip on whether you found the global optimum. The remedy, running many starts and keeping the best, is not optional for these models.

Non-identification hides behind convergence. A model in which two parameters enter only as their *sum* (so they are not separately identified) converged cleanly, reported plausible-looking individual estimates — and its Hessian had *two exactly-zero eigenvalues* (condition number infinite). The optimizer was perfectly happy; the individual estimates were meaningless. Convergence is a *lie* about identification, and only inspecting the Hessian exposes it. This is Chapter 30's identification-versus-estimation distinction reappearing

at the *computational* level: the data may identify a quantity, but the optimizer cannot tell you whether they do — you must check.

Bad scaling wrecks the computation invisibly. A regressor measured in the wrong units (cents instead of euros) worsened the Hessian's condition number by a factor of ~10,000, degrading convergence and standard errors — a purely mechanical problem, invisible in the estimates, fixed by rescaling.

A trustworthy result passes explicit checks. A well-identified model showed a gradient norm of 6×10^{-4} (effectively zero), a positive-definite well-conditioned Hessian, and estimates near the truth — the *checklist* every reported estimate must pass before it is believed.

The chapter has two halves. First, the **reproducible pipeline** — the toolchain and workflow (the R/Apollo/metafor/`targets`/`renv`/Git/Quarto stack your PDF specifies) that makes the computation *repeatable and auditable*. Second, and more important, the **diagnostics** — the discipline of *knowing when not to trust the output*, which is the computational face of the entire book's concern with fragility. And then, at the end, the book closes.

52.1 Prerequisites

From **Chapter 1**: matrix conditioning, definiteness, eigenvalues — the linear algebra of the Hessian. From **Chapters 11–13**: maximum likelihood, the score (gradient), the information matrix (Hessian), and standard errors from its inverse. From **Chapter 30**: identification versus estimation — here at the computational level. From **Chapter 38**: simulated maximum likelihood, draws, and seed sensitivity. From **Chapter 46**: reproducibility, literate programming, and version control — the pipeline half of this chapter.

New notation. *Local maximum* (an optimum that is not the global one), *starting values* (the optimizer's initial guess), ∇ (the gradient/score), H (the Hessian), *condition number* (the ratio of largest to smallest eigenvalue), *convergence flag* (the optimizer's self-report).

52.2 The reproducible pipeline: toolchain and workflow

Before the diagnostics, the *infrastructure* that makes any result repeatable — the concrete realization of Chapter 46's principles.

The toolchain (your PDF's stack). A modern valuation pipeline is built from specialized, interoperable tools:

- **Base R and the data ecosystem** — `R` for statistical computing; `tidyverse` / `data.table` for data cleaning and manipulation. The unglamorous data-preparation stage is where most errors enter and most time is spent.
- **Estimation** — `Apollo` (the purpose-built choice-modelling package for mixed logit, MDCEV, WTP space, and more), `maxLik` / `optim` (general maximum likelihood), `mlogit` / `gmnl` (logit family), `metafor` (meta-analysis), `clubSandwich` / `sandwich` (robust and cluster-robust variance), `boot` (bootstrap).
- **Pipeline orchestration** — `targets` (or `drake`) to define the analysis as a *dependency graph*, so that changing one input automatically re-runs only the affected downstream steps and nothing goes stale.
- **Environment and version control** — `renv` to lock package versions, `Git` for version control, so the exact computational state producing any result is recoverable.
- **Literate output** — `Quarto` / `RMarkdown` (your workflow) to bind code, output, and prose into a document that *executes* to produce the report, rendered via `pandoc`.

The end-to-end workflow. A disciplined pipeline runs, as a reproducible dependency graph:

1. **Import and clean** — raw survey data to a documented, validated analysis dataset (with a data dictionary).

2. **Screen** — apply *pre-specified* data-quality rules (Chapter 51), recording exclusions and their reasons.
3. **Balance and describe** — balance checks across arms (Chapter 32), descriptive statistics, representativeness assessment and weights (Chapter 51).
4. **Estimate** — the models (Chapters 36–48), with the diagnostics of this chapter attached to every estimation.
5. **Robustness** — alternative specifications, the specification curve (Chapter 46), sensitivity analyses.
6. **Simulate** — welfare, forecasts, appraisal Monte Carlo (Chapters 41, 45), with seeds and draw counts fixed and reported.
7. **Tabulate and plot** — tables and figures *generated* by code, never pasted.
8. **Assemble** — the literate document rendering to the final report and appendix.

Why the graph structure matters. Defining the pipeline as a `targets` dependency graph does what discipline alone cannot: it *guarantees* that every reported number reflects the *current* code and data. Change an exclusion rule, and every downstream table, estimate, and figure that depends on it re-runs automatically; nothing is left stale (Chapter 46's literate-programming payoff, extended to the whole project). For your work — where a single change to the calibration or the sample propagates through estimation, welfare, aggregation, transfer, and the CBA — this automatic, auditable propagation is what makes a long pipeline trustworthy rather than a source of silent inconsistency.

> **Active recall 52.A.** What does organizing the analysis as a `targets`-style dependency graph provide that careful manual scripting cannot? *(Answer: it* guarantees consistency* between the reported results and the current state of the code and data — a property manual scripting cannot ensure because it depends on the analyst remembering to re-run every affected step. A dependency graph encodes which outputs depend on which inputs, so when *any* input changes (a corrected data point, a revised exclusion rule, an updated calibration factor), the framework *automatically* re-runs *exactly* the downstream steps that depend on it — estimation, welfare simulation, aggregation, transfer, tables, figures — and *nothing else*, leaving nothing stale. This eliminates the pervasive and dangerous failure where a reported number no longer matches the analysis that supposedly produced it (Chapter 46's stale-result problem), which is especially acute in a *long* pipeline like a valuation-to-CBA chain where one upstream change (a sample or calibration revision) propagates through a dozen stages. It also makes the pipeline *auditable* (the graph documents the full provenance from raw data to final NPV) and *efficient* (only affected steps re-run, not the whole project). Combined with `renv` (locked environment), Git (version history), and Quarto (literate output), it turns the whole project into a single reproducible artefact — the infrastructure that makes a long, assumption-laden chain trustworthy rather than a source of silent inconsistency.)*

52.3 Convergence is not validity: the central warning

Now the diagnostics. The foundational error is trusting the convergence flag, so it must be dismantled first.

What convergence actually means. An iterative optimizer stops when its *internal* criteria are met — the gradient is small, the step size is tiny, or the objective barely changed between iterations. "Convergence: success" means *the algorithm found a point it cannot easily improve from.* It does **not** mean:

- that the point is the *global* optimum (it may be a local one);
- that the model is *identified* (the point may lie on a flat ridge);
- that the *standard errors* are valid (the Hessian may be singular);
- that the *model* is correct (a wrong model can converge fine);
- that the *simulation* was adequate (the noisy objective may have a spurious optimum).

The four failures that hide behind "success." Each is demonstrated below: *local optima* (§52.4), *non-identification* (§52.5), *ill-conditioning* (§52.6), and *simulation noise* (§52.7). In every case the optimizer reports success and the flaw is invisible in the point estimates — visible only in the *diagnostics* the analyst

must actively run.

The disciplining principle. *Never report an estimate on the strength of the convergence flag alone.* The flag is necessary (a non-converged run is definitely untrustworthy) but nowhere near sufficient. Trust requires a *checklist* (§52.8): multiple starts agree, the gradient is near zero, the Hessian is negative-definite (of the log-likelihood) and well-conditioned, the estimates are stable across seeds and draws, and the parameters lie in their valid ranges. Only a result passing all of these deserves to be believed — and reported.

52.4 Local optima and the multiple-starting-values rule

The likelihoods of the models you use are *not globally concave*, so an optimizer can converge to a local optimum that is not the global one.

The problem. A non-concave objective has multiple hills; the optimizer climbs the *nearest* one from its starting point and stops. Different starting points lead to different summits, and the software cannot tell whether the summit it reached is the highest.

Worked Example 52.1 (Local optima, verified). A multimodal objective estimated from six starting values:

Start	Converged optimum	Final objective	Reported
(−2, −1)	(−1.99, −0.99)	−2.101	success
(2, 2)	(1.99, 1.99)	**−2.841 (best)**	success
(1, −3)	(0.99, −2.98)	**−1.302 (worst)**	success
(−3, 3)	(−1.99, −0.99)	−2.101	success
(0, 0)	(−1.99, −0.99)	−2.101	success
(3, −1)	(−1.99, −0.99)	−2.101	success

(Verified.) The six runs reach *three different optima*, with objective values from −2.84 to −1.30 — and **every single one reports "success."** A researcher who ran the optimizer once, from the default start (0,0), would have found the −2.101 optimum and reported it, never knowing that a better one (−2.841) existed. **Taking one run is a coin flip on finding the global optimum.**

The rule. *Estimate from many dispersed starting values and keep the solution with the best (highest) log-likelihood.* Practically:

1. **Use many starts** — dozens for a complex mixed logit or MDCEV, drawn from a dispersed distribution over the parameter space (or informed by simpler-model estimates and pilot priors, Chapter 37).
2. **Compare the log-likelihoods** — if the best few agree, you have likely found the global optimum; if they scatter widely, the surface is treacherous and you need more starts and caution.
3. **Report the practice** — state how many starts were used and whether they agreed; a mixed-logit result from a *single* start is not credible (Chapter 46's reproducibility).
4. **Beware near-ties** — two optima with nearly identical log-likelihoods but *different parameters* signal weak identification (§52.5), not a clean global optimum.

For your models. The EC-MXL and MDCEV are exactly the non-concave, simulation-based likelihoods most prone to local optima. Your estimation protocol *must* include multiple starting values as standard, not as a robustness afterthought — and the number of starts should scale with the model's complexity and the number of random parameters (Chapter 38).

> **Stop and think 52.1.** Your mixed-logit estimation converges from its default starting values with a sensible-looking result. Why is this insufficient, and what must you do? *(Answer: it is insufficient because the mixed-logit likelihood is* not globally concave*, so the optimizer climbs the nearest hill to the default start and stops — reporting "success" whether or not that hill is the*

highest. Verified, the same model from six starts reached optima ranging from −2.84 (global) to −1.30 (a poor local), each declared a success*; a single run from the default would have found a mediocre local optimum and reported it as if it were the answer, with no signal that a better one existed. "Sensible-looking" estimates are no protection — a local optimum can look perfectly plausible. What I must do: (i)* re-estimate from many dispersed starting values* — dozens for a mixed logit, drawn from a dispersed distribution or informed by simpler-model/pilot estimates; (ii) *compare the log-likelihoods* and keep the *best* solution — if the top several starts agree, I have likely found the global optimum; if they scatter, the surface is treacherous and I need more starts and greater caution; (iii) *watch for near-ties in log-likelihood with different parameters*, which signal weak identification (§52.5) rather than a clean optimum; (iv) *report* how many starts were used and whether they agreed. A mixed-logit result from a single start is not credible, and the multiple-starts protocol is a standard requirement of the estimation, not an optional robustness check.)*

52.5 Non-identification masked by convergence: read the Hessian

The most insidious failure: a model that is *not identified* can converge perfectly and report plausible estimates. The Hessian is the only witness.

The problem. If two parameters are not *separately* identified — they enter the likelihood only through some combination (a sum, a ratio, a product) — then the likelihood is *flat* along a direction in parameter space (a ridge). The optimizer lands *somewhere* on the ridge, reports success, and returns *individual* parameter values that are *arbitrary* (any point on the ridge fits equally well). The point estimates look fine; they are meaningless.

The Hessian diagnostic. At a true, well-identified maximum, the Hessian of the log-likelihood is *negative definite* — all eigenvalues negative (equivalently, the Hessian of the *negative* log-likelihood is *positive definite*, all eigenvalues positive). Along a flat ridge, the Hessian has a *zero* (or near-zero) eigenvalue — the curvature vanishes in the unidentified direction. So a *near-singular Hessian* (an eigenvalue near zero, a huge condition number) is the signature of non- or weak identification, *regardless of what the convergence flag says.* And because standard errors come from the *inverse* Hessian (Chapter 13), a singular Hessian means the standard errors are undefined or explosive — another tell.

Worked Example 52.2 (Non-identification behind convergence, verified). A model where two parameters a, b enter only as their sum $a + b$ (true sum 1.2), plus a third parameter c with no effect:

Diagnostic	Value
Convergence flag	**success**
Estimates	$a = 0.568$, $b = 0.568$ ($a + b = 1.136 \approx$ true 1.2), $c = 0.500$
Hessian eigenvalues	**[0.0, 0.0, 1021.2]**
Condition number	**infinite**

(Verified.) The optimizer succeeded and returned plausible-looking individual estimates — but two Hessian eigenvalues are *exactly zero*, exposing two unidentified directions: a and b are not separately identified (only their sum is, which *is* recovered correctly), and c is entirely flat. **The convergence flag is a lie about identification; the Hessian tells the truth.** A researcher who reported $a = 0.57$ and $b = 0.57$ as findings would be reporting noise — the split between them is arbitrary.

The connection to Chapter 30. This is the *computational* face of the identification-versus-estimation distinction. Chapter 30 showed that some quantities are not identified *by any amount of data*; here that failure manifests as a *flat likelihood* and a *singular Hessian.* The optimizer cannot detect it — identification is a property of the *model and data*, not of the algorithm — so the analyst must check the Hessian. **For your**

models, this is not hypothetical: the α–γ ridge in MDCEV (Chapter 40), the scale-versus-preference confound (Chapter 25), and the near-collinearity of random-parameter distributions in a rich EC-MXL (Chapter 38) all produce near-singular Hessians that convergence conceals. Inspecting the Hessian's eigenvalues and condition number is therefore a *mandatory* step, not a diagnostic reserved for when something looks wrong.

> **Active recall 52.B.** Why can a non-identified model converge with plausible estimates, and how does the Hessian reveal the non-identification? *(Answer: when parameters are not* separately* identified — they enter the likelihood only through a combination (sum, ratio, product) — the likelihood is *flat along a direction* in parameter space (a ridge): every point on the ridge fits the data *equally well*. The optimizer, seeking a point it cannot improve from, lands *somewhere* on the ridge and stops, reporting "success" because it genuinely cannot improve the *objective* — it simply cannot move *along* the flat ridge to a better value because there isn't one. The returned *individual* estimates are arbitrary (any split consistent with the identified combination fits identically), yet they look plausible. The Hessian reveals this because it measures the *curvature* of the log-likelihood: at a well-identified maximum the Hessian is *negative definite* (all eigenvalues negative — curved down in every direction), but along a flat ridge the curvature *vanishes* in the unidentified direction, producing a *zero (or near-zero) eigenvalue* and hence a *huge/infinite condition number*. Verified: a model where a and b entered only as $a + b$ converged with plausible $a = b = 0.568$ but had two exactly-zero Hessian eigenvalues (infinite condition number) — exposing that only $a + b$ (correctly recovered as 1.136) is identified. Additionally, since standard errors come from the *inverse* Hessian, a singular Hessian gives undefined/explosive SEs, a second tell. The lesson: convergence is silent about identification (a property of model and data, not the algorithm), so the Hessian's eigenvalues and condition number must be inspected as a *mandatory* step — especially for the MDCEV α–γ ridge, the scale-preference confound, and rich EC-MXL specifications where near-singularity is expected.)*

52.6 Conditioning, scaling, and bounds

Some computational failures are purely mechanical — nothing wrong with the model, only with its numerical representation — and are fixed by simple preparation.

Ill-conditioning from bad scaling. If regressors have wildly different magnitudes (cost in euros, area in square metres, a dummy in {0,1}), the Hessian becomes *ill-conditioned* — a large condition number — which slows or breaks convergence and inflates numerical error in the standard errors (Chapter 1's conditioning). This is not a modelling problem; it is a units problem.

Worked Example 52.3 (Scaling wrecks conditioning, verified). The same model with a regressor in good units versus 100× too large (euros versus cents):

Regressor scaling	Hessian condition number
Well-scaled	1.3
Badly-scaled (×100)	1.3×10^4 **($\approx 10,000\times$ worse)**

(Verified.) A purely cosmetic change of units degraded the conditioning by four orders of magnitude — enough to cause convergence failures or unreliable standard errors, with the *estimates themselves* giving no hint of the problem. **The fix is trivial: rescale all regressors to comparable magnitudes** (standardize, or choose sensible units) *before* estimation. This is standard practice in choice modelling and should be automatic.

Parameter bounds and transformations. Some parameters have *natural constraints* — a variance must be positive, a probability in [0,1], a log-normal scale positive. Estimating them unconstrained can send the optimizer into invalid regions (a negative variance) and break the likelihood. The standard fixes: *reparameterize* (estimate $\ln \sigma$ freely, so $\sigma = e^{\ln \sigma} > 0$ always; estimate a logit-transformed probability), or impose

bounds (a constrained optimizer). Reparameterization is usually cleaner — it turns a constrained problem into an unconstrained one and improves conditioning. But it changes the standard-error interpretation (an SE on $\ln \sigma$, not σ — the delta method, Chapter 13, converts back).

The general lesson. A large class of "the model won't converge" problems are not modelling failures but *numerical hygiene* failures — bad scaling, unbounded constrained parameters, poor starting values — fixed by rescaling, reparameterizing, and better starts. Diagnosing whether a convergence problem is *numerical* (fixable by hygiene) or *structural* (non-identification, misspecification) is the first triage: rescale and reparameterize first; if the problem persists, suspect the model.

52.7 Simulation noise, seeds, and draws

For the *simulated* likelihoods of your models (SML mixed logit, MDCEV welfare), a further computational fragility: the objective is *estimated*, so it is *noisy*, and the noise interacts with the optimizer.

The problem (recapping Chapter 38 at the computational level). A simulated likelihood computed with R draws is a *noisy* approximation of the true likelihood. The noise makes the objective surface *jagged*, so the optimizer can stop at a *noise-induced* spurious optimum, and the result *changes with the random seed*. Convergence on a jagged surface is meaningless — it reflects the draws, not the parameters.

The diagnostics and fixes.

- **Seed sensitivity.** Re-estimate with different seeds; if the estimates move materially, the draws are too few (Chapter 38 verified this: at low draws, results swing with the seed).
- **Draw-count sensitivity.** Re-estimate with increasing R; the estimates should *stabilize*. If they are still moving at your chosen R, increase it.
- **Quasi-random draws.** Use scrambled Halton or Sobol sequences, which cut the simulation noise substantially for the same R (Chapter 38 verified 3–10× reduction), smoothing the surface.
- **Report R, the draw type, and the seed** — they are part of the specification, and the result is not reproducible without them (Chapter 46).

The interaction with the other diagnostics. Simulation noise *compounds* the local-optima and Hessian problems: a jagged surface has *more* spurious local optima (worsening §52.4), and a noisy Hessian is harder to evaluate reliably (worsening §52.5). So the simulation diagnostics are not separate from the others — adequate draws are a *precondition* for the multiple-starts and Hessian checks to be meaningful. Estimate with enough good draws *first*, then run the identification and optimum diagnostics on the smoothed surface.

52.8 The trustworthy-result checklist

Assembling the diagnostics into the discipline that must precede *reporting* any estimate.

Worked Example 52.4 (A result that passes, verified). A well-identified logit at its optimum:

Check	Result	Verdict
Gradient norm	6.0×10^{-4}	≈ 0 — true stationary point ✓
Hessian eigenvalues (of NLL)	[503.6, 803.2, 823.9]	all positive — true minimum ✓
Condition number	1.6	small — well-identified ✓
Estimates vs truth	(0.365, 0.898, −0.625) vs (0.3, 0.9, −0.6)	close — unbiased ✓

(Verified.) This is what a *trustworthy* result looks like: the gradient vanishes (a genuine stationary point), the Hessian is positive-definite (of the negative log-likelihood — a true minimum, so valid standard errors)

and well-conditioned (identified), and the estimates recover the truth. Contrast every earlier example in this chapter, each of which *failed* one of these while reporting "success."

The checklist (run before reporting any estimate).

1. **Convergence flag** — necessary but not sufficient; a non-converged run is discarded.
2. **Multiple starts agree** — the best log-likelihood is reached from several dispersed starts (§52.4).
3. **Gradient near zero** — a genuine stationary point, not a premature stop.
4. **Hessian negative-definite (of log-lik) and well-conditioned** — a true, identified maximum; eigenvalues checked, condition number reasonable (§52.5–§52.6).
5. **Standard errors finite and sensible** — from a non-singular Hessian; no explosive or undefined SEs.
6. **Seed and draw stability** — for simulated models, estimates stable across seeds and increasing R (§52.7).
7. **Parameters in valid ranges** — variances positive, probabilities in [0,1], signs theory-consistent.
8. **Estimates plausible** — magnitudes and signs make economic sense (a WTP of €4,000 or a wrong-signed cost coefficient signals a problem).

Only a result passing *all* of these is trustworthy. **Reporting an estimate that has passed only check 1 — the convergence flag — is the most common computational failure in applied work, and this book's final warning against it.**

> **Stop and think 52.2.** Your EC-MXL converges, the estimates look economically sensible, and the standard errors are reasonable. What remains to be checked before you trust and report it? *(Answer: several things, because "converged + sensible estimates + reasonable SEs" passes only three of the eight checks and can coexist with a wrong result. What remains: (i)* multiple starting values* — the mixed-logit likelihood is non-concave, so I must confirm the best log-likelihood is reached from *several dispersed starts*, not just the default; sensible-looking estimates can be a local optimum (verified: six starts reached optima from −2.84 to −1.30, all "successful"). (ii) *Hessian eigenvalues and condition number* — reasonable-looking SEs are *not* proof of identification; I must confirm the Hessian is negative-definite (of the log-likelihood) and *well-conditioned*, because a near-singular Hessian (from the α–γ ridge, scale-preference confound, or collinear random parameters) signals non-identification that convergence hides (verified: a non-identified model converged with plausible estimates but had zero Hessian eigenvalues). (iii) *Gradient norm* near zero — confirming a genuine stationary point, not a premature stop. (iv) *Seed and draw-count stability* — because the EC-MXL likelihood is *simulated*, I must confirm the estimates are stable across random seeds and do not move as R increases (with scrambled Halton draws); otherwise the "convergence" is on a noise-induced spurious optimum (Chapter 38). (v) *Valid ranges* — variances positive, distributions sensible. Only after *all* of these pass — the full checklist, not the convergence flag plus a plausibility glance — is the result trustworthy and reportable. The economic sensibility of the estimates is reassuring but not decisive: a local optimum or a non-identified split can look perfectly sensible while being wrong.)*

52.9 Failure modes

F1. Trusting the convergence flag. Reporting on "success" alone. *Fix:* the full checklist (§52.3, §52.8).

F2. Single starting value. Reporting a possible local optimum. *Fix:* many dispersed starts; keep the best log-likelihood (§52.4).

F3. Ignoring the Hessian. Non-identification hidden behind convergence. *Fix:* check eigenvalues and condition number (§52.5).

F4. Reporting individual estimates on a ridge. Arbitrary values from a flat direction. *Fix:* detect near-singular Hessian; report only identified combinations or fix the model (§52.5).

F5. Bad variable scaling. Ill-conditioning breaking convergence/SEs. *Fix:* rescale regressors to comparable magnitudes (§52.6).

F6. Unbounded constrained parameters. Optimizer entering invalid regions. *Fix:* reparameterize ($\ln \sigma$, logit) or bound (§52.6).

F7. Too few draws. Noisy objective, seed-dependent spurious optima. *Fix:* increase R; scrambled Halton; verify stability (§52.7).

F8. Unreported computational settings. Non-reproducible results. *Fix:* report starts, seeds, R, draw type, optimizer, environment (§52.2; Chapter 46).

F9. Stale results. Reported numbers not matching current code. *Fix:* dependency-graph pipeline; literate output (§52.2).

F10. Confusing numerical and structural failure. Fixing a misspecification with more iterations, or a scaling problem by changing the model. *Fix:* triage — rescale/reparameterize first, then suspect the model (§52.6).

52.10 Diagnostic questions for your own papers

1. **Checklist.** Has every reported estimate passed the full trustworthiness checklist, not just convergence? *(§52.8, F1.)*
2. **Starts.** How many starting values, and did the best agree? *(§52.4, F2.)*
3. **Hessian.** Are the eigenvalues and condition number reported/checked; are SEs finite? *(§52.5, F3–F4.)*
4. **Scaling.** Are regressors sensibly scaled and constrained parameters reparameterized? *(§52.6, F5–F6.)*
5. **Simulation.** Are estimates stable across seeds and draw counts, with scrambled draws? *(§52.7, F7.)*
6. **Reporting.** Are all computational settings (starts, seeds, R, draws, optimizer, environment) reported? *(§52.2, F8.)*
7. **Pipeline.** Is the analysis a reproducible dependency graph with literate output? *(§52.2, F9.)*
8. **Triage.** Have I distinguished numerical from structural convergence problems? *(§52.6, F10.)*

52.11 Exercises

(A) Comprehension.

A1. Explain why "convergence: success" does not establish that a result is correct.

A2. How does the Hessian reveal non-identification that convergence conceals?

A3. Why must simulated-likelihood estimates be checked for seed and draw stability?

(B) Derivation / reasoning.

B1. Explain why a non-concave likelihood requires multiple starting values.

B2. Explain the link between a zero Hessian eigenvalue, a flat likelihood direction, and non-identification.

B3. Explain why bad variable scaling degrades the Hessian's condition number and how rescaling fixes it.

(C) Research application / critique.

C1. Design the estimation-and-diagnostics protocol for your EC-MXL and MDCEV.

C2. Your co-author reports a mixed-logit result from one converged run. Critique and prescribe.

C3. Your MDCEV won't converge. Write the triage from numerical hygiene to structural diagnosis.

52.12 Solutions

A1. "Convergence: success" means only that the optimizer reached a point it cannot easily improve from — its *internal* stopping criteria (small gradient, tiny step, negligible objective change) were met. It does *not* establish: (i) that the point is the *global* optimum rather than a *local* one (verified: six starts, six "successes," objectives from -2.84 to -1.30); (ii) that the model is *identified* — the point may lie on a flat ridge where individual estimates are arbitrary (verified: a non-identified model converged with zero Hessian eigenvalues); (iii) that the *standard errors* are valid — the Hessian may be singular; (iv) that the *model* is correctly specified — a wrong model converges fine; or (v) for simulated likelihoods, that the *draws* were adequate — a noisy objective can have spurious optima. Convergence is *necessary* (a non-converged run is untrustworthy) but far from *sufficient*; trust requires the full checklist of §52.8.

A2. The Hessian is the matrix of second derivatives of the log-likelihood — it measures the *curvature* of the objective at the optimum. At a well-identified maximum, the log-likelihood curves *downward in every direction*, so the Hessian is *negative definite* (all eigenvalues negative; equivalently the negative-log-likelihood Hessian is positive definite, all eigenvalues positive). Non-identification means the likelihood is *flat along some direction* (a ridge) — the parameters can move along it with no change in fit — so the curvature *vanishes* in that direction, producing a *zero (or near-zero) eigenvalue* and hence an enormous or infinite *condition number*. Verified: a model where a, b entered only as $a + b$ had Hessian eigenvalues [0.0, 0.0, 1021.2] — two exactly-zero eigenvalues exposing two unidentified directions, while the optimizer reported success. Because standard errors are the square roots of the diagonal of the *inverse* Hessian, a singular Hessian also produces undefined or explosive SEs — a second symptom. So inspecting the Hessian's eigenvalues and condition number detects the non-identification that the convergence flag (blind to the shape of the surface) cannot.

A3. A simulated likelihood is computed from a *finite number of random draws* R, so it is a *noisy estimate* of the true likelihood, and the noise depends on the particular draws — hence on the *seed*. Two consequences require checking: (i) the noisy objective surface is *jagged*, so the optimizer can stop at a *spurious, noise-induced* optimum that is an artefact of the draws rather than a feature of the parameters; (ii) the resulting estimates *change with the seed*, so a result from one seed is not reproducible or reliable if the draws are too few. Verified in Chapter 38, at low draw counts the estimates swing materially across seeds and stabilize only as R grows. Therefore one must (a) re-estimate across *seeds* — material movement means too few draws; (b) re-estimate across increasing R — the estimates should converge to a stable value; (c) use *scrambled Halton/Sobol* draws to cut the noise for a given R; and (d) report R, the draw type, and the seed. Crucially, adequate draws are a *precondition* for the other diagnostics — a jagged surface has more spurious local optima and an unreliable Hessian — so the simulation must be smoothed *first*, then the optimum and identification checks run.

B1. A *concave* likelihood has a *single* maximum, so any local optimum is the global one and a single optimization from any start finds it. A *non-concave* likelihood (mixed logit, MDCEV, and most latent-variable/simulation-based models) has *multiple* local maxima — several "hills" of different heights. A gradient-based optimizer climbs the hill *nearest its starting point* and stops at that summit, unable to see whether a higher hill exists elsewhere. So the optimum reached *depends on the starting value*, and a single run finds only the local optimum in its own basin of attraction — which may be far from the global one (verified: from six starts, objectives ranged from -2.84 to -1.30, each a "successful" convergence). To find the *global* optimum with high probability, one must launch the optimizer from *many dispersed starting values* — sampling different basins of attraction — and keep the solution with the *best log-likelihood*. If the top several starts agree, the global optimum is likely found; if they scatter, more starts and caution are needed. There is no algorithmic guarantee of finding the global optimum of a non-concave function, so multiple starts are a *necessity*, not a robustness luxury, for these models.

B2. Consider the log-likelihood $\ell(\theta)$ near a point $\hat{\theta}$. A second-order Taylor expansion gives $\ell(\hat{\theta} + d) \approx \ell(\hat{\theta}) + \nabla\ell' d + \frac{1}{2} d' H d$, where H is the Hessian. At an optimum $\nabla\ell = 0$, so the *change* in the log-likelihood for a small move d is $\frac{1}{2} d' H d$. If H has an eigenvalue $\lambda_k = 0$ with eigenvector v_k, then moving in the direction

$d = v_k$ gives $\frac{1}{2}d'Hd = \frac{1}{2}\lambda_k\|v_k\|^2 = 0$ — *the log-likelihood does not change* along v_k. That is a *flat direction*: the data are equally consistent with all parameter values along v_k, which is exactly *non-identification* — the parameters cannot be distinguished by the likelihood. Verified: a model with two parameters entering only as their sum had two zero eigenvalues (two flat directions), and the individual estimates along those directions were arbitrary while the identified combination $(a+b)$ was recovered. Equivalently: identification requires the likelihood to have a *unique* peak (strictly negative-definite H, curving down in *every* direction); a zero eigenvalue makes the peak a *ridge*, and the optimizer's landing point on that ridge is determined by the starting value and numerical noise, not the data. Hence a zero (or near-zero) Hessian eigenvalue $\iff$ a flat likelihood direction $\iff$ non-identification.

B3. The Hessian's *condition number* is the ratio of its largest to smallest eigenvalue, and it governs how well the optimization and the standard-error computation behave numerically. When regressors have *very different magnitudes* (cost in cents vs a 0/1 dummy), the curvature of the log-likelihood with respect to the large-magnitude parameter is *vastly* different from that with respect to the small-magnitude one — because the same coefficient change produces a much larger utility change for the large regressor. This makes the Hessian's eigenvalues span many orders of magnitude, so the condition number is huge. Verified: rescaling a regressor by 100× worsened the condition number from 1.3 to 1.3×10^4 — ~10,000× worse. Consequences: gradient-based optimizers take tiny steps in the steep directions and struggle in the flat ones (slow or failed convergence), and inverting a badly-conditioned Hessian to get standard errors amplifies numerical error (unreliable SEs). The *fix* is to *rescale the regressors to comparable magnitudes* (standardize to unit variance, or choose sensible units — euros not cents) *before* estimation: this equalizes the curvatures, brings the eigenvalues to similar magnitudes, and drives the condition number toward its well-behaved minimum (verified 1.3). It is a purely mechanical fix to a purely mechanical problem — the model and the estimates are unaffected in principle, only the numerics improve — and it should be automatic.

C1. *Estimation-and-diagnostics protocol for EC-MXL and MDCEV.* **Numerical hygiene first:** rescale all attributes and cost to comparable magnitudes (verified: bad scaling worsens conditioning ~10,000×); reparameterize constrained parameters — estimate $\ln\sigma$ for log-normal scales/standard deviations so they stay positive, logit-transform any bounded parameters — improving conditioning and preventing invalid regions. **Simulation adequacy (precondition):** use *scrambled Halton/Sobol* draws (Chapter 38), and set R by *stability testing* — re-estimate at increasing R and across seeds until the estimates are stable to reporting precision, since a noisy surface undermines every other diagnostic. **Multiple starting values:** estimate from *many dispersed starts* (dozens, informed by simpler-model and pilot estimates), compare the log-likelihoods, and keep the *best* — a single start is not credible for these non-concave likelihoods (verified: six starts, six different "successful" optima); flag near-ties in log-likelihood with different parameters as weak identification. **Hessian/identification checks:** at the best optimum, compute the Hessian, inspect its *eigenvalues* and *condition number* — a near-zero eigenvalue signals non-identification (verified: the MDCEV α–γ ridge and rich EC-MXL specifications are exactly where this bites), in which case fix a profile (Chapter 40) or simplify the random-parameter structure (Chapter 38) rather than reporting arbitrary ridge estimates; confirm standard errors are finite and sensible. **Gradient check:** confirm the gradient norm is ≈ 0 (a genuine stationary point). **Plausibility and ranges:** confirm signs, magnitudes, and valid ranges (positive variances, sensible WTP). **The checklist gate:** report an estimate *only* after it passes *all* of §52.8, not the convergence flag alone. **Reproducibility:** report the number of starts, the seed, R, the draw type, the optimizer, and the environment; run the whole thing as a `targets` dependency graph with `renv`-locked packages and Quarto literate output (§52.2), so every number is regenerable and auditable.

C2. *Critique and prescription for a mixed-logit result from one converged run.* The result is *not credible as reported*, not because it is necessarily wrong but because a single converged run provides *no evidence* it is right. Three specific problems. (i) *Local optimum risk*: the mixed-logit likelihood is non-concave, so one run finds only the optimum in its starting value's basin of attraction — verified, the same model from six starts reached optima from -2.84 to -1.30, all "successful," so a single run is a coin flip on the global optimum, and "sensible-looking estimates" offer no protection (a local optimum can look plausible). (ii) *Unchecked identification*: without inspecting the Hessian, there is no evidence the parameters are identified rather than sitting on a flat ridge (the scale-preference confound and collinear random parameters produce near-singular Hessians that convergence hides — verified). (iii) *Unchecked simulation adequacy*: without

seed/draw-stability evidence, the "convergence" may be on a noise-induced spurious optimum. *Prescription*: re-estimate from *many dispersed starting values* and keep the best log-likelihood, reporting whether the top starts agree; *compute and inspect the Hessian* (eigenvalues, condition number, finite SEs) to confirm identification; *verify seed and draw-count stability* with scrambled Halton draws at adequate R; *confirm the gradient vanishes and the parameters are in valid ranges*; and *report* the number of starts, seed, R, draw type, and optimizer. Only a result passing this full checklist (§52.8) is reportable — a single converged run is the starting point of the diagnostics, not the end.

C3. *Triage for an MDCEV that won't converge — numerical hygiene to structural diagnosis.* Work from the cheapest, most common causes to the deepest. **(1) Scaling (numerical):** check regressor magnitudes; if attributes/cost differ wildly in scale, *rescale* to comparable magnitudes — verified, bad scaling worsens conditioning ~10,000× and routinely breaks convergence; this fixes a large share of "won't converge" problems at zero modelling cost. **(2) Constrained parameters (numerical):** confirm positivity-constrained parameters (satiation γ, scale σ) are *reparameterized* (estimate their logs) so the optimizer cannot wander into invalid regions where the likelihood is undefined. **(3) Starting values (numerical):** poor starts on a non-concave surface cause failures; try *many dispersed starts*, and warm-start from a simpler model (an MNL or a fixed-parameter MDCEV) whose estimates seed the richer one. **(4) Simulation noise (numerical, if mixed MDCEV):** if the likelihood is simulated, too few draws make the surface jagged and convergence erratic — increase R and use scrambled Halton, checking seed stability. **(5) If it still fails, suspect structure:** compute the Hessian at the best point reached and inspect its eigenvalues/condition number. A *near-singular Hessian* points to *non-identification* — most likely the α–γ **ridge** (Chapter 40: satiation captured redundantly by both parameters, verified to be only weakly separately identified), fixed by *imposing a profile* ($\alpha = 0$ or $\gamma = 1$); or an over-parameterized random-coefficient structure (Chapter 38), fixed by *reducing the heterogeneity* to what the data identify. Non-convergence here is the *symptom* of a model asking the data for more than they contain. **(6) Misspecification:** if the model is structurally identified and numerically clean but still fails or gives implausible estimates, reconsider the specification (the outside good, the attribute functional forms, the profile choice). *The triage principle*: rescale and reparameterize *first* (numerical hygiene fixes most cases cheaply); only after exhausting hygiene should you diagnose *structural* problems (non-identification, over-parameterization, misspecification), because treating a scaling problem as a modelling problem — or a modelling problem with more iterations — wastes effort and can entrench an untrustworthy result (§52.6, F10).

52.13 Spaced review prompts

+1 day. From memory: why convergence is not validity; the multiple-starts rule; how the Hessian reveals non-identification; the trustworthiness checklist.

+1 week. Explain why non-concave likelihoods need many starts; the link between a zero Hessian eigenvalue and non-identification; why scaling degrades conditioning; why simulated likelihoods need seed/draw checks.

+1 month. For your EC-MXL and MDCEV, run the full protocol — hygiene, draws, multiple starts, Hessian, gradient, ranges — and gate every reported estimate on the checklist. Close with the spine sentence about whether the software's "success" has been *earned* by the diagnostics or merely *reported* by the optimizer.

52.14 How this chapter connects to the rest of the book — and the book's conclusion

- **Chapter 1** supplies the eigenvalues, definiteness, and conditioning that the Hessian diagnostics use; **Chapters 12–13** supply the maximum-likelihood, score, and information-matrix framework whose

computation this chapter scrutinizes; **Chapter 30** supplies the identification-versus-estimation distinction that reappears here as the flat-likelihood/singular-Hessian problem; **Chapter 38** supplies the simulation and seed/draw analysis; **Chapter 46** supplies the reproducibility discipline that the pipeline half realizes.

- This chapter is the *last link* in the fragility chain of **Chapter 50**: even a correctly-specified, well-identified, honestly-designed study produces a wrong number if the computation fails silently — and the checklist is how that final fragility is controlled.

The book's conclusion. This chapter ends the volume, and the volume has been a single sustained argument. Every conclusion in empirical economics — every WTP, every treatment effect, every net present value — is a claim that survives *only if* a long chain of conditions holds: the mathematics is right (Parts I–III), the estimator is sound and the inference honest (Parts IV–VII), the welfare object is meaningful and the value validly elicited (Parts VIII–X), the appraisal aggregates and discounts defensibly (Part XI), the evidence is synthesized and transferred without over-reach (Part XII), the survey measured what it claimed (Chapter 51), and — this chapter — the computation that produced the number can be trusted. The spine question — *what am I identifying, under which assumptions, and how fragile is the result?* — is the thread through all of it, and the fragility audit of Chapter 50 is the method that answers it for any conclusion.

The discipline the book asks of you is not pessimism. It is *honesty*: to know what your number assumes, to measure how fragile it is, to locate its weakest link, to report it as a conditional and a range rather than a false point of certainty, and to earn — through the diagnostics of this final chapter — the right to believe the output before you ask anyone else to. A number reported with that honesty is worth more than a hundred reported with false confidence, because it tells the decision-maker the one thing they most need to know: *how much to trust it*. That is the whole book, and it is where your research — the finfish valuations, the fish-restoration appraisal, the blue-finance decision they inform — should stand: on conclusions whose fragility is known, disclosed, and, wherever it matters most, reduced.

52.15 Research translation task

For your **entire computational workflow**, produce a diagnostics-and-pipeline protocol:

1. **Pipeline.** The `targets` dependency graph, `renv` lockfile, Git repository, and Quarto literate report — the reproducible infrastructure (Chapter 46).
2. **Numerical hygiene.** Regressor rescaling and constrained-parameter reparameterization as automatic pre-estimation steps.
3. **Simulation.** Scrambled Halton/Sobol draws with R set by seed- and draw-stability testing.
4. **Optimum.** Multiple dispersed starting values, best log-likelihood retained, agreement reported.
5. **Identification.** Hessian eigenvalues and condition number checked for every estimate; near-singularity triggering profile restriction or model simplification.
6. **The checklist gate.** Every reported estimate passes the full §52.8 checklist, not the convergence flag alone.
7. **Close** with the spine sentence: *"Every reported estimate passes the trustworthiness checklist — best of [·] starts (agreeing), gradient ≈ 0, Hessian negative-definite and well-conditioned (condition number [·]), stable across seeds and [·] draws — computed in a `targets` pipeline with `renv`-locked packages and Quarto output, so the software's 'convergence' has been earned by the diagnostics, not merely reported."*

This protocol governs the computation behind every number in your research program — the final fragility, controlled.

52.16 Bibliography

Core. - Train, *Discrete Choice Methods with Simulation* — simulation, draws, and the practical estimation of the models whose diagnostics this chapter covers (§52.4–§52.8). *(Verify edition/chapters.)* - Hess & Palma (2019), "Apollo: A Flexible, Powerful and Customisable Freeware Package for Choice Model Estimation," *Journal of Choice Modelling* — the estimation package and its diagnostics (§52.2). *(Verify volume/pages.)* - Nocedal & Wright, *Numerical Optimization* — the definitive treatment of optimization, convergence, conditioning, and the Hessian (§52.3–§52.6). *(Verify edition.)*

Applied. - Landau (2021), "The targets R Package," *Journal of Open Source Software* — pipeline orchestration (§52.2). *(Verify volume/pages.)* - Ushey & Wickham, `renv` documentation — reproducible environments (§52.2). *(Verify.)* - Viechtbauer (2010), "Conducting Meta-Analyses in R with the metafor Package," *Journal of Statistical Software* (§52.2). *(Verify volume/pages.)*

Advanced. - Chiou & Walker (2007), "Masking Identification of Discrete Choice Models Under Simulation Methods," *Journal of Econometrics* — how simulation conceals non-identification, the core of §52.5. *(Verify volume/pages.)* - Gill & King (2004), "What to Do When Your Hessian is Not Invertible," *Sociological Methods & Research* — the singular-Hessian problem (§52.5). *(Verify volume/pages.)* - Knittel & Metaxoglou (2014), "Estimation of Random-Coefficient Demand Models: Two Empiricists' Perspective," *Review of Economics and Statistics* — how optimizer and starting-value choices change published estimates, the definitive cautionary study (§52.4). *(Verify volume/pages.)*

Entries marked "verify" are pointers; confirm editions, pages, and the Hess–Palma / Chiou–Walker / Gill–King / Knittel–Metaxoglou citations before using them, per the book's no-unverified-citation rule.

End of Chapter 52, and of the book. The fifty-two chapters have traced a single question — what am I identifying, under which assumptions, and how fragile is the result? — from the linear algebra of the first chapter to the convergence diagnostics of the last, and assembled it into a method: enumerate the assumptions, propagate the fragility, locate the weakest link, earn the right to trust the computation, and report the conclusion as honestly conditional. Applied to your research program — the contingent valuation, the discrete choice experiments, the MDCEV, the meta-analysis, and the fish-restoration cost–benefit analysis that draws them together — it is the difference between a number and a number you can defend. Tell me if you would like the completed volume assembled: the files stitched, the master table of contents built, cross-references checked, and the whole rendered through your pandoc/RMarkdown pipeline to a single PDF.

Part XIV Assessment — Computational Diagnostics and Reproducible Research

Covers Chapters 52. Five components, in the order they are best attempted. Allow a full working day; the mock oral is the part that matters most for your viva.

1. Cumulative problem set

P1. For one of your models, report the gradient norm, the Hessian eigenvalues and the condition number at the reported optimum.

P2. Re-estimate from 20 dispersed starting values; report the distribution of log-likelihoods and whether the best is unique.

P3. Vary the number of Halton draws over $\{100, 250, 500, 1000, 2000\}$ and plot the estimated WTP against draws.

P4. Vary the seed at your chosen number of draws and report the spread of the headline estimate.

P5. Rescale your cost variable by 10 and 100 and report what changes in the estimates, the standard errors and the convergence behaviour.

2. Research critique

Take the computational appendix of a paper you have written or read. Assess whether a competent stranger could regenerate every number in it from what is provided. List precisely what is missing.

3. Red-flag diagnosis

A referee is told that a headline result "replicates exactly — we reran the code and got identical numbers." Explain why this is weaker evidence than it sounds, and state what three additional checks would actually establish computational trustworthiness.

4. Mock oral examination

1. What is cosmetic convergence and how would you detect it?
2. How many draws is enough, and how do you know?
3. Your Hessian is not invertible. What now?
4. What does a seed-dependent conclusion mean substantively?
5. Why is "the code runs" not the same as "the result is trustworthy"?

5. Integration mini-project

Rebuild your entire analysis as a `targets` pipeline with `renv`-locked dependencies, Git version control and a Quarto report, such that every table and figure in your dissertation regenerates on one command — and every estimate in it passes the §52.8 trustworthiness checklist before it is allowed into the report.

Appendix A — Notation and Symbols

This book draws on linear algebra, probability, microeconomic theory, discrete-choice econometrics, meta-analysis and cost–benefit appraisal, and those literatures do not share a notation. Several symbols are therefore genuinely overloaded. Where that happens the table says so explicitly, because the collisions are the places a careless reading goes wrong.

Conventions

Scalars are lowercase italic (x, β). Vectors are lowercase bold ($\mathbf{x}$, $\boldsymbol{\beta}$) and are **column** vectors unless stated otherwise. Matrices are uppercase bold ($\mathbf{X}$, $\mathbf{V}$). The transpose is written with a prime, $\mathbf{X}'$ (some literatures write $\mathbf{X}^\top$; the prime is the dominant econometric convention and is used throughout). Estimators carry a hat: $\hat{\beta}$. Expectation is $\mathbb{E}[\cdot]$, variance $\mathrm{Var}(\cdot)$, covariance $\mathrm{Cov}(\cdot,\cdot)$. The identity matrix is $\mathbf{I}$, or $\mathbf{I}_k$ when the size matters.

Symbols in order of first substantial use

Symbol	Meaning	Where
$\mathbf{X}$	data / design matrix, $n \times k$	Ch 1
$\mathbf{X}'\mathbf{X}$	cross-product (second-moment) matrix	Ch 1, 16
$\mathbf{V}$	variance–covariance matrix of the estimates	Ch 1, 17
$\mathbf{L}$	lower-triangular Cholesky factor, $\Sigma = \mathbf{L}\mathbf{L}'$	Ch 1, 9, 22
$\mathbf{H}$	Hessian of the log-likelihood	Ch 1, 2, 11
$\mathcal{J}(\theta)$	Fisher (expected) information; $-\mathbb{E}[\mathbf{H}]$	Ch 1, 11
$\mathbf{s}(\theta)$	score, $\nabla \ell(\theta)$	Ch 2, 11
$\ell(\theta)$	log-likelihood	Ch 11
Φ, ϕ	standard normal CDF and density	Ch 5
Λ	logistic CDF, $1/(1+e^{-z})$	Ch 5, 18, 21
$S(\cdot)$	survival function, $1 - F(\cdot)$	Ch 5, 14, 15
V_j	deterministic (systematic) utility of alternative j	Ch 20–25
ε_j	random utility component (Gumbel in logit)	Ch 20–25
ω_k	willingness to pay for attribute k (WTP-space parameter)	Ch 24, 38
ψ_k	MDCEV baseline marginal utility of good k	Ch 40–42

Symbol	Meaning	Where
γ_k	MDCEV translation / satiation parameter	Ch 40–42
$e(\mathbf{p}, u)$	expenditure function	Ch 3, 34
$V(\mathbf{p}, m)$	indirect utility function	Ch 3, 34
$h(\mathbf{p}, u)$	Hicksian (compensated) demand	Ch 34
$x(\mathbf{p}, m)$	Marshallian (ordinary) demand	Ch 34
CV, EV	compensating and equivalent variation	Ch 3, 34
r	discount rate	Ch 4, 43–45
$\delta,\ \eta,\ g$	Ramsey parameters: pure time preference, elasticity of marginal utility, growth	Ch 44
τ^2	between-study variance in meta-analysis	Ch 47–49
$I^2,\ Q$	heterogeneity share; Cochran's statistic	Ch 47
w_i	meta-analytic weight	Ch 47
Δ	equivalence margin (TOST)	Ch 8, 33, 39
d	standardized effect size	Ch 8

Deliberately overloaded symbols — read the context

These are the collisions. Each is standard in its own literature, and renaming them would make the book harder to read against the sources, so the book keeps them and flags them here.

Symbol	Meaning 1	Meaning 2	How to tell
λ	**Lagrange multiplier** on the budget constraint; the marginal utility of income (Ch 3, 34, 40) also: eigenvalues $\lambda_{\max}, \lambda_{\min}$ (Ch 1, 52)	one of **your two behavioural parameters** in the κ-vs-λ separation problem taken from your study plan (Ch 29, 30, 32)	Meaning 1 appears inside optimization and MDCEV; meaning 2 only in the identification discussion of the cost-vector experiment, always paired with κ subscripted max/min and always inside a matrix discussion
κ	**condition number** of a matrix, $\lambda_{\max}/\lambda_{\min}$ (Ch 1, 52) also: **Cohen's** κ, inter-coder agreement (Ch 47)	the **other behavioural parameter** in the κ-vs-λ separation (Ch 29, 30, 32)	written $\kappa(\mathbf{A})$ with a matrix argument in meaning 1; bare and paired with λ in meaning 2 only in the coding-protocol discussion

Symbol	Meaning 1	Meaning 2	How to tell
α	**marginal utility of income** / cost coefficient in the discrete-choice welfare formulas (Ch 34, 38) also: **test size**, $\alpha = 0.05$ (Ch 7, 8, 33)	**MDCEV satiation exponent** α_k (Ch 40–42)	subscripted by good (α_k) in MDCEV; unsubscripted in the log-sum and WTP formulas appears with $z_{1-\alpha/2}$ or "at the 5% level"
μ	**scale parameter** of the logit, $\mu = 1/\sigma_\varepsilon$ (Ch 5, 21, 25)	**mean** of a distribution, e.g. mean WTP (Ch 5, 15)	in Part V μ multiplying utility is always scale; in Ch 15 μ is the WTP location parameter
σ	standard deviation / error scale	MDCEV likelihood scale (Ch 40)	context
β	preference parameter (utility coefficient)	Type II error rate, so power is $1 - \beta$ (Ch 8)	the Type II usage appears only in power formulas

The practical rule. When you meet one of these in your own writing, subscript or rename it. The book keeps the collisions because they are what the literatures use; a dissertation chapter does not have to.

Appendix B — Editorial Changelog

This volume was assembled iteratively, and the chapters were written before the final chapter order was fixed. A full editorial pass was subsequently run over the assembled manuscript. This appendix records what it changed, so that any reader comparing against an earlier draft can see exactly what moved and why.

1. Architecture reconciled with the manuscript

Volume 0 originally described a planned structure in which Parts VIII–XIV differed substantially from the chapters that were ultimately written — a consumer-theory chapter separate from a welfare-measures chapter, standalone chapters on public-good provision and environmental economics, a separate efficient-design chapter at 49, survey design at 50, and the reproducible pipeline and diagnostics split across 51 and 52. The manuscript as written consolidates these differently: welfare foundations in Chapters 34–35, applied stated preference in 36–39, MDCEV in 40–42, cost–benefit appraisal in 43–46, evidence synthesis in 47–49, the fragility audit and survey design in 50–51, and diagnostics with the pipeline in 52.

The following parts of Volume 0 were rewritten to describe the book that exists: the complete table of contents (§4), the chapter-by-chapter learning objectives for Chapters 34–52 (§5), the dependency map (§6), the study schedule for Blocks 5 and 6 and the "continuous" reading track (§7), and the chapter tags on the high-risk-mistakes list (§10).

2. Cross-references audited

Every cross-reference in the manuscript was extracted and checked against the final chapter order. References written against the earlier numbering were corrected. The main systematic corrections were:

- design and efficiency references pointing to "Chapter 49" now point to **Chapter 37**;
- survey- and questionnaire-design references pointing to "Chapter 50" now point to **Chapter 51**;
- publication-bias, small-sample-correction and dependence references pointing to "Chapter 48" now point to **Chapter 47 §47.6–§47.7**;
- welfare-measure references (CV, EV, WTP–WTA, Willig) pointing to "Chapter 35" now point to **Chapter 34**;
- public-good-provision and free-riding references pointing to "Chapter 37" now point to **Chapter 35**;
- MDCEV likelihood and identification references pointing to "Chapter 41" now point to **Chapter 40**, and exact-CV references pointing to "Chapter 42" now point to **Chapter 41**;
- aggregation-and-benefit-transfer pairs written as "Chapters 36, 44" now read **Chapters 35, 49**;
- prerequisite blocks in Chapters 34–52 citing Chapter 6 for discounting now cite **Chapter 4**; citing Chapters 22–23 for random utility and WTP space now cite **Chapters 20–21 and 24**; citing Chapter 11 or 13 for Monte Carlo and Krinsky–Robb now cite **Chapter 9**; citing Chapter 12 for the likelihood framework now cite **Chapter 11**; citing Chapter 21 for panel and multilevel structure now cite **Chapter 19**; citing Chapter 9 for power and equivalence now cite **Chapter 8**.

3. Numerical corrections

Chapter 1, Worked Example 1.5. The example previously reported a mean simulated WTP of about 4.38 alongside the median of 3.92. Because the cost coefficient is drawn $N(-2, 1)$ and therefore has strictly positive density at zero, the expectation of the ratio does not exist; the simulated mean does not converge and moves substantially with the seed (values from roughly 3.5 to 47 were obtained across seeds at 200,000 draws, while the median and the percentile interval were stable to two decimals). The text now reports the median and interval, states that the mean does not exist, and uses the seed instability as the diagnostic. The associated failure mode (F6), exercise C3 and its solution were updated to match. This brings Chapter 1 into line with §2.11 and §24.2, which already stated the result correctly.

Chapter 15, Worked Example 15.1. The reported yes-proportions at the five bids were inconsistent with the stated data-generating process (a 0.98 acceptance rate at a €5 bid is roughly twelve standard errors from what a lognormal with median €30 and $\sigma_\ell = 0.6$ implies). The proportions are now those of an actual run of the stated design, and reproduce the €25.6 Turnbull lower bound quoted in the surrounding text. The fitted median and mean columns were independently re-verified and left unchanged.

Chapter 24, Worked Example 24.2. The first row of the tail table reported a 99th-percentile WTP of €18.7 at a mean cost coefficient of 0.06; the correct value under the stated parameters is approximately €23, and the value quoted in the 0.03 row (€138.7) was also inconsistent with the €144.0 reported for the identical configuration in Worked Example 24.1. The example has been rebuilt so that **median WTP is held fixed at roughly €20** while the cost coefficient weakens, which isolates the tail effect instead of confounding it with a shift in the centre of the distribution, and all figures were recomputed at four million draws. Dependent sentences elsewhere in the chapter were updated.

Chapter 47, Worked Example 47.5. The dependence example assumed 40 estimates from 18 studies, while Chapter 17 stipulated 120 estimates from the same 18 studies. The two are now consistent at **120 estimates**, and the design effect, standard-error inflation factor and effective sample size were recomputed accordingly (deff = 4.40, inflation 2.10×, roughly 27 effective independent estimates). The chapter opener and solution B3 were updated.

4. Conventions and claims made precise

The compensating-variation sign convention. Definition 3.5 defined CV with the opposite sign to Definition 34.3, and Worked Example 3.2 in fact computed it using Chapter 34's convention rather than the one stated seventy lines above it. The book now uses the expenditure-difference convention throughout ($\mathrm{CV} = e(\mathbf{p}^1, u^0) - m$, positive for a welfare loss, with $-\mathrm{CV}$ the maximum WTP for an improvement), and Definition 3.5 carries an explicit flag naming the competing convention — consistent with the book's rule of naming conventions as conventions.

The stock–flow identity. Section 4.11 asserted that in a well-managed steady state the liquidation value of a standing stock and the capitalized value of its sustainable rent coincide. This is not a theorem; the two coincide only under restrictive conditions, and the fisheries and forestry literatures are substantially about how they diverge. The section now states the accounting rule it actually needs — count the asset or the discounted flow, never both — and flags the equality in the worked example as true by construction.

The Ramsey table. The row previously labelled "Nordhaus-like" used a pure rate of time preference of 3.0%, which is higher than Nordhaus's own calibrations and inconsistent with the 4.5–6% range quoted in the surrounding text. The row is now labelled "descriptive / market-calibrated," which is what the parameters represent.

5. Material added

- **A systematic-review section (§47.1b).** The plan promised PRISMA-compliant search, a coding protocol and inter-coder reliability; the manuscript discussed none of them. Section 47.1b now covers

systematic search and grey literature, pre-specified inclusion criteria, the PRISMA flow, the coding protocol, double-coding and reliability statistics, adjudication, and the correct order of operations for unit, deflation and PPP harmonization.

- **The fourteen part assessments and the capstone.** Volume 0 promised an assessment at the end of every part — cumulative problem set, research critique, red-flag diagnosis, mock oral, integration mini-project — and a capstone re-analysis. These did not exist in the manuscript. All fourteen assessments and the capstone are now present, each tied to the chapters of its part and to the reader's own research program.
- **This appendix and the notation glossary (Appendix A),** the latter added because several symbols are unavoidably overloaded across the literatures the book draws on, and the collisions are worth naming rather than leaving for the reader to discover.

6. What was checked and left unchanged

The following were independently re-derived or recomputed during the editorial pass and found correct: the linear-algebra worked examples of Chapter 1 (inverse, eigenvalues, condition numbers, Cholesky factor, both illustrative draws, and the Bernoulli information); the logit score, Hessian and concavity proof and the marginal-effect and elasticity calculations of Chapter 2; the Cobb–Douglas duality and compensating-variation example of Chapter 3 and the MDCEV miniature that precedes it; every figure in Chapter 4 (annuity factor, NPV, the internal rate of return at 7.7547%, the real/nominal four-way table, the century discount factors and the 321-fold ratio, and the full switching-value table); the Gumbel-to-logistic derivation, the softmax probabilities and both variance decompositions of Chapter 5; the whole of Chapter 8's power, minimum-detectable-effect and TOST arithmetic; the boundary-corrected likelihood-ratio critical values of Chapter 12; the distributional comparison of Chapter 15 including the Turnbull lower bound; Chapter 24's Worked Example 24.1; the welfare orderings, Willig bounds, quasilinear collapse and Small–Rosen log-sum of Chapter 34; the MDCEV utility, Kuhn–Tucker conditions, multiplier formula, likelihood and its multinomial-logit special case in Chapter 40, together with Worked Example 40.1, which was reproduced from an independent implementation of the allocation algorithm; and the heterogeneity statistics, prediction interval and Egger diagnostics of Chapter 47.

The three-tier bibliographies were sampled extensively and the citations checked for existence and correct attribution, including author lists. No fabricated references were found. The "verify" flags on individual entries have been left in place: they mark edition, volume and page details that the reader should confirm before citing, not doubts about whether the work exists.

Capstone — Integrated Re-Analysis

After Part XIV. Allow two weeks. This is the exercise the whole book exists to make possible.

Choose **one** of your own papers — the contingent valuation paper is the natural choice, because it touches the most machinery, but the DCE or the FRD appraisal will work. Re-analyse it from the raw data to the reported conclusion, and produce a document with the following structure.

1. The claim inventory

List every empirical claim the paper makes. Not the results tables — the *claims*: the sentences a reader would quote. There are usually fewer than you expect, and stating them plainly is already diagnostic.

2. For each claim, the spine

For every claim in the inventory, write the four-part statement the book has been drilling since Chapter 1:

- **What economic object am I identifying?** Name it precisely. "The effect of the cost vector" is not precise; "the average difference in stated WTP between the high- and low-cost-vector arms, among respondents who passed the comprehension screen" is.
- **By what identifying variation?** Point to the specific source of variation in the design that pins it down. If you cannot name it, the claim cannot be defended.
- **Under which assumptions?** Every one of them, including the untestable ones, and especially the untestable ones.
- **How fragile is it?** With respect to specification, distribution, incentives, survey design, sampling, computation and interpretation — the seven fragilities of the spine question.

3. The re-analysis

Re-estimate everything, in a reproducible pipeline, with:

- the distributional robustness of Chapter 15 (normal, logistic, lognormal, spike; Turnbull floor);
- the space comparison of Chapter 24 (preference space and WTP space, with fit comparison);
- the scale-versus-preference diagnostics of Chapter 25;
- the identification analysis of Chapter 30 applied to each treatment contrast;
- the exclusion and weighting analysis of Chapter 32, reporting the estimand under each rule;
- the multiplicity and equivalence apparatus of Chapter 33 for every "no difference" claim;
- the welfare gate of Chapter 34 applied to every number that enters a CBA;
- the computational trustworthiness checklist of Chapter 52 for every estimate reported.

4. The divergence report

State honestly where the re-analysis differs from the published paper. Some claims will survive unchanged. Some will need weakening. At least one will probably need withdrawing or reframing. Write this section as though a referee will read it — because the discipline of writing it is the whole point, and the researcher who can write it is the one whose remaining claims are worth believing.

5. The fragility dossier

Produce the Chapter 50 dossier for the paper's single most important claim: the full assumption chain, the fragility rating of each link, the sensitivity analysis for the weakest link, and the conclusion restated as an honestly conditional sentence.

6. One page for your supervisor

Finally, write one page — no equations — explaining what you now know about your own paper that you did not know before, and what you will do differently in the next one.

End of the assessment apparatus. The book's argument is complete; what remains is the work.